Linguistic Phylogenetics

An Introduction and Comprehensive Guide to Phylogenetic Methods for Language Evolution

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Preface

Every language spoken today descends from earlier languages. This is as old an idea as historical linguistics itself. In 1863 August Schleicher drew Indo-European as a branching tree, a few years after Darwin's *Origin of Species* appeared. Darwin, for his part, used the genealogy of languages to make the descent of species intuitive. The two sciences of descent with modification grew up side by side. For most of the twentieth century they then drifted apart. Biologists built a statistical theory of trees, and linguists refined the comparative method largely by hand.

Over the last quarter century the two traditions have met again. Computational phylogenetic methods, built for DNA and morphology, are now used to infer the family trees of thousands of languages. They are used to date the origins of language families, to locate their homelands, and to reconstruct the vocabulary, grammar and culture of extinct speech communities. They are also used to test hypotheses about how and why languages change. The field is known as *linguistic phylogenetics*, or *computational historical linguistics* when it also covers automated cognate detection and reconstruction. It has produced some of the most widely discussed, and most fiercely contested, results in the study of human prehistory.

This book is a graduate textbook. It teaches how phylogenetic methods work, what they have been used to show about the history of languages, and how far their results can be trusted. It assumes no prior knowledge of phylogenetics. Each chapter starts from a linguistic question, builds the intuition behind the methods that address it, works through examples in R, and ends by asking what the methods can and cannot tell us. Chapters also point to methods from evolutionary biology that could be adapted to linguistic problems, and say how far each of these is established, plausible or still speculative.

The more technical material is collected in appendices. These cover derivations, full likelihood formulas, algorithmic detail and implementations written from scratch. Readers can follow the main argument of every chapter without them, and turn to them when they need the details.

Who this book is for

The primary readers are graduate students and researchers in linguistics who want to understand, evaluate or carry out phylogenetic analyses. A reader who knows what a cognate set is, but not what a Bayes factor or a covarion model is, is the reader the chapters are written for.

The book is also meant for three other groups. Biologists and statisticians can use it to learn what is distinctive about language data: cognate sets, borrowing, dialect continua, and written records that serve as "fossils". Anthropologists, archaeologists and geneticists who use language trees as a scaffold for their own questions can use it to learn what those trees can and cannot

support. Instructors can use it for a one- or two-semester graduate course in quantitative historical linguistics or quantitative typology.

Prerequisites and how to use the book

The chapters assume basic probability and some comfort with algebra. Chapters 5–7 and 9–13 also use likelihood, matrices, continuous-time Markov chains and Markov chain Monte Carlo. These ideas are taught where they are needed. Appendix A collects all of this background in one place, with a table showing which sections each chapter relies on. Readers without a quantitative background should read the relevant sections of that appendix alongside the chapters. Some familiarity with R helps with the examples but is not required to follow the argument.

Different readers can take different paths through the book.

- **A core course** might cover Chapters 1–6, 7, 9–10, 15 and 17. This path leads from data to dated trees and to a critical understanding of both.
- **A course in quantitative typology** might cover Chapters 1–3, 5–6, 12–14 and 15.
- **Readers interested mainly in prehistory** can read Chapters 1, 3, 8–11, 15 and 18, using the earlier chapters for reference.
- **Researchers** can go directly to the technical appendices, the decision guide in Section 15.13, and the RevBayes implementations in Appendix H.

How the book is organised

The book has seven parts.

1. **Foundations** (Chapters 1–3). This part covers the history of the idea of language trees, the vocabulary of trees and networks, and the kinds of linguistic data that phylogenetic methods use.
2. **Inference** (Chapters 4–6). This part covers distance methods, parsimony, maximum likelihood and Bayesian inference. It ends with a vocabulary for validating analyses that the rest of the book uses (Section 6.10).
3. **Models of language change** (Chapters 7–8). This part compares models of lexical evolution and treats borrowing, contact and network methods.
4. **Time** (Chapters 9–10). This part covers clock models, rates of lexical replacement and calibration. It also reviews what dated phylogenies have suggested about Indo-European, Austronesian, Bantu, Sino-Tibetan, Pama-Nyungan and other families.
5. **Space** (Chapter 11). This part covers discrete and continuous phylogeography and homeland inference.
6. **Traits** (Chapters 12–14). This part covers ancestral state reconstruction, phylogenetic comparative methods in typology, and the use of language trees to study the evolution of culture.
7. **Problems and prospects** (Chapters 15–18). This part covers critiques and a guide to choosing models, methods from evolutionary biology that linguistics could adopt, a complete workflow with simulated and real data, and a synthesis of what the field has found.

The appendices contain the mathematical preliminaries, four sets of technical notes that accompany the chapters, guides to software and data, the RevBayes implementations of the book's Bayesian models, and a glossary. A subject index closes the printed edition.

Features of each chapter

Every chapter opens with learning objectives and closes with a summary of key points, exercises and annotated further reading. The exercises come at three levels. Conceptual exercises check understanding. Computational exercises use the book's code and data. Research exercises ask readers to justify modelling choices for a realistic problem.

Six kinds of boxes appear throughout the text.

Learning objectives state what a reader should be able to do after working through a chapter.

Key points summarise the most important ideas of a chapter or section.

Case studies describe a published linguistic analysis in some detail.

Boxes headed "From the biologist's toolbox" describe methods from evolutionary biology that have seen little or no use on linguistic data, and say what they could offer.

Technical details boxes point to the appendix where a derivation, algorithm or implementation is given in full.

Caution boxes flag common mistakes and contested assumptions.

The book distinguishes carefully between the strength of different claims. A method *estimates* a quantity under a model. A result *suggests* or *is consistent with* a historical hypothesis. Only rarely does an analysis *demonstrate* one. Chapter 6 also separates several questions that are often run together: whether software computes what a model says, whether the model's parameters can be recovered from data simulated under it, whether inferences are robust when the model is wrong, whether the model is adequate for real data, and whether its parameters correspond to anything historical. The rest of the book uses this vocabulary.

Code, data and companion materials

Code is written in R. The examples use **ape**, **phangorn**, **phytools**, **geiger**, **caper**, **nlme** and **ggtree**, among other packages. Most examples run on small simulated datasets, so the book can be rebuilt offline. Chapter 17 also analyses a real dataset of Polynesian basic vocabulary. It runs the same data through MrBayes, BEAST 2, RevBayes, IQ-TREE and RAxML-NG and compares the results.

The book's source project contains three sets of companion materials:

- `software-runs/`, which holds the input files, commands, logs and diagnostics of every external program run for the book;
- `data/case-study/`, which holds the Polynesian dataset with its licence and citation;
- `revbayes/`, which holds RevBayes implementations of the book's Bayesian models, each checked against simulated data with known parameter values (Appendix H).

Mathematical notation is kept consistent across chapters.

Table 1: Notation used throughout the book.

Symbol	Meaning
τ	a tree topology
$\mathbf{t}$ or $\{t_i\}$	branch lengths (in time or expected changes)

Symbol	Meaning
$\mathbf{Q}$	instantaneous rate matrix of a continuous-time Markov chain
$\mathbf{P}(t) = e^{\mathbf{Q}t}$	transition probability matrix over time t
π	stationary (equilibrium) distribution
D	the data matrix (languages × characters)
θ	model parameters other than the tree
λ, μ	birth (speciation/split) and death (extinction) rates
r or ν	clock rate

Building the book

To build the HTML version, open the project in RStudio and run:

```
bookdown::render_book("index.Rmd", "bookdown::gitbook")
```

The PDF version (`bookdown::pdf_book`) also needs a LaTeX distribution, for example via `tinytex::install_tinytex()`.

Acknowledgements

This book builds on the work of a large and generous research community. Its datasets, software and arguments are cited throughout.

Chapter 1

Language Trees: History and Motivation

The idea that languages form family trees is more than a century and a half old, older in its explicit form than Darwin's tree of life. For most of that time the tree was drawn by expert judgement, and the one serious attempt to compute it, glottochronology, collapsed under its own assumptions. Since about 2000, methods borrowed from evolutionary biology have made it possible to infer language trees, and their dates, from large datasets with explicit statements of uncertainty. This chapter asks why the tree became the central metaphor of historical linguistics, why the first quantitative attempt failed, and what the second attempt does differently. The answers set up the questions that run through the whole book: what a tree of languages represents, what evidence can support it, and how far a statistical model of language change can be trusted.

After reading this chapter you should be able to:

- Outline the history of the family-tree model from Jones and Schleicher to the computational studies of the 2000s, and explain Darwin's use of the linguistic analogy.
- Contrast the tree model with Schmidt's wave model, and say what kind of linguistic evidence favours each.
- Explain why the comparative method subgroups by shared innovations, and relate this principle to Hennig's synapomorphies.
- Derive the glottochronological dating formula from its assumptions and identify which assumption failed empirically.
- Explain why agreement between a computational tree and a traditional classification is not independent validation of either.
- Evaluate where the analogy between biological and linguistic evolution holds and where it breaks down.
- List the questions phylogenetic methods can address, and the one they cannot (whether languages are related at all).

1.1 Before the tree: the discovery of language families

The observation that some languages resemble each other is ancient, but the systematic study of language relationships began in the late eighteenth and early nineteenth centuries. In a famous

lecture delivered in Calcutta in 1786, the British judge and orientalist Sir William Jones remarked that Sanskrit bore to Greek and Latin a stronger affinity, "both in the roots of verbs and in the forms of grammar", than could have been produced by accident, so that no philologer could examine the three without believing them to have sprung from some common source which perhaps no longer existed. He suggested that Gothic, Celtic and Old Persian might belong to the same family. Similar observations had been made before Jones, but his formulation was influential, and its key elements anticipate much of what followed: the comparison of grammar as well as vocabulary, the argument from the improbability of chance resemblance, and the inference of an unattested common ancestor.

In the decades after Jones, scholars such as Rasmus Rask, Jacob Grimm and Franz Bopp laid the foundations of comparative Indo-European grammar. Grimm's formulation of the consonant correspondences between Germanic and the other Indo-European languages (now known as Grimm's law) showed that the differences between related languages were not random but followed systematic patterns. Latin *pater*, *tres* and *cornu* correspond to English *father*, *three* and *horn*; in each case a Latin voiceless stop (*p*, *t*, *k*) corresponds to a Germanic fricative (*f*, *th*, *h*). By the middle of the nineteenth century the Indo-European family was firmly established, and by its end the main branches (Indo-Iranian, Greek, Italic, Celtic, Germanic, Baltic, Slavic, Armenian and Albanian) were recognized; Anatolian and Tocharian were added in the early twentieth century, when their texts were identified and deciphered (Fortson, 2010).

What these early comparativists did not yet have was an explicit model of how the branches related to one another, or of what kind of process produced them. The organizing metaphor of the family (mother and daughter languages, sister languages) was present from the start, but it was August Schleicher who turned it into a diagram.

1.2 Schleicher and the Stammbaum

August Schleicher (1821–1868) was a botanist by inclination and a linguist by profession, and he regarded languages as natural organisms that are born, grow, age and die. He began drawing branching diagrams of Indo-European in the early 1850s (Schleicher, 1853), and his *Compendium* of Indo-European comparative grammar presented a *Stammbaum* ("family tree") in which Proto-Indo-European split successively into a hierarchy of intermediate proto-languages and finally into the attested branches (Schleicher, 1861). Figure 1.1 shows a simplified version.

```
schleicher <- read.tree(text = paste0(
  "((Germanic,(Lithuanian,Slavic)Balto-Slavic)Slavo-Germanic,",
  "(((Greek,Albanian)Greek-Albanian,(Italic,Celtic)Italo-Celtic)",
  "Graeco-Italo-Celtic,(Iranian,Indic)Aryan)Aryo-Graeco-Italo-Celtic)",
  "Indo-European;"))
plot(schleicher, use.edge.length = FALSE, node.depth = 2,
     label.offset = 0.1, cex = 0.95, x.lim = c(-1.1, 5.4))
# each subgroup name sits on its stem branch, on the side away from the parent's
# vertical line (above for upper daughters, below for lower ones); the root name
# sits to the left of the root
lp <- get("last_plot.phylo", envir = .PlotPhyloEnv)
nn <- Ntip(schleicher) + seq_len(Nnode(schleicher))
up <- lp$yy[nn] >= lp$yy[schleicher$edge[match(nn, schleicher$edge[, 2]), 1]]
for (i in seq_along(nn)) {
```

```
  a <- if (i == 1) c(1.08, 0.5) else
    if (up[i]) c(1.03, -0.45) else c(1.03, 1.45)
  nodelabels(schleicher$node.label[i], nn[i], frame = "none", cex = 0.72,
             adj = a, col = "grey30")
}
```

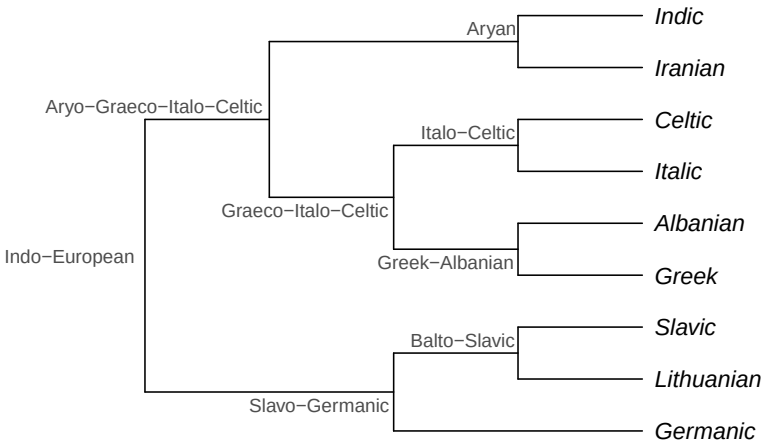


Figure 1.1: A simplified rendering of the Indo-European family tree published by August Schleicher in the early 1860s, with labels translated into English. Schleicher grouped Germanic with Balto-Slavic, and Greek (with Albanian) with Italic and Celtic, as sisters of Indo-Iranian (his "Aryan"). Branch lengths are schematic.

Several features of Schleicher's tree survive in modern phylogenetics. It is a rooted, bifurcating tree whose internal nodes are explicitly reconstructed ancestors. Its branch lengths were meant to suggest time: the earlier a lineage split off, the longer its branch. And it was accompanied by an explicit methodology of reconstruction. Schleicher was the first to reconstruct Proto-Indo-European forms systematically, marking them with an asterisk, and in 1868 he even composed a short fable in his reconstructed language. Other features have not survived. Schleicher's particular higher-level groupings, such as the union of Germanic with Balto-Slavic, are not part of the modern consensus, and the modern view of what subgroups require (shared innovations, discussed below) postdates him.

Schleicher read Darwin's *On the Origin of Species* (Darwin, 1859) in its German translation soon after it appeared, and in 1863 he published an open letter to the zoologist Ernst Haeckel, *Die Darwinsche Theorie und die Sprachwissenschaft* ("Darwinian theory and the science of language"), arguing that Darwin's theory of descent with modification was already familiar, in substance, to linguists, who had long observed the gradual divergence of languages from common ancestors (Schleicher, 1863). The letter is one of the first explicit statements of the parallel between biolog-

ical and linguistic evolution, and it included a tree diagram of Indo-European.

1.2.1 Darwin's linguistic analogy

The influence ran in both directions. Darwin used language repeatedly to make the idea of descent intuitive. In the chapter of the *Origin* on classification, he argued that a genealogical arrangement is the natural way to classify, and illustrated the point with languages: if we possessed a perfect pedigree of mankind, he wrote, a genealogical arrangement of human groups would afford the best classification of the languages spoken throughout the world, and if all extinct languages and all intermediate dialects were included, such an arrangement would be the only possible one (Darwin, 1859). The same chapter compares rudimentary organs to letters retained in the spelling of a word but no longer pronounced. In *The Descent of Man*, Darwin developed the analogy further, observing that "the formation of different languages and of distinct species, and the proofs that both have been developed through a gradual process, are curiously the same" (Darwin, 1871). The geologist Charles Lyell, in a book published in 1863, devoted a chapter to the comparison between the origin of languages and of species (Lyell, 1863). The intellectual history of these exchanges has been traced in detail by Alter (1999), and their relevance to modern phylogenetics by Atkinson and Gray (2005).

It is worth emphasizing that the language-species analogy was, in the nineteenth century, mostly an argument by illustration. Darwin used languages because his readers already accepted that languages had diverged gradually from common ancestors, and he could then ask them to extend the same reasoning to species. Neither Darwin nor Schleicher had a method for inferring trees from data, beyond expert judgement. That would come a century later, and would come from biology.

1.3 Trees versus waves

Schleicher's tree was challenged almost immediately. Johannes Schmidt, one of his students, argued in 1872 that the relationships among the Indo-European branches could not be represented as a tree at all (Schmidt, 1872). Schmidt observed that the branches shared innovations in overlapping, crisscrossing patterns. Slavic shares some features with Indo-Iranian, others with Germanic; Greek shares some with Indo-Iranian, others with Italic. No tree can accommodate all of these groupings as clades. Schmidt proposed instead that innovations spread outward from different centres within an originally continuous dialect area, like waves from stones dropped into a pond, each covering a different, partly overlapping territory. This is the *Wellentheorie* or *wave theory*. Its natural representation is not a tree but a map of overlapping *isoglosses*, the boundaries of the areas in which particular innovations are found (Figure 1.2). Hugo Schuchardt made similar arguments, and the rise of dialect geography at the end of the nineteenth century, which revealed that every feature in a dialect landscape has its own geographic distribution, seemed to confirm that languages do not split cleanly.

```
pos <- data.frame(
  lang = c("Celtic", "Italic", "Germanic", "Balto-Slavic", "Albanian",
           "Greek", "Armenian", "Indo-Iranian", "Tocharian", "Anatolian"),
  x = c(1, 1.5, 3, 5.5, 3.8, 3.5, 6, 8, 9.5, 6),
  y = c(6, 4, 7.5, 7, 4.2, 2.5, 3.2, 4.5, 7.5, 1.2))
# rounded outline around the language labels
```

```
blob <- function(langs, pad, lty) {
  p <- pos[pos$lang %in% langs, ]
  hw <- strwidth(p$lang, cex = 0.8, font = 2) / 2   # half-width of each label
  hh <- strheight(p$lang, cex = 0.8, font = 2) / 2
  ang <- seq(0, 2 * pi, length.out = 25)[-25]
  pts <- do.call(rbind, lapply(seq_len(nrow(p)), function(i) {
    cx <- p$x[i] + c(-1, 1, 1, -1) * hw[i]
    cy <- p$y[i] + c(-1, -1, 1, 1) * hh[i]
    do.call(rbind, lapply(1:4, function(k)
      cbind(cx[k] + pad * cos(ang), cy[k] + pad * sin(ang))))
  }))
  h <- chull(pts)
  polygon(pts[h, 1], pts[h, 2], lty = lty, lwd = 2, border = "grey30")
}
op <- par(mar = c(0, 0, 0, 0))
plot(pos$x, pos$y, type = "n", xlim = c(0, 10.5), ylim = c(0, 8.5),
     axes = FALSE, xlab = "", ylab = "", asp = 1)
blob(c("Balto-Slavic", "Albanian", "Armenian", "Indo-Iranian"), 0.5, 1)
blob(c("Germanic", "Balto-Slavic"), 0.25, 2)
blob(c("Greek", "Armenian", "Indo-Iranian"), 0.28, 3)
blob(c("Celtic", "Italic"), 0.3, 4)
blob("Tocharian", 0.3, 4)
blob("Anatolian", 0.3, 4)
text(pos$x, pos$y, pos$lang, cex = 0.8, font = 2)
legend("bottomleft", lty = 1:4, lwd = 2, col = "grey30", bty = "n",
       cex = 0.8, legend = c("satem", "dative/instrumental plural in *-m-",
                             "augment *e-", "mediopassive in *-r"))
```

```
par(op)
```

The tension between trees and waves has never been resolved, and it runs through the whole of this book. In one view, articulated by the Neogrammarians and their successors, the tree is the correct model of genealogy, but the evidence for it must be carefully filtered: only shared innovations count as evidence for subgroups, and features spread by contact must be identified and set aside (Campbell, 2013). In this view, overlapping isoglosses reflect contact after divergence, or the early stages of the breakup of a dialect continuum, which leave little trace in the final tree. In another view, the wave model is a better representation of how languages actually diversify, especially when a language spreads over an area and its dialects remain in contact as they diverge. Modern proponents of this view, such as Alexandre François and Siva Kalyan, argue that many families, including the Oceanic languages of Vanuatu and much of Polynesia, show patterns of intersecting innovations that no tree can represent, and they have developed quantitative methods ("Historical Glottometry") to measure the strength of overlapping subgroups (François, 2015; Kalyan and François, 2018). Leonard Bloomfield's influential textbook presented both models and the evidence for each (Bloomfield, 1933).

Phylogenetic methods take a pragmatic position. Tree-based methods assume a tree but can be checked for how badly the data violate that assumption. Network methods, discussed in Chapters 2 and 8, represent conflicting signals explicitly. And explicit models of borrowing and of dialect

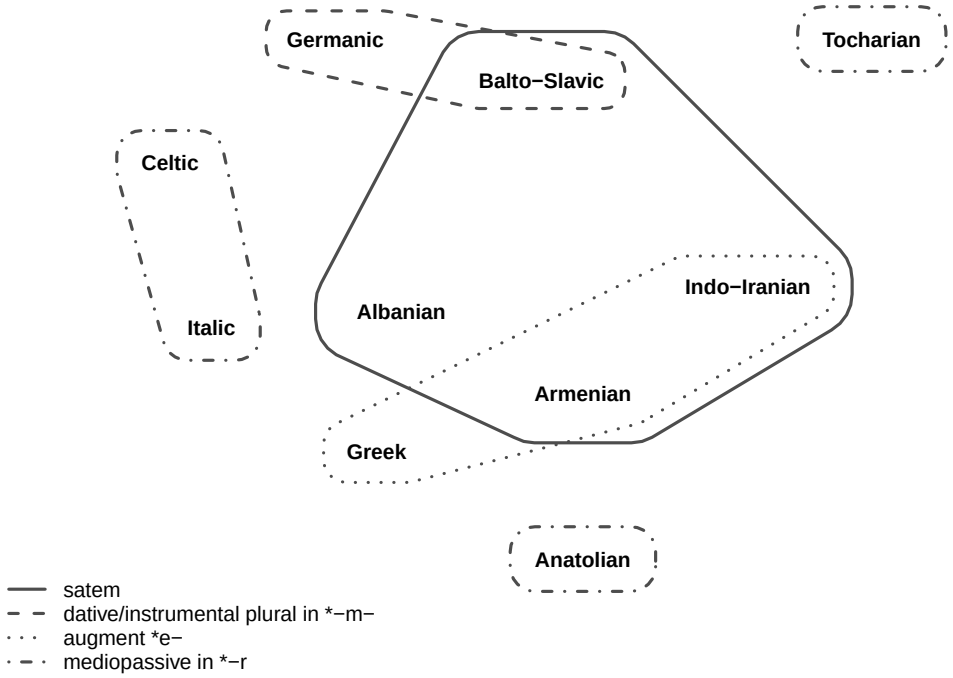


Figure 1.2: A schematic wave-theory diagram of four classic isoglosses among the branches of Indo-European, simplified from the kind of diagram found in handbooks. Satem: the assibilation of the Proto-Indo-European palatal velars. Dative/instrumental plural endings in *-m- rather than *-bh-. The past-tense augment *e-. A mediopassive marked by *-r, found in geographically peripheral branches. The placement of the branches is schematic, and the status of several of these features (especially satem in Albanian) is debated.

continua, still at an early stage, aim to combine the two pictures in a single statistical framework.

1.4 The comparative method and regular sound change

The decisive methodological advance of nineteenth-century linguistics was the *comparative method* and the principle on which it rests, the regularity of sound change. In 1878, Hermann Osthoff and Karl Brugmann, leaders of the group of young scholars in Leipzig known as the *Junggrammatiker* (Neogrammarians), declared that sound changes operate according to laws that admit no exceptions, except where other identifiable factors, such as analogy or borrowing, intervene (Osthoff and Brugmann, 1878). A sound change, on this view, affects every word in which the relevant sound occurs in the relevant environment, at a particular time in a particular speech community. This *regularity hypothesis* was controversial (and is still qualified in various ways), but it made possible a rigorous method for establishing relationships and reconstructing proto-languages (Hoenigswald, 1960; Fox, 1995).

The comparative method works as follows. First, assemble sets of words with similar meanings in the languages being compared. Second, identify recurrent *sound correspondences*: for example, that Latin *p* corresponds to Germanic *f* in word after word (*pater/father*, *piscis/fish*, *pes, pedis/foot*). Third, explain each set of correspondences by reconstructing a proto-sound and a set of regular changes in each daughter language. Fourth, use the reconstructed sounds to reconstruct proto-words, and use the regular changes to distinguish genuine cognates (which obey the correspondences) from chance look-alikes and loanwords (which do not). Fifth, establish subgroups by finding *shared innovations*: changes that a set of languages underwent together and that therefore point to a period of common development. The Germanic consonant shift is such an innovation, shared by all the Germanic languages and by no other branch, and it is part of the evidence that Germanic is a subgroup.

The emphasis on shared innovations has a precise counterpart in biology. In the 1950s and 1960s, the German entomologist Willi Hennig argued that biological classification should be based on genealogy and that only shared derived characters (synapomorphies) are evidence of common ancestry, whereas shared ancestral characters (symplesiomorphies) are not (Hennig, 1966). Hennig's *phylogenetic systematics*, or *cladistics*, became the dominant philosophy of biological classification. The two principles were developed independently, but they express the same logic. A feature inherited from the ancestor of the whole family tells us nothing about subgrouping within it, because it could have been retained in any lineage; only a feature that arose within the family, in the common ancestor of a subset of its languages, identifies that subset as a clade. Brugmann himself stated the principle for linguistic subgrouping in the 1880s (Brugmann, 1884).

The comparative method has been extraordinarily successful. It has established dozens of language families, from Uralic and Austronesian to Bantu and Uto-Aztecan, and reconstructed large parts of their proto-languages. It provides the cognate judgements on which almost all lexical phylogenetic analyses depend (Chapter 3). But it has limits. It is laborious and depends on the expertise of specialists. Its results for the higher-level structure of large families are often uncertain or contested, because the innovations defining deep subgroups have been obscured by later change. It offers no principled way of dating proto-languages. And it gives no explicit measure of uncertainty: two experts examining the same data can reach different subgroupings without any formal way to compare the strength of the evidence for each. These limits are the openings through which quantitative methods entered the field.

The parallel between linguistics and biology has a third partner. In the nineteenth century, the philologist Karl Lachmann and others developed methods for reconstructing the history of manuscript copies of a text, a discipline known as *stemmatics*, by tracing shared copying errors back to lost common exemplars. Shared errors in manuscripts play the same role as shared innovations in languages and synapomorphies in organisms. Phylogenetic software designed for biology has since been applied to manuscript traditions, for instance to the manuscripts of Chaucer's *Canterbury Tales* (Barbrook et al., 1998).

1.5 Lexicostatistics and glottochronology

1.5.1 Swadesh's program

The first systematic attempt to quantify language relationships and to date their divergence was made by the American linguist Morris Swadesh around 1950. His *lexicostatistics* compared languages by the proportion of cognates they shared on a fixed list of basic meanings, the *Swadesh list* (Chapter 3), which was designed to contain vocabulary that every language has and that is rarely borrowed (Swadesh, 1950, 1952, 1955). Pairs of languages sharing a high proportion of cognates were taken to be closely related, and a classification could be built from the matrix of pairwise percentages. This is, in modern terms, a distance method (Chapter 4).

Glottochronology went further, proposing that the percentage of shared cognates could be converted into a date of divergence. The key assumption was that basic vocabulary is replaced at a constant rate. Robert Lees estimated this rate from a small sample of languages with long written histories, comparing older and modern stages such as Old and Modern English or Latin and its Romance descendants, and found that about 81 per cent of the 200-item list was retained per thousand years (Lees, 1953); Swadesh later estimated a retention rate of about 86 per cent per millennium for the 100-item list (Swadesh, 1955).

The resulting dating formula is simple. Suppose that each meaning in a language retains its word over a period of t millennia with probability r^t, where r is the retention rate per millennium, and that two languages diverged t millennia ago and have since replaced vocabulary independently. A meaning will still be expressed by cognate words in both languages if neither has replaced it, which has probability $r^t \times r^t = r^{2t}$. The expected proportion of shared cognates is therefore $c = r^{2t}$, and solving for t gives

$$t = \frac{\ln c}{2 \ln r}. \tag{1.1}$$

For example, two languages sharing 70 per cent of cognates on the 200-item list would, with $r = 0.805$, be estimated to have diverged $\ln 0.7 / (2 \ln 0.805) \approx 0.82$ millennia ago, around 820 years. Figure 1.3 shows the relationship between shared cognates and estimated time depth for the two standard retention rates.

```
glotto_t <- function(c, r) log(c) / (2 * log(r))
c_shared <- seq(0.10, 0.95, by = 0.01)
plot(c_shared, glotto_t(c_shared, 0.805), type = "n", las = 1,
     xlab = "Proportion of shared cognates",
     ylab = "Estimated divergence time (millennia)",
     ylim = c(0, 8.5))
```

```
polygon(c(c_shared, rev(c_shared)),
        c(glotto_t(c_shared, 0.75), rev(glotto_t(c_shared, 0.85))),
        col = "grey88", border = NA)
lines(c_shared, glotto_t(c_shared, 0.805), lwd = 2)
lines(c_shared, glotto_t(c_shared, 0.86), lwd = 2, lty = 2)
legend("topright", c("r = 0.805 (200-item list)", "r = 0.86 (100-item list)",
                     "r between 0.75 and 0.85"), lwd = c(2, 2, 10),
       lty = c(1, 2, 1), col = c("black", "black", "grey88"), bty = "n")
```

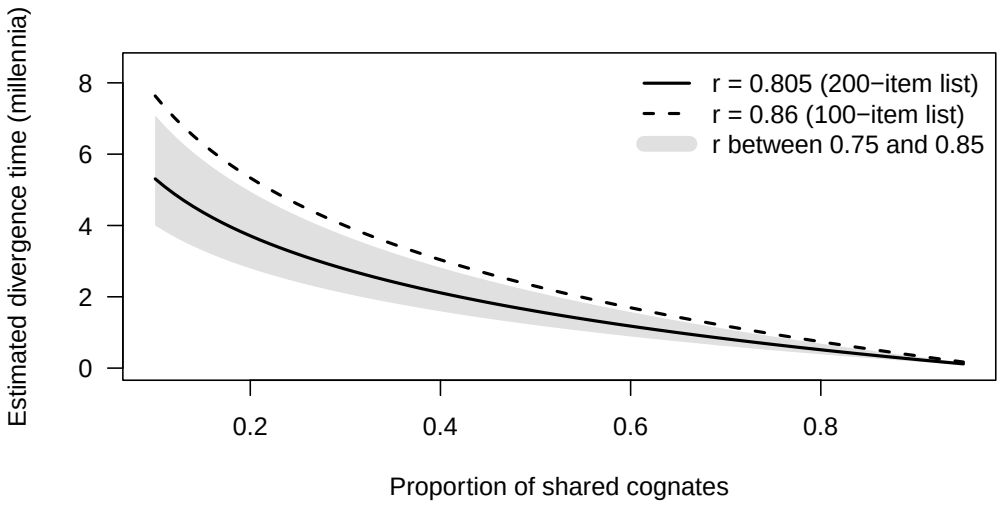


Figure 1.3: Glottochronological divergence time as a function of the proportion of shared cognates, for retention rates of 0.805 (200-item list) and 0.86 (100-item list) per millennium. The shaded band shows the range of dates implied by the 200-item formula if the true retention rate is anywhere between 0.75 and 0.85 per millennium, illustrating the sensitivity of the method to its central assumption.

```
round(glotto_t(c(0.7, 0.5, 0.3), 0.805), 2)
```

```
#> [1] 0.82 1.60 2.78
```

1.5.2 The critique

Glottochronology was taken up enthusiastically in the 1950s, and lexicostatistical classifications and dates were produced for many families. Within little more than a decade, however, it had been largely rejected by mainstream historical linguists. The critique, summarized in reviews such as Hymes (1960) and McMahon and McMahon (2005), had several strands.

The most damaging was empirical. Bergsland and Vogt (1962) tested the constant-rate assumption on languages with documented histories and found large departures from it. Among their examples, Icelandic had retained far more of its Old Norse basic vocabulary than the standard rate predicts, and other cases also departed from the supposed constant; applied naively, the for-

mula would grossly misdate such divergences. Later analyses of Austronesian found that retention rates in basic vocabulary vary very widely, even among closely related languages (Blust, 2000). If rates vary this much, a single universal retention constant cannot exist.

Other criticisms concerned the model. The formula treats all meanings as having the same retention rate, when some (such as pronouns and low numerals) are far more stable than others. It assumes that the two languages diverged at a single moment and then evolved independently, with no borrowing between them. It ignores the possibility that the same meaning might be replaced more than once, or that two languages might independently acquire cognate words for a meaning (for instance, by parallel semantic shift). Its mathematical properties were criticized too: Chrétien (1962) argued that the published derivations were flawed. And because each pair of languages was dated separately, the dates for the various pairs in a family were not necessarily consistent with any single tree.

Further problems concerned the data: the difficulty of defining the meanings on the list unambiguously, the subjectivity of cognate judgements, and the undetected borrowing of basic vocabulary. Lexicostatistics as a classification method survived in some quarters, notably in large-scale classifications of language families (Dyen et al., 1992) and in work on statistical methods in historical linguistics (Embleton, 1986), and a recalibrated form of glottochronology continued to be used by some scholars in the Moscow school of comparative linguistics. But by the 1970s most historical linguists regarded glottochronology as discredited, and quantitative dating of language families as a dead end. McMahon and McMahon (2005) later argued that this verdict ran together two different things. Lexicostatistics, the comparison of cognate proportions on standard meaning lists, makes fewer and weaker assumptions than glottochronology, which adds a universal rate in order to turn those proportions into dates; rejecting both for the failings of the second, in their phrase, risks “throwing out the meaning-list baby with the time-depth bathwater”.

Testing a constant: Bergsland and Vogt (1962). Knut Bergsland and Hans Vogt took glottochronology at its word. If basic vocabulary is lost at a universal constant rate, then languages with long written histories should show that rate when their older and modern forms are compared. They examined several such cases, published their results with commentary from other scholars in *Current Anthropology*, and found that the observed retention rates varied far more than the theory allowed (Bergsland and Vogt, 1962). In a frequently cited example, Modern Icelandic had replaced only a small fraction of the basic vocabulary of Old Norse, so that the glottochronological formula would imply far less elapsed time than the documented history shows. The study is a model of empirical testing of a quantitative method against known history, a strategy that remains essential for modern methods (Chapters 10 and 15). Its lesson is not that languages cannot be dated, but that a method assuming a single rate for all words and all lineages is untenable. The response of modern phylogenetics has been to replace the constant with explicit models of rate variation among meanings and among lineages (Chapters 7 and 9), and to infer rates from calibrations within the family being studied rather than importing a universal constant.

1.6 The computational turn

1.6.1 Phylogenetics grows up in biology

While linguists were abandoning quantitative methods, biologists were building them. Numerical taxonomy in the late 1950s and 1960s introduced distance-based clustering methods such as UPGMA (Sokal and Sneath, 1963). Hennig’s cladistics led to parsimony methods, which seek the

tree requiring the fewest evolutionary changes. Joseph Felsenstein's 1981 paper made maximum likelihood estimation of phylogenies from DNA sequences practical (Felsenstein, 1981), providing a statistical framework in which trees, branch lengths and models of character change could be estimated together and compared. In the late 1990s and early 2000s Bayesian methods, implemented in programs such as MrBayes (Huelsenbeck and Ronquist, 2001) and BEAST (Drummond and Rambaut, 2007; Bouckaert et al., 2014), made it possible to quantify uncertainty about trees directly, to estimate dates with relaxed molecular clocks calibrated by fossils, and to integrate over uncertainty in all the parameters of a model. By 2000 phylogenetics had become one of the most sophisticated areas of statistical biology, with a large body of theory, mature software and an extensive literature on model choice, model adequacy and the effects of model violation (Felsenstein, 2004).

1.6.2 Language data meet phylogenetic methods

The application of these methods to languages began in earnest around the turn of the millennium, though there were precursors. Mathematical and statistical approaches to language classification had continued at the margins of linguistics, and Warnow (1997) had outlined how methods from computational biology could be adapted to linguistic data. Mark Pagel proposed the use of maximum likelihood models of lexical replacement for linguistic phylogenies and for dating (Pagel, 2000). Within a few years, several groups had published phylogenetic analyses of major language families.

Gray and Jordan (2000) applied maximum parsimony to lexical data from 77 Austronesian languages and tested the fit of the resulting trees to the "express train" model, according to which Austronesian speakers expanded rapidly from Taiwan through Island Southeast Asia into the Pacific. Holden (2002) used parsimony on basic vocabulary of 75 Bantu languages and found a tree consistent with the spread of Bantu languages alongside farming. Ringe et al. (2002) developed a character-based method for Indo-European that searched for trees on which as many carefully selected lexical, phonological and morphological characters as possible were compatible. Rexová et al. (2003) applied cladistic analysis to the Dyen et al. lexical data for Indo-European, and Forster and Toth (2003) used network and dating methods on Celtic and Indo-European lexical data.

The paper that brought the field to wide attention was Gray and Atkinson (2003). Its methods and results, and the controversy they provoked, set the agenda for much of what followed.

Gray and Atkinson (2003): dating Indo-European. Russell Gray and Quentin Atkinson took the Dyen et al. lexical database (Dyen et al., 1992), added data for Hittite and the two Tocharian languages, and coded the cognate judgements as a binary matrix of 2,449 cognate sets across 87 languages. They inferred a posterior distribution of trees with Bayesian MCMC under a model of binary character evolution with rate variation across characters, and then dated the trees by penalized-likelihood rate smoothing, a method that allows rates to vary across lineages, using calibrations from historically known splits and dates of attestation. The estimated age of the root of the Indo-European family fell between about 7,800 and 9,800 years before present, an age consistent with the hypothesis that Indo-European spread with farming from Anatolia (Renfrew, 1987) and older than expected under the competing hypothesis of an origin on the Pontic-Caspian steppe in the fifth or fourth millennium BCE (Anthony, 2007). They tested the robustness of the result to alternative assumptions about rates, calibrations and the root. The paper was important not so much for its date, which remains contested (Chapters 10 and 15), as for showing how the main objections to glottochronology could, in principle, be addressed. The constant rate was replaced

by models that allow rates to vary across meanings and lineages; dates were estimated from calibrations within the family rather than a universal constant; the problem of pairwise inconsistency vanished because dates were estimated on trees; uncertainty in the tree was integrated over; and borrowing was dealt with by excluding known loans and by tests of the sensitivity of the results.

1.6.3 A rapid expansion

The decade after 2003 saw a rapid expansion in the range of families, questions and methods. Dated Bayesian phylogenies were published for Austronesian (Gray et al., 2009), Semitic (Kitchen et al., 2009), Bantu (Grollemund et al., 2015), Pama-Nyungan (Bowern and Atkinson, 2012) and many other families, a story told family by family in Chapter 10. New models were developed specifically for language data, most notably the stochastic Dollo model of Nicholls and Gray (2008), which treats cognate classes as born once and then lost (Chapter 7). Phylogeographic models were used to infer the homeland and spread of Indo-European (Bouckaert et al., 2012) and other families (Chapter 11). Rates of lexical change were related to word frequency (Pagel et al., 2007), and evidence was reported that languages change faster at the moments when they split (Atkinson et al., 2008) (Chapter 9). Trees of languages were used to test hypotheses about the co-evolution of grammatical features (Dunn et al., 2011) (Chapter 13), and as a framework for the study of cultural evolution, a program inaugurated by Mace and Pagel (1994) (Chapter 14). Simulations explored how much borrowing tree-building methods can tolerate (Greenhill et al., 2009), and network methods were used to visualize conflicting signals (Bryant et al., 2005; Gray et al., 2010) (Chapter 8). Probabilistic models of sound change made it possible to reconstruct ancestral word forms automatically (Bouchard-Côté et al., 2013) (Chapter 12). And the development of large standardized databases and automated cognate detection methods (Chapter 3) made it possible to infer trees at a global scale (Jäger, 2018).

The debate over Indo-European origins continued throughout this period. Chang et al. (2015) showed that constraining certain ancient languages to be ancestral to modern ones (Latin to the Romance languages, for instance) produced substantially younger root ages for Indo-European, consistent with the steppe hypothesis, which was also gaining support from ancient DNA evidence for large-scale population movements from the steppe into Europe in the third millennium BCE (Haak et al., 2015). Heggarty et al. (2023), using a new, carefully curated dataset and a model in which ancient languages can be sampled ancestors, estimated an origin around 8,000 years ago and proposed a hybrid scenario. These studies, and their critics (Pereltsvaig and Lewis, 2015; Kassian and Starostin, 2025), are discussed in Chapters 10 and 15.

Since about 2020 the field has turned increasingly to examining its own foundations. New dated trees continue to appear, for Kra-Dai (Tao et al., 2023), Japonic (Huisman et al., 2025) and the Philippine languages (King et al., 2024) among others, but much recent work tests the standard models rather than applying them. Simulation-based calibration and posterior predictive checks have been applied to the standard Bayesian set-up for cognate data (King, 2026b); a doctoral dissertation has argued that widely used models of vocabulary evolution overfit linguistic data and that dating methods are not robust on realistically simulated data (Rönchen, 2025); and fully automated alternatives to expert cognate judgements are being evaluated at a global scale (Jäger, 2025). These developments are discussed in Chapters 7, 10 and 15.

Several reviews chart the development of the field from different perspectives (Nichols and Warnow, 2008; Bowern, 2018; Greenhill et al., 2020; Pagel, 2009; Dunn, 2015; Pagel, 2017). Greenhill (2023) groups the applications into two programmes: using language phylogenies to

test hypotheses about human prehistory, and using them as a backbone for studying how aspects of language and culture evolve, by revealing their evolutionary dynamics, modelling the history of particular traits and testing whether traits co-evolve.

1.7 Biology and language: parallels and limits

1.7.1 The parallels

The application of biological methods to language rests on an analogy between the two systems of descent with modification. Atkinson and Gray (2005) set out the parallels systematically, and Table 1.1 extends their comparison. The core of the analogy is that both genes and words are discrete heritable units that are copied (by DNA replication or by language learning) with occasional modification, that lineages split and then evolve independently, and that the resulting pattern of shared and unshared units can be used to reconstruct the history of splitting.

Table 1.1: Parallels between biological and linguistic evolution, after and extending Atkinson and Gray (2005).

Biological evolution	Language evolution
Discrete heritable units: genes, nucleotides, morphological traits	Words, cognate sets, sounds, grammatical features
Homology (similarity due to common descent)	Cognacy
Orthologs (homologs related by speciation)	Cognates in the same meaning slot
Paralogs (homologs related by duplication)	Doublets within one language, e.g. English *shirt* and *skirt*
Mutation	Innovation: sound change, semantic shift, coinage
Natural selection	Social selection, prestige, frequency effects, learnability
Genetic drift	Random sociolinguistic variation and change
Horizontal gene transfer, introgression, hybridization	Borrowing, contact-induced change, language mixing
Species	Language (or dialect)
Speciation	Language split (divergence of dialects into languages)
Allopatric speciation	Divergence after migration or isolation
Population (gene pool)	Speech community
Extinction	Language death (often by shift to another language)
Fossils	Ancient texts and inscriptions
Molecular clock	Glottochronology; relaxed clocks for lexical data
Vestigial organs	Irregular forms and silent letters, e.g. the *b* of *lamb*
Convergent evolution	Chance resemblance, parallel change, areal convergence
Incomplete lineage sorting	Ancestral synonyms or variants retained differently in daughters

Some of these parallels are close and useful. Cognacy is a very good analogue of homology: both

are hypotheses of common descent, both are established by systematic correspondences rather than overall similarity, and both have their analogues of orthology and paralogy. English *shirt* and *skirt* are both descended from the same Proto-Germanic word, *skirt* via a borrowing from Old Norse, which makes them something like paralogs within English. Ancient texts are useful analogues of fossils: Hittite, Mycenaean Greek, Vedic Sanskrit and Old Chinese provide dated observations of ancestral or near-ancestral states that can be used to calibrate trees (Chapter 10). The analogy needs care, though. An attested ancient language is a written record of one variety at one time, and it may be the actual ancestor of later languages, a sister of their ancestor, or one member of a dialect network whose other members went unrecorded. Classical Latin, for instance, is not literally the ancestor of the Romance languages, which descend from spoken varieties that coexisted with it. Which of these situations holds, and whether limited data can distinguish them, affects which varieties to sample, how to code them and how to interpret the tree (Section 3.2 and Chapter 10). And borrowing is a good analogue of horizontal gene transfer, with similar consequences for tree inference.

Other parallels are looser. The social selection of variants, through prestige, identity or ease of learning, is not much like natural selection, and it is debated whether a theory of language change should be cast in explicitly evolutionary terms at all; William Croft's evolutionary theory of language change is one influential attempt (Croft, 2000). Mutations are random with respect to their effects, whereas linguistic innovations may be motivated by phonetic, cognitive or social factors, and a single sound change affects every word in which the relevant sound occurs, a kind of coordinated change across thousands of "sites" that has no close molecular counterpart.

Gene trees inside species trees. In biology, different genes in the same organisms can have different histories, because ancestral populations are polymorphic: two variants of a gene may coexist in an ancestral species and be sorted differently among its descendants, so that the gene tree disagrees with the species tree even without any hybridization. The *multispecies coalescent* models this process of *incomplete lineage sorting* explicitly and estimates species trees from many gene trees. The linguistic analogue is real. Languages often have several words for a meaning, with different stylistic or regional values, and the daughter languages of a polymorphic ancestor may retain different variants. The Latin words for 'eat', for example, included both inherited *edere* and the expressive *manducare* and *comedere*, and the Romance languages selected among them differently: French *manger* continues *manducare* (as does Italian *mangiare*, probably via an early borrowing from Old French), and Spanish *comer* continues *comedere*. Each meaning of the lexicon therefore has its own "gene tree", which can disagree with the language tree because of ancestral synonymy, not only because of borrowing. Treating meanings as loci and applying multispecies-coalescent-style models to lexical data is a promising direction that has, to our knowledge, rarely been explored (Chapter 16).

1.7.2 Where the analogy breaks

The analogy also has important limits, which every user of phylogenetic methods on language data should keep in mind.

Borrowing is pervasive and structured. Horizontal transfer is common in bacteria but comparatively rare in animals and plants, which are the groups for which most phylogenetic methods were developed. In language, borrowing is ubiquitous, and its intensity varies enormously with the social circumstances of contact. Basic vocabulary is more resistant than cultural vocabulary, but it is not immune (Chapter 3). Undetected borrowing between closely related languages, which

share sound systems and so give few clues to the borrowing, is a particularly insidious source of error (Chapter 8).

Languages can have more than one parent. Mixed languages such as Michif, which combines Cree verbs with French nouns, cannot be assigned to a single family in any straightforward way (Bakker, 1997). Creoles and other contact languages raise similar difficulties (Thomason and Kaufman, 1988). Such cases are rare, but they show that the tree model is not universally applicable.

Taxa are not natural units. Biological species are themselves notoriously difficult to define, but languages are arguably harder. The boundary between a language and a dialect is partly political, and dialect continua, in which neighbouring varieties are mutually intelligible but the extremes are not, are common. How many languages are sampled from a continuum, and which ones, affects the inferred tree.

Language trees are not population trees. Languages can be adopted by populations that did not previously speak them, through language shift, and a language can spread with little movement of people, or people can move without changing their language. The history of a language family therefore need not match the genetic history of the people who speak its languages, and inferences from language trees to population history require separate justification (Diamond and Bellwood, 2003).

Characters are not independent and are coded by people. Cognate judgements are hypotheses made by linguists, not observations; they depend on the coder, the sources and the conventions (Chapter 3). A single replacement in a meaning changes several binary characters at once, and a single sound change affects many words.

Time depth is limited. Lexical replacement is fast enough that, beyond a time depth of perhaps eight to ten thousand years, the signal of common descent in basic vocabulary becomes very hard to distinguish from chance (Ringe, 1992; Kessler, 2001). Phylogenetic methods cannot overcome this limit. They can only use the available signal efficiently and report honestly how much uncertainty remains.

Phylogenetic methods assume relatedness; they do not demonstrate it. Given any set of languages, a tree-building program will return a tree, whether the languages are related or not. Questions of whether a set of languages forms a family, as in proposals for "macro-families" such as Nostratic or Altaic, cannot be settled simply by running a phylogenetic analysis and inspecting the tree, because a tree of unrelated languages can look superficially like a tree of related ones. Tests of relatedness require comparison with an explicit null model of chance resemblance (Ringe, 1992; Kessler, 2001). The same objection applies with greater force to *multilateral* or *mass comparison*, the practice, associated above all with Joseph Greenberg's classification of the languages of the Americas (Greenberg, 1987), of inspecting lexical resemblances across many languages at once. Because it collects resemblances without establishing whether they reflect inheritance, borrowing, chance, sound symbolism or nursery forms, its results have been rejected by most historical linguists (Campbell, 2013), and McMahon and McMahon (2005) doubted that any computer program could reproduce them, precisely because the criteria are not explicit.

1.8 What phylogenetic methods can do

With these caveats in mind, phylogenetic methods offer answers, or at least principled estimates with quantified uncertainty, to a wide range of questions in linguistics and neighbouring fields.

1. **Subgrouping.** What is the branching structure of a language family? Which subgroups are well supported, and which are not? Phylogenetic methods complement the comparative method by weighing large amounts of evidence consistently and by quantifying support (Chapters 4 to 6).
2. **Dating.** When did a proto-language begin to diverge, and how long did intermediate stages last? Relaxed-clock models calibrated by historical information yield dates with credible intervals (Chapters 9 and 10).
3. **Homelands and routes.** Where was a proto-language spoken, and by what routes did its descendants spread? Phylogeographic models answer these questions in a probabilistic framework (Chapter 11).
4. **Tempo and mode.** How fast do different parts of language change, and why? Are rates constant, related to word frequency, population size or contact? Does change concentrate at splits (Chapter 9)?
5. **Ancestral states.** What were the vocabulary, sounds and grammar of proto-languages? Phylogenetic methods complement the comparative method by estimating the probabilities of ancestral states and propagating uncertainty about the tree (Chapter 12).
6. **Correlated evolution and universals.** Do grammatical features evolve together, as typological universals predict, and are the dependencies the same in all families? Phylogenetic comparative methods correct for the non-independence of related languages, a problem known as Galton's problem (Chapter 13).
7. **Cultural evolution.** Language trees provide a scaffold for studying the history of non-linguistic traits, from kinship and political organization to religion and folktales (Chapter 14).
8. **Contact.** How much have languages borrowed, from whom, and when? Network methods and explicit models of lateral transfer address these questions (Chapter 8).

What these methods cannot do is replace the expertise on which their data rest. Every lexical phylogeny depends on cognate judgements that embody the comparative method, and every calibration depends on historical and philological knowledge. The most productive work in the field has been done by collaborations between historical linguists who understand the data and quantitative researchers who understand the models.

The express train to Polynesia. The first major phylogenetic test of a specific prehistoric scenario using language data was the analysis of 77 Austronesian languages by Gray and Jordan (2000). The "express train" model, derived from archaeology and linguistics, holds that Austronesian-speaking farmers spread from Taiwan through the Philippines and Indonesia and then rapidly out into the Pacific. The model predicts a particular sequence of splits in the language tree, with the Formosan languages of Taiwan branching off first, followed by successive branches in a geographical progression from west to east. Gray and Jordan built maximum parsimony trees from 5,185 lexical items and showed that the trees fitted the predicted sequence far better than randomly generated trees did. The study was methodologically simple by later standards: it used parsimony, a single most parsimonious tree and a test based on the number of extra steps needed to fit the predicted sequence. The test asked whether the tree was consistent with one predicted sequence, not whether that sequence was the only or the best explanation of the data, and any agreement between such a tree and known Austronesian subgroups is, for the reasons given in Section 1.9, only weak support for the method itself. But the study established the use of language phylogenies for testing hypotheses about human prehistory, and it was followed a decade later by the much larger Bayesian analysis of Gray et al. (2009), which dated the expansion and found evidence of pauses and pulses in the settlement of the Pacific (Chapter 10).

Bantu and the spread of farming. Holden (2002) analysed 92 items of basic vocabulary from 75 Bantu languages with maximum parsimony and found a tree whose structure was consistent with the archaeological evidence for the spread of farming across sub-Saharan Africa, with successive splits tracking a southward and eastward expansion. Like the Austronesian analysis, it was an early example of the use of language phylogenies to address questions about population expansion, and it was soon followed by phylogenetic studies of cultural traits on Bantu trees, such as the analysis of matrilineal descent and cattle-keeping by Holden and Mace (2003). Later work, with larger samples and Bayesian methods, revisited the routes taken by the Bantu expansion around or through the central African rainforest (Chapter 11).

1.9 Phylogenetics and the comparative method

Phylogenetic methods are sometimes presented, by enthusiasts and critics alike, as rivals to the comparative method. It is more accurate to see them as tools that operate on different parts of the same problem, with different strengths.

The comparative method does three things. It establishes that languages are related, by demonstrating regular sound correspondences across a body of vocabulary and morphology that cannot be due to chance or borrowing. It reconstructs the proto-language. And it establishes subgroups by identifying shared innovations. Phylogenetic methods, as normally used, do none of the first two: they take relatedness for granted and take cognate judgements, the main product of the comparative method, as their input. What they offer is a systematic, explicit and quantitative treatment of the third task, together with tasks the comparative method does not attempt, such as dating and the estimation of rates.

On the third task, the two approaches differ in what evidence they weigh most heavily. Traditional subgrouping relies on a relatively small number of carefully chosen innovations, preferably ones that are unusual, unlikely to recur independently and unlikely to spread by contact: an idiosyncratic sound change, a shared morphological irregularity, a peculiar semantic shift. Lexical replacement in basic vocabulary is usually regarded as among the weakest kinds of evidence, because the same replacement can happen independently in different lineages and because words can be borrowed. Lexical phylogenetics, by contrast, uses hundreds or thousands of individually weak characters and relies on a statistical model to extract the signal they share. It is sometimes objected that such analyses ignore the distinction between innovations and retentions. This is not quite right. In a model-based analysis, a character in which several languages share the ancestral state is consistent with many trees and contributes little to grouping them; a character in which they share a derived state supports their being a clade. The distinction between innovation and retention is built into the probabilistic calculation, which integrates over the possible ancestral states, although its success depends on the model and the rooting being adequate. Even then, the model only *estimates* which shared cognates are innovations. Sharing a cognate class is not the same thing as sharing an innovation: a class present in a set of languages may continue a word that was already there in an ancestor much older than the subgroup in question, so its distribution is compatible with several subgroupings, and a single lexical replacement is a much weaker innovation than an irregular sound change or a shared morphological anomaly.

Each approach also has distinctive weaknesses. The comparative method has no formal measure of the strength of evidence, so disagreements between specialists can be hard to adjudicate, and it is vulnerable to the selective use of evidence that supports a favoured hypothesis. Phylogenetic methods are explicit and reproducible, but they can be misled by systematic errors in the data

(undetected loans, inconsistent cognate coding), by misspecified models, and by the treatment of weak characters as if they were independent. The most interesting results in the field are those in which the two kinds of evidence disagree for reasons that can be identified. Agreement between them is welcome, but, as the next subsection explains, it is weaker evidence than it looks.

1.9.1 Agreement with traditional subgroups is not independent validation

Computational studies routinely report that their trees "recover" the subgroups established by the comparative method: Germanic, Slavic and Indo-Iranian in Indo-European, Oceanic and Polynesian in Austronesian. Such agreement is reassuring, and its absence would be alarming. The point is not new: when early computational trees reproduced familiar classifications, critics asked what was gained, and McMahon and McMahon (2005) replied that familiar trees from familiar data still mark progress, because they come with measures of statistical robustness and because a method must pass known cases before it can be trusted on unknown ones. That is fair as far as it goes. But agreement should not be read as independent validation of the model, for three reasons that recur throughout this book.

First, the data are not independent of the classification. Cognate judgements are products of the comparative method. The specialists who coded the cognates usually knew the accepted subgroups, used the same sound correspondences that define them, and sometimes resolved doubtful cases in the light of them. A tree built from those judgements partly returns the knowledge that went into them.

Second, cognate sharing is not shared innovation. The comparative method establishes a subgroup by a few innovations that are unlikely to have arisen twice or to have been borrowed. A lexical phylogeny weighs many shared cognate classes, most of which are not diagnostic innovations at all. Recovering a well-established low-level subgroup, whose members share hundreds of cognates, is therefore an easy test that almost any method passes; the informative cases are the deeper and contested nodes, where agreement is usually weakest.

Third, agreement does not show that the model is adequate. Different models, with different and partly wrong assumptions about rates, borrowing or the root, often recover the same shallow subgroups while disagreeing about dates and deep structure. Topological agreement says little about whether branch lengths, rates or root ages are estimated well.

What would count as stronger evidence? Recovery of relationships known from history rather than from the same comparative evidence (for example, attested dates of separation used out of sample rather than as calibrations), tests on simulated data in which the true history is known and the simulation deliberately violates the model's assumptions, posterior predictive checks of model adequacy, and agreement between analyses of different and independently coded kinds of evidence. Section 6.10 introduces a vocabulary for these different kinds of validation, and later chapters use it whenever a result is presented as a success. Throughout the book, agreement with accepted subgroups is reported as *consistent with* the traditional classification, not as proof that a method has recovered the true history. Increasingly, the boundary between the two is dissolving: computer-assisted methods help specialists find sound correspondences and cognates (Chapter 3), and probabilistic models of sound change on trees bring the core operations of the comparative method within the scope of statistical inference (Chapter 12).

1.9.2 Why the second attempt succeeded

It is worth asking why phylogenetic methods were accepted by many historical linguists, at least as one legitimate approach, when glottochronology had been rejected. Several factors combined. First, the models were better. Rate variation among meanings and among lineages, the central failing of glottochronology, could be modelled explicitly, and the sensitivity of results to model assumptions could be tested. Second, inference was based on the whole dataset and on trees rather than on pairwise percentages, so that dates and subgroupings were mutually consistent. Third, uncertainty was quantified: a Bayesian analysis reports a distribution of trees and a credible interval for each date, rather than a single number. Fourth, the methods came with a large body of statistical theory and well tested software developed by biologists, together with an established culture of validation by simulation (although, as Section 6.10 explains, a method that recovers trees simulated under its own assumptions has passed only the easiest kind of test). Fifth, the data improved, with large, documented, expert-coded databases replacing ad hoc lists. And sixth, the work was often done by interdisciplinary teams including historical linguists, which helped to ensure that the data and the questions were linguistically sound. None of this means that the results are always right; much of this book is concerned with how they can go wrong. But it explains why the debate has shifted from whether quantitative methods can be used at all to how they should be used.

1.10 A map of this book

The remaining chapters are organized as follows.

Chapter 2 introduces the vocabulary of trees, networks and tree space: rooted and unrooted trees, clades and splits, file formats, consensus trees and tree distances. Chapter 3 describes the data used in linguistic phylogenetics, from Swadesh lists and cognate coding to typological databases and automated cognate detection. Together with this chapter, they form the foundations of the book.

Chapters 4, 5 and 6 develop the three main families of inference methods: distance and parsimony methods, maximum likelihood, and Bayesian inference. Each starts from first principles and includes worked examples in R.

Chapters 7 and 8 are about modelling language change: models of lexical evolution, including binary and multistate models, ascertainment correction and the stochastic Dollo model, and the treatment of borrowing and contact, including networks.

Chapters 9 and 10 cover time: clock models and rates of language change, and the calibration and estimation of divergence times, with a family-by-family review of dated phylogenies. Chapter 11 covers space: phylogeographic models and the inference of homelands.

Chapters 12, 13 and 14 concern traits: ancestral state reconstruction, phylogenetic comparative methods for testing hypotheses about linguistic diversity, and the use of language trees to study cultural evolution.

Finally, Chapter 15 discusses critiques, pitfalls and controversies; Chapter 16 catalogues methods from evolutionary biology that have not yet been widely applied to language; Chapter 17 presents a complete worked analysis in R; and Chapter 18 synthesizes what linguistic phylogenetics has found. The appendices begin with the mathematical preliminaries the inference chapters assume

(Appendix A), followed by technical notes that collect the derivations, algorithms and from-scratch implementations behind Chapters 2 to 6 (Appendix B) and the later chapters (Appendices C, D and E). The remaining appendices describe software (F), data resources (G) and a glossary (I).

Each chapter opens with learning objectives and closes with key points and three kinds of exercises: conceptual questions, computational exercises that use the chapter's code or the book's simulated data, and research problems that ask you to justify modelling choices for a realistic scenario.

- The family tree model of language history goes back to Schleicher in the 1850s and 1860s; Darwin used the genealogy of languages as an analogy for the descent of species, and linguists and biologists have borrowed ideas from each other ever since.
- Schmidt's wave theory challenged the tree from the start, pointing to overlapping innovations that no tree can represent; the tension between trees and waves remains central and motivates network methods and models of contact.
- The comparative method, based on regular sound correspondences and subgrouping by shared innovations, is the foundation of historical linguistics and supplies the cognate judgements on which lexical phylogenies rest; its logic parallels Hennig's cladistics.
- Lexicostatistics and glottochronology attempted to classify and date languages from shared cognate percentages under a constant rate of replacement; the constant-rate assumption was refuted empirically, and glottochronology was largely abandoned.
- Since about 2000, phylogenetic methods from biology (parsimony, likelihood and especially Bayesian inference with relaxed clocks) have been applied to linguistic data, addressing the main objections to glottochronology by modelling rate variation, using family-internal calibrations and quantifying uncertainty.
- The analogy between biological and linguistic evolution is close in some respects (cognacy and homology, texts and fossils, borrowing and horizontal transfer) and loose or broken in others (pervasive borrowing, mixed languages, the language/dialect problem, language shift, human-coded characters, limited time depth).
- Phylogenetic methods can estimate subgroupings, dates, homelands, rates, ancestral states and correlated evolution with explicit uncertainty, but they cannot demonstrate relatedness and do not replace expert data.
- Agreement between computational trees and traditional subgroups is not independent validation: cognate judgements come from the comparative method, and shared cognates are not the same as shared innovations.

1.11 Exercises

Conceptual

1. Jones, Schleicher and Darwin all argued from resemblance to common descent. What did the comparative method add to this argument, and why is surface resemblance on its own insufficient?
2. Schmidt pointed to innovations shared in overlapping patterns. Give two historical scenarios that could produce such patterns, one compatible with a tree and one not, and say what further evidence would distinguish them.
3. Explain, using the example of 'hand' in Romance, Germanic and Slavic, why a shared retention does not support a subgroup while a shared innovation does. Why is a shared cognate class in a lexical dataset not automatically a shared innovation?

4. List the assumptions behind Equation (1.1). For each, give a linguistic situation in which it fails, and say which later model (Chapters 7 and 9) relaxes it.
5. A paper reports that its Bayesian tree "recovers all the established subgroups of the family". Write a short referee comment explaining what this does and does not show (Section 1.9).
6. Choose three rows of Table 1.1. For each, argue whether the parallel is close enough to justify borrowing the corresponding biological method unchanged.
7. Four languages share final devoicing, intervocalic voicing of stops and the loss of a vowel-length contrast; two of them also share an irregular suppletive paradigm for 'go'. Rank these features as evidence for subgrouping, using the terms synapomorphy, symplesiomorphy and homoplasy, and say which of them a binary presence/absence coding in a phylogenetic analysis would weight most heavily.

Computational

1. Using `glotto_t()` from the chunk that draws Figure 1.3, compute the divergence time implied by 40, 60 and 80 per cent shared cognates for retention rates of 0.75, 0.805 and 0.86. How much does the uncertainty in r matter at each time depth?
2. Glottochronology assumes that all meanings have the same retention rate. Simulate 200 meanings whose per-millennium replacement rates are drawn with `rgamma(200, shape = 0.5, rate = 0.5 / 0.22)` (a mean rate of about 0.22, roughly the rate implied by $r = 0.805$), compute the expected proportion of shared cognates after $t = 1, \dots, 8$ millennia as `mean(exp(-2 * rates * t))`, and apply `glotto_t()` with $r = 0.805$. Plot estimated against true time. In which direction does rate variation bias the dates, and why?
3. Read Schleicher's tree from the chunk that draws Figure 1.1 and write a Newick string for a modern textbook view of the same branches (treat the deep structure as a polytomy where it is unresolved). Use `ape::read.tree()`, `ape::is.monophyletic()` and `phangorn::RF.dist()` to list which of Schleicher's groupings are absent from the modern tree.
4. Using the sets of branches passed to `blob()` in the chunk that draws Figure 1.2, write each isogloss as a binary character (1 inside, 0 outside) for the ten branches, as a matrix with one row per branch. Check with the four-gamete test of Section 4.13 (all four combinations 00, 01, 10, 11 present) which pairs of isoglosses are incompatible with any tree.

Research

1. You are asked to estimate when a family of 25 languages, with no written records older than 200 years, began to diverge. Write a one-page plan that explains why glottochronology is not acceptable, which sources of calibration you would look for, and what you would report if the only available calibrations are archaeological.
2. A colleague proposes to test a contested macro-family by running a Bayesian phylogenetic analysis on shared cognates and checking whether the putative branches form clades. Explain why this design cannot answer the question, and outline a design that could (see the caution box in Section 1.7 and the references there).
3. A family has a classification based on a handful of sound changes, and a lexical phylogeny that disagrees with it at one deep node. Describe what evidence you would examine, and in what order, to decide which, if either, to trust.

1.12 Further reading

- Atkinson and Gray (2005): The classic overview of the parallels between phylogenetic thinking in biology and historical linguistics, and of the history of their interaction; the natural companion to this chapter.
- Alter (1999), *Darwinism and the Linguistic Image*. An intellectual history of the exchanges between linguistics and evolutionary biology in the nineteenth century, including Schleicher and Darwin.
- Campbell (2013), *Historical Linguistics: An Introduction*. The standard textbook on the comparative method, sound change, subgrouping and borrowing, with a critical discussion of lexicostatistics.
- Fortson (2010), *Indo-European Language and Culture*. A readable introduction to the Indo-European family, its branches and its reconstruction, with discussion of subgrouping.
- McMahon and McMahon (2005), *Language Classification by Numbers*. A balanced assessment of quantitative methods in language classification, from lexicostatistics to early phylogenetic work, by two linguists.
- Bergsland and Vogt (1962): The empirical critique that undermined glottochronology, still worth reading as a model of testing a quantitative method on known cases.
- Gray and Atkinson (2003): The paper that brought Bayesian phylogenetics to historical linguistics; short, and best read together with the critiques discussed in Chapter 15.
- Nichols and Warnow (2008): A tutorial on computational linguistic phylogeny, written for linguists, covering methods and evaluation.
- Wichmann (2017): An encyclopedia overview of genealogical classification that distinguishes internal classification (the comparative method, lexicostatistics and their computational successors) from the much harder task of establishing that languages are related at all.
- Bowern (2018): A concise review of computational phylogenetics in linguistics, its achievements and its open problems.
- Greenhill et al. (2020): An accessible introduction to Bayesian phylolinguistics in a handbook of historical linguistics.

Chapter 2

Trees, Networks and Tree Space

Every method in this book takes linguistic data and returns a tree, a set of trees, or a network. To read such results critically we need a precise vocabulary: what a node, a clade, a root and a branch length mean, and what each corresponds to in historical linguistics (a node is a proto-language, a clade is a subgroup). We also need to know how trees are stored and handled in R, why the number of possible trees makes inference hard, and how a sample of thousands of trees is summarized and compared without hiding its uncertainty. Throughout, R code with **ape** and **phangorn** works on a small Indo-European example.

After reading this chapter you should be able to:

- explain the anatomy of a phylogenetic tree (tips, internal nodes, clades, roots, branch lengths) and relate each part to its counterpart in historical linguistics;
- distinguish cladograms, phylograms and chronograms, crown and stem ages, and hard and soft polytomies;
- read, write, plot, prune, root and query trees in R with **ape** and **phangorn**, from Newick strings, NEXUS files and Glottolog-style classifications;
- explain why the number of possible trees forces heuristic search or sampling;
- represent a tree as a set of splits and test whether a cognate character is compatible with it;
- summarize a sample of trees with consensus and MCC trees, and say what each summary hides;
- choose and compute an appropriate tree distance (RF, quartet, branch-score) for a given comparison, and describe tree shape with balance indices and LTT plots.

2.1 The anatomy of a tree

2.1.1 Graphs, nodes and branches

Mathematically, a phylogenetic tree is a *graph*: a set of *vertices* (in phylogenetics usually called *nodes*) joined by *edges* (usually called *branches*). A graph is a *tree* if it is connected and contains no cycles, so that there is exactly one path between any two nodes. The nodes of degree one, which have only a single branch attached, are the *tips*, *leaves* or *terminal nodes*; these are the observed taxa. In linguistic phylogenetics the taxa are languages, dialects or, occasionally, historical stages of a language (Old English, Middle English, Modern English) or even individual manuscripts. Biologists sometimes call them *operational taxonomic units* (OTUs), a term from numerical taxonomy

(Sokal and Sneath, 1963). The remaining nodes are *internal nodes*. An internal node represents a hypothetical ancestor, sometimes called a *hypothetical taxonomic unit* (HTU). In a language tree an internal node is a proto-language: Proto-Germanic, Proto-Romance (essentially late spoken Latin), Proto-Oceanic, Proto-Bantu.

The *degree* of a node is the number of branches attached to it. In a *binary* (or *bifurcating*, or *fully resolved*) tree every internal node has degree three: one branch leading back towards the ancestor and two leading to descendants. (The root of a rooted tree is the exception, with only two branches.) If an internal node has more than two descendants, it is a *polytomy* (or *multifurcation*). Biologists distinguish two kinds. A *hard polytomy* is a claim about history: three or more lineages really did diverge from one ancestor at effectively the same time, as might happen when a dialect continuum breaks up rapidly into several separately developing varieties. A *soft polytomy* is a statement of ignorance: the data are insufficient to resolve the order of branching. Most polytomies in linguistic classifications are soft. When a handbook lists the primary branches of Indo-European as a flat list of a dozen subgroups, it does not claim that Proto-Indo-European split into twelve daughters simultaneously; it records that the comparative method has not established a consensus higher-order structure. The distinction matters because many algorithms treat polytomies as if they were hard, and some comparative methods require fully resolved trees (Chapter 13).

2.1.2 Clades, subgroups and the logic of shared innovation

A *clade* (or *monophyletic group*) is an ancestor together with all of its descendants. In a rooted tree each internal node defines exactly one clade: the set of tips below it. In linguistics the usual term for a clade is a *subgroup*: Germanic, Slavic, Balto-Slavic, Oceanic, Polynesian and Bantu are all subgroups in this sense. A group that contains an ancestor and only some of its descendants is *paraphyletic*. Paraphyletic groupings are common in traditional classifications, typically because the remaining descendants are so distinctive that they were classified separately. “Formosan”, the cover term for the Austronesian languages of Taiwan, is widely regarded as paraphyletic: on most accounts the Formosan languages form several primary branches of Austronesian, and the Malayo-Polynesian languages spoken outside Taiwan descend from within this diversity, so “Formosan” is in effect everything in Austronesian except Malayo-Polynesian (Blust, 1999). A *polyphyletic* group unites tips that resemble each other because of convergence rather than common descent. Grouping languages as “tone languages” or “click languages” defines sets that are, in large part, polyphyletic.

The phylogenetic notion of a clade has a direct counterpart in the logic of the comparative method. Since the work of the Neogrammarians in the late nineteenth century, historical linguists have held that subgroups must be established by *shared innovations*: changes that occurred once in the common ancestor of the subgroup and were inherited by all its members (Brugmann, 1884; Campbell, 2013). Shared *retentions* of ancestral features do not define subgroups, because any number of independent lineages can retain an old feature. Exactly the same principle was formulated independently in biology by Willi Hennig, who argued that only shared derived characters (*synapomorphies*) support monophyletic groups, whereas shared ancestral characters (*symplesiomorphies*) do not (Hennig, 1966). Chapter 1 discusses this convergence of ideas in more detail. Here we simply note that it gives the two disciplines a common understanding of what a tree is supposed to represent: a hierarchy of nested innovations.

Two clades that are each other’s closest relatives, descending from the two daughters of a single node, are *sister groups*. In the tree in Figure 2.1, Romance and Celtic are sisters (an “Italo-Celtic”

grouping that we use only as an illustration; its status in the real family is debated), as are Greek and Armenian. The *most recent common ancestor* (MRCA) of a set of tips is the youngest node that is an ancestor of all of them.

2.1.3 Roots, rooting and outgroups

A tree is *rooted* if one node is designated as the ancestor of all others; the branches then acquire a direction, from older to younger. An *unrooted* tree specifies only which tips are closer to which, without saying where the ancestor was. Many inference methods, including all time-reversible substitution models used in maximum likelihood (Chapter 5) and standard parsimony (Chapter 4), cannot by themselves detect the position of the root: they return an unrooted tree, and the root has to be placed by some other argument. There are several ways of doing so.

Outgroup rooting is the standard approach in biology. One or more taxa known to lie outside the group of interest (the *ingroup*) are included, and the root is placed on the branch joining the outgroup to the ingroup. Outgroup rooting is difficult in linguistics. Language families are, by definition, the largest units whose relatedness can be demonstrated; there is usually no language outside a family that is demonstrably related to it, so no proper outgroup exists. An unrelated language added to the analysis would share with the ingroup only chance resemblances and would attach essentially at random. Linguists therefore usually root trees in other ways. One is to use prior knowledge of the first split, for example by constraining the Anatolian languages to be the sister of the rest of Indo-European, or by requiring Formosan lineages to branch off first in Austronesian. A second is *clock rooting*: under a molecular-clock model (Chapter 9) the root is the point from which all tips are equidistant in time (after allowing for the ages of ancient tips), and the root position becomes an estimable parameter. A third is the use of *directional models* such as the stochastic Dollo model, under which cognate classes are born once and then lost; this asymmetry carries information about the root (Chapter 7). *Midpoint rooting* places the root at the midpoint of the longest path between two tips. It is a cheap heuristic that works only if rates of change are roughly equal across lineages.

2.1.4 Branch lengths: cladograms, phylograms and chronograms

Trees differ in what, if anything, their branch lengths mean. A *cladogram* conveys only topology: branch lengths are arbitrary and drawn for legibility. A *phylogram* (or *additive tree*) has branch lengths proportional to the amount of change, typically the expected number of changes per character; with lexical data this is roughly the expected number of cognate replacements per meaning. A *chronogram* (or *time tree*) has branch lengths in units of time, such as years or millennia. When all tips are contemporaneous and branch lengths are in time, every root-to-tip path has the same length and the tree is *ultrametric*. Trees containing ancient languages such as Hittite, Tocharian, Latin or Old Irish are chronograms but not ultrametric, because those tips are older than the present; Chapter 10 shows how such dated tips, placed at the time of their attestation, are used to calibrate the clock. A phylogram becomes a chronogram only through a clock model, which converts amounts of change into time by modelling how rates vary (Chapter 9).

Two further concepts recur. The *height* of a node in a chronogram is its age, measured back from the present (or from the youngest tip); the root height is the age of the proto-language at the root. The *stem* of a clade is the branch leading to its MRCA from the parent node, and the *crown* of a clade is the subtree descending from the MRCA of its sampled members. The distinction is not mere pedantry. A date for “Proto-Germanic” can mean either the age of the crown node,

the last common ancestor of all attested Germanic languages, or the age of the stem, when the pre-Germanic lineage separated from its sister. These can differ by thousands of years. Published linguistic dates almost always refer to crown ages, but readers should check. Crown ages also depend on sampling: if the only sampled Germanic languages were English, Dutch and German, the crown age of "Germanic" in the tree would really be the age of West Germanic.

2.1.5 A first look in R

The **ape** package represents a tree as an object of class `phylo` (Paradis and Schliep, 2019; Paradis, 2012). Its core is an *edge matrix* with one row per branch and two columns giving the parent and child node numbers. Tips are numbered 1 to n and internal nodes $n + 1$ upwards, the root being node $n + 1$. A vector of `tip.labels`, an optional `edge.length` vector and optional `node.labels` complete the object. The **phangorn** package (Schliep, 2011) adds tree inference and many utilities, and **phytools** (Revell, 2012) and **ggtree** (Yu et al., 2017) add comparative methods and flexible plotting.

We read a toy Indo-European tree with sixteen modern languages from a Newick string (the format is described in Section 2.2). The topology follows widely accepted low-level subgroups, but the higher-level structure and all branch lengths, measured in millennia, are invented for illustration and are not an estimate of the real history.

```
library(phangorn)
ie_nwk <- paste0(
  "((((((Italian:1.2,Spanish:1.2):0.3,French:1.5)Romance:2.5,",
  "(Irish:2.5,Welsh:2.5)Celtic:1.5)ItaloCeltic:0.5,",
  "((English:1.5,German:1.5):0.5,Swedish:2.0)Germanic:2.5):0.5,",
  "((Russian:1.2,Polish:1.2):1.8,Lithuanian:3.0)BaltoSlavic:2.0):1.5,",
  "(((Greek:4.8,Armenian:4.8):0.4,Albanian:5.2):0.6,",
  "(Hindi:3.8,Persian:3.8)IndoIranian:2.0):0.7)IndoEuropean;")
ie <- read.tree(text = ie_nwk)
ie
```

```
#>
#> Phylogenetic tree with 16 tips and 15 internal nodes.
#>
#> Tip labels:
#>   Italian, Spanish, French, Irish, Welsh, English, ...
#> Node labels:
#>   IndoEuropean, , , ItaloCeltic, Romance, , ...
#>
#> Rooted; includes branch length(s).
```

```
c(tips = Ntip(ie), internal = Nnode(ie), branches = Nedge(ie))
```

```
#>     tips internal branches
#>       16       15       30
```

```
c(rooted = is.rooted(ie), binary = is.binary(ie),
  ultrametric = is.ultrametric(ie))
```

```
#>      rooted      binary ultrametric
#>        TRUE        TRUE        TRUE
```

```
head(ie$edge)
```

```
#>      [,1] [,2]
#> [1,]   17   18
#> [2,]   18   19
#> [3,]   19   20
#> [4,]   20   21
#> [5,]   21   22
#> [6,]   22    1
```

A rooted binary tree with n tips has $n-1$ internal nodes and $2n-2$ branches; an unrooted binary tree has $n-2$ internal nodes and $2n-3$ branches. Our tree, with 16 tips, has 15 internal nodes and 30 branches. Figure 2.1 shows four ways of drawing the same object. The choice of drawing style is cosmetic: rotating the two descendants of any node, or switching from a rectangular to a circular layout, changes nothing about the tree. This simple fact is worth stating because readers often over-interpret the vertical order of tips in a figure, which carries no information, as if adjacent tips were more closely related. In the phylogram, Albanian is drawn next to Armenian, but it is no more closely related to Armenian than to Greek.

Two further misreadings are common (Baum and Smith, 2013). One is *node counting*: the number of nodes on the path between two tips depends on how many other languages happen to be sampled, whereas relatedness is defined only by the recency of the most recent common ancestor. The other is a residue of what Baum and Smith call *ladder thinking*: describing a lineage that branches off near the root, such as Anatolian in Indo-European or a Formosan language in Austronesian, as "basal", "archaic" or "primitive". Every living language has been evolving for exactly as long as the languages of its sister lineage, and lexical conservatism (Icelandic beside English, for instance) is a matter of rate, not of closeness to the ancestor (Section 4.3). In the same way, a cladogram orders in time only nodes that lie on the same root-to-tip path; two nodes in different parts of the tree cannot be ranked by age from the drawing.

```
op <- par(mfrow = c(2, 2), mar = c(1, 1, 2, 1))
plot(ie, cex = 0.75, label.offset = 0.1, main = "Phylogram (chronogram)")
plot(ie, type = "cladogram", use.edge.length = FALSE, cex = 0.75,
     label.offset = 0.1, main = "Cladogram")
# radial layouts: wider margins, and labels may run into them rather than
# being clipped
par(mar = c(1.5, 1.5, 2.5, 1.5), xpd = NA)
plot(unroot(ie), type = "unrooted", cex = 0.62, lab4ut = "axial",
     label.offset = 0.1, main = "Unrooted")
plot(ie, type = "fan", cex = 0.7, label.offset = 0.1, main = "Fan")
```

```
par(op)
```

Because our tree is a chronogram, we can add a time axis and the labels of named internal nodes (Figure 2.2). The function `branching.times()` returns node heights, which for an ultrametric tree are node ages.

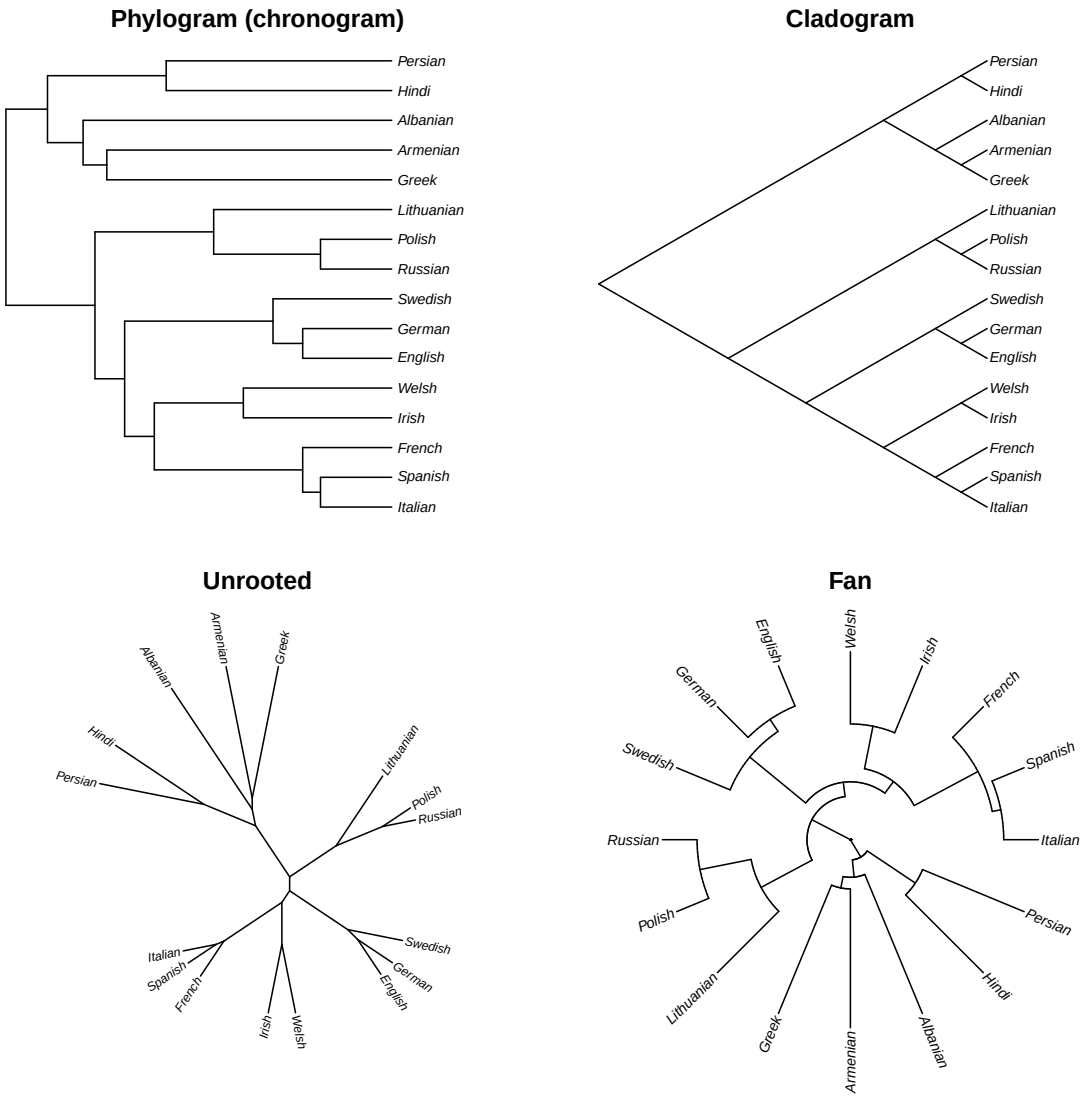


Figure 2.1: The same toy Indo-European tree drawn as a phylogram (branch lengths proportional to time), a cladogram (topology only), an unrooted tree, and a fan. The topology of low-level subgroups is conventional; the deeper structure and the branch lengths are invented for illustration.

```
plot(ie, cex = 0.8, label.offset = 0.08, x.lim = c(-1.3, 8.3))
# each name sits on its stem branch, on the side away from the parent's
# vertical line
lp <- get("last_plot.phylo", envir = .PlotPhyloEnv)
nn <- Ntip(ie) + seq_len(Nnode(ie))
up <- lp$yy[nn] >= lp$yy[ie$edge[match(nn, ie$edge[, 2]), 1]]
for (i in seq_along(nn)) {
  a <- if (i == 1) c(1.08, 0.5) else if (up[i]) c(1.05, -0.4) else c(1.05, 1.4)
  nodelabels(ie$node.label[i], nn[i], frame = "none", adj = a,
             cex = 0.65, col = "grey25")
}
axisPhylo(backward = TRUE)
mtext("Thousands of years before present", side = 1, line = 2.5, cex = 0.8)
```

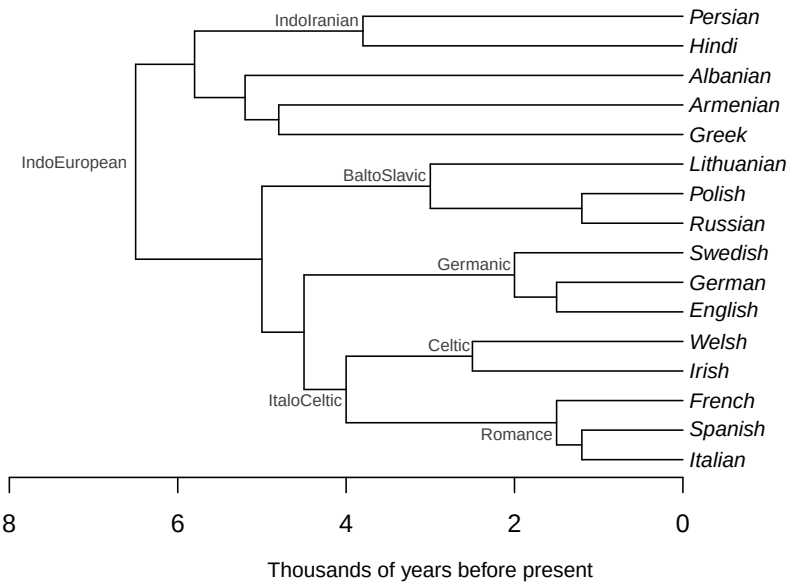


Figure 2.2: The toy Indo-European chronogram with named internal nodes (proto-languages) and a time axis in millennia before present. Node ages are invented for illustration.

```
round(branching.times(ie)[c("IndoEuropean", "Germanic", "Romance")], 2)
```

```
#> IndoEuropean     Germanic      Romance
#>          6.5          2.0          1.5
```

The common manipulations of trees are all one-liners in **ape** and **phangorn**. `drop.tip()` prunes taxa, which is how a large published tree is matched to the languages for which a comparative dataset has data (Chapter 13); `extract.clade()` pulls out a subtree; `getMRCA()` and `is.monophyletic()` query relationships; `root()`, `unroot()` and `midpoint()` change the

rooting; `ladderize()` and `rotate()` change only the drawing order; `di2multi()` collapses very short branches into polytomies; and `multi2di()` resolves polytomies arbitrarily into zero-length branches.

```
is.monophyletic(ie, c("Italian", "Spanish", "French"))
```

```
#> [1] TRUE
```

```
is.monophyletic(ie, c("Italian", "Spanish", "Irish"))
```

```
#> [1] FALSE
```

```
getMRCA(ie, c("English", "Swedish"))   # node number of Proto-Germanic
```

```
#> [1] 24
```

```
ie$node.label[getMRCA(ie, c("English", "Swedish")) - Ntip(ie)]
```

```
#> [1] "Germanic"
```

```
Descendants(ie, getMRCA(ie, c("Russian", "Lithuanian")), type = "tips")
```

```
#> [[1]]
#> [1]  9 10 11
```

```
europe <- drop.tip(ie, c("Hindi", "Persian", "Armenian"))
Ntip(europe)
```

```
#> [1] 13
```

```
germanic <- extract.clade(ie, "Germanic")
germanic$tip.label
```

```
#> [1] "English" "German"  "Swedish"
```

```
# Collapse branches shorter than 0.31 kyr into polytomies
write.tree(di2multi(ie, tol = 0.31))
```

```
#> [1]
"(((((Italian:1.2,Spanish:1.2,French:1.5)Romance:2.5,(Irish:2.5,Welsh:2.5)Celtic:1.
5)ItaloCeltic:0.5,((English:1.5,German:1.5):0.5,Swedish:2)Germanic:2.5):0.5,((Russi
an:1.2,Polish:1.2):1.8,Lithuanian:3)BaltoSlavic:2):1.5,(((Greek:4.8,Armenian:4.8):0
.4,Albanian:5.2):0.6,(Hindi:3.8,Persian:3.8)IndoIranian:2):0.7)IndoEuropean;"
```

```
u <- unroot(ie)
r_hindi <- root(u, outgroup = "Hindi", resolve.root = TRUE)
r_mid <- midpoint(u)
op <- par(mfrow = c(1, 2), mar = c(1, 1, 2, 1))
plot(r_hindi, cex = 0.7, label.offset = 0.1, main = "Rooted on Hindi")
plot(r_mid, cex = 0.7, label.offset = 0.1, main = "Midpoint rooted")
```

```
par(op)
```

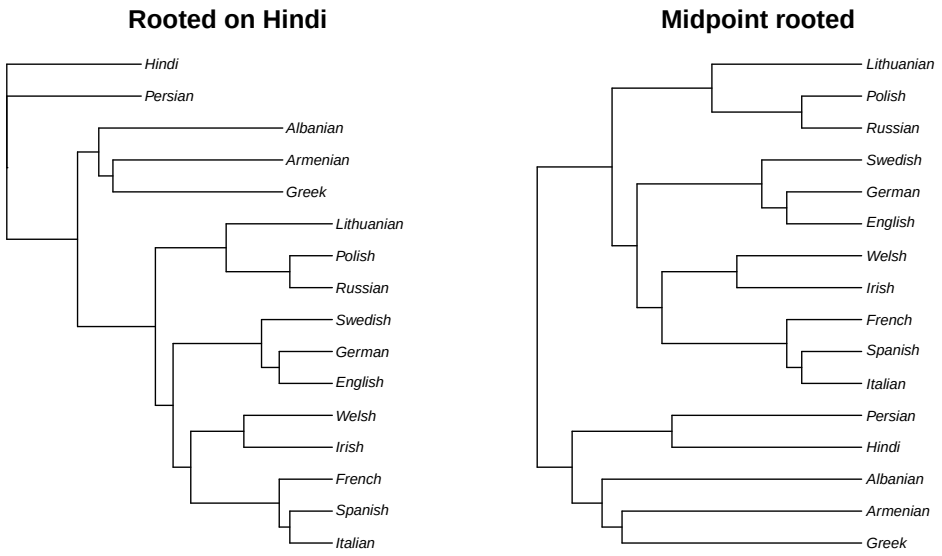


Figure 2.3: Rooting. Left: the unrooted version of the toy tree, rerooted on the branch to Hindi as if it were an outgroup, which misplaces the origin of the family inside Indo-Iranian. Right: midpoint rooting, which recovers the original root here because the tree is ultrametric.

Figure 2.3 illustrates a point that is easily forgotten: an unrooted tree is compatible with as many rooted trees as it has branches, and an unjustified root can produce a very misleading picture of history. Rooted on Hindi, the toy tree implies that the family began in the Hindi lineage and that Persian is the sister of all other branches.

2.2 Storing trees: Newick, NEXUS and their extensions

2.2.1 The Newick format

The *Newick* format represents a tree as nested parentheses. (It is named after a restaurant in Dover, New Hampshire, where a group of authors of phylogenetic software agreed on it in 1986.) Each pair of parentheses encloses the descendants of one internal node, sister subtrees are separated by commas, and the whole string is terminated by a semicolon. A tip label may be followed by a colon and a branch length; an internal node's label and branch length follow its closing parenthesis. Thus `((English:1.5,German:1.5):0.5,Swedish:2.0)Germanic;` describes a three-tip clade named Germanic in which English and German are sisters, each 1.5 units from their common ancestor, which is in turn 0.5 units from the Germanic root. Because the string is a nested list, any rooted tree has many equivalent Newick strings, one for each way of ordering sisters. An unrooted tree is conventionally written with a basal trifurcation, `(A,B,(C,D));`, and many programs will silently treat a string with a basal bifurcation as rooted.

Newick is compact and universal but impoverished. It has no standard way to attach more than one value to a node, so programs have extended it by placing metadata inside square brackets, which Newick treats as comments. BEAST and TreeAnnotator, for example, write node annotations such as `[&height_95%_HPD={4.1,6.9},posterior=0.97]` immediately after the node, which the **treeio** package can parse into a table for plotting with **ggtree** (Wang et al., 2020; Yu et al., 2017). Explicit phylogenetic networks can be written in *Extended Newick*, in which a reticulation node that appears

twice in the string is marked with a shared label such as `#H1` (Cardona et al., 2008). Spaces, commas, colons, parentheses and semicolons in taxon names break Newick parsers; language names such as “Ngadha (Flores)” or “Ainu, Hokkaido” must be sanitized, typically by replacing problematic characters with underscores or by using Glottocodes as identifiers.

2.2.2 NEXUS

The NEXUS format is a modular container for phylogenetic data (Maddison et al., 1997). A NEXUS file begins with `#NEXUS` and is divided into *blocks*, each starting with `BEGIN name;` and ending with `END;`. The TAXA block lists taxon names; the CHARACTERS (or older DATA) block holds the data matrix together with a FORMAT statement declaring the data type, symbols and the codes for missing data and gaps; the TREES block holds one or more Newick trees, usually with a TRANSLATE table that maps short numerical codes to taxon names; and the ASSUMPTIONS and SETS blocks define character partitions, weights and taxon sets. Programs add their own blocks, such as the MRBAYES block of commands or the PAUP block. Linguistic cognate matrices are typically distributed as NEXUS files with `DATATYPE=STANDARD` and `SYMBOLS="01"`, with character sets marking which binary characters belong to which meaning (Chapter 3). BEAST 2 uses its own XML input format, but its tree output (the `.trees` file of posterior samples) is NEXUS. XML-based alternatives such as NeXML were designed to make the semantics of data explicit and machine-verifiable (Vos et al., 2012), but NEXUS remains the lingua franca of linguistic phylogenetics. Two details of the standard matter for linguistic data. Within a matrix, parentheses and braces mean different things: `(01)` records a *polymorphism*, a language that has both states (for instance two synonyms from different cognate classes in a multistate character), whereas `{01}` records *uncertainty* about which single state applies; square brackets enclose comments (Maddison et al., 1997). Programs also differ in which blocks and commands they read: a block that one program uses may be ignored by another or stop it with an error, and many programs treat polymorphism and uncertainty identically, summing over the states in both cases. A matrix that has passed through several programs should be checked for what it still says.

```
tmp <- tempfile(fileext = ".nex")
write.nexus(ie, file = tmp)
cat(readLines(tmp)[-2], sep = "\n")   # line 2 is a time stamp
```

```
#> #NEXUS
#>
#> BEGIN TAXA;
#>  DIMENSIONS NTAX = 16;
#>  TAXLABELS
#>      Italian
#>      Spanish
#>      French
#>      Irish
#>      Welsh
#>      English
#>      German
#>      Swedish
#>      Russian
#>      Polish
#>      Lithuanian
```

```
#>      Greek
#>      Armenian
#>      Albanian
#>      Hindi
#>      Persian
#>  ;
#> END;
#> BEGIN TREES;
#>  TRANSLATE
#>      1   Italian,
#>      2   Spanish,
#>      3   French,
#>      4   Irish,
#>      5   Welsh,
#>      6   English,
#>      7   German,
#>      8   Swedish,
#>      9   Russian,
#>      10  Polish,
#>      11  Lithuanian,
#>      12  Greek,
#>      13  Armenian,
#>      14  Albanian,
#>      15  Hindi,
#>      16  Persian
#>  ;
#>  TREE * UNTITLED = [&R]
((((((1:1.2,2:1.2):0.3,3:1.5)Romance:2.5,(4:2.5,5:2.5)Celtic:1.5)ItaloCeltic:0.5,((
6:1.5,7:1.5):0.5,8:2)Germanic:2.5):0.5,((9:1.2,10:1.2):1.8,11:3)BaltoSlavic:2):1.5,
(((12:4.8,13:4.8):0.4,14:5.2):0.6,(15:3.8,16:3.8)IndoIranian:2):0.7)IndoEuropean;
#> END;
```

```
identical(read.nexus(tmp)$tip.label, ie$tip.label)
```

```
#> [1] TRUE
```

2.2.3 Classification trees with no branch lengths

A great deal of linguistic tree information comes not from phylogenetic inference but from expert classification. The Glottolog database, for instance, provides a genealogical classification of the world's languages into families and subgroups, downloadable as Newick strings (Hammarström et al., 2024). Such trees have three features that distinguish them from inferred phylogenies: they have no branch lengths, they contain many (mostly soft) polytomies, and their internal nodes are named subgroups whose existence the classifiers consider established. When such a tree is used as the scaffold for a comparative analysis (Chapter 13), branch lengths must be invented. A common choice is Grafen's method, which sets the height of each node proportional to the number of tips below it minus one (Grafen, 1989); another is simply setting all branches to unit length. Polytomies can be retained, resolved randomly (and the analysis repeated over many random resolutions), or resolved into zero-length branches. None of these choices is neutral, and their

effects should be checked by sensitivity analysis.

```
germ <- read.tree(text = paste0(
  "(((English,West_Frisian)Anglo_Frisian,",
  "(Dutch,German,Luxembourgish)Continental),",
  "(Swedish,Danish,Norwegian,Icelandic,Faroese)North)Germanic;"))
is.binary(germ)
```

```
#> [1] FALSE
```

```
germ_grafen <- compute.brlen(germ, method = "Grafen")
round(branching.times(germ_grafen), 3)
```

```
#>      Germanic               Anglo_Frisian   Continental         North
#>         1.000         0.444         0.111         0.222         0.444
```

```
germ_resolved <- multi2di(germ, random = TRUE)
c(is.binary(germ_resolved), Nnode(germ), Nnode(germ_resolved))
```

```
#> [1] 1 5 9
```

The Glottolog-style tree above is a simplification for illustration only; real classifications of Germanic differ in detail.

2.3 How many trees are there?

Every phylogenetic method must, implicitly or explicitly, choose among the possible trees for the taxa at hand. The number of those trees grows faster than exponentially. There is only one unrooted binary tree for three tips, three for four tips, fifteen for five, and in general

$$U(n) = 1 \times 3 \times 5 \times \cdots \times (2n-5) = (2n-5)!! \tag{2.1}$$

where !! denotes the double factorial (the product of every second integer). The number of rooted binary trees is

$$R(n) = U(n+1) = (2n-3)!! = \frac{(2n-2)!}{2^{n-1}(n-1)!}. \tag{2.2}$$

These formulas have been known since the nineteenth century and were popularized among systematists by Felsenstein (1978b). For chronograms a further distinction matters: Bayesian tree priors such as the Yule and coalescent processes (Chapter 6) assign probabilities to *ranked* trees (also called *labelled histories*), in which the temporal order of the internal nodes is also specified, and there are more of these again: 18 ranked trees for four tips against 15 rooted ones.

The derivation of equations (2.1) and (2.2) by adding tips one at a time, and the formula for the number of ranked trees, are in Section B.1.1.

Table 2.1 shows how quickly these numbers grow. With 20 languages there are already more than 10^{20} rooted trees; with the 87 Indo-European languages analysed by Gray and Atkinson (2003) the number of rooted trees has more than 150 digits.

```
log10_dfact <- function(m) {     # log10 of (2m - 1)!!, m >= 1
  (lfactorial(2 * m) - m * log(2) - lfactorial(m)) / log(10)
}
log10_unrooted <- function(n) log10_dfact(n - 2)   # (2n - 5)!!
log10_rooted   <- function(n) log10_dfact(n - 1)   # (2n - 3)!!
log10_ranked   <- function(n) (lfactorial(n) + lfactorial(n - 1) -
                                 (n - 1) * log(2)) / log(10)
n <- c(4, 5, 6, 8, 10, 16, 20, 50, 87, 400)
fmt <- function(x) ifelse(x < 15, format(round(10^x), big.mark = ",",
                                         scientific = FALSE),
                          sprintf("%.2f x 10^%d", 10^(x - floor(x)), floor(x)))
tab <- data.frame(n = n,
  unrooted = fmt(log10_unrooted(n)),
  rooted = fmt(log10_rooted(n)),
  ranked = fmt(log10_ranked(n)))
knitr::kable(tab, align = "rrrr",
  col.names = c("Tips (n)", "Unrooted trees", "Rooted trees",
                "Ranked rooted trees"),
  caption = paste("Numbers of distinct binary trees for n labelled tips.",
    "Four hundred tips is roughly the size of the largest Austronesian",
    "and Bantu analyses."))
```

Table 2.1: Numbers of distinct binary trees for n labelled tips. Four hundred tips is roughly the size of the largest Austronesian and Bantu analyses.

Tips (n)	Unrooted trees	Rooted trees	Ranked rooted trees
4	3	15	18
5	15	105	180
6	105	945	2,700
8	10,395	135,135	1,587,600
10	2,027,025	34,459,425	2,571,912,000
16	213,458,046,676,875	6.19 x 10^15	8.35 x 10^20
20	2.22 x 10^20	8.20 x 10^21	5.64 x 10^29
50	2.84 x 10^74	2.75 x 10^76	3.29 x 10^112
87	6.66 x 10^152	1.14 x 10^155	6.60 x 10^236
400	7.32 x 10^981	5.84 x 10^984	7.94 x 10^1614

The practical consequences are profound. For more than a dozen or so taxa, exhaustive evaluation of all trees is impossible, and practical methods rely either on algorithms that build a tree directly (distance methods, Chapter 4), on heuristic searches that move through the space of trees by local rearrangements (parsimony and likelihood, Chapters 4 and 5), or on stochastic sampling (Bayesian MCMC, Chapter 6). Branch-and-bound algorithms can guarantee the optimal parsimony tree for up to a few dozen taxa, but no method can guarantee the global optimum for the hundreds of taxa in modern linguistic datasets. The enormity of tree space also explains why claims that a particular tree is "the" tree should be met with caution. Even a well supported tree is one point

in an astronomically large space, and the posterior probability of any single complete topology is typically tiny, even when most individual clades are strongly supported.

```
nn <- 3:60
plot(nn, log10_ranked(nn), type = "n", xlab = "Number of languages",
     ylab = expression(log[10]~"(number of trees)"), las = 1)
lines(nn, log10_rooted(nn), lty = 1, lwd = 2)
lines(nn, log10_ranked(nn), lty = 2, lwd = 2)
lines(nn, log10_unrooted(nn), lty = 3, lwd = 2)
legend("topleft", c("rooted", "ranked rooted", "unrooted"),
       lty = 1:3, lwd = 2, bty = "n")
```

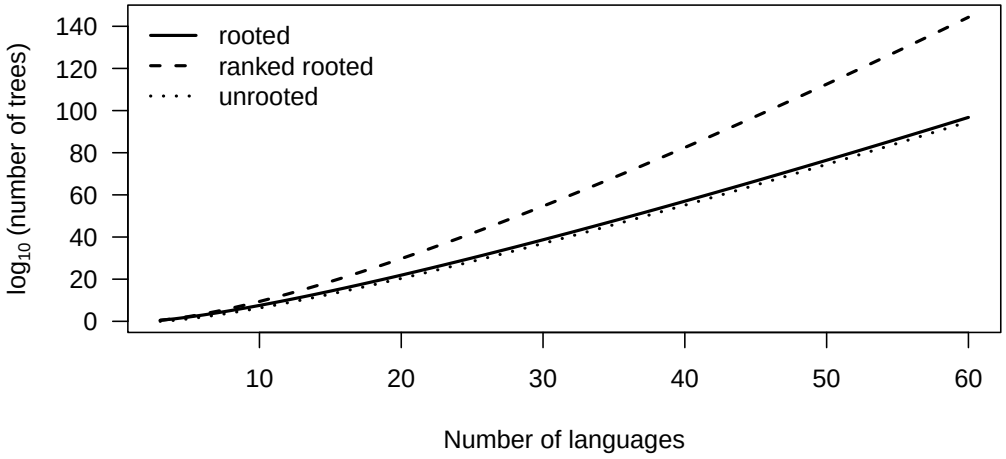


Figure 2.4: The number of rooted binary trees (solid), ranked rooted trees (dashed) and unrooted binary trees (dotted) as a function of the number of tips, on a log scale.

2.4 Splits, clusters and compatibility

2.4.1 Splits as the atoms of trees

Cut any single branch of an unrooted tree and the tips fall into two groups. This bipartition of the taxon set X into two non-empty, disjoint subsets A and $B = X \setminus A$ is called a *split*, written $A \mid B$. Every branch induces exactly one split. Branches leading to tips induce *trivial splits*, which separate one taxon from all the others and are present in every tree; the interesting information is in the *non-trivial splits* induced by internal branches. An unrooted binary tree on n tips has $n-3$ internal branches and therefore $n-3$ non-trivial splits. In a rooted tree the analogous object is the *cluster* (or clade): the set of tips below a node. A rooted binary tree on n tips has $n-2$ non-trivial clusters, excluding the full set and the single tips.

Splits are fundamental because a tree is completely determined by its set of splits. This is the *splits-equivalence theorem*, which goes back to Buneman (1971) and is treated in detail by Semple and Steel (2003). It says that a collection of splits corresponds to a tree if and only if every pair

of splits in the collection is *compatible*, and that the tree is then unique. Two splits $A_1 \mid B_1$ and $A_2 \mid B_2$ are compatible if at least one of the four intersections $A_1 \cap A_2$, $A_1 \cap B_2$, $B_1 \cap A_2$ and $B_1 \cap B_2$ is empty. Intuitively, compatible splits are nested: one side of one split lies entirely within one side of the other. For rooted clusters the condition is simpler: two clusters are compatible if they are disjoint or one contains the other. A set of pairwise compatible clusters forms a *hierarchy*, which is exactly a rooted tree.

The link to linguistics is direct. A binary cognate character, coded 1 for languages that have a reflex of a given cognate set and 0 for those that do not, is itself a split of the languages. If every cognate set arose once and was never borrowed or lost in parallel, the splits defined by the characters would all be compatible with the true tree, and the tree could be read off the data by assembling compatible splits. This is the idea behind *perfect phylogeny* and *character compatibility* methods, which Ringe, Warnow and Taylor applied to Indo-European (Chapter 4) (Ringe et al., 2002). Real data contain incompatible characters: borrowing, parallel semantic change and coding errors create splits that conflict with the tree. A collection of mutually incompatible splits cannot be drawn as a tree, but it can be drawn as a *split network* (Section 2.9 and Chapter 8), in which a set of parallel edges represents each split.

In R, `prop.part()` (ape) returns the clusters of a rooted tree, and `as.splits()` (phangorn) the splits, in a compact representation in which each split is stored by the set of tips on one side.

```
sp <- as.splits(ie)
length(sp)                      # trivial + non-trivial splits
```

```
#> [1] 31
```

```
sp_nontrivial <- sp[lengths(sp) > 1 & lengths(sp) < Ntip(ie) - 1]
length(sp_nontrivial)           # n - 3 non-trivial splits of the unrooted tree
```

```
#> [1] 14
```

```
lapply(sp_nontrivial[1:4], function(s) attr(sp, "labels")[s])
```

```
#> [[1]]
#>  [1] "Italian"    "Spanish"    "French"     "Irish"      "Welsh"
#>  [6] "English"    "German"     "Swedish"    "Russian"    "Polish"
#> [11] "Lithuanian"
#>
#> [[2]]
#> [1] "Italian" "Spanish" "French"  "Irish"   "Welsh"   "English"
#> [7] "German"  "Swedish"
#>
#> [[3]]
#> [1] "Italian" "Spanish" "French"  "Irish"   "Welsh"
#>
#> [[4]]
#> [1] "Italian" "Spanish" "French"
```

Testing compatibility is easy to implement directly from the definition, which shows how a cognate character can be checked against a tree:

```
compatible_splits <- function(A1, A2, X) {
  B1 <- setdiff(X, A1); B2 <- setdiff(X, A2)
  any(c(length(intersect(A1, A2)), length(intersect(A1, B2)),
        length(intersect(B1, A2)), length(intersect(B1, B2))) == 0)
}
X <- ie$tip.label
balto_slavic <- c("Russian", "Polish", "Lithuanian")
# A character shared by the Balto-Slavic and Germanic languages is
# compatible with Balto-Slavic as a subgroup...
compatible_splits(balto_slavic, c(balto_slavic, "English", "German",
                                  "Swedish"), X)
```

```
#> [1] TRUE
```

```
# ...but a character shared by Lithuanian and the Germanic languages,
# excluding the other Slavic languages, is not.
compatible_splits(balto_slavic, c("Lithuanian", "English", "German",
                                  "Swedish"), X)
```

```
#> [1] FALSE
```

The second hypothetical character, if it were real, would have to be explained by borrowing between Baltic and Germanic, by parallel innovation, or by loss in the Slavic languages of a cognate inherited from an older ancestor. Note that the third explanation does not create genuine incompatibility on a rooted tree in which the cognate set predates Balto-Slavic. Whether a pattern is compatible depends on the direction of change, which is one reason why rooted and unrooted analyses can disagree.

2.5 Summarizing sets of trees: consensus and summary trees

Modern analyses rarely produce a single tree. Bootstrap analyses (Chapters 4 and 5) produce hundreds of trees from resampled data; Bayesian analyses produce thousands of trees sampled from the posterior distribution (Chapter 6); parsimony searches often find many equally parsimonious trees. These sets must be summarized, both to report a "best" tree and to convey which parts of it are certain. The many available methods differ in what they assume and in what they discard (Bryant, 2003).

2.5.1 A sample of trees to summarize

To have something to summarize, we simulate 400 binary characters on the toy tree under a symmetric two-state model (Chapter 5), as a crude stand-in for cognate data, and compute 200 bootstrap trees with UPGMA (Chapter 4). Bootstrap trees and Bayesian posterior samples are conceptually different, but for the purpose of illustrating summary methods they behave in the same way: both are collections of trees in which well supported clades recur often and poorly supported ones rarely. Note what kind of simulation this is. The data are generated on an ultrametric tree with no borrowing and independent characters, which are exactly the assumptions UPGMA makes, so whatever the summaries recover below is *parameter recovery* under the method's own

assumptions, in the vocabulary of Section 6.10; it says nothing about how the method fares on real cognate data.

```
set.seed(2024)
sim <- simSeq(ie, l = 400, type = "USER", levels = c("0", "1"), rate = 0.05)
boot_trees <- bootstrap.phyDat(sim, bs = 200,
  FUN = function(x) upgma(dist.hamming(x)))
boot_trees <- .compressTipLabel(boot_trees)
boot_trees
```

```
#> 200 phylogenetic trees
```

2.5.2 Strict, majority-rule and related consensus trees

The *strict consensus tree* contains only those clades (or splits) that appear in every tree of the set. It is the most conservative summary: anything it shows is certain relative to the sample, but a single aberrant tree can collapse it to a near-total polytomy. The *majority-rule consensus tree* contains all clades that appear in more than half of the trees (Margush and McMorris, 1981). Any two clades each present in more than half of the trees must co-occur in at least one tree, so they are compatible, and the set of majority clades always forms a tree. Clade frequencies are usually written on the branches. The *extended majority-rule* (or *greedy*) consensus, used by MrBayes and many bootstrap programs, goes further: after including the majority clades, it adds the most frequent remaining clades that are compatible with those already accepted, in descending order of frequency, until the tree is fully resolved or no compatible clade remains. Minority clades included in this way are not well supported, and their frequencies should be shown so readers can judge. Further variants include the *semi-strict* (or combinable components) consensus, which keeps every clade that conflicts with no clade in any tree, and the *Adams consensus*, which keeps nested structure even when the content of clades differs and can therefore contain groupings that appear in none of the input trees (Adams, 1972).

Consensus trees in this sense describe topology only. Their branch lengths are not defined by the method and must be computed separately, for example as the mean length of the corresponding split across the trees containing it. A majority-rule consensus of time trees is therefore not a chronogram unless node heights are added explicitly.

2.5.3 The maximum clade credibility tree

Bayesian programs for dated trees, and especially TreeAnnotator in BEAST, usually report the *maximum clade credibility* (MCC) tree. The MCC tree is the tree *in the sample* that maximizes the product (in practice the sum of logarithms) of the posterior frequencies of its clades. Because it is one of the sampled trees, it is always fully resolved and has a definite ranked topology, so node heights can be attached in several ways: by keeping the heights of the sampled tree, by averaging heights of each clade over the sampled trees in which it appears (“mean heights” or “median heights”), or by the *common ancestor heights* method, which uses the age of the MRCA of the clade’s tips in every sampled tree, whether or not the clade is monophyletic there. Heled and Bouckaert (2013) compared these options by simulation and found that the commonly used mean-heights option can produce misleading summary trees, including negative branch lengths, when clade support is low, whereas common ancestor heights performed better. The MCC tree is a summary, not the most probable tree: in a large dataset the posterior probability of the MCC

topology itself may be negligibly small. This matters most when the summary tree is reused. A downstream analysis (an ancestral state reconstruction, a homeland inference, a comparative test) run on the MCC tree alone treats one plausible-looking tree as if it were known, and it can be distorted in two ways: clades with modest support are treated as certain, and the MCC tree's node heights, averaged over trees with different topologies, need not correspond to any history the posterior actually contains. Such analyses should be repeated over a sample of trees from the posterior (Chapters 12 and 13).

Other summaries have been proposed. The *maximum a posteriori* (MAP) topology is the topology sampled most often, but in large analyses almost every sampled topology is unique, so the MAP is undefined in practice. *Conditional clade distributions* estimate the probability of a tree from the frequencies of clades conditional on their parent clade, which allows the probabilities of unsampled trees to be estimated and a better point estimate to be found (Larget, 2013). Berling et al. (2025) developed a related tree distribution parameterized by clade probabilities and showed by simulation that its point estimate performs at least as well as standard summary trees and often better, while noting that which summary performs best depends on the number of sampled trees and the size of the problem. *Median trees* minimize the average distance (for example, the Robinson-Foulds distance) to all trees in the sample; the majority-rule consensus is in fact the median tree under the Robinson-Foulds distance, when non-binary trees are allowed. Each of these makes a different trade-off, and the choice between them matters most when the posterior is diffuse, as it often is for the deeper nodes of language trees.

```
strict <- consensus(boot_trees, p = 1, rooted = TRUE)
majority <- consensus(boot_trees, p = 0.5, rooted = TRUE)
mcc <- maxCladeCred(boot_trees)
mcc_support <- prop.clades(mcc, boot_trees, rooted = TRUE) / length(boot_trees)
maj_support <- prop.clades(majority, boot_trees, rooted = TRUE) /
  length(boot_trees)
# plot a tree with room left of the root, then write each clade frequency on
# its stem branch, on the side away from the parent's vertical line
plot_support <- function(tr, sup = NULL, main) {
  depth <- if (is.null(tr$edge.length)) Ntip(tr) else
    max(node.depth.edgelength(tr))
  plot(tr, cex = 0.85, main = main, label.offset = 0.01 * depth)
  if (is.null(sup)) return(invisible())
  lp <- get("last_plot.phylo", envir = .PlotPhyloEnv)
  nn <- Ntip(tr) + seq_len(Nnode(tr))
  pa <- tr$edge[match(nn, tr$edge[, 2]), 1]
  for (i in seq_along(nn)) {
    a <- if (is.na(pa[i])) c(1.15, 0.5) else
      if (lp$yy[nn[i]] >= lp$yy[pa[i]]) c(1.1, -0.35) else c(1.1, 1.35)
    nodelabels(round(sup[i], 2), nn[i], frame = "none", adj = a,
               cex = 0.75, col = "grey20")
  }
}
# the root value sits in the left margin
op <- par(mfrow = c(1, 3), mar = c(1, 2.2, 2, 0.5), xpd = NA)
plot_support(strict, main = "Strict consensus")
plot_support(majority, maj_support, main = "Majority-rule consensus")
```

```
plot_support(mcc, mcc_support, main = "MCC tree")
```

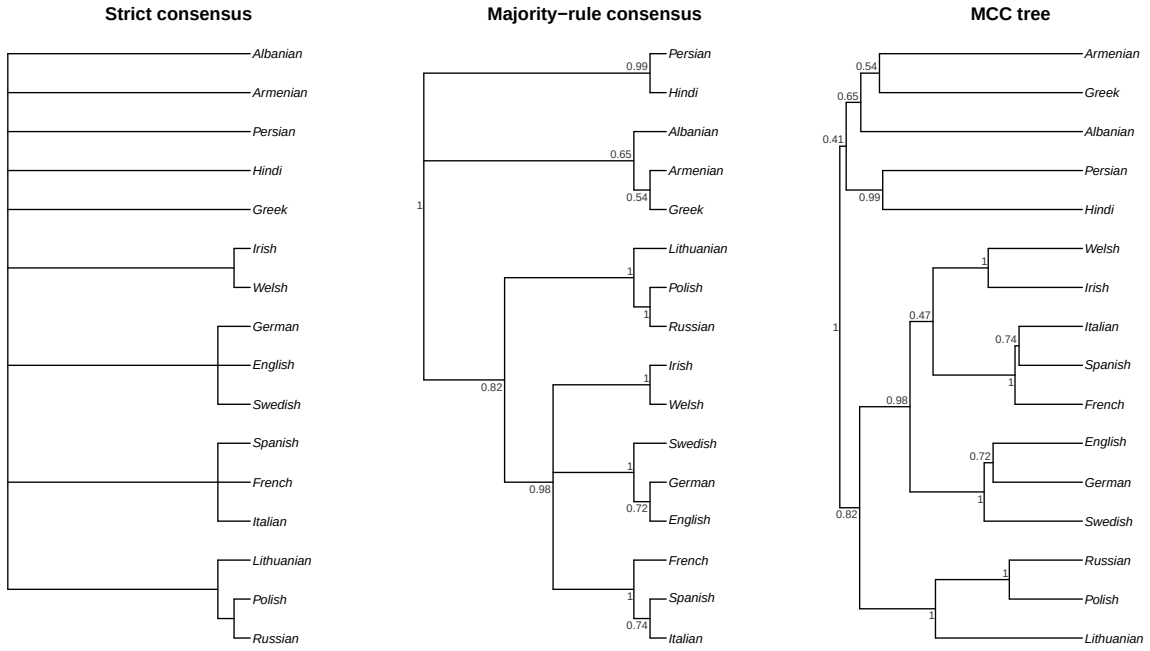


Figure 2.5: Three summaries of the same 200 bootstrap trees. Left: strict consensus, retaining only clades present in all trees. Centre: majority-rule consensus with clade frequencies. Right: maximum clade credibility tree with the frequency of each of its clades among the bootstrap trees.

```
par(op)
c(strict = Nnode(strict), majority = Nnode(majority), mcc = Nnode(mcc))
```

```
#>   strict majority      mcc
#>        6       13       15
```

```
RF.dist(mcc, ie, rooted = TRUE)   # 0 = MCC topology equals the true tree
```

```
#> [1] 0
```

In this simulation the MCC tree recovers the true topology exactly, but the strict consensus resolves only a handful of clades and the majority-rule tree leaves some of the deeper relationships unresolved. Exact recovery here is unsurprising, since the data were generated under the method's assumptions; the pattern of resolution is the more general lesson, and it is typical. Low-level subgroups that share many innovations (Romance, Germanic, Slavic) are found in nearly every tree, while the order of the early splits, here and in real Indo-European, is poorly determined by lexical data.

2.5.4 Visualizing uncertainty directly: DensiTrees and consensus networks

Rather than collapsing a set of trees to one, one can draw all of them. A *DensiTree* plots every tree in the sample on top of the others with partial transparency, so that well supported clades appear as sharp, dark lines and uncertain regions as a blur (Bouckaert, 2010). DensiTrees have

become a standard figure in linguistic phylogenetics, partly because they convey the uncertainty in node ages and topology at a glance. A *consensus network* includes all splits present in more than a chosen proportion of the trees, even when they are incompatible, and draws them as a split network (Holland and Moulton, 2003). Where the trees agree the network is tree-like; where they disagree, boxes appear, and their size reflects how often the conflicting splits occur. Figure 2.6 shows both for our bootstrap sample.

```
op <- par(mfrow = c(1, 2), mar = c(1, 1, 2, 1))
densiTree(boot_trees, alpha = 0.05, consensus = mcc, cex = 0.8,
          scaleX = TRUE, col = "steelblue4")
title("DensiTree")
cnet <- consensusNet(boot_trees, prob = 0.1)
par(xpd = NA)   # labels may run into the margins
plot(cnet, type = "2D", show.edge.label = FALSE, cex = 0.7, direction = "axial",
     edge.width = 1, main = "Consensus network")
```

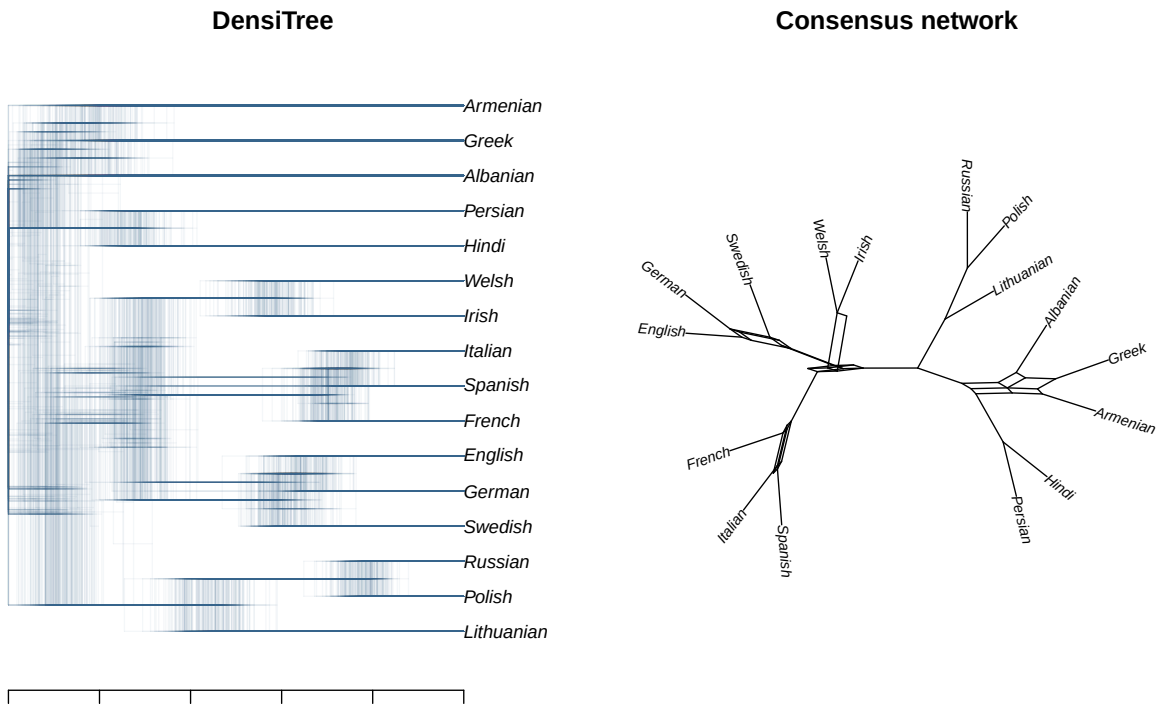


Figure 2.6: Two views of uncertainty in the bootstrap sample. Left: a DensiTree, in which all 200 trees are drawn with transparency. Right: a consensus network of all splits present in at least 10 per cent of the trees; boxes indicate conflicting splits.

```
par(op)
```

Trees as samples rather than single objects. One of the most important consequences of the Bayesian turn in linguistic phylogenetics is that a published "tree" is now usually a posterior sample of thousands of trees. The Austronesian analysis of Gray et al. (2009), for example, produced a posterior distribution of trees for 400 languages from the Austronesian Basic Vocabulary Database. Later comparative studies did not use a single summary tree from this analysis but

repeated their analyses over a sample of posterior trees, so that uncertainty about subgrouping and branch lengths was carried forward into conclusions about cultural evolution. Currie et al. (2010a), for instance, tested competing models of the evolution of political complexity in Austronesian societies across a sample of trees, and Jordan et al. (2009) did the same for post-marital residence. The lesson for readers of the linguistic literature is that a MCC tree in a figure is a convenient picture, but inferences that depend on tree structure should, where possible, integrate over the whole posterior sample (Chapter 13).

2.6 Comparing trees: tree distances

How different are two trees? The question arises constantly: when comparing a phylogeny inferred from lexical data with one inferred from grammatical data, with an expert classification, with a tree from an automated cognate-detection pipeline, or with the true tree in a simulation study. Many metrics exist. They differ in what they measure (topology only, or branch lengths too), whether they apply to rooted or unrooted trees, how they behave for small changes, and how fast they can be computed.

2.6.1 Robinson-Foulds distance

The *Robinson-Foulds* (RF) or *symmetric difference* distance counts the splits that are present in one tree but not the other (Robinson and Foulds, 1981). If $S(\tau)$ is the set of non-trivial splits of tree τ,

$$d_{RF}(\tau_1, \tau_2) = |S(\tau_1) \triangle S(\tau_2)| = |S(\tau_1) \setminus S(\tau_2)| + |S(\tau_2) \setminus S(\tau_1)|. \tag{2.3}$$

For two unrooted binary trees on n tips the maximum is $2(n-3)$, so the *normalized* RF distance divides by this value and ranges from 0 to 1. The rooted version uses clusters instead of splits. RF is fast to compute and easy to interpret, which explains its popularity, but it has well-known weaknesses. It is coarse: moving a single taxon from one end of the tree to the other can destroy every split, giving the maximum distance even though most relationships among the other taxa are preserved. It saturates quickly, so that most pairs of random trees are at or near the maximum distance. And two splits that differ only by the placement of one taxon count as completely different, just like two splits that share nothing. *Generalized RF* distances address this last weakness by giving partial credit to similar splits, for example by matching splits between trees so as to maximize their shared information (Smith, 2020).

2.6.2 Branch-score and path distances

The RF distance ignores branch lengths. The *branch-score distance* of Kuhner and Felsenstein (1994) (often called the Kuhner-Felsenstein or KF distance) includes them: it is the Euclidean distance between the two trees' vectors of split lengths, with a split absent from a tree counted as having length zero. So two trees with the same topology but different branch lengths have KF distance greater than zero, and a short misplaced branch costs little. *Path-length distances* compare instead the matrices of distances between tips, again by a Euclidean distance; the topological version, with path lengths counted in branches, was studied by Steel and Penny (1993). Path distances are more sensitive than RF to how far a taxon has moved, but they are dominated by the tips with the longest paths.

The formulas for the branch-score and path distances, and a from-scratch implementation of the quartet distance below checked against the **Quartet** package, are in Section B.1.2.

2.6.3 Quartet and triplet distances

A *quartet* is a set of four taxa; an unrooted binary tree resolves every quartet $\{a, b, c, d\}$ into exactly one of the three possible topologies $ab \mid cd$, $ac \mid bd$ or $ad \mid bc$. The *quartet distance* is the number (or proportion) of the $\binom{n}{4}$ quartets that are resolved differently in the two trees (Estabrook et al., 1985). The rooted analogue counts *triplets* (rooted three-taxon trees). Quartet distances do not saturate as quickly as RF and degrade gracefully when a single taxon moves, because moving one taxon affects only the quartets that contain it. They are also well suited to comparing a fully resolved inferred tree with a classification containing polytomies, because quartets that the classification leaves unresolved can simply be excluded. This is the basis of the *generalized quartet distance* (GQD) introduced by Pompei et al. (2011) to evaluate automatically inferred language trees against expert classifications, a measure that has since been widely used in computational historical linguistics.

Quartet distances can be computed in $O(n \log n)$ time by sophisticated algorithms, implemented in the R package **Quartet**. Its function `QuartetStatus()` counts, for a pair of trees, the quartets resolved the same way in both (`s`), resolved differently (`d`), and resolved in only one of the two (`r1`, `r2`), out of `Q` quartets in all. The quartet distance between two binary trees is the proportion d/Q; quartets left unresolved by one tree count as disagreements. The GQD instead ignores quartets that the reference classification leaves unresolved, giving $d/(s+d)$ when the inferred tree is fully resolved.

```
library(Quartet)
quartet_distance <- function(t1, t2, generalized = FALSE) {
  qs <- QuartetStatus(t1, cf = t2)   # t2 is the reference
  if (generalized) return(unname(qs[, "d"] / (qs[, "s"] + qs[, "d"])))
  unname((qs[, "d"] + qs[, "r1"] + qs[, "r2"]) / qs[, "Q"])
}
quartet_distance(ie, boot_trees[[1]])
```

```
#> [1] 0.08626
```

```
# A "classification" with a polytomy: collapse the three deepest splits
classification <- di2multi(ie, tol = 0.71)
c(quartet = quartet_distance(boot_trees[[1]], classification),
  GQD = quartet_distance(boot_trees[[1]], classification, generalized = TRUE))
```

```
#> quartet     GQD
#>  0.3099  0.0000
```

Against the classification, the plain quartet distance penalizes the binary bootstrap tree for every quartet that the classification leaves open, whereas the GQD counts only genuine contradictions.

2.6.4 Rearrangement distances

Tree search algorithms move between trees by three standard rearrangements, which are described in detail in Chapter 5. A *nearest-neighbour interchange* (NNI) swaps two subtrees across an internal

branch; each internal branch of an unrooted binary tree admits two NNIs, so a tree has $2(n-3)$ NNI neighbours. A *subtree prune and regraft* (SPR) cuts a subtree off and reattaches it to any other branch; an unrooted binary tree has $2(n-3)(2n-7)$ SPR neighbours (Allen and Steel, 2001). A *tree bisection and reconnection* (TBR) cuts the tree into two subtrees and reconnects them by any pair of branches. Each rearrangement defines a distance: the minimum number of moves needed to transform one tree into the other. These distances have a natural interpretation for borrowing. A single SPR move is the topological signature of one lateral transfer event, because a borrowed character's history is the species tree with one subtree pruned and regrafted elsewhere. Rooted SPR distances are therefore closely connected to the minimum number of reticulations needed to reconcile conflicting trees (Chapter 8). Unfortunately, computing NNI, SPR and TBR distances exactly is NP-hard (Allen and Steel, 2001), so software usually reports approximations or bounds; phangorn's `SPR.dist()` returns an approximation.

2.6.5 Distances in practice

```
other <- boot_trees[[which.max(RF.dist(boot_trees, ie, rooted = TRUE))]]
compare <- function(t1, t2) {
  c(RF = RF.dist(t1, t2),
    nRF = RF.dist(t1, t2, normalize = TRUE),
    KF = KF.dist(t1, t2),
    path = path.dist(t1, t2),
    SPR = unname(SPR.dist(t1, t2)),
    quartet = quartet_distance(t1, t2))
}
dtab <- rbind(`True vs. MCC` = compare(ie, mcc),
              `True vs. one bootstrap tree` = compare(ie, boot_trees[[1]]),
              `True vs. most distant bootstrap tree` = compare(ie, other),
              `True vs. random tree` = compare(ie, rtree(16, tip.label =
                                                 sample(ie$tip.label))))
knitr::kable(round(dtab, 3), caption = paste(
  "Distances between the true toy tree and other trees. RF, nRF",
  "(normalized RF), path and quartet distances use topology only; KF uses",
  "branch lengths and is therefore affected by the scale of the trees (the",
  "bootstrap trees are in units of character differences, not millennia);",
  "the SPR value is an approximation."))
```

Table 2.2: Distances between the true toy tree and other trees. RF, nRF (normalized RF), path and quartet distances use topology only; KF uses branch lengths and is therefore affected by the scale of the trees (the bootstrap trees are in units of character differences, not millennia); the SPR value is an approximation.

	RF	nRF	KF	path	SPR	quartet
True vs. MCC	0	0.000	12.69	0.00	0	0.000
True vs. one bootstrap tree	4	0.154	12.72	9.11	2	0.086
True vs. most distant bootstrap tree	12	0.462	12.69	18.76	3	0.221
True vs. random tree	26	1.000	11.73	31.73	6	0.765

Table 2.2 brings out the differences between metrics. The random tree is at or near the maximum normalized RF distance of 1, while its quartet distance is in the region of the value of about two thirds expected for unrelated trees (two independent random resolutions of a quartet agree with probability one third). The bootstrap trees, by contrast, have small quartet distances even when their RF distances are substantial, because their disagreements concern only a few taxa. The KF distance between the true tree and any bootstrap tree is large even when the topologies agree, because the bootstrap trees measure branch lengths in different units. This is a practical warning: before computing branch-length-sensitive distances, make sure the trees are on a common scale.

How accurate are automatically inferred language trees? Pompei et al. (2011) set out to evaluate trees inferred from ASJP word lists with distance methods against expert classifications for families including Indo-European, Austronesian and several others. Expert classifications are full of polytomies, and the standard RF distance penalizes a binary inferred tree for every split that the classification leaves unresolved. They therefore introduced the generalized quartet distance, which counts only disagreements on quartets that the classification resolves. The metric made it possible to compare, on a common scale, how well different distance measures and tree-building algorithms recovered accepted classifications, and quartet-based scores of this kind have become a standard way of evaluating computational pipelines against Glottolog (see also Jäger, 2018). The approach also has a limitation that users should keep in mind: a good GQD score means agreement with the expert consensus, which is itself a hypothesis, not a known truth. Nor is the agreement an independent test when the inferred tree is built from cognate judgements, which are themselves products of the comparative method that produced the classification, and when cognate sharing (retention as well as innovation) is used in place of the shared innovations on which subgroups are defined (Section 1.9). A good score is consistent with the pipeline capturing genealogical signal; it does not demonstrate it.

2.7 Tree shape and balance

The *shape* of a tree, ignoring tip labels, carries information about the process that generated it. A tree in which every split divides its tips roughly evenly is *balanced*; one in which each split peels off a single tip is maximally unbalanced, a “caterpillar” or “comb”. Tree shape matters in two ways. First, it is evidence about diversification: if some lineages split more often than others (because of differences in population size, geography or social organization), trees become more unbalanced than expected under a model in which all lineages are equally likely to split (Mooers and Heard, 1997). Second, tree shape interacts with tree priors and inference: the Yule and coalescent priors used in Bayesian analyses imply particular distributions of shape, and an analysis may be biased if the true process differs strongly.

2.7.1 Balance indices

The two classic indices are Colless’s and Sackin’s. For each internal node v of a rooted binary tree, let L_v and R_v be the numbers of tips in its left and right subtrees. The *Colless index* (Colless, 1982) is

$$I_C(\tau) = \sum_{v} |L_v - R_v|, \tag{2.4}$$

and the *Sackin index* (Sackin, 1972) is the sum over tips of the number of internal nodes between

the tip and the root, which equals the sum over internal nodes of the number of tips they subtend, $I_S(\tau) = \sum_v (L_v + R_v)$. Both measure imbalance: a perfectly balanced tree minimizes them, a caterpillar maximizes them. Because their values grow with n, they are compared with their expected values under null models. The two standard null models are the *Yule model*, a pure-birth process in which every lineage splits at the same rate (Chapter 6; Section A.8 reviews the Poisson process behind it), and the *proportional-to-distinguishable-arrangements* (PDA) model, under which every labelled rooted topology is equally likely. PDA trees are on average much less balanced than Yule trees. Interestingly, the Kingman coalescent produces the same distribution of labelled topologies as the Yule model; the two differ only in branch lengths. Empirical trees in biology tend to be somewhat less balanced than Yule trees but more balanced than PDA trees (Mooers and Heard, 1997).

In **ape**, the function `balance()` returns L_v and R_v for every internal node, so both indices are one-liners. To compare with null models we simulate Yule-distributed topologies with `rcoal()` and PDA topologies with `RandomTree()` from the **TreeTools** package, which draws a binary topology uniformly at random by inserting each tip on a randomly chosen branch; rooting it on a random branch (`root = TRUE`) gives a uniformly random rooted topology.

```
colless <- function(tree) { b <- balance(tree); sum(abs(b[, 1] - b[, 2])) }
sackin  <- function(tree) sum(balance(tree))
c(Colless = colless(ie), Sackin = sackin(ie))
```

```
#> Colless  Sackin
#>      19      69
```

```
set.seed(1)
n_tips <- 30
colless_yule <- replicate(500, colless(rcoal(n_tips)))
colless_pda  <- replicate(500,
  colless(TreeTools::RandomTree(n_tips, root = TRUE)))
rbind(Yule = summary(colless_yule), PDA = summary(colless_pda))
```

```
#>      Min. 1st Qu. Median   Mean 3rd Qu. Max.
#> Yule   25    56.0   69.0  71.51      85  156
#> PDA    60   127.8  160.5 164.37     195  340
```

A from-scratch implementation of Rémy's algorithm for uniform (PDA) random trees, which reproduces this comparison without **TreeTools**, is in Section B.1.3.

```
brks <- seq(0, max(colless_pda) + 10, by = 10)
hist(colless_pda, breaks = brks, col = NA, border = "grey15", main = "",
     xlab = "Colless index (30 tips)", las = 1, ylim = c(0, 160))
hist(colless_yule, breaks = brks, col = adjustcolor("grey50", 0.6),
     border = "white", add = TRUE)
legend("topright", c("Yule (= coalescent topologies)", "PDA (uniform)"),
       fill = c(adjustcolor("grey50", 0.6), NA), border = c("white", "grey15"),
       bty = "n")
```

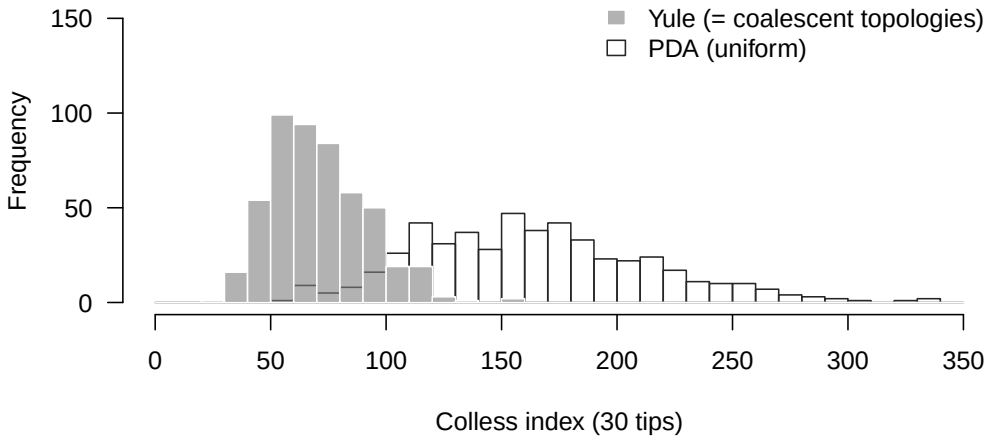


Figure 2.7: Distributions of the Colless imbalance index for 500 random 30-tip trees under the Yule model (grey) and the uniform PDA model (outline). Uniformly random topologies are much less balanced than those produced by a branching process in which all lineages split at the same rate.

2.7.2 Tempo: lineages-through-time and the gamma statistic

Balance describes topology. For chronograms, the distribution of node ages describes the *tempo* of diversification. A *lineages-through-time* (LTT) plot shows the number of lineages present at each point in the past, on a log scale (Nee et al., 1992). Under a pure-birth process with complete sampling the log number of lineages grows approximately linearly; an early burst of splitting followed by a slowdown produces a curve that rises steeply and then flattens, while a recent acceleration (or extinction removing old lineages) produces upturns near the present. The γ statistic of Pybus and Harvey (2000) summarizes this: it is approximately standard normal under a constant-rate pure-birth model, negative values indicating that nodes are concentrated towards the root and positive values that they are concentrated towards the tips. **ape** computes it with `gammaStat()` and draws LTT plots with `ltt.plot()`. Incomplete sampling of languages biases γ downwards, because unsampled lineages are disproportionately the young ones, a point that matters for language families in which only a fraction of the varieties have been documented.

```
set.seed(7)
yule_tree <- phytools::pbtree(n = 40, scale = 1)
coal_tree <- rcoal(40); coal_tree$edge.length <- coal_tree$edge.length /
  max(branching.times(coal_tree))
ltt.plot(yule_tree, log = "y", xlab = "Time (root = -1)", ylab = "Lineages",
         las = 1)
ltt.lines(coal_tree, lty = 2)
legend("topleft", c(sprintf("Yule, gamma = %.2f", gammaStat(yule_tree)),
                    sprintf("Coalescent, gamma = %.2f", gammaStat(coal_tree))),
       lty = 1:2, bty = "n")
```

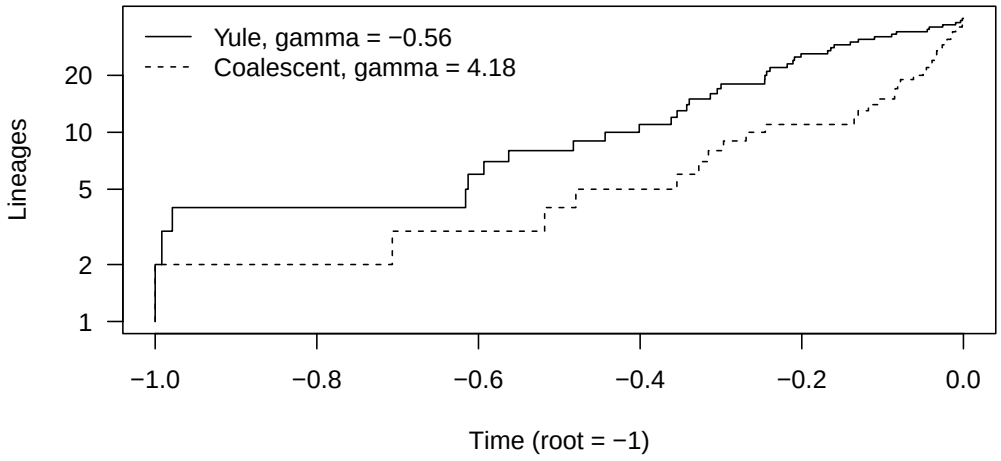


Figure 2.8: Lineages-through-time plots (log scale) for a 40-tip tree from a pure-birth (Yule) process and a 40-tip tree from the Kingman coalescent. The coalescent compresses splits towards the present, giving a strongly positive gamma statistic.

Tree shape has been used in linguistics far less than in biology, but it has been used. The most influential example concerns not balance itself but a related statistic of tree shape, the number of nodes on each root-to-tip path.

Punctuational bursts of lexical change. Atkinson et al. (2008) asked whether languages change faster at the moment they split, for instance because a newly separated community marks its identity through linguistic innovation, or because a founder population undergoes rapid change. Their test exploits tree shape. If change is concentrated at splits, then the total amount of change along a root-to-tip path, the path length in a phylogram, should increase with the number of nodes along that path, even after accounting for elapsed time. Using Bayesian posterior trees for Bantu, Indo-European and Austronesian lexical data, they found a significant positive relationship in all three, and estimated that punctuational bursts at splitting events account for a substantial share of total lexical divergence. The method had been developed for molecular data by Pagel et al. (2006), and requires care, because the number of nodes on a path is affected by sampling density and by artefacts of tree reconstruction (the "node-density effect"). Chapter 9 returns to this result and to the question of what it means for dating.

Tree-shape statistics as evidence about diversification. In evolutionary biology the shape and tempo of trees are routinely used to test hypotheses about diversification: whether clades with a particular trait split faster, whether diversification slowed as ecological niches filled, whether mass extinctions left a signature. Statistics such as Colless's index and γ, LTT plots, and their model-based successors (Chapter 16) have, to our knowledge, rarely been applied systematically to language trees. Yet linguistic questions abound. Do language families of hunter-gatherers have different tree shapes from those of agriculturalists? Do families that expanded into empty or sparsely occupied regions (Polynesian, Bantu) show early-burst signatures that families diversifying in place do not? Are classifications from Glottolog more or less balanced than Bayesian trees of the same families, and what does that say about the classification process? A caution is that

language trees, unlike most species trees, are heavily affected by the decisions about what counts as a separate language (Chapter 3), so shape statistics would have to be shown to be robust to those decisions.

2.8 Tree space

2.8.1 Neighbourhoods, islands and landscapes

Imagine all possible trees for a set of languages as the points of a space, with two trees adjacent if one NNI (or SPR, or TBR) move transforms one into the other. A tree search is then a walk through this space, and the optimality score (parsimony length, likelihood or posterior density) defines a *landscape* over it. The landscape for real data is often rugged: there may be several *islands* of good trees separated by valleys of poor ones, so that a hill-climbing search that starts on one island never finds the others. Bayesian MCMC faces the same problem as poor mixing between modes. The structure of tree space therefore has direct practical implications: it is the reason searches are repeated from many starting trees, why SPR and TBR moves are used in addition to NNI (they make bigger jumps), and why Bayesian analyses run multiple independent chains and check that they agree (Chapter 6).

A useful way to look at a collection of trees is to compute all pairwise distances and project them into two dimensions by multidimensional scaling (MDS), so that trees close in the plot are similar (Hillis et al., 2005). Figure 2.9 does this for our bootstrap trees, using RF distances, and marks each tree according to whether it contains a particular clade (here Greek plus Armenian, a grouping that the simulated data support only weakly). Such plots reveal whether a posterior sample consists of one cloud or several, and whether two analyses (different data, models or software) occupy the same region of tree space.

```
all_trees <- c(boot_trees, ie)
rf <- as.matrix(RF.dist(.compressTipLabel(all_trees), rooted = TRUE))
xy <- cmdscale(rf, k = 2)
has_ga <- sapply(boot_trees, is.monophyletic,
                 tips = c("Greek", "Armenian"))
set.seed(3)
plot(jitter(xy[-201, ], amount = 0.25), pch = ifelse(has_ga, 16, 1),
     col = "grey25", xlab = "MDS axis 1", ylab = "MDS axis 2", las = 1,
     ylim = range(xy[, 2]) + c(-0.5, 0.45 * diff(range(xy[, 2]))))
points(xy[201, 1], xy[201, 2], pch = 4, cex = 2, lwd = 3)
legend("topleft", c("contains Greek + Armenian", "does not", "true tree"),
       pch = c(16, 1, 4), col = c("grey25", "grey25", "black"), bty = "n",
       cex = 0.85)
```

```
mean(has_ga)   # proportion of trees with the clade
```

```
#> [1] 0.54
```

2.8.2 The geometry of tree space

Distances defined by counting moves give tree space the structure of a graph. Billera et al. (2001) showed that the space of trees *with branch lengths* has a much richer, continuous geometry, now

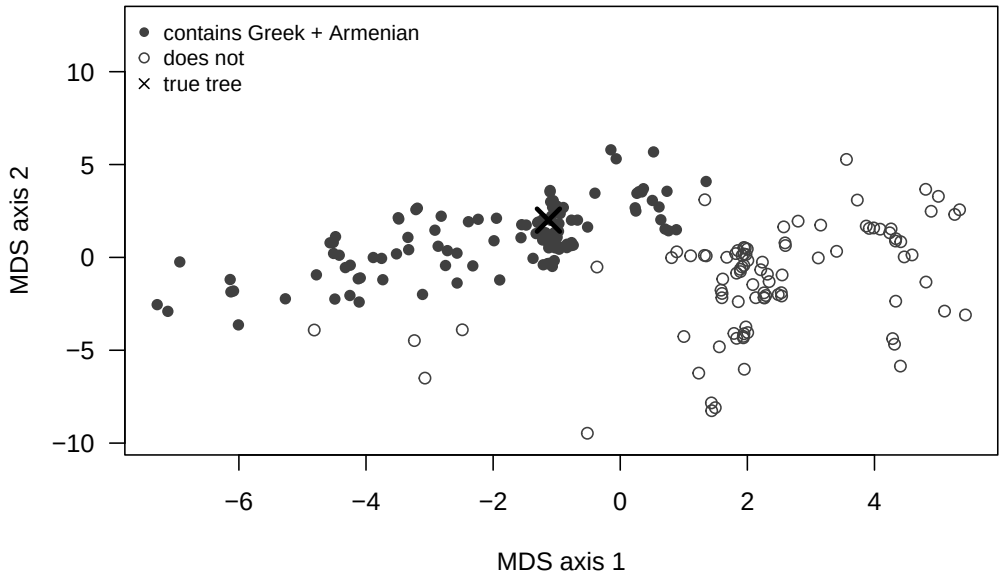


Figure 2.9: Multidimensional scaling of Robinson-Foulds distances among the 200 bootstrap trees. Trees containing the clade Greek + Armenian are shown as filled circles, others as open circles; the true tree is marked with a cross. Points are jittered slightly because many trees share the same topology.

called *BHV tree space*, in which trees with different topologies are connected through trees with polytomies and there is a unique shortest path between any two trees (Owen and Provan, 2011). This makes it possible to define genuinely geometric means and variances of tree samples. For rooted time trees, other metrics have been designed to capture differences in both topology and timing, such as the metric of Kendall and Colijn (2016), which compares the depths of the MRCAs of all pairs of tips.

The construction of BHV tree space from orthants, its geodesics and what they make possible are described in Section B.1.4.

Statistics in tree space. Biologists have developed a toolkit for exploring and testing distributions of trees directly: MDS and clustering of tree samples to find distinct islands or "tree landscapes", geometric (Fréchet) means and medians in BHV space, metrics that weight differences near the root more heavily than differences near the tips, and permutation tests of whether two sets of trees (say, gene trees from two genomic regions) differ more than expected. The **treespace** R package implements several of these (Jombart et al., 2017). To our knowledge, such methods have rarely if ever been used in linguistic phylogenetics, though the questions they answer are familiar ones. Do trees inferred from different semantic domains of the lexicon (body parts, kinship terms, numerals), or from lexical versus structural data, occupy different regions of tree space? Is the posterior from one model (say, covarion) distinguishable from that of another? Do automated and expert cognate codings lead to systematically different trees, or just noisier ones? Comparing whole distributions of trees, rather than two summary trees, is the statistically appropriate way to answer these questions.

2.9 Beyond trees: networks

Trees are not always adequate models of language history. Languages borrow words and structures from their neighbours; dialect continua split gradually, with innovations spreading across them in overlapping waves (Chapter 1); and some languages, such as mixed languages, have two or more parents in a meaningful sense. *Phylogenetic networks* generalize trees to represent such histories. It is useful to distinguish two kinds (Huson and Bryant, 2006; Huson et al., 2010).

Explicit (or *evolutionary*) networks are directed graphs in which some nodes, *reticulation nodes*, have two or more parents. Each reticulation represents an event: hybridization or lateral gene transfer in biology, borrowing, language mixing or dialect convergence in linguistics. A character's history follows one of the trees displayed by the network. Explicit networks make strong historical claims, and inferring them is computationally hard.

Implicit (or *data-display*) networks make no claims about specific events. They are visualizations of conflicting signal. The most important for linguistics are *split networks*, which represent a weighted collection of splits that need not be compatible. Each split is represented by a set of parallel edges whose length reflects the split's weight; compatible splits produce tree-like regions, and incompatible ones produce boxes. Split networks are built from distances by NeighborNet (Bryant and Moulton, 2004), from characters by methods such as median networks, or from tree sets as consensus networks (Figure 2.6). A box in a split network says that the data contain conflicting signals, but not why: borrowing, parallel change, coding error and genuine uncertainty can all produce boxes. NeighborNet analyses have been especially influential in linguistics, both as exploratory tools and as the basis of measures of how tree-like a dataset is (Bryant et al., 2005; Gray et al., 2010). Chapter 8 covers networks, treelikeness and explicit models of borrowing in

detail.

A network is not automatically more realistic than a tree. A NeighborNet or consensus network shows conflict in the data, but a reader can easily over-interpret its boxes as evidence of contact between particular languages. Conversely, a tree can be a useful summary of a history that included some borrowing, provided the borrowing is modest and not concentrated along particular branches. The choice between trees and networks should be driven by the question, by measures of treelikeness, and by explicit models when they are available, not by the aesthetic appeal of one picture or the other.

2.10 Trees in linguistic practice: a short guide

The vocabulary and tools of this chapter appear in every later one. A few practical observations draw them together.

First, the trees reported in linguistic publications come from different sources and should be read differently. An expert classification, such as Glottolog's, is a topology with soft polytomies and no branch lengths; it encodes accepted subgroups, not estimated distances. A distance or parsimony tree is a single point estimate whose support is quantified by bootstrap proportions. A Bayesian MCC tree is a summary of a posterior sample, typically a chronogram with node ages and credible intervals; its clade posterior probabilities refer to the model and data used, and are often much higher than bootstrap proportions for the same clades.

Second, support values are properties of clades, not of trees. A tree in which every clade has posterior probability 0.9 may still have a very low probability as a whole. Statements about the "topology" should always say which clades are meant.

Third, crown ages, stem ages and the effect of language sampling should be kept distinct when interpreting dates.

Fourth, distances between trees should be chosen to fit the question. RF is a reasonable default for topologies; quartet-based measures are better when comparing to classifications with polytomies or when a few rogue taxa move around; branch-length-sensitive measures require trees on a common scale.

Fifth, because tree space is vast and searches are heuristic, every serious analysis should report how its trees were searched or sampled and how convergence was assessed (Chapters 5 and 6).

Visualizing a posterior: Pama-Nyungan. Bowern and Atkinson (2012) used Bayesian phylogenetic methods on a large cognate-coded dataset of Pama-Nyungan languages to investigate the internal structure of the family, which covers most of the Australian continent and whose subgrouping had long been contested. Their results combined strongly supported low-level subgroups, many of which matched those proposed by specialists, with considerable uncertainty about the relationships among them. The match is reassuring but not an independent validation: the cognate judgements were made by specialists working within the same comparative tradition, and shared cognates include retentions as well as innovations (Section 1.9). A follow-up study with a larger sample, which added dates and continuous phylogeography, located the family's origin and traced its spread across Australia (Bouckaert et al., 2018) (Chapter 11). Analyses of this kind illustrate why figures such as DensiTrees and consensus networks, rather than single summary trees, have become common in linguistic phylogenetics: the scientifically interesting statement is often precisely that some parts of the tree are well determined and others are not.

- A phylogenetic tree is a connected acyclic graph whose tips are languages and whose internal nodes are proto-languages; clades correspond to subgroups, and the cladistic principle that clades are defined by shared derived characters mirrors the linguistic principle of subgrouping by shared innovations.
- Trees may be rooted or unrooted; rooting language trees is difficult because true outgroups are rarely available, and is usually done by clock models, directional models, prior constraints or ancient languages.
- Branch lengths may be absent (cladograms), proportional to change (phylograms) or proportional to time (chronograms); dates may refer to crown or stem nodes and depend on sampling.
- Newick strings and NEXUS files are the standard formats; classification trees from Glottolog have polytomies and no branch lengths, which must be invented with care for comparative analyses.
- The number of possible trees grows super-exponentially, so inference relies on heuristics or sampling, and even well supported analyses have low probability for any single complete tree.
- A tree is equivalent to a set of compatible splits; incompatible splits are the raw material of split networks.
- Sets of trees are summarized by strict, majority-rule or extended majority-rule consensus trees or by MCC trees, and visualized with DensiTrees and consensus networks; these summaries discard different information.
- Tree distances (RF, generalized RF, KF branch score, path, quartet, SPR) measure different aspects of tree difference; quartet-based measures are well suited to comparisons with classifications.
- Tree shape (balance, LTT, gamma) carries information about diversification and is underexploited in linguistics.
- A single summary tree is a picture, not a known history: downstream analyses that depend on tree structure should be repeated over a sample of trees, and agreement of a summary tree with an expert classification is not independent validation.

2.11 Exercises

Conceptual

1. A handbook lists twelve primary branches of Indo-European as a flat list. Explain whether this is a hard or a soft polytomy, and why the distinction matters for a comparative method that requires a fully resolved tree.
2. A paper reports that “Proto-Germanic dates to 2,400 years ago” from a tree containing English, Dutch and German only. What exactly has been dated? Distinguish crown and stem ages, and say how adding Gothic or Icelandic would change the answer.
3. Why does an unrooted tree inferred under a reversible model not tell us where a language family began? List three ways in which linguists place the root, and the assumption each makes.
4. Explain, using the splits-equivalence theorem, why a binary cognate character that is incompatible with a tree must have experienced borrowing, parallel change, loss or coding error. Why can loss create apparent incompatibility in an unrooted analysis but not in a rooted one?

5. A study reports the MCC tree of a Bayesian analysis and then reconstructs ancestral word orders on it. What information has been discarded, and how could the reconstruction be distorted even if the MCC topology looks plausible?
6. Two inferred trees have normalized RF distance 1 but quartet distance 0.1. Describe a situation that produces this combination and say which number better describes their similarity.

Computational

1. Using the chapter's tree `ie`, compute the crown age of Germanic and of Balto-Slavic with `getMRCA()` and `branching.times()`. Then drop Swedish with `drop.tip()` and recompute the crown age of Germanic. Explain the change.
2. Root `u <- unroot(ie)` on each of its 16 tips in turn with `root(u, outgroup = tip, resolve.root = TRUE)` and count, with `is.monophyletic()`, in how many of these rootings Romance (Italian, Spanish, French) is a clade. Which rootings break it, and why? Relate the exercise to the statement that an unrooted tree corresponds to as many rooted trees as it has branches.
3. Repeat the simulation of Section 2.5 with `simSeq()` on `ie` for 100, 400 and 1,600 characters. For each, compute the number of nodes of the strict and majority-rule consensus (`consensus()`) and the RF distance of the MCC tree (`maxCladeCred()`, `RF.dist()`) to `ie`. Which kind of validation, in the vocabulary of Section 6.10, does this experiment provide?
4. Convert the Glottolog-style Germanic tree `germ` into three trees with branch lengths: `compute.brlen(germ, method = "Grafen")`, `compute.brlen(germ, 1)` and a random resolution with `multi2di()`. Compare their path distances with `path.dist()` and their quartet distances with the chapter's `quartet_distance()`, and explain why the quartet distance ignores the branch-length choices.
5. Compute the Colless index (`colless()` from the chapter) of 200 Yule trees (`rcoal()`) and 200 uniform trees (`TreeTools::RandomTree(n, root = TRUE)`) for $n = 10$ and $n = 50$, and compare the two distributions with `boxplot()`. How does the separation between the null models change with n?
6. For $n = 4$, 5, 10 and 24 languages, compute the numbers of rooted and unrooted binary topologies with `ape::howmanytrees()` and check them against `log10_unrooted()` and `log10_rooted()` from the chapter. Check the numbers of internal nodes and branches of Section 2.1 on random trees from `rtree()` with `Nnode()` and `Nedge()`, rooted and unrooted. Under a uniform prior on unrooted topologies, what is the prior probability of any one topology for 5 and for 24 languages? Use the result to explain why a posterior probability of 0.3 for a single 24-language topology represents an enormous update, and why reporting clade probabilities is nonetheless more informative.

Research

1. You are asked to compare a Bayesian tree of 120 Bantu languages with the Guthrie zones and with Glottolog's classification. Which tree distance would you use, how would you handle the polytomies in the classifications, and how would you incorporate the uncertainty in the Bayesian posterior rather than comparing a single summary tree? Justify each choice.
2. A colleague proposes to test whether language families of hunter-gatherers have less balanced trees than those of farmers, using Glottolog classifications. Identify the main threats to validity (classification practice, the language/dialect problem, incomplete sampling, polytomies) and design an analysis, including null models and robustness checks, that would

address them.

2.12 Further reading

- Felsenstein (2004), *Inferring Phylogenies*. The standard reference on phylogenetic inference; its early chapters on counting trees, consensus and tree comparison are a lucid introduction to the topics of this chapter.
- Baum and Smith (2013), *Tree Thinking*. An introductory textbook on reading and inferring trees, unusually careful about common misreadings (tip order, node counting, "primitive" lineages, similarity versus relatedness); its early chapters are the natural companion to Section 2.1. Warnow (2017), *Computational Phylogenetics*, is a more algorithmic textbook on how tree-estimation methods are designed and on their statistical properties.
- Semple and Steel (2003), *Phylogenetics*. A rigorous mathematical treatment of trees, splits, compatibility and the splits-equivalence theorem, for readers who want proofs.
- Paradis (2012), *Analysis of Phylogenetics and Evolution with R*. The book-length guide to **ape** and the `phylo` data structure, with worked examples of reading, manipulating and plotting trees.
- Bryant (2003): A clear classification of consensus methods, showing what each assumes and which properties it satisfies.
- Heled and Bouckaert (2013): Explains, with simulations, how summary trees and their node heights are constructed from Bayesian posterior samples and why the choice matters; essential for anyone reporting MCC trees.
- Huson et al. (2010), *Phylogenetic Networks*. A comprehensive introduction to split networks and explicit networks, written by the authors of SplitsTree and Dendroscope.
- Mooers and Heard (1997): A classic review of what tree shape can and cannot reveal about evolutionary processes.
- Pompei et al. (2011): A linguistic application of quartet distances to evaluate inferred language trees against expert classifications.
- Hillis et al. (2005): A short, readable introduction to visualizing tree space with multidimensional scaling.

Chapter 3

Linguistic Data for Phylogenetics

Phylogenetic methods do not operate on languages; they operate on data matrices. Every analysis in this book begins with a table whose rows are languages and whose columns are *characters*, and every result depends on how that table was built: which varieties were chosen as rows, which features as columns, how each cell was judged, and how uncertain those judgements were. Two analyses with identical models but differently built matrices can disagree more than two analyses with different models and the same matrix. This chapter is about the decisions that go into the matrix and about the evidential weight that different kinds of characters can bear. A small hand-typed Indo-European word list, turned into a cognate matrix and an automatic cognate clustering in R, lets us watch each decision in action.

After reading this chapter you should be able to:

- Explain why cognacy, not similarity, is the linguistic counterpart of homology, and how the depth of cognate judgements affects the characters.
- Decide which varieties to sample as taxa, including how to treat attested ancient languages, reconstructed proto-languages and dialect continua.
- Code a word list as binary and multistate cognate characters in R, with loans kept, coded as missing or dropped, and describe the consequences of each choice.
- Identify the main sources of uncertainty in cognate judgements and of dependence between characters, and say which models or analyses account for them.
- Compare the evidential status of lexical, phonological, morphological and syntactic characters, including the Parametric Comparison Method.
- Describe CLDF, Glottolog, Concepticon and Lexibank, and evaluate automated cognate detection with B-cubed scores.

3.1 From words to characters

3.1.1 Taxa, characters and states

In biology a *character* is a heritable feature that varies among taxa, and its values are *character states*. A nucleotide position in an aligned gene is a character with four states (A, C, G, T); the presence or absence of wings is a character with two. The data for a phylogenetic analysis form a matrix D whose entry D_{ij} is the state of character j in taxon i. Almost every model in this book assumes that characters evolve independently along the same tree, so that the probability of the

whole matrix is a product over characters (Chapter 5). The two fundamental requirements on a character are therefore that its states in different taxa be *homologous*, that is, comparable because they descend from a common ancestral state, and that characters be (approximately) independent of one another.

In linguistics the taxa are languages, but even that is not straightforward. A "language" in a dataset is really a particular variety as documented in a particular source, at a particular time, by a particular fieldworker or lexicographer. The term *doculect* has been proposed for this unit (Cysouw and Good, 2013). Two datasets that both contain "Hindi" may contain different doculects, and a dataset may contain several varieties that others would treat as dialects of one language. These decisions affect tree shape, the apparent density of sampling, and the estimates of rates and dates (Chapters 2 and 15).

The characters can be of many kinds. The vast majority of published linguistic phylogenies use *lexical* characters derived from cognate judgements in basic vocabulary, because these are plentiful, comparatively easy to collect for many languages, and relatively resistant to borrowing. Smaller numbers of studies use *structural* (typological) characters, *phonological* characters such as sound changes, or *morphological* characters such as paradigm irregularities. Each kind of data comes with its own notion of homology and its own pitfalls.

3.1.2 Cognacy as homology

The linguistic counterpart of homology is *cognacy*. Two words in different languages are *cognate* if they descend, by ordinary inheritance, from the same word in a common ancestral language. English *two*, German *zwei*, Latin *duo*, Russian *dva* and Sanskrit *dvá* are cognate because they all continue the Proto-Indo-European word for 'two', usually reconstructed as **dwóh*$_1$. Their resemblance in form is a consequence of common descent, not the criterion for it. The criterion is the existence of *regular sound correspondences*: systematic, recurring relationships between the sounds of words with related meanings across the languages, of the kind formulated as sound laws (Campbell, 2013; Hock and Joseph, 1996). English *t* corresponds to German *z* [ts] not only in *two/zwei* but in *ten/zehn*, *tongue/Zunge*, *tooth/Zahn* and *to/zu*, a correspondence explained by the High German consonant shift. Once such correspondences have been established, they allow cognates to be recognized even when they look nothing alike (English *wheel* and Sanskrit *cakrá-* 'wheel' are cognate), and non-cognates to be rejected even when they look identical.

The rejection of look-alikes is as important as the recognition of cognates. English *have* and Latin *habere* 'to have' are similar in sound and meaning but are not cognate; the regular correspondences show that *have* goes back to the same root as Latin *capere* 'to take', while *habere* continues a different root. English *day* and Latin *dies* 'day' are likewise unrelated, as are English *much* and Spanish *mucho*, or English *bad* and Persian *bad* 'bad'. Chance resemblances of this kind are surprisingly frequent: with thousands of words in each language and a limited inventory of sounds, some pairs will match by accident (Ringe, 1992; Kessler, 2001). This is why phylogenetic analyses of lexical data are almost always built on expert cognate judgements grounded in the comparative method, or on automated methods that attempt to emulate it (Section 3.12), rather than on raw similarity of form.

Cognacy is also a matter of depth. English *hound* and German *Hund* are cognate within Germanic; Latin *canis*, Greek *kúōn* and Lithuanian *šuõ* go back to the same Proto-Indo-European root for 'dog'. Whether a dataset places all of these in one cognate set depends on the time depth at which cognacy is judged. Most datasets aimed at a single family code cognacy at the level of the family's

proto-language (so that *Hund* and *canis* are cognate), but there are exceptions, and analyses of subgroups sometimes use shallower judgements. The choice matters because deeper judgements create fewer, larger cognate sets and therefore fewer characters with different distributions across the tree.

3.2 Choosing and coding the taxa

The rows of the matrix deserve as much thought as the columns. Phylogenetic models assume that each tip is a lineage that evolved independently of the others after it split from its sister, and that the tips are sampled in a way the tree prior can describe (Chapter 6). Linguistic taxa fit this picture to varying degrees, and three situations need particular care.

3.2.1 Dialects, languages and sampling density

Whether two varieties count as one language or two is partly a political decision, and dialect continua have no natural cut points (Chapter 1). The choice still has consequences. Sampling many closely related varieties from one branch and one variety from another changes the shape of the tree, shortens some terminal branches and lengthens others, and can bias rates and dates, because the tree prior and the clock model both see the sample as if it were representative. A single variety standing for a whole branch also has a long terminal branch, which makes it vulnerable to long-branch attraction (Section 4.14). A useful rule is to decide the sampling scheme from the question: for the deep structure of a family, sample each major branch with several well-documented and well-separated varieties, preferably including its most divergent ones; for the internal history of one subgroup, sample its dialects densely but think about whether a tree is the right model at all (Chapter 8).

3.2.2 Attested ancient languages: ancestors, sisters or members of a dialect network?

Written records give linguistics something that most of biology lacks: dated observations of earlier stages of the same lineages. But an attested ancient language can stand in three different relationships to the modern languages in a dataset, and they call for different treatments.

- *An ancestor, or nearly so.* The attested variety lies on the path from the root to some modern tips, with no or very little independent change of its own. The Old Norse of medieval Icelandic manuscripts, in relation to Modern Icelandic, is close to this case. In a tree it is a *sampled ancestor*: a tip with a zero-length branch, or a node on a branch.
- *A sister.* The attested variety belongs to a lineage that split off before the common ancestor of the modern languages and then died out. Gothic, the only well-attested East Germanic language, is the sister of the ancestor of North and West Germanic, not the ancestor of either; Hittite and the other Anatolian languages left no descendants at all. Such varieties are tips on side branches that end at their date of attestation.
- *A member of a dialect network.* The attested variety is one standardized or literary variety among many spoken ones, and the modern languages descend from its unrecorded neighbours. Classical Latin is the literary language of a speech community whose spoken varieties gave rise to the Romance languages; the West Saxon of most Old English texts is not the direct ancestor of Modern Standard English, which descends mainly from Midland dialects;

and Classical Arabic stands in a similar relation to the modern Arabic dialects. Here neither "ancestor" nor "sister" is literally true: the attested variety and the real ancestor were near-contemporaries in one network, sharing most of their vocabulary but not all of it.

These possibilities affect sampling, coding and interpretation. A tree prior without sampled ancestors forces a nearly ancestral variety onto a side branch with a length of its own. In dated analyses this can push the root deeper, which is the core of the debate over the "ancestry constraints" of Chang et al. (2015) and the sampled-ancestor analyses of Heggarty et al. (2023) (Sections 10.6 and 10.6.4). Coding raises its own problems. The vocabulary of an ancient language is known from a corpus, not from elicitation. A meaning that is unattested in the surviving texts must be coded as missing (?), never as absent, and the word that a corpus happens to use for a meaning may be a literary or genre-specific choice rather than the everyday basic word. The phonology behind an ancient orthography is itself a reconstruction, so cognate judgements involving ancient forms carry extra uncertainty. And the "date" of an ancient tip is the date range of a corpus that may span centuries, which is better represented by a prior on the tip age than by a fixed number (Section 10.5.1). Ancient languages typically have much more missing data than modern ones, and the way missing data enter the likelihood can influence where they are placed (Section 10.5.3).

Whether the data can distinguish the three situations is an empirical question that deserves to be asked explicitly. Under a fossilized birth-death prior the probability that a tip is a sampled ancestor is, in the vocabulary of Section 6.10, *identifiable under the model*, but with a few hundred lexical characters and much missing data it may not be *practically estimable*: the posterior probability of ancestry can then be driven more by the prior on the sampling process than by the words. Two practical consequences follow. Report the prior as well as the posterior probability that each ancient tip is an ancestor, and check how dates change when ancient varieties are treated as ancestors, as sisters, or are removed and used only as calibrations. And interpret a "sampled ancestor" for a variety such as Classical Latin as a statement that it is very close to the ancestral network, not as a claim that the Romance languages descend from Cicero's prose.

3.2.3 Reconstructed proto-languages are not data

Some databases include reconstructed proto-languages alongside attested ones; the Polynesian dataset used in Chapter 17, for example, contains a Proto-Polynesian "variety". A reconstruction is an inference from the daughter languages, made with the comparative method and usually with a particular subgrouping in mind. Including it as a tip counts the daughters' evidence twice and builds the expected answer into the data. Reconstructions are valuable for checking ancestral state estimates (Chapter 12) and for designing cognate sets, but they should be removed before inferring a tree.

3.3 Basic vocabulary lists

3.3.1 The Swadesh lists

Lexical phylogenetics inherited its data from lexicostatistics, and in particular from the word lists devised by Morris Swadesh (Chapter 1). The idea was to compare languages on a fixed list of *basic* meanings: concepts expected to be lexicalized in every language, independent of culture and environment, and resistant to replacement by borrowing. The original 200-item list (Swadesh, 1952) contains pronouns, body parts, numerals, natural phenomena, basic actions and properties.

Swadesh later reduced it to 100 items (Swadesh, 1955), dropping meanings that proved hard to elicit unambiguously, culturally specific, or prone to borrowing. The Swadesh 100 and 200 lists, and variants of them, remain the most widely used meaning lists in the field.

Swadesh lists have well-known problems. Many items are ambiguous as glosses: 'lie' (recline, or tell a falsehood?), 'bark' (of a tree, or of a dog?), 'dull' (of a knife, or of a person?). Others, such as 'snow' or 'ice', are not lexicalized separately in every language, while some, like 'we', may correspond to two words (inclusive and exclusive) in languages that make that distinction. Lists also exist in many slightly different versions, and two datasets labelled "Swadesh 200" may not contain exactly the same meanings. These issues are not trivial, because the choice of which word fills a meaning slot determines its cognate class. Kassian et al. (2010) provided a detailed semantic specification of the 100-item list to reduce this ambiguity, and IE-CoR has invested heavily in precise meaning definitions for the same reason (Heggarty et al., 2023).

Fieldwork on poorly documented languages makes the problem concrete. Discussing Muno, a language of Papua New Guinea that he documented, Rönchen (2025) notes that its numerals above six are loans from Tok Pisin, that the word for 'dog' is unlikely to be older than the arrival of dogs in New Guinea, that 'we' has inclusive and exclusive (and possibly dual) forms, and that the very short stems of complex verbs make cognates hard to identify. He concludes that word lists should be tailored to the region and family rather than treated as universal, and that the "raw data" of lexical phylogenetics are already products of analysis. The idea is older than modern phylogenetics. McMahon and McMahon (2005) review regionally adapted Swadesh lists, among them lists for Southeast Asia and for Australia that add meanings such as 'armpit' or 'goanna', and suggest keeping a common core while substituting items within the same semantic field; in the Andes, for instance, 'llama' can replace 'cow', whose words are uniformly Spanish loans and so say nothing about relations among the indigenous languages. The price of such flexibility is comparability across families, which matters more for global analyses than for the phylogeny of a single family.

3.3.2 Lists designed from data

Later lists were constructed from empirical evidence about stability or borrowability. The *Leipzig-Jakarta list* of 100 meanings was derived from the World Loanword Database (WOLD), which recorded loanwords for more than 1,400 meanings in 41 languages from around the world (Haspelmath and Tadmor, 2009). The meanings were ranked by a combination of how rarely they were borrowed, how often they were expressed by simple (unanalysable) words, and how old the words tended to be, and the top 100 formed the list (Tadmor et al., 2010). Its overlap with the Swadesh 100 is substantial but not complete; it includes, for instance, 'ant' and 'wing', which are not on the Swadesh 100.

The *ASJP list* of 40 meanings was selected by the Automated Similarity Judgment Program team as the subset of the Swadesh 100 whose words were most stable, judged by how consistently they were shared within known language groups, and which gave the best classification results (Holman et al., 2008). The brevity of the list is what made it possible for ASJP to collect word lists for the great majority of the world's languages (Wichmann, 2022). Other short lists include Dolgopolsky's list of about 15 especially stable meanings, compiled for long-range comparison (Dolgopolsky, 1986), and the 35-item list of S. Yakhontov used in the Moscow school of comparative linguistics. At the other extreme, project-specific lists may be longer: the Austronesian Basic Vocabulary Database uses a list of 210 meanings (Greenhill et al., 2008), and IE-CoR uses 170 meanings

(Heggarty et al., 2023).

Table 3.1: Some basic vocabulary lists used in linguistic phylogenetics.

List	Meanings	Selection basis	Typical use
Swadesh 200	200	intuition: universal, culture-free meanings	lexicostatistics, early phylogenies (e.g. Dyen et al. data)
Swadesh 100	100	subset of the 200 list, removing problematic items	most lexical datasets
Leipzig-Jakarta	100	low borrowability, simple words, age (from WOLD)	borrowing-resistant comparisons
ASJP	40	stability within known groups; classification performance	global-scale automated classification
Dolgopolsky	about 15	stability in long-range comparison	deep (long-range) comparison
ABVD	210	Swadesh-type list adapted for Austronesian	Austronesian phylogenies
IE-CoR	170	Swadesh-type list with precise meaning definitions	Indo-European phylogenies

3.3.3 Stability, borrowability and frequency

Basic vocabulary was chosen in the hope that it would be stable and rarely borrowed. Both hopes are only partly met. Stability varies enormously across meanings. Some meanings, such as low numerals, pronouns and words for 'name', 'eye' or 'water', tend to retain the same word over thousands of years, while others, such as 'dirty', 'bad' or 'to throw', are replaced frequently. Pagel et al. (2007) showed that across Indo-European the rate of lexical replacement in a meaning is strongly and negatively related to how frequently the meaning is used in everyday speech (Chapter 9). Such variation among characters is not a problem in itself, since modern models allow rates to vary among characters (Chapter 7), but it is a strong reason not to assume a uniform rate, as glottochronology did. Borrowing in basic vocabulary is also not negligible. In the WOLD sample, loanwords made up on average roughly a quarter of the vocabulary, with much lower proportions in basic meanings but substantial variation across languages (Tadmor, 2009). English, for instance, borrowed the basic pronoun *they* and words such as *sky* and *skin* from Old Norse.

3.3.4 Concepts, glosses and Concepticon

Because so many lists exist, comparing or merging datasets requires knowing which meaning in one list corresponds to which meaning in another. A gloss such as "bark" or "to lie" is not enough. The Concepticon project addresses this by linking the meanings ("concepts") of hundreds of published lists to a common set of *concept sets*, each with a unique identifier and a definition (List et al., 2016). A dataset whose meanings are linked to Concepticon can be aligned with any other linked dataset automatically, which is one of the foundations of the large-scale data aggregation described in Section 3.11.

3.4 Coding cognates as characters

3.4.1 A toy dataset

To make the coding schemes concrete, we use a small hand-typed word list: eight meanings in twelve Indo-European languages, with orthographic forms, a rough phonetic transcription in a simplified ASCII alphabet loosely modelled on the ASJP transcription system, and expert-style cognate judgements. The cognate judgements follow standard etymological reference works, but a few are judgement calls, discussed below; the dataset is for illustration only.

```
langs <- c("English", "German", "Swedish", "Italian", "Spanish", "French",
           "Russian", "Polish", "Lithuanian", "Irish", "Greek", "Hindi")
wl <- rbind(
  data.frame(Language = langs, Concept = "hand",
    Form = c("hand", "Hand", "hand", "mano", "mano", "main", "ruka", "ręka",
             "ranka", "lámh", "chéri", "hāth"),
    Phon = c("hEnd", "hant", "hand", "mano", "mano", "mE", "ruka", "rENka",
             "ranka", "lav", "xeri", "hat"),
    Cognate = c("A", "A", "A", "B", "B", "B", "C", "C", "C", "D", "E", "F")),
  data.frame(Language = langs, Concept = "two",
    Form = c("two", "zwei", "två", "due", "dos", "deux", "dva", "dwa", "du",
             "dó", "dýo", "do"),
    Phon = c("tu", "cvai", "tvo", "due", "dos", "do", "dva", "dva", "du",
             "do", "Dio", "do"),
    Cognate = "A"),
  data.frame(Language = langs, Concept = "water",
    Form = c("water", "Wasser", "vatten", "acqua", "agua", "eau", "voda",
             "woda", "vanduo", "uisce", "neró", "pānī"),
    Phon = c("wat3r", "vas3r", "vat3n", "akwa", "agwa", "o", "voda", "voda",
             "vandua", "iSk3", "nero", "pani"),
    Cognate = c("A", "A", "A", "B", "B", "B", "A", "A", "A", "A", "C", "D")),
  data.frame(Language = c(langs[1:9], "Irish", "Irish", "Greek", "Hindi"),
    Concept = "dog",
    Form = c("dog", "Hund", "hund", "cane", "perro", "chien", "sobaka",
             "pies", "šuo", "madra", "gadhar", "skýlos", "kuttā"),
    Phon = c("dag", "hunt", "hund", "kane", "pero", "SyE", "sabaka", "pyes",
             "Suo", "madr3", "gair", "skilos", "kuta"),
    Cognate = c("A", "B", "B", "B", "C", "B", "D", "E", "B", "F", "G", "H",
                "I")),
  data.frame(Language = langs, Concept = "mountain",
    Form = c("mountain", "Berg", "berg", "montagna", "montaña", "montagne",
             "gora", "góra", "kalnas", "sliabh", "vounó", "pahāṛ"),
    Phon = c("mauntin", "berk", "bery", "montaNa", "montaNa", "motaN",
             "gora", "gura", "kalnas", "Sliav", "vuno", "pahar"),
    Cognate = c("B", "A", "A", "B", "B", "B", "C", "C", "D", "E", "F", "G")),
  data.frame(Language = langs, Concept = "eat",
    Form = c("eat", "essen", "äta", "mangiare", "comer", "manger", "jest'",
             "jeść", "valgyti", "ith", "tróo", "khānā"),
    Phon = c("it", "es3n", "et3", "manjare", "komer", "maZe", "yest",
```

```
            "yeSc", "valgiti", "i", "troo", "kana"),
    Cognate = c("A", "A", "A", "B", "C", "B", "A", "A", "D", "A", "E", "F")),
  data.frame(Language = langs, Concept = "star",
    Form = c("star", "Stern", "stjärna", "stella", "estrella", "étoile",
             "zvezda", "gwiazda", "žvaigždė", "réalta", "astéri",
             "tārā"),
    Phon = c("star", "Stern", "Sarna", "stela", "estreya", "etual",
             "zvezda", "gvyazda", "Zvaigzde", "relt3", "asteri", "tara"),
    Cognate = c("A", "A", "A", "A", "A", "A", "B", "B", "B", "C", "A", "A")),
  data.frame(Language = langs, Concept = "tooth",
    Form = c("tooth", "Zahn", "tand", "dente", "diente", "dent", "zub", "ząb",
             "dantis", "fiacail", "dóndi", "dāť"),
    Phon = c("tu8", "can", "tand", "dente", "diente", "da", "zup", "zomp",
             "dantis", "fiak3l", "Dondi", "dat"),
    Cognate = c("A", "A", "A", "A", "A", "A", "B", "B", "A", "C", "A", "A"))
)
wl$CogID <- paste(wl$Concept, wl$Cognate, sep = "-")
wl$Loan <- (wl$Language == "English" & wl$Concept == "mountain") |
           (wl$Language == "Russian" & wl$Concept == "dog")
dim(wl)
```

```
#> [1] 97  7
```

```
head(wl, 4)
```

```
#>   Language Concept Form Phon Cognate  CogID  Loan
#> 1  English    hand hand hEnd       A hand-A FALSE
#> 2   German    hand Hand hant       A hand-A FALSE
#> 3  Swedish    hand hand hand       A hand-A FALSE
#> 4  Italian    hand mano mano       B hand-B FALSE
```

This is a *long* (or "tidy") table with one row per word, the format used by CLDF and by tools such as LingPy and EDICTOR. Table 3.2 shows the cognate classes in the more familiar *wide* format, one column per meaning.

```
wide <- tapply(paste0(wl$Form, " (", wl$Cognate, ")"),
               list(factor(wl$Language, levels = langs), wl$Concept),
               paste, collapse = ", ")
knitr::kable(wide[, c("hand", "two", "water", "dog", "mountain", "eat",
                      "star", "tooth")],
  caption = paste(
    "The toy word list: forms with cognate class labels (letters are",
    "local to each meaning). Irish has two synonyms for 'dog'. English",
    "'mountain' (from French) and Russian 'sobaka' (probably from an",
    "Iranian language) are flagged as loanwords."))
```

Table 3.2: The toy word list: forms with cognate class labels (letters are local to each meaning). Irish has two synonyms for ‘dog’. English ‘mountain’ (from French) and Russian ‘sobaka’ (probably from an Iranian language) are flagged as loanwords.

	hand	two	water	dog	mountain	eat	star	tooth
English	hand (A)	two (A)	water (A)	dog (A)	mountain (B)	eat (A)	star (A)	tooth (A)
German	Hand (A)	zwei (A)	Wasser (A)	Hund (B)	Berg (A)	essen (A)	Stern (A)	Zahn (A)
Swedish	hand (A)	två (A)	vatten (A)	hund (B)	berg (A)	äta (A)	stjärna (A)	tand (A)
Italian	mano (B)	due (A)	acqua (B)	cane (B)	montagna (B)	mangiare (B)	stella (A)	dente (A)
Spanish	mano (B)	dos (A)	agua (B)	perro (C)	montaña (B)	comer (C)	estrella (A)	diente (A)
French	main (B)	deux (A)	eau (B)	chien (B)	montagne (B)	manger (B)	étoile (A)	dent (A)
Russian	ruka (C)	dva (A)	voda (A)	sobaka (D)	gora (C)	jest’ (A)	zvezda (B)	zub (B)
Polish	ręka (C)	dwa (A)	woda (A)	pies (E)	góra (C)	jeść (A)	gwiazda (B)	ząb (B)
Lithuanian	ranka (C)	du (A)	vanduo (A)	šuo (B)	kalnas (D)	valgyti (D)	žvaigždė (B)	dantis (A)
Irish	lámh (D)	dó (A)	uisce (A)	madra (F), gadhar (G)	sliabh (E)	ith (A)	réalta (C)	fiacail (C)
Greek	chéri (E)	dýo (A)	neró (C)	skýlos (H)	vounó (F)	tróo (E)	astéri (A)	dóndi (A)
Hindi	hāth (F)	do (A)	pānī (D)	kuttā (I)	pahāṛ (G)	khānā (F)	tārā (A)	dā̃t (A)

The table already shows most of the phenomena discussed in this chapter. ‘Two’ is a single cognate class across all twelve languages: it carries no information about subgrouping. ‘Star’ and ‘tooth’ separate the Slavic languages (and, for ‘star’, Lithuanian) from the rest, an innovation shared within Balto-Slavic. ‘Hand’ has one class each for Germanic, Romance and Balto-Slavic, and separate classes for Irish, Greek and Hindi, while ‘eat’ unites the Germanic and Slavic forms (and Irish *ith*) as retentions of the inherited root against a Romance innovation (Italian *mangiare*, French *manger*, from Latin *manducare*). Italian *mangiare* is itself usually regarded as an early borrowing from Old French, a reminder that loans between closely related languages are easily overlooked; we follow common practice and code it as cognate with the French word without flagging it. The entry for ‘water’ contains a judgement call: Irish *uisce* is usually derived from a formation built on the same root as English *water* and Russian *voda*, and we code it in the same class, but a coder working at a shallower depth, or one who required the same stem formation, might separate it. The entry for Spanish ‘eat’, *comer*, illustrates *partial cognacy*: it continues Latin *comedere*, a compound of the prefix *com-* and *edere* ‘eat’, so it contains a morpheme cognate with English *eat* while the word as a whole is a distinct formation.

3.4.2 Binary presence/absence coding

The most common coding scheme converts each cognate set into a binary character: 1 if the language has a word belonging to that set for the meaning in question, 0 if it has a word for the meaning but from a different set, and ? if the meaning is unknown. A meaning with k cognate classes produces k characters. The resulting matrix is analogous to restriction-site data in molecular biology, and models for its evolution (the binary CTMC, covarion and stochastic Dollo models) are the subject of Chapter 7. Binary coding handles synonyms naturally: Irish *madra* and *gadhar* are both coded 1 in their respective characters.

```
make_binary <- function(wl, loans = c("keep", "missing", "drop")) {
  loans <- match.arg(loans)
  langs <- unique(wl$Language)
  if (loans == "drop") wl <- wl[!wl$Loan, ]
  sets <- unique(wl$CogID)
  M <- matrix("0", length(langs), length(sets),
              dimnames = list(langs, sets))
  M[cbind(wl$Language, wl$CogID)] <- "1"
  for (cc in unique(wl$Concept)) {
    cols <- unique(wl$CogID[wl$Concept == cc])
    ok <- wl$Concept == cc & (loans != "missing" | !wl$Loan)
    # a language with no (inherited) word for a meaning gets "?" for all
    # of that meaning's cognate sets
    M[setdiff(langs, wl$Language[ok]), cols] <- "?"
  }
  M
}
B <- make_binary(wl, loans = "missing")
dim(B)
```

```
#> [1] 12 39
```

```
B[, grep("^(hand|dog)", colnames(B))]
```

```
#>           hand-A hand-B hand-C hand-D hand-E hand-F dog-A dog-B dog-C
#> English    "1"    "0"    "0"    "0"    "0"    "0"    "1"   "0"   "0"
#> German     "1"    "0"    "0"    "0"    "0"    "0"    "0"   "1"   "0"
#> Swedish    "1"    "0"    "0"    "0"    "0"    "0"    "0"   "1"   "0"
#> Italian    "0"    "1"    "0"    "0"    "0"    "0"    "0"   "1"   "0"
#> Spanish    "0"    "1"    "0"    "0"    "0"    "0"    "0"   "0"   "1"
#> French     "0"    "1"    "0"    "0"    "0"    "0"    "0"   "1"   "0"
#> Russian    "0"    "0"    "1"    "0"    "0"    "0"    "?"   "?"   "?"
#> Polish     "0"    "0"    "1"    "0"    "0"    "0"    "0"   "0"   "0"
#> Lithuanian "0"    "0"    "1"    "0"    "0"    "0"    "0"   "1"   "0"
#> Irish      "0"    "0"    "0"    "1"    "0"    "0"    "0"   "0"   "0"
#> Greek      "0"    "0"    "0"    "0"    "1"    "0"    "0"   "0"   "0"
#> Hindi      "0"    "0"    "0"    "0"    "0"    "1"    "0"   "0"   "0"
#>           dog-D dog-E dog-F dog-G dog-H dog-I
#> English    "0"   "0"   "0"   "0"   "0"   "0"
#> German     "0"   "0"   "0"   "0"   "0"   "0"
```

```
#> Swedish    "0"   "0"   "0"   "0"   "0"   "0"
#> Italian    "0"   "0"   "0"   "0"   "0"   "0"
#> Spanish    "0"   "0"   "0"   "0"   "0"   "0"
#> French     "0"   "0"   "0"   "0"   "0"   "0"
#> Russian    "?"   "?"   "?"   "?"   "?"   "?"
#> Polish     "0"   "1"   "0"   "0"   "0"   "0"
#> Lithuanian "0"   "0"   "0"   "0"   "0"   "0"
#> Irish      "0"   "0"   "1"   "1"   "0"   "0"
#> Greek      "0"   "0"   "0"   "0"   "1"   "0"
#> Hindi      "0"   "0"   "0"   "0"   "0"   "1"
```

Two features of binary coding deserve attention. The first is that characters from the same meaning are not independent. If a language has a word from class A for 'hand', it almost certainly does not have a word from class B, so the binary characters for a meaning are negatively correlated, and a single replacement event (say, the Balto-Slavic replacement of the inherited word for 'hand') changes two characters at once: one goes from 1 to 0, another from 0 to 1. Models that treat the characters as independent therefore overcount the evidence from each replacement, a point made forcefully by critics of binary coding and taken up in Section 3.5.2 and Chapter 7.

The second feature is *ascertainment*. A binary cognate matrix never contains a column of all zeros, because cognate sets are only recorded if at least one language has them. Depending on the coding conventions, it may also exclude or include *singletons* (classes found in only one language) and constant columns of all ones (such as 'two' in our data). The likelihood of the data must be corrected for the patterns that could not have been observed, a correction known as *ascertainment bias correction*, which is crucial for estimating branch lengths and dates correctly (Chapter 7).

```
informative <- function(x) {      # at least two languages in each state
  x <- x[x != "?"]
  sum(x == "1") >= 2 && sum(x == "0") >= 2
}
c(characters = ncol(B),
  constant = sum(apply(B, 2, function(x) length(unique(x[x != "?"])) == 1)),
  singletons = sum(apply(B, 2, function(x) sum(x == "1") == 1)),
  parsimony_informative = sum(apply(B, 2, informative)))
```

```
#>            characters              constant            singletons
#>                    39                     2                    22
#> parsimony_informative
#>                    15
```

Of the 39 characters, 22 are singletons and only 15 are parsimony-informative, that is, divide the languages into two groups of at least two each. One of the two constant columns is 'two', which every language shares; the other is the cognate set of the Russian loan *sobaka*, which, once the loan is coded as missing, contains only zeros. A column of zeros is exactly the kind of pattern that a cognate matrix should never contain, and it illustrates how coding decisions interact with ascertainment: such columns must be removed, and the likelihood corrected accordingly.

3.4.3 Treating loanwords

Our function offers three treatments of the two flagged loanwords, which are also the three used in practice. Loans can be *kept*, in which case English *mountain* is coded as sharing a cognate class with the Romance words; this treats a borrowing as if it were inherited and will, if loans are common, pull the borrowing language towards the donor in the tree. Loans can be *coded as missing*, so that the language contributes no information about that meaning; this is the most common choice in Bayesian analyses. Or the loan can be *dropped* from its cognate set while the language is still coded 0 for all other sets of the meaning, which treats the loan as if it were a unique innovation. Figure 3.1 shows the matrix under the "missing" treatment.

```
num <- matrix(suppressWarnings(as.numeric(B)), nrow(B),
              dimnames = dimnames(B))
op <- par(mar = c(5.5, 6.5, 1, 1))
image(seq_len(ncol(num)), seq_len(nrow(num)), t(num[nrow(num):1, ]),
      col = c("grey92", "grey20"), axes = FALSE, xlab = "", ylab = "")
miss <- which(is.na(t(num[nrow(num):1, ])), arr.ind = TRUE)
rect(miss[, 1] - 0.5, miss[, 2] - 0.5, miss[, 1] + 0.5, miss[, 2] + 0.5,
     col = "grey60", density = 30, border = NA)
axis(1, at = seq_len(ncol(num)), labels = colnames(num), las = 2,
     cex.axis = 0.72, tick = FALSE)
axis(2, at = seq_len(nrow(num)), labels = rev(rownames(num)), las = 1,
     cex.axis = 0.8, tick = FALSE)
concept_breaks <- cumsum(rle(sub("-.*", "", colnames(num)))$lengths)
abline(v = concept_breaks + 0.5, col = "white", lwd = 3)
```

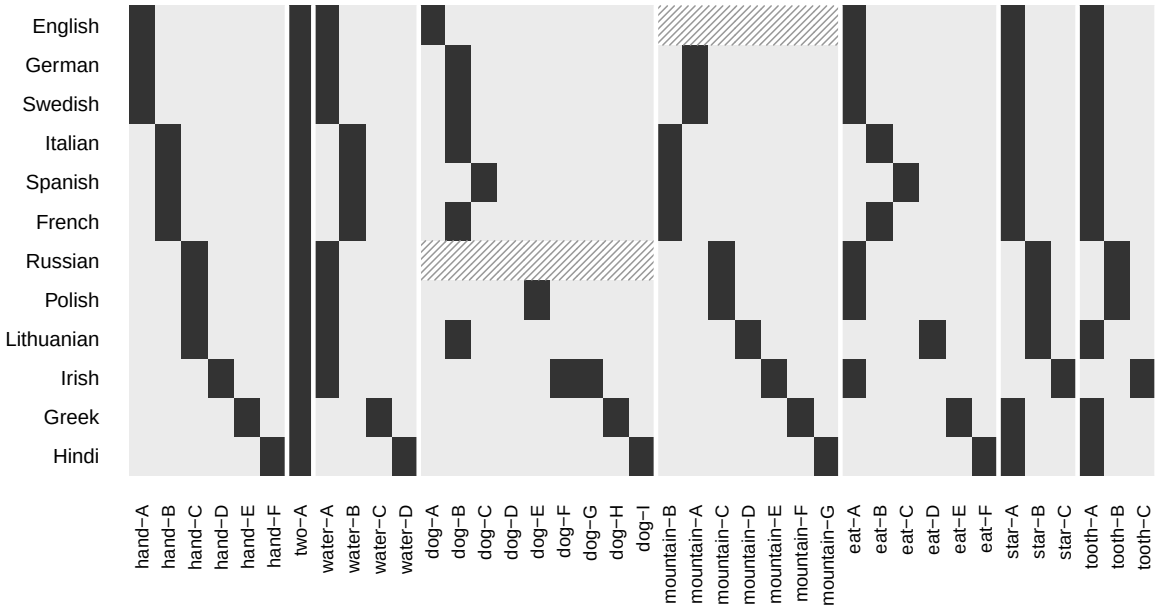


Figure 3.1: The binary cognate matrix for the toy data, with loanwords coded as missing. Each column is a cognate set (grouped by meaning), each row a language. Dark cells are 1 (the language has a word from the set), light cells 0, and hatched grey cells missing.

```
par(op)
```

The matrix can be passed to **phangorn** as a `phyDat` object, from which distances, parsimony scores and likelihoods can be computed (Chapters 4 and 5), and exported to NEXUS for BEAST 2, MrBayes or RevBayes (Chapter 17). Figure 3.2 shows a neighbour-joining tree from Hamming distances between the rows, computed while ignoring missing entries. Even eight meanings are enough to recover several of the main subgroups, though not their relationships.

```
library(phangorn)
pd <- phyDat(B, type = "USER", levels = c("0", "1"), ambiguity = "?")
pd
```

```
#> 12 sequences with 39 character and 30 different site patterns.
#> The states are 0 1
```

```
nex <- tempfile(fileext = ".nex")
write.phyDat(pd, file = nex, format = "nexus")
cat(readLines(nex)[-2][1:9], sep = "\n")
```

```
#> #NEXUS
#> BEGIN DATA;
#>   DIMENSIONS NTAX=12 NCHAR=39;
#>   FORMAT DATATYPE=STANDARD MISSING=? GAP=- INTERLEAVE=NO symbols="0123456789";
#>   MATRIX
#>     English         100000110001000000007777777100000100100
#>     German          100000110000100000000100000100000100100
#>     Swedish         100000110000100000000100000100000100100
#>     Italian         010000101000100000001000000010000100100
```

```
nj_tree <- NJ(dist.hamming(pd))
nj_mid <- midpoint(nj_tree)
# NJ can return a negative terminal branch (here Russian), which is drawn
# pointing left; its label is set to the right of the parent node so that it
# does not sit on top of the branch
depth <- max(node.depth.edgelength(nj_mid))
plot(nj_mid, show.tip.label = FALSE, x.lim = c(0, 1.25 * depth))
lp <- get("last_plot.phylo", envir = .PlotPhyloEnv)
tips <- seq_len(Ntip(nj_mid))
par_x <- lp$xx[nj_mid$edge[match(tips, nj_mid$edge[, 2]), 1]]
text(pmax(lp$xx[tips], par_x) + 0.01 * depth, lp$yy[tips],
     nj_mid$tip.label, adj = 0, cex = 0.8, font = 3)
```

3.4.4 Multistate coding

The alternative to binary coding is *multistate* coding, in which each meaning is a single character and its cognate classes are the states. ‘Hand’ becomes one character with six states. Multistate coding respects the structure of the data: a replacement is a single change of state, and the non-independence of binary characters within a meaning disappears. Its difficulties are practical. The number of states can be large (some meanings have dozens of cognate classes across a big family),

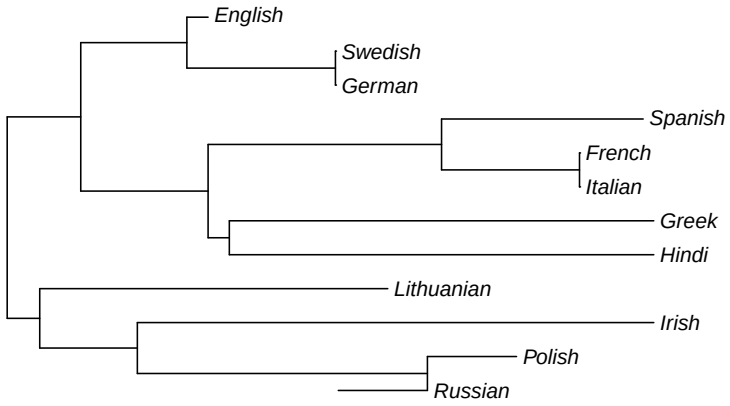


Figure 3.2: Neighbour-joining tree from Hamming distances computed on the toy binary cognate matrix, midpoint-rooted for display. With only eight meanings, the Romance, Germanic and Slavic groupings emerge, but the deeper relationships are not resolved meaningfully.

which makes standard Mk-type models (Chapter 7) parameter-rich and computationally expensive, and the states have no natural order or meaning across characters. Synonyms produce *polymorphic* states (Irish ‘dog’ is F/G below), which many programs handle only as ambiguity (“either F or G”) rather than as genuine polymorphism (“both”). And the model must cope with the fact that new states can appear: a meaning that has six classes in the sample may have had states that are now lost. Studies that have compared binary and multistate coding on the same data, and models designed to bridge them, are reviewed in Chapter 7.

```
multistate <- tapply(wl$Cognate,
                     list(factor(wl$Language, levels = langs), wl$Concept),
                     function(x) paste(sort(unique(x)), collapse = "/"))
noquote(multistate)
```

```
#>            dog eat hand mountain star tooth two water
#> English    A   A   A    B        A    A     A   A
#> German     B   A   A    A        A    A     A   A
#> Swedish    B   A   A    A        A    A     A   A
#> Italian    B   B   B    B        A    A     A   B
#> Spanish    C   C   B    B        A    A     A   B
#> French     B   B   B    B        A    A     A   B
#> Russian    D   A   C    C        B    B     A   A
#> Polish     E   A   C    C        B    B     A   A
#> Lithuanian B   D   C    D        B    A     A   A
#> Irish      F/G A   D    E        C    C     A   A
#> Greek      H   E   E    F        A    A     A   C
```

```
#> Hindi      I   F   F    G       A    A     A   D
```

3.4.5 Root-meaning coding, root coding and partial cognates

The schemes above are *root-meaning* coding: a character is a cognate set *within a meaning slot*. English *hound* is cognate with German *Hund*, but because *hound* is not the basic English word for 'dog', English is coded 0 for that set. Root-meaning coding captures *semantic replacement* (the shift of 'dog' from *hound* to *dog* in English), not the loss of a root from the language. An alternative that has occasionally been used is *root* (or *root-presence*) coding, in which a character records whether a root is present anywhere in the language's lexicon, regardless of meaning. Root presence is much harder to establish consistently, because it requires searching the entire lexicon of every language, and it changes on a different time scale.

Coding schemes differ in a second dimension as well: the *depth* at which cognacy is judged. Two words can descend from the same root without descending from the same word. Latin *cor* and Italian *cuore* 'heart' continue the same lexeme, whereas Spanish *corazón* and Portuguese *coração* continue a derivative of the same root that replaced it in Ibero-Romance. Crossing the two dimensions gives four kinds of lexical character. Under *lexeme-meaning* coding the Spanish and Italian words for 'heart' fall in different classes, and the replacement of *cor* by the derivative counts as an innovation shared by Spanish and Portuguese; under *root-meaning* coding they fall in one class and the replacement is invisible; and *root* and *lexeme* coding that ignore meaning record whether a language has the root or the word anywhere in its lexicon. Romanian *inimă* 'heart', from Latin *anima* 'soul', belongs to another class under every scheme that respects meaning. Shallow (lexeme-level) cognacy gives more characters that resolve recent splits; deep (root-level) cognacy gives characters that change more slowly. Which is appropriate depends on the time depth of the question, and the choice should be stated, since published datasets differ in it.

Two questions define the space of options: does meaning matter, and is cognacy judged on the root or on the whole word (the lexeme)? Crossing them gives four kinds of lexical homology (Table 3.3). Latin *cor*, Italian *cuore* and Spanish *corazón* 'heart' share a root-meaning trait, since all continue the root of *cor* in the meaning 'heart'. But only Latin and Italian share a lexeme-meaning trait: *corazón* goes back to a derivative of *cor* with an additional suffix, not to *cor* itself. Romanian *inimă* 'heart' continues Latin *anima* 'breath, soul', so it shares a lexeme-cognate trait with Latin *anima* but neither kind of meaning-based trait with Latin *cor*. Most phylogenetic databases use root-meaning traits. IE-CoR, for example, judges cognacy at the level of the Proto-Indo-European root regardless of later affixes, so that *corazón* falls in the same cognate set as *cor*; it treats words from different ablaut grades of one root as cognate, but words from roots that differ by a "root extension" as distinct (Heggarty et al., 2023). Each convention is defensible, but they produce different matrices from the same etymologies, and the choice should be reported. Scarborough (2020), writing from inside the IE-CoR project, discusses further cases in which a binary cognate decision is hard to make: competing analyses of what counts as one root, taboo deformation of words such as 'tongue' and 'louse', and contamination between words.

Table 3.3: Four kinds of lexical homology, defined by whether meaning is taken into account and by whether the root or the whole lexeme is compared.

	Meaning relevant	Meaning irrelevant
Root compared	root-meaning trait: same root in the same meaning slot (Latin *cor*, Spanish *corazón* 'heart')	root cognate: same root in any meaning
Lexeme compared	lexeme-meaning trait: the same word in the same meaning slot (Latin *cor*, Italian *cuore*)	lexeme cognate: the same word in any meaning (Latin *anima* 'soul', Romanian *inimă* 'heart')

Partial cognacy arises when words are morphologically complex and only some of their parts are cognate. Spanish *comer* versus English *eat* is one example; others are compounds and derivations, which are pervasive in some families. In the Sino-Tibetan and Hmong-Mien families, and in many languages of Southeast Asia, words for basic meanings are often compounds of two or more morphemes, each with its own history. Coding such words with a single cognate label forces a choice: treat them as cognate if any morpheme is shared ("loose" coding), only if all are shared ("strict" coding), or code each morpheme separately. List (2016) argued that these choices can strongly affect phylogenetic results and proposed coding cognacy at the morpheme level, and Hill and List (2017) illustrated the practical challenges with data from the Burmish languages. Partial cognacy is also an issue in families with rich prefixing or suffixing morphology, such as Bantu (noun-class prefixes) or Austronesian (affixes and reduplication).

The same word list can yield noticeably different matrices depending on whether loans are kept or excluded, how synonyms are handled, at what depth cognacy is judged, how partial cognates are treated, and whether singletons are retained. These decisions are often documented only in supplementary materials, if at all. When comparing published analyses, or reusing a published matrix, check how each was made. A sensitivity analysis that repeats the inference under alternative coding decisions is one of the cheapest and most informative robustness checks available.

The Austronesian Basic Vocabulary Database. The ABVD was set up to collect basic vocabulary for the Austronesian languages in a common format, with a 210-meaning list, source references and cognate judgements made by specialists (Greenhill et al., 2008). It is a model for how lexical data for phylogenetics can be organized: every form is linked to its source, cognate decisions are recorded at the level of individual words rather than only as a final matrix, and loanwords are flagged. Words for 'two', for example, reflect Proto-Austronesian **duSa* across the family (Malay *dua*, Fijian *rua*, Hawaiian *lua*, Māori *rua*), while 'hand' shows how semantic shift complicates coding: the Proto-Austronesian word **lima* meant 'five' and probably also 'hand', and its reflexes mean 'hand' in some languages (Hawaiian *lima*, Fijian *liga*) and only 'five' in others (Malay *lima*), where 'hand' is expressed by a different word (Malay *tangan*). Under root-meaning coding, Malay is coded 0 for the **lima* set in the meaning 'hand' even though it retains the root. The ABVD provided the data for the dated Austronesian phylogeny of Gray et al. (2009), which analysed 400 languages, and for many comparative studies built on it (Chapters 10 and 14).

3.5 Uncertainty and dependence in coded data

A cognate matrix looks like a table of observations, but every cell is the output of a judgement, and the columns are not the independent draws that the models of Chapters 5 to 7 assume. Both

facts matter for what an analysis can claim. This section collects them in one place; Section 3.13 returns to measurement error between coders, and Chapter 15 to the consequences for published results.

3.5.1 How uncertain are cognate judgements?

Uncertainty about cognacy comes in several kinds, which have different consequences:

- *Etymological doubt.* Is a pair of words cognate at all? Short forms, irregular correspondences, sound-symbolic words and large semantic distances all make judgements uncertain, and specialists disagree about individual etymologies.
- *Depth.* At which ancestor is cognacy judged (Section 3.1)? A coder working at the level of Proto-Celtic and one working at the level of Proto-Indo-European will split and lump differently.
- *Segmentation.* Which part of a complex word is compared? Partial cognates (Section 3.4) force a choice between loose, strict and morpheme-level coding.
- *Inheritance versus borrowing.* Loans between close relatives that share sound changes can be indistinguishable from inherited words (Italian *mangiare* in Table 3.2).
- *Slot filling.* Which of several near-synonyms is the basic word for a meaning? The choice determines the cognate class even when every individual etymology is certain.
- *Source and transcription.* Errors in dictionaries, wordlists and their transcription propagate into the judgements.

These uncertainties are not spread evenly. They concentrate in particular meanings (those with many competing synonyms), in poorly documented languages, in ancient languages known only from texts (Section 3.2.2), at deep time depths, and in contact zones. Their effects are therefore partly systematic rather than random: they can shift particular clades and dates rather than simply adding noise.

They are also rarely recorded. CLDF has a field for doubtful cognate judgements, but in the Polynesian dataset used in Chapter 17 none of the 7,518 judgements is flagged as doubtful. That does not mean every judgement was certain; it means the uncertainty was not written down, which is the norm in the field. The options for representing it, roughly in increasing order of ambition, are to flag doubtful judgements; to code ambiguous cells as uncertain (a language is cognate with set A or set B; in NEXUS, the ambiguity `{AB}` in a multistate character); to prepare several alternative codings that differ in the doubtful cases and analyse each; and to treat cognacy itself as a parameter to be inferred jointly with the tree (see the box at the end of Section 3.12). Even the simplest of these, analysing two or three alternative codings and reporting how much the conclusions move, is more informative than a single matrix presented as if it were certain.

The toy data contain one such judgement call: Irish *uisce* ‘water’, which we coded as cognate with English *water* and Russian *voda*. A coder who required the same stem formation would separate it. The code below recodes it as a class of its own and compares the two matrices.

```
wl_alt <- wl
i_uisce <- wl_alt$Language == "Irish" & wl_alt$Concept == "water"
wl_alt$Cognate[i_uisce] <- "E"
wl_alt$CogID[i_uisce] <- "water-E"
B_alt <- make_binary(wl_alt, loans = "missing")
pd_alt <- phyDat(B_alt, type = "USER", levels = c("0", "1"),
```

```
                ambiguity = "?")
d_orig <- as.matrix(dist.hamming(pd))
d_alt <- as.matrix(dist.hamming(pd_alt))
round(rbind(original = d_orig["Irish", c("English", "Russian", "Italian")],
            alternative = d_alt["Irish", c("English", "Russian",
                                           "Italian")]), 3)
```

```
#>             English Russian Italian
#> original      0.231   0.205   0.385
#> alternative   0.275   0.250   0.375
```

```
RF.dist(NJ(dist.hamming(pd)), NJ(dist.hamming(pd_alt)), rooted = FALSE)
```

```
#> [1] 2
```

One judgement in one meaning moves Irish measurably away from Germanic and Slavic. With eight meanings the effect is exaggerated, but in a real dataset of 200 meanings there are typically dozens of such cases, and if they all lean the same way (for instance, because one coder lumped doubtful cases in one branch) their effects add up rather than cancel.

3.5.2 Dependence between characters

Almost every model in this book multiplies the likelihoods of characters, which assumes that they evolve independently given the tree. Linguistic characters violate this in several distinct ways (Table 3.4).

Table 3.4: Sources of dependence between linguistic characters, their consequences, and the main remedies.

Source of dependence	Example	Consequence if ignored	Ways to address it
Binary coding of one meaning	the cognate sets for 'hand'	one replacement counted as two or more changes; clade support overstated	multistate meaning-slot models; resampling meanings rather than cognate sets; meaning-level ascertainment (Chapter 7)
Synonymy within a meaning	Irish *madra* and *gadhar* 'dog'	polymorphism treated as independent presences	polymorphism-aware coding and models; explicit rules for choosing basic words
Regular sound change	Grimm's law in hundreds of Germanic words	many characters change in one event	code the sound change once as a phonological character; models of concerted change (Hruschka et al., 2015)
Borrowing episodes	Old Norse loans in English, including *they* and *skin*	correlated gains pull the borrower towards the donor	flagging and excluding loans; explicit models of contact (Chapter 8)

Source of dependence	Example	Consequence if ignored	Ways to address it
Shared roots across meanings	one word for ‘bark’ and ‘skin’	one change scored in two meanings	choosing meanings that rarely colexify; coding at the level of roots
Logical and functional links among structural features	adjective order without adjectives; verb-object and adposition order	imaginary ancestral states; correlated features counted twice	hierarchical coding of inapplicable features; models of correlated evolution (Chapter 13)

The first source is the one built into the standard coding scheme, and it can be seen directly in the toy matrix. The correlation between two binary characters from the same meaning is typically negative, because a language that has one cognate set for a meaning usually lacks the others, while characters from different meanings are nearly uncorrelated on average:

```
Bn <- matrix(suppressWarnings(as.numeric(B)), nrow(B),
             dimnames = dimnames(B))
variable <- apply(Bn, 2, function(x) length(unique(na.omit(x))) == 2)
Bv <- Bn[, variable]
meaning <- sub("-.*", "", colnames(Bv))
R <- cor(Bv, use = "pairwise.complete.obs")
same <- outer(meaning, meaning, "==") & upper.tri(R)
diff <- !outer(meaning, meaning, "==") & upper.tri(R)
round(c(within_meaning = mean(R[same], na.rm = TRUE),
        between_meanings = mean(R[diff], na.rm = TRUE)), 2)
```

```
#>   within_meaning between_meanings
#>            -0.18             0.00
```

Ignoring dependence has three main consequences. Posterior probabilities and bootstrap values become overconfident, because the analysis believes it has more independent evidence than it has. Model comparison is distorted, because the likelihood of dependent characters overstates how sharply the data discriminate between models. And predictive checks of model adequacy can be misleading unless the simulated data reproduce the same dependence structure. Which lexical models account for which kind of dependence, and which allow polymorphism and recurrent innovation, is compared systematically in Chapter 7.

3.6 Typological and structural data

3.6.1 Structural features as characters

Lexical cognates are not the only heritable features of languages. Grammatical and phonological *structure* is also inherited: whether a language has a dual number, whether verbs agree with objects, whether the adjective precedes the noun, whether there are tones or clicks. Structural features have three apparent advantages for phylogenetics. They do not require cognate judgements, so they can be coded for languages whose lexical relationships are unknown or too remote to establish. They are thought by some to be more stable than vocabulary and hence to carry signal from deeper time. And they can be coded for any language from a grammar, using a fixed questionnaire. That last advantage hides a problem of comparability. The categories of a question-

naire are not the categories of any one grammar: "dual", "article" or "object agreement" are what Haspelmath (2010) calls *comparative concepts*, defined by the typologist for the purpose of comparison, which may correspond only loosely to the descriptive categories of individual languages. Two languages coded identically may therefore have quite different grammars, and a "change" between two states can be an artefact of how a description was mapped onto the questionnaire.

The main databases are the *World Atlas of Language Structures* (WALS), which codes nearly 200 structural features for more than 2,500 languages, though very sparsely (Dryer, 2013), and *Grambank*, which codes 195 features, mostly binary, for more than 2,400 languages, with much denser coverage and explicit coding rules designed with comparative analyses in mind (Skirgård et al., 2023). Other important resources include the AUTOTYP database, which codes structural variables in a more fine-grained, multivariate way, and PHOIBLE, a database of phoneme inventories (Moran, 2019). Grammars Across Time Analyzed (GATA) codes morphological, syntactic and phonological features for 52 languages from two published grammars of the same language written at different times, which makes it possible to compare the same language at two points in time rather than infer change from a tree, although a difference between two descriptions is not always a difference between two stages of the language (Blum et al., 2023).

3.6.2 Structural phylogenetics and its limits

The use of structural features to infer phylogenies was pioneered by Michael Dunn and colleagues for the Papuan languages of Island Melanesia, where the lexical evidence for relationships among the non-Austronesian languages had largely been eroded. Dunn et al. (2005) coded a set of structural features (mostly binary features of phonology, morphology and syntax) and showed that, for the Oceanic (Austronesian) languages of the region, whose relationships are known from lexical evidence, a phylogenetic analysis of the structural features recovered a tree largely consistent with the established classification. Because the structural features were coded without cognate judgements, this agreement is a less circular check than the agreement of a lexical tree with the same classification would be (Section 1.9), although it tests only whether the method works where the answer is already known. They then applied the same method to the Papuan languages and found a phylogenetic signal that, they argued, reflected ancient relationships. A longer methodological study explored the reliability of the approach (Dunn et al., 2008), and Reesink et al. (2009) extended the analysis of structural features across Sahul (New Guinea and Australia) using population-genetic clustering methods. In a different tradition, the *parametric comparison method* uses binary values of syntactic parameters, in the sense of generative grammar, as characters (Longobardi and Guardiano, 2009); it is discussed in Section 3.9.

Structural data have serious limitations as phylogenetic characters, and these are now well understood. Most features have few states, often only two, so independent parallel changes (homoplasy) are common: a feature that changes back and forth a few times across a tree carries little information about it. Many structural features diffuse readily through contact, so they reflect areal convergence as much as descent (Chapter 8). Features are often logically or functionally dependent: in WALS, the order of object and verb and the order of adposition and noun are correlated for reasons of processing or grammaticalization (Chapter 13), and a feature such as "order of adjective and noun" is inapplicable to a language without a distinct adjective class. And rates of change vary widely among features. Dediu and Cysouw (2013) compared seven methods for estimating feature stability and found that they agreed only partly. Greenhill et al. (2017) compared rates of change in Austronesian lexical and structural data within a common framework and found that, contrary to a widespread expectation, structural features were not in general more stable than ba-

sic vocabulary; many changed considerably faster, while a minority were very stable. An earlier study of Austronesian found that structural data carried a weaker and less tree-like signal than lexical data (Greenhill et al., 2010a). The conclusion is not that structural data are useless for phylogenetics, but that they need models that account for homoplasy, dependence and contact, and that claims of deep relationships based on structural similarity alone should be treated with caution (see also Wichmann and Saunders, 2007).

Inapplicable characters. Biologists working with morphological data face a problem familiar to typologists: some characters are logically inapplicable in some taxa. "Tail colour" cannot be coded for an animal without a tail; "order of adjective and noun" cannot be coded for a language without adjectives, and "position of the tone-bearing unit" is meaningless in a language without tone. The standard practice in both fields has been to code such cells as missing data, but this is not the same thing. Missing data say "the state is unknown"; an inapplicable cell says "the character does not exist here", and treating it as missing allows algorithms to reconstruct imaginary tails, or imaginary adjective orders, in ancestors that had none, which can distort the tree. Brazeau et al. (2019) developed a parsimony algorithm that handles inapplicable characters properly, together with a hierarchical approach to coding the controlling ("is there a tail?") and dependent ("what colour is it?") characters. Model-based treatments of character hierarchies have also been developed in morphological phylogenetics. To our knowledge, these methods have rarely if ever been applied to typological data, although typological questionnaires such as Grambank are full of such dependencies. They are a natural import for structural phylogenetics and for ancestral state reconstruction of grammatical features (Chapter 12).

3.7 Phonological characters and sound change

The comparative method identifies subgroups through shared innovations, and the most celebrated innovations are sound changes. Grimm's law, the set of consonant shifts by which Proto-Indo-European **p, *t, *k* became Germanic *f*, *þ* [θ], *h* (Latin *pater* versus English *father*; Latin *tres* versus English *three*), is a defining innovation of Germanic. A sound change can be coded as a binary character (has the language undergone the change or not?), and because sound changes are regular, a single character summarizes evidence from hundreds of words.

Ringe et al. (2002) used phonological characters of this kind, alongside morphological and lexical characters, in their analysis of Indo-European, and treated them as particularly reliable: a well-chosen sound change is unlikely to recur independently in exactly the same form, and the comparative method can usually establish its relative chronology. The difficulty is that sound changes are not immune to homoplasy or diffusion. Common changes, such as the palatalization of velars before front vowels or the loss of final consonants, recur independently in many lineages. Others can spread across a dialect continuum after the dialects have begun to diverge; the "satem" assibilation of Proto-Indo-European palatal velars, shared by Indo-Iranian and Balto-Slavic among others, has been interpreted by some as a shared innovation and by others as an areal phenomenon. Phonological characters are also few in number, which limits their statistical power.

Sound change can also be modelled at a finer grain. Because regular sound change affects many words at once, a single event creates correlated changes across the lexicon. Hruschka et al. (2015) exploited this *concerted evolution* in Turkic: they modelled the evolution of sound correspondences across many cognate sets on a phylogeny and identified episodes in which many words changed together, recovering known regular sound changes from the data. Other studies have treated *phonotactic* properties (the patterns in which sounds combine) as characters and found that they

carry phylogenetic signal in Pama-Nyungan (Macklin-Cordes et al., 2021). And methods for the automatic inference of sound correspondence patterns across many languages have made it possible to code correspondence patterns directly (List, 2019), bringing computational methods closer to the core of the comparative method. Chapter 12 returns to probabilistic models of sound change on trees, which allow the forms of proto-words to be reconstructed.

Two recent studies set sound-level characters against cognates directly. Häuser et al. (2024) compared trees built from automatically inferred sound correspondences with trees built from automatically detected cognates on ten datasets from different families, and found the cognate-based trees closer to gold-standard classifications, by roughly a third in generalized quartet distance on average. Huisman et al. (2025) combined the two kinds of evidence instead, adding binary phonotactic characters (the presence of particular two-segment sequences) as a separate partition to cognate data for Japonic, which they report made parameter estimates more precise. On current methods, phonological characters can add to lexical evidence but have not replaced it. The same coding ideas extend beyond spoken languages: Abner et al. (2024) inferred phylogenies of 19 sign languages from phonologically coded lexical data, obtaining trees consistent in key respects with documented history but challenging some assumed groupings.

3.8 Morphological paradigms

Inflectional morphology has a special place in the history of the comparative method. Antoine Meillet argued that the strongest proof of genealogical relationship is not lexical resemblance but shared morphological *irregularity*, because idiosyncratic patterns such as suppletion are extremely unlikely to arise twice by chance or to be borrowed (Meillet, 1925). The suppletive comparison *good/better/best* in English, matched by German *gut/besser/best*, is a classic example of a shared morphological irregularity within Germanic; the use of an unrelated stem for the comparative is also found elsewhere (Latin *bonus/melior*), but there with different, non-cognate stems.

Morphological characters are attractive for phylogenetics for the same reasons: they are rarely borrowed and, if carefully chosen, rarely homoplastic. Ringe et al. (2002) included morphological characters such as particular inflectional endings and the formation of particular tenses. But they are few, they are difficult to code consistently across languages with very different morphological systems, and they tend to be lost through regularization and analogy, which can make their absence uninformative. Inflectional *paradigms* are also structured objects, with internal dependencies between cells, which simple character coding cannot capture. Systematic phylogenetic use of paradigm structure, for instance by modelling the evolution of inflection classes or principal parts, remains, to our knowledge, rare. Morphological characters are therefore most often used as constraints (for example, to enforce a subgroup that is supported by a strong shared irregularity) or as a check on lexical results, rather than as the main data.

Character choice in Ringe, Warnow and Taylor's Indo-European analysis. Ringe et al. (2002) analysed 24 Indo-European languages, most of them ancient (Hittite, Vedic, Avestan, Classical Greek, Latin, Gothic, Old Irish, Old Church Slavonic, Tocharian A and B and others), using a few hundred characters: mostly lexical, plus a small number of phonological and morphological characters. They selected characters carefully and coded them as multistate characters, and they treated characters believed to be prone to parallel development with particular caution. Their coding of absence is instructive. For the past-tense augment, every language that has it shares one state, but each language that lacks it receives a state of its own, because a shared "absent" value would imply a shared historical change where there may have been none (Goldstein, 2020). Their

method searched for trees on which as many characters as possible were *compatible*, that is, could have evolved without homoplasy (Chapter 4). The analysis showed that no tree was compatible with all characters, and the conflicts pointed to contact between early branches, especially involving Germanic; later work by the same group modelled these conflicts with *perfect phylogenetic networks* (Nakhleh et al., 2005a). The study illustrates an approach to data opposite to that of large-scale lexicostatistical datasets: few languages, carefully curated characters of several types, and expert judgements about which characters are reliable.

3.9 Syntactic characters and the Parametric Comparison Method

Syntax has traditionally been regarded as poor evidence for genealogy: word orders and construction types recur independently, spread through contact, and are hard to compare without shared morphemes. The *Parametric Comparison Method* (PCM) developed by Giuseppe Longobardi, Cristina Guardiano and colleagues challenges this view (Longobardi and Guardiano, 2009; Longobardi et al., 2013; Guardiano and Longobardi, 2026). Its characters are not surface features of the kind listed in WALS but *parameters* in the sense of generative grammar: abstract binary choices, set to $+$ or $-$, that a theory of grammar takes to underlie clusters of surface properties. Most work has concentrated on the syntax of the noun phrase, with parameters such as whether definiteness is grammaticalized by an article, or whether the noun is pronounced before or after certain classes of adjectives. The most ambitious claim so far is that of Ceolin et al. (2020), who coded 94 binary nominal parameters for 69 languages from up to 13 Eurasian families and reported a phylogenetic signal that separates all the families and reproduces their established internal topologies, with what they call “near-perfect” phylogenies little affected by homoplasy. The agreement with established topologies is the kind of evidence that Section 1.9.1 treats with caution, and the cross-family inferences built on it depend on the null model of chance agreement discussed below.

Three features make PCM instructive for this chapter. First, its characters are defined in advance by a list that applies to every language, and coding them needs a grammatical analysis but no etymology, so in principle they can be compared between languages whose lexical relationship has been obscured. Second, PCM codes dependence between characters explicitly. Parameters are linked by implications: if one parameter has a particular value, another may become irrelevant, because the structure it regulates does not arise. PCM records such a parameter as neutralized, with a third symbol 0, and distances between languages are computed only over the parameters that are relevant in both. This is the same problem as the inapplicable characters of the box in Section 3.6, handled at the coding stage, and it is the most systematic treatment of character dependence in any linguistic dataset. Third, because syntactic coding does not use cognate judgements, agreement between a PCM tree and the lexically based classification is a less circular check than the agreement of a lexical tree with the same classification (Section 1.9). The coders know the languages and their accepted subgroups, so it is not fully independent, but the evidence itself is distinct. The PCM studies report trees broadly consistent with the accepted subgrouping of Indo-European, including an analysis of 26 contemporary Indo-European varieties in which, the authors report, horizontal transmission did not obscure the genealogical signal (Longobardi et al., 2013).

The limitations mirror those of structural data generally. The number of parameters is a few dozen to around a hundred, far fewer than lexical characters, and binary characters agree by chance often, so statistical power at deep time depths is limited and must be assessed against a null model of

chance agreement. Parameter values are analyses, not observations: they depend on a theory of grammar and on the analyst's interpretation of the data, which introduces an uncertainty of a different kind from etymological doubt. Parameters can change through contact and can recur independently. And most PCM analyses have used distance methods; a model-based treatment would need models of parameter change that respect the implicational structure, for instance hierarchical models of inapplicable characters or models of correlated evolution (Chapter 13). Guardiano and Longobardi (2026) review the method and its results.

3.10 Comparing kinds of evidence

Lexical, phonological, morphological and syntactic characters are often treated as interchangeable columns in a matrix. They are not. They differ in how many of them exist, in what establishes their homology, in how they depend on each other, in how well their ancestral states are known, and in whether a shared state can be read as a shared innovation. Table 3.5 summarizes the differences.

Table 3.5: The evidential status of four kinds of linguistic characters.

	Lexical (cognate sets)	Phonological (sound changes)	Morphological (paradigms, irregularities)	Syntactic and typological (features, parameters)
Number of characters	hundreds to thousands	tens	tens	tens to a few hundred
Homology criterion	regular sound correspondences	identity of the change and its conditioning	shared morphemes and shared irregularity	identity of a feature or parameter value; no etymology needed
Main dependence	within meaning slots; borrowing episodes; shared roots	relative chronology; one change affects many words	within paradigms; analogical levelling	implicational and functional links between features
Reconstruction uncertainty	which cognate set is ancestral is often unknown	direction usually known from the comparative method	direction usually known, but loss is uninformative	direction usually unknown; states recur
Homoplasy and borrowing	parallel semantic shift; loans, especially between relatives	natural changes recur; some changes diffuse across dialects	rare for irregular forms	common; contact-sensitive
Interpretable as innovations?	weakly: a replacement is an innovation, but rarely diagnostic	yes, if the change is unusual	yes, strongly for shared irregularities	rarely, without a reconstruction of the ancestral grammar

Three of these differences change the model-selection problem, not just the amount of data.

Dependence. For lexical data the main dependence is created by the coding (Section 3.5.2) and can be removed by multistate models of meaning slots. For phonological characters it comes from

relative chronology (one change can only apply after another has created its input) and from the fact that one change affects many words, so a sound change should be coded once rather than as many lexical differences. For morphological and syntactic characters it is part of the grammar itself: features are linked by implications and by functional pressures. Here independence is not a harmless approximation, and the model must either encode the dependence (hierarchical or inapplicable coding, the neutralization of PCM) or estimate it (models of correlated evolution).

Reconstruction uncertainty. Sound changes and morphological innovations usually come with a known ancestral state and direction, because the comparative method reconstructs them. They can be coded as directional characters, analysed with irreversible models or costs (Section 4.10), or used as constraints on the tree. For cognate sets, the birth of a set is directional but which set was ancestral in a meaning is often unknown, which is what stochastic Dollo and reversible binary models handle in different ways (Chapter 7). For structural features the ancestral state is usually unknown and states recur, so reversible models with rate variation among features are the natural choice, and the ancestral state is an output of the analysis rather than an input (Chapter 12).

Interpretable innovations. A shared unusual sound change or a shared morphological irregularity is close to a single historical event and is strong evidence for a subgroup. A shared cognate class is weak evidence individually and strong only in aggregate; a shared typological feature value is weak evidence even in aggregate unless contact and chance can be excluded. This is why Ringe et al. (2002) weighted phonological and morphological characters more heavily, and why Nakhleh et al. (2005b) found that weighting mattered (Chapter 4). In a model-based analysis the equivalent of weighting is *partitioning*: giving each kind of character its own model, rates and, where appropriate, its own assumptions about direction, and checking the fit of each partition separately.

Two cautions apply to combining them. First, phonological and morphological characters are usually *selected* because they are informative innovations, often the very innovations that define the accepted subgroups. A tree built from them will recover those subgroups largely by construction (Section 1.9), and the selection is an ascertainment process that no standard correction describes. Second, rates differ enormously between kinds of characters, and structural features are not in general more stable than basic vocabulary (Greenhill et al., 2017). A combined analysis with a single rate distribution for everything will let the most numerous characters (usually lexical) dominate, and will misplace the rare but informative ones. The choice of data and the choice of model are therefore one decision, not two.

3.11 Databases and data standards

For most of the history of linguistic phylogenetics, datasets were assembled by individual research groups, stored in idiosyncratic spreadsheets and distributed, if at all, as NEXUS matrices in which the words themselves had been discarded. Reanalysis was difficult and errors were hard to trace. Over the last decade this has changed dramatically, thanks to a set of shared standards and reference catalogues.

3.11.1 Reference catalogues: Glottolog and Concepticon

Two catalogues anchor the system. *Glottolog* provides a unique identifier, the *Glottocode*, for every language, dialect and language family (collectively, *languoids*), together with a genealogical classification and an extensive bibliography (Hammarström et al., 2024). Linking every doculect

in a dataset to a Glottocode makes it possible to merge datasets, to attach geographic coordinates and classification information, and to match languages across lexical, structural and cultural databases. *Concepticon* plays the same role for meanings (Section 3.3), and a related catalogue, *CLTS* (Cross-Linguistic Transcription Systems), standardizes phonetic transcriptions by mapping the many transcription conventions used in the literature to a common set of sound descriptions (Anderson et al., 2018).

3.11.2 CLDF and Lexibank

The *Cross-Linguistic Data Formats* (CLDF) specification defines a simple, uniform way to store linguistic datasets as sets of CSV tables with a JSON metadata file (Forkel et al., 2018). A CLDF *Wordlist* has a `FormTable` with one row per word (like our toy `wl` table), a `LanguageTable` linking each language to a Glottocode, a `ParameterTable` linking each meaning to a Concepticon concept set, and optionally a `CognateTable` that records cognate judgements for individual forms, including partial cognates and alignments. A CLDF *StructureDataset* holds typological data such as Grambank or WALS. Because the format is the same for every dataset, the same tools (in R, Python or any language that reads CSV) work on all of them, and datasets can be checked automatically for consistency.

Lexibank is a repository of lexical datasets converted to CLDF with standardized languages, concepts and transcriptions (List et al., 2022a). It aggregates a large number of previously published datasets, from family-specific collections to global resources, and computes from them a range of derived features (for example, phoneme inventories and patterns of colexification). For phylogenetic purposes, Lexibank offers two things: ready-to-use cognate-coded datasets in a common format, and uniformly transcribed word lists on which automated cognate detection can be run. Its second release covers more than 3,100 languages and over one and a half million word forms (Blum et al., 2025). Chapter 17 shows how to read CLDF datasets into R.

3.11.3 Family-specific lexical databases

Many of the major phylogenetic studies rest on databases curated by specialists in one family. A few of the most important are listed here; Appendix G gives a fuller catalogue.

- **Indo-European.** The dataset of Dyen et al. (1992), with cognate judgements on 200 meanings for 95 speech varieties, was the starting point for early phylogenetic work, including that of Gray and Atkinson (2003). It was extended and corrected in the *Indo-European Lexical Cognacy Database* (IELex), used by Bouckaert et al. (2012) and, with further changes, by Chang et al. (2015). The most recent and most carefully designed resource is *IE-CoR* (Indo-European Cognate Relationships), built by a large team of specialists for about 160 languages and 170 rigorously defined meanings, with explicit criteria for meaning selection and cognate coding (Heggarty et al., 2023). It has since been published as an open dataset in its own right, with 4,981 cognate sets and explicit coding of known horizontal transfer (Anderson et al., 2025). IE-CoR was developed through an earlier project database known as CoBL (Cognates in Basic Lexicon).
- **Austronesian.** The ABVD, described above (Greenhill et al., 2008).
- **Bantu.** Several lexical datasets underlie phylogenetic studies of Bantu, including the basic vocabulary data of Grollemund et al. (2015) for over 400 languages.
- **Sino-Tibetan.** The datasets of Sagart et al. (2019) and of Zhang et al. (2019), which differ in language sample, meaning list and coding decisions.

- **Uralic.** UraLex, a basic vocabulary database for Uralic languages with cognate judgements (Syrjänen et al., 2013).
- **Dravidian.** The dataset of Kolipakam et al. (2018), collected in large part through new fieldwork.

3.11.4 Global resources

At the global scale, the *ASJP database* provides 40-item word lists in a uniform, simplified transcription for the large majority of the world's languages (Wichmann, 2022; Brown et al., 2008). It contains no cognate judgements, so it is used with distance methods based on string similarity (Chapter 4) or with automated cognate detection. WALS, Grambank and PHOIBLE provide structural and phonological data at a similar scale, all in CLDF. Because many features in these databases are logically or statistically dependent, Graff et al. (2025b) derived curated global datasets from Grambank and from WALS, AUTOTYP, PHOIBLE and Lexibank that minimize such dependencies, for analyses that treat features as independent characters (Section 3.5.2).

Rebuilding the Indo-European dataset: IE-CoR. Much of the criticism of early Bayesian analyses of Indo-European focused on the data rather than the models: inherited errors in the Dyen et al. data, inconsistencies in meaning selection, undetected loans and the inclusion of languages whose data were of uneven quality (Pereltsvaig and Lewis, 2015; Chang et al., 2015). IE-CoR was designed to answer these criticisms at the source (Heggarty et al., 2023). Each meaning has an explicit definition specifying, for instance, which of several near-synonyms counts as the basic word; each language's data were compiled or checked by a specialist in that branch; cognate sets are documented with reference to etymological sources; and loanwords are flagged. The choice of the basic word follows a written protocol. The target is one lexeme per meaning in every language, because a coder who admits extra synonyms in one language and not in another creates artificial differences between them; near-synonyms are tolerated in at most 8 of the 170 meanings of a fully attested language. Among candidates, the protocol favours the more frequent, earlier-acquired and more general (hypernymous) word, and the unmarked one: the word of everyday register, used literally rather than figuratively, neither intensifying nor attenuating, and free of emotive overtones. So *burn* is preferred to *combust*, *kill* to *slay*, and, in the British English variety targeted, *small* to *little*, whose uses often carry endearment or disparagement. Rules of this kind do not remove judgement, but they make it explicit and repeatable. Ancient languages are included as sampled ancestors or as dated tips (Chapter 10). The resulting analysis supported a "hybrid" scenario for Indo-European origins, with an early phase south of the Caucasus and a later expansion through the steppe; Chapters 10 and 15 discuss the result and the debate. The general lesson concerns data: more than any change in models, the credibility of the newest Indo-European phylogenies rests on the effort invested in defining and documenting the data.

3.12 Automated cognate detection

Expert cognate judgement is slow. Coding a few hundred languages for a 200-meaning list can take years of specialist work, and for most of the world's language families no such coding exists. *Automated cognate detection* (ACD) aims to identify cognates directly from word lists, using computational analogues of the steps of the comparative method: compare words phonetically, identify recurrent correspondences, and cluster words into cognate sets (List, 2014; List et al., 2017).

3.12.1 String similarity and phonetic alignment

The simplest approach compares word forms by *edit distance*. The *Levenshtein distance* between two strings is the minimum number of insertions, deletions and substitutions needed to turn one into the other (Levenshtein, 1966). Divided by the length of the longer string, it gives the *normalized Levenshtein distance* (LDN), which ranges from 0 for identical strings to 1 for strings with nothing in common. The ASJP project defined a distance between languages as the average LDN between words for the same meaning, divided by the average LDN between words for *different* meanings; this *LDND* corrects for chance similarity due to shared phoneme inventories and phonotactics (Wichmann et al., 2010a; Serva and Petroni, 2008). These measures are used directly to build trees by distance methods (Chapter 4), without any cognate judgements at all.

Edit distance treats all substitutions as equally costly, which is linguistically naive: *t* and *d* are more likely to correspond than *t* and *a*. Better methods align words *phonetically*. Early algorithms scored alignments using hand-crafted measures of phonetic similarity between segments (Covington, 1996; Kondrak, 2000). A powerful simplification is to map sounds to *sound classes*, groups of sounds that frequently correspond to each other, following an idea of Dolgopolsky (Dolgopolsky, 1986). Two words can then be judged potentially cognate if their first two consonant classes match (Turchin et al., 2010), or aligned using class-based scoring matrices, as in the SCA method (Sound-Class-Based Phonetic Alignment) (List, 2012b).

3.12.2 LexStat, PMI and machine learning

The most important advance was to make similarity *language-specific*, mirroring the comparative method's reliance on regular correspondences rather than raw similarity. *LexStat* first aligns all word pairs between two languages, then uses a permutation procedure to estimate which segment correspondences occur more often in words with the same meaning than in words with different meanings, and finally rescores all word pairs using these language-pair-specific correspondence scores (List, 2012a). Two words are then judged cognate if their rescored distance falls below a threshold, and words are clustered into cognate sets within each meaning. Jäger (2013) developed a related approach based on *pointwise mutual information* (PMI) between sound segments, estimated from probable cognates across many language pairs, which defines a data-driven substitution matrix for alignment. Later work combined PMI-based alignment scores and other features in supervised classifiers such as support vector machines trained on expert-coded data (Jäger et al., 2017), and explored neural network approaches (Rama, 2016). More recent supervised models adapt transformer architectures from computational biology and improve steadily with the amount of expert-labelled training data (Akavarapu and Bhattacharya, 2024).

The final step, clustering words into cognate sets, can use simple hierarchical clustering (for example UPGMA with a distance threshold) or network partitioning algorithms. List et al. (2017) compared several methods on a benchmark of expert-coded datasets from different families and found that LexStat combined with the Infomap community-detection algorithm performed best overall, while all methods performed worse on families with complex morphology or much contact. Automated cognate detection is implemented in the Python library LingPy, and the EDICTOR web application allows experts to inspect and correct automated judgements (List, 2017), an approach often called *computer-assisted* language comparison.

3.12.3 Evaluating cognate detection: B-cubed scores

How should the output of an ACD method be compared with expert judgements? Cognate sets are clusters, and different methods can produce different numbers of clusters with different labels, so simple accuracy is not meaningful. The standard measure in the field is the family of *B-cubed* scores, originally developed for evaluating coreference resolution (Bagga and Baldwin, 1998) and introduced to cognate detection by Hauer and Kondrak (2011). For each word i in a meaning, let $G(i)$ be the set of words in its gold-standard (expert) cognate set and $C(i)$ the set of words in its automatically inferred cluster. Then

$$P = \frac{1}{N}\sum_{i=1}^{N}\frac{|G(i)\cap C(i)|}{|C(i)|}, \qquad R = \frac{1}{N}\sum_{i=1}^{N}\frac{|G(i)\cap C(i)|}{|G(i)|}, \qquad F = \frac{2PR}{P+R}, \tag{3.1}$$

where N is the number of words. *Precision* P is high when inferred clusters contain few false cognates; *recall* R is high when true cognates are not split apart; and the *F-score* is their harmonic mean. A method that puts every word in its own cluster has perfect precision and poor recall; one that lumps all words for a meaning together has perfect recall and poor precision.

3.12.4 A toy demonstration

We can demonstrate the whole pipeline on our toy data. We compute LDN between the phonetic forms within each meaning, cluster them by average-linkage hierarchical clustering cut at a threshold, and compare the clusters with the expert cognate classes using B-cubed scores. This is far cruder than LexStat, but it shows the logic and the typical pattern of errors.

```
ldn <- function(a, b) adist(a, b) / outer(nchar(a), nchar(b), pmax)
auto_cluster <- function(forms, threshold = 0.5) {
  d <- as.dist(ldn(forms, forms))
  cutree(hclust(d, method = "average"), h = threshold)
}
cluster_all <- function(wl, threshold = 0.5) {
  out <- character(nrow(wl))
  for (cc in unique(wl$Concept)) {
    i <- wl$Concept == cc
    out[i] <- paste(cc, auto_cluster(wl$Phon[i], threshold), sep = "-")
  }
  out
}
bcubed <- function(gold, pred) {
  p <- r <- numeric(length(gold))
  for (i in seq_along(gold)) {
    both <- sum(gold == gold[i] & pred == pred[i])
    p[i] <- both / sum(pred == pred[i])
    r[i] <- both / sum(gold == gold[i])
  }
  P <- mean(p); R <- mean(r)
  c(precision = P, recall = R, F = 2 * P * R / (P + R))
}
```

```
wl$Auto <- cluster_all(wl, threshold = 0.5)
wl[wl$Concept == "hand", c("Language", "Phon", "Cognate", "Auto")]
```

```
#>      Language  Phon Cognate   Auto
#> 1     English  hEnd       A hand-1
#> 2      German  hant       A hand-1
#> 3     Swedish  hand       A hand-1
#> 4     Italian  mano       B hand-2
#> 5     Spanish  mano       B hand-2
#> 6      French    mE       B hand-3
#> 7     Russian  ruka       C hand-4
#> 8      Polish rENka       C hand-4
#> 9  Lithuanian ranka       C hand-4
#> 10      Irish   lav       D hand-5
#> 11      Greek  xeri       E hand-6
#> 12      Hindi   hat       F hand-1
```

```
round(bcubed(wl$CogID, wl$Auto), 3)
```

```
#> precision    recall         F
#>     0.985     0.574     0.725
```

The clustering of 'hand' is nearly perfect, with one instructive error: Hindi *hāth* (here *hat*) is placed with English *hand* and its Germanic cognates because the forms happen to be similar, although they are not cognate. This is a false positive of exactly the kind the comparative method guards against. Across all meanings, precision is high and recall low: the method rarely groups non-cognates, but it fails to recognize cognates that have diverged in form, such as the reflexes of 'two' (*tu*, *cvai*, *dva*, *Dio*) or 'tooth' (*tu8*, *can*, *dente*, *dat*). Figure 3.3 shows how the balance between precision and recall shifts with the clustering threshold. In this tiny dataset the F-score peaks at a high threshold, because many true cognates are phonetically far apart; in real applications the threshold is tuned on expert-coded training data and depends strongly on the method and the family.

```
th <- seq(0.1, 0.95, by = 0.05)
scores <- t(sapply(th, function(t) bcubed(wl$CogID, cluster_all(wl, t))))
matplot(th, scores, type = "l", lty = 1:3, lwd = 2, col = "grey15",
        xlab = "LDN clustering threshold", ylab = "B-cubed score",
        ylim = c(0, 1), las = 1)
legend("bottomright", c("precision", "recall", "F-score"), lty = 1:3,
       lwd = 2, col = "grey15", bty = "n")
```

```
th[which.max(scores[, "F"])]
```

```
#> [1] 0.85
```

```
round(sapply(split(wl, wl$Concept),
             function(x) bcubed(x$CogID, x$Auto)["F"]), 2)
```

```
#>      dog.F      eat.F     hand.F mountain.F     star.F    tooth.F
#>       0.84       0.74       0.88       0.93       0.54       0.56
```

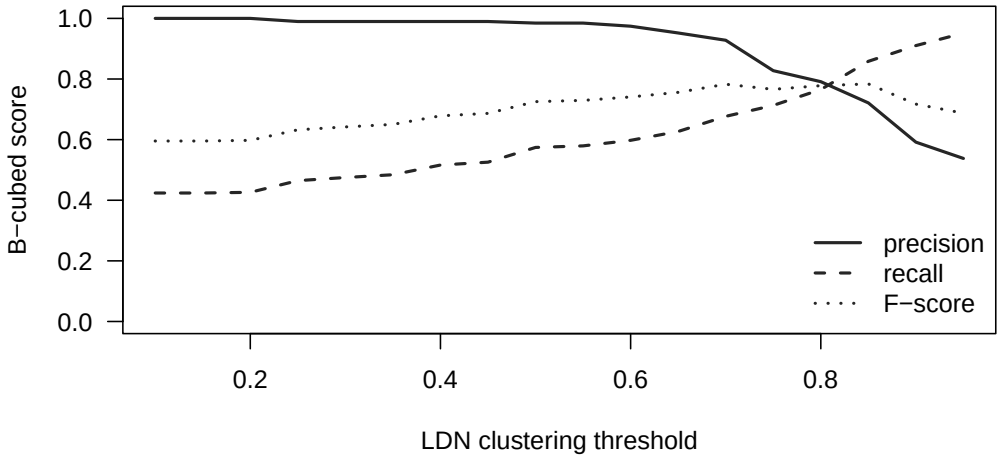


Figure 3.3: B-cubed precision, recall and F-score of the toy cognate clustering as a function of the distance threshold. Low thresholds split true cognate sets (high precision, low recall); high thresholds merge unrelated words (high recall, low precision).

```
#>      two.F    water.F
#>       0.36       0.69
```

The per-meaning scores show that ACD performance varies greatly among meanings. Meanings whose cognates have undergone extensive sound change, like ‘two’ and ‘tooth’, are hard; meanings where the classes are phonetically distinct, like ‘mountain’, are easy. LexStat and PMI-based methods do much better than raw LDN on ‘two’ because they learn, for instance, that German *z* [ts] regularly corresponds to English *t*. But no method recovers all deep cognates from 40 or 200 words, because there is too little data to establish correspondences for rare sounds.

Finally, an LDN-based distance between languages can be computed directly from the word list, without any cognate judgements. For each pair of languages we average, over meanings, the smallest LDN between their words for that meaning (the minimum handles synonyms):

```
D <- matrix(0, length(langs), length(langs), dimnames = list(langs, langs))
for (a in langs) for (b in langs) {
  D[a, b] <- mean(sapply(unique(wl$Concept), function(cc) {
    min(ldn(wl$Phon[wl$Language == a & wl$Concept == cc],
            wl$Phon[wl$Language == b & wl$Concept == cc]))
  }))
}
round(D[1:4, 1:4], 2)
```

```
#>         English German Swedish Italian
#> English    0.00   0.81    0.69    0.79
#> German     0.81   0.00    0.44    0.81
#> Swedish    0.69   0.44    0.00    0.83
#> Italian    0.79   0.81    0.83    0.00
```

```
ldn_nj <- midpoint(NJ(as.dist(D)))
plot(ldn_nj, cex = 0.8,
     label.offset = 0.01 * max(node.depth.edgelength(ldn_nj)))
```

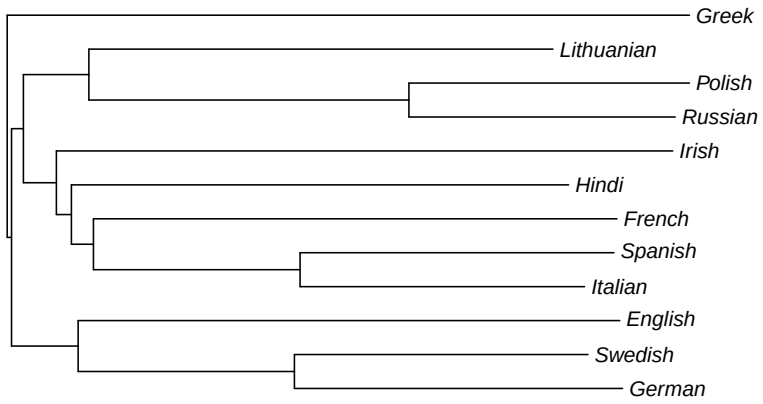


Figure 3.4: Neighbour-joining tree (midpoint-rooted) from average normalized Levenshtein distances between the toy word lists, computed without cognate judgements. String similarity recovers the closest relationships but is affected by chance resemblances and by the loss of similarity through sound change.

3.12.5 How good is automated cognate detection for phylogenetics?

The practical question is not whether ACD reproduces expert judgements exactly but whether the resulting trees are good enough. Several studies have compared phylogenies inferred from automated and expert cognates on the same word lists. Rama et al. (2018) found that trees inferred from automatically detected cognates using PMI-based methods came close to those inferred from expert cognates when both were compared with gold-standard classifications. Jäger (2018) used PMI-based methods to infer trees for most of the world's language families from ASJP data, and evaluated them against Glottolog with quartet-based distances (Chapter 2). Both evaluations measure agreement with expert classifications, which are themselves built from the comparative method; good scores show that automated pipelines reproduce expert judgement, not that either has found the true history (Section 1.9). These results suggest that automated methods are now good enough for exploratory work on poorly studied families and for providing tree samples for comparative analyses, while expert judgements (or expert-corrected automated judgements) remain essential for analyses whose conclusions depend on fine details of subgrouping or dating.

Recent work questions whether cognate sets are the right intermediate representation at all. Jäger (2025) compared inference from expert cognates with two fully automated alternatives on Lexibank data: characters derived from automatic cognate clustering, and characters derived from

multiple sequence alignments of word forms produced with a pair-hidden Markov model. Evaluated against Glottolog classifications and against Grambank data, the alignment-based characters were competitive with both kinds of cognate data within individual families, and on datasets spanning many families they outperformed both, giving trees closer to Glottolog, a better fit to the Grambank features and a clearer phylogenetic signal. This suggests a scalable route to global phylogenies that does not wait for expert annotation, although the evaluation concerns topologies only, not branch lengths or dates. (A revised version of the paper corrected parts of the pipeline and changed all the reported numbers while leaving these qualitative conclusions unchanged, a reminder to cite such work by version.) Other studies point to the limits of automation. Häuser and Stamatakis (2025) found that cognate matrices extracted automatically from a large multilingual dictionary gave trees largely inconsistent with gold-standard trees, and Häuser et al. (2026) concluded that the small size of the available cognate datasets limits both parameter-rich models and machine-learning approaches. Agreement with Glottolog, here as above, is agreement with the comparative method, not with the true history.

Homology uncertainty and joint inference. In molecular phylogenetics, the analogue of cognate coding is sequence alignment, which determines which positions in different sequences are homologous. For decades the standard practice was to fix an alignment and then infer the tree as if the alignment were known, just as linguists fix cognate judgements and then infer the tree. Biologists showed that this ignores alignment uncertainty and can bias the tree, and developed methods that infer alignment and tree jointly, sampling both in a single Bayesian analysis; BAli-Phy is the best-known implementation (Redelings and Suchard, 2005). A similar move is possible in linguistics: rather than fixing cognate sets, a model could treat cognacy (or the alignment of word forms) as uncertain, integrate over plausible cognate partitions, and let the tree inform which ambiguous words are cognate. Such joint inference of cognacy and phylogeny has, to our knowledge, rarely been attempted on real linguistic data, although probabilistic models of word-form evolution on trees (Chapter 12) provide much of the machinery. A step in this direction is the adaptation of the TKF91 insertion-deletion model of molecular evolution to aligned cognate word forms by Goldstein et al. (2026), which infers phylogenies from segmental strings rather than from abstract cognate codes, although it still starts from cognate sets that have been identified and aligned. Gene-family and orthology-inference methods in comparative genomics, which cluster genes into families of homologs while accounting for duplications and transfers, are another underexploited source of ideas for cognate detection (Chapter 16).

3.13 Data quality and disagreement

3.13.1 Sources of error

Every stage of data collection introduces error. *Elicitation and transcription*: the word recorded for a meaning depends on the source (a dictionary, a grammar, a field interview), on how the meaning was explained to the speaker, and on the transcription conventions used. *Meaning selection*: when a language has several near-synonyms for a meaning, the choice of which one to code determines its cognate class, and different coders may choose differently; the problem is acute for meanings such as ‘to say’, ‘dirty’ or ‘to throw’, which have many competing words in many languages. *Cognate judgement*: even specialists disagree about doubtful etymologies, about the depth at which cognacy should be judged, and about how to treat partial cognates. *Loan detection*: borrowings between closely related languages are often undetectable by phonological criteria, because the borrowed word would look the same as an inherited one. Each kind of error has a different effect. Random

coding errors mostly add noise, which inflates branch lengths and can reduce support. Systematic errors, such as undetected loans between particular neighbours or a coder's tendency to lump doubtful cognates in one branch, can bias the topology and dates.

3.13.2 How much do coders disagree?

Remarkably, the reliability of cognate coding has seldom been measured directly. In biology, measurement error in morphological characters is routinely discussed; in linguistics, it is common for a dataset to be coded by a single specialist per family or branch, with no independent recoding. Where datasets for the same family have been compiled independently, comparisons reveal substantial differences. The Indo-European datasets used by Gray and Atkinson (2003), Bouckaert et al. (2012), Chang et al. (2015) and Heggarty et al. (2023) differ in the languages and meanings sampled, in which word was chosen for many meanings, in cognate judgements and in loan flagging, and some of the differences between their results have been attributed to these data differences rather than to models (Pereltsvaig and Lewis, 2015). Heggarty (2010) argued that much information is lost when complex etymological relationships are reduced to a single binary cognate label, and that word-list comparisons should record degrees of relatedness and the sources of uncertainty.

Even before cognates are judged, the choice of which word answers each concept varies between compilers. Snee et al. (2026) measured the consequences with independently compiled wordlists for the same varieties and found that such differences in concept translation can change low-level subgrouping in Bayesian trees noticeably while leaving overall lexical distances highly correlated (Section 15.2).

The B-cubed scores introduced for evaluating ACD are also the natural measure of agreement between two human coders: precision and recall become symmetric measures of how much one coder splits relative to the other. Computing such scores between independent codings of a sample of meanings, and propagating the resulting uncertainty into phylogenetic analyses (for example by analysing several alternative codings and combining the posteriors), would be a straightforward improvement on current practice.

3.13.3 Good practice

A short list of recommendations for building and documenting a linguistic phylogenetic dataset follows from this chapter.

1. Use a well-defined meaning list, linked to Concepticon, with explicit rules for choosing among synonyms.
2. Record the actual word forms with their sources, not only the cognate matrix. Link every language to a Glottocode.
3. Record cognate judgements at the level of individual words (and, where relevant, morphemes), with notes on doubtful cases and on the etymological sources used.
4. Flag known and suspected loans, and report how they were treated in the analysis.
5. Store the data in CLDF and archive them in a public repository with a version number, so that analyses can be reproduced and errors corrected.
6. Report the coding decisions (singletons, synonyms, partial cognates, depth of cognacy, loans) explicitly, and test the sensitivity of key conclusions to the main alternatives.

7. Where possible, have a sample of the data recoded independently, or compare expert codings with automated ones, and report the agreement.
8. Record doubtful judgements (CLDF has a field for them) and prepare alternative codings of the doubtful cases, so that their effect can be measured (Section 3.5).
9. State the role assumed for each ancient variety (ancestor, sister or member of a dialect network), code meanings unattested in its corpus as missing, and exclude reconstructed proto-languages from the taxa (Section 3.2).

Structural phylogenetics in Island Melanesia. The study of Dunn et al. (2005) is worth revisiting from the data perspective. The Papuan languages of Island Melanesia had been classified as unrelated or as forming small groups, because shared vocabulary had been eroded to the point where the comparative method could no longer establish relationships. Dunn and colleagues designed a questionnaire of structural features, coded the Papuan and the neighbouring Oceanic languages, and used the Oceanic languages, whose history is known from lexical evidence, as a control: if structural features recovered the Oceanic tree, they could perhaps be trusted for the Papuan languages too. The design is a model of how a new kind of data should be checked against a known case before being applied to an unknown one. In the vocabulary of Section 6.10 it is a test on real data with a known answer, which is stronger than a simulation under the method's own assumptions; it does not, however, show that the method is robust to the conditions that differ between the control and the target, such as the much greater time depth of the Papuan relationships. The approach also drew criticism, including concerns that structural features are prone to borrowing and that the Papuan signal might reflect contact or areal patterns rather than descent; Chapters 8 and 15 take up these concerns.

Global trees from short word lists. Large-scale projects have used the ASJP database to infer trees for nearly all of the world's language families from 40-item word lists, either with string-distance methods (Holman et al., 2008; Brown et al., 2008) or with automated cognate detection followed by character-based phylogenetic inference (Jäger, 2018). These trees are less accurate than expert-based phylogenies of well studied families, as evaluation against Glottolog shows, but they cover thousands of languages for which no phylogeny exists. They are used, with appropriate caution, as scaffolds for global comparative analyses of linguistic and cultural traits (Chapters 13 and 14). The trade-off between depth of curation and breadth of coverage is one of the central tensions in the field's data.

- A phylogenetic data matrix has languages (strictly, doculects) as rows and characters as columns; models assume that characters are homologous across languages and evolve independently.
- In lexical data, homology is cognacy, established by regular sound correspondences rather than by surface similarity; chance resemblances and loans must be excluded, and the depth at which cognacy is judged matters.
- Basic vocabulary lists (Swadesh 100 and 200, Leipzig-Jakarta, ASJP 40, family-specific lists) differ in size and selection principle; meanings must be precisely defined and linked to Concepticon.
- Cognates can be coded as binary presence/absence characters (one per cognate set) or as multistate characters (one per meaning); binary coding creates non-independence within meanings and requires ascertainment correction, while multistate coding raises problems of large state spaces and polymorphism.
- Loans can be kept, coded as missing or dropped; synonyms, partial cognates and root-meaning versus root coding are further decisions that should be documented and tested.

- Structural (typological) features are easy to collect at scale but have few states, high homoplasy, contact sensitivity and logical dependencies; they are not in general more stable than basic vocabulary.
- Sound changes and morphological irregularities are highly informative shared innovations but are few in number and not immune to diffusion.
- CLDF, Glottolog, Concepticon, CLTS and Lexibank have made data sharing and reuse routine; family databases such as ABVD and IE-CoR set the standard for documentation.
- Automated cognate detection (LexStat, PMI-based methods, machine learning) is evaluated with B-cubed precision, recall and F-score and is now good enough for many phylogenetic purposes, though it performs worst on deep cognates and morphologically complex words.
- Inter-coder reliability is rarely measured; sensitivity analyses over alternative codings should be standard.
- Cognate judgements carry uncertainty of several kinds (etymology, depth, segmentation, borrowing, slot filling) that is concentrated, not random, and is rarely recorded; characters are dependent through coding, sound change, borrowing episodes and grammatical implications, and ignoring this makes support overconfident.
- An attested ancient language may be an ancestor, a sister, or one member of a dialect network; the choice affects sampling, coding and dates, and reconstructed proto-languages should not be used as tips.
- Lexical, phonological, morphological and syntactic characters differ in number, dependence, reconstruction uncertainty and interpretability as innovations; these differences call for different models and, in combined analyses, separate partitions. The Parametric Comparison Method shows how syntactic characters and their dependencies can be coded explicitly.

3.14 Exercises

Conceptual

1. English *day* and Latin *dies* look alike and mean the same thing, but are not cognate; English *wheel* and Sanskrit *cakrá-* look different but are. Explain what kind of evidence decides each case, and why a phylogenetic analysis should not use raw similarity of form.
2. For each of Old English (West Saxon), Gothic and Classical Latin, argue whether it is best treated as an ancestor, a sister or a member of a dialect network relative to the modern languages of its branch, and say how you would code it in a dated analysis.
3. Why does binary coding of a meaning with four cognate classes create dependence between characters? Give one model and one resampling scheme that respect the meaning-slot structure.
4. A combined analysis uses 1,200 lexical characters, 25 sound changes and 30 morphological characters, with one gamma distribution of rates for all of them. What can go wrong, and how would partitioning help?
5. Explain why agreement between a PCM tree and the accepted subgrouping of Indo-European is less circular than agreement between a lexical tree and the same subgrouping, and why it is still not fully independent.
6. List three kinds of uncertainty in cognate judgements that are likely to be systematic rather than random, and the direction in which each might bias a tree.
7. Using Table 3.3, say which kinds of lexical homology link French *cœur*, Spanish *corazón*, Romanian *inimă* and Latin *cor* and *anima*. Then explain what pattern in a Latin-and-Romance

cognate matrix would count as evidence *against* treating Classical Latin as the direct ancestor of the Romance languages (hint: a Latin innovation in a meaning where the Romance languages are more conservative), and which coding convention could hide it.

Computational

1. Use `make_binary()` to build the matrix with loans kept, coded as missing and dropped. For each version compute `dist.hamming()` and an `NJ()` tree, and compare the three trees with `RF.dist()`. Which languages move?
2. Extend the chunk that recodes Irish 'water': also recode Italian *mangiare* as a loan (`wl$Loan`) and Spanish *comer* as cognate with English *eat* (a loose treatment of partial cognacy). Report how the Hamming distances from Spanish and Italian to the Germanic languages change.
3. Build a multistate matrix with the `multistate` table of Section 3.4, resolving the Irish polymorphism once as F and once as G. Convert each to a `phyDat` object with `type = "USER"` and the appropriate `levels`, and compare the parsimony scores of the NJ tree with `parsimony()`.
4. Read `data/case-study/forms.csv` with `read.csv()`. Count how many language-meaning pairs have more than one form (synonyms), how many forms are flagged in the `Loan` column, and which variety is a reconstruction. Using `table()`, find the five meanings with the most synonyms and discuss which coding scheme of this chapter they would strain most.
5. Run `cluster_all()` with thresholds from 0.2 to 0.8 and compute, for each threshold, the RF distance between the NJ tree from the automatic cognates (via `make_binary()` on a copy of `wl` whose `CogID` is replaced by `Auto`) and the NJ tree from the expert cognates.

Research

1. You are designing a dataset for a family of 40 languages that includes four varieties known only from texts between 1,500 and 800 years old, one of them a liturgical standard. Justify which of them you would include as tips, how you would code their unattested meanings and their dates, and which sensitivity analyses you would plan.
2. A colleague proposes to combine Grambank features and lexical cognates in one Bayesian analysis of a family with deep, contested structure. Write a short methods plan that justifies the coding of dependent and inapplicable features, the partitioning of the data, and the checks you would use to decide whether the structural partition adds genealogical signal or contact signal.
3. Your cognate coding for 200 meanings was done by one specialist. Propose a feasible design for measuring and propagating coding uncertainty, including how many meanings to recode independently, how to compare the codings (Section 3.12), and how to report the result.

3.15 Further reading

- Campbell (2013), *Historical Linguistics: An Introduction*. The standard textbook introduction to the comparative method, cognacy, sound change and borrowing; the necessary background for anyone coding cognates.
- Swadesh (1955): The paper that introduced the 100-item list, with Swadesh's own discussion of the criteria for basic vocabulary.
- McMahon and McMahon (2005): A linguist's critical discussion of meaning lists, synonyms, shared roots, regional variants of the Swadesh list and borrowing in basic vocabulary, with

tests of stable and unstable sublists.

- Tadmor et al. (2010): Explains how the Leipzig-Jakarta list was derived from cross-linguistic data on borrowing, and what "basic vocabulary" means empirically.
- Greenhill et al. (2008): Describes the design of the ABVD, a model for documented, reusable lexical databases.
- Heggarty et al. (2023) and its supplementary materials. The most detailed account of how a modern cognate database (IE-CoR) defines meanings and codes cognates.
- Forkel et al. (2018) and List et al. (2022a). Introductions to CLDF and Lexibank, the standards and repository that now underpin most data sharing in the field.
- List et al. (2017): A systematic comparison of automated cognate detection methods on expert-coded benchmark data, with a clear explanation of B-cubed evaluation.
- List (2016): A thoughtful discussion of partial cognacy and the ways in which word-level cognate coding simplifies lexical history.
- Heggarty (2021): A review of the cognacy databases behind Indo-European phylogenetics, their history and their flaws, and of how database design affects results. Scarborough (2020) shows, case by case, why many Indo-European cognate decisions resist a clean binary coding.
- Anderson et al. (2025): The data paper for IE-CoR, documenting meaning definitions, cognate coding and the treatment of loans; the most detailed published account of how a modern cognate database is built.
- Jäger (2025) and Häuser et al. (2024): Two recent comparisons of cognate-based and sound-based characters for tree inference, with opposite conclusions for different kinds of sound-level data.
- Dunn et al. (2008): A detailed methodological exploration of structural phylogenetics, its potential and its pitfalls.
- Greenhill et al. (2017): A comparison of the rates of change of lexical and grammatical features in Austronesian that challenges assumptions about the stability of grammar.
- Longobardi and Guardiano (2009) introduced the Parametric Comparison Method; Guardiano and Longobardi (2026) is a recent review of syntactic methods for language phylogenies by its authors.
- Ringe et al. (2002), read alongside Nakhleh et al. (2005b), is the best worked example of arguing explicitly for the evidential weight of different kinds of characters.

Chapter 4

Distance and Parsimony Methods

How can a table of words, or of cognate judgements, be turned into a tree? The two oldest computational answers are *distance methods*, which first compress the data into a table of pairwise dissimilarities between languages and then fit a tree to that table, and *parsimony methods*, which work directly on characters and prefer the tree that requires the fewest changes. Both were developed in biology in the 1960s and adopted in linguistics long before model-based methods became common, and both remain useful: distance methods because they are fast enough for thousands of languages and can be applied to raw word forms without cognate judgements, and parsimony because its logic of shared innovations is close to the reasoning of the comparative method. This chapter explains how they work, what they assume, and why their weaknesses, above all long-branch attraction, motivate the model-based methods of Chapters 5 and 6.

After reading this chapter you should be able to:

- compute lexicostatistical, Hamming, Jaccard, corrected and Levenshtein-based (LDN, LDND) distances between languages, and explain what each measures and fails to measure;
- state the four-point and three-point conditions and explain why UPGMA, unlike neighbor joining, is misled by unequal rates of change;
- build and compare distance trees in R with `upgma()`, `nj()`, `bionj()` and `fastme.bal()`, and explore distance matrices with MDS and PCA (`cmdscale()`, `isoMDS()`, `prcomp()`) without over-reading them;
- explain the parsimony criterion, compute parsimony scores with Fitch and Sankoff costs, and match cost schemes (Wagner, Camin–Sokal, Dollo) to kinds of linguistic characters;
- search for most parsimonious trees, and interpret consistency and retention indices and bootstrap support, including their limits for cognate data;
- explain character compatibility and the four-gamete test, and their role in the Indo-European work of Ringe, Warnow and Taylor;
- explain long-branch attraction and why agreement with an accepted classification is not, on its own, validation of a method.

4.1 Two ways to build a tree

Any tree-building method has to answer two questions. The first is what information it uses. *Distance methods* reduce a data matrix D of languages by characters to a symmetric matrix of pairwise

distances d_{ij}, one number for each pair of languages, and discard everything else. *Character-based methods*, which include parsimony, likelihood and Bayesian inference, use the full matrix: they ask how each individual character (each cognate set, each sound change, each structural feature) could have evolved along the branches of a candidate tree. The distinction matters because the reduction to pairwise distances loses information. Two pairs of languages can have the same distance for quite different reasons, and a distance matrix cannot record which particular cognate sets are shared by which subsets of languages.

The second question is whether the method follows a fixed recipe or optimizes a criterion. *Algorithmic* (or clustering) methods such as UPGMA and neighbor joining apply a sequence of steps that terminates in a single tree; they never compare alternative trees explicitly. *Optimality-criterion* methods define a score for every possible tree, such as the number of changes required (parsimony), the sum of branch lengths (minimum evolution) or the probability of the data (likelihood), and then search for the tree with the best score. Because the number of trees grows super-exponentially with the number of languages (Chapter 2), optimality-criterion methods require heuristic search, and the tree they return is only as good as the search.

Table 4.1 places the methods of this and the next two chapters in this two-way scheme.

Table 4.1: A classification of tree-building methods by the information they use and by how they arrive at a tree.

	Algorithm	Optimality criterion
Distances	UPGMA, WPGMA, neighbor joining, BIONJ	least squares (Fitch–Margoliash), minimum evolution, balanced minimum evolution (FastME)
Characters	(stepwise addition, used to build starting trees)	maximum parsimony, maximum compatibility, maximum likelihood; Bayesian inference (which samples rather than optimizes)

A third distinction, between methods that rest on an explicit probabilistic model of change and methods that do not, cuts across this table. Distance corrections and the least-squares criterion can be motivated by models; parsimony and compatibility were developed without them, and their statistical behavior was only worked out later (Felsenstein, 1978a, 2004). We return to this point at the end of the chapter.

4.2 From word lists to distances

4.2.1 Lexicostatistical distances from cognate judgements

The simplest linguistic distance is the one that lexicostatistics has used since the 1950s (Swadesh, 1955; Lees, 1953). Take a list of m basic meanings (a Swadesh list, Chapter 3), decide for each meaning whether the words in languages i and j are cognate, and compute the proportion of meanings for which they are:

$$s_{ij} = \frac{\text{number of meanings with cognate words in } i \text{ and } j}{\text{number of meanings attested in both}}, \qquad p_{ij} = 1 - s_{ij}. \tag{4.1}$$

Here s_{ij} is the *proportion of shared cognates* (the "cognate percentage" of the older literature) and p_{ij} is the uncorrected distance. English and German, for example, have cognate words for 'two' (*two*, *zwei*), 'hand' (*hand*, *Hand*), 'night' (*night*, *Nacht*) and 'heart' (*heart*, *Herz*), but not for 'dog' (*dog*, *Hund*: the English cognate of *Hund* is *hound*, which no longer means simply 'dog'). The denominator excludes meanings missing in either language, and some convention is needed for synonyms: a common one counts a pair as cognate if any of the words for the meaning are cognate, another divides credit fractionally. These choices are not innocent, and they are discussed in Chapter 3.

Equivalently, if the cognate data have been recoded as a binary presence/absence matrix (Chapter 3), with one column per cognate set, we can compute a Hamming distance: the proportion of columns in which two languages differ. The two measures are related but not identical. In the binary coding, a meaning for which i and j have different, non-cognate words contributes two mismatches (one for each cognate set), whereas a shared cognate contributes a match for that set *and* matches of "absent" for every cognate set that both languages lack. Since most cognate sets are absent from most languages, binary Hamming distances are dominated by shared absences, which carry little information. For distance work the meaning-based proportion (4.1) is generally preferable; for character-based work the binary matrix is the norm.

A third option keeps the binary matrix but ignores shared absences. Treat each language as the set A_i of cognate sets it possesses. The *Jaccard distance* is

$$d^{\mathrm{J}}_{ij} = 1 - \frac{|A_i \cap A_j|}{|A_i \cup A_j|}, \tag{4.2}$$

the proportion of the cognate sets found in either language that are not found in both. Because only presences enter the formula, the many cognate sets that both languages lack cannot make them look similar. In R, `dist(x, method = "binary")` computes it from a 0/1 matrix. Which value counts as "present" is a decision, not a fact of the data: computed on the 0s rather than the 1s, the same formula measures shared absence, and for structural features coded as presence or absence of a property the choice of polarity changes the distances. More generally, the choice of distance measure can change the classification that results, which is why Wichmann et al. (2010a) evaluated competing measures against expert classifications before adopting one.

Not every dissimilarity is a distance in the mathematical sense. A *metric* is non-negative, symmetric, zero for an object compared with itself, and satisfies the *triangle inequality* $d_{ik} \leq d_{ij} + d_{jk}$. Hamming and Jaccard distances are metrics, and so is every tree metric (Section 4.3). Corrected distances need not be. The corrections (4.3) and (4.4) below stretch large distances more than small ones, so the corrected distance between two languages can exceed the sum of their corrected distances to a third. Neighbor joining runs on any symmetric matrix, but methods that embed distances in a geometric space, such as the multidimensional scaling of Section 4.7, register such violations as distortion.

4.2.2 Correcting distances for multiple replacements

The proportion p_{ij} underestimates how much change has really happened, because a meaning slot can be replaced more than once. If English replaced an inherited word for 'dog' and later replaced the replacement, the second event is invisible: the slot is "non-cognate" either way. As two languages drift apart, p_{ij} therefore rises ever more slowly and eventually saturates. A *corrected distance* undoes this saturation under some model of change.

The classic correction is implicit in glottochronology (the Poisson-process reasoning it uses is reviewed in Section A.8). Suppose each meaning slot in each lineage is replaced at a constant rate r per unit time, events occur as a Poisson process, and every replacement introduces a word that is not cognate with anything else in the data (an *infinite-alleles* assumption: replacements never reintroduce an old cognate and never happen in parallel). Two languages that separated t time units ago are joined by a path of total length $2t$ in the tree. They share a cognate in a given slot if and only if no replacement has occurred anywhere along that path, which under the Poisson model has probability

$$\mathbb{E}[s_{ij}] = e^{-2rt}. \tag{4.3}$$

Inverting gives the corrected, additive distance $\hat{d}_{ij} = -\ln s_{ij}$, which estimates $2rt$, the expected number of replacements per slot along the path between the two languages. Glottochronology took a further step. It wrote the per-millennium retention rate as $\rho = e^{-r}$ and solved for time, giving the formula $t = \ln s_{ij} / (2 \ln \rho)$ associated with Lees (1953) and Swadesh (1955), with a supposedly universal retention rate of roughly 80 to 86 percent per millennium depending on the list. The correction itself is sound under its assumptions. What failed was the assumption of a constant, universal rate, which is a strict molecular clock under another name. Chapter 1 recounts the critiques, and Chapter 9 discusses what replaced the method.

For binary characters a different correction applies. If each character flips between 0 and 1 at the same rate in both directions (a symmetric two-state model, the binary analogue of the Jukes–Cantor model of DNA evolution), then the probability that two languages differ after a path of length d (in expected changes per character) is $p = \tfrac{1}{2}(1 - e^{-2d})$, which saturates at one half rather than one. Inverting,

$$\hat{d}_{ij} = -\tfrac{1}{2} \ln\left(1 - 2p_{ij}\right). \tag{4.4}$$

Chapter 5 derives this formula as a maximum likelihood estimate (Section A.7 introduces the two-state Markov chain behind it). The general lesson is that corrections become large and noisy as p_{ij} approaches its saturation value, so that distances between distantly related languages are estimated with much more error than distances between close relatives. Good distance methods take this into account.

4.2.3 Distances from word forms: Levenshtein, LDN and LDND

Cognate judgements are expensive and require expert knowledge of each family. An alternative that has been popular since the 1990s computes distances directly from the phonetic forms of words. The workhorse is the *Levenshtein distance* or *edit distance* (Levenshtein, 1966): the minimum number of insertions, deletions and substitutions of single segments needed to turn one string into another. The distance between *hant* (German *Hand*, in a broad transcription) and *hEnd* (English) is 2: substitute *a* by *E* and *t* by *d*. It is computed by dynamic programming in time proportional to the product of the two word lengths. Its first linguistic uses were in dialectometry, the measurement of differences between dialects of one language, where most words are cognate and differences of pronunciation are what one wants to measure (Kessler, 1995; Heeringa, 2004).

Raw edit distances favor short words, so in practice they are normalized. Two normalizations are standard in the Automated Similarity Judgment Program (ASJP), which has compiled 40-item word lists in a simplified phonetic alphabet for thousands of languages and dialects (Brown et al., 2008; Holman et al., 2008; Wichmann et al., 2010a):

- *LDN* (Levenshtein distance normalized) divides the edit distance for a meaning by the length of the longer word, then averages over meanings: $\mathrm{LDN}_{ij} = \frac{1}{m}\sum_{k} \mathrm{LD}(w_{ik}, w_{jk}) / \max(|w_{ik}|, |w_{jk}|)$.
- *LDND* (LDN divided) further divides LDN_{ij} by the average normalized distance between words for *different* meanings in the same two languages. The denominator estimates how similar the two languages' words are by chance, given their phoneme inventories and phonotactics. Two languages that both have small inventories and simple CV syllables will have spuriously low LDN; the LDND corrects for this.

Under the LDND, two unrelated languages have an expected value near 1 (their same-meaning words are no more similar than random pairs), and close relatives have values well below 1. Wichmann et al. (2010a) compared these and several other measures against expert classifications and found that LDND performed best among the measures they tested, which is why it became the default in ASJP work.

The following code builds a small, hand-typed word list for ten meanings in ten Indo-European languages, in a broad ASJP-like transcription (capital letters and digits stand for particular sounds, e.g. `3` for schwa, `8` for the "th" of English *three*, `S` for "sh", `C` for "ch"). The transcriptions are simplified and illustrative, not authoritative.

```
words <- read.table(header = TRUE, stringsAsFactors = FALSE, text = "
concept English German Dutch Swedish Italian Spanish French Russian Polish Greek
two     tu      cvai   tve   tvo     due     dos     d3     dva     dva    Dio
three   8ri     drai   dri   tre     tre     tres    trwa   tri     tSi    tria
hand    hEnd    hant   hant  hand    mano    mano    mE     ruka    renka  xeri
night   nait    naxt   naxt  nat     note    noCe    nwi    noC     noc    nixta
water   wat3r   vas3r  wat3r vat3n   akwa    agwa    o      voda    voda   nero
mother  m3D3r   mut3r  mud3r mur     madre   madre   mEr    mat     matka  mitera
name    nem     nam3   nam3  namn    nome    nombre  nom    imya    imye   onoma
tooth   tu8     can    tant  tand    dEnte   dyente  da     zup     zomp   Donti
fish    fiS     fiS    vis   fisk    peSe    pes     pwaso  riba    riba   psari
heart   hart    herc   hart  yEta    kwore   korason k3r    serce   serce  kardia
")
W <- as.matrix(words[, -1]); rownames(W) <- words$concept
langs <- colnames(W)

# normalized Levenshtein distance between two words
ldn_word <- function(a, b) as.numeric(adist(a, b)) / max(nchar(a), nchar(b))

# LDN and LDND between two languages (vectors of words, same meaning order)
ldn_ldnd <- function(x, y) {
  m <- length(x)
  same <- mean(mapply(ldn_word, x, y))
  diff <- mean(unlist(lapply(seq_len(m), function(i)
    sapply(seq_len(m)[-i], function(j) ldn_word(x[i], y[j])))))
  c(LDN = same, LDND = same / diff)
}

n <- length(langs)
LDN <- LDND <- matrix(0, n, n, dimnames = list(langs, langs))
```

```
for (i in 1:(n - 1)) for (j in (i + 1):n) {
  r <- ldn_ldnd(W[, i], W[, j])
  LDN[i, j] <- LDN[j, i] <- r["LDN"]
  LDND[i, j] <- LDND[j, i] <- r["LDND"]
}
round(LDND[1:6, 1:6], 2)
```

```
#>         English German Dutch Swedish Italian Spanish
#> English    0.00   0.56  0.46    0.62    0.80    0.89
#> German     0.56   0.00  0.38    0.57    0.79    0.86
#> Dutch      0.46   0.38  0.00    0.49    0.76    0.81
#> Swedish    0.62   0.57  0.49    0.00    0.74    0.77
#> Italian    0.80   0.79  0.76    0.74    0.00    0.38
#> Spanish    0.89   0.86  0.81    0.77    0.38    0.00
```

The matrix behaves as expected for the closest pairs: German and Dutch, Russian and Polish, and Italian and Spanish have the smallest LDND values. Figure 4.1 shows trees built from the full matrix with two methods described below.

```
library(phangorn)
par(mfrow = c(1, 2), mar = c(1, 1, 2, 1))
upgma_ldnd <- upgma(as.dist(LDND))
plot(upgma_ldnd, main = "UPGMA", cex = 0.8,
     label.offset = 0.01 * max(node.depth.edgelength(upgma_ldnd)))
nj_ldnd <- nj(as.dist(LDND))
# labels may run into the margins rather than be clipped
par(mar = c(1, 2, 2, 2), xpd = NA)
plot(nj_ldnd, type = "unrooted", main = "Neighbor joining", cex = 0.8,
     lab4ut = "axial", label.offset = 0.01 * sum(nj_ldnd$edge.length))
```

Both trees recover the Germanic, Slavic and Ibero-Romance groupings. Neither is a good tree of Indo-European, and with ten words one would not expect one. The instructive failure is French, which neighbor joining attaches near Greek. French has undergone extensive sound change that has shortened many words (Latin *aqua* became *eau*, pronounced o), so its forms are dissimilar to everything, including their Italian and Spanish cognates. The Levenshtein distance has no way to know that *o* and *akwa* continue the same word, because it compares strings, not histories. This is the central limitation of form-based distances, and it is why distances built on cognate judgements, or on alignments with learned sound correspondences, generally outperform raw edit distances (Greenhill, 2011; Jäger, 2013).

Levenshtein-based distances measure *similarity of form*, which mixes three signals: inheritance (cognate words that still look alike), borrowing (loanwords that look alike because they were recently copied), and chance (short words in languages with similar inventories). They are blind to regular sound correspondences, so that heavily eroded cognates such as French *eau* and Spanish *agua* count as maximally different, while unrelated look-alikes count as similar. Greenhill (2011) compared Levenshtein-based classifications of Austronesian languages with expert subgroupings and found that they performed poorly. Form-based distances are useful for exploratory work, for dialectometry where words are mostly cognate, and for very large samples where cognate coding is impossible, but their trees should not be read as genealogies without corroboration.

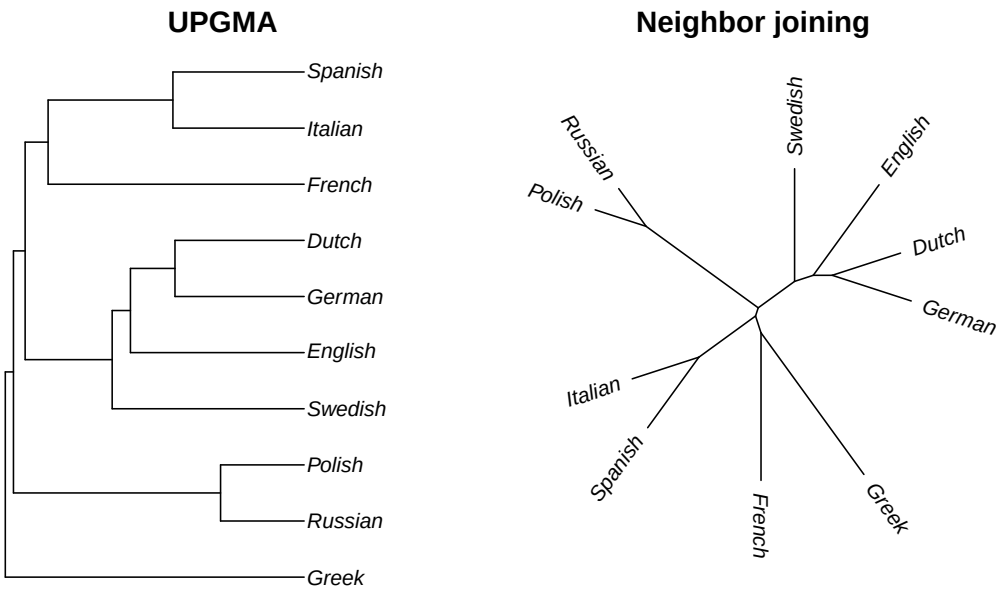


Figure 4.1: Trees inferred from LDND distances for ten Indo-European languages and ten meanings. Left: UPGMA. Right: neighbor joining, drawn unrooted. With only ten short words per language, French is pulled toward Greek by neighbor joining because its heavily eroded forms are dissimilar to everything.

4.2.4 Refinements: alignment-based and sound-class distances

The weakness of plain edit distance can be addressed in two ways. One is to weight substitutions so that phonetically or historically likely changes cost less than unlikely ones: *p* for *f* should be cheaper than *p* for *a*. Early approaches used phonetic feature systems (Kondrak, 2000; Heeringa, 2004). A more powerful approach learns the weights from data. Jäger (2013) estimated pointwise mutual information (PMI) scores between ASJP sound classes from a large set of probably cognate word pairs, used these scores in a Needleman–Wunsch alignment, and derived distances that track genealogy better than LDND. The same PMI machinery was later used to argue for deep macro-groupings (Jäger, 2015), a use that remains controversial because it pushes form-based similarity to time depths where chance resemblance and borrowing are hard to exclude. The second refinement is to cluster words into automatically detected cognate sets first (LexStat, SCA and related methods; Chapter 3) and then compute cognate-based distances or, better, run character-based methods on the resulting matrix.

4.3 Tree metrics: additivity and ultrametricity

Before discussing algorithms, we ask what kind of distance matrix a tree produces. Suppose we have a tree with positive branch lengths, and define d_{ij} as the sum of branch lengths on the path between leaves i and j (the *patristic* distance). Any matrix produced in this way is called a *tree metric* or an *additive* metric. If we could measure distances without error, and if language change were perfectly treelike, our matrix would be additive, and the tree would be recoverable exactly from it.

4.3.1 The four-point condition

Which matrices are additive? Take any four leaves i, j, k, l, and form the three sums of distances that pair them up:

$$d_{ij} + d_{kl}, \qquad d_{ik} + d_{jl}, \qquad d_{il} + d_{jk}. \tag{4.5}$$

If the tree restricted to these four leaves has the split $ij \mid kl$, then the first sum counts only the four terminal paths, while the other two sums each also count the internal path between the two pairs twice. So the first sum is the smallest, and the other two are equal. The *four-point condition* states that a matrix is a tree metric if and only if, for every quartet of leaves, the two largest of the three sums are equal (Buneman, 1974). The condition is necessary by the argument just given; that it is also sufficient, and that the tree is then unique, follows from the work of Buneman (1971) and Buneman (1974; see also Semple and Steel, 2003).

The four-point condition also suggests a measure of how *un*treelike real data are. For each quartet, the difference between the two largest sums, relative to the spread of all three, measures the conflict among the three possible quartet trees. This is the idea behind the *delta score*, which Gray et al. (2010) applied to linguistic data, and behind the NeighborNet split network (Bryant and Moulton, 2004). Both are covered in Chapter 8.

The following code generates an additive matrix from a known five-leaf tree and checks the four-point condition for all five quartets, then applies the same check to the LDND matrix.

```
true5 <- read.tree(text = "((A:2,B:3):2,C:4,(D:1,E:5):3);")
D5 <- cophenetic(true5)[LETTERS[1:5], LETTERS[1:5]]
D5
```

```
#>    A  B  C D  E
#> A  0  5  8 8 12
#> B  5  0  9 9 13
#> C  8  9  0 8 12
#> D  8  9  8 0  6
#> E 12 13 12 6  0
```

```
four_point <- function(D, tol = 1e-8) {
  q <- combn(nrow(D), 4)
  gaps <- apply(q, 2, function(x) {
    i <- x[1]; j <- x[2]; k <- x[3]; l <- x[4]
    s <- sort(c(D[i, j] + D[k, l], D[i, k] + D[j, l], D[i, l] + D[j, k]),
              decreasing = TRUE)
    s[1] - s[2]                       # zero for a perfectly treelike quartet
  })
  c(quartets = ncol(q), violations = sum(gaps > tol), max_gap = max(gaps))
}
four_point(D5)
```

```
#>   quartets violations    max_gap
#>          5          0          0
```

```
four_point(LDND)
```

```
#>   quartets violations    max_gap
```

```
#>   210.0000   210.0000     0.2113
```

The tree-generated matrix satisfies the condition exactly. The LDND matrix from our word lists violates it for essentially every quartet, as any real matrix will. The question in practice is never whether data are perfectly additive but how far they depart from additivity, and whether the departure is noise or a signal of reticulation.

4.3.2 Ultrametricity and the three-point condition

A stronger property holds if the tree is rooted and all leaves are equidistant from the root, as they would be if change accumulated at the same rate in every lineage (a strict clock). Such a tree is *ultrametric*, and its patristic distances satisfy the *three-point condition*: for every triple of leaves, the two largest of d_{ij}, d_{ik}, d_{jk} are equal. Intuitively, if i and j are sisters relative to k, then k is equally far from both of them, because the path from k goes up to its common ancestor with them and then down by equal amounts to each. Every ultrametric is additive, but not conversely.

Ultrametricity is the assumption that glottochronology made about lexical replacement, and it is the assumption that UPGMA makes. When lineages change at different rates, as languages do (Icelandic, for instance, is usually described as lexically conservative compared with English), ultrametricity fails, and methods that assume it can be misled. Additive methods such as neighbor joining do not assume a clock.

4.4 Clustering with UPGMA

The *unweighted pair group method with arithmetic mean* (UPGMA) is the simplest tree-building algorithm and one of the oldest (Sokal and Michener, 1958). It is a form of hierarchical agglomerative clustering:

1. Treat each language as a cluster of size one.
2. Find the pair of clusters A, B with the smallest distance d_{AB}. Join them into a new cluster AB, placing their common ancestor at height $d_{AB}/2$ above the leaves.
3. Compute the distance from the new cluster to every other cluster C as the average over all pairs of leaves, which works out to the size-weighted mean

$$d_{AB,C} = \frac{|A|\, d_{AC} + |B|\, d_{BC}}{|A| + |B|}. \tag{4.6}$$

4. Repeat until one cluster remains.

The result is a rooted, ultrametric tree. If the input matrix is itself ultrametric, UPGMA recovers the tree exactly. If it is not, UPGMA forces it to be, and when the departure comes from unequal rates of change it can produce the wrong topology. (WPGMA weights the two merged clusters equally instead of by size; single and complete linkage use the minimum or maximum pairwise distance. All share UPGMA's clock assumption.)

A four-language example shows how this happens. Let the true tree be $((A:1, B:4):0.5, (C:1, D:6):0.5)$, whose internal path between the two pairs has length 1, and in which A and C are slowly changing languages and B and D are fast. Its patristic distances are $d_{AB} = 5$, $d_{AC} = 3$, $d_{AD} = 8$, $d_{BC} = 6$, $d_{BD} = 11$ and $d_{CD} = 7$. UPGMA joins the closest pair, A and C (distance 3), although they are not sisters. They are close only because both have changed little. The method then computes

$d_{AC,B} = (5+6)/2 = 5.5$ and $d_{AC,D} = (8+7)/2 = 7.5$, joins B next, and finishes with D. The resulting unrooted tree has the split $AC \mid BD$, which is wrong. Shared *conservatism* has been mistaken for shared ancestry. This is exactly the error that the comparative method warns against when it insists on shared innovations rather than shared retentions, and it is a real danger with linguistic data, where some languages are lexically conservative and others innovative.

```
true4 <- read.tree(text = "((A:1,B:4):0.5,(C:1,D:6):0.5);")
D4 <- cophenetic(true4)[c("A", "B", "C", "D"), c("A", "B", "C", "D")]
D4
```

```
#>    A  B C  D
#> A 0  5 3  8
#> B 5  0 6 11
#> C 3  6 0  7
#> D 8 11 7  0
```

```
write.tree(upgma(as.dist(D4)))   # groups A with C: wrong
```

```
#> [1] "(D:4.333333333,(B:2.75,(A:1.5,C:1.5):1.25):1.583333333);"
```

```
# NJ recovers the split AB|CD and the branch lengths
write.tree(nj(as.dist(D4)))
```

```
#> [1] "(D:6,C:1,(A:1,B:4):1);"
```

4.5 Neighbor joining and its relatives

4.5.1 The neighbor-joining algorithm

Neighbor joining (NJ) was introduced by Saitou and Nei (1987) and is still the most widely used distance method. It also builds the tree by successively joining pairs, but it chooses the pair to join so as to correct for unequal rates. Its trick is to compare each distance d_{ij} not with zero but with how far i and j are from everything else. Let n be the current number of clusters and $R_i = \sum_k d_{ik}$ the total distance from i to all others. NJ joins the pair minimizing

$$Q_{ij} = (n-2)\, d_{ij} - R_i - R_j. \tag{4.7}$$

A language that is far from everything (a fast-changing language) has a large R_i, which makes its Q_{ij} values more negative and compensates for its long branch. When i and j are joined into a new node, their branch lengths are computed so that the faster of the two receives the longer branch, the new node replaces them in the distance matrix, and the procedure repeats with $n-1$ clusters until three remain, which are joined to a final central node. The output is an unrooted tree with branch lengths. Rooting it requires an outgroup or a further assumption (Chapter 2).

Applied to the four-language example, the row sums are $R_A = 16$, $R_B = 22$, $R_C = 16$ and $R_D = 26$, and with $n = 4$, $Q_{AB} = 2\times 5 - 16 - 22 = -28$, $Q_{CD} = 2\times 7 - 16 - 26 = -28$, while all other pairs have $Q = -26$. So NJ correctly joins A with B (equivalently, C with D), and its branch-length formula returns $v_A = 1$ and $v_B = 4$, exactly the true values. The Q criterion has seen through the conservatism of A and C.

NJ has two key theoretical properties: it recovers the true tree from any additive matrix, and it is robust to noise, returning the correct topology whenever every distance is within half the length of the shortest internal branch of its true value (Atteson, 1999). It can be understood as a fast greedy approximation to the *balanced minimum evolution* criterion described below (Gascuel and Steel, 2006), and it runs in time $O(n^3)$ for n languages, fast enough for thousands of languages.

The branch-length and distance-update formulas of neighbor joining (equations (B.4) and (B.5)), the history of the Q criterion, and code that computes the Q matrix for a five-language example are in Section B.2.1.

4.5.2 BIONJ and weighted distances

Distances between distantly related languages are noisier than distances between close relatives, because the correction formulas (4.3) and (4.4) amplify sampling error as proportions approach saturation. NJ ignores this. *BIONJ* (Gascuel, 1997) modifies the reduction step (B.5) so that the new distances are weighted averages that minimize their variance, using a simple model in which the variance of an estimated distance is proportional to the distance itself. BIONJ is as fast as NJ and is typically somewhat more accurate in simulations, especially for datasets with long branches. It is available in **ape** as `bionj()` and is used to build starting trees by PhyML (Chapter 5).

4.5.3 Least squares and minimum evolution

Optimality-criterion distance methods first fit branch lengths to a fixed topology and then search among topologies. In the *least-squares* approach (Fitch and Margoliash, 1967; Cavalli-Sforza and Edwards, 1967), branch lengths are chosen to minimize

$$S(\tau) = \sum_{i<j} w_{ij} \left(d_{ij} - \delta_{ij}(\tau) \right)^2 , \tag{4.8}$$

where $\delta_{ij}(\tau)$ is the patristic distance in the fitted tree and w_{ij} are weights. Ordinary least squares sets $w_{ij} = 1$; Fitch and Margoliash (1967) proposed $w_{ij} = 1/d_{ij}^2$, down-weighting large distances, which mirrors the logic of BIONJ. The tree with the smallest $S(\tau)$ is the least-squares estimate.

Minimum evolution (ME) instead fits branch lengths by least squares but scores each topology by the *sum* of its fitted branch lengths, preferring the shortest tree (Edwards and Cavalli-Sforza, 1964; Rzhetsky and Nei, 1992). The rationale is analogous to parsimony: the true tree should require the least total change. The *balanced* version of minimum evolution uses a particular weighting of the tree length that can be computed directly from the distance matrix (Pauplin, 2000). Desper and Gascuel (2002) developed fast tree searches under this criterion using NNI and SPR rearrangements (Chapter 5), implemented in the program FastME (Lefort et al., 2015). Balanced minimum evolution has good statistical properties and, in simulation, is usually at least as accurate as NJ, which can be regarded as a fast greedy approximation to it. In **ape**, `fastme.bal()` and `fastme.ols()` implement balanced and ordinary-least-squares minimum evolution.

4.5.4 Comparing distance methods on simulated cognate data

To compare the methods on data where the truth is known, we simulate a binary cognate matrix on a random tree with unequal rates among lineages, using **phangorn**'s `simSeq()`. The tree has 12 languages, branch lengths in expected changes per character, and 400 characters; characters that

are constant across all languages are removed, as they would be in a real cognate matrix. We then compute corrected binary distances (4.4) and build trees with each method. The Robinson–Foulds (RF) distance (Chapter 2) counts the splits that differ between an inferred tree and the true tree; 0 means perfect recovery.

```
set.seed(11)
tree12 <- rtree(12, tip.label = paste0("L", 1:12))
tree12$edge.length <- tree12$edge.length * 0.4      # unequal rates, no clock
sim <- simSeq(tree12, l = 400, type = "USER", levels = c("0", "1"))
M <- as.character(sim)
M <- M[, apply(M, 2, function(z) length(unique(z)) > 1)]  # drop constants
cog <- phyDat(M, type = "USER", levels = c("0", "1"))
cog
```

```
#> 12 sequences with 377 character and 263 different site patterns.
#> The states are 0 1
```

```
p_ham <- as.matrix(dist.hamming(cog))
d_jc  <- -0.5 * log(1 - 2 * pmin(p_ham, 0.499))       # binary correction
d_jc  <- as.dist(d_jc)

dist_trees <- list(UPGMA   = upgma(d_jc),
                   NJ      = nj(d_jc),
                   BIONJ   = bionj(d_jc),
                   FastME  = fastme.bal(d_jc),
                   OLS_ME  = fastme.ols(d_jc))
sapply(dist_trees, function(tr) RF.dist(tr, tree12))
```

```
#>  UPGMA     NJ  BIONJ FastME OLS_ME
#>     10      6      4      6      6
```

In the vocabulary of Section 6.10, this experiment is of two kinds at once. For the additive methods it is a test of *recovery*: the data are generated by exactly the independent, symmetric binary process that the distance correction assumes. For UPGMA it is a small test of *robustness under misspecification*, because the simulated tree violates UPGMA's clock assumption. Neither says anything about real cognate data, where borrowing, dependence among the cognate sets of a meaning and rate variation among meanings add misspecifications that none of these methods models. On this dataset UPGMA is clearly worse than the additive methods. With a different random seed the ranking among NJ, BIONJ and FastME can change; over many replicates, simulations in the literature generally find balanced minimum evolution and BIONJ slightly ahead of NJ, and all three far ahead of UPGMA whenever rates vary among lineages (Desper and Gascuel, 2002; Gascuel, 1997).

4.6 Distance methods in linguistics

Distance methods were the first quantitative approaches to language classification, and they remain the method of choice for very large samples. Lexicostatistics in the 1950s and 1960s produced cognate percentage matrices for many families, clustered by eye or with simple linkage rules. The program of Embleton (1986) reviewed the statistics of this tradition, and McMahon

and McMahon (2005) gave a book-length treatment of quantitative classification from a linguist's point of view, including distance methods and early network methods.

Dyen et al. (1992) compiled 200-meaning Swadesh lists for 95 Indo-European speech varieties (the "Dyen database"), with cognate judgements, and classified them by lexicostatistical percentages. Their analysis grouped the languages into the familiar branches (Germanic, Romance, Celtic, Slavic, Baltic, Indic, Iranian and so on) but was less decisive about the higher-order structure, a recurring outcome for distance methods applied to deep splits where few cognates are shared. That the familiar branches emerged is consistent with the method capturing genealogical signal, but it is not an independent test: the cognate judgements that went into the percentages were made by the comparative method that defined those branches, and cognate percentages count shared retentions as well as shared innovations (Section 1.9). The Dyen database itself had a long afterlife: it was the core of the dataset used by Gray and Atkinson (2003), and in revised and expanded form it fed into the IELex database and later work (Chapter 3). It was also analyzed with cladistic parsimony by Rexová et al. (2003), discussed below, so that the same data have now been run through almost every method in this book.

The Automated Similarity Judgment Program (ASJP) took the opposite approach to the Dyen database: rather than expert cognate judgements for a few dozen languages, it collected short (40-meaning) lists in a uniform simplified transcription for as many languages as possible, and computed LDND distances automatically (Brown et al., 2008; Holman et al., 2008). Neighbor-joining trees built from these distances recover many established families and subgroups, although they make errors at deeper levels and where borrowing is heavy, and the authors presented them as a tool for preliminary classification rather than as a substitute for the comparative method. Here the input is not cognate judgements but raw forms, so agreement with expert classifications is closer to an independent check than it is for cognate-based trees; but form similarity mixes inheritance with borrowing and chance, so agreement on low-level groups, where all three point the same way, is weak evidence about deeper structure. Holman et al. (2011) went further and calibrated LDND against known divergence dates to produce automated date estimates for many families, effectively a revived glottochronology with an empirically fitted, non-linear calibration curve. These dates have been widely cited and widely criticized, for the same reasons as the original glottochronology: rates of change differ among lineages and over time (Chapter 9).

Form-based distances have also been used on small, well-studied families as a test of method. Serva and Petroni (2008) built an Indo-European tree from normalized Levenshtein distances on Swadesh-list words and reported broad agreement with standard subgrouping; Greenhill (2011), testing Levenshtein distances systematically on Austronesian, concluded that they fail to recover known relationships accurately. The apparent contradiction is partly one of standards: form-based distances usually recover low-level groupings, where many words are recognizably similar, and usually fail on deeper structure, where cognates have diverged in form. "Broad agreement with standard subgrouping" is in any case a weak criterion, since a method can reproduce the uncontroversial low-level groups that any reasonable method finds while being wrong about exactly the higher-level questions it is used to answer. In dialectometry, by contrast, where nearly all words are cognate, Levenshtein-based methods are well established (Kessler, 1995; Heeringa, 2004), typically combined with network or multidimensional scaling displays rather than trees, because dialects form continua (Chapter 8).

Distance methods are also the backbone of global-scale computational work. Jäger (2018) derived character matrices from ASJP word lists for thousands of languages, using automatically inferred cognate classes and sound-class information, and inferred trees from them that are now used as

scaffolds for comparative analyses where expert trees with branch lengths are unavailable (Chapter 13).

Distance methods in biology have moved well beyond plain NJ, in directions that have seen little use in linguistics. *Weighted least-squares and minimum-evolution searches* (FastME) with variance models tailored to the data are standard in molecular work but are rarely, if ever, used on cognate data, where NJ dominates. *Alignment-free distances* based on shared substrings or k-mers (short subsequences), developed for genomes, are a natural fit for word forms, since they avoid alignment altogether; to our knowledge they have rarely been applied to lexical data in a systematic way. *Distance-based phylogenetic placement* places new taxa onto an existing reference tree rather than rebuilding the whole tree, which would suit the incremental way language databases grow (Chapter 16). And *supertree methods* such as matrix representation with parsimony (Baum, 1992; Ragan, 1992) combine many small, overlapping trees into one large tree. These could combine family-level expert trees from different sources into a single global tree with a principled treatment of conflict, a job currently done by hand in Glottolog.

4.7 Ordination: looking at distances without a tree

A distance matrix can be explored before, or instead of, fitting a tree to it. *Ordination* methods place the languages as points in a space of two or three dimensions, so that languages with similar data lie close together. The two most common are multidimensional scaling (MDS), which starts from a distance matrix, and principal component analysis (PCA), which starts from the data matrix itself. Neither has an evolutionary component. There are no ancestors, no branches and no direction of change in an ordination plot, only a low-dimensional snapshot of overall similarity. That is a limitation when the question is genealogical, but it is also a strength: an ordination does not force the data into a tree, so it can show clusters, outliers and continua that a tree-building method would hide. Dialectometry has long used MDS for exactly this reason (Heeringa, 2004; Nerbonne, 2009), and ordinations are a natural companion to the split networks of Chapter 8.

4.7.1 Multidimensional scaling

Classical (or *metric*) MDS, also known as principal coordinates analysis, looks for coordinates whose Euclidean distances reproduce the input distances as closely as possible. It has a closed-form solution. With $\mathbf{D}^{(2)}$ the matrix of squared distances and $\mathbf{J} = \mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}^{\top}$ the centering matrix, form

$$\mathbf{B} = -\tfrac{1}{2}\,\mathbf{J}\,\mathbf{D}^{(2)}\mathbf{J}; \tag{4.9}$$

the coordinates on axis k are the k-th eigenvector of $\mathbf{B}$ scaled by the square root of its eigenvalue. If the distances are Euclidean, all eigenvalues are non-negative and $n - 1$ axes reproduce the matrix exactly. If they are not, as with many corrected and form-based linguistic distances, some eigenvalues are negative and have no geometric interpretation. The share of the eigenvalue sum carried by the first two axes measures how much of the matrix a two-dimensional plot can show (Borg and Groenen, 2005).

Non-metric MDS (Kruskal, 1964) asks for less. It looks for a configuration in which the *rank order* of the plotted distances matches the rank order of the input dissimilarities as closely as possible, and measures the mismatch by *stress*, which is zero for a perfect fit. Kruskal proposed verbal benchmarks (a stress of 20 percent “poor”, 10 percent “fair”, 5 percent “good”, 2.5 percent

"excellent"), but acceptable values depend on the number of objects and dimensions, and the benchmarks are rules of thumb rather than tests. A *scree plot* of stress against the number of dimensions shows where adding a dimension stops paying. In R, classical MDS is `cmdscale()` and non-metric MDS is `isoMDS()` in **MASS**; `isoMDS()` requires all off-diagonal dissimilarities to be positive.

The derivation of Equation (4.9), the equivalence of classical MDS and PCA, the formula for Kruskal's stress, and implementations of both checked against `cmdscale()`, `prcomp()` and `isoMDS()` are in Section B.2.4.

```
library(MASS)
mds_ldnd <- cmdscale(as.dist(LDND), k = 2, eig = TRUE)
round(mds_ldnd$GOF, 2)               # share of the eigenvalues on 2 axes
```

```
#> [1] 0.56 0.56
```

```
stress <- sapply(1:4, function(k)
  isoMDS(as.dist(LDND), k = k, trace = FALSE)$stress)
setNames(round(stress, 1), paste0("k=", 1:4))   # stress in percent
```

```
#>  k=1  k=2  k=3  k=4
#> 37.7  5.2  0.2  0.0
```

For the LDND matrix of our ten Indo-European word lists, a one-dimensional non-metric solution fits poorly, while two dimensions give a stress of about 5 percent, "good" by Kruskal's benchmarks. The left panel of Figure 4.2 shows the classical solution. Germanic, Slavic and a group of Romance languages separate cleanly. Greek sits with Romance, as it did in the neighbor-joining tree, because MDS displays the same form similarities that the tree was built from. An ordination can reveal the structure of a distance matrix, but it cannot repair the measure.

4.7.2 Principal component analysis

PCA works on the data matrix rather than on distances. It finds orthogonal linear combinations of the characters, the *principal components*, that successively capture as much of the variance among languages as possible. Each language receives a *score* on each component, which serves as its coordinate in the plot. Each character receives a *loading* on each component, which shows how strongly it contributes. The proportion of the total variance captured by each component, displayed as a scree plot, shows how many components are worth looking at (Jolliffe and Cadima, 2016). The two methods are closely related. Classical MDS applied to the Euclidean distances between the rows of a data matrix returns exactly the PCA scores, up to the sign of each axis. For a binary matrix the Euclidean distance between two languages is the square root of the number of characters in which they differ, so PCA of a binary cognate matrix is MDS of a Hamming-type distance and inherits its sensitivity to shared absences (Section 4.2). PCA was designed for continuous variables, and on binary data it is best treated as a descriptive device.

```
X <- matrix(as.integer(as.character(cog)), nrow = length(cog),
            dimnames = list(names(cog), NULL))
pca_cog <- prcomp(X)                        # centred, not scaled
round(summary(pca_cog)$importance["Proportion of Variance", 1:4], 2)
```

```
#>  PC1  PC2  PC3  PC4
#> 0.23 0.17 0.12 0.08
```

```
# classical MDS of Euclidean distances reproduces the PCA scores
all.equal(abs(cmdscale(dist(X), k = 2)), abs(pca_cog$x[, 1:2]),
          check.attributes = FALSE)
```

```
#> [1] TRUE
```

```
# characters with the largest absolute loadings on PC1
head(order(abs(pca_cog$rotation[, 1]), decreasing = TRUE))
```

```
#> [1]  66 136  22  60 172 354
```

```
# fix the arbitrary signs of the axes so that the layout is reproducible
xy <- mds_ldnd$points
xy <- sweep(xy, 2, sign(xy["German", ]) * c(-1, 1), "*")
pc <- pca_cog$x[, 1:2]
pc <- sweep(pc, 2, c(sign(pc["L10", 1]), -sign(pc["L1", 2])), "*")
par(mfrow = c(1, 2), mar = c(4, 4, 2, 1), las = 1)
plot(xy, pch = 19, cex = 0.7, asp = 1, xlab = "MDS axis 1",
     ylab = "MDS axis 2", main = "Classical MDS of LDND",
     xlim = range(xy[, 1]) + c(-0.15, 0.12))
pos_mds <- c(English = 3, German = 3, Dutch = 2, Swedish = 4, Italian = 3,
             Spanish = 4, French = 1, Russian = 3, Polish = 1, Greek = 2)
text(xy, rownames(xy), pos = pos_mds[rownames(xy)], cex = 0.75, xpd = NA)
deep <- rownames(pc) %in% c("L1", "L2")
ve <- round(100 * pca_cog$sdev^2 / sum(pca_cog$sdev^2))
plot(pc, pch = ifelse(deep, 17, 19), cex = 0.8,
     xlab = paste0("PC1 (", ve[1], "%)"), ylab = paste0("PC2 (", ve[2], "%)"),
     main = "PCA of simulated cognates", xlim = range(pc[, 1]) + c(-1.5, 1.5))
pos_pca <- setNames(rep(4, nrow(pc)), rownames(pc))
pos_pca[c("L5", "L10", "L2")] <- 2
pos_pca[c("L6")] <- 3
pos_pca[c("L12")] <- 1
text(pc, rownames(pc), pos = pos_pca[rownames(pc)], cex = 0.75, xpd = NA)
```

The PCA of the simulated cognate matrix (right panel) makes a point that is easy to miss. The first component, which carries the most variance, separates two clades nested deep inside the larger of the two subtrees, while the deepest split of the true tree, between L1 and L2 and everything else, shows up only on the second component. Components are ordered by variance, not by time depth: a clade whose members share many cognate sets dominates the first axis wherever it sits in the tree. In the vocabulary of Section 6.10, this is a demonstration of what PCA displays on data simulated on a known tree, not a test of any method's adequacy.

4.7.3 Ordination in linguistics

Skelton (2015) used non-metric MDS as a *preliminary cluster analysis* in a parsimony study of the ancient Greek dialects. Her parsimony trees left the West Greek dialects unresolved, and she asked whether some of the traditional West Greek "dialects", whose boundaries had often been drawn

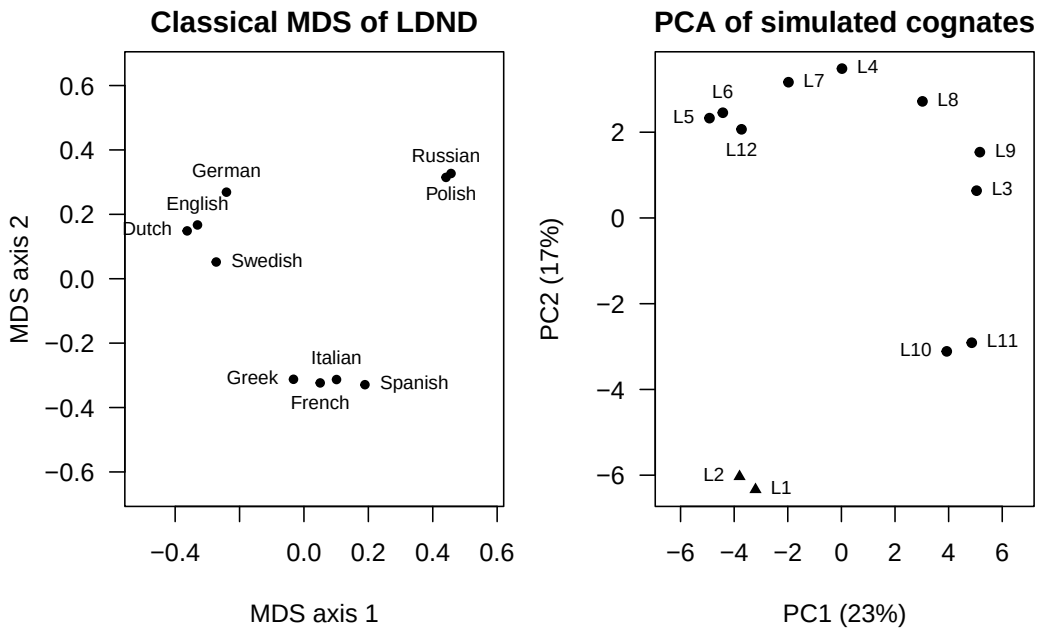


Figure 4.2: Two ordinations. Left: classical MDS of the LDND distances between ten Indo-European word lists. Right: PCA of the simulated binary cognate matrix for twelve languages used earlier in this chapter; triangles mark L1 and L2, which form one side of the deepest split of the true tree. The first component separates two clades nested inside the larger subtree; the deepest split appears only on the second.

along geographic and political lines, were distinct taxa at all. An MDS of the West Greek part of the matrix (with the number of dimensions chosen from a scree plot of stress) placed several dialects close together, and she merged them: Phocian with Locrian, the two Argolic dialects with Laconian, and Theran, Coan and Rhodian. With the merged taxa, maximum parsimony returned a single fully resolved tree. The study raises three general issues. The cut-off for "close enough to merge" was, as she noted, at the investigator's discretion. Merging taxa changes the data that the tree is then inferred from, so a resolved tree after merging is partly a consequence of having fewer, internally homogenized taxa. And the character-weighting analyses of the same paper recovered a different West Greek subgrouping, which she attributed to a dialect continuum within West Greek, for which she judged network methods more appropriate than trees. Ordination is a sensible tool for deciding what the taxa should be, but the decision should be reported as a modelling choice and its consequences tested.

PCA has also been applied to structural data. Norvik et al. (2022) compiled 360 binary features (195 Grambank features and 165 features designed for Uralic) for 35 Uralic languages. In a PCA the full feature set separated the main Uralic subgroups, whereas the Grambank features alone were too coarse to show this family-internal structure. Because structural features also spread by contact, such a separation is consistent with genealogical signal without demonstrating it, and the same paper used model-based admixture analysis to identify areas of historical interaction. For phonetic data, Heggarty et al. (2010) argued that continuous divergence measures, displayed as networks, can show splits and wave-like spread in the same picture (Chapter 8).

Ordination plots are easy to over-read. Proximity can come from shared inheritance, contact or chance, and an ordination cannot tell shared innovations from shared retentions, which is precisely

the distinction that subgrouping needs (Section 4.8). A two-dimensional plot shows only part of the matrix: report the stress or the variance explained, and look at further axes before concluding that two languages are similar. Data that vary smoothly along a continuum, as in a dialect chain, produce curved and arched configurations, and in PCA they can produce components that look like oscillations but are artifacts of the smoothness itself (Shinn, 2023). Ordinations are also sensitive to sampling: densely sampled groups pull the first axes toward their internal differences. Phylogenetic versions of PCA that model the covariance due to shared ancestry exist in biology (Section 16.5.1). Ordinary PCA treats languages as independent observations, which they are not (Chapter 13).

4.8 Parsimony: the principle

Parsimony methods choose the tree on which the observed character states can be explained with the fewest changes. The idea has two roots. One is Willi Hennig's *phylogenetic systematics* (Hennig, 1966), which held that groups should be defined by shared derived states (*synapomorphies*), not by shared ancestral states (*symplesiomorphies*). The other is the computational work of Kluge and Farris (1969), Farris (1970) and Fitch (1971), which turned "fewest changes" into an algorithm. Linguists will recognize Hennig's distinction at once. It is the comparative method's principle that subgroups are defined by *shared innovations*, and that shared retentions of inherited features prove nothing about subgrouping (Ringe et al., 2002). That French, Italian and Spanish all continue Latin *manus* for 'hand' supports a Romance subgroup because the replacement of the inherited Indo-European word for 'hand' by *manus* is an innovation; that English and Icelandic both retain many old Germanic words supports nothing in particular.

The principle, which historical linguists often trace to August Leskien in the 1870s, can also be read as a policy about errors. Counting only shared innovations guards against positing a subgroup that never existed (a false positive). The price is a higher risk of missing subgroups that did exist but left few clear innovations (a false negative), and the comparative method has no explicit way to weigh the two kinds of error or to say how many innovations are enough. Two further difficulties follow. First, a "shared innovation" may be a change that happened once in a common ancestor or the same change happening independently in several lineages, and often the evidence does not settle which. Classifying each innovation as either subgrouping evidence or parallel development is an all-or-nothing decision that can convey more certainty than the evidence supports. Second, whether shared retentions carry any information depends on the inferential framework. Under parsimony, which counts changes, they do not. Under likelihood, the weight of a shared retention depends on the model and the branch lengths and is not zero in general (Sober, 1988, 2015): the retention of a feature that is usually lost is itself an improbable event. Model-based methods (Chapters 5 and 6) replace the binary choice with probabilities of change along each branch.

A few definitions. A *character* is a set of comparable features that can take different *states* in different languages: the meaning 'hand' with states given by cognate classes, a binary cognate set (present or absent), or a structural feature such as basic word order (SOV, SVO, VSO and so on). A change of state along a branch is a *step*. The *length* or *parsimony score* of a tree for a character is the minimum number of steps needed to produce the observed states at the leaves, minimized over all possible assignments of states to the internal nodes; the length for a data matrix is the sum over characters. *Homoplasy* is any change beyond the minimum that a character requires, that is, parallel change or reversal: the same state arising independently in two lineages, or a lineage returning to an earlier state. In linguistic data, homoplasy arises from borrowing, from parallel

semantic shifts (words for 'bark' and 'skin' converging independently), from chance resemblance in coded features, and from coding errors.

The *maximum parsimony* (MP) tree is the tree with the smallest total length. Computing it involves two nested problems: computing the length of a given tree (the *small parsimony problem*), which is easy, and finding the shortest tree among all trees (the *large parsimony problem*), which is NP-hard (Semple and Steel, 2003) and solved heuristically.

4.9 Counting changes on a tree: Fitch's algorithm

4.9.1 The algorithm

For unordered characters with equal costs for all changes, the small parsimony problem is solved by the algorithm of Fitch (1971). It works from the leaves to the root, giving each node a set of candidate states: a leaf gets its observed state, and an internal node gets the *intersection* of its children's sets if they share a state, or otherwise their *union*, at the cost of one step. The number of unions is the parsimony score, and the root set contains every state that some most-parsimonious reconstruction assigns to the root. Because the score of a reversible character does not depend on where the tree is rooted, the algorithm can be applied to an unrooted tree by rooting it arbitrarily.

The formal statement of Fitch's algorithm (equation (B.6)), the top-down pass that recovers ancestral states, and an implementation from scratch checked against **phangorn** are in Section B.2.2.

4.9.2 A worked example: 'hand' in seven Indo-European languages

Consider the meaning 'hand' and the tree (((English, German), Swedish), ((Italian, Spanish), (Russian, Polish))). The cognate classes are the Germanic *hand* class (English *hand*, German *Hand*, Swedish *hand*), the Latin *manus* class (Italian *mano*, Spanish *mano*), and the Slavic *ruka* class (Russian *ruka*, Polish *ręka*). Call them H, M and R.

Working upward: (English, German) gets $\{H\} \cap \{H\} = \{H\}$; adding Swedish, $\{H\}$ again. (Italian, Spanish) gets $\{M\}$; (Russian, Polish) gets $\{R\}$. These two disagree, so their parent gets $\{M, R\}$ and the score rises to 1. At the root, $\{H\}$ and $\{M, R\}$ again have an empty intersection, so the root set is $\{H, M, R\}$ and the score is 2. Two steps is the minimum for any three-state character, so this character is perfectly consistent with the tree. Notice also what parsimony tells us about the ancestral state: the root set contains all three classes, so parsimony cannot decide which, if any, was the Proto-Indo-European word for 'hand'. Here the comparative method knows more. Greek *kheír* and Hittite *keššar* are usually traced to an old Indo-European root for 'hand', which suggests that none of the three classes in our seven languages is the original. Parsimony can only choose among states it sees.

In R, **phangorn**'s `parsimony()` computes Fitch scores for a tree and a `phyDat` object, and `ancestral.pars()` returns the parsimonious ancestral state sets. Here we score the 'hand' character on the tree above and on a wrong tree that groups the Slavic languages with Germanic.

```
tr7 <- read.tree(text =
  "(((English,German),Swedish),((Italian,Spanish),(Russian,Polish)));")
tr7_wrong <- read.tree(text =
  "((((English,German),Swedish),(Russian,Polish)),(Italian,Spanish));")
hand <- c(English = "H", German = "H", Swedish = "H",
```

```
          Italian = "M", Spanish = "M", Russian = "R", Polish = "R")
hand_pd <- phyDat(matrix(hand, ncol = 1, dimnames = list(names(hand), NULL)),
                  type = "USER", levels = c("H", "M", "R"))
c(right_tree = parsimony(tr7, hand_pd, method = "fitch"),
  wrong_tree = parsimony(tr7_wrong, hand_pd, method = "fitch"))
```

```
#> right_tree wrong_tree
#>          2          2
```

On the wrong tree the character still needs only two steps (the H, R and M classes each form a clade on that tree too), so 'hand' cannot distinguish these two trees. A character is *informative* for parsimony only if at least two states each occur in at least two languages; otherwise every tree has the same score for it. For binary cognate data this means a cognate set must be present in at least two and absent from at least two languages. Cognate sets found in a single language (*singletons*, or autapomorphies) are uninformative for parsimony, although they carry information about branch lengths under likelihood (Chapter 5).

4.9.3 Ancestral states: ACCTRAN and DELTRAN

When the top-down pass has a choice, different most-parsimonious reconstructions place the changes on different branches. Two conventions are common. *ACCTRAN* (accelerated transformation) places changes as close to the root as possible, which favors reversals over parallel changes; *DELTRAN* (delayed transformation) places them as close to the leaves as possible, favoring parallelism. For cognate data, where a cognate set that is lost cannot easily be regained, the choice matters for interpretation. ACCTRAN would explain a scattered cognate set as one early gain followed by losses, which is linguistically natural, whereas DELTRAN would favor multiple independent gains, which for true cognates is impossible unless the "gains" are borrowings. In **phangorn**, `acctran()` assigns branch lengths in numbers of changes under ACCTRAN. Chapter 12 returns to ancestral state reconstruction with probabilistic methods.

4.10 Weighted parsimony and its variants

4.10.1 Sankoff's algorithm

Fitch parsimony treats all changes as equally costly. Often they are not. A cost matrix $c(a, b)$ gives the cost of a change from state a to state b, and the *Sankoff algorithm* (Sankoff, 1975) computes the minimum total cost by dynamic programming: working from the leaves to the root, it records for each node and each possible state the cheapest way of explaining the data below that node if the node were in that state. The structure of this recursion is identical to that of Felsenstein's pruning algorithm for likelihoods (Chapter 5), with "minimum" in place of "sum" and "plus" in place of "times", which is why the two are worth learning together.

The Sankoff recursion (equation (B.7)) and a step-by-step evaluation on the 'hand' example with a Dollo-like cost matrix are in Section B.2.3.

4.10.2 Classic cost schemes

Several named variants of parsimony correspond to particular cost matrices (Table 4.2).

Table 4.2: Named parsimony variants and the kinds of linguistic characters to which their assumptions are suited.

Variant	Assumption about change	Cost matrix	Linguistic interpretation
Fitch (Fitch, 1971)	unordered, reversible	$c(a,b) = 1$ for $a \neq b$	cognate classes in a meaning slot; structural features without a natural order
Wagner (Kluge and Farris, 1969; Farris, 1970)	ordered, reversible	$c(a,b) = \lvert a - b \rvert$	ordinal features (e.g. number of cases, size of a vowel inventory in bins)
Camin–Sokal (Camin and Sokal, 1965)	irreversible: $0 \to 1$ only	$c(0,1) = 1$, $c(1,0) = \infty$	unidirectional changes, such as some sound changes or mergers (a merger is not usually undone)
Dollo (Farris, 1977)	each derived state arises once; losses free to recur	gain cost much larger than loss cost	cognate sets: a word is born once but can be lost in many lineages

Dollo parsimony is the natural parsimony model for binary cognate data: a cognate set originates once, when a new word or new meaning arises in one ancestral language, and may then be lost independently in any number of descendants. Dollo's law in the strict sense allows a single gain per character, which makes the root position matter (gains and losses are not symmetric). The probabilistic version of this idea, the *stochastic Dollo* model of Nicholls and Gray (2008), is one of the most important lexical models and is treated in Chapter 7. Camin–Sokal parsimony fits sound changes that are generally regarded as irreversible, such as mergers of phonemes, and was in effect the logic of the phonological characters in the Indo-European work of Ringe et al. (2002).

The following code scores the same tree under Fitch parsimony and under a Dollo-like Sankoff cost matrix in which gaining a cognate set costs ten times as much as losing one. Because the costs are asymmetric, the tree must be rooted first.

```
dollo_cost <- matrix(c(0, 10,
                       1,  0), nrow = 2, byrow = TRUE,
                     dimnames = list(c("0", "1"), c("0", "1")))
dollo_cost                                          # rows: from, columns: to
```

```
#>   0  1
#> 0 0 10
#> 1 1  0
```

```
true_rooted <- root(tree12, "L1", resolve.root = TRUE)
c(fitch        = parsimony(true_rooted, cog, method = "fitch"),
  sankoff_unit = parsimony(true_rooted, cog, method = "sankoff"),
  dollo_like   = parsimony(true_rooted, cog, method = "sankoff",
                           cost = dollo_cost))
```

```
#>        fitch sankoff_unit   dollo_like
```

```
#>          827          827         1565
```

Unit-cost Sankoff reproduces the Fitch score, as it must. The Dollo-like score is much larger, because our simulated data, generated under a symmetric model, contain many characters whose distribution requires multiple gains; real cognate data, generated by a process much closer to Dollo, would be penalized far less.

4.10.3 Weighting characters

Weights can also be applied to whole characters. In linguistics, the strongest tradition of weighting comes from the comparative method itself: some characters are judged more reliable than others. Ringe et al. (2002) gave phonological and morphological innovations, which are rarely borrowed and rarely arise in parallel, greater weight than lexical characters. In biology, *successive weighting* (Farris, 1969) reweights characters by their consistency on a first-round tree and repeats the search, and *implied weighting* (Goloboff, 1993) down-weights characters in proportion to their homoplasy during the search itself, using a concave function of the number of extra steps. Both are based on the idea that characters that conflict with the tree are less reliable. Implied weighting is standard in morphological cladistics and implemented in TNT (Goloboff et al., 2008), but to our knowledge it has rarely if ever been applied to cognate data, where it could be used as an automatic way of down-weighting characters affected by borrowing.

4.11 Searching for the most parsimonious tree

For small numbers of languages the large parsimony problem can be solved exactly. *Exhaustive search* evaluates every tree, which is feasible up to about ten or eleven taxa. *Branch and bound* (Hendy and Penny, 1982) builds trees by adding taxa one at a time and abandons any partial tree whose length already exceeds the best complete tree found so far; it guarantees the optimum and can handle roughly twenty to thirty taxa when the data are clean. Beyond that, heuristics are needed. A typical heuristic builds a starting tree by *stepwise addition* (adding languages one at a time to the position that increases the length least) and then improves it by *branch swapping*: nearest-neighbor interchange (NNI), subtree pruning and regrafting (SPR) and tree bisection and reconnection (TBR), in increasing order of the size of the neighborhood they explore. These rearrangements are described in detail in Chapter 5, where they serve the same purpose for likelihood.

Hill climbing gets trapped in local optima. The *parsimony ratchet* (Nixon, 1999) escapes them by alternating between searches on reweighted data (a random subset of characters is temporarily up-weighted, which perturbs the landscape) and searches on the original data, keeping the best trees found. It is the default in **phangorn**'s `pratchet()` and one of the main search strategies in TNT. Parsimony analyses often return many equally short trees, which are then summarized by a strict or majority-rule consensus tree (Chapter 2).

```
mp_tree <- pratchet(cog, trace = 0, minit = 20, k = 10, all = FALSE)
mp_tree <- acctran(mp_tree, cog)     # branch lengths = number of changes
c(score_MP   = parsimony(mp_tree, cog),
  score_true = parsimony(tree12, cog),
  score_NJ   = parsimony(dist_trees$NJ, cog),
  RF_to_true = RF.dist(mp_tree, tree12))
```

```
#>   score_MP score_true   score_NJ RF_to_true
```

```
#>        821        827        831          2
```

```
# exact search on a subset of 8 languages
cog8 <- subset(cog, paste0("L", 1:8))
exact <- bab(cog8, trace = 0)
c(n_optimal = length(exact), score = parsimony(exact, cog8)[1])
```

```
#> n_optimal     score
#>         1       552
```

One detail of this output deserves comment. The most parsimonious tree (length 821) is *shorter* than the true tree (length 827), yet it differs from the true tree in only a few splits. With this much homoplasy the true history is not the most parsimonious explanation of the data. Parsimony nonetheless gets close to the truth here because the branches are not in the Felsenstein zone described below.

4.12 Fit and support

4.12.1 Consistency and retention indices

How much homoplasy do the data contain? For a character c let m_c be the minimum number of steps it could require on any tree (the number of observed states minus one), s_c the number of steps it requires on the tree in question, and g_c the maximum it could require on any tree (the number of steps on a star tree). The *consistency index* (Kluge and Farris, 1969) and *retention index* (Farris, 1989) for the whole matrix are

$$\mathrm{CI} = \frac{\sum_c m_c}{\sum_c s_c}, \qquad \mathrm{RI} = \frac{\sum_c (g_c - s_c)}{\sum_c (g_c - m_c)}. \tag{4.10}$$

$\mathrm{CI} = 1$ means no homoplasy at all. CI is inflated by uninformative characters and falls as the number of taxa increases (Sanderson and Donoghue, 1989), which makes it hard to compare across datasets. RI measures the proportion of potential synapomorphy that is retained as synapomorphy on the tree; it is unaffected by uninformative characters. The *rescaled consistency index* is their product. Linguistic cognate matrices typically have low CI and moderate RI, reflecting both genuine homoplasy (borrowing, parallel semantic change) and the length of the evolutionary histories involved.

```
c(CI = CI(mp_tree, cog), RI = RI(mp_tree, cog))
```

```
#>     CI     RI
#> 0.4592 0.4994
```

Two cautions apply to linguistic uses of these indices. First, they describe the fit of a matrix to a tree, and their values depend on how the matrix was assembled. A dataset from which characters suspected of borrowing or parallel development have been removed, as in the "screened" Indo-European data of Ringe et al. (2002), will have CI and RI close to one on its best tree almost by construction; the high values then reflect the screening as much as the history. Second, coding conventions affect the indices. In the morphological characters of the same Indo-European dataset, languages that lack an innovation such as the augment are not given a shared "absent" state but each receives a state of its own, because a shared code would assert a shared change. Such

characters carry information only through the languages that share the innovation, and the RI of a single uninformative character, for which $g_c = m_c$, is undefined ($0/0$), which `RI(..., sitewise = TRUE)` reports as `NaN`.

A related question is whether the data contain more hierarchical structure than chance would produce. The *permutation tail probability* test answers it by shuffling the states of each character independently among the languages, which keeps the number of languages in each state but destroys any shared history, and by finding the most parsimonious tree for each shuffled matrix (Faith and Cranston, 1991). If the shortest tree for the real data is shorter than those for nearly all shuffled matrices, the data are more tree-like than random data. Goldstein (2020) applied the test to the screened Indo-European characters of Ringe and Taylor: the most parsimonious tree for the real data was far shorter than the trees for any of 100 permuted datasets. The result shows phylogenetic signal beyond chance, which is unsurprising for a character set screened to exclude likely homoplasy. It does not show that a tree is the right model for the history, since contact also produces non-random structure.

4.12.2 The bootstrap

How confident should we be in each clade? The nonparametric bootstrap, introduced into phylogenetics by Felsenstein (1985a) from the general method of Efron (1979), answers by resampling. Characters are drawn *with replacement* from the data matrix to create a pseudo-replicate of the same size, a tree is inferred from the pseudo-replicate by the same method, and the procedure is repeated hundreds of times. The *bootstrap support* for a clade is the percentage of replicate trees that contain it. The rationale is that characters are treated as independent samples from some distribution of characters; the bootstrap approximates sampling a new dataset from that distribution and asks how often the clade would be recovered.

```
set.seed(3)
bs_trees <- bootstrap.phyDat(cog,
  function(x) pratchet(x, trace = 0, minit = 5, k = 3), bs = 50)
mp_rooted <- root(mp_tree, "L1", resolve.root = TRUE)
# bootstrap support as computed by plotBS(), written on each stem branch on the
# side away from the parent's vertical line so that values never sit on a branch
bs_sup <- plotBS(mp_rooted, bs_trees, type = "none")$node.label
plot(mp_rooted, cex = 0.8,
     label.offset = 0.01 * max(node.depth.edgelength(mp_rooted)))
lp <- get("last_plot.phylo", envir = .PlotPhyloEnv)
nn <- Ntip(mp_rooted) + seq_len(Nnode(mp_rooted))
pa <- mp_rooted$edge[match(nn, mp_rooted$edge[, 2]), 1]
# as plotBS(): root unlabelled, p = 0; values are printed as percentages
for (i in which(!is.na(pa) & bs_sup > 0)) {
  a <- if (lp$yy[nn[i]] >= lp$yy[pa[i]]) c(1.1, -0.35) else c(1.1, 1.35)
  nodelabels(round(100 * bs_sup[i]), nn[i], frame = "none", adj = a, cex = 0.8,
             col = "darkred")
}
```

The interpretation of bootstrap values has been much discussed. In a simulation study, Hillis and Bull (1993) found that under their conditions bootstrap values tended to be conservative, with values around 70 percent or more corresponding to a high probability that the clade was

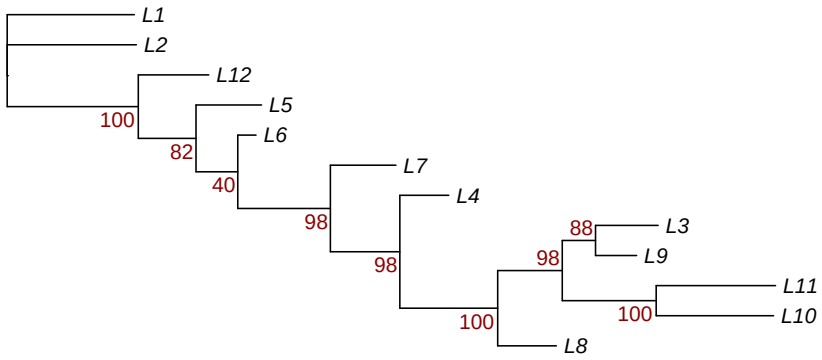


Figure 4.3: Maximum parsimony tree for the simulated 12-language cognate matrix, with bootstrap support (percent of 50 replicates) on the branches.

real. That result has often been turned into a rule of thumb, but it depends on the conditions simulated. Bootstrap support measures the *repeatability* of a result given the method, not the probability that it is true: a method that is consistently wrong, as parsimony can be (Section 4.14), will give strong bootstrap support to the wrong clade. Low support, conversely, has two different readings that the number alone cannot separate: the data may contain too little signal about the clade, or they may contain conflicting signals, for instance from borrowing, that support incompatible groupings. Neither reading licenses the conclusion that the subgroup never existed. Related resampling measures include the *jackknife* (deleting a proportion of characters without replacement) and, specific to parsimony, *Bremer support* or the *decay index* (Bremer, 1988), the number of extra steps needed before a clade disappears from the set of shortest trees.

For linguistic data the independence assumption behind the bootstrap is questionable. In binary-coded cognate data, the cognate sets of one meaning are not independent: if a language has a word from one cognate set for ‘hand’, it usually lacks the other sets for ‘hand’. Resampling individual cognate sets breaks up these dependencies. A more defensible alternative is to resample *meanings*, carrying all the cognate sets of a sampled meaning into the replicate together. Borrowing creates further dependence, since loans often come in batches from the same donor. Curated datasets raise a further problem: their characters were chosen for their reliability, not sampled at random from a population of characters, which is what the bootstrap’s rationale presupposes (Goldstein, 2020). Bootstrap values from linguistic data should be read with these caveats in mind.

4.13 Compatibility and perfect phylogeny

4.13.1 Compatible characters

A different way of using characters, older than parsimony in some respects, starts from the idea that a character with no homoplasy is a perfect witness. A character is *compatible* with a tree

if it can evolve on that tree without homoplasy, that is, with exactly m_c steps (each state arises once, and the languages sharing each state form a connected part of the tree). Two characters are *compatible with each other* if there is some tree with which both are compatible. *Compatibility methods* (Le Quesne, 1969; Estabrook et al., 1976) seek the largest set of mutually compatible characters, and the tree they define, on the theory that characters incompatible with the majority are the ones affected by homoplasy and should be discarded rather than counted.

For binary characters there is a simple test. Two binary characters are compatible if and only if, among the four combinations 00, 01, 10 and 11, at most three occur among the languages. This is the *four-gamete test*, so called because the same condition appears in population genetics as a test for recombination between two sites. If all four combinations occur, no tree can accommodate both characters without a parallel change or reversal. Moreover, for binary characters pairwise compatibility implies joint compatibility: a set of binary characters can all evolve without homoplasy on a single tree if and only if every pair passes the four-gamete test (Estabrook et al., 1976; Semple and Steel, 2003). Finding the largest such set is a maximum clique problem in the graph whose vertices are characters and whose edges join compatible pairs. Maximum clique is NP-hard in general but feasible for moderate numbers of characters.

```
Mb <- matrix(as.integer(as.character(cog)), nrow = length(cog),
             dimnames = list(names(cog), NULL))
informative <- apply(Mb, 2,
                     function(z) min(table(factor(z, levels = 0:1))) >= 2)
Mi <- Mb[, informative][, 1:60]
four_gamete_ok <- function(x, y) length(unique(paste(x, y))) < 4
nc <- ncol(Mi)
Cm <- matrix(TRUE, nc, nc)
for (i in 1:(nc - 1)) for (j in (i + 1):nc)
  Cm[i, j] <- Cm[j, i] <- four_gamete_ok(Mi[, i], Mi[, j])
mean(Cm[upper.tri(Cm)])          # proportion of compatible pairs
```

```
#> [1] 0.3452
```

```
par(mar = c(2, 2, 1, 1))
image(1:nc, 1:nc, Cm * 1, col = c("grey95", "grey25"), axes = FALSE,
      xlab = "", ylab = "", asp = 1)
```

For multistate characters, such as cognate classes within a meaning slot, things are harder. Pairwise compatibility no longer guarantees joint compatibility, and the *perfect phylogeny problem* (does there exist a tree on which every character is compatible?) is NP-hard in general (Bodlaender et al., 1992; Steel, 1992), although it can be solved efficiently when the number of states per character or the number of characters is small (Gusfield, 1991). The number of states also affects what incompatibility can reveal. Hoyal Cuthill (2015) modelled the probability that unordered characters are incompatible and found that it depends non-linearly on the number of states and on the number of taxa. With more possible states, homoplasy actually occurs less often in simulations, yet incompatibility can become more likely, so that homoplasy inferred by parsimony can move in the opposite direction from the true level. Counts of incompatible characters, and homoplasy indices such as the consistency index, are therefore not comparable across data types (binary cognate characters versus meanings with many cognate classes, say) without taking the number of states into account. This is the computational setting of the Indo-European work of

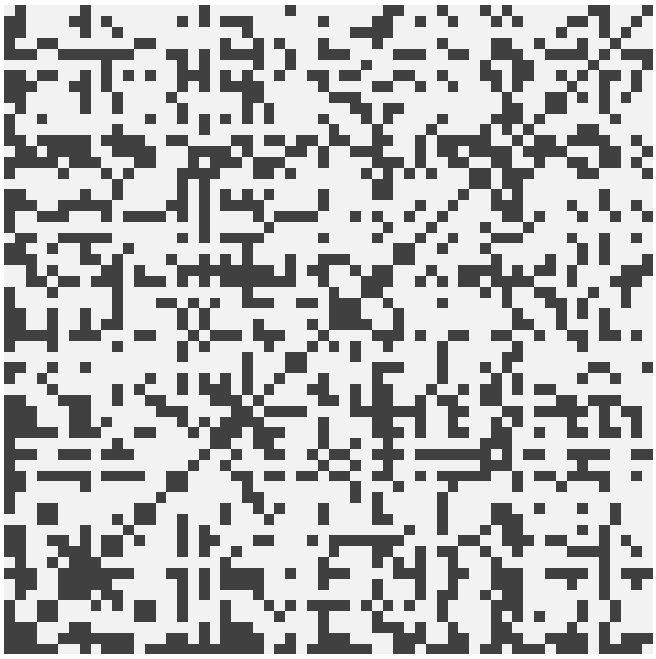


Figure 4.4: Pairwise compatibility among the first 60 informative characters of the simulated cognate matrix (dark cells: compatible pairs, by the four-gamete test). Because the simulated branches are long, much homoplasy has accumulated and many pairs are incompatible.

Ringe, Warnow and Taylor.

Ringe et al. (2002) built a dataset for 24 ancient and modern Indo-European languages, chosen to represent each major branch by its oldest well-attested members, with characters of three kinds: lexical characters (meaning slots with cognate classes as states), morphological characters (inflectional and derivational innovations) and phonological characters (sound changes, coded as present or absent). Crucially, the characters were not a random sample of vocabulary but were screened by experts. Known loanwords were excluded, lexical characters prone to parallel semantic change were set aside, and phonological characters were restricted to changes judged unlikely to occur independently. The authors then sought trees compatible with as many characters as possible, with morphological and phonological characters required to be compatible and lexical characters treated as weaker evidence (a *weighted maximum compatibility* criterion). Their best trees placed Anatolian as the first branch to separate and Tocharian as the next. They found that no tree accommodated all the characters, the main problem being Germanic, which shared characters with more than one other branch. Rather than attribute this to error, they interpreted it as evidence of early contact, which led to the extension below. The study is a model of explicitly argued character selection, and also a demonstration of how much the results depend on it.

Nakhleh et al. (2005a) extended the method to *perfect phylogenetic networks*, which add a small number of contact edges to a tree so that characters incompatible with the tree become compatible with the network, each contact edge representing borrowing between two branches at some period. Applied to the Indo-European data, a network with a few contact edges, notably involving Germanic, accommodated the incompatible characters. This anticipated the explicit network models of Chapter 8. Warnow (1997) gave an accessible account of the mathematical problems involved, and Warnow et al. (2006) proposed a stochastic model of language evolution incorporating homoplasy and borrowing, with the aim of evaluating reconstruction methods under realistic conditions.

Nakhleh et al. (2005b) compared six reconstruction methods on the Indo-European dataset of Ringe, Warnow and Taylor, in screened and unscreened versions and with and without its phonological and morphological characters: UPGMA, neighbor joining, maximum parsimony, maximum compatibility, weighted maximum compatibility and the Bayesian method of Gray and Atkinson. Evaluated against the subgrouping accepted by Indo-Europeanists, UPGMA performed clearly worst. The other five methods all recovered the established subgroups but differed considerably in how they related those subgroups to one another, and only weighted maximum compatibility, which gives extra weight to morphological and phonological characters, returned trees on which all of those characters were compatible. The authors concluded that the differences between methods were due at least in part to the choice of data, that analyses of lexical characters alone were probably less accurate, and that weighting characters by their reliability is an important part of a phylogenetic analysis. The benchmark itself deserves a caveat. The “established subgroups” were established by the comparative method from largely the same body of evidence that was coded as characters, and the screened dataset was built by experts who knew those subgroups; that every method except UPGMA recovered them shows that the methods do not destroy a strong signal, not that they would find the right answer where specialists are uncertain (Section 1.9). The real test is at the contested higher-level nodes, where the methods disagreed. A later simulation study by Barbançon et al. (2013) compared methods on data simulated under models with homoplasy and borrowing, which makes it a test of *robustness under misspecification* for methods that assume neither (Section 6.10). It found that the ranking of the unweighted methods was remarkably consistent across conditions, with maximum parsimony the most accurate, followed by Gray and

Atkinson's Bayesian method, neighbor joining and, last, UPGMA. The accuracy of weighted parsimony and weighted compatibility, by contrast, depended on the weighting scheme and on the amount of homoplasy; with low homoplasy and appropriate weights they were the most accurate methods of all. Nichols and Warnow (2008) review this whole line of work in a tutorial written for linguists.

4.14 Statistical properties and long-branch attraction

Is parsimony *consistent*, that is, does it converge on the true tree as the amount of data grows? Felsenstein (1978a) showed that it is not, in general. Consider four languages A, B, C and D on a tree with the split $AB \mid CD$, in which B and D sit at the ends of long branches (they have changed a great deal) while the other three branches, including the internal branch, are short. On long branches the same state often arises by chance in parallel. So B and D share many states by convergence, and these shared states look to parsimony like synapomorphies supporting BD. The genuine synapomorphies of AB and CD can arise only on the short internal branch and are few. As the number of characters grows, the excess of convergent over genuine shared states becomes ever more certain, and parsimony converges with increasing confidence on the wrong tree $AC \mid BD$, grouping the long branches together. This is *long-branch attraction*, and the region of branch-length space in which it happens is often called the *Felsenstein zone* (Bergsten, 2005).

The code below simulates binary characters on such a tree and records how often maximum parsimony and maximum likelihood (Chapter 5) choose the true quartet, for increasing numbers of characters. In the terms of Section 6.10, the simulation is a test of parameter (here, topology) recovery for ML, whose model is exactly the one that generated the data, and a test of robustness for parsimony, which has no explicit model but whose implicit assumption that change is rare is violated on the long branches.

```
fz <- read.tree(text = "((A:0.05,B:0.7):0.04,(C:0.05,D:0.7):0.04);")
quartets <- lapply(c(AB = "((A,B),(C,D));", AC = "((A,C),(B,D));",
                     AD = "((A,D),(B,C));"), function(s) {
  t <- unroot(read.tree(text = s)); t$edge.length <- rep(0.1, Nedge(t)); t })

one_replicate <- function(nsites) {
  x  <- simSeq(fz, l = nsites, type = "USER", levels = c("0", "1"))
  ps <- sapply(quartets, parsimony, data = x)
  ll <- sapply(quartets, function(t)
    optim.pml(pml(t, x), optEdge = TRUE,
              control = pml.control(trace = 0))$logLik)
  c(MP = names(which.min(ps + runif(3, 0, 1e-6))),    # break ties at random
    ML = names(which.max(ll)))
}
set.seed(8)
lba <- t(sapply(c(100, 500, 2000), function(n) {
  r <- replicate(30, one_replicate(n))
  c(characters = n, MP_correct = mean(r["MP", ] == "AB"),
    ML_correct = mean(r["ML", ] == "AB"))
}))
knitr::kable(lba, caption = paste(
```

```
  "Proportion of 30 simulated datasets in which maximum parsimony (MP)",
  "and maximum likelihood (ML) choose the true quartet, for data simulated",
  "in the Felsenstein zone (two long, non-sister branches)."))
```

Table 4.3: Proportion of 30 simulated datasets in which maximum parsimony (MP) and maximum likelihood (ML) choose the true quartet, for data simulated in the Felsenstein zone (two long, non-sister branches).

characters	MP_correct	ML_correct
100	0	0.4667
500	0	0.8667
2000	0	0.9667

Parsimony almost always chooses the wrong tree here, and more data make it worse, whereas likelihood under the correct model improves with more data. The result for ML should not be over-read: it shows what ML does when its model is right, not how it behaves on real data whose generating process is unknown (Section 5.10). Equivalent behavior affects UPGMA (as we saw) and, to a lesser degree, uncorrected distance methods. Corrected distances with NJ and model-based methods are consistent when their model is adequate, although they too can be misled when the model is badly wrong.

The relationship between parsimony and likelihood is subtle. Tuffley and Steel (1997) showed that maximum parsimony gives the same tree as maximum likelihood under a "no common mechanism" model in which every character has its own independent set of branch lengths. Parsimony is thus a likelihood method under a model with an enormous number of parameters, which explains both its robustness to some kinds of heterogeneity and its inconsistency in the Felsenstein zone (see also Steel and Penny, 2000). Indeed, when rates vary across lineages in different ways for different characters (*heterotachy*), parsimony can sometimes outperform a misspecified likelihood model (Kolaczkowski and Thornton, 2004). The modern consensus in biology is that model-based methods are preferable when an adequate model is available, and that the important question is model adequacy rather than a philosophical choice between parsimony and models (Felsenstein, 2004; Sober, 1988).

What does long-branch attraction look like in linguistics? Long branches arise from fast lexical replacement, from long periods of separate development, and from sparse sampling of a subgroup (a single language representing a whole branch has a long terminal branch). Languages with unusually high replacement rates, isolates or near-isolates attached to large families, and languages with extensive data gaps are candidates. Borrowing produces a related artifact: two lineages that have borrowed from the same donor share states that parsimony reads as synapomorphies. There is little systematic work measuring long-branch attraction in real linguistic analyses, but the general theory suggests that the problem is real for lexical data with strongly heterogeneous rates, and it is one reason to check parsimony results against model-based ones.

4.15 Parsimony in linguistics

Parsimony was the first character-based method widely applied to languages in the era of computational phylogenetics, mostly between the late 1990s and the mid-2000s, before Bayesian methods

became dominant.

Gray and Jordan (2000) applied maximum parsimony to lexical data for 77 Austronesian languages, coded as presence or absence of cognate sets, to test the "express train" model of Austronesian expansion: a rapid movement of Austronesian speakers from Taiwan through Island Southeast Asia into the Pacific. They compared the parsimony tree with the sequence of expansion predicted by the model and measured the fit with a tree-based test, finding a significantly better fit than expected from random trees. The study was one of the first to use a phylogenetic tree inferred from linguistic data to test an explicit prehistoric scenario, and its framework was later carried forward with Bayesian methods and much larger datasets (Gray et al., 2009).

Holden (2002) inferred a maximum parsimony tree for Bantu languages from basic vocabulary, coding each meaning's cognate classes as characters, and found a tree whose structure was broadly consistent with archaeological evidence for the spread of farming in sub-Saharan Africa. She also reported bootstrap support and noted that the deepest splits were weakly supported, a pattern that has recurred in later Bantu work. Holden's analysis was an important early example of combining a language tree with archaeological hypotheses, and later Bayesian analyses of Bantu built on it (Chapter 10). Holden et al. (2005) compared parsimony and Bayesian trees for Bantu.

Rexová et al. (2003) recoded the Dyen et al. Indo-European lexicostatistical data as characters for cladistic analysis and inferred maximum parsimony trees. The analysis recovered the main branches (an expected result given that the cognate coding encodes them; Section 1.9) and proposed higher-order groupings, while showing the instability of the deepest nodes. In a follow-up, Rexová et al. (2006) combined lexical and grammatical characters for Bantu languages in a single parsimony analysis, an early example of combining different kinds of linguistic evidence in one matrix (what biologists call a total-evidence analysis).

Dunn et al. (2005) applied phylogenetic methods to *structural* rather than lexical characters, coding the presence or absence of more than a hundred grammatical and phonological features in Papuan languages of Island Melanesia, whose relationships are too deep to be detected through cognates. As a control, they first applied the same method to Oceanic (Austronesian) languages of the same region, whose relationships are known from the comparative method, and found that structural parsimony recovered much of the known Oceanic structure. Here the control is closer to independent than in lexical studies, because the structural characters were not used to establish the Oceanic subgroups; but success on one family is consistent with, not proof of, a genealogical signal in another, whose contact history may differ. Applied to the Papuan languages, the structural data suggested deep historical signal. The method was examined further by Dunn et al. (2008), who explored its sensitivity to feature selection. Structural phylogenetics raises distinctive problems: structural features have few states, so chance similarity is high; they are prone to convergence and areal diffusion; and their rates of change differ widely (Chapter 13). Wichmann and Saunders (2007) explored the use of typological databases such as WALS for historical classification with several methods and found, broadly, that structural features can carry genealogical signal but must be selected and weighted with care.

What does a parsimony analysis look like when a family did not diversify as a tree? Ringe (2022) describes an early test run by his group on modern West Germanic languages, with Danish and Swedish as outgroup, chosen because these languages are known to have been in contact throughout their history. The best parsimony score that any tree could achieve for the data, the score of a perfect phylogeny, was 262. The best trees found scored between 309 and 335, so every candidate needed more than forty additional changes. The three least bad trees each corresponded

to a defensible grouping, but the next three, only a few steps worse, split the Frisian languages, which is implausible. Ringe takes this pattern, several mediocre trees each bad in a different way, as the signature of non-treelike diversification. He also notes that the analysis cannot tell clean splits followed by undetectable borrowing from a dialect network whose non-adjacent members happen to survive. The example is a useful complement to the consistency index: what matters is not only how much homoplasy the best tree requires but whether the near-optimal trees agree.

Since the late 2000s parsimony has mostly given way to Bayesian methods in published linguistic phylogenetics, for three reasons: its inconsistency in the presence of rate variation; its inability to estimate branch lengths in units that can be converted into time; and the lack of a natural way to integrate it with models of lexical change such as the stochastic Dollo model. It remains useful as a fast first look at a dataset, as a check on model-based results (a clade supported by parsimony, likelihood and Bayesian analysis alike is unlikely to be an artifact of any one method's assumptions, though it can still be an artifact of the data, such as undetected loans that all methods read as shared inheritance), and for structural and phonological characters where the model of change is poorly understood.

Morphological cladistics in biology has developed tools that, to our knowledge, have seen little or no use on linguistic data. *TNT* (Goloboff et al., 2008) searches parsimony trees for thousands of taxa in minutes using sectorial searches, tree drifting and tree fusing, and implements *implied weighting*, which automatically down-weights homoplastic characters (Goloboff, 1993); this could be tested as a way of down-weighting cognate sets affected by undetected borrowing. *Bremer support* (Bremer, 1988) and *partitioned Bremer support* measure how many extra steps a clade costs and which partitions of the data (for instance, lexical vs morphological characters) support or oppose it, a natural way to diagnose conflict between kinds of linguistic evidence. Finally, *quartet-based methods*, which build a tree from many four-taxon subtrees (Strimmer and von Haeseler, 1996), are now central to species-tree estimation from gene trees in phylogenomics (Chapter 16), and could be used to combine trees inferred from different meaning classes or different kinds of characters.

- Distance methods reduce data to pairwise dissimilarities; they are fast and can use raw word forms, but they lose information and are only as good as the distance measure.
- Corrected distances (such as $-\ln s$ under an infinite-alleles model, or the binary Jukes–Cantor correction) undo saturation but become noisy for distant pairs.
- Additive matrices satisfy the four-point condition and determine a unique tree; ultrametric matrices additionally imply a clock.
- UPGMA assumes a clock and groups slowly changing languages together; neighbor joining, BIONJ and balanced minimum evolution do not, and are consistent for additive data.
- Levenshtein-based distances (LDN, LDND, ASJP) measure form similarity, are blind to sound correspondences, and are most reliable for shallow relationships.
- The choice of distance measure (meaning-based, Hamming, Jaccard, corrected) changes the results; corrected and form-based distances need not satisfy the triangle inequality.
- MDS and PCA display similarity without a tree. They are useful for spotting clusters, outliers and continua and for deciding what the taxa are, but they have no evolutionary model, cannot separate innovations from retentions, and order their axes by variance, not by time depth.
- Parsimony prefers the tree with the fewest changes and formalizes the comparative method's reliance on shared innovations; Fitch and Sankoff algorithms compute tree lengths by dynamic programming, and Dollo and Camin–Sokal costs suit cognate sets and irreversible

sound changes.
- CI, RI and bootstrap support measure homoplasy and repeatability, not truth; for cognate data, resampling meanings is more defensible than resampling cognate sets.
- Compatibility and perfect phylogeny methods, used with expertly screened characters by Ringe, Warnow and Taylor, seek trees on which characters evolve without homoplasy and extend naturally to networks.
- Parsimony is inconsistent in the Felsenstein zone (long-branch attraction), which motivates model-based methods.
- Recovering accepted subgroups is a weak test of a method: cognate judgements already embody the comparative method, and cognate sharing is not the same as shared innovation. Simulations under a method's own assumptions test recovery, not robustness or adequacy.

4.16 Exercises

Conceptual

1. Explain why the binary Hamming distance between two languages is dominated by shared absences, and why the meaning-based proportion of equation (4.1) is preferred for distance work.
2. A dataset contains one lexically conservative language from each of two branches. Explain, with reference to the three-point condition, why UPGMA may group them, and why neighbor joining should not.
3. Under which cost scheme in Table 4.2 would you score (a) cognate sets, (b) a phoneme merger, (c) the number of nominal cases, (d) basic word order? Justify each choice and name one way in which the assumption could fail for real linguistic data.
4. Why are singleton cognate sets uninformative for parsimony but informative for likelihood?
5. A clade has 98 percent bootstrap support in a parsimony analysis of binary cognate data. List three reasons why this need not mean that the clade is historically real.
6. A new method recovers every established subgroup of Indo-European. What does this result show, and what does it not show? What further evidence would you want before using the method on a family whose subgrouping is unknown?
7. A colleague runs a non-metric MDS of a matrix of 15 dialects, finds three tight clusters on the first two axes (stress 18 percent), and proposes to merge each cluster into a single taxon before inferring a parsimony tree. What would you check before accepting the merge, and how could the merge itself change the tree and its support values?

Computational

1. Using the word list `W` from the chapter, add one meaning of your own choice (for example 'foot' or 'new') with forms in the same broad transcription (a new row added with `rbind()`), recompute LDND with `ldn_ldnd()`, and rebuild the trees with `upgma()` and `nj()`. Do any groupings change? Compare the trees with `RF.dist()`.
2. Simulate 400 binary characters on `tree12` with `simSeq()` (as in the chapter) for three overall rate multipliers (multiply `tree12$edge.length` by 0.2, 0.4 and 1.0). For each, compute corrected distances and the RF distance to `tree12` of the `upgma()`, `nj()` and `fastme.bal()` trees, and of the `pratchet()` tree. Which methods degrade first as saturation increases?
3. Repeat the long-branch simulation of Section 4.14 with the internal branch lengthened from 0.04 to 0.15 (edit the Newick string of `fz`). How do the proportions of correct parsimony

and ML choices change? Relate your result to the Felsenstein zone.

4. Implement a *meaning-level* bootstrap for the toy word list of Chapter 3 (rerun the code of Section 3.4 to create `wl` and `make_binary()`): resample meanings with replacement, rebuild the binary matrix from the resampled meanings, infer an NJ tree with `NJ(dist.hamming())` for each replicate, and compute support with `prop.clades()`. Compare with the support obtained by resampling individual cognate sets with `bootstrap.phyDat()`.
5. For the simulated matrix `cog`, compute `CI()` and `RI()` on the most parsimonious tree, the NJ tree and the true tree `tree12`. Which tree has the highest CI, and why does that not make it the best estimate of history? Then call `RI(mp_tree, cog, sitewise = TRUE)`: which characters return `NaN`, and why? Explain also why the CI of a matrix can approach but never reach zero.
6. For the toy word list of Chapter 3 (rerun the code of Section 3.4 to create the binary matrix with `make_binary()`), compute Hamming distances with `dist.hamming()` and Jaccard distances with `dist(x, method = "binary")` on the 0/1 matrix. Ordinate both with `cmdscale()` and with `isoMDS()` for $k = 1, \dots, 4$, plot stress against k, and compare the two configurations with the NJ trees from the same distances. Finally, count the triples that violate the triangle inequality in the corrected distances `d_jc` of this chapter and in the Hamming distances `p_ham`.

Research

1. You have ASJP word lists (40 meanings, no cognate judgements) for 300 languages of a poorly documented region, and the task is a preliminary classification to guide fieldwork. Choose a distance measure and tree-building method, explain what the results could and could not tell you, and say how you would check whether apparent groupings reflect contact or chance rather than descent.
2. You are coding 30 morphological and 15 phonological innovations together with 200 lexical meaning slots for 25 ancient and modern languages of one family. Would you analyse them with weighted compatibility, weighted parsimony or a model-based method? Justify the weighting (or its absence), the treatment of irreversible sound changes, and how you would test the sensitivity of the conclusions to these choices.
3. Design a simulation study to assess whether parsimony is affected by long-branch attraction in a family with one isolate-like, fast-changing language. State which aspects of the simulation would test recovery under the method's assumptions and which would test robustness to borrowing and rate variation.

4.17 Further reading

- Felsenstein (2004), chapters 1 to 11, remains the best single account of parsimony, compatibility and distance methods, with their history and statistical properties, written by one of the people who developed them.
- Semple and Steel (2003) gives the mathematical theory (tree metrics, the four-point condition, compatibility and perfect phylogeny) rigorously and concisely.
- McMahon and McMahon (2005) is a book-length introduction to quantitative language classification for linguists, covering lexicostatistics, distance methods and early networks.
- Nichols and Warnow (2008) is a long tutorial on computational linguistic phylogeny by a linguist and a computer scientist, covering compatibility, parsimony, distance and Bayesian

methods, with a clear discussion of character selection.

- Ringe et al. (2002) is essential reading for anyone who wants to see how carefully argued character coding works in practice, and for the case for weighting morphological and phonological characters.
- Nakhleh et al. (2005b) and Barbançon et al. (2013) compare methods on real and simulated linguistic data, respectively.
- Goldstein (2020) is a hands-on tutorial in R (**ape**, **phangorn**) that applies parsimony, distance methods, maximum likelihood, homoplasy indices, permutation tests and the bootstrap to the Ringe-Taylor Indo-European characters, with a balanced discussion of what computational methods can add to traditional subgrouping.
- Wichmann et al. (2010a) and Jäger (2013) describe the development of form-based distances, from LDND to PMI-weighted alignment, and Greenhill (2011) provides the main critique.
- Bergsten (2005) reviews long-branch attraction thoroughly, with many empirical examples from biology.
- Ringe (2022) is a short, candid assessment by one of the pioneers of computational cladistics in linguistics of what these methods can and cannot establish, with the West Germanic example discussed above.
- Johnson (2008), chapter 6, introduces clustering, cladistics and multidimensional scaling of language similarity in R for linguists; Baayen (2008), chapter 5, and Levshina (2015), chapters 17 and 18, cover MDS and PCA with linguistic examples and R code.
- Borg and Groenen (2005) is the standard reference on multidimensional scaling, and Jolliffe and Cadima (2016) a concise modern review of PCA and its variants.

Chapter 5

Maximum Likelihood

Maximum likelihood (ML) is the bridge between the algorithmic methods of Chapter 4 and the Bayesian methods that dominate linguistic phylogenetics today (Chapter 6). It requires an explicit probabilistic model of how characters change along the branches of a tree, and it chooses the tree and parameter values under which the observed data are most probable. Every model-based method in this book, Bayesian ones included, rests on the same machinery: a continuous-time Markov chain that describes change along a branch, the transition probabilities $\mathbf{P}(t) = e^{\mathbf{Q}t}$ that it implies, and Felsenstein's pruning algorithm, which computes the probability of a whole data matrix on a tree efficiently. This chapter explains that machinery with small linguistic examples, shows how ML trees are searched for, supported and compared, and explains why cognate data need a correction that most DNA data do not. Readers who want to refresh the background can consult Sections A.3 (likelihood), A.4 (the gamma distribution), A.6 (the matrix exponential) and A.7 (continuous-time Markov chains) of the mathematical appendix.

After reading this chapter you should be able to:

- explain the difference between a probability and a likelihood, and find the maximum likelihood estimate of the path length between two languages;
- specify the binary model for cognate data by its rate matrix, and read its transition probabilities as a function of branch length;
- compute the likelihood of a cognate character on a small tree by hand with Felsenstein's pruning algorithm, and explain why the algorithm is efficient;
- explain why cognate data are ascertained, what goes wrong if the likelihood is not corrected, and apply a correction in **phangorn**;
- run and interpret a heuristic ML tree search, bootstrap support values and a topology test;
- compare models with likelihood-ratio tests and information criteria, and distinguish the choice of the best candidate model from a check of absolute adequacy.

5.1 The likelihood principle

5.1.1 Probability and likelihood

A *probability model* assigns probabilities to possible data given fixed values of its parameters. If a coin has probability θ of landing heads, the probability of seeing 7 heads in 10 tosses is $\binom{10}{7}\theta^7(1-$

$\theta)^3$. The *likelihood function* reads the same expression the other way around: the data are now fixed, and it is a function of the parameter,

$$L(\theta) = P(D \mid \theta), \tag{5.1}$$

which measures how well each value of θ predicts the data actually observed. The likelihood is not a probability distribution over θ; it does not integrate to one, and only ratios of likelihoods are meaningful. The *maximum likelihood estimate* (MLE) $\hat{\theta}$ is the value that maximizes $L(\theta)$, or equivalently the log-likelihood $\ell(\theta) = \ln L(\theta)$. For the coin, $\hat{\theta} = 7/10$.

The idea goes back to Fisher (1922), and its use as a general principle of inference was argued at length by Edwards (1972). Two features make it attractive. First, under regularity conditions, MLEs are *consistent* (they converge on the true value as the data grow), asymptotically normal, and asymptotically efficient (no consistent estimator has smaller variance in large samples). Second, the log-likelihood provides a natural scale for comparing hypotheses: a difference of about two log-likelihood units corresponds roughly to the boundary of a 95 percent confidence interval for a single parameter, since twice the log-likelihood difference is approximately χ^2-distributed with one degree of freedom, and the 95 percent point of that distribution is 3.84.

The application to trees was pioneered by Joseph Felsenstein, first for discrete characters (Felsenstein, 1973) and then, decisively, for DNA sequences (Felsenstein, 1981). In phylogenetics, the "parameters" include the tree topology τ, the branch lengths $\mathbf{t}$ and the parameters θ of the model of change, and the likelihood is $L(\tau, \mathbf{t}, \theta) = P(D \mid \tau, \mathbf{t}, \theta)$. The topology is an unusual parameter, discrete and with a huge number of possible values, which causes both the computational and the statistical difficulties discussed in this chapter.

5.1.2 A first example: two languages

Consider two languages and N binary cognate characters, and suppose that each character switches between presence (1) and absence (0) at the same rate in both directions. We will derive this model properly in the next section; for now we need one result: if the path between the two languages has length t in expected changes per character, the probability that they are in different states for a character is $p(t) = \frac{1}{2}(1 - e^{-2t})$. If k of the N characters differ, the likelihood is

$$L(t) \propto p(t)^k \, [1 - p(t)]^{N-k} \, . \tag{5.2}$$

This is a binomial likelihood in p, maximized at $\hat{p} = k/N$. Because $p(t)$ is a one-to-one function of t (for $p < 1/2$), the MLE of t is obtained by inverting it: $\hat{t} = -\frac{1}{2}\ln(1 - 2k/N)$, which is precisely the corrected binary distance of equation (4.4). Corrected distances are thus maximum likelihood estimates of pairwise path lengths. What full ML adds is that it estimates all the branch lengths of the tree *jointly* from all the characters, using the information in which languages share which states, not just pairwise counts.

```
N <- 100; k <- 30
ll_t <- function(t) {
  p <- 0.5 * (1 - exp(-2 * t))
  k * log(p) + (N - k) * log(1 - p)
}
t_hat <- -0.5 * log(1 - 2 * k / N)
```

```
t_grid <- seq(0.05, 1.5, length.out = 400)
ll <- ll_t(t_grid)
ci <- range(t_grid[ll > ll_t(t_hat) - 1.92])
par(mar = c(4, 4, 1, 1))
plot(t_grid, ll, type = "n",
     xlab = "path length t (expected changes per character)",
     ylab = "log-likelihood", ylim = c(ll_t(t_hat) - 15, ll_t(t_hat) + 1))
rect(ci[1], -1e6, ci[2], 1e6, col = "grey90", border = NA)
lines(t_grid, ll, lwd = 2)
abline(v = t_hat, lty = 2)
```

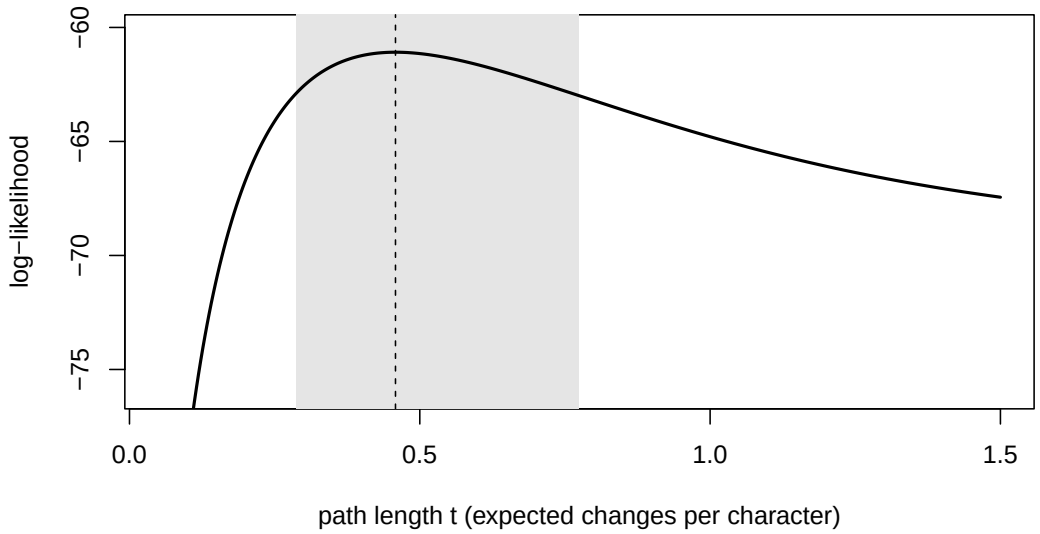


Figure 5.1: Log-likelihood of the path length t between two languages that differ in 30 of 100 binary characters, under the symmetric binary model. The dashed line marks the MLE; the shaded band contains the values within 1.92 log-likelihood units of the maximum (an approximate 95 percent confidence interval).

```
c(t_hat = t_hat, lower = ci[1], upper = ci[2])
```

```
#>  t_hat  lower  upper
#> 0.4581 0.2862 0.7732
```

The interval is asymmetric: the likelihood falls off steeply toward short paths (which would make 30 differences implausible) but slowly toward long ones, because the probability of a difference saturates at one half. Long paths are always estimated less precisely than short ones, a fact that matters for the deep nodes of language trees.

5.2 Continuous-time Markov chains

5.2.1 Definitions

Models of character change along a branch are *continuous-time Markov chains* (CTMCs). A CTMC is a random process $X(t)$ that is in one of a finite number of states $\{1, \dots, k\}$ at each time $t \geq 0$

(for a cognate character, absent or present; for a meaning slot, one of several cognate classes; for word order, SOV, SVO and so on), and that satisfies the *Markov property*: the probability of future states depends only on the current state, not on how the process got there. A language that has lost its inherited word for ‘dog’ does not “remember” how long it had it.

A CTMC is specified by its *instantaneous rate matrix* $\mathbf{Q} = (q_{ij})$. For $i \neq j$, $q_{ij} \geq 0$ is the rate of change from state i to state j: in a short interval of length h, a process in state i moves to j with probability approximately $q_{ij}h$. The diagonal entries are set so that each row sums to zero, $q_{ii} = -\sum_{j \neq i} q_{ij}$, so that $-q_{ii}$ is the total rate of leaving state i. The time spent in state i before leaving is exponentially distributed with rate $-q_{ii}$, and when the process leaves, it jumps to j with probability $q_{ij}/(-q_{ii})$.

5.2.2 From rates to probabilities: $\mathbf{P}(t) = e^{\mathbf{Q}t}$

What we need for likelihood calculations is not the rates but the *transition probabilities* $P_{ij}(t) = P(X(t) = j \mid X(0) = i)$ over a finite time t, collected in the matrix $\mathbf{P}(t)$. The key fact is that the rates determine these probabilities completely. Because the process has no memory, the probability of going from i to j in time $t + h$ is the probability of going from i to some intermediate state in time t and then on to j in time h. Letting h shrink turns this observation into a matrix differential equation whose solution is

$$\mathbf{P}(t) = e^{\mathbf{Q}t} = \sum_{n=0}^{\infty} \frac{(\mathbf{Q}t)^n}{n!}, \tag{5.3}$$

the *matrix exponential*. In practice it is computed from an eigendecomposition of $\mathbf{Q}$. Because the rows of $\mathbf{Q}$ sum to zero, one eigenvalue is always 0; the others are negative and determine how fast the chain forgets its starting state.

The derivation of equation (5.3) from the Chapman–Kolmogorov equation, and its computation by eigendecomposition, are in Section B.3.1. Sections A.6 and A.7 of the mathematical appendix introduce the matrix exponential and continuous-time Markov chains from the beginning.

5.2.3 Stationarity, reversibility and units

If every state can be reached from every other, the chain has a unique *stationary distribution* π, the row vector satisfying $\pi\mathbf{Q} = \mathbf{0}$. As $t \to \infty$, every row of $\mathbf{P}(t)$ converges to π: after a long time the state of the process is independent of where it started. Phylogenetic models usually assume that the process at the root of the tree is already at stationarity, so the root state is drawn from π.

Most phylogenetic models are also *time-reversible*: they satisfy the *detailed balance* condition $\pi_i q_{ij} = \pi_j q_{ji}$ for all i, j, meaning that at stationarity the flow from i to j equals the flow from j to i. A reversible process looks the same run forward or backward in time, which, as we shall see, makes the likelihood independent of where the root is placed.

Finally, rate and time are confounded. Multiplying $\mathbf{Q}$ by a constant and dividing t by the same constant leaves $\mathbf{Q}t$, and so $\mathbf{P}(t)$, unchanged. Without external information (a clock and calibrations, Chapters 9 and 10) only the product is identifiable. By convention $\mathbf{Q}$ is therefore *normalized* so that the expected rate of change at stationarity is one, $-\sum_i \pi_i q_{ii} = 1$, and branch lengths are measured in *expected number of changes per character*. This is the first example in the book of a quantity, the absolute rate of lexical change, that is not identifiable from the character data alone under the model (Section 6.10).

5.2.4 The binary model for cognate data

The workhorse model for cognate data is the two-state CTMC. Let state 0 be "cognate set absent" and state 1 "present", let α be the rate of gain ($0 \to 1$) and β the rate of loss ($1 \to 0$):

$$\mathbf{Q} = \begin{pmatrix} -\alpha & \alpha \\ \beta & -\beta \end{pmatrix}. \tag{5.4}$$

Solving $\pi\mathbf{Q} = \mathbf{0}$ with $\pi_0 + \pi_1 = 1$ gives $\pi_1 = \alpha/(\alpha+\beta)$ and $\pi_0 = \beta/(\alpha+\beta)$. The eigenvalues are 0 and $-(\alpha+\beta)$, and working through the eigendecomposition (Section B.3.1) gives the closed form

$$\mathbf{P}(t) = \begin{pmatrix} \pi_0 + \pi_1 e^{-(\alpha+\beta)t} & \pi_1\left(1 - e^{-(\alpha+\beta)t}\right) \\ \pi_0\left(1 - e^{-(\alpha+\beta)t}\right) & \pi_1 + \pi_0 e^{-(\alpha+\beta)t} \end{pmatrix}. \tag{5.5}$$

Each row starts at the identity when $t = 0$ and decays exponentially toward π. The normalization $\pi_0\alpha + \pi_1\beta = 1$ gives $2\alpha\beta/(\alpha+\beta) = 1$, so that $\alpha + \beta = 1/(2\pi_0\pi_1)$. The normalized model therefore has a single free parameter, π_1, the equilibrium frequency of presence; with $\pi_1 = 1/2$ it reduces to the symmetric model used in the two-language example. The model is reversible: $\pi_0\alpha = \pi_1\beta = \alpha\beta/(\alpha+\beta)$.

This model was introduced for restriction-site data in molecular biology (Felsenstein, 1992), which, like cognate data, record the presence or absence of a feature. It is often called the *restriction-site* or *binary* model in software (MrBayes's `lset coding` options, BEAST 2's binary substitution models, IQ-TREE's `GTR2`), and with $\pi_1 = 1/2$ it is the two-state member of the *Mk* family of models for discrete morphological characters (Lewis, 2001), which generalizes the Jukes–Cantor model (Jukes and Cantor, 1969) to k states with equal rates between all pairs. Table 5.1 lists the binary and multistate models alongside their better-known DNA counterparts (Kimura, 1980; Hasegawa et al., 1985; Tavaré, 1986).

Table 5.1: Substitution models used for linguistic data and their DNA counterparts.

Model	States	Free parameters in $\mathbf{Q}$	Linguistic use
Binary symmetric (JC2, Mk with $k = 2$)	2	0	cognate presence/absence with equal gain and loss
Binary with unequal frequencies (GTR2, restriction site)	2	1 (π_1)	cognate presence/absence, the standard choice
Mk (Lewis, 2001)	k	0	multistate meaning slots or structural features
Asymmetric Mk / GTR on k states	k	up to $k(k-1) - 1$	ordered or directional structural changes
Jukes–Cantor (JC69) (Jukes and Cantor, 1969)	4 (DNA)	0	(biology)
HKY85 (Hasegawa et al., 1985)	4 (DNA)	4 (κ and 3 frequencies)	(biology)
GTR (Tavaré, 1986)	4 (DNA)	8 (5 exchangeabilities, 3 frequencies)	(biology)

The code below builds the normalized binary **Q**, checks the closed form (5.5) against a numerical matrix exponential from the **expm** package (a small piece of *implementation validation*, in the vocabulary of Section 6.10), and plots the transition probabilities.

```
binary_Q <- function(pi1) {
  pi0 <- 1 - pi1
  s <- 1 / (2 * pi0 * pi1)              # alpha + beta after normalization
  matrix(c(-s * pi1, s * pi1,
            s * pi0, -s * pi0), 2, byrow = TRUE,
         dimnames = list(c("0", "1"), c("0", "1")))
}
binary_P <- function(t, pi1) {           # closed form, equation (binaryP)
  pi0 <- 1 - pi1
  e <- exp(-t / (2 * pi0 * pi1))
  matrix(c(pi0 + pi1 * e, pi1 * (1 - e),
           pi0 * (1 - e), pi1 + pi0 * e), 2, byrow = TRUE)
}
Q <- binary_Q(0.3)
Q
```

```
#>         0       1
#> 0 -0.7143  0.7143
#> 1  1.6667 -1.6667
```

```
-sum(c(0.7, 0.3) * diag(Q))              # expected rate at stationarity = 1
```

```
#> [1] 1
```

```
all.equal(unname(expm::expm(Q * 0.4)), binary_P(0.4, 0.3))
```

```
#> [1] TRUE
```

```
tt <- seq(0, 2, length.out = 200)
par(mfrow = c(1, 2), mar = c(4, 4, 2, 1))
for (p1 in c(0.5, 0.2)) {
  P01 <- sapply(tt, function(t) binary_P(t, p1)[1, 2])
  P11 <- sapply(tt, function(t) binary_P(t, p1)[2, 2])
  plot(tt, P11, type = "l", lty = 2, lwd = 2, ylim = c(0, 1),
       xlab = "branch length t", ylab = "P(present at end)",
       main = bquote(pi[1] == .(p1)))
  lines(tt, P01, lwd = 2)
  abline(h = p1, lty = 3)
}
```

5.2.5 Variation in rates among characters

Not all characters change at the same rate. In basic vocabulary, words for ‘two’, ‘name’ or ‘I’ are replaced far more slowly than words for ‘dirty’ or ‘to throw’ (Pagel et al., 2007). A model that forces all characters to share one rate will overestimate change for the slow characters and underestimate it for the fast ones, and it will typically underestimate long branches. The standard

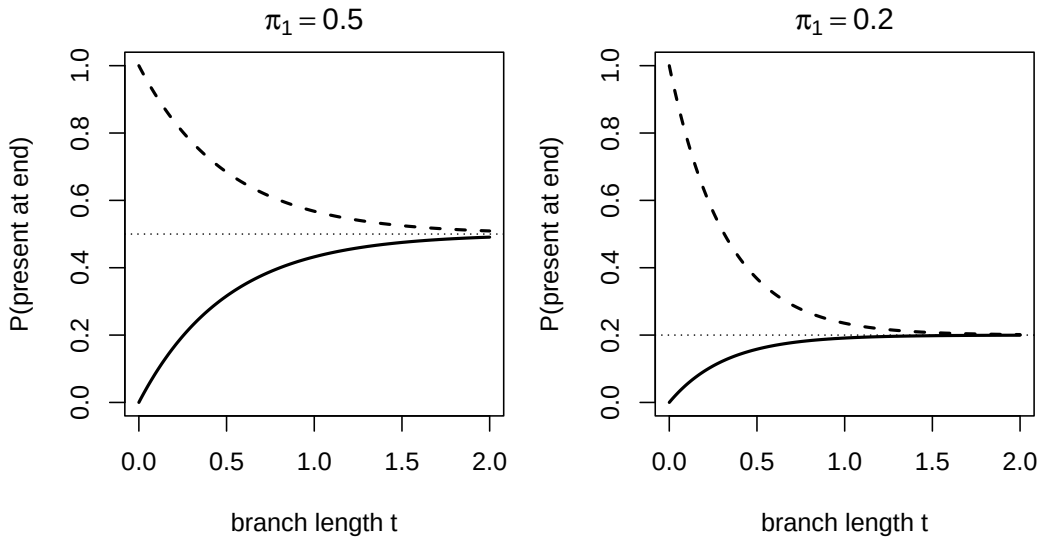


Figure 5.2: Transition probabilities of the normalized binary model as a function of branch length. Left: symmetric model ($\pi_1 = 0.5$). Right: unequal frequencies ($\pi_1 = 0.2$), typical of cognate data in which any given cognate set is absent from most languages. Solid lines start in state 0 (absent), dashed lines in state 1 (present); each converges to the stationary frequency of presence (dotted).

remedy, from DNA work, is to multiply each character's branch lengths by a rate r drawn from a gamma distribution with mean 1 and shape parameter α_Γ (small values mean strong heterogeneity), approximated by a few discrete categories (Yang, 1994). The likelihood of a character is then the average of its likelihoods over the categories:

$$L_i = \frac{1}{K} \sum_{c=1}^{K} P(D_i \mid \tau, r_c \mathbf{t}, \theta), \tag{5.6}$$

where $r_1, \ldots, r_K$ are the mean rates of K equal-probability categories of the gamma distribution and $r_c\mathbf{t}$ denotes all branch lengths multiplied by r_c (Section A.4 describes the gamma distribution and its discretization). The cost is a K-fold increase in computation; $K = 4$ is the usual compromise. The category rates show how strongly a small shape parameter spreads the rates out:

```
library(phangorn)
rates_table <- t(sapply(c(0.2, 0.5, 1, 5),
                        function(a) discrete.gamma(a, k = 4)))
dimnames(rates_table) <- list(paste("shape", c(0.2, 0.5, 1, 5)),
                              paste("category", 1:4))
round(rates_table, 3)
```

```
#>           category 1 category 2 category 3 category 4
#> shape 0.2      0.001      0.034      0.384      3.582
#> shape 0.5      0.033      0.252      0.820      2.894
#> shape 1        0.137      0.477      1.000      2.386
#> shape 5        0.502      0.804      1.083      1.611
```

With shape 0.2, three quarters of the characters are nearly invariable and the fastest quarter carries almost all the change; with shape 5, rates are nearly equal. Cognate data typically yield

small shape estimates, reflecting the very different stabilities of basic meanings. A proportion of invariable characters (+I) is another common option. These extensions, and more linguistically motivated alternatives such as rates that depend on meaning or word frequency and covarion models in which rates change over time, are the subject of Chapter 7. The gamma shape is a parameter that is identifiable and usually well estimated, but its historical interpretation is indirect: it summarizes how unequal the stabilities of the sampled meanings are, not any particular historical event.

5.3 The likelihood of a tree

5.3.1 Assumptions

To go from a model of change along one branch to the probability of a data matrix on a whole tree, three standard assumptions are made:

1. *Independence across characters*: each character evolves independently of the others, given the tree and the model. The likelihood of the data matrix is the product of the likelihoods of its characters.
2. *Markov evolution along the tree*: along each branch, each character evolves according to the CTMC, starting from the state at the branch's upper end. At a split, both daughter lineages inherit the parent's state and then evolve independently.
3. *Homogeneity and stationarity*: the same $\mathbf{Q}$ applies on every branch (up to rate multipliers), and the root state is drawn from π.

All three are idealizations. Binary coding of meaning slots violates independence (Chapter 7); borrowing violates the assumption that lineages evolve independently after they split (Chapter 8); and rates of lexical change vary over time and among lineages (Chapter 9). But the assumptions make the computation tractable, and many of their violations can be relaxed within the same framework.

5.3.2 Summing over unobserved ancestral states

Under these assumptions, if we knew the states at every internal node, the probability of one character would be a product of transition probabilities, one per branch, times the root probability. We do not know the ancestral states, so we sum over all possible assignments. For a rooted tree with leaves 1 to 4, internal nodes x (root) and y, and the topology $((1,2)_y, 3, 4)_x$, with observed states $s_1, \dots, s_4$ and branch lengths $t_1, \dots, t_4$ and t_y,

$$L = \sum_{x} \sum_{y} \pi_x \, P_{xy}(t_y) \, P_{ys_1}(t_1) \, P_{ys_2}(t_2) \, P_{xs_3}(t_3) \, P_{xs_4}(t_4). \tag{5.7}$$

With n leaves and k states there are k^{n-1} terms in this double, triple, ... sum, one for each assignment of states to the $n-1$ internal nodes of a rooted binary tree. For 50 languages and binary characters this is $2^{49} \approx 5.6 \times 10^{14}$ terms per character, so direct summation is out of the question. Summing over all assignments is also what distinguishes likelihood from parsimony, which scores a tree by its single best reconstruction (Chapter 4). If one tree can explain a character by many different histories and another by only one, the first gains likelihood from all of them, just as a total of seven with two dice is six times as probable as a total of two, although each particular throw is equally likely (Holder and Lewis, 2003).

5.3.3 The pulley principle

Because the model is reversible and stationary, the likelihood does not depend on where the root is placed. This is Felsenstein's *pulley principle* (Felsenstein, 1981): one can slide the root along any branch, like a rope over a pulley, without changing the likelihood, as long as the sum of the two branch segments is constant. As a consequence, ML under reversible models estimates *unrooted* trees, and the root has to be placed afterward by an outgroup or by a clock model.

5.4 Felsenstein's pruning algorithm

5.4.1 The recursion

The solution to the explosion of terms is the *pruning algorithm* (Felsenstein, 1981), a dynamic programming method that rearranges the sum (5.7) by pushing each summation as far inside the product as possible. Define for each node v and each state a the *conditional likelihood*

$$L_v(a) = P(\text{observed states at the leaves below } v \mid v \text{ is in state } a).$$

At a leaf, $L_v(a) = 1$ if a is the observed state and 0 otherwise. If the state is missing or ambiguous, $L_v(a) = 1$ for every state compatible with the observation, which is how missing data are handled. For an internal node v with children u and w reached along branches of length t_u and t_w,

$$L_v(a) = \left[\sum_b P_{ab}(t_u)\, L_u(b)\right] \times \left[\sum_c P_{ac}(t_w)\, L_w(c)\right]. \tag{5.8}$$

The first factor is the probability of the data below u given that v is in state a: the process moves from a to some state b along the branch, and then generates the data below u with probability $L_u(b)$. Independence of the two daughter lineages makes the two factors multiply. At the root,

$$L = \sum_a \pi_a L_{\text{root}}(a). \tag{5.9}$$

Visiting nodes in postorder (children before parents), each node requires $O(k^2)$ operations, so the likelihood of one character costs $O(nk^2)$ instead of $O(k^{n-1})$, and the likelihood of m characters costs $O(nk^2m)$. For multifurcating nodes, the bracketed factor is simply repeated for every child. The recursion is the same as that of Sankoff parsimony (equation (B.7)), with sums of products in place of minima of sums.

Two practical refinements matter. Characters with identical patterns of states across the languages have identical likelihoods, so programs compute each distinct *site pattern* once and weight it by its count. And conditional likelihoods become tiny for large trees and can underflow the floating-point range, so programs rescale them at intervals and keep track of the logarithms of the scaling factors.

5.4.2 A worked example by hand

Consider four languages, English, German, Italian and Spanish, a cognate set present in English and German and absent in Italian and Spanish (the pattern 1100), and the unrooted tree with branch lengths English 0.1, German 0.2, Italian 0.1, Spanish 0.3, and internal branch 0.2. We use

the symmetric binary model ($\pi_0 = \pi_1 = 1/2$), for which $P_{\text{same}}(t) = \frac{1}{2}(1 + e^{-2t})$ and $P_{\text{diff}}(t) = \frac{1}{2}(1 - e^{-2t})$. By the pulley principle we can root the tree at the node u that joins English and German; call the other internal node v.

Step 1 (node v). Both Italian and Spanish are in state 0. So $L_v(0) = P_{\text{same}}(0.1) \times P_{\text{same}}(0.3)$ and $L_v(1) = P_{\text{diff}}(0.1) \times P_{\text{diff}}(0.3)$.

Step 2 (node u). Node u has three neighbors: English and German (both in state 1) and v, reached along the internal branch of length 0.2. For $a \in \{0, 1\}$,

$$L_u(a) = P_{a1}(0.1)\, P_{a1}(0.2)\, [P_{a0}(0.2) L_v(0) + P_{a1}(0.2) L_v(1)]\,.$$

Step 3 (root). $L = \frac{1}{2}L_u(0) + \frac{1}{2}L_u(1)$.

The following code carries out exactly these steps and shows the intermediate quantities, then checks the result against **phangorn**'s `pml()`, an independent implementation.

```
Ps <- function(t) 0.5 * (1 + exp(-2 * t))      # same state
Pd <- function(t) 0.5 * (1 - exp(-2 * t))      # different state
Lv <- c("0" = Ps(0.1) * Ps(0.3), "1" = Pd(0.1) * Pd(0.3))
Lu <- c("0" = Pd(0.1) * Pd(0.2) * (Ps(0.2) * Lv["0"] + Pd(0.2) * Lv["1"]),
        "1" = Ps(0.1) * Ps(0.2) * (Pd(0.2) * Lv["0"] + Ps(0.2) * Lv["1"]))
L_hand <- sum(0.5 * Lu)
knitr::kable(data.frame(node = c("v", "v", "u", "u"), state = c(0, 1, 0, 1),
                        conditional_likelihood = signif(c(Lv, Lu), 4)),
             row.names = FALSE,
             caption = "Conditional likelihoods in the worked pruning example.")
```

Table 5.2: Conditional likelihoods in the worked pruning example.

node	state	conditional_likelihood
v	0	0.7042
v	1	0.0204
u	0	0.0088
u	1	0.1011

```
tr4 <- read.tree(
  text = "((English:0.1,German:0.2):0.2,Italian:0.1,Spanish:0.3);")
x4 <- phyDat(matrix(c("1", "1", "0", "0"), ncol = 1,
                    dimnames = list(c("English", "German", "Italian",
                                      "Spanish"), NULL)),
             type = "USER", levels = c("0", "1"))
c(by_hand = L_hand, phangorn = exp(pml(tr4, x4)$logLik))
```

```
#>  by_hand phangorn
#>  0.05498  0.05498
```

The likelihood of this character is 0.055, and **phangorn** agrees. Notice where it comes from: $L_u(1)$ is more than ten times larger than $L_u(0)$, because English and German are both in state 1

and are joined to u by short branches. The algorithm has, in effect, reconstructed that the ancestor of English and German most probably had the cognate, while still averaging over the alternative. The same quantities, normalized, give the marginal posterior probabilities of ancestral states used in ancestral state reconstruction (Chapter 12).

A likelihood is a probability, and a probability can be approximated by a long-run frequency. If the character were to evolve on this tree, with these branch lengths and this model, again and again, the proportion of replicates ending in the pattern 1100 should approach the value just computed. **phangorn**'s `simSeq()` simulates characters on a tree under the same model, so a few hundred thousand simulated characters give an approximation whose Monte Carlo standard error can be stated.

```
set.seed(11)
n_sim <- 2e5
sim4 <- simSeq(tr4, l = n_sim, type = "USER", levels = c("0", "1"))
sim_mat <- as.character(sim4)[c("English", "German", "Italian",
                                "Spanish"), ]
p_hat <- mean(colSums(sim_mat == c("1", "1", "0", "0")) == 4)
c(by_hand = L_hand, simulated = p_hat,
  mc_se = sqrt(p_hat * (1 - p_hat) / n_sim))
```

```
#>   by_hand simulated     mc_se
#> 0.0549836 0.0556450 0.0005126
```

The simulated frequency agrees with the pruning result to within its Monte Carlo error (here the two differ by a little more than one standard error). Such a check costs nothing, uses code that shares nothing with the likelihood calculation except the model, and is an instance of what Section 6.10 calls *implementation validation*. It is hopeless as a way of computing likelihoods for real data, since the probability of any particular matrix of hundreds of characters is astronomically small, but it makes vivid what the pruning algorithm computes: how often the model would produce exactly the data we observed.

A general implementation of the pruning algorithm in R, which handles any tree, all characters at once and missing data, together with its verification against **phangorn** and a numerical check of the pulley principle, is in Section B.3.2.

5.5 Estimating branch lengths and model parameters

For a fixed topology, the ML branch lengths and model parameters are found by numerical optimization. Programs usually optimize one branch at a time, holding the others fixed, and cycle through the branches until the likelihood stops improving. For a single branch, the log-likelihood is a smooth function of its length, and its first and second derivatives can be computed cheaply, so Newton–Raphson iteration converges in a few steps. Model parameters such as π_1 or the gamma shape are optimized with one-dimensional methods such as Brent's method.

To see this at work we need data for which the truth is known. The function below simulates binary characters on a tree: it draws the root state from π and then evolves each character down the tree using the transition probabilities (5.5), optionally with a different rate multiplier for each character. We simulate 600 characters for eight invented languages with $\pi_1 = 0.3$.

```
sim_binary <- function(tree, nchar, pi1, rates = rep(1, nchar)) {
  tree <- reorder(tree, "cladewise")              # parents before children
  ntip <- Ntip(tree); pi0 <- 1 - pi1
  S <- matrix(NA_integer_, ntip + tree$Nnode, nchar)
  S[ntip + 1, ] <- rbinom(nchar, 1, pi1)
  for (e in seq_len(nrow(tree$edge))) {
    p <- tree$edge[e, 1]; ch <- tree$edge[e, 2]
    ex <- exp(-tree$edge.length[e] * rates / (2 * pi0 * pi1))
    prob1 <- ifelse(S[p, ] == 1, pi1 + pi0 * ex, pi1 * (1 - ex))
    S[ch, ] <- rbinom(nchar, 1, prob1)
  }
  out <- S[seq_len(ntip), , drop = FALSE]
  rownames(out) <- tree$tip.label
  out
}
set.seed(5)
langs8 <- c("Ana", "Bel", "Cor", "Dun", "Esk", "Fal", "Gor", "Hal")
tr8 <- rtree(8, rooted = FALSE, tip.label = langs8)
tr8$edge.length <- tr8$edge.length * 0.25
X8 <- sim_binary(tr8, 600, pi1 = 0.3)
pd8 <- phyDat(X8, type = "USER", levels = c(0, 1))
```

Starting from the true topology with every branch set to 0.1, `optim.pml()` optimizes all 13 branch lengths.

```
start8 <- tr8; start8$edge.length <- rep(0.1, Nedge(start8))
fit8 <- optim.pml(pml(start8, pd8, bf = c(0.7, 0.3)), optEdge = TRUE,
                  control = pml.control(trace = 0))
c(logLik_start = pml(start8, pd8, bf = c(0.7, 0.3))$logLik,
  logLik_fitted = fit8$logLik,
  tree_length_fitted = sum(fit8$tree$edge.length),
  tree_length_true = sum(tr8$edge.length))
```

```
#>       logLik_start      logLik_fitted tree_length_fitted
#>          -2399.873          -2334.514              1.590
#>   tree_length_true
#>              1.551
```

The estimated tree length is close to the true value. This is a *parameter-recovery* experiment in the sense of Section 6.10: the data were simulated under exactly the model that is fitted, so its success shows that branch lengths are estimable from 600 characters of this kind, not that the binary model describes real cognate data.

Newton–Raphson optimization of a single branch, a profile of the log-likelihood for one branch (Figure B.1), and a hand-coded optimizer that reaches the same maximum as `optim.pml()` are in Section B.3.3.

5.6 Ascertainment: why cognate data need a correction

The simulated matrix above contains a quiet impossibility. It includes characters that are absent (state 0) in all eight languages. A real cognate matrix can never contain such a column, because a cognate set is only defined by the words that belong to it: if no language in the sample has a word from the set, nobody will ever code it. The data are *ascertained*: we only observe characters that pass a filter, here "present in at least one language". Fitting a model that assumes all patterns are observable to data from which one pattern has been systematically removed makes the data look more variable than they are, and the branch lengths are overestimated.

The fix is to condition the likelihood on the filter. If $L_\varnothing$ is the probability of the all-absent pattern under the current tree and parameters, the probability of a character *given that it is observable* is $L_i/(1 - L_\varnothing)$, and the corrected log-likelihood is

$$\ell_{\text{asc}} = \sum_{i=1}^{m} \ln L_i - m \ln\left(1 - L_\varnothing\right). \tag{5.10}$$

This correction was introduced for restriction sites (Felsenstein, 1992) and, in the form that conditions on characters being *variable*, for morphological data (Lewis, 2001), where it is known as the Mkv model. Corrections of one kind or the other are standard in linguistic analyses (as `ascertained` options in BEAST 2 and BEASTling, `coding=noabsencesites` in MrBayes, `+ASC` in IQ-TREE and RAxML-NG). The two filters differ. Equation (5.10) removes only the all-absent pattern, which matches a cognate matrix that keeps sets shared by every language (such as 'two'); the Mkv correction removes both constant patterns, which matches a matrix from which invariant characters were deleted. A correction must match how the data were actually filtered.

phangorn computes the Mkv-corrected log-likelihood when `pml()` is called with `ASC = TRUE`. To demonstrate the correction we delete all constant characters from the simulated matrix, as a coder who kept only variable cognate sets would, and fit the branch lengths with and without it. In our tests with **phangorn** 2.12.1, `optim.pml()` did not reliably optimize branch lengths under `ASC = TRUE` (it stopped well short of the maximum of the corrected likelihood), so the short function below maximizes the log-likelihood returned by `pml()` directly with R's general-purpose optimizer `optim()`, working on log branch lengths to keep them positive. Checking a program's output against an independent route to the same quantity, as here, is worth doing whenever an option is used for the first time.

```
fit_lengths <- function(tree, data, ASC) {
  negll <- function(log_bl) {
    tree$edge.length <- exp(log_bl)
    -pml(tree, data, bf = c(0.7, 0.3), ASC = ASC)$logLik
  }
  o <- optim(log(tree$edge.length), negll, method = "L-BFGS-B",
             lower = -12, upper = 1.5)
  tree$edge.length <- exp(o$par)
  list(tree = tree, logLik = -o$value)
}
variable <- apply(X8, 2, function(z) length(unique(z)) > 1)
pd8_var <- phyDat(X8[, variable], type = "USER", levels = c(0, 1))
fit_unc <- fit_lengths(start8, pd8_var, ASC = FALSE)
```

```
fit_asc <- fit_lengths(start8, pd8_var, ASC = TRUE)
c(characters_simulated = ncol(X8), characters_kept = sum(variable),
  true = sum(tr8$edge.length),
  uncorrected = sum(fit_unc$tree$edge.length),
  corrected = sum(fit_asc$tree$edge.length))
```

```
#> characters_simulated      characters_kept                 true
#>              600.000             444.000                1.551
#>          uncorrected           corrected
#>                2.313               1.374
```

Without the correction the tree length is inflated substantially; with it, the estimate returns to the neighbourhood of the truth. (In ten further simulated datasets of the same kind, the corrected estimates ranged from about 1.4 to 1.8 around a true value of 1.55, and the uncorrected ones from 2.3 to 2.9.) In the vocabulary of Section 6.10, this is a small *robustness* experiment: the uncorrected analysis is misspecified in one known respect (it ignores how the data were filtered), and we see the size of the resulting bias. The size depends on how many characters are unobservable, which in turn depends on the tree and the frequency of presence. Ascertainment is discussed in more depth, including corrections for singletons and for the meaning-slot structure of lexical data, in Chapter 7.

A hand-coded version of the all-absent correction (5.10), applied to data from which only the all-absent characters have been removed, is in Section B.3.4.

Forgetting the ascertainment correction is one of the most common errors in analyses of cognate data with general-purpose phylogenetic software, which by default assumes that every pattern, including all-absent, could have been observed. The error inflates branch lengths and, in dated analyses, distorts divergence times. Check that the correction matches how the data were collected: if cognate sets present in only one language (singletons) were also excluded, the correction must condition on that too.

5.7 Searching tree space

5.7.1 The problem

For a fixed topology, likelihood optimization is routine. Finding the best topology is not. The number of unrooted binary trees for n languages is $(2n-5)!! = 3 \times 5 \times \cdots \times (2n-5)$ (Chapter 2), over two million for 10 languages and about 2×10^{20} for 20, and finding the ML tree is NP-hard (Chor and Tuller, 2005). All practical ML programs therefore use heuristic search: start from a reasonable tree and improve it by local rearrangements until no rearrangement improves the likelihood.

5.7.2 Starting trees

Good starting trees save time and reduce the risk of ending in a poor local optimum. Common choices are an NJ or BIONJ tree (PhyML), a parsimony tree built by randomized stepwise addition (RAxML, IQ-TREE), or several of each. Starting from many different trees and keeping the best result is the simplest protection against local optima.

5.7.3 Rearrangements: NNI, SPR and TBR

Three families of rearrangements define the "neighbors" of a tree (Figure 5.3):

- *Nearest-neighbor interchange* (NNI). Each internal branch separates the tree into four subtrees A, B, C, D, arranged as $AB \mid CD$. An NNI swaps one subtree from each side, producing $AC \mid BD$ or $AD \mid BC$. Each internal branch gives two neighbors, so an unrooted binary tree with n leaves has $2(n-3)$ NNI neighbors.
- *Subtree pruning and regrafting* (SPR). Cut a branch, detach the subtree below it, and reattach it on any branch of the remaining tree. The number of SPR neighbors grows quadratically in n.
- *Tree bisection and reconnection* (TBR). Cut a branch, splitting the tree in two, and reconnect the two parts by joining any branch of one to any branch of the other. TBR neighborhoods grow roughly cubically in n and contain all SPR and NNI neighbors.

Larger neighborhoods make each step more expensive but allow bigger jumps, and so escape more local optima. Modern programs combine them cleverly. PhyML improves trees with simultaneous NNIs and SPR moves (Guindon and Gascuel, 2003; Guindon et al., 2010); RAxML uses "lazy" SPR moves in which only the branches near the regrafting point are re-optimized (Stamatakis, 2014; Kozlov et al., 2019); and IQ-TREE alternates hill climbing by NNI with random perturbations of the current best tree, keeping a pool of candidate trees (Nguyen et al., 2015). The genetic algorithm in GARLI evolves a population of trees (Zwickl, 2006).

```
par(mfrow = c(1, 3), mar = c(1.5, 1, 3, 1), xpd = NA)  # labels use margins
for (s in c("((A,B),(C,D));", "((A,C),(B,D));", "((A,D),(B,C));")) {
  nni4 <- unroot(read.tree(text = s))
  plot(nni4, type = "unrooted", edge.width = 2, show.tip.label = FALSE,
       main = gsub(";", "", s))
  # labels set just beyond each tip, along the direction of its branch
  lp <- get("last_plot.phylo", envir = .PlotPhyloEnv)
  par_n <- nni4$edge[match(1:4, nni4$edge[, 2]), 1]
  dx <- lp$xx[1:4] - lp$xx[par_n]; dy <- lp$yy[1:4] - lp$yy[par_n]
  len <- sqrt(dx^2 + dy^2)
  text(lp$xx[1:4] + 0.12 * dx / len, lp$yy[1:4] + 0.12 * dy / len,
       nni4$tip.label, cex = 1.2, font = 3)
}
```

5.7.4 Tree search in practice

We now simulate a larger cognate dataset: 14 languages on a clocklike tree, 600 characters with gamma-distributed rates (shape 0.5), and $\pi_1 = 0.3$. For simplicity we keep all simulated characters, including those that would be unobservable, so that **phangorn**'s standard likelihood is appropriate (Section 5.6 explains what changes otherwise). We then search for the ML tree with NNI rearrangements, starting from an NJ tree and from five random trees.

```
set.seed(42)
true14 <- rcoal(14, tip.label = paste0("L", sprintf("%02d", 1:14)))
true14$edge.length <- true14$edge.length * 0.3
site_rates <- rgamma(600, shape = 0.5, rate = 0.5)
X14 <- sim_binary(true14, 600, pi1 = 0.3, rates = site_rates)
```

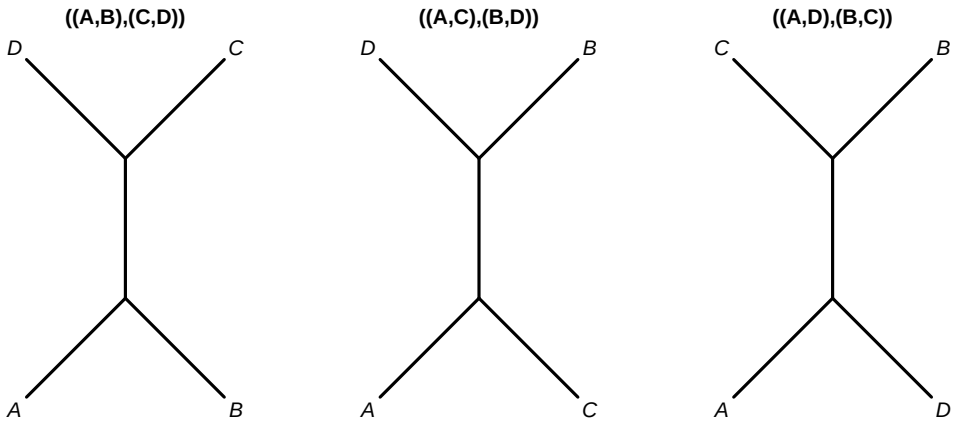


Figure 5.3: The three unrooted topologies that can be formed around one internal branch from four subtrees A to D. An NNI move on the internal branch of any one of them produces the other two.

```
cog14 <- phyDat(X14, type = "USER", levels = c(0, 1))

nj_start <- nj(dist.hamming(cog14))
fit_nj_start <- optim.pml(pml(nj_start, cog14, k = 4), optNni = TRUE,
                          optBf = TRUE, optGamma = TRUE,
                          control = pml.control(trace = 0))
random_starts <- lapply(1:5, function(i) {
  st <- rtree(14, rooted = FALSE, tip.label = names(cog14))
  f <- optim.pml(pml(st, cog14, k = 4), optNni = TRUE, optBf = TRUE,
                 optGamma = TRUE, control = pml.control(trace = 0))
  list(start = st, fit = f)
})
search_table <- data.frame(
  start = c("NJ tree", paste("random tree", 1:5)),
  logLik = c(fit_nj_start$logLik,
             sapply(random_starts, function(r) r$fit$logLik)),
  RF_to_true = c(RF.dist(fit_nj_start$tree, true14),
                 sapply(random_starts,
                        function(r) RF.dist(r$fit$tree, true14))))
knitr::kable(search_table, digits = 1,
  caption = paste("Results of NNI hill climbing from an NJ starting tree",
                  "and from five random starting trees."))
```

Table 5.3: Results of NNI hill climbing from an NJ starting tree and from five random starting trees.

start	logLik	RF_to_true
NJ tree	-1895	0
random tree 1	-2186	16
random tree 2	-2336	18

start	logLik	RF_to_true
random tree 3	-2222	14
random tree 4	-2134	12
random tree 5	-2347	12

The contrast is stark. NNI hill climbing from the NJ tree reaches a high likelihood, whereas every search from a random tree stalls on a local optimum hundreds of log-likelihood units worse, with a quite different topology. Running the NNI search for more iterations does not help: these are genuine local optima of the NNI neighborhood, trees none of whose NNI neighbors is better. A stochastic search that perturbs the tree and restarts hill climbing, in the spirit of IQ-TREE, escapes from the worst of these starting points and ends close to the NJ-start result (Section B.3.5). This is why ML programs build careful starting trees, use larger rearrangements such as SPR, perturb the search, and run several independent searches. Notice also that the best ML tree need not coincide exactly with the true tree; with finite data, short internal branches may be resolved wrongly. The experiment tests the search algorithm, not the model: the data were simulated under the model being fitted.

The stochastic search from the worst random start, with `optim.pml(..., rearrangement = "stochastic")`, is run and compared with the NNI results in Section B.3.5.

5.8 Measuring support

5.8.1 The bootstrap and its fast approximations

The nonparametric bootstrap (Felsenstein, 1985a), introduced in Chapter 4, applies directly to ML: resample characters with replacement, rerun the ML search on each pseudo-replicate, and report the proportion of replicate trees containing each clade. It is expensive, since it multiplies the cost of an ML search by the number of replicates. Several shortcuts are in common use:

- *RELL* (resampling estimated log-likelihoods) avoids re-optimization by resampling the per-character log-likelihoods of a fixed set of candidate trees (Kishino et al., 1990).
- The *ultrafast bootstrap* (UFBoot) of IQ-TREE combines RELL with the candidate trees encountered during the search (Minh et al., 2013; Hoang et al., 2018). Its values are less conservative than standard bootstrap values; its authors suggest treating support of about 95 percent or more as strong.
- The *approximate likelihood-ratio test* (aLRT) for a branch compares the likelihood of the ML tree with that of the best NNI alternative around that branch (Anisimova and Gascuel, 2006). The SH-like variant (SH-aLRT) uses a resampling correction and is widely used in PhyML and IQ-TREE (Guindon et al., 2010). A survey of branch support methods by Anisimova et al. (2011) found these fast likelihood-based measures accurate and powerful.
- The *transfer bootstrap expectation* replaces the all-or-nothing presence of a clade in a replicate by a graded measure of how close the replicate comes to containing it, which is more informative for large trees with a few unstable "rogue" taxa (Lemoine et al., 2018).

```
set.seed(1)
bs14 <- bootstrap.pml(fit_nj_start, bs = 30, optNni = TRUE,
                      control = pml.control(trace = 0))
# bootstrap support as computed by plotBS(), written on each stem branch
```

```
# on the side away from the parent's vertical line
ml_mid <- midpoint(fit_nj_start$tree)
bs_sup <- plotBS(ml_mid, bs14, type = "none")$node.label
plot(ml_mid, cex = 0.8,
     label.offset = 0.01 * max(node.depth.edgelength(ml_mid)))
lp <- get("last_plot.phylo", envir = .PlotPhyloEnv)
nn <- Ntip(ml_mid) + seq_len(Nnode(ml_mid))
pa <- ml_mid$edge[match(nn, ml_mid$edge[, 2]), 1]
for (i in which(!is.na(pa) & bs_sup > 0)) {  # root unlabelled, as plotBS()
  a <- if (lp$yy[nn[i]] >= lp$yy[pa[i]]) c(1.1, -0.35) else c(1.1, 1.35)
  nodelabels(round(100 * bs_sup[i]), nn[i], frame = "none", adj = a,
             cex = 0.75, col = "darkred")              # percent
}
```

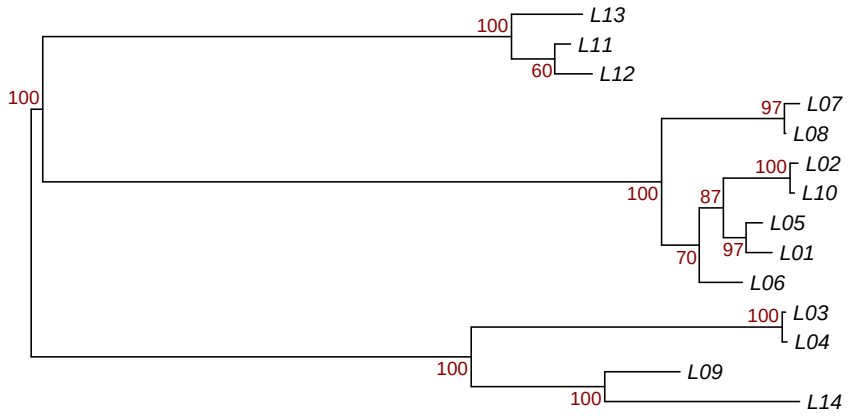


Figure 5.4: ML tree for the simulated 14-language dataset (binary model with estimated frequencies and gamma rate variation), rooted at the midpoint, with standard bootstrap support (percent) from 30 replicates. Real analyses use hundreds or thousands of replicates.

Bootstrap values measure how repeatable a clade is when characters are resampled as if they were independent. For cognate data, whose binary characters within a meaning slot are not independent, the caveats of Section 4.12 apply here too.

5.8.2 Topology tests

Sometimes the question is not which clades are supported but whether a particular alternative tree, for instance a subgrouping proposed by specialists, can be rejected. Likelihood offers formal tests. The *Kishino–Hasegawa* (KH) test compares two trees chosen in advance, using the variance

of per-character log-likelihood differences (Kishino and Hasegawa, 1989). When one of the trees is the ML tree, chosen after looking at the data, the KH test is biased, and the *Shimodaira–Hasegawa* (SH) test corrects for this by comparing a set of candidate trees simultaneously (Shimodaira and Hasegawa, 1999). The SH test is conservative; the *approximately unbiased* (AU) test, based on multiscale bootstrap resampling, is the most widely recommended (Shimodaira, 2002). As an illustration, we compare the ML tree with two alternatives: one that differs from it by a single NNI, and one that differs by several random NNIs. Each alternative gets its own optimized branch lengths.

```
set.seed(7)
alt_near <- rNNI(fit_nj_start$tree, moves = 1)
alt_far  <- rNNI(fit_nj_start$tree, moves = 4)
fit_near <- optim.pml(update(fit_nj_start, tree = alt_near), optEdge = TRUE,
                      control = pml.control(trace = 0))
fit_far  <- optim.pml(update(fit_nj_start, tree = alt_far), optEdge = TRUE,
                      control = pml.control(trace = 0))
c(RF_near = RF.dist(alt_near, fit_nj_start$tree),
  RF_far = RF.dist(alt_far, fit_nj_start$tree))
```

```
#> RF_near  RF_far
#>       2       8
```

```
sh <- SH.test(fit_nj_start, fit_near, fit_far, B = 500)
sh
```

```
#>      Trees  ln L Diff ln L p-value
#> [1,]     1 -1895    0.0000   0.758
#> [2,]     2 -1896    0.4769   0.634
#> [3,]     3 -1986   90.5287   0.000
```

The tree that differs by several rearrangements is decisively rejected. The single-NNI alternative is much closer in likelihood, and its p-value (0.63 in this run) shows how weakly the data can distinguish trees that differ only around one short branch. Rejection of a specialist's proposal by such a test is evidence against it under the model; failure to reject means only that the data cannot tell the trees apart.

Why not simply use the likelihood-ratio test of Section 5.9, comparing the best tree that contains a clade with the best tree overall? Because a topology is not a continuous parameter, and the constrained and unconstrained trees are not nested models in the sense required by the χ^2 approximation. Huelsenbeck et al. (1996) therefore obtained the null distribution of this likelihood ratio by parametric bootstrap, simulating data on the best constrained tree. The same caution applies to a tempting shortcut: comparing a fit with optimized branch lengths on a fixed tree against a fit in which the topology was also rearranged, and reading the difference against a χ^2 distribution with one or two degrees of freedom. The topology search adds no countable parameters, so the degrees of freedom of such a comparison are undefined.

Topology tests are underused in linguistics. Rather than reporting only the best tree and its bootstrap values, one can ask directly whether a traditional subgrouping (say, Italo-Celtic, or a particular internal structure for a Bantu zone) is significantly worse than the ML tree, by constraining the search to trees containing the traditional clade and comparing it with the unconstrained ML tree

with an AU test. IQ-TREE makes this a routine operation. The test inherits the model's assumptions, however: if binary characters from the same meaning slot are treated as independent, or borrowed cognates are coded as inherited, the per-character log-likelihood differences on which it rests are not independent observations, and its p-values will be too small.

5.9 Model selection

5.9.1 Likelihood-ratio tests

Models can be compared by their maximized likelihoods. If model M_0 is a special case of M_1 (it is *nested*: obtained by fixing some of M_1's parameters), then under M_0 the statistic

$$\Lambda = 2\left[\ell(\hat{\theta}_1) - \ell(\hat{\theta}_0)\right] \tag{5.11}$$

is approximately χ^2-distributed with degrees of freedom equal to the difference in the number of free parameters. The binary model with $\pi_1 = 1/2$ is nested in the model with free π_1, which is nested in the model with gamma rate variation. A complication arises when the restricted value lies at the boundary of the parameter space, as when testing for a proportion of invariable characters (restricted value zero); the null distribution is then a mixture of χ^2 distributions (Self and Liang, 1987; Goldman, 1993). Goldman also proposed the parametric bootstrap as a general way to obtain the null distribution of Λ by simulation.

5.9.2 Information criteria

Non-nested models are compared with *information criteria*, which penalize the maximized log-likelihood for the number of free parameters p:

$$\mathrm{AIC} = -2\ell + 2p, \qquad \mathrm{AIC}_c = \mathrm{AIC} + \frac{2p(p+1)}{m - p - 1}, \qquad \mathrm{BIC} = -2\ell + p \ln m, \tag{5.12}$$

where m is the number of characters. AIC (Akaike, 1974) estimates the expected predictive performance of a model; its small-sample correction AIC_c (Hurvich and Tsai, 1989) matters when m is not much larger than p; BIC (Schwarz, 1978) approximates the log marginal likelihood (Chapter 6) and penalizes extra parameters more heavily when data are plentiful. Lower values are better. Note that branch lengths count as parameters, so p includes $2n - 3$ of them for an unrooted tree; this is the same for all models on the same tree and cancels in comparisons. Only differences in AIC matter. As a rough guide, Harmon (2019) treats models within about 4 units of the best as roughly equivalent and those more than 10 units worse as negligible. Differences Δ_i from the best model can also be converted into *Akaike weights*, $w_i \propto \exp(-\Delta_i/2)$, which sum to one over the candidate set and can be used to average a parameter estimate across models instead of conditioning on a single winner (Burnham and Anderson, 2002). Model selection for DNA was automated by ModelTest (Posada and Crandall, 1998), jModelTest (Darriba et al., 2012) and ModelFinder (Kalyaanamoorthy et al., 2017), the last built into IQ-TREE.

```
ctl <- pml.control(trace = 0)
fits <- list()
fits$Binary_equal <- optim.pml(pml(fit_nj_start$tree, cog14),
                               optEdge = TRUE, control = ctl)
```

```
fits$Binary_F <- optim.pml(fits$Binary_equal, optEdge = TRUE, optBf = TRUE,
                           control = ctl)
fits$Binary_F_G4 <- optim.pml(update(fits$Binary_F, k = 4), optEdge = TRUE,
                              optBf = TRUE, optGamma = TRUE, control = ctl)
fits$Binary_F_G4_I <- optim.pml(update(fits$Binary_F_G4, inv = 0.1),
                                optEdge = TRUE, optBf = TRUE,
                                optGamma = TRUE, optInv = TRUE,
                                control = ctl)
model_table <- data.frame(
  model  = names(fits),
  logLik = sapply(fits, logLik),
  p      = sapply(fits, function(f) attr(logLik(f), "df")),
  AIC    = sapply(fits, AIC),
  BIC    = sapply(fits, BIC))
knitr::kable(model_table, digits = 1, row.names = FALSE,
  caption = paste("Comparison of binary models on the simulated",
                  "14-language dataset (fixed topology). p is the number",
                  "of free parameters, including branch lengths. The data",
                  "were simulated with unequal frequencies and gamma rate",
                  "variation."))
```

Table 5.4: Comparison of binary models on the simulated 14-language dataset (fixed topology). p is the number of free parameters, including branch lengths. The data were simulated with unequal frequencies and gamma rate variation.

model	logLik	p	AIC	BIC
Binary_equal	-1984	25	4018	4128
Binary_F	-1918	26	3888	4003
Binary_F_G4	-1895	27	3844	3963
Binary_F_G4_I	-1895	28	3846	3969

```
anova(fits$Binary_F, fits$Binary_F_G4)           # likelihood-ratio test
```

```
#> Likelihood Ratio Test Table
#>   Log lik. Df Df change Diff log lik. Pr(>|Chi|)
#> 1    -1918 26
#> 2    -1895 27         1          46.3      1e-11 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Both criteria favor the models with unequal frequencies and gamma rate variation, which is how the data were simulated, and the likelihood-ratio test strongly rejects the model without rate variation. Adding invariable characters gains little. That the criteria pick out the generating model is again a parameter-recovery result, obtained on data that contain none of the complications of real cognate data. A model chosen this way is the best of the candidates, not necessarily a good model; tests of absolute model adequacy, such as posterior predictive checks, are discussed in Chapters 6 and 15.

5.9.3 Absolute fit and the parametric bootstrap

Comparing candidate models does not tell us whether any of them describes the data well. An old and simple check of absolute fit compares the maximized likelihood of the phylogenetic model with that of an unconstrained *multinomial* model, which treats each of the possible site patterns as having its own probability, estimated by its observed frequency (Goldman, 1993). The multinomial model fits the pattern frequencies perfectly, so the difference in log-likelihood measures how far the phylogenetic model falls short. Because the usual χ^2 approximation is unreliable here (many patterns are rare or unobserved), the null distribution of the difference is obtained by *parametric bootstrap*: simulate many datasets from the fitted phylogenetic model, refit both models to each, and see whether the observed difference is typical of the simulated ones. The same simulation logic can test any feature of the data that the model should reproduce, such as the number of cognate sets shared by exactly two languages, or the distribution of cognate-set sizes. Applied to real data, such checks are what Section 6.10 calls *empirical model adequacy*. Their Bayesian counterpart, posterior predictive simulation, is demonstrated in Chapter 6.

5.10 Statistical properties of maximum likelihood trees

Chapter 4 showed that parsimony can converge on the wrong tree as data accumulate. Does ML do better? Under the model that generated the data, yes. For the standard Markov models, the tree topology and branch lengths are *identifiable*: different trees (with positive internal branch lengths) give different distributions of site patterns, so with enough characters the data determine the tree (Chang, 1996). Combined with the general consistency of maximum likelihood, this means that ML converges on the true tree as the number of characters grows, which is consistent with what the long-branch simulation of Chapter 4 found. Identifiability is a statement about unlimited data; whether a particular split is *practically estimable* from a few hundred cognate sets is a separate question, answered by support values and by simulation.

The guarantee is only as good as the model, and here lies the practical difficulty. If the model is misspecified in a way that matters, ML can be inconsistent too. A classic example is fitting a model without rate variation to data with strong rate variation: the fitted model underestimates how often long branches converge on the same state, attributes the convergence to shared ancestry, and can be drawn into long-branch attraction much like parsimony. Ignoring ascertainment (Section 5.6), ignoring borrowing, and treating dependent characters as independent are all forms of misspecification that real cognate data invite. The modern view in biology is therefore not that ML is right and parsimony wrong, but that model-based inference is only as reliable as the fit of its model, which is why model selection and model checking deserve as much attention as tree search (Steel and Penny, 2000; Felsenstein, 2004).

Two further points matter for interpreting ML trees of languages. First, the ML tree is a *point estimate*: it is the single best tree, and it says nothing by itself about how much better it is than its competitors, which is why support values and topology tests (Section 5.8) are indispensable. A downstream analysis (of homelands, ancestral states or correlated features) that conditions on the ML tree alone ignores this uncertainty; repeating the analysis over bootstrap trees or a Bayesian posterior sample is the remedy (Chapters 6 and 13). Second, under reversible models the ML tree is unrooted and its branch lengths are in expected changes per character, not in years. Converting it to a tree of time requires a clock model and calibrations, which can be done by penalized likelihood (Sanderson, 2002) or, more commonly in linguistics, by Bayesian relaxed-clock methods (Chapters

9 and 10).

5.11 Software

Table 5.5 lists the main ML programs and their support for linguistic data. All accept binary data, and most accept multistate characters under Mk-type models.

Table 5.5: Maximum likelihood software that handles linguistic data.

Program	Binary and multistate models	Search	Support	Notes
IQ-TREE 2 (Minh et al., 2020b)	JC2, GTR2, Mk (MORPH), +ASC, +G, +R, mixtures	NNI with stochastic perturbation	UFBoot, SH-aLRT, standard bootstrap, AU test	ModelFinder; fast and versatile
RAxML-NG (Kozlov et al., 2019)	BIN, MULTI-state, ascertainment corrections	lazy SPR	bootstrap, transfer bootstrap	scales to very large datasets
PhyML 3 (Guindon et al., 2010)	binary and multistate	NNI and SPR	aLRT, SH-aLRT, aBayes, bootstrap	starting trees by BIONJ
PAUP* (Swofford, 2003)	standard discrete models	NNI, SPR, TBR	bootstrap	also parsimony and distance methods
phangorn (R) (Schliep, 2011)	binary, USER-defined, Mk	NNI, stochastic, ratchet	bootstrap, SH test	used throughout this book

A typical IQ-TREE analysis of a binary cognate matrix, with ascertainment correction, gamma rate variation, ultrafast bootstrap and SH-aLRT, would be run as follows (not run here, because IQ-TREE is an external program):

```
# cognates.phy: binary matrix in PHYLIP format, characters 0/1
# (IQ-TREE 3 is called iqtree3; IQ-TREE 2 is called iqtree2)
iqtree3 -s cognates.phy -st BIN -m GTR2+FO+G4+ASC -B 1000 -alrt 1000
# let ModelFinder choose among binary models instead:
iqtree3 -s cognates.phy -st BIN -m MFP+ASC -B 1000
```

and an equivalent RAxML-NG analysis with Lewis's ascertainment correction:

```
raxml-ng --all --msa cognates.phy --model BIN+G+ASC_LEWIS --bs-trees 200
```

We checked these commands with IQ-TREE 3.1.4 and RAxML-NG 2.0.3 on the toy cognate matrix of Chapter 17 (24 languages, 430 variable characters), where each IQ-TREE run took under 15 seconds. ModelFinder chose `GTR2+FO+ASC+R2`. Program options change between versions, so check the documentation of the version you use.

5.12 Maximum likelihood in linguistics

Given its central place in molecular phylogenetics, ML has had a surprisingly small role as the main method of tree inference in linguistics. When model-based methods arrived in the field,

they arrived in Bayesian form: the analysis of Indo-European by Gray and Atkinson (2003) used MrBayes, and most subsequent dating studies used MrBayes, BEAST or TraitLab (Chapters 6 and 10). There are good reasons for this. Dating requires a clock model and calibrations, which are more naturally expressed as prior distributions; linguistic datasets are small enough that MCMC is affordable; and Bayesian methods integrate over tree uncertainty when the trees are used for downstream questions, such as homeland inference or comparative analysis. But the likelihood is the same in both frameworks. Every Bayesian analysis of cognate data computes the pruning algorithm of this chapter millions of times, and the models of Chapter 7 are likelihood models first.

Pagel (2000) was among the first to formulate explicit likelihood models of lexical replacement on language trees. He argued that the fixed retention rate of glottochronology should give way to explicit stochastic models of word replacement fitted by likelihood, in which rates are estimated from the data and allowed to vary, and in which trees and time depths can be estimated together. The chapter appeared in an influential volume on time depth in historical linguistics. The program it outlined was carried forward in the model-based work of the following decade, including the studies of rate variation across meanings discussed in Chapter 9.

Large-scale computational work is where ML-style inference has been most natural in linguistics, because Bayesian MCMC becomes slow for thousands of languages. Jäger (2018) derived binary character matrices from ASJP word lists for a large share of the world's languages, using automatically detected cognate classes and sound-class characters, and inferred phylogenies from them, making the data and trees publicly available. Such trees, often constrained to agree with expert classifications in Glottolog, are used as scaffolds for global comparative analyses when no expert tree with branch lengths exists (Chapter 13). Their quality depends on automated cognate detection, whose consequences for tree inference were assessed by Rama et al. (2018) (Chapter 3). Agreement of such trees with Glottolog is a useful benchmark but not an independent validation, since the classification itself rests on the comparative method and may have been used to constrain the trees (Section 1.9).

Likelihood methods are not restricted to presence or absence of cognates. Bouchard-Côté et al. (2013) developed a probabilistic model of sound change acting on the phonological forms of words along the branches of a fixed tree of Austronesian languages, estimated its parameters by maximizing (an approximation to) the likelihood with a Monte Carlo expectation–maximization algorithm, and used it to reconstruct proto-forms automatically from data on several hundred languages. The reconstructions were evaluated against those made by experts using the comparative method. This is a sequence-level likelihood model of exactly the kind that molecular phylogenetics uses for DNA, applied to words, and it points toward a closer integration of phylogenetics with the comparative method (Chapters 12 and 16).

ML has also been used routinely in linguistics in places where it is less visible: for ancestral state reconstruction on fixed trees (Chapter 12), in phylogenetic comparative methods such as Pagel's test of correlated evolution (Chapter 13), and in method comparisons. As datasets from Lexibank and Grambank grow to thousands of languages, fast ML programs such as IQ-TREE and RAxML-NG are likely to become more important, both as exploratory tools and as a way to obtain starting trees and sanity checks for Bayesian analyses.

Several ML tools that are routine in phylogenomics could be applied to linguistic data with little effort, but to our knowledge have rarely if ever been. *Partition models* and their automatic selection (Lanfear et al., 2012; Kalyaanamoorthy et al., 2017) would let different meaning classes (body

parts, numerals, verbs) or different data types (lexical, structural, phonological) have their own model and rate, chosen by information criteria at ML speed. *Concordance factors* (Minh et al., 2020a) report, for each branch of a tree, the percentage of characters (site concordance factor) or of separately inferred subtrees (gene concordance factor) that support it. Applied to subsets of the vocabulary, they would show how much of the lexicon supports a clade, distinguishing a clade supported weakly by everything from one supported strongly by a few borrowed items. *Heterotachy models* such as GHOST (Crotty et al., 2020) allow different classes of characters to have different branch lengths on the same topology, a natural model for vocabulary strata that changed at different rates in different lineages. And *mixture models* that fit several substitution processes across characters (Pagel and Meade, 2004) are an ML-friendly way to accommodate the heterogeneous processes behind cognate data (Chapter 7).

- The likelihood is the probability of the data as a function of the tree, branch lengths and model parameters; ML chooses the values that maximize it, and corrected binary distances are its two-language special case.
- Character change along branches is modeled by continuous-time Markov chains with rate matrix $\mathbf{Q}$ and transition probabilities $\mathbf{P}(t) = e^{\mathbf{Q}t}$; rate and time are confounded, so branch lengths are measured in expected changes per character.
- The binary model for cognate data has the closed-form $\mathbf{P}(t)$ of equation (5.5), with one free parameter after normalization; gamma-distributed rates accommodate the very different stabilities of meanings.
- Reversible models make the likelihood independent of the root (the pulley principle), so ML trees are unrooted until a clock or other argument places the root.
- Felsenstein's pruning algorithm computes the likelihood in time linear in the number of languages by passing conditional likelihoods from the leaves to the root.
- Cognate data are ascertained (all-absent characters cannot be observed), and the likelihood must be conditioned on the filter actually applied, or branch lengths will be inflated.
- Finding the ML tree is NP-hard; programs use heuristic search with NNI, SPR and TBR moves, good starting trees, perturbations and multiple starts.
- Support is measured by the bootstrap and its fast approximations (UFBoot, SH-aLRT), and alternative topologies can be tested with KH, SH and AU tests, all under the model's independence assumptions.
- Nested models are compared by likelihood-ratio tests, others by AIC, AIC_c or BIC; choosing the best candidate is not the same as showing that it is adequate, and simulations under the fitted model demonstrate parameter recovery, not empirical adequacy.
- In linguistics ML has mostly been overshadowed by Bayesian inference, but it underlies every model-based analysis and is well suited to very large datasets.

5.13 Exercises

Conceptual

1. In the two-language example, the 95 percent likelihood interval for t is asymmetric. Explain why, and say what this implies for the precision of the deepest node ages in a language family compared with the shallowest.
2. Rate and time are confounded in a CTMC. Explain what this means for a tree of English, German and Dutch estimated by ML, and what extra information would be needed to turn its branch lengths into years.

3. A colleague analyses a binary cognate matrix with IQ-TREE using `-m GTR2+G4`, without `+ASC`. Predict the effect on branch lengths and on any dates later derived from them, and explain the mechanism.
4. For each of the three assumptions of Section 5.3, name a linguistic process that violates it and the chapter of this book in which a remedy is discussed.
5. A clade has 70 percent bootstrap support, and an AU test fails to reject a tree that lacks it. Are these results contradictory? What does each say, and what does neither say?
6. The model-selection example recovered the model under which the data were simulated. Why does this not show that the same procedure will choose an adequate model for a real cognate matrix? Use the vocabulary of Section 6.10.

Computational

1. Adapt the code of chunk `likelihood-byhand` (or call `pml()` on one-character `phyDat` objects) to compute the likelihood of each of the 16 possible patterns of a binary character on `tr4`. Generate the patterns with `expand.grid()`, check that the 16 likelihoods sum to one, and explain which patterns are most and least probable.
2. Using `sim_binary()` and `optim.pml()`, simulate 20 datasets on `tr8` for each of 150, 600 and 2,400 characters and record the estimated tree length. Plot the estimates against the number of characters. Which kind of validation does this experiment provide, and which does it not?
3. Repeat the ascertainment experiment of chunk `likelihood-asc-phangorn` with $\pi_1 = 0.1$, 0.3 and 0.5 (change both `sim_binary()` and the `bf` argument inside `fit_lengths()`). How does the bias of the uncorrected analysis change, and why? Then fit the same data with `optim.pml(pml(start8, pd8_var, ASC = TRUE), optEdge = TRUE)` and compare its log-likelihood with that of `fit_lengths()`: what does the comparison tell you about relying on a single program?
4. Rerun the random-start searches of chunk `likelihood-search-data` with `optim.pml(..., rearrangement = "SPR")` and compare the log-likelihoods and Robinson–Foulds distances (`RF.dist()`) with those of the NNI searches.
5. Simulate a 14-language dataset with `sim_binary()` on `true14` with all rates equal (omit the `rates` argument). Fit the four models of chunk `likelihood-models` and compare them with `AIC()`, `BIC()` and `anova()`. Do the criteria still prefer gamma rate variation? Then extract the per-character log-likelihoods of the best model (`fits$Binary_F_G4$siteLik`, one value per distinct site pattern), find the least probable pattern, and describe what it looks like on the tree. What kind of real cognate set would you expect to have such a low likelihood?
6. Optimize the branch lengths of `tr4` for the single character 1100 of chunk `likelihood-byhand` with `optim.pml(pml(tr4, x4), optEdge = TRUE)`. Inspect the estimated branch lengths and the maximized log-likelihood, and explain both (compare $\ln \frac{1}{4}$). Then call `anova()` on the starting and optimized fits and explain why its output cannot be interpreted as a likelihood-ratio test.
7. Rate variation by hand. (a) Approximate the mean rates of the four equal-probability categories of a gamma distribution with shape 0.5 by averaging `qgamma(p, 0.5, 0.5)` over a fine grid of `p` within each quartile, and compare them with `discrete.gamma(0.5, 4)`. (b) For the two-language tree `read.tree(text = "(L1:0.1,L2:0.1);")` and one binary character with states 0 and 1, compute the log-likelihood under the symmetric binary model with these four rates by applying Equation (5.6) to the transition probabilities `Ps()` and `Pd()` of chunk `likelihood-byhand`. (c) Check your answer with `pml(tree, data, k = 4, shape = 0.5)`

on a one-character `phyDat` object. (d) Repeat (b) for shapes between 0.2 and 100, and for a character with the same state in both languages. Explain why a smaller shape parameter lowers the likelihood of the variable character but raises that of the constant one.

Research

1. You have cognate data for 60 Oceanic languages and 210 meanings, coded by a single specialist; singletons were removed and loanwords coded as missing. Design an ML analysis: choose a model and an ascertainment correction (and justify which filter it must condition on), say how support will be measured, and explain how you would test whether a traditional subgroup can be rejected.
2. A handbook proposes an Italo-Celtic subgroup. Plan a constrained-topology test with an AU test in IQ-TREE. What would rejection and non-rejection each mean, and how could the non-independence of binary characters within meaning slots and undetected borrowing between Italic and Celtic affect the validity of the test?
3. For a family in which lexical evidence is eroded, you have 120 binary structural features from Grambank. Argue for or against a partitioned model (for example, phonological versus morphosyntactic features), describe how you would choose among models, and design a parametric-bootstrap check of absolute fit that targets the feature of the data you consider most likely to be misfitted.

5.14 Further reading

- Felsenstein (2004), chapters 16 to 21, is the classic account of likelihood methods, including the pruning algorithm, the pulley principle and the bootstrap.
- Yang (2006) and its successor Yang (2014) give a rigorous statistical treatment of Markov models, likelihood calculation, optimization and model selection, with many worked examples.
- Felsenstein (1981) is short, readable and still worth reading in the original.
- Huelsenbeck and Crandall (1997) review ML estimation of phylogenies and, above all, the formulation of hypotheses as likelihood-ratio tests whose null distributions are obtained by simulation.
- Myung (2003) is a gentle tutorial on maximum likelihood estimation, written for cognitive scientists rather than biologists, for readers who want the statistics without the trees.
- Lewis (2001) introduced the Mk and Mkv models for discrete morphological characters and explains the ascertainment correction clearly; it transfers almost directly to linguistic data.
- Nguyen et al. (2015), Minh et al. (2020b) and Kozlov et al. (2019) describe the search strategies of the two most widely used ML programs.
- Shimodaira (2002) and Anisimova et al. (2011) cover topology tests and branch support measures.
- Schliep (2011) describes the **phangorn** package used for the R examples in this chapter.
- Pagel (2000) is an early statement, by a biologist, of how likelihood models could be applied to linguistic data.
- Cathcart (2018) writes out, for linguists, the two-state transition probabilities, the pruning recursion and the weighting of the root by the stationary distribution, and Goldstein (2020) works through likelihood on toy trees and maximum likelihood trees in **phangorn** with Indo-European data.

Chapter 6

Bayesian Phylogenetic Inference

Since the Indo-European study of Gray and Atkinson (2003), Bayesian inference has been the dominant framework in linguistic phylogenetics. Nearly every dated language tree discussed in this book, and most of the phylogeographic and comparative analyses built on them, comes from a Bayesian program: MrBayes, BEAST and BEAST 2, TraitLab or RevBayes. These programs answer the questions linguists actually ask (how probable is this subgroup, how old is this proto-language) in the form they want, a probability distribution, but they do so by combining the data with priors and by sampling with Markov chain Monte Carlo, and both steps can fail silently. This chapter explains what the programs do, where the prior enters, how to tell whether a sampler has worked, how models are compared, and how to check whether a model is adequate at all. It also introduces the vocabulary of validation and identifiability that the rest of the book uses (Section 6.10). Sections A.2 and A.12 of the mathematical appendix give background on Bayes' theorem and on MCMC.

After reading this chapter you should be able to:

- write down Bayes' theorem for trees, and explain why the marginal likelihood cannot be computed directly but is not needed for sampling;
- explain how topology, branch-length, tree and calibration priors shape a posterior, and check their influence by sampling from the prior;
- describe the Metropolis–Hastings algorithm, tune a proposal, and diagnose convergence and mixing with trace plots, ESS, PSRF and split frequencies;
- summarize a posterior sample of trees with clade probabilities, consensus and MCC trees, and explain why downstream analyses should use the whole sample;
- explain how Bayes factors depend on priors and why the harmonic mean estimator must not be used;
- distinguish implementation validation, parameter recovery, robustness under misspecification, empirical model adequacy and historical interpretation, and classify a given simulation study accordingly;
- design a posterior predictive check for a cognate model.

6.1 Bayes' theorem for trees

6.1.1 From likelihood to posterior

Maximum likelihood (Chapter 5) treats the tree and parameters as fixed unknowns and asks which values make the data most probable. Bayesian inference treats them as random variables and asks how probable each value is *given* the data. The link is Bayes' theorem (Bayes, 1763). Writing τ for the topology, $\mathbf{t}$ for the branch lengths (or node ages) and θ for the remaining model parameters,

$$f(\tau, \mathbf{t}, \theta \mid D) = \frac{f(D \mid \tau, \mathbf{t}, \theta)\, f(\tau, \mathbf{t}, \theta)}{f(D)}. \tag{6.1}$$

The *likelihood* $f(D \mid \tau, \mathbf{t}, \theta)$ is exactly the quantity computed by the pruning algorithm. The *prior* $f(\tau, \mathbf{t}, \theta)$ expresses what is believed about the tree and parameters before seeing the data. The *posterior* $f(\tau, \mathbf{t}, \theta \mid D)$ is the updated belief. The denominator, the *marginal likelihood* or *evidence*,

$$f(D) = \sum_{\tau} \int\!\!\int f(D \mid \tau, \mathbf{t}, \theta)\, f(\tau, \mathbf{t}, \theta)\, d\mathbf{t}\, d\theta, \tag{6.2}$$

is the average probability of the data over everything the prior allows. It is a constant that makes the posterior integrate to one, and it plays a central role in model comparison (Section 6.7).

Bayesian phylogenetic inference was developed independently by several groups in the second half of the 1990s (Rannala and Yang, 1996; Yang and Rannala, 1997; Mau et al., 1999; Larget and Simon, 1999; Li et al., 2000), and it became widely used after the release of MrBayes (Huelsenbeck and Ronquist, 2001; Huelsenbeck et al., 2001).

6.1.2 What the posterior offers

The attraction of the posterior is that it answers the questions linguists actually ask, in the form they want. The posterior probability of a clade (the sum of the posterior probabilities of all trees containing it) is the probability, under the model and prior, that the clade is real: a direct statement of the form "Germanic and Italic form a clade with probability 0.12". A divergence time comes with a credible interval, an interval that contains the true value with stated posterior probability. And, crucially for downstream work, uncertainty in the tree can be carried forward: a homeland analysis (Chapter 11) or a comparative test (Chapter 13) can be repeated over thousands of trees sampled from the posterior, so that conclusions do not depend on a single best tree.

Even with flat priors, the Bayesian and maximum-likelihood answers need not coincide. Maximum likelihood estimates the tree jointly with the substitution and clock parameters, at the single highest point of the likelihood surface. The posterior probability of a tree instead integrates over these *nuisance* parameters, so a tree whose likelihood peak is slightly lower but which fits well across a wide range of parameter values can receive the higher posterior probability: in the phrase of Holder and Lewis (2003), a Bayesian analysis measures the volume under the surface rather than its maximum height. With many characters and few parameters the surface is sharply peaked and the distinction is largely academic. With parameter-rich models and the modest datasets typical of linguistics it can matter.

The price is the prior. Every Bayesian result depends on it, and with the modest amounts of data typical of linguistics, some quantities, especially absolute dates, depend on it heavily. Much of

this chapter is about how to make that dependence visible. The other qualification is in the words "under the model": a posterior probability of 0.99 is a statement about the model as much as about history, which is why Section 6.10 separates the questions of whether the computation is right, whether the model fits, and what its parameters mean.

6.1.3 A posterior over three trees

The simplest phylogenetic use of Bayes' theorem needs no sampling at all. Four languages can be related by only three unrooted topologies, so the sum over topologies in Equation (6.2) has three terms. Take English, German, Italian and Spanish again, six binary cognate characters, the symmetric binary model, and for the moment fix the branch lengths: each language keeps the terminal branch it had in the pruning example of Section 5.4 (English 0.1, German 0.2, Italian 0.1, Spanish 0.3), and the internal branch is 0.2 in all three trees. With a uniform prior of 1/3 on each topology, the posterior is the likelihood times the prior divided by their sum.

```
cog6 <- rbind(English = c(1, 1, 1, 0, 1, 0),
              German  = c(1, 1, 0, 1, 0, 0),
              Italian = c(0, 0, 1, 0, 0, 1),
              Spanish = c(0, 0, 0, 1, 0, 1))
pd6 <- phangorn::phyDat(cog6, type = "USER", levels = c(0, 1))
topos <- c(
  "EG|IS" = "((English:0.1,German:0.2):0.2,Italian:0.1,Spanish:0.3);",
  "EI|GS" = "((English:0.1,Italian:0.1):0.2,German:0.2,Spanish:0.3);",
  "ES|GI" = "((English:0.1,Spanish:0.3):0.2,German:0.2,Italian:0.1);")
loglik3 <- sapply(topos, function(nw)
  phangorn::pml(read.tree(text = nw), pd6)$logLik)
log_joint <- loglik3 + log(1/3)                    # likelihood x prior
log_evidence <- log(sum(exp(log_joint)))           # f(D): a sum of three
data.frame(log_likelihood = loglik3, log_prior = log(1/3),
           posterior = exp(log_joint - log_evidence))
```

```
#>       log_likelihood log_prior posterior
#> EG|IS         -20.26    -1.099   0.86590
#> EI|GS         -22.25    -1.099   0.11851
#> ES|GI         -24.27    -1.099   0.01559
```

Three characters (columns 1, 2 and 6) group English with German and Italian with Spanish, while two (columns 3 and 4) group English with Italian and German with Spanish; the remaining character is a singleton. The posterior nevertheless gives the first topology a probability near 0.87, not 3/5, because the likelihood weighs each pattern by how easily the model can produce it on each tree, given its branch lengths. Three features of this small calculation carry over to every Bayesian phylogenetic analysis. The marginal likelihood is a prior-weighted average of the likelihoods of the hypotheses, here 6.13×10^{-10}. Under a uniform prior the posterior odds of two trees equal their likelihood ratio, so the prior on topologies matters only when it is not uniform, or through priors on other parameters that differ between trees. And the posterior probability of a clade is the sum of the posterior probabilities of the trees that contain it; with four languages each clade occurs in just one tree.

The calculation is conditional on the fixed branch lengths, which a real analysis would integrate over as well (Equation (6.2)). And it is possible only because the trees can be listed. With 20

languages there are about 2.2×10^{20} unrooted topologies (Section B.1.1), each with its own branch lengths to integrate over.

6.1.4 A one-parameter example

Before turning to trees, consider the simplest case from Chapter 5: two languages, $N = 100$ binary characters of which $k = 30$ differ, and a symmetric binary model, so that the only parameter is the path length t between the languages. Its likelihood is equation (5.2). Take an exponential prior with mean 1, $f(t) = e^{-t}$. With a single parameter we can compute the posterior on a grid: evaluate likelihood times prior at many values of t and normalize so the result integrates to one.

```
N <- 100; k <- 30
loglik_t <- function(t, N = 100, k = 30) {
  p <- 0.5 * (1 - exp(-2 * t))
  k * log(0.5 * p) + (N - k) * log(0.5 * (1 - p))  # root frequency 1/2
}
grid <- seq(0.001, 2, length.out = 2000); h <- diff(grid)[1]
prior <- dexp(grid, rate = 1)
lik <- exp(loglik_t(grid) - max(loglik_t(grid)))
post <- prior * lik; post <- post / sum(post * h)
par(mar = c(4, 4, 1, 1))
plot(grid, post, type = "l", lwd = 2, xlab = "path length t",
     ylab = "density", xlim = c(0, 1.5))
lines(grid, prior, lty = 2, lwd = 2)
lines(grid, lik / sum(lik * h), lty = 3, lwd = 2)
legend("topright", c("prior", "likelihood (scaled)", "posterior"),
       lty = c(2, 3, 1), lwd = 2, bty = "n")
```

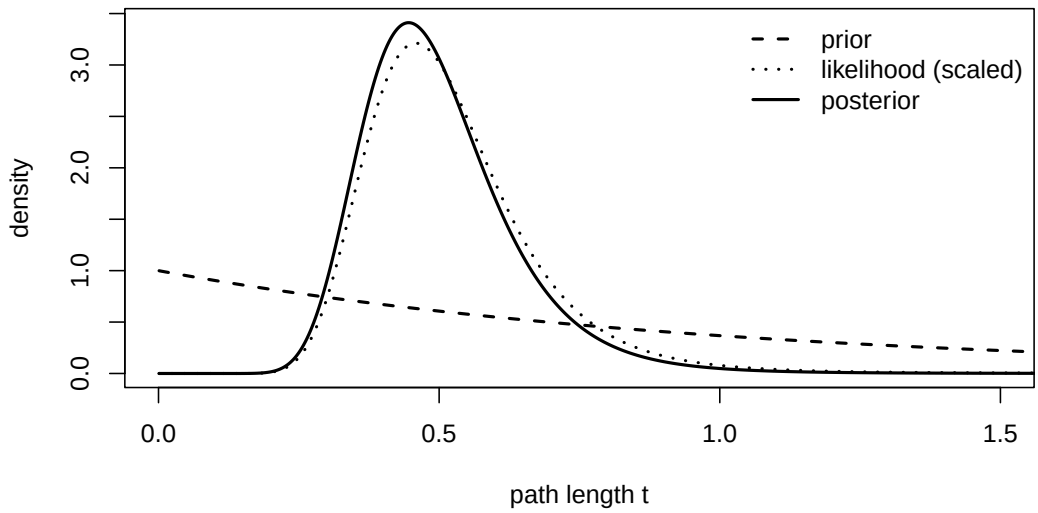


Figure 6.1: Prior (dashed), likelihood (dotted, rescaled) and posterior (solid) for the path length between two languages differing in 30 of 100 binary characters, computed on a grid.

```
c(posterior_mean = sum(grid * post * h), MLE = -0.5 * log(1 - 2 * k / N))
```

```
#> posterior_mean            MLE
#>         0.5013         0.4581
```

Here the data are informative and the posterior is close to the (rescaled) likelihood; the prior pulls the posterior mean only slightly toward smaller values. Grid evaluation works for one parameter and, with effort, for two or three. A tree with 50 languages has 97 branch lengths, a handful of model parameters and about 3×10^{74} possible topologies. No grid can cover that space, and the evidence (6.2) cannot be computed. We need a way to draw samples from the posterior without knowing its normalizing constant. That is what MCMC provides.

6.2 Priors

A Bayesian phylogenetic model has three kinds of prior: on the topology, on branch lengths or node ages, and on the other parameters. In a dated analysis the first two are combined into a single *tree prior*, and the branch lengths in units of change are the product of times and rates, with a separate *clock prior* on rates (Chapter 9).

6.2.1 Priors on topologies and branch lengths for undated trees

For an unrooted, undated tree, the usual topology prior is uniform: all $(2n-5)!!$ labeled topologies are equally probable. Note that a uniform prior on topologies is not uniform on clades: some clades occur in more topologies than others, so a "flat" topology prior implies non-flat prior probabilities for particular groupings. Users can also impose *topological constraints*, giving zero prior probability to trees that violate a grouping established by the comparative method. Constraints are common in linguistic analyses, for example forcing well-established subgroups such as Germanic or Oceanic to be monophyletic. A constrained clade can of course no longer be "recovered" by the analysis, a point worth remembering when results are compared with traditional classifications.

Branch lengths are usually given independent exponential priors. A seemingly harmless choice can be influential. With $2n-3$ branches each given an exponential prior with mean 0.1, the prior on the total tree length is a gamma distribution whose mean grows with the number of languages, and under some conditions this produces posterior tree lengths far too long (Brown et al., 2010; Marshall, 2010). Rannala et al. (2012) analyzed this behavior and, with Zhang et al. (2012), proposed *compound Dirichlet* priors that place a prior on total tree length and divide it among branches with a Dirichlet distribution, which is more robust. MrBayes implements both.

6.2.2 Tree priors for dated trees

When the tree is *time-calibrated*, so that branch lengths are durations, the prior on topology and node ages together comes from a model of how lineages split, die out and are sampled. These *tree priors* were developed for species and pathogens; their linguistic interpretation is that a "birth" is the split of one language into two, a "death" is the extinction of a language without descendants, and "sampling" is the selection of languages into the dataset.

The *Yule process* (Yule, 1925) is the simplest: each lineage splits at rate λ and never goes extinct. Starting with two lineages at the root, while there are j lineages the waiting time to the next split is exponential with rate $j\lambda$. Multiplying these exponential densities over the history of a tree with n leaves shows that the prior density of a particular ranked topology with its node times is $\lambda^{n-2}e^{-\lambda T}$, where T is the total length of all branches. The density depends only on the number of

languages and the total tree length: trees with many long branches are penalized, and the speed of that penalty is set by λ.

The derivation of the Yule density (equation (B.12)) is in Section B.4.1.

The *birth–death process* (Kendall, 1948) adds extinction at rate μ. The tree observed today is the *reconstructed* tree, from which extinct lineages have been pruned, and its probability density was derived by Nee et al. (1994) and in conditioned forms by Gernhard (2008). If only a fraction ρ of the living languages is sampled, the density changes again (Stadler, 2009). Sampling matters in linguistics: a dataset of 100 Bantu languages samples perhaps a fifth of the family, and treating it as complete would distort the inferred branching times. Birth and death rates are also a first example of the difference between being *identifiable* and being *practically estimable* (Section 6.10): with trees of living languages only, extinction is notoriously hard to estimate.

The *coalescent* (Kingman, 1982) describes the genealogy of a sample from a large population looking backward in time, with lineages merging at a rate that depends on population size. Coalescent priors, including flexible demographic versions such as the Bayesian skyline (Drummond et al., 2005), are standard for within-species and pathogen data, and are occasionally used as convenient tree priors in linguistic analyses, although the population-genetic process they describe is a poor match for the splitting of languages.

The *fossilized birth–death* (FBD) process (Stadler, 2010; Heath et al., 2014) adds the sampling of lineages through time, so that dated samples from the past (fossils in biology, ancient languages in linguistics) are part of the tree, either as tips on extinct side branches or as *sampled ancestors* lying directly on the path to living descendants (Gavryushkina et al., 2014). For language data with written records, this is attractive: Latin can be placed as an ancestor (or near-ancestor) of the Romance languages, and Classical Greek, Sanskrit or Old Chinese can be included at their attested dates rather than used only as calibrations. Heggarty et al. (2023) used a sampled-ancestor tree prior for Indo-European. Chapter 10 discusses these models and the controversy over ancestry constraints (Chang et al., 2015).

The code below simulates trees of ten languages under Yule processes with three different birth rates and shows the prior distribution of the root age that each implies. The root age is a quantity that linguists care about, yet even a seemingly uninformative tree prior has a strong opinion about it.

```
library(phytools)
set.seed(10)
root_ages <- lapply(c(0.5, 1, 2), function(b) {
  trs <- pbtree(b = b, n = 10, nsim = 1000)
  sapply(trs, function(tr) max(node.depth.edgelength(tr)))
})
par(mar = c(4, 4, 1, 1))
ymax <- max(sapply(root_ages, function(r) max(density(r, from = 0)$y)))
plot(density(root_ages[[1]], from = 0), lwd = 2, main = "", xlim = c(0, 12),
     ylim = c(0, ymax), xlab = "root age (millennia)")
lines(density(root_ages[[2]], from = 0), lwd = 2, lty = 2)
lines(density(root_ages[[3]], from = 0), lwd = 2, lty = 3)
legend("topright", c("lambda = 0.5", "lambda = 1", "lambda = 2"),
       lty = 1:3, lwd = 2, bty = "n")
```

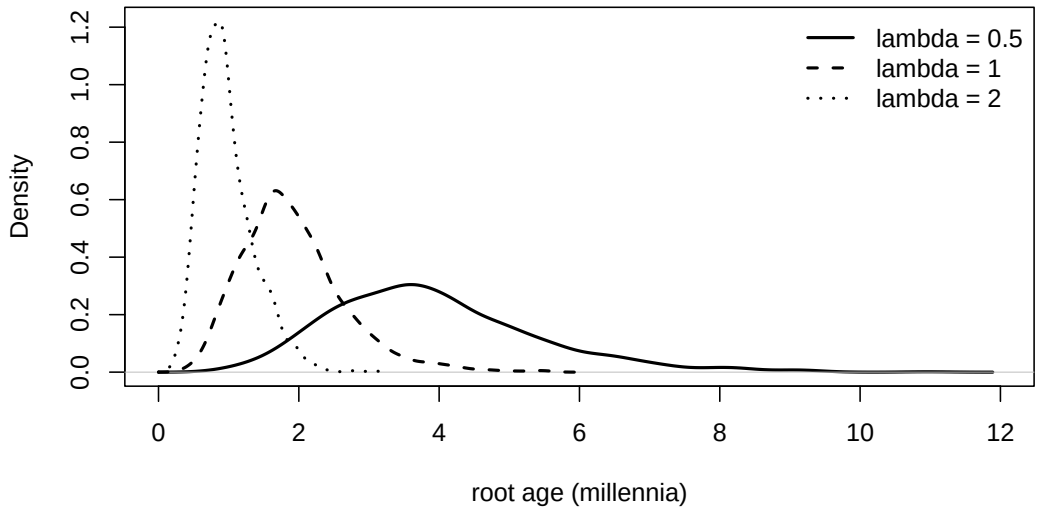


Figure 6.2: Prior distributions of the root age of a ten-language tree under Yule processes with three birth rates (splits per lineage per millennium), from 1,000 simulated trees each.

```
sapply(root_ages, median)
```

```
#> [1] 3.6964 1.8198 0.9069
```

6.2.3 Calibrations and their interaction with the tree prior

In a dated analysis, external information about the ages of some nodes enters as *calibration priors*: for instance, a prior stating that the common ancestor of the Romance languages is between about 1,500 and 2,000 years old, based on the history of Latin. These are combined with the tree prior, and the combination is not simple. The joint prior on node ages is not the product of the tree prior and the calibration densities, because both constrain the same node ages, and the *effective* prior on a calibrated node can differ substantially from the calibration density the user specified (Heled and Drummond, 2012). Heled and Drummond proposed calibrated tree priors that correct for this. The practical lesson, discussed in detail in Chapter 10, is to run the analysis without data (sampling from the prior) and inspect the effective priors on calibrated and uncalibrated nodes.

6.2.4 Priors on model parameters

Substitution and clock parameters also need priors: Dirichlet distributions for state frequencies, exponential or gamma distributions for rates and for the gamma shape parameter of rate variation, lognormal or exponential distributions for the clock rate. Defaults in software are not neutral. A default exponential prior with mean 1 on a clock rate measured in changes per character per year, for example, places almost all its mass on absurdly high rates for lexical data. Every prior should be set with the units of the analysis in mind, and checked. A prior should not be derived from an analysis of the same dataset, since the data would then be counted twice, once in the prior and once in the likelihood (Barido-Sottani et al., 2023). A prior also limits what the sampler can explore. Values outside its support are never proposed, so a hard-bounded uniform prior can exclude reasonable values that a long-tailed lognormal or exponential prior would merely discount. An unbounded uniform prior, such as Uniform$(0, \infty)$ on a clock rate, is improper: it

puts most of its weight on implausibly high values and leaves the marginal likelihood undefined (Section 6.7). When checking a prior, the range between its 5 and 95 percent quantiles is usually more informative than the shape of its density curve.

There is no such thing as an uninformative prior on a tree. A uniform prior on topologies is informative about clades; independent exponential priors on branch lengths are informative about tree length; a Yule or birth–death prior is informative about the root age; and combining calibrations with a tree prior can produce effective priors very different from the ones specified. Always run the analysis once with the data removed (BEAST: "sample from prior"; MrBayes: `mcmc data=no`; RevBayes: by omitting the clamped data) and compare the prior and posterior of every quantity you report.

6.3 Markov chain Monte Carlo

6.3.1 The idea

MCMC constructs a Markov chain, a random walk through the space of trees and parameters, whose long-run distribution is the posterior. If the chain is run long enough, the fraction of time it spends in any region is proportional to that region's posterior probability, and samples taken from it can be used like samples from the posterior: the proportion of sampled trees containing a clade estimates its posterior probability, and the 2.5 and 97.5 percentiles of the sampled root ages estimate a 95 percent credible interval.

6.3.2 The Metropolis–Hastings algorithm

The most important MCMC algorithm is due to Metropolis et al. (1953) and Hastings (1970). Writing x for the current state (tree, branch lengths and parameters) and $\pi(x) \propto f(D \mid x) f(x)$ for the unnormalized posterior, one iteration is:

1. Propose a new state x' by drawing from a *proposal distribution* $q(x' \mid x)$.
2. Compute the acceptance probability

$$\alpha(x, x') = \min\left(1,\ \underbrace{\frac{f(D \mid x')}{f(D \mid x)}}_{\text{likelihood ratio}} \times \underbrace{\frac{f(x')}{f(x)}}_{\text{prior ratio}} \times \underbrace{\frac{q(x \mid x')}{q(x' \mid x)}}_{\text{Hastings ratio}}\right). \tag{6.3}$$

3. With probability α, move to x'; otherwise stay at x (and record x again).

The marginal likelihood $f(D)$ cancels from the ratio, which is why MCMC works without it. Moves that increase the posterior are always accepted; moves that decrease it are accepted with probability equal to the ratio, which lets the chain explore the whole distribution rather than climb to a single peak. The *Hastings ratio* corrects for asymmetric proposals, in which moving from x to x' is more likely than moving back.

Why does this work? The acceptance rule is constructed so that, at equilibrium, the chain moves from x to x' exactly as often as from x' to x (the *detailed balance* condition, the same condition as for reversible CTMCs in Chapter 5). A chain with this property leaves the posterior unchanged once it has reached it, and if it can reach every state from every other, it converges to the posterior from any starting point.

The proof that Metropolis–Hastings satisfies detailed balance, the derivation of the Hastings ratios of the multiplier and tree-scaler proposals, and a demonstration of what happens when a Hastings ratio is forgotten are in Section B.4.2. Section A.12 introduces MCMC from the beginning.

6.3.3 Proposals and their Hastings ratios

The art of MCMC lies in the proposals. For a positive parameter such as a branch length, a common choice is the *multiplier* (or scale) proposal: draw $u \sim \text{Uniform}(0, 1)$, set $m = e^{\lambda(u-1/2)}$, and propose $t' = mt$, where λ is a tuning parameter. Because the proposal is uniform on the log scale rather than on t itself, it is asymmetric, and its Hastings ratio is m. Scaling all B branch lengths of the tree by the same m (a *tree scaler*) has Hastings ratio m^B. Forgetting such terms is the classic bug in hand-written samplers: the chain still runs and looks healthy, but it samples the wrong distribution.

For topologies, proposals include the *nearest-neighbor interchange* (NNI) of Chapter 5 and its stochastic variants, subtree pruning and regrafting (SPR), the LOCAL move of Larget and Simon (1999), which perturbs a path of three branches, and, for time trees, the *narrow* and *wide exchange*, *subtree slide* and *Wilson–Balding* moves of BEAST and MrBayes. Lakner et al. (2008) compared the efficiency of tree proposals and found that some popular moves rarely got accepted on large trees, which led to improved and guided proposals (Höhna and Drummond, 2012). Because the efficiency of a topology move cannot be adjusted through a single step size, it is usual to combine moves that make small rearrangements (NNI) with moves that make large ones (SPR), and to give topology moves much higher weights than moves on single numerical parameters (Barido-Sottani et al., 2023).

The tuning parameter matters. A proposal that takes tiny steps is almost always accepted but moves slowly; one that takes huge steps is almost always rejected. For random-walk proposals on a single continuous parameter, an acceptance rate somewhere around 0.2 to 0.5 is a common target, guided by theoretical results on optimal scaling (Roberts et al., 1997). BEAST and MrBayes tune their proposals automatically during the run. Automatic tuning has its own pitfall: tuning throughout the run can shrink the step sizes around the chain's current position until it is trapped near a local optimum, so tuning mainly during the burn-in is usually preferable (Barido-Sottani et al., 2023). In RevBayes, `mcmc.burnin()` is such a separate tuning run whose samples are not kept, not an instruction to discard part of the main chain (Höhna et al., 2017). Iteration counts are not comparable across programs either. One "generation" of RevBayes by default attempts as many moves as the sum of the move weights, whereas other programs may attempt a single move per iteration, so chain lengths should be compared through effective sample sizes, not through numbers of iterations (Höhna et al., 2017; Barido-Sottani et al., 2023).

6.3.4 A Metropolis–Hastings sampler in R

The following function implements a Metropolis–Hastings sampler for the two-language path length, with the multiplier proposal and its Hastings ratio. The argument `beta` raises the likelihood to a power, which is needed for marginal likelihood estimation (Section B.4.4); with `beta = 1` the sampler targets the ordinary posterior.

```
mh_t <- function(niter, lambda = 1, t_start = 1, beta = 1, prior_rate = 1,
                 N = 100, k = 30) {
  t <- t_start
```

```
  lp <- beta * loglik_t(t, N, k) + dexp(t, prior_rate, log = TRUE)
  out <- numeric(niter); accepted <- 0
  for (i in seq_len(niter)) {
    m <- exp(lambda * (runif(1) - 0.5))           # multiplier proposal
    t_new <- t * m
    lp_new <- beta * loglik_t(t_new, N, k) +
      dexp(t_new, prior_rate, log = TRUE)
    if (log(runif(1)) < lp_new - lp + log(m)) {   # log(m): Hastings ratio
      t <- t_new; lp <- lp_new; accepted <- accepted + 1
    }
    out[i] <- t
  }
  structure(out, acceptance = accepted / niter)
}
```

We run three chains from the same poor starting value ($t = 2$) with tuning parameters that are too small, about right, and too large.

```
library(coda)
set.seed(1)
chains <- lapply(c(0.05, 1, 8),
                 function(l) mh_t(5000, lambda = l, t_start = 2))
post_mean <- sum(grid * post * h)
par(mfrow = c(3, 1), mar = c(3, 4, 2, 1))
for (j in 1:3) {
  plot(chains[[j]], type = "l", ylab = "t", xlab = "",
       main = sprintf("lambda = %s, acceptance = %.2f",
                      c(0.05, 1, 8)[j], attr(chains[[j]], "acceptance")))
  abline(h = post_mean, col = "grey50", lwd = 2)
}
```

The first chain creeps slowly from its starting value and is still far from the bulk of the posterior after hundreds of iterations; its consecutive samples are almost identical. The third chain jumps well when it moves, but it rarely moves, producing long flat stretches. The middle chain mixes well. These differences show up in the *effective sample size* (ESS), the number of independent samples that would carry the same information as the autocorrelated chain (Section 6.5).

```
burnin <- 1:500
data.frame(lambda = c(0.05, 1, 8),
           acceptance = sapply(chains, attr, "acceptance"),
           ESS = sapply(chains, function(x) effectiveSize(mcmc(x[-burnin]))),
           mean = sapply(chains, function(x) mean(x[-burnin])))
```

```
#>   lambda acceptance     ESS   mean
#> 1   0.05     0.9778   4.843 0.7414
#> 2   1.00     0.6372 709.190 0.4955
#> 3   8.00     0.0984 331.314 0.4998
```

Finally we check the well-tuned chain against the exact posterior computed on the grid. Agreement

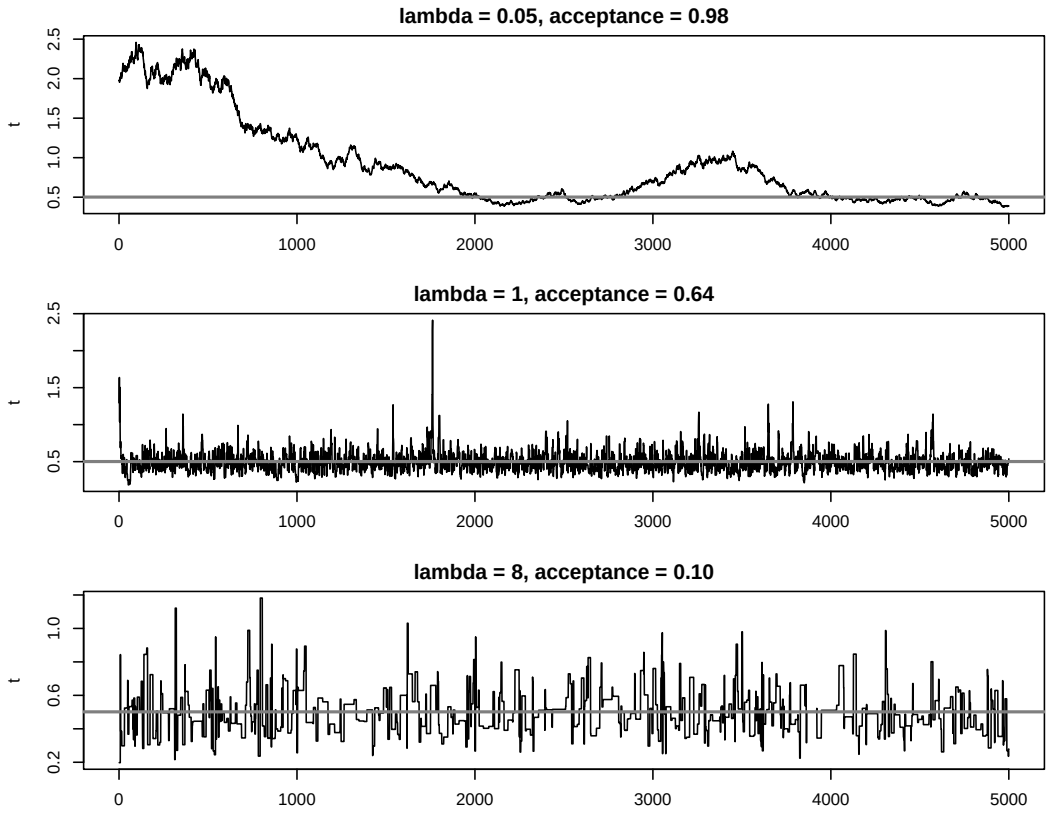


Figure 6.3: Trace plots of three Metropolis-Hastings chains for the two-language path length, started at t = 2, with tuning parameters that are too small (top), reasonable (middle) and too large (bottom). The grey line marks the posterior mean.

with an exact answer obtained by a different route is the simplest form of *implementation validation* (Section 6.10).

```
good <- chains[[2]][-burnin]
par(mar = c(4, 4, 1, 1))
hist(good, breaks = 40, freq = FALSE, main = "", xlab = "path length t",
     col = "grey85", border = "white")
lines(grid, post, lwd = 2)
```

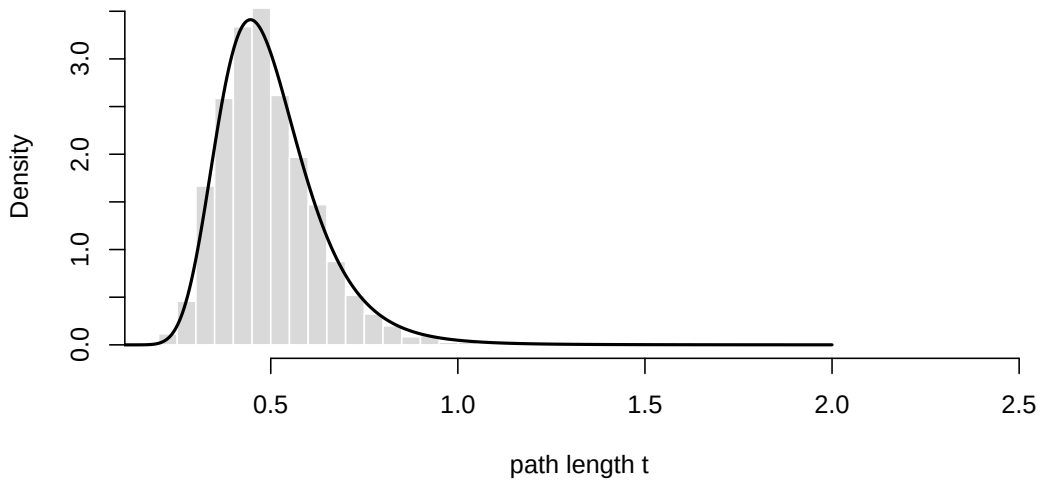


Figure 6.4: Histogram of 4,500 post-burn-in samples from the well-tuned chain, with the exact posterior density from the grid calculation (line).

```
HPDinterval(mcmc(good))      # 95% highest posterior density interval
```

```
#>       lower  upper
#> var1 0.2829 0.7653
#> attr(,"Probability")
#> [1] 0.95
```

6.3.5 Metropolis-coupled MCMC

Tree space is rugged: the posterior can have several peaks separated by valleys of low probability, and a chain that takes small steps can stay on one peak indefinitely. *Metropolis-coupled MCMC* (MC^3) (Geyer, 1991) runs several chains in parallel. One "cold" chain samples the posterior; the others are "heated", sampling from a flattened version of it, with the posterior raised to a power $\beta_i < 1$ (MrBayes uses $\beta_i = 1/(1 + (i-1)\Delta T)$ for chain i and temperature increment ΔT). Heated chains cross valleys easily. Periodically, two chains propose to swap states, with an acceptance probability that preserves the correct target distribution for each, so that the cold chain can inherit a state from a different peak. MrBayes runs four chains (one cold, three heated) by default, and parallel implementations spread them across processors (Altekar et al., 2004). BEAST 2 offers a similar scheme through its coupled MCMC package.

6.3.6 Beyond Metropolis–Hastings

Several other samplers are used in phylogenetics. *Gibbs sampling* (Geman and Geman, 1984) updates one parameter at a time from its full conditional distribution, which is possible when that distribution has a standard form. Updating parameters one at a time, whether by Gibbs or by Metropolis steps, mixes slowly when parameters are strongly correlated, because the chain must creep along a narrow ridge of the posterior; updating correlated parameters jointly ("blocking") or with proposals shaped to the correlation helps (van Ravenzwaaij et al., 2018). In dated trees the clearest case is rate against time: node ages and the mean clock rate trade off almost exactly, which is why BEAST 2 provides an *up–down* move that scales node ages up while scaling the clock rate down (Barido-Sottani et al., 2023). *Reversible-jump MCMC* (Green, 1995) moves between models with different numbers of parameters, so that model uncertainty is integrated over rather than selected; it underlies the averaging over substitution models in bModelTest (Bouckaert and Drummond, 2017) and the reversible-jump analyses of correlated evolution in BayesTraits (Pagel and Meade, 2006). *Hamiltonian Monte Carlo* uses gradients of the posterior to make long, well-directed moves; efficient gradients of phylogenetic likelihoods with respect to branch lengths and rates (Ji et al., 2020) have made it practical for some parameters in BEAST X.

6.4 A small Bayesian tree analysis

To see all the pieces working together on trees, we use a complete, if small, Bayesian phylogenetic sampler. The data are binary cognate characters simulated for six languages on a known tree, with gamma-distributed rates across characters. The model we fit is deliberately simpler than the one that generated the data: a symmetric binary model without rate variation (Section 6.9 asks whether this misspecification can be detected). The priors are uniform on unrooted topologies and independent exponential with mean 0.1 on each branch length. The proposals are a single-branch multiplier (probability 0.5), an NNI on a random internal branch (0.3), and a tree scaler (0.2), each with the Hastings ratio derived in Section B.4.2.

The sampler is about a hundred lines of R, kept in the file `R/toy-tree-sampler.R` of the book's source. It is listed and explained in Section B.4.3; here we load it and look at its output.

```
library(phangorn)
source("R/toy-tree-sampler.R")   # sim_sym(), make_loglik(), run_mcmc(), ...
set.seed(7)
langs6 <- c("Ana", "Bel", "Cor", "Dun", "Esk", "Fal")
true6 <- read.tree(text = paste0("((Ana:0.10,Bel:0.15):0.04,Cor:0.25,",
  "((Dun:0.08,Esk:0.12):0.015,Fal:0.20):0.03);"))
X6 <- sim_sym(true6, 800, rates = rgamma(800, shape = 0.5, rate = 0.5))
c(characters = ncol(X6),
  distinct_patterns = length(unique(apply(X6, 2, paste, collapse = ""))))
```

```
#>        characters distinct_patterns
#>               800                62
```

The likelihood function compresses identical character patterns and runs the pruning algorithm recursively. Before using it, we check it against **phangorn**, an independent implementation.

```
loglik_tree <- make_loglik(X6)
```

```
c(toy = loglik_tree(true6),
  phangorn = pml(true6, phyDat(X6, type = "USER", levels = c(0, 1)))$logLik)
```

```
#>      toy phangorn
#>    -2158    -2158
```

We run two independent chains of 10,000 iterations from different random starting trees, sampling every tenth iteration.

```
set.seed(101); run1 <- run_mcmc(10000, random_start(langs6), loglik_tree)
set.seed(202); run2 <- run_mcmc(10000, random_start(langs6), loglik_tree)
round(rbind(chain1 = run1$acceptance, chain2 = run2$acceptance), 3)
```

```
#>        branch   nni scaler
#> chain1  0.531 0.108  0.291
#> chain2  0.551 0.120  0.313
```

The branch-length moves are accepted about half the time, the NNI moves much less often. Low acceptance of topology moves is typical: most neighbors of a good tree are much worse. In large analyses this is the main reason runs must be long.

6.5 Convergence and mixing

An MCMC sample is only useful if the chain has converged to the posterior and has explored it thoroughly. Neither can be proven, but several diagnostics can reveal failures (Nascimento et al., 2017).

Burn-in. The early part of a chain reflects its starting point, not the posterior. It is discarded; 10 to 25 percent of the run is a common choice, but the right amount is whatever the trace plots show. The burn-in is therefore chosen after sampling, by inspecting the chains, and it pays to be conservative: discarding too much costs only some precision, provided enough samples remain, whereas keeping samples from before convergence biases every summary (van Ravenzwaaij et al., 2018). Starting independent chains from dispersed points, for example from different draws from the prior, makes their eventual agreement more informative.

Trace plots of the log-likelihood, tree length and key parameters against iteration should look like stationary "fuzzy caterpillars", without trends or long excursions, and independent chains should overlap. Figure 6.5 shows them for our two chains.

```
par(mfrow = c(1, 2), mar = c(4, 4, 1, 1))
for (v in c("logLik", "TL")) {
  rng <- range(c(run1$trace[[v]], run2$trace[[v]]))
  plot(run1$trace$iter, run1$trace[[v]], type = "l", ylim = rng,
       xlab = "iteration", ylab = v)
  lines(run2$trace$iter, run2$trace[[v]], col = "grey55")
  rect(0, rng[1] - 1e3, 2000, rng[2] + 1e3, col = rgb(0, 0, 0, 0.08),
       border = NA)
}
```

Effective sample size. Successive MCMC samples are autocorrelated. The ESS estimates how many

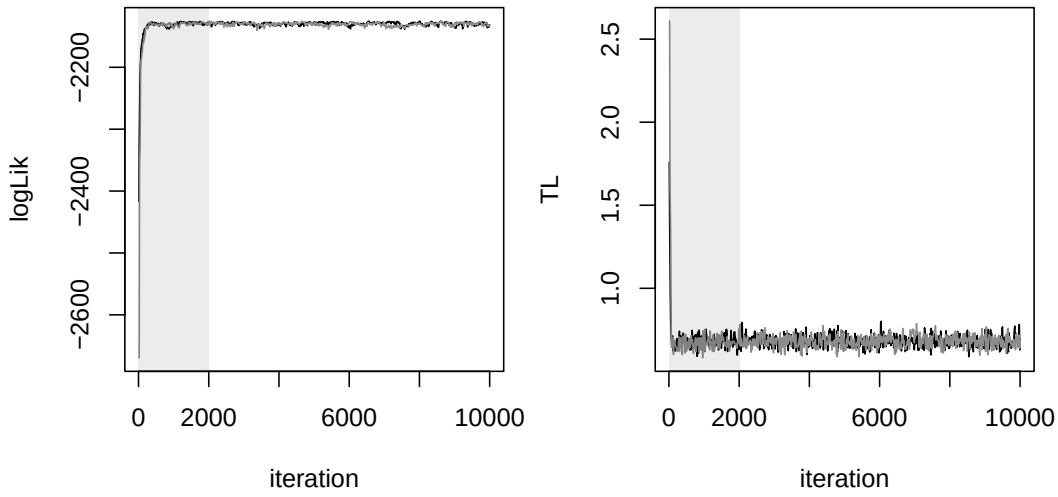


Figure 6.5: Trace plots of the log-likelihood (left) and tree length (right) for two independent chains of the toy tree sampler (black and grey). The first 20 percent of samples, shaded, are discarded as burn-in.

independent samples the chain is worth, using the autocorrelation function. Tracer (Rambaut et al., 2018), the standard tool for inspecting BEAST and MrBayes output, flags parameters with ESS below 200; the R package **coda** (Plummer et al., 2006) computes the same quantities. The threshold of 200 is a convention rather than a theorem (Barido-Sottani et al., 2023). Fabreti and Höhna (2022) instead derive a threshold from the precision required: under approximate normality, an ESS of 625 makes the Monte Carlo standard error of a posterior mean about 1 percent of the width of its 95 percent interval, against about 1.8 percent for an ESS of 200. Higher thresholds are worth their cost for the quantities on which conclusions rest, such as a root age.

Potential scale reduction factor. With several chains, the PSRF of Gelman and Rubin (1992) compares the variance within chains with the variance between them. If the chains have converged to the same distribution, the two agree and PSRF is close to 1; values much above 1 (say, above 1.01 to 1.1, depending on the standard adopted) indicate that the chains have not yet mixed.

Topological convergence. Continuous summaries can look perfect while the chains disagree about the tree. MrBayes reports the *average standard deviation of split frequencies* (ASDSF): for every split that appears in the samples, compute its frequency in each chain, take the standard deviation across chains, and average. As chains converge on the same posterior distribution of trees, the ASDSF goes to zero; values below about 0.01 are often taken as a sign of good convergence, and values up to about 0.05 as acceptable for exploratory work. Graphical tools such as AWTY (Nylander et al., 2008) and RWTY (Warren et al., 2017) plot split frequencies over time and across runs, and Lanfear et al. (2016) proposed ways to estimate the ESS of the topology itself. Whidden and Matsen (2015) showed that tree-space exploration can fail in subtle ways, with chains trapped among sets of trees separated by low-probability bridges. The R package **convenience** (Fabreti and Höhna, 2022) computes an ESS for every split, treating its presence or absence in the sampled trees as a binary trace, and uses Kolmogorov–Smirnov tests to ask whether independent runs have sampled the same distribution.

Monte Carlo error. An ESS translates directly into precision. The Monte Carlo standard error of a posterior mean is roughly the posterior standard deviation divided by the square root of the ESS,

and Tracer reports it next to each mean; tail quantities, such as the bounds of a 95 percent HPD interval for a root age, are estimated much less precisely than the mean from the same samples. It helps to ask three separate questions of every reported number: whether its Monte Carlo error is small relative to the precision with which it is reported, whether a longer run would change it by more than that, and whether independent runs from dispersed starting points reproduce it. A root age reported to the nearest century from a chain whose ESS for the root age is 150 is unlikely to pass the first test.

Tuning is not burn-in. Programs adapt the step sizes of their proposals during an initial phase, aiming at acceptance rates of the kind discussed in Section 6.3. In RevBayes, `mymcmc.burnin(generations = 1000, tuningInterval = 100)` runs such a preliminary, tuning-only chain whose samples are not written to the output files; it does not remove anything from the main run, whose early samples must still be discarded as burn-in when the output is summarized. The `weight` of each move sets how often, on average, it is attempted per iteration, and giving one parameter several scale moves with small and large tuning values can help mixing; `mymcmc.operatorSummary()` reports, after the run, how often each move was tried and accepted. Analyses that run for days should write checkpoints (the `checkpointInterval` and `checkpointFile` arguments of `run()`, and `initializeFromCheckpoint()` to resume), so that a time limit on a computing cluster does not end the run for good.

The code below computes these diagnostics for our two chains.

```
burn <- 1:200                                   # first 20% of 1000 samples
post1 <- run1$trace[-burn, ]; post2 <- run2$trace[-burn, ]
mc <- mcmc.list(mcmc(as.matrix(post1[, c("logLik", "TL")])),
                mcmc(as.matrix(post2[, c("logLik", "TL")])))
effectiveSize(mc)                               # ESS, summed over chains
```

```
#> logLik     TL
#>  112.9  997.1
```

```
gelman.diag(mc)$psrf                            # PSRF
```

```
#>        Point est. Upper C.I.
#> logLik      1.030      1.135
#> TL          1.003      1.018
```

```
split_freqs <- function(trees) {
  table(unlist(lapply(trees, split_keys))) / length(trees)
}
f1 <- split_freqs(run1$trees[-burn]); f2 <- split_freqs(run2$trees[-burn])
all_splits <- union(names(f1), names(f2))
sf <- cbind(chain1 = as.numeric(f1[all_splits]),
            chain2 = as.numeric(f2[all_splits]))
sf[is.na(sf)] <- 0; rownames(sf) <- all_splits
c(ASDSF = mean(apply(sf, 1, sd)))
```

```
#>   ASDSF
#> 0.01273
```

The two chains agree well on the tree length, but the diagnostics for the log-likelihood are less

comfortable: its ESS is modest, its PSRF is a little above 1, and the ASDSF (0.013) is around the conventional threshold of 0.01. By the usual standards these chains are close to convergence but should be run longer before their results are reported. The samples are strongly autocorrelated because we record every tenth iteration of a chain that changes one branch or one NNI at a time. Real analyses run for millions or hundreds of millions of iterations for exactly this reason. Note what convergence diagnostics can and cannot establish: that the sampler has explored the posterior defined by the model, which is a matter of implementation, not that the model is adequate.

When chains do not converge. The cause may lie in the sampler, in the priors or in the fit between model and data, and these are best examined in that order (Barido-Sottani et al., 2023). Every estimated parameter needs at least one move, since a parameter without one never leaves its starting value; moves on poorly mixing parameters can be given more weight; and a sketch of the model's graph (Section 6.11) shows which parameters depend on which. A chain that misbehaves even without data points to the set-up, such as conflicting calibrations or interacting priors; one that behaves under the prior but not with the data points to the data or the model. Running data partitions separately shows whether they support incompatible trees, and parameters whose traces merely reproduce their priors suggest that the data have been over-partitioned. A strict clock, or an informative prior on a weakly identified parameter, may converge where a relaxed clock wanders. An overly simple analysis that runs usually teaches more than an elaborate one that never converges.

6.6 Summarizing the posterior

6.6.1 Topologies and clades

The posterior sample is a set of thousands of trees. The simplest summaries are the posterior probabilities of topologies (when the posterior is concentrated enough for individual topologies to recur) and of clades (or, for unrooted trees, splits).

```
topo <- table(c(post1$topology, post2$topology))
head(sort(topo, decreasing = TRUE) / sum(topo), 4)       # top topologies
```

```
#>
#> Cor+Dun+Esk+Fal | Dun+Esk | Dun+Esk+Fal
#>                                 0.73125
#> Cor+Dun+Esk+Fal | Dun+Esk+Fal | Esk+Fal
#>                                 0.09875
#>     Cor+Dun+Esk+Fal | Cor+Fal | Dun+Esk
#>                                 0.06062
#>         Bel+Cor | Dun+Esk | Dun+Esk+Fal
#>                                 0.02937
```

```
paste(split_keys(true6), collapse = " | ")               # the true tree
```

```
#> [1] "Cor+Dun+Esk+Fal | Dun+Esk | Dun+Esk+Fal"
```

```
round(sort(rowMeans(sf), decreasing = TRUE), 3)          # split probabilities
```

```
#>     Dun+Esk+Fal Cor+Dun+Esk+Fal         Dun+Esk         Esk+Fal
#>           0.926           0.906           0.844           0.143
```

```
#>         Cor+Fal Bel+Dun+Esk+Fal         Bel+Cor     Cor+Esk+Fal
#>           0.073           0.048           0.046           0.013
#>     Cor+Dun+Esk         Dun+Fal
#>           0.001           0.001
```

The true topology is the most probable one, with a posterior probability of about 0.73, and its three splits have the highest split probabilities. But the internal branches of the true tree are short, and the posterior is genuinely uncertain: in particular, the sister of Esk is sometimes Fal rather than Dun. With 800 characters, a posterior probability of around 0.85 for a true clade is what short branches look like. Because the fitted model ignores the rate variation with which the data were generated, this is a small *robustness* result rather than a pure parameter-recovery one: the topology survives this particular misspecification, while, as we shall see, the tree length does not.

A single tree is usually wanted for display. The options are:

- The *majority-rule consensus tree* contains every clade with posterior probability above one half (Chapter 2). It is the default in MrBayes, and under a loss function that penalizes wrongly included and wrongly omitted clades it is the optimal point estimate of the tree (Holder et al., 2008). Its branch lengths are averages over the trees containing each clade, and it may not be identical to any sampled tree.
- The *maximum a posteriori* (MAP) tree is the most frequently sampled topology. It is meaningful only when topologies recur often, which is rare for large trees.
- The *maximum clade credibility* (MCC) tree is the sampled tree that maximizes the product of the posterior probabilities of its clades. It is the default in BEAST's TreeAnnotator and is the tree drawn in most published linguistic analyses. Node heights on an MCC tree are commonly the mean or median heights of the corresponding clades across the sample; Heled and Bouckaert (2013) showed that this can produce misleading summaries and proposed alternatives, such as common-ancestor heights, now available in TreeAnnotator.
- *Agreement subtrees* (Cranston and Rannala, 2007) summarize the posterior by subsets of languages whose relationships recur frequently across the sample, each with a fully resolved subtree. They make visible which languages are responsible for the uncertainty, a useful complement when one or two poorly attested languages wander across the tree.

Node ages are reported with 95 percent *highest posterior density* (HPD) intervals, the shortest intervals containing 95 percent of the sampled values. Visual summaries that overlay the whole posterior sample, such as DensiTree (Bouckaert, 2010), are often more honest than any single tree, especially when the posterior is genuinely uncertain about deep relationships.

```
post_trees <- c(run1$trees[-burn], run2$trees[-burn])
post_trees <- lapply(post_trees,
                     function(t) root(t, "Ana", resolve.root = TRUE))
class(post_trees) <- "multiPhylo"
mcc <- maxCladeCred(post_trees)
par(mfrow = c(1, 2), mar = c(3, 1, 1, 1))
# clade probabilities as computed by plotBS(), written on each stem branch
# on the side away from the parent's vertical line
pp_sup <- plotBS(mcc, post_trees, type = "none", digits = 2)$node.label
plot(mcc, cex = 0.9, label.offset = 0.01 * max(node.depth.edgelength(mcc)))
lp <- get("last_plot.phylo", envir = .PlotPhyloEnv)
```

```
nn <- Ntip(mcc) + seq_len(Nnode(mcc))
pa <- mcc$edge[match(nn, mcc$edge[, 2]), 1]
for (i in which(!is.na(pa) & pp_sup > 0)) {  # root unlabelled, as plotBS()
  a <- if (lp$yy[nn[i]] >= lp$yy[pa[i]]) c(1.1, -0.35) else c(1.1, 1.35)
  nodelabels(pp_sup[i], nn[i], frame = "none", adj = a, cex = 0.9,
             col = "darkred")
}
set.seed(3)
densiTree(post_trees[sample(length(post_trees), 100)], alpha = 0.05,
          consensus = mcc, type = "phylogram", scaleX = TRUE, cex = 0.9)
```

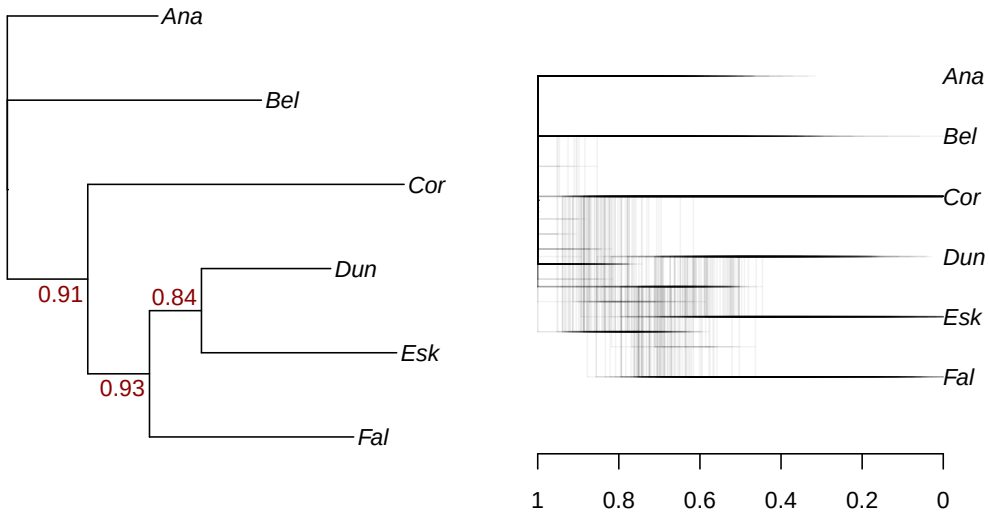


Figure 6.6: Summary of the toy posterior. Left: maximum clade credibility tree (rooted on Ana for display) with clade posterior probabilities. Right: a sample of 100 posterior trees drawn on top of each other, in the style of DensiTree.

A summary tree is a picture, not an input. When the posterior is used for a further question, such as reconstructing an ancestral state, locating a homeland or testing a correlation between features, conditioning on one MCC tree treats its topology and branch lengths as known. This can distort the downstream inference even when the summary topology looks entirely plausible: the MCC tree's node heights are averages over trees in which its clades exist and may correspond to no sampled history, a weakly supported clade on it is treated as certain, and the downstream result may depend precisely on the part of the tree that is uncertain. The remedy is to repeat the downstream analysis over a sample of a few hundred to a few thousand posterior trees and report the distribution of results (Chapters 11, 12, 13 and 17).

6.6.2 Posterior probabilities and bootstrap values

Clade posterior probabilities and bootstrap percentages answer different questions and are not on the same scale. Simulation studies found that posterior probabilities are typically higher than bootstrap values for the same clades (Alfaro et al., 2003), and that when the model is misspecified they can be overconfident (Suzuki et al., 2002). When the true tree contains a very short internal

branch, posterior probabilities can strongly favor one of the three resolutions even though the data contain essentially no information about it, the so-called *star-tree paradox* (Lewis et al., 2005; Yang and Rannala, 2005). Linguistic datasets, with their short internal branches near rapid expansions (Chapter 10), are particularly exposed to this. A posterior probability of 0.95 for a deep clade in a family that expanded rapidly deserves scrutiny, not celebration.

6.7 Marginal likelihoods and Bayes factors

6.7.1 Bayes factors

To compare two models M_0 and M_1 (say, a strict versus a relaxed clock, or a binary model versus a stochastic Dollo model), the Bayesian approach uses their marginal likelihoods, equation (6.2). The *Bayes factor* is their ratio,

$$\mathrm{BF}_{10} = \frac{f(D \mid M_1)}{f(D \mid M_0)}, \tag{6.4}$$

which is the factor by which the data change the prior odds of M_1 against M_0 (Kass and Raftery, 1995). Kass and Raftery suggested interpreting $2 \ln \mathrm{BF}$ as follows: 0 to 2, barely worth mentioning; 2 to 6, positive evidence; 6 to 10, strong; above 10, very strong. Because the marginal likelihood averages the likelihood over the prior, it automatically penalizes models with many parameters spread over regions of low likelihood; it also makes Bayes factors sensitive to the prior, in a way that posterior distributions of parameters often are not. Doubling the width of a vague prior on a parameter that the data determine well barely changes its posterior but halves that model's marginal likelihood.

The marginal likelihood can be read as a prediction made before the data were seen: the probability that the model, with its parameters drawn from their prior, assigned to the data actually observed. A Bayes factor is then a ratio of predictive successes. It measures evidence *relative* to a stated alternative, and the prior is part of each model's prediction rather than a separable nuisance (Morey et al., 2016). Verbal scales are rough conventions. The older scale of Jeffreys, as tabulated in the RevBayes documentation (Höhna et al., 2017), is stated for the Bayes factor itself and calls 3.2 to 10 "substantial", 10 to 100 "strong" and above 100 "decisive" (that is, $\ln \mathrm{BF}$ above 1.16, 2.3 and 4.6); some linguistic studies report $\ln \mathrm{BF}$ on this scale (e.g. Goldstein, 2022a). Neither scale was calibrated on phylogenetic problems, so before comparing reported values check whether a log Bayes factor is natural or base-10 and whether it has been doubled.

Our one-parameter model is small enough to compute the marginal likelihood exactly by numerical integration, which shows this sensitivity directly. We compare three exponential priors on t, with means 1, 10 and 100.

```
ll_max <- loglik_t(-0.5 * log(1 - 2 * k / N))
exact_logZ <- function(prior_rate) {
  f <- function(t) exp(loglik_t(t) - ll_max) * dexp(t, prior_rate)
  log(integrate(f, 0, Inf, rel.tol = 1e-10)$value) + ll_max
}
posterior_mean_t <- function(prior_rate) {
  p <- exp(loglik_t(grid) - ll_max) * dexp(grid, prior_rate)
  sum(grid * p) / sum(p)
}
```

```
rates <- c(1, 0.1, 0.01)
logZ <- sapply(rates, exact_logZ)
data.frame(prior_mean = 1 / rates,
           posterior_mean = sapply(rates, posterior_mean_t),
           log_marginal_likelihood = logZ,
           log_BF_vs_mean_1 = logZ - logZ[1])
```

```
#>   prior_mean posterior_mean log_marginal_likelihood log_BF_vs_mean_1
#> 1          1         0.5013                  -132.1            0.000
#> 2         10         0.5211                  -133.9           -1.835
#> 3        100         0.5233                  -136.1           -4.017
```

The posterior mean of t is practically the same under all three priors, but the log marginal likelihood falls by about two log units each time the prior becomes ten times more diffuse, approaching $\ln 10 \approx 2.3$ as the prior becomes vague relative to the likelihood. Two models that differ only in how vague the prior on one well-determined parameter is can therefore differ in marginal likelihood by a factor of about 50 (four log units), which on the Kass–Raftery scale would count as strong evidence. When Bayes factors are reported, the priors on the parameters that differ between the models must be reported and justified too. This sensitivity is not a peculiarity of Bayes factors. The prior is how a model with free parameters makes predictions at all, and a Bayes factor depends on it only insofar as it changes those predictions (Morey et al., 2016). It follows that improper priors, such as an unbounded uniform prior on a clock rate, leave the marginal likelihood undefined, so models that differ in parameters carrying such priors cannot be compared by Bayes factors. A related caution concerns what counts as data. When ancient languages enter a sampled-ancestor or fossilized birth–death prior (Section 10.6), their ages are part of the tree prior rather than of the likelihood, and May and Rothfels (2023) argue that models of this kind, which conflate likelihood and prior, cannot be compared with conventional marginal-likelihood tools.

Bayes factors and information criteria such as AIC (Section 5.9) can also disagree for a more philosophical reason. Posterior model probabilities treat one of the candidates as the process that generated the data, whereas AIC assumes that every candidate approximates a more complex reality and asks which will predict best; this is one reason why AIC tends to favour richer models (Harmon, 2019). Neither assumption is literally true of language change, which is a further argument for checking absolute adequacy (Section 6.9).

6.7.2 Estimating marginal likelihoods

For real phylogenetic models the marginal likelihood is an integral over the whole parameter space, trees included, and has to be estimated. An estimator proposed by Newton and Raftery (1994), the *harmonic mean* of the likelihoods of the posterior samples, requires only the log-likelihoods already recorded during an ordinary MCMC run. It is consistent in theory, which explains its popularity: it was for years the default in MrBayes and Tracer. In practice it is dreadful. Its variance can be infinite, so it converges extremely slowly and erratically; it is dominated by the rare samples with low likelihood; and because posterior samples rarely visit the regions where the prior has most of its mass, it is almost insensitive to the prior, overestimating the marginal likelihood and favoring parameter-rich models (Lartillot and Philippe, 2006; Xie et al., 2011).

Better methods run a series of MCMC analyses that move gradually from the prior to the posterior,

through *power posteriors* in which the likelihood is raised to a power β between 0 (the prior) and 1 (the posterior). *Path sampling* or thermodynamic integration (Gelman and Meng, 1998; Lartillot and Philippe, 2006) integrates the expected log-likelihood over β; *stepping-stone sampling* (Xie et al., 2011) multiplies together estimated ratios of normalizing constants between successive values of β. A *generalized* stepping-stone method replaces the prior at $\beta = 0$ by a reference distribution close to the posterior, which improves accuracy (Fan et al., 2011), and versions that accommodate tree uncertainty are available in BEAST (Baele et al., 2012). *Nested sampling* (Skilling, 2006) takes a different route, sampling from the prior under a rising likelihood threshold; it was implemented for phylogenetics in BEAST 2 by Maturana Russel et al. (2019) and has the advantage of providing an estimate of its own error.

Applied to the one-parameter example with the three priors above, stepping-stone estimates track the exact values closely, including the drop of about two log units per tenfold widening of the prior, while harmonic-mean estimates barely respond to the prior at all, are biased upward, and vary from run to run. In this one-dimensional example the errors are small; in phylogenetic problems with hundreds of parameters they are large, and they are exactly the errors that lead the harmonic mean to favor the wrong model.

The formulas of the harmonic mean, path sampling and stepping-stone estimators (equations (B.14), (B.15) and (B.16)) and the R code that compares them with the exact values in the one-parameter example are in Section B.4.4.

Do not use the harmonic mean estimator. Some older linguistic studies reported Bayes factors based on it, and some software still prints it by default. Use stepping-stone or path sampling (MrBayes `ss`; the MODEL_SELECTION package in BEAST 2; `powerPosterior` in RevBayes) or nested sampling (BEAST 2's NS package), and check that estimates are stable across independent runs. Remember also that Bayes factors depend on the priors on parameters that differ between the models.

6.8 Prior sensitivity

How much do results depend on the prior? The answer depends on how much information the data carry about each quantity. Figure 6.7 shows the posterior of the path length for two amounts of data and three priors.

```
grid2 <- seq(0.001, 3, length.out = 3000); h2 <- diff(grid2)[1]
priors <- list("Exp(mean 0.1)" = function(t) dexp(t, 10),
               "Exp(mean 10)"  = function(t) dexp(t, 0.1),
               "Unif(0, 5)"    = function(t) dunif(t, 0, 5))
par(mfrow = c(1, 2), mar = c(4, 4, 2, 1))
for (dat in list(c(N = 20, k = 6), c(N = 500, k = 150))) {
  ll <- loglik_t(grid2, dat["N"], dat["k"]); ll <- exp(ll - max(ll))
  dens <- sapply(priors, function(pr) {
    p <- ll * pr(grid2); p / sum(p * h2)
  })
  matplot(grid2, dens, type = "l", lty = 1:3, lwd = 2, col = "black",
          xlab = "path length t", ylab = "posterior density",
          main = paste(dat["N"], "characters"), xlim = c(0, 1.5))
  if (dat["N"] == 20) legend("topright", names(priors), lty = 1:3, lwd = 2,
```

```
                          bty = "n", cex = 0.9, inset = c(0.02, 0))
}
```

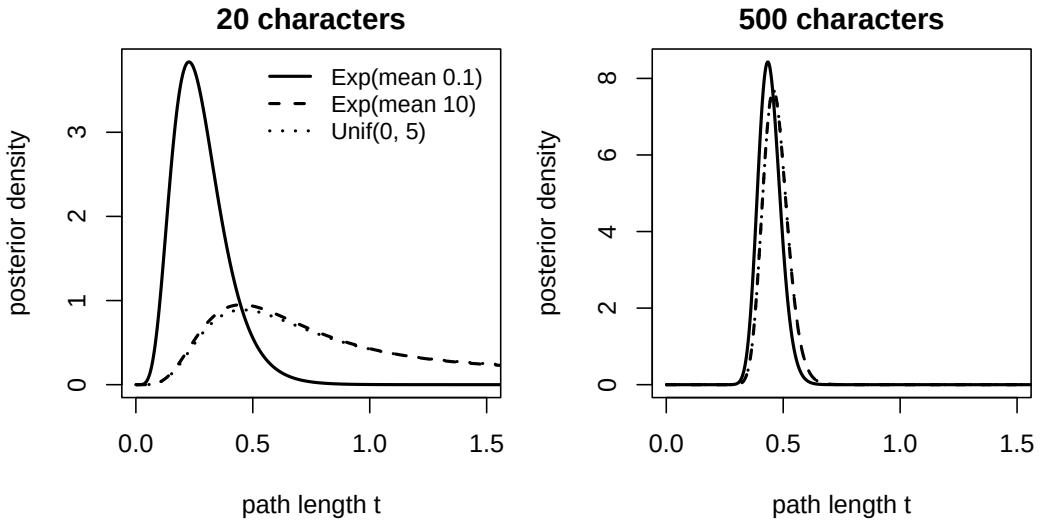


Figure 6.7: Posteriors of the path length between two languages under three priors (exponential with mean 0.1, exponential with mean 10, and uniform on 0 to 5), for 20 characters (left, 6 differences) and 500 characters (right, 150 differences). With few data the prior matters; with many it hardly does.

The same holds for trees, with an important twist: different quantities are informed by the data to very different degrees. Topologies of shallow clades, and the relative lengths of branches, are usually well determined by lexical data. Absolute dates are determined by the data only through the combination of calibrations, clock model and tree prior, so their posteriors can be strongly prior-dependent even in large analyses (Rannala, 2002). Two practical checks follow. First, *sample from the prior* (run without data) and compare prior and posterior for every reported quantity; a posterior that closely resembles its prior has not been informed by the data. Second, *rerun with alternative reasonable priors* (different tree priors, different calibration densities, different clock-rate priors) and report how much the conclusions change. A prior-only run is also a diagnostic: if the chain mixes badly even without data, the problem lies in the set-up rather than in the data (Barido-Sottani et al., 2023). And if a parameter's posterior is indistinguishable from its prior, the same estimate would have been obtained with no data at all; the options are then more data, a model in which the parameter is identifiable, or reporting the estimate explicitly as prior-driven (Höhna et al., 2017). Our toy sampler can do the first check by switching the likelihood off.

```
set.seed(303)
prior_run <- run_mcmc(10000, random_start(langs6), loglik_tree,
                      use_data = FALSE)
TL_prior <- prior_run$trace$TL[-burn]
TL_post  <- c(post1$TL, post2$TL)
par(mar = c(4, 4, 1, 1))
d_post <- density(TL_post)
hist(TL_prior, breaks = 40, freq = FALSE, col = "grey80", border = "white",
     xlim = c(0, 3), ylim = c(0, max(d_post$y)), main = "",
```

```
     xlab = "tree length")
curve(dgamma(x, shape = 9, rate = 10), add = TRUE, col = "grey40", lwd = 2)
lines(d_post, lwd = 2)
```

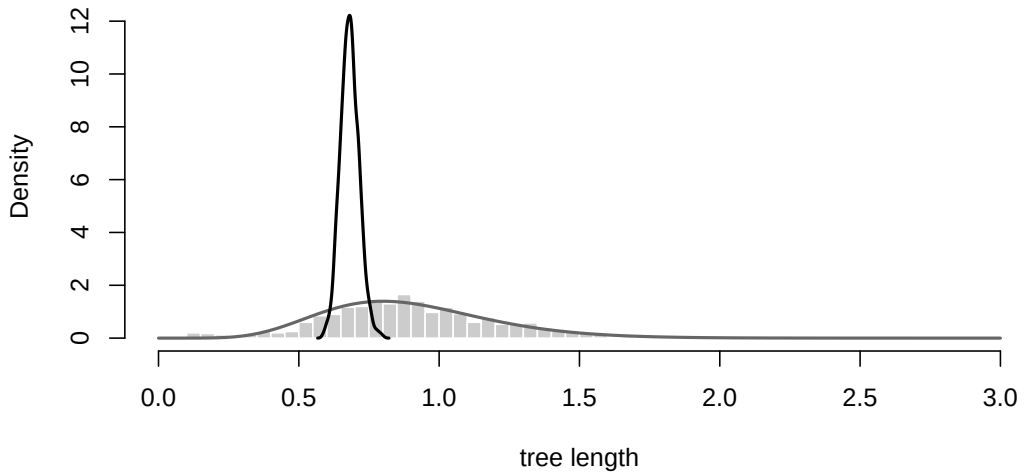


Figure 6.8: Prior and posterior distributions of total tree length in the toy analysis. Grey histogram: samples from the prior-only run; grey curve: the exact prior, a gamma distribution implied by nine independent exponential branch-length priors with mean 0.1; black curve: posterior density.

```
c(prior_mean = mean(TL_prior), posterior_mean = mean(TL_post),
  true_TL = sum(true6$edge.length))
```

```
#>     prior_mean posterior_mean        true_TL
#>         0.9139         0.6812         0.9850
```

The prior-only run reproduces the gamma distribution it should. This is a check of the implementation, including its Hastings ratios: a sampler that forgot the tree scaler's Jacobian would fail it. The posterior is much narrower than the prior and centered elsewhere, so the data dominate here. Notice, though, that the posterior tree length is below the true value. That is not the prior's doing: it is the signature of the misspecified model, which ignores rate variation across characters and therefore underestimates the amount of change on long paths.

6.9 Posterior predictive checks

Marginal likelihoods compare models with each other. They do not say whether any of the models is adequate. *Posterior predictive* checks ask whether data simulated from the fitted model resemble the observed data (Gelman et al., 1996). For each of many posterior samples, simulate a new dataset of the same size under the model with those parameter values, and compute a *test statistic* on both the simulated and the observed data. If the observed statistic lies in the tail of the simulated distribution, the model fails to reproduce that feature of the data. Bollback (2002) introduced the approach to phylogenetics, and it has since been extended to statistics that target the tree itself (Brown, 2014a) and automated in RevBayes (Höhna et al., 2018).

We check our toy model with the number of *invariant* characters (those with the same state in

all six languages) as the test statistic. Rate variation across characters produces an excess of invariant characters, because slow characters rarely change, so this statistic should be sensitive to the misspecification we built in.

```
n_invariant <- function(X) {
  sum(apply(X, 2, function(z) length(unique(z)) == 1))
}
set.seed(5)
idx <- sample(length(post_trees), 200)
sim_stat <- sapply(idx, function(i)
  n_invariant(sim_sym(unroot(post_trees[[i]]), 800)))
obs_stat <- n_invariant(X6)
par(mar = c(4, 4, 1, 1))
hist(sim_stat, breaks = 25, col = "grey80", border = "white", main = "",
     xlab = "number of invariant characters",
     xlim = range(c(sim_stat, obs_stat)))
abline(v = obs_stat, lwd = 3)
```

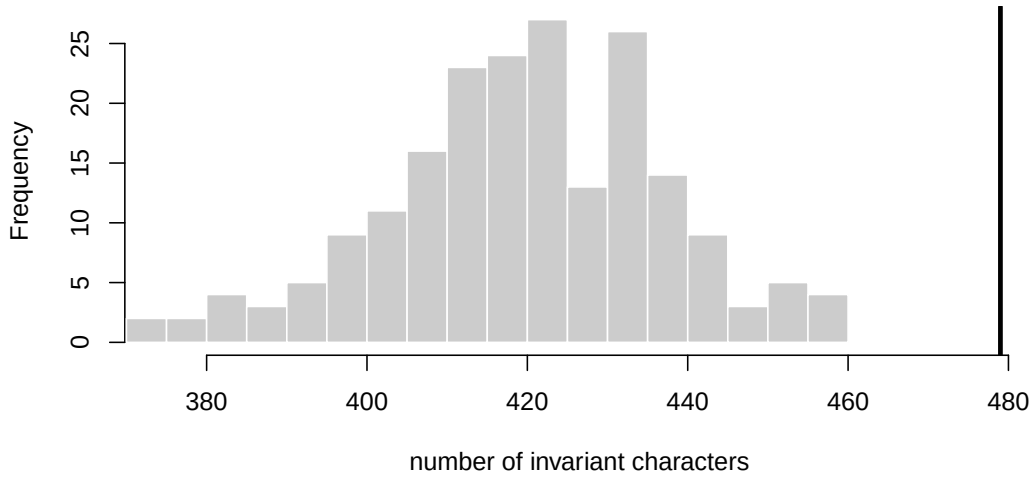


Figure 6.9: Posterior predictive distribution of the number of invariant characters under the fitted model (no rate variation), from 200 posterior samples, with the observed value (vertical line).

```
c(observed = obs_stat, posterior_predictive_p = mean(sim_stat >= obs_stat))
```

```
#>               observed posterior_predictive_p
#>                    479                      0
```

The observed data contain far more invariant characters than the model predicts, and the posterior predictive p-value is essentially zero: the model without rate variation is inadequate, as it should be, since the data were simulated with strong rate variation. Because we know the truth here, the exercise demonstrates that this test statistic has the power to detect this kind of misspecification; it becomes evidence about *empirical* adequacy only when the same check is applied to real cognate data. The most probable topology under this model was nonetheless the true one, which is a common pattern: model misspecification often affects branch lengths and dates more than topology. Posterior predictive checks are still rare in published linguistic phylogenetics, though

they are a natural complement to Bayes factors (Chapters 7 and 15).

6.10 Validating a Bayesian analysis

A posterior distribution is the answer to a question asked under a model. Whether it is also a good answer to the historical question that motivated the analysis depends on several distinct checks, which are easily conflated. A simulation in which a method "recovers the true tree", a chain whose diagnostics look perfect, and a model favoured by a Bayes factor each establish something different, and none establishes that a date or a homeland is historically right. This book uses a fixed vocabulary for these checks, defined here and used in every later chapter.

Five kinds of validation

1. **Implementation validation** asks whether the software computes what the model says it computes. The evidence comes from comparisons with exact results or with independent implementations (the toy likelihood against **phangorn**, the Metropolis–Hastings chain against the grid posterior in this chapter), from *prior-only runs*, which must reproduce the prior that the model specifies (Figure 6.8), and from *simulation-based calibration*, in which parameters are drawn from the prior, data are simulated from them and analysed, and the true values must fall at uniformly distributed ranks in the resulting posteriors (Talts et al., 2018). A linguistic example: before analysing a cognate matrix with a stochastic Dollo model and five calibrations in BEAST 2, run the XML file without data and check that the effective prior on the root age is what the tree prior and calibrations jointly imply. Implementation validation says nothing about whether the model is appropriate for language. King (2026b) applied simulation-based calibration to the standard BEAST 2 set-up for cognate data and found that it is well calibrated only after a change to the default weighting of meaning-partition rates (Section 7.10), a reminder that defaults need validating too.
2. **Parameter recovery** asks whether the model's parameters can be estimated from data of realistic size *when the data are generated under the model's own assumptions*. One simulates, say, 40 languages and 200 meanings under the binary CTMC with a relaxed clock and three calibrations, reanalyses the data, and asks how often the 95 percent interval for the root age contains the true value and how wide it is. This is the easiest test a method can face, because the model is right by construction. Success shows practical estimability; failure is a warning that no amount of real data of that kind will settle the question.
3. **Robustness under misspecification** asks how inferences degrade when the data are generated by processes the model ignores: borrowing between neighbouring languages, bursts of change at splits, rate shifts in particular lineages, unobserved extinction and incomplete sampling, dependence between the binary characters of a meaning slot, coding errors. Greenhill et al. (2009), for example, simulated borrowing on known trees and asked how far topologies and dates inferred under a tree model were affected. Robustness is always robustness *to a specified violation*, of a specified size. Rönchen et al. (2025a), in a manuscript, simulated Indo-European vocabulary evolution with loans and varying rates and reported that published dating methods did not recover the simulated root ages reliably (Section 10.8).
4. **Empirical model adequacy** asks whether the fitted model reproduces relevant features of the real data. Posterior predictive checks (Section 6.9), the parametric bootstrap (Chapter 5) and cross-validation are the main tools; useful statistics for cognate data include the number of cognate classes per meaning, the frequency of singletons, the distribution of cognate-set

sizes and the pattern of pairwise distances. Adequacy is absolute; a Bayes factor, which compares two models, can favour a model that fails every adequacy check. The covarion model, preferred by Bayes factors in many linguistic studies, failed several posterior predictive checks on Indo-European data (King, 2026b).

5. **Historical interpretation** asks whether a parameter corresponds to a historically meaningful quantity. The root age of a dated tree is the age of the most recent common ancestor of the *sampled* varieties under the model; equating it with the break-up of a speech community, the arrival of farming, or an archaeological culture requires a further argument. A phylogeographic root location is the location of that ancestor under an assumed diffusion process, not automatically a homeland (Chapter 11).

A simulation in which data are generated under the fitted model is a parameter-recovery study, and should never be presented as evidence of empirical adequacy. Agreement between an inferred tree and the subgroups accepted by specialists is a further kind of evidence, valuable but not independent, because the cognate judgements on which the tree rests are themselves products of the comparative method (Section 1.9).

Four properties of a parameter

The same distinctions apply to individual parameters. For each quantity reported from an analysis it is worth asking four questions:

- Is it **identifiable under the model**? A parameter is identifiable if different values imply different probability distributions for the data, so that unlimited data would determine it. The rate of lexical change and absolute time are not separately identifiable from character data alone (Section 5.2); only their product is, until calibrations are added.
- Is it **practically estimable with realistic linguistic samples**? An identifiable parameter may still be estimated so imprecisely from 100–200 meanings and a few dozen languages that its posterior hardly differs from its prior. Extinction rates in birth–death tree priors are the standard example.
- Is it **robust to misspecification**? Some estimates (shallow subgroups, for instance) change little under moderate violations of the model; others (absolute dates, rates in particular lineages) can move a great deal.
- Is it **interpretable as a historical quantity**? A gamma shape parameter is identifiable, well estimated and fairly robust, yet it has no direct historical interpretation: it summarizes how unequal the stabilities of the sampled meanings are.

Table 6.1 applies the four questions to some parameters that recur in this book. The entries are typical judgements for lexical analyses, not universal verdicts; each must be checked for the analysis at hand.

Table 6.1: The four properties applied to parameters of typical lexical analyses.

Parameter	Identifiable under the model	Practically estimable	Robust to misspecification	Historical interpretation
Topology of shallow subgroups	yes	usually, with a few hundred cognate sets	fairly; vulnerable to borrowing among close relatives	as subgroups only insofar as shared cognates reflect shared innovations

Parameter	Identifiable under the model	Practically estimable	Robust to misspecification	Historical interpretation
Root age	only with calibrations or a clock prior	yes, but often strongly prior-dependent	sensitive to borrowing, clock model, calibrations and ancestry assumptions	age of the common ancestor of the sampled varieties; links to events need separate argument
Mean clock rate	only with calibrations	yes	sensitive to ascertainment and saturation	average replacement per character per unit time, not per meaning
Yule birth rate λ	yes	roughly	sensitive to sampling fraction and the language/dialect decisions	splitting rate of documented lineages, not of speech communities
Extinction rate μ	in principle	rarely, from living languages alone	poor	depends on what counts as a lineage
Gamma shape of rate variation	yes	yes	absorbs other heterogeneity, such as borrowability	indirect
Phylogeographic root location	yes, under the diffusion model	often with wide uncertainty	sensitive to geographic sampling and the dispersal model	location under the model, not automatically a homeland

In this chapter, the comparison of the Metropolis–Hastings chain with the grid posterior, the check of the toy likelihood against **phangorn**, and the prior-only run of the tree sampler are implementation validation. The toy posterior placing most of its weight on the true topology, although the fitted model ignored the rate variation with which the data were simulated, is a small robustness result for topology, and the underestimated tree length shows a parameter that is not robust to the same violation. The posterior predictive check demonstrates, on data whose generating process we know, that a particular test statistic can detect that violation; it is not itself evidence about the adequacy of any model for real languages.

6.11 Software

Table 6.2 lists the main Bayesian phylogenetic programs used with language data. Appendix F gives fuller descriptions, and Chapter 17 shows how to prepare input files.

Table 6.2: Bayesian phylogenetic software used with linguistic data.

Program	Approach	Strengths for linguistic data
MrBayes 3.2 (Ronquist et al., 2012b)	MC^3, unrooted and clock trees	restriction-site (binary) model with ascertainment coding options; stepping-stone sampling; used by Gray and Atkinson (2003)

Program	Approach	Strengths for linguistic data
BEAST 2 (Bouckaert et al., 2014, 2019)	MCMC on time trees; modular packages	Babel package with binary covarion and stochastic Dollo models; relaxed clocks; sampled ancestors; phylogeography; BEASTling front end (Maurits et al., 2017)
BEAST 1.10 and BEAST X (Suchard et al., 2018; Baele et al., 2025)	MCMC, HMC for some parameters	phylogeography and phylodynamic models; fast likelihood computation
RevBayes (Höhna et al., 2016)	graphical-model language	complete flexibility of model specification; power posteriors; posterior predictive checks
TraitLab	MCMC for stochastic Dollo models	lateral transfer and missing-data models designed for cognate data (Nicholls and Gray, 2008; Ryder and Nicholls, 2011)
BayesTraits (Pagel and Meade, 2006)	MCMC and RJMCMC on given trees	rates, correlated evolution and ancestral states over posterior tree samples
PhyloBayes (Lartillot et al., 2009), ExaBayes (Aberer et al., 2014)	MCMC	profile mixture models (PhyloBayes); very large datasets (ExaBayes); rarely used for language data

RevBayes differs from the other programs in that a model is written out as a *probabilistic graphical model*: a directed acyclic graph whose nodes are constants, stochastic variables with prior distributions, and deterministic functions of other nodes. Observed data enter as stochastic nodes *clamped* to their observed values, which then contribute to the likelihood rather than to the prior, and a block of variables repeated over branches, meanings or partitions forms a *plate* (Höhna et al., 2014, 2017). Drawing the graph before writing the script makes the model's conditional dependencies explicit, and the same graph helps diagnose a poorly mixing parameter by showing what it depends on (Appendix H).

A minimal MrBayes analysis of a binary cognate matrix illustrates the main settings. It uses the restriction-site model with gamma rate variation, corrects for the absence of all-zero characters, runs two independent analyses of four chains each, and estimates the marginal likelihood by stepping-stone sampling. It is shown here but not run, since MrBayes is an external program. We checked it with MrBayes 3.2.9 on the simulated cognate matrix of Chapter 17, with much shorter chains. For unattended runs, add `set autoclose=yes nowarn=yes;` at the start of the block, so that MrBayes does not stop to ask whether to extend the chain or overwrite existing output files.

```
begin mrbayes;
  lset coding=noabsencesites rates=gamma;
  prset brlenspr=unconstrained:gammadir(1,1,1,1);
  mcmc ngen=10000000 samplefreq=1000 nruns=2 nchains=4;
  sump burninfrac=0.25;
  sumt burninfrac=0.25 contype=allcompat;
  ss ngen=5000000 nsteps=50;
end;
```

6.12 Bayesian phylogenetics in linguistics

The detailed results of Bayesian dating studies are reviewed family by family in Chapter 10. Here we focus on what the Bayesian framework contributed methodologically, through some of the studies that shaped the field.

Gray and Atkinson (2003) analyzed 2,449 cognate sets from 87 Indo-European languages, largely from the Dyen database plus Hittite and the two Tocharian languages, coded as binary characters. They used MrBayes with a restriction-site model and gamma rate variation to obtain a posterior sample of trees, then dated each sampled tree with penalized-likelihood rate smoothing (Sanderson, 2002) and 14 calibrations from historically attested events. The estimated root age, roughly 7,800 to 9,800 years before present, was interpreted as supporting the Anatolian farming hypothesis over the steppe hypothesis. Methodologically, the study set the pattern for much of what followed: binary cognate coding, a model-based tree sample rather than a single tree, dating that integrated over topological uncertainty, and an explicit set of robustness checks varying the coding, the calibrations and the treatment of rate variation. Its date and conclusions have been debated ever since (Chapters 10 and 15), and later studies replaced the two-step procedure with fully Bayesian relaxed-clock analyses.

Gray et al. (2009) inferred a dated phylogeny of 400 Austronesian languages from the Austronesian Basic Vocabulary Database using BEAST with a relaxed clock and archaeological and historical calibrations. The root age was estimated at about 5,200 years before present, consistent with an origin in Taiwan, and the tree showed a sequence of "pulses" of rapid diversification separated by "pauses", which the authors linked to the crossing of geographic barriers and to cultural innovations. The analysis illustrates two strengths of the Bayesian time-tree framework: dates and topology are estimated jointly with their uncertainty, and the posterior sample of trees can be used to test hypotheses about the tempo of expansion. A companion study by Greenhill et al. (2010b) assessed how closely such Bayesian trees match the subgroups established by the comparative method and how robust they are to alternative analysis choices. Agreement of this kind is reassuring, but it is not independent validation: the cognate judgements in the database are themselves products of the comparative method, and sharing a cognate is not the same as sharing an innovation (Section 1.9).

Prior choices about ancient languages can move dates substantially. Chang et al. (2015) reanalyzed Indo-European with a more carefully curated dataset and with *ancestry constraints*, priors that place attested ancient languages (such as Latin, Old Irish and Vedic Sanskrit) directly ancestral or nearly ancestral to their modern descendants. With these constraints the root age moved to a range compatible with the steppe hypothesis; without them, their analyses produced older dates. The study made the role of the prior on tree structure unusually visible. Heggarty et al. (2023) subsequently used a sampled-ancestor tree prior, which treats the question of ancestry probabilistically rather than imposing it, on a new Indo-European dataset and obtained an older root, of roughly eight millennia. The disagreement is substantially about priors and about how ancient languages should enter the tree, which is why sampling from the prior and comparing alternatives are essential (Chapter 10).

Bayesian methods have also been central to the development of linguistically motivated models. The stochastic Dollo model of Nicholls and Gray (2008), implemented in TraitLab, was developed for cognate data from the start and was extended to handle missing data and lateral transfer (Ryder and Nicholls, 2011); it is treated in Chapter 7. Bayesian tree samples have been used to estimate the rates of change of individual meanings (Pagel et al., 2007), the internal structure of Pama-

Nyungan (Bowern and Atkinson, 2012), the history of Semitic (Kitchen et al., 2009), Japonic (Lee and Hasegawa, 2011), Dravidian (Kolipakam et al., 2018) and Sino-Tibetan (Sagart et al., 2019; Zhang et al., 2019), and as input to phylogeographic and comparative analyses (Bouckaert et al., 2012). Structural data have been analyzed in a Bayesian framework too (Dunn et al., 2008). BEASTling (Maurits et al., 2017) lowered the barrier to entry by generating BEAST 2 configuration files from simple descriptions of linguistic datasets, and several tutorial introductions aimed at linguists are now available (Greenhill and Gray, 2009; Hoffmann et al., 2021).

Critics have raised several concerns about Bayesian linguistic phylogenetics: that results depend on priors that are hard to justify, that complex software becomes a black box, and that posterior probabilities can look more certain than the data warrant. These criticisms are not arguments against the Bayesian framework so much as arguments for using it well: reporting priors and their effects, sampling from the prior, comparing models with reliable marginal likelihood estimates, checking model adequacy, and making data and configuration files available. Chapter 15 returns to these points.

Bayesian phylogenetics in biology is moving in several directions that linguistics has barely touched. *Sequential Monte Carlo* methods build up trees gradually through a population of weighted partial trees rather than by a random walk (Bouchard-Côté et al., 2012); they can update a posterior when new languages are added without rerunning the whole analysis, which suits growing databases such as Lexibank. *Variational Bayesian phylogenetics* approximates the posterior with an optimized parametric distribution over trees (Zhang and Matsen, 2019), trading some accuracy for great gains in speed. *Gradient-based samplers* such as Hamiltonian Monte Carlo now make it possible to fit models with thousands of rate parameters, for instance a separate rate for every branch or every meaning (Ji et al., 2020). *Bayesian model averaging* by reversible-jump MCMC over substitution models (Bouckaert and Drummond, 2017) would let linguists average over lexical models rather than select one. And systematic *posterior predictive model checking* (Brown, 2014a; Höhna et al., 2018) and simulation-based calibration (Talts et al., 2018) are, to our knowledge, rarely applied to linguistic analyses, though tools exist for both. Each of these could be adopted with existing software.

- Bayesian inference combines the likelihood with priors on the tree and parameters to give a posterior distribution, which provides clade probabilities, credible intervals for dates, and samples of trees that carry uncertainty into downstream analyses.
- Priors are never uninformative on trees: topology, branch-length, tree (Yule, birth–death, coalescent, FBD) and calibration priors all shape the results, and they interact, so the effective prior must be checked by sampling without data.
- The posterior is sampled by MCMC; the Metropolis–Hastings algorithm accepts proposals with a probability that combines likelihood, prior and Hastings ratios, and a forgotten Hastings ratio produces a chain that looks healthy but samples the wrong distribution.
- Proposals must be tuned, MC^3 helps cross valleys in tree space, and convergence must be checked with trace plots, ESS, PSRF and, for topologies, ASDSF across independent runs; convergence shows that the posterior was sampled, not that the model is adequate.
- Posteriors are summarized by consensus or MCC trees with HPD intervals, but downstream analyses should be repeated over a posterior sample, because a single summary tree can distort them even when it looks plausible; clade probabilities can be overconfident when the model is wrong or branches are very short.
- Bayes factors compare models through marginal likelihoods, which depend on the priors and must be estimated by stepping-stone, path or nested sampling, never by the harmonic

mean.
- Posterior predictive checks test whether a model is adequate at all.
- Implementation validation, parameter recovery, robustness under misspecification, empirical model adequacy and historical interpretation are different questions; parameters should be judged for identifiability, practical estimability, robustness and historical interpretability.
- Most published dated language trees are Bayesian, and their most contested conclusions often turn on priors.

6.13 Exercises

Conceptual

1. Why can MCMC sample from a posterior without ever computing the marginal likelihood $f(D)$? In which part of a Bayesian analysis does $f(D)$ nevertheless matter, and why is it hard to compute there?
2. Explain, with one example each, why a uniform topology prior, independent exponential branch-length priors and a Yule tree prior are all informative about quantities a linguist might report.
3. A clade has posterior probability 0.98 in a Bayesian analysis and 70 percent bootstrap support in an ML analysis of the same cognate data. Give three possible reasons for the difference and say how you would distinguish between them.
4. Classify each of the following as implementation validation, parameter recovery, robustness under misspecification, empirical model adequacy or historical interpretation, and justify your answer: (a) a BEAST 2 run without data reproduces the specified calibration densities; (b) data are simulated with 10 percent borrowing between neighbouring languages and reanalysed under a tree model; (c) the number of singleton cognate sets in a real dataset is compared with its posterior predictive distribution; (d) a scholar asks whether an estimated root location corresponds to the homeland of a proto-language; (e) data are simulated under the fitted model and the coverage of 95 percent HPD intervals for the root age is measured.
5. A Bayes factor strongly favours model A over model B, yet model A fails a posterior predictive check. Is this contradictory? What should be reported?
6. Explain how basing a test of correlated evolution on a single MCC tree could lead to a confident but wrong conclusion, even though the MCC topology matches the accepted subgrouping.
7. Write down, for a model in which $y_i \sim \text{Normal}(\mu, \sigma)$, $\mu \sim \text{Normal}(0, 10)$ and $\sigma \sim \text{Half-Normal}(25)$, which terms are the likelihood and which the prior, how many parameters the posterior has, and Bayes' theorem for this model. Then do the same for a binary CTMC on a tree with a uniform topology prior and exponential branch-length priors, and identify which RevBayes node (`~`, `:=`, clamped) each term corresponds to (Section 6.11).

Computational

1. Use `mh_t()` with `N = 20, k = 6` and with `N = 500, k = 150`, each under `prior_rate = 10` and `prior_rate = 0.1`. Compare the posterior means of the chains with those obtained on the grid (adapt chunk `bayesian-sensitivity`), and relate the results to Figure 6.7.
2. Copy `mh_t()` and remove the Hastings term (replace `log(m)` by `0` in the acceptance test). Run the modified sampler and compare its posterior mean and histogram with the exact posterior

from the grid. Which kind of validation detects this bug, and would a trace plot reveal it?

3. Run `run_mcmc()` from four different random starting trees (`random_start(langs6)`) for 2,000 and for 20,000 iterations. Compute ESS (`effectiveSize()`), PSRF (`gelman.diag()`) and ASDSF as in chunk `bayesian-toy-diagnostics`, and describe how the diagnostics change with run length.
4. Call `run_mcmc()` with `prior_rate = 1` (mean branch length 1) instead of 10, both with and without data (`use_data = FALSE`). Compare the posterior tree length and split probabilities with those of chunks `bayesian-prior-run` and `bayesian-toy-summary`. Which quantities are sensitive to the branch-length prior?
5. Write a posterior predictive check with a different test statistic, the number of characters in which exactly one language differs from all the others, using `sim_sym()` and `post_trees` as in chunk `bayesian-ppc`. Does this statistic also detect the missing rate variation?
6. In chunk `bayesian-three-trees`, replace the uniform prior by one that gives probability 0.6 to the topology EI|GS favoured by a hypothetical specialist and 0.2 to each of the others, and recompute the posterior. Then duplicate the data matrix (`cbind(cog6, cog6)`) and repeat both analyses. How strong would the prior on EI|GS have to be to make it the most probable topology, with the original and with the duplicated data?

Research

1. You plan a dated Bayesian analysis of 50 Bantu languages, out of the roughly 500 in the family, with three historical calibrations. Choose and justify a tree prior (Yule, or birth–death with a sampling fraction), a clock model and calibration densities, and write a validation plan that includes at least one check of each of the five kinds defined in Section 6.10.
2. A published study used harmonic-mean estimates to report a Bayes factor of 40 log units in favour of a relaxed clock over a strict clock for a Uralic dataset. Describe how you would re-evaluate this claim, which software options you would use, and which additional checks (prior sensitivity, posterior predictive) would make the model comparison more convincing.
3. For an Indo-European dataset that includes Latin, Old Irish, Vedic Sanskrit and Hittite, compare the consequences of treating the ancient languages as dated tips, as constrained ancestors, or under a sampled-ancestor prior. Plan the prior-only runs and sensitivity analyses that would show whether a conclusion about the root age is driven by these choices rather than by the data. Suppose the analysis also constrains Germanic, Italic, Celtic, Baltic, Indo-Iranian and Tocharian to be monophyletic and Anatolian to be the outgroup. Which of these constraints would you defend, what can the analysis no longer be said to “recover”, and how would you show what the constraints do to the prior on the root age and on the ages of the constrained clades?

6.14 Further reading

- Nascimento et al. (2017) is a short, practical guide to Bayesian phylogenetic analysis, with clear advice on priors and convergence; Holder and Lewis (2003) compares Bayesian and traditional approaches for a general audience.
- van Ravenzwaaij et al. (2018) is a short tutorial on MCMC with R code, covering Metropolis and Metropolis-within-Gibbs samplers, burn-in, starting values, proposal width and correlated parameters; Baum and Smith (2013) give, in their chapter on tree inference, an introduction to likelihood and Bayesian tree inference for readers without a statistical back-

ground.

- Barido-Sottani et al. (2023) is a practical troubleshooting guide to phylogenetic MCMC (moves, tuning, starting values, priors and convergence tools), written for dated trees with fossils but directly applicable to dated language trees; Höhna et al. (2017) introduces RevBayes and its graphical-model approach step by step; Morey et al. (2016) explains what a Bayes factor measures and why its dependence on priors is a consequence of what it measures rather than a flaw.
- Harmon (2019), chapter 2: A short, openly available introduction to likelihood, likelihood-ratio tests, AIC and Akaike weights, Metropolis–Hastings MCMC and Bayes factors, built on one simple example, with a clear account of how AIC and Bayes factors differ.
- Yang (2014), chapters 6 to 8, gives a thorough statistical treatment of Bayesian phylogenetics, MCMC and model comparison.
- Drummond and Bouckaert (2015) is the book-length guide to BEAST 2, covering tree priors, clocks, calibrations and practical analysis.
- Chen et al. (2014) collects chapters by leading researchers on Bayesian phylogenetic methods, algorithms and applications, including marginal likelihood estimation.
- Gelman et al. (2013) is the standard reference on Bayesian data analysis in general, including MCMC diagnostics and posterior predictive checks.
- For readers new to Bayesian statistics, Lambert (2018), Kruschke (2015) and Johnson et al. (2022) are gentle, code-based introductions to priors, likelihoods, MCMC and posterior predictive checking in simple regression models, which make the machinery of this chapter easier to follow before it is applied to trees.
- Xie et al. (2011) and Lartillot and Philippe (2006) explain stepping-stone and path sampling, and why the harmonic mean fails.
- Heled and Drummond (2012) is essential reading on how calibrations and tree priors interact.
- Gray and Atkinson (2003), Gray et al. (2009) and Chang et al. (2015) are the most instructive linguistic applications for seeing how data, models and priors combine.
- Maurits et al. (2017) describes BEASTling and its approach to reproducible linguistic analyses.

Chapter 7

Models of Lexical Evolution

Every likelihood-based or Bayesian analysis of cognate data rests on a stochastic model of how words come into and go out of use along the branches of a tree. The inference machinery of Chapters 5 and 6 is generic: give it a model that assigns a probability to each column of the data matrix, and it returns trees, branch lengths and dates. The substance lies in the model. This chapter asks what the models in current use assume about lexical change, which of those assumptions matter for which historical questions, and how we can tell when a model is failing. Rather than presenting the models one after another, we first set them side by side in a single comparative framework, and then examine each in turn: the reversible binary model and its ascertainment correction, rate variation among meanings and through time, the stochastic Dollo family, models for whole meaning slots, and explicit models of replacement and synonymy.

After reading this chapter you should be able to:

- convert a table of cognate judgements into binary and multistate characters, and explain what each coding assumes about synonyms and about the dependence among cognate classes of the same meaning;
- compare the main models of lexical evolution (Table 7.2) by the process they assume, whether they allow polymorphism and recurrent innovation, how they treat borrowing, and what their parameters can and cannot identify;
- choose the ascertainment correction that matches the way a cognate matrix was filtered, and recognize the most common mismatch between data and software;
- explain why among-character rate variation and covarion switching change estimated branch lengths and dates more than topologies;
- state the stochastic Dollo assumption, the linguistic processes that violate it, and its relation to the reversible binary model;
- identify which models actually account for the dependence among cognate classes of one meaning, and what binary analyses lose by ignoring it;
- design a posterior predictive check with a test statistic relevant to a given question, and distinguish it from a Bayes-factor comparison.

7.1 From cognate sets to characters

Recall from Chapter 3 how a lexical dataset is built. The analyst chooses a list of meanings (say, the 200 meanings of the Swadesh list), collects the most basic word for each meaning in each language, and then sorts the words for each meaning into *cognate sets*: groups of words that descend from a common ancestral word by regular sound change. Table 7.1 shows a small example for the meaning 'hand' in sixteen Indo-European languages. The Germanic languages share the *hand* word, the Romance languages continue Latin *manus*, Russian, Polish and Lithuanian share a Balto-Slavic word, Irish and Welsh share a Celtic word, and Hindi *hāth* and Persian *dast* both continue an Indo-Iranian word (compare Sanskrit *hasta*, Avestan *zasta*). The last pair is a useful reminder that cognacy is a historical judgement based on regular sound correspondences, not an impression of similarity: *hāth* and *dast* look nothing alike, yet they are cognate, while English *hand* and Hindi *hāth* look superficially similar and are not.

```
hand <- data.frame(
  language = c("English", "German", "Dutch", "Swedish", "Latin", "Italian",
               "French", "Spanish", "Russian", "Polish", "Lithuanian", "Irish",
               "Welsh", "Greek", "Hindi", "Persian"),
  form = c("hand", "Hand", "hand", "hand", "manus", "mano", "main", "mano",
           "ruka", "ręka", "ranka", "lámh", "llaw", "cheri",
           "hāth", "dast"),
  class = c("A", "A", "A", "A", "B", "B", "B", "B", "C", "C", "C", "D", "D",
            "E", "F", "F"))
# binary coding: one presence/absence character per cognate class
classes <- sort(unique(hand$class))
binary <- 1 * outer(hand$class, classes, "==")
dimnames(binary) <- list(hand$language, paste0("hand_", classes))
knitr::kable(cbind(hand[, c("form", "class")], binary), row.names = TRUE,
             caption = "Words for 'hand' in sixteen Indo-European languages, their
             cognate classes, and the corresponding binary presence/absence coding
             (one character per cognate class).")
```

Table 7.1: Words for 'hand' in sixteen Indo-European languages, their cognate classes, and the corresponding binary presence/absence coding (one character per cognate class).

	form	class	hand_A	hand_B	hand_C	hand_D	hand_E	hand_F
English	hand	A	1	0	0	0	0	0
German	Hand	A	1	0	0	0	0	0
Dutch	hand	A	1	0	0	0	0	0
Swedish	hand	A	1	0	0	0	0	0
Latin	manus	B	0	1	0	0	0	0
Italian	mano	B	0	1	0	0	0	0
French	main	B	0	1	0	0	0	0
Spanish	mano	B	0	1	0	0	0	0
Russian	ruka	C	0	0	1	0	0	0
Polish	ręka	C	0	0	1	0	0	0
Lithuanian	ranka	C	0	0	1	0	0	0

	form	class	hand_A	hand_B	hand_C	hand_D	hand_E	hand_F
Irish	lámh	D	0	0	0	1	0	0
Welsh	llaw	D	0	0	0	1	0	0
Greek	cheri	E	0	0	0	0	1	0
Hindi	hāth	F	0	0	0	0	0	1
Persian	dast	F	0	0	0	0	0	1

There are two ways to turn such a table into phylogenetic characters. In **multistate coding** the meaning itself is the character, and the cognate classes A to F are its states; each language has one state (or several, if it has synonyms). In **binary coding** each cognate class becomes its own character, scored 1 in languages that have a word from that class for that meaning and 0 in languages that do not. A 200-meaning list for a few dozen languages typically yields several hundred to a few thousand binary characters. Binary coding has dominated Bayesian linguistic phylogenetics since Gray and Atkinson (2003), largely because the binary models are simple, well understood and implemented in all the major software. Multistate coding was used in the parsimony and compatibility work of Ringe et al. (2002) and is returning in several forms.

Whatever the coding, a model of lexical evolution must describe a few basic events. A new cognate class appears in a meaning slot when a language coins a new word, extends an existing word to a new meaning (semantic shift), or borrows a word; in binary terms this is a **gain** (a 0 becoming 1). A cognate class disappears from a meaning slot when the word falls out of use or is displaced; in binary terms, a **loss** (1 to 0). In multistate terms, a gain and a loss usually happen together as a **replacement**, often after a period in which old and new words coexist as synonyms.

7.2 A comparative framework for models of lexical change

The models in use differ less in their mathematics than in the answers they give to six questions about the linguistic process. Each question corresponds to a column of Table 7.2.

1. **What is the unit of observation?** A presence/absence column for one cognate class, or a meaning slot whose value is one (or several) cognate classes.
2. **What generates and removes cognate classes?** Reversible switching between “absent” and “present”, a birth-death process in which each class is born once, or replacement within a slot.
3. **Can a language have more than one word for a meaning (polymorphism)?** Synonymy is common in real word lists, and a model must either allow it, forbid it, or treat it as uncertainty about a single state.
4. **Can the same cognate class arise more than once (recurrent innovation)?** Under a reversible model a lost class can return; under Dollo-type models it cannot. Borrowing, parallel semantic shift and loose coding all produce apparent recurrence.
5. **Are the cognate classes of one meaning dependent?** If ‘hand’ gains a new word, it usually loses the old one. Most binary models ignore this; meaning-slot models build it in.
6. **How is borrowing treated?** Ignored (and absorbed by other parameters), forbidden (so that it becomes misspecification), or modelled.

Two further questions concern inference rather than process: which parameters the data can identify, and which checks reveal that a model is failing. The second part of the comparison

(Table 7.3) summarizes these, together with the software that implements each model. Rows refer to the same models in both parts; the sections below take them in turn.

Table 7.2: A comparison of models of lexical evolution, part 1: the assumed process. Rows 1-7 use binary coding (one column per cognate class); rows 8-10 model whole meaning slots.

Model	Data structure	Assumed process	Polymorphism allowed?	Recurrent innovation allowed?	Dependence among classes of one meaning	Borrowing
1. Binary CTMC, no ascertainment correction	one 0/1 column per cognate class	reversible gain and loss at constant rates; columns independent	yes; any number of 1s per meaning, including none	yes: regain after loss, parallel gains	ignored	not modelled; loans look like extra gains
2. Binary CTMC with ascertainment correction	as 1, after filtering (no all-absent columns; perhaps no constant or singleton columns)	as 1, conditioned on the column being observable	yes	yes	ignored	not modelled
3. Binary CTMC with rate variation	as 2; columns may be grouped by meaning	as 2; each column or meaning has its own rate multiplier, constant over the tree	yes	yes, concentrated in fast columns	rates shared within a meaning if partitioned; states independent	not modelled; loans can mimic fast characters
4. Binary covarion	as 2	as 2, plus a hidden on/off switch; visible change only while “on”	yes	yes, clustered within clades	ignored	not modelled; may absorb clade-specific bursts
5. Stochastic Dollo (with missing data and catastrophes)	0/1 columns; missingness recorded by language and meaning	each class born once (Poisson births) and lost independently; bursts of change at catastrophes	yes; meanings are not represented	no: homoplasy only by parallel loss	ignored	violates the model: multiple origins forced into one deep birth plus many losses

Model	Data structure	Assumed process	Polymorphism allowed?	Recurrent innovation allowed?	Dependence among classes of one meaning	Borrowing
6. Pseudo-Dollo	as 2	three-state chain: not yet born, present, lost; a lost class cannot return	yes	parallel births allowed; no regain after loss	ignored	partly absorbed as parallel births
7. Lateral-transfer stochastic Dollo	as 5	as 5, plus copying of present classes into contemporaneous lineages	yes	apparent recurrence explained by transfer	ignored	modelled explicitly
8. Multistate Mk on meaning slots	one character per meaning; states are the observed cognate classes	symmetric replacement among the K_m observed classes	no; synonyms coded as ambiguity at best	yes: return to any existing class	respected by construction	not modelled; a loan looks like an ordinary change
9. Infinite-states meaning slot	as 8	every replacement creates a new class	no, in the basic form	no, except through an explicit homoplasy term	respected by construction	an explicit term in Warnow et al. (2006); otherwise not modelled
10. Explicit replacement with synonymy	set of cognate classes for each language and meaning	words enter a slot, coexist as synonyms and are lost; loss may depend on competitors	yes, explicitly	optional (semantic shift, borrowing)	modelled explicitly	can be added as a source of new entries

Table 7.3: A comparison of models of lexical evolution, part 2: parameters, identifiability, diagnostics and software. Row numbers refer to Table 7.2.

Model	Principal parameters	Identifiability limits	Diagnostic checks	Software
1	frequency of presence π_1, overall rate, branch lengths	only rate $\times$ time; π_1 and rate biased by the missing all-absent columns	comparison with a corrected fit	any binary ("restriction site") model: MrBayes, BEAST 2, RevBayes, phangorn
2	as 1	only rate $\times$ time; the number of classes lost everywhere is not estimable	does the correction match the filter (singletons, constant columns)? site frequency spectrum	MrBayes (`coding`), BEAST 2 and BEASTling, RevBayes; phangorn and IQ-TREE `+ASC` give the variable-only (Mkv) correction
3	as 2, plus gamma shape α or per-meaning rates	α imprecise with few languages; gamma and invariant components trade off; slow meanings carry little information about their own rates	homoplasy statistics (proportion of one-step characters); per-meaning rates against word frequency	all major packages; meaning-level rates via BEASTling
4	as 3, plus switching rates s_{on}, s_{off}	switching rates weakly identified; trade off against gamma variation and branch lengths (Tuffley and Steel, 1998)	distribution of homoplasy across clades; comparison with a gamma-only fit	BEAST 2; BayesTraits; custom chains in RevBayes
5	birth rate λ (a nuisance), death rate μ, catastrophe rate and severity	only μt; λ factors out; catastrophes trade off against branch lengths	site frequency spectrum; parallel losses; cognates per language	TraitLab; BEAST (Alekseyenko et al., 2008)
6	birth and death rates, root state frequencies	birth rate trades off against root frequencies; only rate $\times$ time	as 5; comparison with covarion fits	Babel package for BEAST 2 (Bouckaert and Robbeets, 2017)
7	λ, μ, transfer rate	transfer trades off against μ and tree height; local transfer hard to separate from shorter branches; cost grows with the number of lineages	prior against posterior transfer rate; simulation with known transfer	TraitLab (Kelly and Nicholls, 2017)
8	per-meaning rate (or gamma), branch lengths	K_m arbitrary, so the meaning of the rate depends on it; invariant meanings need an ascertainment decision	classes per meaning; homoplasy per meaning	MrBayes, BEAST 2, RevBayes, BayesTraits, phangorn

Model	Principal parameters	Identifiability limits	Diagnostic checks	Software
9	per-meaning replacement rates	identifiability of the tree studied by Warnow et al. (2006); slow meanings uninformative	compatibility counts; classes per meaning	compatibility methods (Ringe et al., 2002); multistate pseudo-Dollo as an approximation; custom simulation
10	rate of entry into a slot, loss rate, strength of competition	entry, loss and the sampling of synonyms are confounded; ancestral synonymy hard to separate from borrowing (Chapter 16)	synonyms per slot; predictive checks per meaning	no standard implementation; small state spaces can be specified in RevBayes (Höhna et al., 2016)

Three points stand out. First, all seven binary models treat the cognate classes of a meaning as independent; they differ in what they assume about each column, not in how columns relate. Second, the binary models differ chiefly along two axes, rate heterogeneity (rows 3 and 4) and the single-birth assumption (rows 5 to 7), and these axes are largely independent of each other: a Dollo-type model can be given gamma rates, and a reversible model can be made covarion. Third, most identifiability limits concern time. Every model on a tree with unknown dates identifies only products of rates and times, so the step from branch lengths to years always depends on calibrations or clock priors (Chapters 9 and 10). In the vocabulary of Section 6.10, the rate parameters of these models are identifiable under the model only as rate-time products, are practically estimable mainly in aggregate, and are interpretable as historical quantities only once the coding is fixed: a "loss rate" in a binary model is not the same thing as a "replacement rate" in a meaning-slot model.

7.2.1 Coding choice and dependence within meanings

The coding decision interacts with every row of the table. Binary coding is required by models 1 to 7, single-state multistate coding by models 8 and 9, and set-valued coding (each language may have several classes per meaning) by model 10. Polymorphism is handled differently in each. In binary coding a synonym is simply a second 1 in the same meaning, so binary models accommodate synonymy without effort, which is one of their real strengths. In multistate coding a synonym can only be entered as an ambiguity, which most software reads as "one of these states, we do not know which", a different claim from "both words were in use". Ascertainment differs too. A binary matrix never contains all-absent columns and must be corrected for them (Section 7.4); a meaning-slot matrix has no all-absent pattern, but meanings in which every language has the same class are observed and informative about the rate, unless they have been dropped. Rates mean different things: a binary rate is a rate of gain or loss per cognate class, while a meaning-slot rate is a rate of replacement per meaning, and a single replacement appears in binary coding as one loss and one gain in two different columns.

Which models account for the dependence among the cognate classes of one meaning? Only the meaning-slot models of rows 8 to 10. If each language has exactly one word per meaning, the binary columns of a meaning are a deterministic re-coding of one multistate character: K columns could in principle take 2^K joint values, but only K of them can occur. Independent binary models spread probability over all 2^K values, including languages with no word at all for

'hand', and treat a replacement as two independent events. Models 8 and 9 respect the constraint by allowing only one state per slot, and model 10 represents it explicitly while also allowing synonyms. Partitioning a binary analysis by meaning (row 3) introduces dependence in *rates* among a meaning's columns, but not in their *states*. The stochastic Dollo models (rows 5 to 7) do not represent meanings at all; they treat the vocabulary as a bag of traits, and the meaning structure enters only through the treatment of missing data, in which all classes of an unrecorded meaning are unknown together (Ryder and Nicholls, 2011). The general remedy, familiar from morphology, is to combine dependent characters into one character with a structured state space, as in the correlated-evolution models of Pagel (1994) and the structured Markov models of Tarasov (2019). A meaning-slot model is exactly such a structured character for cognate classes.

What do binary analyses lose by ignoring the dependence? Their likelihood is a *composite* likelihood: a product of terms that are treated as independent although they are not. Composite likelihoods often give reasonable point estimates, which is consistent with the frequent agreement between binary and multistate topologies, but they overstate the information in the data, so that posterior intervals are too narrow. They also weight meanings by the number of cognate classes they produce, so that fast meanings dominate (Section 7.8). The size of these effects on real datasets has been studied less than it deserves (Chapter 15).

The probability that an independent binary model assigns to configurations that are impossible in synonym-free data, and the algebra relating binary and multistate codings, are worked out in Section C.1.9.

The framework also shows which differences between models matter for which question. For **topology**, the binary models usually agree once ascertainment and rate variation are handled, because the topology is driven mainly by shared presences. For **dates and branch lengths**, everything in the table matters: the ascertainment correction, the rate model, the single-birth assumption and the treatment of borrowing all change the inferred amount of change. For **rates of lexical replacement** as historical quantities, meaning-slot models give the more interpretable parameters. For **borrowing** itself, only model 7 among these, and the network models of Chapter 8, estimate it at all.

7.3 The binary continuous-time Markov chain

7.3.1 The model

The simplest model treats each binary cognate character as a two-state continuous-time Markov chain (CTMC), exactly as in Section 5.2.4. In each small interval of time dt, a character in state 0 switches to 1 with probability $q_{01}\,dt$, and a character in state 1 switches to 0 with probability $q_{10}\,dt$. The instantaneous rate matrix is

$$\mathbf{Q} = \begin{pmatrix} -q_{01} & q_{01} \\ q_{10} & -q_{10} \end{pmatrix} = \beta \begin{pmatrix} -\pi_1 & \pi_1 \\ \pi_0 & -\pi_0 \end{pmatrix}, \tag{7.1}$$

where the second form reparameterizes the two rates as an overall rate β and a stationary distribution $\pi = (\pi_0, \pi_1)$ with $\pi_0 + \pi_1 = 1$. (In Chapter 5 the gain and loss rates were called α and β; here $q_{01} = \beta\pi_1$ and $q_{10} = \beta\pi_0$.) The stationary frequency of presence is $\pi_1 = q_{01}/(q_{01} + q_{10})$. It is conventional to scale β so that one unit of branch length corresponds to one expected change per

character, which requires $2\beta\pi_0\pi_1 = 1$. The transition probabilities have the closed form derived in Chapter 5:

$$\mathbf{P}(t) = \begin{pmatrix} \pi_0 + \pi_1 e^{-\beta t} & \pi_1\left(1 - e^{-\beta t}\right) \\ \pi_0\left(1 - e^{-\beta t}\right) & \pi_1 + \pi_0 e^{-\beta t} \end{pmatrix}. \tag{7.2}$$

The intuition is worth spelling out. Think of the chain as experiencing "resampling events" at rate β, at each of which the state is redrawn from the stationary distribution π (a redraw may return the same state). With probability $e^{-\beta t}$ no resampling event has occurred along a branch of length t, and the character keeps its state; with probability $1 - e^{-\beta t}$ at least one has occurred, and the state is a fresh draw from π. As t grows, every row of $\mathbf{P}(t)$ converges to π: after a long enough time, the state of a character tells us nothing about its ancestral state. This is **saturation**, and it sets the horizon beyond which cognate data cannot resolve relationships.

The model was introduced for restriction-site data by Felsenstein (1992), which is why MrBayes and BEAST call it the *restriction site* or *binary* model. When $\pi_0 = \pi_1 = 1/2$ it is the two-state version of the Jukes-Cantor model (Jukes and Cantor, 1969), also known as the Mk model with $k = 2$ (Lewis, 2001). The likelihood of a tree τ with branch lengths $\mathbf{t}$ for a character D_i is computed by Felsenstein's pruning algorithm (Section 5.4), and the likelihood of the whole matrix is the product over characters, which assumes that characters evolve independently:

$$L(\tau, \mathbf{t}, \theta) = \prod_{i=1}^{N} P(D_i \mid \tau, \mathbf{t}, \theta). \tag{7.3}$$

The product in Equation (7.3) is where the independence of cognate classes within a meaning, discussed in Section 7.2.1, is assumed. The function below implements Equation (7.2) with the conventional scaling.

```
binary_P <- function(t, pi1, r = 1) {
  # transition probabilities of the binary CTMC, scaled so that r is the
  # expected number of changes per character per unit time
  pi0 <- 1 - pi1
  beta <- r / (2 * pi0 * pi1)
  e <- exp(-beta * t)
  matrix(c(pi0 + pi1 * e, pi1 * (1 - e),
           pi0 * (1 - e), pi1 + pi0 * e), 2, 2, byrow = TRUE,
         dimnames = list(from = c("0", "1"), to = c("0", "1")))
}
round(binary_P(0.5, pi1 = 0.3), 4)
```

```
#>      to
#> from      0      1
#>    0 0.7912 0.2088
#>    1 0.4871 0.5129
```

Figure 7.1 shows how the probability that a cognate present at the start of a branch is still present at its end, $P_{11}(t)$, decays toward π_1, and how the probability that a cognate absent at the start has

been gained, $P_{01}(t)$, rises toward π_1. When π_1 is small, as it is for real cognate data, most of the action lies in the loss rate.

```
tt <- seq(0, 3, length.out = 200)
cols <- c("#1b6ca8", "#d1495b", "#2e8b57")
pis <- c(0.1, 0.3, 0.5)
plot(NA, xlim = range(tt), ylim = c(0, 1), xlab = "branch length t",
     ylab = "transition probability", las = 1)
for (k in seq_along(pis)) {
  P <- sapply(tt, function(x) binary_P(x, pis[k])[c(4, 3)])  # P11, P01
  lines(tt, P[1, ], col = cols[k], lwd = 2)
  lines(tt, P[2, ], col = cols[k], lwd = 2, lty = 2)
  abline(h = pis[k], col = cols[k], lty = 3)
}
legend("topright", legend = paste("pi1 =", pis), col = cols, lwd = 2, bty = "n")
```

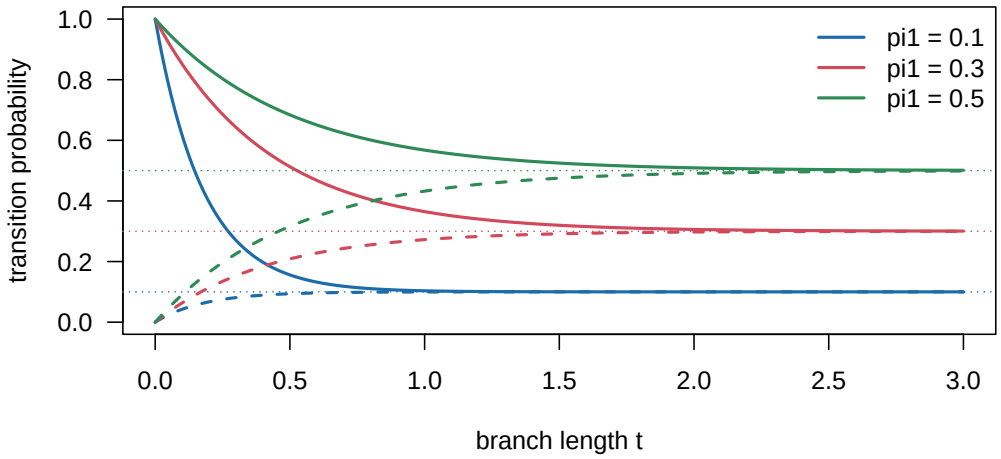


Figure 7.1: Transition probabilities of the binary CTMC as a function of branch length (in expected changes per character), for three values of the stationary frequency of presence. Solid lines: probability that a present cognate is present at the end of the branch. Dashed lines: probability that an absent cognate has been gained. Both converge to the stationary frequency (dotted).

The examples in this chapter use two simulators, `sim_ctmc_binary()` for the reversible binary CTMC (with optional gamma-distributed rates) and `sim_dollo()` for the stochastic Dollo process of Section 7.7. They are loaded silently here.

The code of the two simulators is listed in Section C.1.1; a check of Equation (7.2) against a numerical matrix exponential, and the scaling argument, are in Section C.1.2.

7.3.2 What the binary CTMC assumes about words

The binary CTMC is a convenient fiction, and it pays to be explicit about which parts are fictional. First, it allows a lost cognate to be regained. Once English replaced Old English *wēpan* with *cry* as its basic word for 'weep', the *wēpan* lineage cannot return as the basic word for that meaning except by borrowing or by a parallel semantic shift; under the CTMC, however, a 0-to-1 change at rate q_{01} is always possible. A related oddity is that the model treats absence as a state that can persist for ever in the deep past, whereas a cognate class did not exist at all before its birth. Second, the CTMC treats the many characters derived from the same meaning as independent, although within a meaning they are tightly linked (Section 7.2.1). Third, it assumes one rate for all characters and all branches, unless we add the extensions discussed below.

Why, then, has it worked as well as it has? Part of the answer is that when the stationary frequency π_1 is small, gains are rare relative to losses, and the reversible model behaves almost like an irreversible one; after ascertainment correction, the stochastic Dollo model is the limit of the binary CTMC as $\pi_1 \to 0$ (Section 7.7.4). Part of the answer is that the topology of a tree is driven mainly by shared presences, which the CTMC handles well. And part is that the CTMC is easy to extend with rate variation, relaxed clocks and partitions in standard software. "Has worked" should be read carefully, however. The usual evidence is that inferred trees recover subgroups established by the comparative method, and that agreement is not independent validation: the cognate judgements that make up the data are themselves products of the comparative method, and sharing a cognate class is not the same as sharing an innovation (Chapter 15). Agreement shows that the model extracts the signal that historical linguists put into the data; it does not show that the model's account of the process is right.

7.4 Ascertainment bias and its correction

7.4.1 Why some characters can never be observed

A cognate character is created only when at least one language in the sample has a word belonging to that cognate class. Cognate classes that existed in some ancestral language but died out in every descendant leave no trace, so the column of all zeros, which the CTMC predicts with substantial probability, never appears in the data. Section 5.6 introduced this **ascertainment bias** and showed that ignoring it inflates the estimated tree length. What is specific to lexical data is that cognate matrices are filtered in several different ways, and each way requires its own correction.

The same problem arises in biology whenever characters are selected because they vary. Restriction-site data have no all-absent columns because a site is only scored if it is cut in at least one sample (Felsenstein, 1992); morphologists usually score only characters that vary among the taxa studied (Lewis, 2001); and SNP datasets contain, by construction, only variable sites (Leaché et al., 2015).

7.4.2 The correction formula

The remedy is to condition the likelihood on the fact that each character is observable. Let $\mathcal{U}$ be the set of patterns that could never enter the dataset. The ascertainment-corrected likelihood is

$$L_{\text{asc}}(\tau, \mathbf{t}, \theta) = \prod_{i=1}^{N} \frac{P(D_i \mid \tau, \mathbf{t}, \theta)}{1 - \sum_{u \in \mathscr{U}} P(u \mid \tau, \mathbf{t}, \theta)}. \tag{7.4}$$

The denominator is the same for every character, so the correction amounts to subtracting $N \log\bigl(1 - \sum_u P(u)\bigr)$ from the log-likelihood, and the probability of each unobservable pattern is computed with the same pruning algorithm used for the data. Different choices of $\mathscr{U}$ correspond to different ways of building a dataset:

- **No-absence correction**: $\mathscr{U} = \{\mathbf{0}\}$. This is the natural choice for cognate data in which every cognate class attested in at least one language is retained, including classes attested in one language only (*singletons*) and classes attested in all languages. It corresponds to the `noabsencesites` coding option of MrBayes (Ronquist et al., 2012b), and an equivalent correction is available for binary cognate data in BEAST 2 (Bouckaert et al., 2014; Maurits et al., 2017).
- **Variable-only correction**: $\mathscr{U} = \{\mathbf{0}, \mathbf{1}\}$. This is the Mkv model of Lewis (2001), appropriate when characters present in all languages have also been discarded.
- **Informative-only correction**: $\mathscr{U}$ contains all constant and singleton patterns, appropriate when only parsimony-informative characters were retained.

Each correction is right only for the corresponding way of building the dataset. Applying the variable-only correction to data that still contain all-present columns, for example, is itself a misspecification. In the SNP literature, Leaché et al. (2015) compare the conditional correction of Equation (7.4), which follows Lewis, with alternatives due to Felsenstein that require knowing (or estimating) how many invariant sites were dropped. For cognate data the number of cognate classes that died out everywhere is unknowable: in the terms of Section 6.10, it is not identifiable under any of the models, and the conditional likelihood is the device that lets the analysis proceed without it.

Missing data complicate matters. If a language has no recorded word for a meaning, its entries for all cognate classes of that meaning are unknown, not absent. Most software treats such entries as "?" and sums over both states at that leaf. But the set of unobservable patterns now depends on which languages are missing for each meaning, so a single denominator is no longer exact. Ryder and Nicholls (2011) work through this problem for the stochastic Dollo model and show how to condition correctly when the pattern of missingness varies across meanings.

The `ASC = TRUE` option of `phangorn::pml()` implements the Lewis (variable-only) correction, which excludes both constant patterns. It is correct for data from which all-present as well as all-absent columns were deleted, as in the example in Section 7.5 below, but not for the usual cognate matrix that retains universally shared cognates. Read the documentation of your software to find out which patterns its correction excludes, and make sure your data were filtered in exactly that way. A mismatch between the filtering of the data and the correction applied is one of the most common silent errors in phylogenetic analyses of discrete characters.

The conditional-likelihood argument behind Equation (7.4), the denominators for each filtering scheme and for meaning-specific missing data, and a hand-coded pruning likelihood with a no-absence correction are in Section C.1.3. There, a parameter-recovery simulation (Table C.1) shows that the uncorrected fit overestimates both the frequency of presence and the rate, while the corrected fit recovers them.

7.5 Rate heterogeneity among characters

7.5.1 Why meanings differ in rate

The words for some meanings are replaced quickly and the words for others hardly at all. Across Indo-European, the words for ‘two’, ‘three’, ‘who’ and ‘I’ fall into very few cognate classes, while the words for meanings such as ‘dirty’, ‘to turn’ or ‘to throw’ fall into a dozen or more. Pagel et al. (2007) estimated a separate rate of replacement for each of the 200 Swadesh meanings across 87 Indo-European languages and found that the rates spanned roughly two orders of magnitude. A single-rate model is therefore badly wrong. It forces fast and slow characters onto the same clock, so that the fast characters saturate and the slow ones look anomalously conservative. The usual consequence is that deep branches are underestimated: the model reads saturation as recency.

7.5.2 The gamma model

The standard remedy, borrowed from molecular phylogenetics (Yang, 1994), is to let each character i evolve at its own relative rate ρ_i, a random variable drawn independently from a gamma distribution with mean one and shape parameter α:

$$f(\rho \mid \alpha) = \frac{\alpha^{\alpha}}{\Gamma(\alpha)}\, \rho^{\alpha-1} e^{-\alpha\rho}, \qquad \mathrm{E}[\rho] = 1, \quad \mathrm{Var}[\rho] = 1/\alpha. \tag{7.5}$$

When α is large, all rates are close to one. When $\alpha < 1$, the distribution is L-shaped, with many very slow characters and a long tail of fast ones. The character likelihood becomes an integral over the unknown rate, approximated in practice by K (usually four) equally probable categories, each represented by its mean rate (Section A.4.3). A proportion of invariant characters can be added (the “+I” of models such as “GTR+I+G”), although for cognate data this is rarely sensible: a cognate present in all languages is not invariant in any meaningful sense, and the gamma and invariant components are notoriously hard to separate. Figure 7.2 shows three gamma distributions and their four-category approximations.

```
library(phangorn)
par(mfrow = c(1, 3), mar = c(4, 4, 2, 1))
for (a in c(0.3, 1, 4)) {
  x <- seq(0.01, 4, length.out = 300)
  plot(x, dgamma(x, a, a), type = "l", lwd = 2, col = "#1b6ca8",
       ylim = c(0, 2), xlab = "relative rate", ylab = "density", las = 1,
       cex.axis = 1.15, cex.lab = 1.15, cex.main = 1.15,
       main = bquote(alpha == .(a)))
  rug(discrete.gamma(a, 4), lwd = 3, col = "#d1495b", ticksize = 0.08)
}
```

The shape parameter has a direct linguistic reading. A small α says that a minority of characters accounts for most of the cognate turnover and that many characters change very rarely. Gray and Atkinson’s influential dating of Indo-European (Gray and Atkinson, 2003) used the binary restriction-site model with gamma-distributed rates in MrBayes, and almost every later Bayesian analysis of cognate data has included some form of among-character rate variation.

The number of categories deserves a moment’s thought, because it is easy to mistake for a sub-

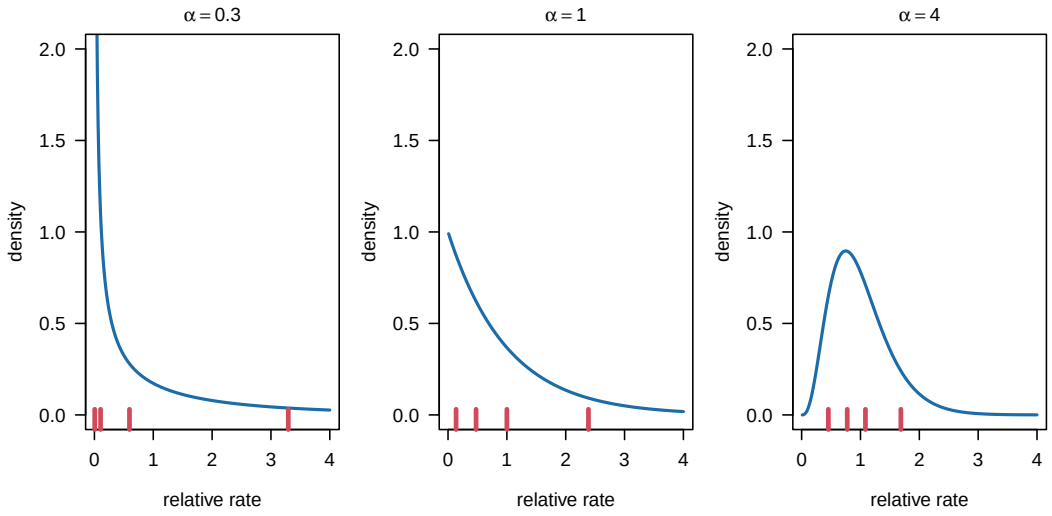


Figure 7.2: Gamma distributions of relative rates for three shape parameters (curves) and the mean rates of the four equiprobable categories used to approximate them (ticks along the bottom of each panel). Small shape parameters imply many nearly frozen characters and a few very fast ones.

stantive choice. Four equiprobable categories are a convention that approximates the continuous gamma well enough for most purposes; the categories are not four classes of meanings. Comparing two, four or eight categories by Bayes factors therefore measures how well the approximation works, not how many "rate classes" the vocabulary has, and once the approximation is adequate the differences are of the order of the Monte Carlo error of the estimates. A genuinely different model is obtained when the category rates and weights are both estimated, the *free-rate* model of Yang (1995), called *FreeRate* by Soubrier et al. (2012) and available in IQ-TREE as `+R`. It can represent rate distributions that the gamma cannot, such as a block of nearly frozen meanings beside a fast-changing remainder. Simulation studies of molecular data show that misspecified rate heterogeneity can bias rate estimates in ways that depend on the time scale (Soubrier et al., 2012), which makes the choice relevant to dating. Script `17_rate_categories.Rev` of Appendix H compares these options on data simulated with continuous gamma rates (Section H.3.18): two categories fit clearly worse than four, eight are indistinguishable from four, and neither free weights nor FreeRate improves on the gamma when the true rates are gamma-shaped. A maximum-likelihood version of the same experiment, which also shows how the estimated tree length creeps up as categories are added, is in Section C.1.4.

7.5.3 Fitting rate-variable models with phangorn

The next chunk simulates data with strong rate variation ($\alpha = 0.4$) on a random 20-language tree, removes both kinds of constant columns so that the Lewis correction implemented in **phangorn** is the right one, and compares a single-rate model with a four-category gamma model by maximum likelihood and AIC (Chapter 5). Tree topology and branch lengths are estimated in both cases. In the vocabulary of Section 6.10, the experiment has two parts: for the gamma model it is a parameter-recovery simulation, since the data were generated under that model's own assumptions; for the single-rate model it tests robustness under misspecification, since the data contain rate variation that the model ignores.

Branch lengths under `ASC = TRUE` in phangorn. In our tests with **phangorn** 2.12.1, `optim.pml()` did not reliably optimize branch lengths for a model fitted with `ASC = TRUE`. Started from a tree already fitted without the correction it could leave the branch lengths unchanged, and from other starting trees it stopped short of the maximum of the corrected likelihood; `update()` on such a fit also silently dropped the correction. Chapter 5 met the same problem (Section 5.6). We therefore use `optim.pml()` for the topology search only and then re-optimize all branch lengths, and the gamma shape where there is one, directly on the corrected log-likelihood with the small helper `refine_asc_edges()` defined in the book's `R/setup.R`. The NNI search relies on the same routines, so its topology is best treated as a good tree rather than a guaranteed optimum. Later versions of **phangorn** may behave differently; comparing the log-likelihood before and after such a refinement is a quick check. IQ-TREE and RAxML-NG (Section 17.13.4) implement the same correction.

```
set.seed(11)
tr20 <- rtree(20, tip.label = paste0("L", 1:20))
Xg <- sim_ctmc_binary(tr20, 2000, pi1 = 0.5, rate = 1, shape = 0.4)
Xv <- Xg[, colSums(Xg) > 0 & colSums(Xg) < nrow(Xg)]   # variable columns only
pd <- phyDat(Xv, type = "USER", levels = c("0", "1"))
start <- NJ(dist.hamming(pd))
ctl <- pml.control(trace = 0)
fit_eq <- optim.pml(pml(start, pd, ASC = TRUE), optNni = TRUE, control = ctl)
fit_g  <- optim.pml(pml(start, pd, ASC = TRUE, k = 4), optNni = TRUE,
                    optGamma = TRUE, control = ctl)
# optim.pml() keeps the NNI topology; branch lengths (and, for the gamma
# model, the shape) are re-optimized on the corrected likelihood (see text)
fit_eq <- refine_asc_edges(fit_eq)
fit_g  <- refine_asc_edges(fit_g, opt_shape = TRUE)
tab <- data.frame(
  model = c("binary, single rate", "binary + Gamma(4)"),
  logLik = c(fit_eq$logLik, fit_g$logLik),
  AIC = AIC(fit_eq, fit_g)$AIC,
  alpha = c(NA, fit_g$shape),
  tree_length = c(sum(fit_eq$tree$edge.length), sum(fit_g$tree$edge.length)),
  RF_to_truth = c(RF.dist(fit_eq$tree, tr20), RF.dist(fit_g$tree, tr20)))
knitr::kable(tab, digits = c(0, 1, 1, 2, 2, 0), caption = "Maximum-likelihood fits
of a single-rate and a gamma rate-variation model to binary data simulated with
strong rate heterogeneity (true shape 0.4, true tree length about 16.8). RF is the
Robinson-Foulds distance to the true topology.")
```

Table 7.4: Maximum-likelihood fits of a single-rate and a gamma rate-variation model to binary data simulated with strong rate heterogeneity (true shape 0.4, true tree length about 16.8). RF is the Robinson-Foulds distance to the true topology.

model	logLik	AIC	alpha	tree_length	RF_to_truth
binary, single rate	-19886	39846	NA	7.63	0
binary + Gamma(4)	-19326	38727	0.43	19.27	4

```
c(true_tree_length = sum(tr20$edge.length))
```

```
#> true_tree_length
#>            16.75
```

The gamma model is strongly preferred by AIC, and the most striking difference lies in the total tree length. The single-rate model, unable to accommodate the fast characters except by treating them as saturated noise, collapses the tree to less than half its true length (46%). The gamma model estimates the shape parameter close to its true value of 0.4 and comes much nearer the true length, though it overshoots it by about 15%; errors of this size in total tree length are not unusual for a dataset of this size, because the shape parameter and the tree length are hard to estimate separately. Applied to a dated analysis, a bias of the first kind would make a family look much younger than it is. The topology, by contrast, is close to the truth under both models in this easy case, differing from it by at most a couple of splits. This is a general pattern: rate heterogeneity matters most for branch lengths and dates, and for topology mainly when branches are long or unequal. The simulation demonstrates the direction of the bias; it does not show how large the bias is for any real dataset, which depends on the true distribution of rates.

7.5.4 Beyond the gamma: partitions and meaning-specific rates

The gamma model treats rates as exchangeable random effects: each character's rate is drawn from the same distribution, without reference to what the character is. An alternative is to use what we know about the characters. All binary characters derived from the same meaning can share a rate, so that the analysis estimates one rate per meaning (a **partitioned** model), or meanings can be grouped into a few rate classes, for example by semantic domain or part of speech. Since all the cognate classes of a meaning are linked by the process of replacement, a per-meaning rate is linguistically more natural than independent per-character rates, and it is the one step toward within-meaning dependence that binary models can take (Section 7.2.1). BEASTling makes it straightforward to request meaning-specific rate variation in BEAST 2 (Maurits et al., 2017), and the practical guide of Hoffmann et al. (2021) discusses the available options. Estimating hundreds of separate rates costs precision, of course, so such models are best combined with a hierarchical prior that shrinks meaning rates toward a common distribution. That brings us back to something like the gamma model, now applied at the level of meanings rather than cognate classes.

Partitioning can also separate data types. Lexical, phonological and grammatical characters change at very different rates and by different mechanisms (Greenhill et al., 2017), and a combined analysis should allow each partition its own substitution model, rate and possibly clock. Chapter 9 returns to rate differences among data types; Chapter 16 discusses total-evidence analyses that combine them.

7.5.5 Word frequency as a rate predictor

A further step is to model rate variation as a function of a measured covariate. The best known example is word frequency. Pagel et al. (2007) related per-meaning rates of cognate replacement in Indo-European to frequencies of use in corpora of four modern Indo-European languages and found that frequency alone accounted for about half of the variation in rates; Calude and Pagel (2011) showed that the frequencies of Swadesh-list meanings are strongly correlated across languages from different families. Chapter 9 discusses this work and the debate over "ultraconserved

words”. From a modelling point of view, the frequency effect suggests a regression model for rates, $\log \rho_m = \beta_0 + \beta_1 \log(\text{frequency}_m) + \varepsilon_m$ for meaning m, embedded within the phylogenetic likelihood, so that frequency information sharpens rate estimates for meanings with few cognate changes. To our knowledge, covariate-dependent rate models of this kind have rarely if ever been built directly into a tree-inference analysis of cognate data; the published studies estimate rates first and relate them to frequency afterwards, which ignores the uncertainty in the estimated rates.

Frequency and the rate of lexical replacement in Indo-European. Pagel et al. (2007) analyzed the Indo-European Swadesh-list data compiled by Dyen and colleagues for 87 languages. Rather than coding each cognate class as a binary character, they treated each meaning as a multistate character, the states being its cognate classes, and estimated its rate of replacement across a Bayesian sample of trees. The rates varied enormously, from meanings expected to retain the same cognate for many millennia to meanings replaced every few centuries. Corpus frequencies predicted these rates well, with frequent words changing slowly, and the relation held within parts of speech as well as across them. The study made two methodological points. First, rate heterogeneity among meanings is not noise but a lawful property of vocabulary, which justifies modelling it explicitly. Second, because rates differ predictably, the rate of a meaning is partly knowable in advance, which could in principle be used to build informative priors for dating. Because the rates were estimated with a multistate model (row 8 of Table 7.2), they are rates of replacement per meaning and are not directly comparable with the per-class loss rates of binary analyses.

7.6 Covarion models: rates that change through time

The gamma model lets characters differ in rate but holds each character’s rate constant over the whole tree. That is unrealistic for words. A meaning can be stable for thousands of years and then enter a period of rapid turnover, for instance when taboo, contact with a prestige language or a change in material culture makes the old word unsuitable. Conversely, a meaning that is volatile in one subfamily may be stable in another. In biology this phenomenon is called **heterotachy**, a change in the rate of a site through time (Lopez et al., 2002). Its classic formulation is the **covarion** (“concomitantly variable codon”) hypothesis of Fitch and Markowitz (1970), who proposed that at any moment only a subset of the sites in a gene are free to vary, and that the set of variable sites changes as the gene evolves.

Tuffley and Steel (1998) turned the idea into a tractable Markov model by giving each site a hidden switch that is either “on” (free to change) or “off” (frozen), and letting the switch itself flip at random. Galtier (2001) and Huelsenbeck (2002) developed related covarion-like models and tests for molecular data. For binary characters the state space has four states, absent and present while on and absent and present while off. The visible state changes only while the switch is on, and since the data record only the visible state, the likelihood sums over the hidden switch at each leaf. The gamma model and the covarion model can be combined, giving characters that differ in mean rate and also switch between active and inactive periods.

The binary covarion model is available for cognate data in BEAST 2 and was used, for example, by Bouckaert et al. (2012) for Indo-European, alongside other substitution models. In BayesTraits, Pagel and Meade implemented a closely related covarion model for discrete characters. One frequently overlooked consequence of the covarion model concerns how fast similarity decays with time, which is what drives date estimates. Figure 7.3 shows the expected proportion of

binary characters at which two languages that separated t units ago differ, under a single-rate model, a gamma model and a covarion model with the same mean rate.

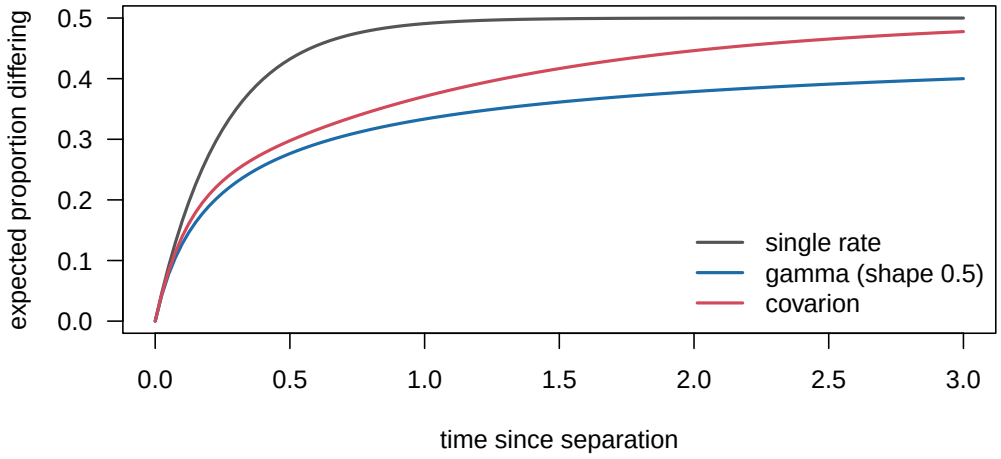


Figure 7.3: Expected proportion of binary characters that differ between two languages as a function of the time since they separated, under a single-rate binary CTMC, a gamma model (shape 0.5) and a covarion model (switching rates 0.5), all with the same mean rate and equal state frequencies. Rate heterogeneity of either kind slows saturation, so that a given level of divergence implies a greater age.

Under a single rate, divergence saturates at one half within about one unit of time. Under the gamma and covarion models it keeps rising slowly for much longer, because some characters (the slow ones, or the ones switched off) retain information about deep splits. Inverting these curves to infer age from divergence gives very different answers in the saturated region, which is why the choice of rate model has consequences for the dating controversies discussed in Chapters 10 and 15. It is also why the rate model is part of what a date estimate depends on: two analyses with the same calibrations but different rate models can place the same root at different ages, and in the saturated region the data alone cannot say which is right.

The covarion rate matrix (Equation (C.1)), its stationary distribution and scaling, the closed-form divergence curves for the single-rate and gamma models, and the code for Figure 7.3 are in Section C.1.5.

Gamma rates and covarion switching are not interchangeable even when they give similar divergence curves. Under the gamma model a character is consistently fast or slow across the whole tree, so the fast characters are fast in every subfamily. Under the covarion model a character's rate is correlated only among closely related lineages. The two models therefore make different predictions about how homoplasy is distributed across clades, which is what allows the data to distinguish them. With few languages, however, that information is thin, and the switching rates are among the least practically estimable parameters in Table 7.3.

7.7 The stochastic Dollo model

7.7.1 Dollo's law and cognate data

The palaeontologist Louis Dollo argued in 1893 that evolution does not retrace its steps: a complex structure, once lost, is not regained in the same form (Dollo, 1893). In phylogenetics, *Dollo characters* are characters that may be gained only once but lost any number of times. Dollo parsimony, which finds the tree requiring the fewest losses given a single gain per character, was formalized by Farris (1977). Cognate classes are almost ideal Dollo characters. A cognate class is, by definition, the set of words descended from one ancestral word, and a word is coined once. It can be lost many times independently, as Latin *ignis* 'fire' was lost in favor of descendants of *focus* across most of Romance, but it cannot arise twice by inheritance. (Borrowing and parallel semantic shift can make a cognate class appear in a meaning slot more than once, which is why the Dollo assumption is not quite exact for meaning-based cognate coding; Section 7.7.5 returns to this.)

7.7.2 The Nicholls-Gray model

Nicholls and Gray (2008) built a stochastic version of Dollo's law, now called the **stochastic Dollo model**, and implemented it in the program TraitLab. The model is a birth-death process for traits (cognate classes) running along the branches of a rooted, dated tree:

- On every lineage, new traits are born as a Poisson process with rate λ per unit time. Each new trait is a new cognate class, never seen before.
- Every trait present in a lineage dies (is lost) independently at rate μ.
- At a speciation event, both daughter lineages inherit all traits present in the parent.

There is no rate of gain for an existing trait: once a trait has died in a lineage, it is gone from that lineage for ever. If the process has run for a long time before the root, the number of traits present at the root is Poisson with mean λ/μ, the stationary size of the meaning-cognate "vocabulary".

The likelihood is derived by thinking about where each trait was born. A trait observed with a particular pattern of presences among the leaves must have been born on a branch above the most recent common ancestor of the languages that have it, and must then have died on exactly the branches needed to produce the pattern. Because births form a Poisson process, the number of traits showing each pattern x is Poisson, with a mean Λ_x obtained by summing, over all the branches on which the trait could have been born, the expected number of births there that survive and then produce exactly x. The probability of producing x from a given node is computed by a pruning recursion that contains only deaths.

Three features of this likelihood deserve comment. First, ascertainment is built in. The all-absent pattern simply does not appear, and the Poisson form accounts automatically for the fact that we see only traits that survived somewhere. Second, the birth rate λ enters only as a multiplier, so that, conditional on the number of observed traits, the pattern probabilities do not depend on λ at all; λ is a nuisance parameter that says how large the vocabulary is, not how fast it changes. Third, as for any model on a dated tree, only the products μt are identifiable from the data alone, so absolute dates require calibrations (Chapter 10).

The full stochastic Dollo likelihood (Equations (C.2) to (C.5)), a hand-coded implementation and a profile likelihood for the death rate (Figure C.1) are in Section C.1.6.

Nicholls and Gray (2008) added two features that matter for real data. They allowed for **catastrophes**, points on branches at which a burst of trait deaths and births occurs, modelling episodes of rapid lexical change. And they handled the way the data had been *registered*, for example the fact that some datasets exclude cognate classes found in only one language, by conditioning the likelihood on the corresponding observability criterion. Ryder and Nicholls (2011) extended the model to handle missing data properly, which required replacing the single observability condition by one that depends on which languages are unrecorded for each meaning. Alekseyenko et al. (2008) developed related machinery for stochastic Dollo likelihoods and made the model available in BEAST, which is the route by which many later analyses accessed it. A further extension to lateral transfer between lineages (Kelly and Nicholls, 2017) is described in Chapter 8.

Dating Proto-Indo-European with the stochastic Dollo model. Nicholls and Gray (2008) applied the stochastic Dollo model to two Indo-European lexical datasets, the Swadesh-list data of Dyen and colleagues and the dataset of Ringe et al. (2002), using historically attested split times as calibrations. Their age estimates for the root were broadly consistent with the earlier estimates of Gray and Atkinson (2003) under the reversible model, and so with an Anatolian rather than a steppe-age origin, although the stochastic Dollo model dispenses with the reversible model's implausible re-gains of lost cognates. Ryder and Nicholls (2011) revisited the problem with an explicit treatment of missing data and found root ages in the same broad range. The interest of these studies lies less in the particular dates than in showing that the conclusion did not hinge on the reversibility assumption criticized in early reactions to Gray and Atkinson (2003). Robustness to one assumption is not robustness to all, however: the analyses shared much of their data, coding and calibration information, so their agreement leaves those choices untested. Chapter 10 discusses the dating debate as a whole, including later analyses that reached different conclusions with different data and constraints (Chang et al., 2015; Heggarty et al., 2023).

7.7.3 How Dollo and reversible data differ

What does the single-birth assumption change in the data themselves? Two diagnostic features are easy to compute. The first is the *site frequency spectrum*: the distribution of the number of languages in which each cognate class is present. The second is homoplasy measured by the Fitch parsimony score of each character on the true tree. Under the stochastic Dollo model a character can need several steps only through parallel losses; under the reversible model parallel gains also occur. The chunks below simulate both kinds of data on the same 12-language tree, with parameters chosen to give a comparable number of cognates per language. This is not a test of either model: it shows what each model predicts, and therefore which statistics could discriminate between them in a check of empirical adequacy (Section 7.10).

```
set.seed(7)
tr <- rcoal(12, tip.label = paste0("L", 1:12))
tr$edge.length <- tr$edge.length / max(node.depth.edgelength(tr))  # root age 1
set.seed(7)
Xd <- sim_dollo(tr, lambda = 100, mu = 0.8)   # stochastic Dollo data
dim(Xd)
```

```
#> [1]  12 430
```

```
set.seed(8)
Xr <- sim_ctmc_binary(tr, 3000, pi1 = 0.25, rate = 1)
```

```
Xr <- Xr[, colSums(Xr) > 0]
sfs <- rbind(Dollo = tabulate(colSums(Xd), 12) / ncol(Xd),
             reversible = tabulate(colSums(Xr), 12) / ncol(Xr))
barplot(sfs, beside = TRUE, names.arg = 1:12, col = c("#d1495b", "#1b6ca8"),
        las = 1, xlab = "number of languages with the cognate",
        ylab = "proportion of characters", legend.text = TRUE,
        args.legend = list(bty = "n", x = "topright"))
```

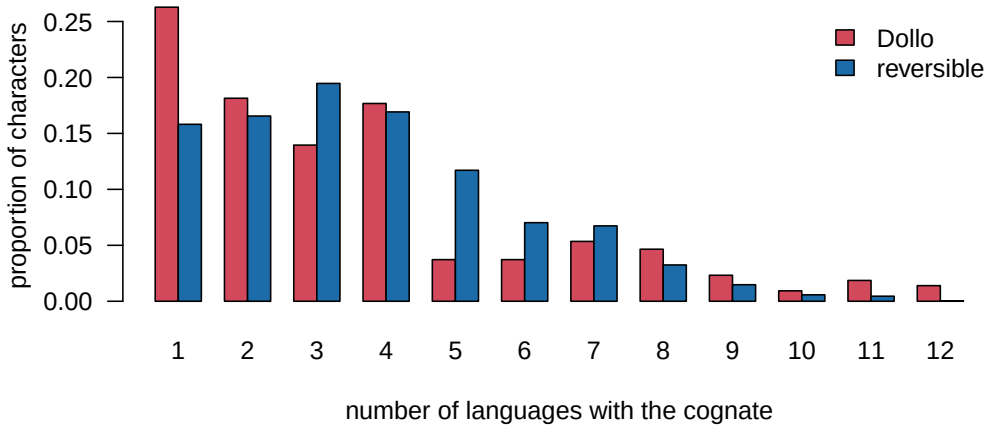


Figure 7.4: Site frequency spectra (the proportion of cognate classes present in k of the 12 languages) for data simulated on the same tree under the stochastic Dollo model and under the reversible binary CTMC. All-absent characters are excluded from both, as in real data.

```
fitch_steps <- function(tree, X) {
  pd <- phyDat(X, type = "USER", levels = c("0", "1"))
  parsimony(tree, pd, site = "site")[attr(pd, "index")]
}
steps <- rbind(Dollo = table(factor(pmin(fitch_steps(tr, Xd), 3), 0:3)),
               reversible = table(factor(pmin(fitch_steps(tr, Xr), 3), 0:3)))
colnames(steps) <- c("0 (all present)", "1", "2", "3+")
round(prop.table(steps, 1), 3)
```

```
#>            0 (all present)     1     2    3+
#> Dollo                0.014 0.767 0.191 0.028
#> reversible           0.000 0.418 0.399 0.183
```

Under the stochastic Dollo model, about three quarters of the characters fit the tree with a single step (one birth, or one loss of a cognate inherited from the root), and characters requiring three or more steps are rare, because they require several independent losses of the same cognate. Under the reversible model on the same tree, fewer than half of the characters fit with a single step and a substantial share need three or more, because the same cognate class is re-gained in unrelated lineages. The spectra differ too: the reversible model produces many more cognates shared by in-

termediate numbers of languages scattered across the tree. These are exactly the kinds of statistics one can use to check which model better describes real data (Section 7.10).

7.7.4 The Dollo model as a limit of the reversible model

There is a precise connection between the two models. Take the reversible binary CTMC with the no-absence ascertainment correction, and let the frequency of presence π_1 go to zero while holding the loss rate fixed at μ. Gains become so rare that any observable character almost certainly arose from a single gain, and, after conditioning on observability, the pattern probabilities of the corrected CTMC converge to those of the stochastic Dollo model. The stochastic Dollo model is thus a boundary case of the ascertainment-corrected reversible model, not a completely different model. This helps explain why the two often give similar topologies and dates for data with low cognate frequencies, and why they diverge when apparent multiple origins are common. Fitting both models to data simulated under each confirms the limit: on Dollo data the reversible model reaches essentially the same likelihood by pushing π_1 toward zero, while on reversible data the Dollo model fits far worse, because it can explain re-gained cognates only by births deep in the tree followed by many parallel losses.

The limit argument and the cross-fitting experiment (Table C.3) are in Section C.1.7.

7.7.5 When the Dollo assumption fails

The stochastic Dollo model assumes that every cognate class in a meaning slot has a single origin. Three linguistic processes violate this. **Borrowing** introduces a cognate class into a lineage from outside, so that the class appears to have been born twice. **Parallel semantic shift** recruits the same inherited word into the same meaning independently in different branches. Shifts such as ‘shell’ or ‘pot’ to ‘head’ (the source of French *tête*, from Latin *testa*) or ‘skin’ to ‘bark’ recur across languages, and when the same inherited word undergoes the same shift in two lineages, the result looks like two births of one cognate class in that meaning slot. **Coding artefacts**, such as root-level cognate judgements that group words sharing a remote etymological root but not a common ancestral word in that meaning, produce the same effect. Under the stochastic Dollo model, a cognate class with multiple origins forces its birth back to a common ancestor of all languages that have it, followed by many losses. That inflates the inferred death rate and pushes births, and possibly root ages, deeper into the past. Under the reversible model, the same pattern is absorbed by extra gains. Chang et al. (2015) draw attention to precisely this problem, which they call *advergence*: the independent development of the same lexical item in the same meaning in different lineages. They argued that advergence is common enough in Indo-European to bias dated analyses, and that it should be handled by careful coding or by constraints. Chapter 8 treats the borrowing case and the lateral-transfer extension of the stochastic Dollo model (row 7 of Table 7.2).

7.7.6 Pseudo-Dollo and other compromise models

The full stochastic Dollo likelihood is awkward to combine with the rest of a modern Bayesian toolkit, because it is not a standard CTMC that can be dropped into the pruning machinery used for relaxed clocks, partitions and rate heterogeneity. Bouckaert and Robbeets (2017) proposed *pseudo-Dollo* models, which approximate the single-birth behaviour of the stochastic Dollo model within a standard Markov-chain framework. The pseudo-Dollo chain has three states: absent because the cognate has not yet been born, present, and absent because it has been lost. Births lead from

the first state to the second, deaths from the second to the third, and nothing leads back from the third; the two absent states look the same in the data. A lost cognate therefore cannot return to a lineage, as in the stochastic Dollo model, but the same cognate can be born independently on two lineages that never had it, so the model tolerates some apparent multiple origins. Bouckaert and Robbeets (2017) also described multistate and covarion variants and implemented the models in the Babel package for BEAST 2. A cheaper compromise, used implicitly in many analyses, is the reversible model with a low frequency of presence, which, as we have just seen, approaches Dollo behaviour. For any of these models, the relevant empirical question is not whether they are true but whether their predictions about the data, such as the site frequency spectrum and the distribution of homoplasy, match what is observed (Section 7.10).

7.8 Multistate meaning-slot models

7.8.1 Meanings as characters

The alternative to binary coding is to treat each meaning as one character whose states are its cognate classes. This is closer to how historical linguists think: a language has *a* word for ‘hand’, and the question is which lineage of words fills the slot. The natural Markov model for K states is the Mk model of Lewis (2001), the K-state generalization of Jukes-Cantor, in which every state changes to every other at the same rate (Equation (C.6) in Section C.1.8). Multistate models were used in the parsimony and compatibility analyses of Ringe et al. (2002) and in the likelihood-based estimates of per-meaning rates by Pagel et al. (2007), and the Mk model is available in BEAST 2, MrBayes and RevBayes. But applying the Mk model to meanings raises several problems that have no exact counterpart in molecular data.

1. **The number of states differs among meanings and is arbitrary.** One meaning may have two cognate classes in the sample and another twenty. The Mk model with K equal to the observed number of classes assumes that the process could only ever have visited those classes, which is false: every replacement creates a new class.
2. **Changes between existing states are implausible.** Under Mk, a language whose word for ‘hand’ belongs to class B can switch directly to class F, the class of a distant relative, as easily as to any other. Linguistically, replacements almost always introduce a *new* class (an innovation), not one already present elsewhere in the family, except by borrowing.
3. **Synonyms (polymorphism).** Languages often have two basic words for a meaning. Multistate models must either pick one, which discards information, or treat the character as ambiguous between the states, which most software handles by summing over them, as if only one of them were the true state.

The model that matches the linguistic process best is an *infinite-states* (or *infinite-alleles*, in population-genetic terms) model in which each replacement creates a brand-new class. Warnow et al. (2006) formulated a stochastic model of this kind for lexical and other linguistic characters, allowing a small amount of homoplasy and borrowing, and studied its identifiability. The infinite-states process is, at the level of cognate classes, a Dollo process: a class is born once and dies once in each lineage where it is replaced. The binary stochastic Dollo model and the infinite-states meaning-slot model are therefore close relatives. The main difference is that the meaning-slot model ties births and deaths together (a birth replaces exactly one existing word), whereas the binary model lets the number of cognates per meaning per language fluctuate freely.

This *Multistate model*, as Rönchen et al. (2024) call it, was formulated two decades ago but has

rarely been used for inference, partly because its likelihood is not standard: the states are cognate classes drawn from an unbounded set, so the pruning algorithm over a fixed rate matrix does not apply directly. Rönchen et al. (2024) gave a general recursive algorithm for the likelihood, generalizing an algorithm of Warnow et al. (2006), and argued that the model represents lexical replacement through time more realistically than the stochastic Dollo model does. Their case study was deliberately small, a dataset of five ancient Indo-European languages (Vedic Sanskrit, Avestan, Ancient Greek, Old Church Slavonic and Old English), because they also identified the model's main weakness. The Multistate model allows neither re-gains nor synonyms, so an undetected loan, or a semantic shift that leaves two cognate classes competing for one slot, makes the data for that meaning contradictory under the model, and the risk of such conflicts grows with the number of languages (Rönchen et al., 2024; Rönchen, 2025). A model that is more realistic on average can be less robust to the errors that real datasets contain, a trade-off that recurs throughout this chapter.

7.8.2 Binary or multistate? A simulation

The simulation below makes the dependence discussed in Section 7.2.1 concrete. We simulate meanings with different rates of replacement under an infinite-states model (row 9 of Table 7.2), convert them to binary characters, and count. In the vocabulary of Section 6.10, this sets up a misspecification: data generated by a meaning-slot process are presented in the form that binary models, which assume independent columns, would analyze.

```
sim_meanings <- function(tree, rates) {
  # infinite-states meaning-slot model: each replacement creates a new class
  tree <- reorder(tree, "cladewise")
  ntip <- Ntip(tree); m <- length(rates)
  st <- matrix(NA_integer_, ntip + tree$Nnode, m)
  st[ntip + 1, ] <- 1L
  next_id <- rep(1L, m)
  for (i in seq_len(nrow(tree$edge))) {
    p <- tree$edge[i, 1]; ch <- tree$edge[i, 2]
    changed <- runif(m) > exp(-rates * tree$edge.length[i])
    st[ch, ] <- st[p, ]
    next_id[changed] <- next_id[changed] + 1L
    st[ch, changed] <- next_id[changed]
  }
  out <- st[1:ntip, , drop = FALSE]
  dimnames(out) <- list(tree$tip.label, paste0("meaning", seq_len(m)))
  out
}
to_binary <- function(M) {
  do.call(cbind, lapply(colnames(M), function(m) {
    cl <- sort(unique(M[, m]))
    b <- 1 * outer(M[, m], cl, "==")
    colnames(b) <- paste0(m, "_", cl)
    b
  }))
}
set.seed(3)
```

```
rates <- rep(c(0.1, 0.5, 2), each = 100)      # slow, medium and fast meanings
M <- sim_meanings(tr, rates)                  # 12 languages, 300 meanings
B <- to_binary(M)
chars_per_meaning <- as.vector(table(sub("_.*", "", colnames(B)))[colnames(M)])
data.frame(rate = c(0.1, 0.5, 2),
           binary_chars_per_meaning =
             as.vector(tapply(chars_per_meaning, rates, mean)))
```

```
#>   rate binary_chars_per_meaning
#> 1  0.1                    1.40
#> 2  0.5                    2.41
#> 3  2.0                    5.74
```

```
range(rowSums(B))      # every language has exactly 300 presences
```

```
#> [1] 300 300
```

```
# within a meaning, binary characters are negatively correlated
fast <- B[, grepl("^meaning201_", colnames(B)), drop = FALSE]
round(cor(fast)[1:4, 1:4], 2)
```

```
#>              meaning201_1 meaning201_2 meaning201_3 meaning201_4
#> meaning201_1         1.00        -0.32        -0.32        -0.21
#> meaning201_2        -0.32         1.00        -0.20        -0.13
#> meaning201_3        -0.32        -0.20         1.00        -0.13
#> meaning201_4        -0.21        -0.13        -0.13         1.00
```

The fast meanings produce several times as many binary characters as the slow ones, and therefore several times the weight in a binary analysis. Within a meaning, the binary columns are perfectly constrained (they sum to one in every language) and negatively correlated. None of this invalidates binary analyses, which have proved robust in many tests, but it does mean that the binary likelihood is a composite likelihood that overstates the information in the data. A practical remedy is to estimate rates per meaning (Section 7.5.4), which at least prevents fast meanings from dictating the clock. A more principled remedy is a model for whole meaning slots.

7.8.3 Explicit models of replacement and synonymy

Neither the binary nor the single-state multistate models describe what a lexical replacement actually looks like. In most documented cases a new word enters a meaning slot (by coinage, semantic shift or borrowing), coexists with the old word for a while, perhaps with a difference in register or nuance, and eventually one of the two falls out of use or shifts to another meaning. English *dog* and *hound*, or the coexistence of inherited and borrowed words for the same meaning in many languages, show slots at different stages of this process. An explicit model of replacement therefore treats each meaning slot in each lineage as a *set* of words: new words enter at some rate, each word is lost at a rate that may increase with the number of competitors in the slot, and a slot is never left empty (row 10 of Table 7.2). Such a model allows polymorphism, represents the dependence among the classes of a meaning, and connects the other models: without competition, and if empty slots are allowed, it approaches the independent binary model, and if synonymy is resolved instantly it becomes the infinite-states model.

Models of this kind raise problems of their own. Word lists usually record only the "most basic" word, so synonymy in the data is partly an artefact of how a list was compiled, and the rate at which words enter a slot is confounded with the sampling of synonyms. Ancestral synonymy sorted differently in different descendants produces patterns that look like borrowing, a problem with a close analogue in the incomplete lineage sorting of population genetics (Section 16.2.1). And the state space of a slot with many possible classes is large, although for a handful of classes it can be written down as a CTMC on subsets and specified in general-purpose software such as RevBayes (Höhna et al., 2016). To our knowledge, no general implementation of such a model is in routine use in linguistic phylogenetics. It is the clearest gap between the models available and the process they are meant to describe.

Two recent studies show why the gap is hard to close. Canby et al. (2024) formulated a parametric model of roughly this kind, in which a meaning slot changes only through semantic shift or borrowing and passes through (possibly brief) periods of polymorphism, and used it to simulate data. Across a wide range of simulated conditions, a simple modification of maximum parsimony that accepts polymorphic characters recovered the tree more accurately than the Bayesian methods based on the gain and loss of cognates with which it was compared. Their model is so far a simulator and a benchmark rather than a model used for inference, and the comparison concerns topology, not dates. From the other direction, Häuser et al. (2026) introduced parameter-rich models for an alternative, non-binary representation of cognate data and found clear signs of overparameterization: the cognate datasets available today are too small, by the standards of molecular phylogenetics, to estimate such models. Richer models of replacement are needed, but they must be parsimonious enough to be estimable from a few hundred meanings.

Root-meaning and meaning-slot cognate coding answer different questions. Under *root-meaning* coding, words are cognate if they share a root and occupy the same meaning slot; under looser *root* coding, words sharing a remote etymological root may be grouped even if the root reached the meaning by different paths. Loose coding manufactures apparent multiple origins (the advergence of Chang et al. (2015)) and violates the Dollo assumption. Check how a dataset was coded before choosing a model, and see Chapter 3 for the coding conventions of the major databases.

7.9 What linguistic studies found when comparing models

Most Bayesian analyses of cognate data published since 2003 have compared at least a few models, typically by Bayes factors computed from marginal likelihoods (Chapter 6). Table 7.5 summarizes a selection of studies in which model comparison or robustness to model choice was a central concern. The descriptions are deliberately brief; readers should consult the papers and their supplementary materials for the exact model specifications, which are often more complex than a single label suggests.

Table 7.5: Selected linguistic studies in which comparison of models of lexical evolution, or robustness to such choices, played a central role.

Study	Family and data	Models or choices compared	Main finding about models
Gray and Atkinson (2003)	Indo-European, Dyen et al. Swadesh data	binary restriction-site model with gamma rates; several codings and calibrations	root date robust to the treatment of borrowings, rate heterogeneity and calibration choices

Study	Family and data	Models or choices compared	Main finding about models
Atkinson et al. (2005)	Indo-European	reversible models with various assumptions; early stochastic Dollo analysis	dates broadly robust to model and data choices
Nicholls and Gray (2008)	Indo-European (Dyen et al.; Ringe et al.)	stochastic Dollo with catastrophes	root age consistent with earlier reversible-model estimates
Ryder and Nicholls (2011)	Indo-European	stochastic Dollo with explicit missing-data model	correct handling of missing data matters; root age in a similar range
Greenhill et al. (2010b)	Austronesian (ABVD)	robustness to model and data subsets; comparison with traditional subgrouping	inferred trees largely agree with subgroups established by the comparative method
Bouckaert et al. (2012)	Indo-European (IELex)	binary CTMC, covarion and stochastic Dollo substitution models; strict and relaxed clocks	conclusions on root age and homeland robust across the substitution models
Chang et al. (2015)	Indo-European (IELex, recoded)	ancestry constraints on ancient languages; treatment of advergence	constraints and coding choices change the date substantially
Kolipakam et al. (2018)	Dravidian	alternative substitution and clock models compared by marginal likelihood	model choice reported alongside a root age of roughly 4,500 years
Rönchen et al. (2024)	Indo-European, five ancient languages	infinite-state Multistate model (Dollo-type) with a new likelihood algorithm	argued to be more realistic than the stochastic Dollo model, but sensitive to undetected loans and semantic shift
King (2026b)	simulated data; IE-CoR subsets (Indo-Iranian; Germanic, Italic and Celtic)	standard BEAST 2 set-up (binary covarion, ascertainment correction, meaning partitions, relaxed clock) checked by simulation-based calibration and posterior predictive simulation	default reweighting of partition rates gave poorly calibrated posteriors; the covarion model did not reproduce several features of real cognate data
Rönchen (2025)	simulated and empirical datasets, including Indo-European	finite-state Markov chains (including covarion) against Dollo-type models	with linguistic ascertainment correction, finite-state chains approximate Dollo behaviour but overfit; better fit did not imply more accurate trees

A few lessons emerge from this literature. First, models with among-character rate variation, and often with covarion structure, are generally favored over a simple single-rate binary model whenever they are compared, as one would expect from the strong rate heterogeneity of basic vocabulary. Second, the topologies inferred under reversible and stochastic Dollo models are

usually similar, while dates can differ more. This fits the limit relationship of Section 7.7.4: the models agree where gains are rare and disagree where multiple origins (borrowing, advergence, coding artefacts) are common. Third, in the Indo-European dating debate the largest differences between studies come not from the substitution model but from data coding, the treatment of ancient languages and the choice of constraints and calibrations (Chang et al., 2015; Heggarty et al., 2023). A substitution model can only be as good as the definition of the characters it is applied to. Fourth, agreement between inferred trees and established subgroups, as reported by Greenhill et al. (2010b), is consistent with the models capturing the main genealogical signal, but it is not independent validation of them, for the reasons given in Section 7.3.2. Fifth, Bayes factors compare models only with each other. A model that wins a Bayes-factor comparison may still be badly inadequate, which is why absolute tests of fit matter. Sixth, and most uncomfortably for current practice, recent work suggests that relative fit can point in the wrong direction. Rönchen (2025) argues, on the basis of a formal analysis and simulation studies reported in manuscripts that form part of his dissertation (Rönchen et al., 2025b), that the ascertainment correction used for cognate data fundamentally changes the behaviour of finite-state Markov chains such as the covarion model: under conditions that he reports to hold for many empirical and realistically simulated datasets, they come to approximate a Dollo-type model, but they overfit, so that a model with higher apparent fit can infer a less accurate tree. If this result holds up in peer-reviewed form and on other data, the Bayes-factor preference for covarion models reported in many of the studies above would say little about which model recovers the history better. Notice, finally, what the table does not contain: we know of no published comparison that sets binary models against a meaning-slot model with synonymy (row 10 of Table 7.2) on the same data.

7.10 Model adequacy

Model *selection* asks which of a set of models is best supported. Model *adequacy* asks whether the chosen model could plausibly have generated the data at all. The standard tool for the second question is the **posterior predictive check** (Gelman et al., 1996), introduced to phylogenetics by Bollback (2002) and reviewed by Brown (2014b) (Section 6.9). Draw parameter values (and trees) from the posterior distribution; simulate a replicate dataset of the same size from the model with those values; compute a test statistic on each replicate and on the real data; and see where the observed statistic falls in the predictive distribution. In a maximum-likelihood setting, the same idea with the fitted values in place of posterior draws is a *parametric bootstrap*. Where the analysis produces a posterior sample of trees, replicates should be simulated on trees drawn from that sample rather than on a single summary tree, so that the check reflects phylogenetic uncertainty.

The choice of test statistic is what makes a check informative, and the diagnostic column of Table 7.3 suggests statistics for each model. For cognate data, useful statistics include the number of cognate classes per language (which tests whether the model captures variation in vocabulary turnover among lineages), the site frequency spectrum, the distribution of Fitch parsimony scores (homoplasy), the number of cognate classes per meaning, and treelikeness statistics such as the delta score (Chapter 8). The code below uses the data simulated with strong rate heterogeneity in Section 7.5 and asks whether the single-rate model and the gamma model reproduce the observed proportion of characters that fit the tree with a single parsimony step. Because the “observed” data are themselves simulated, with a known misspecification of the single-rate model, the example shows that the check has the power to detect that misspecification; it says nothing about the adequacy of either model for any real dataset.

```
one_step_prop <- function(tree, X) {
  pd <- phyDat(X, type = "USER", levels = c("0", "1"))
  mean(parsimony(tree, pd, site = "site")[attr(pd, "index")] == 1)
}
predictive <- function(fit, nrep = 100, shape = NULL) {
  n <- ncol(Xv)
  replicate(nrep, {
    Y <- sim_ctmc_binary(fit$tree, 3 * n, pi1 = 0.5, rate = 1, shape = shape)
    Y <- Y[, colSums(Y) > 0 & colSums(Y) < nrow(Y)][, 1:n]  # same filtering
    one_step_prop(fit$tree, Y)
  })
}
set.seed(21)
obs_eq <- one_step_prop(fit_eq$tree, Xv)
obs_g  <- one_step_prop(fit_g$tree, Xv)
pp_eq <- predictive(fit_eq)
pp_g  <- predictive(fit_g, shape = fit_g$shape)
par(mfrow = c(1, 2), mar = c(4, 4, 2, 1))
hist(pp_eq, xlim = extendrange(c(pp_eq, obs_eq), f = 0.06),
     main = "single-rate model", xlab = "proportion of one-step characters",
     col = "grey85", border = "white")
abline(v = obs_eq, col = "#d1495b", lwd = 3)
hist(pp_g, xlim = extendrange(c(pp_g, obs_g), f = 0.06),
     main = "gamma model", xlab = "proportion of one-step characters",
     col = "grey85", border = "white")
abline(v = obs_g, col = "#d1495b", lwd = 3)
```

```
rbind(single_rate = c(observed = obs_eq, quantile(pp_eq, c(0.025, 0.975))),
      gamma = c(observed = obs_g, quantile(pp_g, c(0.025, 0.975))))
```

```
#>             observed     2.5%   97.5%
#> single_rate   0.1155 0.008507 0.02053
#> gamma         0.1149 0.104167 0.12849
```

The single-rate model fails decisively: the observed data contain several times more perfectly tree-compatible characters than it can generate, because it has no slowly evolving characters. The gamma model, whose estimated shape is close to the true value (Table 7.4), reproduces the observed proportion, which lies inside the central 95% of its predictive distribution. That is expected here, since the data were simulated under a gamma model; with real data even the better of two models may fail some checks. This is how adequacy checks are most useful in practice: not as a pass-or-fail verdict, but as a pointer to the aspect of the data that a model handles least well. Posterior predictive checks of this kind are cheap and illuminating, yet they remain rare in published linguistic phylogenies. Most studies report only Bayes-factor comparisons, if they compare models at all. The most systematic exception is King (2026b), who checked the analysis set-up that has become standard in BEAST 2 in two steps. Simulation-based calibration, an implementation check, showed that the covarion parameters are recovered correctly, but that the default reweighting of partition rates by the number of observed cognates interacts with the ascertainment correction and produces poorly calibrated posteriors; an alternative that treats each meaning as drawing on

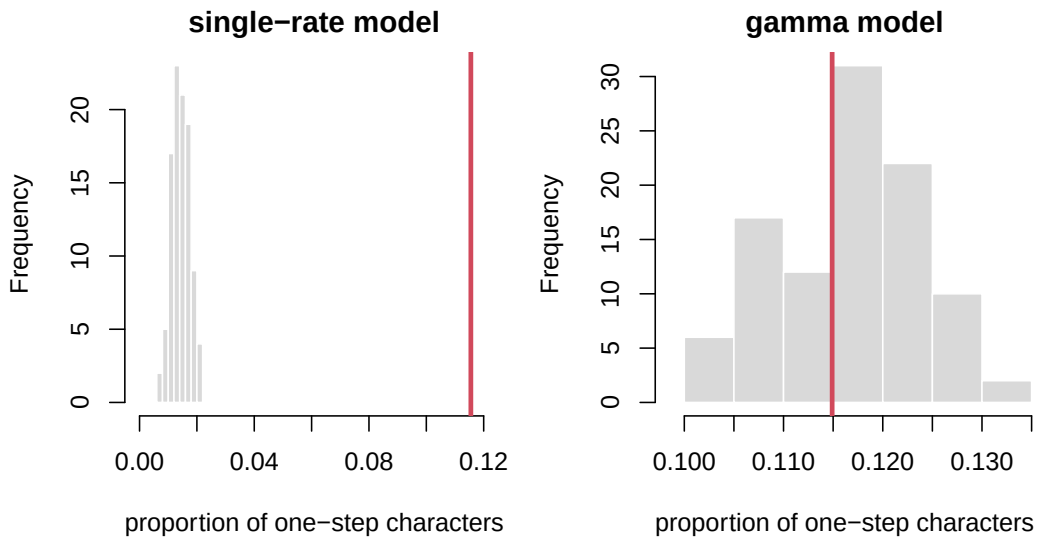


Figure 7.5: Parametric bootstrap check of two models fitted to data simulated with strong rate heterogeneity. Histograms show the proportion of variable characters requiring a single parsimony step on the fitted tree in 100 datasets simulated from each fitted model; the vertical line shows the observed value. The single-rate model cannot reproduce the excess of slowly evolving, perfectly tree-compatible characters; the gamma model reproduces it.

a cognate set of equal size behaved correctly and fitted empirical data better. Posterior predictive simulation on a sample of Germanic, Italic and Celtic languages from IE-CoR, an empirical adequacy check, then showed that data simulated under the fitted covarion model differ from the real data in several statistics of language divergence and cognate distribution, a failure that King attributes most plausibly to semantic shift and to the non-independence of cognate replacements within meanings. The two results answer different questions: the first says that the software computes what the model specifies, the second that the model does not describe lexical evolution well. Chapter 15 argues that adequacy checks should become routine, and Chapter 16 lists the relevant tools.

7.11 What biology offers

The models in common linguistic use are, with the exception of the stochastic Dollo family, simplified versions of models developed for DNA and morphology twenty or more years ago. Molecular phylogenetics has since moved on to much richer descriptions of heterogeneity among sites and lineages. Several of these could be adapted to cognate data with little effort.

Mixture and profile-mixture models. In a mixture model, each character is assumed to evolve under one of several distinct processes, and the likelihood sums over the possibilities with estimated weights; unlike a partitioned model, the assignment of characters to processes need not be known in advance. Pagel and Meade (2004) introduced a general mixture model for gene sequences and discrete characters, implemented in BayesTraits. In protein phylogenetics, the CAT model of Lartillot and Philippe (2004) goes further, letting each site have its own equilibrium profile drawn from a Dirichlet process, so that the number of profile classes is itself inferred. Empirical profile mixtures for proteins (Le et al., 2008) are now standard in deep phylogenetics. For cognate data, the analogue of an equilibrium profile is the frequency of presence π_1 together with the rate.

A mixture over a few (π_1, rate) classes would let the data separate, say, cognates that are long-lived and widespread from cognates that are short-lived and local, without the analyst having to specify which is which. Morphological phylogenetics offers a ready-made version: Wright et al. (2016) let the stationary frequencies of binary characters vary across characters according to a discretized Beta distribution, a mixture of F81 models. There the Beta is usually symmetric, because the labels 0 and 1 of a morphological character are arbitrary; for cognate presence and absence they are not, and the mean frequency of presence should be estimated. Section H.3.17 implements both versions in RevBayes and compares them by marginal likelihood with the binary Jukes–Cantor, F81 and F81 + Γ models on simulated data. **Dirichlet process rates.** Huelsenbeck and Suchard (2007) replaced the parametric gamma distribution of rates with a Dirichlet process prior, which clusters characters into an unknown number of rate classes. Applied at the level of meanings, this would deliver a data-driven classification of meanings into stable and volatile groups, a direct test of the claim that the Swadesh list contains distinct strata of stability. **Heterotachy mixtures.** Kolaczkowski and Thornton (2004) showed that when rates shift across lineages in a site-specific way, standard models can be misled; mixtures of branch-length sets and covarion-type models address this. **Hidden-rate models.** The hidden Markov models of Beaulieu et al. (2013), developed for morphological characters, allow a character to switch between unobserved rate categories, as in the covarion model, but with asymmetric rates of gain and loss in each category. **Structured characters.** The structured Markov models of Tarasov (2019) combine dependent morphological characters into a single character with a constrained state space; applied to the cognate classes of a meaning, the same construction gives a principled meaning-slot model. **Model averaging.** Rather than choosing a single substitution model, bModelTest (Bouckaert and Drummond, 2017) averages over a family of nucleotide models within BEAST 2 by reversible-jump MCMC; an analogous scheme for binary and multistate lexical models would remove the need for separate Bayes-factor comparisons. **Data-driven partitioning.** PartitionFinder (Lanfear et al., 2012) chooses partitioning schemes for molecular data by information criteria, and the same logic could group meanings into rate classes objectively. To our knowledge, none of these approaches, apart from occasional use of BayesTraits mixtures and covarion models, has been applied systematically to linguistic cognate data.

Beyond these, two less obvious directions deserve mention. First, population genetics has a mature theory of the site frequency spectrum under infinite-sites and infinite-alleles models, and of how demography (expansion, bottlenecks) distorts it. The cognate site frequency spectrum computed in Section 7.7 is the linguistic analogue, and spectrum-based tests could help detect expansions or the effects of borrowing. Second, the morphological-phylogenetics literature has grappled with many of the same issues as cognate analysis, including the choice of ascertainment correction, variable numbers of states and correlated characters, and the guide by Wright (2019) is a useful source of ideas for linguists.

7.12 Practical recommendations

Pulling the threads together, the following practices are sound defaults for model choice in lexical phylogenetics.

1. Start from the historical question. Topology, dates, rates of replacement and borrowing are sensitive to different features of the models in Table 7.2; choose and check the model with the target quantity in mind.
2. Match the ascertainment correction to the way the data were filtered. For standard cognate

matrices that retain singletons and universally shared cognates, use the no-absence correction; if singletons were removed, use a correction that excludes them too.
3. Always include among-character rate variation, ideally at the level of meanings rather than independent cognate classes, and check how sensitive dates are to the rate model.
4. Compare at least one reversible model (with gamma and possibly covarion rates) with a Dollo-type model. Agreement between them is reassuring; disagreement points to multiple origins of cognates, which calls for a closer look at borrowing and coding. Do not treat a better marginal likelihood for a finite-state model as evidence that it recovers the tree better (Rönchen, 2025).
5. Be explicit about binary versus multistate coding and about how synonyms and missing entries were handled, and remember that binary analyses do not model the dependence among the classes of a meaning, so that their posterior intervals are likely to be too narrow.
6. Add posterior predictive checks with statistics that matter for your question, such as the site frequency spectrum, homoplasy and cognates per language, simulated over the posterior sample of trees. If meanings are partitioned by number of cognate classes, check how partition rates are weighted, since the default weighting can miscalibrate the posterior (King, 2026b).
7. Report the full model specification (substitution model, rate variation, ascertainment correction, clock and tree prior) so that the analysis can be reproduced.

- Models of lexical evolution differ in their unit of observation, the process that creates and removes cognate classes, whether they allow synonyms and recurrent innovation, whether they represent the dependence among classes of one meaning, and how they treat borrowing (Table 7.2).
- The binary CTMC (restriction-site, Mk2) model treats each cognate class as a presence/absence character evolving independently and reversibly; its transition probabilities have a simple closed form and saturate with time.
- Cognate classes absent from all sampled languages are never observed. The ascertainment correction divides each character's likelihood by the probability of observability; the appropriate correction depends on exactly how the data were filtered.
- Rates of lexical replacement vary by two orders of magnitude among meanings, partly predictably from word frequency. Ignoring this variation shortens trees and biases dates toward the recent past; covarion switching has similar consequences for deep dates.
- The stochastic Dollo model gives each cognate class a single birth followed by independent deaths; its likelihood has built-in ascertainment and is the limit of the corrected reversible model as the frequency of presence goes to zero. Pseudo-Dollo models approximate it within a standard CTMC.
- Multiple origins of a cognate class (borrowing, parallel semantic shift, loose coding) violate the Dollo assumption and can bias dates.
- Only meaning-slot models account for the dependence among the cognate classes of a meaning; binary analyses are composite likelihoods that overweight fast meanings and overstate the information in the data. Explicit models of replacement with synonymy are the clearest gap in current practice.
- Every model identifies only rate-time products from the data, and rate parameters are interpretable as historical quantities only relative to a given coding.
- Relative model fit is not adequacy, and agreement with established subgroups is not independent validation; posterior predictive checks should be routine.

7.13 Exercises

Conceptual

1. A language has two words for ‘stone’, from cognate classes B and D. Explain how this fact is represented under binary coding, under single-state multistate coding, and under the replacement-with-synonymy model of Section 7.8.3. Which representation makes a claim that the data do not support?
2. Using Table 7.2, explain how a borrowed cognate class shared by two distantly related languages is treated by the reversible binary model, the stochastic Dollo model and the pseudo-Dollo model, and predict the direction of the bias in the root age under each.
3. A colleague removes all singleton cognate classes from a matrix “because they are uninformative about topology” and then runs MrBayes with `coding=noabsencesites`. What is wrong, which parameters will be affected, and in which direction?
4. Why does the stochastic Dollo likelihood not depend on the birth rate λ once we condition on the number of observed cognate classes? What, then, does λ tell us historically?
5. Explain the difference between a parameter that is *identifiable under the model* and one that is *interpretable as a historical quantity*, using the binary loss rate and the meaning-slot replacement rate as examples.
6. An analysis recovers all the subgroups of a family accepted by historical linguists. Give three reasons why this does not, by itself, validate the substitution model used.

Computational

1. Use `binary_P()` to find, for $\pi_1 = 0.1$, 0.3 and 0.5, the branch length at which $P_{11}(t)$ comes within 0.01 of π_1. Relate the answer to the saturation horizon of cognate data. Then show that, for two languages separated by a path of length t, the expected proportion of characters on which they agree tends to $\pi_0^2 + \pi_1^2$ as t grows (one half when $\pi_1 = 0.5$), and explain why the raw proportion of agreeing characters between two unrelated languages is therefore not zero.
2. Using `loglik_binary()` and `fit_binary()` from Section C.1.3, repeat the ascertainment simulation with $\pi_1 = 0.05$ and $\pi_1 = 0.5$. How does the bias of the uncorrected fit depend on π_1? Then remove the singleton columns as well and refit with the no-absence correction: what happens to the estimates?
3. Repeat the rate-heterogeneity experiment of Section 7.5 with `sim_ctmc_binary()` and `phangorn::optim.pml()` for true shapes 0.2, 1 and 5. Plot the ratio of estimated to true tree length for the single-rate and gamma models against the true shape. Then refit the shape-0.2 data with `k = 2`, `4` and `8` gamma categories (the `k` argument of `pml()`) and compare the maximized log-likelihoods and estimated shapes: from how many categories on do the results stop changing?
4. Simulate stochastic Dollo data with `sim_dollo()` on the 12-language tree `tr`, then mimic borrowing by copying 30 randomly chosen cognate classes present in language L1 into language L12. Recompute the site frequency spectrum and the distribution of `fitch_steps()`. Which statistic is more sensitive to the borrowing? Which validation category of Section 6.10 does the exercise belong to?
5. With `sim_meanings()` and `to_binary()`, confirm that every simulated language has exactly one 1 per meaning. Then use the function `p_exactly_one()` of Section C.1.9, with the observed frequency of 1s in `B` as π_1, to compute how much probability an independent binary

model gives to the configurations that actually occur, for meanings with 2, 4 and 8 classes.

6. Run `revbayes/analyses/lexical/16_f81_beta_mixture.Rev` (Section H.3.17) from `revbayes/`, one model at a time with `--args <model> 500`, and tabulate the log Bayes factor of each model against the binary Jukes–Cantor model. Then add discrete Gamma(4) rate variation (`fnDiscretizeGamma()` and the `siteRates` argument of `dnPhyloCTMC()`) to the Beta-mixture model as a sixth model. Does it improve the marginal likelihood on these data, which were simulated without rate variation, and by how much does the extra parameter cost? What would, and what would not, follow about lexical change if the same ranking were found for a real cognate matrix?

Research

1. You are preparing a dated analysis of a family of about 40 languages with a 200-meaning list in which about 15 percent of entries have synonyms, some meanings are unrecorded for the poorly documented languages, and known loans have been flagged. Choose a coding, a substitution model, an ascertainment correction and a set of adequacy checks, and justify each choice by reference to Tables 7.2 and 7.3.
2. Design a simulation study to measure how much binary coding overstates posterior certainty about the root age, compared with a meaning-slot model. Specify the generating process (including synonymy), the analyses to be compared and the summary statistics, and say which of the validation categories of Section 6.10 each part of the study addresses.
3. Outline an explicit replacement-with-synonymy model for a single meaning slot with at most three cognate classes, as a CTMC on the non-empty subsets of classes. Write down its rate matrix, say which parameters you expect to be practically estimable from 30 languages, and propose a test statistic for checking it against real data.

7.14 Further reading

- Felsenstein (1992) and Lewis (2001). The original papers on the restriction-site model and the Mk/Mkv models, including the derivation of the ascertainment corrections; still the clearest statements of the logic.
- Yang (1994): The discrete-gamma approximation to among-site rate variation, used in nearly every analysis since.
- Tuffley and Steel (1998): The mathematical formulation of the covarion model, with a discussion of identifiability.
- Nicholls and Gray (2008): The stochastic Dollo model with catastrophes and its application to Indo-European; demanding but rewarding statistically.
- Ryder and Nicholls (2011): How to handle missing data in the stochastic Dollo model; essential reading for anyone with gappy lexical datasets.
- Bouckaert and Robbeets (2017): The pseudo-Dollo models and their multistate and covarion variants.
- Warnow et al. (2006): A stochastic model of linguistic characters with homoplasy and borrowing, with a discussion of identifiability for meaning-slot models.
- Pagel et al. (2007): The demonstration that word frequency predicts rates of lexical replacement; the empirical basis for modelling rate heterogeneity among meanings.
- Chang et al. (2015): A detailed critique of how lexical characters are defined and how advergence and ancient languages affect dating; required reading alongside the model papers.

- Hoffmann et al. (2021): A practical guide to setting up models of lexical evolution in BEAST 2, with discussion of the choices covered in this chapter.
- Bollback (2002) and Brown (2014b). Introductions to posterior predictive model checking in phylogenetics.
- Rönchen et al. (2024): The likelihood of the infinite-state Multistate model, with a frank discussion of its robustness problems; read with the summary of the dissertation (Rönchen, 2025), which argues that finite-state chains such as the covarion model overfit cognate data.
- King (2026b): Simulation-based calibration and posterior predictive checks of the standard BEAST 2 set-up for cognate data; the best current example of implementation validation and empirical adequacy testing for lexical models.
- Canby et al. (2024): A model of lexical change with polymorphism, used to compare parsimony and Bayesian methods on simulated data.
- Wright (2019): A guide to Bayesian analysis of morphological characters, many of whose problems (ascertainment, state spaces, rate variation) have direct lexical analogues.

Chapter 8

Borrowing, Contact and Reticulation

Languages do not evolve in isolation. Speakers of neighboring languages trade, intermarry, conquer one another and shift from one language to another, and words, sounds and grammatical patterns pass between languages in the process. Contact is the feature of language history that most clearly distinguishes it from the textbook picture of biological descent, and it was the basis of the oldest objection to the family tree, Johannes Schmidt's wave theory of 1872. This chapter asks what contact does to phylogenetic inference, how much of it a tree can absorb, and what to use when a tree is the wrong representation. Its central distinction is between *descriptive* networks, which summarize conflicting signal in the data, and *generative* models, which propose a history of descent and contact that could have produced the data; and its central argument is that the choice among trees, networks and wave-like representations should follow from the historical question being asked.

After reading this chapter you should be able to:

- distinguish a tree read as an approximation to a genealogy from a tree read as a statistical summary of variation, and explain why under extensive reticulation there may be no bifurcating history to approximate;
- match a historical question (reconstructing divergence, describing similarity, identifying innovations, estimating a dominant pattern of transmission) to an appropriate representation and method (Table 8.1);
- predict how local and distant borrowing affect topologies, branch lengths and dates under reversible and Dollo-type models;
- compute and interpret delta scores and Q-residuals, and explain why they measure conflict rather than borrowing;
- read a NeighborNet or consensus network correctly as a description of conflicting signal, not a history of contact;
- explain what explicit, generative models of contact (lateral-transfer stochastic Dollo, explicit networks) add, and what they cost;
- state when a wave or linkage model, rather than a tree, is the appropriate description, and what Historical Glottometry measures.

8.1 Borrowing and the tree model

8.1.1 What is borrowed, and how often

Contact linguistics distinguishes several outcomes of language contact (Thomason and Kaufman, 1988; Thomason, 2001). In **borrowing**, speakers of a language adopt material from another language while keeping their own; vocabulary is borrowed first and most easily, and structural features follow only under intense and prolonged contact. In **shift**, a community abandons its language for another and often leaves traces of the old language, a substrate, in the new one. In extreme cases contact produces **mixed languages**, whose grammar and lexicon come from different sources, and **creoles**, which emerge from pidgins in situations of radical social disruption. Each of these processes creates a different kind of departure from treelike descent.

For phylogenetics based on basic vocabulary, the central question is how much of that vocabulary is borrowed. The World Loanword Database project compiled loanword data for a standardized list of meanings in 41 languages (Haspelmath and Tadmor, 2009). Across the whole list, roughly a quarter of the words were loans on average, with enormous variation among languages (Tadmor, 2009). Borrowability also varied strongly by meaning, and Tadmor et al. (2010) used these differences to build the Leipzig-Jakarta list of the 100 meanings least likely to be borrowed. Basic vocabulary is therefore resistant to borrowing but not immune. English took *they*, *sky*, *skin* and *egg* from Old Norse; the numerals of Thai and of many other languages of mainland Southeast Asia are of Chinese origin; and Japanese, Korean and Vietnamese each have a full set of Chinese-derived numerals alongside native ones. Borrowing rates in basic vocabulary vary among regions and social settings, but the rates measured in large samples are generally modest. Bowern et al. (2011) found, for example, that borrowing rates in the basic vocabulary of hunter-gatherer languages from several regions of the world were generally low, contrary to the expectation that hunter-gatherer languages borrow unusually freely.

8.1.2 Known and hidden loans

Linguists handle borrowing first by identifying loans. Most large cognate databases flag known loanwords (Chapter 3), and analysts either exclude them or code them as separate characters that do not unite donor and recipient. Gray and Atkinson (2003), for instance, removed known borrowings from the Indo-European data before analysis. The difficulty is with *undetected* loans. Loans between closely related languages are the hardest to detect, because regular sound correspondences, the main diagnostic, may not distinguish a borrowed word from an inherited one when the languages have undergone few sound changes since their split. Loans from unrelated or distantly related languages are easier to identify. This creates an asymmetry: the borrowings most likely to slip through are those between neighbors and close relatives, and, as we shall see, these are also the ones that phylogenetic methods are least able to detect.

8.1.3 The biological parallel and the ethnogenesis critique

Biologists face the same problem in other guises. In prokaryotes, horizontal (lateral) gene transfer is so pervasive that some have argued the tree of life is better described as a web (Doolittle, 1999). In plants and animals, **hybridization** creates lineages with two parents, and **introgression**, the movement of genes from one species into another by repeated backcrossing, is the closest biological analogue of lexical borrowing. Within species, **recombination** means that different

parts of the genome have different genealogies. Biologists have responded with explicit models of all three processes, which we survey in Section 8.11.

In anthropology, the parallel argument was made forcefully by Moore (1994), who contrasted *phylogenesis*, the branching of cultures from common ancestors, with *ethnogenesis*, their formation by blending, and argued that human history was dominated by the latter. The empirical response has been to measure rather than assume. Collard et al. (2006) compared the fit of cultural and biological datasets to trees using the retention index, and found that cultural data were, on average, no less treelike than biological data. That comparison does not settle any particular case, but it reframes the question: how treelike is a given dataset, and how robust are the conclusions we want to draw from it?

8.2 Trees, networks and historical questions

8.2.1 Two readings of a tree

A tree inferred from linguistic data can be read in two quite different ways, and much confusion about contact comes from sliding between them.

On the **genealogical reading**, the tree is an approximation to a real history in which speech communities separated and then diverged. Internal nodes stand for proto-languages spoken by identifiable (if unknown) communities, branch lengths correspond to elapsed time or amounts of change, and borrowing is a perturbation: a set of lateral events superimposed on a history that is fundamentally branching. Every dating analysis, every reconstruction of a proto-lexicon at an internal node and every homeland inference presupposes this reading. Whether it holds is a matter of empirical adequacy and historical interpretation, in the vocabulary of Section 6.10, not something the inference itself can establish: a posterior probability for a clade is a probability *given* that the history was a tree.

On the **summary reading**, the tree is a statistical summary of variation: the hierarchical arrangement that best fits the similarities in the data, in the way that a cluster dendrogram summarizes a distance matrix. Its internal nodes need not correspond to communities, and its branch lengths need not measure time. Support values measure how consistently the data favor a grouping, not whether the grouping ever existed as a speech community. The summary reading is always available, which is part of its appeal, but it licenses far fewer historical conclusions.

The two readings coincide when the history really was close to a tree, and diverge sharply when it was not. Under extensive reticulation, as in Ross's *linkages* (Section 8.7) or a dialect continuum in which innovations spread across overlapping sets of varieties, there may be no single bifurcating history underlying the data at all. A tree inferred from such data is then not a slightly blurred picture of the true history, with a few borrowings hidden; it is a projection of a different kind of history onto a model that cannot represent it. Its internal nodes do not correspond to proto-languages, its "divergence times" do not date separations, and high support for a clade may reflect nothing more than the fact that some grouping had to be chosen. It is sometimes reasonable to call a tree inferred from overlapping innovations a summary that hides their distribution; it is not reasonable to then treat its nodes as ancestors. Between the two extremes lies the common case of a dominant vertical signal with contact superimposed, for which a tree with lateral edges, or a network with estimated inheritance proportions, is the natural representation.

8.2.2 Descriptive networks and generative models

Two kinds of network are discussed in this chapter, and they answer different questions. **Descriptive networks**, such as split networks, NeighborNets and consensus networks (Section 8.5), are functions of the data: they display the signal in a distance matrix or a set of trees, conflicting signal included. They have no likelihood, make no claim about what happened, and their edges have neither direction nor date. They are to a dataset what a scatterplot is to a regression: indispensable for seeing the data, silent about mechanism. **Generative models of contact** (Section 8.6) specify a history of descent and contact, with parameters such as borrowing rates, donor and recipient lineages, times and inheritance proportions, together with a probability model under which that history could have produced the data. Because they define a probability of the data, they can be fitted, compared, checked against data simulated from them and used to estimate how much borrowing occurred and when. Their cost is the cost of any model: they can be misspecified, they may be computationally demanding, and their parameters may be only weakly identifiable. Some methods fall between the two: perfect phylogenetic networks and minimal lateral networks posit explicit contact edges, but select them by combinatorial or heuristic criteria rather than by a probability model.

8.2.3 Choosing by the historical question

Which representation to use should follow from the historical question being asked, not from a general preference for trees or for waves. Table 8.1 sets out four common questions.

Table 8.1: Four historical questions and the representations and methods that suit them.

Historical question	What is estimated	Appropriate representation	Typical tools	What it cannot deliver
Reconstructing divergence: in what order, and when, did speech communities separate?	a genealogy of communities, with dates	rooted tree; explicit network if lineages merged	Bayesian trees with clocks (Chapters 9, 10); lateral-transfer models; explicit networks	meaningful answers where no separation into communities took place (linkages, continua)
Describing similarity: how similar are the languages, and where does the signal conflict?	a summary of the data	distance matrix, split network, consensus network, dialectometric map	NeighborNet, delta scores and Q-residuals, multidimensional scaling	the direction, timing or cause of contact
Identifying innovations: which changes are shared, and by which groups?	the distribution of innovations established by the comparative method	compatibility analysis, perfect phylogenetic network, glottometric diagram	comparative method; perfect phylogenetic networks; Historical Glottometry	rates, dates, or the probability of parallel change, unless modelled

Historical question	What is estimated	Appropriate representation	Typical tools	What it cannot deliver
Estimating a dominant pattern of transmission: how much is vertical, how much horizontal, and along which tree?	a tree plus the amount and location of transfer	tree with lateral transfer; network with inheritance proportions	lateral-transfer stochastic Dollo; minimal lateral networks; four-taxon tests; network coalescent	a complete history of individual borrowing events

The questions are not exclusive. A typical project begins by describing similarity, to see how treelike the data are; uses the result to decide whether reconstructing divergence is a sensible goal at all; and, if it is, estimates the dominant tree while checking whether the amount of transfer is large enough to change the conclusions. What the table rules out is the use of a method built for one question to answer another: dating the nodes of a NeighborNet, for instance, or reading a box in a split network as a borrowing event.

8.3 How borrowing distorts trees

8.3.1 Two kinds of borrowing

It helps to distinguish two situations. When languages borrow from their close relatives and immediate neighbors (**local** borrowing), the borrowed words reinforce relationships that the tree already shows. They do not create much conflict in the data, but they make related languages look more similar than inheritance alone would, compressing the tree: internal branches become shorter and the whole tree shallower. When languages borrow from distant relatives or unrelated languages (**distant** or **global** borrowing), the borrowed words create characters that unite languages from different clades. Such characters conflict with the tree and, if there are enough of them, can pull the recipient toward the donor, distorting the topology.

The effect on dates depends on the substitution model. Under reversible models and distance methods, both kinds of borrowing reduce the inferred amount of change, so dates tend to be underestimated. Under the stochastic Dollo model (Chapter 7), a cognate class present in two distant languages must have been born above their common ancestor, so distant borrowing can push inferred births, and possibly the root, deeper into the past, while forcing many parallel losses. Borrowing therefore has no single direction of bias; its effect depends on who borrows from whom and on the model used (Table 7.2).

Does horizontal transmission invalidate cultural phylogenies? Greenhill et al. (2009) addressed this question by simulation. They generated binary trait data on known trees and allowed traits to be copied between lineages at a range of borrowing rates, contrasting borrowing restricted to nearby lineages with borrowing between any lineages. They then reconstructed trees from the simulated data with Bayesian phylogenetic methods and compared the results with the true trees. In the vocabulary of Section 6.10, this is a study of robustness under misspecification: the inference model ignores the borrowing that the generating process contains. Their key finding was that the reconstructed topologies were fairly robust to the levels of borrowing thought to be realistic for basic vocabulary, with accuracy declining at higher rates; they also found that borrowing tended to make the estimated trees shallower, which matters for dating. A companion study (Currie et al.,

2010b) asked the analogous question for comparative analyses of continuous cultural traits and also found a reasonable degree of robustness to moderate horizontal transmission. These studies do not show that borrowing is harmless, only that its effects can be quantified, and that they are often smaller than critics had assumed. Their conclusions hold for the simulated scenarios, which assume an underlying tree; they say nothing about histories, such as linkages, in which there is no tree to recover.

8.3.2 A simulation in R

We now run a small experiment of the same kind. Our model is a meaning-slot model: each language has one word for each of 200 meanings; along each lineage, a word is replaced by a newly coined word (a new cognate class) at rate r_{innov}; and it is replaced by the word for the same meaning in another, contemporaneous language at rate r_{borrow}. For *global* borrowing the donor is chosen uniformly among contemporaneous lineages; for *local* borrowing it is chosen with probability proportional to $e^{-d/s}$, where d is the patristic distance between donor and recipient at the time of borrowing and s a scale parameter, so that close relatives are preferred. A third option, *pair*, restricts borrowing to one donor lineage and one recipient lineage within a time window, mimicking a specific episode of contact. The simulation steps through time in small increments, which is crude but transparent. The simulator `sim_borrowing()` and two helpers, `meaning_dist()` (the proportion of meanings for which two languages have non-cognate words) and `to_binary()` (binary presence/absence coding), are loaded silently below.

The code of `sim_borrowing()` and its helpers, with an explanation of the time-stepping scheme and of the glottochronological formula used below, is in Section C.2.1.

```
library(phangorn)
family <- read.tree(text = paste0(
  "((((A1:0.2,A2:0.2):0.2,(A3:0.25,A4:0.25):0.15):0.35,",
  "((B1:0.2,B2:0.2):0.25,(B3:0.3,B4:0.3):0.15):0.3):0.25,",
  "(((C1:0.15,C2:0.15):0.3,C3:0.45):0.25,(C4:0.4,C5:0.4):0.3):0.3);"))
plot(family, edge.width = 2, label.offset = 0.01, cex = 0.9)
axisPhylo(backward = FALSE)
```

We simulate datasets at six levels of borrowing, expressed as the proportion b of all replacement events that are borrowings, holding the total replacement rate fixed at 0.6 per meaning per unit time. For each dataset we infer a neighbor-joining tree from the proportion of non-cognate meanings, compute its normalized Robinson-Foulds distance to the true tree (Chapter 2), compute the mean delta score with the function `quartet_stats()` introduced in the next section, and estimate the root age in the manner of glottochronology: we transform distances with the formula $\hat{t} = -\log(1-d)/(2r)$ appropriate to the model without borrowing, and take the root height of a UPGMA tree built from them (Chapter 4). In the vocabulary of Section 6.10, this is a test of robustness under misspecification: both the tree-building method and the dating formula assume that there is no borrowing.

```
set.seed(42)
r_total <- 0.6
grid <- expand.grid(b = seq(0, 0.5, by = 0.1), mode = c("local", "global"),
                    rep = 1:6, stringsAsFactors = FALSE)
res <- t(sapply(seq_len(nrow(grid)), function(i) {
```

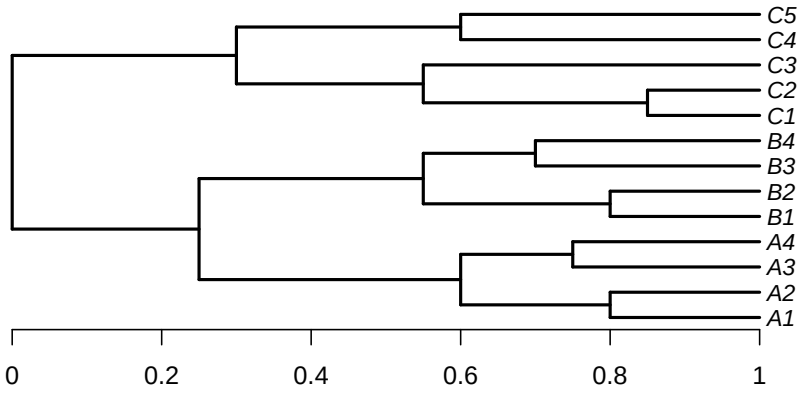


Figure 8.1: The toy language family used in the simulations of this chapter: thirteen languages in three subgroups (A, B, C), with root age 1.

```
  g <- grid[i, ]
  M <- sim_borrowing(family, innov = (1 - g$b) * r_total,
                     borrow = g$b * r_total, mode = g$mode)
  d <- meaning_dist(M)
  qs <- quartet_stats(d)
  t_hat <- -log(pmax(1 - d, 1e-3)) / r_total   # = 2 x estimated divergence time
  c(RF = RF.dist(NJ(d), family, normalize = TRUE), delta = qs$delta,
    Qres = qs$qres, root_age = max(node.depth.edgelength(upgma(t_hat))))
}))
sim <- cbind(grid, res)
summ <- aggregate(cbind(RF, delta, Qres, root_age) ~ b + mode, data = sim,
                  FUN = mean)
knitr::kable(summ, digits = c(1, 0, 3, 3, 4, 3), caption = "Mean results over six
replicate simulations per setting: normalized Robinson-Foulds distance of the
neighbor-joining tree to the true tree, mean delta score, mean Q-residual, and
glottochronological estimate of the root age (true value 1), as a function of the
proportion b of replacements due to borrowing.")
```

Table 8.2: Mean results over six replicate simulations per setting: normalized Robinson-Foulds distance of the neighbor-joining tree to the true tree, mean delta score, mean Q-residual, and glottochronological estimate of the root age (true value 1), as a function of the proportion b of replacements due to borrowing.

b	mode	RF	delta	Qres	root_age
0.0	global	0.000	0.079	0.0017	1.000
0.1	global	0.000	0.082	0.0017	0.898

b	mode	RF	delta	Qres	root_age
0.2	global	0.000	0.095	0.0021	0.744
0.3	global	0.000	0.101	0.0027	0.666
0.4	global	0.000	0.133	0.0035	0.534
0.5	global	0.017	0.136	0.0040	0.454
0.0	local	0.000	0.072	0.0015	1.011
0.1	local	0.000	0.078	0.0016	0.905
0.2	local	0.000	0.075	0.0017	0.840
0.3	local	0.000	0.075	0.0019	0.723
0.4	local	0.000	0.069	0.0016	0.571
0.5	local	0.000	0.068	0.0016	0.500

```
par(mfrow = c(1, 2), mar = c(4, 4, 1, 1))
cols <- c(local = "#1b6ca8", global = "#d1495b")
for (v in c("delta", "root_age")) {
  plot(NA, xlim = c(0, 0.5), ylim = range(c(summ[[v]], if (v == "root_age") 1)),
       xlab = "proportion of replacements by borrowing", las = 1,
       cex.lab = 0.85,
       ylab = if (v == "delta") "mean delta score" else "estimated root age")
  for (m in names(cols)) {
    s <- summ[summ$mode == m, ]
    lines(s$b, s[[v]], type = "b", pch = 19, col = cols[m], lwd = 2)
  }
  if (v == "root_age") abline(h = 1, lty = 3)
  legend(if (v == "root_age") "bottomleft" else "topleft", names(cols),
         col = cols, lwd = 2, pch = 19, bty = "n")
}
```

The results in Table 8.2 and Figure 8.2 illustrate three general points. First, the topology of this well-structured tree survives even heavy borrowing: the neighbor-joining tree is essentially always correct. That is in line with the robustness of topology reported by Greenhill et al. (2009), although our tree, with long internal branches, is an easy case. Second, global borrowing leaves a detectable signature in the data, a rising delta score, while local borrowing leaves almost none. Local borrowing is the most dangerous kind precisely because it is invisible to treelikeness statistics. Third, both kinds of borrowing make the languages look much more similar than their true divergence times imply, and the root age estimated by the glottochronological formula falls steadily as borrowing increases. In a Bayesian analysis with calibrations, part of this bias would be absorbed by the estimated rate, because the calibrating languages would be affected too, but the compression of the tree is real and unevenly distributed. Note what the experiment does and does not show: the borrowing simulated here is superimposed on a true tree, so the experiment speaks to the genealogical reading of Section 8.2 when it holds; it cannot tell us whether a real family had a tree-like history in the first place.

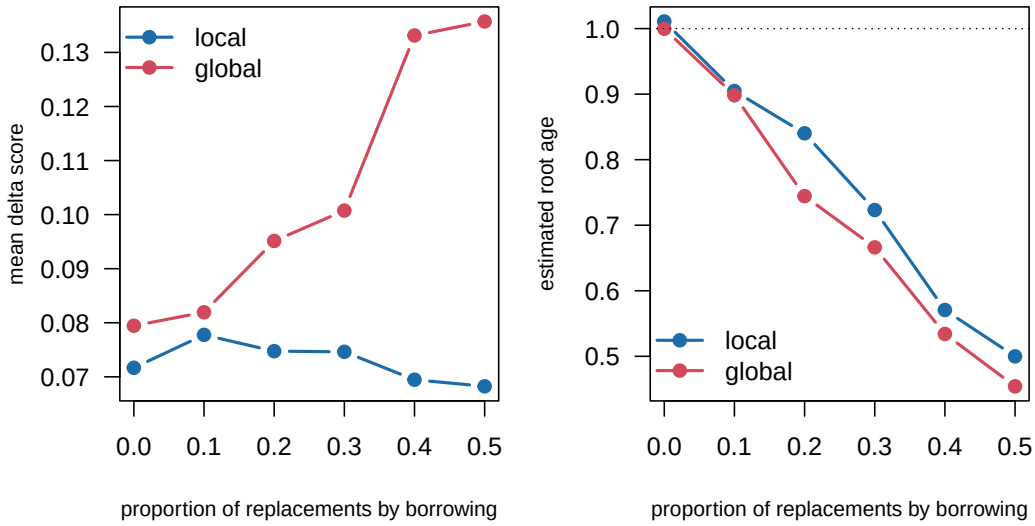


Figure 8.2: Effects of borrowing in the simulation. Left: mean delta score. Right: estimated root age (true value 1, dotted line). Global borrowing increases conflict in the data (higher delta scores), whereas local borrowing hardly changes the delta score; both kinds compress the tree and lead to substantial underestimation of the root age. The topology (not shown) was recovered almost perfectly throughout.

8.4 Measuring treelikeness

8.4.1 Quartets and the four-point condition

How can we tell from the data whether a tree is an adequate summary? The most widely used statistics work with quartets, sets of four languages. Recall from Chapter 4 that a distance matrix fits a tree exactly if and only if it satisfies the **four-point condition**: for every quartet $\{w, x, y, z\}$, of the three sums

$$s_1 = d_{wx} + d_{yz}, \qquad s_2 = d_{wy} + d_{xz}, \qquad s_3 = d_{wz} + d_{xy}, \tag{8.1}$$

the two largest are equal. If the true tree groups w with x and y with z, then s_1 is the smallest sum and $s_2 = s_3$, both equal to s_1 plus twice the length of the internal edge. Real distances never satisfy the condition exactly. The question is how far they depart from it, and whether the departures look like noise or like a systematic conflict.

8.4.2 The delta score

Holland et al. (2002) proposed measuring the departure for each quartet by the **delta score**. Order the sums so that $s_{(1)} \le s_{(2)} \le s_{(3)}$, and define

$$\delta_q = \frac{s_{(3)} - s_{(2)}}{s_{(3)} - s_{(1)}}. \tag{8.2}$$

The delta score is zero when the quartet fits a tree exactly ($s_{(2)} = s_{(3)}$) and one when the quartet is maximally unresolved, with the two largest sums as far apart as the largest and smallest, which

corresponds to a perfect "box" in a split network. Averaging δ_q over all quartets gives a score for the whole dataset; averaging over the quartets that contain a given language gives a **taxon-specific delta score**, which can pick out languages involved in conflicting signal. Because δ_q is a ratio, it does not depend on the overall scale of the distances.

8.4.3 The Q-residual

Gray et al. (2010) introduced the **Q-residual** alongside the delta score in their discussion of tree-likeness in language data. It measures the same departure in absolute rather than relative terms, as the squared difference between the two largest sums,

$$Q_q = \left(s_{(3)} - s_{(2)}\right)^2, \tag{8.3}$$

computed after the distances have been rescaled so that their mean is one, which makes Q-residuals comparable across datasets. The two statistics behave differently when the internal edge of a quartet is short. A nearly unresolved quartet with a short internal edge can have a large delta score, because the denominator $s_{(3)} - s_{(1)}$ is small, even though its absolute departure from treelikeness is tiny. The Q-residual is insensitive to this. For this reason, Gray et al. (2010) recommended reporting both. Implementations differ in details, for example in how distances are normalized and whether a constant factor is included, so values should be compared only when computed in the same way. Both statistics are computed by SplitsTree (Huson and Bryant, 2006); phangorn's `delta.score()` computes delta scores from Hamming distances.

The function `quartet_stats()`, which computes both statistics for every quartet, and a check of its delta scores against `phangorn::delta.score()`, are in Section C.2.2.

Delta scores and Q-residuals measure conflict in the data, not borrowing. Conflict can arise from borrowing, but also from parallel innovation, homoplasy due to rapid change, rate heterogeneity among lineages, coding errors, missing data and simple sampling noise in small datasets. Conversely, as the simulation above showed, borrowing among close relatives produces very little conflict. A low delta score does not prove that a family's history was treelike, and a high score does not prove that it was not.

The next chunk simulates a specific contact episode: during the last 40 percent of the family's history, language B4 borrows heavily from the lineage leading to C5. We compute taxon-specific delta scores with and without the contact. Because the true history is known, this is a check of what a descriptive statistic responds to; in the vocabulary of Section 6.10 it is neither parameter recovery (no model is fitted) nor a test of empirical adequacy.

```
set.seed(5)
M_contact <- sim_borrowing(family, innov = 0.5, borrow = 1, mode = "pair",
                           pair = c("B4", "C5"), window = c(0.6, 1))
set.seed(5)
M_tree <- sim_borrowing(family, innov = 0.5, borrow = 0)
d_contact <- meaning_dist(M_contact); d_tree <- meaning_dist(M_tree)
qs_contact <- quartet_stats(d_contact); qs_tree <- quartet_stats(d_tree)
B_contact <- to_binary(M_contact)            # binary coding, used below
pd_contact <- phyDat(B_contact, type = "USER", levels = c(0, 1))
tab <- rbind(no_contact = round(qs_tree$taxon_delta, 3),
```

```
            contact = round(qs_contact$taxon_delta, 3))
knitr::kable(t(tab), caption = "Taxon-specific delta scores for data simulated on
the toy family without contact and with heavy borrowing into B4 from the lineage of
C5. The recipient of the loans stands out.")
```

Table 8.3: Taxon-specific delta scores for data simulated on the toy family without contact and with heavy borrowing into B4 from the lineage of C5. The recipient of the loans stands out.

	no_contact	contact
A1	0.070	0.118
A2	0.080	0.122
A3	0.101	0.119
A4	0.079	0.117
B1	0.070	0.137
B2	0.073	0.142
B3	0.096	0.154
B4	0.086	0.326
C1	0.076	0.118
C2	0.068	0.117
C3	0.066	0.124
C4	0.076	0.129
C5	0.070	0.149

With contact, the recipient B4 has by far the highest taxon-specific delta score, roughly twice that of any other language. Every other language's score also rises somewhat, because every language belongs to some quartets that contain B4; the next highest values belong to the donor C5 and to B4's former close relatives in subgroup B, whose quartets with B4 are the most disturbed. Taxon-specific scores of this kind have been used to flag languages with mixed histories before an analysis, and to decide whether to exclude them. Wichmann et al. (2011) computed delta scores across a large number of language families from ASJP word lists and explored what predicts how reticulate a family appears.

8.4.4 Other measures

Several other statistics carry information about treelikeness. The **consistency index** and **retention index** from parsimony (Chapter 4) measure how much homoplasy the data require on the best tree; they were the basis of the comparison by Collard et al. (2006). Bootstrap support and posterior probabilities of clades indicate how consistently the data favor particular splits, although they confound conflict with lack of signal. Counts of characters that are incompatible with the tree, as in the compatibility methods of Ringe, Warnow and Taylor (Section 8.6), are directly interpretable for carefully chosen characters. And posterior predictive checks (Section 7.10) can use any of these quantities as test statistics, comparing their observed values with the values expected under a tree model without borrowing. That last use turns a descriptive statistic into a test of the genealogical reading of Section 8.2: if the observed delta score lies far outside what a fitted tree model generates, the tree is at best a summary.

8.5 Split networks: describing conflict

8.5.1 From trees to split systems

A tree can be described by its **splits**: each edge, if cut, divides the languages into two groups (Chapter 2). Two splits are **compatible** if they can appear together on one tree, which is the case when one side of the first split is contained in one side of the second. A tree corresponds to a set of pairwise compatible splits. When data support incompatible splits, for example $\{B4, C5\}$ versus the rest alongside $\{B1, B2, B3, B4\}$ versus the rest, no tree can display them all. A **split network** (or splits graph) displays an arbitrary collection of splits by representing each split not by a single edge but by a band of parallel edges; cutting all the edges of the band separates the two sides of the split. Incompatible splits produce boxes, or more generally parallelograms, and the length of each band represents the weight of the split. A split network is therefore a picture of the *signal* in the data, conflicting signal included, and not a picture of a history. A box between two languages does not say that they were in contact; it says that some characters group them together and others do not. In the terms of Section 8.2, split networks are descriptive: they answer the question "how similar are these languages, and where does the signal conflict?", and no other.

Not every collection of splits can be drawn in the plane without crossing bands. **Circular** split systems, in which the languages can be arranged around a circle so that every split cuts the circle into two arcs, can always be drawn as planar networks. Restricting attention to circular systems keeps the networks readable and the computations fast.

8.5.2 Split decomposition and NeighborNet

Two methods build split networks from distances. **Split decomposition** (Bandelt and Dress, 1992) identifies the splits that are supported by every quartet that spans them, which yields a weakly compatible split system. It is conservative, and with large or noisy datasets it tends to give poorly resolved networks. **NeighborNet** (Bryant and Moulton, 2004) is the method of choice for linguistic data. It is an agglomerative method, like neighbor joining (Chapter 4), but instead of joining two nodes into a cluster at each step it builds up a circular ordering of the languages, allowing a node to be paired with two neighbors rather than one. The ordering defines a circular split system, and the weights of the splits are then estimated by non-negative least squares, so that the patristic distances in the network approximate the observed distances as closely as possible. Splits with zero weight are dropped. If the distances fit a tree, NeighborNet returns that tree; the more conflict there is, the more box-like structure appears.

For binary character data there are also **median networks** (Bandelt et al., 1995), which contain all most parsimonious trees for a set of compatible-in-pairs characters and become very complex when there is much homoplasy, and **median-joining networks** (Bandelt et al., 1999), which reduce that complexity heuristically. Both were developed for mitochondrial DNA within species. Finally, **consensus networks** (Holland and Moulton, 2003; Holland et al., 2004) summarize a collection of trees, such as bootstrap replicates or a posterior sample, by displaying every split that appears in more than a chosen proportion of the trees. Where a consensus tree must discard conflicting splits, a consensus network shows them, which makes it a natural way to display phylogenetic uncertainty as well as conflict. All of these methods are implemented in SplitsTree (Huson and Bryant, 2006), the standard program for split networks, and the main ones are available in phangorn. Huson et al. (2010) give a comprehensive treatment.

8.5.3 NeighborNets of simulated data

The code below draws NeighborNets for the two datasets of the previous section, without contact and with heavy borrowing into B4 from the C5 lineage (Figure 8.3).

```
par(mfrow = c(1, 2), mar = c(0.5, 0.5, 2, 0.5))
nn_tree <- neighborNet(d_tree)
nn_contact <- neighborNet(d_contact)
plot(nn_tree, cex = 0.8, edge.width = 1)
title("no contact", font.main = 1)
plot(nn_contact, cex = 0.8, edge.width = 1)
title("borrowing into B4 from C5", font.main = 1)
```

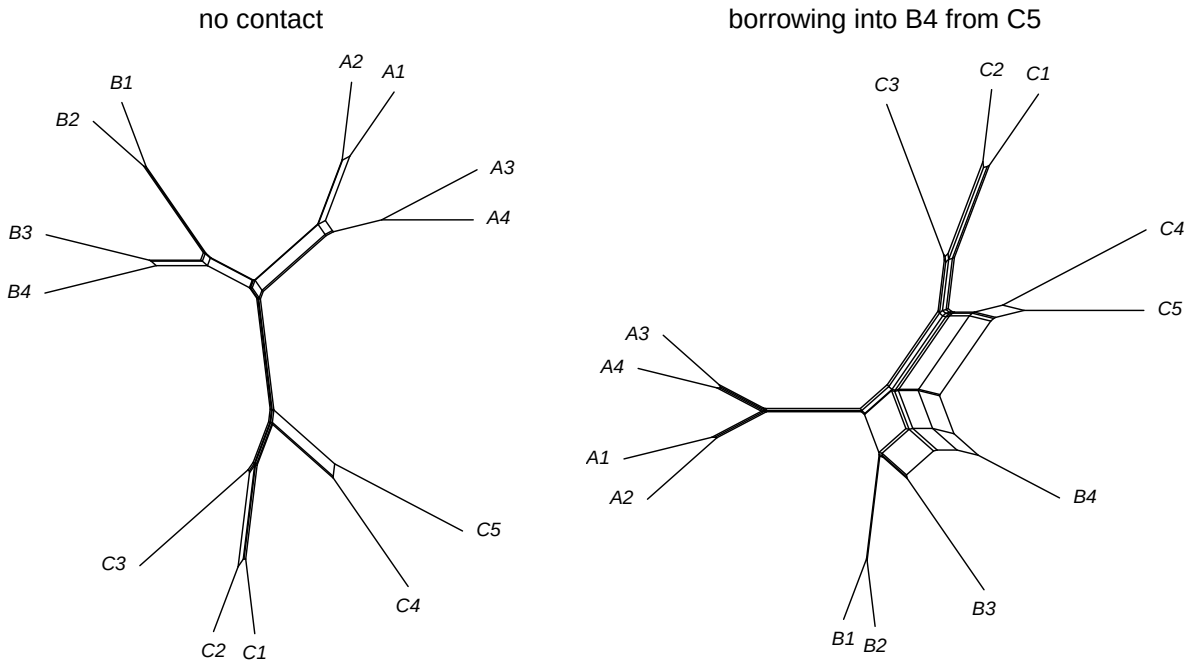


Figure 8.3: NeighborNets of data simulated on the toy family without contact (left) and with heavy borrowing into B4 from the lineage leading to C5 (right). Without contact the network is nearly a tree. With contact, a prominent band of parallel edges represents the split grouping B4 with C5, in conflict with the split grouping B4 with its own subgroup.

The left network is almost a tree: the boxes that appear are small and reflect sampling noise. In the right network, B4 is drawn between subgroup B and C5, and the box that connects them is the visual signature of the conflicting splits. We can quantify the conflict by listing the heaviest splits that are incompatible with the true tree, using a helper function `conflicting_splits()` (listed in Section C.2.3).

```
knitr::kable(conflicting_splits(nn_contact, family), row.names = FALSE,
             digits = 4,
             caption = "The heaviest NeighborNet splits that are absent from the true
             tree, for the data with contact. Each split is labeled by its smaller
             side.")
```

Table 8.4: The heaviest NeighborNet splits that are absent from the true tree, for the data with contact. Each split is labeled by its smaller side.

split	weight
B4,C1,C2,C3,C4,C5	0.0422
B4,C5	0.0258
B4,C4,C5	0.0231

All of the heaviest splits absent from the true tree group B4 with members of subgroup C: with the whole of subgroup C, with C5 alone, and with C4 and C5. This is what the simulated contact predicts. Many of the words B4 acquired from C5 are old words that C5 shares with the rest of its subgroup, so the contact appears not only as a B4-C5 split but also as a split uniting B4 with all of subgroup C. The example also shows the limits of a descriptive network. The network displays the conflict, but nothing in it identifies B4 rather than C5 as the recipient, dates the contact to the last part of the history, or distinguishes contact from, say, parallel innovation. We know these things only because we simulated them. In real applications the true tree is not known, and the analyst must judge which splits are surprising in light of other evidence.

8.5.4 Consensus networks of bootstrap trees

A consensus network over bootstrap trees, or over a posterior sample of trees, shows how the uncertainty about relationships is structured. If the trees disagree only randomly, the network shows many weak, unrelated splits; if they disagree systematically, a few strong conflicting splits appear (Figure 8.4).

```
set.seed(9)
bs <- bootstrap.phyDat(pd_contact, FUN = function(x) NJ(dist.hamming(x)),
                       bs = 200)
cnet <- consensusNet(bs, prob = 0.1)
op <- par(xpd = NA)   # let tip labels at top and bottom extend into margins
plot(cnet, cex = 0.8, edge.width = 1)
```

```
par(op)
```

8.5.5 NeighborNets in linguistics

NeighborNet entered linguistics in the early 2000s and quickly became the standard exploratory tool. Bryant et al. (2005) applied it to Indo-European lexical data and found that, although the major subgroups emerge clearly, the relationships among them are surrounded by conflicting signal. McMahon and McMahon (2005) used NeighborNets throughout their book on quantitative language classification. One of their diagnostics deserves wider use. They divided the 200-item list into a sublist of meanings that are widely reconstructable and rarely replaced and a sublist of meanings that are neither, reasoning that related languages should be more distant on the unstable meanings than on the stable ones. For Quechua and Aymara they found the reverse, and concluded that contact rather than common descent accounts for most of the lexical similarity between the two families (McMahon and McMahon, 2005; McMahon et al., 2005). Comparing trees or networks built from fast and slow subsets of meanings is a cheap way of locating suspected

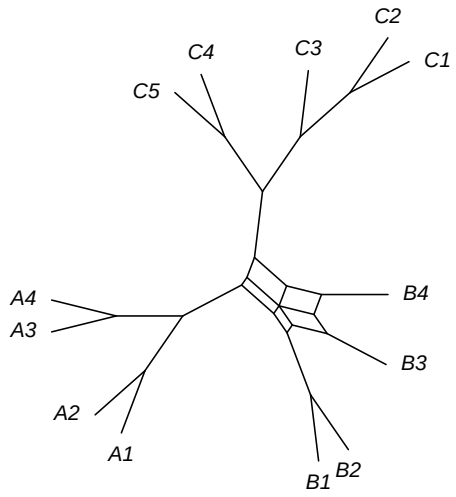


Figure 8.4: Consensus network of 200 bootstrap neighbor-joining trees for the data with contact, showing every split present in at least 10 percent of the bootstrap trees. Edge lengths are proportional to the frequency of the split. The position of B4 is the main source of conflict.

loans before borrowing is modelled explicitly, although it assumes that stability rankings transfer across families. Lehtinen et al. (2014) made a related comparison for Uralic, between networks built from basic vocabulary and from more easily borrowed vocabulary: the networks broadly supported a treelike diversification, but showed conflicting signal behind some ambiguous splits and the effect of borrowing between branches, mostly within and between the Finnic and Saami subgroups. Ben Hamed (2005) analyzed Chinese dialects and showed that their relationships are poorly represented by a tree, consistent with a long history of dialect contact and the influence of standard languages; the theme was developed further by Ben Hamed and Wang (2006). Heggarty et al. (2010) discussed how divergence measures and network analyses can help distinguish tree-like splitting from wave-like diffusion in language histories. Bowern (2010) discussed the relevance of trees and networks for Australian languages, where the extent of borrowing has long been debated. Bakker et al. (2011) used NeighborNets of typological features to argue that creoles form a typologically distinct group (Section 8.9). And Gray et al. (2010) used NeighborNets together with delta scores and Q-residuals to discuss the "shape" of language history across several families.

The shape and fabric of human history. Gray et al. (2010) addressed a long-standing question: is the history of languages and cultures fundamentally treelike or fundamentally reticulate? They argued that the question is empirical and that the answer may differ among datasets and families. They proposed using delta scores and Q-residuals, together with NeighborNet visualizations, to quantify how treelike a dataset is, and introduced the Q-residual for this purpose. Applying these measures to language data, they showed that families vary in their degree of treelikeness, and argued that even fairly reticulate data can contain enough tree signal to support phylogenetic inference, provided the degree of conflict is measured and its consequences assessed. The paper's lasting contribution is methodological: it shifted the debate from assertions about whether "language evolution is a tree" to measurements of how much conflict particular datasets contain. Those measurements describe the data; deciding whether a tree inferred from moderately reticulate data should be read genealogically or only as a summary (Section 8.2) still requires historical argument.

Split networks are routinely over-interpreted. A box in a NeighborNet does not identify a borrowing event, a donor or a recipient, and it carries no information about timing. Boxes can come from sampling error, rate heterogeneity, coding inconsistencies or incomplete data as well as from contact. NeighborNet also always produces a circular split system, so it cannot represent some patterns of conflict at all, and planar drawing places languages on the circle in ways that can suggest spurious relationships between neighbors in the drawing. Use split networks to explore and display conflict, and use generative models or careful linguistic analysis to explain it.

8.6 Explicit models of borrowing

Split networks describe conflict; explicit models try to explain it by positing a tree of descent plus lateral events. The approaches below differ in how far they are *generative* in the sense of Section 8.2. Perfect phylogenetic networks and minimal lateral networks posit explicit contact edges, but choose them by combinatorial or heuristic criteria; the lateral-transfer stochastic Dollo model and the contact-aware models for structural data define a probability for the data, and can therefore estimate borrowing rates, compare hypotheses and be checked against data simulated from the fitted model.

8.6.1 Perfect phylogenetic networks

Ringe, Warnow and Taylor approached Indo-European subgrouping through **compatibility**: a character, such as a shared sound change, a morphological innovation or a cognate class, is compatible with a tree if it can evolve on that tree without homoplasy, that is, with each state forming a connected region. With carefully chosen characters, almost all should be compatible with the true tree (Ringe et al., 2002). The characters that remain incompatible are candidates for borrowing or parallel development. Nakhleh et al. (2005a) formalized this idea as a **perfect phylogenetic network**: a tree plus a small number of contact edges, such that every character is compatible with at least one of the trees displayed by the network. In the Indo-European dataset, a network with only a few contact edges accounted for nearly all characters, and edges involving Germanic were prominent among them. The position of Germanic has long been the most problematic in Indo-European subgrouping. The approach gives explicit, interpretable hypotheses of contact, and because its characters are innovations rather than cognate sets it addresses the question of identifying innovations directly; but it requires characters coded with great care and it does not provide a probability model for weighing alternatives. Warnow et al. (2006) later proposed a stochastic model of language evolution with both homoplasy and borrowing, which could underlie statistical versions of this approach.

8.6.2 Minimal lateral networks

Nelson-Sathi et al. (2011) imported a method from the study of prokaryotic genome evolution (Dagan et al., 2008) to detect hidden borrowing in Indo-European. The approach works as follows. Take a reference tree as given. For each cognate set, consider gain-loss scenarios that allow at most one gain, at most two gains, and so on; under a model allowing more gains, patchily distributed cognates are explained by multiple origins rather than by early origin and many losses. Each model implies a distribution of *ancestral vocabulary sizes*, the number of cognates reconstructed for each ancestral language. The criterion for choosing among models is that ancestral vocabulary sizes should resemble those of the modern languages: a model that allows too few gains produces implausibly large ancestral vocabularies. Under the selected model, cognate sets that require more than one gain are interpreted as candidates for borrowing, and the inferred multiple origins are connected by lateral links, yielding a **minimal lateral network**. The study argued that a non-trivial amount of borrowing had gone undetected in standard Indo-European data. List et al. (2014b) applied the method to Chinese dialect data, and List et al. (2014a) discussed the parallels between lexical borrowing and lateral gene transfer more generally. The main limitations are the dependence on a fixed reference tree, and the fact that the method does not distinguish borrowing from parallel semantic shift or from coding error, all of which create multiple apparent origins.

Networks uncover hidden lexical borrowing. Nelson-Sathi et al. (2011) analyzed the Indo-European Swadesh-list data of Dyen and colleagues, using a reference tree of the major subgroups. Their gain-loss analysis found that models allowing only a single origin per cognate set implied ancestral vocabularies far larger than any modern Indo-European language possesses, so that some cognate sets had to have multiple origins. Interpreting these as borrowings, they built a minimal lateral network linking the branches involved and argued that a non-trivial fraction of the basic vocabulary had been affected by borrowing that had not been flagged by the original coders. The study is a good example of how a method from microbial genomics can be transferred to language with minimal modification. It also illustrates the inferential trade-off: the conclusions rest on the vocabulary-size criterion and on the reference tree, and other researchers have questioned whether all inferred multiple origins are really borrowings.

8.6.3 Lateral transfer in the stochastic Dollo model

The most complete probabilistic treatment of lexical borrowing so far is the extension of the stochastic Dollo model (Chapter 7) by Kelly and Nicholls (2017), row 7 of Table 7.2. In addition to births at rate λ and deaths at rate μ, each trait present in a lineage can be copied into other lineages that exist at the same time, at a rate governed by a borrowing parameter β. A borrowed trait does not displace anything; it simply becomes present in the recipient, where it can subsequently die. The difficulty is computational. Without borrowing, the fate of a trait in different lineages is independent once they have split, which is what makes the pruning recursion of Chapter 7 possible. With borrowing, the presence of a trait in one lineage affects its chance of appearing in others, so the likelihood requires tracking the joint distribution of each trait across all lineages alive at a given time, and the cost grows quickly with the number of contemporaneous lineages. Kelly and Nicholls (2017) showed how to compute the likelihood exactly, given the tree, for moderate numbers of languages, and implemented the model in TraitLab. This allows the borrowing rate to be estimated jointly with the tree and the dates, and the question of whether ignoring borrowing biases date estimates to be addressed within a single model rather than by simulation. Like any rate parameter, the borrowing rate is identifiable under the model only jointly with the death rate and the tree, and borrowing among close relatives, which barely alters the patterns of presence (Section 8.3), will be hard to estimate in practice.

Why borrowing breaks the pruning recursion, and the size of the state space the lateral-transfer likelihood must track, are explained in Section C.2.4.

8.6.4 Contact events on a tree: contacTrees

Neureiter et al. (2022) approached the same goal in BEAST 2 with a model they call *contacTrees*. It infers the phylogenetic tree of a language family together with contact events between its clades, represented as contact edges between lineages that coexist in time, so that the history of an individual cognate class can follow a contact edge instead of the tree. In a simulation study the model recovered the simulated history, including the contact events, and showed that ignoring contact can bias the estimated tree height, rate of change and topology; this is parameter recovery under the model's own assumptions (Section 6.10). Applied to a subset of well-documented Indo-European languages, it recovered known contact events and identified known loanwords. As with the lateral-transfer Dollo model, borrowing parameters are identifiable only jointly with the tree and the rates, and contact between close relatives will be hard to detect, for the reasons given in Section 8.3.

8.6.5 Contact-aware models for structural data

Borrowing affects grammar as well as vocabulary, and several models address structural features. Daumé III (2009) proposed a non-parametric Bayesian model that combines a phylogenetic component with latent geographic "areas" in which features diffuse. Cathcart et al. (2018) modelled the evolution of Indo-European grammatical features with terms representing pressure from geographically neighboring languages. And the sBayes model of Ranacher et al. (2021) identifies contact zones as sets of languages that share features beyond what their family membership and global preferences predict, using a mixture of inheritance, contact and universal components. These are generative models in the sense of Section 8.2, and they move toward the joint modelling of descent and diffusion that the biology of introgression has developed (Section 8.11). Hartmann and Jäger (2024) proposed a related device for trees inferred from lexical data: a distance-based Gaussian

process with latent phylogenetic distances that identifies potential contact zones and estimates how strongly geographic proximity affects clade support and pairwise distances.

8.7 Waves, linkages and Historical Glottometry

8.7.1 The wave model

The oldest alternative to the family tree is the **wave model** (*Wellentheorie*) of Schmidt (1872). Schmidt observed that the Indo-European subgroups share innovations in overlapping patterns: Balto-Slavic shares some features with Indo-Iranian, others with Germanic, and so on, in a way that no single tree can accommodate. He proposed that innovations spread like waves across a dialect continuum, each covering a different, possibly overlapping, area. The wave model was never fully abandoned, and dialectologists have always worked with it, but because it offered no procedure for reconstruction comparable to the tree-based comparative method, it remained largely qualitative.

The Oceanic linguist Malcolm Ross gave the wave picture a sharper typology. He distinguished **families**, which arise when a speech community splits into separate groups that then innovate independently, from **linkages**, which arise when a dialect network diversifies in place, so that innovations spread across overlapping subsets of dialects (Ross, 1988, 1997). In Ross's terms, a family is *innovation-defined*, because each of its subgroups is defined by innovations shared by all and only its members, whereas a linkage is *innovation-linked*, characterized by intersecting innovations. Many Oceanic groups, in Ross's analysis, are linkages rather than families. The distinction is fundamental for phylogenetics: a tree model is appropriate for families, and for linkages it is not merely approximate but misspecified, because there is no sequence of splits for it to approximate (Section 8.2).

8.7.2 Historical Glottometry

François (2015) and Kalyan and François (2018) developed **Historical Glottometry**, a method that takes the wave model seriously as a quantitative framework. Its input is a set of innovations established by the comparative method, such as sound changes, morphological changes and lexical replacements, each with the set of languages that share it. No tree is assumed. Instead, for every subgroup G (any set of languages that share at least one innovation), the method computes several quantities. Let S_i be the set of languages sharing innovation i. Then:

- the number of **exclusively shared innovations** is $\varepsilon_G = |\{i : S_i = G\}|$;
- innovations **support** G if they are shared by all members of G, $G \subseteq S_i$;
- innovations **conflict** with G if they are shared by some members of G together with languages outside it, that is, $S_i \cap G \neq \emptyset$, $S_i \setminus G \neq \emptyset$ and $G \setminus S_i \neq \emptyset$;
- the **cohesiveness** of G is the proportion of relevant innovations that support it,

$$\kappa_G = \frac{|\{i : G \subseteq S_i\}|}{|\{i : G \subseteq S_i\}| + |\{i : S_i \text{ conflicts with } G\}|}; \tag{8.4}$$

- and the **subgroup strength** is $\varsigma_G = \varepsilon_G \kappa_G$, which weights the number of exclusive innovations by how consistently the group behaves as a unit.

Innovations shared by a proper subset of G and by no outsiders are neither supporting nor conflicting: they are compatible with G as a subgroup containing smaller subgroups. A tree is the special case in which no innovation ever conflicts with any subgroup, so that every cohesiveness equals one. The results are displayed in a **glottometric diagram**, a map-like picture in which each subgroup is drawn as an isogloss whose thickness is proportional to its strength. The formulas above follow Kalyan and François (2018) in slightly simplified form, and readers who want to apply the method should consult the original for the treatment of details such as innovations that are uncertain or attested in only part of a subgroup. Glottometry is descriptive in the sense of Section 8.2, but it describes something different from a split network: the distribution of innovations identified by the comparative method, not overall similarity.

The code below applies these definitions to a toy chain of seven languages, V1 to V7, in which innovations spread over overlapping stretches of the chain, as in a dialect continuum.

```
innov <- c(rep(list(c("V1", "V2")), 6), rep(list(c("V2", "V3")), 3),
           rep(list(c("V1", "V2", "V3")), 5), rep(list(c("V3", "V4")), 2),
           rep(list(c("V4", "V5")), 4), rep(list(c("V4", "V5", "V6")), 3),
           rep(list(c("V5", "V6")), 2), rep(list(c("V6", "V7")), 5),
           rep(list(c("V5", "V6", "V7")), 2), rep(list(c("V3", "V4", "V5")), 2),
           list(c("V1", "V2", "V3", "V4")),
           rep(list(c("V4", "V5", "V6", "V7")), 2))
glottometry <- function(innov) {
  key <- sapply(innov, function(s) paste(sort(s), collapse = "+"))
  groups <- unique(innov[!duplicated(key)])
  res <- t(sapply(groups, function(G) {
    eps <- sum(sapply(innov, function(S) setequal(S, G)))
    sup <- sum(sapply(innov, function(S) all(G %in% S)))
    con <- sum(sapply(innov, function(S)
      any(S %in% G) && any(!S %in% G) && any(!G %in% S)))
    c(exclusive = eps, supporting = sup, conflicting = con,
      cohesiveness = sup / (sup + con), strength = eps * sup / (sup + con))
  }))
  out <- data.frame(subgroup = sapply(groups, paste, collapse = "+"), res)
  out[order(-out$strength), ]
}
gm <- glottometry(innov)
knitr::kable(gm, row.names = FALSE, digits = 2,
             caption = "Historical Glottometry for a toy dialect chain: exclusively
             shared, supporting and conflicting innovations, cohesiveness and
             subgroup strength for each subgroup attested by at least one
             innovation.")
```

Table 8.5: Historical Glottometry for a toy dialect chain: exclusively shared, supporting and conflicting innovations, cohesiveness and subgroup strength for each subgroup attested by at least one innovation.

subgroup	exclusive	supporting	conflicting	cohesiveness	strength
V1+V2	6	12	3	0.80	4.80
V6+V7	5	9	5	0.64	3.21

subgroup	exclusive	supporting	conflicting	cohesiveness	strength
V1 + V2 + V3	5	6	4	0.60	3.00
V4 + V5	4	11	7	0.61	2.44
V2 + V3	3	9	10	0.47	1.42
V5 + V6	2	9	11	0.45	0.90
V4 + V5 + V6	3	5	12	0.29	0.88
V5 + V6 + V7	2	4	9	0.31	0.62
V4 + V5 + V6 + V7	2	2	5	0.29	0.57
V3 + V4	2	5	17	0.23	0.45
V3 + V4 + V5	2	2	18	0.10	0.20
V1 + V2 + V3 + V4	1	1	11	0.08	0.08

```
langs <- paste0("V", 1:7)
Xg <- sapply(innov, function(S) as.integer(langs %in% S))
rownames(Xg) <- langs
chain_net <- neighborNet(dist.hamming(phyDat(Xg, type = "USER",
                                             levels = c(0, 1))))
# draw the network without labels, then set each label just outside its tip,
# on the side facing away from the centre of the network
op <- par(mar = c(1.5, 1.5, 1.5, 1.5), xpd = NA)
pn <- plot(chain_net, show.tip.label = FALSE, edge.width = 1)
v <- pn$.plot$vertices[seq_along(chain_net$tip.label), , drop = FALSE]
d <- sweep(v, 2, colMeans(pn$.plot$vertices))
u <- d / sqrt(rowSums(d^2))   # unit vector from the centre towards each tip
off <- 0.6 * strwidth("V0", cex = 0.9, font = 3)   # gap between tip and label
for (i in seq_len(nrow(v)))
  text(v[i, 1] + off * u[i, 1], v[i, 2] + off * u[i, 2], chain_net$tip.label[i],
       cex = 0.9, font = 3, adj = c(0.5 - 0.5 * u[i, 1], 0.5 - 0.5 * u[i, 2]))
```

```
par(op)
```

The strongest subgroups are V1 + V2 and V6 + V7, which have many exclusive innovations and are rarely cut across. Groups such as V1 + V2 + V3 are also strong, despite being crossed by innovations linking V3 to V4. Groups like V3 + V4, with few exclusive innovations and much conflicting evidence, are weak. The table and network (Figure 8.5) together describe a structure that a tree would force into an arbitrary hierarchy: a tree must either join V3 with V1 and V2 or with V4, discarding the evidence for the other grouping. Whichever it chose, its internal node would be a summary of the data, not a proto-language.

The Torres and Banks languages of Vanuatu. The Torres and Banks Islands of northern Vanuatu are home to seventeen closely related Oceanic languages spoken in a chain of small islands. François (2015) and Kalyan and François (2018) assembled several hundred innovations in phonology, morphology and lexicon from comparative work on these languages, and found that their distributions intersect so extensively that no tree can accommodate more than a fraction of them.

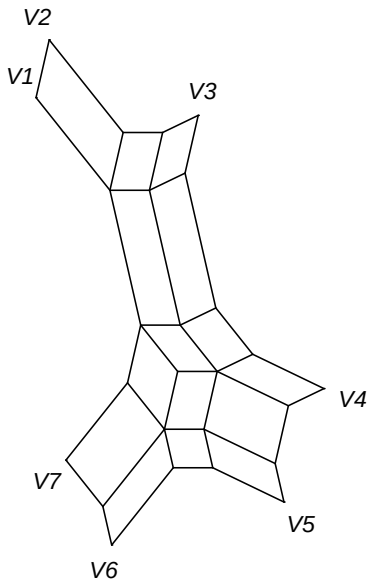


Figure 8.5: NeighborNet of the same toy innovations, coded as binary characters (one per innovation). The chain-like structure of overlapping innovations produces a network with conflicting splits throughout, which no single tree can represent.

Their glottometric analysis showed strong subgroups among geographically adjacent languages and a pattern of overlapping groupings that follows the geography of the islands. They interpreted the result as the signature of a dialect network that diversified in place, with innovations spreading between neighboring communities after the initial settlement: in Ross's terms, a linkage rather than a family. The case is a reminder that the comparative method itself produces evidence that may not be treelike, and that forcing such evidence into a tree can manufacture subgroups that never existed as speech communities.

8.7.3 Critiques and the place of trees

Historical Glottometry has been criticized from the standpoint of tree-based historical linguistics. Jacques and List (2019) argued that many apparent conflicts among innovations can be explained by parallel innovation (especially of common, natural sound changes) or by later diffusion between languages that had already separated, and that distinguishing these possibilities requires more than counting shared innovations. They argued that tree models remain indispensable for reconstruction, and that the choice between tree and wave models should depend on the data rather than on a general preference. Their paper appeared in a special issue of the *Journal of Historical Linguistics* on alternatives to the tree model (Kalyan et al., 2019), together with a reply by Kalyan and François (2019), and with a study by Verkerk (2019) that took a model-based route: the multiple-topologies method of BayesPhylogenies, which lets different characters follow different trees, found evidence for more than one topology in lexical data from Sinitic, Indo-European and Japonic, but not in Austronesian, and Verkerk discusses dialect-chain break-up, borrowing, incomplete lineage sorting and properties of lexical datasets as possible sources of this reticulation. They also pointed out that glottometry, as formulated, treats all innovations as equally informative and does not model the probability of parallel change, which a probabilistic phylogenetic model does.

Defenders of glottometry reply that the method is intended for exactly the situations in which the tree model fails, notably dialect continua that diversified in place, and that it makes explicit what tree diagrams hide. In the terms of Table 8.1, the two sides are partly answering different questions: glottometry describes the distribution of innovations, while tree-based reconstruction asks about the sequence of separations, and the dispute is over which question a particular body of evidence can answer. A synthesis would combine the two: a generative model with both descent and diffusion, in which the relative contributions of each are estimated from the data. The biological models of Section 8.11 show what such models can look like.

One step toward such a synthesis within the tree framework is the *Dialect Chain Tree* of Elgh and Hammarström (2024), an extension of the tree model that represents both tree-like descent and the systematic disintegration of dialect chains, and so formalizes the notion of a linkage in a way that quantitative methods can use. Bayesian case studies point the same way. In the Timor-Alor-Pantar family, Kaiping and Klamer (2022) concluded that dialect-cluster effects played a major role in the formation of the present languages; King et al. (2024) found their Philippine results consistent with a linkage shaped by diffusion after a rapid initial spread; and Auderset et al. (2023) showed that Bayesian methods can suggest new subgroups within the Mixtecan dialect continuum while raising questions about how primary data are gathered and organized.

8.8 Dialect continua and dialectometry

Dialect continua are the clearest case in which treelike descent is the wrong model. Neighboring dialects are mutually intelligible, innovations spread across dialect boundaries, and the boundaries themselves are often gradients rather than lines. Dialectology developed its own quantitative tradition, **dialectometry**, which measures aggregate distances between dialects from large numbers of linguistic features and analyses them with clustering, multidimensional scaling and network methods. Nerbonne (2009) and Wieling and Nerbonne (2015) review the field, including the use of edit-distance measures of pronunciation difference, statistical tests of the relation between linguistic and geographic distance, and the problem of identifying stable dialect groups when different clustering methods disagree (Prokić and Nerbonne, 2008). Dialect data can also be analyzed with the categories of historical linguistics rather than with aggregate distances: Babel et al. (2013) studied the dialectology of Western Numic (Uto-Aztecan) and argued that both descent and diffusion shaped its diversification. Phylogenetic network methods entered dialectology through NeighborNet analyses such as those of Chinese dialects (Ben Hamed, 2005; List et al., 2014b). A recurrent finding is that linguistic distance increases with geographic distance roughly as one would expect from local diffusion, so that a spatially explicit model of diffusion is often a better null model than a tree.

The phylogenetic relevance of dialectometry is twofold. First, it shows that the "language" level at which most phylogenetic analyses operate is itself an abstraction over a continuum, and that the choice of which varieties to sample affects the shape of the result (Chapter 15). Second, its methods for relating linguistic to geographic distance, and for detecting where gradients become boundaries, have counterparts in landscape genetics and phylogeography (Chapter 11) that have scarcely been brought together.

8.9 Creoles and mixed languages

Creoles and mixed languages strain the tree model in the most extreme way, because they do not have a single parent. Thomason and Kaufman (1988) argued that such languages are "not genetically related" to any of their source languages in the traditional sense, since the comparative method's assumption of continuous transmission of the whole linguistic system breaks down. Mixed languages make the point vividly. Michif, spoken by Métis communities in Canada and the northern United States, combines French noun phrases with Plains Cree verbs (Bakker, 1997); other mixed languages combine the grammar of one language with the lexicon of another in different ways (Matras and Bakker, 2003). A tree built from lexical data alone would place Michif with French or with Cree depending on which part of the vocabulary dominated the list, and neither placement would be a meaningful statement about its history.

Creoles raise a related but different question: whether they form a typologically distinct class of languages. Bakker et al. (2011) compared creoles and non-creoles on a set of structural features and used NeighborNet to show that creoles cluster together, apart from both their lexifier languages and other languages, which they took as evidence for a distinct creole typological profile. The interpretation has been contested, partly because the result depends on the choice of features and the sampling of comparison languages. The Atlas of Pidgin and Creole Language Structures (Michaelis et al., 2013) now provides a much larger standardized dataset for questions of this kind. For phylogenetic purposes, the essential point is that such languages require models with multiple parents, such as hybridization networks or admixture graphs, or should be excluded from tree-based analyses with an explicit justification.

8.10 Contact and typological convergence

Contact affects grammar as well as vocabulary, and grammatical convergence among unrelated languages creates **linguistic areas** (*Sprachbünde*). The Balkan languages, belonging to several different branches of Indo-European, share features such as a postposed definite article and the loss of the infinitive; the languages of mainland Southeast Asia, from several families, converge on tone, isolating morphology and classifier systems. For phylogenetic inference from structural data, convergence is a confound of the same kind as lexical borrowing. For comparative studies of typology, it is one half of **Galton's problem**: languages may share features because they share ancestors or because they are neighbors (Chapter 13).

Statistical approaches that separate inheritance from contact in typological data come in several forms. Population-genetic clustering methods such as STRUCTURE have been applied to structural features: Reesink et al. (2009) analyzed the languages of Sahul (New Guinea, Island Melanesia and Australia) and identified components that combine deep inherited signal with areal spread. Dunn et al. (2008) explored the use of structural features to reconstruct deep relationships among Papuan languages of Island Melanesia, where lexical evidence is exhausted, and discussed the danger that contact-induced similarity may mimic deep relationship. The latent-area model of Daumé III (2009), the model of Murawaki and Yamauchi (2018), which estimates for each feature both its vertical stability and its horizontal diffusibility, the areal-pressure model of Cathcart et al. (2018) and the contact-zone mixture model of Ranacher et al. (2021) (Section 8.6) are explicit models for typological data. Chapter 13 discusses spatio-phylogenetic regression models that include both a phylogenetic and a spatial covariance term, which are the most widely used tool for this problem in comparative typology.

Genetics now offers an independent proxy for contact. Graff et al. (2025a) used genetic admixture between speaker populations, and location in a shared geohistorical area, as proxies for contact, and found that language pairs with such contact share noticeably more linguistic features, with a similar increase across world regions and demographic situations. The effect varied strongly among features and was only partly explained by their learnability or social function, and for some features admixture *decreased* sharing, which the authors interpret as a possible signal of divergence under contact (schismogenesis). Results of this kind suggest realistic orders of magnitude for borrowing in simulation studies of structural data.

8.11 Phylogenetic networks from biology

The most striking contrast between linguistics and biology in this area is the sophistication of the explicit network models that biologists now use routinely. Split networks, which dominate linguistic network analysis, are *implicit* networks: they display conflict but do not posit reticulation events. *Explicit* phylogenetic networks are rooted, directed graphs in which some nodes (reticulation nodes) have two parents, each representing a hybridization or introgression event with an estimated time and an estimated proportion of inheritance from each parent (Huson et al., 2010). The implicit/explicit distinction of the network literature is the descriptive/generative distinction of Section 8.2, applied to graphs; explicit networks fitted under a probability model are the generative models that linguistics mostly lacks.

Explicit networks under the multispecies network coalescent. In biology, gene trees differ from the species tree for two main reasons: **incomplete lineage sorting** (ancestral polymorphism sorted differently in different descendant lineages) and **hybridization/introgression**. The multispecies network coalescent combines both: each locus has its own gene tree, generated by a coalescent process within the branches of a species network that includes reticulation nodes. PhyloNet (Than et al., 2008; Wen et al., 2018) infers such networks by maximum likelihood, maximum pseudo-likelihood or Bayesian MCMC; SNaQ, in the PhyloNetworks package (Solís-Lemus and Ané, 2016; Solís-Lemus et al., 2017), uses pseudo-likelihood based on quartet concordance factors and scales to large datasets; and SpeciesNetwork (Zhang et al., 2018a) performs full Bayesian inference of species networks in BEAST 2. The linguistic analogy is close but not exact (Chapter 16 discusses what an explicit model of lexical transmission would need for it to hold). Each meaning's history of cognate replacement is like a gene tree, ancestral synonymy (two words for one meaning in a proto-language, sorted differently in its descendants) is like incomplete lineage sorting, and borrowing of a word into a lineage is like introgression of a locus. A language network with a reticulation node of inheritance proportion γ would say that a fraction γ of the basic vocabulary of a lineage comes from one parent and $1-\gamma$ from the other, which is exactly the kind of statement linguists make about heavily mixed lexicons. To our knowledge, these methods have rarely if ever been applied to lexical data. **Admixture tests and graphs.** The D-statistic (ABBA-BABA test) detects introgression from an excess of discordant patterns of one type over the other in a four-taxon comparison (Green et al., 2010; Durand et al., 2011); Patterson et al. (2012) developed a family of related f-statistics and admixture graphs, and TreeMix (Pickrell and Pritchard, 2012) infers population trees with migration edges from allele frequencies. The four-taxon logic transfers directly to cognate data: for languages (((P1, P2), P3), O), an excess of cognates shared by P2 and P3 over those shared by P1 and P3 indicates contact between P2 and P3, and can be tested formally. **Ancestral recombination graphs.** Within species, the ancestral recombination graph describes how recombination gives different parts of the genome different genealogies (Griffiths and Marjoram, 1997), and modern methods such as ARGweaver infer it from genome sequences

(Rasmussen et al., 2014). The ARG is the natural model for dialect continua, in which different features have different genealogies, but no such model has, to our knowledge, been developed for linguistic data. **Reconciliation.** Gene-tree/species-tree reconciliation with duplication, transfer and loss (DTL) models infers transfers by comparing each gene tree with the species tree; Chapter 16 discusses how it could be applied to the histories of individual meanings.

The obstacles to applying these methods to language are partly practical and partly conceptual. Practically, multispecies-coalescent methods need many independent loci, each with enough information to estimate a gene tree or at least concordance factors. A single meaning has only a handful of cognate classes and carries very little information about its own tree, so the methods would have to work from site patterns or quartet frequencies pooled across meanings rather than from estimated gene trees. Conceptually, borrowing in language often happens between languages that have been separated for a long time, and continuously, rather than in a few discrete pulses, which strains models built around a small number of reticulation events. And an explicit network still presupposes a mostly genealogical history with occasional merging; for linkages and continua, the ancestral recombination graph or a spatial diffusion model is the better analogy. Neither obstacle seems insurmountable, and the payoff would be large: explicit, testable statements about which languages exchanged how much vocabulary, and when.

8.12 Practical guidance

For an analyst working with lexical data from a family in which contact is known or suspected, the following sequence is a reasonable default.

1. **State the historical question.** Decide whether the goal is to reconstruct divergence, describe similarity, identify innovations or estimate a dominant pattern of transmission (Table 8.1), and say whether the tree is to be read genealogically or as a summary.
2. **Flag and examine known loans.** Code them so that they do not unite donor and recipient, and record the decision. Analyze the data both with and without problematic items.
3. **Describe the conflict.** Compute delta scores and Q-residuals for the whole dataset and for each language, and draw a NeighborNet. Look for languages with exceptionally high taxon-specific scores and for strong conflicting splits.
4. **Interpret conflict linguistically.** Ask whether conflicting splits correspond to known contact situations, dialect continua, coding inconsistencies or simply poorly sampled languages. If the data look like a linkage, reconsider whether a dated tree answers any meaningful question.
5. **Assess sensitivity.** Rerun the phylogenetic analysis without the most reticulate languages and without the meanings most prone to borrowing (for example, using the Leipzig-Jakarta list), and compare topologies and dates.
6. **Remember the invisible kind.** Borrowing among close relatives leaves little trace in tree-likeness statistics but can compress trees and bias dates. When it is suspected, consider simulation studies tailored to the family, or explicit lateral-transfer models.
7. **Use descriptive networks to explore, generative models to explain.** Split networks are exploratory displays. Statements about who borrowed from whom, how much and when require generative models or careful comparative analysis.

- A tree can be read as an approximation to a genealogy or as a statistical summary of variation. Dating, ancestral reconstruction and homeland inference require the genealogical reading; under extensive reticulation there may be no bifurcating history for a tree to approximate.

- The choice of representation should follow from the historical question: reconstructing divergence, describing similarity, identifying innovations, or estimating a dominant pattern of transmission (Table 8.1).
- Basic vocabulary is resistant to borrowing but not immune; undetected loans between close relatives are the hardest to find and the hardest for methods to detect.
- Simulation studies suggest that tree topologies are fairly robust to realistic levels of borrowing superimposed on a tree, while branch lengths and dates are more sensitive; borrowing among close relatives compresses trees without creating much conflict.
- Delta scores and Q-residuals measure the departure of quartets from the four-point condition, in relative and absolute terms respectively; they measure conflict, not borrowing as such.
- Split networks (NeighborNet, split decomposition, consensus networks) are descriptive: they display conflicting signal, and their boxes have many possible causes, no direction and no date.
- Generative models of contact (the lateral-transfer stochastic Dollo model, contact-aware models for structural data, explicit networks) define a probability for the data and can estimate how much borrowing occurred, at the price of new assumptions and weakly identified parameters. Perfect phylogenetic networks and minimal lateral networks posit explicit contact edges without a full probability model.
- The wave model, Ross's linkages and Historical Glottometry describe diversification in a dialect network; dialect continua, creoles and mixed languages require models with multiple parents or spatial diffusion.
- Biology offers explicit network models under the multispecies network coalescent, admixture tests and ancestral recombination graphs, almost none of which have been applied to language.

8.13 Exercises

Conceptual

1. Explain the difference between reading a tree as an approximation to a genealogy and reading it as a statistical summary of variation. For each of the following, say which reading it requires: a NeighborNet, a dated MCC tree used to estimate the age of a proto-language, a reconstruction of the proto-lexicon at an internal node, a dendrogram of dialect distances.
2. A NeighborNet shows a large box joining language X with a language of a different subgroup. List four possible causes, and for each say what additional evidence would distinguish it from borrowing.
3. Why is local borrowing nearly invisible to delta scores and Q-residuals, and yet harmful to date estimates?
4. Using Table 8.1, explain why dating the internal nodes of a tree inferred from a linkage is not a meaningful exercise, even if the tree has high posterior support.
5. What does a generative model of contact allow that a descriptive network does not? What does it cost?

Computational

1. With `sim_borrowing()` and `quartet_stats()`, extend the experiment of Section 8.3 to the `local_scale` values 0.05, 0.2 and 1. How does the relation between the borrowing proportion and the mean delta score change as borrowing becomes more strictly local?

2. Using `sim_borrowing(..., mode = "pair")`, simulate contact from C5 into B4 in an early window (`window = c(0.2, 0.5)`) and a late window (`window = c(0.6, 1)`) of the same intensity. Compare the taxon-specific delta scores and the NeighborNets (`neighborNet()`). Which episode is easier to detect, and why?
3. Implement the four-taxon test described in the biobox of Section 8.11 for cognate data: for the binary coding `B_contact` and the quartet (((B3, B4), C5), A1), count the cognate classes present in B4 and C5 but not B3, and those present in B3 and C5 but not B4, and compare the two counts with those obtained from `to_binary(M_tree)`. How would you assess whether a difference is larger than expected by chance?
4. Modify the toy innovations `innov` of Section 8.7 so that the chain becomes a family with two clean subgroups, rerun `glottometry()`, and check that every cohesiveness equals one.

Research

1. You are asked to estimate the age of a group of about 25 closely related languages spoken along a river valley, where dialectologists report gradual transitions between neighboring varieties. Using Table 8.1, decide which historical question can sensibly be asked of the data, and design an analysis that first tests whether the genealogical reading of a tree is defensible.
2. Propose a simulation study that would tell you how well the lateral-transfer stochastic Dollo model distinguishes local borrowing from short branches in a family of 30 languages. Specify the generating scenarios, the quantities to be recorded, and which validation categories of Section 6.10 each scenario addresses.
3. A language is suspected to be mixed, with a lexicon drawn from two families. Compare how a tree analysis, a NeighborNet and an explicit network model with an inheritance proportion γ would represent it, and argue which you would report.

8.14 Further reading

- Thomason and Kaufman (1988): The classic treatment of contact-induced change, mixed languages and the limits of genetic classification.
- McMahon and McMahon (2005): Simulations of borrowing on trees, network analyses of Indo-European, Andean and Australian data, and the use of stable versus unstable sublists to diagnose contact.
- Haspelmath and Tadmor (2009) and Tadmor et al. (2010). The World Loanword Database project and its findings on what is borrowed and how often; the empirical basis for estimating realistic borrowing rates.
- Greenhill et al. (2009): The simulation study of how borrowing affects phylogenetic inference; a model for family-specific simulation studies.
- Gray et al. (2010): The argument for measuring treelikeness, with delta scores, Q-residuals and NeighborNets applied to language data.
- Bryant and Moulton (2004) and Huson and Bryant (2006). The NeighborNet algorithm and the SplitsTree software; Huson et al. (2010) is the comprehensive textbook on phylogenetic networks.
- Nakhleh et al. (2005a): Perfect phylogenetic networks for Indo-European; the clearest statement of the compatibility approach to contact.
- Nelson-Sathi et al. (2011): Minimal lateral networks and hidden borrowing in Indo-

European; a direct transfer from microbial genomics.

- Kelly and Nicholls (2017): The lateral-transfer stochastic Dollo model; technically demanding, but the most complete probabilistic model of lexical borrowing on a tree.
- Neureiter et al. (2022): contacTrees, a BEAST 2 model that infers contact events jointly with the tree, with a simulation study and an Indo-European case study.
- Elgh and Hammarström (2024): The Dialect Chain Tree, a formal extension of the tree model to linkages.
- François (2015) and Kalyan and François (2018). The wave model, linkages and Historical Glottometry, with the Torres and Banks case study; read together with the critique by Jacques and List (2019).
- Solís-Lemus and Ané (2016) and Wen et al. (2018). Entry points to explicit phylogenetic network inference in biology, for readers who want to bring these methods to language.

Chapter 9

Clock Models and Rates of Language Change

A phylogeny inferred from cognate data tells us how much change separates languages, not how much time. Turning change into time requires two things: a model of how the rate of change behaves along the branches of the tree, called a *clock model*, and external information that fixes the scale. This chapter follows the idea from glottochronology, which assumed that every language replaces its basic vocabulary at one universal rate, to the relaxed clocks of modern Bayesian phylogenetics, and then asks what linguistic phylogenetics has learned about the rates themselves. Its central lesson is that lexical data never identify rate and time separately. Every date is a joint product of the data, the clock model, the tree prior and the calibrations, and the first question to ask of any clock model is which of its parameters the data actually inform. Chapter 10 takes up calibration and the dates that have been obtained for particular families.

After reading this chapter you should be able to:

- explain why rate and time are confounded in every substitution model, and state which quantities (branch lengths, relative ages, absolute ages, branch rates) are identifiable under which assumptions;
- derive the glottochronological formula from its assumptions, and explain which assumptions failed and why rate variation among meanings biases its dates downward;
- distinguish strict, uncorrelated, autocorrelated, local and random local clocks, and penalized-likelihood rate smoothing, by what each assumes about rate variation among lineages;
- distinguish rate variation among characters from rate variation among lineages, and interpret a published clock rate in light of how the data were coded;
- date a simulated phylogram under several rate models with `ape::chronos()`, and say which of the chapter's simulations are parameter-recovery checks and which are misspecification experiments;
- evaluate empirical claims about rates of language change (word frequency, punctuational bursts, grammar versus vocabulary, population size) in terms of their design and main confounds.

9.1 Why time is hard

Imagine two languages that have each replaced forty of the meanings on a two-hundred-item basic vocabulary list since they last shared a common ancestor. Did they separate a thousand years ago and change quickly, or three thousand years ago and change slowly? The cognate data alone cannot tell us. Every substitution model introduced in Chapters 5 and 7 depends on time t and rate r only through their product. The transition probability matrix for a branch is

$$\mathbf{P}(b) = e^{\mathbf{Q}b}, \qquad b = r\,t, \tag{9.1}$$

where b is the *branch length*, measured in expected numbers of changes per character. Doubling the rate and halving the time leaves b, and therefore the likelihood, unchanged. Rate and time are said to be *confounded*, or *non-identifiable*, from the data alone.

This is the central problem of all phylogenetic dating, in biology and linguistics alike. It has two parts. First, we need some external information that fixes the absolute scale of time: a *calibration*, such as the date at which a daughter language was first written down, or the archaeologically dated settlement of an island. Calibrations are the subject of Chapter 10. Second, we need a model that tells us how rates are allowed to vary from one branch of the tree to another. Without such a model, every branch could have its own rate, and the branch lengths of a tree estimated without a clock (a *phylogram*) would carry no information at all about the relative ages of its nodes. A clock model is a set of assumptions, ideally probabilistic, about how r varies across the branches of a tree. The strongest assumption is that r is the same everywhere. That is the *strict clock*, and it is where linguistic dating began.

A concrete example shows the stakes. Icelandic and Norwegian both descend from the Old West Norse spoken around the time Iceland was settled in the late ninth century CE. Over the following millennium, Norwegian replaced far more of its basic vocabulary than Icelandic did, partly through contact with Danish and Low German. A model that assumed a single rate for both lineages would place their split at different dates depending on which language it treated as typical. English and Frisian, or the Romance languages descended from a well-documented Latin, provide similar cases where the true time depth is known from written history, and where lineages have demonstrably changed at different speeds. Any credible clock model must accommodate such differences, and any credible dating analysis must say how much they matter.

9.1.1 What the data can and cannot identify

It helps to be precise about what is identified, and by what. In the vocabulary of Section 6.10:

- **Branch lengths** $b_j = r_j t_j$ are *identifiable under the model* from the characters alone, and with a few hundred cognate sets they are *practically estimable*, up to the limits set by saturation on long branches (Section 7.3.1).
- **Relative node ages** are identifiable only under a clock model that ties the rates of different branches together. Under a strict clock, ratios of node ages are identified by the data. Under a relaxed clock, they are identified only in part: how each branch length is divided into rate and time depends on the prior on rates, so relative ages are informed jointly by the data and by that prior.
- **Absolute ages and the absolute rate** are not identifiable from the characters at all. Only calibrations identify them, and their precision has a floor, set by the uncertainty of the

calibrations, that no amount of additional lexical data can lower (the infinite-sites argument of Section 10.1).

- **Branch-specific rates** in a relaxed clock are identifiable as a population, through the variance of the rate distribution, but individual rates are practically estimable only for branches that carry many changes or lie close to a calibration. The rates of single terminal branches are among the most poorly estimated quantities in any dated tree.

Identifiability is not the same as *historical interpretability*. A clock rate in a binary cognate analysis can be estimated precisely and still not correspond to any historically observable event, because it counts gains and losses of cognate classes rather than replacements of words (Section 9.6). A large coefficient of variation among branch rates is estimable, but whether it measures genuine differences in the tempo of change, or unmodelled borrowing, coding inconsistency and missing data, all of which also alter branch lengths, is a question of historical interpretation that the clock model cannot answer by itself. Nor is a parameter that is identifiable under the model necessarily *robust to misspecification*: the simulations in this chapter show how strongly dates depend on the match between the assumed and the real pattern of rate variation.

9.2 Glottochronology: the first strict clock

9.2.1 The model

In the early 1950s Morris Swadesh proposed that a small core of basic vocabulary, the meanings that every human language has a word for and that are rarely borrowed, is replaced at a roughly constant rate in all languages (Swadesh, 1952, 1955). If that were true, the proportion of shared cognates between two related languages would be a clock. Robert Lees made the proposal quantitative (Lees, 1953). He compiled thirteen pairs of language stages separated by known spans of time, for example Old English and Modern English, or Latin and its Romance descendants, and measured how much of a 200-item test list each had retained. He found that, on average, a language retains about 81 percent of its basic vocabulary per thousand years. Swadesh later shortened the list to 100 items, which he argued were more stable, with a retention rate of about 86 percent per millennium (Swadesh, 1955). The enterprise of dating language splits in this way became known as *glottochronology*. The broader activity of classifying languages by shared-cognate percentages is *lexicostatistics*. Chapter 1 tells the history; here we take the model apart.

The model makes three assumptions. First, each meaning on the list is replaced independently of all others. Second, replacement is a memoryless process: in any short interval dt, the word currently expressing a meaning is replaced with probability $k\,dt$, where the *replacement rate* k is the same for every meaning, every language and every period. Third, the two daughter lineages descending from a common ancestor change independently, and a meaning that has been replaced in either lineage no longer yields a shared cognate.

The first two assumptions describe a Poisson process. The probability that the original word for a meaning survives for t millennia is e^{-kt}. Writing $\rho = e^{-k}$ for the retention rate per millennium (Lees' 0.805, or Swadesh's 0.86), the expected proportion of the list retained in one lineage after t millennia is

$$R(t) = e^{-kt} = \rho^{t}. \tag{9.2}$$

Two sister languages share a cognate for a meaning only if both have retained the ancestral word. By the independence assumption, the expected proportion of shared cognates after a split t mil-

lennia ago is

$$C(t) = R(t)^2 = \rho^{2t}. \tag{9.3}$$

Solving for t gives the glottochronological formula:

$$t = \frac{\ln C}{2 \ln \rho}. \tag{9.4}$$

This is the formula that appeared in introductory textbooks for half a century. If two languages share 50 percent cognates on the 200-item list, then $t = \ln 0.5 / (2 \ln 0.805) \approx 1.6$ millennia. On the 100-item list with $\rho = 0.86$, the same 50 percent would imply about 2.3 millennia.

In the language of Chapter 5, glottochronology is a strict molecular clock combined with a very simple substitution model. Each meaning is a character, a split resets nothing, every branch has the same rate k, and distances between languages are converted to time by a fixed transformation. The formula is the linguistic counterpart of using a known substitution rate to convert a corrected genetic distance into a divergence time. Glottochronology also used a distance-based route from data to tree, in the spirit of Chapter 4, rather than a model of the whole tree at once. In the terms of Section 9.1.1, it made absolute time identifiable by fiat: the universal retention rate served as a calibration imported from other languages.

9.2.2 The retention curve in R

The following code draws the retention curves implied by Equations (9.2) and (9.3), and tabulates the dates that Equation (9.4) assigns to a range of shared-cognate percentages.

```
glotto_C <- function(t, rho) rho^(2 * t)           # shared cognates
glotto_t <- function(C, rho) log(C) / (2 * log(rho)) # Equation (glotto)

t_grid <- seq(0, 10, by = 0.05)                      # millennia
plot(t_grid, 0.805^t_grid, type = "l", lwd = 2, ylim = c(0, 1),
     xlab = "Time since split (millennia)", ylab = "Proportion",
     las = 1)
lines(t_grid, glotto_C(t_grid, 0.805), lwd = 2, lty = 2)
lines(t_grid, 0.86^t_grid, lwd = 2, col = "grey55")
lines(t_grid, glotto_C(t_grid, 0.86), lwd = 2, lty = 2, col = "grey55")
abline(h = 0.5, col = "grey85")
legend("topright", bty = "n", lwd = 2, lty = c(1, 2, 1, 2),
       col = c("black", "black", "grey55", "grey55"),
       legend = c("Retained, 200-list", "Shared, 200-list",
                  "Retained, 100-list", "Shared, 100-list"))
```

```
C_vals <- c(0.90, 0.75, 0.60, 0.50, 0.40, 0.25, 0.10, 0.05)
glotto_tab <- data.frame(
  shared = paste0(100 * C_vals, "%"),
  t200 = round(1000 * glotto_t(C_vals, 0.805)),
  t100 = round(1000 * glotto_t(C_vals, 0.86))
)
knitr::kable(glotto_tab, col.names = c("Shared cognates",
  "Years (200-list, 0.805)", "Years (100-list, 0.86)"),
  caption = "Glottochronological dates implied by different proportions of shared
  cognates.")
```

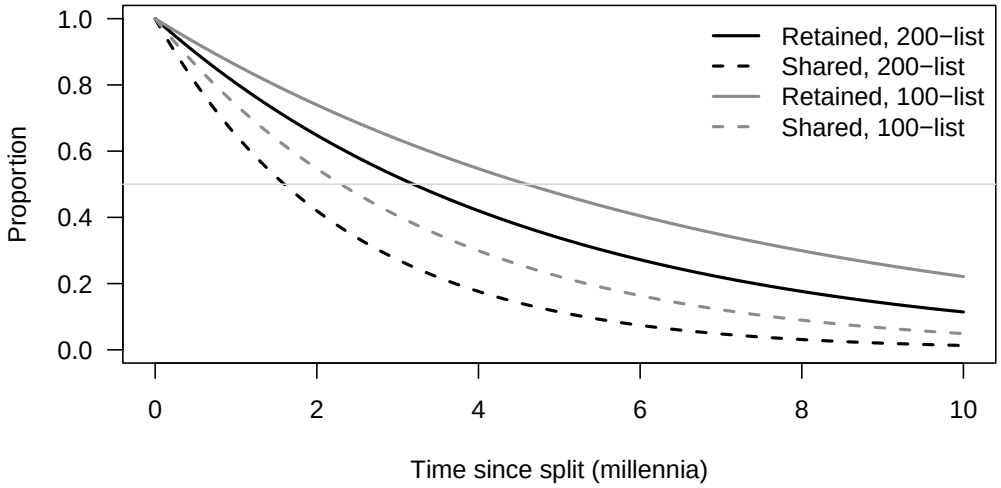


Figure 9.1: Glottochronological retention curves. Solid lines: expected proportion of the list retained by a single language; dashed lines: expected proportion of cognates shared by two sister languages. Black: Lees' 200-item list with a retention rate of 0.805 per millennium; grey: Swadesh's 100-item list with 0.86.

Table 9.1: Glottochronological dates implied by different proportions of shared cognates.

Shared cognates	Years (200-list, 0.805)	Years (100-list, 0.86)
90%	243	349
75%	663	954
60%	1177	1693
50%	1598	2298
40%	2112	3038
25%	3196	4596
10%	5308	7633
5%	6905	9931

Table 9.1 makes one of glottochronology's weaknesses immediately visible. The curve flattens as time increases, so at great time depths a small difference in the percentage of shared cognates translates into a very large difference in date. Going from 10 to 5 percent shared cognates adds well over a millennium on either list. Yet at those depths, a handful of chance resemblances or misjudged cognates can easily make up five percent of a 100-item list.

9.2.3 Sampling error

Even if every assumption of the model held, a 100- or 200-item list is a small sample. If each meaning is retained with probability $C(t)$ independently, the number of shared cognates on a list

of N items is binomially distributed with mean $NC(t)$ and variance $NC(t)(1-C(t))$. The resulting uncertainty in the date can be large. The next chunk computes exact binomial confidence intervals for the shared proportion and transforms them into intervals on the date, for a range of observed counts on the 100-item list.

```
N <- 100
obs <- c(80, 60, 40, 20, 10)
ci <- t(sapply(obs, function(x) binom.test(x, N)$conf.int))
ci_tab <- data.frame(
  shared = obs,
  date = round(1000 * glotto_t(obs / N, 0.86)),
  lower = round(1000 * glotto_t(ci[, 2], 0.86)),
  upper = round(1000 * glotto_t(ci[, 1], 0.86))
)
knitr::kable(ci_tab, col.names = c("Shared (of 100)", "Point date (years)",
  "Lower 95% (years)", "Upper 95% (years)"),
  caption = "Binomial sampling uncertainty alone in glottochronological dates on a
  100-item list, assuming the retention rate is known exactly.")
```

Table 9.2: Binomial sampling uncertainty alone in glottochronological dates on a 100-item list, assuming the retention rate is known exactly.

Shared (of 100)	Point date (years)	Lower 95% (years)	Upper 95% (years)
80	740	449	1144
60	1693	1198	2316
40	3038	2279	3955
20	5336	4083	6850
10	7633	5755	9998

The intervals in Table 9.2 widen dramatically with time depth, and they ignore every other source of error: uncertainty in ρ itself, variation of ρ across lineages, borrowing, and coding mistakes. Even in the earliest discussions of the method, its proponents knew that individual dates carried uncertainties of several centuries. What they did not anticipate was how badly the core assumption of a universal constant rate would fare.

9.2.4 What went wrong

The critique of glottochronology came quickly and from several directions. Here we organize it by the assumption under attack. Excellent longer treatments are Embleton (1986) and McMahon and McMahon (2005).

Rates differ between languages. The most damaging empirical test was by Bergsland and Vogt (1962), who measured retention in cases where the time depth is independently known. Their most famous example was Icelandic. Compared with the Old Norse of its medieval texts, Icelandic had retained far more of the test list than the model predicted, so the formula would have placed the separation of Icelandic from its own ancestor at a small fraction of the documented time depth. They also examined other languages with long written records, including Georgian and Armenian, and found that the amount of retention over comparable spans differed from case to

case. Other test cases in the literature showed the opposite problem, languages that had replaced their vocabulary faster than expected. Blust (2000) later made the same point at the scale of a whole family. Across Austronesian, languages separated by similar time depths, as judged from subgrouping and archaeology, vary enormously in how much basic vocabulary they share, and some lineages are dramatically more conservative than others. Blust concluded that the "universal constant" assumption is simply false for Austronesian.

Rates differ between meanings. Words for some meanings, such as low numerals, pronouns and certain body parts, are replaced far more slowly than words for others, such as 'dirty', 'to throw' or 'to play'. This was clear to Swadesh himself and was one motivation for his shortened list. As we show in Section 9.2.5, rate variation among meanings does not merely add noise. It biases glottochronological dates downward, more and more severely as time depth increases.

The mathematics was shaky. Chrétien (1962) argued that the mathematical models presented in support of glottochronology contained serious errors and that the claimed constancy had not been established. Later work proposed corrected formulations, including models with meaning-specific rates (see the review in Embleton, 1986). Some Russian scholars, notably Sergei Starostin, developed a revised glottochronology that excluded loanwords and modified the decay function. These revisions improved some results but did not resolve the deeper problem of variable rates across lineages.

Borrowing and chance inflate shared percentages. If one language borrows a basic word from its sister, or if both borrow from the same source, the pair will appear to share a cognate that they did not inherit. Chance resemblances, which become proportionally more important as genuine cognates decay, have the same effect. Both biases make languages look closer than they are and dates younger; Section 8.3 quantifies the effect of borrowing by simulation.

The list and its coding matter. Different scholars coded the same languages differently, chose different words when a language has several synonyms for a meaning, and used different versions of the list. Because the formula is very sensitive to the shared percentage at great depths, small coding differences translated into large date differences.

Distances are not trees. Glottochronology computed a date for each pair of languages independently. Dates for different pairs need not be mutually consistent, and nothing in the method ensured that the dated pairs fitted together into a single tree. The information in other languages about the history of a pair was ignored.

Bergsland and Vogt (1962) tested glottochronology on cases where the answer was known. They compared modern languages with their own documented ancestors, so that the time depth was given by the written record. Icelandic, compared with Old Norse, had retained so much of the basic vocabulary list that the formula implied a time depth far shorter than the one documented in writing. Other cases deviated by different amounts, so no single constant could fit them all. Because the test cases were the very kind of data on which the retention constant had been calibrated, the results showed that the constant was not constant. The paper, published with a range of commentaries in *Current Anthropology*, is often cited as the moment glottochronology lost its credibility among historical linguists. It is worth reading with a modern eye: what Bergsland and Vogt documented is exactly the among-lineage rate variation that relaxed clocks were designed to model. It is also an early example of what Section 10.8 calls independent validation: dates checked against historical evidence that played no part in estimating them.

9.2.5 Rate variation among meanings biases dates

Of all these problems, rate heterogeneity among meanings has the most instructive consequences, because it produces a systematic bias rather than random error. Suppose each meaning has its own replacement rate, drawn independently from a gamma distribution with mean $\bar{k}$ and shape parameter α, the model of among-character rate variation introduced in Section 7.5. Fast meanings are lost early, so what remains shared at great depths is dominated by the stable meanings, and shared cognates decay more slowly than the exponential curve of Equation (9.3). A glottochronologist who knows the mean rate $\bar{k}$ exactly and applies Equation (9.4) therefore underestimates every date. The expected estimate is approximately

$$\hat{t} \approx \frac{\alpha}{2\bar{k}} \ln\left(1 + \frac{2\bar{k}t}{\alpha}\right),$$

which is always less than the true time t and grows only logarithmically with it: the deeper the split, the worse the underestimate.

The derivation of the gamma-averaged shared-cognate curve (Equation (C.7)) and of the bias of the glottochronological estimate (Equation (C.8)), a numerical check of both, and the code for Figure 9.2 are in Section C.3.1.

Figure 9.2 makes the point concrete. We take a 200-meaning list with mean retention 0.86 per millennium, draw meaning-specific rates from a gamma distribution with shape $\alpha = 1$, simulate the independent survival of each word along two lineages, and apply the classical formula to the resulting shared percentages. In the vocabulary of Section 6.10, this is a test of *robustness under misspecification*: the data are generated with rate variation among meanings that the glottochronological formula ignores.

With $\alpha = 1$, a moderate level of heterogeneity, a true split of ten millennia is estimated at well under half that. Studies of Indo-European and Austronesian vocabulary suggest that rate variation across meanings is at least this strong (Sections 7.5 and 9.7). This single effect is enough to explain why glottochronology systematically produced shallow dates for old families. It also explains why the remedy is not to abandon dating. Modern phylogenetic models of lexical change include among-character rate variation, typically with the same gamma distribution, and estimate α from the data (Pagel, 2000; Gray and Atkinson, 2003). The bias in Figure 9.2 is then removed in expectation, *provided that* the gamma model describes the real rate variation adequately; Section 7.10 shows how to check whether it does.

It is common to read that phylogenetic dating "is just glottochronology with more computing power." The criticism has a grain of truth: both convert amounts of lexical change into time and both depend on calibration. But the differences are substantive. Modern methods (i) model rate variation across meanings explicitly rather than assuming a single constant; (ii) allow rates to vary across lineages through relaxed clock models; (iii) estimate rates from multiple internal calibrations within the family under study rather than importing a universal constant; (iv) infer a single tree for all languages jointly, using all the data at once; and (v) report uncertainty that integrates over topology, branch lengths, rates and calibrations. Atkinson et al. (2005) and Gray et al. (2011) set out this case in detail. Whether these improvements are enough is an empirical question, and Chapter 10 and Chapter 15 discuss the evidence on both sides.

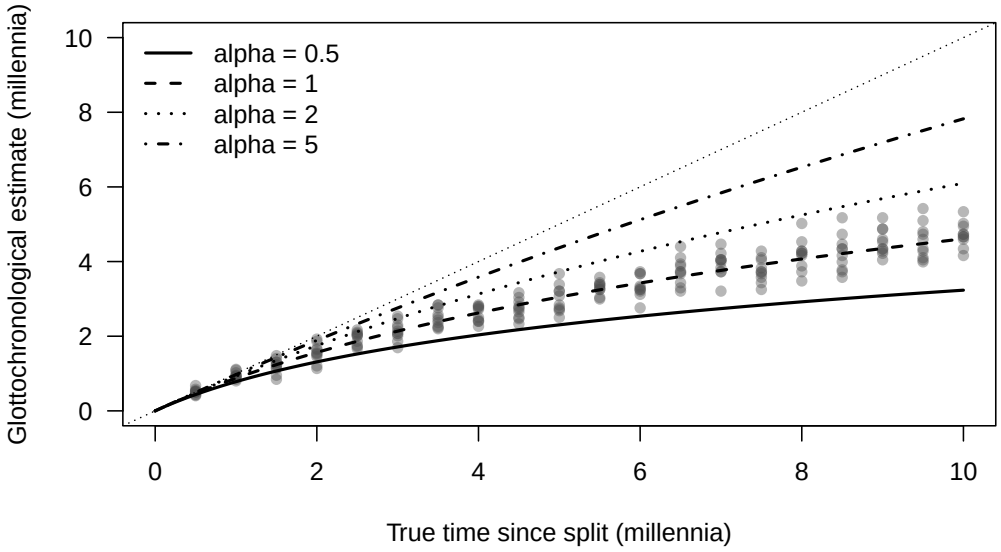


Figure 9.2: Rate heterogeneity among meanings biases glottochronological dates. Each point is one simulated pair of languages (200 meanings, gamma-distributed replacement rates with shape 1, mean retention 0.86 per millennium), dated with the classical formula. Curves show the expected estimate implied by the gamma model for several values of the gamma shape parameter; the diagonal is the true time.

9.2.6 Lexicostatistical dating after glottochronology

Distance-based dating did not disappear. The Automated Similarity Judgment Program (ASJP), which compares phonetically transcribed 40-item word lists for thousands of languages, developed an automated dating method calibrated on dozens of historically or archaeologically dated splits (Holman et al., 2011). Instead of cognate percentages, it uses a normalized Levenshtein distance between word forms (Chapter 4), which is converted to a date through an empirically fitted relationship. ASJP chronology makes no claim that the rate is universal for any single pair. It treats the calibration relationship as a statistical average with substantial scatter, and reports correspondingly wide intervals. The method is valuable for a first approximation of time depths across the world's families, many of which have never been dated phylogenetically. But it inherits several of glottochronology's problems: a pairwise rather than tree-based treatment, and no explicit model of lineage-specific rates. The 40-item ASJP list was itself chosen by ranking meanings according to their stability across language families (Holman et al., 2008), an empirical counterpart of Swadesh's intuition that some meanings change more slowly than others.

The *generalized Bayesian dating* of Rama and Wichmann (2020), discussed in Section 10.9.16, replaces family-specific calibrations by a single regression of known time depths on tree lengths. In the same comparison, an automated version of glottochronology performed worse than both it and ASJP chronology. A generalized calibration of this kind reintroduces, at the level of whole trees, the assumption of a common rate that the relaxed clocks below were designed to avoid, which is why its dates, like those of ASJP, are best read as first approximations.

A cautionary contemporary of the first Bayesian datings is Forster and Toth (2003), who adapted the network methods and dating approach of genetics to a small, hand-coded set of Celtic and Indo-European word forms, including Gaulish inscriptions, and obtained a tentative date of 8100 BCE ($\pm$ 1,900 years) for Proto-Indo-European. McMahon and McMahon (2005) criticized the very small dataset, the unexplained assignment of words to character states and the neglect of borrowing within Celtic, contrasted it with the expert cognate data and broad sampling of Gray and Atkinson (2003), and still declined, in 2005, to recommend any linguistic dating method, arguing that where contact dominates there may be no branching points left to date.

9.3 From glottochronology to molecular clocks

At almost the same time that glottochronology was being discredited in linguistics, biologists proposed the *molecular clock*. Zuckerkandl and Pauling (1965) observed that the number of amino-acid differences between the hemoglobins or cytochromes c of different species increased roughly linearly with the time since the species diverged, as estimated from fossils. The neutral theory of molecular evolution then offered an explanation (Kimura, 1968). If most substitutions are selectively neutral, the rate at which they become fixed in a population equals the mutation rate. Kimura's argument is short and worth knowing, because it has a linguistic analog that we revisit in Section 9.10. In a diploid population of N individuals, $2N\mu$ new neutral mutations arise per generation, and each has a probability $1/(2N)$ of eventually becoming fixed. The substitution rate is the product, $2N\mu \times 1/(2N) = \mu$, independent of population size.

The molecular clock was never exact. Rates differ between genes, between sites within genes, and between lineages. Rodents, with short generation times, accumulate substitutions faster than primates in many genes. The *nearly neutral theory* (Ohta, 1973) showed that if many mutations are slightly deleterious, they behave as neutral in small populations but are purged by selection in

large ones, which makes the rate depend on population size after all. Over the following decades the "clock" became a family of statistical models in which the rate is a quantity to be modeled and estimated, rather than a constant to be assumed. Bromham and Penny (2003) and Kumar (2005) review this history. Ho and Duchêne (2014) and Bromham et al. (2018a) review modern methods, and the textbooks by Felsenstein (2004) and Yang (2014) give the mathematics.

The parallel with linguistics is close but imperfect. Swadesh's constant rate of vocabulary retention resembles the original molecular clock. The linguistic data that undermined glottochronology resemble the data showing rate variation among genes and lineages. And the response in both fields was, eventually, the same: to stop treating the rate as a known universal constant and to model its variation statistically, estimating it from calibrations internal to the group under study. Linguistics made the transition about four decades later than biology, through the work of Gray and Atkinson (2003) and others, by borrowing the biological methods wholesale (Atkinson and Gray, 2005).

9.4 The strict clock

Under a strict clock, every branch of the tree has the same rate r. The tree is *ultrametric*: all tips that are sampled at the same time are equidistant from the root (Chapter 2). Branch lengths in time units, t_j for branch j, are determined by the ages of the nodes at either end. The branch lengths in expected changes per character are $b_j = r\,t_j$. Rather than $2n-3$ free branch lengths for an unrooted tree of n languages, a strict-clock tree has $n-1$ node ages and one rate. Fixing one age by calibration, or setting $r = 1$ to obtain relative ages, leaves $n-2$ free parameters to estimate in addition to the rate. Even a strict clock does not tick like a metronome. Under the CTMC, changes arrive after random waiting times, so two sister lineages evolving at exactly the same rate will usually accumulate different numbers of changes; the strict clock is a statement about expected amounts of change, and a test of it must ask whether the differences between lineages exceed this stochastic scatter.

The strict clock can be tested with a *likelihood ratio test* (Chapter 5). Fit the tree with and without the clock constraint. Twice the difference in maximized log-likelihood is approximately χ^2-distributed with $n-2$ degrees of freedom, the difference in the number of free branch-length parameters (Felsenstein, 2004). With linguistic data the test almost always rejects the strict clock for families of any size, which is unsurprising given the evidence in Section 9.2.4. A simpler diagnostic is the regression of root-to-tip distance on sampling time, discussed in Section 10.5.2. For contemporaneous tips, the coefficient of variation of root-to-tip distances in a rooted phylogram is a rough index of departure from clocklike behavior.

A strict clock nonetheless has its uses. It is the most parsimonious model and has the fewest parameters, so it can be preferable when data are few and the family is shallow. It is also a sensible null model for Bayesian model comparison by marginal likelihoods (Chapter 6). Its single rate still needs a prior, and when not even the order of magnitude of the rate is known a prior that is uniform on the logarithm of the rate (a log-uniform prior, as in the RevBayes scripts of Appendix H) spreads the prior mass evenly over orders of magnitude instead of piling it onto the largest values allowed, as a uniform prior on the rate itself would. Several linguistic studies have compared strict and relaxed clocks with Bayes factors, and the relaxed clock is usually, though not always, preferred.

9.5 Relaxed clocks

A *relaxed clock* allows each branch j to have its own rate r_j. Because the rates are not identifiable from the data branch by branch (Equation (9.1)), they must be tied together by a model. In a Bayesian analysis, the model is a prior distribution on the branch rates. That prior encodes an assumption about how rates vary, and the posterior combines it with calibrations and the tree prior to apportion each branch length b_j between rate and time. Two broad families of models exist. In *uncorrelated* models, rates on different branches are drawn independently. In *autocorrelated* models, rates evolve along the tree, so that closely related lineages tend to have similar rates. Table 9.3 summarizes the models discussed in this section.

Table 9.3: Clock models compared by their assumptions about rate variation among lineages. "What the data identify" assumes calibrations that fix the absolute scale.

Clock model	Assumption about rates among lineages	Main parameters	What the data identify	Typical software
Strict	One rate for all branches	Mean rate	Relative ages; the rate only with calibrations	All
Uncorrelated lognormal (UCLN)	Independent lognormal rate per branch	Mean rate, spread σ	Spread of rates; individual rates poorly, especially on terminal branches	BEAST, BEAST 2, MrBayes, RevBayes
Uncorrelated exponential (UCED)	Independent exponential rate per branch (coefficient of variation fixed at 1)	Mean rate	As UCLN, with the spread fixed in advance	BEAST, BEAST 2
Autocorrelated (geometric Brownian motion)	Log rate drifts along the tree	Mean rate, drift variance ν	Clade-level rate trends, if branches are short relative to the drift	MCMCTree, PhyloBayes, RevBayes
Local clock	A few pre-specified clades have their own constant rates	One rate per clade	Rate differences between the specified clades	BEAST, BEAST 2, RevBayes
Random local clock	An unknown number of rate shifts at unknown places	Shift indicators and multipliers	Number and placement of strong shifts, if any	BEAST, BEAST 2
Penalized likelihood	Rates of adjacent branches should be similar (smoothing parameter λ)	Branch rates, node ages, λ	Point estimates only; depends on λ	r8s, treePL, `ape::chronos()`

The mathematical definitions of the uncorrelated lognormal and geometric Brownian motion clocks (Equations (C.9) and (C.10)), with the derivation of their mean-one parameterization and of the coefficient of variation, are in Section C.3.2. The penalized-likelihood objective function (Equation (C.11)) and the cross-validation of its smoothing parameter are in Section C.3.4.

9.5.1 Uncorrelated relaxed clocks

The uncorrelated relaxed clocks of Drummond et al. (2006) are the default in BEAST and its descendants, and they are the most widely used clock models in linguistic phylogenetics. In the *uncorrelated lognormal* (UCLN) model, the rate of each branch is the mean clock rate $\bar{r}$ multiplied by an independent random factor drawn from a lognormal distribution with mean one. The standard deviation σ of the logarithm of these factors measures how "relaxed" the clock is. A more interpretable quantity is the *coefficient of variation* of the branch rates, $\sqrt{e^{\sigma^2} - 1}$, which is close to σ when σ is small. Values near zero indicate a nearly strict clock. Values around 0.3 to 0.5 are common in molecular data sets, and values above 1 indicate extreme rate heterogeneity. In the *uncorrelated exponential* (UCED) model, rates are drawn from an exponential distribution with mean $\bar{r}$, which implies a coefficient of variation fixed at one. Because the exponential has its mode at zero, the UCED model gives many branches very low rates. It is less often appropriate than the UCLN but is a useful alternative in model comparison.

In practice, BEAST originally implemented these models by discretizing the rate distribution into as many equiprobable categories as there are branches and assigning each branch to a category. Later implementations sample continuous rates or rate quantiles. A summary statistic routinely reported is the *covariance* of rates between parent and child branches. If its posterior distribution spans zero, there is little evidence for autocorrelation, which supports the choice of an uncorrelated model.

The uncorrelated models are attractive because they make weak assumptions and are computationally convenient. They have a less attractive property. Because a rate is drawn independently for every branch, a long terminal branch can be explained either by a long time or by a high rate on that one branch. The model has no way to borrow strength from neighboring branches. Rates and times on terminal branches are therefore *identifiable under the model* only through the prior, which makes them poorly *estimable in practice*, and the posterior can be strongly influenced by calibrations and the tree prior. This matters for ancient languages used as dated tips (Section 10.5): an ancient language on a long terminal branch tells the model little about the rate if that branch is free to have its own rate.

9.5.2 Autocorrelated relaxed clocks

The intuition behind autocorrelated clocks is that the rate of evolution depends on heritable properties of lineages. For organisms, these are traits such as generation time, metabolic rate or population size. For languages, they might be the size and social structure of the speech community, literacy, or the intensity of contact. Such properties change gradually, so a daughter lineage should have a rate similar to its parent's. Thorne et al. (1998) proposed a model of this kind, and Kishino et al. (2001) refined it into the most widely used form, the *geometric Brownian motion* model, in which the logarithm of the rate performs a Brownian motion along the tree with variance ν per unit time. Rates are similar on short branches and can drift far apart on long ones. The Cox-Ingersoll-Ross process, a mean-reverting alternative, has also been used. Lepage et al. (2007) compared a range of autocorrelated and uncorrelated models on molecular data and found that autocorrelated models were often preferred by Bayes factors, a finding that remains debated. To our knowledge, autocorrelated clocks have rarely been compared with uncorrelated ones on linguistic data, though they are available in RevBayes, MCMCTree and PhyloBayes. The drift variance ν is a good example of a parameter that is identifiable in principle but hard to estimate in practice: it is informed only by the pattern of branch lengths across many branches, and with a

few dozen languages its posterior usually remains close to its prior.

9.5.3 Local clocks and random local clocks

Some rate variation is better described as a few discrete shifts than as continuous scatter. A *local clock* model allows different, internally constant rates in specified parts of the tree (Yoder and Yang, 2000). One might give the Polynesian clade within Austronesian its own rate, for example, to test whether rates changed when speakers settled the remote Pacific. Local clocks require the shifts to be specified in advance. The *random local clock* (RLC) of Drummond and Suchard (2010) removes this requirement. Each branch has an indicator for whether the rate changes at its start, a rate multiplier applies to that branch and its descendants until the next change, and the number of changes is given a prior (for example, a Poisson distribution with a small mean). MCMC then averages over the number and placement of rate shifts. When the posterior favors zero shifts, the RLC reduces to a strict clock. When it favors many, it approaches an uncorrelated relaxed clock. The *compound Poisson process* of Huelsenbeck et al. (2000) is a related idea in which rate changes occur as random events along branches. Random local clocks are well suited to a linguistic question discussed below: whether particular lineages, such as those that underwent rapid geographic expansion or intense contact, show discrete rate shifts.

9.5.4 The optimized relaxed clock

The uncorrelated relaxed clock is notoriously slow to mix in MCMC. Branch rates and node ages are strongly correlated in the posterior, because changing a node age while keeping its branch lengths fixed requires compensating changes in the rates of all adjacent branches; this posterior correlation is the computational face of the rate-time confounding of Section 9.1. Douglas et al. (2021) introduced the *optimized relaxed clock* (ORC) package for BEAST 2. It combines several improvements: a parameterization of branch rates by real values rather than by discrete categories, proposals that change node ages and rates jointly so as to keep the genetic distances approximately constant, and adaptive weighting of operators. The model is still the uncorrelated lognormal clock. What changes is how efficiently MCMC explores it, which in practice determines whether an analysis of hundreds of languages converges at all (Chapter 6). Recent linguistic analyses in BEAST 2 increasingly use the ORC.

9.5.5 Penalized likelihood and rate smoothing

Before Bayesian relaxed clocks were widely available, the main tools for dating trees with variable rates were semi-parametric. Sanderson (1997) proposed *nonparametric rate smoothing* (NPRS), which chooses node ages that minimize the squared differences between the rates of adjacent branches. Sanderson (2002) then proposed *penalized likelihood* (PL), which balances the fit of a set of node ages and branch rates to the branch lengths of an estimated phylogram against a penalty on differences between the rates of adjacent branches. A *smoothing parameter* λ sets the balance. With λ very large, the rates are forced to be equal and the method approaches a strict clock. With λ near zero, the rates are free and the ages are poorly determined. Sanderson proposed choosing λ by cross-validation: drop each terminal branch in turn, predict its length from the rest of the tree, and choose the λ that minimizes prediction error.

Gray and Atkinson's influential dating of Indo-European (Gray and Atkinson, 2003) used penalized likelihood, applied to a sample of trees from a Bayesian posterior distribution, with fourteen historically known calibrations. PL remains useful for quick analyses and for very large trees where

full Bayesian dating is impractical. In R, the function `chronos()` in **ape** implements a penalized-likelihood framework with several rate models (Paradis, 2013). Its `"correlated"` model penalizes differences between the rates of adjacent branches. Its `"relaxed"` model penalizes departures of the rate distribution from a gamma distribution, without regard to which branches are adjacent. Its `"discrete"` model assigns branches to a small number of rate categories, and its `"clock"` model forces a single rate. Paradis also proposed an information criterion, ΦIC, for comparing these models. Penalized likelihood gives point estimates; uncertainty must be obtained by repeating the analysis over a sample of trees, as Gray and Atkinson did, and that captures topological and branch-length uncertainty but not calibration uncertainty.

9.5.6 Simulating and estimating clocks in R

To see how the clock models behave, we simulate a time tree, impose three different patterns of rate variation on it, convert the result into a phylogram with stochastic numbers of lexical changes on each branch, and then try to recover the node ages with `chronos()`. We imagine a family of sixteen languages whose root is 6,000 years old, and a dataset in which about 80 lexical changes (cognate gains and losses across the whole list) are expected per millennium per lineage at the mean rate. The three scenarios are a strict clock, an uncorrelated lognormal clock with $\sigma = 0.6$, and an autocorrelated clock with drift variance $\nu = 0.15$ per millennium.

The simulation code (time tree, branch rates under the three clock models and Poisson numbers of changes) is listed in Section C.3.3.

The coefficients of variation of the simulated branch rates are 0, 0.65 and 0.4 for the strict, uncorrelated and autocorrelated scenarios. (A constant of 0.5 is added to each branch's count simply to avoid zero-length branches, which `chronos()` cannot handle.) Figure 9.3 shows the true time tree and the three phylograms. Under the strict clock the phylogram is nearly ultrametric, apart from Poisson noise. Under the uncorrelated clock, individual branches are stretched or shrunk at random. Under the autocorrelated clock, whole clades are stretched or shrunk together, which makes rate variation easier to mistake for differences in time.

Now we date each phylogram, fixing the root at its true age of six millennia, under four of the rate models available in `chronos()`, and compare the estimated node ages with the true ones by the root-mean-square error (RMSE). In the vocabulary of Section 6.10, the strict-clock phylogram dated under the strict-clock model is a *parameter-recovery* check: the data were generated under the model's own assumptions. Every other cell of the resulting table is a test of *robustness under misspecification*, because none of the penalties in `chronos()` is the process that generated the data. None of the cells says anything about *empirical adequacy* for real cognate data.

```
true_ages <- branching.times(time_tree)
models <- c("clock", "correlated", "discrete", "relaxed")
fit_one <- function(ph, model) {
  # root fixed at its true age; lambda = 1 for every model
  cal <- makeChronosCalib(ph, node = "root", age.min = 6, age.max = 6)
  ctrl <- chronos.control(nb.rate.cat = 3)
  suppressWarnings(chronos(ph, lambda = 1, model = model,
                           calibration = cal, quiet = TRUE, control = ctrl))
}
fits <- lapply(phylograms, function(ph)
  setNames(lapply(models, fit_one, ph = ph), models))
```

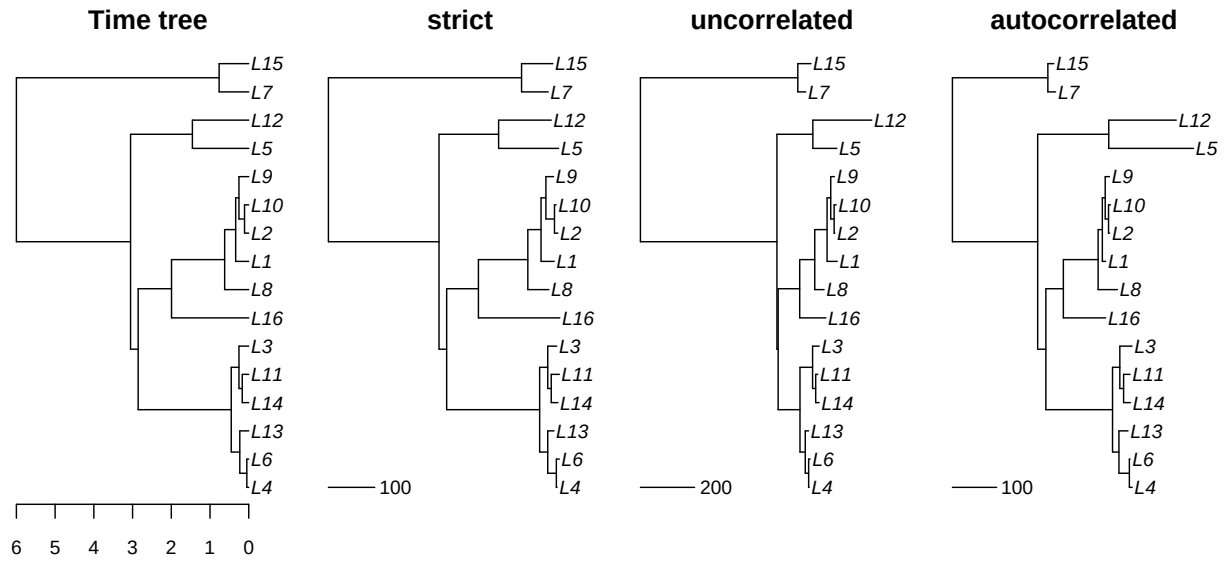


Figure 9.3: A simulated time tree (left) and phylograms produced from it under a strict clock, an uncorrelated lognormal relaxed clock and an autocorrelated (geometric Brownian motion) clock. Phylogram branch lengths are numbers of simulated lexical changes.

```
rmse <- sapply(fits, function(f) sapply(f, function(ch)
  sqrt(mean((branching.times(ch) - true_ages)^2))))
est_unc <- sapply(fits$uncorrelated[c("clock", "correlated")],
                  branching.times)
# the two models give almost identical ages: draw the filled points first
# and the larger open circles on top, keeping both visible
plot(true_ages, est_unc[, "clock"], type = "n", xlim = c(0, 6),
     ylim = c(0, 6), las = 1, xlab = "True node age (millennia)",
     ylab = "Estimated node age (millennia)")
points(true_ages, est_unc[, "correlated"], pch = 16, col = "grey40")
points(true_ages, est_unc[, "clock"], pch = 1, cex = 1.6, lwd = 1.2)
abline(0, 1, lty = 3)
legend("topleft", bty = "n", pch = c(1, 16), col = c("black", "grey40"),
       pt.cex = c(1.6, 1), pt.lwd = c(1.2, 1),
       legend = c("strict clock model", "correlated-rates model"))
```

Table 9.4: Root-mean-square error (millennia) of node ages estimated by chronos under four rate models (rows) for phylograms simulated under three clock scenarios (columns). The root age was fixed at its true value of 6 millennia.

	strict	uncorrelated	autocorrelated
clock	0.03	0.52	0.28
correlated	0.03	0.52	0.29
discrete	0.03	0.52	0.96
relaxed	1.24	1.26	1.31

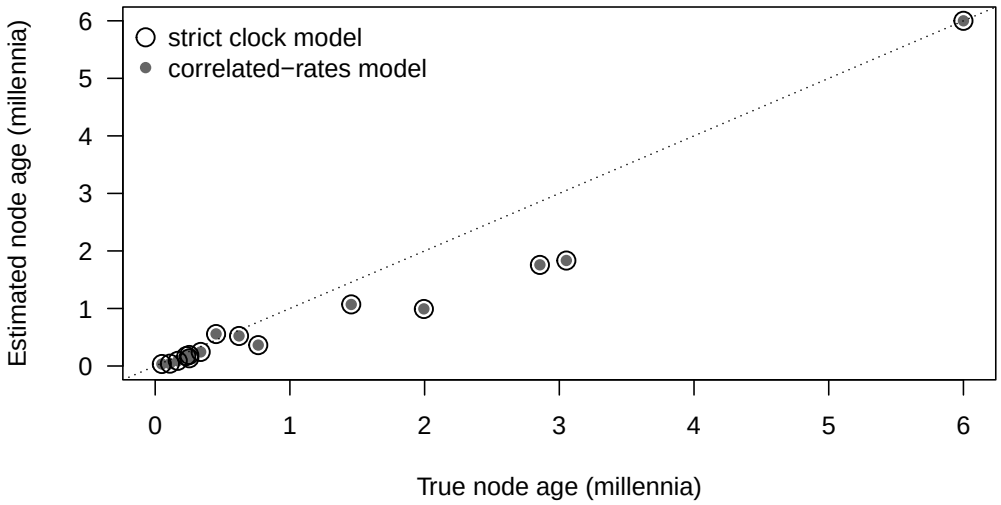


Figure 9.4: Node ages estimated by penalized likelihood (chronos) against the true ages, for the phylogram simulated under the uncorrelated relaxed clock. The diagonal marks perfect estimation.

Table 9.4 is worth studying, because it shows several general lessons on a small scale. When the data really are clocklike, the strict clock model recovers the node ages almost perfectly. When rates vary, every model makes errors of several hundred years even though the root age is known exactly, because rate and time on internal branches can trade off. In this small example, the correlated-rates model, whose penalty is designed for autocorrelated rates, does no better than the strict clock model even when the rates really are autocorrelated: with sixteen languages and a single calibration, the extra flexibility buys little, and the discrete-rates model can do noticeably worse. And the gamma-penalized `"relaxed"` model of `chronos()`, which does not tie the rates of adjacent branches together, is poorly constrained by a single calibration and can perform badly. In a full Bayesian analysis the corresponding uncorrelated model is constrained by the tree prior and, ideally, by several calibrations (Chapter 10). The exact numbers depend on the random seed and on the smoothing parameter λ, which in a real study should be chosen by cross-validation or by comparing ΦIC values (Section C.3.4 shows how). The qualitative message is robust: estimated dates are only as good as the match between the clock model and the real pattern of rate variation, and calibrations do much of the work.

9.5.7 Choosing and checking a clock model

Clock models can be compared like any other part of a phylogenetic model. In Bayesian analyses, the standard approach is to estimate marginal likelihoods by path sampling, stepping-stone sampling or nested sampling and compute Bayes factors (Chapter 6). Bayes factors rank models relative to one another; they do not show that the preferred model is adequate. Two other tools from biology deserve wider use in linguistics. *Clock partitioning* asks whether different subsets of the data share a pattern of rate variation across lineages. ClockstaR (Duchêne et al., 2014) addresses this by comparing branch-length patterns across data partitions and clustering partitions

with similar patterns; linguists might ask whether, for example, verbs and nouns, or vocabulary and grammar, share a clock. *Clock model adequacy* asks not which model is best but whether the chosen model could plausibly have generated the data. Duchêne et al. (2015a) proposed a posterior predictive approach: simulate data sets from the posterior, re-estimate branch lengths, and compare them with branch lengths estimated from the observed data. To our knowledge, neither approach has yet been applied to linguistic data, although both could be run with standard software on existing cognate data sets.

Because rate and time are confounded, a clock model that fits the branch lengths well can still date the tree badly if the calibrations are wrong, and a poorly fitting clock can be partly rescued by good calibrations. Clock adequacy is therefore necessary but not sufficient for trustworthy dates. Section 10.8 discusses how dates can be validated against evidence that played no part in estimating them.

Molecular rates often appear to depend on the timescale over which they are measured. Ho et al. (2005) showed that rates estimated from pedigrees or recently diverged populations are often considerably faster than rates estimated from deep calibrations, because many recent changes are transient polymorphisms or slightly deleterious variants that will be lost before they become fixed. Rates estimated over short spans therefore overestimate long-term rates, and applying them to deep divergences can produce dates that are too young. Language change plausibly has the same structure. Many lexical innovations, slang terms and semantic shifts are visible in a short time window but never become established across the speech community. Rates estimated from very recent historical splits, such as those between colonial and metropolitan varieties, might not transfer to splits several millennia old. Linguistic phylogenetics has paid little attention to the timescale dependence of rates. Two approaches from biology could address it directly: fitting explicit models in which the rate declines with the age of the calibration, and comparing rates estimated from calibrations of different ages in the same family. Both are feasible for well-calibrated families such as Indo-European and Austronesian.

9.6 Clock models in linguistic practice

Most dated language phylogenies published since 2009 were estimated in BEAST or BEAST 2 (Bouckaert et al., 2019), often with model specifications generated by BEASTling (Maurits et al., 2017), or in MrBayes, RevBayes or TraitLab. The typical analysis combines a binary or multistate model of cognate evolution (Chapter 7), gamma-distributed rate variation among cognate sets or meanings, a relaxed clock (most often the uncorrelated lognormal), a birth-death or Yule tree prior, and a set of calibrations. Four aspects of the clock component deserve comment.

First, the clock rate in a cognate analysis has a particular meaning. In a binary analysis, each character is a cognate set and the rate is the expected number of gains and losses per cognate set per unit of time. Because most cognate sets are present in only a few languages, this rate is not directly comparable with a rate of lexical replacement per meaning. Rates per meaning can be derived from multistate models, in which each character is a meaning and its states are cognate classes, or by summing over the cognate sets belonging to a meaning (Section 7.8). When reading a published rate, it is always worth checking which unit it refers to. A binary clock rate is *estimable* but only indirectly *interpretable as a historical quantity*.

Second, the separation between rate variation across *characters* and rate variation across *lineages* is essential. Gamma rate heterogeneity (Sections 7.5 and 9.2.5) lets some cognate sets or meanings

change faster than others at all times and in all lineages. A relaxed clock lets some lineages change faster than others across all characters. Neither substitutes for the other. A model with both makes the assumption that the two kinds of variation combine multiplicatively, so that a fast meaning is fast by the same factor in every lineage. That assumption is untested in most linguistic data and is likely to be violated, for example if contact affects some semantic domains more than others. Covarion models (Section 7.6) relax it partially.

Third, published analyses frequently report the coefficient of variation of branch rates. In linguistic data sets these values are often substantial, consistent with the lineage-level rate variation that undermined glottochronology. The value depends on the data and model, and, as Section 9.1.1 noted, it absorbs every unmodelled process that stretches or shrinks branches, including undetected borrowing (Section 8.3) and missing data in poorly attested languages. Rather than quoting typical figures we encourage readers to check them in the papers reviewed in Chapter 10.

Fourth, the clock interacts strongly with the tree prior and the calibrations. With a relaxed clock and few calibrations, node ages far from any calibration are determined largely by the tree prior and by the assumed distribution of rates. Section 10.3.2 shows how to detect this by running analyses without data (sampling from the prior) and comparing the prior and posterior distributions of node ages.

Specification by default carries its own risks. King (2026b) found, by simulation-based calibration, that the default weighting of meaning-partition rates in BEAST 2 analyses of cognate data interacts with the ascertainment correction and miscalibrates the posterior, whereas an alternative weighting behaved correctly (Section 7.10). Because partition rates scale the expected amount of change on every branch, details of this kind can in principle affect dates as well as substitution parameters.

9.7 Rate variation across meanings

9.7.1 Frequency of use

That some meanings change faster than others has been known since Swadesh. Phylogenetic methods made it possible to measure the rates. Section 7.5 introduced the central result: Pagel et al. (2007) estimated a rate of lexical replacement for each of the 200 Swadesh meanings across 87 Indo-European languages, on a posterior sample of trees so that phylogenetic uncertainty was carried into the rate estimates, and found that frequency of use in modern corpora accounted for roughly half of the variation, with frequent words replaced slowly. Expressed as half-lives (the time over which a word has a 50 percent chance of being replaced), the most stable meanings had half-lives of many millennia, the least stable well under a millennium. Here we add what matters for dating.

The frequency effect is not an isolated finding. The same issue of *Nature* carried a paper by Lieberman et al. (2007) showing an analogous law for the regularization of English irregular verbs: the half-life of an irregular past-tense form scales with the square root of its frequency, so that rare irregular verbs regularize much faster than common ones. Calude and Pagel (2011) showed that the frequencies of the Swadesh meanings are strongly correlated across a sample of 17 languages from several families, which supports treating frequency as a near-universal property of meanings rather than an accident of Indo-European. Monaghan (2014) reported that the age at which children acquire a word also predicts its rate of replacement in Indo-European, with early-

acquired words changing more slowly, though age of acquisition and frequency are themselves correlated. Vejdemo and Hörberg (2016) examined semantic properties of meanings and reported that they help predict rates of replacement over and above frequency. Pagel and Meade (2018) focused on numerals, arguing that number words for the lowest numbers are among the most stable elements of vocabulary, with expected lifetimes of many thousands of years. Work on borrowing, such as the Leipzig-Jakarta list of Tadmor et al. (2010), has shown that the meanings least prone to borrowing overlap strongly with the meanings that are most stable within lineages.

For dating, the practical consequence is that the composition of the meaning list strongly influences the estimated rate. A clock rate is a property of a particular list, coded in a particular way, not of a language family. Different lists, or different subsets of the same list, will give different absolute rates and must be calibrated separately, and a rate estimated in one family cannot be transferred to another unless the lists and coding conventions match. This is one reason why borrowing a "universal" rate, glottochronology's original sin, remains a temptation to resist.

How should frequency be included in a phylogenetic model? So far, most analyses have estimated per-character or per-meaning rates and examined their correlates afterward. Section 7.5 sketched a regression model for meaning rates built into the likelihood. For dating, its attraction is that external data would inform the rate category of each meaning, so that fewer of the data's degrees of freedom are spent estimating among-meaning rate variation and more are available for branch lengths. Such a model is formally similar to the phylogenetic generalized linear models used in phylogeography (Chapter 11). To our knowledge, this has rarely if ever been done in a published dating analysis.

9.7.2 Ultraconserved words and deep time

If some meanings are replaced very slowly, perhaps their words could preserve signals of relationship older than any established language family. Pagel et al. (2013) tested this idea. They used reconstructed proto-words for seven Eurasian language families (Indo-European, Uralic, Altaic, Kartvelian, Dravidian, Chukchi-Kamchatkan and Inuit-Yupik), taken from an etymological database, and asked which meanings had proto-words that were judged cognate across several families. They found that the meanings with such putative cognates were precisely those predicted by frequency to be the most stable, including pronouns ('I', 'thou', 'we', 'ye'), demonstratives and interrogatives, and words such as 'not', 'mother' and 'to give'. They identified 23 "ultraconserved" meanings and inferred a tree of the seven families with a root of around 15,000 years ago, which they interpreted as evidence for a deep Eurasiatic macrofamily.

The paper was strongly criticized. Heggarty (2013) argued that the underlying proto-forms, taken from an etymological database associated with the Moscow school of long-range comparison, are subjective interpretations that cannot be independently validated and that most specialists in reconstruction reject, and that the result was circular: because those reconstructions are built on family-level proto-forms that are themselves weighted toward stable meanings, finding the putative cognates concentrated in stable meanings was, in his words, "correlation in, correlation out". A further concern raised by other critics in the same debate is that frequent meanings tend to have short, phonetically simple words, and short words are also the most likely to show chance resemblances, so the match between predicted stability and observed cross-family similarity does not by itself distinguish inheritance from chance. More generally, the probability of chance resemblance among short words can be substantial, as Ringe (1992) showed in a classic analysis of the statistics of language comparison. Most historical linguists regard the existence of Eurasiatic

as unproven. The episode illustrates a general point about rates. Knowing that some meanings change slowly does not by itself make it possible to detect relationships beyond the time depth at which genuine cognates become indistinguishable from chance similarities. That limit is set by the rate of change of the slowest characters, by the size of the data set, and above all by the reliability of the cognate judgments.

Long-range comparisons combine two hard problems: very low expected numbers of surviving cognates, and a high rate of false positives from chance resemblance, sound symbolism, nursery words (‘mama’, ‘papa’) and borrowing. Phylogenetic models assume that the cognate judgments are correct and say nothing about whether they are. A phylogenetic analysis of doubtful cognate sets yields a tree with the appearance of precision, and its dates inherit every error in the input. Before interpreting a deep date, ask how the cognate sets were identified and how many would be expected by chance.

9.8 Punctuational bursts at language splits

9.8.1 The idea

In the clock models discussed so far, change accumulates in proportion to time. An alternative hypothesis holds that some change happens in bursts associated with lineage splitting. In biology, Webster et al. (2003) and Pagel et al. (2006) reported that, across many phylogenies, lineages that have undergone more speciation events have also accumulated more molecular change, after accounting for time. Venditti and Pagel (2010) reviewed the evidence and the possible mechanisms, such as founder effects and rapid adaptation in new populations. The linguistic analog is intuitive. When a group of speakers separates from its parent community, by migration or by social fission, the new community may innovate rapidly, deliberately or not, as a marker of a new identity. Small founding populations may also lose and replace vocabulary by drift.

The test is simple. Under a gradual process, the expected root-to-tip path length P_i of language i in a phylogram depends only on the time elapsed since the root, which is the same for all contemporaneous languages. If each split adds a burst of change β to the descendant lineages, then

$$\mathbb{E}[P_i] = rT + \beta\, n_i, \tag{9.5}$$

where T is the root age and n_i is the number of nodes on the path from the root to tip i. A positive slope β in the regression of path length on node count, estimated with the phylogenetic non-independence of the tips taken into account, indicates punctuational change. The proportion of total path length attributable to bursts, $\beta n_i / P_i$, measures its importance.

9.8.2 Evidence from language families

Atkinson et al. (2008) applied this test to Bayesian posterior samples of trees for Bantu, Indo-European and Austronesian, all estimated from basic vocabulary. They found a significant positive relationship between path length and number of nodes in all three families. By their estimates, bursts at splits accounted for roughly 10 to 33 percent of the total lexical divergence, depending on the family.

Atkinson et al. (2008) tackled the main technical objection to the path-length test, the *node-density artefact* (Venditti et al., 2006). In phylogenies estimated from data, branch lengths tend to be underestimated in sparsely sampled regions of the tree, because multiple changes on long branches

are not detected. Densely sampled regions have more nodes and also more fully recovered branch lengths, so a positive relationship between path length and node count can arise even without any real bursts. The test of Venditti et al. (2006) looks for a characteristic curvilinear shape in the relationship, and Atkinson and colleagues reported that their data did not show the artefact. Their result has been widely cited as evidence that language change is partly punctuational. It has implications for dating: if change is concentrated at splits, branch lengths depend partly on the number of unsampled or extinct splits along each lineage, which a clock model cannot see. It would be valuable to repeat the analysis on the larger and more carefully coded data sets that have since become available, such as IE-CoR for Indo-European, and with models that build the bursts into the likelihood rather than testing for them after the fact.

Figure 9.5 simulates the test. We generate a pure-birth (Yule) tree of 60 languages, then add lexical change along each branch in proportion to time, plus, in one scenario, a burst of extra change at the start of each branch, so that every split contributes. We then regress root-to-tip path length on the number of nodes. For simplicity the figure uses ordinary least squares, which ignores the shared ancestry of tips. In the vocabulary of Section 6.10, this is a *parameter-recovery* exercise for the test statistic: the data are generated under the very process the test is designed to detect, with true branch lengths and no node-density artefact, so it shows what the test estimates in the most favorable case and nothing about its behavior on real, estimated trees.

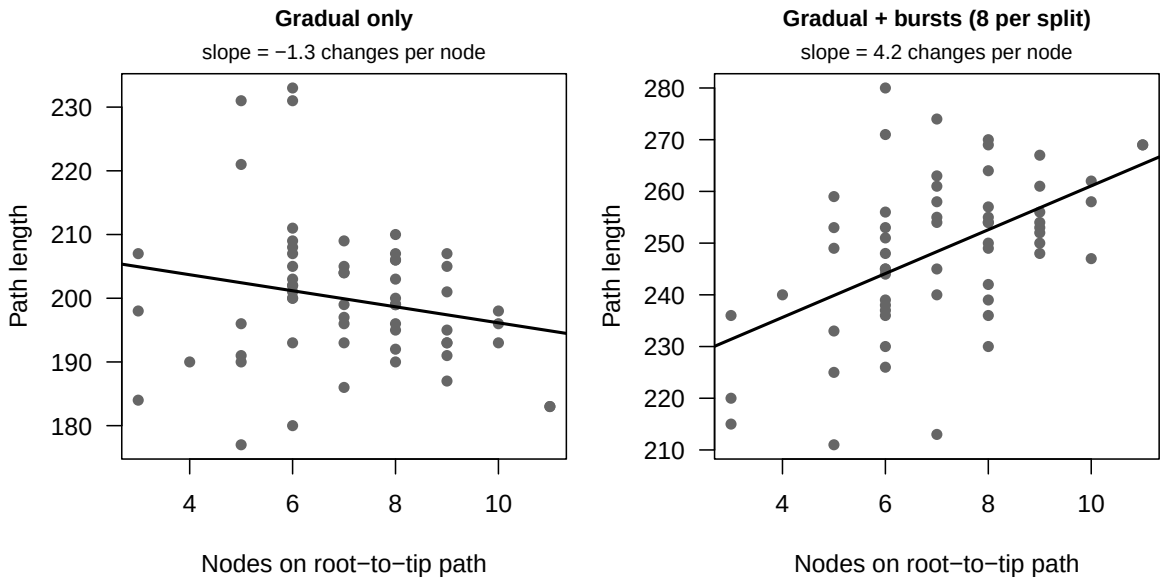


Figure 9.5: Simulated test for punctuational change. Root-to-tip path length (number of lexical changes) against the number of nodes on the path from the root, for a 60-language Yule tree under purely gradual change (left) and with a burst of change at every split (right). Lines are least-squares fits.

A single simulated dataset gives a noisy slope, because tips share ancestral branches and their path lengths are therefore correlated, so the effective sample size is much smaller than the number of languages. Averaging over many replicate datasets on the same tree, with the function `path_lengths()` that simulates one dataset, shows what the test estimates:

```
reps <- sapply(c(gradual = 0, bursts = 8), function(b)
  replicate(200, coef(lm(path_lengths(b) ~ nodes_on_path))[2]))
```

```
round(rbind(mean_slope = colMeans(reps),
            sd_slope = apply(reps, 2, sd)), 2)
```

```
#>            gradual bursts
#> mean_slope   -0.15   8.15
#> sd_slope      1.57   1.98
```

The simulation code for Figure 9.5, and a version of the test that uses phylogenetic generalized least squares to account for the shared ancestry of the tips, are in Section C.3.5.

In the gradual scenario the average slope is close to zero; with bursts it is close to the burst size per split. The spread of slopes across replicates is a reminder that a single family provides one noisy realization of the process, and that significance tests must account for the shared ancestry of languages. Real analyses face the additional difficulties that tree topology and branch lengths are estimated with error, that extinct and unsampled languages leave splits invisible, and that the node-density artefact can mimic the signal. A convincing case for punctuational change in a language family would therefore need a robustness-under-misspecification experiment as well: data simulated *without* bursts, but with saturation and uneven sampling, analyzed in the same way as the real data (Exercise 4 below).

9.9 Vocabulary, grammar and sound systems

Basic vocabulary is only one part of a language. Are grammatical and phonological systems more or less stable? Claims on both sides have a long history. Some typologists have argued that grammatical features such as word order or morphological type are slow to change and preserve deep historical signals. Others have pointed out that grammatical features are few, mostly binary or low-dimensional, prone to convergence through contact, and liable to change back and forth, so that their signal decays quickly.

Greenhill et al. (2017) compared rates of change in grammatical structures and basic vocabulary in a sample of Austronesian languages. They estimated rates for each lexical meaning and each grammatical feature on the same set of trees. On average, grammatical features changed faster than basic vocabulary, although the two distributions overlapped broadly and some grammatical features were among the most stable characters of all. The authors also suggested that grammatical features are more susceptible than basic vocabulary to influence from neighboring languages. And when the authors mapped the lineages with unusually fast or slow change, the two systems showed different regional patterns: the Indonesian and Malay languages, for example, showed elevated rates of lexical change, while languages of the Bismarck Archipelago, in long contact with non-Austronesian languages, showed more structural change. A lineage that was conservative in vocabulary was not necessarily conservative in grammar. The study suggests that grammar and vocabulary evolve as partly independent systems with distinct dynamics.

The design of Greenhill et al. (2017) is a model for comparative rate studies. Lexical cognate data from the Austronesian Basic Vocabulary Database and grammatical features coded for the same languages were analyzed on a common phylogenetic framework, so that differences in rates could not be attributed to different trees. Rates were estimated with explicit models of character evolution and compared as distributions, not just as averages. The authors then looked separately at where on the tree each system showed significantly elevated or reduced rates, and found that the two systems did not share the same lineage-specific patterns, although they did not report

a formal test of correlation. This has a practical implication for dating. A clock calibrated on vocabulary cannot simply be transferred to grammatical data, and combined analyses of lexical and structural data probably need separate clock models for the two partitions.

Measures of the stability of typological features have also been developed outside a full phylogenetic framework. Wichmann and Holman (2009b) estimated the stability of features in the World Atlas of Language Structures by comparing how often related and unrelated languages agree, and Dediu and Cysouw (2013) compared seven different methods for measuring feature stability, finding moderate but not perfect agreement among them. Phylogenetic estimates of rates of change for structural features on Grambank-scale data (Chapter 3) are an active area of research.

Sound change poses different problems. Regular sound changes affect whole classes of words at once, so their effect on cognate-based characters is not independent across words. Hruschka et al. (2015) proposed a phylogenetic model in which sound correspondences evolve by "concerted evolution", in the sense used for gene families in molecular biology, and applied it to Turkic. They showed that regular sound changes could be detected as correlated changes of many characters at the same point in the tree, and estimated when particular changes occurred. To our knowledge, systematic comparisons of the rates of phonological change with lexical and grammatical rates on the same trees remain rare. Phonological inventory data (Chapter 3) and sound-change databases offer obvious opportunities. A first example of such a comparison is the study of Mixtec tone by Auderset (2024). Using a database of tonal and segmental sound changes in 42 Mixtec languages, she measured phylogenetic signal with the D statistic and estimated rates with a hidden Markov model over a posterior sample of trees, and found that most tone changes carry phylogenetic signal and that tones do not, in general, change faster than segments, contrary to the common assumption that tone is inherently unstable. At the level of word forms, Cathcart (2024) used phylogenetic models of homologous words to ask why sequences of identical consonants are rare, and found that such words arise less often than others but, once present, do not die out more often, although they are replaced in basic meaning functions more frequently.

9.10 Population size and rates of change

9.10.1 Theory

Does the size of a speech community affect how fast its language changes? Two opposing intuitions exist. One holds that small, tightly knit communities can tolerate and transmit idiosyncratic innovations, while large communities, with many weak ties and many adult learners, favor simplification and slow down idiosyncratic change (Trudgill, 2011). Another, derived from population genetics, holds that random drift is stronger in small populations, so that small communities should both lose and fix variants faster (Nettle, 1999). In the neutral model of Section 9.3, these effects cancel: the number of innovations grows with population size while the chance that any one of them spreads through the community falls. The cancellation is exact only if innovations are neutral, if every speaker is equally likely to innovate, and if the community mixes randomly. None of these conditions is likely to hold for languages. Models of language change with social structure, prestige-biased transmission or conformist learning predict a range of relationships between population size and rate. Population size might also affect the *kind* of change: Reali et al. (2018) used models and experiments to argue that larger populations favor simpler grammars and larger vocabularies.

9.10.2 Evidence

Nettle (1999) argued from simulation and comparative data that the rate of linguistic change is not constant and is faster in smaller groups. Wichmann and Holman (2009a) tested for a relationship between population size and rates of lexical change in a large global sample of lexical data, and reported little evidence of a strong effect. Their study used distances between related languages rather than phylogenetic branch lengths.

Bromham et al. (2015) brought phylogenetic methods to the question. Using a Bayesian tree of Polynesian languages, they compared the amount of lexical change along phylogenetically independent pairs of related languages that differ in population size. They distinguished the *gain* of new words from the *loss* of existing cognates. Languages with larger populations gained new words at a faster rate, and languages with smaller populations lost words at a faster rate. Both patterns fit population-genetic expectations: more speakers produce more innovations, and in small populations existing words are more easily lost by drift. Greenhill et al. (2018) extended the test to Indo-European, Austronesian and Bantu. They found less consistent support for the relationships in these larger and more heterogeneous families. The question of whether population size systematically affects rates of lexical change therefore remains open. Current population sizes are a noisy proxy for population sizes over the history of a lineage, and families differ in how much their histories have been shaped by expansion, contact and language shift.

The paired-comparison design used by Bromham et al. (2015) is a standard tool in comparative biology for testing rate correlates. Each pair consists of two lineages that share a common ancestor and differ in the trait of interest, here population size. The lineages of a pair are compared on the amount of change accumulated since their split, measured as root-to-tip or ancestor-to-tip distance on the tree. Because each pair shares the same span of time and the pairs do not overlap on the tree, the comparisons are phylogenetically independent. The approach does not require a clock model, since both members of a pair have experienced the same amount of time; it sidesteps the rate-time confounding of Section 9.1 by comparing rates *relative* to each other rather than absolutely. It does require care in measuring the trait: in Bromham et al.'s study, population sizes were current speaker numbers, and the authors discussed the assumption that relative sizes have been reasonably stable over the history of the pairs. The Polynesian setting is especially favorable, because island populations are relatively isolated and their settlement history is well constrained by archaeology.

9.11 Rate heterogeneity among lineages

The variation in rates among lineages that defeated glottochronology is now routinely estimated by relaxed clocks, and it is of interest in its own right. Which lineages are fast and which are slow, and why? Several kinds of explanation have been proposed, though few have been tested systematically.

Contact and language shift. Lineages that have absorbed many speakers of other languages, or that have been in intense contact, often show high rates of lexical replacement. Norwegian and English, both heavily influenced by contact, are classic examples within Germanic. Since borrowed vocabulary should ideally be flagged and removed in cognate coding (Chapter 3), contact may affect rates through undetected loans or through contact-induced restructuring that is not itself borrowing.

Founder effects and expansion. Gray et al. (2009) found in Austronesian that the expansion across the Pacific proceeded in a series of pulses and pauses (Chapter 10). Rates of lexical change may themselves vary with phases of expansion, as the punctuational analysis of Section 9.8 suggests.

Literacy and standardization. Written standard languages with long literary traditions are sometimes held to change more slowly, as the conservatism of Icelandic suggests. But Icelandic is also an isolated island community with an explicit purist language policy, so literacy alone cannot explain its conservatism. Literacy may also have a different effect on vocabulary than on grammar or phonology.

Population size and social structure, discussed above.

Disentangling these factors requires analyses that estimate branch rates on a tree and then relate them to lineage properties, or better, that build the properties into the clock model. Two cautions apply to the first approach. Estimated branch rates are among the least well-identified quantities in a dated tree (Section 9.1.1), so a regression of branch rates on lineage properties should be repeated over the posterior sample of trees and rates rather than run once on a summary tree. And branch rates absorb every process that changes branch lengths, so a "fast" lineage may simply be one with undetected loans or many missing entries.

Several families of methods developed for molecular rates could address the causes of rate variation among language lineages directly.

Rate-trait correlation models. Coevol (Lartillot and Poujol, 2011) jointly models the evolution of substitution rates and continuous traits (such as body mass or generation time) as a multivariate Brownian motion along the tree, and estimates the correlation between them while integrating over divergence times. Linguistic analogs of the traits might be log population size, degree of multilingualism or a measure of contact intensity, reconstructed along the tree. To our knowledge, no linguistic analysis has used a joint rate-trait model of this kind.

Clocks with covariates. Instead of drawing branch rates independently, one can model the log rate of each branch as a linear function of branch-level predictors plus a random effect. This approach parallels the generalized linear models used for phylogeographic transition rates (Chapter 11). Predictors could include whether a branch led to an island colonization, a change of subsistence, or a period of documented contact.

Comparative rate studies. Lanfear et al. (2010) review the extensive biological literature on why rates differ between species, covering generation time, metabolic rate, population size and life history, and the statistical designs (sister-pair comparisons, phylogenetic regression of branch rates) used to test them. Many of these designs could be transferred directly to language families with existing dated phylogenies.

Episodic and random local clocks for shifts. Random local clocks (Drummond and Suchard, 2010) identify the branches on which rates shift without prior specification. Applied to language trees, they could test whether rate shifts coincide with known historical events, such as island settlement or language shift, more often than expected by chance.

9.12 What biology offers, and what remains to be done

Gathering the threads of this chapter, the following methods from evolutionary biology seem especially promising for the study of linguistic rates, ranked roughly by how easily they could be

applied to existing data:

1. *Clock model comparison and adequacy.* Routine comparison of strict, uncorrelated, autocorrelated and random local clocks by marginal likelihoods, together with posterior predictive tests of clock adequacy (Duchêne et al., 2015a). These require no new data or software.
2. *Clock partitioning.* Testing whether vocabulary partitions (by part of speech, frequency band or semantic domain) and structural partitions share a clock (Duchêne et al., 2014). The results of Greenhill et al. (2017) suggest they may not. Clock models for discrete morphological characters, which raise many of the same problems as linguistic characters (few characters, strong rate variation among them, ascertainment), are reviewed by Lee (2020).
3. *Time-dependent rates.* Testing whether rates estimated from shallow calibrations are higher than those from deep calibrations (Ho et al., 2005), and modeling the dependence explicitly.
4. *Rate-trait correlation and covariate clocks.* Joint inference of rates and lineage traits (Lartillot and Poujol, 2011), or log-linear models for branch rates.
5. *Models with bursts in the likelihood.* Instead of testing for punctuational change after the fact, models that include a per-node burst parameter, or in which the clock rate depends on the speciation process, can estimate the size of bursts while accounting for unobserved splits.
6. *Meaning-level predictors of rates.* Building frequency, age of acquisition and other covariates into among-character rate models, so that external data inform the rates of individual meanings.

All of these are within reach with current software and data, and several would bear directly on the reliability of the dates discussed in the next chapter.

- Cognate data determine branch lengths as products of rate and time. Branch lengths are identifiable from the data; relative ages only under a clock model; absolute ages and rates only with calibrations; individual branch rates are poorly estimable in practice.
- Glottochronology is a strict clock with a universal rate. It fails because rates vary among lineages and among meanings. Rate variation among meanings biases glottochronological dates downward, increasingly so at greater depths.
- Modern clock models let rates vary: uncorrelated relaxed clocks (UCLN, UCED), autocorrelated clocks (geometric Brownian motion), local and random local clocks, and penalized-likelihood smoothing. The optimized relaxed clock improves MCMC efficiency rather than changing the model.
- Rate variation across characters (gamma) and across lineages (relaxed clock) are different things and are usually both needed. A binary clock rate counts gains and losses of cognate classes and is not a rate of word replacement.
- In simulation, dates are recovered well only when the clock model matches the generating process; parameter recovery under the model's own assumptions is not evidence of adequacy for real data.
- Frequency of use is the best-established predictor of the rate at which a meaning's word is replaced, so a clock rate is a property of a particular list and coding. Claims that ultraconserved words reveal relationships of 15,000 years or more remain highly controversial.
- Path-length tests suggest that part of lexical change in several families occurs in bursts at language splits, though the node-density artefact must be ruled out.
- In Austronesian, grammatical features appear on average to change faster than basic vocabulary, and the two systems show different lineage-specific patterns of fast and slow change.
- Evidence on population size and rates is mixed. A Polynesian study found faster gain of words in larger populations and faster loss in smaller ones, but the pattern is less consistent

in other families.
- Rate-trait models, covariate clocks, clock partitioning, time-dependent rates and clock adequacy tests are available in biology and largely unused in linguistics.

9.13 Exercises

Conceptual

1. Explain why doubling every branch rate and halving every node age leaves the likelihood of a cognate matrix unchanged. List the components of a Bayesian dating analysis that break this symmetry, and say for each whether it carries historical evidence or modelling assumptions.
2. Two languages share 60 percent of their cognates on a 100-item list. Explain why a glottochronological date and a Bayesian relaxed-clock date for the same pair could differ even if both use the same mean rate of replacement. Which assumptions differ, and which difference would you expect to matter most at this time depth?
3. A study reports a posterior mean coefficient of variation of 0.8 for the branch rates of a binary cognate analysis. List three processes other than genuine differences in the tempo of lexical replacement that could produce such a value, and describe a check that would help distinguish each from real rate variation.
4. Contrast an uncorrelated lognormal clock, an autocorrelated clock and a random local clock as hypotheses about why the Polynesian languages might have changed faster than the Formosan languages. What would each predict about the distribution of rates over the Austronesian tree?
5. Why is the clock rate estimated from binary cognate characters not directly comparable with the per-meaning replacement rate estimated by Pagel et al. (2007)? Describe how you would convert one into the other, and what extra information you would need.
6. Why is a positive regression slope of root-to-tip path length on node count not by itself evidence for punctuational change? Name the main artefact and one way of addressing it.

Computational

1. Using `glotto_t()` and `binom.test()` as in Section 9.2.3, compute the point date and 95 percent interval for 30 shared cognates on Lees' 200-item list (retention 0.805) and for 15 shared cognates on Swadesh's 100-item list (retention 0.86). Which list gives the more precise date, and why?
2. Rerun the simulation behind Figure 9.2 (code in Section C.3.1) with shape parameters $\alpha = 0.5$ and $\alpha = 5$, and with a 40-item list (`n_mean <- 40`). Report how the bias and the scatter of the estimates change at a true depth of 6 millennia. Which part of the result is due to model misspecification and which to sampling error?
3. Using the simulation code of Section C.3.3, raise `sigma` to 1 and the number of languages to 40, and recompute Table 9.4 with `chronos()` for `lambda` values of 0.1, 1 and 10. How sensitive is the ranking of the four rate models to λ? Use `attr(fit, "PHIIC")$PHIIC` to see which λ the information criterion would choose.
4. Simulate the node-density artefact. Starting from `path_lengths()` (Section C.3.5) with `burst = 0`, transform each branch's count x to a saturated value $K(1 - e^{-x/K})$ with $K = 30$ before computing path lengths, so that long branches are underestimated, and repeat the regression

of Section 9.8.2 over 200 replicates. Does a spurious positive slope appear? Which kind of validation is this?

5. Fit `chronos()` with `model = "clock"` and `model = "correlated"` to the `"autocorrelated"` phylogram of Section 9.5.6, with the root calibrated to between 5 and 7 millennia instead of fixed at 6 (`makeChronosCalib(ph, node = "root", age.min = 5, age.max = 7)`). How far does the estimated root age move within the window, and what does this tell you about calibration dependence?

Research

1. You plan to date a family of 60 languages for which two medieval attestations and one archaeologically dated island settlement are available. Choose a clock model and justify the choice; specify which prior-only runs, parameter-recovery simulations and misspecification simulations you would perform before interpreting the root age; and say which results would make you abandon the clock model you chose.
2. Design a study testing whether intense contact raises rates of lexical replacement. Define the predictor at the level of branches, choose between a two-step analysis (estimate branch rates, then regress) and a covariate clock, and explain how you would prevent undetected loans, which also lengthen branches, from producing a spurious association.
3. Propose a test of time-dependent rates of lexical change in Indo-European or Austronesian, using calibrations of different ages. What pattern of results would support time dependence, what alternative explanations would produce the same pattern, and how would you distinguish them?

9.14 Further reading

- McMahon and McMahon (2005): A balanced and readable account of lexicostatistics and glottochronology, their critiques, and the transition to quantitative phylogenetic methods, written by linguists for linguists.
- Embleton (1986): The most thorough technical treatment of the statistics of glottochronology and its revisions. Essential for anyone who wants to understand exactly which assumptions failed.
- Bergsland and Vogt (1962): The classic empirical test of glottochronology on languages with known histories, with published commentaries. Worth reading as a primary source.
- Atkinson et al. (2005): A direct response to the charge that phylogenetic dating is merely glottochronology revisited, explaining how modern models address its failures.
- Ho and Duchêne (2014): A clear review of molecular-clock methods for biologists that is accessible to linguists, covering strict, relaxed and local clocks and calibration.
- Drummond et al. (2006): The paper that introduced uncorrelated relaxed clocks, which underlie most linguistic dating analyses. The introduction gives the intuition clearly.
- Pagel et al. (2007): The study linking word frequency to rates of lexical replacement in Indo-European, with Lieberman et al. (2007) on irregular verbs.
- Atkinson et al. (2008): A short paper on punctuational bursts of change at language splits, with supplementary material explaining the method and the node-density test.
- Greenhill et al. (2017): A comparison of rates in vocabulary and grammar on the same trees, with careful attention to methodological issues.
- Bromham et al. (2015): A test of population size effects on lexical rates, illustrating the

paired-comparison design of comparative biology.

Chapter 10

Calibration and Divergence Times: Family by Family

Chapter 9 showed that a clock model can turn a tree measured in amounts of change into a tree measured in relative time, but that the data alone never fix the absolute scale. This chapter is about the last step, anchoring relative time to the calendar, and about what that step has produced. Calibration is where historical linguistics, philology and archaeology enter a dating analysis most directly, and it is also where most disputes about published dates turn out to lie: which events can be dated, how securely, which node of the tree they belong to, and whether an attested ancient language is an ancestor, a sister or merely a witness. The first half of the chapter develops the theory of calibration and of validating dates; the second reviews the published dated phylogenies of the world's language families. Chapter 15 takes up the broader controversies, and Chapter 11 the closely related question of where families originated.

After reading this chapter you should be able to:

- explain, using the infinite-sites argument, why calibrations determine the precision of every date in a tree and not only its time scale;
- choose and justify a calibration density for a piece of historical evidence, and explain why minimum ages are usually more secure than maximum ages;
- distinguish specified from effective priors on node ages, and diagnose the difference by sampling from the prior;
- describe the ways an attested ancient language can enter a dating analysis (node calibration, dated tip, sampled ancestor, constrained ancestor), and explain how the choice affects taxon selection, coding and dates;
- assess whether a published date has been independently validated, distinguishing evidence used as calibration from evidence held out, and recognizing circular agreement;
- summarize the published dated phylogenies of the major families and identify for each the main source of disagreement.

10.1 The anatomy of a dating analysis

A Bayesian dating analysis has five components, and every published date is the product of all of them:

1. **Data** D: usually a matrix of cognate sets coded as binary characters, sometimes multistate meaning-classes or structural features (Chapter 3).
2. **A substitution model** describing how characters change: for example a binary CTMC with gamma rate variation or a stochastic Dollo model (Chapter 7).
3. **A clock model** describing how rates vary across branches: strict, relaxed or local (Chapter 9).
4. **A tree prior** describing the distribution of topologies and node ages before the data are seen: Yule, birth-death, coalescent or fossilized birth-death (Chapter 6).
5. **Calibrations** linking some node ages, or tip ages, to absolute time.

Components 2 to 4 form what Warnock and Wright (2020) call the *tripartite model* of Bayesian divergence-time estimation, the substitution, clock and tree models, each of which can be specified and checked separately; their short introduction, written for palaeobiologists, is a useful companion to this chapter.

The joint posterior is

$$p(\tau, \mathbf{t}, \mathbf{r}, \theta \mid D) \propto P(D \mid \tau, \mathbf{t}, \mathbf{r}, \theta)\; p(\mathbf{r} \mid \theta)\; p(\tau, \mathbf{t} \mid \theta, \mathscr{C})\; p(\theta), \tag{10.1}$$

where $\mathbf{t}$ are node ages, $\mathbf{r}$ branch rates, θ the remaining parameters, and $\mathscr{C}$ the calibrations. The data enter only through the likelihood, which, as Chapter 9 emphasized, depends on rates and times only through their products. Everything that separates rate from time comes from the clock prior $p(\mathbf{r} \mid \theta)$ and the time prior $p(\tau, \mathbf{t} \mid \theta, \mathscr{C})$.

A simple thought experiment shows how much this matters. Suppose we had an unlimited number of cognate sets, so that every branch length $b_j = r_j t_j$ was known exactly. Under a strict clock, the node ages would then be known up to a single scale factor, the rate. All the remaining uncertainty would come from the calibrations: if the calibration on the root allowed ages between 5,000 and 7,000 years, then every node age would have a correspondingly proportional range. Adding more data would not narrow the interval at all. This *infinite-sites* argument, developed by Yang and Rannala (2006) and Rannala and Yang (2007), shows that in dating, unlike in topology estimation, uncertainty does not vanish as data accumulate. With a relaxed clock, the limit is wider still, because each branch has its own rate. Plotting the width of posterior intervals against their mean for all nodes in a tree (an *infinite-sites plot*) reveals how close an analysis is to this limit. If the points fall on a straight line, more data will not help, and only better calibrations can (dos Reis et al., 2016).

The derivation of the infinite-sites limit for a strict clock, with a short R illustration of the infinite-sites plot, is in Section C.4.1.

Linguistic datasets are very far from infinite. A typical basic vocabulary list has 100 to 200 meanings, yielding a few hundred to a few thousand cognate sets, many of which are present in only a few languages. In practice, therefore, both data and calibrations limit precision. Yet the infinite-sites argument contains a lesson that is easily forgotten. Calibrations do not just fix the time scale; they determine the precision of every date in the tree, and errors in them propagate to every node.

10.1.1 What is identified, and by what

Section 9.1.1 classified the quantities in a dated tree by whether the lexical data identify them. Table 10.1 extends that classification to the quantities that dating analyses report, using the four

questions of Section 6.10: is the quantity identifiable under the model, practically estimable with realistic linguistic data, robust to misspecification, and interpretable as a historical quantity?

Table 10.1: What a dating analysis can and cannot identify. Calibrations are assumed to fix the absolute time scale; "practically estimable" refers to typical basic-vocabulary datasets of a few dozen to a few hundred languages.

Quantity	Identified by	Practically estimable?	Most sensitive to	Historical interpretation
Topology and branch lengths	The characters	Usually, for well-sampled families	Borrowing, coding, saturation	Shared cognates are not shared innovations (Chapter 3); topology is a summary of lexical similarity under the model
Relative node ages	Characters plus clock prior	Yes under a strict clock; partly under relaxed clocks	Clock model, among-character rate variation	Order and spacing of lineage splits
Absolute node ages	Calibrations (with clock and tree priors)	Only as precisely as the calibrations allow	Calibration placement and densities; effective priors	The time of the "split" depends on what a split of a dialect continuum means
Mean clock rate	Calibrations plus characters	Yes, given good calibrations	List composition, coding (binary versus multistate)	A rate of cognate-class gain and loss on a given list, not a universal rate of change
Individual branch rates	Clock prior plus characters	Poorly, especially on terminal branches	Missing data, undetected loans	Ambiguous: tempo of change or unmodelled processes
Tree-prior parameters (speciation, extinction, sampling)	Node-age distribution	Weakly; often prior-dominated	Taxon sampling scheme	Rarely interpretable as rates of language diversification
Ancestor or sister status of an ancient language	Absence of unique changes in the ancient language	Poorly when the language is fragmentarily attested	Missing data, literary registers	Whether the attested variety, or a close relative, gave rise to the descendants

Two lessons follow from the table. First, the quantities that attract the most attention, absolute root ages, are precisely those that the lexical data do not identify: they are borrowed from the calibrations through the clock model. A root age is therefore only as good as the weakest of three things: the calibrations, the match between clock model and real rate variation, and the extrapolation from calibrated nodes to the root. Second, the quantities that the data identify best, topology

and branch lengths, are summaries of lexical similarity under the substitution model. Agreement between an inferred topology and the subgroups of the comparative method is welcome, but it is not independent validation of either, since the cognate judgments are themselves products of the comparative method, and it says nothing at all about whether the dates are right.

10.2 Kinds of calibration information

Biologists calibrate trees primarily with fossils, whose ages are determined by radiometric dating of the rocks in which they are found. Linguists have no direct equivalent of radiometric dating for languages. Instead, they use several indirect sources of absolute time, each with characteristic strengths and weaknesses. Table 10.2 summarizes them.

Table 10.2: Kinds of calibration information available for language families.

Source	Example	What it constrains	Main risks
Documented historical events	Settlement of Iceland in the late ninth century CE; Dutch colonization of the Cape from the mid-seventeenth century	Upper (and sometimes lower) bound on the split of a colonial variety from its source	Dialect differentiation may predate or postdate the migration; continued contact
Written attestation of an ancient language	Hittite texts from the second millennium BCE; the fourth-century Gothic Bible	A minimum age for the clade containing that language; or a dated tip	Attestation is not origin; literary languages may not be ancestral to spoken descendants
Disappearance of a common language	Latin as the last common stage before the Romance split	Age of a clade's last common ancestor	When exactly a dialect continuum became separate languages is a matter of definition
Archaeological correlates	Lapita-associated settlement of western Polynesia; first settlement of New Zealand	Ages of clades assumed to correspond to an archaeological expansion	Circularity if the dated tree is then used to test the archaeological hypothesis; language and material culture need not move together
Datable loanword strata	Latin loans in Germanic; Chinese loans in neighboring languages	Relative order of events and contact dates	Indirect; often constrain contact rather than splits

Three general points apply. First, a calibration is a statement about a *node* in the tree: the last common ancestor of a specified set of languages. Its validity depends on the clade being correctly identified. If a calibration is attached to the most recent common ancestor of the Romance languages, but the tree places a non-Romance language within that clade, the calibration constrains the wrong event. In BEAST and similar programs, calibrated clades are often constrained to be monophyletic for exactly this reason.

Second, most historical evidence provides *minimum* ages more securely than *maximum* ages. A written text proves that a language existed and was already distinct by the time it was written. It says nothing about how long before that the language had been separating from its relatives. Maximum ages are typically inferred from negative evidence, such as the absence of a population

from a region before a certain date, which is weaker. The asymmetry is the same as with fossils, and it shapes the choice of calibration distribution (Section 10.3).

Third, the split between two languages is not an instantaneous event. Dialects diverge gradually, and the date at which two varieties became “separate languages” depends on contact, mutual intelligibility and social identity. A tree model treats a split as a point in time. The calibration therefore represents a modeling idealization, and its uncertainty should be wide enough to reflect that.

10.3 Node calibrations and their densities

10.3.1 Calibration distributions

In a Bayesian analysis, a calibration is a probability density on the age of a node. Several standard forms are used (Ho and Phillips, 2009):

- A **uniform** density between a minimum a and a maximum b, with zero probability outside. This is a *hard-bounded* calibration. It says that all ages in the interval are equally plausible and all ages outside it are impossible.
- A **soft-bounded uniform**, in which most of the probability lies between a and b but small tails, often 2.5 percent on each side, extend beyond them (Yang and Rannala, 2006). Soft bounds allow the data to override a calibration that conflicts with the rest of the evidence, which hard bounds do not.
- An **offset lognormal** or **offset exponential**, in which the minimum age a is treated as a hard lower bound and the time by which the true age exceeds it has a lognormal or exponential distribution. These forms express the idea that the true age is probably not much older than the minimum but could be considerably older. Their parameters are best chosen by deciding where the bulk of the probability should lie, for example by placing the 97.5 percent quantile at the oldest plausible age, rather than by guessing the parameters directly.
- A **normal** density centered on a best estimate, appropriate when an event is believed to be dated to within a symmetric range, as with some historical events. Normal calibrations were common in early linguistic BEAST analyses.

Figure 10.1 compares these shapes for a hypothetical node whose minimum age, from the earliest written attestation of one of its descendants, is 2,500 years.

The offset-lognormal density (Equation (C.12)), the construction of soft bounds, functions that choose the parameters of offset lognormal and exponential calibrations from a minimum and an upper quantile, and the code for Figure 10.1 are in Section C.4.2.

Which shape is right? There is no general answer. Biologists have argued for years about how to translate paleontological knowledge into calibration densities, and have developed best-practice guidelines for justifying them (Parham et al., 2012). The same discipline would benefit linguistics. For each calibration, a study should state the clade it applies to, the evidence for its minimum and maximum, the reason for the chosen distribution and its parameters, and a reference to the historical or archaeological source. Most published linguistic analyses provide the first two; fewer justify the shape of the density. The choice matters most for the tail on the old side, because that is where the evidence is weakest and where an offset exponential and an offset lognormal with the same median can differ by centuries.

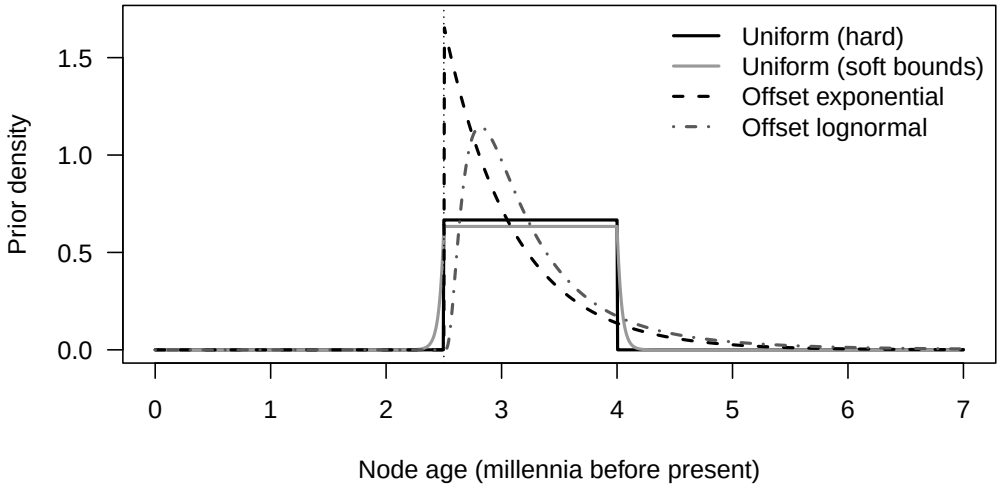


Figure 10.1: Four calibration densities for a node with a minimum age of 2.5 millennia (dotted line): a hard-bounded uniform (2.5 to 4 millennia), a soft-bounded uniform with exponential tails, an offset exponential and an offset lognormal. All have similar central ranges but very different assumptions about how old the node could be.

10.3.2 How calibrations interact

A calibration density is what the analyst *specifies*. It is not necessarily what the analysis *uses*. The prior on node ages actually applied in an analysis, the *effective prior* or *marginal prior*, can differ substantially from the specified calibration densities, for three reasons.

Truncation by ordering. A node cannot be older than its parent. If two calibrated nodes are nested and their calibration densities overlap, the joint prior is truncated to the region where the ancestor is older than the descendant. Both marginal distributions shift: the ancestor toward older ages, the descendant toward younger ones.

Interaction with the tree prior. In BEAST, the joint prior on node ages is typically constructed by multiplying the tree prior density by the calibration densities. This product is not, in general, a distribution whose marginals equal the calibration densities. The tree prior pulls node ages in its own direction, and uncalibrated nodes are also affected. Heled and Drummond (2012) derived *calibrated tree priors* whose marginals do match the specified calibrations for the Yule process, and explained the difference between the multiplicative and conditional constructions. Yang and Rannala (2006) use a conditional construction in MCMCTree, in which the calibrations define the marginal densities of calibrated nodes and the birth-death process defines the distribution of the others conditional on them.

Interaction with the clock prior. The prior on rates also contributes. Even the prior on the mean clock rate, combined with the root calibration and the tree prior, implies a prior on the total amount of change expected in the tree.

Inoue et al. (2010) and Warnock et al. (2012) demonstrated with molecular examples that effective priors can differ greatly from specified ones and that the differences can drive posterior dates. The

diagnostic is simple. Run the analysis without data, sampling from the prior alone, and compare the resulting marginal prior distributions of calibrated and uncalibrated node ages with what was intended. BEAST 2 and RevBayes both make this easy. A prior-only run is also the first step of *implementation validation* in the sense of Section 6.10: if the software's prior does not match the analyst's understanding of it, nothing downstream can be trusted.

The following simulation illustrates truncation. A root has a normal calibration with mean 6 and standard deviation 0.5 millennia, and a descendant node has a uniform calibration between 4 and 7 millennia. Sampling from the product of the two densities and imposing the order constraint, by rejection, gives the effective joint prior.

```
set.seed(3)
m <- 2e5
root <- rnorm(m, 6, 0.5)
desc <- runif(m, 4, 7)
ok <- desc < root                          # ordering constraint
op <- par(mfrow = c(1, 2), mar = c(4, 4, 2, 1))
hist(root[ok], breaks = 60, freq = FALSE, col = "grey85", border = "white",
     main = "Root", xlab = "Age (millennia)", xlim = c(3.5, 8), las = 1)
curve(dnorm(x, 6, 0.5), add = TRUE, lwd = 2)
hist(desc[ok], breaks = 60, freq = FALSE, col = "grey85", border = "white",
     main = "Descendant node", xlab = "Age (millennia)", xlim = c(3.5, 8),
     las = 1)
segments(4, 1/3, 7, 1/3, lwd = 2); segments(c(4, 7), 0, c(4, 7), 1/3, lwd = 2)
```

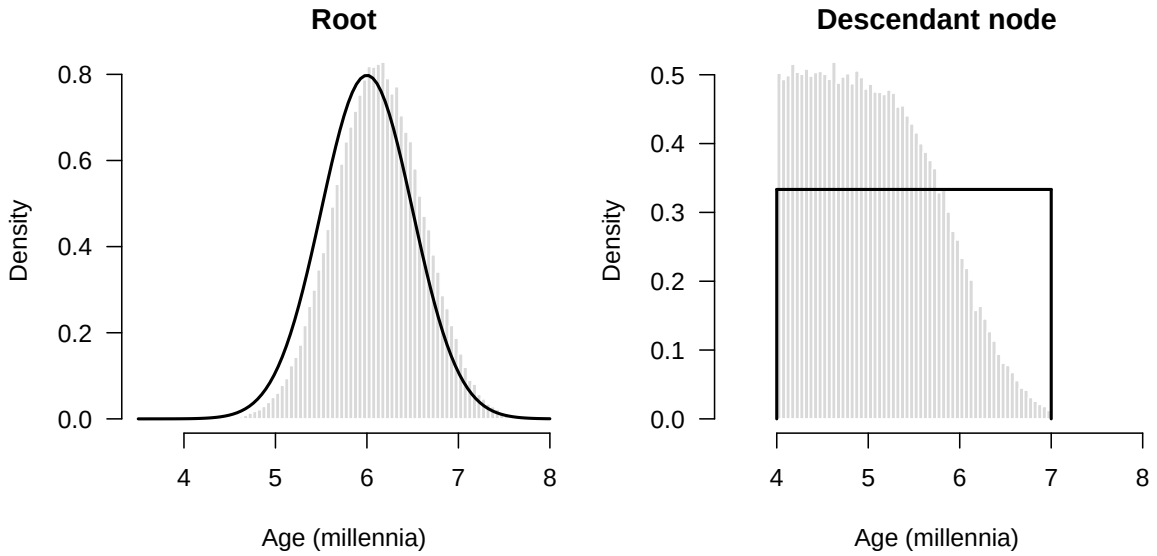


Figure 10.2: Effective versus specified priors. Histograms show the effective marginal priors on the root (left) and a calibrated descendant node (right) after the two calibrations are combined and the ordering constraint (descendant younger than root) is imposed. Lines show the densities the analyst specified.

```
par(op)
round(c(mean_root_specified = 6, mean_root_effective = mean(root[ok]),
```

```
        mean_desc_specified = 5.5, mean_desc_effective = mean(desc[ok])), 2)
```

```
#> mean_root_specified mean_root_effective mean_desc_specified
#>                6.00                6.12                5.50
#> mean_desc_effective
#>                5.06
```

The root's effective prior is shifted older than specified, and the descendant's is no longer uniform: ages near 7 millennia have become improbable, because they would usually exceed the root. With more calibrations and a tree prior, such interactions multiply. None of this is a flaw in the software. It is a consequence of the logic of probability, and the analyst's task is to check that the effective prior expresses what she believes.

The exact effective densities for this example, obtained by integration and used to check the rejection sampler, and the general form of the multiplicative and conditional constructions, are in Section C.4.3.

Always sample from the prior. A posterior date that sits close to the effective prior tells you that the data are uninformative about that node and the answer was determined by the assumptions. A posterior that sits close to a *specified* calibration may still be strongly shaped by the tree prior or by other calibrations. Report effective priors alongside posteriors for the key nodes, especially the root. Reviewers of linguistic dating studies should ask for this, as reviewers in molecular dating now routinely do (Brown and Smith, 2018).

10.4 The tree prior

The tree prior specifies the distribution of node ages before the data are seen. Three classes are common (Chapter 6). A *Yule* or pure-birth process assumes that lineages split at a constant rate λ and never go extinct. A *birth-death* process adds an extinction rate μ and a sampling proportion ρ, the fraction of extant languages included. A *coalescent* prior, derived from population genetics, is designed for samples from a single population and is rarely appropriate for language families, although it has occasionally been used.

The tree prior matters most for the nodes that are far from any calibration and for the root, if the root is not itself calibrated. A Yule prior tends to place nodes relatively evenly in time, whereas a birth-death prior with high extinction concentrates them near the present. A coalescent prior concentrates them near the present even more. Under weak data and few calibrations, the choice can shift root ages noticeably. Rama (2018) compared three tree priors (uniform, fossilized birth-death and coalescent) on five Indo-European datasets and found that the choice shifted the inferred root date by up to about three millennia, and changed the placement of Anatolian and Tocharian near the root, while well-established subgroups were recovered under all three priors. Ritchie and Ho (2019), testing four lexical datasets, found origin-time estimates to be fairly robust to the choice of tree prior, though less so than in comparable analyses of genetic data. They also found that speciation priors fitted better than a constant-size coalescent prior at all the taxonomic scales they examined, but that datasets with narrower taxonomic and geographic sampling fitted the expectations of a birth-death model less well. The sampling proportion ρ also matters. If a family is represented by a small fraction of its languages, a birth-death prior that assumes complete sampling will place internal nodes in the wrong places. Diversified sampling, in which one language per subgroup is chosen, violates the assumption of random sampling made by most birth-death

priors, and biology has developed priors for this case. The parameters of the tree prior are themselves seldom well estimated from a single family (Table 10.1), so it is the *choice* of prior, not the estimated speciation and extinction rates, that usually matters for dates.

10.5 Tip-dating with ancient languages

10.5.1 Dated tips

Indo-European, Semitic, Sinitic and a few other families include languages known from texts that are centuries or millennia old: Hittite, Tocharian, Latin, Ancient Greek, Vedic Sanskrit, Gothic, Old Irish, Akkadian, Classical Chinese. Such languages can be used in two ways. In *node dating*, they provide minimum ages for the clades containing them, and they may be excluded from the data matrix entirely. In *tip dating*, they are included in the data matrix as tips whose ages are fixed, or given a prior distribution, at the dates of their attestation. Because the date of a text is itself often uncertain by decades or centuries, the second option is the safer one: in palaeontology, fixing uncertain fossil ages at point values has been shown to bias estimates of divergence times and topology (Barido-Sottani et al., 2019, 2020). Their lexical data then directly inform the rate of change: a language that stopped changing 3,000 years ago should have accumulated fewer changes since the root than a living language. The idea comes from the study of *measurably evolving populations* in virology (Drummond et al., 2003), where viruses sampled at different times provide a direct measure of the rate.

Tip dating is attractive because the ages of attested languages are among the most secure pieces of chronological information available. They do not require assumptions about when a split happened, only about when a text was written. Its limitations are also clear. Ancient languages are often attested in limited corpora, so many meanings of a basic vocabulary list are missing. They may be literary or archaic registers rather than the spoken vernacular of their time. And they raise the question of whether they are ancestors of later languages, discussed in Section 10.6.

10.5.2 Root-to-tip regression

A quick check of temporal signal in heterochronous data is the *root-to-tip regression*, popularized in virology by TempEst (Rambaut et al., 2016). If the tree is rooted correctly and the clock is roughly strict, the root-to-tip distance d_i of tip i in a phylogram is approximately

$$d_i \approx r\,(T - a_i), \tag{10.2}$$

where T is the root age and a_i is the age of tip i (zero for living languages). The slope of the regression of d_i on a_i estimates $-r$, and the point where the line crosses zero distance estimates T. The scatter around the line indicates departure from a strict clock. In virology, this plot is used to detect outliers and to check whether there is enough temporal signal to estimate rates from the tip dates alone. Because the tips share ancestral branches, the usual regression p-values are invalid. A complementary test randomizes the tip ages among tips and checks whether the observed slope is stronger than in the randomized datasets, the *date-randomization test* (Duchêne et al., 2015b).

We simulate a family of 24 languages with a root at 6,000 years, of which eight are known only from ancient texts, with lexical change under a mildly relaxed clock (`het_phylo`, with tip ages `tip_age`). In the vocabulary of Section 6.10, this is a *parameter-recovery* exercise: the data follow the nearly clocklike process that the regression assumes, so success here shows only that the method works when its assumptions hold.

The code that simulates the heterochronous family is listed in Section C.4.4, together with the derivation of Equation (10.2).

```
d <- node.depth.edgelength(het_phylo)[1:n]          # root-to-tip distances
fit <- lm(d ~ tip_age)
root_hat <- unname(coef(fit)[1] / -coef(fit)[2])    # where d reaches zero
plot(tip_age, d, pch = ifelse(tip_age > 0, 16, 1), xlim = c(0, 7),
     ylim = c(0, max(d) * 1.05), las = 1,
     xlab = "Tip age (millennia before present)",
     ylab = "Root-to-tip distance (changes)")
abline(fit, lwd = 2); abline(v = 6, lty = 3)
```

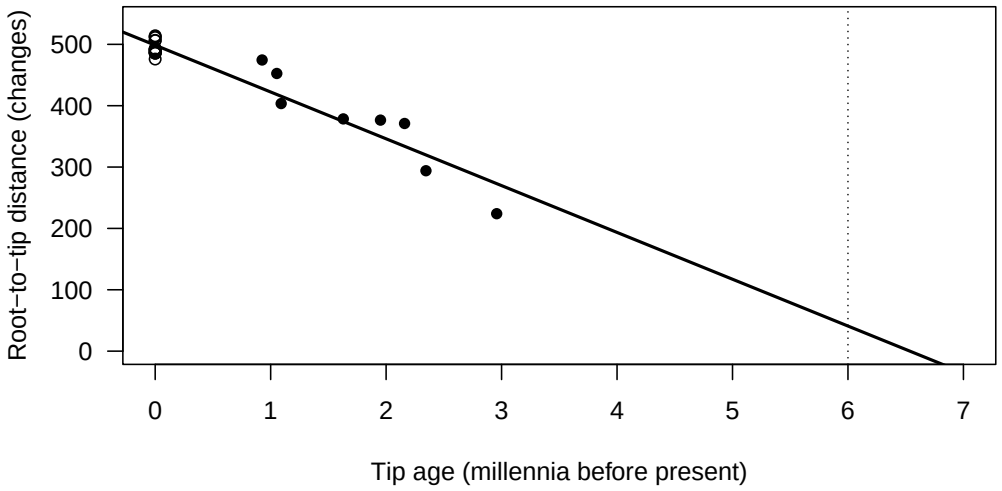


Figure 10.3: Root-to-tip regression for a simulated family of 24 languages, eight of them ancient (filled circles, plotted at their attestation ages). The regression line crosses zero distance near the root age; the dotted line marks the true root age of 6 millennia.

```
# date-randomization test: permute tip ages among tips
perm_slopes <- replicate(500, coef(lm(d ~ sample(tip_age)))[2])
c(rate_per_kyr = unname(-coef(fit)[2]), root_age = root_hat,
  randomization_p = mean(perm_slopes <= coef(fit)[2]))
```

```
#>    rate_per_kyr        root_age randomization_p
#>          76.290           6.536           0.000
```

The estimated rate is close to the simulated mean of 80 changes per millennium, the estimated root age is within about half a millennium of the truth (somewhat too old here, partly because of rate variation and the small constant the simulation adds to every branch), and the date-randomization test shows that the temporal signal is far stronger than would be expected by chance. In real linguistic data, root-to-tip plots are rarely shown, but they should be. They reveal ancient languages that are too divergent or not divergent enough for their age, which can indicate coding problems,

missing data, misplacement in the tree, or an ancient language that is an ancestor rather than a sister of its relatives.

10.5.3 Missing data in ancient languages

Ancient languages are often poorly attested, and missing data interacts with the model of lexical evolution in subtle ways. If an unattested meaning is treated as absent rather than unknown, the language appears to have lost cognates it may in fact have had, which makes it look more divergent and pushes it away from its relatives. Ryder and Nicholls (2011) developed a stochastic Dollo model that handles missing data explicitly, and applied it to dating Proto-Indo-European, building on Nicholls and Gray (2008). Their analyses showed that careful treatment of missing data matters for root ages. More generally, the way ancient languages are coded, and the way their placement in the tree is treated, became central issues in the debate over Indo-European dates (Section 10.9.1).

10.6 Ancient languages: tips, ancestors or sisters?

10.6.1 What an attested language is

Ancient languages are not simply fossils in the biological sense, and the analogy, useful as it is, hides a problem that historical linguists have long recognized. An attested stage of a language can stand in at least four different relations to the later languages in a dataset:

- it can be (very close to) the **direct ancestor** of some of them, as Old Norse is of Icelandic;
- it can be a **sister lineage** with no surviving descendants, as Hittite, Tocharian and Gothic are;
- it can be a **close relative of the real ancestor**, a different variety of the same speech community. Most Old English texts are in West Saxon, whereas Modern Standard English continues mainly Anglian dialects; the Romance languages continue spoken Latin rather than the Classical Latin of the texts; the Middle Indo-Aryan languages are not the descendants of Classical Sanskrit in any simple sense;
- it can be a **witness to a dialect network** rather than to a single lineage, if its descendants or sisters arose by the diversification of a continuum in which innovations spread across overlapping areas [Garrett (2006); Section 8.7].

Which of these holds is itself an inference, made with the comparative method and philology, and it is often uncertain. It matters for three stages of an analysis. For **taxon selection**, it determines which ancient varieties to include (a vernacular-like variety is more informative than a literary standard, and two stages of one lineage may be better than one), and whether an ancient language should enter the data matrix at all or only provide a node calibration. For **character coding**, it determines how to treat register-specific vocabulary, archaisms and gaps in the corpus: a basic word unattested in a small corpus must be coded as unknown, not absent (Section 10.5.3), and a literary word that the spoken language did not use creates a spurious unique change. For **interpretation**, it determines what a date means: if an ancient language sits on a side branch, the split between it and its relatives must be older than its attestation, and a model that cannot place it on the line of descent will push that split, and with it the root, back in time.

10.6.2 The fossilized birth-death process

Paleobiology faced the same problem as linguistics: how to combine dated extinct taxa and living taxa in a single, coherent prior. The answer was the *fossilized birth-death* (FBD) process (Stadler, 2010; Heath et al., 2014). In the FBD model, lineages split at rate λ, go extinct at rate μ, and are sampled as fossils at rate ψ while alive. Living lineages at the present are sampled with probability ρ. The process defines a prior on trees that includes both the living tips and the dated fossil samples. Because fossils are sampled from lineages that may have continued to evolve, a fossil can be a *sampled ancestor*: a sample lying directly on a lineage that has living descendants, rather than on a side branch that went extinct. Sampled ancestors appear in the tree as tips with zero-length branches, and specialized MCMC moves are needed to add and remove them (Gavryushkina et al., 2014).

For languages, the analogy is direct but, as the previous section showed, imperfect. An attested ancient language is a "fossil" sampled from a lineage at a known time. Latin is the standard example of a plausible ancestor; Hittite and Tocharian have no living descendants. A standard birth-death prior forces every ancient language to sit at the tip of its own branch, which must be at least slightly longer than zero. If the ancient language is really an ancestor, the model must explain the absence of its unique changes by accumulating a long branch elsewhere, and this can distort the rates and dates. The FBD prior with sampled ancestors allows the ancient language to sit directly on the lineage leading to its descendants, if the data support it.

The FBD process brings two further benefits. It provides a coherent prior on the times of all nodes given the sampled ancient languages, instead of combining ad hoc calibration densities with a tree prior and suffering the interactions described in Section 10.3.2. And it allows *total-evidence dating*, in which the ancient languages contribute both their dates and their data, as fossils contribute both stratigraphic ages and morphology in paleobiology (Pyron, 2011; Ronquist et al., 2012a; Zhang et al., 2016). Table 10.3 compares the four ways of using an attested language.

Table 10.3: Four ways of using an attested ancient language in a dating analysis.

Treatment	What is assumed	What the ancient language's data contribute	Typical consequence for dates	Main risk
Node calibration only	The language belongs to a specified clade, attested by a given date	Nothing; only its date enters, as a minimum age	Depends on the density chosen for the clade age	Treating attestation as origin; wrong clade
Dated tip, standard tree prior	The language is a sister lineage that ended at attestation	Its cognates, which inform both topology and rate	If it is really ancestral, splits are pushed older than they should be	Forced side branches; missing data read as change
Dated tip, sampled-ancestor (FBD) prior	It may be an ancestor or a sister; the data decide	As above, plus evidence on ancestry	Intermediate; depends on how well ancestry is identified	Ancestry weakly identified with fragmentary data

Treatment	What is assumed	What the ancient language's data contribute	Typical consequence for dates	Main risk
Ancestry constraint	The language is (nearly) the ancestor of specified descendants	As above, with its position fixed	Younger splits if it is in fact a sister	Encodes a philological judgment as a certainty

10.6.3 Ancestor or sister: what the data can tell

How well can the lexical data distinguish an ancestor from a sister on a short side branch? The evidence for ancestry is negative: an ancestral language should show no lexical changes of its own, no cognate classes gained or lost that its descendants lack. Suppose an ancient language is attested for n meanings of the list, each replaced at rate k per millennium, and sits on a side branch of length x millennia. It is then expected to show about nkx unique replacements. If none is observed, the data are consistent with any side branch shorter than about $3/(nk)$ millennia, the familiar "rule of three" for a Poisson count of zero. With a retention of 0.86 per millennium ($k \approx 0.15$), a language attested for all 200 meanings excludes side branches longer than about a century, but one attested for 50 meanings excludes only those longer than about four centuries, and one attested for only 20 meanings cannot exclude a side branch of almost a millennium. Conversely, a few apparent unique changes can arise from a literary register, a scribal tradition or a regional variety rather than from a separate line of descent, and the model has no way to tell these apart. In the terms of Section 6.10, ancestry is *identifiable under the model* but often not *practically estimable* from fragmentarily attested languages, and the posterior probability that an ancient language is an ancestor is correspondingly sensitive to coding and to the prior on sampled ancestors.

The calculation behind these figures, as a short R function with a table for several list sizes, is in Section C.4.6.

10.6.4 Ancestry constraints and sampled ancestors in linguistics

Chang et al. (2015) argued that much of the disagreement between phylogenetic and traditional estimates of the age of Indo-European arose from the way ancient languages were treated. Several ancient languages in their dataset are widely accepted as ancestral, or very close to ancestral, to particular modern languages, such as Latin to the Romance languages and Vedic Sanskrit to many modern Indo-Aryan languages. When these were constrained to be ancestors, the estimated root age fell considerably. Heggarty et al. (2023) later used the sampled-ancestor framework in BEAST 2, which estimates rather than imposes ancestry, on a newly compiled Indo-European dataset. Their results and the debate they generated are discussed in Section 10.9.1. Seen through Table 10.3, the two studies made different choices about how much of the philological judgment to impose and how much to leave to the data, and the previous section shows why the data alone may not settle the question. A dating study with ancient languages should therefore report its root age under more than one of the four treatments, and say which ancient languages it regards as ancestors and on what philological grounds.

Goldstein (2024) examined these issues in a single, well-documented case, the diversification of Latin into the Romance languages. The study addresses three problems that recur in Bayesian

dating: sensitivity to the model, the dependence of root-age estimates on uncertain interior-node ages, and the relationship between ancient languages and their modern counterparts. Its results support a gradualist account of the formation of Romance that most likely began after 300 CE, and they are consistent with treating Classical Latin as a sampled ancestor of the Romance languages, that is, as lying on the branch that leads to them. Because the history of Romance is unusually well known, analyses of this kind are among the few places where a dating method's behavior can be compared with independent historical evidence.

Several developments of the FBD framework in paleobiology have not, to our knowledge, been used with language data. *Skyline* versions of the birth-death and FBD processes let the rates of splitting, extinction and sampling change through time in piecewise-constant intervals (Stadler et al., 2013). For languages, a time-varying sampling rate is especially relevant: the probability that a language is attested in writing was essentially zero before the invention of writing and has increased since. A model with constant sampling through time misrepresents this. *Stratigraphic range* models treat a taxon known over an interval of time, rather than at a single date, as a single lineage observed repeatedly. Languages attested over several centuries in a sequence of texts, such as Latin, Greek or Chinese, are natural candidates. *Total-evidence dating with morphological clocks* (Zhang et al., 2016; Gavryushkina et al., 2017) combines data partitions with different clock models, which is directly relevant for analyses that combine lexical and structural data for ancient and modern languages (Chapter 9). Finally, *calibration cross-validation*, in which each calibration in turn is left out and its node age predicted from the others, can detect inconsistent calibrations (Near et al., 2005); Section 10.8 shows how simple it is to apply.

10.7 Calibration pitfalls

Many disputes over linguistic dates turn out, on close inspection, to be disputes over calibrations. The following problems recur.

Circularity. If a calibration is derived from an archaeological hypothesis, the dated tree cannot then be used to test that hypothesis. Using the date of Lapita pottery to calibrate the age of Oceanic, and then claiming that the dated tree supports a Lapita-associated spread of Oceanic languages, would be circular. Careful studies avoid this by using historical calibrations unrelated to the hypothesis under test, or by checking that the result is robust when the relevant calibrations are removed.

Attestation is not origin. The date of the earliest text in a language is a minimum age for the clade that contains it, not an estimate of the clade's age. Treating it as the age of the split, for example with a narrow normal distribution centered on it, biases dates downward.

Wrong clade. A calibration must be attached to the correct node. If historical evidence dates the separation of two dialect groups but the calibration is applied to the most recent common ancestor of a larger clade, or the clade is not monophyletic in the tree, the calibration constrains the wrong event. The miscalibration scenario in the R example below shows the consequences.

Overconfident calibrations. Narrow calibration densities express certainty that the evidence rarely supports. Because calibrations determine the precision of all dates (Section 10.1), overconfident calibrations yield overconfident dates throughout the tree.

Unintended interactions. As shown in Section 10.3.2, several calibrations together, combined with a tree prior, can produce an effective prior that nobody intended.

Extrapolation. If all calibrations are shallow, for example within the last two millennia, the root age is estimated by extrapolating the clock model far beyond the range of the calibrations. Any misspecification of the clock, for instance the time-dependence of rates discussed in Section 9.5.7, then affects the root most strongly.

Undetected borrowing and parallel change. Borrowing between sister languages makes them appear more similar and the nodes that join them younger (Section 8.3). Chang et al. (2015) used the term *advergence* for the process by which languages in contact become more similar after their split, which can bias dates downward. Undetected borrowing from outside the family can have the opposite effect, making a language appear more divergent.

Model misspecification beyond the clock. The substitution model, the treatment of ascertainment bias and the model of rate variation among characters all affect branch lengths and therefore dates. Chapter 7 discusses these; Chapter 15 gathers the evidence on how much they matter in practice.

The following example explores some of these pitfalls with **ape**'s `chronos()` on a simulated family of 20 languages (`cal_phylo`, a phylogram with lognormally varying rates) with a true root age of 6,000 years. We date the same phylogram under six calibration schemes, including one in which a calibration has been attached to the wrong, younger node with a minimum age that belongs to an older event. In the vocabulary of Section 6.10, the misattributed scheme is a test of *robustness under misspecification* of the calibrations, the other schemes a comparison of how much each kind of calibration constrains the dates when the calibrations are correct.

```
calib <- function(node, lo, hi)
  data.frame(node = node, age.min = lo, age.max = hi, soft.bounds = FALSE)
schemes <- list(
  "Root fixed at 6" = calib(n + 1, 6, 6),
  "Root between 5 and 7" = calib(n + 1, 5, 7),
  "Three internal (+/-15%)" = calib(cal_nodes, 0.85 * tc, 1.15 * tc),
  "Root window + three internal" = calib(c(n + 1, cal_nodes),
                                         c(5, 0.85 * tc), c(7, 1.15 * tc)),
  "One shallow internal only" = calib(cal_nodes[3], 0.85 * tc[3],
                                      1.15 * tc[3]),
  "Three internal, one misattributed" = calib(cal_nodes,
      c(0.85 * tc[1:2], 1.6 * tc[3]), c(1.15 * tc[1:2], 2 * tc[3]))
)
res <- t(sapply(schemes, function(cal) {
  ch <- suppressWarnings(chronos(cal_phylo, lambda = 1, model = "correlated",
                                 calibration = cal, quiet = TRUE))
  est <- branching.times(ch)
  c(root = unname(est[as.character(n + 1)]),
    rmse = sqrt(mean((est - true_ages[names(est)])^2)))
}))
knitr::kable(round(res, 2), col.names = c("Estimated root age (millennia)",
  "RMSE of all node ages (millennia)"),
  caption = "Root age and node-age error under six calibration schemes for the same
  simulated phylogram (true root age 6 millennia).")
```

Table 10.4: Root age and node-age error under six calibration schemes for the same simulated phylogram (true root age 6 millennia).

	Estimated root age (millennia)	RMSE of all node ages (millennia)
Root fixed at 6	6.00	0.19
Root between 5 and 7	6.73	0.34
Three internal (+/-15%)	6.27	0.25
Root window + three internal	6.27	0.25
One shallow internal only	6.35	0.26
Three internal, one misattributed	7.39	0.72

The simulation of `cal_phylo` and the choice of calibrated nodes are listed in Section C.4.5.

Several lessons emerge from Table 10.4. A fixed root age gives good estimates of the other nodes, but only because we told the method the right answer. A root window lets the root age drift within the window, here toward its older end. Internal calibrations that bracket the true ages constrain the root reasonably well even without a root calibration. In this run, a single shallow calibration also happens to give a reasonable root age, but this is luck; with other seeds or stronger rate variation, extrapolating from a single shallow node can go badly wrong. And a single misattributed calibration pushes the whole tree older, with the error spreading to nodes far from the one that was miscalibrated. A penalized-likelihood analysis like this one gives point estimates only. A Bayesian analysis would show the same effects in the posterior distributions, along with the uncertainty.

10.8 Validating dates

How do we know whether a published date is right? The answer depends on which of five different questions is being asked, and the vocabulary of Section 6.10 keeps them apart.

1. **Implementation validation.** Does the software compute what the model says? Prior-only runs that reproduce the intended effective priors (Section 10.3.2), and agreement between independent implementations of the same model, answer this question. They say nothing about whether the model is appropriate.
2. **Parameter recovery.** Are the dates practically estimable from data of this size? Simulating cognate data on trees with the family's size, calibrations and rate variation, under the analysis model, and checking that the root age is recovered with intervals of the stated coverage, answers this question. Success shows that the data *could* determine the dates if the model were right.
3. **Robustness under misspecification.** How much do dates change when the data are generated by processes the model ignores? Simulations with borrowing (Section 8.3), rate shifts and bursts (Section 9.8), ancestral languages placed on side branches (Section 10.6.3) and realistic missing data in ancient languages answer this question.
4. **Empirical adequacy.** Could the fitted model have generated data like the real data? Posterior predictive checks (Section 7.10), tests of clock adequacy (Section 9.5.7) and root-to-tip outliers (Section 10.5.2) address this question.
5. **Independent historical validation.** Do the dates agree with evidence that played no part in estimating them? This is the question most readers have in mind, and it has two forms. *Calibration cross-validation* leaves each calibration out in turn, re-estimates the tree, and asks

whether the omitted node's age is predicted correctly from the others (Near et al., 2005). *External comparison* sets the estimated dates against archaeological, genetic or documentary evidence that was not used as a calibration.

External comparison is only as independent as the evidence. An archaeological date used to calibrate a tree cannot then validate it, and a root age that is "consistent with" an archaeological horizon is weak support if the credible interval is wide enough to be consistent with its competitors too. Agreement between an inferred topology and the subgroups of the comparative method is not independent validation of the topology, still less of the dates, because the cognate judgments that went into the analysis are themselves products of the comparative method, and because shared cognates include shared retentions, whereas subgroups are defined by shared innovations. The appropriate hierarchy of certainty is that a dated phylogeny *estimates* a root age under its assumptions; agreement with independent evidence *suggests* or *is consistent with* a historical scenario; only convergent agreement among several kinds of independent evidence, robust to the main misspecifications, comes close to *demonstrating* one.

Published examples of the first and third kinds of evidence are recent. King (2026a) applied *posterior* simulation-based calibration, which simulates datasets from parameter values drawn from the posterior for a given empirical dataset, to a tip-dated analysis of Indo-European vocabulary in BEAST 2. The posteriors were well calibrated, which suggests that the dating machinery is not biased by implementation problems; but datasets simulated from the posterior did not make node ages any more precise than the original analysis, in line with theoretical limits on the identifiability of node ages (Section 10.1.1). This is implementation validation, not evidence that the model is right. Rönchen et al. (2025a), in a manuscript forming part of a doctoral dissertation (Rönchen, 2025), tested robustness under misspecification instead. They simulated Indo-European vocabulary evolution with a richer model than the inference models (loans, a topology close to the consensus of historical linguists and varying rates of change) and reported that the methods of Bouckaert et al. (2012) and Chang et al. (2015) did not reliably recover the simulated root age: their inferences varied with model assumptions and with the randomness of the evolutionary process, and their credible intervals substantially understated the errors. The two studies are complementary rather than contradictory. A sampler can be well calibrated for its own model while the model is misspecified for the data, and only the second kind of study can show how much that matters for dates.

Calibration cross-validation is easy to apply and catches the misattribution pitfall of the previous section. The function below leaves each calibration out in turn, re-dates the tree with `chronos()`, and records the predicted age of the omitted node. We apply it to the "root window plus three internal" scheme, once with all calibrations correct and once with the third internal calibration misattributed as in Table 10.4.

```
loo_calib <- function(phylo, cal, model = "correlated", lambda = 1) {
  # leave each calibration out, re-date, predict the omitted node's age
  out <- t(sapply(seq_len(nrow(cal)), function(i) {
    ch <- suppressWarnings(chronos(phylo, lambda = lambda, model = model,
                                   calibration = cal[-i, ], quiet = TRUE))
    c(predicted = unname(branching.times(ch)[as.character(cal$node[i])]))
  }))
  data.frame(node = cal$node, min = cal$age.min, max = cal$age.max,
             predicted = as.vector(out),
             outside = as.vector(out) < cal$age.min |
```

```
                    as.vector(out) > cal$age.max)
}
good <- calib(c(n + 1, cal_nodes), c(5, 0.85 * tc), c(7, 1.15 * tc))
bad <- calib(c(n + 1, cal_nodes), c(5, 0.85 * tc[1:2], 1.6 * tc[3]),
             c(7, 1.15 * tc[1:2], 2 * tc[3]))
loo <- rbind(cbind(scheme = "all correct", loo_calib(cal_phylo, good)),
             cbind(scheme = "third misattributed", loo_calib(cal_phylo, bad)))
knitr::kable(loo, digits = 2, row.names = FALSE,
  caption = "Leave-one-calibration-out cross-validation: the age of each calibrated
  node predicted from the remaining calibrations, compared with its calibration
  window.")
```

Table 10.5: Leave-one-calibration-out cross-validation: the age of each calibrated node predicted from the remaining calibrations, compared with its calibration window.

scheme	node	min	max	predicted	outside
all correct	21	5.00	7.00	6.27	FALSE
all correct	22	3.68	4.97	4.08	FALSE
all correct	34	2.36	3.20	2.48	FALSE
all correct	24	1.55	2.10	2.08	FALSE
third misattributed	21	5.00	7.00	7.39	TRUE
third misattributed	22	3.68	4.97	4.88	FALSE
third misattributed	34	2.36	3.20	3.70	TRUE
third misattributed	24	2.92	3.65	2.07	TRUE

With correct calibrations, every omitted node is predicted inside its window. With the misattributed calibration, most of the calibrations fail: the misattributed node is predicted far younger than the window it was given, and other calibrated nodes are predicted older than theirs, because the misattributed calibration, still in place when they are left out, stretches the whole tree. Cross-validation therefore shows that the calibrations are mutually inconsistent under the clock model, but it does not by itself say which one is wrong. The largest discrepancy, here at the misattributed node, is the natural place to start, and the historical evidence behind each calibration must decide. Most language families have only a handful of calibrations, so cross-validation costs a few extra runs.

10.9 Family-by-family review

The rest of this chapter reviews published dated phylogenies, family by family. For each family we describe the main studies, the data and calibrations they used, their root-age estimates, and the debates they provoked. Dates are given as years before present (BP), as reported by the authors and rounded; readers should consult the original papers for exact values and intervals, which depend on the model and data subset. We have included only studies we are confident exist, and where we are unsure of a precise figure we say so rather than guess. Chapter 11 covers the homeland inferences that many of these studies also made.

Two cautions apply throughout. First, each root age is an *estimate* conditional on the data, coding, clock model, tree prior and calibrations of the study, and the main sources of disagreement between studies of the same family are usually to be found among these (Table 10.1). Second, several studies report that their trees recover subgroups established by the comparative method. That agreement is reassuring about data quality, but it is not independent validation of the trees, since the cognate judgments come from the same comparative tradition, and it is no validation at all of the dates (Section 10.8). When a study links its dates to an archaeological horizon, the useful question is whether that horizon was also used as a calibration.

10.9.1 Indo-European

No family has been dated more often, or with more controversy, than Indo-European. Two hypotheses about its origin have dominated the debate. The *steppe* or *Kurgan* hypothesis places Proto-Indo-European in the Pontic-Caspian steppe around 6,000 to 6,500 years ago, and associates its spread with mobile pastoralists who used horses and wheeled vehicles (Anthony, 2007). A key argument is linguistic paleontology. Proto-Indo-European can be reconstructed with a vocabulary for wheels, axles and wagons, and wheeled vehicles are not attested archaeologically before the late fifth or fourth millennium BCE. The *Anatolian* hypothesis associates Indo-European with the spread of farming from Anatolia into Europe, beginning around 8,000 to 9,500 years ago (Renfrew, 1987). The two hypotheses predict root ages that differ by two to three millennia, which makes dating potentially decisive.

Gray and Atkinson (2003) analyzed 87 Indo-European languages using the cognate judgments of Dyen et al. (1992) for the Swadesh 200-meaning list, supplemented with Hittite and the two Tocharian languages, for a total of 2,449 cognate sets coded as binary characters. They estimated trees with MrBayes under a binary restriction-site model with gamma rate variation, then dated a sample of trees by penalized-likelihood rate smoothing (Sanderson, 2002), using 14 calibrations from historically documented events and attestations. The estimated root age fell between about 7,800 and 9,800 years BP, consistent with the Anatolian hypothesis and older than the steppe hypothesis predicts. They tested the robustness of this result to several of their assumptions, including the choice of calibrations and the treatment of the data. The paper, in *Nature*, launched the modern debate about phylogenetic dating in linguistics.

Subsequent work examined the robustness of the result. Atkinson et al. (2005) analyzed the Indo-European dataset of Ringe, Warnow and Taylor (Chapter 4) under both a reversible model and a stochastic Dollo model, tested both methods on simulated data with borrowing, and answered several criticisms of the earlier analysis of the Dyen data. Nicholls and Gray (2008) introduced a stochastic Dollo model for cognate data, in which each cognate set arises once and can then be lost many times (Chapter 7), and found root ages broadly compatible with Gray and Atkinson's. Ryder and Nicholls (2011) extended the model to handle missing data and also estimated a root age broadly in line with the earlier results.

Bouckaert et al. (2012) combined dating with phylogeography in a single Bayesian model. They used cognate data for 103 ancient and modern Indo-European languages, treated the ancient languages as dated tips, and modeled the geographic spread of the family as a random walk on the map (Chapter 11). The analysis favored an Anatolian homeland over a steppe homeland, and the root age, which overlapped the range estimated by Gray and Atkinson, fitted an agricultural expansion from Anatolia beginning roughly 8,000 to 9,500 years ago. The paper was criticized on several grounds, including its geographic model, its treatment of language contact and its data

(Pereltsvaig and Lewis, 2015). Its methodological contribution, the integration of time, geography and lexical evolution in one framework, has been widely adopted.

Chang et al. (2015) re-examined the question with a revised dataset, building on IELex, and argued that the earlier analyses had a systematic bias. In their unconstrained analyses, ancient languages such as Latin, Vedic Sanskrit and Old Irish were placed as side branches rather than as ancestors of their modern relatives. They argued that this, combined with *advergence* (convergent similarity through contact) among some modern languages, inflated the root age. When they imposed *ancestry constraints*, requiring the ancient languages to be ancestral or nearly ancestral to their modern descendants, the estimated root age fell to roughly 6,000 years BP (medians of about 5,900 to 6,500 years across their main analyses), consistent with the steppe hypothesis. The paper made the treatment of ancient languages, and the question of whether they are ancestors or sisters, central to the debate.

These analyses appeared during the same years in which ancient DNA transformed the archaeology of Indo-European. Haak et al. (2015) reported genomic evidence for a massive migration of people related to steppe pastoralists of the Yamnaya culture into central Europe around 5,000 years ago, and argued that this migration was a source for at least some Indo-European languages in Europe. Ancient DNA does not date languages, but it identifies population movements to which language spreads can plausibly be linked, and it shifted many scholars toward the steppe hypothesis, at least for the European branches. Pereltsvaig and Lewis (2015) provide a critique of the phylogenetic approach from a linguistic perspective, defending the steppe hypothesis. Garrett (2006) argued that several Indo-European subgroups arose through convergence among already-differentiated dialects rather than through clean tree-like splits, which complicates any attempt to date them as nodes of a tree.

Heggarty et al. (2023) assembled a new Indo-European dataset, IE-CoR, with cognate judgments for around 160 languages and 170 meanings, compiled by specialists in each branch under explicit and consistent protocols. They analyzed it in BEAST 2 with a sampled-ancestor model, so that ancient languages could be ancestors if the data supported it, rather than having that relationship imposed or forbidden. The estimated root age was around 8,100 years BP. The authors proposed a "hybrid" model in which Indo-European originated south of the Caucasus, with one branch later reaching the steppe and spreading from there into Europe, a scenario intended to reconcile the phylogenetic dates with the ancient DNA evidence for steppe migrations. The study is the most thorough phylogenetic analysis of Indo-European to date, but it did not end the debate. The interpretation of the root date, the coding of particular cognates and the handling of ancestry continue to be discussed.

The IE-CoR data have since been published as a stand-alone, openly documented dataset of 160 languages and 170 meanings, with 4,981 cognate sets, explicit coding of known horizontal transfer and calibration data for each language (Anderson et al., 2025), so the analysis can now be re-examined independently. The first critiques have addressed the data and the topology as much as the dates. Kassian and Starostin (2025) acknowledged that the lexical data are much better curated than in earlier studies, but argued that cognate coding in places still fails to separate true cognacy from areal diffusion and from independent derivations from the same root, that both the consensus topology and the dates conflict with historical evidence in several respects, and that a list of 170 meanings may be too short to resolve the topology of so many languages robustly. Two simulation studies bearing on the reliability of Indo-European dating are discussed in Section 10.8.

Indo-European dating has also been used to test methodological choices. Rama (2018) compared tree priors on several Indo-European datasets, and found that the choice affected the inferred root age. The general lesson of the Indo-European literature is that the main sources of disagreement have been calibrations, the treatment of ancient languages, the data, and the interpretation of the results in light of archaeology and genetics, more than the inference software. Chapter 15 returns to this controversy.

10.9.2 Austronesian

Austronesian is one of the largest language families in the world, with well over a thousand languages spread from Madagascar to Easter Island and from Taiwan to New Zealand. Its expansion is one of the clearest cases of a language family spread by seafaring migration. The standard hypothesis places its homeland in Taiwan, with a subsequent spread through the Philippines and Island Southeast Asia into the Pacific, associated with the "out of Taiwan" or "express train" model of Austronesian settlement (Bellwood, 2005; Diamond and Bellwood, 2003).

Gray and Jordan (2000) applied maximum parsimony to 77 Austronesian languages and found a tree consistent with the express-train model, with Formosan languages at the base and a sequence of splits following the proposed route of settlement. The Austronesian Basic Vocabulary Database (Greenhill et al., 2008) then made much larger analyses possible.

Gray et al. (2009) analyzed basic vocabulary for 400 Austronesian languages in a Bayesian framework with a relaxed clock and calibrations from archaeology and history. The tree placed the root in Taiwan around 5,200 years BP, consistent with the archaeological evidence. More strikingly, the dated tree revealed a pattern of *pulses and pauses*. After the initial split, there was a long pause of roughly a millennium before Malayo-Polynesian expanded from Taiwan into the Philippines, followed by a rapid pulse of expansion through Island Southeast Asia and into the Pacific. A second pause occurred in western Polynesia before the rapid settlement of eastern Polynesia. The authors linked the pauses to the challenges of crossing new sea gaps, such as the Bashi Channel between Taiwan and the Philippines, and the pulses to technological or social innovations such as improved outrigger canoes. The study is an example of how a dated tree can reveal the tempo of a prehistoric expansion, not just its starting date.

Greenhill et al. (2010b) examined the robustness of the Austronesian topology to borrowing, coding choices and missing data, and found that the main subgroups were robust. Greenhill and Gray (2009) addressed common misconceptions about Bayesian phylogenetic methods among Austronesianists. Some of the archaeological dates used in Austronesian work have themselves been revised. For example, high-precision radiocarbon dating has placed the initial settlement of East Polynesia, including New Zealand, later than many earlier estimates (Wilmshurst et al., 2011), which matters for calibrations of the youngest Polynesian clades.

King et al. (2024) turned from the age of the family to the internal history of Malayo-Polynesian in the Philippines. They analyzed 202 doculects from the ABVD (147 Philippine languages and an outgroup of 55 Formosan ones), with 7,565 cognate sets in 185 meanings whose coding had been revised by specialists, under a covarion model with a relaxed clock in BEAST 2. They did not estimate the age of the family: the root had a narrow prior centered on 5,200 years BP, taken from Gray et al. (2009), so all their dates are conditional on that value. Under this prior, the Philippine languages began to diverge about 3,400 years BP (HPD interval roughly 2,600 to 4,200). The main results concern topology and process. The languages of northern Sulawesi (Sangiric and Minahasan) were strongly supported as sister to all other Philippine languages, and the

Batanic languages nested within northern Luzon, a pattern the authors read as incompatible with a simple north-to-south dispersal out of Taiwan and as consistent with a rapid initial expansion followed by south-to-north spread. The trees carried a pervasive geographical signal, which they interpreted as evidence that diffusion across chains of neighboring languages dominated after the initial spread; the unstable position of the Gorontalo-Mongondow languages was plausibly due to later contact in Sulawesi. Language shifts to Austronesian in "Negrito" populations were dated to various points within the last two millennia, with no sign of elevated rates of lexical change on the branches where they occurred. Methodologically, the study examined which cognate sets supported each competing position of an unstable group, rather than reporting clade probabilities alone. As with other analyses of subgrouping, its agreement or disagreement with traditional subgroups is evidence about the cognate data under the model, not an independent test of either.

At a smaller scale, Smith and Rama (2022) dated a phylogeny of 87 Austronesian languages of Borneo with four historical calibrations. The inferred root was shallower than archaeological estimates for the settlement of the island, which they related to a possible linguistic leveling event, and a phylogeographic analysis placed the homelands of several subgroups along Borneo's major rivers.

10.9.3 Bantu

The Bantu languages, a branch of Niger-Congo with several hundred languages spoken across central, eastern and southern Africa, spread within the last few millennia from a homeland near the border of present-day Nigeria and Cameroon. Their expansion has long been associated with the spread of farming, and, later, iron working. The main debates concern the route: whether the eastern Bantu languages descend from populations that moved around the central African rainforest to the north and east, or through it.

Holden (2002) applied maximum parsimony to basic vocabulary for 75 Bantu languages and found that the tree was consistent with the spread of farming. Currie et al. (2013) used Bayesian phylogeographic methods on Bantu language trees to reconstruct the locations of ancestral populations.

Grollemund et al. (2015) analyzed basic vocabulary for over 400 Bantu languages in a Bayesian framework, combined with a phylogeographic model. They inferred that the expansion began in the mid-Holocene, on the order of five millennia ago, and that the major movements of Bantu-speaking populations tracked savannah habitats, crossing the rainforest zone mainly along savannah corridors that opened within it, such as the Sangha River Interval, rather than moving through closed forest. They also argued that the rate of expansion was slower where populations had to adapt to new habitats, such as the rainforest itself. The study linked both the route and the pace of a language expansion to habitat.

Koile et al. (2022) revisited the question with a different phylogeographic model and a dataset of Bantu languages, and found support for an early route *through* the rainforest, contrary to the conclusion of Grollemund et al. (2015). Their time frame for the expansion was broadly similar. The disagreement shows how sensitive phylogeographic conclusions can be to the spatial model and the choice of languages, even when the dates broadly agree (Chapter 11). Grollemund et al. (2015) place the start of the expansion at roughly 5,000 years ago, while Koile et al. (2022) estimate the root at about 5,100 years BP and the migration through the forest at about 4,400 years BP, well before the savannah corridor opened.

Lower-level studies have since refined the picture regionally. Gunnink et al. (2022) inferred a

lexicon-based phylogeny of 34 Southern Bantu languages in which all derive from a single direct ancestor; because archaeology indicates several earlier migrations of Bantu speakers into southern Africa, they argued that the languages of the first migrants were lost and that the ancestor of the present Southern Bantu languages emerged in southern Africa during the second millennium CE. Gunnink et al. (2025) placed the previously unclassified Yeyi in Wider Eastern Bantu as a lineage of its own, closest to Narrow Eastern Bantu rather than to any of its present neighbors, and suggested an early migration into southern Africa. Pacchiarotti et al. (2024) showed for part of West-Coastal Bantu how a lexical phylogeny and a diagnostic sound change can disagree about subgrouping, because the sound change itself may have spread through lexical diffusion and contact.

10.9.4 Sino-Tibetan

Sino-Tibetan includes Chinese (Sinitic) and several hundred Tibeto-Burman languages spoken from the Himalayas to Southeast Asia. Its homeland and age have long been debated. Hypotheses range from northern China, associated with millet farming in the Yellow River basin, to the eastern Himalayas or Sichuan.

In 2019 two independent dated phylogenies of Sino-Tibetan were published within weeks of each other. Sagart et al. (2019) analyzed lexical data for 50 Sino-Tibetan languages, with root-meaning cognate coding and a relaxed clock, and estimated a root age of roughly 7,200 years BP. They linked the origin of the family to millet farmers of northern China, in the late Cishan and early Yangshao cultures. Zhang et al. (2019) analyzed 109 Sino-Tibetan languages and estimated a root age of roughly 5,900 years BP, also placing the homeland in northern China and associating the origin with the Yangshao culture, with later westward and southward movements linked to subsequent Neolithic cultures such as Majiayao. The two studies agree on a northern Chinese origin linked to millet agriculture but differ in their root dates by more than a millennium, and in some details of the internal structure. The difference is a useful reminder of how much dates depend on data, coding and calibrations, even when the general scenario agrees.

A third continental analysis, by Zhang et al. (2020), used a larger and linguistically more diverse sample. It again found a primary split between Sinitic and Tibeto-Burman but placed the initial divergence earlier, at approximately 8,000 years before present, which the authors linked to the onset of millet agriculture in the Yellow River region. Three analyses with different data and teams thus agree on the general scenario while their root ages span roughly two millennia. Later work has concentrated on parts of the family that the continental studies sampled thinly. Wu et al. (2022) extended a subset of an existing Sino-Tibetan dataset with new vocabularies for the Kho-Bwa and Hrusish clusters and neighboring languages of the Tibet-Arunachal area and inferred dated phylogenies for their internal and external relations, and Dhakal et al. (2024) collected, transcribed and cognate-coded new data for eight South-Western Tibetic languages, including partial cognates, for a Bayesian analysis. Small studies built on primary data of this kind are what any future continental analysis will rest on.

10.9.5 Dravidian

The Dravidian family has about 80 languages, spoken mainly in southern India, with outliers such as Brahui in Pakistan. Kolipakam et al. (2018) collected new basic vocabulary data for 20 Dravidian languages and estimated a root age of approximately 4,500 years BP. The estimate is consistent with some linguistic and archaeological proposals. They also examined the internal structure of the family, including the relationships among its main subgroups, and discussed the possible re-

lationship of the Dravidian languages to the archaeological cultures of South Asia, including the Indus civilization, a question the phylogenetic results alone cannot resolve.

10.9.6 Uralic

Uralic includes Finnish, Estonian, Hungarian, the Saami languages and the Samoyedic languages of Siberia, among others. Syrjänen et al. (2013) analyzed the internal classification of Uralic using basic vocabulary and phylogenetic methods, and the UraLex dataset grew out of this work. Honkola et al. (2013) produced a dated Uralic phylogeny, with a root in the mid-Holocene, and argued that the timing of major splits in the family coincided with climatic and cultural changes in northern Eurasia. We do not quote a precise root date from this study; readers should consult the original. Uralic is a family with relatively few calibrations and a long history of debate about its homeland and internal structure, so dated phylogenies are particularly sensitive to calibration choices.

10.9.7 Turkic

The Turkic languages extend from Turkey to Siberia and China. Savelyev and Robbeets (2020) analyzed basic vocabulary for Turkic languages in a Bayesian framework and estimated the root age at roughly 2,100 years BP, with the earliest split separating the Bulgharic branch (represented today by Chuvash) from Common Turkic. The young age reflects the well-known shallowness of the family, whose languages are closely related. Hruschka et al. (2015) used Turkic as a test case for their model of regular sound change as concerted evolution (Chapter 9). Egorov et al. (2022) reanalyzed the internal structure of the family with strictly controlled word lists, explicit treatment of derivational drift and several algorithms (distance, Bayesian and parsimony). They confirmed Chuvash as the first outlier, found the rest of the family to diversify in a multifurcation into eight branches, and argued that Siberian Turkic is an areal rather than a genealogical unit; the subtitle of their paper names the limitation that intensive contact imposes on tree-based analyses of the family.

10.9.8 Transeurasian

The *Transeurasian* or *Altaic* hypothesis groups Turkic, Mongolic and Tungusic, with or without Koreanic and Japonic, into a single family. Most historical linguists regard this grouping as unproven or as reflecting long-term contact rather than common descent. Robbeets and Bouckaert (2018) applied Bayesian phylogenetic methods to a dataset of putative cognates across the five groups. Robbeets et al. (2021) then combined linguistic phylogenetics with archaeology and ancient DNA. They estimated a root age of roughly 9,000 years BP and associated the family's origin with millet farmers of the West Liao River region in northeastern China, with subsequent spreads linked to agriculture. The study was widely publicized and heavily criticized. Critics argued, in a widely discussed preprint, that the cognate sets were not established by the comparative method and included many doubtful comparisons, that the archaeological and genetic data did not support the proposed scenario, and that the triangulation framework could not compensate for weaknesses in the linguistic data (Tian et al., 2022). The debate is a reminder that a phylogenetic analysis cannot establish the genealogical unity of a family whose cognates are themselves in dispute. The individual branches are on firmer ground. For Tungusic, whose unity is not in doubt, Oskolskaya et al. (2022) analyzed 254 basic-vocabulary items for 21 doculects; their trees are consistent with two classifications proposed earlier, and they estimate the age of Proto-Tungusic at between the

8th century BCE and the 12th century CE (95% HPD), an interval whose width reflects the scarcity of calibrations.

10.9.9 Semitic

Kitchen et al. (2009) analyzed basic vocabulary for 25 Semitic languages, including ancient languages such as Akkadian, in a Bayesian framework. They estimated that Semitic originated around 5,750 years BP in the Levant, and that Ethiosemitic languages resulted from a single introduction of Semitic from southern Arabia into the Horn of Africa around 2,800 years BP. Semitic is unusual in having ancient languages attested from the third millennium BCE, which makes it a promising family for tip-dating and sampled-ancestor analyses. To our knowledge, a dated phylogeny of Afroasiatic as a whole, the much older family to which Semitic belongs, has not been attempted with comparable methods, which is unsurprising given the difficulty of establishing cognates at that time depth.

10.9.10 Pama-Nyungan

Pama-Nyungan is the largest family of Australian languages, covering most of the continent. Bowern and Atkinson (2012) used Bayesian phylogenetics to investigate the internal structure of the family, which had long been debated, and found support for many established subgroups. Bouckaert et al. (2018) then produced a dated phylogeography of over 300 Pama-Nyungan languages. They estimated a root age of roughly six millennia and placed the origin near the Gulf of Carpentaria in northern Australia, with the family spreading across the continent from there. The result is significant because Pama-Nyungan speakers were hunter-gatherers. Its spread cannot be explained by agriculture, and it challenges the view that large language families spread mainly with farming (Diamond and Bellwood, 2003). The Chirila database of Australian languages supports much of this work.

10.9.11 Japonic and Ainu

Lee and Hasegawa (2011) analyzed 59 Japonic languages and dialects and estimated a root age of roughly 2,200 years BP. They interpreted this as support for the hypothesis that Japonic arrived in the Japanese archipelago with rice farmers from the Korean peninsula during the Yayoi period, rather than descending from the languages of the earlier Jomon hunter-gatherers. Lee and Hasegawa (2013) applied similar methods to Ainu dialects and discussed the results in relation to the archaeology of Hokkaido and neighboring regions.

Huisman et al. (2025) revisited Japonic with a new dataset of 48 Japanese (including Old Japanese) and 33 Ryukyuan lects for 256 concepts, analyzing cognate characters together with binary phonotactic characters (the presence of particular two-segment sequences) in separate partitions on a shared tree. They estimated the Japanese-Ryukyuan split at around 400 to 500 BCE, consistent with the Yayoi-period scenario, and dated the ancestor of the Ryukyuan languages to around the ninth century CE, when the first traces of cereal farming appear in the northern Ryukyus. Old Japanese was a dated tip and the age of Southern Ryukyuan was calibrated, so the agreement of the deeper dates with the archaeological record is conditional on these choices.

10.9.12 South American families

South America has a large number of language families, many of them poorly documented, and phylogenetic dating is still in its early stages there. Walker and Ribeiro (2011) applied Bayesian phylogeography to Arawakan, one of the most widespread families of lowland South America, to investigate its homeland and dispersal. Walker et al. (2012) studied the Tupi family with a simpler, distance-based tree built from 40-item ASJP word lists, mapped onto geography. Within Tupi, the Tupi-Guarani branch, which includes Guarani and the extinct Tupinamba, has been studied in more detail. Michael et al. (2015) produced a Bayesian phylogenetic classification of Tupi-Guarani, and Ferraz Gerardi et al. (2023) revisited its chronology, comparing phylogenetic dates with the archaeological record of the region. A recurrent difficulty for these families is the scarcity of reliable calibrations: there are few written records before European contact, and the correlation of archaeological cultures with language groups is often uncertain. We do not quote root dates for these studies. Birchall et al. (2016) combined the comparative method with computational phylogenetic analysis for the small Chapacuran family of the Bolivian–Brazilian border region, an example of how the two can be used together where documentation is thin. Michael and Chousou-Polydouri (2019) review the computational classifications of South American families and argue that character-based methods should be preferred to distance-based ones, and that adequate comparative datasets are the main bottleneck.

Two recent studies prepare the ground for dated analyses. Michael et al. (2023) identified twelve calibration points for Arawakan from archaeological sites in zones into which Arawakan peoples expanded, together with more recent calibrations from the first historical mentions of ethnonyms and early lexical sources. Ferraz Gerardi et al. (2025) produced a Bayesian classification of the Je family (14 languages, 303 concepts) with well-supported Northern, Central and Southern subgroups.

10.9.13 Uto-Aztecan

Uto-Aztecan extends from the western United States to Central America and includes Nahuatl, the language of the Aztecs. A long-standing debate concerns whether the family originated in the south, in Mexico, with maize farmers, as proposed by Jane Hill and others, or in the north, among hunter-gatherers, with maize farming adopted later by some branches. Wheeler and Whiteley (2015) analyzed the family with a parsimony-based sequence optimization method and examined its biogeography. Greenhill et al. (2023) produced a dated Bayesian phylogeny of Uto-Aztecan and argued for a relatively recent northern origin, roughly four millennia ago, which they interpreted as evidence against a southern, farming-driven origin. As with other families, the conclusion depends on the calibrations and on the relationship between language spread and subsistence, which remain debated.

10.9.14 Kra-Dai

Tao et al. (2023) compiled a lexical dataset of 100 Kra-Dai (Tai-Kadai) languages, with 646 cognate sets based on the Swadesh 100-meaning list, and analyzed it in BEAST 2 under a covarion model with a relaxed lognormal clock, a birth-death skyline tree prior and node calibrations from historical records. The estimated root age was about 4,000 years BP (mean 4,041; 95% HPD 2,741 to 5,550). A discrete phylogeographic analysis over a sample of posterior trees favored an origin in the Guangxi-Guangdong area of southern China, with later dispersal toward Mainland Southeast Asia. The authors linked the diversification to demic diffusion and the spread of agriculture under

late Holocene climate change, drawing also on genetic and archaeological data; as elsewhere, the association is an interpretation of the dates, not a test of them. Buckley et al. (2025) compared phylogenies of Kra-Dai languages and of the looms used by their speakers (Chapter 14).

10.9.15 Families with few or no dated phylogenies

Many large and historically important families have not, to our knowledge, been the subject of a dated Bayesian phylogeny that is comparable in scope to the studies above, or have been analyzed only in preliminary or unpublished work. These include Austroasiatic (including Vietnamese, Khmer and the Munda languages of India), Hmong-Mien, Mayan, Oto-Manguean, Trans-New Guinea, most branches of Niger-Congo beyond Bantu, Nilo-Saharan (whose unity is itself contested), Afroasiatic as a whole, Nakh-Daghestanian, and most North American families other than Uto-Aztecan. Some of these, such as Mayan, have rich traditions of lexicostatistical and comparative work, including glottochronological estimates, and ancient written records (Classic Maya inscriptions) that could provide tip dates. Others, such as Trans-New Guinea, pose difficult problems of data and classification. These families represent clear opportunities for future work, with the caveat that many depend on primary documentation and cognate judgments that do not yet exist. Some gaps have begun to close. Kra-Dai now has a dated phylogeny (Section 10.9.14); the Mixtecan branch of Oto-Manguean has been subgrouped with Bayesian methods (Auderset et al., 2023); and the Timor-Alor-Pantar family of Papuan languages has been analyzed with a systematic comparison of models (Kaiping and Klamer, 2022). For Kartvelian (South Caucasian), Gavashelishvili et al. (2023) inferred a dated phylogeny whose topology agrees with the accepted one but whose mean split dates are earlier than in previous studies, and combined it with the ranges of animals and plants whose names reconstruct to Proto-Kartvelian, archaeological cultures and ancient genomes to place the homeland in the Colchis region of the South Caucasus. Both of the last two studies stress how dialect-chain histories complicate tree inference (Chapter 8).

10.9.16 Global-scale dating

At the global scale, Holman et al. (2011) applied ASJP chronology (Chapter 9) to estimate the ages of many of the world's language families. The approach uses a single empirically calibrated relationship between lexical similarity and time, so it inherits the limitations of distance-based dating, but it provides rough estimates for families that have never been dated phylogenetically, and a common yardstick for comparing them. Rama and Wichmann (2020) proposed a *generalized Bayesian dating* method in the same spirit: trees are inferred from automatically detected cognates in 40-item ASJP lists, and node ages are predicted from tree lengths by a gamma regression fitted to a set of known historical time depths, so that no family-specific priors or calibrations are needed. Applied to 116 families, it gave dates broadly similar to those of published family-specific studies, but in tests against known dates its errors were somewhat larger than those of ASJP chronology, although the difference between the methods was not statistically significant.

10.10 Summary of dated studies

Table 10.6 summarizes the studies reviewed above. Root ages are approximate and are given only where we are confident of the value reported by the authors; intervals and exact point estimates should be taken from the original publications. Figure 10.4 displays the same root ages.

```
studies <- data.frame(
  family = c("Indo-European", "Indo-European", "Indo-European",
             "Indo-European", "Austronesian", "Bantu", "Bantu",
             "Sino-Tibetan", "Sino-Tibetan",
             "Dravidian", "Uralic", "Turkic", "Transeurasian (contested)",
             "Semitic", "Pama-Nyungan", "Japonic", "Uto-Aztecan",
             "Austronesian (Philippines)", "Kra-Dai", "Japonic"),
  study = c("Gray and Atkinson (2003)", "Bouckaert et al. (2012)",
            "Chang et al. (2015)", "Heggarty et al. (2023)",
            "Gray, Drummond and Greenhill (2009)", "Grollemund et al. (2015)",
            "Koile et al. (2022)", "Sagart et al. (2019)",
            "Zhang et al. (2019)",
            "Kolipakam et al. (2018)", "Honkola et al. (2013)",
            "Savelyev and Robbeets (2020)", "Robbeets et al. (2021)",
            "Kitchen et al. (2009)", "Bouckaert, Bowern and Atkinson (2018)",
            "Lee and Hasegawa (2011)", "Greenhill et al. (2023)",
            "King et al. (2024)", "Tao et al. (2023)",
            "Huisman, McLean and Wu (2025)"),
  data = c("87 languages, 2,449 cognate sets (Dyen et al. data)",
           "103 ancient and modern languages, cognates",
           "Revised IELex cognates, ancestry constraints",
           "IE-CoR, about 160 languages, 170 meanings, sampled ancestors",
           "ABVD, 400 languages", "Over 400 languages, basic vocabulary",
           "Bantu basic vocabulary, phylogeography",
           "50 languages, root-meaning cognates",
           "109 languages, lexical cognates",
           "20 languages, new basic vocabulary data", "Uralic basic vocabulary",
           "Turkic basic vocabulary", "98 languages, putative cognates",
           "25 languages incl. ancient", "Over 300 languages, phylogeography",
           "59 languages and dialects", "Uto-Aztecan basic vocabulary",
           "202 doculects (ABVD), 7,565 cognate sets",
           "100 languages, 646 cognate sets",
           "81 lects, 256 concepts, lexical and phonotactic"),
  root_age = c("about 7,800-9,800", "about 8,000-9,500", "about 6,000",
               "about 8,100", "about 5,200", "expansion from about 5,000",
               "about 5,100", "about 7,200", "about 5,900",
               "about 4,500", "not reported here", "about 2,100",
               "about 9,000", "about 5,750", "roughly 6,000", "about 2,200",
               "roughly 4,000", "not estimated (root prior 5,200)",
               "about 4,000", "about 2,400 (400-500 BCE)"),
  interpretation = c("Anatolian farming",
    "Anatolian farming (with phylogeography)",
    "Steppe", "Hybrid: south of the Caucasus, later steppe branch",
    "Taiwan; pulses and pauses",
    "Savannah corridors through the rainforest",
    "Early route through the forest, about 4,400 BP",
    "Northern China, millet farmers",
    "Northern China, Yangshao", "South Asia",
```

```
    "Splits linked to climate and culture", "Bulgharic split first",
    "West Liao River millet farmers", "Levant; Ethiosemitic from Arabia",
    "Gulf of Carpentaria; hunter-gatherers",
    "Yayoi farmers from Korean peninsula", "Northern origin",
    "Philippine split about 3,400 BP; south-to-north pattern",
    "Guangxi-Guangdong; dispersal to Mainland Southeast Asia",
    "Yayoi; Proto-Ryukyuan about ninth century CE")
)
knitr::kable(studies, col.names = c("Family", "Study", "Data",
  "Root age (years BP)", "Main interpretation"),
  caption = "Summary of dated phylogenetic studies of language families reviewed in
  this chapter. Root ages are approximate; consult the original publications for
  exact estimates and credible intervals.")
```

Table 10.6: Summary of dated phylogenetic studies of language families reviewed in this chapter. Root ages are approximate; consult the original publications for exact estimates and credible intervals.

Family	Study	Data	Root age (years BP)	Main interpretation
Indo-European	Gray and Atkinson (2003)	87 languages, 2,449 cognate sets (Dyen et al. data)	about 7,800-9,800	Anatolian farming
Indo-European	Bouckaert et al. (2012)	103 ancient and modern languages, cognates	about 8,000-9,500	Anatolian farming (with phylogeography)
Indo-European	Chang et al. (2015)	Revised IELex cognates, ancestry constraints	about 6,000	Steppe
Indo-European	Heggarty et al. (2023)	IE-CoR, about 160 languages, 170 meanings, sampled ancestors	about 8,100	Hybrid: south of the Caucasus, later steppe branch
Austronesian	Gray, Drummond and Greenhill (2009)	ABVD, 400 languages	about 5,200	Taiwan; pulses and pauses
Bantu	Grollemund et al. (2015)	Over 400 languages, basic vocabulary	expansion from about 5,000	Savannah corridors through the rainforest
Bantu	Koile et al. (2022)	Bantu basic vocabulary, phylogeography	about 5,100	Early route through the forest, about 4,400 BP
Sino-Tibetan	Sagart et al. (2019)	50 languages, root-meaning cognates	about 7,200	Northern China, millet farmers
Sino-Tibetan	Zhang et al. (2019)	109 languages, lexical cognates	about 5,900	Northern China, Yangshao
Dravidian	Kolipakam et al. (2018)	20 languages, new basic vocabulary data	about 4,500	South Asia

Family	Study	Data	Root age (years BP)	Main interpretation
Uralic	Honkola et al. (2013)	Uralic basic vocabulary	not reported here	Splits linked to climate and culture
Turkic	Savelyev and Robbeets (2020)	Turkic basic vocabulary	about 2,100	Bulgharic split first
Transeurasian (contested)	Robbeets et al. (2021)	98 languages, putative cognates	about 9,000	West Liao River millet farmers
Semitic	Kitchen et al. (2009)	25 languages incl. ancient	about 5,750	Levant; Ethiosemitic from Arabia
Pama-Nyungan	Bouckaert, Bowern and Atkinson (2018)	Over 300 languages, phylogeography	roughly 6,000	Gulf of Carpentaria; hunter-gatherers
Japonic	Lee and Hasegawa (2011)	59 languages and dialects	about 2,200	Yayoi farmers from Korean peninsula
Uto-Aztecan	Greenhill et al. (2023)	Uto-Aztecan basic vocabulary	roughly 4,000	Northern origin
Austronesian (Philippines)	King et al. (2024)	202 doculects (ABVD), 7,565 cognate sets	not estimated (root prior 5,200)	Philippine split about 3,400 BP; south-to-north pattern
Kra-Dai	Tao et al. (2023)	100 languages, 646 cognate sets	about 4,000	Guangxi-Guangdong; dispersal to Mainland Southeast Asia
Japonic	Huisman, McLean and Wu (2025)	81 lects, 256 concepts, lexical and phonotactic	about 2,400 (400-500 BCE)	Yayoi; Proto-Ryukyuan about ninth century CE

```
plot_df <- data.frame(
  label = c("Indo-European: Gray & Atkinson 2003",
            "Indo-European: Bouckaert et al. 2012",
            "Indo-European: Chang et al. 2015",
            "Indo-European: Heggarty et al. 2023",
            "Austronesian: Gray et al. 2009", "Bantu: Grollemund et al. 2015",
            "Bantu: Koile et al. 2022", "Sino-Tibetan: Sagart et al. 2019",
            "Sino-Tibetan: Zhang et al. 2019",
            "Dravidian: Kolipakam et al. 2018",
            "Turkic: Savelyev & Robbeets 2020",
            "Transeurasian: Robbeets et al. 2021",
            "Semitic: Kitchen et al. 2009",
            "Pama-Nyungan: Bouckaert et al. 2018",
            "Japonic: Lee & Hasegawa 2011",
            "Uto-Aztecan: Greenhill et al. 2023",
            "Kra-Dai: Tao et al. 2023",
            "Japonic: Huisman et al. 2025"),
  mid = c(8.8, 8.75, 6.0, 8.1, 5.2, 5.0, 5.1, 7.2, 5.9, 4.5, 2.1, 9.0, 5.75,
          6.0, 2.2, 4.0, 4.04, 2.4),
  lo = c(7.8, 8.0, 5.9, NA, NA, NA, 4.64, NA, NA, NA, NA, NA, NA, NA, NA, NA,
         2.74, NA),
```

```
  hi = c(9.8, 9.5, 6.5, NA, NA, NA, 5.77, NA, NA, NA, NA, NA, NA, NA, NA, NA,
         5.55, NA)
)
plot_df <- plot_df[order(plot_df$mid), ]
y <- seq_len(nrow(plot_df))
op <- par(mar = c(4, 17, 1, 1))
plot(plot_df$mid, y, xlim = c(0, 10.5), yaxt = "n", ylab = "", las = 1,
     xlab = "Approximate root age (thousands of years BP)", type = "n")
abline(v = seq(0, 10, 2), col = "grey90")
has_range <- !is.na(plot_df$lo)
segments(plot_df$lo[has_range], y[has_range],
         plot_df$hi[has_range], y[has_range], lwd = 5, col = "grey60")
points(plot_df$mid[!has_range], y[!has_range], pch = 16)
axis(2, at = y, labels = plot_df$label, las = 1, cex.axis = 0.75)
```

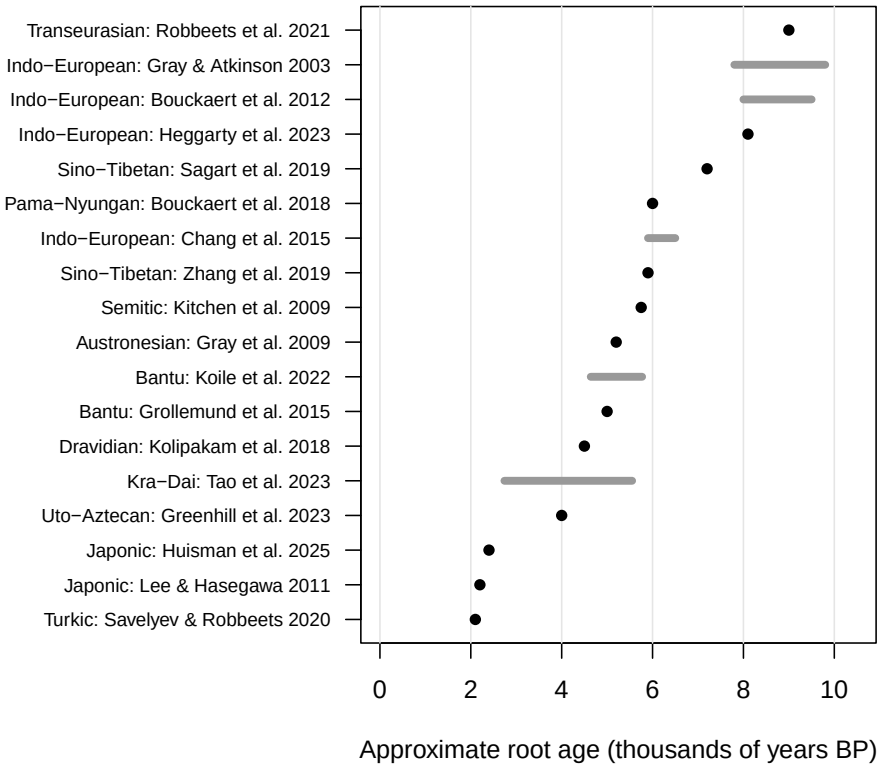


Figure 10.4: Approximate root-age estimates from the dated phylogenies in the summary table. Bars show ranges reported by the authors where these are given as ranges in the text; points show approximate central estimates. Studies whose root age we report only qualitatively are omitted.

```
par(op)
```

10.11 What the dates have in common

Looking across families, several patterns stand out, though each comes with caveats.

Most dated families are younger than about ten millennia. This partly reflects the method's limits. Cognate-based methods require cognates that can be identified, and beyond roughly ten millennia, very few basic vocabulary cognates survive in a recognizable form (Chapter 9). It also reflects selection: the families that have been dated are those with well-established cognate data, which tends to mean families of moderate depth.

Many root ages fall in the Holocene and coincide roughly with the spread of agriculture. Indo-European (under the Anatolian interpretation), Austronesian, Bantu, Sino-Tibetan, Japonic and, if it exists, Transeurasian have all been linked to farming expansions, in line with the *farming/language dispersal hypothesis* (Bellwood, 2005; Diamond and Bellwood, 2003). But the counterexamples are important. Pama-Nyungan spread among hunter-gatherers, and the steppe interpretation of Indo-European links it to pastoralism rather than to the first farmers. Uto-Aztecan, on the analysis of Greenhill et al. (2023), also does not fit a farming-driven origin. The coincidence of dates with farming expansions is, moreover, partly a matter of study design: where archaeological dates of farming dispersals have served as calibrations, agreement with the farming hypothesis is built in rather than discovered (Section 10.8).

Expansions proceed in pulses and pauses. The Austronesian pattern of long pauses followed by rapid pulses has been influential, and similar questions can be asked of Bantu and Pama-Nyungan. The tempo of expansion is a question that dated phylogenies are especially well placed to answer, and one that cannot be addressed with root ages alone.

The main disagreements concern calibrations and data, not software. The Indo-European debate has turned on the treatment of ancient languages, the reliability of cognate judgments, and the interpretation of dates in relation to archaeology and genetics. The two Sino-Tibetan studies differ in data and coding. The Transeurasian debate concerns whether the cognates are real. These are problems of linguistic evidence, and they show that dating methods cannot be separated from the comparative method on which the data rest.

Few published dates have been independently validated in the strict sense. Calibration cross-validation, prior-only runs reported alongside posteriors, and comparisons with evidence held out of the analysis remain the exception in linguistic dating. Until they become routine, the dates in Table 10.6 are best read as estimates under stated assumptions, whose robustness is still being established.

Molecular dating offers a model for how linguistic dating could become more systematic. Large-scale molecular dating projects now publish their calibrations in standardized form, with explicit justification of minimum and maximum ages, the specimens and stratigraphic evidence on which they rest, and the phylogenetic placement of each calibrating fossil (Parham et al., 2012). They routinely sample from the prior to report effective priors (Warnock et al., 2012), use infinite-sites plots to judge whether calibrations or data limit precision (dos Reis et al., 2016), perform cross-validation of calibrations (Near et al., 2005), and compare node dating, tip dating and FBD analyses of the same data. Linguistics could adopt each of these practices immediately. A shared, versioned database of linguistic calibrations, linked to Glottolog and documenting the historical evidence for each, would make dating analyses more transparent and more comparable across studies.

- Calibrations fix the absolute time scale and also determine the precision of all dates. Even unlimited data cannot reduce date uncertainty below the floor set by calibration uncertainty.
- Absolute ages and rates are not identified by lexical data; they are borrowed from calibrations through the clock model. Topology and branch lengths are identified by the data, but they summarize lexical similarity and are not validated by agreement with traditional

subgroupings.

- Linguistic calibrations come from documented history, attestation of ancient languages, archaeological correlates and datable loans. Each provides minimum ages more securely than maximum ages, and each has characteristic risks, including circularity.
- Calibration densities (uniform, soft-bounded, offset lognormal or exponential, normal) express different assumptions. Their interactions with each other and with the tree prior produce effective priors that may differ from those specified. Always sample from the prior.
- Ancient languages can serve as dated tips. Root-to-tip regression and date-randomization tests check temporal signal. Missing data in ancient languages must be handled explicitly.
- An attested language may be an ancestor, a sister, a relative of the real ancestor, or a witness to a dialect network. Node calibration, dated tips, sampled-ancestor priors and ancestry constraints encode different assumptions, and ancestry is poorly identified for fragmentarily attested languages, so dates should be reported under more than one treatment.
- Validation of dates has five distinct senses; only calibration cross-validation and comparison with evidence held out of the analysis count as independent historical validation.
- Dated phylogenies exist for Indo-European, Austronesian, Bantu, Sino-Tibetan, Dravidian, Uralic, Turkic, Semitic, Pama-Nyungan, Japonic, Kra-Dai, Uto-Aztecan and several South American families. Root ages range from about two millennia (Turkic, Japonic) to roughly eight or nine millennia (Indo-European under several analyses, and the contested Transeurasian).
- Indo-European dates remain disputed. The disagreements concern calibrations, ancient languages, data and interpretation more than methods.
- Many large families remain undated with modern methods, including Austroasiatic, Mayan, Trans-New Guinea, Afroasiatic and most of Niger-Congo.

10.12 Exercises

Conceptual

1. Using the infinite-sites argument, explain why doubling the number of cognate sets in a dating analysis may leave the width of the root-age interval almost unchanged. What kinds of new evidence would narrow it?
2. The earliest text in language X dates from 1200 BCE. Explain why a normal calibration centered on 3,200 BP for the clade containing X is inappropriate, and propose and justify an alternative density.
3. Distinguish the specified and the effective prior on a node age. How would you obtain the effective prior in BEAST 2 or RevBayes, and what would you conclude if the posterior for the root were almost identical to it?
4. Classical Latin, the spoken Latin of the late Empire, and Proto-Romance are three different objects. Which, if any, should be treated as a sampled ancestor of the Romance languages in a dating analysis, and how would treating Classical Latin as an ancestor rather than as a sister affect the estimated split times within Romance?
5. A dated tree's root age agrees with the archaeological date of a farming expansion. Under what conditions is this agreement independent support for a farming-driven spread, and under what conditions is it circular?
6. Why does the recovery of traditional subgroups by a dated phylogeny not validate its dates? What would?

Computational

1. Using the functions of Section C.4.2 (`soft_unif()`, `offset_lnorm_par()` and `offset_exp_rate()`), construct an offset-lognormal, an offset-exponential and a soft-bounded uniform calibration for a node with minimum age 2.5 millennia and 97.5 percent upper bound 4 millennia. Plot the three densities and compute the probability each assigns to ages above 5 millennia.
2. Extend the rejection sampler of Section 10.3.2 to three nested nodes: a root with a normal calibration (mean 6, standard deviation 0.5), a descendant with a uniform calibration between 4 and 7, and a node below that with a uniform calibration between 3.5 and 5 millennia. Report the effective means and compare the marginal of the middle node with the two-node case.
3. Rerun the heterochronous simulation of Section 10.5.2 (code in Section C.4.4) with rate variation `sdlog = 0.6` and with only three ancient languages. How do the estimated rate, root age and date-randomization *p*-value change? Which kind of validation does each version represent?
4. Apply `loo_calib()` (Section 10.8) to a scheme in which the *first* internal calibration is misattributed (multiply its window by 1.4), and repeat with `model = "clock"`. Is the misattributed calibration always identified? What happens to the predictions for the correctly calibrated nodes?
5. Using `max_side_branch()` from Section C.4.6, find the number of attested meanings an ancient language needs before a side branch of 200 years can be excluded, for retention rates of 0.80 and 0.90 per millennium.
6. With RevBayes and the simulated twelve-language data of Appendix H (true root age 6 millennia), run `revbayes/analyses/time/01_strict_clock_calibrations.Rev` and `02_ucln_relaxed_clock.Rev` as distributed, then again with the uniform root prior `dnUniform(0.5, 30.0)` replaced by a truncated normal, `dnNormal(mean = 6.5, sd = 2.0, min = 5.0, max = 10.0)`, and once more without the data, by removing the `dnPhyloCTMC` node and its `clamp()` as script `14_effective_prior_calibrations.Rev` does, which samples the effective prior. Tabulate the prior and posterior root ages and the ages of the two calibrated clades (`age_C1`, `age_C2`) for all runs. Which clock model is more sensitive to the root prior, and why? Finally, run `09_clock_model_comparison_ss.Rev` for both models on both data sets (`--args strict clock_ucln`, and so on) and say whether the Bayes factors pick the generating clock model.

Research

1. You are planning a dated analysis of Semitic that includes Akkadian, Ugaritic, Biblical Hebrew, Classical Arabic and Ge'ez together with modern languages. For each ancient language, decide whether to use it as a node calibration, a dated tip on a side branch, a possible sampled ancestor, or a constrained ancestor; justify each decision philologically; and design the sensitivity analyses you would report.
2. Design a validation plan for a dated Bantu phylogeny in which archaeology is the only source of calibrations. Which archaeological evidence would you use for calibration and which would you hold out for validation, and how would you prevent the hypothesis under test (for example, the route of the expansion) from influencing both?
3. Draft a specification for a shared database of linguistic calibrations. Which fields would each record need (clade, evidence, minimum and maximum ages, density and its justification,

sources, Glottocodes), how would uncertainty about clade membership be recorded, and how would the database support calibration cross-validation across studies?

10.13 Further reading

- Ho and Phillips (2009): A clear and practical discussion of calibration densities and how to choose them, written for biologists but directly applicable to languages.
- Heled and Drummond (2012): The technical account of why calibration densities and tree priors interact, and how to construct priors with the intended marginals.
- Heath et al. (2014) and Gavryushkina et al. (2014): The fossilized birth-death process and its sampled-ancestor implementation, the natural framework for ancient languages.
- Brown and Smith (2018): A thoughtful essay on how to interpret divergence-time estimates and their uncertainty, with lessons that apply to linguistics.
- Warnock and Wright (2020): A short book on the substitution, clock and tree components of Bayesian dating, with an overview of the most widely used models for each; a good bridge between this chapter and the molecular literature.
- Gray and Atkinson (2003): The paper that launched the modern debate about dating Indo-European with phylogenetic methods. Short, with an informative supplement.
- Chang et al. (2015): The most detailed argument for a steppe-compatible Indo-European root, with an extended discussion of ancestry constraints and advergence.
- Heggarty et al. (2023): The most comprehensive phylogenetic analysis of Indo-European so far, with sampled ancestors and a new dataset, and the “hybrid” hypothesis.
- Gray et al. (2009): The Austronesian study of pulses and pauses, showing what a dated tree can reveal beyond a root age.
- King et al. (2024): A Bayesian analysis of the Philippine languages that shows how to scrutinize unstable clades cognate by cognate, and how pervasive diffusion can shape a lexical tree.
- Dunn (2025) and Heggarty and Powell (2025): Two recent handbook chapters on linguistic dating and on Bayesian phylogenetics for prehistory, written for archaeologists; the second is frank about past failings of the approach.
- Kassian and Starostin (2025): A critique of the IE-CoR analysis from the lexicostatistical tradition, focusing on cognate coding and list length.
- Bouckaert et al. (2018): A dated phylogeography of Pama-Nyungan, a family that spread without agriculture.
- Pereltsvaig and Lewis (2015): A book-length critique of phylogenetic dating of Indo-European from a linguistic perspective; read together with the replies to it.

Chapter 11

Phylogeography and Homelands

Where was Proto-Indo-European spoken? Did the ancestors of the Bantu languages cut straight through the Central African rainforest, or skirt around it? Where did the Pama-Nyungan languages, which today cover most of Australia, begin? Questions about homelands (in the German tradition, the *Urheimat*) and routes of dispersal are among the oldest in historical linguistics and among the most consequential for prehistory, because they connect language families to archaeological cultures, to the spread of farming and to population movements visible in ancient DNA. *Phylogeography*, the joint modeling of genealogy and geography, turns these questions into statistical inference: the location of each language is treated as a character that evolves along the branches of a dated tree. This chapter asks what such models can establish, and what they cannot. A posterior distribution over the location of a proto-language is easy to obtain; showing that the assumed process of movement describes how speech communities actually spread is much harder.

After reading this chapter you should be able to:

- explain the classical homeland arguments (linguistic paleontology, the diversity principle, migration theory) and why the Brownian homeland estimate is a phylogenetically weighted average of present-day locations;
- specify a discrete phylogeographic model (a rate matrix on regions, with BSSVS or a GLM) and a continuous one (Brownian motion or a relaxed random walk), and say what each parameter means;
- show by simulation why undirected diffusion models cannot detect a directional expansion from contemporaneous languages, and how ancient languages restore identifiability;
- distinguish parameter recovery, robustness under misspecification and empirical model adequacy in phylogeography, and carry out a simple adequacy check;
- interpret reconstructed locations and diffusion rates as statements about language lineages under a model, and judge which parameters can be read as historical quantities;
- carry a sample of trees through a phylogeographic analysis and explain why a single maximum clade credibility tree can mislead.

11.1 The homeland question

A language family is a set of languages descended from a single ancestor, the *proto-language*. That proto-language was once spoken by a real community, at a real time, in a real place. Locating that

place is not idle curiosity. If Proto-Indo-European was spoken in Anatolia around nine thousand years ago, the family's spread is plausibly linked to the spread of farming into Europe, as argued by Renfrew (1987). If it was spoken on the Pontic-Caspian steppe some six thousand years ago, the spread is plausibly linked to the mobile pastoralist cultures associated with horse riding and wheeled vehicles, as argued in different ways by Mallory (1989) and Anthony (2007). Each scenario makes different predictions about the archaeological record, about population genetics and about the vocabulary we should be able to reconstruct. The same is true for every large family: the location of the Proto-Austronesian homeland (Taiwan, on most accounts) anchors the story of the settlement of the Pacific; the location of the Bantu homeland (the Grassfields region on the border of present-day Nigeria and Cameroon, on most accounts) anchors the story of the spread of farming across sub-Saharan Africa.

Historical linguists developed three classic families of argument for homelands before phylogenetic methods existed.

Linguistic paleontology. If a word for 'beech', 'salmon', 'snow' or 'wheel' can be reconstructed for the proto-language, then its speakers presumably knew the referent, and the homeland must lie in a region and period where the referent existed. Linguistic paleontology has a long and contentious history in Indo-European studies, where the reconstructed vocabulary for wheeled vehicles ('wheel', 'axle', 'shaft', 'to convey in a vehicle') has been used to argue that Proto-Indo-European cannot predate the invention of the wheel in the fourth millennium BCE (Anthony, 2007). The argument's weakness is semantic drift and parallel borrowing: words can change meaning, and words for new technologies can spread among already-separated languages, so that a "reconstructible" word may in fact be a wanderwort that diffused after the breakup.

The diversity or "center of gravity" principle. The region in which a family shows its greatest *genealogical* diversity (the most first-order branches, or the deepest splits, packed into the smallest area) is likely to be the homeland, because that is where the family has been in place longest. Sapir (1916) articulated the principle for North American families, and Dyen (1956) formalized it as a *migration theory* that seeks the homeland requiring the fewest migrations to explain the present distribution of subgroups. On this argument, the concentration of the primary branches of Austronesian on Taiwan, and of the deepest branches of Bantu in the northwest of the Bantu area, points to those regions as homelands.

Geographical typology. Nichols (1992) contrasted *spread zones*, regions in which one family repeatedly expands and replaces others, leaving low genealogical diversity (the Eurasian steppe is her central example), with *residual zones*, mountainous or peripheral regions where diversity accumulates (the Caucasus is the classic case). Spread zones complicate the diversity principle, because the area of greatest diversity may be a refuge rather than a point of origin (Nichols, 1997).

All three approaches are informal, and none comes with a measure of uncertainty. Phylogeographic models offer an explicit probabilistic framework in which the tree, its branch lengths in time, the process by which locations change and the uncertainty in all of these are estimated together. As we will see, though, the most widely used phylogeographic models are formalizations of the diversity principle more than alternatives to it, and they inherit some of its weaknesses.

11.2 Diversity-based homeland methods

Before turning to explicit models of movement, it is worth looking more closely at quantitative versions of the diversity principle, because they are cheap, they are widely applied, and they

provide a useful baseline against which to judge more elaborate models.

The idea has a parallel in the *age-area hypothesis* of early twentieth-century biogeography: the area where a group has lived longest is where it has had most time to diversify. In linguistics, Sapir (1916) reasoned that the region of greatest linguistic differentiation within a stock marks its historical center of gravity. Dyen (1956) turned this into a procedure. Given a classification and a map, consider each candidate homeland and count the number of migrations needed to bring every subgroup to its present location; the candidate requiring the fewest moves is preferred. The principle is essentially parsimony applied to geography, and it is a direct ancestor of the discrete parsimony reconstructions discussed below.

Wichmann et al. (2010b) proposed a fully quantitative version using lexical distances from the Automated Similarity Judgment Program (ASJP; Chapter 3). For each language in a family, they computed a diversity score that combines the lexical distances between that language and the other members of the family with the geographical distances between them, so that a language surrounded by many nearby but lexically divergent relatives receives a high score. The homeland is placed at the location of the language with the highest score. Applied to a large number of families, the method reproduces many homelands that specialists accept on other grounds, and its simplicity makes it a useful first look at families for which no phylogeographic analysis exists. That agreement should not be over-read as independent validation, however: many of the specialists' homelands were themselves argued from the distribution of subgroups, that is, from the diversity principle applied to classifications built by the comparative method, so method and benchmark share much of their evidence. The method also makes its assumptions transparent: it will place the homeland wherever divergent languages are most tightly packed, and nowhere else.

At a much larger scale, Atkinson (2011) argued that phoneme inventory size decreases with distance from Africa, as a serial founder effect would predict, and that the point of origin that best explains the global distribution of inventory sizes lies in southern or central Africa. The analysis is a diversity-based homeland method applied to all languages at once, and it was sharply criticized on statistical and conceptual grounds (Cysouw et al., 2012): phoneme inventories are not neutral markers, they change too fast to preserve a signal across tens of millennia, and the statistical evidence depends on the choice of origin model and of data source. It illustrates both the attraction and the danger of diversity methods: they always return an answer, and nothing in the method itself tells us whether the answer reflects history.

Diversity methods share three weaknesses.

1. **Replacement and spread zones.** If a family's original core has been overrun by later expansions (of the family itself, or of another family), the zone of greatest surviving diversity may lie far from the homeland. Turkic, now spoken from Siberia to the Balkans, illustrates how later movements can obscure an earlier distribution.
2. **Sampling.** Diversity scores depend on which languages are included. Well-documented regions with many described varieties will look more diverse than poorly documented ones, independently of history. The distinction between "language" and "dialect" (Chapter 15) directly changes the count.
3. **No model of time.** Diversity is not converted into age, and a region with many shallowly divergent languages can outscore a region with a few deeply divergent ones.

The phylogenetic methods developed below tackle the third weakness head-on and give a principled handle on the second. The first, replacement, is a limit on what the surviving languages can tell us, and no model applied only to present-day languages can fully overcome it.

A useful mental model for the rest of this chapter: under the simplest continuous phylogeographic model (Brownian motion on a fixed tree), the estimated homeland is a *weighted average of the present-day locations*, with weights determined by the tree. Languages on long, isolated branches (for example, the sole survivor of a deep primary branch) receive more weight than languages in large, shallow clades, whose many members largely duplicate one another's information. This is a formal version of the intuition behind the diversity principle, and it explains both why phylogeographic and diversity-based methods often agree and why both fail in the same way when a family has expanded steadily in one direction.

11.3 The phylogeographic framework

Phylogeography treats the location of each language as a character that evolves along the branches of the tree. When a lineage splits, both daughters begin at the location of their ancestor; along each branch the location changes according to a stochastic process; at the tips we observe the present (or attested) locations. Inference then asks which ancestral locations, and which parameters of the movement process, best explain the observed tip locations given the tree.

The term itself comes from biology, where Avise et al. (1987) coined "phylogeography" for the study of the geographic distribution of genealogical lineages within and among closely related species, originally using mitochondrial DNA. The model-based, Bayesian version used in linguistics descends more directly from viral phylodynamics, where the questions (where did an epidemic start, and how did it spread?) closely parallel the homeland question.

Formally, let D_L denote the linguistic data (for example a cognate matrix), D_G the geographical data (tip locations), τ the tree with branch lengths in time, θ_L the parameters of the lexical substitution and clock models, and θ_G the parameters of the geographic process. If we assume that geography and vocabulary evolve independently given the tree, the joint posterior factorizes as

$$p(\tau, \theta_L, \theta_G \mid D_L, D_G) \propto p(D_L \mid \tau, \theta_L)\, p(D_G \mid \tau, \theta_G)\, p(\tau)\, p(\theta_L)\, p(\theta_G). \tag{11.1}$$

The first factor is the familiar phylogenetic likelihood of Chapter 5, computed under a lexical model from Chapter 7; the second is the geographic likelihood, the topic of this chapter. In principle the geography also informs the tree, since a candidate tree that groups languages spoken thousands of kilometers apart will have a lower geographic likelihood. In practice the geographic data are a single character (or a pair of coordinates) against hundreds or thousands of lexical characters, so the tree is driven overwhelmingly by the lexicon.

Two analysis strategies are common. In *joint* analyses, as in Bouckaert et al. (2012), the tree, the dates and the geographic history are sampled together in a single Markov chain Monte Carlo (MCMC) run. In *two-step* or *conditional* analyses, a posterior sample of dated trees is first obtained from the lexical data, and then the geographic model is fitted on each tree in the sample, with the results averaged over trees. The second strategy is cheaper and makes it easy to compare many geographic models, at the cost of ignoring the (usually small) information that geography provides about the tree. Both approaches integrate over phylogenetic uncertainty. A third shortcut, fitting the geographic model to a single summary tree, does not, and Section 11.10 shows what it can hide.

The geographic factor $p(D_G \mid \tau, \theta_G)$ depends on how location is represented. If we divide the map into K regions and code each language by its region, location is a discrete character and the models of Chapter 5 apply with little modification. If we record coordinates, location is a

continuous character and we need a model of diffusion in the plane or on the sphere. We take these in turn.

11.4 Discrete phylogeography

11.4.1 Locations as states of a Markov chain

Suppose the region occupied by a family is divided into K areas: for Bantu, perhaps the Cameroon Grassfields, the western rainforest, the Great Lakes region and the southern savannahs. Each language is assigned to one area. We model the history of a lineage's location as a continuous-time Markov chain with K states and an instantaneous rate matrix $\mathbf{Q} = \{q_{ij}\}$, where q_{ij} for $i \neq j$ is the rate at which a lineage currently in area i moves to area j, and the diagonal elements are $q_{ii} = -\sum_{j \neq i} q_{ij}$ so that rows sum to zero. Exactly as for nucleotides or cognate states, the probability that a lineage starting in area i is in area j after time t is the (i, j) entry of

$$\mathbf{P}(t) = e^{\mathbf{Q}t}. \tag{11.2}$$

The likelihood of the observed tip areas given the tree is computed by Felsenstein's pruning algorithm (Chapter 5), exactly as for any other discrete character. The posterior probability that the root (the proto-language) was in area i is proportional to the prior probability of area i at the root times the likelihood of the tip areas given that the root was in i. The same idea extended to internal nodes yields ancestral area probabilities for every proto-language in the tree (Chapter 12 treats ancestral state reconstruction in general).

The pruning recursion written out for regions, Equation (D.1), is in Section D.1.1.

The rate matrix can be parameterized with different degrees of freedom:

- **Equal rates (ER)**: a single rate for all moves, one parameter.
- **Symmetric (SYM)**: $q_{ij} = q_{ji}$, giving $K(K-1)/2$ parameters. This is the model introduced for viral phylogeography by Lemey et al. (2009).
- **Asymmetric or all-rates-different (ARD)**: $q_{ij} \neq q_{ji}$ in general, giving $K(K-1)$ parameters. Asymmetric rates allow, for example, more movement out of a homeland than into it; an asymmetric extension of the BSSVS approach described below was introduced by Edwards et al. (2011).
- **Constrained or adjacency models**: rates between non-adjacent areas fixed at zero, so that lineages must move through intermediate areas. This is the simplest form of landscape awareness.

The number of parameters grows quadratically with K, while the data are a single character with one observation per language. With ten areas an asymmetric model has ninety rates to be estimated from, perhaps, a hundred tip states. The parameters are therefore weakly identified, and the prior matters a great deal. This motivates BSSVS.

A word on parsimony. Before likelihood models were available, ancestral areas were reconstructed by Fitch parsimony (Chapter 4), which finds the assignment of areas to internal nodes requiring the fewest moves. This is Dyen's migration theory in algorithmic form. Gray and Jordan (2000) used parsimony mapping of this sort, among other analyses, to examine whether Austronesian subgroups were arranged on the tree in the sequence predicted by a Taiwan-to-Pacific "express train". Parsimony ignores branch lengths, so it cannot tell a move that happened on a long branch

(unsurprising) from one on a short branch (surprising), and it gives no measure of uncertainty; the likelihood framework addresses both.

11.4.2 Bayesian stochastic search variable selection

Lemey et al. (2009) proposed to let the data decide which dispersal routes are needed at all. Each off-diagonal rate is written as the product of a rate $r_{ij} > 0$ and a binary indicator $\delta_{ij} \in \{0, 1\}$:

$$q_{ij} = \delta_{ij}\, r_{ij}, \qquad \sum_{i<j} \delta_{ij} - (K-1) \sim \text{Poisson}(\ln 2). \tag{11.3}$$

MCMC then samples over the indicators as well as the rates, switching routes on and off. The prior on the number of active routes is a Poisson distribution with a small mean ($\ln 2$, so that there is prior probability one half on no routes beyond the minimum) plus an offset of $K-1$, the minimum number of symmetric connections needed for every area to be reachable from every other. The posterior mean of δ_{ij} is the posterior inclusion probability of route $i \leftrightarrow j$, and the support for each route is summarized as a Bayes factor: the ratio of the posterior odds that the route is included to its prior odds. Routes with Bayes factors above about 3 are conventionally said to be "supported" and those above 10 or 100 "strongly" or "decisively" supported, following the usual Jeffreys-style scales (Chapter 6). The resulting maps of well-supported routes have become a staple of viral phylogeography. BSSVS is in effect a search over the space of graphs connecting the areas, and it answers a question that linguists rarely ask explicitly: not only where the homeland was, but which pathways of dispersal are required to explain the present distribution.

Because the prior inclusion probability of a route depends on the number of regions, the same posterior inclusion probability means different things in analyses with different K (Section 11.13.7). A "supported route" is also a route the *model* needs, given its other assumptions; it is not an attested path of migration.

The route Bayes factor, Equation (D.2), and a function that computes it from BSSVS output are in Section D.1.2.

11.4.3 Markov jumps and rewards

Ancestral area probabilities at nodes tell us where proto-languages were spoken, but they do not tell us *when* lineages moved or *how often*. Two lineages could both start and end in the same area, one having stayed put and one having left and returned. Minin and Suchard (2008) showed how to compute the posterior expected number of transitions of each type (*Markov jumps*) and the expected time spent in each state (*Markov rewards*) along each branch, conditional on the data at the tips. The same quantities can be approximated by stochastic character mapping (Section 12.6), which simulates complete histories of movement on the tree consistent with the tip data. Markov jump counts answer questions such as "how many times did Bantu lineages enter the rainforest from the savannah?" or "how much of the history of Oceanic was spent in Near Oceania before the move into Remote Oceania?" They are among the most useful and least used outputs of discrete phylogeography in linguistics.

11.4.4 Explaining dispersal: phylogeographic generalized linear models

BSSVS asks *which* routes are needed. A natural next question is *why* some routes have high rates. Lemey et al. (2014), studying the global spread of influenza H3N2, proposed modeling the log

rates as a linear function of predictors:

$$\log q_{ij} = \sum_{k=1}^{P} \beta_k \, \delta_k \, x_{ijk}, \tag{11.4}$$

where x_{ijk} is the (usually standardized, log-transformed) value of predictor k for the pair of areas (i, j) (for viruses, air-passenger flux, geographic distance, population sizes at origin and destination), β_k its coefficient, and δ_k an indicator for whether the predictor is included, again sampled by stochastic search. The posterior inclusion probability of each δ_k measures support for that predictor as a driver of dispersal, and β_k measures its effect size. The GLM replaces $K(K-1)$ free rates with P coefficients, which makes the model far better identified when P is small.

For language families the candidate predictors are obvious and interesting: great-circle distance between area centroids, elevation difference, the presence of a navigable river or a sea crossing, ecological similarity (do both areas belong to the same biome?), agricultural suitability, and archaeological measures such as the presence of a given pottery tradition. To our knowledge, phylogeographic GLMs of this kind have rarely if ever been applied to language data, though they would directly address hypotheses such as "the Bantu expansion followed habitats suitable for cereal cultivation" or "Austronesian lineages dispersed preferentially along coastlines". We return to this in the toolbox section. A coefficient from such a model measures an association between region attributes and rates of *lineage* relocation; as with any regression on observational data (Section 13.11), correlated predictors and unmodeled history limit its causal reading.

Discrete phylogeography is sensitive to how the map is cut up and to how many languages are sampled from each area. The number of tips in an area acts, in effect, as evidence that lineages spend a lot of time there, and the root area is pulled toward over-sampled areas. In viral phylogeography this bias is well documented. For languages, sampling is shaped by documentation history (which regions have dictionaries and wordlists) and by the language/dialect decisions of the data compilers, rather than by any process related to the history of the family. Always report how many languages fall in each area, try alternative area schemes, and consider subsampling to equalize area sizes as a robustness check.

11.5 Continuous phylogeography

Discretization throws away information: two languages in the same area may be a few kilometers or a thousand kilometers apart. Continuous phylogeography models the coordinates themselves.

11.5.1 Brownian diffusion

The simplest model of movement in the plane is *Brownian motion* (BM), introduced into comparative biology for continuous traits by Felsenstein (1985b) and discussed at length in Chapter 13. Let $\mathbf{x}_v = (x_v, y_v)$ be the location of node v, measured in some planar projection (for example, kilometers east and north of an origin). Under bivariate BM, the location of a child node c given its parent a, separated by a branch of length t_c, is

$$\mathbf{x}_c \mid \mathbf{x}_a \sim \mathcal{N}_2(\mathbf{x}_a, \; t_c \, \Sigma), \qquad \Sigma = \begin{pmatrix} \sigma_x^2 & \rho \, \sigma_x \sigma_y \\ \rho \, \sigma_x \sigma_y & \sigma_y^2 \end{pmatrix}. \tag{11.5}$$

Displacements along different branches are independent, their expected value is zero (no preferred direction), and their variance grows linearly with time. The matrix Σ is the *diffusion matrix*: σ_x^2 and σ_y^2 are the rates of spread in the east-west and north-south directions, and ρ captures any tendency to move along a diagonal axis (for example, along a northeast-southwest coastline). Its inverse is often called the *precision matrix*, and it is the precision matrix on which Bayesian software usually places its prior.

Because sums of independent normal displacements are normal, the tip locations have a joint multivariate normal distribution whose covariance is built from the *phylogenetic covariance matrix* $\mathbf{C}$: the entry C_{ij} is the time from the root to the most recent common ancestor of tips i and j, and C_{ii} is the root-to-tip time of tip i. Two languages that shared a long stretch of history before they split are expected to lie close together, because they share all the displacements on that stretch. The maximum-likelihood estimate of the root location, considering one coordinate at a time, is the generalized least squares (GLS) mean of the tip locations:

$$\hat{x}_{\text{root}} = \left(\mathbf{1}^{\top}\mathbf{C}^{-1}\mathbf{1}\right)^{-1}\mathbf{1}^{\top}\mathbf{C}^{-1}\mathbf{x}, \qquad \operatorname{Var}(\hat{x}_{\text{root}}) = \sigma_x^2\left(\mathbf{1}^{\top}\mathbf{C}^{-1}\mathbf{1}\right)^{-1}. \tag{11.6}$$

This is the formal statement of the remark in the note above: the estimated homeland is a weighted average of the tip locations, with weights proportional to $\mathbf{1}^{\top}\mathbf{C}^{-1}$, which down-weight clusters of closely related languages. Three consequences follow immediately. First, the homeland estimate is pulled toward the middle of the present distribution; for most trees the weights are all positive, and the estimate then lies within the convex hull of the tip locations. Second, the *uncertainty* of the root depends on the tree and on σ^2 but not on where the tips are; a family with deep, star-like structure gives a more precise root than one with a single long stem leading to a recent radiation. Third, a directional expansion, in which the family moved steadily from one edge of its range, cannot be distinguished from undirected diffusion if all languages are contemporaneous, because adding a constant drift to every lineage simply shifts all tips by the same amount, which is absorbed into the root location. We return to this identifiability problem below and illustrate it in the R section.

The joint distribution of all tip coordinates, Equation (D.3), the derivation of Equation (11.6) from it, and a numerical check that the GLS mean equals the maximum-likelihood root computed by **phytools** are in Section D.1.3.

In a Bayesian implementation such as BEAST, the internal node locations are either integrated out analytically or sampled, priors are placed on Σ^{-1} (commonly a Wishart prior) and on the root location, and the posterior distributions of the locations of all internal nodes are summarized as maps of highest posterior density (HPD) regions, one per node or per time slice.

11.5.2 The relaxed random walk

Strict BM assumes that all lineages diffuse at the same rate. Languages, like viruses, plainly do not: some lineages stay put for millennia while others move hundreds of kilometers in a few generations, as during the Austronesian settlement of Remote Oceania or the Turkic and Mongolic expansions across the steppe. Lemey et al. (2010) introduced the *relaxed random walk* (RRW), the spatial analogue of the relaxed molecular clock (Chapter 9). Each branch b receives its own rate scaler ϕ_b, drawn independently from a distribution with mean (or median) one, and the displacement along that branch is

$$\mathbf{x}_c \mid \mathbf{x}_a, \phi_b \sim \mathcal{N}_2\!\left(\mathbf{x}_a,\ \phi_b\, t_c\, \Sigma\right). \tag{11.7}$$

The choice of distribution for ϕ_b matters. A lognormal distribution gives moderately heavy-tailed displacements. If instead the *precision* scaler $1/\phi_b$ is drawn from a gamma distribution with shape and rate both equal to $\nu/2$, then, after integrating over ϕ_b, the displacement along a branch follows a bivariate t distribution with ν degrees of freedom. With $\nu = 1$ this is the *Cauchy RRW*, whose very heavy tails allow occasional long-distance jumps alongside mostly local movement. For language families with a mixture of local drift and long-distance migration, heavy-tailed RRWs are usually preferred over strict BM by marginal likelihood comparisons.

The posterior output of an RRW analysis supports several summary statistics. The *weighted diffusion coefficient* is the sum over branches of squared displacements divided by four times the sum of branch durations, an average rate of spread in, for example, km^2 per year. The *dispersal velocity* is the sum of great-circle distances covered by branches divided by their total duration (km per year). The *wavefront* is the maximum distance from the root location reached by any lineage at a given time, which traces the leading edge of an expansion. Software such as SPREAD (Bielejec et al., 2011) and SpreaD3 (Bielejec et al., 2016) animates node locations over time on a map. All three statistics are computed from reconstructed node locations and dated branches, so they inherit every assumption of the model and of the dating; Section 11.9 asks what they can mean historically.

11.5.3 Diffusion on a sphere

Both BM and the RRW, as usually implemented, treat latitude and longitude as if they were coordinates on a flat plane. At the scale of a single country this is harmless. At the scale of Indo-European, Austronesian or the world, it is not. A degree of longitude spans about 111 km at the equator but only about 56 km at 60 degrees north, so planar diffusion on unprojected coordinates exaggerates east-west movement at high latitudes, and the Pacific, which spans the 180th meridian, poses an obvious discontinuity. Bouckaert (2016) introduced a model of diffusion on the sphere, implemented in the BEAST 2 package GEO_SPHERE, in which the location at the end of a branch is drawn from a spherical distribution centered on the parent's location, with angular spread that increases with branch length and diffusion rate. The analogue of the normal distribution on the sphere is approximated by distributions such as the von Mises-Fisher distribution, and distances are great-circle distances. Spherical diffusion removes the distortions of the planar approximation and makes "whole world" phylogeography of large families coherent, at some computational cost. The R section includes a small calculation showing how badly planar distances on raw longitude and latitude can mislead at high latitudes.

11.5.4 Landscape-aware and constrained models

Speakers do not diffuse uniformly across open plains. They cross mountains slowly, if at all; they move along rivers and coasts; and over most of human history they could not cross open ocean without seafaring technology. Several modeling strategies have been used to incorporate landscape.

- **Tip and node constraints.** Tip locations can be given as polygons (the area where a language is spoken) rather than points, with the precise location sampled within the polygon during MCMC, as in Bouckaert et al. (2012). Internal nodes can be constrained to land.
- **Heterogeneous rates by terrain.** Bouckaert et al. (2012) also fitted a *landscape-aware* variant in which movement across water was penalized relative to movement over land, so that

the walk could cross seas only at a reduced rate. Rate heterogeneity over the landscape is the continuous-space counterpart of an adjacency-constrained discrete model.

- **Post hoc landscape tests.** In viral phylogeography, the SERAPHIM approach (Dellicour et al., 2016) takes the posterior node locations from an RRW analysis and tests whether branch durations are better explained by distances computed on environmental "cost" rasters (elevation, forest cover, rivers) than by straight-line distances. This is a natural tool for the Bantu rainforest debate and, to our knowledge, has rarely if ever been used on linguistic data.

Landscape awareness also helps with a problem specific to continuous models: an unconstrained random walk happily places proto-languages in the middle of the Black Sea or the Coral Sea. Such reconstructions are usually harmless artifacts of averaging (the posterior mean of a bimodal distribution with modes on either shore can fall in the water), but they signal that the summary statistic, not only the model, needs care. Reporting HPD regions rather than point estimates is the minimal remedy.

11.5.5 Trend, ancient languages and identifiability

The third consequence of Equation (11.6) deserves emphasis. Suppose a family began at the western edge of its present range and expanded steadily eastward, as Bantu did (on most accounts) from the Grassfields toward East and southern Africa, or as Austronesian did from Taiwan into the Pacific. A Brownian model with a *trend* (or drift) term would write

$$\mathbf{x}_c \mid \mathbf{x}_a \sim \mathcal{N}_2(\mathbf{x}_a + \boldsymbol{\mu}\, t_c,\; t_c\, \Sigma)\,. \tag{11.8}$$

If every language is observed at the present, all root-to-tip times are equal to the root age T, and the expected location of every tip is $\mathbf{x}_{\text{root}} + \boldsymbol{\mu} T$. Only this sum is identifiable: the data cannot distinguish a western homeland with eastward drift from a central homeland without drift. Undirected models will then place the homeland near the (weighted) center of the present distribution, and their credible intervals will be far too narrow to contain the true homeland.

Two kinds of information break the deadlock. One is *non-contemporaneous tips*: historically attested ancient languages (Hittite, Tocharian, Vedic Sanskrit, Old Irish; Old Javanese; Ge'ez, Akkadian) placed at their attested locations and dates provide observations at different times and therefore information about $\boldsymbol{\mu}$, exactly as fossils do for trait trends in biology. With tips of different heights h_i, the expected location of tip i is $x_{\text{root}} + \mu h_i$, and x_{root} and μ can be estimated jointly by GLS regression of tip location on tip height. The other is an *informative prior* on the root location or on the direction of movement, derived for instance from archaeology; this solves the identifiability problem by assumption rather than by data and must be reported as such. The R section of this chapter demonstrates both the failure of undirected models under a trend and the way ancient tips restore identifiability.

A homeland estimated under an undirected diffusion model from contemporary languages alone is not evidence *against* a peripheral homeland. If a family expanded directionally, the model cannot see it, and the reconstructed root will sit near the middle of the range with misleadingly tight credible regions. Before comparing homeland hypotheses with Bayes factors, ask whether the model could, even in principle, have favored a peripheral homeland given these data.

11.6 Linguistic applications

Phylogeographic methods arrived in linguistics with a single high-profile paper on Indo-European and have since been applied to most of the large, well-documented families. This section reviews the main studies family by family, emphasizing what was modeled, what was found and what remains contested; Table 11.1 summarizes them. Dates of the corresponding trees are discussed in Chapter 10.

11.6.1 Indo-European

Bouckaert et al. (2012) combined a cognate dataset for about a hundred ancient and contemporary Indo-European languages with a geographic model in a joint Bayesian analysis in BEAST. Each language was represented by a polygon covering the area where it is (or was) spoken, and locations diffused along the branches of the dated tree under a relaxed random walk; a landscape-aware variant penalized movement across water. The root of the tree was dated to roughly 8,000 to 9,500 years ago, and the posterior distribution of the root location was centered on Anatolia. To compare the two dominant homeland hypotheses, the authors computed the posterior probability that the root lay in an Anatolian region versus a Pontic-Caspian steppe region and reported decisive Bayes-factor support for Anatolia. The study introduced phylogeography to a wide linguistic audience and was immediately controversial. Critics argued that the tree contained subgroupings and dates at odds with the comparative method, that the diffusion model could not represent the kind of directional, punctuated migrations that the steppe hypothesis posits, and that the maps themselves contained implausible reconstructions (Pereltsvaig and Lewis, 2015). Subsequent work changed the debate. Chang et al. (2015) constrained known ancestor-descendant relationships (for example, Latin as ancestral to the Romance languages) and obtained dates more consistent with the steppe hypothesis, while ancient DNA showed a massive population movement from the steppe into central Europe around 5,000 years ago (Haak et al., 2015). Heggarty et al. (2023), with a new, carefully curated cognate dataset and sampled ancestors, obtained a root age around 8,000 years and proposed a “hybrid” scenario with an initial homeland south of the Caucasus and a later secondary spread of some branches via the steppe. Phylogeographic modeling, by itself, did not settle the question; it became one strand in an argument now dominated by archaeogenetics.

The Indo-European case illustrates three general lessons. First, the geographic inference cannot be better than the tree and the dates underneath it: if the root is old, an Anatolian homeland is easier to accommodate, and vice versa. Second, because the RRW is undirected, a reconstructed central homeland is the default outcome, and the analysis has little power to detect a homeland at the edge of the range (Section 11.5.5). Third, the Bayes factor comparing two homeland regions is a ratio of posterior to prior odds for the root falling in each region, and it depends on how the regions are drawn and how large they are. Such Bayes factors are informative only to the extent that the model could have discriminated between the hypotheses. The wider controversy is reviewed in Section 15.9.

11.6.2 Pama-Nyungan

Pama-Nyungan covers about seven-eighths of the Australian continent and includes roughly three hundred languages, a large proportion of them with historical data collected only in the last century and a half. Earlier work by Bowern and Atkinson (2012) established a Bayesian lexical phylogeny for the family. Bouckaert et al. (2018) then combined an expanded cognate dataset

with continuous phylogeographic models, including spherical diffusion and models in which rates of movement could differ across the landscape. Their reconstruction placed the origin of Pama-Nyungan in the Gulf Plains region of northern Australia, roughly six thousand years ago, and traced a subsequent spread across the continent. The study is important for two reasons beyond its conclusions. Pama-Nyungan is a family whose expansion is not plausibly tied to agriculture, so it tests whether large-scale language spreads require a farming engine. And because the Australian landscape offers sharp ecological contrasts (deserts, tropical north, temperate south-east), it is a natural laboratory for landscape-aware diffusion models.

11.6.3 Bantu

The Bantu expansion, which carried several hundred closely related languages from the area of present-day Cameroon across most of sub-Saharan Africa over the last few thousand years, is the classic case of a farming-related language dispersal and the most intensively studied with phylogeographic methods. A central question is whether the early expansion moved *through* the Central African rainforest or *around* it, through savannah corridors to the east and south.

Currie et al. (2013) reconstructed the spread of Bantu languages using a phylogeny of Bantu lexical data and phylogeographic models of ancestral locations, comparing scenarios in which early Bantu lineages moved south through the rainforest with ones in which they moved east along its northern margin before turning south. Their results favored an early movement south through the western part of the rainforest region (the so-called "late split" scenario, in which the eastern Bantu languages descend from a lineage that first moved south) over an early east-west divergence along the northern margin. Grollemund et al. (2015), with a much larger lexical dataset covering several hundred Bantu languages, argued that habitat shaped both the route and the pace of the expansion: their analyses suggested that Bantu speakers tracked suitable habitat, following a savannah corridor that opened in the rainforest during a period of climatic change, and that rates of dispersal changed when populations entered new habitats. Koile et al. (2022) revisited the question with continuous phylogeographic models in BEAST and concluded, as their title states, that the evidence supports a route through the rainforest rather than one dependent on a late savannah corridor, with early movement through the forest predating the climatic opening. The conflicting conclusions reflect differences in language sampling, tree inference, diffusion models and the way habitat history enters the analysis, and the debate is continuing, in close dialogue with archaeology and paleoclimatology (Bostoen et al., 2015).

The Bantu debate shows how phylogeography interacts with external evidence. A diffusion model on its own reconstructs ancestral locations; to test whether habitat mattered, one needs either a model in which dispersal rates depend on habitat (a landscape-aware or GLM-type model) or a post hoc test of reconstructed movements against reconstructed environments. Both approaches are well developed in viral phylogeography and only beginning to be used for languages. The debate also shows that a route reconstructed through the forest is a statement about where the model places the ancestors of today's languages; whether speakers of Bantu languages made the journey, or the languages spread partly through shift, is a further question (Section 11.9).

11.6.4 Austronesian

Austronesian, with over a thousand languages spread from Madagascar to Rapa Nui, is the family in which phylogenetic thinking about homelands and dispersal sequences has the longest history. Gray and Jordan (2000) used parsimony on a lexical tree to test whether the branching order of

Austronesian matches an "express train" spread from Taiwan through Island Southeast Asia into the Pacific, and found it broadly consistent. Gray et al. (2009) inferred a dated Bayesian tree of 400 Austronesian languages, rooted in Taiwan at roughly 5,200 years ago, and showed that the sequence of splits implies a series of *pulses and pauses*: rapid expansions after the colonization of the Philippines and after the move into Remote Oceania, separated by long standstills, for example before the settlement of Polynesia. Although this study did not fit an explicit continuous diffusion model, it is phylogeographic in substance: the geography of the tips was used to interpret the topology and branch lengths as a settlement sequence. Austronesian is also the family where a continuous random walk is least appropriate, because most movement consisted of long sea crossings between islands. Discrete models on island groups, heavy-tailed RRWs and explicitly island-structured models are natural candidates; DEC-type biogeographic models with founder-event speciation (see the toolbox below) were designed for exactly this kind of island system.

Two recent studies examine the first stages of the spread beyond Taiwan at a finer scale. King et al. (2024) found that the languages of northern Sulawesi branch first among the Philippine languages and that the Batanic languages nest within northern Luzon, a topology that, read geographically, suggests a south-to-north rather than a north-to-south pattern of dispersal within the Philippines; the pervasive geographical signal in their trees also suggests that diffusion between neighbors has shaped the lexical data (Chapter 10). This is a geographic interpretation of a topology, not a fitted model of movement, and it depends on the stability of a few deep branches. Smith and Rama (2022) fitted a phylogeographic model to 87 Austronesian languages of Borneo and found that the inferred homelands of four of the eight major subgroups supported a riverine model, in which subgroups first developed in communities along the island's major rivers.

11.6.5 Arawakan and Tupian

South America's lowland families pose the homeland question in a landscape structured by rivers. Walker and Ribeiro (2011) applied Bayesian phylogeographic methods to a lexical dataset for the Arawak family, one of the most widely dispersed families of the Americas, reconstructing the location of the root and the dispersal history of its branches, and comparing these with archaeological hypotheses about a riverine expansion of Arawak-speaking societies. Walker et al. (2012) took a simpler, distance-based approach to the Tupi family: a neighbor-joining tree built from 40-item ASJP word lists, with each internal node placed at the average location of its descendants. The result was consistent with the traditional linguistic homeland hypothesis, which places Proto-Tupi in west-central Brazil, in and around present-day Rondônia in southwestern Amazonia, where the earliest-diverging branches of the family are found. Because that hypothesis rests on the same diversity reasoning, the agreement is a consistency check rather than independent confirmation. Both studies are early examples of linguistic phylogeography outside Eurasia and Oceania, and both work with families whose documentation is uneven, so they also illustrate how strongly the sample of languages shapes the reconstruction.

11.6.6 Other families

Several of the dated family phylogenies discussed in Chapter 10 include geographic interpretations. Kitchen et al. (2009) dated Semitic to the Early Bronze Age, roughly 5,750 years ago, and interpreted the tree, together with the location of its primary branches, as supporting an origin in the Levant and a later introduction of Ethiosemitic into the Horn of Africa from southern Arabia. For Sino-Tibetan, Sagart et al. (2019) and Zhang et al. (2019) both associated the root with north-

ern China and with Neolithic millet-farming cultures of the Yellow River basin, although they differed in their dates. Robbeets et al. (2021) combined linguistic phylogenetics, archaeology and genetics to argue that the Transeurasian languages (Japonic, Koreanic, Tungusic, Mongolic and Turkic, a grouping that many linguists do not accept as a genealogical unit) spread with millet farming from the West Liao River region. Each of these claims rests at least as much on dating and on the identity of the first branches as on a formal model of movement.

Tao et al. (2023) applied discrete phylogeography to Kra-Dai: they assigned 100 languages to five regions and reconstructed ancestral areas over 1,000 trees resampled from the posterior, comparing several dispersal models by Bayes factors. The reconstruction favored an origin in the Guangxi-Guangdong area of southern China about four millennia ago, with later dispersal toward Mainland Southeast Asia. The analysis carries tree uncertainty through the reconstruction, as recommended in Section 11.10, but with five large regions it can locate the homeland only coarsely.

Homeland arguments can also rest on comparisons of whole trees rather than on a diffusion model. Sicoli and Holton (2014) coded typological features for Na-Dene and Yeniseian languages and used network methods and Bayes factors to compare trees representing two dispersal scenarios; they concluded that the putative Dene-Yeniseian family is better explained by a radiation out of Beringia, with a back-migration into central Asia, than by a migration from Asia into North America. Yanovich (2020) re-examined the data and argued that, analyzed more carefully, they do not support Beringia as the homeland, discussing along the way several general pitfalls in drawing geographic conclusions from phylogenetic analyses. The exchange illustrates that a homeland "supported" by a model comparison is only as good as the coding of the data and the set of scenarios compared, and that a putative family whose unity is itself debated adds a further layer of uncertainty.

Because phylogeographic models assume that languages move only by descent along the tree, horizontal contact can distort them (Chapter 8). Yang et al. (2024) proposed a tree-free alternative, *language velocity field estimation*, which infers dispersal trajectories and centers from the geographic distribution of linguistic similarity without a phylogeny. They checked it on simulated and empirical data and applied it to about 700 languages of four agricultural families and groups, finding dispersal centers close to proposed Neolithic homelands. Whether such methods identify the same quantity as a root location on a tree is an open question of historical interpretation.

Table 11.1: Selected phylogeographic and geographically interpreted phylogenetic studies of language families. "RRW" = relaxed random walk; "Parsimony/topology" = geographic interpretation of tree topology without an explicit diffusion model. Results are summarized loosely; see the original papers for credible intervals and caveats.

Family	Study	Approach	Homeland or main result
Indo-European	Bouckaert et al. (2012)	Joint Bayesian, RRW with landscape variant	Anatolia, root roughly 8,000 to 9,500 years ago
Indo-European	Heggarty et al. (2023)	Bayesian dating with sampled ancestors, geographic interpretation	Hybrid: initial homeland south of the Caucasus, later steppe spread of some branches
Pama-Nyungan	Bouckaert et al. (2018)	Continuous, including spherical and landscape-aware models	Gulf Plains, roughly 6,000 years ago

Family	Study	Approach	Homeland or main result
Bantu	Currie et al. (2013)	Phylogeny plus ancestral location reconstruction	Early southward movement through western rainforest favored
Bantu	Grollemund et al. (2015)	Phylogeny plus ancestral location and habitat	Habitat shaped route and pace; savannah corridor
Bantu	Koile et al. (2022)	Continuous phylogeography in BEAST	Route through the rainforest supported
Austronesian	Gray and Jordan (2000); Gray et al. (2009)	Parsimony/topology, dated tree	Taiwan root; pulses and pauses in Pacific settlement
Arawakan	Walker and Ribeiro (2011)	Bayesian phylogeography	Root location and riverine dispersal examined
Tupian	Walker et al. (2012)	Distance-based (ASJP neighbor-joining) tree mapped onto geography	Consistent with a homeland in west-central Brazil (Rondônia)
Semitic	Kitchen et al. (2009)	Dated tree, geographic interpretation	Levant origin, Early Bronze Age
Sino-Tibetan	Sagart et al. (2019); Zhang et al. (2019)	Dated trees, geographic interpretation	Northern China, Neolithic
Kra-Dai	Tao et al. (2023)	Discrete phylogeography over posterior trees	Guangxi-Guangdong, about 4,000 years ago
Austronesian (Borneo)	Smith and Rama (2022)	Dated tree, Bayesian phylogeography	Riverine homelands for several subgroups
Austronesian (Philippines)	King et al. (2024)	Dated tree, geographic interpretation	South-to-north pattern; strong diffusion signal

11.7 How good are linguistic homeland reconstructions?

Viral phylogeographers can sometimes check their reconstructions against epidemiological records of where and when an outbreak began. Linguists rarely have such ground truth. Two strategies are available: validate on families or subfamilies whose history is known from written records (Romance from Latin, the Arabic dialects, the spread of English), and validate on simulated data where the truth is known by construction. The vocabulary of Section 6.10 keeps the second strategy honest. A simulation in which locations diffuse under Brownian motion and are then analyzed under Brownian motion tests *parameter recovery*: it shows that the method works when its assumptions hold. A simulation in which locations move in ways the model ignores (directional expansion, jumps, replacement, biased sampling) tests *robustness under misspecification*. Neither is evidence of *empirical model adequacy*, which asks whether the fitted model reproduces features of the real data (Section 11.8), and none of the three establishes *historical interpretation* (Section 11.9).

Neureiter et al. (2021) took the simulation route, and their study is a test of robustness under misspecification. They simulated language expansions under a range of scenarios, including directional migrations and expansions into empty territory, generated phylogenies and tip locations from them, and then reconstructed root locations and migration histories with the relaxed random walk models standard in linguistic phylogeography. Their central finding was that the RRW performs reasonably when movement really is undirected, but that it systematically underestimates

the distances traveled and biases the reconstructed homeland toward the center of the present-day distribution when expansions are directional. They also examined real families and argued that several published reconstructions are likely affected by this bias. The study is a direct demonstration, with realistic simulations, of the identifiability argument of Section 11.5.5, and it is required reading for anyone interpreting a reconstructed homeland.

The simulations in the R section below reproduce the core of this result with a few lines of code: when the true process includes a consistent directional trend, the 95% confidence interval of the Brownian root estimate covers the true homeland in only a small fraction of replicates. Validation against known histories has its own limits. Where a reconstruction agrees with a homeland accepted by specialists, the agreement is reassuring but not independent, because the accepted homeland usually rests on the same subgrouping (itself a product of the comparative method) and on the same diversity reasoning that the model formalizes.

11.8 Model adequacy: what the data can reveal

Model adequacy asks a different question from model comparison. Marginal likelihoods can tell us that a Cauchy RRW fits better than strict BM; they cannot tell us that either fits *well*. The general tool is the posterior predictive check of Section 6.9: simulate new tip locations from the fitted model (on trees and with parameters sampled from the posterior), compute a summary statistic for each simulated dataset and for the observed data, and ask whether the observed value is typical of the simulated ones. Useful statistics for phylogeography include:

- the distribution of standardized displacements (for example, independent contrasts of the coordinates; Section 13.5), whose tails reveal jumps that strict BM cannot produce;
- the relationship between patristic distance and geographic distance among pairs of languages, which diffusion models predict to be roughly linear in squared distance;
- the extent and shape of the present range, and the proportion of reconstructed nodes falling in water or other impossible terrain;
- for discrete models, the number of distinct regions in each major clade, or the parsimony score of the location character, compared with their predictive distributions.

Adequacy checks have a hard limit, and it is the most important point of this section. A check can only detect misspecification that leaves a trace in the observed data. A directional trend applied to contemporaneous languages leaves no such trace: it shifts every tip by the same amount (Section 11.5.5), so data simulated with and without the trend have exactly the same distribution once the root location is free. The same is true of an early branch that went extinct, or of a homeland that was later overrun by another family. Passing an adequacy check is therefore necessary but not sufficient for trusting a reconstructed homeland. The R section (Section 11.13.3) shows a simple check that detects a heavy-tailed relaxed walk in a minority of datasets of realistic size, and a directional expansion no more often than it raises false alarms. Information that can detect a trend has to come from outside the present-day locations: ancient languages at their attested places and dates, archaeology, or ancient DNA.

11.9 Interpreting reconstructions: lineages, communities and rates

What does a posterior distribution over the location of a proto-language actually say? Under the model, it is the distribution of the location of a *lineage* at the moment of a split, given the tip

locations, the tree and the assumed process. Three steps separate this from the historical statement that a speech community lived in a particular place.

First, a proto-language was spoken over an area, usually by a dialect network rather than a single village, and splits were usually gradual (Chapter 8). A point location at a node is a summary of that area under a model in which a lineage has a single position at each instant. Polygons for tips help on the data side, but the node is still a point.

Second, a posterior over an ancestral location is not evidence that the diffusion process describes how speech communities moved. Any model returns a posterior, and its width is determined largely by the model's assumptions; under Brownian motion it does not even depend on where the tips are (Equation (11.6)). Evidence that the process is adequate has to come from elsewhere: from adequacy checks (Section 11.8), from comparison with process models that represent the alternatives (directional expansion, waves of advance, long-distance jumps, replacement), and from independent evidence. A narrow credible region around a homeland is a statement about the model's confidence, not about the historical process.

Third, the model describes the movement of language lineages, which is not the same as the movement of people. A lineage can "move" because its speakers migrated, because a neighboring population shifted to it (so that the language moved and the people did not), or because a community was relocated and documented at its new home (Section 11.11). The diffusion model adds all of these together as displacement along a branch. Linking reconstructed lineage movements to population movements requires independent evidence, such as ancient DNA, which in the Indo-European case has proved decisive for exactly this reason (Haak et al., 2015).

Rates need particular care. The diffusion rate σ^2 is identifiable under the model when the tree has branch lengths in time, but it is estimated *relative to those branch lengths*. If all branch lengths are multiplied by a constant, as happens when the root age changes, the reconstructed locations are unchanged and σ^2 is divided by the same constant. Diffusion rates, weighted diffusion coefficients and dispersal velocities therefore inherit the full uncertainty of the dating (Chapter 10), and two studies that agree on the homeland can disagree on velocity simply because they date the root differently. The R section demonstrates this exactly (Section 11.13.6). Even with the dates fixed, a velocity computed from reconstructed node locations is a property of the fitted model: it averages over long periods without movement and short periods of rapid spread, and under directional expansion it is underestimated (Neureiter et al., 2021). It is not the speed at which people walked.

Table 11.2 summarizes the main quantities in the four respects distinguished in Section 6.10.

Table 11.2: Identifiability and interpretability of the main phylogeographic quantities, for a single family of a few dozen to a few hundred languages.

Quantity	Identifiable under the model?	Practically estimable?	Robust to misspecification?	Interpretable as a historical quantity?
Root location (BM or RRW)	Yes, given the tree, if there is no trend	Yes, with wide intervals	No: biased by directional expansion, sampling density and displacement	As the point location of the root lineage under the model; not the center of a population

Quantity	Identifiable under the model?	Practically estimable?	Robust to misspecification?	Interpretable as a historical quantity?
Diffusion rate σ^2	Only relative to the time scale of the tree	Yes, given dates	Sensitive to dating, displacement and sampling	Average squared lineage displacement per unit time; not a migration speed
Directional trend μ	Not from contemporaneous tips; yes with ancient tips	Only with several well-dated, well-located ancient languages	Depends on ancient locations being representative	Net direction of lineage displacement
Branch rate scalers ϕ_b (RRW)	Weakly	Mostly prior-driven for single branches	They absorb many unmodeled processes	Flags unusual branches; does not explain them
Discrete rates q_{ij} and BSSVS routes	With constraints or variable selection	Weakly for more than a few regions	Sensitive to area definitions and to sampling per area	Routes the model needs; not attested migration paths
GLM coefficients β_k	Better than free rates	Yes, with few predictors	Confounded predictors share credit	Associations between region attributes and lineage relocation

11.10 Phylogenetic uncertainty: homelands over a sample of trees

Both the joint and the two-step strategies of Section 11.3 integrate over trees. The shortcut of fitting a geographic model to a single maximum clade credibility (MCC) tree does not, and it can mislead in four ways, even when the MCC topology looks plausible.

1. **It treats uncertain structure as known.** The MCC tree is the sampled tree whose clades have the highest product of posterior frequencies (Section 2.5). Clades with modest support appear in it as if they were certain, and the order of the first splits, which matters most for the homeland, is often among the least certain parts of a language tree.
2. **Its branch lengths are summaries.** Node heights on an MCC tree are taken from one sampled tree or averaged across trees with different topologies, and may correspond to no history in the posterior (Heled and Bouckaert, 2013). Quantities that depend on branch lengths, such as diffusion rates, are then computed on a tree that no analysis actually inferred.
3. **Averaging and summarizing do not commute.** Most phylogeographic outputs are nonlinear functions of the tree. The average of a reconstruction over trees is not the reconstruction on an average tree.
4. **Intervals are too narrow.** An analysis on one tree ignores the between-tree component of uncertainty altogether.

In phylogeography these effects are uneven. Because the Brownian homeland is a weighted average of tip locations, it is fairly robust to moderate topological uncertainty. Rates and velocities are not, because they scale directly with the dates, and discrete reconstructions of ancestral regions can vary considerably from tree to tree. The R section (Section 11.13.6) shows all three patterns on a mock posterior sample. In a two-step analysis the remedy is simple: fit the geographic model to each of 100 or more trees drawn from the posterior and pool the results, either by concatenating posterior samples (Bayesian) or by combining estimates and their variances across trees (Section

13.12.1).

11.11 Sampling bias, displacement and replacement

Three features of linguistic data make phylogeography harder for languages than for viruses or most organisms.

Sampling bias. The languages included in a dataset are not a random sample of the family. They over-represent well-documented regions, national and standard languages, and the decisions of compilers about what counts as a separate language rather than a dialect (Section 15.8). In continuous phylogeography, densely sampled regions attract the root because the model treats each tip as an independent observation of the end point of a diffusion process. Kalkauskas et al. (2021) showed in simulations that biased spatial sampling distorts estimates of both diffusion rates and ancestral locations under the standard models, and Guindon and De Maio (2021) proposed ways of modeling the spatial sampling process explicitly, so that the absence of samples from a region is treated as informative only to the extent that the region was searched. Both papers concern pathogens and organisms, but their conclusions transfer directly to languages, for which the sampling process is, if anything, more uneven. A related development in viral phylogeography is the use of individual travel histories to correct for unsampled locations (Lemey et al., 2020); the linguistic analogue would be to use historical records of where a speech community lived before its documented location.

Displacement. The location assigned to a language is usually its location at the time of documentation, often the nineteenth or twentieth century. Many speech communities have moved within the historical period, through voluntary migration, forced relocation or the reorganization of colonial boundaries. Coding such languages at their current location distorts the reconstruction near the tips and, through the weighting of Equation (11.6), the root. Datasets such as Glottolog give a single point per language, which is a convenience, not a historical claim. For ancient languages, the attested location is that of the surviving texts, which may be far from the core area of the speech community (as with administrative languages of empires).

Replacement and language shift. As Section 11.9 stressed, phylogeography models the movement of *languages*: a language can spread by shift without substantial migration, while people can move without their language surviving. Replacement removes branches from the tree entirely: if the earliest offshoots of a family were spoken in what is now the homeland of another family and have since died out, the surviving branches carry no information about them. The spread of Turkic across Anatolia, the spread of Hungarian in the Carpathian basin and the spread of English are cases where historical records tell us that the present distribution says little about earlier ones. In such cases the diffusion model is misspecified in a way that no amount of data from the surviving languages can detect.

A practical checklist for linguistic phylogeography: (1) report the number of languages per region and test robustness to subsampling; (2) use polygons or uncertainty radii rather than single points when the location of a speech community is uncertain or historically mobile; (3) include ancient languages at their attested locations and dates whenever possible, because they provide the only direct information about directional trends; (4) compare undirected, heavy-tailed and landscape-aware models with marginal likelihoods (Chapter 6), and report the sensitivity of the root to the model; (5) run posterior predictive checks, and say which plausible misspecifications they could not have detected; (6) analyze a sample of trees rather than one summary tree, and report

rates together with the dates they depend on; (7) treat reconstructed movement as movement of linguistic lineages, not of populations, unless independent evidence links them.

11.12 From the biologist's toolbox

Historical biogeography and phylodynamics have developed a large repertoire of geographic models beyond the CTMC and the random walk. Most have, to our knowledge, rarely if ever been applied to languages.

DEC and BioGeoBEARS. The *dispersal-extinction-cladogenesis* (DEC) model of Ree et al. (2005) and Ree and Smith (2008) treats the geographic *range* of a lineage as a set of areas, not a single area. Along branches, ranges expand by dispersal into a new area (rate d) and contract by local extinction from one area (rate e), giving a CTMC on the $2^K - 1$ non-empty subsets of K areas. At each speciation event, a widespread ancestral range is divided between the daughters according to explicit *cladogenetic* scenarios: one daughter inherits a single area while the other inherits the rest (peripheral isolation), the range is split into two disjoint parts (vicariance), or one daughter inherits a subset nested inside the ancestor's range (sympatric subset). The R package BioGeoBEARS (Matzke, 2014) implements DEC alongside likelihood versions of other biogeographic models and adds a "+J" parameter for *founder-event speciation*, in which a daughter lineage jumps to a new area at the moment of splitting; the +J extension is powerful for island systems but has been criticized as a model whose likelihood comparisons with DEC are not straightforward (Ree and Sanmartín, 2018). For languages, ranges spanning several areas are the norm (a language spoken across a dialect continuum), and the cladogenetic scenarios map naturally onto linguistic processes: vicariance onto the splitting of a dialect continuum by an intrusive population, peripheral isolation onto the separation of an outlying dialect, and founder events onto island colonization in Austronesian. The Bayesian implementation of Landis et al. (2013) handles many areas efficiently, which matters for families spread over dozens of regions. To our knowledge DEC-type models have rarely if ever been used on language data.

GeoSSE. The *geographic state-dependent speciation and extinction* model (Goldberg et al., 2011) couples range evolution to diversification: speciation and extinction rates may differ between areas, and range evolution and diversification are estimated jointly from the tree. For languages it would address a long-standing question, whether some environments promote language diversification (mountains, tropical forests, islands), without the bias that arises when ancestral areas are reconstructed under a model that ignores area-dependent diversification. The usual caveats about state-dependent diversification models apply (Chapter 16).

Structured coalescent and MASCOT. Discrete-trait phylogeography treats location as a character mapped onto a tree generated independently of it. For densely sampled populations, the *structured coalescent* models the tree itself as arising from subpopulations connected by migration, so that where lineages coalesce depends on population sizes and migration rates. Approximations such as BASTA (De Maio et al., 2015) and the marginal approximation implemented in the BEAST 2 package MASCOT (Müller et al., 2018) make the structured coalescent practical and are less sensitive to sampling bias than discrete-trait models. For languages, the coalescent framework is most natural at the shallow end, among dialects or closely related varieties sampled from a few regions, where the "population" of speakers and the number of distinct varieties are both meaningful.

Phylodynamic GLMs and landscape phylogeography. The GLM of Equation (11.4) (Lemey

et al., 2014) and the landscape tests of SERAPHIM (Dellicour et al., 2016) together offer a direct way to ask what drove a language family's spread: distance, terrain, rivers, ecological similarity, agricultural suitability. A Bantu analysis with rainforest cover, rainfall and river networks as predictors of dispersal rates, or an Austronesian analysis with distance across open sea as a predictor, would move linguistic phylogeography from describing expansions to explaining them.

Beyond these, several other biological methods deserve mention. *Isolation-by-distance* and spatial population genetics models describe gradual differentiation in a continuum, a closer analogue to dialect geography than a branching tree, and connect to the wave-model approaches of Chapter 8. *Spatially explicit simulations* of expansion on a landscape, combined with approximate Bayesian computation, would allow linguists to fit models of directional, wave-of-advance expansion that the standard diffusion models cannot represent. And *phylogenetic models with jumps* (Lévy processes), used for trait evolution, offer an alternative to the Cauchy RRW for representing rare long-distance moves such as island colonization.

11.13 Phylogeography in R

Full Bayesian phylogeography of real language families is done in BEAST 2 (with the GEO_SPHERE and BEAST_CLASSIC packages) or BEAST X (Appendix F). The basic models, however, can be explored in R with **ape** and **phytools** (Paradis and Schliep, 2019; Revell, 2024), and doing so makes their behavior, including their failures, easy to see. All data below are simulated.

11.13.1 Simulating a language family in space

We simulate a dated tree of 30 languages with a root age of 6 (read: 6,000 years, so branch lengths are in millennia) under a pure-birth (Yule) process, and then let the location of each lineage diffuse along the branches under bivariate Brownian motion starting from a known homeland at the origin $(0, 0)$. Coordinates are in units of 100 km east and north of the homeland, and the diffusion matrix has variance 0.36 per millennium in each direction with a modest positive correlation. Figure 11.1 draws the resulting tree in geographic space.

```
library(phytools)
set.seed(11)
tree <- pbtree(n = 30, scale = 6)                 # root age 6 millennia
tree$tip.label <- paste0("L", 1:30)
Sigma <- matrix(c(0.36, 0.10, 0.10, 0.36), 2)  # diffusion matrix per kyr
XY <- sim.corrs(tree, vcv = Sigma, anc = c(0, 0), internal = TRUE)
tips <- XY[tree$tip.label, ]
colnames(tips) <- colnames(XY) <- c("east", "north")
plot(NA, xlim = range(XY[, 1]), ylim = range(XY[, 2]), asp = 1,
     xlab = "East (100 km)", ylab = "North (100 km)")
for (i in seq_len(nrow(tree$edge))) {
  a <- tree$edge[i, 1]; d <- tree$edge[i, 2]
  segments(XY[a, 1], XY[a, 2], XY[d, 1], XY[d, 2], col = "gray50")
}
points(tips, pch = 21, bg = "gray80", cex = 1.2)
points(0, 0, pch = 23, bg = "red", cex = 1.8)
```

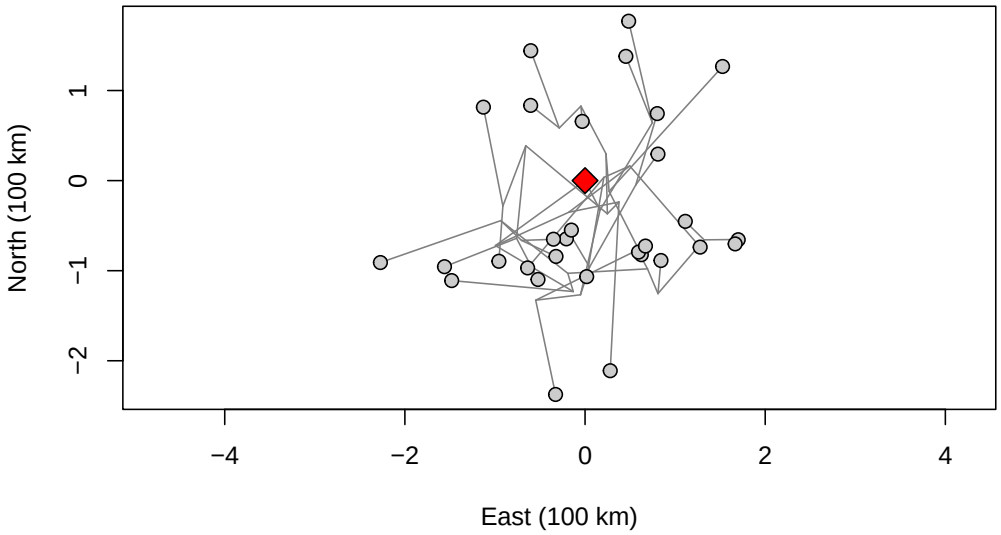


Figure 11.1: A simulated language family diffusing in space under bivariate Brownian motion. Lines are the branches of the tree drawn in geographic space (from simulated internal node locations), gray points are the 30 present-day languages, and the red diamond is the true homeland at the origin. Units are hundreds of kilometers.

11.13.2 Estimating the homeland

Under BM the maximum-likelihood ancestral locations can be computed coordinate by coordinate with `fastAnc()` (equivalently `ace(..., method = "REML")` in **ape**), which also returns 95% confidence intervals. The ML estimate of the full diffusion matrix comes from `evol.vcv()`. We compare the phylogenetic estimate of the homeland with two naive alternatives: the unweighted centroid of the present-day languages, and a simple diversity-based estimate that places the homeland at the language whose five geographically nearest neighbors are, on average, most distantly related to it (in the spirit of the diversity principle, though much cruder than Wichmann et al. (2010b)).

```
root <- as.character(Ntip(tree) + 1)
anc_e <- fastAnc(tree, tips[, "east"],  CI = TRUE)
anc_n <- fastAnc(tree, tips[, "north"], CI = TRUE)
est <- rbind(
  truth     = c(0, 0),
  phylo_ML  = c(anc_e$ace[root], anc_n$ace[root]),
  centroid  = colMeans(tips))
# crude diversity-based estimate
geo  <- as.matrix(dist(tips))
pat  <- cophenetic(tree)[rownames(tips), rownames(tips)]
divers <- sapply(rownames(tips), function(l) {
  nb <- names(sort(geo[l, ]))[2:6]       # five nearest neighbours
  mean(pat[l, nb])
})
```

```
est <- rbind(est, diversity = tips[which.max(divers), ])
colnames(est) <- c("east", "north")
round(est, 2)
```

```
#>           east north
#> truth     0.00  0.00
#> phylo_ML  0.09 -0.15
#> centroid  0.06 -0.36
#> diversity 0.67 -0.73
```

```
rbind(east = anc_e$CI95[root, ], north = anc_n$CI95[root, ])
```

```
#>         [,1]  [,2]
#> east  -1.521 1.708
#> north -1.697 1.390
```

```
round(evol.vcv(tree, tips)$R.single, 3)          # ML diffusion matrix
```

```
#>        east north
#> east  0.456 0.084
#> north 0.084 0.417
```

The phylogenetic estimate is close to the truth, its confidence intervals cover the origin, and the estimated diffusion matrix is close to the simulated one. This is a check of *parameter recovery*: the data were simulated under the model used to analyze them. Note that the confidence interval is wide, extending roughly 150 km either side of the estimate in each direction: even with a perfectly known tree and a correctly specified model, thirty languages pin down the homeland only roughly. Note also that the centroid is not much worse than the phylogenetic estimate here; under BM both are averages of the tip locations, differing only in their weights (Section D.1.3 computes the weights explicitly).

11.13.3 When the model is wrong: relaxed walks and directional expansion

How reliable is the BM root under realistic violations of its assumptions? The function below simulates one family under one of three scenarios and records whether the 95% confidence interval of the Brownian root estimate (east coordinate only) contains the true homeland, as well as the error of the phylogenetic and centroid estimates. The scenarios are: (1) BM, the model used for inference (parameter recovery); (2) a relaxed random walk, with branch-specific lognormal rate multipliers; (3) BM with a directional trend, simulating an expansion that moves on average 40 km east per millennium (`mu = 0.4`) for six millennia. Scenarios 2 and 3 test robustness under misspecification. For each replicate the function also runs a simple adequacy check. Under BM on the true tree, the standardized independent contrasts of the tip coordinates (`pic()`; Section 13.5) are independent normal variables, so their kurtosis should look like the kurtosis of 29 draws from a normal distribution; the check flags a dataset whose contrasts are more heavy-tailed than 95% of such draws.

```
kurt <- function(u) mean(u^4) / mean(u^2)^2      # about 3 for normal data
set.seed(99)
kurt_null <- replicate(4000, kurt(rnorm(29)))    # 29 contrasts for 30 tips
```

```
one_rep <- function(scenario, n = 30, T = 6, sig2 = 0.36, mu = 0.4) {
  tr <- pbtree(n = n, scale = T)
  sim_tr <- tr
  if (scenario == "RRW") {
    phi <- rlnorm(nrow(tr$edge), 0, 1)
    sim_tr$edge.length <- tr$edge.length * phi / mean(phi)
  }
  x <- fastBM(sim_tr, a = 0, sig2 = sig2,
              mu = if (scenario == "trend") mu else 0)
  f <- fastAnc(tr, x, CI = TRUE)
  k <- kurt(pic(x, tr))                          # adequacy statistic
  c(coverage       = as.numeric(f$CI95[1, 1] < 0 & f$CI95[1, 2] > 0),
    error_phylo    = abs(unname(f$ace[1])),
    error_centroid = abs(mean(x)),
    ci_width       = unname(diff(f$CI95[1, ])),
    flagged        = as.numeric(mean(kurt_null >= k) < 0.05))
}
set.seed(1)
cov_tab <- sapply(c("BM", "RRW", "trend"),
                  function(s) rowMeans(replicate(150, one_rep(s))))
knitr::kable(round(t(cov_tab), 2), caption = "Performance of the Brownian homeland
estimate (east coordinate) over 150 simulated families per scenario. Coverage is the
proportion of replicates whose 95% confidence interval contains the true homeland;
errors and widths are in units of 100 km; flagged is the proportion of replicates in
which the contrast-kurtosis adequacy check rejects Brownian motion at the 5% level.")
```

Table 11.3: Performance of the Brownian homeland estimate (east coordinate) over 150 simulated families per scenario. Coverage is the proportion of replicates whose 95% confidence interval contains the true homeland; errors and widths are in units of 100 km; flagged is the proportion of replicates in which the contrast-kurtosis adequacy check rejects Brownian motion at the 5% level.

	coverage	error_phylo	error_centroid	ci_width	flagged
BM	0.96	0.47	0.53	2.61	0.03
RRW	0.95	0.54	0.58	2.68	0.26
trend	0.09	2.41	2.46	2.64	0.03

Table 11.3 makes the point of Section 11.5.5 quantitatively. Under BM the intervals have close to their nominal 95% coverage, and the adequacy check rejects about as often as it should (its false alarm rate). Moderate rate heterogeneity among branches (the RRW scenario), with the tree held fixed, does surprisingly little damage to the root estimate; the adequacy check detects it in only about 26% of replicates, because 29 contrasts carry little information about the shape of their distribution. But under a directional trend the coverage collapses: the estimate is displaced toward the center of the present distribution by roughly μT, here about 240 km, and the confidence interval, whose width does not depend on the tip locations, is no wider than before. The phylogenetic and centroid estimates fail together, because both are averages of the tips. This is the bias found in realistic simulations by Neureiter et al. (2021). And the adequacy check is blind to it: it flags the trend scenario no more often than pure BM, because a trend shifts all contemporaneous

tips equally and leaves the contrasts untouched. The misspecification that damages the homeland most is the one the data cannot reveal (Section 11.8).

11.13.4 Ancient languages restore identifiability

Now suppose some of the languages are ancient, attested at earlier times. We mimic this by shortening ten randomly chosen terminal branches, so that those tips are "sampled" between a few centuries and a few millennia before the present, and simulate locations with the same eastward trend. With tips at different heights h_i (distance from the root), the expected location of tip i is $x_{\text{root}} + \mu h_i$, so we can estimate both the homeland and the trend by GLS regression of location on tip height, using the phylogenetic covariance matrix. We compare this with the undirected BM estimate, which ignores the trend. This is a parameter-recovery experiment for the drift model, and a robustness experiment for the undirected one.

```
gls_drift <- function(tr, x) {
  C  <- vcv(tr)[names(x), names(x)]
  h  <- diag(C)                                  # root-to-tip heights
  X  <- cbind(root = 1, trend = h)
  Ci <- solve(C)
  beta <- solve(t(X) %*% Ci %*% X, t(X) %*% Ci %*% x)
  drop(beta)
}
ancient_rep <- function(n = 30, n_anc = 10, mu = 0.4) {
  tr <- pbtree(n = n, scale = 6)
  term <- which(tr$edge[, 2] <= n)
  pick <- sample(term, n_anc)
  tr$edge.length[pick] <- tr$edge.length[pick] * runif(n_anc, 0.05, 0.5)
  x <- fastBM(tr, a = 0, sig2 = 0.36, mu = mu)
  c(bm_root = unname(fastAnc(tr, x)[1]), gls_drift(tr, x))
}
set.seed(2)
anc_res <- t(replicate(150, ancient_rep()))
round(rbind(mean = colMeans(anc_res), sd = apply(anc_res, 2, sd)), 2)
```

```
#>      bm_root  root trend
#> mean    2.13 -0.23  0.45
#> sd      0.83  1.76  0.28
```

The undirected BM estimate is again biased eastward, while the drift model recovers both the homeland (true value 0) and the trend (true value 0.4) on average, albeit with substantial uncertainty from replicate to replicate. The lesson for linguistics is concrete: ancient languages placed at their attested dates and locations are not just useful for calibrating the clock (Chapter 10); they are the main source of information about the *direction* of an expansion. The simulation assumes that the ancient tips were located where their lineages were; an administrative language attested far from its speakers' core area (Section 11.11) would mislead the drift model too.

11.13.5 Discrete phylogeography: regions, rates and Markov jumps

For discrete phylogeography we divide the map into four regions arranged in a ring (R1 to R4, with R1 adjacent to R2 and R4, and so on) and simulate the region of each lineage on the same tree, starting in R1, under a symmetric CTMC in which lineages move only between adjacent regions, quickly between R1 and R2 and slowly elsewhere. We then fit three models with `fitMk()`: equal rates, symmetric rates, and a *ring* model in which only adjacent regions exchange lineages (rates between R1 and R3, and between R2 and R4, fixed at zero). The ring model, which here happens to be the true model family, is the simplest example of a geographically constrained rate matrix, a hand-built version of what BSSVS learns from the data.

```
regions <- c("R1", "R2", "R3", "R4")
Qtrue <- matrix(0.05, 4, 4, dimnames = list(regions, regions))
Qtrue["R1", "R3"] <- Qtrue["R3", "R1"] <- 0     # non-adjacent regions
Qtrue["R2", "R4"] <- Qtrue["R4", "R2"] <- 0
Qtrue["R1", "R2"] <- Qtrue["R2", "R1"] <- 0.4   # a fast corridor
diag(Qtrue) <- 0; diag(Qtrue) <- -rowSums(Qtrue)
set.seed(8)
area <- sim.Mk(tree, Qtrue, anc = "R1")
table(area)
```

```
#> area
#> R1 R2 R3 R4
#> 14 12  3  1
```

```
ring <- matrix(c(0, 1, 0, 2,
                 1, 0, 3, 0,
                 0, 3, 0, 4,
                 2, 0, 4, 0), 4, byrow = TRUE,
               dimnames = list(regions, regions))
fits <- list(ER   = fitMk(tree, area, model = "ER"),
             SYM  = fitMk(tree, area, model = "SYM"),
             ring = fitMk(tree, area, model = ring))
knitr::kable(data.frame(
  model = names(fits),
  k     = sapply(fits, function(f) attr(logLik(f), "df")),
  logL  = round(sapply(fits, logLik), 2),
  AIC   = round(sapply(fits, AIC), 2)),
  row.names = FALSE,
  caption = "Discrete phylogeographic models fitted to simulated region data: number
  of free rate parameters (k), log-likelihood and AIC.")
```

Table 11.4: Discrete phylogeographic models fitted to simulated region data: number of free rate parameters (k), log-likelihood and AIC.

model	k	logL	AIC
ER	1	-35.39	72.78
SYM	6	-27.66	67.32
ring	4	-27.66	63.32

The marginal posterior probabilities of each region at the root and at the other internal nodes are obtained with `ancr()` (Chapter 12 explains marginal reconstruction in detail). Figure 11.2 shows them as pie charts.

```
best <- fits[[which.min(sapply(fits, AIC))]]
anc_area <- ancr(best)
round(anc_area$ace[1, ], 3)                # root (homeland) probabilities
```

```
#>    R1    R2    R3    R4
#> 0.546 0.448 0.005 0.001
```

```
cols <- setNames(c("#1b9e77", "#d95f02", "#7570b3", "#e7298a"), regions)
plotTree(tree, fsize = 0.7, ftype = "i", offset = 0.5)
nodelabels(pie = anc_area$ace, piecol = cols, cex = 0.4)
tiplabels(pie = to.matrix(area[tree$tip.label], regions), piecol = cols,
          cex = 0.3)
legend("bottomleft", legend = regions, pch = 21, pt.bg = cols, pt.cex = 1.5,
       bty = "n")
```

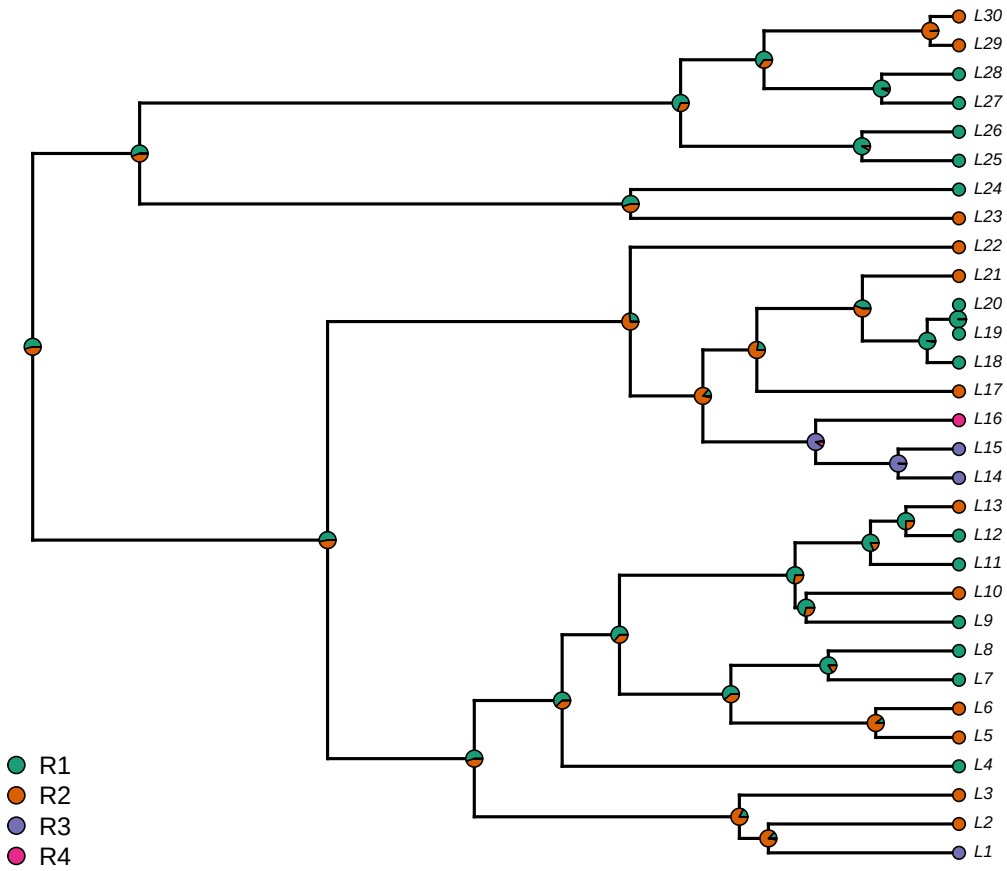


Figure 11.2: Marginal ancestral region probabilities under the best-fitting discrete model, shown as pie charts at internal nodes. Tip colors give the observed regions of the 30 simulated languages.

The ring model is preferred by AIC (Table 11.4), as it should be. Even in this idealized setting,

with the true tree and a correctly specified model, the root probability is split between the true homeland R1 and its neighbor R2: because lineages move quickly along the R1-R2 corridor, the data cannot tell which end of the corridor the family started from. A single discrete character carries limited information about deep nodes, and this is the typical situation in linguistic discrete phylogeography. Finally, stochastic character mapping (`make.simmap()`; Section 12.6) samples complete histories of movement and lets us count the expected number of moves between each pair of regions, the Markov jump counts of Section 11.4.3.

```
set.seed(7)
Qhat <- as.Qmatrix(best)
maps <- make.simmap(tree, area, Q = unclass(Qhat)[regions, regions],
                    nsim = 50, pi = "fitzjohn", message = FALSE)
jumps <- describe.simmap(maps)$count
round(colMeans(jumps), 2)                       # expected moves by type
```

```
#>     N R1,R2 R1,R3 R1,R4 R2,R1 R2,R3 R2,R4 R3,R1 R3,R2 R3,R4 R4,R1 R4,R2
#> 29.68 13.78  0.00  0.00 12.22  2.14  0.00  0.00  0.14  1.20  0.00  0.00
#> R4,R3
#>  0.20
```

Most of the roughly thirty expected moves are back-and-forth exchanges along the fast R1-R2 corridor, with a handful of moves into R3 and R4. Counts of this kind are what one would report for questions such as how often lineages entered or left a given habitat.

11.13.6 Homelands and rates over a sample of trees

So far the tree was known. We now carry the analysis over a mock posterior sample of 100 trees, built by perturbing the true tree: three random nearest-neighbor interchanges, lognormal noise on every branch length, and a root age drawn around six millennia with a standard deviation of about 20% on the log scale, roughly the dating uncertainty of a well-calibrated family. (A real analysis would use trees sampled by BEAST or MrBayes from the lexical data; Section 17.11 shows how to read them.) On each tree we estimate the homeland's east coordinate with its standard error, the east-west diffusion rate (by REML, as the mean squared standardized contrast), and the probability that the root lay in region R1 under the equal-rates discrete model. We then do the same on the MCC tree of the sample, computed with `maxCladeCred()` from **phangorn**, which returns the sampled tree with the highest clade credibility, branch lengths included.

```
library(phangorn)
set.seed(5)
post <- lapply(1:100, function(i) {
  t <- rNNI(tree, moves = 3)                        # topological noise
  t$edge.length <- t$edge.length * rlnorm(nrow(t$edge), 0, 0.3)
  t <- force.ultrametric(t, method = "extend", message = FALSE)
  age <- 6 * rlnorm(1, 0, 0.2)                      # root-age uncertainty
  t$edge.length <- t$edge.length * age / max(node.depth.edgelength(t))
  t })
class(post) <- "multiPhylo"
mcc <- maxCladeCred(post)
per_tree <- function(t) {
```

```
  x <- tips[t$tip.label, "east"]
  f <- fastAnc(t, x, CI = TRUE)
  c(root_age = max(node.depth.edgelength(t)),
    east     = unname(f$ace[1]),
    se_east  = unname(diff(f$CI95[1, ])) / (2 * 1.96),
    sig2     = mean(pic(x, t)^2),                    # REML diffusion rate
    p_R1     = ace(area[t$tip.label], t, type = "discrete",
                   model = "ER")$lik.anc[1, "R1"])
}
res <- t(sapply(post, per_tree))
on_mcc <- per_tree(mcc)
op <- par(mfrow = c(1, 3), mar = c(4.2, 4.2, 1, 1))
hist(res[, "east"], breaks = 15, col = "gray80", border = "white",
     main = "", xlab = "homeland, east (100 km)")
abline(v = 0, lty = 2); abline(v = on_mcc["east"], col = "red", lwd = 2)
plot(res[, "root_age"], res[, "sig2"], pch = 16, col = "gray50",
     xlab = "root age (kyr)", ylab = "diffusion rate (east)")
points(on_mcc["root_age"], on_mcc["sig2"], pch = 18, col = "red", cex = 2)
hist(res[, "p_R1"], breaks = 15, col = "gray80", border = "white",
     main = "", xlab = "P(root in R1)")
abline(v = on_mcc["p_R1"], col = "red", lwd = 2)
```

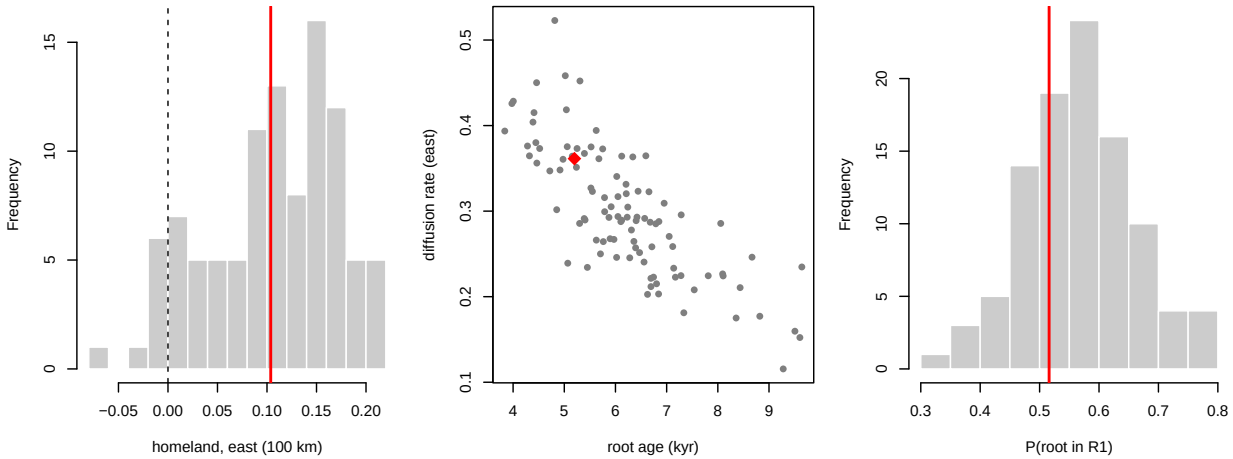


Figure 11.3: A two-step phylogeographic analysis over a mock posterior sample of 100 trees. Left: estimates of the homeland's east coordinate on each tree. Center: diffusion rate against root age, one point per tree. Right: probability that the root lay in region R1 under the equal-rates discrete model. Red marks the analysis of the MCC tree alone; the dashed line in the left panel is the true homeland.

```
par(op)
```

Pooling across trees, with the total variance of the homeland estimate computed as the average within-tree variance plus the between-tree variance (the rule explained in Section 13.12.1), gives the following summaries, compared with the MCC tree alone:

```
m <- nrow(res)
pooled_se <- sqrt(mean(res[, "se_east"]^2) + (1 + 1 / m) * var(res[, "east"]))
q <- function(v, p) unname(quantile(v, p))
round(rbind(
  MCC_tree   = on_mcc[c("east", "se_east", "sig2", "p_R1")],
  over_trees = c(mean(res[, "east"]), pooled_se, mean(res[, "sig2"]),
                 mean(res[, "p_R1"])),
  lower_95   = c(q(res[, "east"], 0.025), NA, q(res[, "sig2"], 0.025),
                 q(res[, "p_R1"], 0.025)),
  upper_95   = c(q(res[, "east"], 0.975), NA, q(res[, "sig2"], 0.975),
                 q(res[, "p_R1"], 0.975))), 3)
```

```
#>              east se_east  sig2  p_R1
#> MCC_tree    0.104   0.606 0.361 0.516
#> over_trees  0.105   0.581 0.300 0.566
#> lower_95   -0.017      NA 0.167 0.381
#> upper_95    0.205      NA 0.451 0.771
```

```
cor(res[, "root_age"], res[, "sig2"])
```

```
#> [1] -0.782
```

```
p_true <- ace(area, tree, type = "discrete", model = "ER")$lik.anc[1, "R1"]
```

Figure 11.3 shows the uneven effects described in Section 11.10. The homeland barely moves from tree to tree: the between-tree spread is small compared with the within-tree standard error, so the MCC tree and the pooled analysis agree. The diffusion rate is a different matter. It varies by more than a factor of two across the sample, mostly because the root age varies (the two are strongly negatively correlated), and the single MCC value sits wherever the dates of that one sampled tree happen to put it. A rate or velocity reported from the MCC tree would carry none of this uncertainty. The discrete root probability, finally, ranges widely across trees, and its value on the MCC tree is one draw from that range. On the true tree the same model gives 0.31, a reminder that the posterior, here as in real analyses, need not be centered on the true tree.

The dependence of rates on the time scale can be made exact. Rescaling every branch of the true tree by the same factor, as a different root age would, leaves the reconstructed homeland unchanged and divides the rate by that factor:

```
rate_by_age <- t(sapply(c(4, 6, 9), function(T) {
  tr <- tree
  tr$edge.length <- tr$edge.length * T / 6
  c(root_age  = T,
    root_east = unname(fastAnc(tr, tips[, "east"])[1]),
    sig2_east = mean(pic(tips[, "east"], tr)^2))
}))
round(rate_by_age, 3)
```

```
#>      root_age root_east sig2_east
#> [1,]        4     0.093     0.708
#> [2,]        6     0.093     0.472
```

```
#> [3,]        9     0.093     0.315
```

The diffusion rate is identifiable only relative to the dates of the tree (Table 11.2).

11.13.7 Bayes factors for BSSVS routes

BSSVS itself requires MCMC in BEAST, but turning its output into route Bayes factors is simple arithmetic, and one consequence is worth knowing before reading any published route map. With four regions, three of the six possible routes are needed for connectivity, so the prior inclusion probability of each route is high (about 0.62), and even a posterior inclusion probability of 0.81 yields only a modest Bayes factor of about 2.7. With twenty regions and 190 possible routes the prior inclusion probability is about 0.1, and the same posterior inclusion probability yields a Bayes factor of about 37. Bayes factors from BSSVS are therefore not comparable across analyses with different numbers of regions, and a route can look "strongly supported" partly because the map was cut into many regions.

The function `bssvs_bf()`, which computes route Bayes factors from posterior inclusion probabilities read from a BEAST log, and the worked numbers above, are in Section D.1.2.

11.13.8 Planar versus spherical distance

Finally, a small calculation shows why diffusion on unprojected longitude and latitude is a problem at continental scales (Section 11.5.3). We compute the great-circle (haversine) distance between pairs of points separated by ten degrees of longitude at different latitudes, and compare it with the "planar" distance implied by treating degrees as equal units everywhere.

```
haversine <- function(lon1, lat1, lon2, lat2, R = 6371) {
  rad <- pi / 180
  dlat <- (lat2 - lat1) * rad; dlon <- (lon2 - lon1) * rad
  a <- sin(dlat / 2)^2 + cos(lat1 * rad) * cos(lat2 * rad) * sin(dlon / 2)^2
  2 * R * asin(sqrt(a))
}
lat <- c(0, 20, 40, 50, 60, 70)
sph <- haversine(0, lat, 10, lat)
knitr::kable(data.frame(latitude = lat,
                        planar_km = round(10 * 111.2),
                        great_circle_km = round(sph),
                        ratio = round(10 * 111.2 / sph, 2)),
             caption = "Distance between two points ten degrees of longitude apart at
              various latitudes: planar approximation on raw degrees versus
              great-circle distance.")
```

Table 11.5: Distance between two points ten degrees of longitude apart at various latitudes: planar approximation on raw degrees versus great-circle distance.

latitude	planar_km	great_circle_km	ratio
0	1112	1112	1.00
20	1112	1045	1.06
40	1112	851	1.31

latitude	planar_km	great_circle_km	ratio
50	1112	714	1.56
60	1112	555	2.00
70	1112	380	2.93

As Table 11.5 shows, at the latitude of the Pontic-Caspian steppe (around 50 degrees north) the planar approximation overstates east-west distances by more than half; at the latitude of northern Scandinavia or Siberia it roughly doubles or triples them. A planar random walk on raw coordinates therefore treats east-west movement at high latitudes as far more costly than it is, which can bias homeland and route reconstructions for Eurasian families. Projecting coordinates or, better, using spherical diffusion avoids the problem.

- Classical homeland methods (linguistic paleontology, the diversity principle, migration theory) are informal; phylogeography makes the assumptions about movement explicit and quantifies uncertainty under them.
- In discrete phylogeography, location is a character evolving under a CTMC on regions. BSSVS selects the dispersal routes needed to explain the data and yields route-specific Bayes factors that depend on the number of regions; Markov jumps count movements; GLMs relate dispersal rates to predictors such as distance and habitat.
- In continuous phylogeography, coordinates diffuse along branches under Brownian motion or a relaxed random walk. Under BM, the ML homeland is a phylogenetically weighted average of present-day locations.
- Undirected diffusion models cannot detect directional expansions from contemporaneous languages. They place the homeland near the center of the range with overconfident uncertainty, and adequacy checks on present-day locations cannot reveal the problem. Ancient languages at their attested times and places are the main remedy.
- A posterior over an ancestral location is a statement about language lineages under a model. It is not evidence that the model describes how speech communities moved; that requires adequacy checks, alternative process models and independent evidence.
- Diffusion rates and velocities are identifiable only relative to the dates of the tree and inherit their uncertainty. Analyses over a sample of trees show this; analyses on a single MCC tree hide it.
- Linguistic sampling bias, the displacement of speech communities and the replacement of early branches are serious and often undetectable sources of error.
- DEC/BioGeoBEARS, GeoSSE, the structured coalescent and phylodynamic GLMs offer largely unexploited tools for modeling ranges, area-dependent diversification and the drivers of linguistic dispersal.

11.14 Exercises

Conceptual

1. A study reports a Bayes factor of 50 in favor of an Anatolian rather than a steppe homeland for a language family. List the information you would need before accepting this as evidence about the homeland (consider the sizes of the two regions, the prior on the root location, the dates of the tree, and whether the model could have favored a peripheral homeland at all).

2. Explain in your own words why adding the same eastward drift to every lineage cannot be detected from contemporaneous languages, and why ancient languages help. What must be assumed about where the ancient languages are attested?
3. A dataset contains 40 languages from one region and 5 from another, because the first region has been far better documented. Explain how this affects a discrete and a continuous phylogeographic reconstruction, and propose two robustness analyses.
4. A paper reports that a family expanded at "1.2 km per year". Which modeling choices and which external inputs does this number depend on, and what would you need to know before reading it as the speed at which people moved?
5. Distinguish demic diffusion from language shift. For each, describe what the movement of a lineage along a branch of the language tree corresponds to historically, and say whether a diffusion model can tell the two apart.

Computational

1. Using `one_rep()`, vary the trend `mu` over 0, 0.1, 0.2, 0.4 and 0.8 and plot the coverage of the Brownian root interval against μT. At what displacement does coverage fall below 80%? Does the adequacy check (`flagged`) ever respond?
2. Modify `ancient_rep()` so that the ten ancient languages all belong to one clade (choose terminal branches below a single node with `getDescendants()`). How do the bias and standard deviation of the drift estimates change? Repeat with `n_anc` equal to 2, 5 and 20.
3. With the mock posterior `post`, compute for each tree the dispersal distance implied by the REML rate over the root age, $\sqrt{2\hat{\sigma}^2 T}$, and compare its 95% range across trees with the single value from the MCC tree. Then double the standard deviation of the root-age noise used to build `post` and repeat.
4. Fit the ring model of Section 11.13.5 with `fitMk()` to the first 20 trees in `post`, and record how often it is preferred to the ER model by AIC. Then pool Markov jump counts across trees by running `make.simmap()` with `nsim = 10` on each tree and combining the `describe.simmap()` counts.
5. Add noise to the tip locations in `one_rep()` (for example `x + rnorm(n, 0, 0.5)`, mimicking displaced speech communities) and record coverage and the adequacy check. Which kinds of misspecification does the contrast-kurtosis check detect, and why?

Research

1. You are planning a phylogeographic study of a South American family with 70 languages, a third of which were relocated during the twentieth century and a handful of which are known only from colonial-era sources. Design the analysis: how you would code locations (points, polygons, historical locations), which diffusion models you would compare, which adequacy checks you would run, how you would use the colonial-era sources, and what you would and would not claim about the movements of speech communities.
2. Design a phylogeographic GLM to test whether the Bantu expansion followed habitats suitable for cereal cultivation. Specify the regions, the predictors and how they would be measured for the relevant period, the analysis over trees, and what result would count as evidence that habitat shaped dispersal rather than merely correlating with it.
3. For Austronesian, argue for or against a continuous relaxed random walk, a discrete model on island groups, and a DEC + J biogeographic model. For each, state which historical quantities it could estimate and which it would misrepresent.

11.15 Further reading

- Lemey et al. (2009): The paper that introduced Bayesian discrete phylogeography with BSSVS. Written for virologists, but the model description is clear and general.
- Lemey et al. (2010): The relaxed random walk for continuous phylogeography. Required reading before running a continuous analysis in BEAST.
- Bouckaert et al. (2012): The landmark linguistic application to Indo-European. Read it together with the critique of Pereltsvaig and Lewis (2015) and the later work of Chang et al. (2015) and Heggarty et al. (2023).
- Neureiter et al. (2021): A simulation study of how well relaxed random walks recover linguistic homelands and migrations. It shows where they fail.
- Bouckaert (2016): Spherical diffusion and its implementation in BEAST 2. It explains why planar models break down at large scales.
- Grollemund et al. (2015) and Koile et al. (2022): Two contrasting phylogeographic analyses of the Bantu expansion. Together they are an excellent case study in how modeling choices shape conclusions.
- Wichmann et al. (2010b): The quantitative diversity-based approach. It is a useful baseline for any phylogeographic study.
- Ree and Smith (2008) and Matzke (2014): DEC and BioGeoBEARS, the core of model-based historical biogeography and a promising and largely untried framework for languages.
- Lemey et al. (2014): Phylogeographic GLMs that relate dispersal rates to predictors. This is a template for testing hypotheses about the causes of language spread.
- Nichols (1992): The classic linguistic treatment of spread zones and residual zones. It gives the geographical intuitions any phylogeographic model should respect.

Chapter 12

Ancestral State Reconstruction

Historical linguistics is, at its core, a science of reconstruction. The comparative method recovers the sounds, words and grammatical structures of languages that were never written down, and Proto-Indo-European, Proto-Austronesian and Proto-Bantu are known to us only through such reconstruction. Evolutionary biology has its own, largely independent, tradition of *ancestral state reconstruction* (ASR): given a tree and the states of a character at its tips, infer the states at the internal nodes. The two traditions use the same word for different operations. This chapter develops the biological toolkit (parsimony, likelihood and Bayesian reconstruction, and stochastic character mapping), shows how its answers depend on the model, the root and the tree, and asks what it can add to the linguist's own methods of reconstruction, and what it cannot replace.

After reading this chapter you should be able to:

- distinguish the reconstruction of coded character states, the reconstruction of historical linguistic forms (proto-forms and sound changes) and the reconstructions of the comparative method, and say what kind of evidence each uses;
- compute and interpret parsimony, marginal and joint reconstructions, and explain why marginal and joint answers can disagree;
- assess how root priors, rate models and hidden rate heterogeneity affect a reconstruction, and check which quantity a software function reports;
- use stochastic character mapping to count changes and explain why the counts exceed parsimony counts;
- reconstruct a named ancestor over a sample of trees, and explain why a reconstruction on a single MCC tree can mislead;
- evaluate published reconstructions of the proto-lexicon, of grammar and of sound systems, including the limits of their agreement with expert reconstructions as evidence.

12.1 What is an ancestral state?

Consider the word for 'five' across the Austronesian family. In Malay and Indonesian it is *lima*, in Tagalog *lima*, in Fijian *lima*, in Hawaiian *lima* (where it also means 'hand' or 'arm'), in Malagasy *dimy*. Comparative Austronesianists reconstruct Proto-Austronesian **lima* 'five; hand', and the reconstruction is about as secure as any in the field. Now consider a less clear case: a meaning for which some branches use one cognate set, some another, and a few a third. Which word did the

proto-language use? Or consider a structural feature: Proto-Oceanic is usually reconstructed with basic subject-verb-object order, but many Oceanic languages of New Guinea have verb-final order, apparently under the influence of neighboring Papuan languages. Which order did the common ancestor of a particular subgroup have, and how many times did the change happen?

Questions of this kind have the same logical structure. There is a tree, which for the moment we take as known; there is a *character* (a meaning slot with cognate classes as its states, a binary feature such as presence of grammatical gender, a multistate feature such as basic word order); and there are observed states at the tips. ASR infers the states at internal nodes, or more generally the history of the character along the branches.

Two clarifications prevent common misunderstandings.

First, an internal node of a phylogeny is not a language in the everyday sense. It represents the last common ancestor of a set of lineages, an idealized moment in a speech community that was certainly internally variable. A reconstructed state at a node is a statement about the variety from which the descendants inherited their state, not a claim that a uniform proto-language existed. The comparative method makes the same idealization, which is why reconstructed proto-languages are often described as "uniform" or "idealized" systems.

Second, the inference is conditional on a model. Even parsimony, which looks model-free, embodies a model: one in which change is rare and every change costs the same. All probabilistic reconstructions depend on the tree, its branch lengths, the rate matrix and the root prior. A probability of 0.9 for SOV order at the root of a family is the probability *under the assumed model*, and a different but equally defensible model can produce a different answer. Much of this chapter is about how to make these dependencies explicit and how to propagate uncertainty about them. For the same reason some authors prefer to speak of ancestral state *estimation* rather than reconstruction, since the latter suggests a certainty that the inference does not have (Goldstein, 2020).

12.2 Three kinds of reconstruction

The word "reconstruction" covers at least three different operations in the literature that this chapter reviews, and much confusion, in both directions, comes from treating them as one.

Reconstruction by the comparative method. A comparative reconstruction such as Proto-Austronesian **lima* is a hypothesis about a form, embedded in a system: a set of cognates, a set of regular sound correspondences, a proto-phoneme for each correspondence set, and sound laws that derive each daughter form from the proto-form. Its evidence is the *regularity* of the correspondences across many words, and its strength comes from that system-wide consistency, not from the distribution of the form over a tree. It is not tied to a dated node, it rarely comes with a probability, and its status is debated: some linguists read reconstructed forms as approximations of real speech, others as formulas summarizing the correspondences (Fox, 1995; Campbell, 2013). It also produces the data that the other two kinds of reconstruction take as given. Cognate sets, in particular, are judgments of the comparative method (Chapter 3), and reconstructed proto-languages are themselves hypotheses rather than observations (Section 3.2.3).

Model-based reconstruction of linguistic forms. A second family of methods reconstructs the forms themselves, or the sound changes that relate them, with explicit probabilistic models: stochastic transducers on the branches of a tree, or concerted changes in sound correspondences

(Sections 12.11 and 12.12). These methods take cognate sets and usually a tree as input and output strings or sound laws. They are closer to the comparative method in what they produce, and closer to phylogenetics in how they reason.

Phylogenetic reconstruction of character states. Most of this chapter concerns a third operation. A *character* is coded from the data, and the reconstruction is a probability distribution over its states at a node of a tree. If the character is the meaning slot 'hand' with cognate classes as states, the output is "cognate class 3 was the word for 'hand' at this node with probability 0.8". This presupposes the cognate classes and says nothing about the form of the word. If the character is a typological feature such as basic word order, the states are categories defined for modern languages in databases such as WALS or Grambank, and the reconstruction presupposes that the proto-language can be described in the same terms, which is not guaranteed (a proto-language might have had a word-order system that fits none of the modern categories well).

Table 12.1 summarizes the contrast. The three operations are complementary. Character-state reconstruction is quick, quantitative and comparable across families, but it reconstructs codes, not languages. Comparative reconstruction reconstructs languages, but without probabilities. Model-based reconstruction of forms aims at both, and is the least developed. Section 12.13 returns to the relationship, including the question of what agreement between them does and does not show.

Table 12.1: Three kinds of reconstruction in historical linguistics.

	Comparative method	Model-based reconstruction of forms	Phylogenetic character-state reconstruction
What is reconstructed	Proto-forms, proto-phonemes, sound laws, constructions	Proto-forms as strings; sound changes on branches	Probabilities of coded states at tree nodes
Main evidence	Regularity of correspondences across many words	Aligned cognate forms and a tree	Distribution of states over the tips of a tree
Role of the tree	Mainly primary subgrouping	Full tree, often fixed	Full topology and branch lengths
Output	Starred forms, sometimes with alternatives	Forms with model-based uncertainty	Posterior probabilities under a model
Presupposes	Cognacy and regularity (established within the method)	Cognate sets, alignments	Cognate sets or typological codes from earlier analysis

12.3 Parsimony reconstruction

12.3.1 Fitch and Sankoff

The oldest algorithmic approach is *maximum parsimony*: choose the assignment of states to internal nodes that minimizes the number (or total cost) of changes on the tree. For unordered characters with equal costs, Fitch (1971) gave a two-pass algorithm. In the *downpass* (from tips to root), each node receives a set of candidate states: the intersection of its children's sets if they overlap, and otherwise their union, at the cost of one change. The number of unions is the parsimony score of the character on the tree. An *uppass* then refines the sets into the most parsimonious recon-

structions (MPRs) at each node: the set of states that appear in at least one most-parsimonious assignment.

Sankoff (1975) generalized this to arbitrary cost matrices, in which a change from state i to state j costs c_{ij}. Sankoff parsimony accommodates ordered characters (changes only between adjacent states of a scale, as for a series of vowel heights), irreversible characters (Dollo-like costs, in which gaining a state is expensive and losing it cheap, a natural model for cognate presence; Chapter 7) and directional sound changes (a cost matrix in which $p > f$ is cheap and $f > p$ expensive). Its recursion has exactly the structure of the likelihood pruning algorithm of Chapter 5: replace "minimum" by "sum", "plus" by "times" and costs by transition probabilities, and Sankoff's algorithm becomes Felsenstein's.

The Fitch and Sankoff recursions, Equations (D.4) and (D.5), and a hand-coded Fitch downpass checked against **phangorn** are in Section D.2.1.

When several reconstructions are equally parsimonious, conventions are needed to choose among them. *ACCTRAN* (accelerated transformation) places changes as close to the root as possible, favoring reversals over parallel changes; *DELTRAN* (delayed transformation) places them as close to the tips as possible, favoring parallel changes (Swofford and Maddison, 1987). For linguistic characters the choice is substantive. ACCTRAN treats a feature shared by two non-sister subgroups as inherited and later lost in between; DELTRAN treats it as independently innovated twice. These correspond to two of the classical options historical linguists weigh when a feature has a patchy distribution: retention with losses, or parallel innovation (or contact).

The convention also changes how certain a reconstruction looks. **phangorn**'s `ancestral.pars()` offers `type = "MPR"`, which returns at every node all the states that appear in at least one most-parsimonious reconstruction, and `type = "ACCTRAN"`, which commits to one of them. Plotted as pie charts, ACCTRAN reconstructions therefore look more decisive than the data warrant, and MPR is the honest summary of what parsimony alone can say. A second trap is circularity. A character that was used to infer the tree is not independent evidence about its own history on that tree, because its distribution has already shaped the topology. Reconstructing a morphological innovation on a tree inferred from a character set that includes it will tend to make the reconstruction look cleaner than it should. Characters that are the object of a reconstruction are best left out of the tree inference, or the sensitivity of the result to their exclusion checked.

12.3.2 What parsimony ignores

Parsimony is fast, intuitive and close to the reasoning of the comparative method, but it has well-known limitations (Cunningham et al., 1998). It ignores branch lengths, so a change on a branch representing five thousand years of separation counts the same as one on a branch of fifty years. It gives no measure of uncertainty beyond the existence of multiple equally parsimonious solutions. And it is biased when rates of change are high: the minimum number of changes is then systematically less than the actual number, and reconstructions at deep nodes become unreliable in ways that parsimony itself cannot detect. These are the same considerations that favor likelihood over parsimony for tree inference (Chapter 4), and they are, if anything, more serious for ASR, because a single character carries far less information than a whole data matrix.

12.4 Likelihood reconstruction: marginal and joint

12.4.1 The Mk model

Probabilistic ASR starts from a continuous-time Markov model of character change, the same machinery used for cognate data (Chapter 7) and for discrete phylogeography (Chapter 11). For a character with k states, an instantaneous rate matrix $\mathbf{Q}$ specifies the rate q_{ij} of change from state i to state j, and $\mathbf{P}(t) = e^{\mathbf{Q}t}$ gives transition probabilities over a branch of length t. For morphological characters in biology this is known as the *Mk model* (Lewis, 2001); the general likelihood framework for discrete characters on trees, including tests of correlated evolution, goes back to Pagel (1994). As in Chapter 11, the rate matrix can be equal-rates (ER: one rate), symmetric (SYM: $k(k-1)/2$ rates) or all-rates-different (ARD: $k(k-1)$ rates), or can be constrained by hand (for example, an ordered model with changes only between adjacent states).

For basic word order with three states (SOV, SVO, VSO), an ARD model has six rates. It is natural to suspect that changes are asymmetric: many typologists have argued that SOV to SVO is common and the reverse rare, and that VSO tends to arise from SVO. A fitted ARD model on a large enough sample of trees could test this, which is what Maurits and Griffiths (2014) set out to do (Section 12.10).

The likelihood of the tip data given the tree τ and parameters θ (the rates and the root distribution π) is computed by Felsenstein's pruning algorithm. The algorithm computes, for every node v and state i, the *conditional likelihood* $L_v(i)$: the probability of the tip states below v given that v is in state i. The likelihood of the whole tree is then $L(\theta) = \sum_i \pi_i \, L_{\text{root}}(i)$. Parameters are estimated by maximizing $L(\theta)$, and reconstruction proceeds in one of two ways.

The pruning recursion for $L_v(i)$, Equation (D.6), and an implementation in R that reproduces the log-likelihood reported by `fitMk()`, are in Section D.2.2.

12.4.2 Marginal reconstruction

The *marginal* reconstruction at node v is the probability distribution of the state at v, summing over all possible states at every other node:

$$P(X_v = i \mid D, \hat{\theta}) = \frac{\sum_{\text{assignments with } X_v = i} P(D, \text{assignment} \mid \hat{\theta})}{L(\hat{\theta})}. \tag{12.1}$$

At the root this is simply $\pi_i L_{\text{root}}(i) / \sum_j \pi_j L_{\text{root}}(j)$, which follows directly from the pruning pass. For other nodes, one must combine the information from below v (its conditional likelihoods $L_v(i)$) with the information from the rest of the tree (the "outside" likelihood $U_v(i)$, computed in a second, root-to-tips pass), so that $P(X_v = i \mid D) \propto L_v(i) \, U_v(i)$. An equivalent and conceptually simple approach is to re-root the tree at each node in turn and use the root formula; this re-rooting algorithm is implemented in several R functions, and the resulting node probabilities are the empirical Bayes reconstructions introduced for molecular sequences by Yang et al. (1995).

Two points about marginal reconstruction are often confused. First, the conditional likelihood $L_v(i)$ normalized over i is *not* the marginal posterior at v, except at the root: it ignores everything outside the subtree. Second, the marginal probabilities computed at the maximum-likelihood estimate $\hat{\theta}$ are *empirical Bayes* posterior probabilities (Yang et al., 1995; Koshi and Goldstein, 1996): they treat the estimated rates as if they were known exactly. With small trees and single characters,

the uncertainty in $\hat{\theta}$ can be large, and empirical Bayes reconstructions are overconfident. The R section shows the first point concretely.

In R, `ape::ace(..., type = "discrete")` returns marginal posterior probabilities in its `lik.anc` component under its default setting `marginal = FALSE`. Setting `marginal = TRUE` returns, despite the name, the scaled conditional likelihoods of each node given only its descendants, which are not marginal reconstructions except at the root. The **phytools** function `ancr()` returns marginal reconstructions by default and joint reconstructions with `type = "joint"`. Always check which quantity a function reports before interpreting it, and say which one you report.

12.4.3 Joint reconstruction

The *joint* reconstruction is the single assignment of states to *all* internal nodes that maximizes the joint probability of the data and the assignment:

$$(\hat{x}_{n+1}, \ldots, \hat{x}_{2n-1}) = \arg\max_{x_{n+1},\ldots,x_{2n-1}} \pi_{x_{\text{root}}} \prod_{\text{branches } (a \to c)} P_{x_a x_c}(t_c), \tag{12.2}$$

where tip states are fixed at their observed values. A naive search over all k^{n-1} assignments is infeasible for large trees, but Pupko et al. (2000) gave a dynamic-programming algorithm, the max-product analogue of pruning, which finds the best assignment in time linear in the number of nodes.

The Pupko algorithm, Equation (D.7), and a hand-coded version that agrees exactly with `ancr(..., type = "joint")` are in Section D.2.3.

Marginal and joint reconstructions answer different questions. Marginal probabilities say what we believe about one node at a time, and they are the right quantity when interest centers on a particular proto-language (“what was the basic word order of Proto-Oceanic?”). The joint reconstruction is a single coherent history and is the right quantity when the reconstruction will be used as a whole, for example to count the minimum number of changes it implies. The two can disagree: the state with the highest marginal probability at a node need not be the state it takes in the most probable joint assignment, and the R section shows an example. Neither reports the uncertainty of an entire history; that is what stochastic mapping is for.

12.4.4 The root prior

The root distribution π has a large and often unappreciated effect on reconstructions near the root, which is usually where linguistic interest lies. Common choices are:

- **Equal (flat)**: $\pi_i = 1/k$. The default in many programs.
- **Stationary**: π equal to the equilibrium distribution of $\mathbf{Q}$. This assumes the process had been running long before the root, which is natural for a family that is itself part of a deeper history.
- **Estimated**: π treated as free parameters, which is rarely well identified.
- **FitzJohn**: $\pi_i \propto L_{\text{root}}(i)$, weighting root states by their own conditional likelihoods, proposed by FitzJohn et al. (2009) in the context of diversification models and now widely used.
- **Fixed**: the root state set to a particular value, for example to test a hypothesis by comparing likelihoods with the root fixed to each state.

With an asymmetric **Q**, the stationary distribution can favor states that are rare at the tips, and a stationary root prior can then dominate the reconstruction at the root. Reporting the sensitivity of the root reconstruction to the root prior is good practice.

Programs differ in their defaults, and the default is not always visible in a script: `fitMk()` in **phytools** uses the flat prior unless told otherwise, whereas RevBayes uses the stationary distribution of **Q** when no root frequencies are given. Harmon (2019) adds a third source of root probabilities, outside information such as attested ancient languages or an outgroup, and notes that the flat and stationary choices coincide only under equal rates. An *estimated* root distribution needs a particular warning. For a single character with a uniform prior on π, the marginal likelihood and the root reconstruction are exactly those of the flat prior, and the posterior of π hardly moves from its prior, because one character supplies a single draw from the root distribution. Root frequencies become estimable only when many characters share them.

12.4.5 Constraints and priors that decide the answer

A rate matrix with zeros in it does more than express a hypothesis: it can settle the reconstruction before the data are consulted. Cathcart (2018) made the point with the study of "dative sickness" in Germanic by Dunn et al. (2017), which modelled how the case of the subjects of particular verbs changed among accusative (A), dative (D) and nominative (N). The model the authors preferred allowed only the changes A → D, A → N and D → N, the directions of the hypothesized drift, and its reconstructions agreed with the philological ones, which they read as support for it. Under that model, however, a nominative subject can be reconstructed at a node only if every descendant language has one, and wherever an accusative subject is attested at a tip, the root must have had an accusative subject. The agreement was therefore largely built into the constraints, and the constrained model was not the one favored by the likelihood. A weaker form of the hypothesis, in which the disfavored changes are rare rather than impossible, can be tested by comparing the estimated rates; Cathcart's reanalysis of the published data found the rates in the favored directions to be higher on average, but not significantly so. Agreement between a reconstruction and expert judgement is a useful sanity check. It is not evidence for the model that produced the reconstruction, and it should play no part in choosing among models (compare Section 12.13).

Constraints are easier to defend when they rest on evidence from outside the character being modelled. Goldstein (2022a) fixed the Indo-European root at "no articles" and excluded the loss of articles, because the earliest attested Indo-European languages lack articles and each class of articles descended from a common ancestral article is confined to a single major branch (Section 13.7). His rate estimates are conditional on that judgement; a model that allows rare losses could test it.

Softer modelling choices act in the same way, only less visibly. Whether a character is given one rate or separate rates of gain and loss can change reconstructions substantially (Mooers and Schluter, 1999). Irreversibility is hard to detect from the states of living languages alone. Harmon (2019) shows a case in which a model with different rates of gain and loss fitted no better than a one-rate model, although the biological expectation of asymmetry was strong, and notes that outside knowledge would have to enter as an informative prior. Historical linguistics has much knowledge of this kind about the direction of change, and using it is one of the natural meeting points of the two traditions (Section 12.13). The price is that the prior then contributes to the answer, and its contribution has to be reported.

The Indo-European *augment*, a past-tense verbal prefix found in Greek, Armenian and Indo-

Iranian, shows how these choices interact with subgrouping. Whether it was present in Proto-Indo-European remains uncertain, and Goldstein (2020) uses it to show that the dependence runs both ways. To treat a shared feature as a subgrouping innovation one must know its ancestral state, and to reconstruct an ancestral state one must know the subgrouping. Phylogenetic ASR makes the second half of this circle explicit, but it does not break it. For a feature attested in only a few branches, the reconstruction depends on the tree, on the asymmetry of gain and loss, and on the root treatment, and the data alone may not decide between "inherited and lost" and "gained more than once".

Section D.2.6 computes the root reconstruction of a rare, augment-like feature exactly under flat, stationary, estimated and fixed roots, and shows why an estimated root frequency for one character reproduces the flat prior. The RevBayes version, with marginal likelihoods for seven root treatments and rate priors, is in Section H.5.21.

12.5 Bayesian ASR: integrating over trees and parameters

Empirical Bayes reconstruction conditions on one tree and one set of rates. A fully Bayesian reconstruction integrates over both:

$$P(X_v = i \mid D) = \iint P(X_v = i \mid D, \tau, \theta)\; p(\tau, \theta \mid D)\; d\tau\, d\theta. \tag{12.3}$$

In practice the integral is approximated by averaging over a posterior sample of trees (usually from an earlier analysis of the lexical data) and over a posterior sample of rates for the character of interest, obtained by MCMC on each tree or jointly across trees. Huelsenbeck and Bollback (2001) compared empirical and hierarchical Bayesian approaches and showed that ignoring uncertainty in the tree and the parameters can give overconfident reconstructions. Pagel et al. (2004) developed the approach implemented in BayesTraits, in which a reversible-jump MCMC samples both the rates and the model structure (which rates are equal, which are zero) across a sample of trees.

A complication specific to integrating over trees is that a node of interest may not exist in every tree. If the question is "what was the word order of Proto-Germanic?", the relevant node is the most recent common ancestor (MRCA) of the Germanic languages, which exists in every tree that contains those languages but may differ in its descendants, its age and its position from tree to tree. Pagel et al. (2004) proposed reconstructing the state at the MRCA of a specified set of tips in each tree, and also showed how to *fix* (or "fossilize") the state of a node and compare the marginal likelihoods of models with the node fixed to different states, which gives a Bayes factor for competing reconstructions. This is particularly useful when a linguistic hypothesis concerns a specific proto-language ("Proto-Indo-European had three genders") rather than a node in a particular tree.

The MRCA approach also exposes a subtlety. If the named languages do not form a clade in some trees, their MRCA in those trees is a deeper node, the ancestor of the named group *and* of other languages, and the reconstruction there answers a different historical question. A reconstruction averaged over trees then mixes answers to two questions, and it should be reported together with the posterior probability that the group is a clade.

Tree uncertainty can matter as much as parameter uncertainty, and the general reasons why a single summary tree misleads in downstream analyses are set out in Section 11.10. For ASR two of them are especially important. A reconstruction on the maximum clade credibility (MCC) tree treats every clade in it as certain, so the answer for "Proto-X" is reported even when X is a clade

in only half of the posterior trees. And reconstructions are nonlinear functions of the tree, so the reconstruction on a summary tree is not the average reconstruction. The R section shows both on a small example (Section 12.15.7).

12.6 Stochastic character mapping

Node reconstructions say nothing directly about what happened *along* the branches: how many times a character changed, when, and in which direction. *Stochastic character mapping* (Nielsen, 2002; Huelsenbeck et al., 2003) answers these questions by sampling complete histories of the character from their posterior distribution given the tip data, the tree and the model. Each sampled history ("map") specifies the state at every point on every branch. Averaging over many maps gives posterior distributions of the number of changes of each type, the time spent in each state, and the states at nodes.

A map is sampled in three steps: a root state is drawn from its marginal posterior; the states of the other nodes are drawn one by one from the root toward the tips, each conditional on its parent's state and on the data below it; and, finally, a history along each branch is simulated conditional on the states at its two ends. The first two steps draw a joint assignment of node states from its exact posterior; the third fills in the changes along the branches.

The conditional distribution used in the second step, Equation (D.8), and the rejection and uniformization methods for the third, are in Section D.2.4.

The method was popularized by the SIMMAP software (Bollback, 2006) and is implemented in R as `make.simmap()` in **phytools**. In the original formulation, the rates are either fixed at their maximum-likelihood values (empirical Bayes) or sampled from their posterior; **phytools** supports both.

Stochastic maps answer many questions that node reconstructions cannot. How many times did SOV order change to SVO in the history of a family, and is that number larger than the number of changes in the opposite direction? Did a feature change mostly on long branches or in bursts? How much of a family's history was spent with a particular state, and did changes in one character tend to follow changes in another (as a test of correlated evolution; see Chapter 13)? The expected number of changes is also computable analytically (Minin and Suchard, 2008), but maps are more flexible because any function of the history can be summarized.

The posterior expected number of changes from a stochastic map is usually larger than the parsimony count, often considerably. Parsimony counts the minimum number of changes needed; stochastic mapping counts the expected number under a model that allows multiple changes along a branch, including changes that are later reversed. When historical linguists say that a feature "changed only once" in a family, they are usually making a parsimony argument. A model-based count can be an instructive check.

12.7 Heterogeneous and hidden-rate models

The standard Mk model assumes that the rates of change are the same on every branch of the tree. Linguistic characters often violate this. Basic word order has been stable for millennia in some lineages (the Japonic and Koreanic languages have been verb-final throughout their recorded history), while it has changed repeatedly in others (in the history of English, from mixed OV/VO

with verb-second in Old English to rigid SVO). A model that assumes a single rate everywhere will underestimate change in the labile lineages and overestimate it in the stable ones, and reconstructions at deep nodes can be badly affected.

Hidden-rate models (Beaulieu et al., 2013) address this by augmenting each observed state with a hidden rate category. With two categories, "slow" and "fast", a binary character with observed states 0 and 1 has four hidden states (0-slow, 1-slow, 0-fast, 1-fast). Each category has its own rates of change between the observed states, and lineages switch between categories at further rates. Only the observed state is known at the tips; the likelihood sums over hidden categories. The model is a phylogenetic version of the covarion model (Chapter 7), and Boyko and Beaulieu (2021) generalized it into a broad family of hidden Markov models for comparative data, implemented in the R package **corHMM**. **phytools** provides a similar model through `fitHRM()`.

The block structure of the hidden-rate rate matrix, Equation (D.9), is in Section D.2.5.

Hidden-rate models are attractive for linguistics because they formalize the intuition that some lineages are "conservative" and others "innovative" for a given feature, without having to specify in advance which ones. Their parameters are, however, hard to estimate from a single character on a tree of modest size, and model comparison (by AIC or marginal likelihood) should be used to check that the extra complexity is warranted.

A different kind of heterogeneity arises when several characters are related. Basic word order is often coded as a single character with six states (SOV, SVO, VSO, VOS, OVS, OSV), but it can also be decomposed into binary characters (OV versus VO, subject-initial or not) whose evolution may be correlated. Models of correlated discrete characters (Pagel, 1994) are the subject of Chapter 13. Here it is enough to note that the choice of coding changes the reconstruction, because it changes the model.

12.8 How reliable are ancestral reconstructions?

Ancestral reconstructions are estimates, and their accuracy depends on several factors that can be stated in general terms.

- **Depth and rate.** Information about a node decays with the time separating it from the tips, relative to the rate of change. A character that changes on average once every few thousand years retains information about nodes a few thousand years deep but very little about nodes ten thousand years deep. At the limit, the posterior at the root converges to the root prior. Studies in biology have repeatedly found that reconstructions of fast-evolving characters at deep nodes are unreliable (Schluter et al., 1997; Cunningham et al., 1998). A simple check is to compare the fitted Mk model with a model that ignores the tree and draws each language's state independently from the observed frequencies. When the rate is so high that the likelihood surface has a long flat ridge, the two likelihoods coincide, and the character carries no information about its history; Harmon (2019) calls this the "total garbage" test (Section D.2.7).
- **Tree shape and sampling.** A node supported by many descendant lineages that are independently informative (a star-like radiation) is reconstructed more precisely than a node with a single long stem. Unsampled lineages, including extinct ones, do not contribute information. Dense sampling of one subgroup can, as in phylogeography, pull reconstructions toward that subgroup's states when the model is misspecified.

- **Tree uncertainty.** A reconstruction at a node is only as secure as the node itself. If the clade below the node of interest is poorly supported, or if the dates of the tree are uncertain, the reconstruction varies from tree to tree (Section 12.5).
- **Model adequacy.** Reconstructions are only as good as the model. Rate heterogeneity among lineages, asymmetric rates, correlations with other characters and contact-induced change all violate the simple Mk model, and violations tend to produce overconfident reconstructions rather than visibly uncertain ones.
- **Diversification.** If the state of a character affects the rate at which lineages split or die out, reconstructions that ignore this are biased toward the state associated with higher diversification (Maddison, 2006). For languages, a feature associated with expanding, rapidly splitting populations (perhaps a feature of the languages of agricultural expansions) could be overrepresented at the tips and wrongly reconstructed as ancestral.

In the vocabulary of Section 6.10, simulations in which characters evolve under an Mk model and are reconstructed under the same model test only parameter recovery; reconstructions are far more often wrong because the model is misspecified than because the algorithm fails. There is a small but instructive literature that tests ASR against known histories. In biology, Hillis et al. (1992) generated a known phylogeny by serial propagation of bacteriophage in the laboratory, and Oakley and Cunningham (2000) used the same system to show that reconstruction of continuous ancestral states was much less accurate than estimation of evolutionary correlations from the same data. In linguistics, the natural test cases are families with a documented ancestor: reconstructing Latin from the Romance languages, Sanskrit or Middle Indo-Aryan from modern Indo-Aryan languages, or Classical Chinese from modern Sinitic varieties. Such tests are a recurring theme in work on automated reconstruction (Section 12.12) and in evaluations of lexical ASR (Section 12.9).

Attested ancient languages are valuable for ASR in a second way. Biologists have shown that including fossil observations at internal positions in a tree improves the accuracy of ancestral reconstructions of continuous traits (Slater et al., 2012). The linguistic analogue is to include ancient languages as tips (or as sampled ancestors; Chapter 10) whenever they are available, rather than reconstructing the deep history from modern languages alone.

12.9 Reconstructing the proto-lexicon

The most direct linguistic application of ASR is to the vocabulary of proto-languages. There are two ways to pose the question, corresponding to the two ways of coding lexical data described in Chapter 3.

Semasiological reconstruction: was this word present? Under binary cognate coding, each character is a cognate set, and its state in a language is 1 if the language has a reflex of the set (in the relevant meaning) and 0 otherwise. ASR then gives, for each cognate set, the probability that it was present in a proto-language. Under the stochastic Dollo model (Chapter 7), each cognate set is born exactly once on the tree, so a set present in two primary branches must have been present at the root; the reconstruction of the proto-lexicon under such a model follows almost directly from the inferred tree and the pattern of presence. Under reversible binary models, in contrast, a cognate set present in two primary branches might instead have been gained twice, and the root reconstruction depends on the relative rates of gain and loss.

Onomasiological reconstruction: which word expressed this meaning? Under multistate coding, each character is a meaning slot (say ‘hand’) and its states are the cognate classes used for

that meaning. ASR then gives the probability that each cognate class was the word for ‘hand’ in the proto-language. This is the question the comparative method typically answers when it lists a reconstructed proto-form with its gloss.

The two framings can give inconsistent answers. If presence/absence characters for the competing cognate classes of a single meaning are reconstructed separately, nothing forces their probabilities to sum to one: the reconstruction may imply that the proto-language had no word for ‘hand’ or several. The multistate framing builds in the constraint that one word fills the slot, though it has its own unrealistic assumption that a language has exactly one word per meaning (synonymy and polysemy are common). The R section demonstrates the inconsistency on a toy example.

Jäger and List (2018) made the first systematic evaluation of ASR methods for onomasiological reconstruction. They compared parsimony (including Sankoff parsimony with different coding schemes) and likelihood-based methods with binary and multistate coding, applying them to multilingual wordlists from families for which expert reconstructions of the proto-language vocabulary exist, and they scored each method by how often it recovered the expert’s choice of word for each meaning. The study showed that ASR can recover a substantial proportion of expert onomasiological reconstructions automatically, and that both the method and the coding choice affect the result. It also clarified the conceptual difference between asking whether a cognate set existed and asking which cognate set filled a meaning slot, which earlier work had often blurred. The design is a test of agreement with the comparative method rather than of truth: the cognate sets fed to the algorithms and the expert reconstructions used to score them come from the same scholarly tradition, so agreement shows that ASR reproduces expert reasoning from the distribution of cognates, not that both are right.

Lexical ASR has been used, implicitly or explicitly, in many studies that use language trees to reconstruct culture. If a word for ‘pig’, ‘taro’ or ‘canoe outrigger’ can be reconstructed with high probability for a node, the referent was presumably known to its speakers. This is linguistic paleontology (Chapter 11) given a probabilistic footing, and it inherits the same problem of semantic change: the probability that a cognate set existed at a node is not the probability that it had the same meaning there. The Austronesian Basic Vocabulary Database (Greenhill et al., 2008) and comparative dictionaries such as Blust and Trussel’s Austronesian Comparative Dictionary are the obvious resources for such work, because their cognate sets can be compared with the reconstructions of specialists.

12.10 Reconstructing grammar

Structural features, such as basic word order, the presence of grammatical gender, the alignment of case marking or the position of adpositions, are natural targets for ASR. Their states are discrete and well defined in typological databases such as WALS and Grambank (Chapter 3), and the questions are of wide interest: was Proto-Indo-European verb-final? Did the ancestor of all languages have SOV order?

12.10.1 Basic word order

Gell-Mann and Ruhlen (2011) assembled a tree of the world’s languages from published classifications and argued, from the distribution of basic word orders across it, that the ancestral order of all (or nearly all) attested languages was SOV, with subsequent drift toward SVO and from there to VSO and VOS. Their reconstruction relied on informal, essentially parsimony-like reasoning over

a deep and largely conjectural tree, and it was criticized for that reason. It nevertheless posed the question in a form that probabilistic ASR can address.

Maurits and Griffiths (2014) addressed it with Bayesian phylogenetic methods. They modeled basic word order (the six orders of subject, object and verb) as a character evolving under a continuous-time Markov chain, with a single set of transition rates shared across many language families, and they integrated over trees for each family generated from published classifications combined with priors on branch lengths. Pooling families allowed the rates to be estimated from far more transitions than any one family provides. They then computed posterior distributions over ancestral word orders at the roots of the families. Their results favored SOV as the ancestral order in many of the families examined, broadly in line with the qualitative claim of Gell-Mann and Ruhlen (2011), but with considerable uncertainty that depended on the family and on the assumptions about the trees. The study is a model of careful ASR: it is explicit about the trees, it pools information across families by estimating a single set of transition rates shared by all of them (a choice its authors supported by model comparison), and it reports uncertainty rather than a single answer.

The approach of Maurits and Griffiths (2014) assumes that word order evolves under the same process in every family. The opposite assumption, that each family has its own process, underlies the analysis of Dunn et al. (2011), who fitted separate models of correlated evolution between word-order features in four families and found that the dependencies between features differ across families (Chapter 13). Between these two extremes lie hierarchical models in which family-specific rates are drawn from a shared distribution, an approach that, to our knowledge, has rarely been applied to ancestral word order but that is standard in other areas of Bayesian statistics.

12.10.2 Structural phylogenetics

In Island Melanesia, the Papuan (non-Austronesian) languages of the Solomon Islands and the Bismarck Archipelago share almost no demonstrable cognate vocabulary, and the comparative method cannot establish whether they form one or several families. Dunn et al. (2005) proposed *structural phylogenetics*: coding over one hundred abstract structural features (properties of phonology, morphology and syntax) for these Papuan languages and for neighboring Oceanic (Austronesian) languages, and analyzing them with maximum parsimony and the NeighborNet network method. For the Oceanic languages, whose subgrouping is known from the comparative method, the structural tree recovered much of the accepted structure, which served as a check on the method. For the Papuan languages, the structural tree suggested a deep genealogical signal of the kind that lexical methods could not detect, though the authors were careful to note that structural similarity can also reflect contact. A longer methodological treatment (Dunn et al., 2008) explored the sensitivity of the results to coding and methods. Structural phylogenetics is relevant to ASR in two ways. The structural trees provide a scaffold for reconstructing the grammars of ancestral nodes. And the approach raises the question of how fast and how independently structural features change, which ASR on trees with known structure can help answer. A related analysis by Reesink et al. (2009) used population-genetic clustering (STRUCTURE) on structural features of the languages of Sahul and is discussed in Chapter 16.

12.10.3 Indo-European morphosyntax

Indo-European offers the richest test bed for grammatical ASR, because its early history is documented in Hittite, Vedic Sanskrit, Ancient Greek, Latin and other ancient languages, and because

its proto-grammar has been reconstructed by the comparative method in great detail. Several studies have combined dated Indo-European trees with grammatical features to reconstruct the history of particular constructions. Widmer et al. (2017) studied the recursion of noun phrases (whether modifiers such as possessors can themselves contain further modifiers in the same construction type) and argued, from Bayesian phylogenetic analyses of 55 Indo-European languages, that although particular NP construction types can gain or lose the ability to embed recursively within a few centuries, it is extremely unlikely that recursive NP embedding was ever lost altogether at any stage of the family: when one construction type becomes unavailable for recursion, an unconstrained alternative is very likely to develop. Recursion in this sense behaves as a strong diachronic bias, which the authors suggest may be universal. Carling and Cathcart (2021) reconstructed the evolution of a large set of Indo-European grammatical features with phylogenetic methods and compared their reconstructions of Proto-Indo-European grammar with those of the comparative tradition, finding substantial agreement for many features and informative disagreements for others. Here too, agreement is partly built in: the feature codes for ancient languages and the comparative reconstructions draw on the same philological analyses, so the disagreements are at least as informative as the agreements. Cathcart et al. (2018) showed that areal (contact) pressure in addition to inheritance shapes the evolution of grammatical features in Indo-European, a reminder that ASR under a purely vertical model can mistake contact-induced convergence for inheritance. Calude and Verkerk (2016) studied the typology and diachrony of higher numerals across Indo-European with phylogenetic comparative methods. Goldstein (2022a) estimated that indefinite articles arose faster in Indo-European languages that already had a definite article (Section 13.7). Shirtz et al. (2021) compared models for the evolution of negative existential constructions in 106 Indo-European languages and found that the unidirectional Negative Existential Cycle alone fits worse than models that allow other and multiple transitions between strategies, while noting that the skewed distribution of types in the sample limits what such models can resolve.

A recurring finding of this literature is that grammatical features are not uniformly more stable than vocabulary. Using a large set of structural and lexical characters across Austronesian, Greenhill et al. (2017) found that rates of change of grammatical features vary widely, with some changing faster than basic vocabulary, and that grammatical features are more susceptible to the influence of neighboring languages. The implication for ASR is that structural reconstructions at deep nodes should not be assumed reliable merely because “grammar is conservative”.

Recent studies have tried to model grammatical change more realistically. Ebert et al. (2025) replaced a single word-order feature per language with object-verb order in 46 parallel sentences from a corpus of 36 Indo-European languages. A mixture model grouped the sentence contexts into three classes with distinct dynamics: contexts with nominal objects changed slowly, pronominal objects changed fast, and complement clauses had a strong bias toward VO. Stochastic character mapping reconstructed VO order for nominal objects in Proto-Indo-European, while the fast rates for pronominal objects did not allow a reliable reconstruction at all, a clear illustration of the depth-and-rate limit of Section 12.8. For argument marking, Phillips and Bowern (2022) used a Brownian-motion model to trace splits in ergative marking across Pama-Nyungan and found evidence for an ergative-accusative split traceable to Proto-Pama-Nyungan, and Hong et al. (2026) modelled ergative, non-ergative and split alignment of case marking in over 200 languages from eleven families. Their best model required two evolutionary regimes that cut across families, one favoring non-ergative and one favoring split-ergative alignment, an example of the heterogeneous-rate models of Section 12.7 applied to grammar.

12.11 Reconstructing sound systems and sound change

Sound systems are the domain where the comparative method is most powerful and where phylogenetic ASR has been least used. The comparative method reconstructs proto-phonemes from regular correspondences, using the directionality of sound change (lenition is more common than fortition, palatalization before front vowels is common and the reverse rare) to decide between alternative reconstructions. Several kinds of phylogenetic work bear on this.

Phonological characters on trees. Phoneme inventories can be coded as presence/absence of each segment, and their evolution modeled on trees. The database BDPROTO compiles reconstructed proto-language inventories and has been used to study diachronic trends in inventory structure (Moran et al., 2021). Macklin-Cordes et al. (2021) showed that phonotactic properties (which sequences of sounds are permitted) carry phylogenetic signal in Pama-Nyungan, so that they can in principle be treated as characters for both tree inference and ASR. Note that the presence of a phoneme in an inventory is a character state, not a reconstruction of the sound system: an ASR saying that a node "had /f/" with probability 0.7 does not say which words contained it or by which correspondences it is known.

Sound changes as characters. In the computational cladistics of Ringe et al. (2002) (Chapter 4), sound changes such as Grimm's Law were coded as characters, on the assumption that a complex sound change is very unlikely to happen twice independently. In ASR terms, these are characters with strongly asymmetric (nearly irreversible) rates. They also illustrate the dependence on the comparative method: whether a language underwent Grimm's Law is itself established by comparative reconstruction.

Regular sound change as concerted evolution. Hruschka et al. (2015) proposed a Bayesian phylogenetic model in which sound correspondences across a large set of Turkic cognates are treated as characters that can change together, so that a regular sound change appears as a concerted shift in many characters at once on a single branch. The model detects branches on which such concerted changes occurred, an automated version of the comparative method's identification of sound laws, and it adapts the biological concept of *concerted evolution* (the homogenization of gene copies within a lineage) to the regularity of sound change.

Directional models. The asymmetries that historical phonologists rely on (for example, that /p/ > /f/ is far more common than /f/ > /p/) are naturally expressed as asymmetric rate matrices. Cross-linguistic databases of attested sound changes could in principle supply informative priors for such matrices, a possibility that, to our knowledge, remains largely unexploited. Garrett and Johnson (2013) develop a typology of asymmetric sound changes grounded in biases of motor planning, aerodynamics, gestural mechanics and perceptual parsing; a typology of this kind is what an informative prior on a directional rate matrix would have to encode, and it would also say which zero rates are defensible (Section 12.4.5).

Irregular reflexes and dialect layers. Where sound change looks irregular, probabilistic models can help to separate its sources before any tree-based reconstruction is attempted. Cathcart (2022) applied a Bayesian model based on a hierarchical Dirichlet process to the irregular reflexes of West Iranian (Persian, Kurdish, Balochi and others). He argued that the usual explanation, borrowing from some stage of Persian, is too simple: Persian may not be the only donor, irregularity is not always introduced by lexical borrowing, and some apparent irregularities may reflect different outcomes in different conditioning environments.

How a rare sound spreads: bilabial trills in Malekula. Bilabial trills are among the rarest

speech sounds in the world's languages, yet several Oceanic languages of Malekula Island in Vanuatu have them. King et al. (2025) built a lexical phylogeny of the Malekula languages from the Vanuatu Voices database and then reconstructed, on that phylogeny, the presence of a trill in fourteen separate meanings whose words contain one in at least some languages. Treating each word as its own character, rather than coding "has a trill" once per language, let them follow the sound through the lexicon. They found a surprising degree of dynamism: trills spread gradually through the vocabulary, were frequently lost and regained, and their distribution was consistent with frequent borrowing between neighboring languages. The authors suggest that the low functional load of the trills, and speakers' attachment to them as a marker of identity, explain the rapid change and the borrowing. The study shows how ASR on a lexical tree can address the diffusion of a sound change, a question the comparative method usually treats qualitatively. It also inherits the caveats of this chapter: fourteen characters on one tree of a small island give limited power, and reconstructed gains and losses are conditional on the tree, which may itself have been affected by contact.

12.12 Automated reconstruction of word forms

The most ambitious form of linguistic ASR reconstructs not states of a coded character but the actual phonological shapes of proto-words, the second kind of reconstruction in Section 12.2.

Bouchard-Côté et al. (2013) developed a probabilistic model in which each branch of a language tree carries a stochastic string transducer, a model of sound change that rewrites a parent word form into a child word form by context-dependent substitutions, insertions and deletions. The parameters of the transducers are shared across words on a branch (so that sound change is regular in the model) but can differ between branches, and the unobserved proto-forms at internal nodes are inferred jointly with the parameters by approximate inference. Applied to a large set of Austronesian languages drawn from the Austronesian Basic Vocabulary Database, the system reconstructed proto-forms for hundreds of cognate sets. The automated reconstructions were compared with the manual reconstructions in Blust's comparative work on Austronesian, and a large majority came within a single character edit of the expert's reconstruction. The authors also used the model to test the *functional load hypothesis*, the long-standing idea that sound contrasts that distinguish many words are less likely to be lost through merger, and found support for it. Earlier work by the same group (Bouchard-Côté et al., 2007) had developed the approach on Romance, where the reconstructions can be checked against attested Latin.

Later work has explored neural network models trained to map descendant forms to ancestral forms. Meloni et al. (2021) trained sequence-to-sequence models to reconstruct Latin forms from their Romance reflexes, and Ciobanu and Dinu (2018) addressed a similar task with other machine-learning methods. List et al. (2022b) proposed a framework that reconstructs proto-forms from aligned cognate sets using sound correspondence patterns and trimmed alignments, designed to be fast, transparent and evaluable on datasets in the CLDF format (Chapter 3). The same sound-correspondence logic can be run forwards. Bodt and List (2022) predicted the most likely forms of words missing from their Western Kho-Bwa (Sino-Tibetan) data from the correspondences in related cognate sets, preregistered the predictions, and then compared them with forms elicited in later fieldwork, an unusually direct out-of-sample test of a reconstruction-based model. Lu et al. (2024) reduced the need for training data with a semisupervised architecture that learns from a few cognate sets with known proto-forms and many without, and that builds in a principle of the comparative method: a reconstructed form must be transformable back into its reflexes.

These methods differ from the character-state ASR of the rest of this chapter in an important way. Neural approaches typically require training data consisting of known ancestral forms, so they learn to imitate expert reconstructions (or attested ancestors) and are best seen as tools for extending or checking reconstructions, not as independent discovery procedures. The model of Bouchard-Côté et al. (2013), in contrast, is a generative model of sound change on a tree that needs no proto-forms for training, only the cognate sets and the tree. Its close agreement with expert reconstructions is therefore stronger evidence than agreement from a supervised model. It is still not fully independent: the cognate sets it starts from were established by the same comparative scholarship whose reconstructions it is scored against, and the hardest part of the comparative method, deciding which words are cognate, has been done in advance. The Romance case, where Latin is attested, is the cleaner test. To our knowledge, joint inference of trees, cognate sets, alignments and proto-forms in a single model, the linguistic equivalent of statistical alignment in biology (Redelings and Suchard, 2005), remains an open goal. A partial step is the TKF91-based model of Goldstein et al. (2026), which infers trees directly from aligned cognate word forms and can be used to study segmental stability and the evolution of phonological inventories.

12.13 ASR and the comparative method

How does phylogenetic ASR relate to the reconstructions of traditional historical linguistics? The comparative method (Fox, 1995; Campbell, 2013; Ringe and Eska, 2013) reconstructs proto-forms by (1) establishing cognate sets, (2) identifying recurrent sound correspondences, (3) reconstructing for each correspondence set the proto-sound that best explains its reflexes, using knowledge of the directionality and naturalness of sound changes, and (4) assembling the reconstructed sounds into proto-words. Grammar is reconstructed by analogous reasoning on paradigms and constructions, with internal reconstruction (the use of irregularities within one language) as a complementary tool.

Skirgård (2024) made the comparison explicit for grammar. Using Grambank data, she compared reconstructions of the grammar of ancestral Oceanic languages from the historical-linguistic literature with reconstructions by maximum parsimony and maximum likelihood, and found a high degree of agreement, with a tendency for classical reconstruction to ignore branch lengths. She argued that taking branch lengths into account is more conceptually sound, and that the most promising path combines computational methods with qualitative knowledge. Several points of contact and contrast stand out; Table 12.2 summarizes them.

- **Both are tree-based, but differently.** The comparative method uses the tree mainly through the principle that a feature attested in two primary branches is likely to be inherited, which is a parsimony argument. Phylogenetic ASR uses the full topology and the branch lengths.
- **The comparative method uses directionality heavily, ASR rarely does.** The comparative method's knowledge of what is a likely sound change is, in modeling terms, a strongly informative prior on an asymmetric rate matrix. Most phylogenetic ASR uses symmetric or uninformative models. Bringing typological knowledge of change into ASR through informative priors is a natural meeting point.
- **ASR quantifies uncertainty; the comparative method rarely does.** Reconstructions marked with a question mark, or alternative reconstructions discussed in the text, are the comparative method's expression of uncertainty. ASR provides probabilities, but probabilities that are conditional on a model whose adequacy has to be argued.

- **They work at different levels.** The comparative method reconstructs forms, sounds and constructions; most phylogenetic ASR reconstructs coded states of characters (Section 12.2). Automated reconstruction of word forms (Section 12.12) begins to close the gap.
- **The input to ASR is the output of the comparative method.** Cognate judgments, the basis of lexical phylogenetics, are themselves made by the comparative method (or by automated approximations to it; Chapter 3). Phylogenetic ASR does not replace the comparative method, it builds on it.

The last point has a consequence for how agreement is read. When a phylogenetic reconstruction agrees with the comparative method's reconstruction of the same proto-language, as in the evaluations of Jäger and List (2018), Carling and Cathcart (2021) and Bouchard-Côté et al. (2013), the agreement shows that the two procedures extract the same answer from overlapping evidence. It is not independent confirmation, because the cognate sets, the subgrouping that shapes the tree and often the coding of ancient languages come from the comparative tradition itself, and because cognate sharing is not the same as shared innovation. The most informative cases are the disagreements, which point either to a weakness of the model (contact, rate heterogeneity, the coding) or to a place where the comparative reconstruction rests on an assumption that the distributional evidence does not support.

Frequency heuristics, explicit trees and tree samples. When the tree is uncertain, Indo-Europeanists have often decided whether a form goes back to the proto-language by counting. A long-standing *lower-bound heuristic*, already in Meillet's *Introduction*, accepts a word as Proto-Indo-European if it is attested in at least three (preferably non-contiguous) branches and borrowing is not suspected; a *majority-rule heuristic* reconstructs the state found in most daughters; an *Anatolian criterion* asks for attestation in Anatolian and at least one other branch. Goldstein (2022b) argues that all three amount to reconstruction on a consensus tree with an unresolved node, and that this cannot work. On each fully resolved tree compatible with the consensus, the reconstruction is fixed by the topology and the optimality criterion, not by the count of daughters. The lower-bound rule can only place a form at the root and ignores the possibility that it arose at a shallower node; it treats forms attested in one branch as innovations, although any of them could be an archaism; and the threshold of three has no stated justification. Hale (2007) makes a related argument against majority rule: if a flat tree stands for a set of equally possible subgroupings, the number of daughters with a feature is no evidence about the proto-language, and counting daughters lets the sociopolitical status of varieties, which decides what gets documented as a separate "language", influence the reconstruction.

Olander (2018) proposes a *cladistic approach* instead. A feature shared by several subgroups is first reconstructed only to their lowest common node in an explicit, binary tree, and pushing it further back requires a separate argument that it was lost in the other lineages. On his working tree, the augment of Greek, Armenian and Indo-Iranian becomes an innovation of the node that unites these branches with Balto-Slavic, followed by loss in Balto-Slavic. A feature attested only in a deeply embedded subgroup, such as Baltic, becomes an unlikely archaism, because many independent innovations would be needed elsewhere. The approach makes reconstructions explicit, but it makes them conditional on one tree. Goldstein's alternative is Bayesian ASR over a sample of trees (Section 12.5). Reconstructing presence and absence over 100 trees from an analysis in the framework of Chang et al. (2015), he obtained a posterior probability above 0.95 for the conjunction **k^{w}e* 'and' at the root, in line with the traditional reconstruction, but only 0.56 for the *absence* of the augment, which leaves its Proto-Indo-European status open (compare Section 12.4.5). Both results are conditional on a binary presence/absence coding and a simple Markov

model, and the augment result shows what the heuristics conceal: uncertainty in the tree becomes uncertainty in the reconstruction.

Table 12.2: Ancestral state reconstruction in biology and the comparative method in linguistics.

Aspect	Phylogenetic ASR	Comparative method
Unit of reconstruction	Coded character states, sometimes sequences	Sounds, word forms, morphemes, constructions
Use of the tree	Full topology and branch lengths	Mainly primary branching and subgrouping
Model of change	Explicit CTMC, usually symmetric	Implicit, strongly directional knowledge of likely changes
Uncertainty	Posterior probabilities (model-conditional)	Qualitative (question marks, alternatives)
Contact	Usually ignored	Identified by irregular correspondences, cultural knowledge
Validation	Simulation, known histories	Attested ancestors, internal consistency

12.14 From the biologist's toolbox

Ancestral sequence resurrection. Molecular biologists reconstruct ancestral protein sequences and then synthesize and test them in the laboratory, to study the functional properties of ancient molecules (Thornton, 2004). The linguistic analogue of resurrection would be the experimental testing of reconstructed proto-forms (for example, psycholinguistic tests of how reconstructed proto-words are perceived), which is unusual but not unthinkable. More practically, the statistical methods used in molecular ASR, including joint reconstruction with insertions and deletions and reconstruction that accounts for alignment uncertainty, are directly relevant to automated reconstruction of word forms.

The threshold model. Many linguistic features are discrete manifestations of an underlying continuum: a language is coded as "having" a phonemic tone contrast or not, as "having" grammatical gender or not, but the underlying systems vary continuously in their extent. The *threshold model* of quantitative genetics treats a discrete character as determined by an unobserved continuous *liability* that evolves under Brownian motion; the character changes state when the liability crosses a threshold (Felsenstein, 2012). ASR under the threshold model reconstructs the liability at internal nodes (Revell, 2014), and the model naturally produces the kind of "lineage-specific propensity" seen in linguistic features (lineages near the threshold flip often, lineages far from it rarely). To our knowledge, the threshold model has rarely if ever been applied to linguistic features; Sections 13.9.3 and 16.5.2 discuss it further.

Structured Markov models. For characters whose states are logically hierarchical (a feature can only take a value if another feature is present: a language can only have gender agreement on verbs if it has gender), naive coding as independent characters is incoherent. Tarasov (2019) developed *structured Markov models* that combine dependent characters into a single rate matrix respecting these logical dependencies. Grambank-style data, with many logically dependent features, would benefit directly.

Model averaging by reversible jump. Rather than choose one rate model for a character, BayesTraits samples over all ways of setting rates equal or to zero by reversible-jump MCMC

(Pagel and Meade, 2006), so that ASR averages over model uncertainty. This is available to linguists and used occasionally, but much ASR in the linguistic literature still reports reconstructions from a single chosen model.

Fossils and sampled ancestors. Integrating fossil observations improves ancestral reconstruction of continuous traits (Slater et al., 2012). Attested ancient languages placed as tips or sampled ancestors play the same role for linguistic characters, and should be included whenever available.

Polymorphic characters. Languages often have more than one value for a feature: several basic word orders with no dominant one, or two competing words for the same meaning. Biologists model polymorphism explicitly as an intermediate state (for example, the polymorphic Mk models of **phytools**, `fitpolyMk()`, described in Revell and Harmon (2022)). Coding "no dominant order" or synonymy as polymorphism rather than as missing data, or as an arbitrary choice of one state, would be more faithful to the data.

12.15 Ancestral state reconstruction in R

We work with a toy dataset: a hand-built, ultrametric tree of fourteen hypothetical languages in four subgroups (A to D), with branch lengths in millennia and a root age of 5,000 years, and the basic word order of each language (SOV, SVO or VSO). The data are invented, but the pattern is realistic: subgroup A is mostly verb-final, subgroups B and D mostly SVO, and subgroup C mostly VSO (Figure 12.1, which also shows the node numbers used below).

```
library(phytools)
library(phangorn)
tree <- read.tree(text = paste0(
  "((((A1:1,A2:1):1.5,(A3:1.2,A4:1.2):1.3):1,",
  "((B1:0.8,B2:0.8):1.2,B3:2):1.5):1.5,",
  "(((C1:1,C2:1):1,C3:2):1.5,((D1:0.5,D2:0.5):2,(D3:1.5,D4:1.5):1):1):1.5);"))
wo <- c(A1 = "SOV", A2 = "SOV", A3 = "SOV", A4 = "SVO", B1 = "SVO", B2 = "SVO",
        B3 = "SOV", C1 = "VSO", C2 = "VSO", C3 = "SVO", D1 = "SVO", D2 = "SVO",
        D3 = "SOV", D4 = "SVO")
wo <- factor(wo[tree$tip.label])
cols <- setNames(c("#1b9e77", "#d95f02", "#7570b3"), levels(wo))
plotTree(tree, offset = 0.6, fsize = 0.9)
tiplabels(pie = to.matrix(wo, levels(wo)), piecol = cols, cex = 0.4)
nodelabels(frame = "none", cex = 0.7, adj = c(1.2, -0.4))   # node numbers
legend("topleft", legend = levels(wo), pch = 21, pt.bg = cols, pt.cex = 1.5,
       bty = "n")
```

12.15.1 Parsimony

phangorn computes the parsimony score and the most-parsimonious reconstruction (MPR) sets at every node.

```
pd <- phyDat(matrix(as.character(wo), dimnames = list(names(wo), "wo")),
             type = "USER", levels = levels(wo))
pscore <- parsimony(tree, pd)
pscore
```

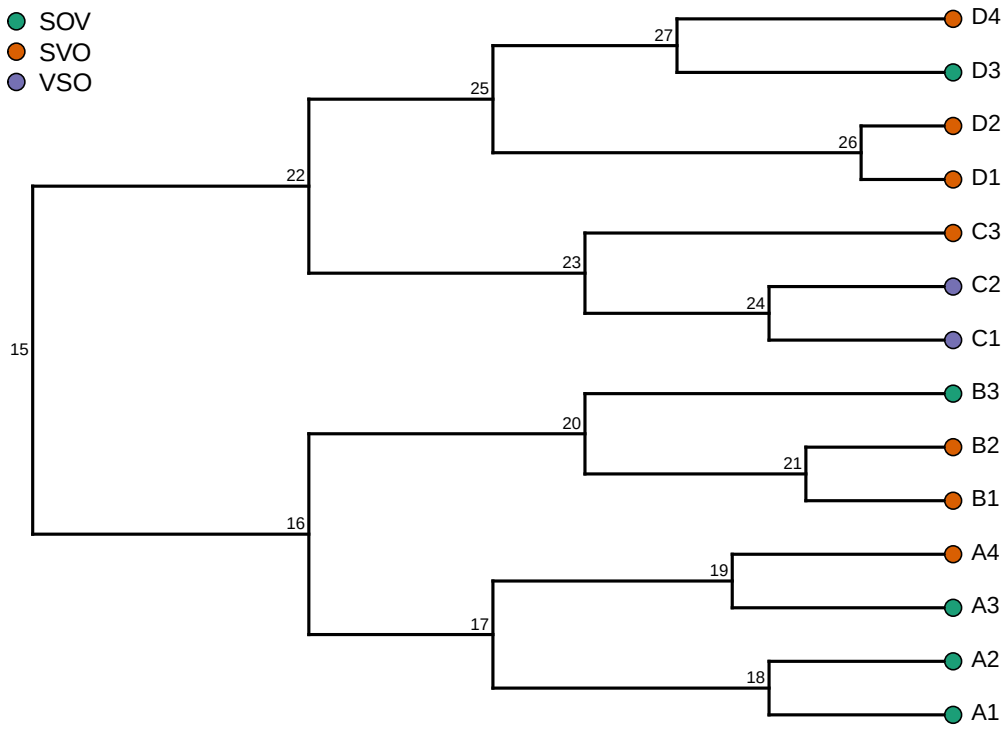


Figure 12.1: A toy tree of fourteen hypothetical languages in four subgroups with their basic word orders. Branch lengths are in millennia.

```
#> [1] 5
```

```
mpr <- ancestral.pars(tree, pd, type = "MPR")
root <- Ntip(tree) + 1
round(t(sapply(root:(root + 3), function(v) mpr[[v]][1, ])), 2)
```

```
#>      [,1] [,2] [,3]
#> [1,]  0.5  0.5    0
#> [2,]  0.5  0.5    0
#> [3,]  0.5  0.5    0
#> [4,]  1.0  0.0    0
```

The minimum number of changes is five. At the root, parsimony cannot choose between SOV and SVO (each receives weight 0.5 in the MPR sets), and it says nothing about how likely either is.

12.15.2 Fitting Mk models

We fit equal-rates, symmetric and all-rates-different Mk models with `fitMk()` and compare them by AIC (Table 12.3).

```
fits <- list(ER  = fitMk(tree, wo, model = "ER"),
             SYM = fitMk(tree, wo, model = "SYM"),
             ARD = fitMk(tree, wo, model = "ARD"))
knitr::kable(data.frame(
  model = names(fits),
```

```
  k     = sapply(fits, function(f) attr(logLik(f), "df")),
  logL  = round(sapply(fits, logLik), 3),
  AIC   = round(sapply(fits, AIC), 2)), row.names = FALSE,
  caption = "Mk models of word-order change fitted to the toy data.")
```

Table 12.3: Mk models of word-order change fitted to the toy data.

model	k	logL	AIC
ER	1	-13.76	29.53
SYM	3	-12.36	30.72
ARD	6	-11.97	35.94

```
fitER <- fits$ER
Q <- unclass(as.Qmatrix(fitER))[levels(wo), levels(wo)]
round(Q, 3)
```

```
#>        SOV    SVO    VSO
#> SOV -0.343  0.171  0.171
#> SVO  0.171 -0.343  0.171
#> VSO  0.171  0.171 -0.343
```

With fourteen languages, the simplest model wins: the data cannot support six separate rates. This is typical of single-character analyses on small trees and is a reason to pool information across families, as Maurits and Griffiths (2014) did.

12.15.3 Marginal reconstruction with ape and phytools

We now compare the marginal reconstructions at every node from `ancr()` and from `ace()` with its default settings, and the output of `ace()` with `marginal = TRUE`, at node 16, the ancestor of subgroups A and B.

```
marg <- ancr(fitER)$ace
ace_default  <- ace(wo, tree, type = "discrete", model = "ER")$lik.anc
ace_marginal <- ace(wo, tree, type = "discrete", model = "ER",
                    marginal = TRUE)$lik.anc
node <- 16                                      # ancestor of subgroups A and B
rbind(ancr_marginal      = marg[as.character(node), ],
      ace_default        = ace_default[node - Ntip(tree), ],
      ace_marginal_TRUE  = ace_marginal[node - Ntip(tree), ]) |> round(3)
```

```
#>                     SOV   SVO   VSO
#> ancr_marginal     0.494 0.367 0.139
#> ace_default       0.494 0.367 0.139
#> ace_marginal_TRUE 0.533 0.324 0.143
```

```
round(marg[1, ], 3)                             # the root
```

```
#>   SOV   SVO   VSO
#> 0.352 0.418 0.230
```

The `ancr()` and default `ace()` results agree: these are the marginal posteriors. The `marginal = TRUE` result is different. It equals the scaled conditional likelihoods of node 16 given only the languages below it (Section D.2.2 verifies this by hand), which ignore the information from subgroups C and D on the other side of the root and therefore overstate the support for SOV at node 16. This is the caution of Section 12.4.2 made concrete.

12.15.4 Joint reconstruction

`ancr(..., type = "joint")` returns the most probable joint assignment of states to all nodes. We compare it, node by node, with the state that has the highest marginal probability.

```
jt <- ancr(fitER, type = "joint")
nodes <- as.character(root:(root + tree$Nnode - 1))
data.frame(node = nodes,
           joint = as.character(jt$ace[nodes]),
           marginal_best = colnames(marg)[apply(marg, 1, which.max)],
           marginal_prob = round(apply(marg, 1, max), 2), row.names = NULL)
```

```
#>    node joint marginal_best marginal_prob
#> 1    15   SVO           SVO          0.42
#> 2    16   SVO           SOV          0.49
#> 3    17   SVO           SOV          0.64
#> 4    18   SOV           SOV          0.94
#> 5    19   SVO           SOV          0.58
#> 6    20   SVO           SVO          0.49
#> 7    21   SVO           SVO          0.94
#> 8    22   SVO           SVO          0.52
#> 9    23   SVO           VSO          0.46
#> 10   24   VSO           VSO          0.90
#> 11   25   SVO           SVO          0.64
#> 12   26   SVO           SVO          0.99
#> 13   27   SVO           SVO          0.59
```

```
c(joint_logP = as.numeric(jt$logLik), logL = as.numeric(logLik(fitER)))
```

```
#> joint_logP       logL
#>     -17.45     -13.76
```

Compare the joint and marginal columns. At the ancestor of subgroup A (node 17), the state with the highest marginal probability is SOV, at about 0.64, but the single most probable joint history assigns SVO there. That history makes the tree as a whole more probable, even though SOV is the better bet for node 17 considered alone. The same happens at nodes 16 and 19, and at node 23 (the ancestor of subgroup C) for VSO. Note also that the joint log-probability (about -17.4) is much lower than the log-likelihood (about -13.8): the latter sums over all histories, the former is the probability of one history.

12.15.5 Root priors

How much does the root prior matter here? We refit the ER model with a flat root prior and with the FitzJohn prior.

```
root_with <- function(pi) ancr(fitMk(tree, wo, model = "ER", pi = pi))$ace[1, ]
rbind(flat = root_with("equal"), fitzjohn = root_with("fitzjohn")) |> round(3)
```

```
#>            SOV   SVO   VSO
#> flat     0.352 0.418 0.230
#> fitzjohn 0.347 0.530 0.123
```

Under the ER model the stationary distribution is flat, so the stationary prior would coincide with the flat prior. The FitzJohn prior sharpens the reconstruction toward the state that is already favored, SVO. With an asymmetric rate matrix the differences can be much larger.

12.15.6 Stochastic character mapping

We sample 200 stochastic maps under the fitted ER model and summarize them (Figure 12.2).

```
set.seed(12)
pi_flat <- rep(1 / 3, 3)
maps <- make.simmap(tree, wo, Q = Q, pi = pi_flat, nsim = 200, message = FALSE)
summ <- describe.simmap(maps)
round(colMeans(summ$count), 2)        # expected number of changes by type
```

```
#>       N SOV,SVO SOV,VSO SVO,SOV SVO,VSO VSO,SOV VSO,SVO
#>   10.97    2.63    1.03    2.71    1.44    1.31    1.85
```

```
round(summ$ace[1:4, ], 2)             # node frequencies, first four nodes
```

```
#>     SOV  SVO  VSO
#> 15 0.32 0.43 0.26
#> 16 0.50 0.39 0.12
#> 17 0.66 0.26 0.09
#> 18 0.96 0.03 0.02
```

```
par(mfrow = c(1, 2))
plot(maps[[1]], colors = cols, fsize = 0.8, ftype = "i")
# plotSimmap leaves zero margins; restore room for the axes
par(mar = c(4.5, 4.5, 1, 1))
hist(summ$count[, "N"], breaks = seq(-0.5, max(summ$count[, "N"]) + 0.5, 1),
     main = "", xlab = "Total number of changes", col = "gray80",
     border = "white")
abline(v = pscore, lty = 2)
```

The frequencies of states at nodes across maps approximate the marginal posteriors computed above, as they should, since the maps are sampled from the same posterior. The expected total number of changes, about eleven, is more than twice the parsimony minimum of five, because the model allows changes that are later reversed and multiple changes on long branches. Asking how often SOV became SVO, and how often the reverse happened, is now a matter of reading off the columns of `summ$count`.

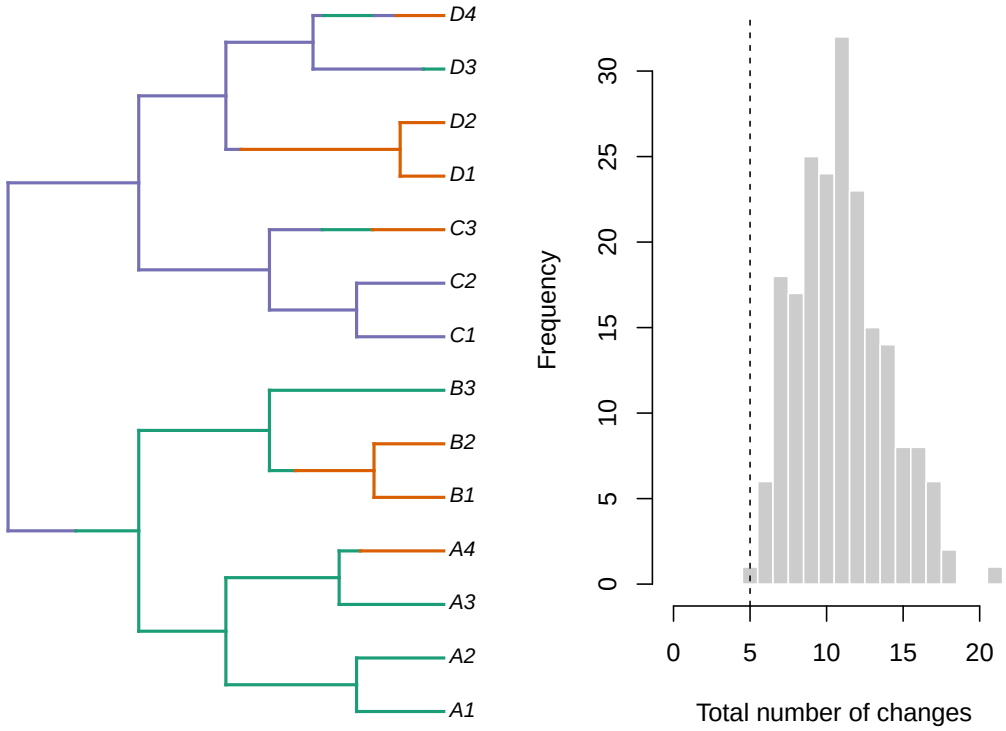


Figure 12.2: One stochastic character map of word order sampled under the fitted equal-rates model (left), and the posterior distribution of the total number of changes across 200 maps (right). The dashed line marks the parsimony minimum of five changes.

12.15.7 Reconstructing a named ancestor over a sample of trees

In a real analysis the tree is uncertain. Suppose the lexical data leave the position of language B3 unresolved: in 60% of the posterior trees it is the sister of B1 and B2, as in the toy tree, and in 40% it branches off the stem of subgroup A instead. We build a mock posterior sample of 100 trees with this mixture, add lognormal noise to every branch length, and ask the question a linguist would ask: what was the word order of Proto-B, the ancestor of B1, B2 and B3? Following Pagel et al. (2004) we reconstruct, on each tree, the state at the MRCA of those three languages, and we record whether they form a clade. We then compare the answer with the one obtained on the MCC tree of the sample (Figure 12.3).

```
set.seed(3)
A <- paste0("A", 1:4); B <- c("B1", "B2", "B3")
alt <- drop.tip(tree, "B3")                              # B3 moved next to A
alt <- bind.tip(alt, "B3", where = getMRCA(alt, A), position = 0.5)
trees <- lapply(1:100, function(i) {
  t2 <- if (runif(1) < 0.6) tree else alt
  t2$edge.length <- t2$edge.length * rlnorm(nrow(t2$edge), 0, 0.2)
  t2 })
class(trees) <- "multiPhylo"
mcc <- maxCladeCred(trees)
recon <- function(t2) {
  a <- ace(wo[t2$tip.label], t2, type = "discrete", model = "ER")$lik.anc
  c(B_clade = is.monophyletic(t2, B),
    protoB_SOV = unname(a[getMRCA(t2, B) - Ntip(t2), "SOV"]), a[1, ])
}
res <- t(sapply(trees, recon))
on_mcc <- recon(mcc)
round(rbind(MCC_tree = on_mcc, mean_over_trees = colMeans(res)), 3)
```

```
#>                 B_clade protoB_SOV   SOV   SVO   VSO
#> MCC_tree           1.00      0.348 0.345 0.419 0.235
#> mean_over_trees    0.65      0.444 0.355 0.431 0.215
```

```
round(tapply(res[, "protoB_SOV"], res[, "B_clade"], mean), 3)
```

```
#>     0     1
#> 0.511 0.408
```

```
par(mfrow = c(1, 2), mar = c(4.5, 4.5, 1, 1))
stripchart(res[, "protoB_SOV"] ~ factor(res[, "B_clade"], c(1, 0),
                                        c("B a clade", "B not a clade")),
           method = "jitter", pch = 16, col = "gray40", vertical = TRUE,
           ylab = "P(SOV) at MRCA of B1, B2, B3", ylim = c(0, 1))
abline(h = on_mcc["protoB_SOV"], col = "red", lwd = 2)
boxplot(res[, levels(wo)], col = cols, ylab = "P(root state)", ylim = c(0, 1))
```

The MCC tree contains the majority topology, in which subgroup B is a clade, and it reports a single probability of SOV for Proto-B. Over the sample, three things are hidden by that single number. First, Proto-B exists as a clade in only about two-thirds of the trees; in the others, the

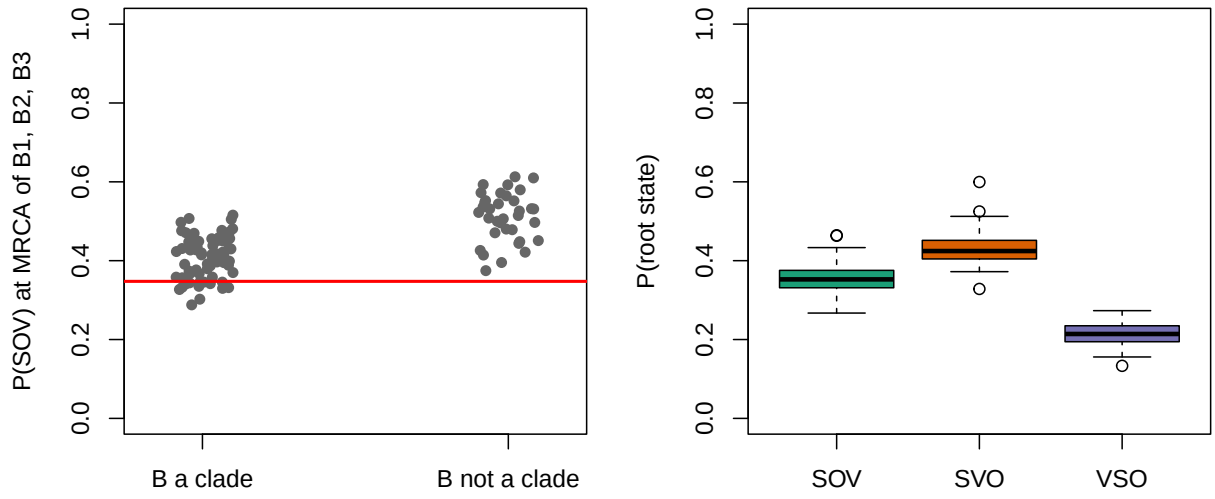


Figure 12.3: Reconstructions over a mock posterior sample of 100 trees in which language B3 is placed with subgroup B in 60% of trees and with subgroup A in 40%. Left: probability of SOV at the MRCA of B1, B2 and B3 on each tree, by whether the three form a clade; the red line is the value on the MCC tree. Right: marginal probability of each word order at the root across the 100 trees.

MRCA of the three B languages is the ancestor of subgroups A and B together, which is a different proto-language, and the reconstruction there answers a different question. Second, the probability of SOV is systematically higher in those trees, because B3 is then a close relative of the SOV-speaking subgroup A, so the average over trees differs from the MCC answer. Third, even among trees with the same topology, branch-length uncertainty spreads the reconstruction over a wide range. The root reconstruction varies less, but still noticeably. A reconstruction reported from a single summary tree would hide all of this; reporting the reconstruction over trees together with the probability that the named group is a clade does not.

12.15.8 Hidden rates

A hidden-rate model with two rate categories can be fitted with `fitHRM()`. We compare it with the ordinary ER model.

```
invisible(capture.output(
  hrm <- fitHRM(tree, wo, model = "ER", ncat = 2, niter = 2)))
c(ER_logL = as.numeric(logLik(fitER)), HRM_logL = as.numeric(logLik(hrm)),
  ER_AIC = AIC(fitER), HRM_AIC = AIC(hrm))
```

```
#>  ER_logL HRM_logL   ER_AIC  HRM_AIC
#>   -13.76   -13.76    29.53    33.53
```

The hidden-rate model does not improve the likelihood, so its extra parameters are not justified (its AIC is worse by about four units): with fourteen languages there is no detectable rate heterogeneity. The same analysis on a large tree with stable and labile lineages could tell a different story. For hidden Markov models with more structure, the **corHMM** package is the standard tool. The

call below fits a comparable model with two rate categories and sums the marginal reconstructions over the hidden categories. In this fit all the probability falls in one rate category, so the summed reconstructions coincide with those of the ordinary equal-rates model: the second category is not used, which is another way of seeing that the data contain no signal of rate heterogeneity.

```
library(corHMM)
dat <- data.frame(lang = names(wo), wo = as.character(wo))
fit_hmm <- corHMM(phy = tree, data = dat, rate.cat = 2, model = "ER",
                  node.states = "marginal")
```

```
#> State distribution in data:
#> States:  1   2   3
#> Counts:  5   7   2
#> Beginning thorough optimization search -- performing 0 random restarts
#> Finished. Inferring ancestral states using marginal reconstruction.
```

```
fit_hmm$loglik
```

```
#> [1] -13.76
```

```
# fit_hmm$states has one column per (observed state, rate category) pair,
# (1,R1), (2,R1), (3,R1), (1,R2), ...; states are numbered in sorted order.
# Sum over the two rate categories:
n_obs <- ncol(fit_hmm$states) / 2
anc <- fit_hmm$states[, 1:n_obs] + fit_hmm$states[, n_obs + 1:n_obs]
colnames(anc) <- sort(unique(dat$wo))
round(head(anc, 3), 3)   # the first row is the root
```

```
#>        SOV   SVO   VSO
#> [1,] 0.352 0.418 0.230
#> [2,] 0.494 0.367 0.139
#> [3,] 0.642 0.264 0.094
```

12.15.9 Lexical reconstruction: one slot or many characters?

Finally, a lexical example. Suppose the fourteen languages use three cognate classes, labeled k, m and t, for the meaning ‘hand’. We reconstruct the proto-word for ‘hand’ in two ways: as one multistate character (the onomasiological framing), and as three separate presence/absence characters, one per cognate class.

```
hand <- c(A1 = "k", A2 = "k", A3 = "k", A4 = "k", B1 = "m", B2 = "m", B3 = "k",
          C1 = "t", C2 = "t", C3 = "m", D1 = "m", D2 = "t", D3 = "k", D4 = "m")
hand <- factor(hand[tree$tip.label])
multi <- ancr(fitMk(tree, hand, model = "ER"))$ace[1, ]
binary <- sapply(levels(hand), function(cl) {
  b <- setNames(as.integer(hand == cl), names(hand))
  ancr(fitMk(tree, b, model = "ARD"))$ace[1, "1"]
})
round(rbind(multistate = multi, binary_presence = binary), 3)
```

```
#>                       k     m     t
#> multistate      0.414 0.328 0.259
#> binary_presence 0.457 0.500 0.506
```

```
c(sum_multistate = sum(multi), sum_binary = sum(binary))
```

```
#> sum_multistate     sum_binary
#>          1.000          1.462
```

The multistate reconstruction distributes a probability of one across the three candidate words, favoring k. The three binary reconstructions, each fitted separately, each give a probability of around one half that the cognate class was present in the proto-language, and their sum is well above one: taken at face value, they imply that the proto-language most likely had more than one word for ‘hand’. Neither answer is “right” in general, since synonymy exists, but the comparison shows that the framing of the question is part of the model, as Jäger and List (2018) emphasized. Neither answer, either, says what the word for ‘hand’ sounded like: that is the business of the comparative method and of the form-based models of Section 12.12.

- “Reconstruction” covers three operations: the comparative method’s reconstruction of forms and systems, model-based reconstruction of word forms and sound changes, and phylogenetic reconstruction of coded character states. The third presupposes the outputs of the first and reconstructs codes, not languages.
- Ancestral state reconstruction infers the states of characters at internal nodes, or their complete histories, from tip data and a tree. Every reconstruction is conditional on a model.
- Parsimony (Fitch, Sankoff) minimizes changes and is close to the reasoning of the comparative method, but ignores branch lengths and gives no probabilities.
- Under Mk models, marginal reconstruction gives the posterior of each node separately, joint reconstruction gives the single most probable history, and the two can disagree. Check whether software reports marginal posteriors or subtree-only conditional likelihoods.
- Rates fixed at zero, informative priors on the direction of change and the root treatment can settle a reconstruction before the data are consulted. Report them, and do not count agreement with expert reconstructions as support for a constrained model.
- Root priors, tree uncertainty and parameter uncertainty can change deep reconstructions substantially. Reconstruct named ancestors over a sample of trees, and report how often the named group is a clade; a single MCC tree hides both the spread and the cases in which the node does not exist.
- Stochastic character mapping samples complete histories. It gives expected numbers and directions of changes, which typically exceed parsimony counts.
- Hidden-rate, threshold and structured Markov models represent lineage-specific propensities and logical dependencies among features. They are largely untried in linguistics.
- Agreement between phylogenetic and comparative reconstructions is not independent confirmation, because the inputs to phylogenetic ASR come from the comparative method. Disagreements are often more informative.

12.16 Exercises

Conceptual

1. For the Austronesian word for ‘five’, state what each of the three kinds of reconstruction in Section 12.2 would produce, what evidence it would use, and what it would presuppose.
2. Explain why the state with the highest marginal probability at a node need not appear in the most probable joint reconstruction. Which quantity would you report for “the word order of Proto-Oceanic”, and which for “the number of word-order changes in Oceanic”?
3. A paper reports that Proto-X was SOV with posterior probability 0.92, computed on the MCC tree of a Bayesian analysis in which X has a posterior clade probability of 0.55. What further information would you ask for, and how might the answer change?
4. A study finds that phylogenetic reconstructions of Proto-Indo-European grammar agree with the comparative method for 80% of features. Explain why this is not independent confirmation of either, and what kind of evidence would be.
5. Why can a stationary root prior dominate a root reconstruction under an asymmetric rate matrix? Give a linguistic example in which a stationary prior would be inappropriate.
6. Explain the difference between semasiological and onomasiological lexical reconstruction, and why binary presence/absence reconstructions of competing cognate classes need not sum to one.
7. A handbook reconstructs a word for Proto-Indo-European because it is attested in three non-contiguous branches of Nuclear Indo-European. Sketch two fully resolved trees, both compatible with an unresolved Nuclear Indo-European node, on which the same distribution implies (a) a Proto-Indo-European origin with losses and (b) an origin at a shallower node. What does an “Anatolian plus one” rule assume about the topology, and what does it imply for a form attested only in Anatolian?

Computational

1. Refit the toy word-order data with an ordered model in which changes occur only between SOV and SVO and between SVO and VSO (an index matrix passed to `fitMk(model = ...)`). Compare its AIC with the ER model and the marginal reconstruction at the root and at node 16 with `ancr()`.
2. Use `make.simmap()` with `Q = "mcmc"` (sampling rates from their posterior) instead of the fixed ML rates, and compare the distribution of the total number of changes with Figure 12.2. What does the difference show about empirical Bayes reconstruction?
3. In Section 12.15.7, change the proportion of trees in which B3 sits with subgroup B from 0.6 to 0.9 and to 0.4. Record the MCC answer, the mean over trees, and the proportion of trees in which B is a clade. When does the MCC tree change, and how abruptly?
4. Simulate a binary character on the toy tree with `sim.Mk()` under an asymmetric rate matrix and reconstruct the root with `ancr(fitMk(..., pi = "equal"))`, `pi = "fitzjohn"` and the stationary prior (`pi = "estimated"`, which in **phytools** sets the root prior to the stationary distribution of the fitted rate matrix). Repeat 100 times and record how often each prior gives the true root state the highest probability. Which kind of validation, in the vocabulary of Section 6.10, is this?
5. Simulate a character with `sim.Mk()` on a tree in which one subgroup evolves ten times faster than the rest (multiply its branch lengths before simulating, but reconstruct on the original tree). Compare the AIC of `fitMk()` and `fitHRM()`, and the reconstruction at the root under

each.

6. With `pd` and `tree` from Section 12.15.1, compute `ancestral.pars(tree, pd, type = "MPR")` and `ancestral.pars(tree, pd, type = "ACCTRAN")` and draw both with `plotAnc()`. At how many internal nodes does MPR leave more than one state? Compare those nodes with the marginal likelihood reconstruction from `ancr()`, and say which of the three summaries best conveys the uncertainty.

Research

1. You want to reconstruct the presence of grammatical gender at the root of a family of 40 languages, of which three are attested ancient languages. Design the analysis: coding (including languages with marginal gender systems), the rate models and root priors to compare, the treatment of tree uncertainty, the use of the ancient languages, and the checks you would run against the comparative literature, including what you would conclude from disagreements.
2. A project proposes to reconstruct the vocabulary of material culture (crops, tools, animals) for a proto-language to locate its homeland. Write a methods plan that separates the three kinds of reconstruction in this chapter, addresses semantic change and borrowing, and specifies what probability statements the results would and would not support.
3. You are asked whether the Indo-European augment was present in Proto-Indo-European, using a posterior sample of Indo-European trees and a single binary character. Specify the rate models, rate priors and root treatments you would compare (Section 12.4.5), how you would report the sensitivity of the answer to them, and what kind of non-phylogenetic evidence could justify an informative prior on the rates of gain and loss.

12.17 Further reading

- Yang et al. (1995) and Pupko et al. (2000): The foundational papers for empirical Bayes marginal reconstruction and for the joint reconstruction algorithm, written for molecular sequences but fully general.
- Pagel (1999b): A clear introduction to likelihood-based reconstruction of discrete characters and its uncertainty.
- Pagel et al. (2004): Bayesian ASR integrating over trees and parameters, with the "fossilization" approach to testing hypotheses about particular ancestors.
- Huelsenbeck et al. (2003): The key paper on stochastic character mapping of morphological characters. It includes tests of character correlation from mapped histories.
- Beaulieu et al. (2013): Hidden-rate models for discrete characters, with a biological case study that has many linguistic analogues.
- Jäger and List (2018): The best introduction to lexical ASR in linguistics. It covers the semasiological and onomasiological framings and evaluates methods against expert reconstructions.
- Maurits and Griffiths (2014): A careful Bayesian reconstruction of ancestral word order across families. Read it alongside Gell-Mann and Ruhlen (2011).
- Skirgård (2024): A systematic comparison of classical and computational reconstructions of Oceanic grammar, with a clear discussion of what branch lengths add.
- Nunn (2011), chapters 3 and 4: An accessible account of parsimony, likelihood and Bayesian reconstruction of ancestral states, with examples from anthropology.
- Olander (2018) and Goldstein (2022b): Two Indo-Europeanist arguments that reconstruc-

tion cannot be separated from the tree, the first reconstructing on one explicit binary tree, the second over a posterior sample of trees.

- King et al. (2025): ASR applied to the spread of a rare sound through the lexicon, with borrowing between neighbors as the likely mechanism.
- Dunn et al. (2005): Structural phylogenetics of Papuan and Oceanic languages. It shows the promise and the pitfalls of reconstructing deep history from grammar.
- Bouchard-Côté et al. (2013): Automated reconstruction of Austronesian proto-forms with a generative model of sound change. It is the most complete bridge so far between phylogenetic ASR and the comparative method.
- Revell and Harmon (2022): A practical, R-based textbook covering Mk models, ASR, stochastic mapping, hidden rates and threshold models with **phytools**.
- Harmon (2019), chapters 7 and 8: A concise, openly available introduction to the Mk model and its extensions, the treatment of the root, and fitting by maximum likelihood and MCMC.
- Cathcart (2018): A short exposition for linguists of the mathematics behind continuous-time Markov models, pruning, MCMC and stochastic mapping, with a sharp discussion of how constrained models can predetermine ancestral reconstructions.

Chapter 13

Phylogenetic Comparative Methods in Linguistics

Linguistic typology has always been a comparative science. Its central questions are about patterns across languages. Do languages with object-verb order tend to have postpositions? Are tone languages more common in humid climates? Do languages spoken by large populations have simpler morphology? Every such question runs into the same obstacle. Languages are not independent data points, because they descend from common ancestors and borrow from their neighbors. Anthropologists have called this *Galton's problem* since the 1880s. This chapter is about the statistical tools that biologists built to solve the same problem for species, *phylogenetic comparative methods* (PCMs), and about how to use them well in quantitative typology: how to design a language sample, which tree and which branch lengths to use, how to carry tree uncertainty into the results, and what a phylogenetically controlled association can and cannot say about causes.

After reading this chapter you should be able to:

- explain Galton's problem, demonstrate by simulation why ordinary regression and contingency tests on related languages give inflated false positive rates, and say what genealogical sampling does and does not fix;
- design a language sample for a comparative question, and compute the effective number of independent observations that a sample provides under a phylogenetic model;
- describe the Brownian motion, Ornstein-Uhlenbeck and Pagel models of trait evolution, measure phylogenetic signal, and interpret the results as patterns rather than processes;
- fit and interpret phylogenetic regressions (PGLS, phylogenetic logistic regression, phylogenetic mixed models) and Pagel's test for correlated evolution with standard R packages;
- assess the sensitivity of a result to branch lengths (including Glottolog topologies with arbitrary branch lengths) and to tree uncertainty, by repeating the analysis over a sample of trees and pooling the results;
- separate phylogenetic and spatial dependence, and state which confounders a phylogenetic control removes and which it leaves untouched;
- critically read the debates over word-order universals and over environmental and social effects on language structure.

```
suppressPackageStartupMessages({
  library(ape)       # trees, vcv(), pic(), corBrownian(), corPagel(), ace()
```

```
  library(phytools)  # pbtree(), fastBM(), phylosig(), fitPagel(), sim.Mk()
  library(geiger)    # fitContinuous(), rescale()
  library(nlme)      # gls()
  library(caper)     # comparative.data(), pgls(), phylo.d()
  library(phylolm)   # phylolm(), phyloglm(), rbinTrait()
})
```

13.1 Why comparison needs phylogeny

13.1.1 The logic of cross-linguistic correlation

Typologists look for regularities in the distribution of structural features across the languages of the world. The most famous example is the set of word-order generalizations of Greenberg (1963). One of them is Universal 4: "With overwhelmingly greater than chance frequency, languages with normal SOV order are postpositional." Later typologists, above all Dryer (1992), tested and refined these generalizations on much larger samples. The result was a list of *word-order correlations*: pairs of features whose values co-occur more often than chance. Languages with object-verb (OV) order, like Japanese, Turkish or Hindi, tend to have postpositions (*Tokyo ni* 'Tokyo to'). They also tend to put genitives before nouns and relative clauses before the noun they modify. Languages with verb-object (VO) order, like English, Indonesian or Swahili, tend to have prepositions and postnominal relative clauses.

How should we test whether such a correlation is real? The simplest approach is to sample languages, tabulate the two features in a 2×2 contingency table, and apply a χ^2 test or a logistic regression. For a continuous variable, such as the size of a phoneme inventory or the number of speakers, we would use an ordinary least-squares (OLS) regression. Every one of these tests assumes that the observations are *independent and identically distributed*. That is, knowing the value for one language tells us nothing extra about the value for another, beyond what the predictors already say.

For languages this assumption is false, and badly so. The Romance languages are verb-object and prepositional because Latin was (or more precisely, because late spoken Latin was). They did not each arrive at this combination independently. If we count French, Spanish, Portuguese, Italian, Romanian, Catalan and fifty Romance dialects as separate data points, we inflate our apparent sample size enormously. We are counting one historical event many times over. The same is true at deeper levels. Hundreds of Bantu languages share SVO order and prepositions from Proto-Bantu. Hundreds of Austronesian languages share a set of structural features from Proto-Austronesian or Proto-Malayo-Polynesian. Languages also influence each other through contact. The languages of the Balkans, of Mainland Southeast Asia, or of the Ethiopian highlands share structural features that cut across family boundaries. This two-fold dependence, through descent and through diffusion, is the central methodological problem of comparative typology.

13.1.2 Galton's problem

The problem was first stated in anthropology, not linguistics. In 1888 Edward Tylor presented to the Royal Anthropological Institute a statistical study of the "adhesions" between customs across several hundred societies. He looked, for instance, at whether the custom of avoidance between a man and his mother-in-law co-occurred with particular rules of residence after marriage (Tylor,

1889). Francis Galton, who chaired the meeting, raised an objection in the discussion. Many of the societies might have shared their customs by borrowing or by common descent, so they were not independent instances. Tylor's "adhesions" might reflect history rather than any functional connection between customs. The objection became known as *Galton's problem*. Naroll (1961) reviewed early solutions in anthropology, and the problem has shaped cross-cultural research ever since.

Galton's problem has two faces. The first is *statistical*. Non-independent observations carry less information than independent ones, so standard errors computed under independence are too small and p-values are too optimistic. The second is *inferential*. Even a correctly estimated association may reflect a historical accident rather than a causal or functional link. If an ancestral language happened to have two features, and its many descendants retained both out of inertia, then the features will be associated across the descendants. Yet nothing about one feature need favor the other. Biologists know this second face well. A trait inherited from a common ancestor is called a *homology*, and a shared trait that arose independently is called a *homoplasy* or convergence. Only independent origins (convergent evolution) provide evidence that two traits are functionally linked.

Mace and Pagel (1994) brought the biological solution to anthropology in an influential paper in *Current Anthropology*. Their argument was simple. Comparative biologists had faced Galton's problem for species and had developed methods to solve it by modeling evolution along a phylogeny. Anthropologists, and by extension linguists, could do the same, using language trees as estimates of the historical relationships among societies. This chapter is about those methods and how linguists have used them. Chapter 14 follows the same program into cultural evolution.

13.1.3 Sampling-based solutions in typology

Typologists recognized the problem long before they adopted phylogenetic methods. Their response was to design samples that reduce non-independence (Rijkhoff et al., 1993). Three strategies stand out.

The first is *genealogical stratification*. The idea is to sample at most one language per family, or per subgroup of a given age, so that no historical lineage is counted twice. Dryer (1989) made this operational with the notion of the *genus*: a group of languages whose relatedness is fairly obvious without systematic comparative analysis, with a time depth comparable to the subfamilies of Indo-European (Germanic, Romance, Slavic), perhaps 2,500 to 4,000 years. Dryer counted genera rather than languages. A genus was scored as having a feature if most of its languages had it. The genera were then grouped into large *macroareas* (Africa, Eurasia, Southeast Asia and Oceania, Australia and New Guinea, North America, South America). A correlation was accepted as a universal tendency only if it held in a majority of genera in every macroarea, or nearly every one (Dryer, 1989, 1992). The macroarea criterion deals with the second face of Galton's problem, contact, at a very coarse scale. It asks whether a pattern recurs in regions that have had little historical contact.

The second strategy is *diversity sampling*. Here the goal is to maximize the range of structural types represented, for instance by sampling in proportion to the internal diversity of families (Rijkhoff et al., 1993; Miestamo et al., 2016). This is useful for discovering what is possible, but less useful for testing correlations.

The third strategy, more recent, is to replace counting with modeling of the processes that pro-

duce the distribution. Maslova (2000) pointed out that the present-day distribution of types is a snapshot of a dynamic system. If languages change from type to type at certain rates, then the cross-linguistic frequency of a type reflects the stationary distribution of that process rather than a timeless preference. A universal is then a claim about transition rates, which is exactly what phylogenetic models of discrete characters estimate. Cysouw (2011) makes the link explicit. For a binary feature with change probabilities p_{AB} and p_{BA}, the stable frequency of type A is $p_{BA}/(p_{AB} + p_{BA})$, whatever the time unit. The converse does not hold: very different pairs of rates give the same stable frequency, so present-day frequencies alone cannot identify the rates. And reading the observed frequencies as stable frequencies assumes that the system has reached its stationary distribution; in a young family the observed frequencies may still reflect the ancestral state (a founder effect) rather than the stable state. Bickel (2011) and Bickel (2013) developed the *Family Bias method*. It asks whether families tend to be biased in the same direction, toward the same value of a feature, more often than chance. For large families the bias can be estimated directly from the daughter languages. For small families and isolates it is extrapolated from the large families. A distributional universal, on this view, is a consistent bias across families that diversify independently. The method is explicitly diachronic in spirit. It treats each family as an independent experiment in language change, while making no attempt to model the internal tree of each family.

A related approach uses *mixed-effects models*. Family and area are treated as random effects, which lets observations within the same family or area be correlated (Jaeger et al., 2011). This was an important step toward modeling dependence directly. But a random effect for "family" treats all languages in a family as equally related to each other. It cannot represent the fact that Spanish and Portuguese are much closer to each other than either is to Hindi. A phylogenetic model can.

Macklin-Cordes and Round (2022) make the link between the two traditions explicit. They argue that phylogenetic comparative methods address the same problem as balanced sampling, statistical non-independence due to genealogy, but whereas sampling can never achieve independence and discards most of the data, phylogenetic methods model the dependence and use all of it. Their paper introduces phylogenetic signal, typological averages that are sensitive to genealogy and the treatment of tree uncertainty for typologists, with a case study of the laminal contrast in Pama-Nyungan.

13.1.4 What a phylogeny adds

All sampling solutions throw away information. If we keep one language per genus, we discard every other language, and with it every within-genus change. Yet those changes are exactly the independent evolutionary events that tell us whether two traits evolve together. Suppose the change from OV to VO has happened twenty times within Indo-European alone. We would like to know whether the adposition order changed along with it. A one-language-per-genus sample cannot tell us. Stratified samples also depend on an arbitrary time threshold, and they treat relatedness as all-or-nothing: Germanic and Romance count as independent genera although they are related, only less closely than Spanish and Portuguese.

Phylogenetic comparative methods take the opposite approach. They keep all the languages and model the dependence explicitly. Relatedness is represented by a phylogeny with branch lengths, and trait evolution by a stochastic process running along the branches. The expected similarity between two languages then follows from how much history they share. Two languages that split 500 years ago are expected to be very similar. Two languages that split 6,000 years ago are

expected to be much less so, and two languages from different families not at all (unless contact has intervened). The phylogeny thus works as a *covariance structure*. A regression or correlation test that uses it gives each language the weight its independent information deserves.

A simulation makes the point concrete. We generate a tree for 60 languages, scaled so that the root lies 6,000 years in the past, and let two continuous traits (say, the proportion of suffixing morphology and the mean word length in a corpus) evolve along it completely *independently* of each other. We then test for a correlation that, by construction, does not exist, 500 times over, and count how often each method declares a significant correlation at the 5% level. A valid test should do so about 5% of the time. In the vocabulary of Section 6.10, this checks the *robustness under misspecification* of OLS, whose assumption of independent languages the data violate; PGLS is applied to data simulated under its own assumptions.

```
set.seed(13)
tree <- pbtree(n = 60, scale = 6)              # pure-birth tree, depth 6 (kyr)
tree$tip.label <- sprintf("L%02d", 1:60)
H <- nodeHeights(tree)

# assign every tip to the lineage that crosses a given time slice
cut_tree <- function(tree, height) {
  H <- nodeHeights(tree)
  crossing <- which(H[, 1] < height & H[, 2] >= height)
  grp <- integer(Ntip(tree))
  for (k in seq_along(crossing)) {
    d <- getDescendants(tree, tree$edge[crossing[k], 2])
    grp[d[d <= Ntip(tree)]] <- k
  }
  setNames(grp, tree$tip.label)
}
genus <- cut_tree(tree, height = 6 - 3.5)      # 3.5 kyr before the present
n_genera <- length(unique(genus))

cols <- hcl.colors(n_genera, "Dark 3")
plot(tree, cex = 0.6, label.offset = 0.05, tip.color = cols[genus],
     edge.width = 1.2)
abline(v = 6 - 3.5, lty = 2, col = "grey40")
axisPhylo(cex.axis = 0.8)
mtext("thousands of years before present", side = 1, line = 2.5, cex = 0.8)
```

The tree in Figure 13.1 has 10 lineages crossing the 3,500-year line. These are our “genera”. We compare three procedures: an OLS regression on all 60 languages, an OLS regression on a stratified sample with one language per genus, and a phylogenetic generalized least squares (PGLS) regression that uses the tree’s covariance structure (Section 13.6). For speed, the PGLS fit is computed by “whitening” the data with the Cholesky factor of the inverse covariance matrix, which is exactly what PGLS does internally.

```
C <- vcv(tree)                                 # BM covariance matrix
W <- chol(solve(C))                            # whitening matrix: W'W = C^-1
one_per_genus <- tapply(seq_along(genus), genus, function(i) i[1])
```

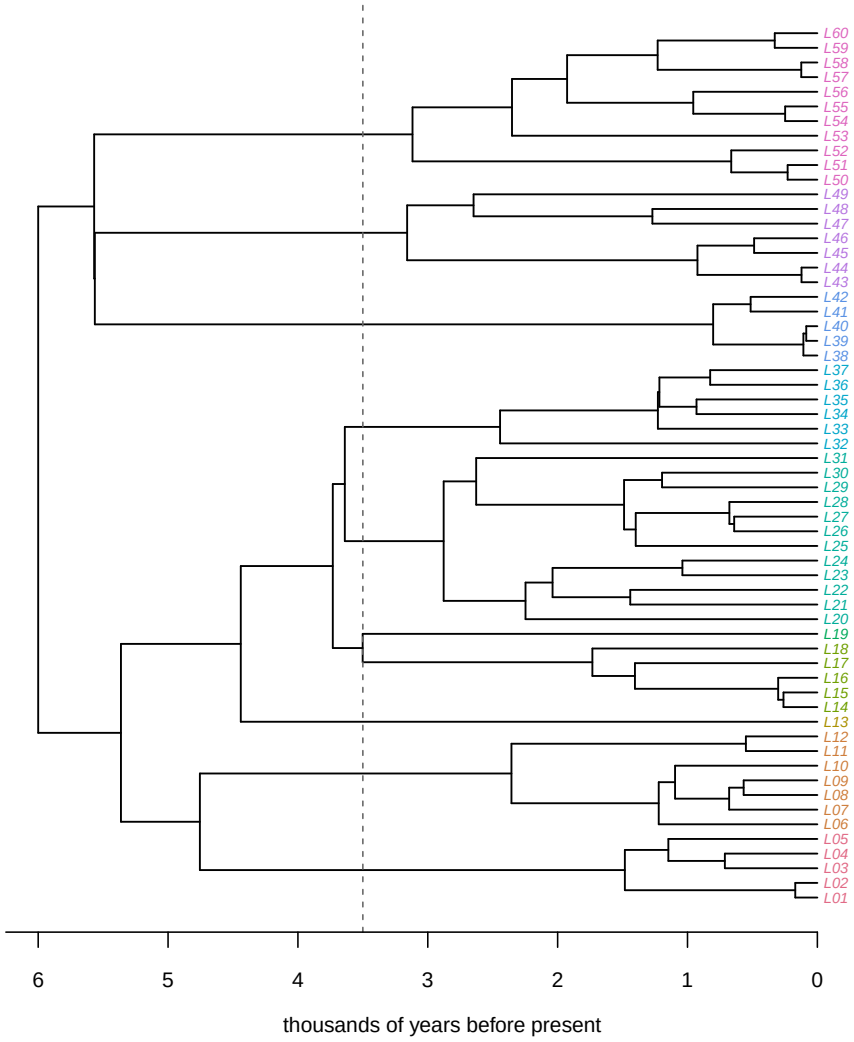


Figure 13.1: A simulated family of 60 languages with a root age of 6,000 years. Colors mark the 'genera' obtained by cutting the tree 3,500 years before the present, a Dryer-style grouping used below for genealogically stratified sampling.

```
nrep <- 500
pvals <- matrix(NA, nrep, 3, dimnames = list(NULL, c("OLS", "Genus", "PGLS")))
for (r in seq_len(nrep)) {
  x <- fastBM(tree)                              # trait 1, Brownian motion
  y <- fastBM(tree)                              # trait 2, independent of trait 1
  g <- one_per_genus
  pvals[r, "OLS"]   <- summary(lm(y ~ x))$coefficients[2, 4]
  pvals[r, "Genus"] <- summary(lm(y[g] ~ x[g]))$coefficients[2, 4]
  Xw <- W %*% cbind(1, x); yw <- W %*% y         # transformed (whitened) data
  pvals[r, "PGLS"]  <- summary(lm(yw ~ Xw - 1))$coefficients[2, 4]
}
fpr <- colMeans(pvals < 0.05)
round(fpr, 3)
```

```
#>   OLS Genus  PGLS
#> 0.406 0.082 0.044
```

```
ci <- sapply(fpr, function(p) p + c(-1, 1) * 1.96 * sqrt(p * (1 - p) / nrep))
bp <- barplot(fpr, ylim = c(0, max(ci) * 1.15),
              col = c("grey70", "grey50", "grey30"),
              names.arg = c("OLS, all languages", "OLS, one per genus", "PGLS"),
              ylab = "false positive rate", border = NA)
arrows(bp, ci[1, ], bp, ci[2, ], angle = 90, code = 3, length = 0.05)
abline(h = 0.05, lty = 2)
```

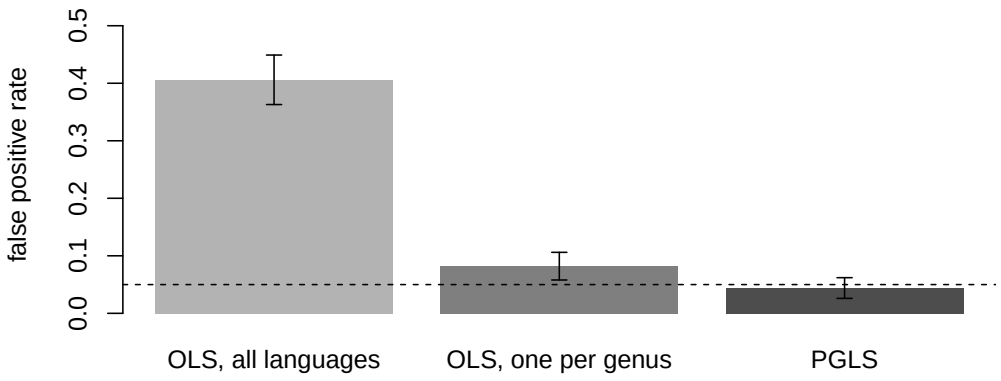


Figure 13.2: Galton's problem in a simulation. Two traits evolved independently by Brownian motion on the simulated 60-language tree. Bars show the proportion of 500 replicates in which each test reported a significant correlation at the 5% level, with 95% binomial intervals. The dashed line marks the nominal 5% rate.

The result (Figure 13.2) is stark. Ordinary regression on all languages finds a "significant" correlation between two unrelated traits in about 41% of replicates, roughly 8 times the nominal rate. Stratified sampling helps but does not fully solve the problem, with a false positive rate of about 8%: the genera are still related through the deeper part of the tree, and the sample is much smaller. PGLS keeps all the languages and achieves the nominal rate (4%). Nothing in the simulation is unusual. Pure-birth trees and Brownian motion are, if anything, a mild scenario compared with real language families, in which a single deep split, such as that between OV and VO branches, can create a strong association that rests on just one historical event.

Galton's problem is just as acute for discrete features. Below, two binary features evolve independently on the same tree, each with a low rate of change, as typological features often do. For a symmetric two-state process with rate q, a branch of length t ends in a different state from the one it started in with probability $\frac{1}{2}(1 - e^{-2qt})$, so we visit the branches from the root toward the tips and flip the state with this probability. We then apply Fisher's exact test to the 2×2 table of the two features.

```
sim_binary <- function(tree, q) {
  tree <- reorder(tree, "cladewise")            # parents before children
  state <- integer(Ntip(tree) + tree$Nnode)
  state[Ntip(tree) + 1] <- rbinom(1, 1, 0.5)    # root drawn from stationarity
  p_flip <- 0.5 * (1 - exp(-2 * q * tree$edge.length))
  flips <- rbinom(nrow(tree$edge), 1, p_flip)
  for (e in seq_len(nrow(tree$edge)))
    state[tree$edge[e, 2]] <- abs(state[tree$edge[e, 1]] - flips[e])
  setNames(state[seq_len(Ntip(tree))], tree$tip.label)
}
p_fisher <- replicate(500, {
  a <- sim_binary(tree, 0.15); b <- sim_binary(tree, 0.15)
  if (length(unique(a)) < 2 || length(unique(b)) < 2) NA else
    fisher.test(table(a, b))$p.value
})
fpr_discrete <- mean(p_fisher < 0.05, na.rm = TRUE)
fpr_discrete
```

```
#> [1] 0.184
```

Here too the naive test rejects the true null hypothesis of independence far more often than it should: in about 18% of replicates rather than 5%. The phylogenetic alternative, Pagel's test for correlated evolution, is developed in Section 13.7.

Dryer's genera and macroareas. Dryer (1992) tested several dozen putative word-order correlations on a database of some 625 languages grouped into genera and six macroareas. A pair of features counted as correlated only if the preferred combination was found in more genera than the dispreferred combination in each macroarea (or in all but one). The analysis confirmed some of Greenberg's correlations, such as those involving adpositions, genitives and relative clauses, and rejected others. The most famous rejected case is adjective-noun order, which Greenberg had expected to correlate with verb-object order and which turned out not to. Dryer's procedure was a pragmatic, transparent and remarkably successful answer to Galton's problem. Its limitations are those discussed above. It counts each genus once, whatever its internal history. It treats the genus threshold as a sharp boundary. And it says nothing about the dynamics of change that pro-

duced the distribution. It nonetheless remains the benchmark against which phylogenetic results on word order are judged, and it played a central role in the debate that followed Dunn et al. (2011) (Section 13.8).

13.2 Designing a language sample

Phylogenetic methods do not make sampling irrelevant. They change what sampling is for. A sampling design for a comparative study should answer five questions before any data are coded.

1. **What is the target of inference?** A claim about "the languages of the world" today, a claim about the *possible* languages, and a claim about the *processes* of change (a universal as a bias in transition rates; Section 13.1.3) call for different samples. For the first, the sampling frame is a list such as Glottolog (Hammarström et al., 2024); for the third, the unit of replication is the family or the independent change, not the language.
2. **How much independent information will the sample contain?** Under a phylogenetic model the information in a sample is not its number of languages. A useful summary is the *effective sample size* for estimating a mean, $n_{\text{eff}} = \mathbf{1}^{\top}\mathbf{R}^{-1}\mathbf{1}$, where $\mathbf{R}$ is the phylogenetic correlation matrix (the covariance matrix $\mathbf{C}$ of Section 13.3.1 scaled to unit diagonal). It equals the number of languages when they are independent and approaches the number of deep, independent lineages when languages cluster in shallow clades. For tests of association, what matters is the number of independent *changes* in the predictor, which can be smaller still.
3. **Does the predictor vary within families?** An association estimated from a sample in which a predictor varies only *between* families rests on as many data points as there are families, whatever the number of languages. Variation within families is what lets a phylogenetic model see coupled change.
4. **How are languages, dialects and families delimited?** Splitting one language into five dialects adds five tips to a shallow clade. PCMs down-weight them, but only to the extent that the tree represents their closeness correctly. Lumping varieties into one tip, or splitting them, changes $\mathbf{C}$, and decisions should be made before seeing the results (Section 15.8).
5. **Which languages are missing, and why?** Typological databases over-represent well-described languages, which are often larger, written, or spoken in regions with a long tradition of description. If documentation is related to the variables under study (population size and morphological complexity are a clear example), restricting the analysis to documented languages can create or mask associations (Section 13.11.4).

The code below makes point 2 concrete. It computes n_{eff} for random subsamples of the 60-language tree, under Brownian motion ($\lambda = 1$) and under a weaker phylogenetic dependence ($\lambda = 0.5$; Section 13.3.4), and for the one-language-per-genus sample.

```
neff <- function(tr, lambda = 1) {
  R <- cov2cor(vcv(tr))                          # phylogenetic correlations
  R <- lambda * R + (1 - lambda) * diag(nrow(R))
  sum(solve(R))                                  # 1' R^-1 1
}
set.seed(14)
neff_tab <- t(sapply(c(5, 10, 20, 40, 60), function(k) {
  v <- replicate(50, {
```

```
    tr <- keep.tip(tree, sample(tree$tip.label, k))
    c(neff(tr), neff(tr, 0.5)) })
  c(languages = k, n_eff_BM = mean(v[1, ]), n_eff_lambda_0.5 = mean(v[2, ]))
}))
round(neff_tab, 1)
```

```
#>      languages n_eff_BM n_eff_lambda_0.5
#> [1,]         5      3.5              4.1
#> [2,]        10      4.8              6.3
#> [3,]        20      6.2              9.1
#> [4,]        40      6.8             11.2
#> [5,]        60      7.0             12.2
```

```
genus_tree <- keep.tip(tree, tree$tip.label[one_per_genus])
c(genera = n_genera, n_eff_genera = round(neff(genus_tree), 1))
```

```
#>       genera n_eff_genera
#>         10.0          5.8
```

Under Brownian motion, sixty languages from this family are worth about seven independent observations of the family mean, and most of that value is already reached with twenty languages. The one-per-genus sample is worth less than its number of genera, because the genera are themselves related. With weaker dependence the value of extra languages is greater. Real families are more unbalanced than this pure-birth tree, and a global sample is a forest of families of very different sizes, so it is worth computing n_{eff} for the actual sample and tree before an analysis. Two practical consequences follow. Adding languages from an already well-sampled clade buys little; adding families or isolates buys a lot, even though each contributes little information about rates of change within lineages. For estimating rates, the balance tips the other way: following Maslova (2000), Cysouw (2011) suggests that a 100-language sample built as pairs of closely related languages from 50 groups, or as about ten densely sampled groups, says more about change than 100 languages from different genealogical units. And reporting results family by family alongside pooled results shows whether a global association is carried by a few large families.

Modeling dependence also relaxes the classical demand for a balanced, probability-based sample. Guzmán Naranjo and Becker (2022) combined a phylogenetic regression term with a two-dimensional Gaussian process for areal effects and argued that, with both controls in the model, strict probability sampling is not required, provided the sample is a variety sample large enough to cover different families and areas. Their test case is instructive: an apparent effect of VO order on the choice between prefixing and suffixing disappeared once genealogical and areal effects were both controlled, and the areal effects were as important as the genealogical ones. The design questions above do not go away, but they shift from "which languages may I include?" to "how much independent information about my question does this sample contain, and where does it come from?"

13.3 Models of trait evolution

Every phylogenetic comparative method rests on a model of how traits change along the branches of a tree. For discrete characters the standard models are the continuous-time Markov chains

(CTMCs) introduced in Chapter 5 and developed for cognate data in Chapter 7. For continuous characters the workhorse is Brownian motion and its relatives. Continuous characters are less common in linguistics than in biology, but far from rare. Examples include phoneme inventory size, the ratio of consonants to vowels, mean word length, measures of morphological synthesis or fusion, entropy-based measures of word-order freedom, lexical rates of change, and even latitude and longitude (Chapter 11). Many discrete features can also be modeled as continuous *liabilities* that cross a threshold (Section 13.9.3).

13.3.1 Brownian motion

Brownian motion (BM) is a random walk in continuous time. Let $X(t)$ be the value of a trait at time t. Under BM, the change over any short interval dt is normally distributed with mean zero and variance $\sigma^2\,dt$:

$$X(t + dt) - X(t) \sim \mathcal{N}(0, \sigma^2\,dt), \tag{13.1}$$

and changes over non-overlapping intervals are independent. The parameter σ^2 is the *evolutionary rate*. Over a branch of length t, the change is $\mathcal{N}(0, \sigma^2 t)$. The process has no tendency to return to any particular value. It simply diffuses, so the variance among independent lineages grows linearly with time.

BM is often described as a model of "neutral drift", and for genetic drift in quantitative traits it can be derived from population genetics. But its usefulness does not depend on that interpretation. BM also arises when a trait is pushed around by many small, independent influences whose direction varies randomly over time: shifting social preferences, contact with changing neighbors, analogical leveling here and sound change there. It is the natural null model for "change happens, but in no particular direction."

Now place BM on a tree. Each lineage inherits its ancestor's value at the moment of splitting and then wanders independently. Two languages i and j share the path from the root to their most recent common ancestor. Along that shared path, they have experienced the same changes. After the split, their changes are independent. Hence the covariance between their trait values equals σ^2 times the length of the shared path, s_{ij}. The variance of a single language equals σ^2 times its distance from the root, T_i. Collecting these into a matrix, the vector of tip values $\mathbf{x} = (x_1, \ldots, x_n)^\top$ is multivariate normal:

$$\mathbf{x} \sim \mathcal{N}_n\!\left(\mu\mathbf{1},\ \sigma^2\mathbf{C}\right), \qquad C_{ij} = s_{ij}, \quad C_{ii} = T_i, \tag{13.2}$$

where μ is the value at the root and $\mathbf{1}$ is a vector of ones. The matrix $\mathbf{C}$ is the *phylogenetic variance-covariance matrix*. It is the single most important object in this chapter. Everything a Gaussian PCM knows about the tree is contained in it. In R, `ape::vcv()` computes it. For a four-language tree in which A and B split 1,000 years ago, C and D 2,000 years ago, and the two pairs 3,000 years ago:

```
toy <- read.tree(text = "((A:1,B:1):2,(C:2,D:2):1);")
vcv(toy)
```

```
#>   A B C D
```

```
#> A 3 2 0 0
#> B 2 3 0 0
#> C 0 0 3 1
#> D 0 0 1 3
```

Read the matrix row by row. A and B share 2 units of history (the branch from the root to their ancestor), so their covariance is 2. A and C share nothing, so their covariance is 0. Each tip has variance 3, the full depth of the tree. The maximum-likelihood estimate of the root value μ is then a weighted mean of the tip values, and it is also the ML estimate of the ancestral state at the root (Chapter 12). It down-weights clusters of closely related languages, since they carry largely redundant information: a family with 400 Bantu languages and a single isolate does not get a root estimate that is 400/401 Bantu.

The Brownian likelihood, Equation (D.10), and the maximum-likelihood estimates of μ and σ^2, Equation (D.11), are in Section D.3.1.

13.3.2 The Ornstein-Uhlenbeck process

Brownian motion lets a trait wander without bound. Many linguistic quantities, however, plausibly stay within a range. Phoneme inventories rarely shrink below about a dozen segments or grow beyond about a hundred. The proportion of a lexicon that is borrowed is bounded by 0 and 1. The *Ornstein-Uhlenbeck* (OU) process adds a deterministic pull toward an optimum, or *attractor*, θ:

$$dX(t) = \alpha\,[\theta - X(t)]\,dt + \sigma\,dW(t), \tag{13.3}$$

where $dW(t)$ is the increment of a standard Brownian motion and $\alpha \geq 0$ is the strength of the pull. When $\alpha = 0$, OU reduces to BM. When $\alpha > 0$, the trait fluctuates around θ with stationary variance $\sigma^2/(2\alpha)$. The *phylogenetic half-life*, $t_{1/2} = \ln 2/\alpha$, is the expected time for a lineage to move halfway from its ancestral value toward the optimum. Comparing $t_{1/2}$ with the depth of the tree tells us whether the pull is weak (half-life longer than the tree) or strong (half-life much shorter, in which case the phylogeny barely matters). Under OU, the covariance between two languages decays exponentially with the time separating them, rather than linearly as under BM. OU was introduced to comparative biology by Hansen (1997) as a model of stabilizing selection toward an adaptive optimum. Butler and King (2004) extended it to multiple optima on different parts of the tree, for example one optimum for languages in one ecological or social setting and another for languages in a different setting. Such multi-optimum models are a natural way to formalize hypotheses that a feature is "adapted" to some environment. A linguistic version might specify one optimum for languages of large, open societies and another for languages of small, close-knit ones.

The OU covariance between two tips, Equation (D.12), and the early-burst covariance, Equation (D.13), are in Section D.3.2.

OU models are easy to fit and easy to over-interpret. With small trees, likelihood-ratio tests favor OU over BM much more often than they should. Measurement error, or a few unusual languages, can also mimic an OU process. And the "stabilizing selection" reading of α is only one of several processes that produce the same pattern (Cooper et al., 2016; Ho and Ané, 2014a). For trees of a few dozen languages, α and σ^2 are often weakly identifiable. Report the half-life with its

uncertainty, check the result by simulation, and avoid talking about "selection" unless there is independent reason to.

13.3.3 Early burst and related models

The *early burst* (EB) model, also called ACDC (accelerating-decelerating), lets the rate of BM change exponentially through time: $\sigma^2(t) = \sigma_0^2 \, e^{rt}$. With $r < 0$, change is fast early in the history of the clade and slows down later. This is the signature expected under adaptive radiation, when a lineage enters new ecological space and its descendants quickly diversify into vacant niches (Blomberg et al., 2003; Harmon et al., 2010).

Linguistic analogues are easy to imagine. A language family expanding into a new and diverse landscape, such as Austronesian into the Pacific or Bantu across sub-Saharan Africa, might diversify structurally faster in its early phases than later. Punctuational bursts of lexical change at language splits (Atkinson et al., 2008), discussed in Chapter 9, are a different phenomenon, linked to splitting events rather than to clade age, but they raise the same general question: is the rate of change constant through time? In biology, Harmon et al. (2010) found that early bursts are actually rare in comparative data on body size and shape. That result is a useful warning against assuming them.

13.3.4 Pagel's tree transformations: lambda, kappa and delta

Pagel (1997) and Pagel (1999a) introduced three transformations of the tree, or equivalently of $\mathbf{C}$. They are a flexible way to test how well BM on the given tree describes the data. Each transformation is controlled by a single parameter, and each has a natural interpretation.

Lambda (λ) multiplies all off-diagonal elements of $\mathbf{C}$ by λ and leaves the diagonal unchanged:

$$\mathbf{C}(\lambda) = \lambda \, \mathbf{C} + (1 - \lambda) \, \mathrm{diag}(\mathbf{C}). \tag{13.4}$$

When $\lambda = 1$ we have pure BM on the original tree. When $\lambda = 0$ the tree becomes a "star" in which all languages are independent. Intermediate values shrink the internal branches relative to the terminal ones. Lambda is the most widely used measure of *phylogenetic signal* (Section 13.4). Note that λ can exceed 1 slightly, up to the value at which $\mathbf{C}(\lambda)$ ceases to be a valid covariance matrix. Most software restricts it to $[0, 1]$.

Kappa (κ) raises every branch length to the power κ. When $\kappa = 1$ the tree is unchanged. When $\kappa = 0$ all branches have equal length, so change is proportional to the number of splits rather than to time. This is a model of *punctuational* or *speciational* change, in which evolution is concentrated at splitting events. Kappa is therefore directly relevant to the linguistic hypothesis of punctuational bursts discussed in Chapter 9.

Delta (δ) raises the node heights (the distances from the root) to the power δ. With $\delta < 1$, the deep parts of the tree are stretched relative to the shallow parts: change is concentrated early, as in an early burst. With $\delta > 1$, change is concentrated toward the tips, as expected under accelerating evolution or for traits that evolve fast and lose their deep phylogenetic signal.

These transformations matter twice over for linguists. As models, they describe how a trait evolved. As tools, they let the data adjust branch lengths that were chosen arbitrarily, which is how they

are often used with Glottolog-based trees (Section 13.12.2). Figure 13.3 shows the effect of each transformation on the tree from Figure 13.1, using `geiger::rescale()`.

```
# cex = 1 keeps full-size titles despite five panels
op <- par(mfrow = c(1, 5), mar = c(1, 0.5, 2.5, 0.5), cex = 1)
plot(tree, show.tip.label = FALSE, main = "original", cex.main = 0.9)
plot(rescale(tree, "lambda", 0.3), show.tip.label = FALSE,
     main = expression(lambda == 0.3), cex.main = 0.9)
plot(rescale(tree, "kappa", 0), show.tip.label = FALSE,
     main = expression(kappa == 0), cex.main = 0.9)
plot(rescale(tree, "delta", 3), show.tip.label = FALSE,
     main = expression(delta == 3), cex.main = 0.9)
plot(rescale(tree, "OU", alpha = 0.5), show.tip.label = FALSE,
     main = expression(OU ~ alpha == 0.5), cex.main = 0.9)
```

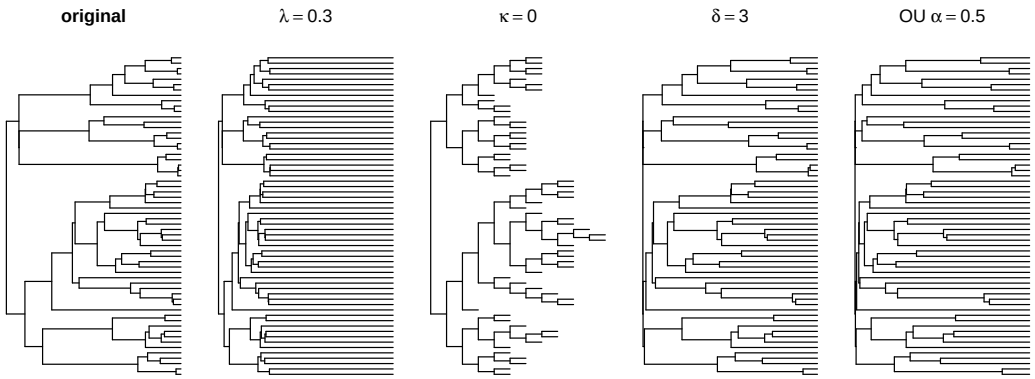


Figure 13.3: Pagel's branch-length transformations applied to the simulated language tree. From left to right: the original tree; lambda = 0.3 (internal branches shortened); kappa = 0 (all branches equal, a speciational model); delta = 3 (change concentrated near the tips); and an Ornstein-Uhlenbeck transformation with alpha = 0.5.

```
par(op)
```

A *traitgram*, or *phenogram*, plots the tree with the trait value on the vertical axis and time on the horizontal axis. It is a good way to see the difference between BM and OU. Under BM, lineages fan out, and their spread grows with time. Under OU with a strong pull, they stay within a band around the optimum, and the spread barely depends on how long ago two languages split (Figure 13.4).

```
set.seed(21)
x_bm <- fastBM(tree, sig2 = 1, internal = TRUE)
x_ou <- fastBM(tree, sig2 = 1, alpha = 1.5, theta = 0, internal = TRUE)
op <- par(mfrow = c(1, 2), mar = c(4, 4, 2, 1))
phenogram(tree, x_bm, ftype = "off", spread.labels = FALSE,
          xlab = "time from root (kyr)", ylab = "trait",
          main = "Brownian motion")
phenogram(tree, x_ou, ftype = "off", spread.labels = FALSE,
```

```
          xlab = "time from root (kyr)", ylab = "trait",
          main = "Ornstein-Uhlenbeck")
```

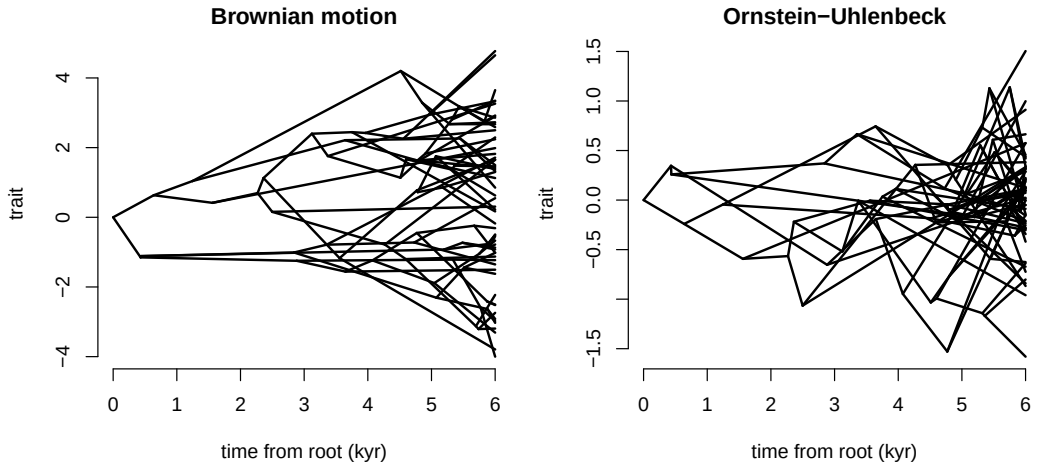


Figure 13.4: Traitgrams of a continuous trait simulated on the same tree under Brownian motion (left) and an Ornstein-Uhlenbeck process with a strong pull toward an optimum of zero (right; alpha = 1.5, half-life about 0.46 thousand years). Internal node values are the simulated ancestral values.

```
par(op)
```

13.3.5 Fitting and comparing models

All of these models can be fitted by maximum likelihood and compared with information criteria such as AIC or its small-sample correction AICc (Chapter 5). The function `geiger::fitContinuous()` (Pennell et al., 2014) fits BM, OU, EB, lambda, kappa, delta and a "white noise" model in which languages are independent draws from a single normal distribution. Below we fit five models to the BM and OU traits just simulated.

```
fit_models <- function(x) {
  models <- c("BM", "OU", "EB", "lambda", "white")
  sapply(models, function(m)
    fitContinuous(tree, x, model = m, control = list(niter = 10))$opt$aicc)
}
tips <- tree$tip.label
aicc <- rbind(BM_trait = fit_models(x_bm[tips]),
              OU_trait = fit_models(x_ou[tips]))
delta_aicc <- t(apply(aicc, 1, function(a) a - min(a)))
knitr::kable(round(delta_aicc, 1),
             caption = "Differences in AICc from the best model (0 = best) for traits
             simulated under Brownian motion and under an Ornstein-Uhlenbeck
             process.")
```

Table 13.1: Differences in AICc from the best model (0 = best) for traits simulated under Brownian motion and under an Ornstein-Uhlenbeck process.

	BM	OU	EB	lambda	white
BM_trait	0.0	2.1	2.2	2.2	64.3
OU_trait	18.2	0.0	20.4	6.3	4.2

With 60 tips, the model comparison usually recovers the generating process. You will find, though, that it is quite easy to construct cases where it does not. Re-running the simulation with different seeds is instructive. Model selection among continuous models is a good exploratory tool. It is a weak basis for strong claims about evolutionary mechanism.

13.3.6 Discrete characters

Most typological features are discrete. Word order is OV or VO. A language has or lacks tone, grammatical gender, a dual number or a definite article. As in previous chapters, such characters are modeled by a CTMC with rate matrix $\mathbf{Q}$. The simplest version is the *Mk* model of Lewis (2001). Its variants include the equal-rates (ER), symmetric (SYM) and all-rates-different (ARD) models, as well as ordered models in which changes can only happen between adjacent states. Chapter 12 uses these models for ancestral state reconstruction. Here they will serve as the building blocks of Pagel's test for correlated evolution (Section 13.7). One feature of the Mk model matters for comparative analysis. Its equilibrium frequencies, the stationary distribution π of $\mathbf{Q}$, are what the "preferences" of typology look like in a dynamic view (Maslova, 2000). A strongly preferred type is a type that languages change *into* readily and *out of* reluctantly.

13.4 Phylogenetic signal

Phylogenetic signal is the tendency of related languages to resemble each other more than languages drawn at random from the same tree (Blomberg et al., 2003; Münkemüller et al., 2012). It is the first thing to measure in any comparative analysis. It tells us how strongly Galton's problem affects a feature, and so whether phylogenetic correction matters. It is also of interest in its own right. Features with strong signal are stable, inherited properties of lineages. Features with weak signal are labile, easily borrowed or easily innovated. The *stability* of linguistic features, which typologists have long discussed (Wichmann and Holman, 2009b; Dediu and Cysouw, 2013), is closely related to phylogenetic signal. The two are not the same thing, though, as we will see.

13.4.1 Pagel's lambda

The most common measure of signal for continuous traits is Pagel's λ (Equation (13.4)), estimated by maximum likelihood. A value near 1 means that the trait covaries among languages as expected under BM on the tree. A value near 0 means that the tree tells us nothing about trait similarity. A likelihood-ratio test of $\lambda = 0$ against the ML value tests for the presence of signal. A test of $\lambda = 1$ tests for departure from BM.

13.4.2 Blomberg's K, Moran's I and the D statistic

Blomberg et al. (2003) proposed the statistic K, which compares how much of a trait's variance lies between clades rather than within them with the amount expected under BM on the tree. $K = 1$ is the BM expectation. $K < 1$ means that relatives resemble each other *less* than expected under BM; $K > 1$ means that they resemble each other more. Significance is assessed by permuting trait values across the tips. In biology, Blomberg et al. (2003) famously found that behavioral traits have lower K than morphological ones, being "more labile". The analogous claim in linguistics, that some kinds of structure are more stable than others, is explored below.

Moran's I (Moran, 1950), borrowed from spatial statistics, measures autocorrelation with respect to a matrix of weights. Gittleman and Kot (1990) proposed using phylogenetic proximity as the weights. Moran's I has the advantage that the *same* statistic can be computed with geographic weights, which makes it convenient for comparing phylogenetic and spatial autocorrelation of a feature. It is not tied to an explicit model of evolution, though, and it depends on the choice of weights.

For binary features, Fritz and Purvis (2010) proposed the statistic D, which compares the observed differences between sister clades with their distribution under random reshuffling of the trait and under a threshold model in which the binary trait arises from a Brownian liability. $D \approx 1$ indicates a random (phylogenetically unstructured) distribution, $D \approx 0$ a distribution as clumped as expected under the Brownian threshold model, and $D < 0$ an even more clumped one. It is implemented as `caper::phylo.d()`.

The formulas for K, Moran's I and D, Equations (D.14), (D.15) and (D.16), and code computing Moran's I and D for simulated traits, are in Section D.3.3.

13.4.3 Signal in R

The code below estimates λ and K for the BM trait simulated above, and for a "trait" that is pure noise.

```
noise <- setNames(rnorm(Ntip(tree)), tips)
lam <- function(v) phylosig(tree, v, method = "lambda", test = TRUE)$lambda
sig <- rbind(
  BM    = c(lambda = lam(x_bm[tips]),
            K = phylosig(tree, x_bm[tips], method = "K")),
  noise = c(lambda = lam(noise), K = phylosig(tree, noise, method = "K")))
round(sig, 3)
```

```
#>       lambda     K
#> BM     1.012 0.953
#> noise  0.095 0.114
```

```
phylosig(tree, x_bm[tips], method = "K", test = TRUE, nsim = 999)$P
```

```
#> [1] 0.001001
```

```
phylosig(tree, noise, method = "K", test = TRUE, nsim = 999)$P
```

```
#> [1] 0.6557
```

The BM trait has λ close to 1 and K close to or above 1, and the noise trait has values near 0.

Phylogenetic signal is a *pattern*, not a process. Revell et al. (2008) showed by simulation that high signal can arise under processes with high or low rates, and that low signal can arise from a strong pull toward an optimum (OU), from fast evolution, or from measurement error. "This feature has high λ, so it evolves slowly" does not follow. Under BM, λ is 1 whatever the rate. Stability, in the sense of a low rate of change per unit time, is better estimated directly as a rate parameter in a CTMC or BM model on a dated tree (Dediu, 2011; Greenhill et al., 2017). And signal estimated on a tree with arbitrary branch lengths (Section 13.12.2) depends on those branch lengths.

13.4.4 Phylogenetic signal in linguistic features

How strong is phylogenetic signal in grammatical features? This question goes back to old debates about "stability" in typology, and phylogenetic methods have given new answers. Wichmann and Holman (2009b) proposed a stability metric computed from WALS (Dryer, 2013), and Dediu and Cysouw (2013) compared it with six other stability estimates, most of them non-phylogenetic but including the Bayesian phylogenetic rates of Dediu (2011). The methods agree only partially, although some features come out as consistently stable and others as consistently unstable across methods.

Dediu (2011): stability of linguistic features on language trees. Dediu (2011) used Bayesian phylogenetic methods to estimate rates of change for WALS features on samples of trees for a number of language families. He treated the estimated rate of change as a measure of instability and, by integrating over tree uncertainty, obtained a ranking of features by stability. He then used the estimates to revisit an earlier hypothesis linking the distribution of tone to population frequencies of particular alleles of two brain-related genes (Dediu and Ladd, 2007). The analysis is a good example of a common design: a large set of typological features analyzed one at a time on samples of trees for several families. It also shows a characteristic limitation. Many families contribute little variation for many features, and the trees themselves are uncertain, so the per-feature estimates are noisy.

The largest analysis to date is that of Grambank, a database of 195 binary and multistate grammatical features coded for more than 2,400 languages (Skirgård et al., 2023). Skirgård and colleagues fitted Bayesian *spatiophylogenetic* models (Section 13.10). These decompose the variance of each feature into a phylogenetic component, based on a global language tree, and a spatial component, based on geographic distances. Across features, the phylogenetic component was on average larger than the spatial one. In other words, inheritance from ancestors explained more of the present-day distribution of grammatical features than proximity to neighbors did, although the strength of the phylogenetic effect varied considerably between features and the spatial effect was, on average, small. The paper is discussed further in Section 13.8.4.

Greenhill et al. (2017) took a different approach. They estimated rates of change for structural (grammatical) features and for basic vocabulary on the same Austronesian trees. Contrary to a widespread expectation, grammatical features were not more stable than vocabulary: on average, most grammatical features changed faster than items of basic vocabulary, although both kinds of data showed a wide range of rates and each contained a core of highly stable features. Both rates and signal, then, are properties of individual features, not of "grammar" or "lexicon" as a whole. Other studies point the same way. Dediu and Levinson (2012) estimated stability profiles of structural features within several families and found both tendencies shared across families and family-specific ones. Hübler (2022), defining stability as high phylogenetic signal combined with

a low rate of change, measured both for 171 structural features in Japonic, Koreanic, Mongolic, Tungusic and Turkic languages; signal and rate were correlated, and features of argument marking, derivation and valency came out among the most stable. Wichmann (2015) reviews the older and newer approaches to stability in typology. Chapter 9 discusses rates of change in more detail.

13.5 Phylogenetically independent contrasts

The first general solution to Galton's problem for continuous traits was Felsenstein's method of *phylogenetically independent contrasts* [PIC; Felsenstein (1985b)]. It is still the most intuitive. The insight is that although the trait values of languages are not independent, the *differences* between sister lineages are. Under BM, the change from the ancestor of Spanish and Portuguese to Spanish is independent of the change from that ancestor to Portuguese. It is also independent of every change elsewhere in the tree. So the difference between Spanish and Portuguese is an independent evolutionary "experiment". So is the difference between French and Italian, and so is the difference between the ancestor of the Ibero-Romance languages and the ancestor of French and Italian, once suitably estimated.

The algorithm works from the tips toward the root, computing a difference between each pair of sister lineages, standardized by the square root of the branch lengths that separate them, and replacing each pair by an estimated ancestral value. A tree with n tips yields $n - 1$ contrasts, which are independent and identically distributed with mean zero under BM. To test for correlated evolution of two traits, we compute contrasts for each and regress the contrasts of y on those of x *through the origin*. There is no intercept because the sign of each contrast is arbitrary.

PIC was revolutionary because it turned a problem with unknown dependencies into a standard regression on independent data. It was quickly adopted in biology (Harvey and Pagel, 1991; Garland et al., 1992). It has two limitations. It assumes BM (although branch lengths can be transformed first). And it is awkward to extend to multiple predictors, categorical predictors or non-normal data. Both limitations are removed by PGLS, of which PIC is a special case.

The contrast algorithm step by step, Equation (D.17), and a numerical check that the contrast slope equals the PGLS slope under BM, are in Section D.3.4.

```
set.seed(31)
x <- fastBM(tree)
y <- 2 + 0.6 * x + fastBM(tree, sig2 = 0.5)       # true slope 0.6
u_x <- pic(x, tree); u_y <- pic(y, tree)
fit_pic <- lm(u_y ~ u_x - 1)                       # regression through origin
coef(summary(fit_pic))
```

```
#>     Estimate Std. Error t value  Pr(>|t|)
#> u_x   0.6405     0.1079   5.937 1.729e-07
```

13.6 Phylogenetic generalized least squares

13.6.1 The model

Phylogenetic generalized least squares (PGLS) is ordinary linear regression with a non-diagonal residual covariance matrix (Grafen, 1989; Martins and Hansen, 1997). The model is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}_n(\mathbf{0}, \sigma^2 \mathbf{V}), \tag{13.5}$$

where $\mathbf{y}$ is the response for n languages, $\mathbf{X}$ is the $n \times p$ design matrix (a column of ones and the predictors), and $\mathbf{V}$ is a phylogenetic correlation structure. Under BM, $\mathbf{V} = \mathbf{C}$. Under Pagel's lambda model, $\mathbf{V} = \mathbf{C}(\lambda)$. Under OU, $\mathbf{V}$ has the OU covariance. The coefficients are estimated by generalized least squares, which is equivalent to transforming ("whitening") the data so that the transformed residuals are independent with equal variance, and then running ordinary least squares on the transformed data. This is exactly what we did in the Galton simulation. Independent contrasts are one particular whitening transformation, which is why the PIC slope through the origin is identical to the PGLS slope under BM (Rohlf, 2001; Blomberg et al., 2012).

The GLS estimator and its variance, Equations (D.18) and (D.19), the whitening transformation, Equation (D.20), and the profile likelihood of λ, Equation (D.21), are in Section D.3.5; the general theory of GLS is in Section A.11.

A few points about Equation (13.5) are worth stressing, because they are often misunderstood.

1. The phylogenetic structure is in the *residuals*, not in the variables (Revell, 2010). A strongly inherited predictor does not by itself call for PGLS. If the predictor explains all the phylogenetic structure in the response, the residuals may show no signal, and estimating λ jointly with the regression will reveal this (Section 13.6.2). Conversely, testing the *response* for signal and skipping PGLS if none is found is bad practice.
2. The predictors are treated as fixed and measured without error, as in any regression. Error in the predictor (for example, poorly estimated speaker numbers) biases the slope toward zero.
3. The *intercept* is the estimated value at the root of the tree for a language with all predictors equal to zero. It is not the mean of the sample.
4. PGLS handles any number of continuous or categorical predictors and their interactions, exactly as in ordinary regression. The only change is $\mathbf{V}$.

13.6.2 Estimating lambda jointly

In practice we rarely know how strong the phylogenetic dependence of the residuals is. The standard solution is to estimate Pagel's λ jointly with $\boldsymbol{\beta}$ and σ^2 by maximum likelihood, with $\mathbf{V} = \mathbf{C}(\lambda)$ (Freckleton et al., 2002). For each value of λ the regression coefficients have a closed form, so the likelihood can be maximized over λ alone, which is fast. The estimated λ then tells us where the analysis sits between OLS ($\lambda = 0$) and PGLS under BM ($\lambda = 1$). Its confidence interval tells us how well the data pin this down. With a few dozen languages, the interval is often wide.

13.6.3 PGLS in R

There are three common routes to PGLS in R, and it is useful to know all of them. The `nlme::gls()` function (Pinheiro and Bates, 2000) accepts correlation structures from **ape** such as `corBrownian()`, `corPagel()` and `corMartins()` (an OU structure). The **caper** package (Orme et al., 2013) provides `pgls()`, which estimates λ, κ and δ and offers profile plots and diagnostics. The **phylolm** package (Ho and Ané, 2014b) fits the same models with a linear-time algorithm, which matters for trees with thousands of tips such as global language phylogenies.

We simulate a scenario of the kind discussed in Section 13.11. A predictor, `logpop`, stands in for the logarithm of speaker population. A response, `complexity`, stands in for an index of morphological complexity. Their relationship has a true slope of -0.3, and the residual variation has phylogenetic signal $\lambda = 0.7$. Both variables are inherited. Population size, too, has a history: languages of a large, expanding branch tend to be large.

```
set.seed(43)
logpop <- fastBM(tree, sig2 = 0.4) + 9
resid  <- fastBM(rescale(tree, "lambda", 0.7), sig2 = 0.15)
complexity <- 5 - 0.3 * logpop + resid
langs <- data.frame(lang = tips, logpop = logpop[tips],
                    complexity = complexity[tips], row.names = tips)
head(langs, 4)
```

```
#>     lang logpop complexity
#> L01  L01  7.750      1.556
#> L02  L02  7.913      2.066
#> L03  L03  7.399      1.435
#> L04  L04  6.747      2.269
```

First, the three models fitted with `nlme::gls()`: OLS (via `gls()` without a correlation structure), PGLS under BM, and PGLS with λ estimated.

```
m_ols <- gls(complexity ~ logpop, data = langs, method = "ML")
m_bm  <- gls(complexity ~ logpop, data = langs, method = "ML",
             correlation = corBrownian(1, phy = tree, form = ~lang))
m_lam <- gls(complexity ~ logpop, data = langs, method = "ML",
             correlation = corPagel(0.5, phy = tree, form = ~lang))
tab <- t(sapply(list(OLS = m_ols, BM = m_bm, lambda = m_lam), function(m)
  c(summary(m)$tTable["logpop", c("Value", "Std.Error", "p-value")],
    AIC = AIC(m))))
lambda_hat <- coef(m_lam$modelStruct$corStruct, unconstrained = FALSE)
knitr::kable(tab, digits = 3,
             caption = "Slope of complexity on log population under OLS, PGLS with
             Brownian motion, and PGLS with Pagel's lambda estimated by maximum
             likelihood (simulated data; true slope -0.3, true lambda 0.7).")
```

Table 13.2: Slope of complexity on log population under OLS, PGLS with Brownian motion, and PGLS with Pagel's lambda estimated by maximum likelihood (simulated data; true slope -0.3, true lambda 0.7).

	Value	Std.Error	p-value	AIC
OLS	-0.125	0.084	0.140	170.9
BM	-0.342	0.141	0.018	160.2
lambda	-0.262	0.127	0.044	151.5

The estimated λ is 0.81, close to the true value of 0.7, and the lambda model is clearly preferred by AIC. Notice what happens to the slope. In this replicate OLS gives an estimate of -0.13 with $p = 0.14$. The real effect is masked by the phylogenetically structured noise, because a few large clades

happen to have residuals that run against the trend. The lambda model, which down-weights redundant information from those clades, recovers a slope of -0.26 with $p = 0.044$. This is a useful corrective to a common belief. Phylogenetic correction is often thought of as a conservative procedure that can only remove "significant" results. In fact, it makes estimates more *efficient* as well as more honest, and it can reveal an effect that OLS misses as easily as it can remove a spurious one. Which estimate is closest to the truth in a single replicate is partly a matter of luck. What is not luck is that OLS standard errors and p-values are systematically wrong when residuals have phylogenetic structure (Figure 13.2).

The same lambda model fitted with **caper** gives identical estimates, and **caper** also reports likelihood-ratio tests of $\lambda = 0$ and $\lambda = 1$ and a confidence interval for λ:

```
cd <- comparative.data(phy = tree, data = langs, names.col = lang, vcv = TRUE)
m_caper <- pgls(complexity ~ logpop, data = cd, lambda = "ML")
summary(m_caper)$coefficients
```

```
#>             Estimate Std. Error t value Pr(>|t|)
#> (Intercept)   4.3347     1.2991   3.337 0.001484
#> logpop       -0.2624     0.1273  -2.061 0.043754
```

```
m_caper$param.CI$lambda[c("opt", "ci.val", "bounds.p")]
```

```
#> $opt
#> lambda
#> 0.8116
#>
#> $ci.val
#> [1] 0.4838 0.9657
#>
#> $bounds.p
#> [1] 3.661e-06 1.046e-03
```

```
par(mar = c(7, 4, 1, 1))   # room for the two-line subtitle below the axis
plot(pgls.profile(m_caper, "lambda"))
```

Finally, **phylolm** fits the lambda model and also an OU model for the residuals, which we can compare by AIC:

```
p_lam <- phylolm(complexity ~ logpop, data = langs, phy = tree,
                 model = "lambda")
p_ou  <- phylolm(complexity ~ logpop, data = langs, phy = tree,
                 model = "OUrandomRoot")
rbind(lambda = c(coef(p_lam), AIC = AIC(p_lam)),
      OU     = c(coef(p_ou),  AIC = AIC(p_ou)))
```

```
#>        (Intercept)  logpop   AIC
#> lambda       4.335 -0.2624 151.5
#> OU           3.700 -0.2110 149.1
```

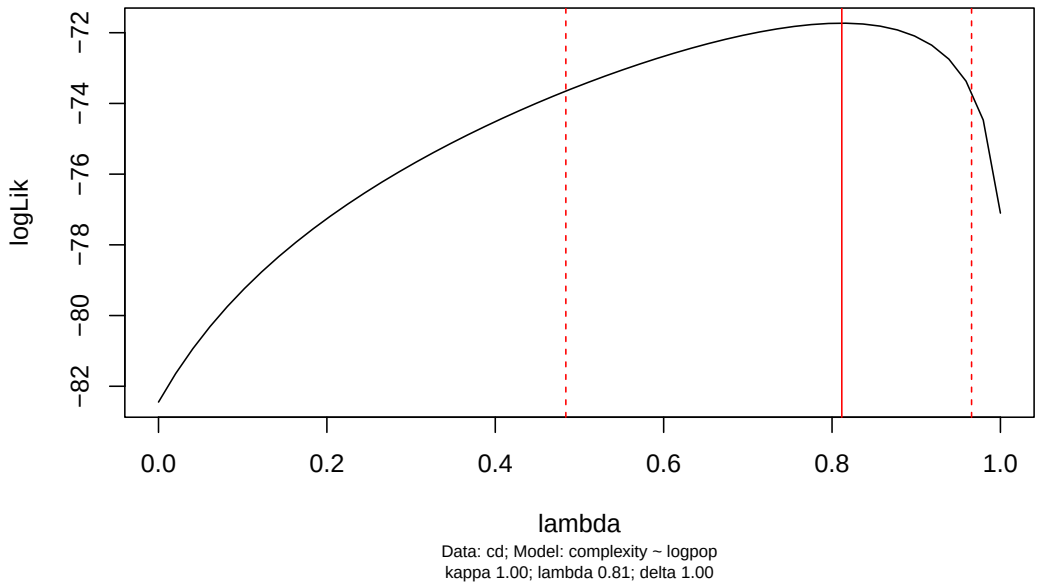


Figure 13.5: Profile log-likelihood of Pagel's lambda for the PGLS of complexity on log population, computed with caper. The vertical lines mark the ML estimate and its 95% confidence interval.

13.6.4 Diagnostics

PGLS has the same diagnostics as ordinary regression, applied to the *normalized* (whitened) residuals. These are what should be approximately independent and normal. In **nlme**, `residuals(m, type = "normalized")` returns them. **caper** offers `plot()` methods for its model objects. Outlying languages deserve particular attention. In a phylogenetic regression a single language on a long branch, such as an isolate or the only representative of a deep branch, can be highly influential, because nothing else in the sample shares its history. Re-fitting without such languages, and without each large clade in turn, is cheap and often revealing.

Correlated predictors need a diagnostic of their own. Many candidate predictors in linguistic typology travel together: speaker population, the proportion of adult second-language learners and the area a language is spoken over; humidity, temperature and latitude. Inherited predictors are correlated for a second reason, because closely related languages share values of all of them. When two predictors carry nearly the same information, a regression that includes both can estimate their joint effect but not how to divide it between them. Each coefficient is then a *partial* effect, the association with one predictor holding the other fixed, and its standard error grows with the correlation. The simulation below makes the point with two inherited predictors that are almost interchangeable and contribute equally to the response.

```
set.seed(7)
x1 <- fastBM(tree, sig2 = 0.4)
x2 <- x1 + fastBM(tree, sig2 = 0.05)          # x2 tracks x1 closely
y  <- 1 + 0.3 * x1 + 0.3 * x2 +
  fastBM(rescale(tree, "lambda", 0.7), sig2 = 0.15)
d2 <- data.frame(lang = tips, x1 = x1[tips], x2 = x2[tips], y = y[tips])
round(cor(d2$x1, d2$x2), 3)
```

```
#> [1] 0.991
```

```
fit_lam <- function(f) gls(f, data = d2, method = "ML",
  correlation = corPagel(0.5, phy = tree, form = ~lang))
fits <- list(x1 = fit_lam(y ~ x1), x2 = fit_lam(y ~ x2),
             both = fit_lam(y ~ x1 + x2))
coef_se <- function(m, v) if (v %in% rownames(summary(m)$tTable))
  summary(m)$tTable[v, c("Value", "Std.Error")] else c(NA, NA)
tab_col <- t(sapply(fits, function(m)
  c(coef_se(m, "x1"), coef_se(m, "x2"), AIC = AIC(m))))
colnames(tab_col) <- c("slope x1", "SE x1", "slope x2", "SE x2", "AIC")
knitr::kable(tab_col, digits = 2, caption = paste(
  "PGLS (Pagel's lambda) with two nearly collinear inherited predictors,",
  "each with a true slope of 0.3 (simulated data)."))
```

Table 13.3: PGLS (Pagel's lambda) with two nearly collinear inherited predictors, each with a true slope of 0.3 (simulated data).

	slope x1	SE x1	slope x2	SE x2	AIC
x1	0.51	0.11	NA	NA	163.8
x2	NA	NA	0.44	0.10	165.6
both	0.55	0.41	-0.04	0.37	165.8

Fitted alone, either predictor has a clear positive slope, because it stands in for both. Fitted together, the standard errors are about four times larger, one slope is close to zero, and neither is distinguishable from the other; by AIC the two-predictor model is no better than either single-predictor model, although it is the true model. None of this is a failure of PGLS. It is what the data can say: languages in which x_1 is high and x_2 low, which alone could separate the two effects, hardly exist in the sample. The practical consequences are to report the correlations among predictors (and their phylogenetic signal), to interpret coefficients as partial effects, to compare models with and without each predictor rather than reading single p-values, and to recognize that choosing between correlated predictors is a substantive question about mechanism (Section 13.11.4) that the regression cannot settle on its own.

Population size and rates of language change. Several studies have used phylogenetic regressions to ask whether languages spoken by larger populations change faster or slower. Bromham et al. (2015) compared pairs of closely related Polynesian languages that differ in population size, an explicitly phylogenetic "sister-pair" design that echoes independent contrasts. They found that languages with larger populations gained new words faster, while smaller populations lost words faster. Greenhill et al. (2018) extended the sister-pair design to 153 pairs of closely related languages in Indo-European, Austronesian and Bantu, comparing numbers of word gains and losses between sisters and using branch lengths from published dated trees to correct for divergence time. They found that the Polynesian pattern did not generalize consistently: smaller Indo-European languages showed significantly higher rates of word loss, but there was no significant pattern in Austronesian or Bantu. These studies illustrate a recurring theme. The response variable (a rate) is itself an output of a phylogenetic analysis, so it inherits that analysis's uncertainty and assumptions. And the effect sizes are modest relative to the noise. Chapter 9 discusses them as findings about rates. Here they matter as examples of design: sister pairs are a sampling design

that guarantees variation within lineages (Section 13.2).

13.7 Correlated evolution of discrete characters

13.7.1 Pagel's model

Most of the classic questions of typology concern discrete features. For these, the standard phylogenetic tool is the method of Pagel (1994). The idea is to model two binary characters jointly as a single four-state character, and to compare a model in which the two evolve independently with one in which the rate of change in each depends on the current state of the other.

Let X be the order of object and verb (0 = OV, 1 = VO) and Y the order of adposition and noun (0 = postpositions, 1 = prepositions). The joint character has four states: $(0,0)$ = OV with postpositions, $(0,1)$ = OV with prepositions, $(1,0)$ = VO with postpositions and $(1,1)$ = VO with prepositions. The first and last are the "harmonic" combinations of Greenberg's Universal 4 and its mirror image. Change happens in continuous time, and only one character changes at a time. Under the *dependent* model, every one of the eight permitted transitions has its own rate:

$$\mathbf{Q}_{\text{dep}} = \begin{pmatrix} \cdot & q_{12} & q_{13} & 0 \\ q_{21} & \cdot & 0 & q_{24} \\ q_{31} & 0 & \cdot & q_{34} \\ 0 & q_{42} & q_{43} & \cdot \end{pmatrix}, \tag{13.6}$$

with rows and columns in the order $(0,0), (0,1), (1,0), (1,1)$ and diagonal elements chosen so that rows sum to zero. Under the *independent* model, each character has its own gain and loss rates whatever the state of the other character, so the eight rates collapse to four (for example, $q_{13} = q_{24}$: the rate at which OV becomes VO does not depend on adposition order). Both models are fitted by maximum likelihood with the pruning algorithm on the four-state character (Chapter 5), and compared with a likelihood-ratio statistic

$$\text{LR} = 2\,[\ell(\hat{\mathbf{Q}}_{\text{dep}}) - \ell(\hat{\mathbf{Q}}_{\text{ind}})], \tag{13.7}$$

which is referred to a χ^2 distribution with 4 degrees of freedom. In a Bayesian implementation, such as BayesTraits (Pagel and Meade, 2006), the comparison is made with a Bayes factor. The log marginal likelihoods of the two models are estimated, for example by stepping-stone sampling (Chapter 6), and their difference is interpreted on the scale of Kass and Raftery (1995). Values of $2\log \text{BF}$ above about 2 are often treated as "positive" evidence, above 6 as "strong" and above 10 as "very strong".

The independent rate matrix, Equation (D.22), and the constraints that turn the dependent model into it are in Section D.3.6.

The dependent model says more than "the two features are correlated." It describes the *dynamics* of the correlation, and its rate parameters can be read directly. If q_{24} (OV-with-prepositions to VO-with-prepositions) is much larger than q_{13} (OV-with-postpositions to VO-with-postpositions), then a language with prepositions is more likely to switch to VO than a language with postpositions. In other words, adposition order *conditions* change in verb-object order. A pair of large rates

leading out of the disharmonic states and small rates leading out of the harmonic ones means that disharmonic combinations are transient. They are visited, but quickly abandoned. This is just the kind of claim about pathways of change that diachronic typologists have made (Maslova, 2000; Croft et al., 2011).

More specific hypotheses can be tested by constraining particular rates. For example:

- *Contingent change*: does X change faster when $Y = 1$ than when $Y = 0$? Test $q_{13} = q_{24}$ (and/or $q_{31} = q_{42}$) against the unconstrained dependent model.
- *Direction of dependence*: is the dependence of X on Y, or of Y on X, or both? The function `fitPagel()` in **phytools** offers the arguments `dep.var = "x"`, `"y"` or `"xy"` for this.
- *Temporal order*: which change usually comes first? The ratio of rates out of the two disharmonic states suggests whether languages typically pass through OV-with-prepositions or through VO-with-postpositions on the way between the harmonic types.

In BayesTraits, *reversible-jump* MCMC can sample over all patterns of equality constraints among the rates, so that the posterior concentrates on the best-supported simplified model (Pagel and Meade, 2006).

Correlated grammaticalization of articles in Indo-European. Goldstein (2022a) coded the presence of definite and indefinite articles in 94 Indo-European languages as a single four-state character and fitted a dependent model in RevBayes over a sample of 100 trees from the ancestry-constrained posterior of Chang et al. (2015). Because the earliest attested languages lack articles, the root was fixed at "no articles" and losses were excluded, so the model is irreversible. The estimated rate of gaining an indefinite article was more than an order of magnitude higher in lineages that already had a definite article (posterior medians 0.00064 and 0.000036, with non-overlapping 95% HPD intervals), while the rate of gaining a definite article did not clearly depend on the presence of an indefinite one (overlapping intervals). A model with separate rates of indefinite-article gain was preferred over one with a single rate by a log Bayes factor of 5.31. The results are consistent with a trajectory from no articles, to a definite article, to both, and they show how an implicational universal can be restated as a claim about conditional rates of change (compare Cysouw, 2011). Three caveats apply, and they illustrate Section 13.7.3. The result rests on many events, not one: the author counts at least 17 independent origins of definite and 15 of indefinite articles. The hyperpriors on the numbers of transitions, which the author describes as strong, were informed by counts of article origins in the same languages. And contact, which plausibly helped spread articles in parts of Europe, violates the assumption that change on different branches is independent.

Cathcart et al. (2020) used a related design for Indo-Iranian, asking whether the development of sortal numeral classifiers is linked to the loosening of obligatory plural marking, as a diachronic reading of the Greenberg–Sanches–Slobin generalization predicts, rather than being due only to contact with neighbouring non-Indo-European languages. Estimating transition rates on 65 languages, with reconstructions constrained by historically attested states, they found broad support for the diachronic reading but no evidence for a strong bias against the synchronic co-occurrence of classifiers and obligatory plural marking, and a stronger effect in Iranian than in Indo-Aryan.

13.7.2 Pagel's test in R

We simulate a family of 100 languages and let VO order and prepositions evolve under a dependent model. In this model the harmonic states (OV with postpositions, VO with prepositions) are stable,

with low rates of change out of them. The disharmonic states are unstable, with rates of leaving them ten times as high.

```
states <- c("OV|Po", "OV|Pr", "VO|Po", "VO|Pr")
Qdep <- matrix(0, 4, 4, dimnames = list(states, states))
Qdep["OV|Po", c("OV|Pr", "VO|Po")] <- 0.2       # leaving a harmonic state: slow
Qdep["VO|Pr", c("VO|Po", "OV|Pr")] <- 0.2
Qdep["OV|Pr", c("OV|Po", "VO|Pr")] <- 2   # leaving a disharmonic state: fast
Qdep["VO|Po", c("VO|Pr", "OV|Po")] <- 2
diag(Qdep) <- -rowSums(Qdep)

set.seed(7)
tree100 <- pbtree(n = 100, scale = 3)
joint <- sim.Mk(tree100, Qdep)
VO <- setNames(ifelse(grepl("^VO", joint), "VO", "OV"), names(joint))
Pr <- setNames(ifelse(grepl("Pr$", joint), "Pr", "Po"), names(joint))
table(VO, Pr)
```

```
#>     Pr
#> VO   Po Pr
#>   OV 55  6
#>   VO  7 32
```

```
traits <- data.frame(VO = factor(VO), Pr = factor(Pr), row.names = names(VO))
plotTree.datamatrix(tree100, traits, fsize = 0.45, yexp = 1.1, sep = 1.5,
                    colors = list(c(OV = "#E69F00", VO = "#56B4E9"),
                                  c(Po = "#009E73", Pr = "#CC79A7")))
legend("topleft", legend = c("OV", "VO", "Postp.", "Prep."), pch = 15,
       bty = "n",
       col = c("#E69F00", "#56B4E9", "#009E73", "#CC79A7"), cex = 0.8, ncol = 2)
```

`fitPagel()` fits the independent and dependent models and computes the likelihood-ratio test. We use `method = "ace"`, which calls `ape::ace()` internally and is much faster than the default for this size of problem.

```
fit_xy <- fitPagel(tree100, VO, Pr, method = "ace")
fit_xy
```

```
#>
#> Pagel's binary character correlation test:
#>
#> Assumes "ARD" substitution model for both characters
#>
#> Independent model rate matrix:
#>         OV|Po   OV|Pr   VO|Po   VO|Pr
#> OV|Po -0.4320  0.2021  0.2299  0.0000
#> OV|Pr  0.3043 -0.5342  0.0000  0.2299
#> VO|Po  0.3596  0.0000 -0.5617  0.2021
#> VO|Pr  0.0000  0.3596  0.3043 -0.6639
#>
```

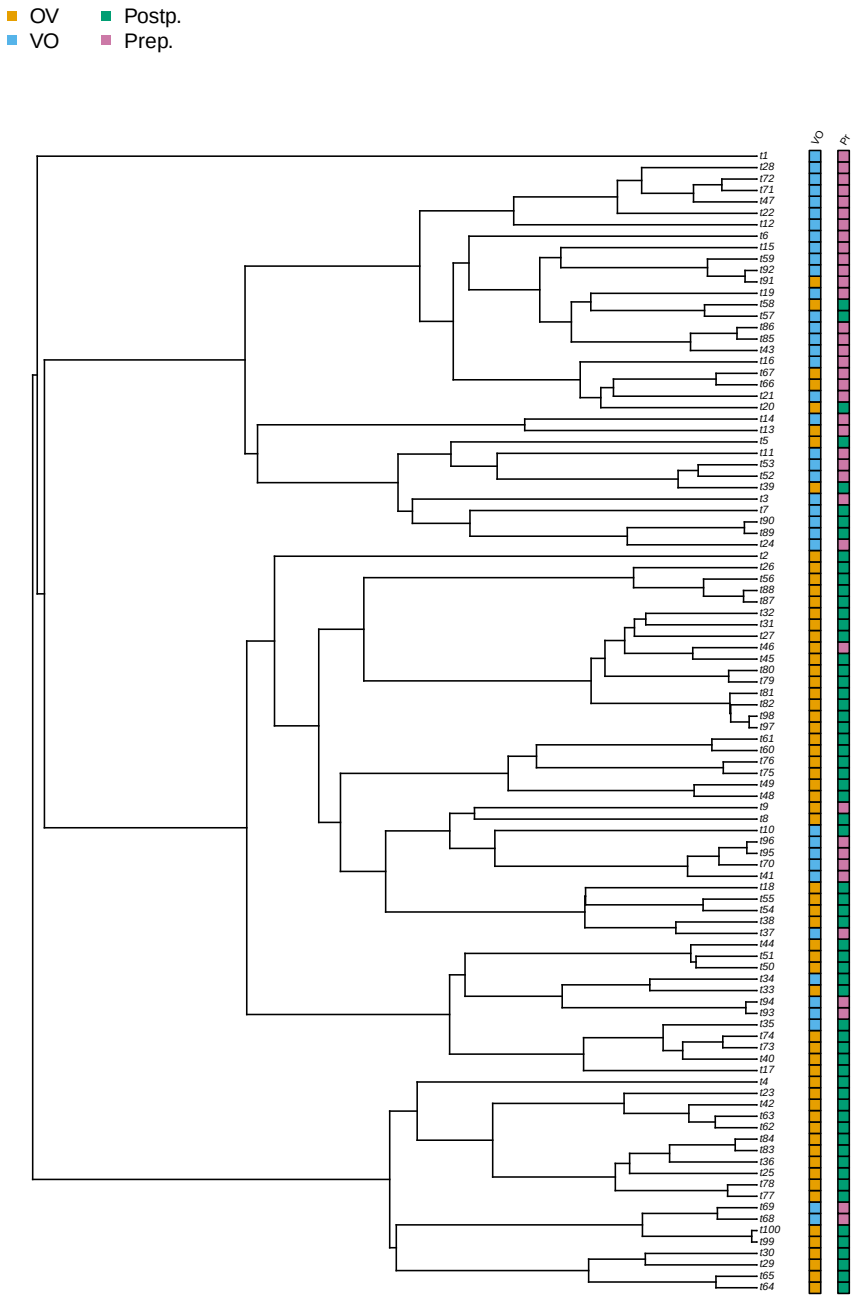


Figure 13.6: Simulated verb-object order (left column: OV or VO) and adposition order (right column: postpositions or prepositions) for 100 languages evolving under a dependent model in which the harmonic combinations are stable.

```
#> Dependent (x & y) model rate matrix:
#>         OV|Po   OV|Pr   VO|Po   VO|Pr
#> OV|Po -0.3169  0.1659  0.1509  0.0000
#> OV|Pr  0.6849 -4.4520  0.0000  3.7671
#> VO|Po  1.7550  0.0000 -2.3991  0.6440
#> VO|Pr  0.0000  0.5455  0.2866 -0.8321
#>
#> Model fit:
#>             log-likelihood   AIC
#> independent         -86.78 181.6
#> dependent           -72.45 160.9
#>
#> Hypothesis test result:
#>   likelihood-ratio:  28.6606
#>   p-value:  9.16161e-06
#>
#> Model fitting method used was ace
```

The dependent model is strongly preferred, and the estimated rate matrix recovers the structure of the generating process. The rates leading out of the disharmonic states (`OV|Pr` and `VO|Po`) are large, and those leading out of the harmonic states are small. We can also ask whether the dependence runs in one direction only:

```
# dep.var = "x": VO depends on Pr; "y": Pr depends on VO
fit_x <- fitPagel(tree100, VO, Pr, method = "ace", dep.var = "x")
fit_y <- fitPagel(tree100, VO, Pr, method = "ace", dep.var = "y")
aic <- c(independent = fit_xy$independent.AIC, VO_depends = fit_x$dependent.AIC,
         Pr_depends = fit_y$dependent.AIC, both = fit_xy$dependent.AIC)
round(aic - min(aic), 1)
```

```
#> independent  VO_depends  Pr_depends        both
#>        24.5         0.0         3.7         3.8
```

The independent model is decisively rejected. The differences among the dependent models are much smaller. In this replicate, a model in which only VO order depends on adposition order fits as well as the full model, or better, even though the generating process had dependence in both directions. This is typical. Detecting *that* two features are coupled requires far less data than establishing *how* they are coupled, and claims about the direction of dependence deserve correspondingly more caution.

For contrast, here is the same test on two features that evolve independently on the same tree. It correctly finds no evidence for dependence:

```
set.seed(8)
Q1 <- matrix(c(-0.5, 0.5, 0.5, -0.5), 2, 2,
             dimnames = list(c("a", "b"), c("a", "b")))
f1 <- sim.Mk(tree100, Q1); f2 <- sim.Mk(tree100, Q1)
fit_null <- fitPagel(tree100, f1, f2, method = "ace")
c(LR = fit_null$lik.ratio, p = fit_null$P)
```

```
#>     LR      p
```

```
#> 1.3428 0.8541
```

13.7.3 What Pagel's test can and cannot show

Maddison and FitzJohn (2015) identified a deep problem with all tests of this kind, which they called the "unsolved challenge" for phylogenetic correlation tests of categorical characters. A strongly significant result can arise from a *single* evolutionary event. Imagine that one branch deep in the tree saw both a change from OV to VO and a change from postpositions to prepositions, and that both states were then inherited by a large clade. The two characters will be perfectly associated across the tips, and the dependent model will fit far better than the independent one. The reason is that under the independent model it is improbable for both characters to change on the *same* branch and nowhere else. Yet a single co-occurrence is weak evidence for a functional link. It is one data point, not a hundred. The code below shows how easily this happens.

```
# pick a clade of about 40 languages; both characters change once, on its stem
n <- Ntip(tree100)
nodes <- (n + 2):(n + tree100$Nnode)
clade_size <- sapply(nodes, function(nd) sum(getDescendants(tree100, nd) <= n))
stem <- nodes[which.min(abs(clade_size - 40))]
members <- getDescendants(tree100, stem)
in_clade <- seq_len(n) %in% members
c1 <- setNames(ifelse(in_clade, "VO", "OV"), tree100$tip.label)
c2 <- setNames(ifelse(in_clade, "Pr", "Po"), tree100$tip.label)
table(c1, c2)
```

```
#>     c2
#> c1   Po Pr
#>   OV 54  0
#>   VO  0 46
```

```
fit_single <- fitPagel(tree100, c1, c2, method = "ace")
c(LR = fit_single$lik.ratio, p = fit_single$P)
```

```
#>       LR        p
#> 16.30159  0.00264
```

The test is "significant" at conventional levels, although the data contain exactly one co-occurring pair of changes. The problem is not a bug in the software. It reflects the question the test answers, which is whether the dependent model describes the data better than the independent one. The test does not ask how many independent instances of correlated change the data contain. Maddison and FitzJohn (2015) proposed that analyses report the number of independent origins of the associated states and treat results driven by one or two events with great skepticism. Uyeda et al. (2018) made a broader version of the same point. Phylogenetic comparative models describe macroevolutionary patterns, and singular historical events, such as one shift on one branch, can dominate both continuous and discrete analyses. They recommended explicitly modeling such events, for instance with models that allow shifts in rate or state on particular branches.

Several other caveats apply with special force to language:

1. *Hidden heterogeneity.* If rates of change vary across lineages for reasons unrelated to the other character (for instance, because some branches experienced intense contact), Pagel's test can

mistake that heterogeneity for dependence. Hidden-rate models (Chapter 12), which add unobserved rate categories, help diagnose this (Beaulieu et al., 2013; Boyko and Beaulieu, 2021).

2. *Contact.* The models assume that change happens independently on each branch. Borrowing of word-order patterns from neighbors violates this. A pair of features may be associated because both spread through the same contact zone.
3. *Coding.* Typological features are often coded from WALS-style categories with a "no dominant order" value that must be dropped or merged. These coding decisions can affect the result.
4. *Tree and branch lengths.* The rates are in units of the tree's branch lengths. Results on trees with arbitrary branch lengths (Section 13.12.2) need cautious interpretation.

13.7.4 Pathways of acquisition

Some linguistic hypotheses are not about correlation in general but about the *order* in which two features arise. Implicational hierarchies such as the sequence of basic color terms of Berlin and Kay (1969), and grammaticalization clines in which one function of a construction develops before another, predict that a language acquires feature Y only after it has acquired feature X. Pagel's four-state coding expresses such a claim directly. If change is restricted to gains and the root is fixed at $(0, 0)$, four rates remain: q_{12} and q_{13} for gaining Y or X first, and q_{24} and q_{34} for completing the pair. The strict hierarchy "X before Y" sets $q_{12} = q_{24} = 0$; a weaker version, in which Y can arise alone but does not lead on to X, sets only $q_{24} = 0$. These hypotheses can be compared by marginal likelihood, or the model structure can itself be sampled. Reversible-jump MCMC can let each rate be exactly zero with some prior probability, so that the posterior probability of a zero rate measures the evidence that a route is closed, and a categorical *model indicator* can switch among a small set of pathway models within one analysis. Both can be run over a posterior sample of trees, so that the answer does not depend on one tree. Haynie and Bowern (2016) used reversible-jump MCMC over a posterior sample of Pama-Nyungan trees in this way to study the gain and loss of color terms (Section 14.10).

Two cautions apply. The gain-only coding with a fixed root makes the state of the root an assumption rather than a result, so the analysis cannot also be used to reconstruct the root (Section 12.4.5). And the Maddison-FitzJohn problem of the previous section applies with full force: a route can look closed simply because the data contain few lineages in which it could have been taken. Simulated data are a useful check on whether the data are informative at all. In the RevBayes implementation of Section H.5.22, which analyses 30 simulated feature pairs over a sample of 100 trees, the posterior probability that the absent route was closed was above 0.99, the three open routes were never closed, and the model indicator put a probability above 0.99 on the generating pathway. These are parameter-recovery results in the sense of Section 6.10, obtained with 30 replicate pairs evolving under the same process; with a single pair of features, as in most linguistic applications, the same analysis is likely to be much less decisive.

The study of articles described in Section 13.7 also illustrates three practical points about pathway models. First, each rate is best read as a *change in a background state*: q_{24} and q_{13} in Equation (13.6) describe the same change in the first character, made with and without the second, so the hypothesis "the second feature promotes the first" becomes the inequality $q_{24} > q_{13}$, which a posterior sample of the two rates addresses directly. *Semi-dependent* models, in which only one character's rates depend on the state of the other, sit between the independent and the fully dependent model and are the natural way to ask which way the conditioning runs. Second, the

gain-only restriction and the root state $(0, 0)$ can rest on the comparative method rather than on convenience, as they do for the articles (Section 12.4.5); the rate estimates are then conditional on that philological judgement. Third, priors on the rates of a gain-only model are easiest to set in terms of the expected number of changes on the whole tree: an exponential prior whose mean is c divided by the tree length says that roughly c changes of that kind would be expected over the history of the sample if every lineage remained able to make them, a quantity on which a specialist can usually express an opinion. Because such counts are easily informed by the same languages that are being analysed, as the caveats to the article study note, the sensitivity of the conclusions to c should be reported.

13.8 Word-order universals on language trees

13.8.1 Dunn, Greenhill, Levinson and Gray (2011)

"Evolved structure of language shows lineage-specific trends in word-order universals." Dunn et al. (2011) applied Pagel's method, in its Bayesian form in BayesTraits, to eight word-order features in four large language families: Austronesian, Bantu, Indo-European and Uto-Aztecan. The features were the orders of adjective and noun, adposition and noun, demonstrative and noun, genitive and noun, numeral and noun, relative clause and noun, object and verb, and subject and verb. For every pair of features in every family, they compared the dependent and independent models by Bayes factors, integrating over samples of trees estimated from lexical cognate data. The striking result was that the networks of strongly supported dependencies differed from family to family. Several dependencies were supported in one family but not in others, and the familiar Greenbergian correlations were not recovered as a uniform set across all four. The authors concluded that word-order dependencies are largely *lineage-specific*. On this view the structure of a language reflects the cultural evolutionary history of its lineage, with each family following its own pathways, rather than universal cognitive or processing constraints. The paper was published in *Nature* and received wide attention well beyond linguistics.

The study was a landmark in two ways. It was the first large-scale attempt to test typological universals with explicit models of diachronic change on dated trees. And it moved the question from "which combinations are frequent?" to "which changes are coupled?" It also provoked one of the most substantial methodological debates in modern typology, much of it collected in a 2011 issue of *Linguistic Typology*.

13.8.2 The responses

Dryer (2011) argued that the study did not test the correlations as typologists had formulated them. The correlations are claims about the languages of the world as a whole, established across genera and macroareas. Four families are a small and unrepresentative sample of that whole. Within a single family, many features show little variation: most Bantu languages are VO and prepositional, for example. A test within that family therefore has little power, and failure to find a dependency there is weak evidence against it. Dryer also argued that the strength of the evidence for a correlation should be judged by how often the preferred combinations arise across independent lineages worldwide, which is what his genus-based method measures. His own data, he noted, show the correlations clearly.

Croft et al. (2011) situated the debate within diachronic typology. Greenbergian universals have long been interpreted as constraints on pathways of change, so a model of change on trees is the

right kind of tool. But a test of *universal* constraints needs to pool information across families, and needs to model the transitions that the constraints are about. They also raised statistical concerns about drawing conclusions from family-specific results based on few independent changes.

Levy and Daumé III (2011) put a similar point sharply in their title: "computational methods are invaluable for typology, but the models must match the questions". A universal is a hypothesis that the same dependency holds in all families. That hypothesis should be tested with a model in which the families *share* parameters, or at least a hierarchical model in which family-specific parameters are drawn from a common distribution. Separate analyses of four families cannot distinguish "the dependency is absent in this family" from "this family provides too little information". A model with shared parameters across families would have far more power to detect a universal tendency, and would test it directly.

Bickel (2011), also in the *Linguistic Typology* discussion, argued for distributional methods that treat families as independent replicates of diachronic processes, of which his Family Bias method is an example. Cysouw (2011) examined what the transition probabilities estimated by such models mean and how they relate to observed frequencies. Because the independent model is nested in the dependent one, he noted, the test can show that dependence is not supported but cannot give strong support for independence, so most of the weak results are absence of evidence rather than evidence of absence. Combining the family-level Bayes factors with Fisher's method (under a frequentist calibration of Bayes factors that is itself debatable), he found support across the families for four classical pairings, among them adposition order with object–verb order. For that pair the stable distributions implied by the published rates are similar in Austronesian and Indo-European; in Austronesian, postpositions with OV order occur in only about 6 percent of the sampled languages but have a stable frequency of about 24 percent, a difference he attributes to the founder effect of a young family. Longobardi and Roberts (2011) responded from the perspective of generative syntax, questioning which kind of universals, if any, the study could bear on. Levinson et al. (2011) replied to the critics. They maintained that if universal dependencies exist, they should be detectable in the history of individual families, and that the family-specific approach makes explicit what claims of universality predict.

With hindsight, the debate settled some issues and sharpened others. Few would now dispute that cross-linguistic correlations should be studied as properties of change, or that phylogenetic models provide the natural framework for doing so. Few would dispute either that analyses confined to a handful of families, each with limited variation, lack the power to establish the *absence* of a universal tendency. In the terms of Section 13.2, the debate was largely about sampling: about what the target of inference was and how many independent changes each family supplied. The real open question is whether the tendencies are weak and universal, strong but lineage-specific, or some mixture of the two. Hierarchical models that let rates vary across families around a shared mean can answer it, because they estimate both the shared tendency and the between-family variance.

13.8.3 Later work

Several later studies took up this agenda. Maurits and Griffiths (2014) used Bayesian phylogenetic models of basic word order (the six orders of subject, object and verb) on trees for many families to estimate ancestral word orders and transition rates. They found evidence that SOV is the most likely ancestral state for many families and that changes among word orders follow characteristic pathways (see Chapter 12). Jäger and Wahle (2021) developed what they called *phylogenetic*

typology. They used posterior samples of trees inferred for each family from ASJP lexical data, in the manner of Jäger (2018), with the Glottolog classification as a constraint (Hammarström et al., 2024), and fitted CTMC models of the kind discussed above to pairs of WALS word-order features, pooling information across 175 lineages. In contrast to the impression left by the 2011 debate, they reported support for a number of the classical word-order correlations once data from many lineages were combined in a single phylogenetic model. Not every proposed correlation was supported, though. Their work shows that the answer depends heavily on the scope of the sample and on whether information is pooled across families, which is exactly what Levy and Daumé III (2011) had argued.

13.8.4 Grambank

Grambank and the relative weight of genealogy and geography. Skirgård et al. (2023) introduced Grambank, a database of 195 grammatical features coded for more than 2,400 languages by a large team, with explicit coding protocols and sources. One of their analyses addressed the old question of whether languages are more similar because of shared descent or because of shared geography. For each feature they fitted a Bayesian model with three latent components: a phylogenetic effect, with covariance derived from a global language phylogeny; a spatial effect, with covariance decaying with geographic distance; and residual variation. Models of this type are called *spatiophylogenetic models*, and they were fitted with the integrated nested Laplace approximation [INLA; Rue et al. (2009)]. Averaged across features, phylogeny accounted for more variance than space, and the relative contributions varied widely between features. The study also illustrated the cost of language endangerment for typology. Simulated loss of the currently endangered languages would reduce the structural diversity documented in Grambank only moderately at the global scale, but much more severely in some regions, so that the loss would be strikingly uneven. Methodologically, the paper marks the point at which spatiophylogenetic models, long used in ecology, became a standard tool in typology. Section 13.10 shows a small example.

Grambank has since become the reference dataset for testing typological universals and claims about social and environmental effects on grammar with phylogenetic and spatial controls, as in the analysis of Shcherbakova et al. (2023) discussed in Section 13.11.

Testing 191 universals at once. Verkerk et al. (2025b) used Grambank to test 191 proposed grammatical universals with Bayesian analyses that account for both genealogical descent and geographic proximity. About a third of the proposed universals received statistical support, most of them word-order universals and hierarchical universals, the two types that have been most prominent in earlier work. Evolutionary analyses suggested that languages tend to change in ways that converge on the preferred patterns, which the authors interpret as evidence that shared cognitive and communicative pressures repeatedly push languages toward similar solutions. Two features of the design matter for interpretation. Testing many universals at once guards against the selective reporting that has plagued single-universal studies, but the result for each universal depends on how its features are coded in Grambank and on the global tree used as a scaffold. And support for a universal under a spatiophylogenetic model is evidence of a statistical tendency in change, not of a particular cognitive mechanism.

Other Grambank-based studies have moved from correlation toward explicit causal models. Shcherbakova et al. (2024) analyzed case marking, verbal indexing and word order in 1,705 languages with phylogenetic causal graph methods and concluded that verb-final order tends to give rise to case marking and, more surprisingly, that case marking tends to lead to flexible

word order. Because Grambank's features were designed for coverage rather than statistical independence, Graff et al. (2025b) derived curated versions of Grambank and of other structural databases in which logically dependent and strongly correlated features are reduced, for analyses that treat features as independent.

13.9 Phylogenetic regression for non-Gaussian responses

Pagel's test handles pairs of discrete features. Many questions, though, call for a *regression*: a binary or count response, with several predictors, some continuous. Does the probability that a language has tone depend on humidity, controlling for latitude? Does the presence of grammatical gender depend on population size and on the number of neighboring languages? PGLS is not appropriate for binary responses. Three families of methods fill the gap: phylogenetic logistic regression, phylogenetic generalized linear mixed models, and the threshold model.

13.9.1 Phylogenetic logistic regression

Ives and Garland (2010) developed a *phylogenetic logistic regression* in which the binary response evolves along the tree as a two-state Markov process. The rates of this process are chosen so that the expected value of the response for language i follows the usual logistic form,

$$\Pr(y_i = 1) = \mu_i = \frac{1}{1 + e^{-\mathbf{x}_i^{\top}\boldsymbol{\beta}}}, \tag{13.8}$$

while a parameter α controls how fast the process "forgets" its past. When α is large, the state of a language is almost independent of its ancestors, and the model approaches ordinary logistic regression. When α is small, related languages are strongly correlated, and the effective sample size shrinks. Estimation is by maximum penalized likelihood. The penalty, due to Firth (1993), reduces the small-sample bias of logistic regression and keeps estimates finite when the data are nearly separated, a common situation when a feature is almost fixed within large clades. The method is implemented as `phylolm::phyloglm()` (Ho and Ané, 2014b).

Below we simulate a binary response (call it `tone`, present or absent) whose probability depends on a continuous predictor (call it `humid`, a standardized humidity index). Both variables are phylogenetically structured. We then compare ordinary and phylogenetic logistic regression.

```
set.seed(11)
humid <- rTrait(n = 1, phy = tree100, model = "BM",
                parameters = list(sigma2 = 0.4))
tone  <- rbinTrait(n = 1, phy = tree100, beta = c(-0.5, 1), alpha = 0.5,
                   X = cbind(1, humid))
dtone <- data.frame(lang = tree100$tip.label, humid = humid, tone = tone,
                    row.names = tree100$tip.label)
table(dtone$tone)
```

```
#>
#>  0  1
#> 64 36
```

```
fit_glm <- glm(tone ~ humid, data = dtone, family = binomial)
fit_plr <- phyloglm(tone ~ humid, data = dtone, phy = tree100,
                    method = "logistic_MPLE")
rbind(glm      = coef(summary(fit_glm))["humid", c(1, 2, 4)],
      phyloglm = coef(summary(fit_plr))["humid", c(1, 2, 4)])
```

```
#>          Estimate Std. Error Pr(>|z|)
#> glm        0.6739     0.2603 0.009628
#> phyloglm   0.7440     0.3124 0.017230
```

```
fit_plr$alpha
```

```
#>
#> 0.7874
```

The comparison mirrors the Gaussian case. The ordinary logistic regression treats the 100 languages as independent observations, and its standard error is correspondingly optimistic. The phylogenetic model estimates α and adjusts the standard errors to the amount of independent information in the data. In real analyses the two often disagree not only in precision but in the sign or magnitude of the effect. That happens when a few large clades with particular values of both variables drive the ordinary regression.

13.9.2 Phylogenetic generalized linear mixed models

The most general and flexible framework is the *phylogenetic generalized linear mixed model* (PGLMM). It comes from quantitative genetics, where a *pedigree* plays the role of the phylogeny (Lynch, 1991; Housworth et al., 2004; Hadfield and Nakagawa, 2010). The model has a linear predictor with a random effect for each language whose covariance is proportional to the phylogenetic correlation matrix:

$$g\big(\mathbb{E}[y_i]\big) = \eta_i = \mathbf{x}_i^\top \boldsymbol{\beta} + a_i + e_i, \qquad \mathbf{a} \sim \mathcal{N}(\mathbf{0}, \sigma_a^2 \mathbf{A}), \quad e_i \sim \mathcal{N}(0, \sigma_e^2), \tag{13.9}$$

where g is a link function (identity for Gaussian, logit or probit for binary data, log for counts), $\mathbf{A}$ is the phylogenetic correlation matrix ($\mathbf{C}$ scaled to unit diagonal for an ultrametric tree), and e_i is a language-level residual. For a Gaussian response, the *phylogenetic heritability*

$$H^2 = \frac{\sigma_a^2}{\sigma_a^2 + \sigma_e^2} \tag{13.10}$$

is equivalent to Pagel's λ (Housworth et al., 2004). For non-Gaussian responses it is defined on the latent scale. The PGLMM framework has several advantages that matter for linguistic data:

- *Multiple observations per language.* The model separates between-language variation (phylogenetic and non-phylogenetic) from within-language variation. This is useful for corpus data, for speaker-level data in sociolinguistics, or for multiple sources coding the same language.
- *Several random effects at once.* A spatial random effect, a random effect for macroarea, and a random effect for the coding source can sit alongside the phylogenetic one (Section 13.10).

- *Multi-response models.* Two or more features can be modeled jointly, with phylogenetic and residual correlations between them estimated separately (Hadfield and Nakagawa, 2010). A phylogenetic correlation means that the features have coevolved over the long term. A residual correlation means that they covary among languages in ways not attributable to shared history, as expected under contact or a recent functional coupling. This decomposition is directly relevant to the debate over lineage-specific versus universal dependencies, and to our knowledge it has rarely if ever been used in typology.
- *Bayesian inference.* The two standard implementations are **MCMCglmm** (Hadfield, 2010) and **brms** (Bürkner, 2017), the latter an interface to Stan. Bayesian inference makes it natural to integrate over a sample of trees (Section 13.12.1).

Ecologists have developed closely related models for communities of species (Ives and Helmus, 2011), and the frequentist **phyr** package fits PGLMMs quickly. In linguistics, mixed models with random effects for family and area (Jaeger et al., 2011; Bentz and Winter, 2013) can be seen as coarse PGLMMs with a block-diagonal $\mathbf{A}$. Every language within a family is assumed to be equally related to every other.

The following code fits a Gaussian PGLMM to the complexity data from Section 13.6.3 with **MCMCglmm**, using a short chain. The phylogeny enters as the inverse of the relatedness matrix, computed by `inverseA()`.

```
library(MCMCglmm)
set.seed(5)
Ainv <- inverseA(tree, nodes = "TIPS", scale = TRUE)$Ainv
prior <- list(G = list(G1 = list(V = 1, nu = 0.002)),
              R = list(V = 1, nu = 0.002))
m_mc <- MCMCglmm(complexity ~ logpop, random = ~lang,
                 ginverse = list(lang = Ainv),
                 data = langs, prior = prior, nitt = 23000, burnin = 3000,
                 thin = 20, verbose = FALSE)
summary(m_mc)$solutions
```

```
#>             post.mean l-95% CI u-95% CI eff.samp pMCMC
#> (Intercept)    4.4432   1.8352  7.28648     1000 0.002
#> logpop        -0.2747  -0.5443 -0.03021     1000 0.046
```

```
H2 <- m_mc$VCV[, "lang"] / rowSums(m_mc$VCV)
hpd <- HPDinterval(H2)
c(posterior_mean_H2 = mean(H2), lower95 = hpd[1, 1], upper95 = hpd[1, 2])
```

```
#> posterior_mean_H2           lower95           upper95
#>            0.8325            0.6122            0.9999
```

The posterior mean of the slope is close to the ML estimate from PGLS. The phylogenetic heritability plays the role of λ, and its posterior shows how uncertain that quantity is with 60 languages. For real analyses, chains should be much longer. Convergence should be checked as described in Chapter 6, and the sensitivity of variance components to their priors should be examined. Weakly informative or parameter-expanded priors are generally preferable to the inverse-gamma prior used here.

The same model in **brms**, with notes on the sampler settings it needs, is in Section D.3.7.

13.9.3 The threshold model

Many discrete linguistic features plausibly reflect an underlying continuous tendency. The degree of reliance on suffixation, the strength of a preference for head-final structures, or the degree of grammaticalization of a construction may vary continuously while being coded as present or absent. The *threshold model* of quantitative genetics, introduced to comparative biology by Felsenstein (2005; Felsenstein, 2012), formalizes this idea. Each language has an unobserved *liability* l_i that evolves by Brownian motion on the tree. The observed state is 1 when the liability exceeds a threshold (conventionally 0) and 0 otherwise:

$$y_i = \begin{cases} 1 & \text{if } l_i > 0, \\ 0 & \text{otherwise,} \end{cases} \qquad \mathbf{l} \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \mathbf{C}). \tag{13.11}$$

Multistate ordered characters use several thresholds. The model has properties that Markov models lack. A lineage whose liability lies close to the threshold may flip back and forth quickly, while one far from it is very stable. The probability of change therefore depends on the lineage's history, and not only on its current state. Correlations between two discrete characters, or between a discrete and a continuous one, are simply correlations between liabilities (Felsenstein, 2012). Ancestral states can be reconstructed on the liability scale (Revell, 2014), which is implemented in `phytools::ancThresh()` and `threshBayes()`. A threshold PGLMM is a probit PGLMM with the residual variance fixed; in **MCMCglmm** it is available as `family = "threshold"`. To our knowledge, threshold models have been used only rarely on linguistic data, although many typological features seem natural candidates for them. They deserve a place alongside Mk models in typological work (see also Section 16.5.2).

A threshold model for the simulated tone data fitted with **MCMCglmm**, and the conversion between its probit-scale coefficients and the logistic coefficients above, are in Section D.3.8.

From the biologist's toolbox: multivariate and shift models. Comparative biology has moved well beyond single-trait BM. Several families of methods could be applied directly to typological and lexical data. (1) *Multivariate BM and OU models*, implemented in **mvMORPH** (Clavel et al., 2015), estimate evolutionary covariance matrices among several continuous traits. With them one can ask, for example, whether measures of fusion and exponence coevolve, and whether that coupling differs between families. (2) *Phylogenetic factor analysis* (Tolkoff et al., 2018) reduces many traits to a few latent factors that evolve along the tree. It could be used to find coevolving "modules" of grammatical features in Grambank-scale data. (3) *Rate-shift models* detect branches or clades where the rate of evolution changes. Examples are the reversible-jump approach of Eastman et al. (2011) and the variable-rates model of Venditti et al. (2011). Applied to typology, they would identify lineages with bursts of structural change, for instance after intense contact or language shift. (4) *Multi-optimum OU models*, with regimes specified in advance (Butler and King, 2004; Beaulieu et al., 2012) or detected from the data (Uyeda and Harmon, 2014; Khabbazian et al., 2016), could test whether languages in different social or ecological settings are drawn toward different optima. (5) *Hidden-rate and structured Markov models* for discrete characters (Beaulieu et al., 2013; Boyko and Beaulieu, 2021) allow the dependence between features to switch on and off across the tree. They are a principled way to represent "lineage-specific" dependencies, which is the very issue raised by Dunn et al. (2011). (6) *Measurement-error and within-species variation models* (Ives et al., 2007) could represent dialect variation and coding uncertainty. Most of these methods are available in R and run on the same tree and data objects used in this chapter. To our

knowledge, few if any have been applied to linguistic data so far.

13.10 Space and phylogeny together

Galton's problem has two faces: descent and diffusion. The methods so far address only descent. Languages that are geographically close share features through contact. They also tend to be closely related, which makes the two sources of similarity hard to separate. A phylogenetic regression that ignores space will attribute some areal similarity to the tree, or leave it in the residuals. A spatial regression that ignores phylogeny will make the opposite mistake.

The natural generalization is a model with separate covariance components for phylogeny and space:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{a} + \mathbf{s} + \mathbf{e}, \quad \mathbf{a} \sim \mathcal{N}(\mathbf{0}, \sigma_p^2 \mathbf{C}),\ \mathbf{s} \sim \mathcal{N}(\mathbf{0}, \sigma_s^2 \mathbf{S}),\ \mathbf{e} \sim \mathcal{N}(\mathbf{0}, \sigma_e^2 \mathbf{I}), \tag{13.12}$$

where $\mathbf{S}$ is a spatial correlation matrix, for instance $S_{ij} = \exp(-d_{ij}/\rho)$ with d_{ij} the geographic distance and ρ a range parameter. Freckleton and Jetz (2009) developed a version of this model for comparative biology. In ecology it is usually fitted in a Bayesian framework with INLA or MCMC. The proportions $\sigma_p^2/(\sigma_p^2+\sigma_s^2+\sigma_e^2)$ and $\sigma_s^2/(\sigma_p^2+\sigma_s^2+\sigma_e^2)$ measure the relative importance of descent and diffusion. These are the quantities reported by Skirgård et al. (2023). Mixed models with crossed random effects for family and macroarea (Jaeger et al., 2011) are a coarse version of the same idea.

Below we simulate languages whose locations are themselves inherited along the tree, as they would be if languages spread by migration from a homeland (Chapter 11). A trait then receives a phylogenetic component, a spatial component and noise, each contributing a known share of the variance. Because the true shares are known, this is a parameter-recovery experiment.

```
set.seed(99)
lon <- fastBM(tree, sig2 = 1) + rnorm(Ntip(tree), 0, 0.3)
lat <- fastBM(tree, sig2 = 1) + rnorm(Ntip(tree), 0, 0.3)
D <- as.matrix(dist(cbind(lon, lat)))
S <- exp(-D / 1)                                  # spatial correlation, range 1
L_s <- t(chol(S + diag(1e-8, nrow(S))))
a_phy <- fastBM(tree, sig2 = 1 / max(diag(C)))  # phylogenetic part, variance 1
s_sp  <- drop(L_s %*% rnorm(Ntip(tree)))            # spatial part, variance 1
trait <- 0.8 * a_phy + 0.8 * s_sp + rnorm(Ntip(tree), 0, 0.5)
true_share <- c(0.64, 0.64, 0.25) / sum(c(0.64, 0.64, 0.25))
round(true_share, 2)                                # phylo, space, noise
```

```
#> [1] 0.42 0.42 0.16
```

The model of Equation (13.12) can be fitted in **MCMCglmm** with two random effects for the same languages, one with the inverse phylogenetic correlation matrix and one with the inverse spatial correlation matrix. The range ρ is held fixed at its true value for simplicity; real analyses estimate it or compare several values.

```
dsp <- data.frame(lang = tips, lang_sp = tips, y = trait[tips])
Sinv <- as(solve(S + diag(1e-6, nrow(S))), "CsparseMatrix")
```

```
dimnames(Sinv) <- list(tips, tips)
px <- list(V = 1, nu = 1, alpha.mu = 0, alpha.V = 1000)  # parameter-expanded
prior_sp <- list(G = list(G1 = px, G2 = px), R = list(V = 1, nu = 0.002))
set.seed(1)
m_sp <- MCMCglmm(y ~ 1, random = ~lang + lang_sp, data = dsp, prior = prior_sp,
                 ginverse = list(lang = Ainv, lang_sp = Sinv),
                 nitt = 33000, burnin = 3000, thin = 30, verbose = FALSE)
share <- m_sp$VCV / rowSums(m_sp$VCV)
colnames(share) <- c("phylo", "space", "noise")
round(rbind(posterior_mean = colMeans(share), t(HPDinterval(share))), 2)
```

```
#>                phylo space noise
#> posterior_mean  0.38  0.57  0.06
#> lower           0.00  0.07  0.00
#> upper           0.90  1.00  0.31
```

The posterior means are in the right region, and the model correctly attributes most of the variance to the two structured components rather than to noise. But the 95% intervals for the phylogenetic and spatial shares are very wide: with 60 languages whose locations are themselves inherited, the two components are partly confounded, and the data cannot apportion the variance between them precisely. Real analyses use larger samples, estimate the range parameter, and report full posterior distributions of the variance shares, as here. The example shows the key logic: two non-independence structures can be modeled simultaneously and their contributions compared.

A maximum-likelihood fit of the same model written from scratch, and its comparison with phylogeny-only and space-only models by AIC, are in Section D.3.9.

Separating phylogeny from space is fundamentally difficult when related languages are also neighbors, which is the normal situation. In such cases, the data contain little information about which of the two structures is responsible for similarity, and the variance shares can depend heavily on priors, on the range parameter and on the tree's branch lengths. Large families that have spread over wide areas (Austronesian, Bantu, Indo-European), and contact zones that cut across families, are where the two can be told apart.

13.11 Environment, society and language: causal claims

Since about 2010, a growing literature has claimed that properties of languages are shaped by the physical or social environment of their speakers. Examples include humidity and tone, altitude and ejectives, population size and morphology, and the proportion of adult learners and case marking. These claims are interesting because they cut against a long-standing assumption in linguistics: that language structure is independent of the environment, apart from the lexicon. They are also exactly the kind of claim that Galton's problem threatens most. Environmental and social variables are strongly spatially and phylogenetically structured. A few large families (Niger-Congo, Austronesian, Sino-Tibetan) dominate global samples, so a handful of historical accidents can produce a worldwide correlation. Ladd et al. (2015) review such correlational studies, both of linguistic features with one another and of features with properties of speakers and environments, and trace much of the scepticism of linguists to differences in assumptions and methods between large-scale quantitative work and traditional historical and theoretical research. This section re-

views the most prominent cases, and then asks what phylogenetic control contributes to a causal argument.

13.11.1 Tone and humidity

Climate, vocal folds and tonal languages. Everett et al. (2015) observed that languages with complex tone systems (three or more tones) are concentrated in humid regions, such as sub-Saharan Africa, Southeast Asia and parts of New Guinea and the Americas. They are rare in very dry or cold regions. The authors argued for a physiological mechanism. Dry air desiccates the vocal folds and makes precise control of pitch harder, so that over many generations the transmission of complex tone would be slightly less successful in arid climates. They tested the pattern on a large sample of languages, using resampling procedures designed to control for relatedness and area, and found that it held across several families and regions. The paper generated a lively exchange. In the inaugural issue of the *Journal of Language Evolution*, Everett et al. (2016) restated and extended the case for climatic adaptation in a target article that drew a series of commentaries. Hammarström (2016) argued that once genealogical and areal dependencies are properly controlled, the evidence for a desiccation effect largely disappears. Collins (2016) argued, with simulations, that the global pattern can be explained by the spread of tone through contact in a few large areas, which happen to be humid. Everett and colleagues replied to the commentaries in the same issue. Roberts (2018) used the debate as a case study for a broader methodological argument: claims of linguistic adaptation should be tested in multiple, independent ways, combining large-scale correlational studies with experimental and causal-inference approaches, rather than relying on a single regression.

The tone-humidity debate illustrates a general structure. A global correlation exists. Its interpretation depends on whether the effective number of independent observations is large (many families and areas showing the pattern) or small (a few large tonal areas). Phylogenetic and spatial models are the formal way to settle that question. Their answer depends on how well the trees and spatial structure are represented, and on the statistical power left once dependencies are accounted for.

13.11.2 Population size and morphological complexity

The linguistic niche hypothesis. Lupyan and Dale (2010) analyzed more than 2,000 languages in WALS and found that languages spoken by larger populations, over larger areas and with more neighboring languages tend to use less inflectional morphology. They rely more on lexical and syntactic strategies. Lupyan and Dale proposed the *linguistic niche hypothesis*. Languages adapt to the social environments in which they are learned and used, and "esoteric" niches (small, close-knit communities with few adult learners) favor complex morphology, while "exoteric" niches (large populations with many adult learners) favor simplification. Related work by Bentz and Winter (2013), using mixed-effects models with random effects for family and area, found that languages with a larger proportion of second-language speakers are less likely to have nominal case. Bentz et al. (2015) reported a related effect on lexical diversity (fewer distinct word forms in languages with more non-native speakers). Koplenig (2019) reanalyzed similar data and found that the number of speakers was associated with some structural measures, but that the proportion of non-native speakers was not, once relatedness and area were controlled. The most comprehensive test to date is that of Shcherbakova et al. (2023). They used Grambank data and Bayesian spatiophylogenetic models of grammatical complexity measures, with population size and the proportion of L2 speakers as predictors. They found no support for the claim that "societies of strangers" speak

less complex languages, which is also the title of their paper.

This line of research shows how conclusions can change as methods for handling non-independence improve: from global correlations, to mixed models with family and area effects, to explicit spatiophylogenetic models on new data. It also illustrates that "complexity" is not one variable. Different operationalizations (number of inflectional categories, fusion, informativity, number of case markers) can behave differently, and choosing among them after seeing the data inflates false positives just as surely as ignoring phylogeny does.

13.11.3 Other environmental and demographic claims

Several other claims follow the same pattern.

- *Phoneme inventories and distance from Africa.* Atkinson (2011) reported that phoneme inventory size declines with distance from a putative origin in southwestern Africa. He interpreted this as a serial founder effect during the human expansion out of Africa, analogous to the decline of genetic diversity. Critics argued that the analysis did not adequately control for relatedness and area, that phoneme counts are not comparable across sources, and that the result is sensitive to modeling choices (Jaeger et al., 2011; Cysouw et al., 2012).
- *Altitude and ejectives.* Everett (2013) argued that ejective consonants are more common in languages spoken at high altitude, proposing an aerodynamic mechanism. Later analyses questioned the robustness of the pattern once areal and genealogical structure was taken into account (Urban and Moran, 2021).
- *Bite configuration and labiodentals.* Blasi et al. (2019) argued that labiodental sounds (such as f and [v]) became more common after the Neolithic, because softer diets preserved the overbite and overjet of adolescence into adulthood. The argument combined biomechanical modeling, cross-linguistic distributional analysis with controls for relatedness, and historical reconstruction within Indo-European.
- *Future tense and saving.* Chen (2013) reported that speakers of languages that grammatically mark future time save less, smoke more and exercise less. Roberts et al. (2015) re-tested the claim with explicit control for linguistic relatedness and geography, using phylogenetic and mixed-model approaches. They found that the relationship was considerably weaker, and not robust across all methods, once historical non-independence was taken into account.
- *Spurious correlations.* Roberts and Winters (2013) made the general point vividly. They showed that cultural and linguistic traits correlate with all kinds of unrelated variables in cross-national data, and that such correlations are expected from shared history and geography alone. A related argument in the cross-cultural literature, Bromham et al. (2018b), showed that correlations between pathogen load and cultural traits are highly sensitive to controls for relatedness, proximity and covariation among predictors.
- *Ecology and diversity.* Hua et al. (2019) examined the ecological correlates of language diversity worldwide, analyzing the density of languages rather than their structure. They too had to deal with spatial autocorrelation and with the confounding of climate, topography and history.

13.11.4 What phylogenetic control does and does not do

It helps to state the causal logic explicitly, in the language of directed acyclic graphs (DAGs) (Pearl, 2009). Galton's problem is a problem of *confounding*. Shared history (H) affects both a predictor X and a response Y, because a proto-language passed on its values of both, so X and Y are

associated across its descendants even if neither affects the other: $X \leftarrow H \rightarrow Y$. A phylogenetic regression controls for H to the extent that the tree and the evolutionary model capture it. In effect, it compares changes in X with changes in Y along independent branches, a within-lineage comparison analogous to a fixed-effects design in econometrics. The estimand is therefore an *evolutionary* association: whether lineages that changed in X also tended to change in Y.

Several other routes from X to Y remain open, and Table 13.4 lists the most important. Contact creates associations among neighbors whatever their ancestry. Environmental variables such as climate are not inherited at all (speakers move, climates change), and they are structured in space, so they can confound a regression on any other spatially structured variable. A single deep event can produce an association that looks strong but rests on one observation (Section 13.7.3). The process that determines which languages are documented, and so enter the sample, can itself create associations: if large, written languages are more likely to be described in detail, and description affects how complexity is coded, conditioning on documentation conditions on a common effect of population size and coded complexity. And with observational data the direction of causation is not given: population size might affect morphology, or the history that produced a large population might also have produced a particular kind of morphology.

Table 13.4: Sources of cross-linguistic association other than a causal effect of X on Y, and what addresses each.

Source of association	Example	What addresses it
Shared inheritance	Bantu languages share SVO and prepositions from Proto-Bantu	Phylogenetic covariance (PGLS, PGLMM), if the tree and model are adequate
Contact and areal diffusion	Tone spreading through Mainland Southeast Asia	Spatial covariance, area effects, explicit contact models
Shared environment that is not inherited	Humidity and latitude both structured in space	Measuring and including the environmental covariates; spatial models
A single historical event	One branch changes both features, then radiates	Counting independent origins; dropping large clades; shift models
Selective documentation	Well-described languages are larger and coded in more detail	Modeling missingness; restricting to uniformly coded data; sensitivity analyses
Reverse causation	Language structure affecting group size or spread	Temporal order on dated trees; directional models of coevolution
Measurement and coding choices	Several operationalizations of “complexity”	Pre-specification; reporting all codings; multiple-testing control

Three practical conclusions follow. First, phylogenetic control is necessary but not sufficient: it removes one confounder, the one represented by the tree. Second, the strength of an argument depends on the number of independent changes in X that the sample contains (Section 13.2), not on the number of languages. Third, the most persuasive studies combine several lines of evidence, as advocated by Roberts (2018): large-sample comparative analyses with phylogenetic and spatial controls, within-family tests where the predictor varies, experiments on learning and production, and mechanistic models. Databases such as CHIELD, which records causal hypotheses in evolutionary linguistics as explicit causal graphs (Roberts et al., 2020), help make such hypotheses

explicit and comparable. Dated trees add one thing that a cross-sectional sample cannot: the order of changes. Directional versions of Pagel's test (Section 13.7) ask whether changes in X tend to precede changes in Y; temporal precedence is weaker than causation, but it rules out some causal stories.

13.11.5 Phylogenetic path analysis

When the hypotheses involve several variables, a useful framework is *phylogenetic confirmatory path analysis* (von Hardenberg and Gonzalez-Voyer, 2013). It combines PGLS with the *d-separation* test of Shipley (2000). A causal hypothesis is written as a DAG. Every DAG implies a set of conditional independencies: pairs of variables that should be independent given certain others. For example, the DAG "population size → proportion of L2 speakers → morphological complexity" implies that population size and complexity are independent once the L2 proportion is controlled for. Each implied independence is tested with a PGLS regression, which yields a p-value p_k, and the results are combined with Fisher's statistic

$$C = -2 \sum_{k=1}^{K} \ln p_k \sim \chi^2_{2K}. \tag{13.13}$$

A small C (large p) means that the DAG is consistent with the data. Competing DAGs can be ranked with an information criterion based on C. The R package **phylopath** (van der Bijl, 2018) automates the procedure. The steps are simple enough to carry out directly, which also makes the logic visible. We simulate data under a mediation DAG and compare it with a "common cause" DAG in which population size affects both L2 proportion and complexity directly.

```
set.seed(77)
pop   <- fastBM(tree, sig2 = 0.3)                # log population
l2    <- 0.7 * pop + fastBM(tree, sig2 = 0.1)    # L2 proportion (latent scale)
morph <- -0.6 * l2 + fastBM(tree, sig2 = 0.1)    # morphological complexity
pdat  <- data.frame(lang = tips, pop = pop[tips], l2 = l2[tips],
                    morph = morph[tips])

pgls_p <- function(formula, term) {   # p-value of one term in a BM PGLS
  m <- gls(formula, data = pdat,
           correlation = corBrownian(1, tree, form = ~lang))
  summary(m)$tTable[term, "p-value"]
}
# DAG A (mediation): pop -> l2 -> morph.
# Implied independence: morph _||_ pop | l2
p_A <- pgls_p(morph ~ l2 + pop, "pop")
# DAG B (common cause): l2 <- pop -> morph.
# Implied independence: morph _||_ l2 | pop
p_B <- pgls_p(morph ~ pop + l2, "l2")
C_stat <- -2 * log(c(A_mediation = p_A, B_common_cause = p_B))
cbind(C = C_stat, p = pchisq(C_stat, df = 2, lower.tail = FALSE))
```

```
#>                     C         p
#> A_mediation     0.9843 6.113e-01
```

```
#> B_common_cause 26.3115 1.934e-06
```

The mediation DAG, which generated the data, is consistent with them: its implied independence holds. The common-cause DAG is rejected, because complexity is *not* independent of L2 proportion once population size is controlled for. The logic extends to larger graphs, in which each DAG implies several independencies, and to PGLS with λ estimated or to phylogenetic logistic regression for binary variables. Its limits are those of any DAG-based method. The candidate DAGs must be specified in advance, unmeasured confounders (including every row of Table 13.4 except the first) can create spurious dependencies, and several DAGs may imply the same independencies (they are then *Markov equivalent* and cannot be distinguished by the data).

13.12 Practicalities

13.12.1 Integrating over tree uncertainty

All the analyses so far have treated the tree as known. In practice, language trees are estimated, often with considerable uncertainty in topology and branch lengths (Chapter 6). There are three common ways to handle this.

1. *Use a single summary tree*, such as a maximum clade credibility tree. This is simple, but it ignores uncertainty and can give overconfident results; Section 11.10 explains why a summary tree can mislead even when its topology looks plausible, and Section 12.15.7 shows an example for a named ancestor.
2. *Repeat the analysis on a sample of trees* from the posterior and summarize the distribution of estimates. For frequentist analyses, the estimates can be combined with the rules developed for multiple imputation (Rubin, 1987), whose use for phylogenetic uncertainty in comparative analyses was set out by Nakagawa and de Villemereuil (2019). If $\hat{\beta}_m$ and s_m are the estimate and its standard error on tree m of M, the pooled estimate and its total variance are
$$\bar{\beta} = \frac{1}{M}\sum_{m=1}^{M}\hat{\beta}_m, \qquad T = \underbrace{\frac{1}{M}\sum_{m=1}^{M}s_m^2}_{\text{within trees}} + \underbrace{\left(1+\frac{1}{M}\right)\frac{1}{M-1}\sum_{m=1}^{M}(\hat{\beta}_m-\bar{\beta})^2}_{\text{between trees}}. \tag{13.14}$$
The between-tree term is what a single-tree analysis leaves out.
3. *Integrate over trees within a Bayesian analysis.* BayesTraits does this by sampling trees from the posterior sample as part of the MCMC (Pagel et al., 2004). In **MCMCglmm** or **brms** one can fit the model to each tree in turn and pool the posterior samples.

Below we create a mock "posterior sample" of 100 trees by applying random nearest-neighbor interchanges and branch-length noise to the true tree. We then fit the BM PGLS of complexity on log population to each tree, combine the results with Equation (13.14), and compare them with an analysis of the MCC tree of the sample alone.

```
set.seed(2025)
tree_sample <- phangorn::rNNI(tree, moves = 4, n = 100)
tree_sample <- lapply(tree_sample, function(t) {
  t$edge.length <- t$edge.length * rlnorm(length(t$edge.length), 0, 0.25); t })
class(tree_sample) <- "multiPhylo"
gls_bm <- function(tr, y, x) {        # fast BM GLS via whitening
```

```
  Wt <- chol(solve(vcv(tr)[names(y), names(y)]))
  f <- lm(Wt %*% y ~ Wt %*% cbind(1, x) - 1)
  coef(summary(f))[2, 1:2]
}
y_c <- setNames(langs$complexity, tips); x_c <- setNames(langs$logpop, tips)
est <- t(sapply(tree_sample, gls_bm, y = y_c, x = x_c))
M <- nrow(est)
pooled <- mean(est[, 1])
total_se <- sqrt(mean(est[, 2]^2) + (1 + 1 / M) * var(est[, 1]))
mcc_s <- phangorn::maxCladeCred(tree_sample)
tab_unc <- rbind(true_tree  = gls_bm(tree, y_c, x_c),
                 MCC_tree   = gls_bm(mcc_s, y_c, x_c),
                 over_trees = c(pooled, total_se))
cbind(tab_unc, p = 2 * pnorm(-abs(tab_unc[, 1] / tab_unc[, 2])))
```

```
#>            Estimate Std. Error       p
#> true_tree   -0.3416     0.1406 0.01510
#> MCC_tree    -0.2660     0.1401 0.05752
#> over_trees  -0.3369     0.1470 0.02192
```

```
hist(est[, 1], breaks = 20, col = "grey80", border = "white", main = "",
     xlab = "PGLS slope (Brownian motion)")
abline(v = tab_unc["true_tree", 1], lty = 2)
abline(v = tab_unc["MCC_tree", 1], col = "red", lwd = 2)
abline(v = pooled, lwd = 2)
```

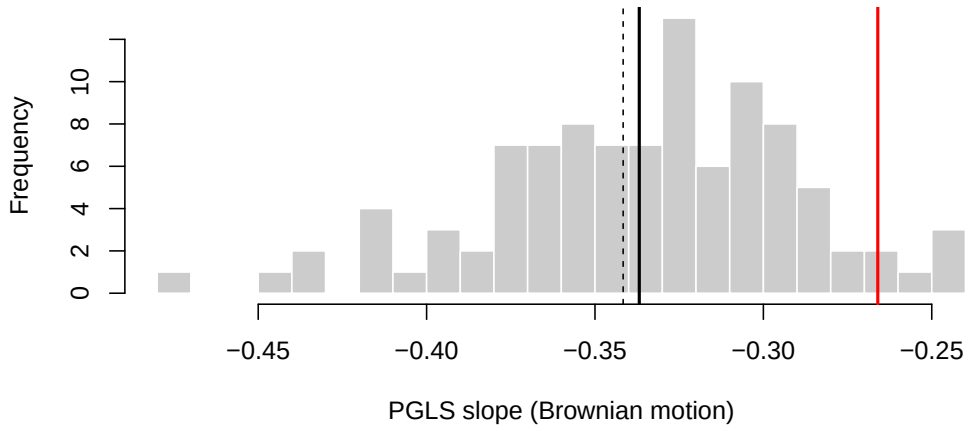


Figure 13.7: Distribution of PGLS slope estimates for complexity on log population across 100 trees representing phylogenetic uncertainty. The dashed line marks the estimate on the true tree, the red line the estimate on the MCC tree, and the solid black line the pooled estimate.

In this example the topological and branch-length uncertainty is moderate, so the slope estimates

vary somewhat across trees, and the pooled standard error is somewhat larger than that of a single-tree analysis. The MCC tree gives just one of the slopes in the histogram (here -0.27, against a pooled estimate of -0.34), with a standard error that reflects none of their spread. In this replicate the MCC tree happens to lie in the tail of the distribution, and its p-value falls on the other side of the conventional threshold from the pooled one. The direction of the error is a matter of chance: on another sample the MCC tree could just as well make a weak result look strong. With real posterior samples the variation can be much larger, particularly when the deep structure of a family is uncertain. Pooling is also a diagnostic. If the estimates on different trees disagree in sign, the data do not support a conclusion about the association whatever the pooled p-value says, and the trees on which the result depends deserve inspection.

13.12.2 Global analyses on Glottolog topologies with arbitrary branch lengths

Typological questions are often global, and no dated phylogeny covers all the world's languages. The usual workaround is to take the classification in Glottolog (Hammarström et al., 2024), convert it to a tree, and assign *arbitrary* branch lengths. Common choices are to make all branches equal in length, or to use Grafen's method (Grafen, 1989), in which a node's height is proportional to the number of tips below it, minus one, raised to a power. Families are then joined at a root, either as a "star" or under a deep node of arbitrary age. This produces a usable covariance structure, but its consequences need to be understood:

- Glottolog classifications are *topologies*, often with many polytomies, and they reflect the uneven state of comparative research. Well-studied families have deep, detailed subgrouping, while poorly studied ones are flat. The covariance structure therefore partly encodes the history of scholarship, and the same is true of the cognate-based trees used elsewhere, whose cognate judgments come from the comparative method.
- With arbitrary branch lengths, rates and transition probabilities are in arbitrary units, and the strength of phylogenetic signal depends on the branch-length scheme.
- Joining families at a common root implies covariance among them, unless the families are treated as independent (block-diagonal $\mathbf{C}$). A star of families is usually the more defensible choice, because relationships among families are by definition not demonstrated.
- Equal branch lengths treat every split as equally deep. Grafen's method makes large clades deep and small ones shallow, which is closer to what dated trees usually show but has no historical basis for any particular family.

Three strategies reduce the arbitrariness. The first is to let the data adjust the branch lengths through a transformation estimated jointly with the model (λ, κ or δ; Section 13.3.4). The second is to replace the topology of individual families by dated trees where they exist (Chapter 10), stitched into the global structure. The third is to use global phylogenies estimated from lexical data, such as those of Jäger (2018), which used ASJP word lists with Glottolog constraints and give data-driven branch lengths, at the cost of strong modeling assumptions. Whichever option is chosen, sensitivity analyses across branch-length schemes are essential. The code below estimates λ for the BM trait simulated earlier, first on the true tree, then on several "Glottolog-like" versions of it, with short internal branches collapsed into polytomies and branch lengths replaced by arbitrary schemes.

```
glotto <- di2multi(tree, tol = 0.25)                    # collapse short branches
glotto$edge.length <- NULL                              # keep topology only
schemes <- list(
```

```
  true           = tree,
  equal_branches = compute.brlen(glotto, 1),
  grafen_1       = compute.brlen(glotto, method = "Grafen", power = 1),
  grafen_0.5     = compute.brlen(glotto, method = "Grafen", power = 0.5))
brlen_tab <- t(sapply(schemes, function(tr) {
  ps <- phylosig(tr, x_bm[tips], method = "lambda")
  c(internal_nodes = tr$Nnode, lambda = ps$lambda, logL = ps$logL)
}))
knitr::kable(brlen_tab, digits = 2,
             caption = "Pagel's lambda for the same simulated trait estimated on the
             true tree and on topology-only trees with arbitrary branch lengths.")
```

Table 13.5: Pagel's lambda for the same simulated trait estimated on the true tree and on topology-only trees with arbitrary branch lengths.

	internal_nodes	lambda	logL
true	59	1.01	-99.72
equal_branches	48	1.00	-110.61
grafen_1	48	0.84	-112.91
grafen_0.5	48	1.03	-110.70

The estimates of λ and the log-likelihoods differ among branch-length schemes, even though the trait and the topology (apart from the collapsed branches) are the same. Polytomies themselves are not a problem for GLS, since a polytomy is just a block of equal covariances. But arbitrary branch lengths change the relative weight of shallow and deep structure, and so they change both the estimated signal and the effective sample size. The consequences for a regression are shown next: the PGLS of complexity on log population, with λ estimated, on each scheme.

```
brlen_pgls <- t(sapply(schemes, function(tr) {
  f <- phylolm(complexity ~ logpop, data = langs, phy = tr, model = "lambda")
  cf <- summary(f)$coefficients
  c(slope = cf["logpop", "Estimate"], se = cf["logpop", "StdErr"],
    p = cf["logpop", "p.value"], lambda = f$optpar)
}))
round(brlen_pgls, 3)
```

```
#>                 slope    se     p lambda
#> true           -0.262 0.127 0.044  0.812
#> equal_branches -0.241 0.133 0.076  0.933
#> grafen_1       -0.244 0.142 0.093  0.826
#> grafen_0.5     -0.250 0.139 0.076  0.854
```

The slope is similar on every scheme, but its standard error and p-value move enough that a conventional significance threshold is crossed on some schemes and not on others. A result that holds on one branch-length scheme only should be reported as fragile.

13.12.3 Data issues

Several practical issues recur in comparative analyses of linguistic data:

- *Matching.* Language names, ISO 639-3 codes and Glottocodes must be matched between the trait database and the tree. Dialects and closely related varieties may map onto one tip or several.
- *Missing data.* WALS is famously sparse: most features are coded for a minority of languages. Dropping languages with missing values is simple, but it is unbiased only if the data are missing at random with respect to the variables of interest. Data are often missing for poorly documented languages, which tend to be small and spoken in particular regions. Phylogenetic imputation, which predicts missing values from relatives, is possible within Bayesian PGLMMs. It should be reported as such.
- *Coding.* Multistate features are often collapsed to binary ones. Categories such as "no dominant order" or "both orders" must be handled consistently. Results can depend on these choices, which should be decided before the analysis and checked for sensitivity afterward.
- *Sample composition.* Large families dominate, and isolates and small families contribute little information about rates of change, however well the model handles dependence. A global analysis can be driven by a few families. Reporting family-level results alongside the global ones helps reveal this (Section 13.2).

13.13 Pitfalls

We end with a list of common errors, many of them illustrated earlier in this chapter.

1. **Ignoring phylogeny.** Ordinary regressions and contingency tests on samples of related languages give inflated false positive rates (Figure 13.2). Genus-based sampling helps but does not eliminate the problem.
2. **Confusing languages with information.** Sixty languages may be worth a handful of independent observations (Section 13.2). Compute the effective sample size and count the independent changes in the predictor.
3. **Testing the variables instead of the residuals.** The need for phylogenetic correction depends on the residuals of the model, not on the phylogenetic signal in the response or predictor alone (Revell, 2010). Estimate λ jointly with the regression.
4. **Counting events as if they were languages.** A significant Pagel test or PGLS can rest on one or two historical events, such as a single deep split (Maddison and FitzJohn, 2015; Uyeda et al., 2018). Report the number of independent origins, inspect the data on the tree, and do a sensitivity analysis that drops large clades.
5. **Treating the tree as known.** Integrate over a sample of trees and pool the results, or at least check robustness across a few plausible trees; do not rely on the MCC tree alone.
6. **Arbitrary branch lengths.** Results on Glottolog-style trees depend on the branch-length scheme. Analyze several schemes and report all of them.
7. **Ignoring contact.** A tree models descent only. Areal diffusion is a second source of non-independence that phylogenetic models do not address. Add spatial components, area random effects, or at least check whether residuals are spatially autocorrelated.
8. **Over-interpreting model parameters.** A preference for OU does not demonstrate stabilizing selection (Cooper et al., 2016). A high λ does not demonstrate slow change (Revell et al., 2008). A dependent model in Pagel's test does not demonstrate a functional link.

9. **Confusing correlation with causation.** Phylogenetic control removes one confounder (Table 13.4). Causal claims about environment and language need multiple lines of evidence (Roberts, 2018).
10. **Lack of power.** Within single families, features often vary little, and failure to reject independence says little about the absence of a dependency. Simulate data under the alternative to estimate power before interpreting a null result.
11. **Forking paths.** With many features, codings, trees and model variants, some combination will be "significant." Pre-register analyses where possible, correct for multiple comparisons, and report all analyses performed.

- Galton's problem, the non-independence of related and neighboring languages, inflates false positive rates of naive cross-linguistic tests, often dramatically. Sampling one language per genus reduces but does not remove the problem.
- Phylogenetic methods change the purpose of sampling rather than removing it. The information in a sample is measured by its effective sample size and by the number of independent changes in the predictor, which can be far smaller than the number of languages.
- Gaussian phylogenetic comparative methods rest on a covariance matrix $\mathbf{C}$ derived from the tree under Brownian motion. OU, early-burst and Pagel's λ, κ and δ models modify $\mathbf{C}$ in interpretable ways.
- Phylogenetic signal (λ, K, Moran's I, D) measures how strongly related languages resemble each other. It is a pattern, not a rate, and it should be estimated on residuals when the goal is regression.
- Independent contrasts and PGLS are equivalent under Brownian motion. PGLS generalizes to multiple predictors, estimated λ and other covariance structures.
- Pagel's 1994 test compares independent and dependent four-state Markov models for pairs of binary features. It describes the dynamics of coupled change, but it can be driven by single events.
- Dunn et al. (2011) found lineage-specific word-order dependencies in four families. The responses stressed limited power within families and the need for models that pool information across them. Later global analyses found support for a number of classical correlations.
- Phylogenetic logistic regression, PGLMMs (MCMCglmm, brms) and threshold models extend phylogenetic regression to binary and other non-Gaussian responses. Spatiophylogenetic models add a spatial component, as in the Grambank analyses.
- Phylogenetic control removes the confounding represented by the tree, and nothing else. Contact, non-inherited environment, single events, selective documentation and reverse causation need their own designs; causal claims require multiple lines of evidence.
- Tree uncertainty and arbitrary branch lengths can move a result across a significance threshold. Repeat analyses over a sample of trees and several branch-length schemes, and pool with Rubin's rules.

13.14 Exercises

Conceptual

1. Explain the two faces of Galton's problem with a linguistic example of each, and say which face genealogical stratification addresses and which it does not.
2. A colleague argues that because a feature shows no phylogenetic signal ($\hat{\lambda} \approx 0$), a regression with that feature as response needs no phylogenetic correction. What is wrong with this

argument?
3. Sixty Bantu languages and five isolates are coded for a feature. Which contributes more information about (a) the mean frequency of the feature and (b) the rate at which it changes? Relate your answer to the effective sample size.
4. Explain why a significant Pagel test can arise from a single historical event, and describe two analyses that would reveal whether this has happened.
5. For the claim that humidity favors tone, draw a DAG that includes shared history, contact, latitude and documentation, and say which paths a spatiophylogenetic regression blocks and which remain open.
6. Why does joining all families under a common root in a Glottolog-based tree imply something about their history? When would a block-diagonal covariance be preferable?

Computational

1. Rerun the Galton simulation (`pcms-galton-sim`) with the traits simulated under an OU process (`fastBM(tree, alpha = 2, theta = 0)`) and with a tree of 200 languages from `pbtree()`. How do the false positive rates of OLS and PGLS change? Which kind of test, in the vocabulary of Section 6.10, is each version?
2. Use `neff()` to compare the effective sample size of the 60-language tree under `lambda = 1`, `0.5` and `0.1`, and of a "global" sample made by binding ten copies of a 6-language tree under a star root (`bind.tree()` or `read.tree()` with a polytomy). Explain the difference.
3. Using `tree_sample`, compute the pooled slope and standard error for the lambda model with `phylolm(model = "lambda")` instead of the BM GLS. Then increase the topological noise to `moves = 15` and recompute. How does the between-tree share of the total variance change?
4. Apply `fitPagel()` to the two characters of `pcms-pagel-maddison` after dropping the 40-language clade with `drop.tip()`. Then simulate correlated evolution with `sim.Mk()` under `Qdep` on a tree of 30 languages and estimate the power of the test over 100 replicates.
5. Refit the spatiophylogenetic model of Section 13.10 with the spatial range set to 0.3 and 3 instead of 1 (recompute `S` and `Sinv`), and compare the posterior variance shares. What does this tell you about reporting such shares?
6. In chunk `pcms-collinear`, raise the independent part of `x2` from `sig2 = 0.05` to `0.2` and `1` and refit the three models with `fit_lam()`. Track the correlation between the predictors, the standard errors of the two slopes in the joint model and the AIC differences. At what correlation can the two effects be told apart with 60 languages? Repeat with 20 simulated datasets per setting and report how often the joint model is preferred by AIC.

Research

1. You plan to test whether languages with many L2 speakers have fewer case markers, using Grambank. Write a design: the sampling frame and exclusions, the tree (global or family trees; branch lengths), the treatment of tree uncertainty, spatial controls, the coding of the response, the handling of missing data and documentation bias, and the sensitivity analyses you will pre-register.
2. A reviewer asks you to re-examine a claimed universal on a single well-described family of 80 languages. Argue what such an analysis can and cannot show, design a hierarchical model that pools this family with others, and specify what result would count as evidence for a lineage-specific rather than a universal dependency.
3. Choose one environmental claim reviewed in Section 13.11 and design a multi-pronged test

in the spirit of Roberts (2018), combining a comparative analysis with at least one other kind of evidence.

13.15 Further reading

- Harvey and Pagel (1991), *The Comparative Method in Evolutionary Biology*. The classic statement of the problem and its early solutions. It is still the best conceptual introduction to why comparative data need phylogenies.
- Harmon (2019), *Phylogenetic Comparative Methods: Learning from Trees*. A free, rigorous textbook covering BM, OU, discrete characters and diversification models, with clear derivations.
- Revell and Harmon (2022), *Phylogenetic Comparative Methods in R*. A hands-on guide by the author of **phytools**, covering nearly every method in this chapter with code.
- Garamszegi (2014) (ed.), *Modern Phylogenetic Comparative Methods and Their Application in Evolutionary Biology*. A collection of chapters on PGLS (Symonds and Blomberg, 2014), PGLMMs, path analysis and model checking.
- Nunn (2011), *The Comparative Approach in Evolutionary Anthropology and Biology*. Accessible and attentive to human data, with many examples from anthropology.
- Mace and Pagel (1994), "The comparative method in anthropology." The paper that brought phylogenetic thinking to cross-cultural comparison. It remains a good introduction for social scientists.
- Dunn et al. (2011) and the 2011 *Linguistic Typology* discussion (Dryer, 2011; Croft et al., 2011; Levy and Daumé III, 2011; Bickel, 2011). Essential reading on what phylogenetic tests of universals can and cannot show.
- Jäger and Wahle (2021), "Phylogenetic typology." A clear exposition of CTMC-based correlation tests on global trees and a re-examination of word-order universals.
- Ladd et al. (2015), "Correlational studies in typological and historical linguistics." A review written for linguists of correlations among features and between features and the environment, of why such work meets skepticism, and of the methods used to deal with relatedness and contact.
- Verkerk et al. (2025b): A test of 191 proposed universals on Grambank with spatiophylogenetic models, the largest such test to date.
- Macklin-Cordes and Round (2022): An introduction to phylogenetic comparative methods written for typologists, framed as the natural successor to balanced sampling.
- Guzmán Naranjo and Becker (2022), "Statistical bias control in typology." Phylogenetic and Gaussian-process areal controls for unbalanced typological samples, with an argument that such controls make strict probability sampling unnecessary.
- Nakagawa and de Villemereuil (2019): A general method, based on Rubin's rules, for carrying phylogenetic uncertainty (and uncertainty in species data) through a comparative analysis.
- Skirgård et al. (2023): The Grambank paper, with spatiophylogenetic analyses of nearly 200 features. A model for large-scale comparative typology.
- Uyeda et al. (2018), "Rethinking phylogenetic comparative methods." A critical review of what PCMs can infer, with recommendations that apply directly to linguistic data.
- Bromham (2024), "Solving Galton's problem." A practical survey, written for linguists, of solutions to non-independence among languages that need only widely available data: phylogenetic analysis, sister-pair comparisons and spatially structured models.

- Huntington-Klein (2022), *The Effect*. An introduction to research design and causal inference with directed acyclic graphs, written for the social sciences; useful background for Section 13.11.

Chapter 14

Language Trees as Scaffolds for Cultural Evolution

Language trees are not only about languages. Because languages are usually passed from parents to children, a language phylogeny is also an estimate of the population history of the people who speak those languages, and that history can serve as a scaffold for studying how people organize kinship and marriage, how they govern themselves, what they believe, which stories they tell, what they make and what they eat. This research program, often called *cultural phylogenetics* or the study of *cultural macroevolution*, uses the statistical machinery of Chapters 12 and 13. Its central assumption is also its central risk: that a cultural trait was transmitted along the same paths as the language. This chapter reviews what the program has found, and asks throughout when a language tree is a good model of a cultural trait's history and when it is not.

After reading this chapter you should be able to:

- state the logic of cultural phylogenetics and the assumptions about transmission on which it rests;
- distinguish linguistic inheritance from cultural transmission, explain when a language tree is a reasonable proxy for population history, and identify processes (language shift, diffusion, sex-biased transmission) that decouple cultural traits from language;
- describe the classic applications (kinship, political complexity, religion, folktales, material culture, color and number terms) and the design each used;
- reconstruct ancestral cultural states and test constrained models and coevolution with Mk models and Pagel's test, over a sample of trees rather than a single tree;
- evaluate the robustness of phylogenetic comparative methods to horizontal transmission, and explain why its effect depends on who borrows from whom;
- compare language trees with genetic trees and interpret incongruence.

```
suppressPackageStartupMessages({
  library(ape)
  library(phytools)
})
```

14.1 The cultural phylogenetics program

14.1.1 From Galton's problem to phylogenetic anthropology

Cross-cultural comparison has been a central method of anthropology since the nineteenth century. It has also been haunted by Galton's problem since the day it was first stated (Chapter 13). When Tylor reported associations between customs across hundreds of societies, Galton asked how many of those societies were really independent (Tylor, 1889). For most of the twentieth century, anthropologists responded by sampling. The best-known solution is the *Standard Cross-Cultural Sample* (SCCS) of Murdock and White (1969). It selected 186 societies from Murdock's *Ethnographic Atlas* (Murdock, 1967) so that each represented a distinct "cultural province" and was as independent as possible from the others. The SCCS was a major achievement, and it remains in use. But it shares the limitations of genealogical sampling in typology. It discards most of the data. It treats relatedness as all-or-nothing. And it does not use the information contained in the history of change within groups of related societies.

Mace and Pagel (1994) argued in *Current Anthropology* that comparative biology had already solved this problem for species. Their solution was to treat related societies as related species and to analyze their traits with methods that model evolution along a phylogeny. Where would the phylogeny come from? For human societies the best available source is language. Language classifications and, increasingly, quantitative language phylogenies provide an explicit and independently derived hypothesis of population history. By the early 2000s Ruth Mace, Clare Holden, Mark Pagel, Russell Gray and their colleagues had turned this proposal into an active research program. Its first phase is summarized in Mace and Holden (2005) and in the edited volume of Mace et al. (2005). Later reviews include Mace and Jordan (2011), Gray and Watts (2017) and Mesoudi (2011).

14.1.2 The core logic

The program rests on a simple argument with three steps.

1. *Languages are mostly transmitted vertically*, from one generation to the next within a community. A language phylogeny therefore approximates the history of population splits.
2. *Many cultural traits are also transmitted mostly vertically*, within the same communities and along the same channels of upbringing and social learning as language. They are therefore expected to show phylogenetic signal on the language tree.
3. If both of these hold, then *the methods of Chapter 13 and Chapter 12 apply*. We can reconstruct the cultural traits of ancestral societies, estimate rates and pathways of cultural change, and test hypotheses about the coevolution of cultural traits and about their relationship with the environment, all while controlling for shared history.

Each step can fail, and Section 14.2 and the critiques at the end of this chapter are largely about when and how. But the program has two strong advantages. First, language trees are estimated from data (basic vocabulary) that are largely independent of the cultural traits under study, which avoids circularity. Second, the trees are often dated (Chapter 10), so cultural change can be placed in absolute time, sometimes reaching back thousands of years before written records.

14.1.3 What questions can be asked?

Borgerhoff Mulder et al. (2006) and Gray et al. (2007) gave useful classifications of the questions

that cultural phylogenetics can address. With minor adaptation, they are:

- **Ancestral states.** What was the post-marital residence of Proto-Austronesian society? Did Proto-Indo-European speakers practice dowry or bridewealth? Did the tellers of the earliest Indo-European folktales already know "The Smith and the Devil"?
- **Rates and stability.** Which traits change quickly and which slowly? Are some states "absorbing" or very stable once reached?
- **Pathways and sequences.** Does cultural change follow fixed sequences? Do societies move from simple to complex political organization one step at a time? Do changes in residence precede changes in descent?
- **Coevolution.** Do two traits evolve together, and if so, which one drives the other? Did the adoption of cattle herding lead to the loss of matrilineal descent?
- **Environment and adaptation.** Is a trait associated with an environmental variable once shared history is controlled for?
- **Modes of transmission.** Is a trait transmitted mostly vertically, so that it tracks the language tree, or horizontally, so that it tracks geography?

14.2 Linguistic inheritance and cultural transmission

The first step of the core logic deserves a closer look, because "a language tree is a tree of populations" is an approximation whose quality varies from case to case, and because the second step asks more of it than the first.

14.2.1 What a language tree estimates

A lexical phylogeny estimates the history of *language lineages*: the sequence in which a speech tradition split into traditions that ceased to share innovations in basic vocabulary (Chapters 3 and 7). It is built from one channel of transmission, the learning of a first language, mostly in childhood and mostly from caregivers and peers within the community. When communities split and stay apart, and when children learn the language of their parents' community, that channel tracks the history of the communities, and the language tree is a good estimate of population history. This is the situation the program assumes, and in some families and periods (the settlement of Remote Oceania, where new islands were colonized by small groups who carried their language, is the clearest case) it is close to the truth.

Two things make the approximation imperfect even in good cases. The tree describes the history of a language, which can be passed to people who were not descended from its earlier speakers. And a population at any moment is a network of communities exchanging spouses, goods and ideas, so that its "splits" are rarely clean. A language tree is therefore best read as one estimate of population history, made with one kind of evidence, rather than as the population history itself (Chapter 11 makes the same point about the movement of lineages and of people).

14.2.2 When cultural traits travel without language

Cultural traits can follow paths that language does not. Four situations recur.

- **Language shift.** A community adopts the language of a neighboring or dominant group and keeps much of its own culture: its kinship system, its crafts, its rituals. On the language tree, the community now sits inside the clade of the language it adopted, and its traditional

practices appear as innovations within that clade, or as evidence of convergence, when in fact they are retentions from a different history. Elite dominance and the spread of lingua francas produce the same decoupling on a larger scale.

- **Diffusion without language change.** Crops, technologies, religions and ritual complexes spread among neighbors who keep their own languages. A trait that diffuses in this way tracks geography and contact, not the language tree. Basketry in Native California, which tracks proximity more closely than linguistic affiliation (Jordan and Shennan, 2003), is a well-documented case.
- **Different channels of transmission.** Different traits are learned from different people. A first language is learned in childhood; a craft may be learned through apprenticeship, possibly outside the community; ritual knowledge may be restricted to initiated specialists; a post-marital residence rule determines which sex moves between communities at marriage. Under patrilocal residence, for example, women move between communities, so traits learned from mothers may circulate across community boundaries more than traits learned from fathers. Each channel has its own history, and only some of them run parallel to the transmission of language.
- **Ethnogenesis.** New groups form by the merging of old ones, and their culture is a mosaic of traits with different origins (Moore, 1994). A single branching tree may then describe none of the traits well.

14.2.3 Every trait has its own history

A useful way to think about these cases comes from the relationship between gene trees and species trees in biology (Section 16.2.1). Each gene has its own genealogy, which may differ from the history of the populations that carry it, because ancestral variation is sorted differently in different lineages and because genes flow between populations. In the same way, each cultural trait has its own history of transmission, and the language tree is one such history: that of the trait "language". The cultural phylogenetic program assumes that the trait of interest shares that history. Two implications follow. First, even under perfectly vertical transmission, a trait that was variable within an ancestral community (several residence patterns coexisting, say) can sort into descendant communities in a way that does not match the language tree, the cultural analogue of incomplete lineage sorting. Second, the assumption of a shared history is testable, trait by trait, and should be tested rather than assumed.

14.2.4 Diagnosing co-transmission

Several diagnostics are available, none conclusive on its own.

- **Phylogenetic signal** of the trait on the language tree (Section 13.4) shows that related societies are similar, but in most regions related societies are also neighbors, so signal alone does not distinguish inheritance from diffusion among neighbors.
- **Phylogenetic versus spatial models** (Section 13.10) compare the two explanations directly, as da Silva and Tehrani (2016) did for folktales and Towner et al. (2012) for western North American societies. Their power depends on the existence of related societies that are far apart and unrelated societies that are close together.
- **Treelikeness** statistics (Chapter 8) measure whether the trait data, taken together, have tree-like structure at all (Collard et al., 2006).
- **Alternative scaffolds**, such as genetic trees where they exist (Section 14.13), show whether

conclusions depend on taking the language tree as the population history.

The robustness of comparative methods to horizontal transmission is taken up again in Section 14.14, with a simulation.

14.3 Methods in brief

The methods used in cultural phylogenetics are those of the previous two chapters. They are applied to cultural rather than linguistic traits, and almost always on samples of trees from Bayesian analyses of lexical data. Four features of the typical analysis deserve mention.

First, most cultural traits are coded as discrete characters with two to five states. Descent may be matrilineal, patrilineal, bilateral or ambilineal. Residence may be matrilocal (uxorilocal), patrilocal (virilocal), ambilocal or neolocal. Political organization may be acephalous, a simple chiefdom, a complex chiefdom or a state. These are analyzed with continuous-time Markov models, the Mk family introduced in Chapter 12, whose rate matrices can be constrained to test hypotheses about pathways of change. A model in which political complexity can only increase or decrease one level at a time is one such constraint.

Second, the dominant software for most of the history of the field has been BayesTraits (Pagel et al., 2004; Pagel and Meade, 2006). BayesTraits integrates over a posterior sample of trees, uses reversible-jump MCMC to explore constraints on rate matrices, and implements Pagel's test for correlated evolution (Chapter 13). Ancestral states are estimated as posterior probabilities averaged over trees. Nodes absent from some trees are handled by the "most recent common ancestor" approach, which reconstructs the state of the MRCA of a specified set of tips in each tree (Section 12.5).

Third, the questions about directionality are typically addressed by comparing the dependent model's transition rates. In a four-state model for two binary traits, a large rate from "cattle present, matrilineal" to "cattle present, non-matrilineal", with a much smaller rate for the same descent change in the absence of cattle, indicates that cattle make the loss of matriliny more likely. *Flow diagrams* summarizing which transitions are most probable are a common way of presenting these results.

Fourth, integrating over trees matters at least as much here as in linguistic applications. Cultural traits are typically coded for fewer societies than linguistic features, rates are poorly estimated, and asymmetric models can place estimates at the boundary of the parameter space (a rate of exactly zero), where a reconstruction on any single tree becomes deceptively certain. Section 14.15.2 shows how a reconstruction that is certain on the MCC tree can be far from certain over the posterior.

14.4 Kinship, descent, residence and marriage

Kinship has been a core topic of anthropology since the nineteenth century, and it was the first major area of application for cultural phylogenetics. Social anthropology had produced a body of evolutionary hypotheses about kinship that had never been tested statistically. One example is the claim that matrilineal descent is incompatible with the ownership of large livestock. Another is the "Main Sequence Theory", which holds that changes in residence drive changes in descent and then in kin terminology.

Cattle and the loss of matriliny in Bantu-speaking Africa. An old anthropological dictum holds that "the cow is the enemy of matriliny": matrilineal descent is rare among peoples who keep cattle. Holden and Mace (2003) tested this idea on a phylogeny of Bantu and related languages, which Holden (2002) had earlier built from basic vocabulary using maximum parsimony. They used Pagel's correlated-evolution method with ethnographic data on descent and cattle keeping from the *Ethnographic Atlas*. The dependent model fitted significantly better than the independent one. The estimated rates showed that societies that acquired cattle were much more likely to lose matrilineal descent than societies without cattle. Changes in the other direction, such as acquiring cattle as a consequence of patrilineal descent, were not supported in the same way. The study set the template for many later analyses: an anthropological hypothesis about the coevolution of two traits, a language tree, Pagel's test, and an interpretation of the rate matrix as evidence about the direction of causation. An earlier study by the same group (Holden and Mace, 1997) used a similar approach to show that the cultural practice of dairying and the genetic trait of adult lactose digestion coevolved across populations, with the evidence suggesting that dairying came first.

Matrilocal residence in early Austronesian societies. Jordan et al. (2009) estimated the history of post-marital residence across Austronesian societies on a posterior sample of trees from lexical data. They coded residence from ethnographic sources and reconstructed ancestral states with Bayesian Mk models. The reconstruction indicated that matrilocal (uxorilocal) residence was the most probable state at the deepest nodes of the tree, including the ancestors of the Malayo-Polynesian subgroup. Shifts toward patrilocality occurred later, in many lineages independently. The result was relevant to debates about the social organization of the early Austronesian expansion. It also fitted with arguments, based partly on genetic data, that the early Austronesian expansion involved female-centered residence and sex-biased patterns of migration. It also illustrates a general methodological point. A trait's current frequency (most Austronesian societies today are not matrilocal) can be a poor guide to its ancestral state. The asymmetry in transition rates, with matrilocality lost more readily than gained, matters as much as the distribution at the tips.

Comparable studies have addressed other families and other aspects of kinship. Fortunato et al. (2006) used Bayesian ancestral state estimation on Indo-European trees to examine the long-standing idea that dowry developed out of an earlier system of bridewealth. Fortunato and Jordan (2010) compared the histories of marital residence in Indo-European and Austronesian societies. Their reconstructions pointed to patrilocal (virilocal) residence as the most likely ancestral state in Indo-European, in contrast with the matrilocal reconstruction for Austronesian. Opie et al. (2014) analyzed descent and residence together on Bantu trees. They found that the order of changes predicted by Main Sequence Theory, in which residence changes first and descent follows, was not well supported, and that kinship systems could change along multiple pathways. Guillon and Mace (2016) asked a related question of kin terminology itself. Using a sample of 500 Bayesian trees of Bantu languages and *Ethnographic Atlas* data for 83 societies, they reconstructed the history of cousin terminology and found limited support for the hypothesis that it coevolved with descent and post-marital residence, concluding that the shaping of kinship terminology is multifactorial. Walker et al. (2011) reconstructed marriage practices among hunter-gatherer societies on several language phylogenies. Minocher et al. (2019) analyzed marriage patterns, such as polygyny and monogamy, in a globally representative sample of societies using a supertree of human populations (Duda and Zrzavý, 2016) that combined linguistic and genetic information. They concluded that both socio-ecological factors and population history shape marriage systems. At an even more abstract level, Passmore and Jordan (2020) examined kinship terminologies (the

ways languages classify relatives, such as whether a father's brother and a mother's brother are called by the same term). They found little evidence for universal evolutionary pathways across language families, a result reminiscent of the lineage-specific word-order dependencies of Dunn et al. (2011) (Chapter 13). Kinship terminologies are a special case for the question of Section 14.2: they are both cultural and linguistic, transmitted with the vocabulary that expresses them, so they are among the cultural traits most likely to share the language tree's history.

14.5 Political complexity and social stratification

The rise and fall of political complexity. Do human societies evolve toward greater political complexity in a series of small steps, or can they jump from simple to complex forms? And is the trajectory one-way? Currie et al. (2010a) addressed these questions with data on the political organization of 84 Austronesian societies. They coded each society's political organization on a four-level scale: acephalous (no political authority beyond the local community), simple chiefdom, complex chiefdom, and state. The analysis used a posterior sample of Austronesian language trees and compared Markov models with different constraints on the rate matrix, implemented in BayesTraits. In the best-fitting model, political complexity both rose and fell one level at a time; a close second allowed *increases* only one level at a time but let *decreases* skip levels. Under both models, societies did not jump from acephalous organization to a state without passing through the intermediate levels. Whether they could also collapse from complex to simple organization in one step was left open, since the two models received similar support. The study is a clean example of how a constraint on a rate matrix can embody a substantive theory, here the theory of sequential social evolution associated with mid-twentieth-century anthropology. It is also an example of how phylogenetic methods can test such a theory without assuming that all modern societies lie on a single "ladder" of progress.

Several later studies built on this design using Austronesian data. Watts et al. (2016) examined the role of ritual human sacrifice in the evolution of social stratification across a large sample of Austronesian cultures. They concluded that human sacrifice both promoted the emergence of stratified societies and made them less likely to revert to egalitarian organization. This was consistent with the "social control" hypothesis, in which elites used ritualized killing to deter challenges to their authority. Sheehan et al. (2018) found evidence that intensive agriculture involving long-term modification of the land (terracing, irrigation) coevolved with sociopolitical hierarchy. Kushnick et al. (2014) analyzed the evolution of land tenure norms and found evidence for a sequential pattern of change. Together, these studies made Austronesian the best-studied family in cultural macroevolution. This is partly because of the size and quality of its language trees (Gray et al., 2009) and partly because of the Pulotu database (Section 14.12).

The same logic has recently been applied at a global scale. Opie and Atkinson (2025) mapped trait data for 868 cultures onto a language tree representing relationships among cultures worldwide and tested competing accounts of state formation. Their Bayesian analyses indicated that intensive agriculture was as likely to be a consequence of state formation as a cause, whereas grain cultivation most likely preceded it and predicted the later emergence of taxation; writing more likely arose in tax-raising societies. The authors themselves note that a causal reading is limited by the assumptions of the phylogenetic model and by small samples for some traits, and a global language tree is a coarse scaffold for cultures whose traits often crossed linguistic boundaries (Section 14.2).

14.6 Religion

The relationship between religion and social complexity has been debated since Durkheim and Weber. A prominent recent hypothesis holds that belief in *moralizing high gods* (powerful supernatural beings concerned with human morality) helped large societies to cooperate by making people believe that their behavior was being monitored and judged. If so, moralizing gods should *precede* the evolution of political complexity.

Supernatural punishment and political complexity in Austronesia. Watts et al. (2015a) tested this idea with data from the Pulotu database (Watts et al., 2015b) for a large sample of Austronesian cultures, analyzed on a sample of language trees. They distinguished moralizing high gods from a broader category of *broad supernatural punishment*. The latter is the belief that supernatural agents, not necessarily high gods, punish moral transgressions, for example through ancestral spirits or impersonal forces. Using Pagel's method and analyses of the order of change, they found that broad supernatural punishment tended to precede, and to facilitate, increases in political complexity. Moralizing high gods, by contrast, did not precede political complexity in the Austronesian data; where they appeared, they tended to follow it. The authors concluded that the role of religion in the evolution of complex societies in Austronesia was played by diffuse beliefs in supernatural punishment, not by moralizing high gods specifically.

Botero et al. (2014) took a global perspective. They analyzed the occurrence of moralizing high gods across several hundred societies in relation to ecological, social and political variables, with statistical controls for shared history and geographic proximity. Belief in moralizing high gods was more common in societies facing environmental hardship and in societies with more complex political organization. Later work based on the Seshat database (Section 14.12) brought the question to large-scale historical societies. That literature lies outside phylogenetic methods, because historical states and empires do not form a tree in any simple sense. But it shows how the same hypotheses can be tested on different kinds of historical data. Religion is also the clearest case of a cultural domain that often spreads independently of language: the world religions crossed language boundaries on a vast scale, so phylogenetic analyses of belief are most defensible for traditional religious systems in regions, such as much of the Pacific, where conversion is recent and documented.

14.7 Folktales and oral traditions

Folktales are an appealing target for phylogenetic analysis. They are transmitted orally from generation to generation, they vary in recognizable ways, and folklorists have catalogued them systematically. The Aarne-Thompson-Uther (ATU) index classifies thousands of international tale types, such as ATU 333 "Little Red Riding Hood" or ATU 330 "The Smith and the Devil". Folklorists of the historic-geographic (Finnish) school had already tried to reconstruct the history of tales from the distribution of their variants. Phylogenetic methods offer a more formal version of that enterprise.

Tehrani (2013) analyzed 58 variants of the tale types "Little Red Riding Hood" (ATU 333) and "The Wolf and the Kids" (ATU 123) from Europe, Africa and Asia. He coded the variants for plot elements and characters and used parsimony, Bayesian and network methods. The analysis showed that the two tale types form distinct but related clusters. East Asian versions, which had been suggested as the source of the European tales, appeared instead to have evolved by blending elements of both tale types (ATU 333 and ATU 123), and most African tales grouped with ATU 123.

The study also introduced a recurrent methodological concern. Folktales can be retold, combined and borrowed across linguistic boundaries, so network methods (Chapter 8) are needed alongside trees.

The ancient roots of Indo-European folktales. da Silva and Tehrani (2016) took a different approach. Instead of building trees of tales, they mapped the presence and absence of 275 ATU "tales of magic" onto a phylogeny of Indo-European languages, treating each tale type as a binary trait. They first removed tale types whose distribution was better explained by spatial proximity than by linguistic relatedness, which suggested horizontal transmission. They then used Bayesian ancestral state reconstruction to estimate the probability that each remaining tale was present at various nodes of the Indo-European tree. Several tales were reconstructed with high probability at deep nodes. "The Smith and the Devil", in which a smith sells his soul to an evil being in exchange for power and then outwits it, was reconstructed as probably present in Proto-Indo-European society. That would make it several thousand years old, consistent with its theme of metalworking in a society that had metallurgy. The analysis is a model of how to handle horizontal transmission explicitly: it tested co-transmission with language trait by trait (Section 14.2) before using the language tree. It is also a reminder that reconstructed ancestral states are estimates whose meaning depends on the tree, the dating and the coding of the traits.

14.8 Material culture

Archaeologists adopted phylogenetic methods at about the same time as anthropologists, but mostly for a different purpose. Instead of mapping traits onto language trees, they built trees of the artifacts themselves. O'Brien et al. (2001) applied cladistic methods to Paleoindian projectile points from the southeastern United States, treating attributes of shape and manufacture as characters and point types as taxa. Tehrani and Collard (2002) analyzed the designs of Turkmen textiles. They asked whether similarities among the weaving traditions of different groups reflected branching descent (*phylogenesis*) or blending through contact and intermarriage (*ethnogenesis*). They found that branching explained much of the variation, but that blending was also detectable. Matthews et al. (2011) later showed, using Iranian tribal textiles, that different classes of textile characters can have different transmission histories. Some track one history and others another, so a single tree may not describe all of a craft tradition.

The relationship between material culture and language has been tested directly. Jordan and Shennan (2003) compared basketry traditions among Native Californian groups with their linguistic relationships and with geographic distance. They found that basketry similarity tracked geographic proximity more closely than linguistic affiliation, suggesting substantial transmission of craft techniques across linguistic boundaries. This result is a useful counterweight to the assumption that cultural traits follow language trees. The extent to which they do is an empirical question, to be answered trait by trait. Collard et al. (2006) compared the treelikeness of a set of cultural datasets (artifacts, languages and other cultural traits) with that of biological datasets, using the retention index (Chapter 4). Cultural datasets turned out to be on average no less treelike than biological ones. That finding undercut the claim that culture is too reticulate for phylogenetic analysis. It did not mean, though, that every cultural dataset is treelike, and a treelike dataset of artifacts need not follow the *language* tree.

Two recent studies compare artifact or ritual histories with language trees directly. Buckley et al. (2025) inferred Bayesian phylogenies of Kra-Dai languages and of the looms used by their speakers in southwest China and Southeast Asia. The Hlai of Hainan were outgroups in both phylogenies,

and closely related languages tended to go with similar loom types, but the two histories differed at deeper levels and in their overall patterns of change. Learmouth et al. (2024) coded 90 traits of adolescent initiation, mortuary practice and rock-art motifs for about 100 Pama-Nyungan societies and found a strong association between the language phylogeny and initiation ritual, but not mortuary ritual or rock art. Both results support the principle stated in Section 14.2: co-transmission with language has to be established trait by trait.

14.9 Music

Music has been studied with phylogenetic and population-genetic methods mainly through the lens of population structure. Rzeszutek et al. (2012) analyzed the structure of musical diversity across a sample of the world's populations using methods borrowed from population genetics, such as analysis of molecular variance. They found that most musical diversity lies within populations rather than between them, as is also true of genetic diversity. Brown et al. (2014) compared musical, genetic and linguistic variation among indigenous populations of Taiwan and found significant correlations between musical and genetic structure. In Central Africa, Le Bomin et al. (2016) analyzed musical traits of Gabonese populations against their linguistic classification and argued that vertical transmission played the key role in shaping musical diversity. Music presents particular problems of coding (which features define a musical tradition?) and of sampling (recordings exist for some communities but not others). It is one of the less developed areas of cultural phylogenetics, and one with considerable room for growth.

14.10 Color and number terms

Some cultural traits are themselves linguistic: the semantic systems that languages use to carve up domains of experience. Color terms and numerals are the classic examples. They sit at the boundary between linguistic typology and cultural evolution, and because they are expressed in the vocabulary, they are among the cultural traits most likely to share the history of the language tree.

The evolution of color term systems in Pama-Nyungan. Berlin and Kay (1969) proposed that basic color term systems evolve in a fixed sequence. Languages with two terms distinguish dark/cool from light/warm. A third term is red. The fourth and fifth are green and yellow, in either order. Blue, brown and the remaining terms come later. This was an evolutionary claim, but for decades it was tested only on synchronic distributions. Haynie and Bowern (2016) tested it directly by reconstructing color systems on a Bayesian phylogeny of Pama-Nyungan languages from Australia, using data on the basic color terms of each language. They found that color systems change in both directions. Languages gain terms, but they also lose them, and the reconstructed ancestral systems did not always fit the expectation that systems only grow. Many transitions were consistent with the Berlin-Kay sequence, but the sequence was better described as a set of preferred pathways than as a strict ladder of increasing complexity. The study shows how phylogenetic methods turn a claim about "stages" into testable statements about transition rates, with the possibility of reversals. Section 13.7.4 describes how such pathway hypotheses can be coded and compared.

Zhou and Bowern (2015) applied similar methods to numeral systems in Pama-Nyungan. They modeled the upper limit of the numeral system (the highest number expressed by a simple numeral term) and its change over time. They found that languages gain and lose higher numerals,

and they quantified the considerable uncertainty in reconstructed ancestral states. Both studies are notable for their focus on Australia, a continent long assumed to be dominated by diffusion, where Bayesian phylogenies of Pama-Nyungan (Bowern and Atkinson, 2012) made such analyses possible.

Verkerk et al. (2025a) extended this kind of analysis to numeral systems with restricted ranges or non-decimal anchors in families of lowland South America (Arawakan, Pano-Tacanan, Tucanoan and Tupian) and New Guinea (Nuclear Torricelli, Nuclear Trans New Guinea, Ramu and Sepik). Their reconstructions suggest that the proto-languages of these families likely had restricted or anchor-2 systems, with shifts to anchor 5 in particular groups, although small family sizes limited statistical power.

14.11 Subsistence and economy

Subsistence strategies have deep consequences for social organization and demography. They also provide a bridge between cultural phylogenetics and the farming/language dispersal hypothesis (Bellwood and Renfrew, 2002; Diamond and Bellwood, 2003). Under that hypothesis, several of the world's largest language families, including Austronesian, Bantu and perhaps Indo-European, spread with early farmers. Holden (2002) argued that the shape of the Bantu language tree was consistent with the spread of farming across sub-Saharan Africa. The coevolution of dairying and lactose digestion (Holden and Mace, 1997) and of cattle keeping and descent (Holden and Mace, 2003) has already been described. More broadly, Mathew and Perreault (2015) analyzed the behavioral repertoires of 172 small-scale societies in western North America. They found that variation was better explained by cultural phylogeny, measured by language relationships, than by ecological variables. This suggests that social learning and inheritance of traditions, rather than independent adaptation to local environments, is the main mode of human behavioral adaptation. That result has implications for every claim, in linguistics and in anthropology, that a trait is "adapted" to an environment (Chapter 13). Subsistence is also a domain in which traits clearly can spread without language, as crops and domestic animals did across many linguistic boundaries, so the relative weight of language-tree inheritance and diffusion has to be estimated, not assumed.

14.12 Databases

Cultural phylogenetics depends on systematically coded cross-cultural data linked to language phylogenies. The main resources are listed in Table 14.1. Appendix G gives details on access.

Table 14.1: Major databases used in cultural phylogenetics.

Database	Content	Coverage	Link to languages
Ethnographic Atlas (Murdock, 1967)	Coded ethnographic variables on kinship, marriage, subsistence, settlement, politics	Over a thousand societies worldwide	Via D-PLACE, matched to Glottolog languoids
Standard Cross-Cultural Sample (Murdock and White, 1969)	Many coded variables for a sample chosen to minimize non-independence	186 societies	Via D-PLACE

Database	Content	Coverage	Link to languages
D-PLACE (Kirby et al., 2016)	Integrates the Ethnographic Atlas, SCCS, Binford's hunter-gatherer data and others with environmental data and language phylogenies	More than a thousand societies	Each society linked to a Glottolog languoid; includes published phylogenies
Pulotu (Watts et al., 2015b)	Supernatural beliefs and practices, plus social and environmental variables	Austronesian cultures	Linked to the Austronesian Basic Vocabulary Database and its trees
Seshat (Turchin et al., 2015)	Historical and archaeological data on social complexity, warfare, religion and more	Past polities from the Neolithic onward	Not tree-based; polities in time and space
Grambank, WALS, ABVD	Linguistic data used as traits or to build trees	See Chapter 3	Glottolog

The *Ethnographic Atlas* of George Peter Murdock, published in installments in the journal *Ethnology* and compiled as a book in 1967, remains the backbone of most global analyses. Its codes are coarse, and they refer to an "ethnographic present" that varies from the sixteenth to the twentieth century, depending on when the sources were written. D-PLACE (Database of Places, Language, Culture and Environment) made these data far more usable (Kirby et al., 2016). It linked each society to a Glottolog language code, to geographic coordinates, to environmental variables, and to published language phylogenies. That made it possible to run phylogenetic and spatial analyses of ethnographic data without reassembling the data by hand. The link from society to language is itself an assumption of the kind discussed in Section 14.2: a society is placed on the language tree by the language its members speak. Pulotu (Watts et al., 2015b) was built specifically for phylogenetic analyses of religion in Austronesia, with careful coding protocols and sources. Seshat (Turchin et al., 2015) covers historical polities rather than ethnographic societies. Analyses of Seshat data have proposed that a single dimension of social complexity structures much of the variation among past societies (Turchin et al., 2018). Seshat is not organized around language trees, though, and its analyses rely on spatial and temporal rather than phylogenetic controls.

Ethnographic data were collected by many observers, at different times, with different interests and biases. Coding decisions compress complex and often contested descriptions into a few categories. Missionary or colonial accounts may describe a society that was already transformed by contact. Before treating a coded variable as a trait to be reconstructed back thousands of years, read the coding definitions, check the sources for at least a sample of societies, and consider how much of the variation might reflect the observers rather than the observed.

14.13 Language trees and genetic trees

If languages are transmitted vertically, and if populations split and remain isolated, then language trees and genetic trees should be similar. Cavalli-Sforza et al. (1988) famously juxtaposed a tree of human populations based on classical genetic markers with a grouping of languages into large

phyla and argued that the two corresponded remarkably well. Their argument was later developed at book length (Cavalli-Sforza et al., 1994). The correspondence was contested. Some of the linguistic groupings, such as macro-families like "Nostratic" and "Amerind", were not accepted by most linguists. The genetic trees were based on limited data. And the method of comparison was visual. But the question remains important. Where language and genes agree, we can be more confident that both reflect population history. Where they disagree, the mismatch is itself a clue to processes such as *language shift* (a population adopting another group's language), *elite dominance* or *sex-biased admixture*, the same processes that decouple cultural traits from language (Section 14.2).

Modern studies compare genetic and linguistic histories with explicit statistics. Hunley et al. (2008) examined Northern Island Melanesia, where Oceanic (Austronesian) and non-Austronesian ("Papuan") languages are spoken side by side. They found that genetic and linguistic relationships were partially congruent, with the relationship between the two differing between the language groups. de Filippo et al. (2012) tested hypotheses about the Bantu expansion by comparing linguistic and genetic distances among Bantu-speaking populations. They found a correlation consistent with a demic expansion, in which people moved, and not merely languages. Creanza et al. (2015) compared worldwide phonemic and genetic variation, finding that phonemic differences between populations reflect their geographic and genetic relationships at regional scales. At the global scale, Barbieri et al. (2022) assembled genetic and linguistic data for a large number of populations. They identified many cases where a population's genetic affinities and its language family do not match, which they interpreted mostly as language shift. Matsumae et al. (2021) compared genetic, linguistic and cultural variation among language families of northeast Asia.

A simple and useful tool for comparing two trees on the same set of populations is the *tanglegram*: the two trees are drawn facing each other, and lines connect the same population in each. Nodes are rotated to minimize crossings, so that the remaining crossings indicate genuine incongruence. In R, `phytools::cophylo()` does the rotation and plotting. Tree distances such as the Robinson-Foulds distance (Chapter 2) and Mantel tests on the two patristic distance matrices quantify the congruence. Below we simulate a "gene tree" that differs from a "language tree" by a few rearrangements and plot both.

```
set.seed(3)
ltree <- pbtree(n = 30, scale = 5)
ltree$tip.label <- sprintf("S%02d", 1:30)
gtree <- phangorn::rNNI(ltree, moves = 3)            # minor incongruence
# language shift: in the language tree, S07 is moved next to a distant society
far <- names(sort(cophenetic(ltree)["S07", ], decreasing = TRUE))[1]
ltree_shift <- drop.tip(ltree, "S07")
pos <- which(ltree_shift$tip.label == far)
ltree_shift <- bind.tip(ltree_shift, "S07", where = pos, position = 0.3)
cp <- cophylo(ltree_shift, gtree, rotate = TRUE, print = FALSE)
```

```
#> Rotating nodes to optimize matching...
#> Done.
```

```
plot(cp, fsize = 0.6, link.lwd = 1.2, link.lty = "solid",
     link.col = ifelse(cp$assoc[, 1] == "S07", "firebrick", "grey60"))
```

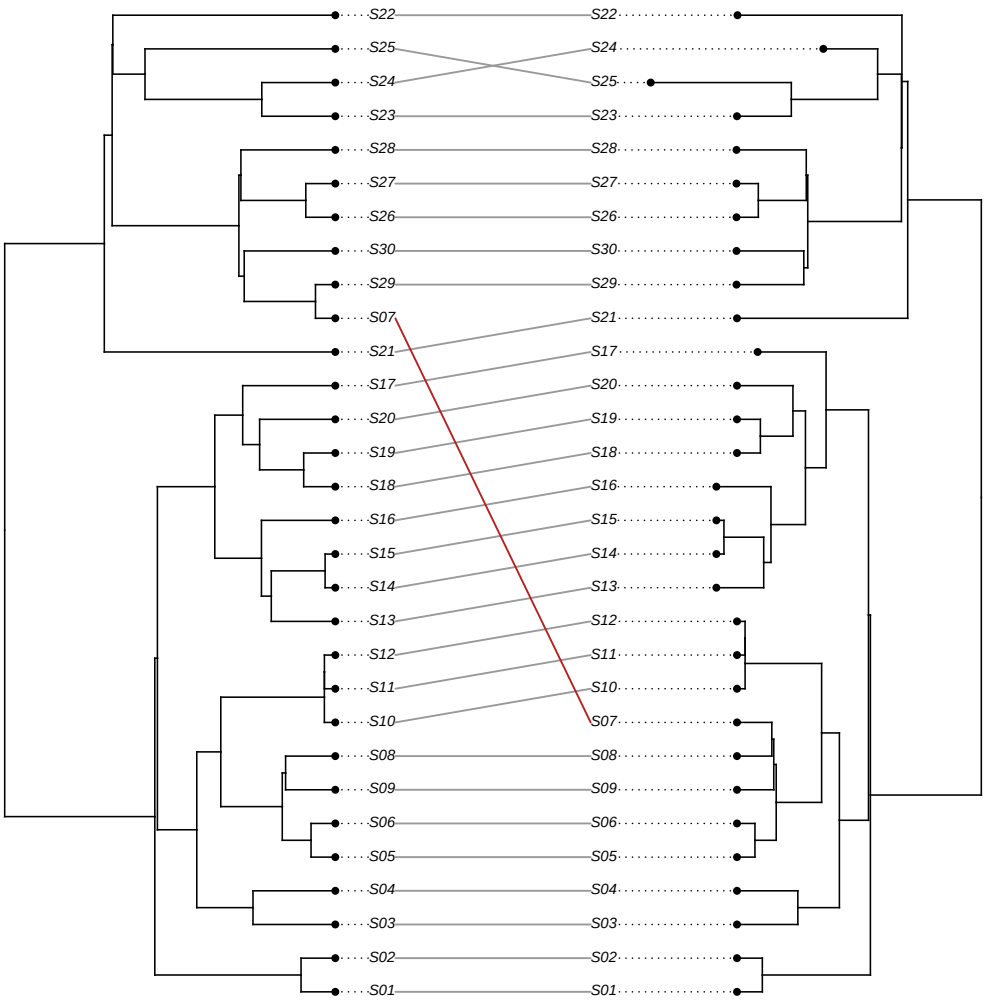


Figure 14.1: A tanglegram comparing a simulated language tree (left) with a simulated population genetic tree (right) for the same 30 societies. The genetic tree differs by a few nearest-neighbor interchanges and one language shift: society S07 has adopted a language from a distant clade.

```
gtree_ord <- cophenetic(gtree)[ltree_shift$tip.label, ltree_shift$tip.label]
c(RF_distance = phangorn::RF.dist(ltree_shift, gtree),
  max_RF = 2 * (Ntip(gtree) - 3),
  Mantel_p = mantel.test(cophenetic(ltree_shift), gtree_ord, nperm = 999)$p)
```

```
#> RF_distance      max_RF    Mantel_p
#>      26.000      54.000       0.001
```

The two trees are clearly related: the Mantel test rejects independence of the two distance matrices. But the Robinson-Foulds distance shows that a number of splits differ. The language shift of S07 shows up as a single long crossing line, the kind of local incongruence that tanglegrams reveal at a glance. For a cultural analysis, the question is which tree S07's traditions followed: if its residence rule or its crafts were retained through the shift, a reconstruction on the language tree would treat them as innovations in a clade to which, historically, they never belonged.

From the biologist's toolbox: cophylogenetic methods. Biologists studying hosts and parasites, or plants and pollinators, have developed a toolkit for comparing two trees whose tips are linked. *Distance-based tests* such as ParaFit (Legendre et al., 2002) and PACo (Balbuena et al., 2013) ask whether the two trees are more congruent than expected by chance. They also identify which individual links contribute most to the congruence or incongruence. *Event-based reconciliation* methods, implemented in programs such as Jane (Conow et al., 2010), explain the incongruence between two trees by a minimum-cost combination of cospeciation, duplication, host switching and loss. In a linguistic setting, "cospeciation" corresponds to the joint splitting of populations and languages, and "host switching" to language shift. Reconciliation would then give a principled way of estimating how often language shift has occurred and where. A related set of methods compares gene trees with species trees under the multispecies coalescent, which separates incongruence due to incomplete lineage sorting from that due to gene flow (Chapter 16). To our knowledge, these methods have rarely if ever been applied to language-gene comparisons. Most studies in that area still rely on visual comparison, Mantel tests or correlations of distance matrices; Section 16.4.5 discusses them further.

14.14 Critiques

Cultural phylogenetics has been controversial from the start. The critiques fall into four groups.

14.14.1 Horizontal transmission and ethnogenesis

The most fundamental objection is that culture does not evolve like a tree. Human societies exchange ideas, goods, spouses and practices across boundaries, and new groups often form by the merger of old ones. Terrell (1988) contrasted the image of history as a family tree with history as an "entangled bank", in the Pacific above all. Moore (1994) argued that *ethnogenesis*, the formation of new ethnic groups by fusion and recombination, has been as important in human history as branching (*phylogenesis*), and that cladistic methods, which assume branching, are therefore inappropriate. Boyd et al. (1997), in a careful theoretical discussion, distinguished several scenarios. In some, cultures are tightly integrated "hierarchical" units that are transmitted together. In others, cultures are collections of independent traits with separate histories. A tree may be meaningful in the first kind of scenario but not in the second.

Defenders of the program have responded in three ways. The first is empirical: measure treelikeness rather than assume it. Cultural datasets are often about as treelike as biological ones (Collard et al., 2006), and Chapter 8 describes the tools (delta scores, Q-residuals, NeighborNet) for doing so. The second is to separate traits that track the tree from traits that track geography, as da Silva and Tehrani (2016) did for folktales. The third is to test the robustness of comparative methods to horizontal transmission by simulation. Currie et al. (2010b) simulated continuous cultural traits evolving on trees with varying amounts of horizontal transmission between neighboring societies. They found that phylogenetic comparative methods remained reasonably accurate under the levels of borrowing they considered realistic, so that horizontal transmission did not by itself invalidate the approach. Nunn et al. (2010) carried out similar simulations comparing different methods for cross-cultural comparison. Towner et al. (2012) developed a method that analyzes vertical (phylogenetic) and horizontal (spatial-neighbor) transmission simultaneously, using neighbor graphs. Applied to western North American societies, it found that the relative importance of the two modes varied considerably among traits. Simulation can also cut the other way. Lukas et al. (2021) built a simulation tool that varies the size of the sample, missing data, the shape of the tree and the rate, number of variants and mode of transmission of a trait, and showed that for two example datasets particular combinations of these properties can produce high rates of both false-positive and false-negative inferences. Evans et al. (2021) distil the debate into two principles: draw on several lines of evidence when deciding whether, and which, phylogenetic methods suit a cross-cultural dataset, and consider that different cultural traits may have had different histories in space and time.

The simulation below explores the same question in a deliberately simple setting. In the vocabulary of Section 6.10, it is a test of the *robustness under misspecification* of PGLS: the data are generated with horizontal transmission that the model ignores. Two independent traits evolve on the tree. Then each society "borrows" a fraction h of its trait values from a single contact partner. We consider two scenarios for who the partner is. In the first, each society's partner is its closest relative on the tree, as in a dialect continuum or a recently expanded family whose members still live side by side. In the second, the partner is the society's nearest neighbor in a landscape where locations are unrelated to the tree, as in a region where groups of different origins live intermingled. In each scenario we measure how often OLS and PGLS on the language tree report a false correlation between the two traits.

```
set.seed(12)
htree <- pbtree(n = 60, scale = 5)
htree$tip.label <- sprintf("S%02d", 1:60)
Wh <- chol(solve(vcv(htree)))                          # PGLS whitening matrix
pgls_p <- function(y, x) {
  f <- lm(Wh %*% y ~ Wh %*% cbind(1, x) - 1)
  coef(summary(f))[2, 4]
}
patr <- cophenetic(htree); diag(patr) <- Inf
partner_relative <- apply(patr, 1, which.min)          # closest relative
geo <- as.matrix(dist(matrix(runif(120), 60, 2))); diag(geo) <- Inf
partner_geographic <- apply(geo, 1, which.min)    # nearest neighbor in space
run_scenario <- function(nn, h_levels = c(0, 0.1, 0.2, 0.3), nrep = 300) {
  borrow <- function(v, h) (1 - h) * v + h * v[nn]     # partial copying
  rel <- mean(patr[cbind(1:60, nn)])                   # mean distance to partner
  t(sapply(h_levels, function(h) {
```

```
    p <- replicate(nrep, {
      x <- borrow(fastBM(htree), h); y <- borrow(fastBM(htree), h)
      c(ols = summary(lm(y ~ x))$coefficients[2, 4], pgls = pgls_p(y, x))
    })
    c(partner_distance = rel, borrowing = h,
      FPR_OLS = mean(p["ols", ] < 0.05), FPR_PGLS = mean(p["pgls", ] < 0.05))
  }))
}
sim_h <- rbind(run_scenario(partner_relative), run_scenario(partner_geographic))
knitr::kable(sim_h, digits = 2,
             caption = "False positive rates of OLS and PGLS for two independent
             traits as horizontal transmission increases. Top four rows: each
             society borrows from its closest relative. Bottom four rows: each
             society borrows from its nearest geographic neighbor, with locations
             unrelated to the tree. The first column gives the mean patristic
             distance (thousands of years, summed along both lineages) between a
             society and its contact partner; 300 replicates per row.")
```

Table 14.2: False positive rates of OLS and PGLS for two independent traits as horizontal transmission increases. Top four rows: each society borrows from its closest relative. Bottom four rows: each society borrows from its nearest geographic neighbor, with locations unrelated to the tree. The first column gives the mean patristic distance (thousands of years, summed along both lineages) between a society and its contact partner; 300 replicates per row.

partner_distance	borrowing	FPR_OLS	FPR_PGLS
0.80	0.0	0.55	0.07
0.80	0.1	0.53	0.05
0.80	0.2	0.55	0.07
0.80	0.3	0.56	0.11
7.39	0.0	0.54	0.04
7.39	0.1	0.55	0.17
7.39	0.2	0.56	0.43
7.39	0.3	0.54	0.57

Without borrowing, PGLS has close to the nominal 5% false positive rate, while OLS has a greatly inflated one. What happens next depends on who borrows from whom. When societies borrow from their closest relatives, borrowing adds similarity where the tree already predicts it, and PGLS degrades only slowly. When they borrow from unrelated neighbors, even modest borrowing creates pairs of similar societies that the tree treats as nearly independent. The PGLS false positive rate then rises quickly, and at the highest level of borrowing simulated it is no better than that of OLS. The contrast helps in reading the broadly reassuring conclusion of Currie et al. (2010b). Phylogenetic methods can be robust to horizontal transmission, but "moderate" borrowing is not a fixed quantity. Its effect depends on how contact is structured relative to the tree. Borrowing among close relatives is largely absorbed by the phylogenetic covariance. Borrowing across distantly related lineages is not. The remedy, when borrowing across lineages is suspected, is to

model it explicitly, for example with a spatial covariance component as in Chapter 13, or with neighbor-graph models (Towner et al., 2012).

Section D.4.1 shows that the partial copying used here transforms the phylogenetic covariance matrix in a simple way, and that a GLS that uses the transformed matrix restores the nominal false positive rate when the contact network is known.

14.14.2 Correlation, causation and confounding

Cultural traits covary with each other and with the environment in complex ways. Many published correlations between cultural traits and ecological variables have turned out to be fragile once relatedness, proximity and covariation among predictors are controlled for. Bromham et al. (2018b) showed this for claims linking pathogen load to cultural and political variables. Bromham et al. (2021) examined the hypothesis that spicy cuisines in hot countries are an adaptation to reduce foodborne infection. They found little support for it once the confounding effects of shared history, geography and covariation with other variables were considered. These studies make the same argument as the linguistic examples in Section 13.11.4: controlling for phylogeny is necessary but not sufficient for causal inference.

14.14.3 Which tree?

A language tree is a hypothesis about the history of languages. It is not automatically the history of the cultural traits of their speakers, or of the speakers' genes (Section 14.2). A society that shifted its language, such as many groups in the Americas or in Africa, brings its non-linguistic traditions into a new linguistic lineage. The language tree will then misrepresent the history of those traditions. Elite dominance, intermarriage and migration can all decouple language from culture. Cultural phylogeneticists respond by using language trees as one of several possible scaffolds, comparing results across trees (including genetic trees, where available), and testing whether traits show phylogenetic signal on the tree before relying on it. A further question is *which language tree*: a posterior sample of trees carries uncertainty that a single summary tree hides, and Section 14.15.2 shows how much it can matter.

14.14.4 The data and their interpretation

Finally, there are critiques of the data themselves. Ethnographic codes are coarse, refer to different time periods, and reflect the biases of their sources. Societies are not bounded units, and the same "society" may be described differently by different observers. Reconstructions of ancestral cultural states, such as matrilocal Proto-Austronesians or a Proto-Indo-European "Smith and the Devil", can easily be over-interpreted as narratives about the lives of real ancestral communities. They carry the full uncertainty of the tree, the dating, the model and the coding, and they assume that the trait was transmitted with the language. There are also ethical questions. Reconstructions of the "original" culture of a people may be used in political or legal contexts, and descendant communities may have their own views about how their history is represented. None of these critiques is fatal to the program. Together, they explain why its best practitioners combine phylogenetic analyses with careful source criticism, explicit treatment of uncertainty and modest interpretation (Gray et al., 2007, 2010).

14.15 Worked examples in R

This section reproduces the logic of three classic designs on simulated data: an ancestral state reconstruction of post-marital residence, in the spirit of Jordan et al. (2009); a test of sequential evolution of political complexity, following Currie et al. (2010a); and a test of coevolution between cattle keeping and descent, following Holden and Mace (2003). The data are simulated so that the true answers are known, and the trait is simulated on the language tree itself, so these are parameter-recovery exercises that assume the co-transmission discussed in Section 14.2. The residence example is then repeated over a sample of trees.

14.15.1 Ancestral residence

We simulate a tree of 50 societies with a root age of 5,000 years. Residence evolves under a model in which matrilocal societies (M) become patrilocal (P) at a much higher rate than the reverse, and the root state is matrilocal.

```
set.seed(101)
atree <- pbtree(n = 50, scale = 5)
atree$tip.label <- sprintf("S%02d", 1:50)
Qres <- matrix(c(-0.3, 0.3, 0.05, -0.05), 2, 2, byrow = TRUE,
               dimnames = list(c("M", "P"), c("M", "P")))
res <- sim.history(atree, Qres, anc = "M", message = FALSE)$states
table(res)
```

```
#> res
#>  M  P
#> 17 33
```

```
fit_er  <- ace(res, atree, type = "discrete", model = "ER")
fit_ard <- ace(res, atree, type = "discrete", model = "ARD")
c(AIC_ER = AIC(fit_er), AIC_ARD = AIC(fit_ard))
```

```
#>  AIC_ER AIC_ARD
#>   48.35   46.37
```

```
fit_ard$rates                    # index 1 = P -> M, index 2 = M -> P
```

```
#> [1] 0.0000 0.3333
```

```
fit_ard$lik.anc[1, ]             # marginal probabilities at the root
```

```
#> M P
#> 1 0
```

Although two-thirds of the modern societies are patrilocal, the reconstruction places matrilocality at the root with high probability. This is because the asymmetric model has learned that matrilocality is easily lost and rarely regained. The pattern of patrilocal societies scattered across many clades, with matrilocal ones also present in several clades, is best explained by an ancestral matrilocal state with repeated independent losses. Figure 14.2 shows the reconstruction.

```
cols_res <- c(M = "#D55E00", P = "#0072B2")
plot(atree, cex = 0.65, label.offset = 0.08, no.margin = TRUE)
nodelabels(pie = fit_ard$lik.anc, piecol = cols_res, cex = 0.3)
tiplabels(pch = 15, col = cols_res[res], cex = 1)   # tip states on top of pies
legend("bottomleft", legend = c("matrilocal", "patrilocal"), pch = 15,
       col = cols_res, bty = "n", cex = 0.8)
```

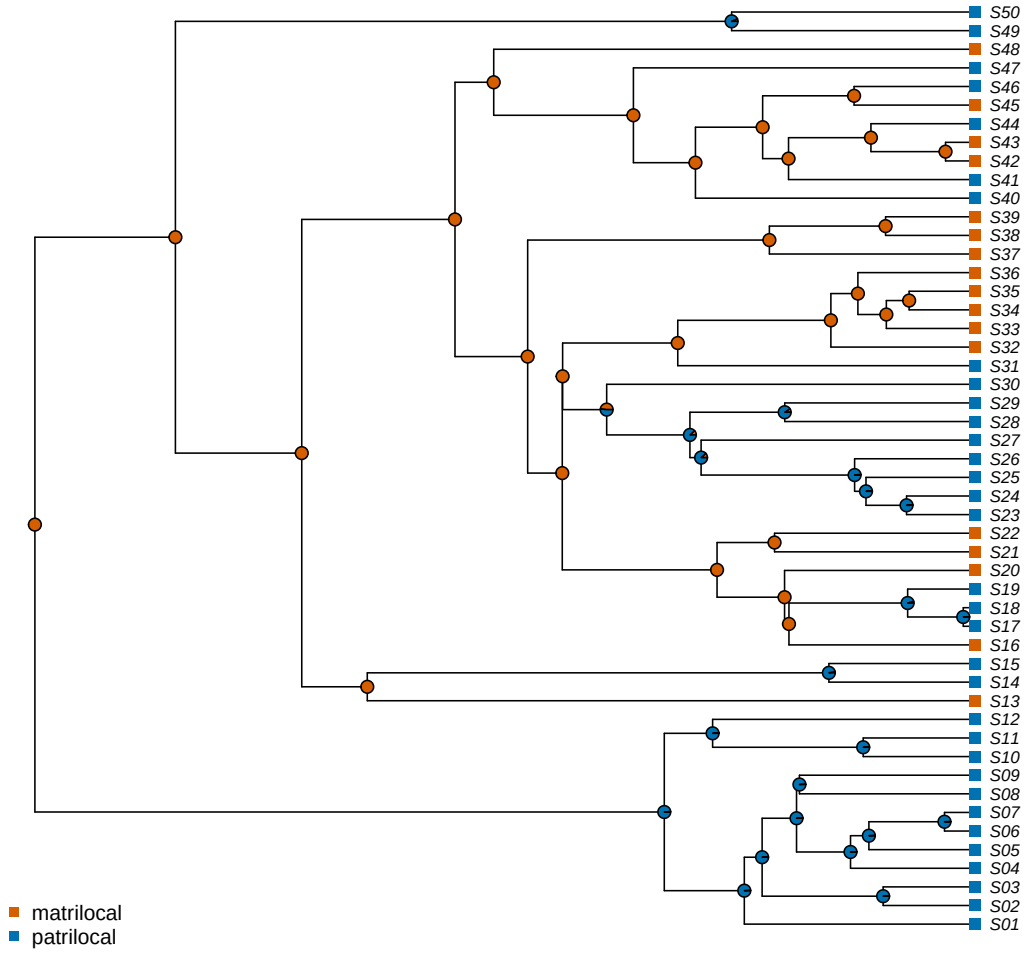


Figure 14.2: Maximum-likelihood marginal reconstruction of post-marital residence (M = matrilocal, P = patrilocal) on a simulated tree of 50 societies under an asymmetric (all-rates-different) Markov model. Pie charts show the probability of each state at internal nodes.

Two cautions apply. First, the ARD model's estimate of the rate of regaining matrilocality may be at or near zero. With 50 societies, the data say little about rare transitions, and an estimate of zero should not be read as "impossible". Second, the reconstruction at the root is sensitive to the model of the root state. Here `ace()` uses equal prior weights on the two states. Alternatives, such as weighting by the stationary distribution or by the method of FitzJohn et al. (2009) (available as `pi = "fitzjohn"` in `phytools::fitMk()`), can give different answers when rates are asymmetric. Chapter 12 discusses these choices.

14.15.2 Ancestral residence over a sample of trees

The first caution has a consequence that only becomes visible over a sample of trees. When a rate is estimated at zero, the reconstruction on that tree becomes certain: if matrilocality can never be regained, every matrilocal society must descend from a matrilocal ancestor, and the root must be matrilocal. Whether the estimate hits zero depends on the details of the tree. We repeat the reconstruction on a mock posterior sample of 50 trees, built from the true tree by random nearest-neighbor interchanges and branch-length noise, and on the MCC tree of that sample.

```
set.seed(102)
post_a <- phangorn::rNNI(atree, moves = 6, n = 50)
post_a <- lapply(post_a, function(t) {
  t$edge.length <- t$edge.length * rlnorm(nrow(t$edge), 0, 0.3); t })
class(post_a) <- "multiPhylo"
root_M <- sapply(post_a, function(t)
  ace(res[t$tip.label], t, type = "discrete", model = "ARD")$lik.anc[1, "M"])
mcc_a <- phangorn::maxCladeCred(post_a)
on_mcc_a <- ace(res[mcc_a$tip.label], mcc_a, type = "discrete",
                model = "ARD")$lik.anc[1, "M"]
round(c(MCC_tree = on_mcc_a, mean_over_trees = mean(root_M),
        share_trees_M_favored = mean(root_M > 0.5)), 3)
```

```
#>              MCC_tree       mean_over_trees share_trees_M_favored
#>                 1.000                 0.696                 0.660
```

On the MCC tree, as on the true tree, the root is matrilocal with probability 1. Over the sample, matrilocality is favored at the root in only 66% of trees, and the average probability is 0.7: on many trees the asymmetric model settles on the opposite boundary solution and reconstructs a patrilocal root with near certainty. Both kinds of certainty are artifacts of fitting a rate at zero to a few dozen societies. Averaging over trees, and over rates in a Bayesian analysis, as BayesTraits does, replaces them with a more honest statement of uncertainty. A conclusion reported from a single summary tree would have hidden the instability entirely (Section 11.10 discusses the general problem).

14.15.3 Sequential evolution of political complexity

We now simulate a four-level political complexity trait (0 = acephalous, 1 = simple chiefdom, 2 = complex chiefdom, 3 = state) on a tree of 84 societies. Increases happen only one step at a time, while decreases can skip levels. We then compare three models, each specified as an index matrix of rate parameters for `ape::ace()` (zero means a transition is not allowed, and equal integers mean equal rates):

- *stepwise*: increases and decreases one level at a time;
- *up stepwise, down any*: increases one level at a time, decreases by any number of levels (the generating model);
- *any jump*: any transition allowed, with one rate for increases and one for decreases.

```
lev <- c("0", "1", "2", "3")
Qpol <- matrix(0, 4, 4, dimnames = list(lev, lev))
for (i in 1:3) Qpol[i, i + 1] <- 0.4                    # stepwise increases
```

```
for (i in 2:4) Qpol[i, 1:(i - 1)] <- 0.15        # decreases may skip levels
diag(Qpol) <- -rowSums(Qpol)
set.seed(5)
ptree <- pbtree(n = 84, scale = 5)
pol <- sim.Mk(ptree, Qpol, anc = "0")
table(pol)
```

```
#> pol
#>  0  1  2  3
#> 33 23 14 14
```

```
idx <- function(m) matrix(m, 4, 4, byrow = TRUE, dimnames = list(lev, lev))
models <- list(
  stepwise          = idx(c(0,1,0,0,  2,0,1,0,  0,2,0,1,  0,0,2,0)),
  up_step_down_any  = idx(c(0,1,0,0,  2,0,1,0,  2,2,0,1,  2,2,2,0)),
  any_jump          = idx(c(0,1,1,1,  2,0,1,1,  2,2,0,1,  2,2,2,0)))
pol_fits <- lapply(models, function(m)
  ace(pol, ptree, type = "discrete", model = m))
aic_pol <- sapply(pol_fits, AIC)
round(aic_pol - min(aic_pol), 2)
```

```
#>         stepwise up_step_down_any         any_jump
#>             4.65             0.00             7.51
```

The generating model ("up stepwise, down any") is preferred. The fully stepwise model is penalized because it cannot accommodate the collapses that skip levels, and the "any jump" model because it allows increases that never happen. With real data, Currie et al. (2010a) used BayesTraits to compare a larger set of such constrained models, integrating over trees and using Bayes factors. The logic was the same.

14.15.4 Cattle and matrilineal descent

Finally, we simulate two binary traits, cattle keeping (C or noC) and descent (Mat or Pat), under a dependent model. In it, the loss of matrilineal descent is twelve times faster in cattle-keeping societies than in others. We then use Pagel's test to compare the independent model with a model in which only descent depends on cattle, and with the full dependent model.

```
st <- c("noC|Mat", "noC|Pat", "C|Mat", "C|Pat")
Qc <- matrix(0, 4, 4, dimnames = list(st, st))
Qc["noC|Mat", "C|Mat"] <- 0.2;   Qc["noC|Pat", "C|Pat"] <- 0.2    # gain cattle
Qc["C|Mat", "noC|Mat"] <- 0.05;  Qc["C|Pat", "noC|Pat"] <- 0.05   # lose cattle
Qc["noC|Mat", "noC|Pat"] <- 0.05; Qc["noC|Pat", "noC|Mat"] <- 0.05
# cattle erode matriliny
Qc["C|Mat", "C|Pat"] <- 0.6;     Qc["C|Pat", "C|Mat"] <- 0.02
diag(Qc) <- -rowSums(Qc)
set.seed(9)
ctree <- pbtree(n = 80, scale = 5)
cs <- sim.Mk(ctree, Qc, anc = "noC|Mat")
cs <- setNames(as.character(cs), names(cs))
```

```
cattle  <- setNames(sapply(strsplit(cs, "|", fixed = TRUE), `[`, 1), names(cs))
descent <- setNames(sapply(strsplit(cs, "|", fixed = TRUE), `[`, 2), names(cs))
table(cattle, descent)
```

```
#>       descent
#> cattle Mat Pat
#>    C    15  29
#>    noC  32   4
```

```
fit_dep_y <- fitPagel(ctree, cattle, descent, method = "ace", dep.var = "y")
fit_dep   <- fitPagel(ctree, cattle, descent, method = "ace")
c(AIC_independent = fit_dep_y$independent.AIC,
  AIC_descent_depends = fit_dep_y$dependent.AIC,
  AIC_both_depend = fit_dep$dependent.AIC)
```

```
#>     AIC_independent AIC_descent_depends     AIC_both_depend
#>               139.5               123.9               125.8
```

```
round(fit_dep_y$dependent.Q, 3)
```

```
#>          C|Mat C|Pat noC|Mat noC|Pat
#> C|Mat   -0.627 0.627   0.000   0.000
#> C|Pat    0.000 0.000   0.000   0.000
#> noC|Mat  0.190 0.000  -0.256   0.066
#> noC|Pat  0.000 0.190   0.786  -0.976
```

The model in which descent depends on cattle keeping is preferred. In its estimated rate matrix, the rate from `C|Mat` to `C|Pat` (loss of matriliny with cattle) is much larger than the rate from `noC|Mat` to `noC|Pat` (loss of matriliny without cattle). This mirrors the conclusion of Holden and Mace (2003). The usual caveats of Pagel's test apply (Chapter 13). Results can be driven by a few historical events, and the direction of dependence is harder to establish than its existence. In the real Bantu case, a further caveat is that the spread of cattle, the spread of patrilineal descent and the spread of particular Bantu subgroups may all reflect a single historical process that the tree only partly captures. And cattle, unlike descent rules, are a trait that can spread between neighbors without any change of language, so the language tree may describe the history of descent better than that of cattle keeping.

From the biologist's toolbox: trait-dependent diversification. All of the analyses in this chapter treat the tree as given and ask how traits evolve on it. Biologists also ask the reverse question: does a trait affect the rate at which lineages split or go extinct? The *BiSSE* model (Maddison et al., 2007) and its relatives (MuSSE for multistate traits, QuaSSE for continuous ones, HiSSE with hidden states (Beaulieu and O'Meara, 2016)) estimate state-dependent speciation and extinction rates jointly with the trait's transition rates. For language, the analogous question is whether a cultural trait, such as farming, political centralization or a writing system, affects the rate at which language lineages split or disappear. That question is central to the farming/language dispersal hypothesis and to discussions of language endangerment. To our knowledge, such models have rarely if ever been fitted to language trees. Two serious caveats apply. First, Rabosky and Goldberg (2015) showed that BiSSE-type models often detect spurious trait-dependent diversification when rates vary for other reasons. Hidden-state models and careful null models are therefore

essential. Second, language trees are typically sampled very incompletely and non-randomly with respect to traits, which biases diversification estimates unless sampling is modeled explicitly.

- Cultural phylogenetics uses language trees as estimates of population history and treats cultural traits as characters evolving along them. This addresses Galton's problem for cross-cultural comparison.
- A language tree estimates the history of one channel of transmission, the learning of a first language. It approximates population history when communities split and children learn their parents' language, and it can misrepresent the history of cultural traits that travel by shift, diffusion, intermarriage or specialist learning.
- Every cultural trait has its own history, and co-transmission with language must be tested trait by trait: by phylogenetic signal, by comparing phylogenetic and spatial models, by treelikeness, and by comparison with other scaffolds such as genetic trees.
- Classic applications include the coevolution of cattle keeping and matrilineal descent in Bantu-speaking Africa, matrilocal residence in early Austronesian societies, the sequential rise of political complexity in Austronesia, the role of supernatural punishment and human sacrifice in social evolution, the deep history of Indo-European folktales, and the evolution of color and number terms in Pama-Nyungan.
- The key tools are Mk models with constrained rate matrices (for pathways and sequences), Pagel's test and its directional variants (for coevolution), and Bayesian ancestral state reconstruction integrated over posterior trees. Reconstructions on a single tree can be deceptively certain, especially when rates are estimated at zero.
- D-PLACE, Pulotu and Seshat link coded cultural data to languages, phylogenies, environments and time. The quality and time depth of the underlying ethnographic data limit what can be inferred.
- Language and genetic trees can be compared with tanglegrams, tree distances, Mantel tests and, potentially, cophylogenetic reconciliation. Mismatches point to language shift and admixture.
- The effect of horizontal transmission on phylogenetic methods depends on who borrows from whom: borrowing among close relatives is largely absorbed, borrowing across lineages is not and should be modeled explicitly.

14.16 Exercises

Conceptual

1. Explain why a language tree can be a good estimate of the history of post-marital residence rules in one region and a poor one in another. Give a concrete scenario for each.
2. A society shifted to the language of its neighbors three centuries ago but kept its matrilineal descent. Describe how this would appear in an ancestral state reconstruction of descent on the language tree, and how it could mislead a test of coevolution between descent and cattle keeping.
3. Why are kinship terminologies and color terms more likely than crafts or religious practices to share the history of the language tree? What does this imply for which cultural domains are best suited to cultural phylogenetics?
4. Explain the analogy between incomplete lineage sorting of genes and the sorting of cultural variation that existed within an ancestral community. What pattern on the language tree would it produce?

5. In the horizontal transmission simulation, borrowing from close relatives barely affects PGLS while borrowing from unrelated neighbors does. Explain why, in terms of the covariance matrix the model assumes.

Computational

1. Rerun `cultural-traits-residence-trees` with `model = "ER"` and with `phytools::fitMk(..., pi = "fitzjohn")` followed by `ancr()`. How do the spread of root probabilities across trees and the MCC value change? Explain the role of boundary estimates.
2. Modify `run_scenario()` so that each society borrows from its nearest geographic neighbor, but with locations simulated by Brownian motion on the tree (`fastBM()` for each coordinate) rather than uniformly. Where do the false positive rates fall between the two scenarios of the chapter?
3. Using `ltree_shift` and `gtree`, simulate a binary cultural trait on the genetic tree with `sim.Mk()` and estimate its phylogenetic signal on both trees (for example with the log-likelihood of `fitMk()` against a star tree from `stree()`). Does the trait's signal identify the tree on which it evolved?
4. Repeat the political complexity analysis on 30 societies instead of 84 (`pbtree(n = 30, ...)`) 50 times, and record how often the generating model is preferred by AIC. What does this suggest about the sample sizes needed for such tests?

Research

1. You want to test whether the adoption of plough agriculture led to the loss of matrilineal descent in a region where several language families meet. Design the study: the scaffold (language tree, genetic tree, or both), the treatment of language shift and diffusion, the comparison of phylogenetic and spatial models, the integration over trees, and the coding of the ethnographic data.
2. A project proposes to reconstruct the "original" religious practices of a language family's ancestors from ethnographic data and to present the results to descendant communities. Write a methods and ethics plan that addresses co-transmission with language, uncertainty in the trees and the reconstructions, the quality of colonial-era sources, and the communication of the results.

14.17 Further reading

- Mace and Pagel (1994), "The comparative method in anthropology." The founding statement of the program, with commentary from critics.
- Mace and Holden (2005), "A phylogenetic approach to cultural evolution." A short, clear review of the early results in *Trends in Ecology and Evolution*.
- Mace et al. (2005) (eds.), *The Evolution of Cultural Diversity: A Phylogenetic Approach*. The first edited volume on the subject, covering language, kinship, material culture and methods.
- Gray et al. (2007), "The pleasures and perils of Darwinizing culture (with phylogenies)." A thoughtful defense of the program that takes the critiques seriously.
- Borgerhoff Mulder et al. (2006), "Cultural macroevolution and the transmission of traits." A review of the questions, methods and evidence on modes of transmission.

- Boyd et al. (1997): A careful theoretical discussion of when cultures can be expected to have tree-like histories, and when their traits have separate histories.
- Currie et al. (2010a) and Watts et al. (2016). Two exemplary studies from Austronesia showing how constrained Markov models and correlated-evolution tests answer questions about social evolution.
- da Silva and Tehrani (2016): A model of explicit handling of horizontal transmission in an ancestral state analysis.
- Kirby et al. (2016): The D-PLACE paper, essential for anyone planning a global cross-cultural analysis.
- Currie et al. (2010b): The key simulation study of how horizontal transmission affects phylogenetic comparative methods.
- Gray and Watts (2017), "Cultural macroevolution matters." A programmatic overview of the field and its prospects.

Chapter 15

Critiques, Pitfalls and Controversies

Linguistic phylogenetics has been controversial from the moment it began producing dates. Some of the criticism has been misdirected, aimed at glottochronology rather than at the models that replaced it, or at a caricature of Bayesian inference. Much of it has been sharp and productive, and has forced practitioners to clean their data, model borrowing and rate variation, justify their calibrations and test whether their models fit. The central question of this chapter is how a reader, or an author, can tell a robust result from an artefact of data, model or priors. We organize the critiques by where in an analysis an error can enter, work through the most prominent controversy in the field, the dispute over Indo-European origins, and end with a decision guide for choosing and checking models and a checklist for evaluating published work. Our aim is to be fair to both sides: critics have often been right about specific weaknesses, and practitioners have often been right that the weaknesses were fixable and that informal judgement has weaknesses of its own.

After reading this chapter you should be able to:

- distinguish stochastic from systematic error and explain why posterior probabilities and HPD intervals measure only the former;
- identify, for a published analysis, the stage (data, model, priors and calibrations, sampling, inference, interpretation) at which each criticism of it applies;
- explain why random coding error degrades trees gradually while concentrated borrowing produces confident wrong subgroups, and why agreement with accepted subgroups is not independent validation;
- classify a validation exercise as implementation validation, parameter recovery, robustness under misspecification, empirical adequacy or historical interpretation, and say what each can and cannot show;
- choose models, codings, priors and diagnostics for a realistic dataset, and state which of its parameters are identifiable, practically estimable, robust and historically interpretable;
- evaluate a published linguistic phylogeny with the checklist of Table 15.5.

15.1 A framework for thinking about error

Before turning to specific critiques it helps to have a vocabulary for the kinds of things that can go wrong. Statisticians distinguish two broad classes of error. *Stochastic error* (or sampling error) arises because the data are finite. A 200-item word list contains only so much information, and

two analyses of independent 200-item lists from the same languages will give somewhat different trees. Stochastic error shrinks as data accumulate, and it is what bootstrap proportions, posterior probabilities and highest posterior density (HPD) intervals are designed to measure. *Systematic error* (or bias) arises because the model used to analyze the data is wrong in some way that matters. Systematic error does not shrink with more data; it often gets worse, because the analysis becomes more and more confident in the wrong answer. A phylogenetic analysis that ignores a strong, consistent signal of borrowing between two neighboring languages will, given enough characters, place them together with high support regardless of their true history.

The distinction matters because most of the uncertainty summaries reported in linguistic phylogenetics, such as posterior clade probabilities or the 95% HPD interval on a root age, measure only stochastic error *conditional on the model being correct*. A posterior probability of 1.0 for a clade means that, if the model, the priors and the data coding are all correct, the data leave no doubt. It does not mean the clade is certainly real. Many of the disputes covered in this chapter are, at bottom, arguments about systematic error that cannot be settled by pointing at narrow credible intervals.

A useful way to organize the critiques is by the stage of the analysis at which the error enters:

1. **Data.** Are the cognate judgements correct? Was the word list appropriate? Were loanwords identified? Are the data coded in a way that respects the assumptions of the model?
2. **Model.** Does the substitution model capture how words are gained and lost? Is rate variation among meanings and among lineages modeled? Is borrowing modeled or at least shown not to matter?
3. **Priors and calibrations.** Do the priors on the tree, the branch lengths and the clock rate drive the results? Are calibrations independent of the hypothesis being tested?
4. **Sampling.** Which languages were included, and how were languages distinguished from dialects?
5. **Inference.** Did the MCMC converge? Were tree searches thorough? (These issues are treated in Chapters 5 and 6, and we mention them only briefly here.)
6. **Interpretation.** Do the results actually bear on the historical question asked? Is a date for the root of a tree the same thing as a date for the dispersal of a speech community? Is a phylogeographic "homeland" a place where people lived or a statistical artifact of sampling?

We will see that the most heated controversies typically involve errors at several of these stages at once, and that critics and practitioners frequently talk past each other because they locate the problem at different stages.

Detecting systematic error requires evidence from outside the fitted model, and that evidence comes in kinds that should not be confused. Section 6.10 distinguishes implementation validation (does the software compute what the model says?), parameter recovery (does the method recover the truth from data simulated under its own assumptions?), robustness under misspecification (does it still do so when the simulated data include borrowing, rate shifts or coding errors that the model ignores?), empirical model adequacy (can the fitted model reproduce the real data?) and historical interpretation (does a parameter correspond to something that happened?). The simulations in this chapter are labelled accordingly. Most published validation of linguistic phylogenetics consists of the first two kinds, which are necessary but easy to pass.

Throughout this chapter we use "practitioners" for researchers who build and publish computational phylogenies of languages and "critics" for those who have challenged them. The labels are a convenience. Many of the sharpest critiques have come from practitioners themselves, and

several prominent critics have also published computational analyses.

15.2 Data quality and cognate judgement

15.2.1 What a cognate judgement is

Almost all lexical phylogenies rest on *cognate judgements*: decisions that two words in different languages descend from a common ancestral word by regular inheritance. English *hand*, German *Hand*, Dutch *hand* and Swedish *hand* belong to one cognate set; French *main*, Italian *mano* and Spanish *mano* (from Latin *manus*) belong to another; Russian *ruka* and Polish *ręka* to a third. In a binary presence/absence matrix (Chapter 3), each of these sets becomes a character, and each language is coded 1 if its word for ‘hand’ belongs to the set and 0 otherwise.

A cognate judgement is not an observation in the way that a nucleotide is. It is an inference, made by a specialist, from evidence that includes regular sound correspondences, morphological analysis, semantic plausibility and knowledge of borrowing history. Different specialists can and do disagree. The judgement may also depend on the depth of the analysis: two words may be cognate at the level of a root but not at the level of the full stem (English *two* and Latin *duo* share a root; English *tooth* and Latin *dens, dent-* also share a root, but with different derivational morphology). Chapter 3 discusses root-meaning versus stem-level coding and partial cognates in detail.

There are four main ways a cognate judgement can go wrong:

- **False positives from chance resemblance.** Words that look alike by accident are judged cognate. With short words and small phoneme inventories, chance resemblances are common, which is why the comparative method insists on systematic sound correspondences rather than similarity.
- **False positives from undetected borrowing.** A loanword is coded as an inherited cognate. English *mountain* (from Old French *montagne*) is cognate with French *montagne* in a broad sense, but the relationship is due to borrowing, not inheritance, and coding it as a shared cognate would wrongly pull English toward French.
- **False negatives from sound change and semantic shift.** Genuinely cognate words are so changed that they are not recognized. Armenian *erku* ‘two’ is regularly descended from the same Proto-Indo-European form as Latin *duo*, but the correspondence is not obvious without knowledge of Armenian historical phonology.
- **Inconsistency.** Different coders, or the same coder at different times, apply different standards. One may split a cognate set into two because of a derivational suffix while another lumps them.

15.2.2 The empirical record

How serious are these errors in practice? Several lines of evidence suggest that they are real but, in well-curated datasets of well-studied families, not overwhelming. Datasets have repeatedly been revised. The lexical database of Dyen et al. (1992), which was used by Gray and Atkinson (2003), was assembled from a lexicostatistical tradition in which cognate decisions were sometimes made quickly across many languages, and later Indo-European datasets (the Indo-European Lexical database, IELex, and its successor IE-CoR, used by Heggarty et al. (2023)) involved extensive recoding, the addition of ancient languages and explicit documentation of judgements.

Practitioners have argued that the main results of earlier studies survived these revisions; critics have argued that the revisions show how much depended on contestable choices. Both points can be true.

In less-studied families the problem is more severe. Where regular sound correspondences have not been worked out in detail, cognate judgements must rely more heavily on resemblance, and the risk of false positives (from chance and borrowing) rises. This is a genuine limit on what phylogenetic methods can do in such families. A tree inferred from impressionistic cognate judgements cannot be better than those judgements, however sophisticated the model.

Automated cognate detection (Chapter 3) offers consistency and reproducibility but introduces its own error profile. Evaluations against expert judgements typically report B-cubed F-scores well below 1, and the errors are not random: algorithms are more likely to miss cognates with extensive sound change and to group look-alike loans with inherited words (List et al., 2017). Rama et al. (2018) found that trees inferred from automatically detected cognates could come reasonably close to trees from expert judgements on several benchmark datasets, which is encouraging, but it should not be read as license to skip expert checking where expertise exists.

Two recent studies widen the picture of what counts as data error. Snee et al. (2026) compared wordlists for the same varieties compiled independently by different authors, seven wordlists from five families, and found that differences in *concept translation*, the choice of the word that answers a given concept before any cognate judgement is made, left lexical distances highly correlated but produced Bayesian trees whose normalized quartet distances ranged from 0.02 to 0.44, with the largest effects on low-level subgrouping. And Kassian and Starostin (2025) argue that even an expert, documented database such as IE-CoR can in places conflate cognacy with areal diffusion or with independent derivation from the same root.

The interplay of data revision and results can be followed in the Indo-European literature. Gray and Atkinson (2003) analyzed a matrix of roughly 2,400 cognate sets for 87 languages, based on the lexicostatistical lists of Dyen et al. (1992), supplemented by ancient languages including Hittite and the Tocharian languages. Critics pointed to errors in the underlying lists, including misidentified cognates, inconsistent treatment of synonyms and undetected loans (see, for example, Pereltsvaig and Lewis, 2015). Later analyses used successively revised datasets: Bouckaert et al. (2012) used an expanded IELex-based dataset with about 100 languages, Chang et al. (2015) used a further curated version, and Heggarty et al. (2023) assembled the new IE-CoR database with explicit, documented cognate decisions for about 160 languages. The root ages estimated by these studies differ (see Section 15.9), but the differences have been attributed mainly to modeling choices, in particular how ancient languages are placed on the tree, rather than to cognate coding. The lesson is double-edged: careful recoding did not overturn the broad topology of Indo-European, but neither did it settle the dating controversy, which turned out to depend on the model.

15.2.3 Simulation: how much coding error can a tree tolerate?

A simple simulation helps build intuition. We simulate a “true” tree of 16 languages, generate binary cognate characters along it under a continuous-time Markov model with gamma-distributed rate variation among characters (Chapter 7), and then corrupt the data by flipping a random fraction of the cells, as if a careless coder had misjudged them. We then reconstruct trees with neighbor-joining (NJ) and maximum parsimony (MP) and measure their normalized Robinson-Foulds (RF) distance to the true tree (Robinson and Foulds, 1981), where 0 means identical topolo-

gies and 1 means no shared splits. In the vocabulary of Section 6.10, this is a test of robustness under misspecification, with a simple and favourable kind of misspecification: errors that fall at random.

```
library(phangorn)
set.seed(1501)
n_lang <- 16
true_tree <- phytools::pbtree(n = n_lang, scale = 1,
                              tip.label = sprintf("L%02d", 1:n_lang))

# Simulate binary cognate characters with discrete-gamma rate variation.
# Characters absent from every sampled language are dropped, because a
# cognate set with no attested members can never be observed.
sim_cognates <- function(tree, n_char = 500, rate = 1.5, shape = 0.5, k = 4) {
  cat_rates <- discrete.gamma(shape, k)
  blocks <- lapply(cat_rates, function(r) {
    as.character(simSeq(tree, l = n_char %/% k, type = "USER",
                        levels = c("0", "1"), rate = rate * r))
  })
  m <- do.call(cbind, blocks)[tree$tip.label, , drop = FALSE]
  m[, colSums(m == "1") > 0, drop = FALSE]
}

as_phydat <- function(m) phyDat(m, type = "USER", levels = c("0", "1"))

infer_trees <- function(m) {
  d <- as_phydat(m)
  nj_tree <- NJ(dist.hamming(d))
  mp_tree <- optim.parsimony(nj_tree, d, trace = 0)
  list(nj = nj_tree, mp = mp_tree)
}

rf_norm <- function(a, b) RF.dist(a, b, normalize = TRUE, rooted = FALSE)

# Coder error: flip a random fraction of matrix cells
flip_cells <- function(m, err) {
  hit <- matrix(runif(length(m)) < err, nrow(m))
  m[hit] <- ifelse(m[hit] == "1", "0", "1")
  m
}
dim(sim_cognates(true_tree))
```

```
#> [1]  16 452
```

```
err_grid <- c(0, 0.02, 0.05, 0.10, 0.15, 0.20)
n_rep <- 10
res_err <- do.call(rbind, lapply(err_grid, function(e) {
  do.call(rbind, lapply(seq_len(n_rep), function(r) {
    tr <- infer_trees(flip_cells(sim_cognates(true_tree), e))
```

```
    data.frame(error = e, NJ = rf_norm(tr$nj, true_tree),
               MP = rf_norm(tr$mp, true_tree))
  }))
}))
summ <- aggregate(cbind(NJ, MP) ~ error, data = res_err, FUN = mean)
lo <- aggregate(cbind(NJ, MP) ~ error, data = res_err, FUN = min)
hi <- aggregate(cbind(NJ, MP) ~ error, data = res_err, FUN = max)

off <- 0.002
plot(summ$error - off, summ$NJ, type = "b", pch = 19, ylim = c(0, 1),
     xlab = "Proportion of matrix cells miscoded",
     ylab = "Normalized RF distance to true tree", las = 1)
segments(summ$error - off, lo$NJ, summ$error - off, hi$NJ)
lines(summ$error + off, summ$MP, type = "b", pch = 17, lty = 2)
segments(summ$error + off, lo$MP, summ$error + off, hi$MP, lty = 2)
legend("topleft", legend = c("Neighbor-joining", "Maximum parsimony"),
       pch = c(19, 17), lty = c(1, 2), bty = "n")
```

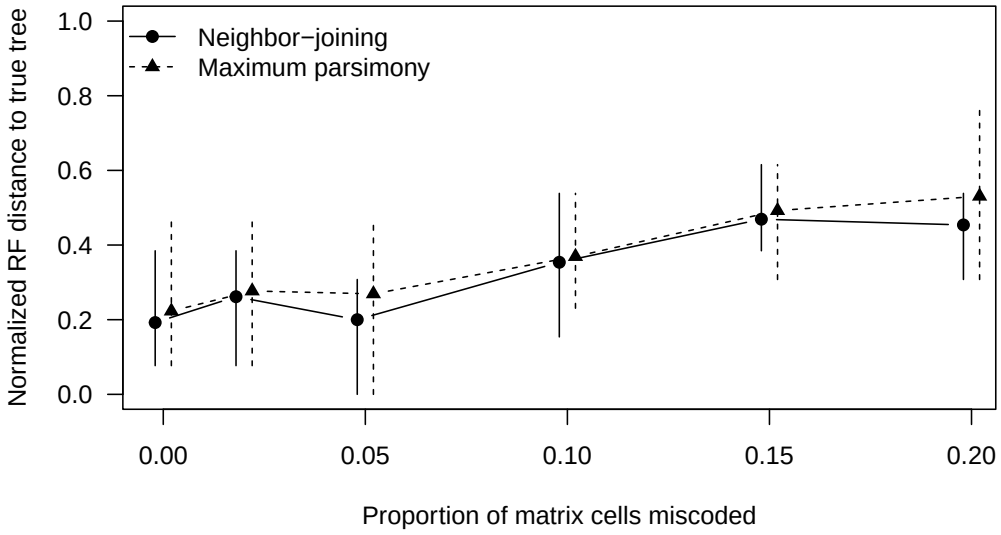


Figure 15.1: Effect of random coding error on tree accuracy. Each point is the mean normalized Robinson-Foulds distance between the true 16-language tree and trees inferred by neighbor-joining (NJ) and maximum parsimony (MP), over 10 simulated datasets; vertical bars span the minimum and maximum. Random errors degrade accuracy gradually.

```
round(summ, 3)
```

```
#>   error    NJ    MP
#> 1  0.00 0.192 0.223
#> 2  0.02 0.262 0.277
```

```
#> 3  0.05 0.200 0.269
#> 4  0.10 0.354 0.369
#> 5  0.15 0.469 0.492
#> 6  0.20 0.454 0.531
```

Figure 15.1 shows the typical pattern. Even with no coding error the inferred trees are not perfect, because some internal branches of the true tree are short and the finite data contain little information about them. As random error increases, accuracy declines gradually; a few percent of miscoded cells does little harm, while a fifth of cells miscoded destroys much of the signal. The important qualifier is *random*. Random errors add noise that is spread evenly across the tree, so they mainly erode support for short branches. Systematic errors, such as consistently failing to recognize loans from one particular donor, behave very differently, as we will see in Section 15.4.

Inter-coder disagreement should be measured and reported, not just acknowledged. Where two or more specialists have coded the same material, agreement statistics (for example, pairwise agreement on cognate set membership, or B-cubed scores computed with one coder as reference) and a sensitivity analysis that reruns the phylogeny on each coder’s matrix are far more informative than a statement that the data were “carefully checked”.

15.3 List choice, meaning slots and semantic coding

15.3.1 Which meanings?

Lexical phylogenies are usually built from *basic vocabulary*: meanings such as ‘hand’, ‘water’, ‘two’, ‘die’ and ‘I’ that every language expresses and that are thought to be relatively resistant to borrowing. The classic lists are those of Morris Swadesh, especially the 100-item list of Swadesh (1955) and its 200-item predecessor. The Leipzig-Jakarta list (Tadmor et al., 2010) was derived empirically from the World Loanword Database (Haspelmath and Tadmor, 2009) by selecting meanings with low borrowing rates across a world-wide sample of 41 languages. The ASJP project uses a 40-item subset of the Swadesh list. Chapter 3 describes these lists.

The choice of list matters in three ways. First, different meanings change at very different rates. Words for ‘two’, ‘three’, ‘who’ and ‘I’ are replaced extremely slowly in Indo-European, while words for meanings such as ‘dirty’ or ‘turn’ are replaced much more often (Pagel et al., 2007). A list weighted toward fast meanings carries more information about recent splits and less about deep ones. Second, meanings differ in borrowability; a list that includes culturally salient items will contain more loans. Third, the meanings on the list must be translatable into every language in the sample without forcing the coder into arbitrary choices. ‘Ice’ or ‘snow’ may lack a clear equivalent in tropical languages, and ‘to wash’ may correspond to several verbs depending on what is washed.

Classic glottochronology made list choice especially consequential. It assumed a universal constant retention rate r per millennium for basic vocabulary and estimated the time since two languages separated as

$$t = \frac{\ln C}{2 \ln r}, \tag{15.1}$$

where C is the proportion of shared cognates (Lees, 1953). Bergsland and Vogt (1962) showed that the constant failed badly: in the case they examined most closely, Icelandic had retained far more of its basic vocabulary than Norwegian over the same period, so the formula would have

dated the split between them very wrongly. Modern models do not assume a universal constant (Chapter 9), but they inherit a softer version of the problem. Rates are estimated from calibrations within the family, and the estimate depends on which meanings are included.

The failure of glottochronology is a cautionary tale that practitioners and critics both invoke, for opposite purposes. Critics such as Pereltsvaig and Lewis (2015) argue that Bayesian dating is "glottochronology in disguise", inheriting its flaws. Practitioners reply that the specific flaws identified by Bergsland and Vogt (1962) and Hoijer (1956), namely a universal constant rate, identical rates across meanings, no account of borrowing and a single distance-based summary of the data, are exactly what modern models relax: rates vary among meanings (gamma rate heterogeneity) and among lineages (relaxed clocks), calibrations are family-internal and expressed as distributions, and the full character data rather than a pairwise percentage are analyzed (Atkinson et al., 2005; Gray et al., 2011). The fair summary is that modern models do address the specific objections, but they replace a strong assumption (a universal rate) with weaker assumptions (rates drawn from a parametric distribution, calibrations correctly specified) that are still assumptions and still need testing.

15.3.2 Synonyms, polysemy and the "one word per meaning" problem

A word list asks, for each meaning, "what is the word?" Natural languages often answer with several words. English has *big* and *large*, *little* and *small*, *dog* and *hound* (with different nuances). The coder must decide whether to record all synonyms, only the most basic one, or only the most frequent one, and those decisions differ among datasets. Recording multiple synonyms increases the number of 1s in a language's row and can make it look more similar to several relatives at once. Recording only one discards information and requires a judgement of "basicness" that is hard to make consistently across languages documented by different fieldworkers.

Polysemy creates a mirror-image problem. In many languages one word covers both 'hand' and 'arm', or both 'tree' and 'wood'. If 'hand' and 'arm' are both on the list, the same lexeme appears twice and its history is counted twice. Swadesh was aware of this and dropped some such pairs when he shortened his list, but others remain; Kessler (2001) calls the general difficulty the "shared roots problem", and McMahon and McMahon (2005) show how acute it is in languages that derive many basic meanings from a few roots. Semantic shift creates a third problem: English *hound* and German *Hund* are cognate, but only the German word fills the 'dog' slot. Whether such cases count as cognate presences depends on whether coding is by meaning slot (the usual practice) or by root regardless of meaning (root-meaning coding; see Chapter 3). Neither choice is wrong, but they encode different evolutionary processes, and a model of lexical replacement within meaning slots assumes the former.

15.3.3 Binary coding and the non-independence of characters

Most Bayesian analyses convert meaning-slot data to a binary matrix in which each cognate set is a separate character. This is convenient, because binary models are simple and well understood (Chapter 7), but it violates a central assumption of standard phylogenetic likelihoods: that characters evolve independently given the tree.

Consider the meaning 'hand' in a sample of Germanic, Romance and Slavic languages with three cognate sets (the *hand*, *manus* and *ruka* sets). In the binary coding each language receives three characters for 'hand', one per set. If a language has exactly one word for 'hand', exactly one of

its three characters is 1. The characters are therefore linked by a constraint. For meaning m with K_m cognate classes and binary characters $x_{m1}, \dots, x_{mK_m}$, every language with data for m satisfies

$$\sum_{k=1}^{K_m} x_{mk} \geq 1, \tag{15.2}$$

with equality if synonyms are excluded. A binary model that treats the K_m characters as independent assigns positive probability to configurations that violate this constraint (for instance, a language with no word for 'hand' at all) and treats the loss of one cognate and the gain of another as two independent events, when in reality a replacement is a single event that changes two characters at once.

Chapter 7 compares the models that respect this constraint with those that do not (Section 7.2.1): binary likelihoods are composite likelihoods, which often give reasonable topologies but overstate the information in the data and weight meanings by their number of cognate classes. For the critic and the reviewer, the practical consequences are two. Posterior intervals from binary analyses, including those on dates, are likely to be too narrow, so an apparent conflict between two published dates may be smaller than their intervals suggest. And branch lengths, and therefore dates, can shift when the same data are recoded by meaning. Both can be checked: rerun the analysis with multistate coding, or bootstrap whole meanings rather than single characters (Chapter 17 shows how). Explicit models of replacement with synonymy, in which cognates are born, coexist and die within a meaning slot (Nicholls and Gray, 2008; Ryder and Nicholls, 2011), remain one of the open modelling problems of the field. Recent work shows both the promise and the difficulty: the infinite-state Multistate model respects the one-word-per-slot constraint but becomes fragile when loans or semantic shifts create competing words (Rönchen et al., 2024), a model with explicit polymorphism has so far been used to simulate data rather than to infer trees (Canby et al., 2024), and richer representations quickly become overparameterized for datasets of realistic size (Häuser et al., 2026) (Section 7.8.3).

15.3.4 Ascertainment: the characters you cannot see

A further subtlety concerns which characters can be observed at all. A cognate set is defined by its members, so a word that existed in the ancestor of a family but was lost in every sampled descendant leaves no trace, and the data matrix contains no all-zero columns. If the likelihood ignores this *ascertainment bias*, the model underestimates the rate of change (Lewis, 2001). The standard correction conditions the likelihood on the character being observable. For N characters,

$$L(\tau, \theta) = \prod_{i=1}^{N} \frac{P(D_i \mid \tau, \theta)}{1 - P(\mathbf{0} \mid \tau, \theta)}, \tag{15.3}$$

where $P(\mathbf{0} \mid \tau, \theta)$ is the probability of the all-absent pattern. Chapter 7 derives the corrections for other filtering schemes (Section 7.4). The point for evaluating published work is that the correction must match how the data were actually assembled. If singletons or constant characters were removed, the denominator must exclude those patterns too, and a mismatch shifts branch lengths and dates. Modern software (BEAST 2, RevBayes, MrBayes) implements several corrections, but the analyst has to choose the one that matches the data, and a reader can check the choice only if both the filter and the correction are reported.

Whether singletons and invariant characters were removed, and whether the ascertainment correction matches that choice, should be stated explicitly in every methods section. A mismatch rarely changes topology much, but it can shift branch lengths and therefore dates by an amount comparable to the HPD intervals that are reported.

15.4 Borrowing, contact and treelikeness

The oldest objection to language trees is that languages do not only split; they also exchange material. Johannes Schmidt's wave model of 1872 was a response to exactly this problem (Chapter 1), and the debate between *Stammbaum* and *Wellentheorie* has never really ended. Contemporary versions of the objection come in three strengths.

The weak version says that undetected loans add noise to cognate data. This is uncontroversially true and is addressed, imperfectly, by flagging known loans and excluding them (Chapter 3). The intermediate version says that undetected borrowing biases tree inference in systematic ways: it pulls borrowing languages together, shortens the branches between them and makes dates too young. The strong version says that for some families, or some periods in the history of all families, the process of diversification is not tree-like at all. Dialect continua diversify by the accumulation of overlapping innovations, so there is no tree to find (François, 2015; Kalyan and François, 2018).

The intermediate version was tested directly by Greenhill et al. (2009), who simulated cognate data on known trees with varying rates of borrowing, either between any pair of languages or only between geographic neighbors, and then reconstructed the trees with Bayesian methods. They found that, at the levels of borrowing they considered realistic for basic vocabulary, topologies were recovered reasonably well, while estimated dates tended to become younger as borrowing increased. This is a useful result, but it should not be over-generalized: the simulation assumes a particular model of borrowing, and real borrowing is often concentrated in time (intense contact episodes) and in space (a dominant neighbor), which is closer to the worst case.

Our own small simulation makes the distinction between diffuse and concentrated borrowing vivid. Starting from the same 16-language tree, we either (a) let each character, with probability b, be copied from a randomly chosen language to another randomly chosen language, or (b) let each character, with probability b, be copied from one fixed donor (`L04`) into one fixed recipient (`L01`) that belongs to a different subgroup. The total amount of borrowing is the same in both scenarios; only its concentration differs. Like the previous simulation, this is a test of robustness under misspecification: the data violate the tree model's assumption that lineages evolve independently after they split. Unlike random error, the violation here is systematic.

```
recipient <- "L01"; donor <- "L04"
true_clade <- c("L01", "L02", "L03")   # the recipient's real subgroup
borrow_fixed <- function(m, b) {
  hit <- runif(ncol(m)) < b
  m[recipient, hit] <- m[donor, hit]
  m
}
borrow_random <- function(m, b) {
  for (j in which(runif(ncol(m)) < b)) {
    pair <- sample(rownames(m), 2)
    m[pair[2], j] <- m[pair[1], j]
```

```
  }
  m
}
b_grid <- c(0, 0.1, 0.2, 0.3, 0.4)
res_b <- do.call(rbind, lapply(b_grid, function(b) {
  do.call(rbind, lapply(seq_len(n_rep), function(r) {
    base <- sim_cognates(true_tree)
    t_fix <- infer_trees(borrow_fixed(base, b))$nj
    t_rnd <- infer_trees(borrow_random(base, b))$nj
    data.frame(b = b,
               diffuse = rf_norm(t_rnd, true_tree),
               concentrated = rf_norm(t_fix, true_tree),
               sisters = is.monophyletic(t_fix, c(recipient, donor)))
  }))
}))
sb <- aggregate(cbind(diffuse, concentrated, sisters) ~ b, data = res_b,
                FUN = mean)
op <- par(mfrow = c(1, 2), mar = c(4.2, 4.2, 1, 1))
plot(sb$b, sb$diffuse, type = "b", pch = 19, ylim = c(0, 0.6), las = 1,
     xlab = "Proportion of characters borrowed (b)",
     ylab = "Normalized RF distance")
lines(sb$b, sb$concentrated, type = "b", pch = 17, lty = 2)
legend("topleft", c("Diffuse", "Concentrated"), pch = c(19, 17),
       lty = 1:2, bty = "n")
plot(sb$b, sb$sisters, type = "b", pch = 17, lty = 2, ylim = c(0, 1),
     las = 1, xlab = "Proportion of characters borrowed (b)",
     ylab = "P(recipient and donor sisters)")
```

```
par(op)
round(sb, 3)
```

```
#>     b diffuse concentrated sisters
#> 1 0.0   0.215        0.215     0.1
#> 2 0.1   0.177        0.208     0.5
#> 3 0.2   0.200        0.246     0.9
#> 4 0.3   0.200        0.277     1.0
#> 5 0.4   0.231        0.262     1.0
```

The left panel of Figure 15.2 shows that the overall RF distance barely registers either kind of borrowing: in both scenarios most of the tree is still recovered. The right panel shows why that summary is misleading. Under concentrated borrowing, once something like a fifth of the characters have been copied, the recipient is placed as the sister of its donor in essentially every replicate, a confident and wrong result. A global accuracy score averages a severe local error over many correct splits. The general point is that “the tree is robust to borrowing” and “this particular subgrouping is an artifact of borrowing” are compatible statements, and claims about specific contested subgroups (Italo-Celtic, Balto-Slavic-Germanic links, the position of Albanian or Armenian) require local tests rather than global ones.

How can such artifacts be detected? Network methods such as NeighborNet display conflicting

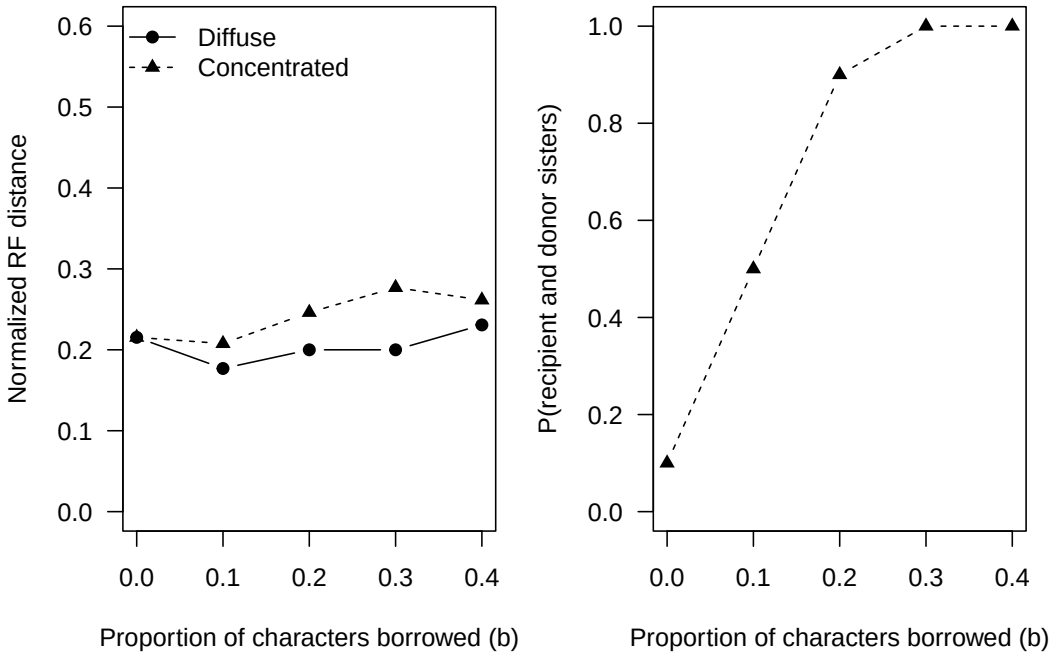


Figure 15.2: Diffuse versus concentrated borrowing. Left: normalized Robinson-Foulds distance between the true and the neighbor-joining tree when a proportion b of characters is borrowed between random pairs of languages (diffuse) or always from L04 into L01 (concentrated). Right: proportion of replicates in which the recipient L01 and the donor L04 are inferred as sister languages, which they are not in the true tree.

signal rather than forcing a tree, and treelikeness statistics such as delta scores and Q-residuals quantify how much conflict there is (Gray et al., 2010). Explicit models of lateral transfer (Kelly and Nicholls, 2017) and contact-aware phylogenetic models (Neureiter et al., 2022) go further by inferring borrowing events jointly with the tree. These are the subject of Chapter 8. The point to make here is that treelikeness is an empirical property that can and should be measured for each dataset, not assumed in either direction.

A well-known exchange in the Austronesian literature shows how the borrowing critique and its rebuttal can talk past each other. Donohue et al. (2012) argued that the basic-vocabulary cognate analysis of Austronesian by Gray et al. (2009) fails to distinguish shared retentions from shared innovations and, where geography and accepted subgrouping diverge, groups languages by geographic and social proximity, reflecting borrowing and diffusion, rather than by subgroup. Greenhill and Gray (2012) replied that the argument misunderstood how Bayesian phylogenetic analyses of Austronesian basic vocabulary had been carried out and what their results showed. Readers of the exchange will notice that the two sides differ less about whether borrowing occurs than about how much of it survives into curated basic-vocabulary datasets after known loans are excluded, and about whether a tree with uncertain internal branches is a defensible summary of a history that is partly reticulate. Questions of that kind can, in principle, be answered with explicit models of borrowing and with posterior predictive checks (Section 15.12), and much of the later literature has moved in that direction.

15.5 Model misspecification

All models are wrong, and the useful question is whether a given model is wrong in ways that matter for the inference at hand. The substitution models used for cognate data (Chapter 7) make several assumptions that are known to be violated to some degree.

Rate homogeneity across characters. Different meanings change at different rates, and within a meaning different cognate sets may have different gain and loss dynamics. Gamma-distributed rates capture some of this variation, and the improvement in fit is typically large (we will see an example in Section 15.12). But a gamma distribution with a single shape parameter is a convenient parametric form, not a law of nature. Meaning-specific rates or mixture models may fit better.

Stationarity and time-reversibility. Standard models assume that the process of gain and loss is the same throughout the tree and that the equilibrium frequency of cognate presence is constant. In reality, a cognate set is born once (when a new word arises or a meaning shift occurs) and then lost independently in descendant lineages. This is the logic of the stochastic Dollo model (Nicholls and Gray, 2008), and a reversible model applied to such data treats some patterns as parallel gains that are in fact single births followed by losses. Because reversible and Dollo-type models account for the same patterns with different numbers of events, they can yield different branch lengths and root ages for the same data, and comparing them is a simple check on how far the dates depend on this assumption.

Independence among lineages. Models assume that, once two lineages have split, their subsequent changes are independent. Borrowing (Section 15.4) and parallel semantic development violate this. Parallel development is underappreciated: the same semantic shift (for instance, from ‘see’ to ‘know’, or from ‘skin’ to ‘bark’) can happen independently in related languages because they share the semantic structure that makes the shift natural. Such parallel shifts create homoplasy that looks like shared inheritance.

Clock assumptions. Relaxed-clock models assume that rates vary among branches according to a parametric distribution (lognormal, exponential, or autocorrelated). Punctuational change at splits (Atkinson et al., 2008), rate changes associated with population bottlenecks or contact, and rate differences between very old and very young parts of a tree are all imperfectly captured. Chapter 9 reviews the evidence.

The difficulty is that model misspecification is not visible in the posterior distribution itself. A misspecified model can produce a well-converged MCMC, a sharp posterior and narrow HPD intervals. Detecting it requires looking outside the model, through model comparison (marginal likelihoods and Bayes factors, Chapter 6), which tells us which of several models is *least wrong*, and through model adequacy tests, which tell us whether even the best model is *good enough*. The second kind of test, an assessment of empirical adequacy, is much less common in linguistic phylogenetics and is discussed in Section 15.12 and Chapter 16. A third kind of evidence comes from simulation with deliberate violations of the model (robustness under misspecification), which shows how large a violation must be before a conclusion changes.

There is a further complication. The phrase "least wrong" presupposes that the model with the best relative fit is also the one whose inferences are most accurate. Rönchen (2025) argues that for cognate data this need not hold: finite-state Markov chains such as the covarion model, combined with the ascertainment correction used for cognate data, can fit better than Dollo-type models and still infer less accurate trees, because they accommodate the data with an incorrect account of how it arose (Rönchen et al., 2025b). The argument rests partly on work that had not been peer reviewed at the time of writing, but it echoes a distinction drawn by Hammarström et al. (2019): inferring a language tree is explanatory rather than predictive modelling, and measures of predictive fit do not guarantee explanatory accuracy. If the argument holds, model comparison by marginal likelihood needs to be complemented by simulation from realistic generating processes before a winning model is trusted.

15.6 Priors that drive results

15.6.1 What priors do

Bayesian inference combines the likelihood with prior distributions on every unknown:

$$p(\tau, \mathbf{t}, \theta \mid D) \propto P(D \mid \tau, \mathbf{t}, \theta)\, p(\mathbf{t} \mid \tau, \lambda, \mu)\, p(\tau)\, p(\theta), \tag{15.4}$$

where $p(\mathbf{t} \mid \tau, \lambda, \mu)$ is the tree prior (often a Yule or birth-death process; Chapter 6) and $p(\theta)$ collects priors on substitution and clock parameters. When data are informative about a quantity, the prior matters little. When data are weakly informative, the prior can dominate.

Divergence times are a case in which the data are inherently weakly informative. Cognate data tell us about the *amount* of change along each branch, which is the product of rate and time. Separating rate from time requires calibrations, and between calibrations the tree prior and the clock prior do much of the work. This is not a peculiarity of linguistics; it is well known in molecular dating (dos Reis et al., 2016). Rannala (2002) discussed how parameters that are not separately identifiable from the data have posteriors that remain sensitive to their priors no matter how much data is collected.

15.6.2 Documented sensitivities

Several kinds of prior sensitivity have been documented or plausibly argued for linguistic data.

- **Tree priors and ages.** The choice between Yule, birth-death and coalescent tree priors affects the distribution of node ages, especially when the tree prior is in tension with calibrations. Rama (2018) compared three tree priors across five Indo-European datasets and reported that the inferred dates depended on the choice of tree prior, which is a warning against treating any single date as robust.
- **Branch-length priors.** In unrooted, non-clock analyses, the prior on branch lengths can strongly influence the total tree length, a phenomenon studied in biology under the label "tail paradox" (Rannala et al., 2012). This is relevant to studies that infer topologies first and dates later.
- **Calibration priors.** When several calibration densities are specified, their joint effect is not simply their product, because they interact through the tree prior. The *effective* prior on a calibrated node can differ markedly from the density the analyst specified (Heath et al., 2014). In biology this is a well-known trap; in linguistics it is rarely checked.
- **Clock-rate priors.** Diffuse priors on the clock rate interact with the calibrations and the tree prior. If the calibrations are few and wide, the posterior root age can be pulled toward wherever the tree prior places mass.

Always run the analysis once with the data removed ("sampling from the prior"). Every Bayesian package used in linguistics supports this. Compare the prior distribution of the root age, of key clade ages, and of clade probabilities with their posteriors. If the posterior root age looks very much like the prior, the data are not driving the result. If clade probabilities under the prior are far from uniform (which happens when calibrations or monophyly constraints are present), interpret posterior clade probabilities relative to them.

15.7 Calibration choices

Calibrations convert relative branch lengths into absolute times. In linguistics they come from three sources: historically attested splits (for example, the settlement of Iceland in the late ninth century as a maximum age for the Icelandic-Norwegian split), ancient languages with known dates used as tips, and archaeological or genetic evidence linked to a linguistic event by argument. The first two are usually uncontroversial in principle, though the details matter: when exactly did Latin and the ancestor of the Romance languages diverge, given that Classical Latin is a literary register and not the direct ancestor of spoken Romance? The third is controversial precisely because the link between an archaeological culture and a language is an inference.

Three pitfalls deserve emphasis.

Circularity. If the hypothesis being tested is that a language family spread with agriculture, then calibrating the tree with archaeological dates for the spread of agriculture is circular. The circularity can be subtle: a calibration on an internal node that is itself justified by the farming hypothesis will push the root age toward the farming date. Good practice is to use only calibrations that are independent of the hypothesis and to show that results hold when contested calibrations are removed.

Calibrations as point values or as hard bounds. A historical "date" for a split is usually a date for an event that implies the split had happened (a terminus ante quem) or had not yet happened

(a terminus post quem). Encoding such information as a narrow normal distribution centered on the event date misrepresents it. Maurits et al. (2020) review how linguistic calibrations have been justified and argue for explicit, documented reasoning for each calibration, in the spirit of the best-practice guidelines developed for fossil calibrations in biology.

Ancestors placed as sisters. Ancient languages such as Latin, Vedic Sanskrit, Old Irish or Classical Armenian are, in many cases, close to the direct ancestors of later languages. Standard tree models treat every sampled language as a tip, so an ancient language must be placed as a sister of its descendants with a branch of its own. The model then has to explain the differences between, say, Latin and its sister Proto-Romance node as change on Latin's own branch, which can distort rate estimates and therefore dates. This was the central technical argument of Chang et al. (2015), discussed below, and it motivated the use of sampled-ancestor models in Heggarty et al. (2023).

15.8 Sampling of languages, and the language-dialect problem

15.8.1 Which languages?

Every phylogeny is a tree of the languages that were sampled, not of the family as a whole. Sampling affects inference in several ways that are well understood in biology. Adding taxa can break up long branches and reduce long-branch attraction (Zwickl and Hillis, 2002), so a sparse sample of a diverse family is more error-prone than a dense one. Root ages and the placement of the root depend on which early-diverging lineages are included; omitting an isolate-like branch can make the family look younger. Phylogeographic inference (Chapter 11) is especially sensitive, because the inferred homeland is pulled toward wherever the sampled languages cluster. And diversification analyses (Chapter 16) require explicit assumptions about the proportion of extant languages sampled.

In linguistics there is an additional complication: the sample of *documented* languages is not random. Languages spoken by larger, more accessible or politically prominent communities are better documented; extinct languages are known only if they were written; and some subgroups have been studied far more intensively than others. A sample of Indo-European that includes 15 Romance and 3 Indo-Iranian varieties says more about Romance than about Indo-Iranian, and analyses of rates or homelands can be distorted by such imbalance.

15.8.2 Languages, dialects and the units of analysis

Biologists argue about species concepts; linguists argue about what counts as a language. The usual criteria, mutual intelligibility, a shared standard or a shared name, are graded and often conflict. Glottolog distinguishes "languoids" at many levels precisely to avoid forcing a single answer. For phylogenetic analysis the problem has a specific form. If a dialect continuum is sampled as several "languages", the tree model is asked to impose a branching structure on what may be a network of overlapping innovations. If it is sampled as a single language, internal variation is ignored and the chosen variety may be unrepresentative.

There is no general solution, but there are better and worse practices. The units of analysis should be justified, ideally with reference to the question asked (a homeland analysis may need geographic coverage, a dating analysis may need well-documented, clearly distinct languages). Sensitivity analyses that swap one variety for another, or that merge closely related varieties, can show

whether the result depends on the choice. Chapter 16 describes species-delimitation methods from biology that could make these decisions more explicit.

15.9 The Anatolia versus Steppe controversy

No dispute illustrates the issues of this chapter better than the debate over where and when Proto-Indo-European was spoken. It involves every stage of an analysis: data, models, priors, calibrations, sampling and interpretation. It is also a case in which independent evidence, ancient DNA, arrived in the middle of the debate and changed its terms.

15.9.1 The two hypotheses

The **Steppe** (or Kurgan) hypothesis, associated with Marija Gimbutas and developed in detail by Mallory (1989) and Anthony (2007), places Proto-Indo-European in the Pontic-Caspian steppe north of the Black and Caspian Seas around 6,500 to 5,500 years ago, and links the dispersal of the family to mobile pastoralist societies of the Yamnaya horizon and their successors. The **Anatolian** hypothesis of Renfrew (1987) places Proto-Indo-European in Anatolia around 9,000 to 8,000 years ago and links the dispersal to the spread of farming into Europe.

The two hypotheses make different predictions about the age of the root of the Indo-European tree. Under the Steppe hypothesis the root (more precisely, the split between Anatolian and the rest) should be no older than about 6,500 to 7,000 years; under the Anatolian hypothesis it should be about 8,000 years or older.

15.9.2 The linguistic paleontology argument

The traditional linguistic argument for the Steppe hypothesis is *linguistic paleontology*. Proto-Indo-European can be reconstructed with a vocabulary for wheeled vehicles, including words for ‘wheel’, ‘axle’, ‘thill’ (shaft) and ‘to convey in a vehicle’. Archaeological evidence for wheeled vehicles appears only in the second half of the fourth millennium BCE. If the vocabulary was inherited from the common ancestor rather than borrowed or independently coined, then Proto-Indo-European (at least the stage ancestral to the non-Anatolian branches) cannot have broken up much earlier than about 5,500 years ago (Anthony, 2007; Anthony and Ringe, 2015). Proponents of the older dates reply that some of the wheel words may be derived independently in different branches from older roots with general meanings (‘turn’, ‘carry’), that semantic change from an older meaning is possible, and that the Anatolian branch, which may have split first, has little of this vocabulary, so the argument constrains only the later “core” of the family.

15.9.3 The phylogenetic dates

Gray and Atkinson (2003) applied Bayesian phylogenetic dating to the cognate data described above and obtained a root age of roughly 7,800 to 9,800 years before present, consistent with the Anatolian hypothesis. Bouckaert et al. (2012) extended the analysis with a larger dataset and continuous phylogeography, again obtaining an old root and inferring an Anatolian homeland. These papers were widely publicized and widely criticized.

The criticisms ranged across the whole framework of this chapter. Pereltsvaig and Lewis (2015), in a book-length critique, argued that the cognate data contained errors and inconsistencies, that

the phylogeographic model of diffusion could not represent known historical migrations and produced maps that conflicted with documented history, that the method could not accommodate linguistic paleontology, and that the dates were driven by assumptions rather than data. Other critics focused on the treatment of ancient languages and calibrations, and on whether a roughly clock-like rate of lexical replacement is plausible over thousands of years.

Chang et al. (2015) provided the most influential technical response from within the computational framework. They pointed out that several ancient languages in the dataset (for instance, Latin relative to Romance, Vedic Sanskrit relative to Indic, Old Irish relative to Modern Irish) are close to being direct ancestors of later languages. In an unconstrained tree model these languages are placed as sisters of their descendants, and the model must "explain" their differences from later languages as change on branches of their own. Chang and colleagues imposed *ancestry constraints*, forcing the ancient languages to sit at or very near the ancestral nodes of their descendants. With these constraints, and a curated version of the data, the root age dropped to roughly 6,000 years before present (medians of about 5,900 to 6,500 years across their main analyses), consistent with the Steppe hypothesis. Without the constraints, their analysis reproduced the older dates. The paper is a model of how to criticize a phylogenetic result from inside the method: it identified a specific modeling assumption, explained mechanically why it would bias dates, and showed that relaxing it changed the result.

15.9.4 Ancient DNA

From 2015 onward, ancient genome studies transformed the archaeology of the question. Haak et al. (2015) and Allentoft et al. (2015) reported a massive movement of people with ancestry related to the Yamnaya steppe populations into central and northern Europe in the third millennium BCE, associated with the Corded Ware culture. Later work traced steppe-related ancestry into South Asia in the second millennium BCE (Narasimhan et al., 2019). These findings were widely read as strong support for the Steppe hypothesis, at least for the spread of Indo-European languages into Europe north of the Alps and into South Asia, because they demonstrated a population movement of the right date and direction for which there had previously been only indirect evidence.

Ancient DNA does not, however, directly identify languages. Genes and languages can be transmitted separately: populations can shift language without much genetic change, and languages can spread with relatively small numbers of migrants. The genetic evidence for steppe ancestry in Anatolia, where the Hittite, Luwian and Palaic languages were spoken in the second millennium BCE, was weak, which created a puzzle for a simple steppe origin of the whole family. Lazaridis et al. (2022), analyzing ancient genomes from the "Southern Arc" (Anatolia, the Caucasus, the Balkans and neighboring regions), proposed that the earliest stage of the family (sometimes called Indo-Anatolian) may have been spoken south of the Caucasus, with the steppe a secondary center from which the non-Anatolian branches dispersed.

Heggarty et al. (2023) assembled the IE-CoR database, with explicit cognate judgements for about 160 Indo-European languages and a standardized meaning list, and analyzed it with a sampled-ancestor model in which ancient languages may lie directly on ancestral branches rather than being forced to be either tips or exact ancestors. They estimated a root age of roughly 8,000 years before present and proposed a "hybrid" scenario: an early homeland south of the Caucasus, with the branches that later spread across Europe having passed through the steppe. In a sense the result reconciles the two traditions: an early date compatible with the older phylogenetic estimates and

a role for the steppe compatible with the ancient DNA. It has nonetheless been challenged, both on the treatment of specific ancient languages and on the general question of whether lexical data can determine such deep dates with the claimed precision. Kassian and Starostin (2025), for example, criticize parts of its cognate coding and argue that 170 meanings are too few for a robust topology of so many languages, and a simulation study reported in a manuscript by Rönchen et al. (2025a) found that the earlier dating methods of Bouckaert et al. (2012) and Chang et al. (2015) did not recover simulated Indo-European root ages reliably (Section 10.8). The debate is, at the time of writing, unresolved.

15.9.5 Lessons

What should a reader take from this controversy? Several lessons generalize.

First, dating results depend on modeling choices in ways that are not captured by HPD intervals. The difference between the unconstrained and constrained analyses of Chang et al. (2015) is larger than the width of either analysis's credible interval. Model sensitivity analysis is therefore not optional.

Second, the treatment of ancient languages is a central technical problem for linguistic dating, and the move from tips to ancestry constraints to sampled-ancestor models is an example of the field improving its models in response to criticism.

Third, independent lines of evidence are essential, but they need to be integrated with care. Ancient DNA tells us about population movements, archaeology about material culture, and lexical phylogenies about language splits. Mapping one onto another requires bridging assumptions that should be stated explicitly. The best recent work in this area explicitly triangulates rather than treating any one line as decisive.

Fourth, the original critics and the original practitioners were each partly right. The critics were right that the early dates were sensitive to how ancient languages were treated and that the data had problems. The practitioners were right that the method could be improved from within, and that linguistic paleontology rests on its own assumptions about semantic stability and borrowing.

15.10 Phylogenetics and the comparative method

15.10.1 Shared innovations and shared retentions

The traditional comparative method of historical linguistics establishes subgroups by *shared innovations*: changes that occurred once in the common ancestor of a subgroup and were inherited by all its members (Campbell, 2013). Shared retentions, features inherited unchanged from the protolanguage of the whole family, are not evidence of subgrouping, because any branch could have retained them. This principle is identical to the central principle of Hennigian cladistics in biology, which distinguishes shared derived characters (synapomorphies) from shared ancestral characters (symplesiomorphies) (Hennig, 1966). The convergence was noted by many authors and is discussed by Atkinson and Gray (2005).

Critics of lexical phylogenetics have often argued that cognate presence/absence data conflate innovations and retentions. When two languages share a cognate for 'hand', is that because both retained the Proto-Indo-European word or because their common ancestor innovated it? A binary matrix does not say. Practitioners reply that probabilistic models handle this automatically: the

likelihood of a character pattern on a tree depends on where changes occur, so a cognate shared by a subgroup and absent elsewhere is evidence for that subgroup to the extent that a single gain (or a single loss elsewhere) is a better explanation than several independent events. In other words, the model infers which shared states are innovations rather than requiring the analyst to decide in advance. The argument is correct as far as it goes. But it depends on the model having a reasonable representation of where in the tree states are likely to change, and in particular on the root state and the gain/loss asymmetry. A stochastic Dollo model, in which each cognate is born exactly once, embodies the logic of innovation more directly than a reversible model does.

15.10.2 Quality of characters versus quantity

A deeper difference concerns what counts as good evidence. The comparative method privileges a small number of high-quality innovations: unusual, irreversible sound changes or morphological innovations that are unlikely to arise independently or to be borrowed. One irregular paradigm shared by two languages can outweigh hundreds of lexical similarities. Even shared innovations differ in weight. Phonetically natural and frequent changes, such as final devoicing, intervocalic voicing, palatalization before front vowels or the loss of a vowel-length contrast, recur independently in separate branches and so have little subgrouping value, and sound changes can diffuse between languages that have already separated (Campbell, 2013). The best-defined subgroups therefore rest on several innovations that are neither likely to recur nor likely to spread, a criterion that model-based analyses express only indirectly, through rates and, where it is modelled, borrowing. Lexical phylogenetics, in contrast, typically uses many characters of individually modest quality and relies on their aggregate signal.

The reasoning behind a high-quality innovation is itself probabilistic, as Ringe (2022) makes explicit for Germanic. Seven interrelated sound changes, including Grimm's and Verner's laws and the shift of stress to the first syllable, occurred in the prehistory of every well-attested Germanic language. Assigning each a rough probability of recurring independently in a lineage, from parallels elsewhere in Indo-European (between 0.1 and 0.5), and multiplying gives about 10^{-4} for all seven. Dividing further by the 24 possible orders of the four changes whose relative chronology is fixed gives roughly 4×10^{-6}. Since the estimates are crude but bounded, the conclusion that the changes happened once, in a common ancestor, is robust to errors in any one of them. The argument shows what makes the comparative method's best evidence strong: the joint improbability of many changes, and of their order, under independent origin. It also shows why such evidence is rare. Ringe stresses that Germanic is unusually lucky, and that many proposed subgroups have nothing comparable; for those, the aggregate signal of many weaker characters, combined with phonological and morphological characters where available, is what computational methods can add. In the same chapter he argues that computational cladistics yields only one line of evidence among several, and is sharply skeptical of lexical dating (Chapter 10), a view that sets out the critics' case clearly whether or not one shares it.

Ringe et al. (2002) and Nakhleh et al. (2005a) developed character-based methods that sit between these traditions: they use a mix of lexical, phonological and morphological characters, weight characters by their expected reliability, and search for trees (or networks) on which the high-quality characters are compatible. Their analyses of Indo-European recovered the well-established subgroups and highlighted where contact must be invoked. Nakhleh et al. (2005b) compared several methods on the same Indo-European dataset and found broad agreement on established subgroups but disagreement on the higher-order structure, with outcomes sensitive to the treatment of characters. This line of work makes an important point: the choice between "the comparative method"

and "phylogenetics" is a false dichotomy. The question is which characters, which models and which weights best capture the evidence that historical linguists already recognize as diagnostic.

15.10.3 What phylogenetics cannot do without the comparative method

It is worth stating plainly that lexical phylogenetics depends on the comparative method. Cognate judgements, the raw material of most analyses, are products of it. Without regular sound correspondences, there is no principled way to separate inherited from borrowed or chance-similar words, and automated cognate detection algorithms themselves model sound correspondences. Phylogenetic methods therefore do not replace the comparative method; they offer a different way of aggregating and weighing its outputs, and a framework for quantifying uncertainty and estimating quantities (dates, rates, locations) that the comparative method does not attempt to estimate. Jacques and List (2019) make a complementary argument from the other direction: that tree models are indispensable for linguistic reconstruction itself, and that the alternatives (wave models without explicit trees) do not remove the need for them.

This dependence creates a conundrum, which Goldstein (2020) states plainly for Indo-European. If a computational analysis contradicts the traditional subgrouping, it is dismissed as wrong; if it agrees, it is dismissed as unnecessary. His way out is to give computational methods two modest roles alongside the comparative method rather than in place of it (Bowern, 2017): a confirmatory role, since it is worth knowing whether an explicit algorithm reproduces the field's conclusions from its own data, and a role in breaking deadlocks where the comparative method has produced no consensus, such as the order in which the non-Anatolian branches separated. He adds two warnings that apply throughout this chapter. Sophistication and precision are not accuracy: a narrow credible interval for the age of Proto-Indo-European is no closer to the truth for being narrow. And no method can make up for flawed or conflicting data. Subgrouping and reconstruction are also mutually dependent, since a shared innovation can be recognized only once the ancestral state is known (Section 12.4.5).

A useful exercise for any lexical phylogeny is to list the classical shared innovations that define the subgroups it recovers (or fails to recover). If the phylogeny contradicts a subgroup supported by strong morphological and phonological innovations, the most likely explanation is a problem with the lexical data or the model, not the discovery that the subgroup is false. If the phylogeny confirms well-established subgroups and adds resolution where the comparative method is silent, it may be adding information, but the confirmation itself is not independent validation. Cognate judgements are products of the comparative method, and the same scholars often established both the cognates and the subgroups; moreover, cognate sharing is not the same as shared innovation, since shared retentions also produce shared cognates. Agreement shows that the phylogeny is *consistent with* the classification, not that the model has captured the true history.

15.11 Black boxes, software and reproducibility

Modern phylogenetic analyses are run in large and complex programs (BEAST 2, RevBayes, MrBayes, TraitLab, IQ-TREE; see Appendix F), with input files that may run to thousands of lines of XML or script and dozens of interacting settings. Critics have described these analyses as "black boxes" whose results cannot be evaluated by linguists who lack the technical training to inspect them. There is some justice in this. A reader of a published paper typically sees a tree, a date and a map, and must take on trust a large number of decisions about priors, models, operators,

burn-in and convergence.

The answer is not to avoid complex software but to make analyses transparent and reproducible. Several developments help. Tools such as BEASTling (Maurits et al., 2017) generate BEAST 2 configurations from short, human-readable descriptions, which makes the modeling decisions visible. Standard data formats, especially the Cross-Linguistic Data Formats (Forkel et al., 2018) and the Lexibank collection (List et al., 2022a), make datasets shareable and comparable. General principles for data stewardship (Wilkinson et al., 2016) and field-specific calls for reproducible research and data citation in linguistics (Berez-Kroeker et al., 2018) set expectations that journals increasingly enforce.

A fully reproducible linguistic phylogeny should provide at least the raw data with cognate judgements and sources, the scripts that convert raw data into analysis matrices, the complete input files for the inference software with version numbers, the random seeds, the MCMC logs or tree samples, and the scripts that produce figures and tables. It should also provide enough description that a reader can understand *why* each choice was made, not only what it was.

Default settings are decisions. A prior that was the default in one version of a program may not be the default in the next, and a default chosen for viral sequence data may be inappropriate for cognate data. Report every prior and model setting, including the ones you did not change.

15.12 Validation: simulation, known histories and model adequacy

How do we know whether a phylogenetic method works on language data? The five kinds of evidence of Section 6.10 answer different questions, and a convincing analysis draws on several. *Implementation validation* checks the software: a run without data should return the prior, two independent programs should agree on the same model and data, and simulation-based calibration (Talts et al., 2018) checks the sampler. *Parameter recovery* simulates data under the model's own assumptions and asks whether the method recovers the truth; passing it shows that a method can work, not that it does work on real data. *Robustness under misspecification* simulates data with the complications the model ignores. *Empirical adequacy* asks whether the fitted model reproduces the real data. *Historical interpretation* asks whether an estimated quantity, such as a node age or a homeland, corresponds to a real event. The three strategies below map onto these kinds.

15.12.1 Simulation

Simulation studies generate data on known trees under known processes and measure how well methods recover the truth. They provide parameter recovery when the simulation model is the inference model, and robustness tests when it is not. Their strength is that the truth is known; their weakness is that the data are only as realistic as the simulation model. The simulations of Greenhill et al. (2009) on borrowing and of Barbançon et al. (2013), who compared a range of distance, parsimony, compatibility and Bayesian methods on data simulated under models with borrowing and homoplasy, are examples in linguistics. The small simulations in this chapter are toy versions. Simulation studies are most informative when the simulation model differs from the inference model in realistic ways, because a method will always do well on data generated by its own model.

Such studies have recently been turned on the standard pipelines themselves. King (2026b) used simulation-based calibration to show that the default weighting of meaning-partition rates in

BEAST 2 analyses of cognate data produces poorly calibrated posteriors, an implementation-level problem that no single analysis would reveal, and Rönchen et al. (2025a) simulated Indo-European histories with loans and rate variation to test the robustness of published dating methods (Section 10.8).

15.12.2 Families with known history

The second strategy applies methods to cases where the history is known from independent evidence. In biology the classic example is the experimental phylogeny of Hillis et al. (1992), who generated a known phylogeny of bacteriophage lineages in the laboratory and tested whether reconstruction methods recovered it. Linguistics has no laboratory, but it has documented histories: the Romance languages with Latin as attested ancestor, the Germanic languages with their documented medieval stages, the Indic languages with Sanskrit and Middle Indic, and the Sinitic varieties with written Chinese. These cases can be used to check whether methods recover documented subgroups and dates, with two caveats. Well-documented families are not a random sample and may be easier than undocumented ones. And the check is independent only for what was documented independently of the coded data: attested dates of separation used out of sample are independent evidence, while subgroups established by the same comparative work that produced the cognate judgements are not. A related strategy is cross-validation of calibrations: remove one calibration at a time and ask whether the analysis predicts the removed date. Agreement increases confidence that the clock model is capturing something real.

15.12.3 Posterior predictive model adequacy

The third strategy asks whether the fitted model could plausibly have produced the observed data. In a posterior predictive check (Gelman et al., 1996; Bollback, 2002), one draws parameter values (including trees) from the posterior, simulates new datasets under the model with those values, and compares a test statistic T computed on the simulated datasets with the same statistic computed on the observed data. The posterior predictive p-value is

$$p_{\text{ppp}} = \Pr\left(T(D^{\text{rep}}) \geq T(D) \mid D\right), \tag{15.5}$$

and values very close to 0 or 1 indicate that the model fails to reproduce that aspect of the data. In a maximum-likelihood setting the same logic is implemented as a parametric bootstrap: simulate from the fitted model, refit or simply recompute the statistic, and compare (Goldman, 1993).

The choice of test statistic is crucial. It should capture an aspect of the data that matters for the inference and that the model might plausibly fail to reproduce. For cognate data, natural choices include the distribution of cognate-set sizes (how many languages share each cognate set), the number of cognate sets per meaning, the proportion of characters that are incompatible with the tree, and statistics of pairwise distances.

King (2026b) gives a worked linguistic example. Posterior predictive datasets simulated from a covarion model fitted to Germanic, Italic and Celtic data from IE-CoR differed from the observed data in several statistics of this kind, a failure he attributes most plausibly to semantic shift and to the non-independence of replacements within meanings (Section 7.10).

Model adequacy testing is a well-developed area in evolutionary biology. Bollback (2002) introduced posterior predictive checks for substitution models; Brown (2014b) and Brown and Thomson (2018) review predictive approaches and argue that adequacy, not just relative fit, should

be routinely assessed; and Duchêne et al. (2015a) developed posterior predictive tests of molecular clock models, asking whether the branch lengths implied by the clock are consistent with the branch lengths supported by the sequence data alone. A direct linguistic analogue would test whether the relaxed clock used to date a language family reproduces the branch-specific amount of lexical change estimated without a clock. To our knowledge, such clock adequacy tests have rarely if ever been reported for language families, though nothing prevents their use: the necessary simulation machinery exists in BEAST 2 and RevBayes. Chapter 16 returns to this.

15.12.4 Simulation: a parametric bootstrap check of rate homogeneity

We illustrate model adequacy with a simple parametric bootstrap. Because the "observed" data are themselves simulated, with a known violation of one of the models, the example shows that the check has the power to detect that violation; it is not evidence about the empirical adequacy of either model for any real dataset. We simulate cognate data on our 16-language tree with strong rate variation among characters (gamma shape 0.3), remove invariant characters (as is common practice), and fit two models by maximum likelihood with an ascertainment correction for the removed patterns: an equal-rates binary model and a model with four discrete gamma rate categories. Because `optim.pml()` in **phangorn** 2.12.1 did not reliably optimize branch lengths under `ASC = TRUE` in our tests, branch lengths and the gamma shape are then re-optimized with `refine_asc_edges()`, as explained in Section 7.5. As a test statistic we use the *cognate-set size spectrum*: the proportion of characters present in exactly j languages, for $j = 1, \dots, 15$. Rate variation leaves a characteristic signature in this spectrum. Slow characters tend to be shared by almost all languages, fast characters tend to be found in only one or two, and a homogeneous model cannot produce both extremes at once.

```
set.seed(1502)
adeq_data <- sim_cognates(true_tree, n_char = 600, shape = 0.3)
# remove invariant characters (present everywhere, or absent everywhere)
n_present <- colSums(adeq_data == "1")
adeq_data <- adeq_data[, n_present > 0 & n_present < n_lang]
d_obs <- as_phydat(adeq_data)
start <- NJ(dist.hamming(d_obs))
ctrl <- pml.control(trace = 0)
fit_eq <- optim.pml(pml(start, d_obs, ASC = TRUE), optNni = TRUE,
                    control = ctrl)
fit_gam <- optim.pml(pml(start, d_obs, ASC = TRUE, k = 4), optNni = TRUE,
                     optGamma = TRUE, control = ctrl)
# re-optimize branch lengths (and shape) on the corrected likelihood
fit_eq <- refine_asc_edges(fit_eq)
fit_gam <- refine_asc_edges(fit_gam, opt_shape = TRUE)
data.frame(model = c("Equal rates", "Gamma (4 categories)"),
           logLik = c(fit_eq$logLik, fit_gam$logLik),
           AIC = c(AIC(fit_eq), AIC(fit_gam)),
           gamma_shape = c(NA, fit_gam$shape))
```

```
#>                  model logLik  AIC gamma_shape
#> 1          Equal rates  -3478 7014          NA
#> 2 Gamma (4 categories)  -3388 6836      0.1674
```

```
size_spectrum <- function(m) {
  cs <- colSums(m == "1")
  cs <- cs[cs > 0 & cs < nrow(m)]
  tabulate(cs, nbins = nrow(m) - 1) / length(cs)
}
n_obs <- ncol(adeq_data)
replicate_spectrum <- function(fit, nrep = 200) {
  t(replicate(nrep, {
    sim <- as.character(simSeq(fit, l = 3 * n_obs))
    cs <- colSums(sim == "1")
    sim <- sim[, cs > 0 & cs < nrow(sim), drop = FALSE]
    size_spectrum(sim[, seq_len(min(n_obs, ncol(sim))), drop = FALSE])
  }))
}
obs_spec <- size_spectrum(adeq_data)
rep_eq <- replicate_spectrum(fit_eq)
rep_gam <- replicate_spectrum(fit_gam)

band <- function(reps, col) {
  q <- apply(reps, 2, quantile, probs = c(0.025, 0.975))
  j <- seq_len(ncol(reps))
  polygon(c(j, rev(j)), c(q[1, ], rev(q[2, ])), col = col, border = NA)
}
j <- seq_along(obs_spec)
# headroom for the legend
plot(j, obs_spec, type = "n",
     ylim = c(0, max(obs_spec, rep_eq, rep_gam) * 1.25),
     xlab = "Number of languages sharing the cognate set (j)",
     ylab = "Proportion of cognate sets", las = 1)
band(rep_eq, adjustcolor("grey40", 0.35))
band(rep_gam, adjustcolor("steelblue", 0.35))
points(j, obs_spec, pch = 19)
legend("topright", legend = c("Observed", "Equal-rates model (95%)",
                              "Gamma model (95%)"),
       pch = c(19, 15, 15), col = c("black", adjustcolor("grey40", 0.5),
                                    adjustcolor("steelblue", 0.6)),
       bty = "n", pt.cex = c(1, 2, 2))
```

```
# A one-number summary: proportion of singleton cognate sets (j = 1)
c(observed = obs_spec[1],
  p_equal_rates = mean(rep_eq[, 1] >= obs_spec[1]),
  p_gamma = mean(rep_gam[, 1] >= obs_spec[1]))
```

```
#>      observed p_equal_rates       p_gamma
#>       0.06186       0.00000       0.23500
```

Two things stand out. The equal-rates model is rejected decisively by the likelihood comparison, as expected; but more instructively, Figure 15.3 shows *how* it fails. It predicts too few singletons

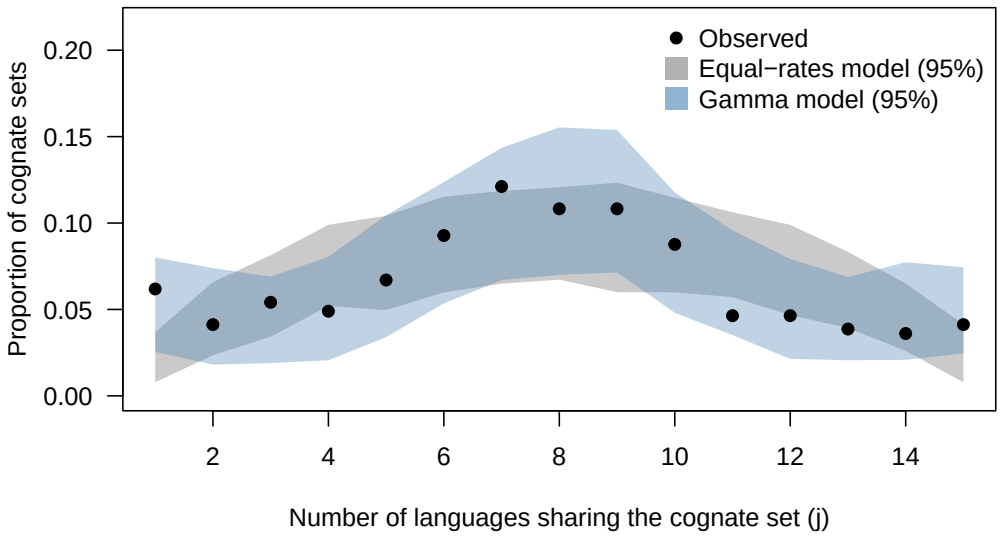


Figure 15.3: Parametric bootstrap check of model adequacy. Points show the observed proportion of cognate sets shared by exactly j of 16 languages. Shaded bands show the central 95% range of the same statistic over 200 datasets simulated from each fitted model (with invariant characters removed). The equal-rates model cannot reproduce the excess of rare and near-universal cognate sets.

(cognate sets found in only one language) and, less clearly, too few near-universal cognate sets, together with a broader hump of intermediate sizes; the observed proportion of singletons lies well outside its 95% band, and the one-number summary gives a parametric bootstrap p-value of essentially zero. The band of the gamma model contains every observed value. Of course, in this simulation the gamma model is the true model, and in real data even the better model may fail some adequacy tests. The point of adequacy testing is not to declare a model true but to discover which aspects of the data it cannot explain, and to judge whether those aspects matter for the question at hand. For a dating study, a model that cannot reproduce the observed number of rare cognates is suspect, because rare cognates carry much of the information about recent branch lengths.

15.13 Choosing a model: a decision guide

The critiques of this chapter are easier to state than to act on. A researcher who has assembled a dataset needs to know which models to fit, which assumptions to test, which results the data can support, and what to do about synonyms, loans, gaps and doubtful cognates. This section gathers the answers that the book gives piecemeal into four tables, and then works through one realistic case. The tables are a starting point, not a recipe. They reflect the identifiability and robustness results discussed in Chapters 7 to 10, and Table 7.2 remains the reference for how the models of lexical change differ in their assumptions.

15.13.1 Data situations

Table 15.1 starts from the data. Each row is a common situation; the columns give the codings and models worth considering, the main decisions about priors and calibrations, and the risk that most deserves a check.

Table 15.1: Common data situations and the models, codings and priors to consider for each.

Situation	Codings and models to consider	Priors and calibrations	Main risk to check
About 40 languages, lexical cognates, one family	binary with matched ascertainment correction and gamma rates; covarion and Dollo-type alternatives; meaning partitions	Yule or birth-death tree prior; relaxed clock; only calibrations independent of the hypothesis	dates driven by priors; short deep branches
Several historically attested varieties	ancient varieties as dated tips; sampled-ancestor (FBD) prior	tip dates as intervals; compare ancestors, sisters and calibration-only treatments	ancient written registers placed as long side branches
Extensive missingness	code unrecorded meanings as missing, never as absent; per-meaning ascertainment	as above; check that missingness does not shorten branches	poorly documented languages pulled to the root or to each other

Situation	Codings and models to consider	Priors and calibrations	Main risk to check
Suspected contact	flag known loans; tree models plus a network view; lateral-transfer or contact-aware models where available	none specific; avoid calibrating on contested contact nodes	a confident subgroup that is an artefact of borrowing
Small family (fewer than about 15 languages)	simple binary or multistate models; few free parameters	informative, justified priors; report prior and posterior side by side	posterior dominated by priors; unidentifiable rate variation
Structural data only	Mk models with ordered or hidden-rate variants; correlated-evolution models for dependent features	broad priors; no absolute dating unless strongly calibrated	logical dependence among features; contact and convergence
Global or multi-family samples	family-level trees (Glottolog or published posteriors) as random effects; spatial terms	branch lengths by rule only as a last resort	arbitrary branch lengths; non-independent families
Dialect continuum	split networks; glottometry; tree models only as summaries	none for dating	imposing a tree on overlapping innovations
No expert cognate judgements	automated cognates as provisional; compare thresholds	as for the corresponding expert case	systematic errors of the detection algorithm

Three rows deserve a comment. For ancient varieties, the choice among treating them as tips, as sampled ancestors or only as calibrations is itself a result to report (Sections 3.2.2 and 10.6). For structural data, many Grambank and WALS features are logically dependent, and independent Mk models overcount the evidence (Section 3.6). For global samples, the tree is a covariance structure for comparative analysis rather than an object of inference, and Chapter 13 discusses the consequences of using classifications with arbitrary branch lengths (Section 13.12.2).

15.13.2 Assumptions to test, and how

Table 15.2 lists the assumptions that most often matter, with a diagnostic for each and what to do if it fails. The diagnostics are of different kinds in the vocabulary of Section 6.10: prior-only runs and comparisons between programs are implementation checks; predictive checks assess empirical adequacy; simulations from fitted models with deliberate violations test robustness.

Table 15.2: Assumptions to test, diagnostics, and responses when an assumption fails.

Assumption	Diagnostic	If it fails
Treelike signal	delta scores, Q-residuals, NeighborNet	report the network; model contact; restrict claims to robust clades
No borrowing in a contested subgroup	lexical *D* statistic; rerun without suspected loans	model contact; treat the subgroup as unresolved

Assumption	Diagnostic	If it fails
Rate variation among characters is captured	site frequency spectrum; posterior predictive homoplasy	gamma or partitions by meaning; covarion
Clock model fits	clock adequacy check; root-to-tip spread	relaxed or local clocks; report dates as model-dependent
Single birth of each cognate class	compare Dollo-type and reversible fits; parallel losses	check coding (root versus root-meaning); pseudo-Dollo
Independence within meanings	rerun with multistate coding; meaning-level bootstrap	report wider intervals; meaning-slot models
Ascertainment correction matches the filter	compare filter and correction; refit with the other correction	match them and rerun
Priors do not drive the result	prior-only run; prior against posterior for key quantities	stronger data, or report the result as prior-dependent
Calibrations are independent of the hypothesis	leave-one-out of calibrations; cross-validation of dates	drop dependent calibrations
Ancient varieties correctly placed	compare tips, sampled ancestors and calibration-only runs	report the range of dates
Model is adequate for the target quantity	posterior predictive check aimed at that quantity	improve the model or narrow the claim
MCMC converged	ESS, multiple runs, ASDSF or PSRF	run longer, change moves, simplify the model

15.13.3 What the data can identify

Not every parameter that a program reports is supported by the data. Table 15.3 assesses common quantities for a dataset like the one in the next subsection, using the four properties of Section 6.10: identifiable under the model, practically estimable with a realistic sample, robust to misspecification, and interpretable as a historical quantity. Table 6.1 in Chapter 6 gives general judgements for the same four properties; the entries here are specific to the scenario, and like those they are judgements based on the results reported in this book, not theorems.

Table 15.3: Identifiability of common quantities for a family of about 40 languages with a few hundred cognate characters.

Quantity	Identifiable under the model	Estimable with about 40 languages	Robust to misspecification	Historical interpretation
Well-supported clades	yes	yes	mostly; not to concentrated borrowing	subgroups, if cognates reflect inheritance
Order of the earliest splits	yes	often not (short branches)	weak	yes, if resolved
Root position	only with a clock or outgroup	partly	weak (rate variation, Dollo violations)	yes
Relative branch lengths	yes	yes	sensitive to ascertainment and coding	amount of change, not time

Quantity	Identifiable under the model	Estimable with about 40 languages	Robust to misspecification	Historical interpretation
Absolute dates	only through calibrations and priors	wide intervals	sensitive to clock, calibrations, ancient tips	yes, conditional on calibrations
Rate per meaning	yes	roughly, for fast meanings	moderately	replacement rate, if coding is by meaning
Gamma shape	yes	usually	trades off with covarion and invariant terms	none directly
Covarion switching rates	yes	rarely	weak	none directly
Birth and death rates of the tree prior	partly	no	weak (sampling)	not language formation and death
Borrowing rate in lateral-transfer models	yes	weakly	depends on the borrowing model	yes, for the coded vocabulary
Ancestral states and homelands	yes	broad	sensitive to tree and model	conditional on bridging assumptions

15.13.4 Handling data problems

Finally, Table 15.4 collects the practical decisions about the data that every lexical analysis faces. The recommended default is the choice we would make before seeing results; the sensitivity analysis is the alternative that should be run and reported.

Table 15.4: Data problems: options, a default and the sensitivity analysis to report.

Problem	Options	Default	Sensitivity analysis
Synonyms	record all; keep the most basic; keep one at random	record all (binary coding handles them)	one word per meaning
Known loans	delete; code as missing; keep as separate characters	code the meaning as missing in the borrowing language if no native word exists	keep loans as cognates of the source
Suspected but uncertain loans	treat as inherited; treat as loans	flag; analyze both ways	the other treatment
Unrecorded meanings	missing; absent	missing (?) for all classes of the meaning	drop languages or meanings with most gaps
Uncertain cognacy (“doubtful”)	lump; split; missing	follow the expert, but flag	lumped and split matrices
Partial cognates	full-word cognacy; morpheme-level cognacy	the dataset’s convention, stated	the other convention where available

Problem	Options	Default	Sensitivity analysis
Root versus root-meaning coding	either	root-meaning (one meaning slot)	root coding, compared with a Dollo-type model
Binary versus multistate coding	binary; multistate; set-valued	binary with ascertainment correction	multistate on meanings
Singletons and constant characters	keep; remove	keep, with matching correction	remove, with matching correction
Reconstructed proto-languages	include as tips; exclude	exclude (Section 3.2.3)	none: reconstructions are not data

15.13.5 A worked scenario: forty languages, three ancient varieties and suspected contact

Consider a researcher with lexical data for 40 languages of one family, coded for 200 meanings by expert cognate judgement. Three of the varieties are historically attested, from texts about 1,000, 2,000 and 2,800 years old. About a fifth of the language-meaning cells are empty, but unevenly: a dozen languages are documented from short lists of 80 to 120 meanings. Two subgroups have been in contact for several centuries, and the specialists disagree about whether one language belongs to the first subgroup or the second. The questions are the topology of the family, the position of the disputed language, and the age of the root.

Data. The researcher records every synonym, codes unrecorded meanings as missing, and codes the 120 or so forms flagged as loans as missing in the languages that borrowed them, keeping a second matrix in which they are cognates of their source. Cognate sets marked as doubtful are lumped in one matrix and split in another. The three ancient varieties are included with their gaps; the reconstructed protolanguage is left out. The binary matrix keeps singletons, and the ascertainment correction conditions on observed characters only (no all-absent columns), applied per meaning so that languages with short lists are not penalized (Section 7.4).

Exploration. Before any tree is inferred, the researcher computes the proportion of missing cells and of singletons per language, delta scores and a NeighborNet, and lexical D statistics (Section 16.4.1) for quartets that place the disputed language with each candidate subgroup. A language with many singletons and few recorded meanings is checked for coding problems; a significant D in the contact quartets warns that the disputed placement may reflect borrowing.

Models and priors. The core model is a binary model with gamma rate variation and the matched ascertainment correction. Alternatives are a covarion model and a Dollo-type model, compared by marginal likelihood (Section 6.7), with meaning partitions if the software allows. The tree prior is birth-death with the three ancient varieties under a sampled-ancestor (fossilized birth-death) prior, with their dates entered as intervals. A relaxed clock is used, because 40 languages with uneven documentation will not change at equal rates. Calibrations are restricted to events independent of the root-age question, encoded as bounds rather than narrow distributions (Section 15.7). Before the data are added, the whole analysis is run once without data, and the prior distributions of the root age and of the disputed clade are saved for comparison.

What to expect. Table 15.3 sets expectations. Well-attested subgroups should be recovered, and their support is consistent with, but not independent confirmation of, the expert classification, because the cognate judgements come from the same comparative work. The earliest splits may

remain unresolved. The root age will have a wide interval that depends on the clock, the calibrations and the treatment of the ancient varieties, so the researcher plans to report it as a range across those choices. Gamma shape, covarion rates and the birth and death rates of the tree prior are nuisance parameters, not results.

Validation. The validation plan uses all five kinds of evidence. *Implementation*: the prior-only run, and the main analysis repeated in a second program (Chapter 17 runs the same data through five). *Parameter recovery*: data simulated from the posterior, with the real pattern of missingness, are reanalyzed to check that the topology and root age are recovered. *Robustness*: the same simulations are repeated with borrowing between the two contact subgroups at the level suggested by the flagged loans, to see whether borrowing alone can move the disputed language; Section E.1.1 gives code for such a study. *Empirical adequacy*: posterior predictive checks of singletons per language, classes per meaning and the D statistics. *Historical interpretation*: each ancient variety in turn is treated as undated and its age predicted, and the root age is compared with archaeological evidence that was not used to calibrate.

Reporting. The paper reports the topology with clade support, the disputed placement under each treatment of loans and doubtful cognates, the root age under each defensible combination of clock, calibrations and treatment of ancient varieties, the prior-only results, and the predictive checks, together with the data and input files (Chapter 17). If the disputed language moves when loans are recoded, or if the D statistic is large, the honest conclusion is that the lexical data cannot settle its position.

Code for a robustness study of the kind described in the worked scenario, with 40 simulated languages, uneven missingness and borrowing between two subgroups, is in Section E.1.1.

A smaller real example of several of these steps is the Polynesian case study of Section 17.14, where a strict clock rooted the tree on the side of the long-branched East Polynesian languages and contact placed East Uvean with Tongan.

15.14 Constructive guidance

The critiques reviewed in this chapter do not show that linguistic phylogenetics is unreliable in general. They show that its results are conditional on choices about data, models and priors, and that those choices must be made explicitly, justified, and tested. Several broad recommendations follow.

Treat data as a product of scholarship. Document every cognate judgement with sources, flag known loans rather than silently deleting them, record synonyms consistently, and, where possible, have more than one expert code the data and report agreement. Use automated cognate detection to check expert judgements and to speed up coding, not as a substitute for expertise where expertise exists.

Measure treelikeness before assuming a tree. Report delta scores or Q-residuals, look at a split network, and, for contested subgroups, test specifically whether the support could be due to contact between the languages involved.

Compare models and check adequacy. Report marginal likelihoods for alternative substitution and clock models where computationally feasible, and run at least one posterior predictive or parametric bootstrap check that targets a feature of the data relevant to the question.

Sample from the prior. Show the prior distribution of key quantities alongside the posterior. Run sensitivity analyses that drop contested calibrations one at a time and that vary the tree prior.

Justify calibrations. For each calibration, state the historical evidence, the event it constrains (split versus attestation), the type of bound (minimum, maximum, or both), and why the chosen distribution represents that evidence. Avoid calibrations that depend on the hypothesis being tested.

Treat ancient languages carefully. Use sampled-ancestor models or ancestry constraints where ancient languages may be ancestral, and report how dates change under alternatives.

Report uncertainty honestly. HPD intervals are conditional on the model. Where results depend on modeling choices, report the range across defensible choices, not only the result of the preferred model.

Make everything reproducible. Share data in CLDF, scripts, input files, logs and seeds.

Engage with the comparative method. Relate results to established shared innovations and explain disagreements.

The checklist in Table 15.5 condenses these recommendations into questions that authors can answer in a methods section or supplement, and that reviewers and readers can use to evaluate a study.

Table 15.5: A checklist for designing, reporting and evaluating a phylogenetic analysis of language data. The last column points to chapters where each topic is treated in detail.

Stage	Question	Chapter
Data	Are sources and cognate judgements documented for every form?	3
Data	Were known loanwords flagged, and how were they treated in the analysis?	3, 8
Data	How were synonyms and polysemous forms handled, consistently across languages?	3
Data	Was inter-coder agreement measured, or a second coder's matrix analyzed?	3
Data	Were invariant and singleton characters removed, and does the ascertainment correction match?	7
Model	Were alternative substitution models (reversible, Dollo, covarion) compared?	7
Model	Is rate variation among characters and among lineages modeled?	7, 9
Model	Is treelikeness reported (delta, Q-residual, split network)?	8
Model	Was the possibility of contact tested for any contested subgroup?	8
Priors	Is the prior distribution of key quantities shown (analysis without data)?	6
Priors	Are results robust to the choice of tree prior and clock-rate prior?	6, 10
Calibration	Is each calibration justified with evidence, event type and bound type?	10

Stage	Question	Chapter
Calibration	Are calibrations independent of the hypothesis being tested?	10
Calibration	Are ancient languages treated as possible ancestors (sampled ancestors or constraints)?	10
Sampling	Is the language sample justified, and is its imbalance across subgroups discussed?	10, 11
Sampling	Are units (language versus dialect) justified, with sensitivity to alternative choices?	15, 16
Inference	Is convergence documented (ESS, multiple runs, ASDSF/PSRF)?	6
Validation	Was model adequacy assessed with a posterior predictive or parametric bootstrap test?	15, 16
Validation	Do results recover documented subgroups and dates (or cross-validated calibrations)?	10, 15
Reporting	Are data, scripts, input files, software versions and seeds publicly available?	17
Reporting	Are all priors and settings reported, including defaults?	17
Interpretation	Are claims about people, places and cultures linked to the tree with explicit bridging assumptions?	11, 14

- Posterior probabilities and HPD intervals measure uncertainty conditional on the model; most controversies in linguistic phylogenetics concern systematic error, which these summaries do not capture.
- Cognate judgements are inferences from the comparative method. Random coding errors degrade trees gradually; systematic errors such as undetected, concentrated borrowing produce confident wrong subgroups that global accuracy scores hide.
- Binary coding of meaning slots creates dependent characters and overly narrow intervals, and ascertainment corrections must match how characters were collected.
- Dates depend on tree priors, clock priors, calibrations and the treatment of ancient languages; prior-only runs and sensitivity analyses are essential.
- The Anatolia-Steppe controversy shows critics and practitioners each identifying real problems, the method improving through ancestry constraints and sampled ancestors, and ancient DNA reframing the question without settling the linguistic one.
- Validation comes in five kinds: implementation validation, parameter recovery, robustness under misspecification, empirical adequacy and historical interpretation. Agreement with accepted subgroups is consistent with a model but is not independent validation, because the cognates and the subgroups come from the same comparative method.
- Model choice should start from the data situation and the target quantity; for each quantity, ask whether it is identifiable, practically estimable, robust and historically interpretable (Section 15.13).

15.15 Exercises

Conceptual

1. A paper reports a clade with posterior probability 1.0 and a root age with a 95% HPD interval of 6,200 to 7,100 years. List three kinds of systematic error that these numbers do not reflect, and for each name a check that would reveal it.

2. Explain why the recovery of well-established subgroups by a lexical phylogeny is weak evidence that the model is correct. What kinds of agreement with traditional scholarship would count as more independent evidence?
3. Two studies of the same family report root ages whose 95% intervals do not overlap. Using the stages of Section 15.1, list the possible sources of the disagreement and say which of them a reader could check from the published methods sections.
4. Why does concentrated borrowing between two particular languages do more damage to a specific subgroup than the same amount of diffuse borrowing does to the tree as a whole? Relate your answer to Figure 15.2.
5. For each simulation in this chapter, say which kind of validation it provides, and what it cannot show about real language data.
6. Summarize the argument of Chang et al. (2015) in terms of the stages of Section 15.1. Which stage did the argument locate the problem in, and why did fixing it change the dates more than the HPD intervals suggested was possible?

Computational

1. Using `sim_cognates()`, `flip_cells()` and `infer_trees()`, replace random coding error by a *systematic* error: flip only cells of characters present in the subgroup `L01-L03` (find its members with `true_clade`). Compare the RF distance and the recovery of that subgroup (`is.monophyletic()`) with the random-error results of Figure 15.1.
2. Extend the borrowing simulation with `borrow_fixed()` so that the donor is the recipient's closest relative outside its own subgroup, and then a distant language. For which donor does borrowing create a false sister relationship more easily, and why?
3. In the adequacy example, replace the test statistic by the number of characters that need more than one change on the true tree (per-pattern scores from `parsimony(true_tree, as_phydat(m), site = "site")`, weighted by the pattern counts in `attr(as_phydat(m), "weight")`). Does this statistic separate the equal-rates and gamma models as clearly as the size spectrum?
4. Rerun `issues-adequacy-fit` with `n_char = 150` and with `n_char = 1200`. How does the power of the size-spectrum check change? What does this imply for small linguistic datasets?
5. Load the Polynesian data of Section 17.14 (`data/case-study/`) and compute the cognate-set size spectrum (`size_spectrum()`) of the binary matrix with and without forms flagged as loans. Which part of the spectrum changes?

Research

1. You have lexical data for 25 Oceanic languages, two of them recorded only in nineteenth-century word lists of about 60 meanings, and good evidence of intense contact among four of the languages. Using the tables of Section 15.13, justify your choice of coding, model, tree prior, clock and calibrations, list the assumptions you would test, and say which conclusions you would and would not expect the data to support.
2. A colleague wants to date a family of 12 languages with structural (Grambank) data only and one archaeological calibration that is itself linked to the expansion hypothesis being tested. Write a referee's assessment of the plan, proposing an alternative design.
3. Design a validation programme for a published dating result in a family you know, specifying one study of each kind (implementation, parameter recovery, robustness, empirical adequacy, historical interpretation), the data and software each would need, and the out-

come that would weaken the published result.

15.16 Further reading

- Atkinson and Gray (2005): A clear account of the parallels between biological and linguistic phylogenetics and of where they break down; a good starting point for readers from either side.
- Pereltsvaig and Lewis (2015): The most detailed book-length critique of Bayesian phylogeography and dating of Indo-European. Not every argument has held up, but it is essential reading for understanding the critics' case.
- Chang et al. (2015): A model of constructive criticism from inside the method: it identifies a specific modeling problem with ancient languages and shows how fixing it changes the dates.
- Heggarty et al. (2023): The current state of the art in Indo-European data curation and sampled-ancestor dating, and a statement of the hybrid hypothesis. Heggarty (2021) reviews the cognacy databases behind earlier Indo-European phylogenies and traces many inconsistencies in published results to them.
- Ringe (2022): A short assessment, by a pioneer of computational cladistics in linguistics, of what these methods can and cannot establish, with a critical view of lexical dating.
- Anthony and Ringe (2015): A concise review of the linguistic and archaeological case for the Steppe hypothesis, including linguistic paleontology.
- Greenhill et al. (2009): The standard simulation study of borrowing and phylogenetic inference; read it alongside Chapter 8.
- Nakhleh et al. (2005b) and Barbançon et al. (2013). Comparisons of reconstruction methods on real and simulated Indo-European-like data.
- Jacques and List (2019): A spirited defense of tree models within historical linguistics, with discussion of when they are and are not appropriate.
- Brown and Thomson (2018): A review of model adequacy testing in evolutionary biology that translates directly to language data.
- Maurits et al. (2020): Practical guidance on justifying calibrations for dating language families.
- King (2026b): Simulation-based calibration and posterior predictive checks applied to the standard BEAST 2 analysis of cognate data; a model for the validation programme of this chapter.
- Rönchen (2025): A doctoral dissertation, with its summary freely available, arguing that finite-state models of vocabulary evolution overfit cognate data and that state-of-the-art dating methods are not robust on realistically simulated Indo-European data; read with Hammarström et al. (2019) on explanatory versus predictive modelling.
- Kassian and Starostin (2025): A critique of the most recent Indo-European phylogeny from the lexicostatistical tradition.
- List (2024): A list of open problems in computational historical linguistics, revisited after five years, most of which remain unsolved.

Chapter 16

The Biologist's Toolbox: Methods Waiting for Linguistics

Almost every analysis in the preceding chapters rests on one simplification: a language family is a single tree, and each character changes along it independently, by a Markov process whose rules are the same everywhere. Each critique in Chapter 15 attacks one part of that simplification. Synonyms and ancestral variation mean that different meanings may have different histories. Borrowing means that some histories leave the tree. Rates differ among meanings and lineages for reasons the model does not represent. Structural features may evolve under rules that differ between families. And the tree itself is the outcome of a process of splitting and extinction that the tree prior describes only crudely. Evolutionary biology has met each of these problems in its own data and built models for them. This chapter asks which of those models can be carried over to languages, and on what conditions. The shared methodological problem is that a biological method is a stochastic model of how its data arise. Transferring it is justified only if a linguistic process satisfies the same assumptions, or if we can show that the violations do not matter for the question asked. For each method, then, we say which assumptions survive the transfer, which are violated, and what simulation or observation could show that the adaptation works.

After reading this chapter you should be able to:

- state the four conditions (mapping, process, data, identifiability) under which a biological method can be transferred to language data, and apply them to a method not covered here;
- classify a proposed method as established in linguistics, a plausible extension or a speculative analogy, and justify the classification by naming the assumptions it preserves and violates;
- define a concept-level genealogy in an explicit model of lexical transmission, and explain when its marginal genealogy approximates the multispecies coalescent and when it does not;
- distinguish incomplete lineage sorting from borrowing using the symmetry of discordant patterns, and name the departures that would invalidate the test;
- explain why extinction rates and diversification shifts are weakly identifiable from trees of living languages, and how incomplete sampling biases diversification statistics;
- design a validation programme for a proposed method that separates parameter recovery, robustness under misspecification, empirical adequacy and historical interpretation.

16.1 How to read this chapter

Transferring a method from biology to linguistics is never just a matter of feeding new data into old software. Every phylogenetic method embodies a model of how its data arise, and the model has to make sense for language. Four questions are worth asking of any candidate method.

1. **What is the mapping?** Which linguistic entity plays the role of the gene, the individual, the population, the species and the fossil? The standard analogy (word or cognate set for gene, language for species, borrowing for horizontal gene transfer, ancient written language for fossil) is a starting point (Atkinson and Gray, 2005), but each method requires a more specific mapping, and some mappings break down.
2. **Do the process assumptions hold?** A coalescent model assumes that lineages within a population merge backward in time at a rate set by population size. A birth-death model assumes that lineages split and go extinct independently. Do speech communities and their words behave like that, at least approximately?
3. **Are the data available?** Many biological methods need data that linguists rarely collect in usable form: multiple sampled individuals per population, aligned sequences, or dated extinct taxa with character data.
4. **Is the quantity of interest identifiable?** Some biological methods estimate quantities, such as extinction rates from trees of living species, that are only weakly identifiable even with large biological datasets. Language trees are usually smaller. In the vocabulary of Section 6.10, a parameter can be identifiable under the model and still fail to be practically estimable with realistic linguistic samples, robust to misspecification, or interpretable as a historical quantity.

We classify each method into one of three tiers. **Tier A (established in linguistics)**: the method, or a close linguistic variant, has been applied in several published studies, and its assumptions have been discussed for language data. **Tier B (plausible extension)**: the mapping is direct and the core process assumptions are approximately satisfied, but the method has been applied rarely or not at all, and adaptation is needed. **Tier C (speculative analogy)**: the mapping itself is uncertain, or the process assumptions are known to fail for language, so the method needs a new model of the linguistic process before it can be used, not just new data. The tier says nothing about how interesting a method is; some of the most interesting ideas are in tier C. Separately, we keep a **status** label that records use: *applied* (several linguistic applications known to us), *barely* (one or a few exploratory or partial applications) or *not applied* (none known to us). These labels reflect our reading of the literature at the time of writing and should be revised as the field moves.

For each method we close with a short assessment: the tier, the assumptions that are preserved and violated, and a *research programme*, that is, a proposal together with the evidence that would show it works. The evidence is described in the terms of Section 6.10. Parameter recovery on data simulated under the method's own model is necessary but easy; it shows only that the method can work. Robustness under misspecification asks whether it still works when the data are simulated with the linguistic complications the model ignores. Empirical adequacy asks whether the fitted model reproduces the real data. Historical interpretation asks whether a parameter corresponds to something that happened. A proposal that offers only the first kind of evidence has not shown that the adaptation is effective.

Rather than list methods one after another, we organize them around five substantive questions about language history: how ancestral variation is sorted among descendants (Section 16.2); what determines rates of lexical and sound change (Section 16.3); how borrowing and language shift

modify genealogies (Section 16.4); whether feature evolution follows shared or lineage-specific mechanisms (Section 16.5); and how languages diversify (Section 16.6). Table 16.2 at the end gathers the assessments.

Several of the methods in this chapter are also mentioned in the "From the biologist's toolbox" boxes of earlier chapters, for example the structured coalescent in Chapter 11, hidden-rate models in Chapter 12 and profile mixture models in Chapter 7. This chapter collects them and goes into more depth on methods that do not fit naturally elsewhere.

16.2 Question 1: How is ancestral variation sorted?

A protolanguage was not a uniform code. It had synonyms, dialects and competing variants, and its daughters inherited this variation unevenly. A cognate shared by two languages may therefore be a shared retention of one of several ancestral variants, not a shared innovation, even when no borrowing occurred. Biology calls the corresponding phenomenon *incomplete lineage sorting*. This section asks whether the biological theory of sorting, the multispecies coalescent, can be given a linguistic foundation, and then turns to a related problem: how variation within and between languages should be partitioned when choosing the units of analysis.

16.2.1 The multispecies coalescent and incomplete lineage sorting

How it works. In biology, different genes in the same set of species often have different histories. The reason is that species are populations, and populations carry genetic variation. When an ancestral species splits into two daughters, both inherit a pool of gene copies, and which copy ends up fixed in which daughter is partly a matter of chance. Looking backward in time, the gene copies sampled from different species do not merge (coalesce) exactly at the species split; they merge somewhere deeper in the ancestral population. If the time between two successive species splits is short relative to the population size, the copies may fail to coalesce within that interval, and the gene tree may then have a topology different from the species tree. This phenomenon is called *incomplete lineage sorting* (ILS) (Maddison, 1997; Pamilo and Nei, 1988).

The *multispecies coalescent* (MSC) is the probability model that describes gene trees inside a species tree (Rannala and Yang, 2003; Degnan and Rosenberg, 2009). Within each branch of the species tree, lineages coalesce according to Kingman's coalescent (Kingman, 1982): when k lineages are present, the waiting time until the next coalescence (backward in time) is exponentially distributed with rate

$$\frac{k(k-1)}{2} \tag{16.1}$$

when time is measured in *coalescent units*, that is, generations divided by twice the effective population size, $t/(2N_e)$ for diploids (for the haploid models used below, t/N_e). Lineages that have not coalesced by the top of a branch pass into the ancestral branch, where they may coalesce with lineages coming from the sister branch.

The simplest consequence can be worked out by hand. Take three species with species tree $((A,B),C)$, and let T be the length, in coalescent units, of the internal branch between the (A,B) split and the root. The lineages from A and B coalesce within that branch with probability $1-e^{-T}$, in which case the gene tree is $((A,B),C)$. With probability e^{-T} they fail to coalesce, and all three lineages enter the root population, where each of the three possible pairs is equally likely to

coalesce first. The probability that the gene tree matches the species tree is therefore

$$P\big(((A,B),C)\big) = 1 - \tfrac{2}{3}\,e^{-T}, \tag{16.2}$$

and each of the two discordant gene trees has probability $\frac{1}{3}e^{-T}$. Two features of this result matter for what follows. First, the two discordant topologies are equally frequent under ILS alone. Second, for very short internal branches the matching topology is only slightly more frequent than either discordant one. For four or more species, there exist species trees, with several short internal branches, for which the most probable gene tree is *not* the species tree, a region of tree space called the *anomaly zone* (Degnan and Rosenberg, 2006). In the anomaly zone, taking a majority vote over gene trees gives the wrong species tree.

Methods for inferring species trees under the MSC fall into two broad families. *Full-likelihood* or Bayesian methods, such as BPP, *BEAST and StarBEAST2 (Heled and Drummond, 2010; Ogilvie et al., 2017), co-estimate gene trees and the species tree from sequence data.* Summary* methods, such as ASTRAL (Mirarab et al., 2014; Zhang et al., 2018b), take estimated gene trees as input and find the species tree that agrees with the largest number of gene-tree quartets, which is a statistically consistent estimator under the MSC. A third family bypasses gene trees altogether: SNAPP (Bryant et al., 2012) integrates over all possible gene trees analytically for unlinked biallelic markers, and SVDquartets (Chifman and Kubatko, 2014) infers quartets from site-pattern frequencies. A crucial theoretical result is that simply concatenating all genes and analyzing them as if they shared a single tree can be statistically inconsistent under the MSC: with enough data, concatenation can converge on the wrong tree (Kubatko and Degnan, 2007; Edwards, 2009).

The linguistic mapping. The natural linguistic analogue of a gene is a *meaning slot* (a concept such as 'fire' or 'water'), and the analogue of a gene copy is the word that fills that slot in a given speech community. The analogue of genetic polymorphism is *lexical variation within a speech community*: synonyms, competing variants, dialectal alternatives and register differences. Language change is in this sense a population process. Variants compete, and over time one of them may go to fixation while the others are lost. The parallel between lexical variants and alleles under drift has been developed formally by Reali and Griffiths (2010), and Newberry et al. (2017) used time-series methods from population genetics to distinguish selection from drift in the history of English grammatical variants.

Given this mapping, ILS has a clear linguistic meaning. If an ancestral language had two words for the same meaning, and its daughters each retained one of them, then which daughters share a word is determined partly by chance and need not reflect subgrouping. A classic example is the vocabulary for 'fire' in Indo-European. Proto-Indo-European is reconstructed with (at least) two words for fire: **h₁n̥gʷnis*, continued in Latin *ignis* and Sanskrit *agní-*, and **péh₂ur*, continued in Hittite *paḫḫur*, Greek *pŷr* and English *fire*. Latin and Sanskrit are not each other's closest relatives, and neither are Greek and English. The distribution of the two roots reflects the sorting of ancestral variation, and anyone who coded it naively would find characters that conflict with the accepted tree. In the terms of the traditional comparative method, these characters are *shared retentions*, not shared innovations, and the MSC is a probabilistic model of exactly why shared retentions are poor evidence of subgrouping (Section 15.10).

The analogy also clarifies what standard linguistic phylogenies do. A binary cognate matrix analyzed under a single tree is a *concatenated* analysis in the MSC sense: all meaning slots are assumed to share one history. The theory just cited implies that if ILS is common and internal branches are short, as they often are in rapid expansions like those of Bantu or Austronesian, concatenation

could in principle mislead. Whether lexical ILS is common enough to matter is an open empirical question. Several linguists have pointed out that sorting of ancestral variation can produce non-tree-like patterns that might otherwise be attributed to contact (see Jacques and List, 2019, for a discussion within the comparative method). The distinction matters, because ILS and borrowing leave different statistical footprints (Section 16.4.1).

The existence of competing synonyms does not, however, establish that their histories follow Kingman's coalescent. The coalescent is the limit of a particular population-genetic process, and whether words behave like genes in that process is a substantive question, not a matter of analogy. The next section therefore specifies a process.

16.2.2 A model of lexical transmission

Consider one speech community, idealized as N speakers, and one concept. Each speaker uses a word for the concept, and each word belongs to a cognate class. Time runs in generations of learners. In each generation every learner acquires the word for the concept from the previous generation in one of three ways:

- **copying** (probability $1-u-m$): the learner adopts the word of a *model* speaker chosen from its own community, under a copying rule that may be neutral (every speaker equally likely to be copied) or biased;
- **innovation** (probability u): the learner uses a word that belongs to no existing class, by coinage or by semantic shift from another concept, so that a new cognate class is born;
- **borrowing** (probability m): the learner copies a model from a different, contemporaneous speech community.

A cognate class is lost when no speaker uses it any longer. A language split is a split of the community into two that no longer copy from each other, except through borrowing. With neutral copying this is the Wright-Fisher model of population genetics (Wright, 1931) with infinitely many possible alleles, and the alleles are cognate classes; a version with overlapping generations is the Moran model (Moran, 1950). Related models of variant competition in a speech community have been studied by Baxter et al. (2006), whose utterance selection model has correspondences with the Wright-Fisher model but also significant differences; by Reali and Griffiths (2010), who showed that iterated learning by Bayesian learners can be equivalent to drift with mutation; and, in cultural evolution more broadly, by Bentley et al. (2004).

Within this model a *concept-level genealogy* can be defined precisely. Every word token in the present generation was copied from a model token in the previous generation, which was copied from an earlier one, and so on. Following these copying links backward from a sample of tokens, one per sampled language (or several, if synonyms or several speakers are sampled), gives a tree: the genealogy of the concept. Innovation does not start a new genealogy. It is a mutation on an existing lineage that changes the class of the token without changing its ancestry, so cognate classes are the *partition* of the sampled tokens induced by innovation events on the genealogy: two tokens belong to the same class exactly when no innovation occurred on the path that connects them. Replacement of the dominant word in a language is the spread to fixation of one of these innovations. Borrowing is a copying link that crosses from one community to another, which in population genetics is migration. When several classes coexist in a language, sampling several tokens per language (several speakers, or the synonyms listed in a dictionary) corresponds to sampling several individuals per species, the setting for which the MSC was designed.

Three consequences follow. They are derived in Section E.2.2.

1. **When the coalescent holds.** If copying is neutral and exchangeable, the community is well mixed, its size changes slowly and N is large, the genealogy of a sample of tokens from one community converges to Kingman's coalescent with time measured in units of N generations. If, in addition, communities split cleanly and never borrow, the genealogy of one token per language follows the multispecies coalescent, and the cognate classes within each community follow Ewens' sampling formula. With borrowing, the genealogy follows a coalescent with migration, not the MSC.
2. **What the data reveal.** For a single concept the data are a partition of the languages into a handful of cognate classes, not a resolved tree. Per-concept "gene trees" cannot be estimated from cognate data, so summary methods that start from estimated gene trees, such as ASTRAL, have little to work with. Methods that integrate over genealogies, such as SNAPP-like likelihoods or site-pattern methods, are the natural route.
3. **What the replacement rate is.** Under neutral copying an innovation fixes with probability $1/N$, and about Nu innovations arise per generation, so the dominant class is replaced at rate u per generation, whatever the size of the community. This is the neutral-theory result of population genetics, and it is the process that the replacement models of Chapter 7 assume implicitly. Synonymy persists, and sorting matters, when $\theta = 2Nu$ is not small and internal branches of the language tree are short in units of N generations.

The coalescent approximation fails, in different ways, when the copying rule is not neutral or the community is not well mixed. Table 16.1 summarizes the main departures.

Table 16.1: Departures from neutral transmission and what they do to the concept-level genealogy. "Symmetry" is the equal frequency of the two discordant patterns used to separate sorting from borrowing (Section 16.4.1).

Departure	Linguistic example	Genealogy	Symmetry
Conformity (frequency-dependent copying)	learners follow the majority usage	close to Kingman, but rare classes are purged, so classes are not Ewens-distributed	preserved
Novelty bias (anti-conformity)	fashionable new words	balancing selection: deep genealogies, long-lived synonym pairs	preserved
Prestige (a few speakers copied by many)	an influential speaker, a written standard	multiple mergers (a Λ-coalescent); smaller effective size; concepts linked	preserved
Directional selection on variants	shorter or more frequent forms win	not Kingman; sweeps reduce variation	preserved if sisters are alike
Structure within a community	dialects, social networks	structured coalescent	can fail if ancestral dialects differ
Borrowing between sampled languages	loans from a neighbour	coalescent with migration	fails (the signature of contact)
Semantic shift and polysemy	'skin' > 'bark'; one word for 'hand' and 'arm'	lineages jump between concepts; concepts are not independent loci	preserved, but loci are linked

Departure	Linguistic example	Genealogy	Symmetry
Founder effects at splits	a small group settles an island	a burst of coalescence at the split	preserved

Three points in the table deserve emphasis. First, the symmetry on which the test for borrowing rests needs only that the two sister lineages are exchangeable, not that copying is neutral, so it survives most social biases; it fails under borrowing, which is what it is meant to detect, and under structure in the ancestral community, which is a genuine alternative explanation. Second, the *amount* of discordance does not survive the biases. Conformity, prestige and founder effects all speed up sorting, so a full MSC analysis would infer internal branches that are too long in coalescent units, or equivalently "population sizes" that are too small. The effective size of a speech community is identifiable under the model, but it is not interpretable as a number of speakers; at best it measures how fast variation is sorted. Third, directional selection matters empirically: Blythe and Croft (2012) argued that the S-shaped curves typical of documented changes arise only if some variants are favoured over others, and Newberry et al. (2017) found selection in some English changes and drift in others. The neutral model is a null model, not a description.

16.2.3 Simulation: concept trees under the multispecies coalescent

We wrote a small MSC simulator, `sim_msc_tree()`. Given a language tree with branch lengths in coalescent units, it starts with one lineage per tip and, moving from the tips toward the root, lets lineages coalesce within each branch according to Equation (16.1). The output is one concept-level tree: the genealogy of the words filling one meaning slot, under the conditions of point 1 above. Section 16.4.1 uses it to generate concept trees with and without borrowing.

The code of `sim_msc_tree()` and a check of the simulator against Equation (16.2) (Figure E.1, an implementation validation) are in Section E.2.1.

16.2.4 Simulation: does the token model produce a coalescent?

The MSC simulator assumes the coalescent. To ask whether the lexical transmission model *produces* one, we simulate the model forward in time. A community of 30 speakers evolves for 400 concepts at once, with innovation probability $u = 0.01$ per learner and concept. After a burn-in that brings the variation to equilibrium, we tag every token with the speaker who used it at a reference time and then follow the tags forward. Two tokens sampled t generations later carry the same tag exactly when their genealogies coalesced within those t generations, the event that decides whether two sister languages that separated t generations after the reference time are sorted for that concept. Under the coalescent the probability is $1 - e^{-t/N}$. We compare three copying rules: neutral copying; a *conformist* rule, under which a learner who meets two models that agree adopts their word half of the time; and a *prestige* rule, under which, in one generation in ten, half of the learners copy a single speaker for every concept. In the vocabulary of Section 6.10, the neutral run checks that the approximation holds where theory says it should, and the other two test the *robustness* of the coalescent description to misspecification.

The functions behind this simulation, the derivation of the coalescent limit of the token model, and a check of the neutral simulator against Ewens' sampling formula are in Section E.2.2.

```
set.seed(1607)
rules <- c(neutral = "neutral", conformist = "conformist",
           prestige = "prestige")
token_sims <- lapply(rules, run_bias)
Tg <- token_sims$neutral$curve[, "T"]
plot(Tg, 1 - exp(-Tg), type = "l", lty = 2, ylim = c(0, 1), las = 1,
     xlab = "Generations since the reference time / N",
     ylab = "P(two tokens share an ancestor)")
cols <- c(neutral = "black", conformist = "steelblue",
          prestige = "firebrick")
for (r in names(rules))
  lines(Tg, token_sims[[r]]$curve[, "coal"], col = cols[r], lwd = 2)
legend("bottomright", c("coalescent", names(rules)), bty = "n",
       lty = c(2, 1, 1, 1), lwd = c(1, 2, 2, 2),
       col = c("black", cols))
```

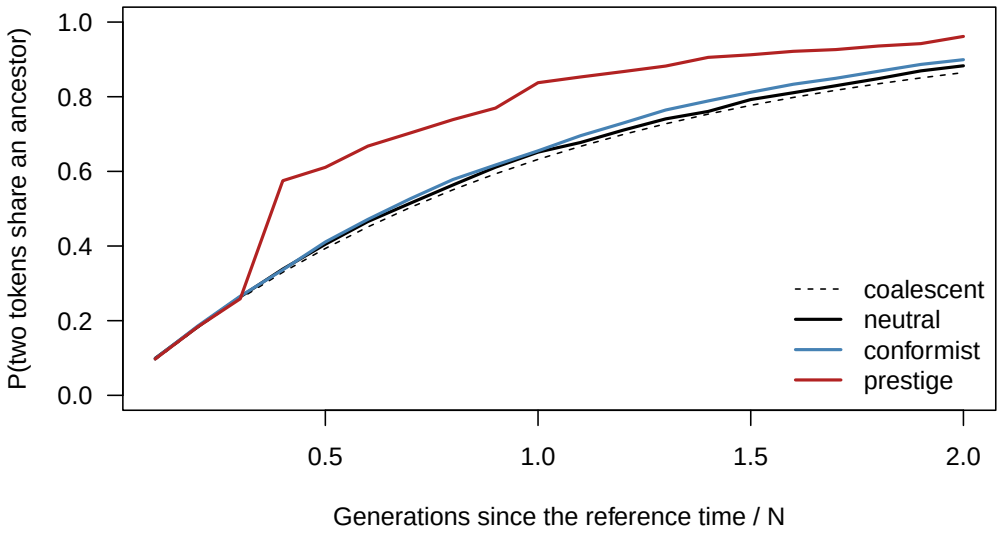


Figure 16.1: A forward model of lexical transmission compared with the coalescent. Each curve is the probability that two word tokens sampled from a community of 30 speakers descend from the same token t generations earlier, averaged over 400 concepts; the horizontal axis is t/N, in coalescent units. The dashed curve is the coalescent prediction. Neutral and conformist copying follow it closely; prestige copying (half of the learners copying one speaker in one generation in ten) produces jumps, the multiple mergers of a Lambda-coalescent, and much faster sorting.

```
# cognate classes per concept among the 30 speakers, and Ewens' expectation
theta <- 2 * 30 * 0.01
round(c(sapply(token_sims, `[[`, "classes"),
        ewens = sum(theta / (theta + 0:29))), 2)
```

```
#>    neutral conformist   prestige      ewens
#>       3.05       1.58       2.62       2.97
```

Figure 16.1 separates two things that a coalescent analysis of cognate data assumes: a genealogy, and a process of innovation on it. Under neutral copying the genealogy follows the coalescent curve, and the number of cognate classes per concept among the 30 speakers agrees with Ewens' formula (the last two printed values). Under the conformist rule the genealogy hardly changes, but the variation does. There are about half as many classes per concept, because conformity purges rare innovations before they can spread. Cognate data generated this way would look like data from a neutral model with a much lower innovation rate, although the genealogy is nearly Kingman. Under the prestige rule the genealogy itself departs from Kingman. Sorting happens in jumps, when one speaker is copied by half the community, and on average much faster than the number of speakers implies. Because the jumps happen in all concepts at once, the concepts are no longer independent loci, and any test that treats them as independent will overstate its confidence. Both biased rules still treat sister communities alike, which is why the symmetry test of Section 16.4.1 still applies to them. One run with 30 speakers is an illustration, not a study; the research programme below asks for the systematic version.

Assessment: the multispecies coalescent. *Tier B for site-pattern tests and for simulation studies of sorting; tier C for full-likelihood species-tree inference with interpretable population sizes. Barely applied. Preserved:* concepts as loci, language splits as population splits, ancestral synonymy as polymorphism, and, under neutral and exchangeable copying, a concept genealogy that is a coalescent (Section 16.2.2). *Violated:* social biases change the amount of sorting and, under prestige, the shape of the genealogy; borrowing adds migration; semantic shift and polysemy link loci; and word lists usually record one "most basic" word per meaning, which is not a random sample of tokens, so within-language variation is censored by the way the data are collected. *What would show it works:* (i) parameter recovery of the language tree from cognate data simulated under the neutral token model; (ii) robustness of the same analysis to conformity, prestige, dialect structure and borrowing, extending the simulation above; (iii) empirical adequacy, with posterior predictive distributions of the number of classes per concept and of synonyms per language that match datasets which record synonyms; (iv) historical interpretation, checking inferred ancestral polymorphism against the synonymy that philology documents for written protolanguages such as Latin.

Research programme. Three projects follow. First, a simulation study, with the token model rather than the coalescent alone, asking how much lexical sorting would be needed for concatenated cognate analyses of rapid radiations to enter something like the anomaly zone, and whether realistic social biases make that more or less likely. Second, a species-tree analysis in which each meaning slot is treated as a multistate locus and multiple varieties (dialects, or multiple sources for the same language) are treated as sampled "individuals" within a language "species". This would allow within-language variation, which is usually discarded, to inform estimates of ancestral variation and to correct for sorting. StarBEAST-type models would need adaptation for cognate-class data, but SNAPP's treatment of biallelic markers (Bryant et al., 2012) suggests a starting point for binary cognate data. Its output should be read with Table 16.1 in mind: the inferred population sizes measure the rate of sorting, not numbers of speakers. Third, the lexical D statistic of Section 16.4.1, which needs much weaker assumptions than either.

Biologists have learned to treat discordance among genes as signal rather than noise. Discordance tells us about ancestral population sizes, about the timing of splits relative to those sizes, and, when asymmetric, about gene flow. The linguistic analogue would treat disagreement among meaning

slots as information about the amount of lexical variation in protolanguages, about how rapidly a family radiated, and about contact. Standard binary cognate analyses discard this information by forcing all slots onto one tree. Two practical tools from biology are especially easy to adapt: site concordance factors, which report, for each branch of a tree, the proportion of informative characters that support it, and the D statistic and its relatives for testing whether discordance is symmetric.

16.2.5 Species delimitation and the language-dialect problem

How it works. Species delimitation methods ask how many species are present in a sample and which individuals belong to which. Three families of methods are prominent. The *generalized mixed Yule-coalescent* (GMYC) model (Pons et al., 2006) fits a single-locus ultrametric tree with a model in which branching events deeper than a threshold time follow a Yule (speciation) process and branching events shallower than the threshold follow a coalescent (within-species) process. The threshold, estimated by maximum likelihood, separates between-species from within-species branching. The *Poisson tree processes* (PTP) model (Zhang et al., 2013) does the same with branch lengths measured in substitutions rather than time. Multilocus methods such as BPP (Yang and Rannala, 2010) and Bayes factor delimitation (Leaché et al., 2014) compare MSC models in which different groupings of individuals are treated as species. An important caution comes from Sukumaran and Knowles (2017), who showed that MSC-based delimitation detects population structure, which need not coincide with species boundaries: any isolated population will be "delimited" given enough data.

The linguistic mapping. The language versus dialect problem (Section 15.8) is a delimitation problem. Its practical form is: given lexical or phonological data from many local varieties (for instance, from a dialect atlas), how many distinct "languages" should be recognized for the purposes of a phylogenetic analysis, and which varieties belong to each? The GMYC logic transfers in outline. Within a language, varieties differ by a slow accumulation of small changes and remain in contact, so their tree should look like a coalescent tree with many short, shallow branches; between languages, the tree should look like a branching process. Sukumaran and Knowles's caution also transfers: any method that identifies discrete clusters in dialect data will find structure, and whether that structure corresponds to "languages" in a sociolinguistic sense (mutual intelligibility, a shared standard) is a separate question. Related work exists: Wichmann (2019) proposed using a threshold on lexical distances computed from ASJP word lists to distinguish languages and dialects, and dialectometry has a long tradition of clustering local varieties (Section 8.8).

Assessment: species delimitation. *Tier C; not applied. Preserved:* the idea that branching within and between units follows different processes. *Violated:* dialects of one language are in continuous contact, so variation within a language is structured by geography and borrowing rather than by the panmictic coalescence the model assumes (Table 16.1); there is no single-locus genealogy of varieties to fit; and the target, a "language", is defined socially as much as genealogically. *What would show it works:* run on dialect atlases with independent judgements of language boundaries, the method reproduces those boundaries better than a simple distance threshold, and the result is robust to changes in the density of sampling.

Research programme. Fit a GMYC-type model to dated trees of dialect varieties within a few well-documented language areas, such as dialect atlases for Germanic or Romance varieties, where independent judgements of language boundaries exist, and compare the inferred thresholds with those boundaries and with ISO 639-3 and Glottolog classifications. Then apply the calibrated

approach to areas where language boundaries are disputed. A second proposal is methodological: use delimitation as a preprocessing step for family-level phylogenies, so that the choice of which varieties to include as tips becomes an explicit, reproducible decision rather than an ad hoc one.

16.3 Question 2: What determines rates of lexical and sound change?

Rates of lexical replacement differ by orders of magnitude among meanings, and frequency of use predicts much of the difference (Pagel et al., 2007); whether they also depend on the size of speech communities is disputed (Section 9.10). The token model of Section 16.2.2 shows why these questions are about process. If copying is neutral, a meaning is replaced at its innovation rate u, independently of community size, so rate differences among meanings must come from differences in how often new words are coined, shifted in or borrowed. If copying is biased, for instance against rare variants or in favour of frequent forms, the replacement rate depends on the strength of the bias and, through the fixation probability, on community size. The methods in this section do not settle the question by themselves, but they supply the tests that any answer needs: whether the rate models we use fit at all, how ancient languages can inform rates directly, and how change in the forms of words, rather than in their presence or absence, can be modelled.

16.3.1 Model adequacy and posterior prediction

How it works. Model *selection* tells us which of several models fits best; model *adequacy* asks whether the best model is good enough to have produced the data (Brown, 2014b; Brown and Thomson, 2018). The main tool is posterior predictive simulation: draw trees and parameters from the posterior, simulate datasets under the model, and compare test statistics of the simulated data with those of the observed data (Bollback, 2002). Biologists have developed adequacy tests for substitution models, for clock models (asking whether the clock-constrained branch lengths are consistent with the branch lengths estimated without a clock; Duchêne et al. (2015a)), for tree priors, and for comparative models.

The linguistic mapping. Chapter 15 gives a worked example with the cognate-set size spectrum (Section 15.12), and Chapter 7 another with homoplasy (Section 7.10). Other natural test statistics for language data include the number of cognate classes per meaning slot; the number of cognate sets shared by each pair of languages; the number of characters requiring more than one gain on the tree; the asymmetry statistics of Section 16.4.1, which would reveal unmodelled contact; the number of synonyms per language, which the token model predicts; and, for clocks, the distribution of root-to-tip amounts of change among languages sampled at the same time.

The first systematic application to the standard linguistic set-up is that of King (2026b), who combined simulation-based calibration of the BEAST 2 implementation with posterior predictive simulation of a covarion model fitted to Indo-European data, using eight statistics of language divergence and cognate distribution. The model fell short on several of them (Section 7.10). King (2026a) applied *posterior* simulation-based calibration, which starts from the posterior for a given dataset rather than from the prior, to tip-dated Indo-European analyses (Section 10.8).

Assessment: model adequacy. *Tier B; first systematic applications (King, 2026b). Preserved:* everything. Posterior predictive checks assume only that the model can be simulated, which every model in this book can. *At risk:* the choice of test statistic. A statistic that the model fits by con-

struction (for example, the number of cognates per language under a model with branch-specific rates) has no power, and a battery of many statistics needs a correction for multiple testing. *What would show it works:* each statistic detects a known misspecification in simulated data (borrowing, rate shifts, Dollo violations) with useful power while not rejecting correctly specified models more often than its nominal rate; applied to published datasets, it identifies failures that change a substantive conclusion when the model is improved.

Research programme. Develop a standard battery of posterior predictive tests for cognate data, implement it as a post-processing tool that reads BEAST 2 or RevBayes output and a CLDF dataset, and apply it to the main published datasets (Indo-European, Austronesian, Bantu, Sino-Tibetan, Pama-Nyungan) under the models used in their published analyses. The result would be a map of which aspects of lexical evolution current models capture and which they miss. A clock adequacy test of the type developed by Duchêne et al. (2015a) would be particularly valuable for the debated dating results of Chapter 10.

16.3.2 Total evidence and tip-dating with the fossilized birth-death process

How it works. In biology, fossils were long used only to calibrate nodes: a fossil assigned to a clade sets a minimum age for that clade. This approach discards the fossils' own character data and requires the analyst to translate a fossil into a calibration density, which is notoriously difficult. *Total-evidence dating* instead includes fossils as tips with their own morphological characters, analyzed jointly with molecular and morphological data from living species (Pyron, 2011; Ronquist et al., 2012a). The *fossilized birth-death* (FBD) process provides a coherent tree prior for such analyses (Stadler, 2010; Heath et al., 2014): lineages split at rate λ, go extinct at rate μ, and are sampled as fossils at rate ψ, with extant species sampled with probability ρ. A key feature of the FBD process is that a fossil can be a *sampled ancestor*, lying directly on a branch of the tree rather than at the end of its own side branch (Gavryushkina et al., 2014). Total-evidence dating under the FBD process is implemented in MrBayes, BEAST 2 and RevBayes (Zhang et al., 2016).

The linguistic mapping. The mapping is unusually direct. Ancient written languages are fossils: they are sampled at a known time in the past and carry character data (cognates, structural features, sound changes). Many of them are close to direct ancestors of later languages, which is exactly the situation sampled-ancestor models were built for (Section 10.6). For the question of rates, dated ancient tips are the most direct evidence there is: the amount of change between an ancient language and its descendants, over a known interval, is a measured rate. Total evidence corresponds to combining lexical cognate data with structural (typological) and phonological characters, each in its own partition with its own model and possibly its own clock.

Ancient languages have long been included as dated tips in Bayesian analyses (for instance in Bouckaert et al. (2012)), and Chang et al. (2015) showed why their position as possible ancestors matters (Section 15.9). Heggarty et al. (2023) used a sampled-ancestor model for Indo-European. Analyses combining lexical and structural characters in a single partitioned Bayesian model remain uncommon (an analysis of Japonic that adds phonotactic characters as a separate partition to cognate data is a recent example of combining data types; Huisman et al. (2025)), although character-based analyses combining lexical, phonological and morphological characters have a long history (Ringe et al., 2002), and structural characters have been used on their own (Dunn et al., 2005, 2008).

Assessment: tip-dating and sampled ancestors. *Tier A for ancient languages as dated tips and sampled ancestors (applied); tier B for full total-evidence analyses with several partitions (barely applied).*

Preserved: dated tips with character data, the possibility of ancestry, and a process of sampling through time. *Violated:* "fossilization" of languages depends on writing, so the sampling rate ψ is zero for most of prehistory and clustered in a few literate regions; written registers (Classical Latin, Vedic Sanskrit) are not samples of the spoken ancestral variety; and the FBD assumption that extinction and sampling are independent of the characters does not hold when the languages that were written down are those of states and empires. *What would show it works:* recovery of known ancestor-descendant relations and documented dates when one ancient language at a time is withheld and predicted (cross-validation, an empirical check), and robustness of the root age when the sampling rate is allowed to vary through time.

Research programme. A full total-evidence analysis of a family with a rich written record: lexical cognates, structural features (for instance from Grambank) and regular sound changes, each as a separate partition, analyzed under an FBD prior in which the fossil sampling rate ψ is allowed to change through time. A skyline FBD with $\psi = 0$ before the invention of writing in the relevant region, and a positive, possibly increasing, rate afterward, would represent the fact that languages can only be "fossilized" in texts. Fragmentarily attested ancient languages (Gaulish, Phrygian, Thracian, many Anatolian and Italic languages) could be included with mostly missing data, as poorly preserved fossils are in biology, and their placement would be estimated rather than imposed. The rates estimated from the ancient tips could then be compared across partitions, a direct test of whether grammar and vocabulary change at different rates (Section 9.9).

16.3.3 Sequence-level models of sound change

How it works. In molecular evolution, the unit of change is the site, and models such as GTR describe substitution rates between nucleotides. For sound change the natural unit is the *sound correspondence* or the *sound law*: a context-sensitive rewrite rule (for instance, Latin *ct* > Italian *tt*) that applies across the lexicon. Biology offers two partial models of such coordinated change. One is *concerted evolution*, in which members of a gene family evolve together because of gene conversion, so that changes spread across copies. The other is the modelling of *correlated sites*, as in models of RNA stems in which paired positions change together.

The linguistic mapping. Bouchard-Côté et al. (2013) developed a probabilistic model of sound change in which each branch of a language tree carries a set of context-dependent sound changes, parameterized by phonological features, and used it to reconstruct Proto-Austronesian word forms automatically from hundreds of modern languages. A large majority of the automated reconstructions were very close to those made by expert historical linguists. The model built on earlier probabilistic approaches to diachronic phonology (Bouchard-Côté et al., 2007). Hruschka et al. (2015) modelled regular sound change in Turkic as "concerted evolution": they treated each sound change as a correlated event affecting many words at once and identified the changes from the data.

The Austronesian reconstruction of Bouchard-Côté et al. (2013) is a good illustration of what sequence-level models can do. The input was word forms (not cognate judgements) from the Austronesian Basic Vocabulary Database for several hundred languages, together with a fixed tree. The model specified, for each branch, a probability distribution over context-sensitive edits to each sound, with parameters shared across the lexicon so that the same change applied regularly. Inference alternated between sampling ancestral word forms and re-estimating the sound-change parameters. Beyond reconstructing proto-forms, the authors used the model to test a long-standing hypothesis about sound change: that phonemic contrasts carrying a high "functional load" (dis-

tinguishing many words) are less likely to be lost through merger. They found support for it. The work shows that a probabilistic treatment of sound change can combine the regularity principle of the comparative method with the uncertainty quantification of phylogenetics.

Assessment: sequence-level sound-change models. *Tier B; barely applied, but with strong results. Preserved:* change along branches of a tree, inheritance of forms, uncertainty about ancestral states. *Violated, deliberately:* the independence of sites, which regular sound change contradicts; the models replace it with lexicon-wide parameters, which is their strength. *At risk:* analogy, borrowing and irregular change, which break regularity; and the fixed tree, which hides the dependence of reconstructions on the topology. *What would show it works:* reconstruction accuracy against expert proto-forms that were withheld from model development, and, for tree inference, recovery of documented subgroups from forms alone in families whose subgrouping rests on sound laws rather than on the same cognate sets used for the comparison.

Research programme. The obvious next step is to estimate the tree jointly with the sound changes, rather than conditioning on a fixed tree, and to compare the support for contested subgroups provided by regular sound changes (which the comparative method treats as the strongest evidence of subgrouping) with that provided by lexical cognates. A Bayesian model in which shared sound changes are characters with a strong prior against parallel occurrence, and lexical cognates are characters with weaker priors, would formalize the traditional hierarchy of evidence.

16.3.4 Joint alignment and phylogeny of word forms

How it works. Molecular phylogenetics usually proceeds in two steps: align the sequences, then infer the tree from the alignment, treating the alignment as known. But the alignment is itself an inference, and errors in it propagate to the tree. *Statistical alignment* treats insertions and deletions as part of an explicit evolutionary model, beginning with the TKF91 model of Thorne et al. (1991), in which single-residue insertions and deletions occur as a birth-death process along each branch. BAli-Phy (Redelings and Suchard, 2005; Suchard and Redelings, 2006; Redelings, 2021) uses MCMC to sample alignments, trees and model parameters jointly, so that the posterior distribution of trees integrates over alignment uncertainty. Later models relax the assumption, shared by TKF91, that insertions and deletions occur at the same rate everywhere; Zhai and Bouchard-Côté (2017), for instance, model variation in indel rates within the Poisson indel process.

The linguistic mapping. Word forms are short sequences of sounds, and cognate words can be aligned segment by segment. Latin *octo*, Italian *otto*, Spanish *ocho* and French *huit* ('eight') can be aligned so that corresponding sounds (Latin *ct*, Italian *tt*, Spanish *ch*, French *it*) fall in the same column, and insertions or deletions (such as the loss of final vowels in French) appear as gaps. Phonetic alignment algorithms such as those in LingPy (List, 2012b) produce such alignments by progressive methods analogous to ClustalW. A joint model would sample alignments and trees together, which matters most for families with extensive sound change, where alignment is genuinely uncertain. The analogy is imperfect in an important way. Because sound change is largely *regular*, the "sites" of different words are not independent, and a model in which each word evolves independently by substitutions and indels ignores the most informative feature of linguistic data. Statistical alignment for language therefore needs to be coupled with models of regular sound change. Wheeler and Whiteley (2015) treated Uto-Aztecan word forms as sequences and used direct optimization, a parsimony-based method in which alignment and tree are inferred together, to infer the phylogeny and biogeography of the family; probabilistic joint alignment in the BAli-Phy sense has, to our knowledge, rarely if ever been applied to word forms. Two recent

studies come close from different sides. Goldstein et al. (2026) adapted the TKF91 model itself to aligned cognate word forms, so that the tree is inferred from segmental strings rather than from cognate codes, although the alignments are fixed in advance. And Jäger (2025) derived phylogenetic characters from multiple sequence alignments of word forms produced with a pair-hidden Markov model, without expert cognate judgements; these characters were competitive with expert cognates within families and outperformed both expert and automatically detected cognates on datasets spanning many families.

Assessment: joint alignment and phylogeny. *Tier B; barely applied. Preserved:* sequences related by descent with substitutions and indels; uncertainty in homology. *Violated:* independence among words (regular correspondences), a substitution process over an alphabet of sounds that is the same everywhere in the tree, and the assumption that every compared form is homologous, which cognate judgement must first establish. *What would show it works:* posterior alignments that match expert alignments in a well-studied family, and trees from forms that agree with trees from cognate data on the parts where both are well supported, without the agreement being forced by sharing the same cognate sets.

Research programme. Encode words as sequences of sound classes (as in LingPy's SCA classes) and run BAli-Phy on cognate sets from a well-studied family with a custom substitution matrix over sound classes, with a shared tree across cognate sets. Compare the posterior alignments with expert alignments and the posterior trees with trees from cognate-coded data. The main aim is not to replace cognate-based phylogenies but to quantify how much phylogenetic information resides in the form of cognates beyond their presence or absence.

16.4 Question 3: How do borrowing and language shift modify genealogies?

Chapter 8 treats borrowing as a problem for tree inference and surveys the models linguists have built for it. Biology suggests a change of perspective. In the token model, a loan is a copying link that crosses from one speech community to another, so borrowing changes the genealogy of a concept, not the language tree. Language shift, in which a population adopts another language, changes the relation between the genealogy of languages and the genealogy of their speakers. The methods below differ in the unit whose genealogy they follow: a concept (the D statistic and network coalescent models), a family of related words (reconciliation), a language's whole profile of features (clustering), or the pairing of populations with languages (cophylogeny).

16.4.1 Borrowing versus sorting: distinguishing the footprints

Discordant concept trees can arise from sorting, from borrowing, or from both. Biology faces exactly the same problem with ILS and hybridization or introgression, and the solution exploits the symmetry noted after Equation (16.2). Under ILS alone, the two discordant topologies for any rooted triplet are equally frequent. Borrowing (or introgression) from one particular lineage into another makes one discordant topology more frequent than the other. This is the logic of the ABBA-BABA or D statistic used to detect Neanderthal admixture in modern humans (Green et al., 2010; Durand et al., 2011; Patterson et al., 2012).

To see this in a linguistic setting, we take a five-language tree $((A, B), (C, (D, E)))$ and let a proportion b of meaning slots undergo borrowing: the word of language B is replaced by a loan from

language C at a recent time. In the concept tree, this is represented by pruning B and regrafting it onto the lineage of C at the time of contact. We then count, for the triplet $\{A, B, C\}$, how often each of its three rooted topologies appears. Because the concept trees are generated by the coalescent, this simulation shows what the test can detect when the coalescent holds (parameter recovery, in the vocabulary of Section 6.10); Table 16.1 and the token simulation above indicate which departures would leave its symmetry intact.

```
lang5 <- read.tree(text = "((A:1.5,B:1.5):0.6,(C:1.2,(D:0.6,E:0.6):0.6):0.9);")

borrow_word <- function(g, recipient = "B", donor = "C", t_contact = 0.3) {
  g2 <- drop.tip(g, recipient)
  bind.tip(g2, recipient, where = which(g2$tip.label == donor),
           position = t_contact, edge.length = t_contact)
}
triplet_topology <- function(g) {
  tr <- keep.tip(g, c("A", "B", "C"))
  if (is.monophyletic(tr, c("A", "B"))) "AB"
  else if (is.monophyletic(tr, c("B", "C"))) "BC" else "AC"
}
b_grid <- c(0, 0.1, 0.2, 0.3, 0.4, 0.5)
n_concepts <- 300
trip <- t(sapply(b_grid, function(b) {
  topo <- replicate(n_concepts, {
    g <- sim_msc_tree(lang5)
    if (runif(1) < b) g <- borrow_word(g)
    triplet_topology(g)
  })
  table(factor(topo, levels = c("AB", "AC", "BC"))) / n_concepts
}))
T_ABC <- 0.6   # internal branch of the triplet in coalescent units
matplot(b_grid, trip, type = "b", pch = c(19, 17, 15), lty = 1,
        col = c("black", "grey50", "steelblue"), ylim = c(0, 1), las = 1,
        xlab = "Proportion of meaning slots borrowed from C into B (b)",
        ylab = "Frequency among concept trees")
abline(h = c(1 - 2/3 * exp(-T_ABC), 1/3 * exp(-T_ABC)), lty = 3)
legend("topright", c("((A,B),C)  true", "((A,C),B)", "((B,C),A)"),
       pch = c(19, 17, 15), col = c("black", "grey50", "steelblue"),
       lty = 1, bty = "n")
```

```
round(cbind(b = b_grid, trip), 3)
```

```
#>        b    AB    AC    BC
#> [1,] 0.0 0.670 0.160 0.170
#> [2,] 0.1 0.587 0.137 0.277
#> [3,] 0.2 0.470 0.163 0.367
#> [4,] 0.3 0.417 0.143 0.440
#> [5,] 0.4 0.333 0.110 0.557
#> [6,] 0.5 0.337 0.090 0.573
```

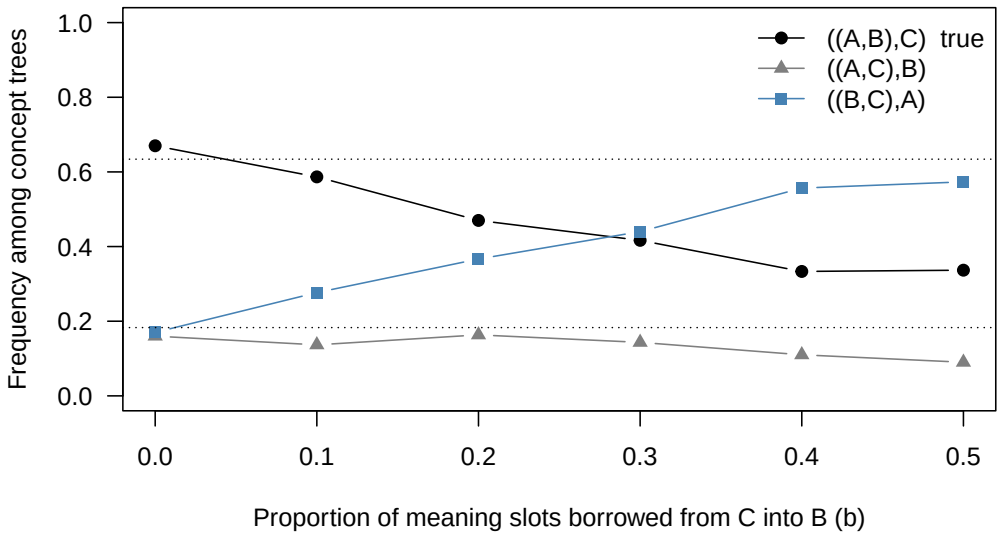


Figure 16.2: Frequencies of the three rooted topologies for the triplet A, B, C among 300 simulated concept trees, as a function of the proportion b of meaning slots in which B borrowed its word from C. With b = 0 the two discordant topologies, (A,C) and (B,C), are equally frequent, as expected under incomplete lineage sorting alone; borrowing breaks the symmetry. Dotted lines show the analytical expectations under sorting alone.

Figure 16.2 shows that without borrowing, about a third of concept trees are discordant for this triplet, and the two discordant topologies are about equally common, as Equation (16.2) predicts. As borrowing increases, the (B, C) topology becomes more frequent than (A, C), and by $b \approx 0.4$ it overtakes the true topology. The asymmetry between the two minor topologies, not the overall amount of discordance, is the signature of contact.

Concept trees are not observed, but the same logic applies to cognate patterns. For four languages with an established tree $(((P_1, P_2), P_3), O)$, call a meaning an ABBA pattern if P_2 and P_3 share a cognate class that P_1 and the outgroup O do not have, with P_1 and O sharing a different class, and a BABA pattern if the roles of P_1 and P_2 are reversed. The lexical D statistic is

$$D = \frac{n_{\mathrm{ABBA}} - n_{\mathrm{BABA}}}{n_{\mathrm{ABBA}} + n_{\mathrm{BABA}}}, \tag{16.3}$$

which has expectation zero under sorting alone and is positive if P_3 lent words to P_2 (or borrowed from it). Its standard error should be estimated by a block jackknife over meanings, or over semantic fields if meanings are linked by polysemy or by prestige effects of the kind shown in Figure 16.1. The derivation of the null expectation, which needs only that P_1 and P_2 evolve under the same process after their split, is in Section E.2.3.

Assessment: the lexical D statistic. *Tier B; not applied. Preserved:* the symmetry of discordant patterns under sorting, which requires exchangeability of the sister languages but not neutral copying or Kingman genealogies (Table 16.1). *Violated or at risk:* structure in the ancestral speech community (if the ancestors of P_2 and P_3 were neighbouring dialects, $D \neq 0$ without later borrowing, as ancestral structure can mimic admixture in genetics); unequal sampling or coding of P_1 and P_2 (a better-documented language has more synonyms recorded and so more chances to share); parallel semantic shift; and an uncertain reference tree. *What would show it works:* power and false-positive rates under the token model with conformity, prestige and ancestral structure; and, empirically, detection of documented contact (Norse and English, French and English, Tongan and East Uvean) together with non-detection in quartets where no contact is known.

Research programme. Compute lexical D statistics from existing CLDF datasets for quartets in which contact is documented, and for control quartets where it is not, with a block jackknife over meanings for standard errors. The comparison gives an empirical false-positive rate. Then simulate cognate data under the token model with ancestral dialect structure to find out how large a D value ancestral structure alone can produce, which sets a threshold for interpreting real values.

16.4.2 Explicit phylogenetic networks

How it works. Split networks such as NeighborNet (Bryant and Moulton, 2004) display conflicting signal, but they do not represent an explicit history of reticulation events (Huson and Bryant, 2006). *Explicit* phylogenetic networks contain reticulation nodes with two parents, representing hybridization or introgression. PhyloNet (Than et al., 2008; Wen et al., 2018) infers networks under a multispecies network coalescent, which combines ILS with reticulation. SNaQ (Solís-Lemus and Ané, 2016) estimates networks from gene-tree quartet concordance factors by pseudolikelihood, scaling to many genes. SpeciesNetwork (Zhang et al., 2018a) performs full Bayesian inference of species networks in BEAST 2. The D statistic and f-statistics (Durand et al., 2011; Patterson et al., 2012) test specific admixture hypotheses from site-pattern counts.

The linguistic mapping. Chapter 8 discusses split networks, minimal lateral networks, models of lateral transfer (Kelly and Nicholls, 2017), contact-aware tree models (Neureiter et al., 2022) and perfect phylogenetic networks (Nakhleh et al., 2005a), all applied to language data. What biology adds is the integration of reticulation with sorting: a network coalescent model would distinguish discordance caused by sorting of ancestral variation from discordance caused by contact, as in the simulation above. It would also allow a reticulation node to carry an inheritance probability, the proportion of the lexicon contributed by each parent, which is a natural parameter for mixed languages and for heavily restructured contact languages. Once such a network is available, comparative methods can be run on it: Bastide et al. (2018) extended Brownian-motion models of continuous trait evolution, and the phylogenetic covariance they imply, from trees to networks with reticulations, so that a trait inherited partly from each parent of a contact event can be modelled directly.

Assessment: network coalescent models. *Tier A for split networks and linguistically motivated network models (applied); tier B for network-coalescent methods such as PhyloNet, SNaQ and SpeciesNetwork (barely or not applied). Preserved:* lineages that split and occasionally merge; a coalescent within lineages. *Violated:* borrowing in language is usually piecemeal and continuous over centuries rather than a single hybridization event, so a reticulation node with one inheritance probability is a coarse summary; the coalescent assumptions of Section 16.2.2; and the quartet concordance factors SNaQ uses are estimated from gene trees that cognate data do not resolve (point 2 in that section). *What would show it works:* recovery of documented contact episodes as reticulations, with inheritance probabilities close to the proportion of loans counted in etymological dictionaries, and no reticulations in subgroups without known contact.

Research programme. Fit SNaQ to quartet concordance factors computed directly from cognate data for a family with known episodes of intense contact, such as the Germanic languages in contact with Norse and French, or a Sinitic dataset, and ask whether the method recovers the documented contact episodes as reticulations with plausible inheritance probabilities. In parallel, compute lexical D statistics for the same quartets as a simpler test.

16.4.3 Reconciliation: duplication, transfer and loss

How it works. Gene families evolve within species by four kinds of events: speciation (the gene copy splits when the species does), duplication (a new copy arises within a genome), horizontal transfer (a copy jumps from one lineage to another) and loss. *Reconciliation* methods take a gene tree and a species tree and find the mapping of gene-tree nodes onto species-tree branches that explains their differences with the fewest or most probable events. Parsimony-based reconciliation assigns costs to duplications, transfers and losses (DTL) and finds the minimum-cost scenario, efficiently for most cost schemes (Tofigh et al., 2011; Bansal et al., 2012). Probabilistic methods such as ALE (Szöllősi et al., 2013a,b) model DTL as a birth-death-like process along the species tree and integrate over uncertainty in the gene tree by using a sample of gene trees, estimating per-family rates of duplication, transfer and loss. A distinctive feature of transfer inference is that it can use timing constraints: a transfer can only occur between lineages that coexisted, so the species tree must be dated or at least partially ordered, and transfers to and from extinct or unsampled lineages must be modelled (Szöllősi et al., 2013b).

The linguistic mapping. The parallel between borrowing and horizontal gene transfer is part of the standard analogy (List et al., 2014a). Reconciliation sharpens it. Consider an etymological family: all the words descended from one Proto-Indo-European root across the family, possibly

with different meanings and derivational forms. Its history is a "gene tree" of word forms. Speciation events correspond to language splits, losses to the disappearance of the word, duplications to the creation of a second word from the same root within one language (for instance, by derivation or by the coexistence of an inherited form and a learned form), and transfers to borrowing. English provides a clean example: *shirt* is the inherited Old English reflex of a Germanic word, while *skirt* is a loan from Old Norse of the cognate Norse word. In gene-family terms, English contains a native copy (an *ortholog* of German *Schürze*, 'apron', in a broad sense) and a transferred copy (a *xenolog*). Reconciliation of the root-level tree with the language tree would identify the Norse-to-English transfer and date it to the period when the two lineages were in contact.

Related linguistic work exists. Minimal lateral networks (Nelson-Sathi et al., 2011) infer borrowing by finding the smallest number of lateral links needed to explain patchy cognate distributions on a tree, which is close in spirit to parsimony DTL reconciliation for presence/absence data. Partial cognacy and the study of word families (List, 2016) provide the gene-family-like structures that reconciliation needs. Information-theoretic methods for inferring directional lexical flow have also been developed (Dellert, 2019).

Nelson-Sathi et al. (2011) analyzed the Indo-European data of Dyen et al. (1992) with a method developed for detecting lateral gene transfer in prokaryote genomes. For each cognate set, they asked how many separate origins on a reference tree would be needed to explain its distribution if borrowing were not allowed, and then sought the minimal network of lateral links that would make the observed distributions of cognates consistent with realistic numbers of origins. The analysis suggested that a non-trivial fraction of cognate sets in a curated basic-vocabulary list may involve undetected borrowing. Like any parsimony-based method it depends on the reference tree and on the assumption that patchy distributions call for lateral links rather than multiple losses, and its estimates should be read in that light. It remains one of the clearest examples of a biological method designed for horizontal transfer being carried over to language data with little modification.

Assessment: DTL reconciliation. *Tier B; not applied in the formal sense. Preserved:* a word family evolving inside a language tree by inheritance, loss and transfer, with transfers constrained to contemporaneous lineages. *Violated:* "duplication" in language (derivation, doublets, learned borrowings from an ancestral literary language such as Latin into French) is a mixture of processes with different dynamics; word-family trees must be estimated from short forms with regular sound change, so they are much less certain than gene trees; and semantic change moves words between meaning slots, which reconciliation does not model. *What would show it works:* precision and recall of inferred transfers against loans documented in etymological dictionaries, in a family whose borrowing history is well known.

Research programme. Build etymological "gene trees" for a few hundred well-studied roots in a family with good etymological dictionaries (Germanic, Romance, Indo-Iranian or Turkic), using aligned reflexes and a simple sequence or character model to obtain samples of word trees. Reconcile them with a dated language tree using a probabilistic DTL method. The test of the approach is validation: compare the inferred transfers with loans documented in etymological dictionaries, estimating precision and recall. If performance is good, apply the method to families where borrowing histories are poorly documented. A conceptual payoff would be a principled distinction between inherited, derived (duplicated) and borrowed members of word families.

16.4.4 Population-genetic clustering: STRUCTURE and its relatives

How it works. STRUCTURE (Pritchard et al., 2000) and ADMIXTURE (Alexander et al., 2009) model each individual's genotype as drawn from a mixture of K ancestral populations, each with its own allele frequencies, and estimate both the population allele frequencies and each individual's mixture proportions. They do not assume a tree: they ask how variation is partitioned among clusters and which individuals are admixed.

The linguistic mapping. Treat each language as an individual and each linguistic feature (a typological feature or a cognate class) as a locus with alleles. STRUCTURE then partitions the languages into K clusters of feature profiles and estimates how much of each language's profile derives from each cluster. Unlike a tree, this can represent languages whose features are drawn from more than one source, the signature of contact or of shared areal influence.

Reesink et al. (2009) applied STRUCTURE to structural (typological) features of a large sample of languages of Sahul, the Pleistocene continent comprising Australia and New Guinea, together with neighbouring Island Melanesia. Because the deep relationships among the many language families of this region cannot be established by the comparative method, they asked instead whether the distribution of structural features could be explained by a small number of ancestral "populations" of features. The analysis identified clusters that partly matched known families and partly cut across them, with admixture-like profiles suggesting areal diffusion of features. The study is the clearest example of a population-genetic clustering method transferred to language data with an explicit argument about why a tree model would be inappropriate for the question. Similar clustering approaches have since been used on lexical data in other settings, including Tasmanian (Bowern, 2012) and Finnish dialects (Syrjänen et al., 2016). Cathcart (2020) applied the same mixed-membership logic to sound change in Indo-Aryan, a family often described as a set of overlapping dialect zones rather than a tree. Sound changes between Old Indo-Aryan and the modern languages were extracted automatically from a large etymological dataset, and each language was modelled as a mixture of dialect components with their own distributions over sound-change outcomes. The analysis found evidence for cohesive dialect groups, and under one of the two priors compared (a logistic-normal prior that lets similar sound changes share information) the components were distributed largely in the core–periphery pattern of the old Inner–Outer hypothesis. The result depends on the prior, and the etymologies themselves are products of the comparative method, so agreement with a hypothesis framed in the same tradition is not independent confirmation of it.

Assessment: STRUCTURE-type clustering. *Tier A; applied in a small number of studies. Preserved:* mixture of sources in a profile, without a tree. *Violated:* the genetic model assumes Hardy-Weinberg and linkage equilibrium within clusters, which have no linguistic meaning, and typological features are strongly correlated with each other, so the clusters must be read as a descriptive decomposition, not as ancestral populations. *What would show it works:* clusters that are stable across K and across feature subsets, and that recover documented linguistic areas and known mixed languages better than ordinary clustering of the same data.

Research programme. Use spatially explicit clustering methods, which add a geographic prior to the cluster memberships, on Grambank data for regions of intense contact (mainland Southeast Asia, the Caucasus, Amazonia), and compare the inferred clusters with linguistic areas proposed in the literature. Combine the results with phylogenetic controls so that clusters reflecting shared descent are distinguished from clusters reflecting contact.

16.4.5 Cophylogeny of languages and genes

How it works. Hosts and their parasites, or plants and their pollinators, often have phylogenies that partly mirror each other. *Cophylogenetic* methods ask how closely two trees correspond and what events explain the mismatches. Event-based methods, starting with TreeMap (Page, 1994) and continuing with Jane (Conow et al., 2010), reconcile a "parasite" tree with a "host" tree by mapping it with four kinds of events: cospeciation, duplication (the parasite splits within one host), host switching, and loss, each with a cost. Global-fit methods ask whether the two trees are more congruent than expected by chance: ParaFit (Legendre et al., 2002) tests the association between the patristic distance matrices of the two trees given the links between their tips, and PACo (Balbuena et al., 2013) does so by Procrustes superimposition of their principal coordinates.

The linguistic mapping. The comparison of language trees with population genetic trees has a long history (Cavalli-Sforza et al., 1988). In cophylogenetic terms, the population tree is the host and the language tree is the parasite. Cospeciation corresponds to a population and its language splitting together; host switching corresponds to *language shift*, in which a population adopts the language of another; duplication corresponds to a language split without a corresponding genetic split; and loss corresponds to the disappearance of a lineage's language without trace. Studies comparing genetic and linguistic relationships, such as that of Hunley et al. (2008) for Northern Island Melanesia and the global analysis of Barbieri et al. (2022), have documented both concordance and frequent mismatches.

We simulate a language tree for 12 populations, derive a genetic tree that mostly follows it but differs by two random local rearrangements, and then introduce two *language shifts*: populations `PopB` and `PopK` each adopt the other's language. ParaFit tests the global congruence of the two trees, and its per-link statistics can flag the associations that contribute least to it. Because the shifts are built into the simulation, this is a parameter-recovery exercise for the test, not evidence that it works on real populations.

```
set.seed(1605)
lang_tree <- pbtree(n = 12, scale = 1,
                    tip.label = paste0("Lang", LETTERS[1:12]))
gene_tree <- lang_tree
gene_tree$tip.label <- sub("Lang", "Pop", gene_tree$tip.label)
gene_tree <- rNNI(gene_tree, moves = 2)      # genetic noise and admixture

links <- cbind(gene_tree$tip.label, sub("Pop", "Lang", gene_tree$tip.label))
links[links[, 1] == "PopB", 2] <- "LangK"    # language shifts
links[links[, 1] == "PopK", 2] <- "LangB"

A <- unclass(table(factor(links[, 1], levels = gene_tree$tip.label),
                   factor(links[, 2], levels = lang_tree$tip.label)))
pf <- parafit(cophenetic(gene_tree), cophenetic(lang_tree), A, nperm = 199,
              test.links = TRUE, correction = "cailliez", silent = TRUE)

invisible(capture.output(
  cp <- cophylo(gene_tree, lang_tree, assoc = links, rotate = TRUE)))
plot(cp, link.lty = "solid", link.col = adjustcolor("steelblue", 0.7),
     fsize = 0.8)
```

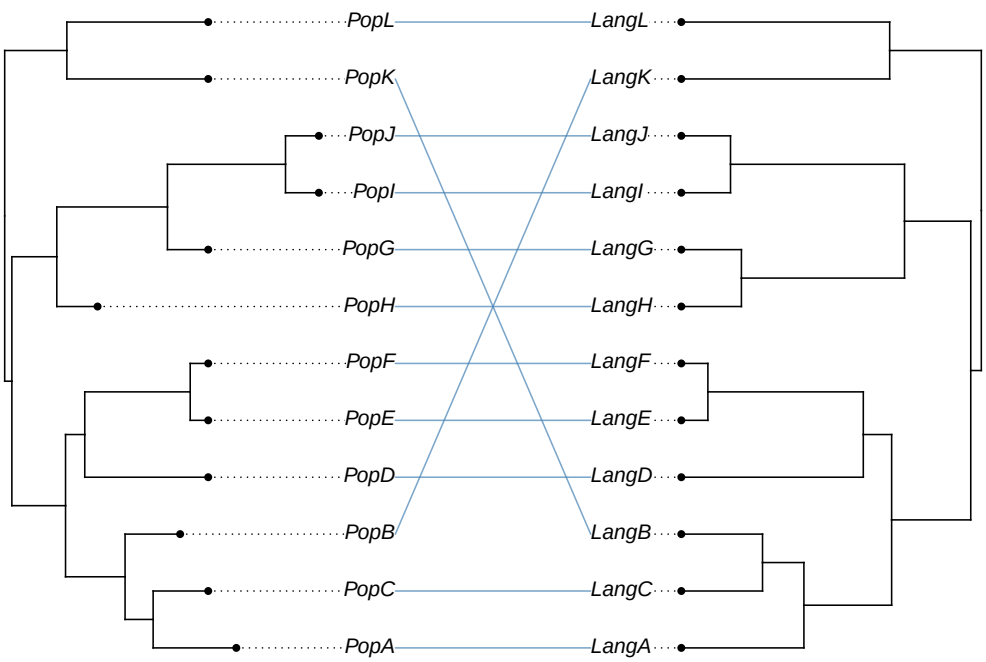


Figure 16.3: Cophylogeny of a simulated population (genetic) tree, left, and a language tree, right. Lines link each population to the language it speaks. Two populations, PopB and PopK, have shifted language, producing crossing links.

```
link_tab <- data.frame(population = rownames(A)[pf$link.table[, "Host"]],
                       language = colnames(A)[pf$link.table[, "Parasite"]],
                       F1 = round(pf$link.table[, "F1.stat"], 2),
                       p = pf$link.table[, "p.F1"])
list(global_p = pf$p.global,
     weakest_links = head(link_tab[order(link_tab$F1), ], 4))
```

```
#> $global_p
#> [1] 0.005
#>
#> $weakest_links
#>    population language   F1     p
#> 2        PopB    LangK 0.62 0.785
#> 11       PopK    LangB 1.94 0.465
#> 8        PopH    LangH 2.95 0.385
#> 12       PopL    LangL 4.93 0.295
```

The global test indicates that the trees are more congruent than expected by chance (with 199 permutations the smallest attainable p-value is 0.005). The per-link statistics then single out the links that contribute least to the congruence, and in this simulation the two weakest links are the two simulated language shifts, `PopB`-`LangK` and `PopK`-`LangB`. With only 12 tips the individual link tests have little power, none of them is significant on its own, and in real data a link may contribute weakly because of admixture or sampling rather than language shift.

Assessment: cophylogeny. *Tier B; barely applied. Preserved:* two genealogies linked at their tips,

with events that explain mismatches. *Violated:* human populations exchange genes continuously, so a population "tree" is a poor summary of genetic history; language shift is often partial, so the links between populations and languages are not one-to-one; and the two trees are estimated with very different uncertainties, which event-based methods ignore unless they integrate over samples of both. *What would show it works:* recovery of language shifts documented historically, with event costs calibrated by simulation, and robustness of the inferred shifts across posterior samples of both trees.

Research programme. Apply event-based reconciliation (with costs chosen to reflect the relative plausibility of co-divergence, language shift and loss) to paired genetic and linguistic trees for regions with good data on both, such as Island Southeast Asia and the Pacific, Africa south of the equator, or the Caucasus. Calibrate the event costs with simulation, and validate them against historically documented language shifts. Run the reconciliation over posterior samples of both trees rather than on two summary trees. The output would be explicit hypotheses of language shift, each tied to a pair of branches and a time window. Visual tanglegrams and Mantel tests between genetic and linguistic distances are common (Chapter 14), but formal event-based reconciliation of language and population trees is, to our knowledge, rare.

16.5 Question 4: Shared or lineage-specific mechanisms of feature evolution?

Do grammatical features evolve under the same rules in every family, or does each lineage have its own dynamics? The question is behind the debate over word-order universals (Section 13.8): Dunn et al. (2011) found family-specific dependencies, while Jäger and Wahle (2021), with a larger sample, found that a model with the same dynamics in every lineage fitted better. Biology offers two ways of making the question sharper. One reduces many correlated features to a few latent dimensions and asks whether those dimensions, and their rates, are shared. The other treats discrete features as thresholds on continuous latent variables, so that "the same mechanism" becomes "the same correlation between liabilities". Both turn a question about many transition rates into a question about a few interpretable parameters, which can then be allowed to vary among families in a hierarchical model and compared with a model in which they are shared.

16.5.1 Phylogenetic factor analysis and multivariate models

How it works. Many traits are measured together, and their covariation partly reflects shared ancestry. Ordinary principal component analysis treats species as independent and can produce components that mainly reflect phylogenetic structure (Revell, 2009). *Phylogenetic factor analysis* (PFA) models a matrix of traits $\mathbf{Y}$ (species by traits) as

$$\mathbf{Y} = \mathbf{FL} + \mathbf{E}, \tag{16.4}$$

where the latent factors $\mathbf{F}$ evolve along the tree by Brownian motion, $\mathbf{L}$ is a loading matrix linking factors to observed traits, and $\mathbf{E}$ is residual variation (Tolkoff et al., 2018). Extensions handle discrete traits through latent liabilities and large numbers of traits efficiently (Hassler et al., 2022).

The linguistic mapping. Typological databases such as Grambank and WALS record hundreds of structural features per language. Many are correlated: head-marking and dependent-marking features, word-order features, and features related to morphological complexity cluster together. Grambank analyses have combined phylogenetic and spatial controls when studying individual

features and summary scores (Skirgård et al., 2023), and the question of whether typological features form a few latent dimensions has a long history. PFA would recover such dimensions while accounting for phylogeny, and it would also estimate how fast each latent dimension evolves. Fitting it separately to several families and comparing the loadings is a direct test of whether the dimensions, and hence the mechanisms that link features, are shared.

Assessment: phylogenetic factor analysis. *Tier B; not applied. Preserved:* correlated traits evolving along a tree, with the correlation carried by a few latent factors. *Violated:* binary features are linked to the factors through liabilities whose Brownian motion is an assumption, not an observation; contact makes neighbouring languages similar in ways the tree does not explain, so a spatial term is needed; and Grambank features are coded with logical dependencies (a feature that only applies if another is present), which create correlations that are artefacts of coding. *What would show it works:* recovery of known factor structure in simulated binary data with realistic missingness and contact; stability of the loadings across families and feature subsets; and factors that predict held-out features better than a model without them.

Research programme. Apply PFA, through its liability extension for binary features, to Grambank features for a family with a well-supported dated tree (Austronesian or Bantu), and ask whether the latent factors correspond to classical typological dimensions (for example, head direction or synthesis). Repeat the analysis on a second family, compare the loadings, and compare the rate of evolution of each factor with lexical rates estimated on the same tree. Fit the model over a posterior sample of trees, not a single summary tree.

16.5.2 Threshold models

How it works. The *threshold model* of quantitative genetics, brought into phylogenetics by Felsenstein (2005) and Felsenstein (2012), treats a discrete character as the visible expression of a continuous, unobserved *liability* that evolves by Brownian motion along the tree. When the liability crosses a threshold, the discrete state changes. Correlated evolution between two discrete characters is modelled as correlation between their liabilities; discrete and continuous characters can be analyzed together; and ordered multistate characters correspond to several thresholds. Ancestral states can be reconstructed under the model (Revell, 2014), and Bayesian implementations are available in phytools (Revell, 2012).

The linguistic mapping. Many typological features are coarse categorical summaries of underlying gradient properties. Basic word order is coded as a single category, but corpora show that languages differ in how rigid or flexible their orders are; a "case system" feature may summarize a gradient degree of morphological marking. The threshold model fits this situation better than a Markov model in which categories switch abruptly. It also offers an alternative to the discrete correlated-evolution test of Pagel (1994) used by Dunn et al. (2011): two word-order features would be correlated if their liabilities are correlated, with a single correlation coefficient that is easier to interpret, and easier to compare across families, than a matrix of transition rates. Chapter 13 (Section 13.9.3) shows a fit in R.

Assessment: threshold models. *Tier B; not applied to linguistic questions beyond illustration. Preserved:* discrete states as coarse readings of a continuous variable. *Violated:* the liability is unobserved, so its Brownian motion is an assumption that cannot be checked directly; for features without an obvious gradient basis (presence of a grammatical category) the liability has no linguistic interpretation; and threshold models predict that change between distant categories passes through the intermediate ones, which may not hold. *What would show it works:* where corpus

measures of the gradient property exist (for instance word-order entropy), the inferred liabilities should correlate with them, and the model should predict held-out categorical codings at least as well as Markov models.

Research programme. Re-analyze the word-order features of Dunn et al. (2011) under bivariate threshold models and compare lineage-specific correlations with the results of the original discrete analysis and its critiques (Chapter 13). A hierarchical version, with family-specific correlations drawn from a shared distribution, would turn the debate over universals into an estimate of how much the correlations vary among families. A second project would use corpus-based measures of word-order entropy where available as continuous data alongside categorical codings, linking the two through the liability.

16.6 Question 5: How do languages diversify?

Why do some families have thousands of languages and others a handful, and did families split faster during expansions than afterwards? These are questions about the process that generated the tree itself, which in Bayesian dating is represented only by the tree prior. Biology has a large toolkit for them, together with hard-won knowledge of how easily it misleads. A second question is how diversification proceeded: by clean splits, or by the gradual break-up of a dialect network. Answering either requires the tree to be well sampled, which motivates the last method of the section, the placement of poorly documented languages.

16.6.1 Diversification models

How it works. A dated phylogeny of living species carries information about the history of speciation and extinction that produced it. The simplest model is the *birth-death process*: each lineage splits at rate λ and goes extinct at rate μ, independently of all other lineages. The *net diversification rate* is $r = \lambda - \mu$ and the *turnover* (or relative extinction) is $\epsilon = \mu/\lambda$. Starting from n_0 lineages, the expected number of lineages after time t grows exponentially,

$$E[N(t)] = n_0\, e^{(\lambda-\mu)t}. \tag{16.5}$$

The phylogeny we observe is the *reconstructed* tree, which contains only lineages with living descendants. Its branching times follow a known distribution that depends on both λ and μ, not only on their difference (Nee et al., 1994; Nee, 2006). This is what makes it possible, in principle, to estimate extinction from a tree of living species: lineages that arose recently have had less time to go extinct, so with $\mu > 0$ the reconstructed tree shows an excess of recent branching, the "pull of the present". A *lineages-through-time* (LTT) plot, the number of lineages in the reconstructed tree against time on a logarithmic scale, makes this visible. Under a pure-birth (Yule) process the expected LTT plot is a straight line; extinction bends it upward near the present; a slowdown of diversification, as might occur when a radiation fills available space, bends it downward. The γ statistic of Pybus and Harvey (2000) summarizes the shape: negative values indicate that internal nodes are concentrated toward the root (early bursts or slowdowns), positive values that they are concentrated toward the tips.

Many extensions exist (Morlon, 2014). Time-varying models let λ and μ change through time, for example in piecewise-constant intervals, as in the *birth-death skyline* (Stadler et al., 2013). Lineage-varying models let rates differ among clades: BAMM (Rabosky, 2014) identifies discrete

rate shifts on the tree, while ClaDS (Maliet et al., 2019) lets every lineage inherit a slightly perturbed rate from its parent. *State-dependent speciation and extinction* (SSE) models tie rates to a character: in BiSSE (Maddison et al., 2007), species in state 0 or 1 have their own λ and μ, and transitions between states occur at rates q_{01} and q_{10}. MuSSE generalizes to several states, GeoSSE to geographic ranges (Goldberg et al., 2011), and HiSSE (Beaulieu and O'Meara, 2016) adds hidden states to distinguish genuine trait-dependent diversification from rate variation that merely happens to correlate with the trait. Most of these models are implemented in R packages such as diversitree (FitzJohn, 2012).

Three cautions from the biological literature are essential. First, SSE models have high false-positive rates when applied naively: with a single large clade that happens to have both a particular trait and a high diversification rate, BiSSE will often find trait-dependent diversification even if the trait is irrelevant (Rabosky and Goldberg, 2015). HiSSE and character-independent null models were developed in response. Second, BAMM's treatment of rate shifts and its priors have been criticized (Moore et al., 2016), and its results should be interpreted cautiously. Third, and most fundamentally, Louca and Pennell (2020) showed that for any tree of extant species, there are infinitely many time-varying birth-death histories that yield exactly the same likelihood. Only certain combinations of parameters (such as the "pulled" diversification rate) are identifiable from extant trees alone. Inferences of extinction rates and of the timing of rate changes from extant trees must therefore rest on strong assumptions, unless extinct lineages are sampled. Incomplete sampling is another major issue. If only some extant species are in the tree, the estimates must account for the sampling fraction (Stadler, 2009). Random sampling of tips tends to remove recent nodes disproportionately, which mimics a slowdown in diversification.

The linguistic mapping. The mapping is direct: a language family tree is a reconstructed tree, language splits are speciation events, and language death is extinction. Language extinction is, moreover, extremely common, both historically (most languages of the past left no descendants) and today. Diversification models could address questions such as these: Did families diversify faster during agricultural expansions than afterwards? Do island colonizations drive bursts of splitting? Are there clades with exceptionally high or low rates of language formation? Does a trait, such as a subsistence mode, a writing tradition, or a mountainous environment, affect the rates at which languages split or die? The Bayesian tree priors used in linguistic dating are themselves birth-death models (Chapter 6), so every BEAST analysis with a birth-death prior implicitly estimates λ and μ. These estimates are rarely reported or interpreted, and for good reason: they are sensitive to the prior, to calibrations, and to the sampling fraction. Atkinson et al. (2008) related the number of nodes on root-to-tip paths to the amount of lexical change, which is a diversification-related question but concerns rates of change rather than rates of splitting. Work on the "pulses and pauses" of Austronesian settlement (Gray et al., 2009) is also, in effect, a claim about time-varying diversification in a geographic setting.

We simulate a family of 80 languages under a birth-death process with $\lambda = 0.6$ and $\mu = 0.3$ splits and deaths per lineage per millennium, keeping only languages that survive to the present. We then fit a birth-death model by maximum likelihood with `ape::birthdeath` and a pure-birth model with `ape::yule`. This is parameter recovery under the model's own assumptions, with a single replicate.

```
set.seed(1603)
family <- pbtree(b = 0.6, d = 0.3, n = 80, extant.only = TRUE)
root_age <- max(nodeHeights(family))
```

```
bd_fit <- birthdeath(family)
yule_fit <- yule(family)
eps_hat <- unname(bd_fit$para["d/b"])
r_hat <- unname(bd_fit$para["b-d"])
lambda_hat <- r_hat / (1 - eps_hat)

data.frame(
  quantity = c("root age (millennia)", "net diversification r",
               "turnover mu/lambda", "lambda", "mu", "Yule lambda",
               "AIC birth-death", "AIC Yule"),
  estimate = c(root_age, r_hat, eps_hat, lambda_hat, lambda_hat * eps_hat,
               yule_fit$lambda, bd_fit$dev + 2 * 2,
               -2 * yule_fit$loglik + 2),
  truth = c(NA, 0.3, 0.5, 0.6, 0.3, NA, NA, NA)
)
```

```
#>                quantity  estimate truth
#> 1  root age (millennia)    9.9396    NA
#> 2 net diversification r    0.2678   0.3
#> 3    turnover mu/lambda    0.6730   0.5
#> 4                lambda    0.8192   0.6
#> 5                    mu    0.5513   0.3
#> 6           Yule lambda    0.4983    NA
#> 7       AIC birth-death -277.0538    NA
#> 8             AIC Yule -271.9175    NA
```

The estimates are of the right order, but the individual rates are imprecise even with 80 languages and a correctly specified model; in this replicate both rates, and hence the turnover, are overestimated, and printing `bd_fit` shows wide profile-likelihood confidence intervals. This is typical. Net diversification is estimated much more precisely than extinction: in the vocabulary of Section 6.10, the extinction rate is identifiable under the model but barely estimable in practice from a tree of this size.

```
sub_family <- keep.tip(family, sample(family$tip.label, 25))
op <- par(mfrow = c(1, 2), mar = c(4.2, 4.2, 1, 1))
lt_full <- ltt(family, plot = FALSE)
lt_sub <- ltt(sub_family, plot = FALSE)
plot(lt_full$times - max(lt_full$times), lt_full$ltt, type = "s", log = "y",
     xlab = "Time before present (millennia)", ylab = "Lineages (log scale)",
     las = 1)
lines(lt_sub$times - max(lt_sub$times), lt_sub$ltt, type = "s", lty = 2)
# log-linear growth from 2 lineages at the root to 80 at the present
lines(c(-root_age, 0), c(2, Ntip(family)), lty = 3)
legend("topleft", c("80 languages", "25 sampled"), lty = 1:2, bty = "n")

g_full <- replicate(100, gammaStat(pbtree(n = 80)))
g_sub <- replicate(100, {
  tr <- pbtree(n = 80)
```

```
  gammaStat(keep.tip(tr, sample(tr$tip.label, 25)))
})
br <- seq(floor(min(g_full, g_sub)), ceiling(max(g_full, g_sub)), by = 0.5)
hist(g_full, breaks = br, col = adjustcolor("grey50", 0.5), border = "white",
     main = "", xlab = expression(gamma ~ "statistic"), ylim = c(0, 35),
     las = 1)
hist(g_sub, breaks = br, col = adjustcolor("steelblue", 0.5),
     border = "white", add = TRUE)
abline(v = -1.645, lty = 2)
```

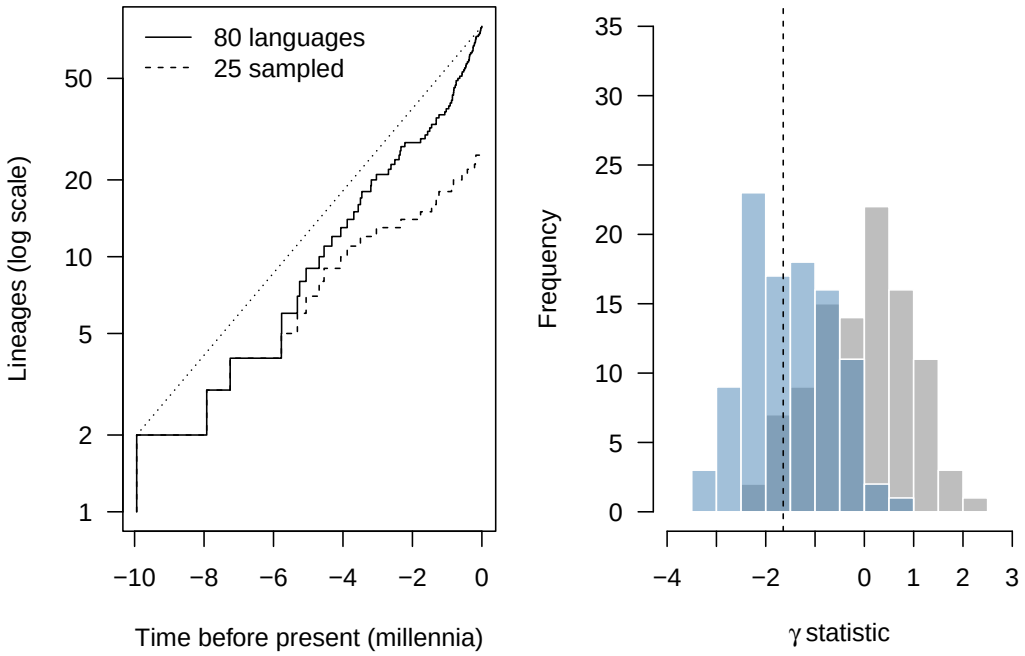


Figure 16.4: Left: lineages-through-time plot for the simulated 80-language family (solid) and for a random subsample of 25 languages (dashed), on a log scale. The dotted line joins 2 lineages at the root to 80 at the present: the straight line expected on this scale if lineages accumulated at a constant rate with no extinction. Right: distribution of Pybus and Harvey's gamma statistic over 100 pure-birth trees of 80 languages, analyzed with all languages (grey) or with a random 25 languages (blue). Incomplete sampling alone pushes gamma toward negative values, mimicking a slowdown in diversification.

```
par(op)
c(gamma_full_tree = gammaStat(family), gamma_subsample = gammaStat(sub_family),
  prop_sig_full = mean(g_full < -1.645), prop_sig_sub = mean(g_sub < -1.645))
```

```
#> gamma_full_tree gamma_subsample   prop_sig_full    prop_sig_sub
#>          2.5823          0.1454          0.0400          0.4300
```

Figure 16.4 illustrates both the pull of the present (the full tree's LTT curve bends upward near the present, as expected with extinction) and the sampling artifact: subsampling languages flattens the recent part of the curve and makes γ negative, suggesting a slowdown where there was none.

The right panel is a robustness test in the sense of Section 6.10: the data violate the assumption of complete sampling, and the statistic fails. Since almost every language phylogeny is a sample (many languages in a family are undocumented or omitted), this artifact is a serious concern for any linguistic diversification analysis. And because a single tree hides the uncertainty in node ages, any such statistic should be computed over a posterior sample of trees, not on one summary tree.

Language samples are rarely random. Well-documented subgroups are oversampled, and analysts often choose one representative per dialect cluster, which removes shallow nodes systematically and produces an apparent slowdown in diversification toward the present. Before interpreting an LTT plot, a γ statistic or a fitted extinction rate for a language family, state the sampling fraction for each clade (Glottolog counts are a reasonable starting point) and use methods that account for it.

Assessment: diversification models. *Tier B; barely applied. Preserved:* lineages that split and die. *Violated:* language "speciation" is gradual and often a matter of classification (Section 16.2.5), so the timing and even the number of splits depend on the lumping and splitting of varieties; extinction and sampling are not independent of traits (languages of small, unwritten communities are both more likely to die and less likely to be documented); and the extant-tree non-identifiability of Louca and Pennell (2020) applies in full. *What would show it works:* recovery of simulated rate shifts under realistic, non-random sampling of tips and with dialect lumping; agreement between rate changes inferred from the tree and independently dated archaeological phases that were not used as calibrations; and, for SSE models, rejection of character-independent nulls by more than the false-positive rate found in simulation.

Research programme. Fit a birth-death skyline model to a well-sampled, densely dated family, such as Austronesian or Bantu, with time intervals chosen a priori to correspond to archaeologically defined phases, and test whether splitting rates were higher during the expansion phases. Use GeoSSE with island versus mainland as a geographic state to test whether island colonization elevates splitting rates in Austronesian, using HiSSE-style hidden states and character-independent null models to guard against false positives. Throughout, specify sampling fractions per clade from Glottolog counts, report only identifiable combinations of parameters, in line with Louca and Pennell (2020), and repeat the analysis over the posterior sample of trees. A complementary project would use dated ancient languages as sampled extinct lineages (Section 16.3.2), which improves the identifiability of extinction.

16.6.2 Phylodynamics

How it works. *Phylodynamics* is the study of how epidemiological, ecological and evolutionary processes jointly shape pathogen phylogenies (Grenfell et al., 2004; Volz et al., 2013). Its core insight is that the shape of a dated phylogeny reflects the population dynamics of the organisms it describes. Coalescent-based methods estimate the history of effective population size: the Bayesian skyline (Drummond et al., 2005), skyride (Minin et al., 2008) and skygrid (Gill et al., 2013) reconstruct $N_e(t)$ as a piecewise or smooth function by relating the rate of coalescence at each time to the population size at that time (fewer lineages coalesce per unit time in a larger population). Birth-death-based methods estimate time-varying transmission (birth), recovery (death) and sampling rates, and from these the effective reproduction number, using the birth-death skyline (Stadler et al., 2013). Structured versions estimate migration between subpopulations (Kühnert et al., 2016). The methods are implemented in BEAST 2 and BEAST X.

The linguistic mapping. Two mappings are possible, at two scales. At the level of a language family, the spread of a family across a landscape resembles an epidemic spreading through a host population: lineages (languages) "infect" new territory and split, and some "recover" (die). The birth-death skyline then estimates time-varying rates of language formation and loss, as discussed above, and structured birth-death models with geographic demes could estimate rates of movement between regions in a way that is coupled to diversification, unlike the CTMC phylogeography of Chapter 11. At the level of a single language or dialect continuum, coalescent methods could, in principle, estimate the history of the "effective size" of a speech community from the genealogy of its lexical variants, if multiple idiolects or local varieties were sampled. Section 16.2.2 shows why that effective size would measure the speed of sorting rather than the number of speakers.

Assessment: phylodynamics. *Tier B for structured birth-death models of family expansion; tier C for coalescent skylines of speech communities. Not applied in the formal sense. Preserved:* lineages that branch, move and die at rates that change through time. *Violated:* languages are sampled by documentation, not at random through time; the coalescent effective size of a speech community is not a population size (Table 16.1); and token genealogies cannot be read off word lists. *What would show it works:* for family-level models, agreement of inferred rate changes with archaeological proxies that were not used as calibrations; for community-level skylines, recovery of known demographic histories (for instance documented immigration into a city) from densely sampled sociolinguistic data.

Research programme. Treat a densely sampled expansion (Bantu is the obvious candidate, given its large datasets and archaeological record) with a structured birth-death model in which geographic regions (for instance, rainforest versus savanna) are demes, and ask whether rates of splitting and of movement between regions differed. The results could be compared directly with archaeological proxies for population growth, such as summed radiocarbon date distributions, providing an independent test. The pulses-and-pauses analysis of Austronesian (Gray et al., 2009) and phylogeographic work on Bantu share the spirit of this proposal without its formal model.

16.6.3 Simulation-based inference: ABC and deep learning

How it works. Some models are easy to simulate from but have likelihoods that are impossible to compute, for instance agent-based models of speakers in space, models of borrowing on a dynamic contact network, or the token model of Section 16.2.2 with social biases. *Approximate Bayesian computation* (ABC) avoids the likelihood: draw parameters θ from the prior, simulate a dataset D^{sim}, compute summary statistics $S(\cdot)$, and accept θ if

$$\rho\big(S(D^{\text{sim}}), S(D)\big) < \varepsilon \tag{16.6}$$

for a distance ρ and tolerance ε. The accepted values approximate the posterior (Beaumont et al., 2002; Csilléry et al., 2010). The quality of the approximation depends critically on the choice of summary statistics. *Deep learning* offers a way to learn summaries, or to bypass them. Neural networks trained on simulated alignments can infer quartet topologies (Suvorov et al., 2020; Zou et al., 2020), and networks trained on simulated trees can estimate epidemiological and diversification parameters from tree shape (Voznica et al., 2022; Lambert et al., 2023). The trained network then acts as a fast amortized estimator: once trained, it can be applied to many datasets at negligible cost.

The linguistic mapping. Simulation-based inference fits language data well in situations where the generative process is understood but the likelihood is intractable. Examples include the wave

model, in which innovations spread over overlapping areas of a dialect continuum; models in which borrowing probability depends on geographic distance and population size; and models combining language splits with demographic processes. Deep learning has been used in computational historical linguistics mainly for cognate detection rather than tree inference (for example, Rama, 2016).

Assessment: simulation-based inference. *Tier B for ABC with explicit process models; tier C for deep-learning estimators trained on simulated language data. Barely applied. Preserved:* whatever the simulator assumes; the method imposes no likelihood. *At risk:* everything rests on the simulator, so the inference is only as good as the model of tree-like or wave-like change it encodes; summaries chosen for convenience may discard the information that separates the hypotheses; and a network trained on simulations will confidently classify real data that resemble none of them. *What would show it works:* calibration of posterior intervals on held-out simulations, classification accuracy on simulations from models *not* used in training (a robustness test), and correct classification of families whose mode of diversification is known historically.

Research programme. Use ABC or a neural classifier to decide between tree-like and wave-like diversification for a given dataset. Simulate cognate data under a tree model with varying borrowing rates and under a spatially explicit wave model of overlapping innovations, compute summaries (delta scores, Q-residuals, the shape of the split network, the D statistics of Section 16.4.1), and train a classifier or ABC procedure to identify which process generated the data and with what parameters. Validate it on simulated data and on families where historical evidence about the mode of diversification exists, such as Oceanic linkages (François, 2015) or Romance. A second project would fit the token model itself by ABC to datasets that record synonyms, estimating innovation rates and the strength of conformity from the distribution of synonyms per meaning.

16.6.4 Phylogenetic placement

How it works. When a reference tree for a group of organisms is well established and a new sequence (often a short environmental read) needs to be classified, re-inferring the whole tree is wasteful. *Phylogenetic placement* holds the reference tree fixed and evaluates the likelihood of attaching the new sequence to each branch. pplacer (Matsen et al., 2010) and the evolutionary placement algorithm EPA and its successor EPA-ng (Berger et al., 2011; Barbera et al., 2019) do this efficiently for millions of sequences. For each branch e, the method optimizes the attachment position and the length of the pendant branch, and converts the resulting likelihoods L_e into *likelihood weight ratios*,

$$w_e = \frac{L_e}{\sum_{e'} L_{e'}}, \tag{16.7}$$

which express how confidently the new sequence can be placed. A spread of weight across several neighbouring branches signals that the data cannot pin down the placement.

The linguistic mapping. Many languages are known only from short word lists: in archival materials, in the notes of early explorers, or in recent but brief documentation. Others are extinct and known from a handful of glosses. Glottolog lists a number of languages as unclassified, and many more have classifications based on limited evidence. Placement offers a principled way to classify such languages onto a reference tree inferred from well-documented relatives, with an explicit measure of uncertainty. For diversification analyses it is also a way to raise the sampling fraction. And it scales: global lexical datasets such as ASJP (Jäger, 2018) could be extended by placing new lists on a fixed tree without re-running a global analysis.

We simulate binary cognate data for 13 languages, delete 85% of the data for one of them (the "query" language, as if it were known only from a short list), infer a reference tree from the other 12 languages by maximum likelihood, and then place the query on each branch in turn. The data follow the placement model exactly, so this is parameter recovery; it shows what the method can do, not that it works on archival word lists.

```
set.seed(1604)
full13 <- pbtree(n = 13, scale = 1,
                 tip.label = c(sprintf("L%02d", 1:12), "Query"))
m13 <- as.character(simSeq(full13, l = 500, type = "USER",
                           levels = c("0", "1"), rate = 1.2))
m13 <- m13[, colSums(m13 == "1") > 0]
m13["Query", runif(ncol(m13)) < 0.85] <- "?"     # 85% missing
d13 <- phyDat(m13, type = "USER", levels = c("0", "1"), ambiguity = "?")
ref_names <- setdiff(full13$tip.label, "Query")
ref_fit <- optim.pml(pml(NJ(dist.hamming(subset(d13, ref_names))),
                         subset(d13, ref_names)),
                     optNni = TRUE, control = pml.control(trace = 0))
ref_tree <- midpoint(ref_fit$tree)   # root for display only

place_on_edge <- function(i, pendant = c(0.02, 0.05, 0.1, 0.2, 0.4, 0.8)) {
  child <- ref_tree$edge[i, 2]
  cand <- bind.tip(ref_tree, "Query", where = child,
                   position = ref_tree$edge.length[i] / 2, edge.length = 0.1)
  q_edge <- which(cand$edge[, 2] == which(cand$tip.label == "Query"))
  max(sapply(pendant, function(pl) {
    cand$edge.length[q_edge] <- pl
    pml(cand, d13)$logLik
  }))
}
edge_ll <- sapply(seq_len(nrow(ref_tree$edge)), place_on_edge)
lwr <- exp(edge_ll - max(edge_ll)); lwr <- lwr / sum(lwr)

true_sis <- full13$tip.label[Siblings(full13,
                                      which(full13$tip.label == "Query"))]
labs <- ifelse(ref_tree$tip.label %in% true_sis,
               paste0(ref_tree$tip.label, " *"), ref_tree$tip.label)
pal <- colorRampPalette(c("grey80", "darkorange", "firebrick"))(101)
shown <- ref_tree
shown$tip.label <- labs
plot(shown, edge.color = pal[round(100 * lwr / max(lwr)) + 1],
     edge.width = 1 + 6 * lwr / max(lwr), cex = 0.9,
     label.offset = 0.02 * max(node.depth.edgelength(shown)))
add.scale.bar(cex = 0.8)
```

```
top <- order(lwr, decreasing = TRUE)[1:3]
data.frame(edge_to = sapply(top, function(i) paste(
             ref_tree$tip.label[unlist(Descendants(ref_tree,
                                ref_tree$edge[i, 2], "tips"))],
```

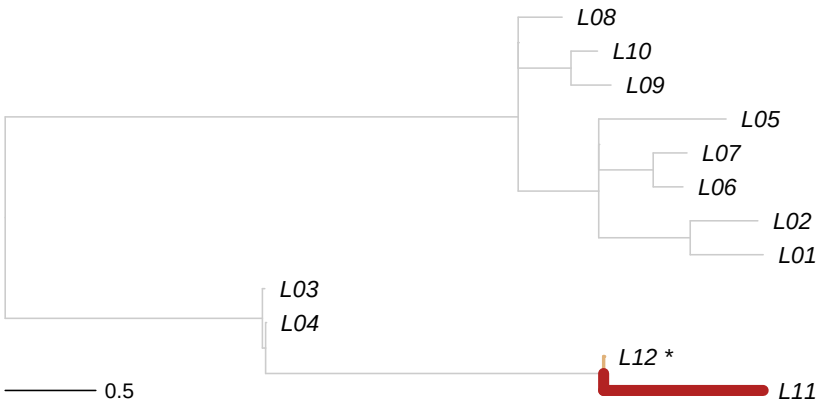


Figure 16.5: Likelihood-based phylogenetic placement of a poorly documented language. The reference tree of 12 languages is fixed; edges are colored and thickened in proportion to the likelihood weight ratio of placing the query language on them. The asterisk marks the query's true sister in the simulation.

```
            collapse = ",")),
          LWR = round(lwr[top], 3))
```

```
#>   edge_to   LWR
#> 1     L11 0.833
#> 2     L12 0.164
#> 3 L11,L12 0.002
```

```
c(observed_characters = sum(m13["Query", ] != "?"), true_sister = true_sis)
```

```
#> observed_characters         true_sister
#>                "66"              "L12"
```

With only about 15% of its characters observed, the query is still placed in the right part of the tree (Figure 16.5): almost all of the weight falls on the two branches leading to `L11` and `L12`, which are adjacent to the true attachment point (the query's true sister is `L12`). The LWRs show that the data identify the small clade but cannot fully resolve the exact attachment point, which is the honest answer given so little data. A real analysis would also optimize the attachment point along the branch and integrate over uncertainty in the reference tree; pplacer's Bayesian mode does the latter.

Assessment: phylogenetic placement. *Tier B; not applied in the likelihood or Bayesian sense*, although distance-based automated classification of languages from short word lists has a long history in the ASJP tradition. *Preserved:* the assumptions of ordinary tree inference, applied to

one new tip. *Violated or at risk:* archival lists are not random samples of a modern word list (they over-represent salient and borrowed vocabulary), transcriptions are inconsistent, so cognate judgements are less certain; a query with many loans from one neighbour will be placed next to the donor (Section 15.4); and the reference tree is treated as known. *What would show it works:* a calibration study in which well-classified languages are reduced to lists of archival size and quality, placed, and scored for how often the highest-weight placement is correct, with LWRs that are calibrated (placements with weight 0.9 correct about 90% of the time).

Research programme. Build reference trees for several large families from Lexibank data, and place every language that Glottolog lists as unclassified within (or possibly within) those families, as well as archival word lists of extinct languages. Report LWRs and, crucially, run the calibration study just described before interpreting any placement.

16.7 Summary table

Table 16.2 gathers the assessments. "Question" refers to the five sections of this chapter, "tier" to the classification of Section 16.1, and the last two columns give the assumption most likely to fail for language data and the kind of evidence that would show the adaptation works. The status labels reflect our reading of the literature and should be revised as the field develops.

Table 16.2: Methods from evolutionary biology and their linguistic prospects. Question: the section of this chapter. Tier: A = established in linguistics; B = plausible extension; C = speculative analogy. Status: Applied = several linguistic applications; Barely = one or a few exploratory or partial applications; Not applied = none known to us.

Method	Question	Tier	Assumption at risk	Validating test	Status
Multispecies coalescent (full)	Sorting	C	neutral, unbiased copying	recovery under biased token model	Barely
Lexical D statistic	Borrowing	B	no ancestral dialect structure	known contact vs control quartets	Not applied
Species delimitation	Sorting	C	no contact among varieties	dialect atlases, known boundaries	Not applied
Model adequacy checks	Rates	B	statistics with power	power on simulated violations	Barely
Tip-dating, sampled ancestors	Rates	A	sampling independent of history	predict withheld ancient tips	Applied
Total evidence	Rates	B	one tree for all partitions	rates match ancient tips	Barely
Sound-change models	Rates	B	regularity; fixed tree	withheld expert proto-forms	Barely
Joint alignment	Rates	B	words evolve independently	match expert alignments	Barely
Split networks	Borrowing	A	none (descriptive)	known contact appears as boxes	Applied
Network coalescent	Borrowing	B	contact as discrete events	documented contact recovered	Barely

Method	Question	Tier	Assumption at risk	Validating test	Status
DTL reconciliation	Borrowing	B	word trees well estimated	dictionary loans: precision, recall	Not applied
STRUCTURE, ADMIXTURE	Borrowing	A	clusters only descriptive	known areas recovered	Applied
Cophylogeny	Borrowing	B	populations form a tree	documented shifts recovered	Barely
Phylogenetic factor analysis	Mechanisms	B	Brownian liabilities	predict held-out features	Not applied
Threshold models	Mechanisms	B	gradient basis of features	liabilities match corpus data	Not applied
Birth-death, skyline	Diversification	B	random sampling of languages	shifts under realistic sampling	Barely
SSE models	Diversification	B	trait-independent sampling	beat trait-independent nulls	Not applied
BAMM, ClaDS	Diversification	B	rates identifiable	shifts under realistic sampling	Not applied
Structured birth-death	Diversification	B	random sampling through time	match archaeological proxies	Not applied
Coalescent skylines	Sorting	C	effective size = speakers	known demographic histories	Not applied
ABC	Diversification	B	realistic simulator	accuracy on untrained models	Barely
Deep learning	Diversification	C	simulations cover reality	accuracy on untrained models	Not applied
Phylogenetic placement	Diversification	B	archival lists like modern ones	calibrated weights, reduced lists	Not applied

Two patterns stand out. The tier A methods are those whose linguistic use requires few assumptions about the process (split networks, clustering) or whose mapping is unusually direct (ancient languages as fossils). Most of the promising methods are in tier B, where the mapping is sound but the decisive tests have not been run. And every row of the last-but-one column asks for more than parameter recovery: the validating evidence is either robustness to a specified linguistic complication or agreement with a historically documented case that the method did not use.

- A biological method is a stochastic model of how data arise; it transfers to language only if the mapping, the process assumptions, the available data and the identifiability of its parameters all survive. We classify methods as established in linguistics (A), plausible extensions (B) or speculative analogies (C).
- In an explicit model of lexical transmission, a concept-level genealogy is the copying history of word tokens, and cognate classes are the partition induced by innovations on it. Under neutral, exchangeable copying without borrowing, the genealogy of one token per language approximates the multispecies coalescent; conformity, prestige, selection, dialect structure and borrowing break the approximation in different ways.
- The symmetry of discordant patterns, the basis of the lexical *D* statistic, survives most social biases but not borrowing or ancestral dialect structure. The amount of discordance does not survive them, so coalescent "population sizes" of speech communities measure the speed of

sorting, not numbers of speakers.
- Rates of change are best informed by dated ancient languages, sound-level models and adequacy checks; under neutral copying the replacement rate equals the innovation rate, independent of community size.
- Network coalescent models, reconciliation, clustering and cophylogeny follow the genealogies of concepts, word families, feature profiles and populations through borrowing and language shift.
- Diversification models are directly applicable in form but suffer from non-identifiability of extinction, false positives and strong sensitivity to the non-random sampling of languages.
- Every proposal needs validation beyond parameter recovery: robustness under realistic misspecification, empirical adequacy and agreement with documented histories.

16.8 Exercises

Conceptual

1. Latin *ignis* and Sanskrit *agní-* continue the same Proto-Indo-European word for 'fire'. Using the multispecies coalescent, explain why this shared cognate is weak evidence for a Latin-Sanskrit subgroup. What pattern across many meanings would turn such retentions into evidence about subgrouping or contact?
2. In the token model, the dominant word for a meaning is replaced at the innovation rate u under neutral copying, whatever the size of the community. Explain why, and say what changes under conformist copying. What does this imply for the debate over population size and rates of lexical change?
3. Why does the symmetry of ABBA and BABA patterns survive conformity and prestige but not structure in the ancestral speech community? Describe a concrete linguistic history that would produce $D \neq 0$ without any borrowing after the languages separated.
4. A study reports a rate of language extinction estimated from a dated tree of living Bantu languages. For that parameter, discuss in turn whether it is identifiable under the model, practically estimable, robust to misspecification and interpretable as a historical quantity.
5. Classify the structured coalescent used in phylogeography (Chapter 11) as a tier A, B or C method for language data. Name the assumptions it preserves and violates, and the evidence that would move it up a tier.
6. The placement simulation in this chapter recovers the true sister of the query language. Explain why this does not show that placement works for archival word lists, and what kind of study would.

Computational

1. Run `run_bias("neutral", N = n)` for `n` = 10, 20, 40 and 80, and plot each curve against `1 - exp(-T)`. Compare the mean number of classes per concept with Ewens' expectation `sum(theta / (theta + 0:(n - 1)))` for `theta = 2 * n * 0.01`. How large must the community be for the coalescent approximation to be good? Which kind of validation is this?
2. Vary the strength of conformity by calling `run_bias("conformist", conform = c)` for `c` in 0, 0.25, 0.5 and 0.9 (the extra argument is passed through `learn()` to `choose_models()`). Plot the number of classes per concept against `c`. If these data were analyzed with a neutral model, what innovation rate would it estimate?

3. With `sim_msc_tree()`, `borrow_word()` and `triplet_topology()`, compute $D = (n_{BC} - n_{AC})/(n_{BC} + n_{AC})$ for 300 concepts at $b = 0$, 0.05 and 0.1. Repeat 200 times for each b and estimate the false-positive rate at $b = 0$ and the power at $b = 0.05$ of a test that rejects when $|D|\sqrt{n_{BC} + n_{AC}} > 1.96$.
4. Repeat the birth-death fit for 100 trees simulated with `pbtree(b = 0.6, d = 0.3, n = 80, extant.only = TRUE)`, recording the turnover estimated by `birthdeath()`. Repeat after keeping a random 25 tips with `keep.tip()`, and after keeping the 25 tips of the largest clades. Compare the three distributions.
5. In the placement simulation, vary the proportion of missing characters for the query from 50% to 95%, and record the likelihood weight ratio of the edge leading to the true sister. Over 50 replicates per level, are placements with a weight above 0.8 correct about 80% of the time or more?

Research

1. You suspect that some conflict in a Bantu lexical dataset reflects sorting of ancestral synonymy rather than contact. Design a study: what data you would need (for example, synonyms recorded per meaning, or several varieties per language), which simulations you would run under the token model (which biases, which borrowing scenarios), which statistics you would compute, and what result would persuade a sceptic.
2. A colleague proposes to use GeoSSE to test whether mountainous terrain slows language splitting in the Caucasus. Identify the threats to validity (lumping of varieties, non-random sampling, extinction, contact, tree uncertainty) and justify each modelling choice you would make, including the null models and the treatment of the posterior sample of trees.
3. Propose an adaptation of DTL reconciliation for Romance word families, in which Latin learned borrowings coexist with inherited reflexes (French *frêle* and *fragile*). Define the events of the model, say which assumptions of the biological method fail, and describe the validation data you would use.

16.9 Further reading

- Degnan and Rosenberg (2009): A clear review of gene-tree discordance and the multispecies coalescent; read it with the "fire" example of this chapter in mind.
- Reali and Griffiths (2010) and Newberry et al. (2017). Two complementary ways of treating linguistic variants as alleles, one theoretical and one empirical.
- Baxter et al. (2006) and Blythe and Croft (2012). Explicit models of variant competition in a speech community, and the case that documented changes require selection as well as drift; the starting point for any coalescent theory of words.
- Morlon (2014): An accessible review of diversification methods; pair it with Louca and Pennell (2020) on the limits of what extant trees can tell us.
- Rabosky and Goldberg (2015): Essential reading before running any state-dependent diversification analysis.
- Szöllősi et al. (2013b): The probabilistic reconciliation framework for transfers, including transfers from unsampled lineages, which has close analogues in borrowing from extinct languages.
- Bouchard-Côté et al. (2013): The most ambitious sequence-level model of sound change applied to a whole language family.

- Heath et al. (2014) and Gavryushkina et al. (2014). The fossilized birth-death process and sampled ancestors, the framework behind current treatments of ancient languages.
- Brown and Thomson (2018): A review of model adequacy testing that translates directly to cognate data.
- Matsen et al. (2010): The standard reference for likelihood and Bayesian phylogenetic placement.
- Reesink et al. (2009): A model for how to transfer a population-genetic method to language data with an explicit justification.

Chapter 17

A Complete Workflow in R

The previous chapters introduced the methods of linguistic phylogenetics one at a time. A real study has to chain them together. Every stage, from the word list to the published tree and the comparative analysis built on it, takes the output of the stage before and inherits its assumptions. This chapter asks how the pieces fit into one reproducible pipeline, and how one can tell whether that pipeline works. We first run it on a simulated family of 24 languages, where the true tree is known and every estimate can be scored against it. We then run it on real Polynesian data, where the answer is not known and the data are far less tidy. Along the way we prepare input files for MrBayes, BEAST 2, BEASTling and RevBayes, read the stored output of runs of those programs, and carry tree uncertainty into a downstream analysis. Every chunk of R code that is not marked as illustrative runs as printed, in well under a minute.

After reading this chapter you should be able to:

- turn a long-format cognate table into a binary presence/absence matrix, scoring unrecorded meanings as missing, and explain the character dependencies and the ascertainment bias that this coding creates;
- infer distance, parsimony and maximum-likelihood trees with **phangorn**, compare binary models by AIC, and score each tree against a known tree with the normalized Robinson-Foulds distance;
- export a matrix as NEXUS, PHYLIP and a BEAST 2 data block, and identify the parts of MrBayes, BEAST 2, BEASTling and RevBayes input files;
- read posterior tree samples and parameter logs into R, check convergence with ESS and PSRF, and summarize the trees as MCC trees, consensus trees and DensiTree plots;
- repeat a comparative analysis over a posterior sample of trees, and explain why a single summary tree can distort the result;
- build a character matrix from a real CLDF dataset, with explicit decisions about reconstructions, loans and language identifiers;
- state which kind of validation (Section 6.10) each analysis in the chapter provides, and explain why agreement with expert subgroups is not independent validation.

17.1 The shape of a phylogenetic project

A phylogenetic analysis of a language family is a pipeline. Each stage takes the output of the previous one, and each makes assumptions that the next stage inherits. Table 17.1 lists the stages we will work through, the questions each answers, the tools we use for the toy data, and the tools one would typically use in a real study.

Table 17.1: The stages of a linguistic phylogenetic analysis, with the tools used for the toy data in this chapter and typical tools for a real study.

Stage	Question	In this chapter	In a real study
1. Data assembly	Which languages, which meanings, which forms?	simulation	Lexibank, IE-CoR, ABVD, fieldwork; Glottolog for language identity
2. Cognate coding	Which forms are related by descent?	known by construction	expert judgment, EDICTOR, LingPy/LexStat
3. Character matrix	How are cognate sets turned into characters?	presence/absence coding in R	R, Python (`pycldf`, `lingpy`), BEASTling
4. Exploration	Is the signal tree-like? Are some languages odd?	distances, delta scores, NeighborNet	the same, plus SplitsTree
5. Tree inference	What is the best-supported tree?	UPGMA, NJ, parsimony, ML (phangorn)	IQ-TREE, RAxML-NG, MrBayes, BEAST 2, RevBayes
6. Support	How sure are we about each clade?	nonparametric bootstrap	bootstrap, UFBoot, posterior clade probabilities
7. Dating	When did the splits happen?	input files for MrBayes, BEAST 2, BEASTling and RevBayes; stored output of short runs	relaxed clocks and calibrations in BEAST 2 or RevBayes
8. Summaries	What does the sample of trees say?	MCC tree, consensus, DensiTree	TreeAnnotator, phangorn, ggtree, DensiTree
9. Down-stream use	What do the trees imply about other traits?	PGLS over a tree sample	BayesTraits, MCMCglmm, brms, RevBayes
10. Reporting	Can someone else reproduce it?	seeds, versions, files	archived data and code (CLDF, Zenodo)

Two features of this pipeline deserve emphasis before we begin. The first is that the most consequential decisions are made early, during data assembly and cognate coding (Chapter 3), and they are rarely revisited once the expensive inference has started. A Bayesian analysis of a large family can run for weeks. An error in a cognate judgment, or a dialect included twice under two names, will be faithfully propagated into every tree in the posterior. The second is that real projects are iterative. Exploratory analyses usually reveal problems in the data, such as a language with far too many singletons that turns out to reflect a coding error or a heavily borrowed lexicon. These problems are fixed before the main inference, and the cycle is repeated.

A tidy project layout pays for itself many times over. A common arrangement keeps raw data (for example a cloned CLDF dataset) in `data-raw/`, derived matrices in `data/`, scripts numbered in the

order they are run (`01-build-matrix.R`, `02-ml-tree.R`, and so on), BEAST or RevBayes input files in `analyses/`, and figures in `figures/`. Raw data are never edited by hand: every transformation is done by a script, so the whole analysis can be rerun from scratch. The **renv** package can record the exact versions of the R packages used, and the git commit hash or version number of each CLDF dataset should be recorded alongside the results.

17.2 Packages and reproducibility

We use **ape** for basic tree handling (Paradis and Schliep, 2019), **phangorn** for distance, parsimony and likelihood inference and for summarizing tree sets (Schliep, 2011), **phytools** for simulation and comparative methods (Revell, 2012, 2024), **ggtree** for plotting (Yu et al., 2017), and **nlme** for phylogenetic generalized least squares (Pinheiro and Bates, 2000). All are available from CRAN, except **ggtree** and its companion **treeio** (Wang et al., 2020), which are distributed through Bioconductor.

```
suppressPackageStartupMessages({
  library(ape)
  library(phangorn)
  library(phytools)
  library(ggtree)
  library(ggplot2)
  library(nlme)
})
set.seed(17)  # every random step below depends on this seed
pkgs <- c("ape", "phangorn", "phytools", "ggtree", "ggplot2", "nlme")
data.frame(package = pkgs,
           version = vapply(pkgs, function(p)
             as.character(packageVersion(p)), ""),
           row.names = NULL)
```

```
#>    package version
#> 1      ape   5.8.1
#> 2 phangorn  2.12.1
#> 3 phytools   2.5.2
#> 4   ggtree   4.2.0
#> 5  ggplot2   4.0.3
#> 6     nlme 3.1.170
```

Setting the random seed once, at the top of the script, makes the whole chapter reproducible: rerunning it produces exactly the same tree, data and results. In a real analysis one should also report the seeds used by external programs (BEAST 2 and MrBayes both print theirs at startup) and the versions of every program, since default priors and operators change between releases.

17.3 Step 1: Simulating a language family

Why simulate at all? There are two reasons, one practical and one methodological. The practical reason is that simulated data are small, free of licensing restrictions, and available offline. The methodological reason is more important. For a real language family we never know the true

tree, so we cannot tell whether a method has succeeded or failed. For simulated data we know the truth by construction, and can measure how close each method comes. Simulation studies of this kind have a long history in phylogenetics and have been used to good effect in linguistics. Barbançon et al. (2013) simulated the evolution of lexical and morphological characters on known trees, with and without borrowing, and compared the accuracy of distance, parsimony, compatibility and Bayesian methods. Greenhill et al. (2009) used simulation to ask how much borrowing it takes to mislead a Bayesian tree analysis. The workflow in this chapter is a miniature version of the same logic, with one important difference. Our simulator stays close to the assumptions of the inference models: there is a tree, there is no borrowing, and meanings evolve independently. In the vocabulary of Section 6.10, Steps 4 to 9 are therefore tests of *parameter recovery*. They show whether the pipeline can recover a history when its assumptions hold. They are not evidence of *empirical adequacy* for real language data, and they say nothing about *robustness under misspecification* unless the simulation deliberately breaks the models' assumptions, as Section 17.15 and the exercises suggest.

17.3.1 The tree

We assume that the family diversified under a *Yule* (pure-birth) process (Yule, 1925): each lineage splits into two at a constant rate λ, and no lineage ever dies out. The Yule process is the simplest of the tree priors discussed in Chapter 6, and it is a common default in BEAST 2 analyses of language families. When there are k lineages, the waiting time until the next split is exponentially distributed with rate $k\lambda$. We simulate a tree with 24 tips with `phytools::pbtree()` and rescale it so that the root lies 6 time units before the present. It is convenient to think of the unit as a millennium, so the family has a time depth of 6,000 years, comparable to Austronesian or Bantu (Chapter 10).

```
n_lang <- 24
true_tree <- pbtree(b = 1, n = n_lang, scale = 6)
true_tree$tip.label <- sprintf("L%02d", seq_len(n_lang))
true_tree <- reorder(true_tree, "cladewise")
c(tips = Ntip(true_tree), root_age = max(node.depth.edgelength(true_tree)),
  ultrametric = is.ultrametric(true_tree))
```

```
#>        tips    root_age ultrametric
#>          24           6           1
```

To see later whether the inferred trees recover the major subgroups of the family, we define "subgroups" as the clades descended from the lineages that were alive at a given moment in the past. Cutting the true tree 3,000 years after the root (that is, 3,000 years before the present) and collecting the descendants of each lineage alive at that time gives a set of subgroups, much as a historical linguist speaks of the primary or secondary branches of a family.

```
subgroups_at <- function(tree, cut_time) {
  H <- node.depth.edgelength(tree)   # distance of each node from the root
  e <- tree$edge
  crossing <- which(H[e[, 1]] < cut_time & H[e[, 2]] >= cut_time)
  grp <- character(Ntip(tree))
  for (k in seq_along(crossing)) {
    desc <- getDescendants(tree, e[crossing[k], 2])
```

```
    grp[desc[desc <= Ntip(tree)]] <- LETTERS[k]
  }
  setNames(grp, tree$tip.label)
}
subgroup <- subgroups_at(true_tree, cut_time = 3)
table(subgroup)
```

```
#> subgroup
#>  A  B  C  D
#> 12  7  1  4
```

Figure 17.1 shows the true tree, with time running from the root on the left to the present on the right, and with tips colored by subgroup.

```
okabe_ito <- c("#E69F00", "#56B4E9", "#009E73", "#D55E00", "#CC79A7",
               "#0072B2", "#999999", "#000000", "#F0E442")
sg_levels <- sort(unique(subgroup))
sg_cols <- setNames(okabe_ito[seq_along(sg_levels)], sg_levels)
tip_info <- data.frame(label = names(subgroup), subgroup = subgroup)
p_true <- ggtree(true_tree) %<+% tip_info +
  geom_tippoint(aes(color = subgroup), size = 1.6) +
  geom_tiplab(aes(color = subgroup), size = 3, offset = 0.08,
              show.legend = FALSE) +
  scale_color_manual(values = sg_cols, name = "Subgroup") +
  theme_tree2() +
  labs(x = "Time since the root (thousands of years)") +
  xlim(0, 6.6)
p_true
```

17.3.2 Lexical replacement

Next we let vocabulary evolve along the tree. Our model is the simplest one that still captures the essentials of lexical change in a basic vocabulary list. Each of M meanings (concepts such as 'hand', 'water' or 'two') is expressed in the root language by a single word, which founds cognate class 1 for that meaning. Along each branch, the word for meaning m is replaced by a new, unrelated word at rate r_m per millennium. Replacements occur as a Poisson process, so the probability that the word survives a branch of length t unchanged is

$$P(\text{no replacement on a branch of length } t) = e^{-r_m t}. \tag{17.1}$$

This is precisely the assumption of glottochronology (Swadesh, 1955; Lees, 1953), applied separately to each meaning and each branch. Every replacement creates a cognate class that has never existed before, so a class can be born only once and, once lost, never comes back. In the terms of Chapter 7, cognate classes evolve under a perfectly Dollo-like process with no homoplasy.

Real meanings differ greatly in their rates of replacement. Words for 'two', 'who' or 'I' are replaced far more slowly than words for 'dirty' or 'to throw', and frequency of use is a good predictor of the difference (Pagel et al., 2007). We capture this by drawing the rate of each meaning from a gamma distribution with mean $\bar{r}$ and shape α,

$$r_m \sim \text{Gamma}(\alpha,\ \alpha/\bar{r}), \qquad \mathbb{E}[r_m] = \bar{r}, \quad \text{Var}[r_m] = \bar{r}^2/\alpha. \tag{17.2}$$

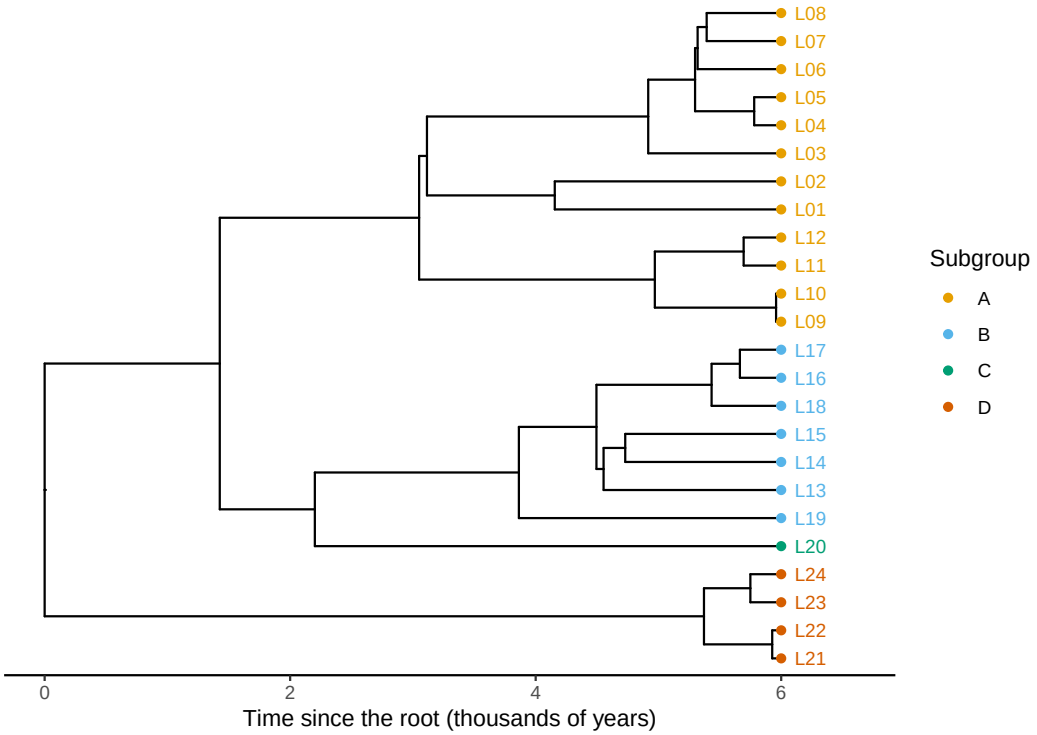


Figure 17.1: The simulated true tree of 24 languages, with a time depth of 6,000 years. Tip labels are colored by subgroup, that is, by the lineage alive 3,000 years before the present from which each language descends.

With $\alpha = 1$ the rates are exponentially distributed, so a few meanings change very fast and many change slowly. We set the mean rate to $\bar{r} = 0.1$ replacements per meaning per millennium. Over a 6,000-year tree this produces a realistic mix of very stable and very unstable meanings. Finally, each language-meaning cell is deleted with a small probability, to mimic the missing data that are ubiquitous in real word lists.

```
simulate_cognates <- function(tree, n_meanings = 110, mean_rate = 0.1,
                              gamma_shape = 1, p_missing = 0.03) {
  tree <- reorder(tree, "cladewise")       # parents are visited before children
  n_tip <- Ntip(tree)
  rates <- rgamma(n_meanings, shape = gamma_shape,
                  rate = gamma_shape / mean_rate)
  states <- matrix(NA_integer_, n_tip + tree$Nnode, n_meanings)
  states[n_tip + 1, ] <- 1L       # the root has class 1 for every meaning
  last_id <- rep(1L, n_meanings)            # highest class number used so far
  for (i in seq_len(nrow(tree$edge))) {
    parent <- tree$edge[i, 1]
    child  <- tree$edge[i, 2]
    s <- states[parent, ]
    k <- rpois(n_meanings, rates * tree$edge.length[i])  # replacements
    hit <- k > 0
    last_id[hit] <- last_id[hit] + k[hit]   # only the last replacement survives
    s[hit] <- last_id[hit]
    states[child, ] <- s
  }
  tips <- states[seq_len(n_tip), , drop = FALSE]
  concepts <- sprintf("C%03d", seq_len(n_meanings))
  cog <- data.frame(
    Language_ID   = rep(tree$tip.label, times = n_meanings),
    Parameter_ID  = rep(concepts, each = n_tip),
    Cognateset_ID = paste0(rep(concepts, each = n_tip), "-", as.vector(tips))
  )
  keep <- runif(nrow(cog)) > p_missing      # random missing data
  list(cognates = cog[keep, ], rates = setNames(rates, concepts))
}
sim <- simulate_cognates(true_tree)
cog <- sim$cognates
head(cog, 8)
```

```
#>   Language_ID Parameter_ID Cognateset_ID
#> 1         L01         C001        C001-2
#> 2         L02         C001        C001-1
#> 3         L03         C001        C001-1
#> 4         L04         C001        C001-1
#> 5         L05         C001        C001-1
#> 6         L06         C001        C001-1
#> 7         L07         C001        C001-1
#> 8         L08         C001        C001-1
```

The output is a *long* table with one row per word: a language, a meaning (`Parameter_ID`) and the cognate class to which the word belongs (`Cognateset_ID`). We have deliberately used the column names of the CLDF standard (Forkel et al., 2018), in which a Lexibank dataset stores its words in a `FormTable` and its cognate judgments in a `CognateTable` (Section 17.13). Code written for this toy table therefore works, almost unchanged, on real data.

```
n_classes <- tapply(cog$Cognateset_ID, cog$Parameter_ID,
                    function(x) length(unique(x)))
summary(as.vector(n_classes))
```

```
#>    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
#>    1.00    2.00    3.00    4.04    5.00   15.00
```

Each meaning is now represented by between 1 and 15 cognate classes across the family. Meanings with a single class are those whose word was never replaced anywhere on the tree; meanings with many classes are the fast-evolving ones.

Our simulator leaves out most of what makes real lexical data difficult. There is no borrowing, so every shared cognate reflects common descent. There are no synonyms, so each language has exactly one word per meaning. There is no semantic shift that brings an old cognate class into a new meaning, no parallel innovation, no partial cognacy, and no coding error. Real data violate all of these assumptions (Chapters 3, 8 and 15). The results below are therefore a best case. It is good practice to rerun a pipeline on simulated data that *do* include these complications, and to see which conclusions survive.

17.4 Step 2: From cognate table to character matrix

Phylogenetic programs need a matrix of languages by characters. As discussed in Chapter 3, the dominant coding in lexical phylogenetics is *binary presence/absence coding*. Each cognate class becomes a character, scored 1 in the languages that have a word belonging to the class for the relevant meaning, and 0 in the languages whose word for that meaning belongs to some other class. If a language has no recorded word for the meaning at all, every character belonging to that meaning is scored as missing (`?`), not as absent. Getting this last point right matters: scoring an unrecorded meaning as 0 would assert that the language lacks the cognate, and would make poorly documented languages look artificially divergent.

```
cognates_to_binary <- function(cog) {
  langs <- sort(unique(cog$Language_ID))
  sets  <- sort(unique(cog$Cognateset_ID))
  set_concept <- cog$Parameter_ID[match(sets, cog$Cognateset_ID)]
  m <- matrix("0", length(langs), length(sets), dimnames = list(langs, sets))
  m[cbind(match(cog$Language_ID, langs), match(cog$Cognateset_ID, sets))] <- "1"
  attested <- table(factor(cog$Language_ID, levels = langs),
                    factor(cog$Parameter_ID, levels = unique(set_concept)))
  for (cc in colnames(attested)) {           # unattested meaning -> all '?'
    absent <- attested[, cc] == 0
    if (any(absent)) m[absent, set_concept == cc] <- "?"
  }
  attr(m, "concept") <- set_concept
```

```
  m
}
bin <- cognates_to_binary(cog)
dim(bin)
```

```
#> [1]  24 444
```

```
table(bin)
```

```
#> bin
#>    ?    0    1
#>  306 7793 2557
```

The matrix has 24 languages and 444 binary characters, and about 3% of its cells are missing. Because the function uses only the three columns `Language_ID`, `Parameter_ID` and `Cognateset_ID`, it handles synonyms (two cognate sets for the same language and meaning) without modification: both characters are simply scored 1. It does assume that cognate set identifiers are unique across meanings; in real data it is safest to paste the meaning identifier onto the cognate set identifier, as our simulator does.

It is worth looking at how the characters are distributed across the family, because the distribution determines what a phylogenetic method can learn from them.

```
n_present <- colSums(bin == "1")
table(cut(n_present, breaks = c(0, 1, 2, 5, 10, n_lang - 1, n_lang),
          labels = c("1 (singleton)", "2", "3-5", "6-10",
                     paste0("11-", n_lang - 1), paste0(n_lang, " (all)"))))
```

```
#>
#> 1 (singleton)             2           3-5          6-10         11-23
#>           173            48            95            40            79
#>      24 (all)
#>             9
```

About 39% of the characters are *singletons*, present in a single language. Singletons tell us nothing about the branching order, since they do not group any languages together, but they are informative about branch lengths: a language with many singletons has replaced much of its vocabulary since it split from its relatives. A handful of characters are present in every language; these belong to meanings whose word was never replaced.

Notice what is *never* observed: a character that is absent from every language. A cognate class that has disappeared from every sampled language leaves no trace and cannot be coded. This is the ascertainment bias discussed in Chapter 7. It is the reason MrBayes and RevBayes offer a `noabsencesites` coding option and BEAST 2 analyses of cognate data include an ascertainment correction (Lewis, 2001; Nicholls and Gray, 2008). We return to it in Step 6.

Figure 17.2 displays the whole matrix, with languages ordered as on the true tree and characters grouped by meaning. Closely related languages share long runs of identical columns.

```
tip_order <- true_tree$tip.label[
  true_tree$edge[true_tree$edge[, 2] <= n_lang, 2]]
```

```
num <- matrix(match(bin[tip_order, ], c("0", "1", "?")), nrow = n_lang,
              dimnames = list(tip_order, colnames(bin)))
op <- par(mar = c(2, 4, 1, 1))
image(t(num[n_lang:1, ]), col = c("grey92", "grey20", "#D55E00"),
      axes = FALSE, useRaster = TRUE)
axis(2, at = seq(0, 1, length.out = n_lang), labels = rev(tip_order),
     las = 1, cex.axis = 0.6, tick = FALSE)
mtext("cognate classes (grouped by meaning)", side = 1, line = 0.5, cex = 0.8)
```

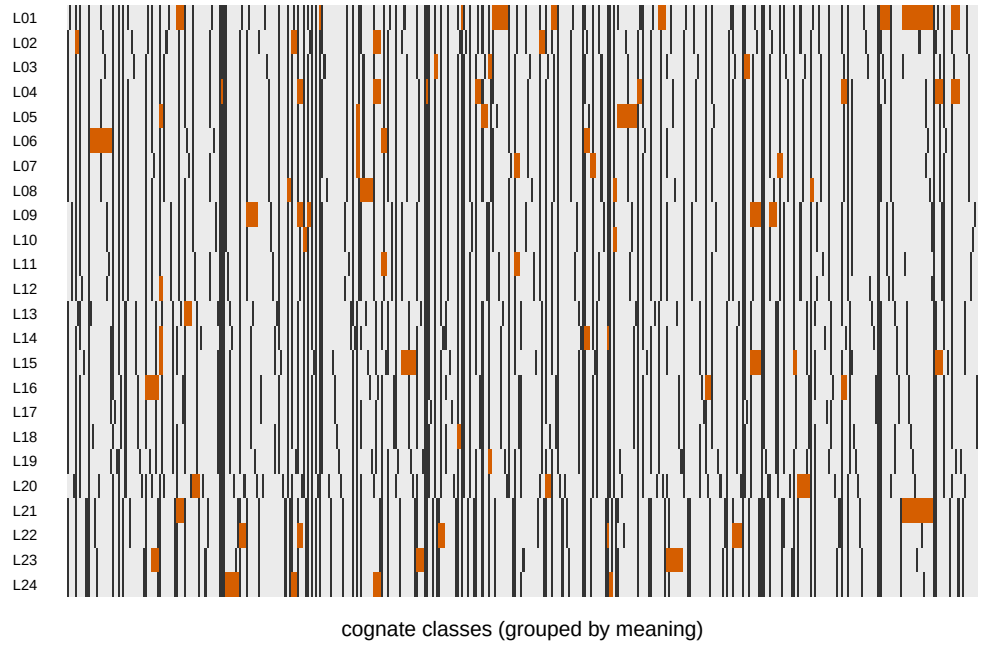


Figure 17.2: The binary cognate matrix. Rows are languages, ordered as in the true tree; columns are cognate classes, grouped by meaning. Dark cells are presences, light cells absences and orange cells missing data.

```
par(op)
```

Finally, we convert the character matrix into a `phyDat` object, the data format used by **phangorn**. We declare a user-defined alphabet with the two states 0 and 1, and tell **phangorn** that `?` means "either state".

```
cog_data <- phyDat(bin, type = "USER", levels = c("0", "1"), ambiguity = "?")
cog_data
```

```
#> 24 sequences with 444 character and 299 different site patterns.
#> The states are 0 1
```

phangorn compresses identical columns into *site patterns*, which is why it reports fewer patterns than characters. All the likelihood calculations below are done once per pattern and weighted by the number of times each pattern occurs.

17.5 Step 3: Exploring the data

Before inferring a tree it is wise to look at the data in ways that do not presuppose one. The most natural summary for lexical data is the *lexicostatistical distance* between two languages: the proportion of meanings, among those recorded for both, for which they do *not* share a cognate. If S_{ij} is the number of meanings for which languages i and j share a cognate and N_{ij} the number of meanings recorded for both,

$$d_{ij} = 1 - \frac{S_{ij}}{N_{ij}}. \tag{17.3}$$

This is the quantity that classical lexicostatistics computed by hand (Chapter 1). It is computed directly from the long cognate table.

```
lexical_distance <- function(cog) {
  langs <- sort(unique(cog$Language_ID))
  n <- length(langs)
  D <- matrix(0, n, n, dimnames = list(langs, langs))
  by_lang <- split(cog, cog$Language_ID)
  for (i in seq_len(n - 1)) for (j in (i + 1):n) {
    a <- by_lang[[langs[i]]]
    b <- by_lang[[langs[j]]]
    common <- intersect(a$Parameter_ID, b$Parameter_ID)
    shared <- intersect(a$Cognateset_ID[a$Parameter_ID %in% common],
                        b$Cognateset_ID[b$Parameter_ID %in% common])
    n_shared_meanings <- length(unique(sub("-[0-9]+$", "", shared)))
    D[i, j] <- D[j, i] <- 1 - n_shared_meanings / length(common)
  }
  as.dist(D)
}
D_lex <- lexical_distance(cog)
summary(as.vector(D_lex))
```

```
#>    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
#> 0.00971 0.31719 0.47573 0.42855 0.56636 0.64583
```

The closest pair of languages differ in about 1% of their recorded meanings, the most distant pair in about 65%. Under the glottochronological model of Equation (17.1) with a single rate r, the expected distance between two languages separated by a total path length T would be $1 - e^{-rT}$, a curve that flattens as T grows. With rates varying among meanings it flattens even more, because the fast meanings are quickly saturated while the slow ones are still informative. This is one reason why converting lexicostatistical distances directly into dates is unreliable (Chapter 9).

How tree-like are these distances? Chapter 8 introduced two diagnostics. The *delta score* of Holland et al. (2002) measures how far each quartet of languages departs from a perfect tree and averages over quartets. Values near 0 indicate tree-like data, and Gray et al. (2010) found considerably higher values in some language datasets than in others. A *NeighborNet* (Bryant and Moulton, 2004) displays conflicting signal as boxes in a split network. Both are available in **phangorn**.

```
round(c(mean = delta.score(cog_data, arg = "mean"),
        sd   = delta.score(cog_data, arg = "sd")), 3)
```

```
#>  mean    sd
#> 0.128 0.199
```

```
nnet <- neighborNet(D_lex)
op <- par(mar = rep(1.5, 4), xpd = NA)
pn <- plot(nnet, type = "2D", show.tip.label = FALSE, edge.width = 1)
net_labels(pn, nnet$tip.label,        # label placement: see the appendix
           col = sg_cols[subgroup[nnet$tip.label]])
```

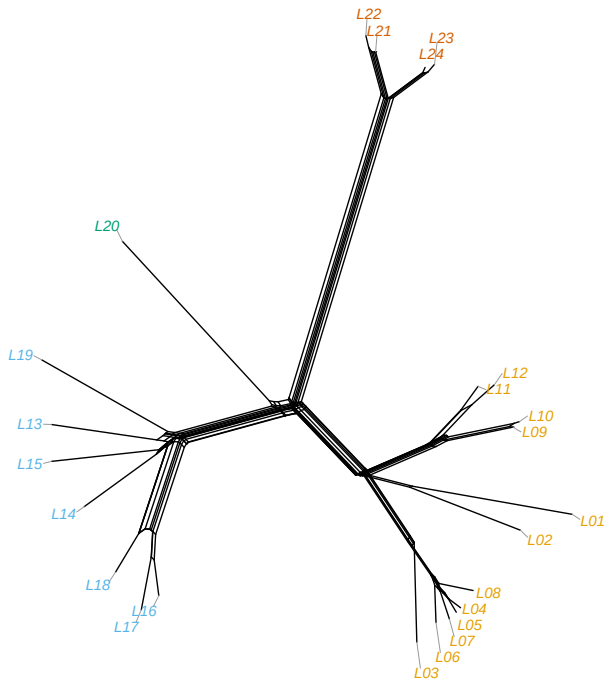


Figure 17.3: NeighborNet of the lexicostatistical distances, with tips colored by subgroup. Because there is no borrowing in the simulation, conflicting signal is weak and the network is close to a tree.

```
par(op)
```

The mean delta score is low and the network in Figure 17.3 has only small boxes, as expected for data simulated without borrowing. The residual conflict comes from short internal branches on which few or no replacements occurred, and from the random noise of a finite word list. In real data, a language that sits at the center of large boxes in a NeighborNet, or that takes part in an unusually large share of the non-tree-like quartets (the `arg = "all"` option returns the score of every quartet, which can be averaged per language), is a candidate for extensive borrowing or coding problems and deserves a closer look before tree inference.

The figures of this chapter use two small plotting helpers: `net_labels()`, which places tip labels around a split network without overlaps, and `support_layers()`, which places support values next to the nodes of a **ggtree** plot. They are defined in hidden chunks, and their code is printed in Section E.3.1.

17.6 Step 4: Distance trees

The quickest route from a distance matrix to a tree is a clustering algorithm (Chapter 4). UPGMA repeatedly joins the two closest clusters and assumes that distances are ultrametric, that is, that every language has changed at the same rate since the root. Neighbor-joining (Saitou and Nei, 1987) relaxes this assumption and produces an unrooted tree with branch lengths that may differ among lineages. We compute both from the lexicostatistical distances, and also a neighbor-joining tree from the Hamming distances between rows of the binary matrix, the proportion of cognate characters (rather than meanings) in which two languages differ.

```
tree_upgma <- upgma(D_lex)
tree_nj    <- NJ(D_lex)
tree_nj_ham <- NJ(dist.hamming(cog_data))
```

To judge these trees we need a measure of distance from the truth. The *Robinson-Foulds* (RF) distance (Robinson and Foulds, 1981) counts the bipartitions (splits) found in one tree but not the other (Chapter 2). For two fully resolved unrooted trees with n tips, each has $n-3$ nontrivial splits, so the RF distance ranges from 0 to $2(n-3)$, and dividing by this maximum gives a normalized distance between 0 (identical topologies) and 1 (no split in common):

$$\mathrm{RF}_{\mathrm{norm}}(\tau_1, \tau_2) = \frac{|\Sigma(\tau_1) \triangle \Sigma(\tau_2)|}{2(n-3)}, \tag{17.4}$$

where $\Sigma(\tau)$ is the set of nontrivial splits of tree τ and $\triangle$ denotes the symmetric difference. We also count how many of the true subgroups with more than one member are recovered as clades. Since several of the methods produce unrooted trees, we ask whether each subgroup forms a clade under some rooting, which is what `ape::is.monophyletic()` checks for an unrooted tree.

```
truth_unrooted <- unroot(true_tree)
multi_groups <- names(which(table(subgroup) > 1))
score_tree <- function(tree) {
  rf <- RF.dist(unroot(tree), truth_unrooted, normalize = TRUE)
  groups_ok <- vapply(multi_groups, function(g)
    is.monophyletic(tree, names(subgroup)[subgroup == g]), logical(1))
  c(RF_normalized = round(rf, 3),
    subgroups_recovered = sum(groups_ok), subgroups_total = length(groups_ok))
}
rbind(UPGMA = score_tree(tree_upgma),
      `NJ (lexical)` = score_tree(tree_nj),
      `NJ (Hamming)` = score_tree(tree_nj_ham))
```

```
#>              RF_normalized subgroups_recovered subgroups_total
#> UPGMA                0.143                   3               3
#> NJ (lexical)         0.238                   3               3
#> NJ (Hamming)         0.238                   3               3
```

On these data the distance methods already recover most of the true tree, including its subgroups. This is not surprising: the data were generated under a clock-like process with no borrowing, which is exactly the setting in which UPGMA's assumptions hold. With real data, where some lineages change faster than others, UPGMA can be badly misled, and NJ is generally preferred

among distance methods (Chapter 4). The remaining differences from the truth lie on the short internal branches, where few replacements happened and the data carry little information either way.

17.7 Step 5: Maximum parsimony

Parsimony seeks the tree that requires the fewest changes to explain the data (Chapter 4). **phangorn** searches tree space with the parsimony ratchet (Nixon, 1999), which alternates between searches on the original data and on randomly reweighted data in order to escape local optima. We then assign branch lengths by the ACCTRAN criterion, which places changes as close to the root as possible, and report the tree length together with the *consistency index* (CI) and *retention index* (RI), two classical measures of how much homoplasy the data contain on the tree.

```
tree_mp <- pratchet(cog_data, trace = 0, minit = 50, all = FALSE)
tree_mp <- acctran(tree_mp, cog_data)
c(length = parsimony(tree_mp, cog_data),
  CI = round(CI(tree_mp, cog_data), 3), RI = round(RI(tree_mp, cog_data), 3))
```

```
#>  length      CI      RI
#> 560.000   0.768   0.862
```

```
score_tree(tree_mp)
```

```
#>       RF_normalized subgroups_recovered     subgroups_total
#>                0.19                3.00                3.00
```

The data were simulated without homoplasy, so why is the CI below 1? The answer lies in the coding. Under binary presence/absence coding, a single replacement event on one branch produces *two* changes: the old cognate class goes from 1 to 0, and the new class goes from 0 to 1. Moreover, a cognate class that is lost independently in two lineages (because the meaning was replaced separately in each) produces two 1-to-0 changes for the same character, which parsimony counts as homoplasy even though no cognate class was ever born twice. This dependence among the binary characters that encode a single meaning is one of the well-known costs of presence/absence coding (Chapters 7 and 15), and the stochastic Dollo models of Chapter 7 were designed with precisely this situation in mind.

17.8 Step 6: Maximum likelihood

17.8.1 Fitting and comparing models

A likelihood analysis requires an explicit model of character change. For binary cognate characters, the standard choice is a two-state continuous-time Markov chain with rate matrix

$$\mathbf{Q} = \begin{pmatrix} -\pi_1 & \pi_1 \\ \pi_0 & -\pi_0 \end{pmatrix}, \tag{17.5}$$

scaled so that the expected number of changes per unit branch length is 1, where π_0 and π_1 are the stationary frequencies of absence and presence. If $\pi_0 = \pi_1 = 1/2$ the model is the binary analogue of the Jukes-Cantor model, sometimes called the Mk model with $k = 2$ (Lewis, 2001).

We compare three models: equal frequencies; estimated frequencies; and estimated frequencies with gamma-distributed rate variation among characters, using four discrete rate categories. Each model is fitted with an NNI tree search starting from the NJ tree, and the models are compared by AIC (Chapter 5).

```
ctrl <- pml.control(trace = 0)
fit_start <- pml(tree_nj, cog_data)
fit_mk  <- optim.pml(fit_start, optNni = TRUE, control = ctrl)
fit_mkf <- optim.pml(fit_mk, optNni = TRUE, optBf = TRUE, control = ctrl)
fit_mkfg <- optim.pml(update(fit_mkf, k = 4), optNni = TRUE, optBf = TRUE,
                      optGamma = TRUE, control = ctrl)
model_table <- data.frame(
  model = c("equal frequencies", "estimated frequencies",
            "estimated frequencies + gamma"),
  logLik = sapply(list(fit_mk, fit_mkf, fit_mkfg), logLik),
  df = sapply(list(fit_mk, fit_mkf, fit_mkfg),
              function(f) attr(logLik(f), "df")))
model_table$AIC <- -2 * model_table$logLik + 2 * model_table$df
model_table$dAIC <- model_table$AIC - min(model_table$AIC)
model_table$logLik <- round(model_table$logLik, 1)
model_table$AIC <- round(model_table$AIC, 1)
model_table$dAIC <- round(model_table$dAIC, 1)
knitr::kable(model_table, caption = paste(
  "Comparison of binary substitution models by AIC.",
  "The degrees of freedom include the branch lengths."))
```

Table 17.2: Comparison of binary substitution models by AIC. The degrees of freedom include the branch lengths.

model	logLik	df	AIC	dAIC
equal frequencies	-2552	45	5195	198.6
estimated frequencies	-2452	46	4996	0.0
estimated frequencies + gamma	-2453	47	5000	3.3

```
best_fit <- fit_mkf
round(c(pi_absent = fit_mkfg$bf[1], pi_present = fit_mkfg$bf[2],
        gamma_shape = fit_mkfg$shape), 3)
```

```
#>   pi_absent  pi_present gamma_shape
#>       0.753       0.247      99.999
```

```
score_tree(best_fit$tree)
```

```
#>       RF_normalized subgroups_recovered     subgroups_total
#>               0.238               3.000               3.000
```

Estimating the stationary frequencies improves the fit enormously, as Table 17.2 shows. This makes sense: in a presence/absence matrix most cells are 0, because each language has only one of the several cognate classes for each meaning, and a model that assumes equal frequencies of

0 and 1 is badly misspecified. The estimated frequency of presence, about 0.25, is close to the overall proportion of 1s among the non-missing cells.

Adding gamma rate variation, by contrast, does not improve the fit enough to justify the extra parameter, and the gamma shape parameter is estimated to be very large (100, close to the upper bound of the optimizer). A large shape parameter means that the binary characters show little rate variation, even though we simulated strong rate variation among *meanings*. The reason is again the coding. A fast meaning does not produce a few fast-changing characters; it produces many cognate classes, each of which is born once and then lost, so each individual character changes only once or twice on the tree. Rate variation among meanings is thus partly converted into variation in the number of characters per meaning rather than into variation in rates among characters. Models that work at the level of meanings, such as multistate models or binary models with meaning-level partitions, address this mismatch (Chapter 7). In what follows we use the model preferred by AIC, the binary model with estimated frequencies and no rate variation, as our best-fitting model. With real cognate data, which are messier than ours, among-character rate variation is usually strongly supported.

17.8.2 Ascertainment correction

The likelihoods above treat the observed characters as if they were a random sample of all possible characters, but, as noted in Step 2, a cognate class absent from every language can never be observed. **phangorn** implements the Lewis correction for unobservable *constant* characters through the argument `ASC = TRUE` (Lewis, 2001). This corrects for the absence of both all-0 and all-1 columns, so to use it consistently we must first remove the few characters that are present in every language. Because `optim.pml()` in **phangorn** 2.12.1 did not reliably optimize branch lengths under `ASC = TRUE` in our tests, we use it only for the topology search and then re-optimize the branch lengths with `refine_asc_edges()` (Section 7.5). The frequency of presence is held at its estimate from the full matrix; in our trials `optBf = TRUE` did not reliably update it under the correction either, and a direct joint maximization drove it to about 0.01, close to the stochastic Dollo limit of the reversible model (Section 7.7.4), as one might expect for data in which every cognate class arises only once.

```
constant <- apply(bin, 2, function(x) all(x[x != "?"] == "1"))
bin_var <- bin[, !constant]
cog_var <- phyDat(bin_var, type = "USER", levels = c("0", "1"), ambiguity = "?")
fit_asc <- optim.pml(pml(best_fit$tree, cog_var, ASC = TRUE, bf = best_fit$bf),
                     optNni = TRUE, control = ctrl)
fit_asc <- refine_asc_edges(fit_asc)   # branch lengths on corrected likelihood
c(constant_removed = sum(constant), logLik = round(logLik(fit_asc), 1))
```

```
#> constant_removed           logLik
#>               14            -2210
```

```
score_tree(fit_asc$tree)
```

```
#>       RF_normalized subgroups_recovered     subgroups_total
#>               0.238               3.000               3.000
```

This log-likelihood is not comparable with those of Table 17.2, because it is computed on a different dataset. Removing constant characters and correcting for them is the closest **phangorn**

comes to the `noabsencesites` correction of MrBayes and RevBayes, which conditions only on the absence of all-0 columns. For binary cognate data the latter is the more appropriate correction, since a cognate class present in all languages is perfectly observable. Here the correction leaves the topology unchanged but shortens the tree considerably, from a total length of 1.65 to 0.97 expected changes per character: without it, the model reads the shortage of all-absent characters as evidence of change and inflates the branch lengths. In Bayesian dating analyses this matters directly, because branch lengths become inferred dates. As Chapter 7 explains, the correction can also be applied per meaning, since each meaning necessarily has at least one cognate class present in some language.

17.8.3 Rooting

The binary model of Equation (17.5) is time-reversible, so the likelihood does not depend on where the root is placed, and maximum-likelihood trees are unrooted. To interpret a tree historically we must root it. With real data, the usual options are an outgroup (a language or subgroup believed to have split off first, such as Anatolian for Indo-European or the Formosan languages for Austronesian, both of which are themselves debated choices), midpoint rooting, which places the root halfway along the longest path between two tips and assumes a roughly clock-like process, or a clock model that estimates the root jointly with the tree (Chapter 9). Here we use midpoint rooting, which works well because the data were simulated under a strict clock.

17.9 Step 7: Bootstrap support and comparison with the truth

The nonparametric bootstrap (Felsenstein, 1985a) resamples characters with replacement to create pseudo-replicate datasets of the same size, re-estimates the tree from each, and reports for each clade of the best tree the proportion of replicate trees that contain it. We use 50 replicates, enough for illustration; published analyses typically use 100 to 1,000. Each replicate repeats the NNI search under the best-fitting model.

```
bs_trees <- bootstrap.pml(best_fit, bs = 50, optNni = TRUE, control = ctrl)
ml_tree <- plotBS(midpoint(best_fit$tree), bs_trees, type = "none", p = 0)
bs_support <- as.numeric(ml_tree$node.label)
if (max(bs_support, na.rm = TRUE) <= 1) bs_support <- 100 * bs_support
bs_support[1] <- NA          # the root has no meaningful support
ml_tree$node.label <- ifelse(is.na(bs_support), "", round(bs_support))
summary(bs_support)
```

```
#>    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NAs
#>      20      53      92      79     100     100       1
```

Half of the internal nodes have support of at least 92%, and 10 of the 22 nodes with a support value reach 95% or more.

Bootstrapping characters in a cognate matrix raises a subtle issue. The binary characters belonging to one meaning are not independent, so resampling them individually breaks up meanings and can overstate support. A more faithful scheme resamples whole meanings, with all their characters, which is straightforward to implement by resampling the `Parameter_ID` values of the long table and rebuilding the matrix. We leave this as an exercise; the code in Step 2 does all the work.

Figure 17.4 shows the midpoint-rooted ML tree with bootstrap support at the internal nodes and tips colored by their true subgroup.

```
p_ml <- ggtree(ml_tree) %<+% tip_info +
  geom_tippoint(aes(color = subgroup), size = 1.6) +
  geom_tiplab(aes(color = subgroup), size = 3, offset = 0.004,
              show.legend = FALSE) +
  support_layers(2.6) +       # support labels: see the appendix
  scale_color_manual(values = sg_cols, name = "Subgroup") +
  theme_tree2() + labs(x = "Expected changes per character")
p_ml + xlim(0, max(node.depth.edgelength(ml_tree)) * 1.15)
```

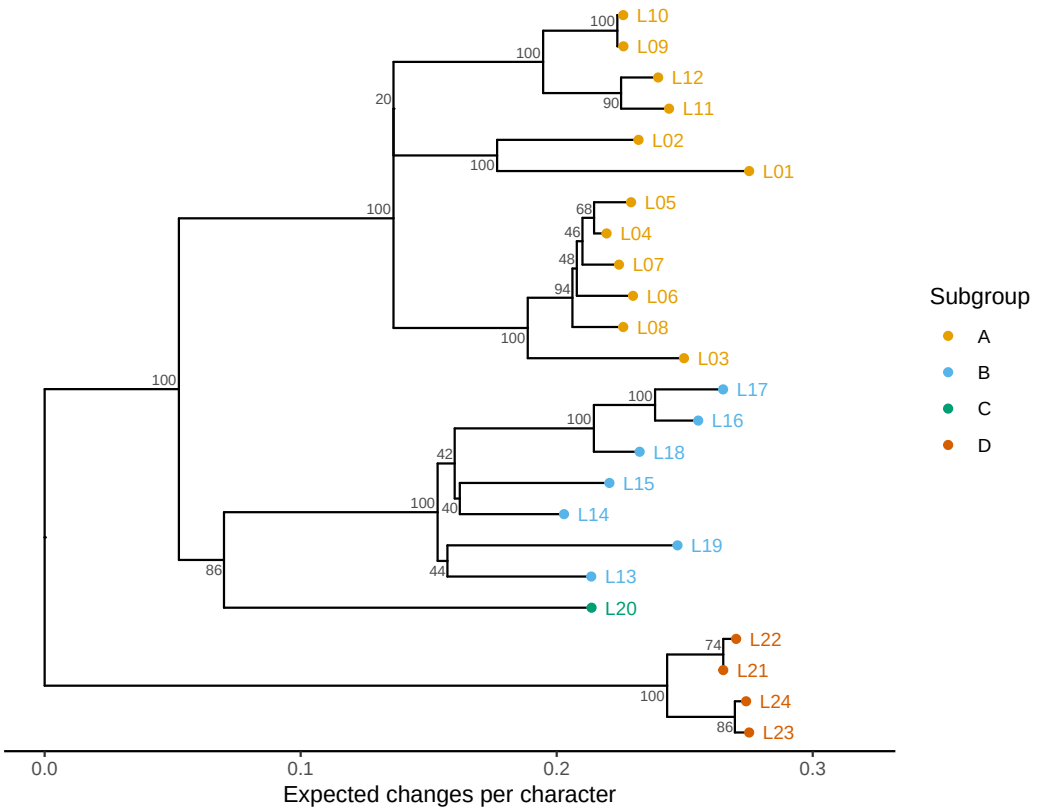


Figure 17.4: Maximum-likelihood tree (binary model with estimated frequencies), midpoint-rooted, with bootstrap support values (percent of 50 replicates). Tip colors show the true subgroups. Branch lengths are in expected changes per character.

We can now compare all the estimates with the true tree (Table 17.3).

```
comparison <- rbind(
  UPGMA = score_tree(tree_upgma),
  `NJ (lexical distance)` = score_tree(tree_nj),
  `NJ (Hamming distance)` = score_tree(tree_nj_ham),
  `Maximum parsimony` = score_tree(tree_mp),
  `ML, equal frequencies` = score_tree(fit_mk$tree),
  `ML, estimated frequencies` = score_tree(fit_mkf$tree),
```

```
  `ML, frequencies + gamma` = score_tree(fit_mkfg$tree),
  `ML, ascertainment-corrected` = score_tree(fit_asc$tree))
knitr::kable(comparison, caption = paste(
  "Normalized Robinson-Foulds distances between each estimated tree and",
  "the true tree (0 = identical), and the number of true multi-language",
  "subgroups recovered as clades."))
```

Table 17.3: Normalized Robinson-Foulds distances between each estimated tree and the true tree (0 = identical), and the number of true multi-language subgroups recovered as clades.

	RF_normalized	subgroups_recovered	subgroups_total
UPGMA	0.143	3	3
NJ (lexical distance)	0.238	3	3
NJ (Hamming distance)	0.238	3	3
Maximum parsimony	0.190	3	3
ML, equal frequencies	0.238	3	3
ML, estimated frequencies	0.238	3	3
ML, frequencies + gamma	0.238	3	3
ML, ascertainment-corrected	0.238	3	3

Two lessons from Table 17.3 generalize well beyond this toy example. First, when the data are clean and the process is close to clock-like, all methods do reasonably well and differ mainly in how they resolve the shortest internal branches. Second, the differences that remain are concentrated precisely on those branches, which are also the ones with low bootstrap support. The bootstrap is doing its job: low support flags the parts of the tree where the data are not decisive, and where different methods are free to disagree. Do not be surprised if a more sophisticated model does not win on any particular simulated dataset. The truth is a single random draw, and with only a few hundred characters the sampling error of any method is substantial. Meaningful comparisons of methods require many replicate simulations, as in the studies of Barbançon et al. (2013) and Rama et al. (2018). And all of these comparisons are parameter recovery on data that fit the methods' assumptions closely. A method that wins here has passed the easiest test, not shown that it fits real cognate data.

A tanglegram (Figure 17.5) is the most direct visual comparison of two trees: the true tree is drawn on the left, the ML tree on the right, and lines connect the same language in both. **phytools** rotates nodes to minimize line crossings.

```
co <- cophylo(true_tree, ml_tree, rotate = TRUE, print = FALSE)
```

```
#> Rotating nodes to optimize matching...
#> Done.
```

```
plot(co, fsize = 0.65, link.type = "curved", link.lwd = 1.2,
     link.col = adjustcolor("#0072B2", 0.6))
```

Barbançon et al. (2013) carried out what remains one of the most careful simulation comparisons of phylogenetic methods for linguistic data. They simulated lexical and morphological characters

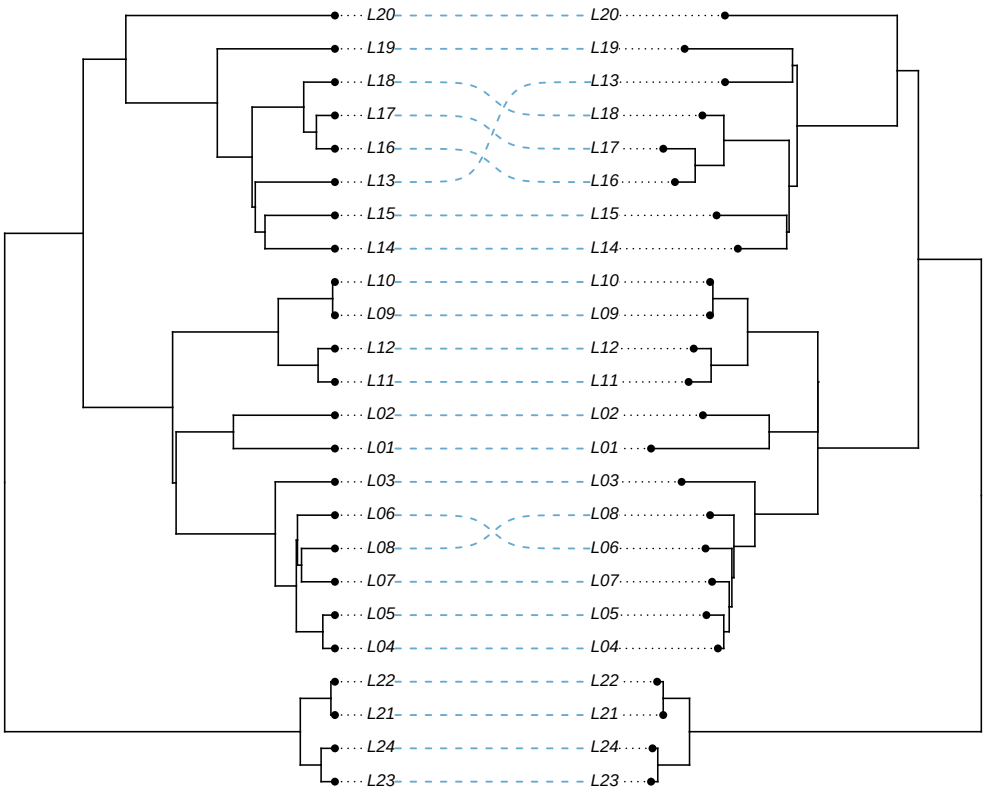


Figure 17.5: Tanglegram of the true tree (left) and the midpoint-rooted ML tree (right). Crossing lines indicate topological differences.

on model trees, with up to three contact edges, under a range of conditions, including different degrees of deviation from a lexical clock, rate variation among characters, homoplasy and borrowing. They then compared the accuracy of six methods: UPGMA, neighbor-joining, maximum parsimony, weighted maximum parsimony, weighted maximum compatibility and the Bayesian method of Gray and Atkinson, run in MrBayes. Among their findings was that UPGMA, the method closest to classical lexicostatistics, was the least accurate, and that its error grew the most as the clock assumption was relaxed. The ranking of the unweighted methods (maximum parsimony first, then Gray and Atkinson's method, then neighbor-joining) was remarkably consistent across conditions, whereas the performance of the weighted methods depended on the weighting scheme and the amount of homoplasy. The study is a model for how to evaluate a pipeline: simulate under conditions that match what is known about real language change, vary those conditions systematically, and measure accuracy against the known truth.

17.10 Step 8: Exporting data for Bayesian software

Bayesian dating analyses with relaxed clocks, calibrations and stochastic Dollo or covarion models are rarely run in R. They are run in dedicated programs: MrBayes (Ronquist et al., 2012b), BEAST 2 (Bouckaert et al., 2014, 2019), RevBayes (Höhna et al., 2016) or TraitLab (Nicholls and Gray, 2008), often with BEASTling (Maurits et al., 2017) to write the BEAST 2 input file. R is nevertheless the natural place to prepare their inputs, because the matrix, the language metadata and the calibrations already live there. In this section we write the files these programs need into a temporary directory. The external programs are not part of the book's build. We ran each of them on exactly these files, with shorter chains than a real analysis would use, and kept the input files, a thinned copy of the output, the scripts and the run logs in the folder `software-runs/` of the book's repository. Its file `run_all.sh` repeats every run, and its `README.md` lists the program versions, run times and diagnostics. Appendix F lists where to obtain the programs.

```
out_dir <- file.path(tempdir(), "workflow-export")
dir.create(out_dir, showWarnings = FALSE)
```

17.10.1 NEXUS

NEXUS (Chapter 2) is the lingua franca of phylogenetic software. A NEXUS data block declares the number of taxa and characters, the data type and the symbols for missing data, followed by the matrix. Two data types are relevant for cognate data. MrBayes uses `DATATYPE=RESTRICTION` for binary characters, which enables its ascertainment options; most other programs read `DATATYPE=STANDARD` with symbols `01`. It is also useful to record which characters belong to which meaning, as *character sets* (`CHARSET`), so that analyses can be partitioned by meaning or apply ascertainment correction per meaning. The following function writes both.

```
write_nexus_binary <- function(m, file, datatype = c("restriction", "standard"),
                               extra = NULL) {
  datatype <- match.arg(datatype)
  fmt <- if (datatype == "restriction") {
    "FORMAT DATATYPE=RESTRICTION MISSING=? GAP=-;"
  } else {
    "FORMAT DATATYPE=STANDARD SYMBOLS=\"01\" MISSING=? GAP=-;"
  }
```

```
  w <- max(nchar(rownames(m)))
  rows <- sprintf("    %-*s  %s", w, rownames(m),
                  apply(m, 1, paste, collapse = ""))
  concept <- attr(m, "concept")
  r <- rle(concept)
  ends <- cumsum(r$lengths)
  starts <- ends - r$lengths + 1
  # spaces around '-': RevBayes's NEXUS reader rejects "1-4"
  charsets <- sprintf("    CHARSET %s = %d - %d;", r$values, starts, ends)
  lines <- c("#NEXUS", "",
             "BEGIN DATA;",
             sprintf("  DIMENSIONS NTAX=%d NCHAR=%d;", nrow(m), ncol(m)),
             paste0("  ", fmt),
             "  MATRIX", rows, "  ;", "END;", "",
             "BEGIN ASSUMPTIONS;", charsets, "END;")
  if (!is.null(extra)) lines <- c(lines, "", extra)
  writeLines(lines, file)
  invisible(file)
}
nexus_std <- file.path(out_dir, "toy_cognates_standard.nex")
write_nexus_binary(bin, nexus_std, datatype = "standard")
show_file <- function(file, n = 12, width = 72) {
  x <- readLines(file, n = n)
  x <- ifelse(nchar(x) > width, paste0(substr(x, 1, width), " ..."), x)
  cat(x, sep = "\n")
}
show_file(nexus_std)
```

```
#> #NEXUS
#>
#> BEGIN DATA;
#>   DIMENSIONS NTAX=24 NCHAR=444;
#>   FORMAT DATATYPE=STANDARD SYMBOLS="01" MISSING=? GAP=-;
#>   MATRIX
#>     L01  0100101000100000100000100101010000000001010001000100?????100010 ...
#>     L02  1000??100010000001000010010100010000000101000100100000100100010 ...
#>     L03  1000101000100000001000100101000010000010010001010000000100100010 ...
#>     L04  1000101000100000100000100101010000000010010001010000000100100010 ...
#>     L05  1000101000100000100000100101010000000001001000??1000000100100010 ...
#>     L06  100010100011????????????100101010000000010010001010000000100010010 ...
```

It is always worth reading an exported file back in to check that nothing was lost or garbled on the way.

```
check <- read.nexus.data(nexus_std)
c(languages = length(check), characters = length(check[[1]]),
  identical = all(sapply(names(check), function(l)
    all(check[[l]] == bin[l, ]))))
```

```
#>  languages characters  identical
#>         24        444          1
```

17.10.2 MrBayes

MrBayes reads commands from a `mrbayes` block appended to the NEXUS file. The block below specifies a binary (restriction-site) model with stationary frequencies estimated under a flat Dirichlet prior, gamma rate variation, and the `noabsencesites` ascertainment correction for the unobservable all-absent characters. It sets up a relaxed clock with independent gamma rates (IGR), a uniform prior on node ages, and one calibration. Since this is a toy, we invent the calibration by pretending that subgroup D is known, from written records, to have begun to diverge within half a millennium of its true age. We generate the block in R, so the language names and the calibration are filled in automatically.

```
calib_tips <- names(subgroup)[subgroup == "D"]
calib_age <- unname(branching.times(true_tree)[
  as.character(getMRCA(true_tree, calib_tips))])
mb_block <- c(
  "BEGIN MRBAYES;",
  "  set autoclose=yes nowarn=yes seed=17 swapseed=17;",
  "  [ binary model with ascertainment correction and gamma rates ]",
  "  lset coding=noabsencesites rates=gamma ngammacat=4;",
  "  prset statefreqpr=dirichlet(1,1);",
  "  [ relaxed clock; time unit = 1,000 years ]",
  "  prset brlenspr=clock:uniform clockvarpr=igr;",
  "  prset clockratepr=lognormal(-2.3,1.0);",
  "  prset treeagepr=uniform(3,12);",
  sprintf("  constraint subgroupD = %s;", paste(calib_tips, collapse = " ")),
  sprintf("  calibrate subgroupD = uniform(%.1f,%.1f);",
          max(calib_age - 0.5, 0.1), calib_age + 0.5),
  "  prset topologypr=constraints(subgroupD) nodeagepr=calibrated;",
  "  mcmc ngen=5000000 samplefreq=1000 printfreq=10000 nchains=4 nruns=2;",
  "  sump relburnin=yes burninfrac=0.25;",
  "  sumt relburnin=yes burninfrac=0.25 contype=allcompat;",
  "END;")
nexus_mb <- file.path(out_dir, "toy_cognates_mrbayes.nex")
write_nexus_binary(bin, nexus_mb, datatype = "restriction", extra = mb_block)
cat(mb_block, sep = "\n")
```

```
#> BEGIN MRBAYES;
#>   set autoclose=yes nowarn=yes seed=17 swapseed=17;
#>   [ binary model with ascertainment correction and gamma rates ]
#>   lset coding=noabsencesites rates=gamma ngammacat=4;
#>   prset statefreqpr=dirichlet(1,1);
#>   [ relaxed clock; time unit = 1,000 years ]
#>   prset brlenspr=clock:uniform clockvarpr=igr;
#>   prset clockratepr=lognormal(-2.3,1.0);
#>   prset treeagepr=uniform(3,12);
#>   constraint subgroupD = L21 L22 L23 L24;
```

```
#>   calibrate subgroupD = uniform(0.1,1.1);
#>   prset topologypr=constraints(subgroupD) nodeagepr=calibrated;
#>   mcmc ngen=5000000 samplefreq=1000 printfreq=10000 nchains=4 nruns=2;
#>   sump relburnin=yes burninfrac=0.25;
#>   sumt relburnin=yes burninfrac=0.25 contype=allcompat;
#> END;
```

Running `mb toy_cognates_mrbayes.nex` would perform two independent runs of four Metropolis-coupled chains each (Chapter 6), report the average standard deviation of split frequencies between the runs as a convergence diagnostic, and write the sampled trees to `.t` files and the parameters to `.p` files. MrBayes 3.2.9 reads the file as written; it skips the `ASSUMPTIONS` block, and in batch mode it exits when the block is done. For the book we shortened the run to 1 million generations with two chains per run (`mcmc ngen=1000000 samplefreq=500 nchains=2 nruns=2`), which took 6 to 11 minutes on one core, depending on the load of the machine. The average standard deviation of split frequencies fell to 0.006, and every PSRF was below 1.12. But the effective sample sizes of the tree height, the clock rate and the state frequencies were only 20 to 40. These parameters are strongly correlated with each other and mix slowly, which is why a real analysis needs the longer run of the block above. The priors here are placeholders. In a real analysis each would need to be justified, and the analysis rerun under alternatives to check that the conclusions do not depend on them (Chapter 15). Note in particular that constraining subgroup D to be monophyletic is a strong assumption that is only justified if the subgroup is securely established by the comparative method.

17.10.3 BEAST 2

BEAST 2 reads a single XML file that describes the data, the model, the priors, the MCMC operators and the output. The file is normally generated with the graphical program BEAUti or, for linguistic data, with BEASTling, rather than written by hand. Writing it from scratch is error-prone, because a typical file for a relaxed-clock covarion analysis runs to many hundreds of lines. It is nevertheless useful to know its structure, and to be able to generate the data part from R. Each language becomes a `sequence` element whose `value` attribute is that language's row of the matrix. For the ascertainment correction, we put a dummy all-absent column in front of the matrix and mark it with the alignment's `ascertained`, `excludefrom` and `excludeto` attributes. BEAST 2 then leaves column 0 out of the likelihood and conditions on characters not being absent from every language, the equivalent of `noabsencesites` in MrBayes.

```
seq_lines <- sprintf(
  '    <sequence id="seq_%s" taxon="%s" totalcount="2" value="0%s"/>',
  rownames(bin), rownames(bin), apply(bin, 1, paste, collapse = ""))
beast_data <- c(paste('<data id="cognates" spec="Alignment" dataType="binary"',
                      'ascertained="true" excludefrom="0" excludeto="1">'),
                seq_lines, "</data>")
writeLines(beast_data, file.path(out_dir, "toy_cognates_beast_data.xml"))
cat(substr(beast_data[1:4], 1, 72), "    ...", sep = "\n")
```

```
#> <data id="cognates" spec="Alignment" dataType="binary" ascertained="true
#>     <sequence id="seq_L01" taxon="L01" totalcount="2" value="00100101000
#>     <sequence id="seq_L02" taxon="L02" totalcount="2" value="01000??1000
#>     <sequence id="seq_L03" taxon="L03" totalcount="2" value="01000101000
```

```
#>       ...
```

The rest of a BEAST 2.7 file for these data specifies the restriction-site model with estimated frequencies and gamma rate variation, a strict clock, a Yule tree prior, the subgroup-D calibration and the root-age prior of the MrBayes block. The skeleton below shows how the file is organized. It has four parts: the *state* (everything the MCMC samples: the tree, the birth rate, the clock rate, the gamma shape and the frequencies); the *posterior*, the product of the priors and the likelihood; the *operators*, which propose new values for each part of the state; and the *loggers*, which write the parameter trace and the sampled trees. The full file, about 140 lines long, is `software-runs/beast2/toy_beast.xml` in the book's repository, and it runs as it stands. Class names changed in BEAST 2.7, when the core packages were moved to the `beast.base` namespace, so XML written for BEAST 2.6 or earlier does not run in 2.7 without editing.

```
<beast version="2.7" namespace="beast.base.core:beast.base.inference:...">
  <data id="cognates" spec="Alignment" dataType="binary" ascertained="true"
        excludefrom="0" excludeto="1"> ... </data>
  <run id="mcmc" spec="MCMC" chainLength="3000000">
    <state id="state"> ... </state>            <!-- 1. what is sampled -->
    <distribution id="posterior" spec="CompoundDistribution">
      <distribution id="prior" ...> ... </distribution>      <!-- 2a -->
      <distribution id="likelihood" ...> ... </distribution> <!-- 2b -->
    </distribution>
    <operator ... /> ...                       <!-- 3. proposals -->
    <logger ... > ... </logger>                <!-- 4. output files -->
  </run>
</beast>
```

A longer annotated excerpt of the BEAST 2.7 file, with the priors, the calibration, the substitution model and the operators written out, is in Section E.3.2.

The strict clock is a `StrictClockModel` element inside the likelihood. For a relaxed clock it is replaced by an uncorrelated lognormal clock (`UCRelaxedClockModel`) or by the optimized relaxed clock of the ORC package. For a covarion model, the substitution model becomes `BinaryCovarion`.

17.10.4 BEASTling

BEASTling (Maurits et al., 2017) was written to spare linguists the work of writing BEAST 2 XML. The user writes a short configuration file in a simple `key = value` format, and BEASTling generates a complete and internally consistent XML file, including ascertainment correction, rate variation among features, calibrations and monophyly constraints. Language identities can be matched to Glottolog (Hammarström et al., 2024), which allows constraints such as "keep every Glottolog subgroup monophyletic" to be stated in a single line. BEASTling reads long-format data with one row per language and feature, which is exactly what our cognate table is, so we export it with a `Value` column holding the cognate class.

```
beastling_data <- data.frame(Language_ID = cog$Language_ID,
                             Parameter_ID = cog$Parameter_ID,
                             Value = sub("^C[0-9]+-", "", cog$Cognateset_ID))
write.csv(beastling_data, file.path(out_dir, "toy_cognates_long.csv"),
          row.names = FALSE, quote = FALSE)
```

```
head(beastling_data, 4)
```

```
#>   Language_ID Parameter_ID Value
#> 1         L01         C001     2
#> 2         L02         C001     1
#> 3         L03         C001     1
#> 4         L04         C001     1
```

A configuration for a covarion analysis with a relaxed clock and one calibration might look as follows. We generate it in R so that the calibrated clade and its age range match the MrBayes block above; BEASTling expects calibration ages in years. Note the `uniform()` around the age range. A bare range such as `129-1129` would be read as the 95% interval of a *normal* distribution, not as the bounds of a uniform one. The option names below are those of BEASTling 1.5.1, the most recent release.

```
beastling_conf <- c(
  "[admin]",
  "basename = toy_family",
  "",
  "[MCMC]",
  "chainlength = 10000000",
  "",
  "[model lexicon]",
  "model = covarion",
  "data = toy_cognates_long.csv",
  "rate_variation = True",
  "",
  "[clock default]",
  "type = relaxed",
  "",
  "[calibration]",
  sprintf("%s = uniform(%d, %d)", paste(calib_tips, collapse = ","),
          round(1000 * max(calib_age - 0.5, 0.1)),
          round(1000 * (calib_age + 0.5))))
writeLines(beastling_conf, file.path(out_dir, "toy_family.conf"))
cat(beastling_conf, sep = "\n")
```

```
#> [admin]
#> basename = toy_family
#>
#> [MCMC]
#> chainlength = 10000000
#>
#> [model lexicon]
#> model = covarion
#> data = toy_cognates_long.csv
#> rate_variation = True
#>
#> [clock default]
```

```
#> type = relaxed
#>
#> [calibration]
#> L21,L22,L23,L24 = uniform(129, 1129)
```

Running `beastling toy_family.conf` writes `toy_family.xml`, a 775-line BEAST 2 file with one likelihood per meaning, each with its own ascertainment correction. BEASTling leaves out the meanings that have only one cognate class, since they carry no information. There is a catch. BEASTling has not been updated since version 1.5.1 (2021). It writes XML for BEAST 2.6 and does not run under Python 3.14. BEAST 2.7 cannot read the old class names (it stops with "Class could not be found"), so the file must either be run with BEAST 2.6 or have its class names translated to the 2.7 namespace. We did the latter with a short script, `software-runs/beastling/beast26_to_27.py`, ran BEASTling 1.5.1 under Python 3.12, and ran the converted file with `beast toy_family_beast27.xml` in BEAST 2.7.7, which needs the BEASTLabs package for BEASTling's starting tree. The covarion model with a relaxed clock is much slower than the strict-clock analysis of the previous section, about 14 minutes per million states on one core, and a 500,000-state test run was far from converged (an ESS of 5 for the posterior). A real analysis of this kind needs tens of millions of states.

17.10.5 RevBayes

RevBayes (Höhna et al., 2016) takes a different approach from BEAST 2. The model is written as a script in the Rev language, which specifies a probabilistic graphical model one variable at a time. This makes RevBayes the most flexible of the four programs: any parameter can be given any prior, and new models can be assembled from existing pieces without writing Java or C++. The script below specifies an unrooted analysis of the binary matrix under the Mk model with gamma rate variation, conditioned on variable characters (the Mkv correction of Lewis (2001)), with two independent runs. It follows the structure of the RevBayes tutorials on discrete morphological characters. The excerpt below shows the data and the model; the moves, the monitors and the summary of the tree sample are left out.

```
# excerpt of toy_cognates.Rev (run with: rb toy_cognates.Rev)
data <- concatenate(readDiscreteCharacterData("toy_cognates_standard.nex"))
...
# tree prior: uniform topology, exponential branch lengths
topology ~ dnUniformTopology(taxa)
br_lens[i] ~ dnExponential(10.0)            # in a loop over the branches
phylogeny := treeAssembly(topology, br_lens)

# substitution model: symmetric two-state Mk model with gamma rates
Q <- fnJC(2)
alpha ~ dnExponential(1.0)
site_rates := fnDiscretizeGamma(alpha, alpha, 4)

# likelihood, conditioned on variable characters (Mkv)
seq ~ dnPhyloCTMC(tree = phylogeny, Q = Q, siteRates = site_rates,
                  type = "Standard", coding = "variable")
seq.clamp(data)
...
```

```
mymcmc = mcmc(mymodel, monitors, moves, nruns = 2, combine = "mixed")
mymcmc.run(generations = 10000)
```

The complete script, as it was run, is printed in Section E.3.3.

Two details of the script needed care in RevBayes 1.4.1. Because the NEXUS file defines a `CHARSET` for each meaning, `readDiscreteCharacterData()` returns 110 separate matrices, one per meaning, and `concatenate()` joins them again. The 14 characters present in every language that records the meaning are incompatible with `coding = "variable"`, and RevBayes drops them with a warning. The two runs of 10,000 iterations, each with about 76 moves per iteration, took 8 minutes on one core. After a 25% burn-in, every parameter had an ESS above 200 in each run, and the maximum a posteriori tree differed from the true tree in 4 of its 21 internal splits.

Replacing the symmetric Mk model with an asymmetric binary model, the `coding` option with an ascertainment scheme suited to presence/absence data, or the unrooted tree prior with a birth-death prior and a relaxed clock, are each changes of a few lines. The RevBayes tutorials on dating, clock models and model comparison show how.

An exported file is only as good as the matrix that went into it, and a Bayesian analysis is only as good as its priors. Three checks are worth making every time. First, re-import the exported file and compare it with the original matrix (as we did for the NEXUS file above). Second, run the analysis once with the data removed (BEAST 2 has a "sample from prior" option; in MrBayes use `mcmc data=no`; in RevBayes, simply omit the `clamp`) and inspect the resulting distributions of dates and rates. This prior-only run is part of implementation validation: it shows the prior that the program actually uses, which is not always the one the analyst wrote down. The marginal prior on the root age, after calibrations and tree prior have interacted, can be very different from what was intended (Chapter 10). Third, run at least two independent chains from different starting points and confirm that they agree.

17.11 Step 9: Working with posterior samples of trees

A Bayesian analysis produces not one tree but a sample of thousands of trees from the posterior distribution, together with a trace of every parameter. Summarizing this sample is a task in its own right. We use the output of the complete BEAST 2 analysis of Section 17.10.3: the binary model with ascertainment correction and gamma rate variation, a strict clock, a Yule prior, the subgroup-D calibration and a uniform prior between 3,000 and 12,000 years on the root age. We ran BEAST 2.7.7 twice, with seeds 1 and 2, for 3 million states each, sampling every 2,000 states. Each run took about seven minutes on one processor core. A real analysis would run longer, but here every parameter reached an effective sample size above 500 (see below). The book's repository keeps the two parameter logs in full and every tenth sampled tree (151 trees per run), so the code below runs offline.

17.11.1 Reading tree files and diagnosing convergence

The first steps are to read the tree files, discard the burn-in (the initial part of the chain, before it reached the region of high posterior probability), and check convergence. BEAST 2 and MrBayes write trees in NEXUS format, which `ape::read.nexus()` reads, and parameter traces as tab-delimited text. RevBayes writes its trees as a table with one Newick tree per row, which `read.tree(text = ...)` can parse.

```
beast_dir <- "software-runs/beast2"
runs <- c("run1", "run2")
read_trees <- function(run) {
  trees <- read.nexus(file.path(beast_dir, run, "toy_thinned.trees"))
  trees[-seq_len(floor(0.1 * length(trees)))]    # discard the first 10%
}
posterior <- do.call(c, lapply(runs, read_trees))
read_log <- function(run) {
  trace <- read.table(file.path(beast_dir, run, "toy.log"), header = TRUE,
                      comment.char = "#")
  trace[-seq_len(floor(0.1 * nrow(trace))), ]
}
traces <- lapply(runs, read_log)
pars <- c("posterior", "TreeHeight", "clockRate", "freqParameter.2",
          "gammaShape", "mrca.age.subgroupD.")
ess <- sapply(traces, function(tr) coda::effectiveSize(tr[, pars]))
colnames(ess) <- runs
psrf <- coda::gelman.diag(coda::mcmc.list(lapply(traces, function(tr)
  coda::mcmc(tr[, pars]))), autoburnin = FALSE)$psrf[, 1]
round(cbind(ess, PSRF = psrf), 2)
```

```
#>                       run1   run2 PSRF
#> posterior           1351.0 1351.0    1
#> TreeHeight          1351.0 1475.2    1
#> clockRate            743.2  602.1    1
#> freqParameter.2      516.5  492.0    1
#> gammaShape          1351.0  759.8    1
#> mrca.age.subgroupD. 1332.7 1351.0    1
```

```
c(trees = length(posterior),
  root_age_median = median(unlist(lapply(traces, `[[`, "TreeHeight"))))
```

```
#>           trees root_age_median
#>          272.00            3.65
```

Every parameter has an ESS of several hundred in each run. The potential scale reduction factor (PSRF) of Gelman and Rubin (1992) compares the variance within and between the two runs; values close to 1 indicate that they have converged to the same distribution. The posterior median of the root age, about 3,600 years, is well below the true 6,000 years. With a strict clock, the absolute dates rest on the calibrations and priors, and our one young calibration, together with the prior on the clock rate, pulls the whole tree toward the lower end of the root-age prior. This is a reminder that dates are only as good as the calibrations (Chapter 10). In the terms of Section 6.10, the absolute root age is identifiable under the model only through the calibration and the priors. With a single young calibration it is not practically estimable from these data, even though the true value lies inside the 95% interval.

TreeAnnotator, distributed with BEAST 2, summarizes the tree sample as a maximum clade credibility tree (Section 17.11.2) and annotates each node with the median and 95% highest posterior density (HPD) interval of its age. We ran it on the two runs combined, after a 10% burn-in, with

median node heights:

```
treeannotator -burnin 0 -height median combined.trees toy_mcc.tree
```

treeio reads the annotated tree, and Figure 17.6 draws the HPD intervals as bars. We draw the bars with `geom_segment()` from the stored interval limits. **ggtree**'s `geom_range()` gives bars of the right width, but it centers them on the node, which misplaces skewed intervals such as the root's.

```
mcc_beast <- treeio::read.beast(file.path(beast_dir, "toy_mcc.tree"))
p_hpd <- revts(ggtree(mcc_beast))           # x axis: time before the present
nodes <- subset(p_hpd$data, !isTip)
hpd <- do.call(rbind, nodes$height_0.95_HPD)
nodes$lower <- -hpd[, 2]
nodes$upper <- -hpd[, 1]
p_hpd +
  geom_segment(data = nodes, aes(x = lower, xend = upper, y = y, yend = y),
               color = "#0072B2", alpha = 0.4, linewidth = 2) +
  geom_tiplab(size = 3, offset = 0.06) + theme_tree2() +
  labs(x = "Thousands of years before present")
```

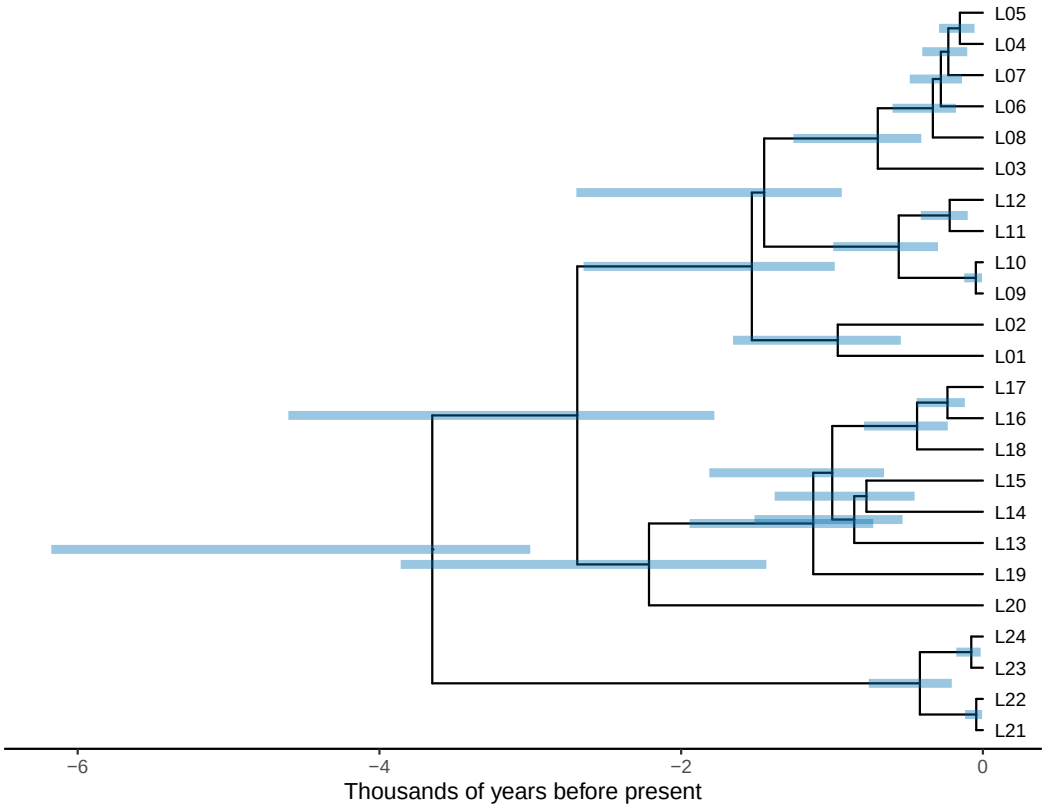


Figure 17.6: The maximum clade credibility tree from TreeAnnotator, with median node ages and 95 percent HPD intervals of node ages (bars), from the BEAST 2 analysis.

The *effective sample size* (ESS) of a parameter estimates how many independent draws the autocorrelated MCMC sample is worth (Chapter 6). Tracer (Rambaut et al., 2018) reports it for BEAST

and MrBayes logs, and the R package **coda** (Plummer et al., 2006) computes it from any trace. A common rule of thumb asks for an ESS of at least 200 for every parameter of interest. We can see what ESS measures with a simulated trace that starts far from its stationary distribution and is strongly autocorrelated, as an unconverged log-likelihood trace typically is (Figure 17.7).

```
n_iter <- 2000
ar_part <- as.numeric(arima.sim(list(ar = 0.95), n = n_iter, sd = 2))
loglik_trace <- -2400 + ar_part - 300 * exp(-(seq_len(n_iter)) / 60)
burn <- seq_len(0.1 * n_iter)
op <- par(mar = c(4, 4, 1, 1))
plot(loglik_trace, type = "l", col = "grey30", xlab = "sample",
     ylab = "log-likelihood")
rect(0, min(loglik_trace) - 10, max(burn), max(loglik_trace) + 10,
     col = adjustcolor("#D55E00", 0.15), border = NA)
```

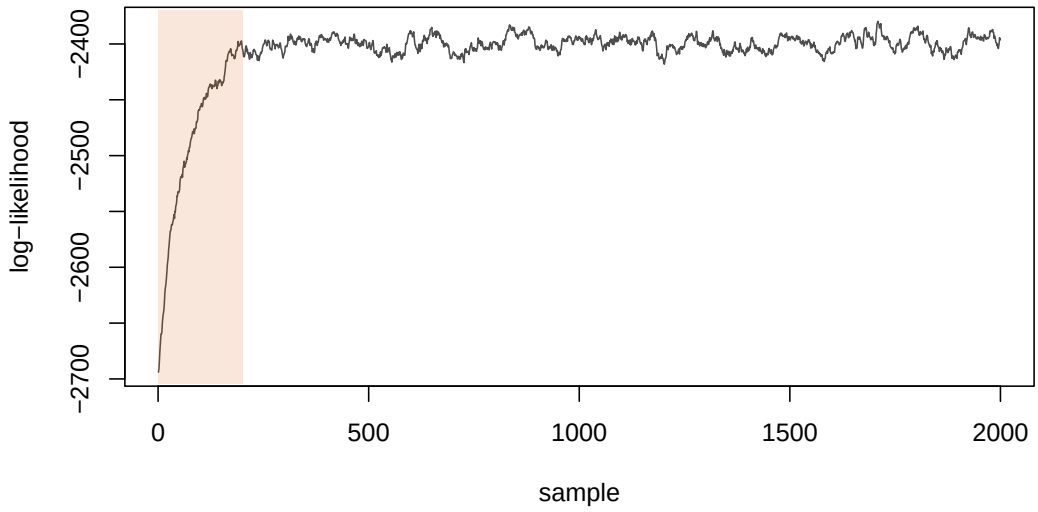


Figure 17.7: A simulated MCMC trace of the log-likelihood. The first 10 percent of the samples (shaded) are a burn-in phase that has not yet reached the stationary distribution.

```
par(op)
round(c(ESS_all = unname(coda::effectiveSize(loglik_trace)),
        ESS_after_burnin = unname(coda::effectiveSize(loglik_trace[-burn]))))
```

```
#>          ESS_all ESS_after_burnin
#>               18               42
```

Two thousand samples are worth only a few dozen independent ones, because each sample is highly correlated with the previous one. Including the burn-in makes things look even more misleading: the trend at the start inflates the variance and distorts the estimate. The remedy for a low ESS is to run the chain longer, to thin less aggressively, or to improve the mixing with better proposals, not to report the estimates anyway. For tree topologies, which have no single numerical value, convergence is assessed by comparing split frequencies between independent runs (the ASDSF reported by MrBayes, Chapter 6) or with tools such as **RWTY** (Warren et al., 2017). The R package **convenience** (Fabreti and Höhna, 2022) combines checks for continuous

parameters and for splits and reports whether a run passes them, and Barido-Sottani et al. (2023) give a practical guide to diagnosing runs that do not.

17.11.2 Summary trees

There is no single best way to summarize a posterior sample of trees (Chapter 2). The *maximum clade credibility* (MCC) tree is the tree in the sample that maximizes the product of the posterior probabilities of its clades; it is the default in BEAST's TreeAnnotator and in **phangorn**'s `maxCladeCred()` (Heled and Bouckaert, 2013). The *majority-rule consensus* tree contains exactly those clades found in more than half the trees, and is the default in MrBayes. Each internal node of either tree is labeled with its *clade credibility*, the proportion of posterior trees that contain the clade.

```
mcc <- maxCladeCred(posterior)
clade_cred <- prop.clades(mcc, posterior, rooted = TRUE) / length(posterior)
mcc$node.label <- sprintf("%.2f", clade_cred)
mcc$node.label[1] <- ""                          # the root is always present
mrc <- consensus(posterior, p = 0.5, rooted = TRUE)
c(MCC_internal_nodes = mcc$Nnode, consensus_internal_nodes = mrc$Nnode,
  MCC_RF_to_truth = RF.dist(unroot(mcc), truth_unrooted, normalize = TRUE))
```

```
#>       MCC_internal_nodes consensus_internal_nodes
#>                  23.0000                  21.0000
#>          MCC_RF_to_truth
#>                   0.1429
```

The MCC tree is fully resolved, because it is one of the sampled trees, whereas the majority-rule consensus collapses every clade with support below 50% and is therefore less resolved. Which to report depends on the purpose. The MCC tree is convenient for plotting dates, but it can contain clades with very low support, and its node heights must be summarized separately (TreeAnnotator offers mean, median or common-ancestor heights; Chapter 6). Figure 17.8 shows the MCC tree with clade credibilities.

```
p_mcc <- ggtree(mcc) %<+% tip_info +
  geom_tippoint(aes(color = subgroup), size = 1.6) +
  geom_tiplab(aes(color = subgroup), size = 3, offset = 0.08,
              show.legend = FALSE) +
  support_layers(2.5) +
  scale_color_manual(values = sg_cols, name = "Subgroup") +
  theme_tree2()
revts(p_mcc) + scale_x_continuous(labels = abs, limits = c(-6.2, 0.8)) +
  labs(x = "Thousands of years before present")
```

17.11.3 Visualizing uncertainty

A summary tree hides the uncertainty that the posterior sample expresses. A *DensiTree* (Bouckaert, 2010) draws every tree in the sample on top of the others with transparent lines, so that well-supported parts of the tree appear as sharp, dark branches and uncertain parts as a blurred cloud (Figure 17.9). The original DensiTree program is distributed with BEAST 2; **phangorn** provides an R implementation.

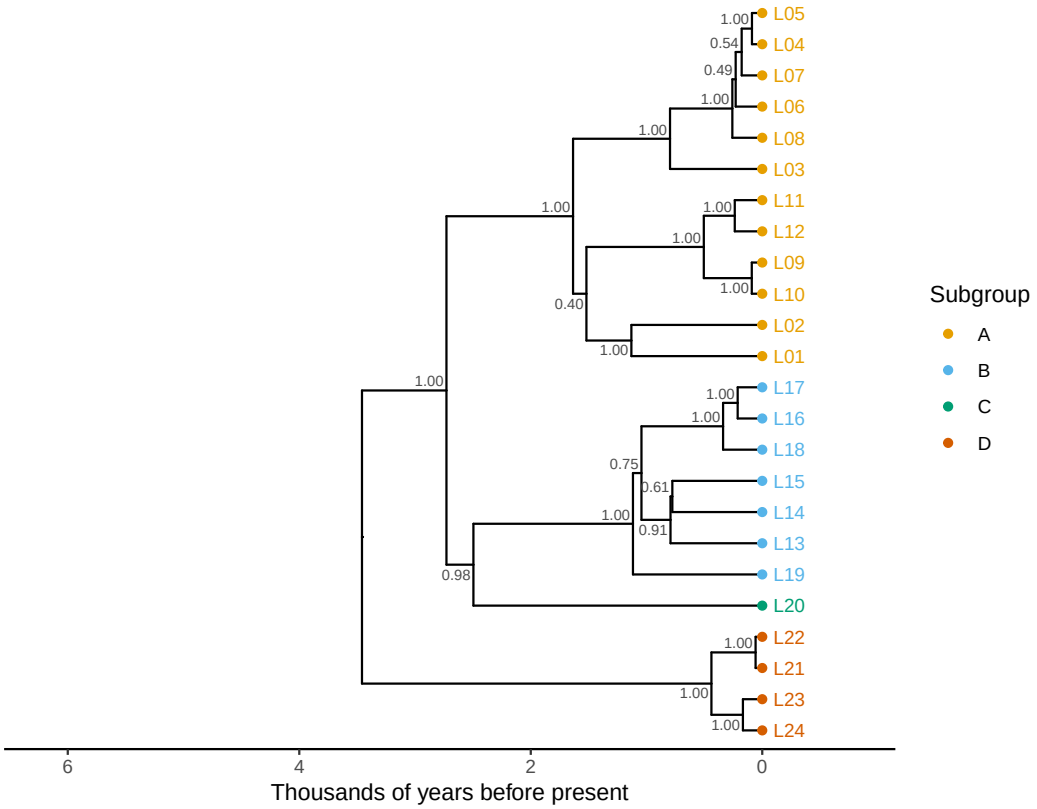


Figure 17.8: Maximum clade credibility tree of the BEAST 2 posterior sample, computed with phangorn, with clade credibilities at internal nodes and tips colored by true subgroup. The time axis is in thousands of years before the present.

```
densiTree(posterior, consensus = mcc, alpha = 0.03, col = "#0072B2",
          cex = 0.7, type = "phylogram")
```

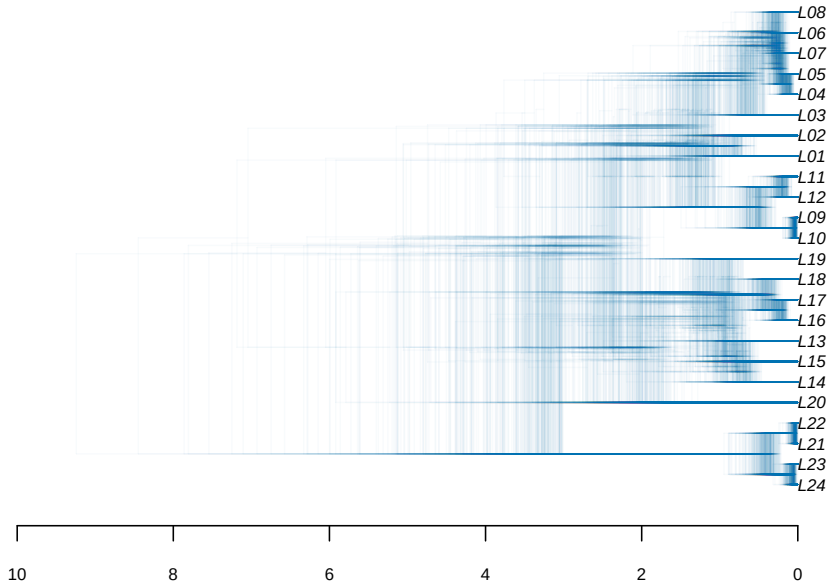


Figure 17.9: DensiTree plot of the BEAST 2 posterior sample of trees. Sharp regions indicate well-supported clades and node heights; blurred regions indicate uncertainty.

The sample can also be compared with the truth tree by tree. The distribution of RF distances between the posterior trees and the true tree shows how much topological uncertainty remains.

```
rf_post <- sapply(posterior, function(tr)
  RF.dist(unroot(tr), truth_unrooted, normalize = TRUE))
summary(rf_post)
```

```
#>    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
#>  0.0476  0.1429  0.1429  0.1647  0.1905  0.2857
```

The stored output of the other programs can be scored in the same way. Table 17.4 compares the summary tree of each program with the true tree, using the function `score_tree()` of Step 4. For the two dated analyses it also gives the median root age.

```
sr <- "software-runs"
mb_con <- treeio::read.mrbayes(
  file.path(sr, "mrbayes", "toy_mrbayes_run.nex.con.tre"))
summary_trees <- list(
  `MrBayes, consensus` = mb_con@phylo,
```

```
  `BEAST 2, MCC (TreeAnnotator)` = mcc_beast@phylo,
  `RevBayes, MAP tree` =
    read.nexus(file.path(sr, "revbayes", "output", "toy_map.tre")),
  `IQ-TREE, ML tree` =
    read.tree(file.path(sr, "iqtree", "ch17_gtr2.treefile")),
  `RAxML-NG, ML tree` =
    read.tree(file.path(sr, "raxml-ng", "ch17.raxml.bestTree")))
root_age <- function(td, col) {          # median age of the root node
  d <- treeio::get.data(td)
  as.numeric(d[[col]][d$node == Ntip(td@phylo) + 1])
}
programs <- data.frame(t(sapply(summary_trees, score_tree)),
  root_age = c(root_age(mb_con, "age_median"),
               root_age(mcc_beast, "height_median"), NA, NA, NA))
knitr::kable(programs, digits = 2, caption = paste(
  "Summary trees from the external programs compared with the true tree:",
  "normalized Robinson-Foulds distance, true subgroups recovered, and",
  "median root age in thousands of years (true value 6)."))
```

Table 17.4: Summary trees from the external programs compared with the true tree: normalized Robinson-Foulds distance, true subgroups recovered, and median root age in thousands of years (true value 6).

	RF_normalized	subgroups_recovered	subgroups_total	root_age
MrBayes, consensus	0.14	3	3	4.08
BEAST 2, MCC (TreeAnnotator)	0.14	3	3	3.65
RevBayes, MAP tree	0.19	2	3	NA
IQ-TREE, ML tree	0.19	3	3	NA
RAxML-NG, ML tree	0.19	3	3	NA

All five programs recover the same tree up to a few splits. The differences from the truth are again on the short internal branches. Agreement among programs is a weak form of implementation validation. The programs were written independently, so a gross error in one of them would show up as a disagreement. But they share models, conventions and file formats, and they did not run identical models here, so agreement cannot show that any of them is exactly right. MrBayes and BEAST 2 date the root at about 4,000 and 3,600 years, with 95% intervals that reach 7,600 and 7,000 years. Both are pulled toward the lower bound of the root-age prior, for the reason given above.

17.12 Step 10: A comparative analysis over a sample of trees

Language trees are increasingly used not as ends in themselves but as scaffolds for testing hypotheses about other traits, as Chapters 13 and 14 describe. The basic problem is Galton's problem: related languages are not independent observations, so a correlation across languages can arise

from shared inheritance alone. Phylogenetic comparative methods solve this by modeling the expected covariance among languages given the tree. And since the tree is itself uncertain, a careful analysis repeats the comparative analysis over a sample of trees and reports the spread of results.

We illustrate with a continuous predictor x and a continuous response y. Think of x as the logarithm of the speaker population and y as some numerical typological index, although nothing hinges on the interpretation. We simulate both by Brownian motion on the true tree (Chapter 13), with a true regression slope of 0.6:

$$y = \beta_0 + \beta_1 x + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2 \mathbf{C}), \tag{17.6}$$

where $\mathbf{C}$ is the matrix of shared branch lengths between pairs of languages implied by the tree. Phylogenetic generalized least squares (PGLS) fits this model by generalized least squares (Grafen, 1989); ordinary least squares (OLS) wrongly assumes $\mathbf{C} = \mathbf{I}$.

```
set.seed(2)   # a separate seed, so this section does not depend on earlier ones
x <- fastBM(true_tree, sig2 = 1)
y <- 0.6 * x + fastBM(true_tree, sig2 = 0.5)
traits <- data.frame(lang = names(x), x = x, y = y[names(x)])
fit_ols <- lm(y ~ x, data = traits)
fit_pgls_true <- gls(y ~ x, data = traits,
                     correlation = corBrownian(1, true_tree, form = ~lang))
rbind(OLS = coef(summary(fit_ols))["x", c(1, 2, 4)],
      `PGLS, true tree` = coef(summary(fit_pgls_true))["x", c(1, 2, 4)])
```

```
#>                 Estimate Std. Error Pr(>|t|)
#> OLS               0.7062     0.1791 0.000692
#> PGLS, true tree   0.5418     0.1840 0.007497
```

Now we repeat the PGLS on every tree in the posterior sample, as one would with a real posterior. We also estimate Pagel's λ (Pagel, 1999a) for the response on every tree, as a measure of phylogenetic signal (Chapter 13).

```
pgls_one <- function(tr) {
  m <- try(gls(y ~ x, data = traits,
               correlation = corBrownian(1, tr, form = ~lang)), silent = TRUE)
  if (inherits(m, "try-error")) return(c(slope = NA, se = NA, p = NA))
  cf <- coef(summary(m))["x", ]
  c(slope = cf[["Value"]], se = cf[["Std.Error"]], p = cf[["p-value"]])
}
pgls_post <- as.data.frame(t(sapply(posterior, pgls_one)))
lambda_post <- sapply(posterior, function(tr)
  phylosig(tr, setNames(traits$y, traits$lang), method = "lambda")$lambda)
round(rbind(
  slope = quantile(pgls_post$slope, c(0.025, 0.5, 0.975), na.rm = TRUE),
  lambda = quantile(lambda_post, c(0.025, 0.5, 0.975))), 3)
```

```
#>         2.5%   50% 97.5%
#> slope  0.371 0.491 0.595
#> lambda 0.999 1.007 1.018
```

```
c(share_p_below_0.05 = mean(pgls_post$p < 0.05, na.rm = TRUE))
```

```
#> share_p_below_0.05
#>             0.9412
```

```
ggplot(pgls_post, aes(x = slope)) +
  geom_histogram(bins = 15, fill = "grey70", color = "white") +
  geom_vline(xintercept = 0.6, linetype = "dashed") +
  geom_vline(xintercept = coef(fit_ols)[["x"]], linetype = "dotted") +
  geom_vline(xintercept = coef(fit_pgls_true)[["x"]]) +
  labs(x = "PGLS slope estimate", y = "Number of trees") +
  theme_minimal(base_size = 11)
```

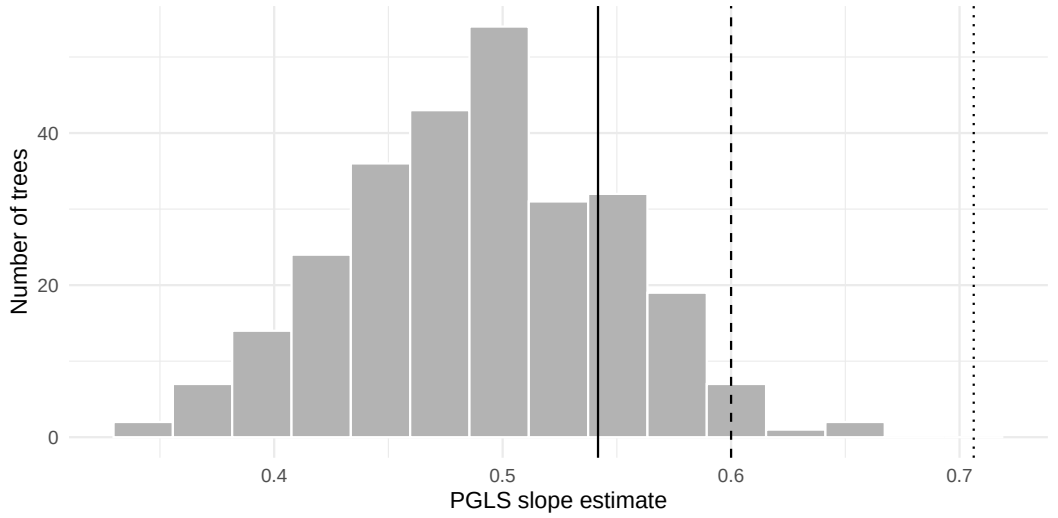


Figure 17.10: Distribution of PGLS slope estimates across the trees of the BEAST 2 posterior sample. The dashed line is the true slope (0.6), the dotted line the OLS estimate, and the solid line the PGLS estimate on the true tree.

Three things are worth noticing in these results. First, phylogenetic signal in the response is strong (λ close to 1 on every tree), as expected for a trait that evolves by Brownian motion, so ignoring the tree is not an option. Second, the OLS and PGLS point estimates (0.71 and 0.54 against a true value of 0.6) need not differ much in any one dataset. The main problem with OLS is that it treats the 24 languages as independent data points, whereas much of their variation is shared within subgroups. Its standard errors are therefore too small, and with phylogenetically structured data it rejects true null hypotheses far more often than its nominal error rate (Chapter 13). Third, the spread of PGLS estimates across the tree sample is a direct measure of how much phylogenetic uncertainty matters for the conclusion. Here 94% of the trees yield a significant positive slope, so the conclusion is robust to tree uncertainty. In a real study the spread is often wider, and it should be reported. This is also why a downstream analysis should not be run on the MCC tree alone. The MCC tree is one tree from the sample, with summary node heights; using it as if it were known treats every clade, including those with low credibility, as certain, and it can shift or narrow the estimate even when its topology looks plausible. A fully Bayesian alternative integrates over trees within a single analysis. BayesTraits, for example, samples a new tree from the posterior

at each step of its chain (Pagel et al., 2004). Alternatively, one can fit a Bayesian phylogenetic mixed model separately on each tree and pool the posterior samples, as in the following code with **MCMCglmm** (Hadfield, 2010). Twenty trees and short chains keep it fast; a real analysis would use more of both.

```
# a Bayesian phylogenetic mixed model over 20 trees of the posterior sample
library(MCMCglmm)
set.seed(4)
prior <- list(G = list(G1 = list(V = 1, nu = 0.002)),
              R = list(V = 1, nu = 0.002))
some_trees <- posterior[round(seq(1, length(posterior), length.out = 20))]
draws <- lapply(some_trees, function(tr) {
  Ainv <- inverseA(tr, nodes = "TIPS")$Ainv
  m <- MCMCglmm(y ~ x, random = ~lang, ginverse = list(lang = Ainv),
                data = traits, prior = prior, nitt = 13000, burnin = 3000,
                thin = 10, verbose = FALSE)
  m$Sol[, "x"]
})
quantile(unlist(draws), c(0.025, 0.5, 0.975))  # slope, integrating over trees
```

```
#>    2.5%     50%   97.5%
#> 0.09289 0.50572 0.91922
```

17.13 Step 11: From toy data to real data

Everything above runs on a simulated table with three columns. Replacing it with real data is mostly a matter of reading the right files and joining them correctly. Real linguistic datasets are increasingly published in the Cross-Linguistic Data Formats (CLDF) (Forkel et al., 2018), a standard built on plain CSV files plus a JSON metadata file that describes their contents. Lexibank (List et al., 2022a) distributes more than a hundred wordlists in CLDF, with concepts linked to Concepticon (List et al., 2016) and languages linked to Glottolog (Hammarström et al., 2024); Grambank (Skirgård et al., 2023) distributes structural data in the same format. Appendix G describes these and other resources.

17.13.1 The CLDF data model

A CLDF dataset is a directory of tables, each of a standard type, plus a metadata file (for example `cldf-metadata.json` or `Wordlist-metadata.json`) that declares which file holds which table and which columns hold which standard properties. Table 17.5 lists the tables most relevant to phylogenetics.

Table 17.5: The CLDF tables most often used in phylogenetic work.

CLDF component	Usual file	Key columns	Content
`LanguageTable`	`languages.csv`	`ID`, `Name`, `Glottocode`, `Latitude`, `Longitude`	one row per language variety
`ParameterTable`	`parameters.csv`	`ID`, `Name`, `Concepticon_ID`	one row per concept (wordlists) or feature (structural data)

CLDF component	Usual file	Key columns	Content
`FormTable`	`forms.csv`	`ID`, `Language_ID`, `Parameter_ID`, `Value`, `Form`, `Segments`	one row per word form
`CognateTable`	`cognates.csv`	`ID`, `Form_ID`, `Cognateset_ID`	one row per cognate judgment
`ValueTable`	`values.csv`	`ID`, `Language_ID`, `Parameter_ID`, `Value`, `Code_ID`	one row per language-feature value (structural data)
`CodeTable`	`codes.csv`	`ID`, `Parameter_ID`, `Name`	the possible values of each categorical feature

Because every table refers to the others by identifiers (`Language_ID`, `Parameter_ID`, `Form_ID`), building the input for `cognates_to_binary()` amounts to two joins. The code below is not run when the book is built, because it requires a downloaded dataset. We tested it on `walworth-polynesian` (version 1.1), a Lexibank wordlist of 31 Polynesian languages with expert cognate judgments. There it produces a matrix of 31 languages and 1,456 cognate characters in a few seconds.

```
# not run: requires a Lexibank dataset with cognate judgments, e.g.
#   git clone https://github.com/lexibank/<dataset>.git
# (tested with <dataset> = walworthpolynesian, release v1.1)
library(readr)
library(dplyr)
cldf_dir  <- "<dataset>/cldf"
forms     <- read_csv(file.path(cldf_dir, "forms.csv"), show_col_types = FALSE)
cognates  <- read_csv(file.path(cldf_dir, "cognates.csv"),
                      show_col_types = FALSE)
languages <- read_csv(file.path(cldf_dir, "languages.csv"),
                      show_col_types = FALSE)

cog_real <- cognates |>
  inner_join(forms, by = c("Form_ID" = "ID")) |>
  transmute(Language_ID, Parameter_ID,
            # make cognate set identifiers unique within meanings
            Cognateset_ID = paste(Parameter_ID, Cognateset_ID, sep = "-")) |>
  distinct()

bin_real <- cognates_to_binary(cog_real)

# label languages by Glottocode, to join with other datasets later;
# check the identifiers before anything is built from the matrix
glotto <- setNames(languages$Glottocode, languages$ID)
ids <- unname(glotto[rownames(bin_real)])
if (anyNA(ids))
  stop("no Glottocode for: ",
       paste(rownames(bin_real)[is.na(ids)], collapse = ", "))
if (anyDuplicated(ids))
  stop("varieties sharing a Glottocode: ",
```

```
      paste(unique(ids[duplicated(ids)]), collapse = ", "))
rownames(bin_real) <- ids
data_real <- phyDat(bin_real, type = "USER", levels = c("0", "1"),
                    ambiguity = "?")
```

Several practical issues arise with real data that the toy example hides. Some datasets list more than one variety per Glottocode, so labeling by Glottocode can create duplicate row names; decide in advance whether to keep one variety per language or to use the dataset's own identifiers. Some cognate judgments are flagged as doubtful (a `Doubt` column in the `CognateTable`), or as loans (a `Loan` column in the `FormTable` of some datasets), and a principled decision is needed about whether to exclude them, code them as missing, or run the analysis both ways (Chapter 8). Datasets also differ in their concept lists, so combining two datasets means matching concepts through their Concepticon identifiers rather than their names. The R package **rcldf** reads a CLDF dataset directly from its metadata file and returns its tables as data frames, which saves hard-coding file names.

```
# not run: rcldf is available from GitHub (SimonGreenhill/rcldf);
# tested with rcldf 1.6.3
ds <- rcldf::cldf("<dataset>/cldf/cldf-metadata.json")
names(ds$tables)   # "FormTable", "LanguageTable", "CognateTable", ...
head(ds$tables$FormTable)
```

17.13.2 When there are no cognate judgments

Many Lexibank datasets contain word forms segmented into sounds but no expert cognate judgments. For these, automated cognate detection (Chapter 3) can supply provisional cognate sets. The standard tool is the LexStat algorithm in the Python library LingPy (List et al., 2017), which learns language-pair-specific sound correspondences and clusters words within each meaning. The following Python code (not run) reads a Lexibank dataset, runs LexStat, and writes a table with a cognate set identifier for each word, which can then be read into R and passed to `cognates_to_binary()`.

```
# not run: Python with lingpy installed (pip install lingpy);
# tested with lingpy 2.6.14
from lingpy import Wordlist, LexStat
wl = Wordlist.from_cldf("<dataset>/cldf/cldf-metadata.json")
wl.output("tsv", filename="wordlist")          # writes wordlist.tsv
lex = LexStat("wordlist.tsv")
lex.get_scorer(runs=10000)       # permutation test for correspondences
lex.cluster(method="lexstat", threshold=0.55, ref="cogid")
lex.output("tsv", filename="wordlist-cognates")
```

We ran this code on the Polynesian dataset used in Section 17.14. The permutation test is the slow step, and the whole run took about 20 minutes on one processor core. LexStat grouped the 7,518 words into 1,897 cognate sets, where the experts recognize 1,456. Compared with the expert judgments, the automatic sets had a B-cubed precision of 0.95 and a recall of 0.79. The algorithm rarely groups words that the experts keep apart, but it splits some expert cognate sets, typically where sound change has made related words look dissimilar.

How much does it matter whether cognates are judged by experts or by an algorithm? Rama

et al. (2018) addressed this question directly. They inferred cognate sets automatically for several language families with expert cognate judgments available, built binary matrices from both the automatic and the expert cognates, inferred Bayesian trees from each, and compared the trees with expert classifications from Glottolog. Trees from automatically detected cognates came reasonably close to those from expert cognates, although expert judgments generally remained better. The study suggests a practical division of labor: automated detection makes it possible to analyze families for which no expert cognate coding exists and to check expert codings for inconsistencies, but it is not yet a substitute for expert judgment where that is available. Jäger (2018) pushed the automated approach to a global scale, inferring cognate classes and trees from the ASJP word lists for thousands of languages.

17.13.3 Structural data and Glottolog trees

Grambank (Skirgård et al., 2023) codes nearly two hundred grammatical features for more than two thousand languages, mostly as binary presence/absence features. Its CLDF `ValueTable` can be reshaped into a matrix and analyzed with exactly the tools of Steps 4 to 7 (with the caveats about structural data discussed in Chapter 3), or, more commonly, used as trait data in a comparative analysis over a lexical tree. Since both Grambank and Lexibank identify languages by Glottocode, joining them is straightforward.

```
# not run: requires a clone of https://github.com/grambank/grambank
# (tested with release v1.0.3)
library(tidyr)
gb_values <- read_csv("grambank/cldf/values.csv", show_col_types = FALSE)
gb <- gb_values |>
  select(Language_ID, Parameter_ID, Value) |>
  pivot_wider(names_from = Parameter_ID, values_from = Value)
gb_mat <- as.matrix(gb[, -1])
rownames(gb_mat) <- gb$Language_ID            # Grambank uses Glottocodes
gb_mat[is.na(gb_mat)] <- "?"                  # unrecorded -> missing

# keep the languages in both the lexical tree and Grambank
shared <- intersect(rownames(gb_mat), mcc_real$tip.label)
# mcc_real: a summary tree from a real analysis, tips labeled by Glottocode
tree_gb <- keep.tip(mcc_real, shared)
trait <- gb_mat[shared, 1]                     # any one feature as a trait

# multistate features can be split into binary indicators before analysis
```

We tested this code on Grambank 1.0.3, with a neighbor-joining tree of the 31 Polynesian languages of `walworthpolynesian`, labeled by Glottocode, standing in for `mcc_real`. It builds a matrix of 2,467 languages by 195 features, and 26 of the 31 Polynesian languages are in Grambank.

For comparisons across many families, no dated lexical tree exists, and analysts often fall back on the Glottolog classification as a topology, with branch lengths assigned by a rule such as Grafen's method (`ape::compute.brlen()`), or on a global tree inferred from ASJP data. Chapter 13 discusses the consequences of such arbitrary branch lengths. Reading Glottolog's Newick trees requires some care, because the node labels contain characters with special meaning in Newick; community R packages such as **glottoTrees** help with parsing, pruning and combining them.

17.13.4 Scaling up the inference

For a few dozen languages, **phangorn** is fast enough for maximum-likelihood analysis. For hundreds of languages, dedicated programs are much faster. IQ-TREE (Minh et al., 2020b) and RAxML-NG (Kozlov et al., 2019) both support binary characters, ascertainment correction and fast bootstrap approximations. Both read the PHYLIP format, which **phangorn** writes directly.

```
phylip_file <- file.path(out_dir, "toy_cognates.phy")
write.phyDat(cog_var, file = phylip_file, format = "phylip")
show_file(phylip_file, n = 3)
```

```
#> 24 430
#> L01 0100101000000001000001001010100000000010100010001000????100010000000 ...
#> L02 1000??1000000000100001001010001000000010100010010000010010001000000 ...
```

```
# run in a shell, not in R: maximum likelihood with a binary model,
# estimated state frequencies, gamma rates, ascertainment correction
# (variable characters only) and 1000 ultrafast bootstrap replicates
# (IQ-TREE 3 is called iqtree3; IQ-TREE 2 is called iqtree2)
iqtree3 -s toy_cognates.phy -m GTR2+FO+G4+ASC -B 1000

# the corresponding RAxML-NG analysis with 100 standard bootstrap replicates
raxml-ng --all --msa toy_cognates.phy --model BIN+G+ASC_LEWIS --bs-trees 100
```

The resulting tree files are Newick, read with `ape::read.tree()`, and bootstrap tree sets can be summarized with the same functions we used above.

BEASTling (Maurits et al., 2017) was designed to make the workflow of this chapter easier for linguists. Its authors observed that writing BEAST 2 XML by hand was a serious barrier, and that the files were rarely published, which made analyses hard to reproduce. BEASTling instead takes a short human-readable configuration file and a data file, and generates the XML. Because the configuration file is short, it can be printed in full in a paper's supplementary material, and anyone with the same data and BEASTling version can regenerate the identical XML. BEASTling also integrates Glottolog, so that languages can be selected by family or macro-area and Glottolog subgroups can be imposed as monophyly constraints, and it supports models for lexical and typological data in the same analysis. The approach illustrates a general principle: the more of a pipeline that is specified in short, declarative, version-controlled files, the easier it is to reproduce and to vary.

Lexibank (List et al., 2022a) is an attempt to do for lexical data what GenBank did for DNA sequences. It aggregates more than a hundred published wordlists, converts each to CLDF with a reproducible conversion script (written with the `cldfbench` and `pylexibank` Python packages), links every concept to Concepticon and every language to Glottolog, and standardizes phonetic transcriptions so that words can be segmented into sounds automatically. The result is a single collection of wordlists covering thousands of language varieties, in which every dataset is versioned, archived on Zenodo and citable. For the workflow of this chapter the consequence is that Step 1 (data assembly) and much of Step 2 (matrix building) can be done with a few lines of code for any dataset in the collection, and that an analysis can cite the exact version of the data it used.

17.14 A case study with real data: Polynesian

We now run the same pipeline on a real language family. The data are the Polynesian wordlist of Walworth (2018) in its Lexibank version (Walworth, 2024), released under a CC BY 4.0 licence. It records Blust's list of 210 basic meanings for 30 Polynesian languages and a Proto-Polynesian reconstruction, with expert cognate judgments. Polynesian is a good first family: it is small, its history is short (the Polynesian homeland in Tonga and Samoa was settled roughly three thousand years ago) (Gray et al., 2009), and its main subgroups are well established. The four CLDF tables the analysis needs, about 1.5 MB, are stored in the book's repository in `data/case-study/`, together with the Glottolog classification of each language (Hammarström et al., 2024), so the section runs offline.

17.14.1 From CLDF to a character matrix

The code is that of Section 17.13.1, with two decisions specific to this dataset. The Proto-Polynesian "language" is a reconstruction made with the comparative method from the daughter languages. It is not independent evidence, so we leave it out. And 213 forms are flagged as loans in the `Loan` column of the `FormTable`. We drop them, so that a meaning whose only recorded word in a language is a loan is scored as missing for that language (Chapter 8).

```
library(readr)
library(dplyr)
cs_dir <- "data/case-study"
read_cs <- function(f) read_csv(file.path(cs_dir, f), show_col_types = FALSE)
forms_p     <- read_cs("forms.csv")
cognates_p  <- read_cs("cognates.csv")
languages_p <- read_cs("languages.csv")
cog_poly <- cognates_p |>
  inner_join(forms_p, by = c("Form_ID" = "ID")) |>
  filter(Language_ID != "Polynesian",     # Proto-Polynesian, a reconstruction
         !Loan) |>                        # forms flagged as loans
  transmute(Language_ID, Parameter_ID,
            Cognateset_ID = paste(Parameter_ID, Cognateset_ID, sep = "-")) |>
  distinct()
bin_poly <- cognates_to_binary(cog_poly)
# identifiers: the dataset's language IDs, which are readable names;
# check them and their Glottocodes before building anything from the matrix
glotto_poly <- setNames(languages_p$Glottocode,
                        languages_p$ID)[rownames(bin_poly)]
stopifnot(!anyDuplicated(rownames(bin_poly)), !anyNA(glotto_poly),
          !anyDuplicated(glotto_poly))
data_poly <- phyDat(bin_poly, type = "USER", levels = c("0", "1"),
                    ambiguity = "?")
c(languages = nrow(bin_poly), meanings = length(unique(cog_poly$Parameter_ID)),
  characters = ncol(bin_poly), missing = round(mean(bin_poly == "?"), 3),
  singletons = sum(colSums(bin_poly == "1") == 1))
```

```
#>  languages   meanings characters    missing singletons
#>     30.000    210.000   1396.000      0.071    769.000
```

The Glottolog classification gives the subgroups we will look for. We take every Glottolog subgroup below Polynesian that contains at least two of the 30 languages.

```
gl <- read_cs("glottolog_classification.csv")
lineage <- strsplit(setNames(gl$Classification, gl$Glottocode)[glotto_poly],
                    " > ")
names(lineage) <- rownames(bin_poly)
groups <- unique(unlist(lapply(lineage, `[`, -1)))
members <- lapply(setNames(groups, groups), function(g)
  names(lineage)[vapply(lineage, function(l) g %in% l, logical(1))])
members <- members[lengths(members) >= 2 & lengths(members) < nrow(bin_poly)]
lengths(members)
```

```
#>                                    Nuclear Polynesian
#>                                                    28
#>                              Vanuatu-Loyalty Outliers
#>                                                     3
#>                                           Mele-Futuna
#>                                                     2
#>          Northern Outlier Polynesian-East Polynesian
#>                                                    16
#> Solomons Northern Outlier Polynesian-East Polynesian
#>                                                    15
#>  Central Northern Outlier Polynesian-East Polynesian
#>                                                    14
#>                                        East Polynesian
#>                                                    12
#>                               East Polynesian Proximal
#>                                                     9
#>                  Central Northern Outlier Polynesian
#>                                                     2
#>                                East Polynesian Distal
#>                                                     3
#>                                   Far East Polynesian
#>                                                     2
#>                    Southern East Polynesian Proximal
#>                                                     8
#>                                               Tongic
#>                                                     2
#>                                             Ellicean
#>                                                     3
#>                                            Pukapukic
#>                                                     2
#>                                     Tahitian-Austral
#>                                                     3
#>                                              Austral
#>                                                     2
```

These are nested. Tongic (Tongan and Niuean) and Nuclear Polynesian (all the rest) form the pri-

mary split. Within Nuclear Polynesian, the East Polynesian languages, from Hawaiian to Rapanui and Māori, form the most famous subgroup. They descend from the language of the settlers who reached East Polynesia less than a thousand years ago (Wilmshurst et al., 2011). Several intermediate groupings link East Polynesian with Polynesian Outliers spoken in the Solomon Islands and Micronesia.

17.14.2 Exploration and maximum likelihood

The delta score and the NeighborNet of Step 3 show how tree-like the data are (Figure 17.11).

```
d_poly <- dist.hamming(data_poly)
delta_poly <- c(mean = delta.score(data_poly, arg = "mean"),
                sd   = delta.score(data_poly, arg = "sd"))
round(delta_poly, 3)
```

```
#>  mean    sd
#> 0.337 0.271
```

```
nnet_poly <- neighborNet(d_poly)
tip_col <- ifelse(nnet_poly$tip.label %in% members[["Tongic"]], "#D55E00",
           ifelse(nnet_poly$tip.label %in% members[["East Polynesian"]],
                  "#0072B2", "grey20"))
op <- par(mar = c(1.5, 1.5, 3, 1.5), xpd = NA)
pn_poly <- plot(nnet_poly, type = "2D", show.tip.label = FALSE)
net_labels(pn_poly, nnet_poly$tip.label, col = tip_col)   # helper (appendix)
```

```
par(op)
```

We then fit the binary model by maximum likelihood, with and without gamma rate variation, starting from a neighbor-joining tree, and assess the NJ tree with 100 bootstrap replicates.

```
tree_nj_poly <- NJ(d_poly)
bs_nj_poly <- bootstrap.phyDat(data_poly,
                               function(x) NJ(dist.hamming(x)), bs = 100)
fit_poly <- optim.pml(pml(tree_nj_poly, data_poly), optNni = TRUE,
                      optBf = TRUE, control = ctrl)
fit_poly_g <- optim.pml(update(fit_poly, k = 4), optNni = TRUE, optBf = TRUE,
                        optGamma = TRUE, control = ctrl)
round(c(AIC_no_gamma = AIC(fit_poly), AIC_gamma = AIC(fit_poly_g),
        gamma_shape = fit_poly_g$shape, pi_present = fit_poly_g$bf[2]), 3)
```

```
#> AIC_no_gamma    AIC_gamma  gamma_shape   pi_present
#>    23270.788    22922.139        2.584        0.180
```

A nonparametric bootstrap of the ML search in **phangorn** takes a few seconds per replicate on these data, which is too slow for the book's build. We therefore ran the same model in IQ-TREE on the variable characters, with the ultrafast bootstrap. The PHYLIP file is written as in Step 11, and the output is stored in `software-runs/case-study/`.

```
poly_var <- bin_poly[, !apply(bin_poly, 2, function(x) all(x[x != "?"] == "1"))]
```

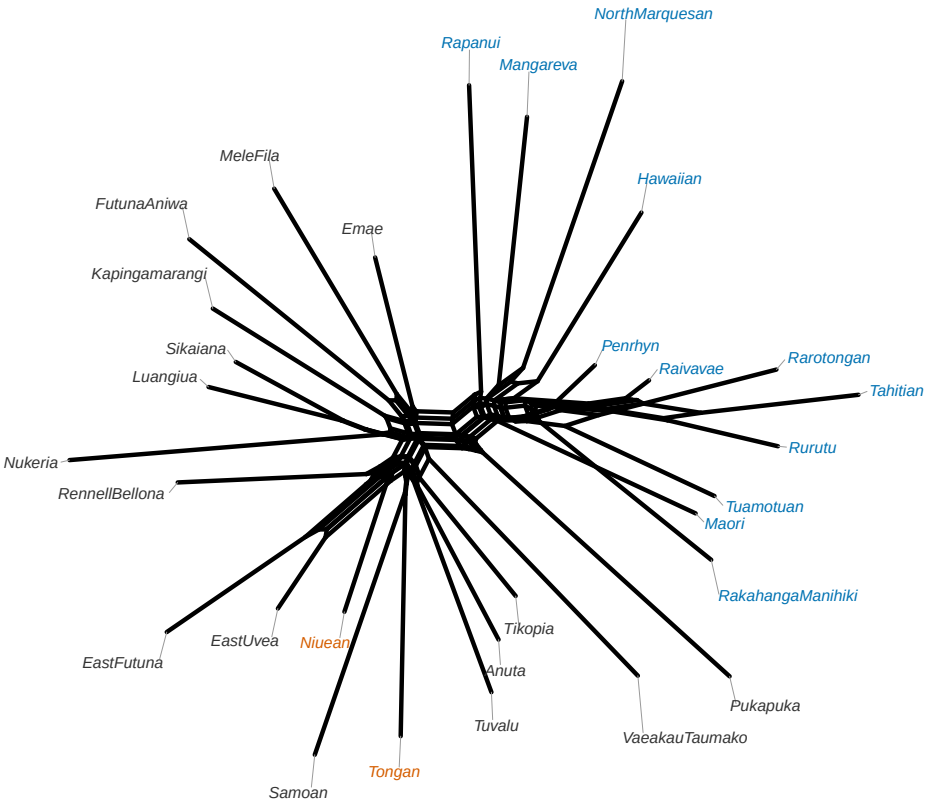


Figure 17.11: NeighborNet of the Hamming distances between the 30 Polynesian languages. Tongic languages (Tongan, Niuean) and East Polynesian languages are marked in color.

```
write.phyDat(phyDat(poly_var, type = "USER", levels = c("0", "1"),
                    ambiguity = "?"),
             file = file.path(out_dir, "poly.phy"), format = "phylip")
seq_poly <- sprintf(
  '    <sequence id="seq_%s" taxon="%s" totalcount="2" value="0%s"/>',
  rownames(bin_poly), rownames(bin_poly),
  apply(bin_poly, 1, paste, collapse = ""))
writeLines(c(paste('<data id="cognates" spec="Alignment" dataType="binary"',
                   'ascertained="true" excludefrom="0" excludeto="1">'),
             seq_poly, "</data>"),
           file.path(out_dir, "poly_beast_data.xml"))
```

```
iqtree3 -s poly.phy -m GTR2+FO+G4+ASC -B 1000 -T 1 -seed 17 --prefix poly
```

```
iq_poly <- read.tree(file.path("software-runs", "case-study", "iqtree",
                               "poly.treefile"))
iq_poly <- root(iq_poly, outgroup = "Tongan", resolve.root = TRUE,
                edgelabel = TRUE)    # Tongic is not monophyletic here
tip_info_poly <- data.frame(
  label = rownames(bin_poly),
  group = ifelse(rownames(bin_poly) %in% members[["Tongic"]], "Tongic",
          ifelse(rownames(bin_poly) %in% members[["East Polynesian"]],
                 "East Polynesian", "other Nuclear Polynesian")))
ggtree(iq_poly) %<+% tip_info_poly +
  geom_tippoint(aes(color = group), size = 1.4) +
  geom_tiplab(aes(color = group), size = 3, offset = 0.002,
              show.legend = FALSE) +
  support_layers(2.4, drop = c("", "Root")) +   # plotting helper (appendix)
  scale_color_manual(values = c(Tongic = "#D55E00",
                                `East Polynesian` = "#0072B2",
                                `other Nuclear Polynesian` = "grey20"),
                     name = NULL) +
  theme_tree2() + xlim(0, max(node.depth.edgelength(iq_poly)) * 1.3) +
  labs(x = "Expected changes per character")
```

17.14.3 A Bayesian analysis

For the Bayesian analysis we adapted the BEAST 2 file of Section 17.10.3 (`software-runs/case-study/poly_beast.xml`): the same binary model with ascertainment correction and gamma rates, and a Yule prior. We used no calibrations. The strict clock has its rate fixed at 1, so node heights are in expected changes per character, a relative time scale. Two runs of 1.5 million states each, sampled every 1,000 states, took about 13 minutes each on one processor core. Every parameter reached an ESS above 600 in each run. As before, the book keeps every tenth sampled tree.

```
poly_dir <- file.path("software-runs", "case-study", "beast2")
read_poly <- function(run, file) file.path(poly_dir, run, file)
post_poly <- do.call(c, lapply(runs, function(run) {
  tr <- read.nexus(read_poly(run, "poly_thinned.trees"))
```

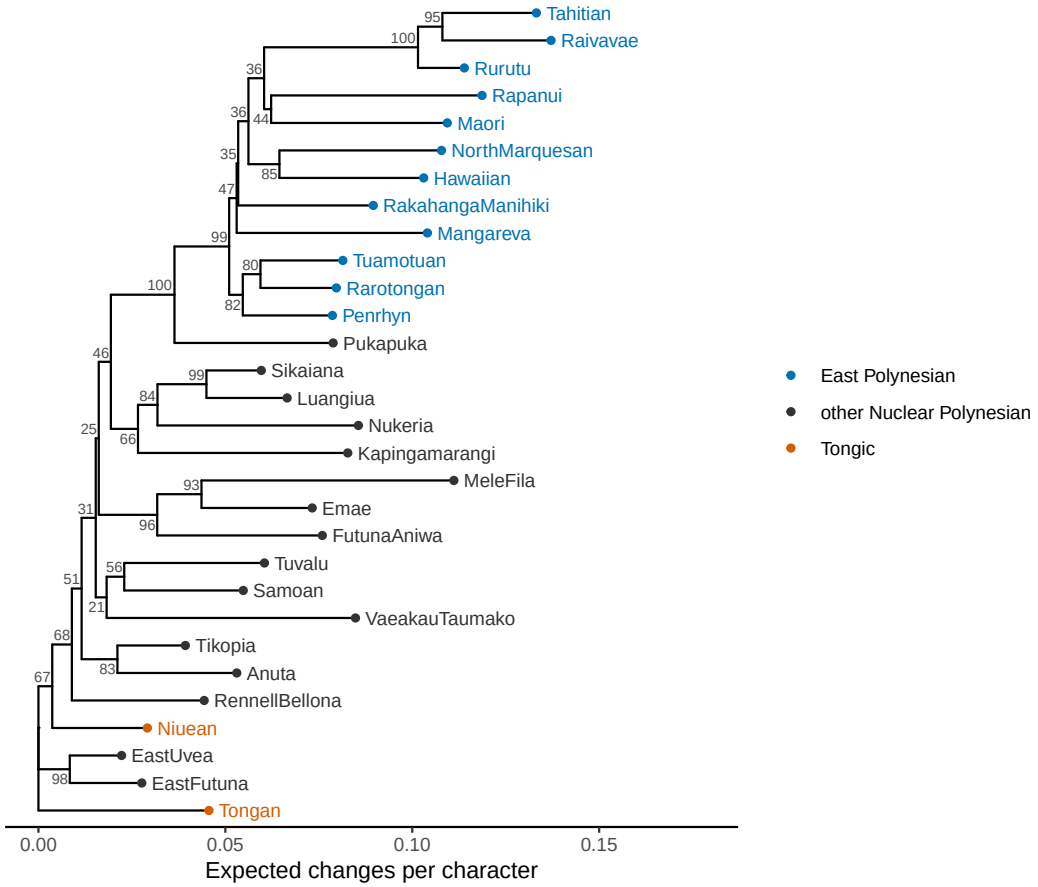


Figure 17.12: Maximum-likelihood tree of the 30 Polynesian languages from IQ-TREE (binary model with estimated frequencies, gamma rates and ascertainment correction), rooted on Tongan, with ultrafast bootstrap support (percent) at internal nodes. Colors as in the NeighborNet.

```
  tr[-seq_len(floor(0.1 * length(tr)))]
}))
traces_poly <- lapply(runs, function(run) {
  tr <- read.table(read_poly(run, "poly.log"), header = TRUE,
                   comment.char = "#")
  tr[-seq_len(floor(0.1 * nrow(tr))), ]
})
pars_poly <- c("posterior", "TreeHeight", "gammaShape", "freqParameter.2")
round(sapply(traces_poly, function(tr)
  coda::effectiveSize(tr[, pars_poly])))
```

```
#>                 [,1] [,2]
#> posterior        821  815
#> TreeHeight       919  977
#> gammaShape       931  907
#> freqParameter.2 1088 1198
```

```
mcc_poly <- maxCladeCred(post_poly)
cred <- prop.clades(mcc_poly, post_poly, rooted = TRUE) / length(post_poly)
mcc_poly$node.label <- ifelse(seq_along(cred) == 1, "", sprintf("%.2f", cred))
```

```
ggtree(mcc_poly) %<+% tip_info_poly +
  geom_tippoint(aes(color = group), size = 1.4) +
  geom_tiplab(aes(color = group), size = 3, offset = 0.002,
              show.legend = FALSE) +
  support_layers(2.4) +   # plotting helper (appendix)
  scale_color_manual(values = c(Tongic = "#D55E00",
                                `East Polynesian` = "#0072B2",
                                `other Nuclear Polynesian` = "grey20"),
                     name = NULL) +
  theme_tree2() + xlim(0, max(node.depth.edgelength(mcc_poly)) * 1.3) +
  labs(x = "Expected changes per character since the root")
```

17.14.4 Which subgroups are recovered?

Table 17.6 checks each Glottolog subgroup against the NJ and ML trees, with the NJ bootstrap support, and gives its posterior probability in the BEAST 2 sample.

```
iq_unrooted <- unroot(iq_poly)
recovered <- function(tree, g) is.monophyletic(unroot(tree), g)
support <- function(trees, g, rooted = FALSE)
  mean(vapply(trees, function(tr)
    if (rooted) is.monophyletic(tr, g) else recovered(tr, g), logical(1)))
poly_table <- data.frame(
  languages = lengths(members),
  NJ = vapply(members, function(g) recovered(tree_nj_poly, g), logical(1)),
  NJ_bootstrap = vapply(members, function(g) support(bs_nj_poly, g), 0),
  ML = vapply(members, function(g) recovered(iq_unrooted, g), logical(1)),
```

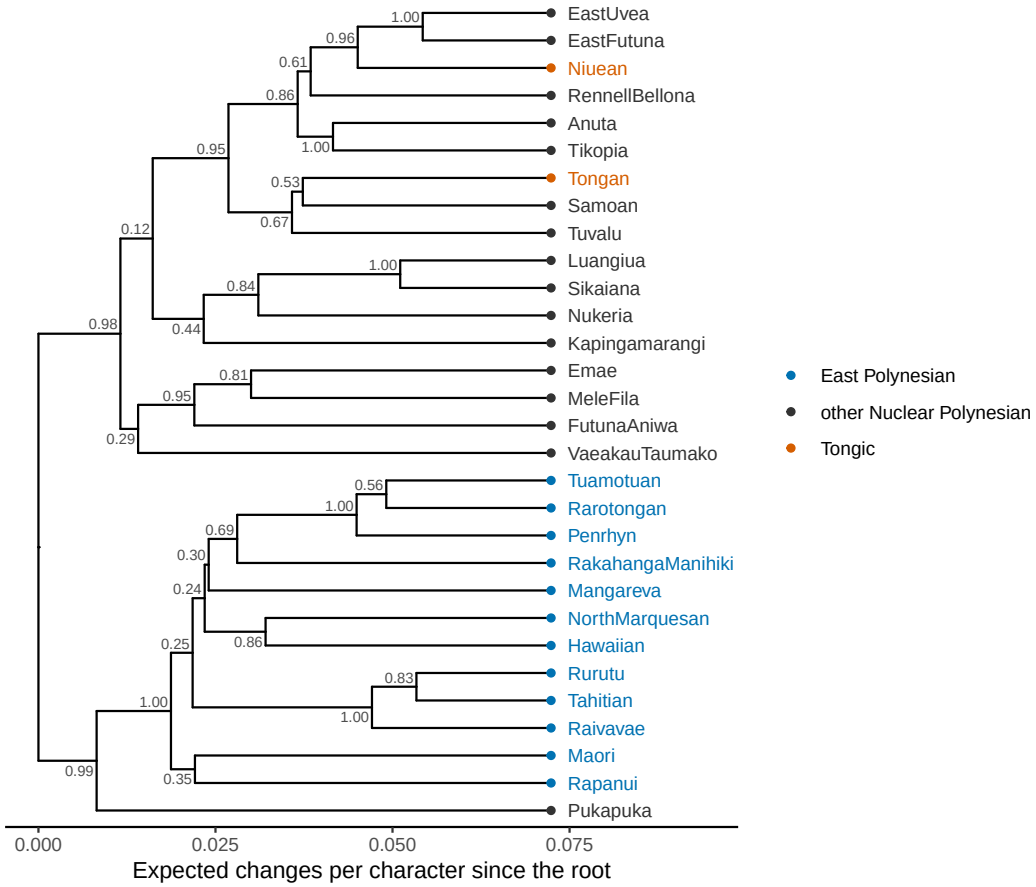


Figure 17.13: Maximum clade credibility tree of the BEAST 2 posterior sample for Polynesian, with clade credibilities at internal nodes. The horizontal axis is in expected changes per character, measured from the root; there are no calibrations. Colors as in the NeighborNet.

```
  BEAST_posterior = vapply(members, function(g)
    support(post_poly, g, rooted = TRUE), 0))
knitr::kable(poly_table, digits = 2, caption = paste(
  "Glottolog subgroups with at least two of the 30 languages: whether the",
  "NJ and IQ-TREE ML trees contain them, their NJ bootstrap support, and",
  "their posterior probability in the BEAST 2 sample (rooted clades)."))
```

Table 17.6: Glottolog subgroups with at least two of the 30 languages: whether the NJ and IQ-TREE ML trees contain them, their NJ bootstrap support, and their posterior probability in the BEAST 2 sample (rooted clades).

	languages	NJ	NJ_bootstrap	ML	BEAST_posterior
Nuclear Polynesian	28	FALSE	0.05	FALSE	0.00
Vanuatu-Loyalty Outliers	3	TRUE	0.41	TRUE	0.95
Mele-Futuna	2	TRUE	0.74	FALSE	0.16
Northern Outlier Polynesian-East Polynesian	16	FALSE	0.00	FALSE	0.00
Solomons Northern Outlier Polynesian-East Polynesian	15	FALSE	0.00	FALSE	0.00
Central Northern Outlier Polynesian-East Polynesian	14	FALSE	0.00	FALSE	0.00
East Polynesian	12	TRUE	0.51	TRUE	1.00
East Polynesian Proximal	9	FALSE	0.02	FALSE	0.00
Central Northern Outlier Polynesian	2	FALSE	0.01	FALSE	0.00
East Polynesian Distal	3	FALSE	0.12	FALSE	0.02
Far East Polynesian	2	FALSE	0.03	FALSE	0.10
Southern East Polynesian Proximal	8	FALSE	0.01	FALSE	0.00
Tongic	2	FALSE	0.05	FALSE	0.01
Ellicean	3	FALSE	0.01	FALSE	0.00
Pukapukic	2	FALSE	0.01	FALSE	0.00
Tahitian-Austral	3	TRUE	1.00	TRUE	1.00
Austral	2	FALSE	0.00	FALSE	0.03

Compared with the simulation, the real data are much less tidy. The mean delta score is 0.34, against 0.13 for the simulated family. The NeighborNet shows large boxes in the center, where the Outliers, Samoan, Tuvaluan and the Tongic languages meet. Rate variation among characters, which the simulated data hardly showed, is strongly supported: adding gamma rates lowers the AIC by 349 units. Support is also lower. Many internal nodes of the ML tree have ultrafast bootstrap values well below 95%, and the methods disagree with each other on about half of their splits.

Only 3 of the 17 Glottolog subgroups appear in the ML tree. One of them is East Polynesian, the best-established subgroup in the family, which has a posterior probability of 1 in the BEAST 2 sample. The deeper structure is not recovered. Tongan and Niuean do not form a Tongic clade. East Uvean and East Futunan are placed next to Niuean or Tongan rather than among the other

Nuclear Polynesian languages, and in the BEAST 2 tree Tongan groups with Samoan. This is the expected signature of contact. East Uvean, in particular, borrowed heavily from Tongan, and borrowings that the coders did not recognize as loans look like shared inheritance (Chapter 8). Pukapuka joins East Polynesian, whereas Glottolog places it with Samoan and Tuvaluan, which is again consistent with later contact. In the strict-clock BEAST 2 analysis, the root falls between East Polynesian (with Pukapuka) and all other languages, with a posterior probability of 0.99, instead of between Tongic and Nuclear Polynesian. The East Polynesian languages have long branches. A strict clock interprets that extra change as extra time and places the root on their side. A relaxed clock, an outgroup or a prior on the root position would be needed to root this tree credibly (Chapter 9). The intermediate groupings that link East Polynesian with particular Outliers rest on few shared innovations, and a basic-vocabulary list of 210 meanings is unlikely to resolve them.

These comparisons need a caveat. Agreement with the expert subgroups is not independent validation of the methods. The cognate judgments are themselves products of the comparative method, and the Glottolog subgrouping comes from the same method, applied by overlapping communities of scholars to overlapping data. Moreover, a shared cognate is not the same thing as a shared innovation: two languages can share a cognate because both retained it. A tree that reproduces the classification may partly reproduce the judgments that went into the coding, and a disagreement may reflect either the method or the coding. In the vocabulary of Section 6.10, the Polynesian analysis is a first step toward *empirical adequacy* (the delta score and the NeighborNet show where the tree model fits poorly) and toward *historical interpretation*, not a validation of the methods. The simulation is the only setting in which we know the right answer. The real data show where a pipeline that works well on clean data runs into the problems discussed in Chapters 8 and 15: contact, rate variation among lineages, and short internal branches.

17.15 Validating the pipeline

The analyses of this chapter provide different kinds of evidence, and it helps to name them with the vocabulary of Section 6.10.

- **Implementation validation** asks whether the software computes what the model says. Three checks in this chapter are of this kind: reading the exported NEXUS file back in (Step 8), running similar models in five independent programs (Table 17.4), and running each Bayesian analysis without data to see the prior it actually uses. The second is the weakest, for the reasons given in Step 9.
- **Parameter recovery** asks whether the pipeline recovers a history simulated under its own assumptions. Steps 4 to 9 on the toy family are tests of this kind. Strictly, the simulator and the models differ slightly: the simulator is Dollo-like and varies rates among meanings, whereas the inference models are reversible and binary. The data are still much closer to the models than any real dataset.
- **Robustness under misspecification** asks how the results change when the data violate the model in realistic ways, through borrowing, rate shifts in one lineage, coding error or missing languages. The simulations of this chapter do not test it; the second computational exercise shows how to begin, and Chapter 15 gives examples.
- **Empirical model adequacy** asks whether the fitted model could have produced the real data. AIC and Bayes factors measure only relative fit. For the Polynesian data, the delta score and the NeighborNet already show aspects of the data that a tree model does not capture,

and the posterior predictive checks described below would make the question formal.

- **Historical interpretation** asks whether the clades, dates and rates correspond to historical events. For Polynesian, the uncalibrated BEAST 2 tree has no time scale at all, and its root position reflects the clock model as much as the history.

Simulation is also the most powerful way to validate a real analysis, beyond parameter recovery on a toy. Take the tree and parameters estimated from the real data and simulate many new datasets from them, with the same numbers of languages, meanings and missing cells. Then run the whole pipeline on each and check whether it recovers what was simulated. If a method systematically misplaces a particular subgroup, or overestimates the root age, on data simulated from a known history, it will do the same on real data. Simulation can also introduce violations of the model deliberately, such as borrowing between particular languages, faster change in one lineage, or coding errors, to see which conclusions are robust to them (Greenhill et al., 2009). That turns a parameter-recovery study into a test of robustness under misspecification. The function `simulate_cognates()` above, extended with borrowing events, is all that is needed to begin (see the second computational exercise).

Evolutionary biologists have developed two formal versions of this idea that have only recently been applied systematically to linguistic phylogenetic pipelines (King, 2026b,a). *Simulation-based calibration* (Talts et al., 2018) checks that a Bayesian program is computing the posterior correctly: draw parameters (including a tree) from the prior, simulate data from them, run the inference, and record where the true value falls among the posterior samples. If the software and the model implementation are correct, these ranks are uniformly distributed, and departures from uniformity reveal bugs or poor mixing. It has become a standard test for new phylogenetic models in packages for BEAST 2 and RevBayes. *Posterior predictive simulation* checks whether the model is adequate for the data: simulate datasets from parameters drawn from the posterior, compute summary statistics such as the number of singletons per language, the distribution of cognate-class sizes or the delta score, and compare them with the observed values (Chapter 15). A model that cannot reproduce the observed number of singletons, for example, is probably mis-describing the process of lexical replacement. Both checks are straightforward to implement with the tools of this chapter, and both would make published linguistic phylogenies considerably more convincing. Applied to the standard BEAST 2 set-up for cognate data, the first revealed that the default weighting of meaning-partition rates miscalibrates the posterior, and the second that the covarion model fails to reproduce several features of Indo-European cognate data (King, 2026b). A third practice worth borrowing is the continuously updated, fully scripted pipeline exemplified by Nextstrain in pathogen phylodynamics (Hadfield et al., 2018), in which a single command rebuilds every analysis from the raw data whenever the data change.

17.16 A reporting checklist

A phylogenetic analysis is reproducible only if a reader can reconstruct every step from the data to the figures. Table 17.7 collects the information that should be reported, drawing on the stages of this chapter. Chapter 15 gives a complementary checklist focused on the substantive pitfalls.

Table 17.7: Information to report for a reproducible linguistic phylogenetic analysis.

Stage	What to report
Languages	how languages were selected; Glottocodes; how dialects and duplicates were handled
Meanings	the concept list (with Concepticon IDs) and why it was chosen
Cognates	who coded them and how; treatment of doubtful judgments, loans and synonyms; for automatic coding, the method, thresholds and evaluation
Matrix	coding scheme (binary, multistate, root-meaning); numbers of characters, singletons and missing cells; the data file itself
Model	substitution model, rate variation, ascertainment correction, partitioning; how the model was chosen
Tree prior and clock	tree prior, clock model, all priors with their parameters; calibrations with their sources and justification
Inference	software and version; chain length, sampling frequency, number of independent runs, burn-in; seeds
Convergence	ESS for all parameters; agreement between runs (e.g., ASDSF); prior-only run
Summaries	how the summary tree was built (MCC, consensus; node heights); credible intervals
Robustness	sensitivity analyses (alternative priors, models, calibrations, language samples); model adequacy checks
Archiving	data in CLDF or NEXUS, input files (XML, Rev, NEXUS), configuration files, scripts, and output tree samples, deposited in a public archive

Archiving the posterior tree sample deserves special mention. Downstream comparative studies (Step 10) need the full sample, not just the summary tree, and many published language phylogenies are now collected in public repositories for exactly this reason (Appendix G). Following the FAIR principles (Wilkinson et al., 2016), the archived data should be findable, accessible, interoperable and reusable: in an open format, with a persistent identifier, and with enough metadata to be understood without the paper.

17.17 Session information

For completeness, here are the R version and platform used to build this chapter. In a real project, the output of `sessionInfo()` (or `renv::snapshot()`) belongs in the supplementary material.

```
c(R = R.version.string, platform = R.version$platform)
```

```
#>                                   R                            platform
#> "R version 4.6.0 (2026-04-24 ucrt)"               "x86_64-w64-mingw32"
```

A phylogenetic analysis is a pipeline: data assembly, cognate coding, matrix building, exploration, inference, support, dating, summaries, downstream analysis and reporting. The early decisions, about languages, meanings, cognates and loans, have the largest effects and are the hardest to undo. Binary presence/absence coding must score unrecorded meanings as missing; it creates dependencies among characters and an ascertainment bias that the models must address. On sim-

ulated, clock-like data without borrowing, distance, parsimony, likelihood and Bayesian methods all recover most of the tree and disagree mainly on short, poorly supported branches. That is parameter recovery, the easiest kind of test, not evidence that the models fit real data. R is the natural place to prepare input files for MrBayes, BEAST 2, BEASTling and RevBayes and to analyze their output; prior-only runs and independent runs belong to every Bayesian analysis. Absolute dates rest on calibrations and priors. Downstream analyses should be repeated over a posterior sample of trees, not run on one summary tree. Real data, such as the Polynesian wordlist, are less tree-like than simulated data, and their agreement with expert subgroups is not independent validation, because cognate judgments and subgroupings both come from the comparative method.

17.18 Exercises

Conceptual

1. In `cognates_to_binary()`, a meaning that is not recorded for a language is scored `?` in every character belonging to that meaning, not 0. What would happen to a poorly documented language if it were scored 0 instead, and at which step of the pipeline would the error first become visible?
2. The toy data contain no homoplasy, yet the consistency index of the parsimony tree is below 1 and the gamma shape parameter of the ML model is estimated to be very large. Explain both results from the binary coding, and say how coding each meaning as one multistate character would change them.
3. Say which kind of validation, in the vocabulary of Section 6.10, each of the following provides: (a) the RF distances of Table 17.3; (b) the agreement of five programs in Table 17.4; (c) a prior-only run of the BEAST 2 file; (d) the recovery of East Polynesian; (e) a posterior predictive check of the number of singletons per language. Which of them could show that a model is wrong for real data?
4. The BEAST 2 posterior median root age of the toy family is about 3,600 years, the true value is 6,000 years, and the 95% interval reaches 7,000 years. Is this a failure of the method? Explain the roles of the calibration, the clock-rate prior and the root-age prior, and what a prior-only run would show.
5. In the strict-clock analysis of Polynesian, the root falls between East Polynesian and the other languages. Explain why long branches in one subgroup can move the root of a strict-clock tree, and name two kinds of evidence, independent of the lexical data, that could justify a different root.

Computational

1. *A meaning-level bootstrap.* Write a function that samples the meanings of `cog` (`unique(cog$Parameter_ID)`) with replacement, renames each copy of a meaning so that its cognate set identifiers stay unique (for example by pasting a copy number onto `Parameter_ID` and `Cognateset_ID`), rebuilds the matrix with `cognates_to_binary()` and `phyDat()`, and infers a tree with `NJ(dist.hamming())`. Repeat 100 times. Compare the support for the clades of `tree_nj_ham` with that from the character bootstrap `bootstrap.phyDat(cog_data, function(x) NJ(dist.hamming(x)), bs = 100)`, using `prop.clades()`. Which clades lose support, and why?
2. *Robustness to borrowing.* Write a function `borrow_cognates(cog, recipient, donor, b)` that, for a random fraction `b` of the meanings, gives the recipient the donor's `Cognateset_ID`.

Choose a recipient and a donor from different subgroups (`subgroup`). For `b` = 0, 0.1, 0.2 and 0.3, rebuild the matrix, compute `delta.score()`, infer NJ and ML trees (`NJ()`, `optim.pml()` with `optBf = TRUE`), score them with `score_tree()`, and check whether recipient and donor come out as sisters with `is.monophyletic()`. Which kind of validation is this, and which conclusion of Step 7 does it qualify?

3. *Loans in the Polynesian data.* Rebuild the Polynesian matrix of chunk `poly-read` without the `!Loan` filter, so that the 213 forms flagged as loans are treated as inherited. Compare `delta.score()`, the NJ tree (`NJ(dist.hamming())`) and the recovery of the Glottolog subgroups in `members` (with the function `recovered()` of chunk `poly-compare`) with and without the loans. Does East Uvean move? What does the result suggest about loans that were *not* flagged?
4. *A poorly documented language.* Simulate a new dataset with `simulate_cognates(true_tree)`, then delete 80% of the rows of one language from the long table, as if it were known only from a short list. Rebuild the matrix, fit the ML model with `optim.pml()`, and compare `score_tree()` with and without the deletion. Use `bootstrap.pml()` with 20 replicates and `prop.clades()` to see whether the position of the sparse language is still supported.
5. *One tree or many.* Fit the PGLS of Step 10 on the MCC tree alone (`pgls_one(mcc)`) and compare its slope and standard error with the spread over `posterior`. Then, for the Polynesian sample `post_poly`, compute the posterior probability of each Glottolog subgroup with `support()` and compare it with the clade credibilities shown on `mcc_poly`.

Research

1. You have a Lexibank-style dataset of 45 Romance varieties with expert cognate judgments, including Latin, Old French and Old Spanish, about 20% missing meanings, and documented borrowing from Germanic, Arabic and from French into other Romance varieties. Design the pipeline: the coding of synonyms and loans, the treatment of the attested older varieties (tips, sampled ancestors or calibrations only), the calibrations, the models to compare, and the checks you would run. Which steps of this chapter change, and what would you simulate to show that your conclusions are robust to the borrowing?
2. A manuscript reports a single BEASTling run with default settings on 80 languages, a single MCC tree, and an ancestral-state reconstruction on that tree. Using Table 17.7, write the core of a referee report: which additional runs (prior-only, independent chains, alternative clock and tree priors, model adequacy checks) and which archived files you would ask for, and why each matters for the conclusions.
3. You want to date the Polynesian tree. Propose calibrations from the archaeology of the settlement of West Polynesia, East Polynesia and New Zealand, and for each say which node it constrains and whether it is a minimum or a maximum bound. Explain the risk of circularity if the dated tree is then used to test hypotheses about the settlement chronology, and say how you would handle the contact between Tongan and its neighbors and the rooting problem of Section 17.14.4.

17.19 Further reading

- Paradis (2012), *Analysis of Phylogenetics and Evolution with R*. A thorough guide to **ape** and related packages, covering tree manipulation, distance and likelihood methods, and comparative analysis; the best book-length companion to the R code in this chapter.

- Schliep (2011) and the **phangorn** vignettes. The vignettes on trees, likelihood and networks work through examples that parallel Steps 4 to 7, including model comparison and bootstrapping.
- Yu et al. (2017) and the online book *Data Integration, Manipulation and Visualization of Phylogenetic Trees* by Guangchuang Yu. A comprehensive guide to plotting and annotating trees with **ggtree**, including posterior trees read with **treeio**.
- Maurits et al. (2017): Describes BEASTling's design, its models and its integration with Glottolog, with worked examples of linguistic analyses in BEAST 2.
- Drummond and Bouckaert (2015), *Bayesian Evolutionary Analysis with BEAST*. The standard textbook for BEAST 2, explaining its models, priors and output; essential before running a dated analysis.
- The RevBayes tutorials (https://revbayes.github.io/tutorials/). Step-by-step scripts for morphological (discrete) characters, clock models, dating and model comparison that adapt directly to cognate data.
- Forkel et al. (2018) and List et al. (2022a). The papers introducing CLDF and Lexibank, explaining the data model and the conversion workflow behind the datasets used in Section 17.13.
- Barbançon et al. (2013): A model simulation study for evaluating phylogenetic methods on linguistic characters, and a template for validating one's own pipeline.
- King (2026b): Simulation-based calibration and posterior predictive checks of a standard BEAST 2 analysis of cognate data; the natural next step after the validation section of this chapter.
- Jäger (2019): A wide-ranging survey of computational historical linguistics, from automated cognate detection to phylogenetic inference, that places the steps of this chapter in their research context.

Chapter 18

Synthesis: What Linguistic Phylogenetics Has Found

A quarter of a century after the first Bayesian and likelihood-based language trees, it is fair to ask what the enterprise has actually established. The answer is not a list of results but a list of results *together with the conditions under which they hold*. Every finding in this chapter rests on particular kinds of data (usually basic-vocabulary cognate sets, sometimes structural features), on particular modelling assumptions (tree-like descent, a clock, a diffusion process, independent evolution of characters), and on evidence that is more or less independent of those assumptions. Some findings survive large changes in data and models and agree with archaeology, genetics and the comparative method; others depend on a single chain of assumptions and have moved with each new analysis. This chapter gathers the substantive findings scattered through the book, states for each the assumptions and evidence on which it rests, and sorts them into what is broadly accepted, what is supported but debated, and what remains contested.

After reading this chapter you should be able to:

- distinguish broadly accepted, supported-but-debated and contested findings in linguistic phylogenetics, and give one example of each;
- state, for a given generalization (the Holocene horizon, pulse-pause expansion, frequency-dependent replacement rates), the data, model assumptions and kinds of evidence under which it holds;
- explain why the recovery of accepted subgroups by a computational tree is not independent validation of the method;
- evaluate a published root-age or homeland claim by identifying its calibrations, its treatment of ancient languages, its sampling and its diffusion or clock model;
- classify the evidence offered for a method's reliability using the validation vocabulary of Section 6.10;
- explain why comparative claims (about universals, culture or demography) should be evaluated over a posterior sample of trees rather than a single summary tree.

18.1 Taking stock

The modern phase of linguistic phylogenetics is usually dated to a handful of papers published around the turn of the millennium. Gray and Jordan (2000) used maximum parsimony on Austronesian basic vocabulary to test the "express train" model of Austronesian expansion from Taiwan. Ringe et al. (2002) applied compatibility methods to a carefully coded set of lexical, phonological and morphological characters for Indo-European. Gray and Atkinson (2003) used Bayesian inference with rate-smoothed dating to estimate the age of Indo-European. In the years since, the field has grown in three directions at once. Datasets have grown from dozens of languages to hundreds, and, with automated methods, to thousands (Jäger, 2018). Models have grown from simple binary substitution processes to stochastic Dollo models, covarion models, relaxed clocks, sampled ancestors and spatial diffusion (Chapters 7 to 11). And the questions asked have widened from "what is the tree?" to "when, where, how fast, and why?".

It helps to sort the findings into five kinds, which differ greatly in how secure they are.

1. **Classification**: which languages group together, and in what order the groups split.
2. **Chronology**: when the family and its subgroups began to diverge.
3. **Geography**: where the family originated and by which routes it spread.
4. **Process**: how language change itself proceeds, including its tempo, its regularities across meanings and features, and its relation to population and society.
5. **Coevolution**: how languages and other cultural or biological traits have changed together, using language trees as a scaffold.

The five kinds form a chain of inference, and each link adds assumptions. Classification rests mainly on the cognate data and the substitution model. Chronology adds a clock model, calibrations and a tree prior. Geography adds a model of movement and assumes that present locations carry information about past ones. Process and coevolution findings add models of trait evolution and usually take the tree, its dates and sometimes its locations as given. It is therefore no surprise that classification is generally the most robust kind of finding and coevolution the least: an error at any earlier link propagates forward. In the vocabulary of Section 6.10, the parameters of the later links (an extinction rate, a homeland, a correlation between traits) may be identifiable under the model and yet hard to estimate with realistic samples, sensitive to misspecification, or difficult to interpret as historical quantities.

18.2 Language trees are recoverable, within limits

The most basic finding is that computational methods applied to basic-vocabulary cognate data produce trees that largely agree with the subgroupings established by the comparative method. For Indo-European, every major analysis recovers the well-established subgroups (Germanic, Celtic, Italic, Balto-Slavic, Indo-Iranian, Greek, Armenian, Albanian, Anatolian, Tocharian) and the closer relationships within them (Gray and Atkinson, 2003; Bouckaert et al., 2012; Chang et al., 2015; Heggarty et al., 2023). For Austronesian, the Bayesian tree of Gray et al. (2009) matched the main outlines of the subgrouping developed by Austronesianists, including the primary diversity of the Formosan languages of Taiwan and the nested structure of Malayo-Polynesian, Oceanic and Polynesian. The same is true for Bantu (Grollemund et al., 2015), Pama-Nyungan (Bowern and Atkinson, 2012) and other families where the phylogenetic results can be compared with expert classifications.

What exactly does this agreement show? It is tempting to read it as independent validation: two methods, applied to the same history, give the same answer. It is not independent, for two reasons. First, the cognate judgements that make up the data matrix are themselves products of the comparative method, made by specialists who usually know the accepted subgrouping, so a tree inferred from those judgements partly reproduces the expertise that went into them. Second, cognate *sharing* is not the same thing as shared *innovation*. The comparative method subgroups on innovations; a binary cognate matrix records retentions and innovations alike, and the model must sort them out (Section 15.10). Agreement therefore shows that the phylogenetic summary of the cognate data is *consistent with* a classification derived, in part, from the same evidence. It does not demonstrate that the method would recover the true history where the comparative method has not already done the work. The Polynesian case study of Chapter 17 makes the point concretely: the tree recovers East Polynesian, the best-established subgroup, but not Tongic, and the disagreements fall where contact and rate variation are suspected (Section 17.14.4).

Stronger evidence of reliability comes from settings where the answer is known independently of the coding. Simulation supplies such settings. Greenhill et al. (2009) simulated borrowing at different rates and with different geographic constraints, and found that tree topologies were robust to the levels of borrowing they considered typical of basic vocabulary, while dates were less robust and tended to be too young, with errors growing as borrowing became more frequent and less geographically restricted. In the vocabulary of Section 6.10, this is a test of *robustness under misspecification*, and it is only as informative as the borrowing model used to simulate the data: concentrated borrowing between particular neighbours, which is closer to the worst case, can produce confident wrong subgroups even when global accuracy looks high (Section 15.4). The finding is therefore best stated with its conditions: *for curated basic-vocabulary lists, in families whose subgroups have been established by the comparative method and from which known loans have been removed, phylogenetic trees reproduce the major subgroups, and simulation suggests that diffuse borrowing at moderate levels does not overturn the topology*. The basic vocabulary lists used in phylogenetics were, after all, designed to minimize borrowing.

Where phylogenetic and traditional classifications disagree, the disagreements cluster at the deepest levels of the tree: the order of the first splits within a family, or the position of a branch that separated early and has few shared innovations with any other. These are precisely the parts of the tree where the comparative method is itself uncertain, where short internal branches carry little signal, and where posterior clade probabilities are low. Within Indo-European, for example, the relative order of the early splits after Anatolian and Tocharian, and whether Italic and Celtic form a clade, have remained difficult for both approaches. Phylogenetic methods have not overturned the comparative method's classifications; they have, at best, quantified uncertainty that philologists already recognized, and occasionally offered support to one of several competing hypotheses.

Trees are, moreover, not the whole story. Measures of treelikeness such as delta scores and Q-residuals vary considerably among families and subgroups (Gray et al., 2010), and some groups, such as dialect continua and the languages of long-standing contact zones, are poorly described by a tree at all (Chapter 8). Gray et al. (2010) argued that the shape of human history is better thought of as a mixture of tree-like and reticulate fabric, with the balance differing between families and periods. Historical glottometry (Kalyan and François, 2018) and the broader critique of the tree model by François (2015) make the same point from within historical linguistics: diversification often proceeds through the gradual break-up of dialect networks, in which innovations spread across overlapping areas, rather than through clean splits. A tree inferred from such data is not

wrong, but it is a summary that hides the overlapping innovations.

Gray et al. (2009) analyzed basic vocabulary from 400 Austronesian languages with a Bayesian relaxed-clock model and calibrations drawn from archaeology and history. The tree placed the root in Taiwan at about 5,200 years ago, in agreement with the archaeological "out of Taiwan" model and with linguists' subgrouping. Its most striking finding concerned the tempo of the expansion. Mapping the dated tree onto geography revealed a sequence of *pauses*, during which the expansion stalled, followed by *pulses* of rapid dispersal. A pause in Taiwan before the move into the Philippines, and another before the settlement of Polynesia, coincided with the major water gaps that had to be crossed and, the authors argued, with the development of new technologies and social strategies (such as improved sailing technology) that made the next pulse possible. The study showed that a dated language phylogeny can be used to address the dynamics of a prehistoric expansion, not just its starting point. Its conclusions rest on the calibrations (some of them archaeological, so part of the agreement with archaeology is built in), on the relaxed-clock model, and on the assumption that the geography of present-day languages reflects the order of settlement.

18.3 Dates: the Holocene horizon

The second widely cited generalization is that most of the world's large and well-studied language families have root ages within the Holocene, the last 11,700 years. Table 18.1 collects some of the most widely cited estimates, which are discussed in detail in Chapter 10. The numbers are approximate central estimates; each has a wide credible interval, and some are contested.

Table 18.1: Selected root-age estimates from Bayesian phylogenetic studies. Values are approximate central estimates or ranges reported by the authors; see Chapter 10 for details and credible intervals.

Family	Study	Approximate root age (years before present)	Homeland or association
Indo-European	Gray and Atkinson (2003)	7,800–9,800 (range)	consistent with the Anatolian farming hypothesis
Indo-European	Bouckaert et al. (2012)	8,000–9,500 (range)	Anatolia (phylogeographic inference)
Indo-European	Chang et al. (2015)	about 6,000 (5,900–6,500 across main analyses)	consistent with the steppe hypothesis
Indo-European	Heggarty et al. (2023)	about 8,100	hybrid model: origin south of the Caucasus, later spread of some branches via the steppe
Austronesian	Gray et al. (2009)	about 5,200	Taiwan
Semitic	Kitchen et al. (2009)	about 5,750	the Levant, Early Bronze Age
Dravidian	Kolipakam et al. (2018)	about 4,500	South Asia
Sino-Tibetan	Sagart et al. (2019)	about 7,200	northern China, millet farmers

Family	Study	Approximate root age (years before present)	Homeland or association
Sino-Tibetan	Zhang et al. (2019)	about 5,900	northern China, Late Neolithic
Japonic	Lee and Hasegawa (2011)	about 2,200	consistent with agricultural (Yayoi) spread
"Transeurasian'	Robbeets et al. (2021)	about 9,000	West Liao River region, millet farmers (family itself contested)
Pama-Nyungan	Bouckaert et al. (2018)	roughly 6,000	Gulf Plains of northern Australia
Kra-Dai	Tao et al. (2023)	about 4,000	Guangxi-Guangdong area of southern China
Japonic	Huisman et al. (2025)	about 2,400 (Japanese-Ryukyuan split, 400–500 BCE)	consistent with a Yayoi-period spread

Two things stand out. First, the estimates cluster between roughly 2,000 and 10,000 years. This is consistent with the long-held view that the comparative method cannot reach back much further than about eight to ten thousand years, because lexical replacement erodes the signal of common ancestry until it becomes indistinguishable from chance resemblance (Nichols, 1992). But the clustering is partly a selection effect and should not be over-interpreted. Phylogenetic dating is applied to families that the comparative method has already recognized, and families older than the comparative method's ceiling are, by construction, not recognized as families. Every estimate also depends on calibrations (usually historical or archaeological events of the last few thousand years), on a clock model that extrapolates rates from those calibrations to the root, and on the treatment of ancient languages. The defensible statement is that *the recognized families that have been dated have Holocene root ages under the calibrations and clock models used*, not that language families in general are no older than the Holocene. Phylogenetic methods have not convincingly broken the ceiling. The most prominent attempt, the claim by Pagel et al. (2013) that a set of "ultraconserved" words supports a Eurasiatic superfamily about 15,000 years old, was strongly criticized for relying on reconstructed proto-forms of uneven quality and on similarity judgements that could reflect chance (Heggarty, 2013). Global trees inferred from automated cognate judgements (Jäger, 2018) are useful for comparative studies, but their deepest nodes are not interpreted as evidence for macrofamilies.

Second, for the few families with several independent estimates, the estimates differ by more than their nominal credible intervals would suggest. The two Sino-Tibetan studies, using different data and calibrations, differ by more than a millennium, although they agree on a northern Chinese origin. For Indo-European, the estimates range from about 6,000 to almost 10,000 years. The differences arise not from errors in the MCMC but from different data (which meanings, which cognate judgements, which languages), different models (in particular the treatment of ancient languages as tips or as ancestors) and different calibrations (Chapters 10 and 15). A credible interval is conditional on the data and the model; it does not capture uncertainty about the choice of data and model. Robust chronological conclusions require multiple analyses with different, independently justified choices, and a root age should be reported as an estimate under stated assumptions rather than as a discovery. Recent simulation work sharpens the point. The Bayesian samplers used for tip-dating appear to be well calibrated for their own models (King, 2026a), but on Indo-European histories simulated under a richer model, with loans and rate variation, published dating methods

did not reliably recover the simulated root age and their intervals understated the error (Rönchen et al., 2025a).

The Indo-European controversy is the most thoroughly studied case of competing dates, and it illustrates how much rests on modelling choices; Section 15.9 treats it in detail. The Anatolian hypothesis (Renfrew, 1987) links the spread of Indo-European to the spread of farming from Anatolia beginning about 9,000 years ago; the steppe hypothesis (Anthony, 2007) links it to the expansion of pastoralists from the Pontic-Caspian steppe about 6,000 to 5,000 years ago. The analyses of Gray and Atkinson (2003) and Bouckaert et al. (2012) estimated the older date, and the second also inferred an Anatolian homeland using a phylogeographic model. Chang et al. (2015) argued that ancient languages such as Latin, Old Irish and Vedic Sanskrit are very close to the ancestors of their modern descendants, and that treating them as distant cousins inflates the estimated rate of change and therefore the root age. Constraining these ancient languages to be ancestral, they obtained a root age of about 6,000 years (medians of roughly 5,900 to 6,500 years across their main analyses), compatible with the steppe hypothesis. Meanwhile, ancient DNA revealed a massive migration from the steppe into Europe around 5,000 years ago (Haak et al., 2015), which many took as decisive for the steppe hypothesis, at least for the European branches. Heggarty et al. (2023) then assembled a new, carefully curated cognate database (IE-CoR) with explicit protocols for meaning and cognate coding, modelled ancient languages as sampled ancestors, and estimated a root age of about 8,100 years. They proposed a hybrid scenario, in which the family originated south of the Caucasus and some branches later spread through the steppe, compatible with ancient DNA evidence for population movements from the south (Lazaridis et al., 2022). The debate is not resolved: the IE-CoR data are now openly available for reanalysis (Anderson et al., 2025), critics from the lexicostatistical tradition have questioned parts of its cognate coding and the adequacy of a 170-meaning list (Kassian and Starostin, 2025), and critics of the whole phylogenetic approach have questioned its foundations (Pereltsvaig and Lewis, 2015). What the episode demonstrates is that the root age is not robust to defensible changes in the treatment of ancient languages and in data design. It has also forced every side to make its assumptions explicit, and the convergence of linguistic, archaeological and genetic evidence on more nuanced scenarios is itself a product of the controversy.

18.4 Homelands and expansions: farming, and not only farming

The *farming/language dispersal hypothesis* holds that many of the world's large language families spread with the expansion of early agricultural populations, whose demographic growth allowed them to spread into areas occupied by hunter-gatherers (Bellwood and Renfrew, 2002; Diamond and Bellwood, 2003; Bellwood, 2005). Linguistic phylogenetics offers some of the most direct tests of this idea, because dated trees can be compared with the archaeological chronology of farming. The test is only as independent as the calibrations: where a tree is calibrated with archaeological dates that are themselves tied to the farming hypothesis, agreement with that hypothesis is partly circular (Section 15.7).

The results are mixed in an informative way. Several families have root ages and homelands consistent with agricultural expansions. Austronesian originated in Taiwan at about the time that rice and millet farming appear in the archaeological record of the region (Gray et al., 2009). The Sino-Tibetan studies both locate the origin among millet farmers of northern China (Sagart et al., 2019; Zhang et al., 2019). The Bantu expansion, whose phylogeny and routes were reconstructed by Currie et al. (2013), Grollemund et al. (2015) and Koile et al. (2022), is closely associated with

the spread of farming (and later iron working) across sub-Saharan Africa. Japonic is dated to about the time of the arrival of wet-rice agriculture in the Japanese archipelago (Lee and Hasegawa, 2011). Robbeets et al. (2021) argued that a Transeurasian family spread with millet farming from northeast China, combining linguistic, archaeological and genetic evidence in what they called "triangulation". The Anatolian version of the Indo-European homeland is itself a farming dispersal hypothesis.

But the farming model is not universal, and some of the most compelling phylogeographic results concern expansions that had nothing to do with agriculture. Pama-Nyungan, which covers most of Australia, spread among hunter-gatherers; Bouckaert et al. (2018) inferred a mid-Holocene origin in the Gulf Plains region of northern Australia and an expansion that was not driven by farming. Semitic originated in the Levant in the Early Bronze Age, millennia after the origins of farming in the region (Kitchen et al., 2009). The steppe version of Indo-European origins links it to mobile pastoralism. And the Transeurasian proposal remains contested at a more basic level, since many specialists doubt that the Turkic, Mongolic, Tungusic, Koreanic and Japonic families are related at all. A fair summary is that the dated trees are consistent with agriculture having been one important driver of language expansions, but that other demographic and social processes, including pastoralism, technological innovation, and changes in social organization, have also produced large language families.

The phylogeographic methods behind these homeland inferences (Chapter 11) deserve their own caveats. They assume that the present locations of languages reflect their past movements reasonably well, so that a family's homeland lies near the centroid of its deepest lineages. Language replacement, long-distance migration and the extinction of entire branches (as happened to many early Indo-European languages of the steppe and Anatolia) can mislead them. Sampling matters too: a homeland inference can be pulled toward the region with the densest sampling of languages. A posterior distribution over the root location, moreover, is a statement under the diffusion model; it does not show that the model describes how speech communities actually moved. In the vocabulary of Section 6.10, the root location may be identifiable under the model without being straightforwardly interpretable as a historical homeland. Homeland estimates should be computed over the posterior sample of dated trees rather than on a single summary tree, since uncertainty in topology and dates propagates into the location. These problems do not invalidate phylogeographic inference, but they mean that homeland estimates are only as good as the coverage of the family, ancient and modern, and they gain greatly in credibility when supported by independent archaeological or genetic evidence.

18.5 Pulses, pauses and routes

Dated phylogenies have been used to infer not just when expansions began but how they proceeded. The pulse-pause pattern found for Austronesian (Gray et al., 2009) has its counterparts elsewhere. In the Bantu expansion, Grollemund et al. (2015) found that the spread of Bantu-speaking populations was shaped by habitat: populations tended to track their preferred savanna environments, moving more slowly through the rainforest, and the opening of savanna corridors during a dry period appears to have facilitated the expansion southward. The route through or around the central African rainforest has long been debated; Koile et al. (2022) used a phylogeographic analysis to support a route through the rainforest rather than around it. For Arawakan in lowland South America, Walker and Ribeiro (2011) inferred an origin in western Amazonia and an expansion along river systems. For the Tupi family, Walker et al. (2012) used a distance-based

phylogeny and geography to examine its spread across lowland South America.

These studies share a common logic. A dated tree with locations at the tips allows the reconstruction of ancestral locations along every branch, and thus of the speed and direction of movement through time. Environmental features such as coastlines, rivers, mountains and vegetation zones can then be tested as influences on the movement. This is one of the areas where linguistic phylogenetics has borrowed most directly from biology, in particular from viral phylogeography, and where the borrowing has been most productive. The inferred tempo and routes are, however, conditional on three things: the resolution of the dates (a "pause" is only detectable if it is longer than the uncertainty in the node ages that bound it), the diffusion model (random diffusion on a sphere cannot easily capture movement along a coast or a river, and landscape-aware models remain underused; Chapter 11), and the sampling of languages across the region. The debate over the Bantu route shows that different diffusion models and datasets can support different routes. Pulse-pause and route findings are therefore best described as supported for the best-studied cases, Austronesian and Bantu, and still debated in their details.

In biology, *comparative phylogeography* asks whether several co-distributed lineages responded to the same historical event at the same time, for example whether many species expanded simultaneously after the last glacial maximum. Hierarchical approximate Bayesian computation was developed to test for such synchronous divergence or expansion across taxa (Hickerson et al., 2006). Linguistic phylogenetics has so far studied families one at a time. A comparative phylogeographic analysis could ask, for instance, whether the expansions of several families in the same region (in mainland Southeast Asia, in Amazonia, or in the Sahel) were synchronous and thus plausibly driven by a common climatic or technological cause. To our knowledge this has rarely if ever been attempted with language data, although the dated trees needed for it are increasingly available.

18.6 The tempo of lexical change

Some of the most general and theoretically interesting findings of linguistic phylogenetics concern the process of language change itself. Because they are estimated as parameters of models fitted on trees, they inherit the assumptions of both the tree and the model of change.

18.6.1 Rates vary predictably among meanings

The best-supported finding about lexical change is that the rate at which words are replaced varies enormously among meanings, and that this variation is not random. Pagel et al. (2007) estimated replacement rates for the 200 meanings of the Swadesh list across Indo-European and found that rates varied roughly a hundredfold. The words for numerals, pronouns and some other very frequent meanings were replaced extremely slowly, while words for meanings such as 'dirty', 'to turn' or 'to throw' changed rapidly. Frequency of use in everyday speech was a strong predictor of the rate, with frequently used words changing more slowly, and the pattern was similar across the Indo-European languages examined. Frequencies of use for the same meanings are themselves strikingly similar across unrelated languages (Calude and Pagel, 2011), which suggests that the relationship reflects general properties of language use rather than a peculiarity of Indo-European. Further work has found that semantic properties, such as how concrete a meaning is and how many related meanings it is connected to, also predict replacement rates (Vejdemo and Hörberg, 2016).

The conditions are worth separating. That rates vary among meanings is supported in every family studied and by the large improvement in fit that rate-heterogeneity models provide (Chapter 7). The *specific* relationship with frequency rests most heavily on Indo-European, on frequency counts from a few modern corpora projected back in time, and on a fixed meaning list in which meaning slots are treated as independent; its extension to other families is suggested rather than demonstrated. Rate estimates for individual meanings are also affected by how synonyms and borrowings are coded (Section 15.3). Practically, the finding explains why glottochronology's assumption of a single, universal rate of replacement failed (Chapter 9), and why modern models must include rate variation among meanings. Theoretically, it is consistent with the view that language change is shaped by the cognitive and social processes of language use, and it parallels the finding in molecular evolution that functionally important genes evolve more slowly.

18.6.2 Change is partly punctuational

Biologists have long debated whether evolutionary change accumulates gradually or is concentrated at speciation events. The same question can be asked of languages, and dated trees make it testable.

Atkinson et al. (2008) tested whether languages accumulate lexical change gradually or whether change is concentrated at the moments when languages split. If splitting itself causes change, for example through founder effects or through speakers emphasizing differences to mark a new identity, then languages on lineages with more splitting events should have accumulated more change, a pattern known in biology as the node-density effect. Analyzing Bantu, Indo-European and Austronesian, they found evidence of such punctuational bursts, which they estimated to account for roughly 10 to 33 percent of the total lexical divergence. The finding suggests that some lexical change is tied to the social processes of language splitting rather than to the passage of time alone, and it has implications for dating, since a clock model that ignores punctuational change will misattribute some change to time (Chapter 9). The inference rests on counting nodes along root-to-tip paths, and node counts depend on how densely each part of the tree is sampled and on errors in tree reconstruction, which can themselves produce an apparent node-density effect. The result is therefore supported, but the artefact has to be ruled out anew in each dataset.

18.6.3 Grammar is not more stable than vocabulary

A long tradition in historical linguistics holds that grammatical structure is more stable than vocabulary, and therefore a better guide to deep relationships. Phylogenetic analyses have tested this claim directly and mostly failed to support it. Greenhill et al. (2010a) compared the phylogenetic signal and tempo of structural and lexical data in Austronesian, and Greenhill et al. (2017) carried out a larger and more detailed comparison of rates of change in grammatical features and basic vocabulary across Austronesian languages. They found that grammatical features, taken as a class, were not more stable than basic vocabulary; many structural features changed faster than the average basic vocabulary item, and the lineages with unusually fast change in one system were often not the ones with fast change in the other. The two systems appear to respond to different evolutionary pressures. The comparison is conditional on how features are coded (binary or multistate, and which features are included) and on a tree inferred from the lexicon, and its strongest evidence comes from one family. Studies of the stability of individual typological features, using a variety of methods, have found that some features are much more stable than others (Wichmann and Holman, 2009b; Dediu, 2011; Dediu and Cysouw, 2013). Agreement among methods about

which features are stable is only partial, however, which is itself an important caution against using any single "stable" feature as evidence of deep relationship. The finding that some structural features retain phylogenetic signal at considerable time depths has nonetheless encouraged their use in difficult cases, such as the structural phylogenetics of the Papuan languages of island Melanesia (Dunn et al., 2005).

18.7 Universals: lineage-specific or universal?

Typologists have long sought universal dependencies between grammatical features, such as Greenberg's correlation between verb-object order and the order of adposition and noun. Phylogenetic comparative methods (Chapter 13) offer a way to test such dependencies while accounting for the non-independence of related languages. The most influential test, by Dunn et al. (2011), used Bayesian models of correlated evolution on trees of four families (Austronesian, Bantu, Indo-European and Uto-Aztecan) and found that the dependencies between word-order features differed from family to family. They concluded that word-order correlations are largely lineage-specific, the product of cultural evolution within families rather than of universal cognitive constraints. The paper prompted vigorous responses. Critics pointed out that four families provide a small and possibly unrepresentative sample, that the power to detect dependencies within single families is limited, and that the results were not as inconsistent with Greenberg's correlations as the paper suggested (Dryer, 2011).

Later work has qualified the original conclusion. Using a much larger sample of languages and families, phylogenies for every lineage and explicit model comparison, Jäger and Wahle (2021) found credible correlations for many of the classic word-order pairs, in line with Dryer's "verb patterners". They also found that a model with the same dynamics in every lineage fitted better than a lineage-specific one, so their data gave no clear support for lineage-specific dependencies. Reconstruction of ancestral word orders has also suggested systematic tendencies in the direction of change (Maurits and Griffiths, 2014). The emerging picture is that both views capture part of the truth. There are cross-linguistic tendencies in how features co-evolve, but they are tendencies rather than laws, they can be overridden by lineage-specific histories, and detecting them requires large, phylogenetically diverse samples. The largest test so far, of 191 proposed universals on Grambank with models that account for both descent and geography, found support for about a third of them, mostly word-order and hierarchical universals, and evolutionary analyses suggesting that languages tend to change toward the preferred patterns (Verkerk et al., 2025b). Two assumptions run through all of these analyses: that the trees (for most families, classifications with arbitrary branch lengths) are adequate scaffolds, and that contact between languages can be neglected or modelled separately. Such results should be checked over posterior samples of trees, or at least over alternative branch-length schemes, because a single summary tree can distort inferred rates and dependencies (Section 17.12). The release of Grambank (Skirgård et al., 2023) has made large-scale tests much easier. Analyses of Grambank data have also shown that shared descent accounts for a substantial part of the variation in grammatical structure, more than spatial proximity, which reinforces the case for taking phylogeny into account in any typological analysis.

18.8 Population, society and environment

A set of high-profile studies has asked whether the properties and rates of change of languages depend on the populations that speak them or on their environments. Results here are less secure than for the themes above, and several have been contested. They sit at the far end of the chain of inference described in Section 18.1: they depend on trees, on measures of population and environment that are usually modern proxies for past conditions, and on controls for contact and areal clustering.

Population size and rates of change. Theory from population genetics suggests that the rate of change in a population can depend on its size, though in different directions depending on whether change is driven by drift or by innovation. Bromham et al. (2015) analyzed Polynesian languages and found that larger populations gained new words faster, while smaller populations lost existing words faster. Greenhill et al. (2018) extended the analysis to Indo-European, Austronesian and Bantu, and did not find a consistent relationship between population size and rates of lexical change across the three families. The effect of population size on rates is therefore not established as a general law.

Population size and complexity. Lupyan and Dale (2010) found, in a large non-phylogenetic sample from WALS, that languages with more speakers tend to have simpler inflectional morphology, and proposed that this reflects the influence of adult second-language learners. The claim has been widely discussed. Using Grambank data and phylogenetically informed models, Shcherbakova et al. (2023) found no evidence that languages spoken in larger societies, or by more non-native speakers, have less grammatical complexity. This is a clear example of how adding phylogenetic control can overturn an influential correlational finding.

Environment. Everett et al. (2015) reported that languages with complex tone systems are rare in very dry climates, and proposed a physiological mechanism involving the effect of dry air on the vocal folds. The finding was challenged on several grounds, including the clustering of tonal languages in a few families and areas and the difficulty of separating environmental effects from shared history and contact (Hammarström, 2016). More generally, Roberts and Winters (2013) showed that cross-linguistic data, analyzed without proper control for history and geography, readily produce strong but spurious correlations with arbitrary cultural variables. The lesson for the field is methodological: claims about environmental or demographic causes of linguistic features require phylogenetic and spatial controls, robustness checks across families and across trees, and, ideally, a mechanism that can be tested independently.

Endangerment. Phylogenetic and comparative methods are also being used to understand the threats to linguistic diversity. Bromham et al. (2022) modelled the predictors of language endangerment worldwide, finding that factors such as the density of roads and the level of schooling in a region are associated with higher endangerment, and predicting a substantial loss of languages within the coming century unless action is taken. These are associations in observational data and projections under the fitted model, not demonstrated causes. Skirgård et al. (2023) emphasized that the loss of endangered languages would remove much of the structural diversity currently documented in some regions.

18.9 Language trees as scaffolds for cultural evolution

Language trees have become the standard scaffold for studying the evolution of culture in human societies, because languages are usually transmitted along the same lines as many other cultural traits (Chapter 14). The program was launched by Mace and Pagel (1994), who argued that the comparative method of evolutionary biology could solve Galton's problem in anthropology. Since then, studies using language phylogenies have reconstructed the ancestral states and evolutionary dynamics of a wide range of traits. Holden and Mace (2003) showed that the adoption of cattle in Bantu-speaking societies was associated with the loss of matrilineal descent. Jordan et al. (2009) inferred that early Austronesian societies were probably matrilocal, and Fortunato and Jordan (2010) compared the evolution of marital residence in Indo-European and Austronesian. Currie et al. (2010a) found that political complexity in Austronesian societies rose, and in the best-fitting model also fell, in small sequential steps, with a model that allowed larger drops fitting almost as well. Watts et al. (2016) reported that ritual human sacrifice was associated with the emergence and maintenance of social stratification in Austronesian societies, and Watts et al. (2015a) found that broad supernatural punishment, but not moralizing high gods, tended to precede political complexity in the same societies. da Silva and Tehrani (2016) used Indo-European trees to argue that some folktales, such as "The Smith and the Devil", may go back to Proto-Indo-European times.

The collective message of this work, summarized by Gray and Watts (2017), is that cultural evolution has macroevolutionary regularities that can be studied with the same quantitative tools as biological macroevolution: some cultural traits are strongly conserved along lineages, others change rapidly; some changes are directional, others reversible; and traits co-evolve in ways that can be tested. Each such result is conditional on three assumptions: that the cultural trait was transmitted along the language lineages rather than horizontally between neighbours, that the trait was coded consistently across societies described by different ethnographers, and that the language tree, with its dates, is a reasonable scaffold. The better studies carry tree uncertainty through by repeating the analysis over a posterior sample of trees; analyses on a single summary tree can overstate the confidence of an ancestral state or of the order of two changes, even when the summary topology looks plausible (Section 17.12). With these qualifications, the program has turned speculative narratives about cultural history into testable hypotheses.

18.10 What the methods themselves have taught us

Beyond its substantive findings, linguistic phylogenetics has learned several lessons about its own methods, which recur throughout this book.

Data matter more than models. The largest differences between published analyses of the same family stem from the data: which languages are included, which meanings are used, how cognates are judged and coded, and how ancient languages are treated. The re-analysis of Indo-European with the IE-CoR database (Heggarty et al., 2023) was motivated as much by data design as by modelling. Investments in carefully curated, openly available data, of the kind represented by Lexibank, Grambank and IE-CoR (Appendix G), have arguably done more for the reliability of results than any new model.

Priors and calibrations shape dates. Dates depend strongly on calibrations, tree priors and clock models, and on how these interact (Chapters 9 and 10). Sensitivity analyses and prior-only runs should be routine, and the uncertainty arising from defensible alternative choices should be reported alongside the credible intervals of a single preferred analysis.

Model adequacy is rarely checked. Most published analyses choose among a handful of models by marginal likelihood or information criteria, but few ask whether even the best model can reproduce key features of the data, such as the distribution of cognate-class sizes or the number of singletons per language. Posterior predictive checks of *empirical model adequacy* are well developed in biology and readily applicable (Sections 15.12 and 17.15). The first systematic check of the standard set-up found that the covarion model fails to reproduce several features of Indo-European cognate data, and that a default setting for partition rates miscalibrated the posterior (King, 2026b).

Better fit is not always better history. A related lesson, still under debate, is that the model with the best marginal likelihood need not recover the history best. Rönchen (2025) argues that finite-state models such as the covarion model, combined with the ascertainment correction used for cognate data, overfit, so that they can fit better than Dollo-type models and still infer less accurate trees (Rönchen et al., 2025b). Whether or not this result holds generally, it reinforces the case for judging models by simulation from realistic generating processes as well as by relative fit.

Different kinds of validation answer different questions. The trust the field places in its methods rests on several kinds of evidence that are easily conflated (Section 6.10). Most simulation studies (Barbançon et al., 2013; Greenhill et al., 2009) test *parameter recovery* (data simulated under the model's own assumptions) or *robustness under misspecification* (data simulated with borrowing or rate variation that the model ignores), and they are only as realistic as the simulator. Agreement with families of "known" history is weaker evidence than it appears, because the known history and the cognate coding both come from the comparative method; the cleanest out-of-sample tests use historically attested dates or splits that were not used as calibrations. *Empirical model adequacy* on real data, and the *historical interpretation* of estimated parameters, are the least developed. What these studies have taught is where methods fail (short internal branches, concentrated contact, misplaced ancient languages) as much as where they succeed.

Openness and reproducibility pay off. The adoption of shared data standards (CLDF) and tools that generate analysis files from short configuration files (BEASTling) has made analyses easier to reproduce and extend, and has made it possible to compare results across families. Chapter 17 gives a practical template, and its software runs show that even published tools need maintenance if analyses are to remain reproducible.

18.11 A scorecard

Table 18.2 summarizes the findings discussed in this chapter. For each, it names the main kind of evidence, the assumptions on which the finding depends, and a status. *Broadly accepted* means that the finding has been replicated with different data and methods, survives reasonable changes to models and priors, and agrees with independent evidence. *Supported but debated* means that the main analyses support it, but that it rests on assumptions that have been seriously questioned or on few families. *Contested* means that analyses disagree, or that the original claim has not survived later tests (noted in parentheses). The ratings are necessarily judgements, and readers should revise them as the evidence changes; Section 15.13 shows how to check the corresponding assumptions in a new analysis.

Table 18.2: A scorecard of findings in linguistic phylogenetics: the main evidence, the assumptions on which each finding depends, and a summary judgement of its status.

Finding	Evidence	Main assumptions	Status
Major subgroups recoverable from basic vocabulary	many families; simulation	expert cognates; few loans; tree-like history	broadly accepted (not independent of the comparative method)
Deepest splits poorly resolved	low support; method disagreement	short internal branches	broadly accepted
Replacement rates vary greatly among meanings	all families studied	meaning-slot coding; rate model	broadly accepted
Rates predicted by frequency of use	mainly Indo-European	modern frequencies; meaning list	supported but debated
Dated families have Holocene roots	recognized families only	calibrations; clock; which families are recognized	broadly accepted, with a selection effect
Precise root ages of particular families	competing analyses	ancient languages; priors; data design	contested
Several families spread with farming	dates and homelands vs archaeology	independent calibrations; diffusion model	supported but debated
Pulses and pauses shaped by geography	Austronesian, Bantu	date resolution; diffusion model	supported but debated
Part of lexical change punctuational	three families	node counts reflect real splits	supported but debated
Grammar not more stable than vocabulary	mainly Austronesian	feature coding; lexical tree	supported but debated
Word-order dependencies purely lineage-specific	four families, then larger samples	power; tree quality; contact	contested (strong form not supported)
Population size sets rates of lexical change	Polynesian; three families	population proxies; trees	contested (not consistent)
Larger societies have simpler grammars	WALS; Grambank with phylogeny	complexity measures; controls	contested (not supported with phylogeny)
Climate affects tone	global correlation	controls for history and contact	contested
Macrofamilies beyond 10,000 years detectable	reconstructed forms; global trees	quality of reconstructions; chance	contested (not established)
Standard lexical models adequate for cognate data	posterior predictive checks; simulation	test statistics; realism of simulators	contested (covarion fails several checks; overfitting argued)

Two patterns stand out in the scorecard. The most secure findings are those closest to the data, about classification and about the relative rates of change of different meanings, while the most contested are those that depend on long chains of inference, from absolute dates to causal claims about society and environment. And the claims that have weakened over time have usually done so because later work added phylogenetic or spatial controls, larger and more diverse samples, or

more careful data curation. That trajectory is a sign of a healthy field: its strongest claims are being tested, and the ones that survive are becoming more credible. Even the first row, however, carries a qualification that should travel with it whenever it is cited: agreement with the comparative method is consistency between two analyses of overlapping evidence, not an independent check of either.

18.12 Open questions and the next decade

What will the next decade bring? The following questions seem likely to drive research, and each connects to methods from biology that are only beginning to be used in linguistics (Chapter 16, which grades each method by how far its transfer to language data has been tested).

How deep can we go? The time ceiling of the comparative method is a property of lexical data, not of the methods themselves. Progress beyond it, if it is possible at all, will probably require different kinds of data: structural features with demonstrably slow rates of change, sound-level models that can distinguish regular correspondences from chance resemblance, or combinations of data types in total-evidence analyses. Models of sequence-level sound change are emerging. Bouchard-Côté et al. (2013) reconstructed ancestral word forms automatically with probabilistic models of sound change, Hruschka et al. (2015) modelled regular sound changes as events of concerted evolution, and Goldstein et al. (2026) inferred trees directly from aligned cognate word forms with an insertion-deletion model borrowed from molecular evolution. Scaling such models up to phylogenetic inference is a major open challenge.

How can borrowing be modelled explicitly? Most analyses still treat borrowing as noise, either by excluding known loans or by hoping that its effect is small. Models that include lateral transfer explicitly, such as the stochastic Dollo model with lateral transfer of Kelly and Nicholls (2017), and Bayesian models that detect contact events on a tree (Neureiter et al., 2022), are beginning to change this. A mature theory would infer the tree and the contact events jointly, estimate their timing, and test hypotheses about which kinds of vocabulary are borrowed under which social conditions. Biological methods for reticulate evolution, from gene-tree/species-tree reconciliation to phylogenetic networks, offer a rich toolbox (Chapters 8 and 16).

Can linguistic, genetic and archaeological histories be integrated formally? "Triangulation" across disciplines is currently mostly qualitative: separate analyses are compared after the fact (Robbeets et al., 2021). Joint models that relate language trees to genetic population histories, and that estimate where and when they diverge, would allow formal tests of whether languages spread by migration or by language shift. Large databases that link genetic and linguistic data for the same populations (Barbieri et al., 2022) make this increasingly feasible, and the mismatches between genetic and linguistic histories are as informative as the matches.

What drives diversification? Why do some families have thousands of languages while others have a handful? Biological diversification models that relate speciation and extinction rates to traits and environments (Chapter 16) have barely been applied to languages. Applying them requires care, since language "extinction" and "speciation" are cultural processes with their own dynamics, since sampling of extinct languages is usually very incomplete, and since many diversification parameters are not identifiable from trees of living lineages alone. But they could turn longstanding questions about the drivers of linguistic diversity into quantitative tests.

How can phylogenies inform the preservation of diversity? Conservation biology uses phylogenies to prioritize the protection of evolutionarily distinct species. The same logic could help

identify languages whose loss would remove the most unique linguistic history.

Conservation biologists measure the evolutionary history represented by a set of species with *phylogenetic diversity*, the total length of the branches of the tree that connect them (Faith, 1992). The EDGE approach ("evolutionarily distinct and globally endangered") combines a species' evolutionary distinctiveness, measured from its share of the tree, with its extinction risk to set conservation priorities (Isaac et al., 2007). The analogy with endangered languages is direct. A language isolate, or the last speaker community of an ancient branch of a family, represents far more unique history than one of many closely related dialects, and global trees and endangerment data are now available (Jäger, 2018; Bromham et al., 2022). Phylogenetic diversity measures and EDGE-like scores could thus inform the allocation of scarce documentation resources. To our knowledge such measures have rarely been applied systematically to languages, and their use would raise interesting questions about how to treat isolates and about which branch lengths (lexical, structural, or temporal) best capture "linguistic history".

Will automation transform the field? Automated cognate detection, phonetic alignment and data standardization have already made it possible to analyze thousands of languages (Jäger, 2018; List et al., 2017). As these methods improve, the bottleneck will shift from data assembly to data quality and to the interpretation of results. The central challenge will be to combine the scale of automated methods with the precision of expert philological work, rather than to replace one with the other. Recent results point both ways. Characters derived from automatic alignments of word forms can rival expert cognates within families and outperform them across families, at least for topology (Jäger, 2025); but the cognate datasets available are small, which limits parameter-rich models and machine learning (Häuser et al., 2026), and trees can change with seemingly innocuous upstream choices such as how concepts are translated into words (Snee et al., 2026).

How should results be evaluated? As analyses become larger and more complex, the field needs agreed standards for reporting, for validating models, and for archiving data and posterior samples. The most important development of the next decade may not be a new model but the routine use of simulation-based validation, posterior predictive checks and transparent sensitivity analyses, so that the reliability of a published phylogeny can be judged by its readers rather than taken on trust.

Linguistic phylogenetics began as an attempt to answer old questions about the history of language families with new tools. It has answered some of them, provisionally and under stated assumptions, and has shown that others are harder than they looked. Its most lasting contribution may turn out to be the habit of stating historical hypotheses as explicit probabilistic models, which can be tested, compared, and improved.

Findings in linguistic phylogenetics form a chain (classification, chronology, geography, process, coevolution); each link adds assumptions, so security generally decreases along the chain. Basic-vocabulary trees reproduce the major subgroups established by the comparative method, but this agreement is not independent validation: the cognate judgements come from the comparative method, and cognate sharing is not shared innovation. Disagreements concentrate on short, deep branches where both approaches are uncertain. The recognized families that have been dated have Holocene root ages under the calibrations and clock models used, a pattern that partly reflects which families can be recognized at all; estimates for the same family can differ by more than their credible intervals because of differences in data, models and the treatment of ancient languages. Dated trees are consistent with many families having spread with farmers, but others, notably Pama-Nyungan, spread among hunter-gatherers or pastoralists, and inferred homelands

and routes are conditional on diffusion models, sampling and the resolution of dates. Rates of lexical replacement vary about a hundredfold among meanings; the link with frequency rests mainly on Indo-European; punctuational change and the relative instability of grammar are supported but debated. Claims about universals, population size and environment have often weakened once phylogeny is properly controlled, and comparative and cultural analyses should be run over posterior samples of trees. Simulation studies are mostly tests of parameter recovery or robustness; empirical model adequacy and historical interpretation remain the least developed forms of validation.

18.13 Exercises

Conceptual

1. A paper reports that its Bayesian tree of a family "recovers all the subgroups recognized by specialists, validating the method". Explain why this is not independent validation, and describe one test that would be more nearly independent.
2. Many large language families have estimated root ages within the Holocene. List the assumptions (calibrations, clock model, treatment of ancient languages, tree prior) and the selection effect on which this generalization depends, and say which of them could be relaxed to test it.
3. Choose one row of Table 18.2 whose status is "supported but debated". What new data or analysis would move it to "broadly accepted", and what result would move it to "contested"?
4. Classify each of the following with the vocabulary of Section 6.10: (a) recovering a simulated tree from data generated under the inference model; (b) recovering a simulated tree from data generated with concentrated borrowing; (c) predicting the attested date of a split that was not used as a calibration; (d) a posterior predictive check of the number of singletons per language.

Research

1. A team proposes to test whether a family spread with farming by comparing its dated root and homeland with the archaeological spread of agriculture in the region. Design the analysis so that the comparison is not circular: which calibrations would you allow, how would you treat ancient languages, how would you carry tree and date uncertainty into the homeland estimate, and what result would count against the hypothesis?
2. You want to know whether larger speech communities replace basic vocabulary faster. Using the studies discussed in Section 18.8, justify a choice of families, population measures, trees (with their uncertainty) and controls for contact, and say which parameters you expect to be practically estimable with the available samples.
3. Pick a published claim about cultural coevolution on a language tree. Identify the assumptions about transmission, coding and the tree on which it rests, and propose a robustness analysis (alternative trees, posterior samples, alternative codings) that would show whether the claim survives.

18.14 Further reading

- Gray et al. (2010): A short, influential essay on the balance between tree-like and reticulate

history in human populations, with a framework for measuring treelikeness that remains useful.

- Bowern (2018): A concise review in the *Annual Review of Linguistics* of computational phylogenetics in linguistics, covering data, methods, results and controversies.
- Greenhill et al. (2020): An accessible handbook chapter on Bayesian phylolinguistics by three of its leading practitioners, explaining the main methods and results.
- Gray and Watts (2017): A review of the cultural macroevolution program, explaining why language trees matter for understanding the evolution of human societies and summarizing results from Austronesian and other families.
- Heggarty et al. (2023) and Chang et al. (2015). Read together, the two most methodologically careful analyses of Indo-European origins show how data design and model choice shape the results.
- Pagel et al. (2007) and Greenhill et al. (2017). Two landmark studies of the tempo of language change, one on lexical rates and frequency, the other on grammar and vocabulary.
- Dunn et al. (2011) and Jäger and Wahle (2021). The debate over lineage-specific versus universal word-order dependencies, from the original provocation to the large-scale reassessment.
- Bellwood (2005): The fullest statement of the farming/language dispersal hypothesis, essential background for the dates and homelands discussed in this chapter.
- Nichols (1992): A classic study of linguistic diversity in space and time, whose discussion of the time limits of the comparative method frames the question of how deep phylogenetic methods can reach.
- List (2024): Ten open problems of computational historical linguistics, revisited in the light of recent progress; a useful agenda to set against this chapter's list.
- Rönchen (2025) and King (2026b): Two recent evaluations of the standard models of vocabulary evolution, by simulation and by posterior predictive checks, whose conclusions any future analysis will need to address.

Appendix A

Mathematical Preliminaries

This appendix collects the mathematics that the chapters take for granted. It is written for readers who met probability in an introductory statistics course, perhaps some years ago, and who have not studied linear algebra or stochastic processes. Each section starts with the intuition, gives the few formulas the chapters rely on, shows a short piece of R code to run and modify, and ends with a note on where in the book the material is needed. Table A.1 shows which sections each chapter draws on; a sensible strategy is to start a chapter and to come here when an idea is unfamiliar.

The sections form two strands. One is statistical (Sections A.1–A.4 and A.12): how a probability model is turned into an inference. The other concerns the models themselves: how a cognate class or a word-order type changes along a branch (Sections A.5–A.7), how lineages split and die out (Section A.8), and how a continuous quantity such as a homeland coordinate wanders over time (Sections A.9–A.11). Every phylogenetic analysis combines the two: a model of change gives the probability of the data, and a method of inference turns that probability into estimates.

Notation used in this appendix and in the book. $P(A)$ is the probability of an event A, and $f(x)$ is a probability density. $P(A \mid B)$, read "the probability of A given B", is a conditional probability. $X \sim \text{Poisson}(\lambda)$ means "X has a Poisson distribution with parameter λ". $\mathrm{E}[X]$, $\mathrm{Var}(X)$ and $\mathrm{Cov}(X, Y)$ are expectation, variance and covariance. Logarithms are natural logarithms (base e), whether written ln or log. Lower-case bold letters ($\mathbf{x}$) are vectors, upper-case bold letters ($\mathbf{Q}$, $\mathbf{C}$) are matrices, $\mathbf{x}^\top$ is a transpose, $\mathbf{I}$ is the identity matrix and $\mathbf{1}$ is a vector of ones. A hat marks an estimate: $\hat{\theta}$ is an estimate of θ.

Table A.1: Which preliminaries each chapter uses. **x**: needed to follow the chapter's key equations; o: useful background. Column codes: Prob = Section A.1, Bayes = A.2, Lik = A.3, Dist = A.4, Mat = A.5, Exp = A.6, CTMC = A.7, Pois = A.8, BM = A.9, MVN = A.10, GLS = A.11, MCMC = A.12.

Chapter	Prob	Bayes	Lik	Dist	Mat	Exp	CTMC	Pois	BM	MVN	GLS	MCMC
1 History and motivation	o			o				o				
2 Trees and tree space	o				o			**x**				
3 Linguistic data	o			o								
4 Distance and parsimony	**x**		o		o		o	o			o	

Chapter	Prob	Bayes	Lik	Dist	Mat	Exp	CTMC	Pois	BM	MVN	GLS	MCMC
5 Maximum likelihood	x	o	x	x	x	x	x	o				
6 Bayesian inference	x	x	x	x	o	o	o	x				x
7 Lexical models	x	o	x	x	x	x	x	x				o
8 Borrowing and networks	x		o	o	o		o					
9 Clocks and rates	x	x	o	x			o	x	x			o
10 Divergence times	x	x	o	x				x				o
11 Phylogeography	x	x	x	x	x	x	x	o	x	x	x	o
12 Ancestral states	x	x	x	o	o	o	x		o	o		o
13 Comparative methods	x	o	x	o	x	o	x		x	x	x	o
14 Cultural traits	x	o	o				x		o	o	x	o
15 Critiques and pitfalls	x	x	x	x			o	o				x
16 The biologist's toolbox	x	o	x	x	o	o	x	x	o			o
17 A workflow in R	o	x	o	x					o	o	x	x
18 Synthesis	o	o										

A.1 Probability and conditional probability

A probability is a number between 0 and 1 attached to an event. It can be read as a long-run frequency (the fraction of meanings on a 200-item list that are replaced within a millennium, over very many comparable cases) or as a degree of belief (how strongly the evidence supports the hypothesis that Anatolian was the first branch to split from the rest of Indo-European). Bayesian inference (Chapter 6) uses both readings. The mathematics is the same in either case and rests on three rules:

1. $0 \le P(A) \le 1$, and an event that is certain has probability 1.
2. If A and B cannot both happen, $P(A \text{ or } B) = P(A) + P(B)$.
3. As a consequence, $P(\text{not } A) = 1 - P(A)$.

A.1.1 Random variables, expectation and variance

A *random variable* is a quantity whose value is determined by chance. The number of meanings, out of 200, for which two languages have cognate words is a *discrete* random variable: it takes whole-number values, and its distribution is given by the probabilities $P(X = x)$, which sum to 1. The time since two languages split is a *continuous* random variable. Its distribution is given by a *probability density* $f(x)$, and probabilities are areas under the density:

$$P(a < X < b) = \int_a^b f(x)\,dx.$$

A density is not a probability: it can exceed 1, and the probability of any single exact value, such as a split exactly 4,312 years ago, is zero. The *cumulative distribution function* $F(x) = P(X \le x)$ is a probability in both cases.

The *expectation* (mean) $\mathrm{E}[X] = \sum_x x\,P(X = x)$ or $\int x f(x)\,dx$ is the probability-weighted average of X. The *variance* $\mathrm{Var}(X) = \mathrm{E}\big[(X - \mathrm{E}[X])^2\big]$ measures spread, and its square root is the *standard deviation*. Expectation is linear, $\mathrm{E}[aX + bY] = a\mathrm{E}[X] + b\mathrm{E}[Y]$, and the variance of a sum of *independent* variables is the sum of their variances. For dependent variables the *covariance* $\mathrm{Cov}(X, Y) = \mathrm{E}\big[(X - \mathrm{E}[X])(Y - \mathrm{E}[Y])\big]$ enters too: $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X, Y)$. The *correlation* is the covariance divided by the product of the two standard deviations.

A.1.2 Joint, marginal and conditional probability

Consider two typological features: the order of object and verb (OV or VO) and the type of adposition (postpositions or prepositions). The R code below gives counts for 100 languages, invented in the spirit of Greenberg's word-order universals. The *joint* probability $P(\mathrm{OV}, \mathrm{postpositions})$ is the proportion of languages with both properties, 0.45. A *marginal* probability such as $P(\mathrm{OV})$ sums the joint probabilities over the other feature: $0.45 + 0.05 = 0.50$. The *conditional* probability of postpositions given OV is the proportion of OV languages that have postpositions:

$$P(A \mid B) = \frac{P(A, B)}{P(B)}, \qquad \text{equivalently} \qquad P(A, B) = P(A \mid B)\,P(B). \tag{A.1}$$

Here $P(\mathrm{postpositions} \mid \mathrm{OV}) = 0.45/0.50 = 0.90$, whereas $P(\mathrm{postpositions} \mid \mathrm{VO}) = 0.08/0.50 = 0.16$. The two features are therefore not *independent*. Two events are independent when knowing one does not change the probability of the other, $P(A \mid B) = P(A)$, or equivalently $P(A, B) = P(A)\,P(B)$.

```
counts <- matrix(c(45, 5, 8, 42), nrow = 2, byrow = TRUE,
                 dimnames = list(order = c("OV", "VO"),
                                 adposition = c("Postp", "Prep")))
joint <- counts / sum(counts)       # joint probabilities
rowSums(joint)                      # marginal P(order)
```

```
#>  OV  VO
#> 0.5 0.5
```

```
joint / rowSums(joint)              # conditional P(adposition | order)
```

```
#>      adposition
#> order Postp Prep
#>    OV  0.90 0.10
#>    VO  0.16 0.84
```

```
# under independence the joint would be the product of the marginals
outer(rowSums(joint), colSums(joint))
```

```
#>    Postp  Prep
#> OV 0.265 0.235
#> VO 0.265 0.235
```

A catch runs through the whole book: the 100 languages are not 100 independent draws. Related languages share both features because they inherited them, so a sample dominated by one large family can make two features look associated when the association reflects a handful of historical events. This is Galton's problem, and the tools for it are phylogenetic covariance matrices (Section

A.10), generalized least squares (Section A.11) and models of correlated change on trees (Chapter 13).

A.1.3 The law of total probability

Often we can compute the probability of an event B only conditionally on some unknown variable. If $A_1, A_2, \ldots$ are mutually exclusive possibilities that cover every case, then

$$P(B) = \sum_i P(B \mid A_i)\, P(A_i). \tag{A.2}$$

In phylogenetics the unknowns are the states of ancestral languages. We never observe whether Proto-Germanic had a word of a given cognate class for ‘dog’, but we can compute the probability of the daughters *given* each possible ancestral state and average over those states. Felsenstein’s pruning algorithm (Chapter 5) organizes this sum efficiently over all internal nodes of a tree.

A.1.4 Conditional independence

Two events can be dependent and yet independent once a third variable is known. This *conditional independence* explains why related languages resemble each other. Suppose a proto-language had a word of a certain cognate class with probability $P(A = 1) = 0.6$. Each of two daughters keeps the class with probability 0.8 if the ancestor had it, and acquires it (by parallel development or borrowing) with probability 0.1 if not. Given the ancestor’s state the daughters change independently, $P(D_1 = 1, D_2 = 1 \mid A = a) = P(D_1 = 1 \mid A = a)\, P(D_2 = 1 \mid A = a)$. Without that knowledge they are dependent: seeing the word in one daughter makes it more likely that the ancestor had it, and so that the other daughter has it too.

```
p_A <- c(absent = 0.4, present = 0.6)     # P(A = 0), P(A = 1)
p_D <- c(absent = 0.1, present = 0.8)     # P(D = 1 | A = 0), P(D = 1 | A = 1)
p_D1 <- sum(p_D * p_A)                    # law of total probability
p_both <- sum(p_D^2 * p_A)                # independent given A, then sum
c(P_D1 = p_D1, P_both = p_both, product_of_marginals = p_D1^2)
```

```
#>                P_D1              P_both product_of_marginals
#>              0.5200              0.3880              0.2704
```

The probability that both daughters have the word, 0.388, is well above the product of the marginal probabilities, 0.270. All statistical dependence between related languages arises in this way: daughters evolve independently along their own branches but start from the same ancestor. Phylogenetic models also assume conditional independence *among characters*: given the tree and parameters, the probability of a data matrix is the product of the probabilities of its columns. For language data this is not always reasonable; cognate classes within one meaning compete for the same slot (Chapter 7).

A.1.5 Working on the log scale

With thousands of characters, the product of their probabilities is smaller than the smallest number a computer can store and is rounded to zero. Software therefore works with logarithms: the log of a product is the sum of the logs. When probabilities must be *added* on the log scale, the *log-sum-exp* trick avoids the same problem: $\ln(e^a + e^b) = m + \ln(e^{a-m} + e^{b-m})$ with $m = \max(a, b)$.

```
p <- rep(0.3, 2000)                # 2000 characters, each with probability 0.3
prod(p)                            # underflows to zero
```

```
#> [1] 0
```

```
sum(log(p))                        # the log-likelihood is perfectly usable
```

```
#> [1] -2408
```

```
log_sum_exp <- function(a) { m <- max(a); m + log(sum(exp(a - m))) }
log_sum_exp(c(-2000, -2001))       # ln(e^-2000 + e^-2001) without underflow
```

```
#> [1] -2000
```

Where this is used in the book. Conditional probability and the law of total probability are the basis of the likelihood calculation in Chapter 5 and of every model in Chapters 7 to 12. Marginal and joint ancestral reconstructions (Chapter 12) are conditional probabilities of ancestral states given the data. Galton's problem and the non-independence of languages are central to Chapters 13 and 14, and the dependence among characters within a meaning is taken up in Chapters 7 and 15.

A.2 Bayes' theorem

Writing the joint probability of H and D in two ways with Equation (A.1), $P(H \mid D)\,P(D) = P(D \mid H)\,P(H)$, and dividing by $P(D)$ gives *Bayes' theorem*:

$$P(H \mid D) = \frac{P(D \mid H)\,P(H)}{P(D)}, \qquad P(D) = \sum_{H'} P(D \mid H')\,P(H'). \tag{A.3}$$

Read H as a hypothesis and D as data. $P(H)$ is the *prior* probability, $P(D \mid H)$ the *likelihood* (how probable the data are if H is true) and $P(H \mid D)$ the *posterior* probability. The denominator, the *marginal likelihood* or *evidence*, comes from the law of total probability (Equation (A.2)) and makes the posterior probabilities sum to one.

A.2.1 A loanword example

Historical linguists reason this way informally all the time. A word in one language resembles its equivalent in a neighbouring, related language: inherited or borrowed? Suppose experience with this semantic field suggests that 30% of such resemblances are loans, $P(\text{loan}) = 0.3$. The data: the word does *not* show the sound correspondence regular in inherited vocabulary. Inherited words are sometimes irregular through analogy or taboo, say with probability 0.1; a word borrowed after the sound changes usually is, say with probability 0.7. (The numbers are illustrative.) Then

$$P(\text{loan} \mid \text{irregular}) = \frac{0.7 \times 0.3}{0.7 \times 0.3 + 0.1 \times 0.7} = \frac{0.21}{0.28} = 0.75.$$

The irregularity raises the probability of borrowing from 0.3 to 0.75, but does not make it certain. Dividing Equation (A.3) for H_1 by the same equation for H_0 cancels $P(D)$ and gives the *odds* form:

$$\underbrace{\frac{P(H_1 \mid D)}{P(H_0 \mid D)}}_{\text{posterior odds}} = \underbrace{\frac{P(D \mid H_1)}{P(D \mid H_0)}}_{\text{Bayes factor}} \times \underbrace{\frac{P(H_1)}{P(H_0)}}_{\text{prior odds}} . \tag{A.4}$$

Here the prior odds of a loan are $3/7$, the Bayes factor is $0.7/0.1 = 7$, and the posterior odds are 3, a probability of $3/4$. The Bayes factor measures how far the data shift the odds, wherever they started. When the hypotheses are whole models with their own parameters, it is a ratio of marginal likelihoods, each averaged over that model's prior (Chapter 6).

A.2.2 Bayes' theorem for continuous parameters

For a continuous parameter θ, such as a rate of word replacement, the theorem applies to densities:

$$f(\theta \mid D) = \frac{f(D \mid \theta)\, f(\theta)}{f(D)}, \qquad f(D) = \int f(D \mid \theta)\, f(\theta)\, d\theta, \tag{A.5}$$

summarized as "posterior $\propto$ likelihood $\times$ prior" ($\propto$ means "proportional to"; $f(D)$ does not depend on θ). Rearranged, the equation says that the ratio of posterior to prior density at any value θ is $f(D \mid \theta)/f(D)$: the data raise the plausibility of exactly those values that predicted the data better than the model did on average (Morey et al., 2016). Estimation and model comparison thus rest on the same evidence. In simple cases this ratio, evaluated at a point value nested in a larger model, equals the Bayes factor in favour of that point value (the Savage–Dickey density ratio).

Suppose we want the probability p that a basic-vocabulary item survives a millennium. In a language documented at two stages a thousand years apart, 81 of 100 meanings still use a word of the same cognate class (an invented but realistic figure). Treating meanings as independent trials, the likelihood is binomial (Section A.4), $f(D \mid p) \propto p^{81}(1-p)^{19}$. With a beta prior, $f(p) \propto p^{a-1}(1-p)^{b-1}$, the product is $p^{81+a-1}(1-p)^{19+b-1}$, so the posterior is Beta$(a+81, b+19)$. Such a *conjugate* prior is rare in phylogenetics, which is why the book relies on Markov chain Monte Carlo (Section A.12), but it makes the logic transparent. The code also computes the posterior on a grid, a brute-force method that works for any one-parameter model.

```
k <- 81; N <- 100
a <- 1; b <- 1                                  # flat Beta(1, 1) prior
post_mean <- (a + k) / (a + b + N)
cred <- qbeta(c(0.025, 0.975), a + k, b + N - k)
c(mean = post_mean, lower = cred[1], upper = cred[2])
```

```
#>   mean  lower  upper
#> 0.8039 0.7219 0.8746
```

```
# the same posterior on a grid: prior x likelihood, then normalize
p <- seq(0.001, 0.999, by = 0.001)
unnorm <- dbeta(p, a, b) * dbinom(k, N, p)
post <- unnorm / sum(unnorm)
c(grid_mean = sum(p * post))
```

```
#> grid_mean
#>    0.8039
```

```
# an informative Beta(20, 20) prior matters more when data are few
qbeta(c(0.025, 0.975), 20 + 8, 20 + 2)        # 8 of 10 retained
```

```
#> [1] 0.4221 0.6933
```

```
qbeta(c(0.025, 0.975), 20 + 81, 20 + 19)      # 81 of 100 retained
```

```
#> [1] 0.6446 0.7923
```

The 95% *credible interval* is a direct probability statement: given the model, prior and data, p lies between the limits with probability 0.95. (A confidence interval does not mean this; see Section A.3.2.) The last two lines show a general pattern. The prior Beta(20, 20), centred on 0.5, pulls the estimate strongly toward 0.5 with 10 meanings but much less with 100. Posteriors become less sensitive to the prior as data accumulate; marginal likelihoods, and hence Bayes factors, remain sensitive to how widely the prior is spread (Chapter 6).

Where this is used in the book. Bayes' theorem is the foundation of Chapter 6 and of most analyses in Chapters 9 to 12, where priors on rates, tree shapes and calibrations do much of the work. Bayes factors appear in model comparison (Chapter 6), in tests of dispersal routes (Chapter 11) and of correlated evolution (Chapters 13 and 14). The sensitivity of results to priors is a recurring theme of Chapters 10, 15 and 17.

A.3 Likelihood and maximum likelihood

The *likelihood* of a parameter value θ is the probability of the observed data computed as if θ were true:

$$L(\theta) = P(D \mid \theta), \tag{A.6}$$

regarded as a function of θ with the data held fixed. As a function of the data, $P(D \mid \theta)$ sums to one; as a function of θ it need not, and it is not a probability distribution for θ. Only *ratios* matter: $L(\theta_1)/L(\theta_2) = 4$ means that the data are four times as probable under θ_1 as under θ_2, so constant factors not involving θ can be dropped.

A.3.1 A worked example: dating a split from shared cognates

Two languages share cognates for $k = 110$ of $N = 200$ meanings. Assume, as glottochronology did, that every meaning's word is replaced at the same constant rate r in every lineage, independently. A word survives time t in one lineage with probability e^{-rt} (Section A.8), so a cognate pair survives in both with probability $p(t) = e^{-2rt}$, and the likelihood of the time since the split is binomial:

$$L(t) = \binom{N}{k} e^{-2rtk} \left(1 - e^{-2rt}\right)^{N-k}. \tag{A.7}$$

The *maximum likelihood estimate* (MLE) is the value of t that makes the data most probable; it is easiest to find by maximizing the *log-likelihood* $\ell = \ln L$. For the binomial probability itself, $\ell(p) = k \ln p + (N - k) \ln(1 - p) + \text{const}$, and setting $\ell'(p) = k/p - (N - k)/(1 - p)$ to zero

gives $\hat{p} = k/N$. MLEs are *invariant* under reparameterization (the MLE of a function of p is that function of $\hat{p}$), so solving $e^{-2r\hat{t}} = k/N$ gives

$$\hat{t} = -\frac{\ln(k/N)}{2r},$$

the glottochronological formula of Equation (9.4). Glottochronology was maximum likelihood estimation under a very strong model; its problems (Chapter 9) lie in the assumptions of a universal constant rate, independent meanings and no borrowing, not in the arithmetic.

A.3.2 Uncertainty from the shape of the likelihood

A sharply peaked log-likelihood means that the data pin the parameter down. The curvature at the peak, the *observed information* $I(\hat{\theta}) = -\ell''(\hat{\theta})$, gives an approximate standard error $1/\sqrt{I(\hat{\theta})}$. A more robust *likelihood interval* contains the values whose log-likelihood is within 1.92 units of the maximum (half of 3.84, the 95% point of the χ^2_1 distribution). It is an approximate *confidence interval*: if the model is true, intervals constructed this way contain the true value in about 95% of repeated data sets, a statement about the procedure rather than about any one interval.

```
N <- 200; k <- 110
r <- -log(0.805)                          # 80.5% retention per millennium
loglik <- function(t) dbinom(k, N, exp(-2 * r * t), log = TRUE)
fit <- optimize(loglik, c(0.01, 10), maximum = TRUE)
t_hat <- fit$maximum
c(numerical = t_hat, formula = -log(k / N) / (2 * r))
```

```
#> numerical   formula
#>     1.378     1.378
```

```
# standard error from the curvature (numerical second derivative)
h <- 1e-4
d2 <- (loglik(t_hat + h) - 2 * loglik(t_hat) + loglik(t_hat - h)) / h^2
se <- 1 / sqrt(-d2)
# likelihood interval: where the log-likelihood has dropped by 1.92
drop_fn <- function(t) loglik(t) - (fit$objective - qchisq(0.95, 1) / 2)
ci <- c(uniroot(drop_fn, c(0.01, t_hat))$root,
        uniroot(drop_fn, c(t_hat, 10))$root)
round(c(t_hat = t_hat, se = se, lower = ci[1], upper = ci[2]), 3)
```

```
#> t_hat    se lower upper
#> 1.378 0.147 1.109 1.688
```

```
tt <- seq(0.8, 2.2, length.out = 200)
par(mar = c(4, 4, 1, 1))
plot(tt, sapply(tt, loglik), type = "l", xlab = "Time since split (millennia)",
     ylab = "Log-likelihood")
abline(h = fit$objective - 1.92, lty = 2)
abline(v = t_hat, col = "grey50")
```

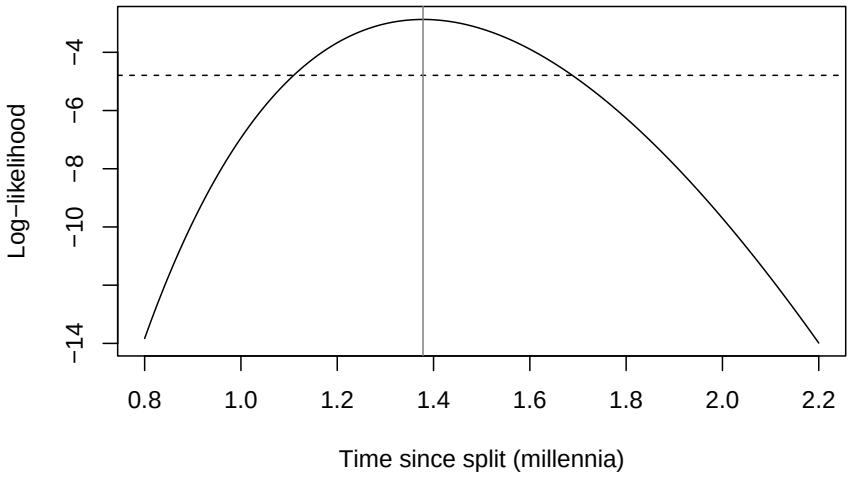


Figure A.1: Log-likelihood of the time since the split of two languages that share 110 cognates out of 200 meanings, with the replacement rate fixed at 0.217 per millennium. The dashed line lies 1.92 log-units below the maximum; the times at which the curve crosses it are the ends of an approximate 95 per cent likelihood interval. The grey line marks the maximum likelihood estimate.

The estimate is about 1.38 millennia with a standard error of about 0.15. That is *sampling* uncertainty under the model, much smaller than the uncertainty about the model itself, such as whether the rate really is 0.217 per millennium in these lineages.

A.3.3 Comparing models

Two models are *nested* when one is a special case of the other, as equal gain and loss rates are a special case of unequal ones. If the smaller model is true and the sample large, the *likelihood ratio statistic* $\Lambda = 2[\ell(\hat{\theta}_1) - \ell(\hat{\theta}_0)]$ has approximately a χ^2 distribution with degrees of freedom equal to the number of extra parameters. The approximation fails when the smaller model puts a parameter on the boundary of its range (testing whether a variance is zero, say), and tree topologies are not nested models at all, so comparing trees needs the special tests of Chapter 5. For non-nested models, AIC $= -2\ell + 2p$ and BIC $= -2\ell + p \ln n$, with p parameters and n observations, trade fit against complexity (Equation (5.12)).

When the model is correct and the sample large, MLEs are *consistent* (they converge to the true value) and asymptotically as precise as any estimator. In phylogenetics "large" usually means many characters; the number of languages, and of independent historical events, is fixed by history. When the model is wrong, the MLE converges to the value that brings the model closest to the truth, which need not be historically meaningful.

Where this is used in the book. Likelihood is the subject of Chapter 5 and the engine of every model-based method in the book: models of lexical change and their comparison (Chapter 7), glottochronology as an estimator (Chapter 9), maximum likelihood ancestral states (Chapter 12), and fitting and comparing Brownian, Ornstein-Uhlenbeck and discrete-character models in Chapters 13, 14 and 16. In Bayesian analyses the same likelihood is combined with a prior (Section A.2).

A.4 Probability distributions used in the book

A handful of standard distributions do almost all the work in phylogenetic models: for counts, waiting times, positive rates, proportions and unbounded quantities (Table A.2). In R each distribution `xxx` has four functions: `dxxx` (probability or density), `pxxx` (cumulative distribution), `qxxx` (quantiles) and `rxxx` (random draws).

Table A.2: The distributions used in the book. The R name is the suffix of the `d`, `p`, `q` and `r` functions; base R has no Dirichlet functions, but Dirichlet draws are easily made from gamma draws.

Distribution	Values	Mean	R name	Typical use in the book
Binomial(N, p)	$0, 1, \dots, N$	Np	`binom`	Number of meanings retained or shared
Poisson(λ)	$0, 1, 2, \dots$	λ	`pois`	Number of changes on a branch; number of cognate classes born
Exponential(λ)	$x > 0$	$1/\lambda$	`exp`	Waiting time until a replacement or a split; priors on rates and branch lengths
Gamma(α, β)	$x > 0$	α/β	`gamma`	Rate variation among characters; priors on rates
Lognormal(μ, σ)	$x > 0$	$e^{\mu+\sigma^2/2}$	`lnorm`	Branch rates in relaxed clocks; calibration densities
Beta(a, b)	$0 < x < 1$	$a/(a+b)$	`beta`	Proportions: retention probabilities, state frequencies
Dirichlet$(\alpha_1, \dots, \alpha_K)$	proportions summing to 1	$\alpha_k / \sum_j \alpha_j$	(via `gamma`)	Frequencies of several states; relative rates
Normal(μ, σ^2)	any real number	μ	`norm`	Brownian increments; coordinates; calibration densities

A.4.1 Counts: binomial and Poisson

If each of N independent trials succeeds with the same probability p, the number of successes K has the *binomial* distribution

$$P(K = k) = \binom{N}{k} p^k (1-p)^{N-k}, \qquad \mathrm{E}[K] = Np, \qquad \mathrm{Var}(K) = Np(1-p). \tag{A.8}$$

Retained and shared meanings were modelled this way in Sections A.2 and A.3. Both assumptions are questionable for vocabulary: some meanings ('two', 'I') are far more stable than others ('dirty', 'to throw'). When p varies across meanings, counts vary more than the binomial predicts (*overdispersion*), and the remedy is to let rates vary across characters, usually with a gamma distribution (below).

The *Poisson* distribution describes counts of independent events occurring at a constant average

rate, with no upper limit:

$$P(K = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \qquad \mathrm{E}[K] = \mathrm{Var}(K) = \lambda. \tag{A.9}$$

It is the limit of the binomial for large N and small p with $Np = \lambda$, and sums of independent Poisson counts are Poisson. The number of replacements of a meaning's word along a branch, and the number of cognate classes born on a branch in the stochastic Dollo model (Chapter 7), are Poisson counts (Section A.8).

A.4.2 Waiting times: the exponential distribution

If events happen at constant rate λ, the waiting time T until the next one has the *exponential* density

$$f(t) = \lambda e^{-\lambda t}, \qquad P(T > t) = e^{-\lambda t}, \qquad \mathrm{E}[T] = 1/\lambda, \qquad \mathrm{Var}(T) = 1/\lambda^2. \tag{A.10}$$

The *survival function* $P(T > t) = e^{-\lambda t}$ is the retention curve of glottochronology, and the *half-life* is $\ln 2/\lambda$. The exponential is *memoryless*: $P(T > s + t \mid T > s) = P(T > t)$, so a word that has survived 2,000 years is as likely to be replaced next century as one borrowed last year. That is a strong assumption; entrenchment with age, or the fragility of recent loans, would make waiting times non-exponential. Exponential distributions also serve as priors for branch lengths and rates when only a rough scale is known.

A.4.3 Rates: the gamma distribution and discrete-gamma categories

The *gamma* distribution with shape α and rate β has density

$$f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \qquad \mathrm{E}[X] = \frac{\alpha}{\beta}, \qquad \mathrm{Var}(X) = \frac{\alpha}{\beta^2}, \tag{A.11}$$

where $\Gamma(\alpha)$ is the gamma function ($\Gamma(n) = (n-1)!$ for whole numbers). For whole-number α, X is the sum of α exponential waiting times with rate β. Parameterizations differ between programs (R's `dgamma()` accepts `rate` $= \beta$ or `scale` $= 1/\beta$; BEAST and RevBayes have their own conventions), so check before typing in a prior.

The gamma distribution's main use in the book is *rate variation among characters*. Each character gets a rate multiplier ρ from a gamma distribution with $\alpha = \beta$, so the mean multiplier is 1 and the variance $1/\alpha$ (Equation (7.5)). Small α (< 1) means strong heterogeneity, with most characters very slow and a few very fast; large α means similar rates for all.

Because integrating the likelihood over a continuous rate distribution is expensive, Yang (1994) approximated it with K equally probable categories (usually $K = 4$): the gamma is cut at its quantiles into K slices of probability $1/K$, and each slice is represented by its mean. Since $x f(x;\alpha,\beta) = (\alpha/\beta)\, f(x;\alpha+1,\beta)$, the slice means come from the cumulative distribution function of a gamma with shape $\alpha + 1$:

$$r_c = K\left[G(b_c;\alpha+1,\alpha) - G(a_c;\alpha+1,\alpha)\right], \qquad c = 1,\ldots,K, \tag{A.12}$$

where G is the gamma cumulative distribution function and $[a_c, b_c]$ the c-th slice. A character's likelihood is the average of its likelihoods with branch lengths multiplied by each r_c (Equation (5.6)). Some programs use rescaled slice medians instead, one reason why likelihoods from different programs are not always comparable.

```
discrete_gamma <- function(alpha, K = 4) {
  cuts <- qgamma(seq(0, 1, length.out = K + 1), shape = alpha, rate = alpha)
  K * diff(pgamma(cuts, shape = alpha + 1, rate = alpha))  # slice means
}
rates <- sapply(c(0.5, 1, 5), discrete_gamma)
colnames(rates) <- paste("alpha =", c(0.5, 1, 5))
round(rates, 3)
```

```
#>      alpha = 0.5 alpha = 1 alpha = 5
#> [1,]       0.033     0.137     0.502
#> [2,]       0.252     0.477     0.804
#> [3,]       0.820     1.000     1.083
#> [4,]       2.894     2.386     1.611
```

```
colMeans(rates)             # each set of category rates averages to 1
```

```
#> alpha = 0.5   alpha = 1   alpha = 5
#>           1           1           1
```

With $\alpha = 0.5$ the slowest quarter of characters change at about 3% of the average rate and the fastest at almost three times it; with $\alpha = 5$ the categories are much closer together.

A.4.4 Multiplicative variation: the lognormal distribution

A positive X is *lognormal* if $\ln X \sim \mathcal{N}(\mu, \sigma^2)$. Its median is e^{μ} and its mean $e^{\mu+\sigma^2/2}$, so a mean of 1 needs $\mu = -\sigma^2/2$. It arises when many small independent effects multiply, is skewed to the right, and has coefficient of variation $\sqrt{e^{\sigma^2} - 1} \approx \sigma$ for small σ. In the uncorrelated lognormal relaxed clock (Chapter 9) each branch draws its rate from a lognormal, and σ measures the departure from a strict clock. Lognormal densities are also popular calibration priors (Chapter 10); programs differ in whether μ is given on the log scale or the mean on the real scale, a common source of error.

A.4.5 Proportions: the beta and Dirichlet distributions

The *beta* distribution describes a proportion p between 0 and 1:

$$f(p) = \frac{p^{a-1}(1-p)^{b-1}}{B(a,b)}, \qquad \mathrm{E}[p] = \frac{a}{a+b}, \tag{A.13}$$

where $B(a,b)$ is a normalizing constant. Beta$(1,1)$ is uniform, and larger $a+b$ concentrates the distribution around its mean. It serves as a prior for retention probabilities (Section A.2.2) and for the frequency of the "present" state in binary cognate models.

The *Dirichlet* distribution generalizes the beta to K proportions summing to 1, with density proportional to $\prod_k x_k^{\alpha_k - 1}$; each proportion alone is Beta$(\alpha_k, \sum_j \alpha_j - \alpha_k)$. Dirichlet$(1, \ldots, 1)$ is flat, and larger α_k cluster the proportions more tightly. The book uses Dirichlet priors for the stationary frequencies of multistate characters (say SOV, SVO and VSO) and for relative rates in a general rate matrix. A Dirichlet draw is a set of independent gamma draws with shapes α_k, each divided by their sum.

```
rdirichlet <- function(n, a) {
  g <- matrix(rgamma(n * length(a), shape = a), ncol = length(a),
              byrow = TRUE)
  g / rowSums(g)                 # each row is one set of proportions
}
set.seed(1)
flat <- rdirichlet(5000, c(SOV = 1, SVO = 1, VSO = 1))
tight <- rdirichlet(5000, c(SOV = 10, SVO = 10, VSO = 10))
rbind(mean_flat = colMeans(flat), sd_flat = apply(flat, 2, sd),
      mean_tight = colMeans(tight), sd_tight = apply(tight, 2, sd))
```

```
#>               [,1]    [,2]    [,3]
#> mean_flat  0.33164 0.33456 0.33380
#> sd_flat    0.23394 0.23577 0.23641
#> mean_tight 0.33429 0.33279 0.33293
#> sd_tight   0.08584 0.08503 0.08598
```

A.4.6 The normal distribution

The *normal* (Gaussian) density with mean μ and variance σ^2 is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]. \tag{A.14}$$

About 95% of its probability lies within 1.96 standard deviations of the mean. Sums of independent normals are normal, and by the *central limit theorem* the sum of many small independent effects is approximately normal, which is why the normal describes Brownian increments (Section A.9) and why MLEs are approximately normal in large samples. Normal densities, sometimes *truncated* or *offset* to start at a minimum age, are also calibration priors (Chapter 10). The multivariate version is the subject of Section A.10. Figure A.2 shows how shape parameters change four of these distributions.

```
op <- par(mfrow = c(2, 2), mar = c(4, 4, 1.5, 1), cex = 0.8)
cols <- c("black", "grey45", "grey70")
panel <- function(x, dens, xlab, main, labels, where = "topright") {
  matplot(x, dens, type = "l", lty = 1, lwd = 2, col = cols,
          ylim = c(0, min(max(dens), 4)), xlab = xlab, ylab = "Density",
          main = main)
  legend(where, labels, col = cols, lwd = 2, bty = "n")
}
x <- seq(0.005, 3, length.out = 400)
panel(x, sapply(c(0.5, 1, 5), function(a) dgamma(x, a, rate = a)),
      "Rate multiplier", "(a) Gamma", paste("alpha =", c(0.5, 1, 5)))
panel(x, sapply(c(0.2, 0.5, 1), function(s) dlnorm(x, -s^2 / 2, s)),
      "Branch rate", "(b) Lognormal", paste("sigma =", c(0.2, 0.5, 1)))
p <- seq(0.001, 0.999, length.out = 400)
panel(p, cbind(dbeta(p, 1, 1), dbeta(p, 2, 2), dbeta(p, 8, 2)),
      "Proportion", "(c) Beta", c("Beta(1, 1)", "Beta(2, 2)", "Beta(8, 2)"),
```

```
      where = "topleft")
kk <- 0:8
plot(kk, dpois(kk, 2), type = "h", lwd = 6, col = "grey60",
     xlab = "Number of changes", ylab = "Probability", main = "(d) Poisson")
points(kk, dbinom(kk, 200, 0.01), pch = 19)
```

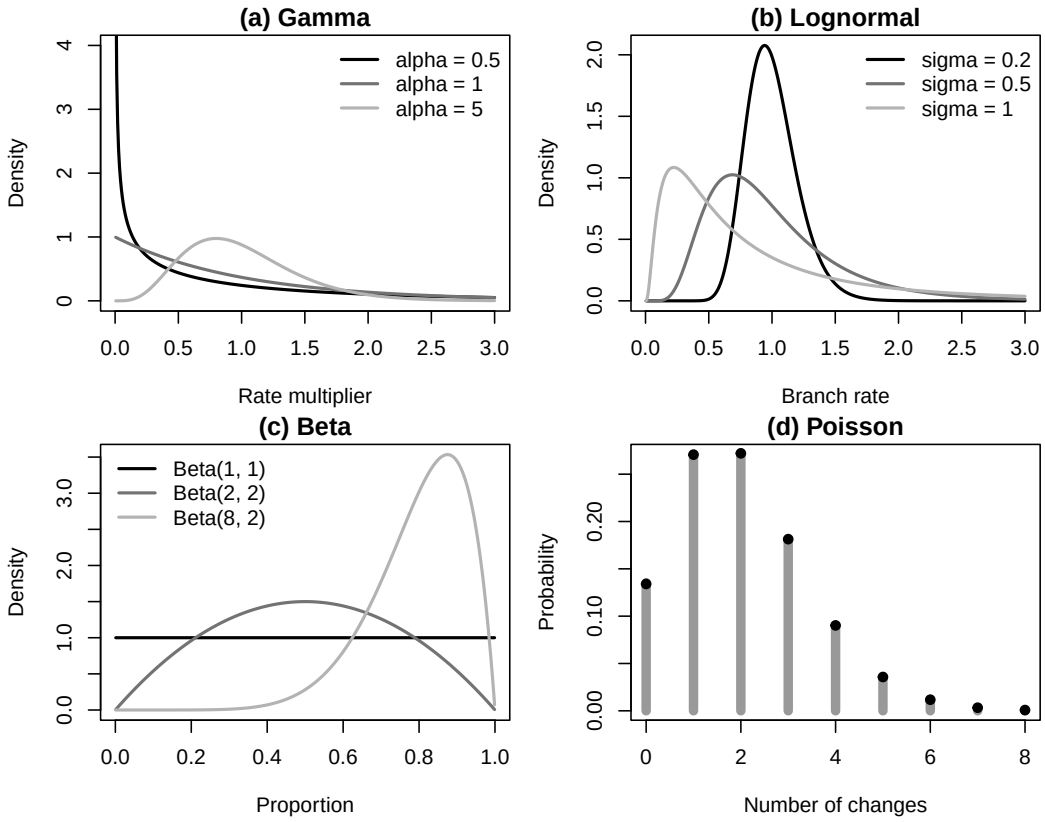


Figure A.2: Four distributions used in the book. (a) Gamma densities with mean one, as used for rate variation among characters: a small shape parameter means that most characters are slow and a few are very fast. (b) Lognormal densities with mean one and different values of sigma, as used for branch rates in relaxed clocks. (c) Beta densities for a proportion. (d) The Poisson distribution with mean 2 (bars) and the binomial distribution with N = 200 and p = 0.01 (points), which it approximates.

```
par(op)
```

Where this is used in the book. Gamma rate heterogeneity appears in Chapters 5, 7, 9 and 15; lognormal and exponential priors on rates and branch lengths in Chapters 6, 9 and 10; calibration densities in Chapter 10; Poisson counts in the stochastic Dollo model (Chapter 7) and in the prior on the number of dispersal routes (Chapter 11); beta and Dirichlet priors in Chapters 6, 7 and 17.

A.5 Vectors, matrices and eigendecomposition

A matrix is a rectangular table of numbers with rules for combining tables. A data matrix has a row per language and a column per character; a rate matrix $\mathbf{Q}$ or transition matrix $\mathbf{P}(t)$ has a row and a column per character state; a covariance matrix $\mathbf{C}$ has a row and a column per language.

A.5.1 Vectors and matrix products

A *vector* is a list of numbers, written as a column $\mathbf{x}$ or a row $\mathbf{x}^\top$. An $m \times n$ matrix $\mathbf{A}$ has entries A_{ij} in row i and column j; its *transpose* $\mathbf{A}^\top$ swaps rows and columns. The product of an $m \times n$ matrix $\mathbf{A}$ and an $n \times p$ matrix $\mathbf{B}$ is the $m \times p$ matrix with entries

$$(\mathbf{AB})_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}, \tag{A.15}$$

combining row i of $\mathbf{A}$ with column j of $\mathbf{B}$. In general $\mathbf{AB} \neq \mathbf{BA}$. The *identity matrix* $\mathbf{I}$ (ones on the diagonal, zeros elsewhere) leaves any matrix unchanged.

A linguistic example shows why the definition is natural. Suppose that each century a language with basic order SOV, SVO or VSO keeps or changes its order, with M_{ij} the probability of moving from order i to order j; each row of $\mathbf{M}$ sums to 1. If the row vector $\mathbf{p}_0$ holds the probabilities of the three orders now, then by the law of total probability (Equation (A.2)) the probability of order j a century later is $\sum_i (\mathbf{p}_0)_i M_{ij}$, so $\mathbf{p}_1 = \mathbf{p}_0\mathbf{M}$ and after n centuries $\mathbf{p}_n = \mathbf{p}_0\mathbf{M}^n$. This is a *discrete-time* Markov chain; Section A.7 develops the continuous-time version that the book uses.

A.5.2 Inverses, determinants and positive definiteness

A square matrix $\mathbf{A}$ has an *inverse* $\mathbf{A}^{-1}$ if $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$. For a 2×2 matrix,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix},$$

and $ad - bc$ is the *determinant*; the inverse exists exactly when the determinant is not zero. The determinant measures how much the matrix stretches volumes and appears in the multivariate normal density (Section A.10). In R, `solve(A)` returns the inverse and `solve(A, b)` solves $\mathbf{Ax} = \mathbf{b}$; the second is faster and more accurate, and should be used for expressions such as $\mathbf{C}^{-1}\mathbf{x}$.

A matrix is *symmetric* if $\mathbf{A}^\top = \mathbf{A}$, and *positive definite* if also $\mathbf{x}^\top\mathbf{Ax} > 0$ for every non-zero $\mathbf{x}$. Covariance matrices are positive definite (unless some variable is an exact linear combination of others), because $\mathbf{x}^\top\mathbf{Ax}$ is the variance of a weighted sum. A positive definite matrix has a *Cholesky decomposition* $\mathbf{A} = \mathbf{L}\mathbf{L}^\top$ with $\mathbf{L}$ lower triangular, a kind of square root used to simulate correlated variables (Section A.10) and to remove correlation from data (Section A.11). R's `chol()` returns the upper triangular factor $\mathbf{L}^\top$. The *Kronecker product* $\mathbf{A} \otimes \mathbf{B}$ (`kronecker(A, B)`) replaces each entry A_{ij} by the block $A_{ij}\mathbf{B}$; Chapter 11 uses it to combine covariance among languages with covariance between latitude and longitude (Equation (D.3)).

A.5.3 Eigenvalues and eigenvectors

Multiplying a vector by a square matrix $\mathbf{A}$ usually changes both its length and its direction. For special vectors, the *eigenvectors*, it changes only the length:

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}, \tag{A.16}$$

and the stretching factor λ is the *eigenvalue*. If an $n \times n$ matrix has n independent eigenvectors, the columns of $\mathbf{V}$, with eigenvalues on the diagonal of $\mathbf{\Lambda}$, then

$$\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}, \qquad \text{and so} \qquad \mathbf{A}^n = \mathbf{V}\mathbf{\Lambda}^n\mathbf{V}^{-1}, \tag{A.17}$$

because the inner factors $\mathbf{V}^{-1}\mathbf{V}$ cancel. This *eigendecomposition* turns raising a matrix to a power (or, in Section A.6, exponentiating it) into doing so to each eigenvalue. Symmetric matrices have real eigenvalues and perpendicular eigenvectors, so $\mathbf{V}^{-1} = \mathbf{V}^\top$. The eigenvalues sum to the *trace* (the sum of the diagonal) and multiply to the determinant.

For a probability matrix such as $\mathbf{M}$, the largest eigenvalue is 1 and the others are smaller in absolute value, so their powers shrink and $\mathbf{p}_0\mathbf{M}^n$ forgets its starting point. It converges to a *stationary distribution* $\boldsymbol{\pi}$ with $\boldsymbol{\pi}\mathbf{M} = \boldsymbol{\pi}$; transposing shows that $\boldsymbol{\pi}$ is an eigenvector of $\mathbf{M}^\top$ with eigenvalue 1. The second-largest eigenvalue sets how fast the start is forgotten.

```
states <- c("SOV", "SVO", "VSO")
M <- matrix(c(0.97, 0.03, 0.00,      # per-century change probabilities
              0.01, 0.97, 0.02,      # (illustrative values)
              0.00, 0.05, 0.95),
            nrow = 3, byrow = TRUE, dimnames = list(states, states))
p <- c(SOV = 1, SVO = 0, VSO = 0)    # the proto-language is SOV
for (n in 1:30) p <- p %*% M         # 30 centuries
round(p, 3)
```

```
#>        SOV   SVO   VSO
#> [1,] 0.461 0.438 0.101
```

```
# the same via the eigendecomposition M = V Lambda V^{-1}
E <- eigen(M)
round(E$values, 4)
```

```
#> [1] 1.0000 0.9656 0.9244
```

```
M30 <- E$vectors %*% diag(E$values^30) %*% solve(E$vectors)
round(c(1, 0, 0) %*% M30, 3)
```

```
#>       [,1]  [,2]  [,3]
#> [1,] 0.461 0.438 0.101
```

```
# the stationary distribution: the eigenvector of t(M) with eigenvalue 1
v <- Re(eigen(t(M))$vectors[, 1])
round(v / sum(v), 3)
```

```
#> [1] 0.192 0.577 0.231
```

After 30 centuries an SOV proto-language has a 46% chance of still being SOV under these illustrative rates, and the long-run distribution favours SVO. Nothing about real word-order change should be read into the numbers; the point is that the chain's long-run behaviour is encoded in its eigenvalues and eigenvectors.

Where this is used in the book. Rate and transition probability matrices are used in Chapters 5, 7, 11, 12, 13 and 14. Distance matrices and the matrices of least-squares tree fitting appear in Chapter 4. Covariance matrices, their inverses, determinants and Cholesky factors are central to Chapters 11 and 13.

A.6 The matrix exponential

The exponential function solves the simplest equation of change: if $dy/dt = a\,y$, then $y(t) = y(0)\,e^{at}$. Retained vocabulary declines this way when each word is replaced at rate r: $dR/dt = -rR$, so $R(t) = e^{-rt}$. As a power series,

$$e^{x} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

For a character with several states and a matrix of rates $\mathbf{Q}$ (Section A.7), the transition probabilities satisfy $d\mathbf{P}(t)/dt = \mathbf{P}(t)\mathbf{Q}$ with $\mathbf{P}(0) = \mathbf{I}$ (Chapter 5). The solution is the *matrix exponential*, the same series with matrices in place of numbers:

$$\mathbf{P}(t) = e^{\mathbf{Q}t} = \mathbf{I} + \mathbf{Q}t + \frac{(\mathbf{Q}t)^2}{2!} + \frac{(\mathbf{Q}t)^3}{3!} + \cdots. \tag{A.18}$$

Intuitively, $\mathbf{I}$ corresponds to no change, $\mathbf{Q}t$ to one change, $(\mathbf{Q}t)^2/2$ to two in succession, and so on, so the sum accumulates all routes through intermediate states. That is why $P_{ij}(t)$ can be positive when the direct rate q_{ij} is zero: a language can get from SOV to VSO by way of SVO.

A.6.1 Computing the matrix exponential

If $\mathbf{Q} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$ (Equation (A.17)), every power in the series is $\mathbf{V}\mathbf{\Lambda}^n\mathbf{V}^{-1}$, and the series collapses to

$$e^{\mathbf{Q}t} = \mathbf{V}\ \mathrm{diag}\left(e^{\lambda_1 t}, \ldots, e^{\lambda_K t}\right)\mathbf{V}^{-1}. \tag{A.19}$$

Phylogenetic software relies on this formula, because one eigendecomposition gives $\mathbf{P}(t)$ for every branch length. For reversible rate matrices (Section A.7.3) the eigenvalues are real and the computation is stable; for general matrices, "scaling and squaring" methods based on $e^{\mathbf{A}} = (e^{\mathbf{A}/2^s})^{2^s}$ are preferred (Moler and Van Loan, 2003). The function `matexpo()` in ape computes the matrix exponential directly. Three pitfalls are worth knowing:

1. In R, `exp(Q * t)` exponentiates each *entry* separately. That is not the matrix exponential, and its rows do not even sum to one: a common, silent bug.
2. $e^{\mathbf{A}+\mathbf{B}} = e^{\mathbf{A}}e^{\mathbf{B}}$ only if $\mathbf{AB} = \mathbf{BA}$. It holds for $\mathbf{Q}s$ and $\mathbf{Q}t$, giving the *Chapman-Kolmogorov* equation $\mathbf{P}(s+t) = \mathbf{P}(s)\mathbf{P}(t)$: a branch's probabilities are the product of those of its two halves.

3. Rate and time enter only through $\mathbf{Q}t$. Doubling all rates and halving all times gives the same probabilities, so the data measure only the amount of change, the starting point of Chapter 9.

```
states <- c("SOV", "SVO", "VSO")
Q <- matrix(c(-0.3,  0.3,  0.0,      # rates per millennium (illustrative);
               0.1, -0.3,  0.2,      # no direct SOV <-> VSO change
               0.0,  0.5, -0.5),
            nrow = 3, byrow = TRUE, dimnames = list(states, states))
t <- 2
expm_series <- function(A, n_terms = 40) {
  out <- term <- diag(nrow(A))
  for (n in 1:n_terms) {
    term <- term %*% A / n           # A^n / n!
    out <- out + term
  }
  out
}
E <- eigen(Q)
P_eigen <- E$vectors %*% diag(exp(E$values * t)) %*% solve(E$vectors)
P_ape <- matexpo(Q * t)
dimnames(P_ape) <- dimnames(Q)
round(P_ape, 4)                      # note P[SOV, VSO] > 0
```

```
#>        SOV    SVO    VSO
#> SOV 0.5831 0.3565 0.0604
#> SVO 0.1188 0.6838 0.1974
#> VSO 0.0503 0.4934 0.4562
```

```
c(series = max(abs(expm_series(Q * t) - P_ape)),
  eigen = max(abs(P_eigen - P_ape)))
```

```
#>    series     eigen
#> 2.776e-16 1.110e-16
```

```
rowSums(exp(Q * t))                  # the wrong, entrywise exponential
```

```
#>   SOV   SVO   VSO
#> 3.371 3.262 4.086
```

```
# Chapman-Kolmogorov: P(1 + 1) = P(1) P(1)
max(abs(matexpo(Q * 1) %*% matexpo(Q * 1) - P_ape))
```

```
#> [1] 3.331e-16
```

Where this is used in the book. The matrix exponential is Equation (5.3) in Chapter 5, and it computes the branch probabilities of every discrete-character model in the book: binary and multistate cognate models (Chapter 7), clock models (Equation (9.1)), discrete phylogeography (Chapter 11), ancestral state reconstruction (Chapter 12) and tests of correlated evolution (Chapters 13 and 14).

A.7 Continuous-time Markov chains

A *continuous-time Markov chain* (CTMC) describes a character that is in one of K states at each moment and can jump between states at any time. The *Markov property* says that the chance of a jump in the next instant depends only on the current state, not on how long the character has been there or how it got there: the chance of losing a word next century depends on whether it is present now, and on nothing else in its history. That is an idealization. Differences in stability between characters can be absorbed by rate variation (Section A.4.3); dependence on a character's own history, such as the fragility of a recent loan, is a genuine violation.

A.7.1 The rate matrix

A CTMC is specified by a *rate matrix* $\mathbf{Q}$. For $i \neq j$, $q_{ij} \geq 0$ is the instantaneous rate from state i to j: in a short interval h, a character in state i moves to j with probability about $q_{ij}h$. The diagonal makes each row sum to zero, so $q_i = -q_{ii}$ is the total rate of leaving i. Equivalently, and this is also a simulation algorithm, a character entering state i stays for an exponential time with rate q_i (Section A.4.2) and then moves to j with probability q_{ij}/q_i. The transition probabilities are $\mathbf{P}(t) = e^{\mathbf{Q}t}$ (Section A.6), with $P_{ij}(t)$ the probability of state j at time t given state i at time 0.

A.7.2 The two-state chain in closed form

The most important CTMC in linguistic phylogenetics has two states: a cognate class is absent (0) or present (1). With gain rate α and loss rate β,

$$\mathbf{Q} = \begin{pmatrix} -\alpha & \alpha \\ \beta & -\beta \end{pmatrix}. \tag{A.20}$$

The transition probabilities need no matrix algebra. Let $p(t) = P_{11}(t)$. In a short interval h a present class is lost with probability βh and an absent one gained with probability αh, so

$$p(t+h) \approx p(t)(1-\beta h) + \big(1-p(t)\big)\alpha h, \qquad \text{hence} \qquad \frac{dp}{dt} = \alpha - (\alpha+\beta)\,p(t).$$

With $p(0) = 1$ the solution is $p(t) = \frac{\alpha}{\alpha+\beta} + \frac{\beta}{\alpha+\beta}e^{-(\alpha+\beta)t}$ (check by differentiating). Writing $\pi_1 = \alpha/(\alpha+\beta)$ and $\pi_0 = \beta/(\alpha+\beta)$, and arguing likewise from state 0,

$$\mathbf{P}(t) = \begin{pmatrix} \pi_0 + \pi_1 e^{-(\alpha+\beta)t} & \pi_1\left(1 - e^{-(\alpha+\beta)t}\right) \\ \pi_0\left(1 - e^{-(\alpha+\beta)t}\right) & \pi_1 + \pi_0 e^{-(\alpha+\beta)t} \end{pmatrix}. \tag{A.21}$$

This is Equation (5.5); Chapter 7 writes the same matrix as an overall rate times the stationary frequencies (Equation (7.1)), the overall rate playing the role of $\alpha+\beta$. The eigenvalues of $\mathbf{Q}$ are 0, which gives the long-run probabilities, and $-(\alpha+\beta)$, the speed at which the chain forgets its starting state.

The formula carries a lesson about time depth. As t grows, both rows approach (π_0, π_1): over branches much longer than $1/(\alpha+\beta)$ the character is *saturated*, and its present state says almost nothing about its earlier state. That is why very deep relationships are hard to recover (Chapter 15). With $\alpha = 0$ a class can only be lost, and $P_{11}(t) = e^{-\beta t}$ is the retention curve of glottochronology; the stochastic Dollo models of Chapter 7 combine this pure-loss process with a single birth of each cognate class.

A.7.3 Stationarity and reversibility

A *stationary distribution* $\boldsymbol{\pi}$ is left unchanged by the process: $\boldsymbol{\pi}\mathbf{Q} = \mathbf{0}$, equivalently $\boldsymbol{\pi}\mathbf{P}(t) = \boldsymbol{\pi}$ for every t. If every state can be reached from every other, it is unique and is the chain's long-run distribution; models usually draw the root state from it. The expected number of changes per unit time at stationarity is $-\sum_i \pi_i q_{ii}$ ($2\alpha\beta/(\alpha+\beta)$ for two states). Rate matrices are usually *normalized* to make this 1, so that branch lengths are in expected changes per character.

A stationary chain is *reversible* if it satisfies *detailed balance*:

$$\pi_i \, q_{ij} = \pi_j \, q_{ji} \quad \text{for all } i, j. \tag{A.22}$$

At equilibrium the flow of probability from i to j equals the flow back, so the process run backwards looks statistically like the process run forwards. There are three consequences. The likelihood does not depend on where the root is placed (the "pulley principle" of Chapter 5), so reversible models estimate unrooted trees and the root must come from a clock or an outgroup. Every reversible rate matrix can be written $q_{ij} = s_{ij}\pi_j$ with symmetric "exchangeabilities" $s_{ij} = s_{ji}$, the parameterization of general time-reversible models. And its eigenvalues are real, which makes Equation (A.19) stable.

Every two-state chain is reversible, since $\pi_0\alpha = \pi_1\beta = \alpha\beta/(\alpha+\beta)$. Multistate chains need not be: a hypothetical word-order cycle SOV → SVO → VSO → SOV with no reverse changes has a stationary distribution but violates detailed balance, and unidirectional grammaticalization clines (lexical item → function word → affix) are further still from reversibility. Such directional models carry some, usually little, information about the position of the root. The simplest multistate model, the *Mk model*, gives every change among K states the same rate; its closed form is Equation (C.6).

```
p_two_state <- function(t, alpha, beta) {
  s <- alpha + beta; pi1 <- alpha / s; pi0 <- beta / s; e <- exp(-s * t)
  matrix(c(pi0 + pi1 * e, pi1 * (1 - e),
           pi0 * (1 - e), pi1 + pi0 * e), nrow = 2, byrow = TRUE)
}
alpha <- 0.05; beta <- 0.25            # gain and loss per millennium
Q2 <- matrix(c(-alpha, alpha, beta, -beta), nrow = 2, byrow = TRUE)
all.equal(p_two_state(3, alpha, beta), matexpo(Q2 * 3))
```

```
#> [1] TRUE
```

```
stationary <- function(Q) {            # left eigenvector with eigenvalue 0
  e <- eigen(t(Q))
  v <- Re(e$vectors[, which.min(abs(e$values))])
  v / sum(v)
}
is_reversible <- function(Q) {         # detailed balance: F = t(F)
  F <- diag(stationary(Q)) %*% unname(Q)   # F[i, j] = pi_i q_ij
  isTRUE(all.equal(F, t(F)))
}
cyc <- matrix(c(-1, 1, 0,  0, -1, 1,  1, 0, -1), nrow = 3, byrow = TRUE)
rbind(stationary(Q2), c(pi0 = beta, pi1 = alpha) / (alpha + beta))
```

```
#>         pi0    pi1
#> [1,] 0.8333 0.1667
#> [2,] 0.8333 0.1667
```

```
c(two_state = is_reversible(Q2), word_order_Q = is_reversible(Q),
  cycle = is_reversible(cyc))
```

```
#>    two_state word_order_Q        cycle
#>         TRUE         TRUE        FALSE
```

The word-order matrix `Q` of Section A.6 is reversible because changes occur only between neighbours in a line, and every chain of that shape satisfies detailed balance; the cycle is not. Finally, a simulation using the holding-time description confirms that over a long period a cognate class is present for a fraction π_1 of the time.

```
simulate_ctmc <- function(Q, start, t_max) {
  state <- start; time <- 0
  times <- 0; path <- start
  repeat {
    time <- time + rexp(1, -Q[state, state])     # holding time
    if (time > t_max) break
    jump <- Q[state, ]; jump[state] <- 0
    state <- sample(seq_len(nrow(Q)), 1, prob = jump)  # where to go
    times <- c(times, time); path <- c(path, state)
  }
  list(times = c(times, t_max), states = path)
}
set.seed(4)
sim <- simulate_ctmc(Q2, start = 2, t_max = 20000)  # state 2 = present
durations <- diff(sim$times)
c(changes = length(sim$states) - 1,
  time_present = sum(durations[sim$states == 2]) / 20000,
  pi1 = alpha / (alpha + beta))
```

```
#>      changes time_present          pi1
#>    1570.0000       0.1586       0.1667
```

Where this is used in the book. Continuous-time Markov chains are the models of character change in Chapters 5 and 7, and they underlie the distance corrections of Chapter 4, the covarion and borrowing models of Chapters 7 and 8, discrete phylogeography (Chapter 11), ancestral state reconstruction (Chapter 12), Pagel's test of correlated evolution (Chapters 13 and 14) and state-dependent diversification (Chapter 16).

A.8 Poisson processes and birth-death processes

A.8.1 The Poisson process

Follow one meaning, say 'bird', down a lineage and mark each moment its word is replaced. If replacements happen at a constant rate λ, independently of one another and of the past (in a short interval h, one event with probability about λh), the marks form a *Poisson process*. Three

properties follow, and they explain most of the book's formulas involving rates:

1. The number of events in an interval of length t has a Poisson distribution with mean λt (Equation (A.9)).
2. The waiting times between successive events are independent and exponential with rate λ (Equation (A.10)).
3. Counts in non-overlapping intervals are independent.

The probability of *no* replacement in time t, $e^{-\lambda t}$, is the retention probability of Section A.3.1. Independent processes add (*superposition*): replacement by internal change at rate λ_1 and by borrowing at rate λ_2 together form a Poisson process with rate $\lambda_1 + \lambda_2$. Keeping each event with probability p (*thinning*) gives rate $p\lambda$. With a time-varying rate $\lambda(t)$ counts are still Poisson, with mean $\int \lambda(t)\,dt$, the idea behind clock models (Chapter 9).

While a character sits in state i of a Markov chain, the events that would move it form a Poisson process with rate q_i (Section A.7.1). In the Mk model one can imagine events at a constant rate whatever the state, each choosing a new state at random, possibly the current one. Some events are then invisible and others undo earlier ones, so observed differences between two languages grow more slowly than the number of events and eventually level off. Hence the distance corrections of Chapter 4.

```
set.seed(7)
lambda <- 0.2; t_max <- 10   # replacements per millennium; 10 millennia
count_events <- function() {
  time <- 0; n <- 0
  repeat {
    time <- time + rexp(1, lambda)      # exponential waiting times
    if (time > t_max) break
    n <- n + 1
  }
  n
}
n <- replicate(5000, count_events())
c(mean = mean(n), var = var(n), expected = lambda * t_max)
```

```
#>     mean      var expected
#>    2.000    2.021    2.000
```

```
c(no_change_sim = mean(n == 0), no_change_exact = exp(-lambda * t_max))
```

```
#>   no_change_sim no_change_exact
#>          0.1350          0.1353
```

The mean and variance of the counts are both close to $\lambda t = 2$, as for a Poisson distribution, and the proportion with no replacement matches $e^{-\lambda t}$.

A.8.2 Birth-death processes

In a *birth-death process* applied to languages, each lineage independently splits at rate λ (a speech community diverging in two) and dies out at rate μ (a language lost without descendants). With n lineages alive, the wait until the next event is exponential with rate $n(\lambda + \mu)$, and the event is a

split with probability $\lambda/(\lambda+\mu)$. From one lineage, the expected number alive at time t is

$$\mathrm{E}[N(t)] = e^{(\lambda-\mu)t}. \tag{A.23}$$

The *net diversification rate* $\lambda-\mu$ sets the growth, and the *turnover* μ/λ measures how much extinction accompanies it (Kendall, 1948). In the *Yule* (pure-birth) process, $\mu = 0$; with j lineages it waits an exponential time with rate $j\lambda$ for the next split, and multiplying these densities gives the Yule tree prior of Equation (B.12).

Reconstructed trees contain only lineages surviving to the present, since extinct languages without written records leave no trace in a cognate matrix. Such a *reconstructed tree* is a biased sample of the history: recent lineages have had little time to go extinct, giving an excess of recent splits, the *pull of the present* (Nee et al., 1994). On a lineages-through-time (LTT) plot with a logarithmic axis, the reconstructed tree grows with slope near $\lambda-\mu$ in the deep past and steepens toward λ near the present. Extensions add a *sampling fraction* ρ of living languages and a rate ψ at which lineages leave dated samples such as ancient written languages, which gives the fossilized birth-death process of Chapter 10. Looking backwards instead gives the *coalescent*, in which each pair of k lineages in a population merges at the same rate, for a total rate $\binom{k}{2}$ in units scaled by population size; Chapter 16 uses it for the histories of individual words within a language tree.

These processes are cruder for languages than for species. Language splits are rarely instantaneous (a dialect continuum can take centuries to break apart), rates differ between regions and periods, and present-day language endangerment makes current extinction rates unlike those of the past. Birth-death tree priors are best read as convenient assumptions about tree shape, whose influence on dates Chapters 6 and 10 examine.

```
set.seed(11)
full <- rlineage(birth = 0.4, death = 0.2, Tmax = 15)  # includes extinct
recon <- drop.fossil(full)                             # survivors only
c(lineages_ever = Ntip(full), surviving = Ntip(recon))
```

```
#> lineages_ever     surviving
#>            66            36
```

```
op <- par(mfrow = c(1, 2), mar = c(4, 4, 1.5, 1), cex = 0.8)
plot(full, show.tip.label = FALSE, main = "(a) Complete history")
ltt.plot(full, log = "y", main = "(b) Lineages through time",
         xlab = "Time before the present", ylab = "Number of lineages")
ltt.lines(recon, lty = 2)
```

```
par(op)
```

Where this is used in the book. The Poisson process underlies retention and glottochronology (Chapters 1 and 9), distance corrections (Chapter 4), the stochastic Dollo model (Chapter 7) and the strict and relaxed clocks of Chapter 9. Birth-death processes provide the measures of tree shape in Chapter 2, the tree priors of Chapters 6 and 10, including the fossilized birth-death process, and the diversification models of Chapter 16, which also introduces the coalescent.

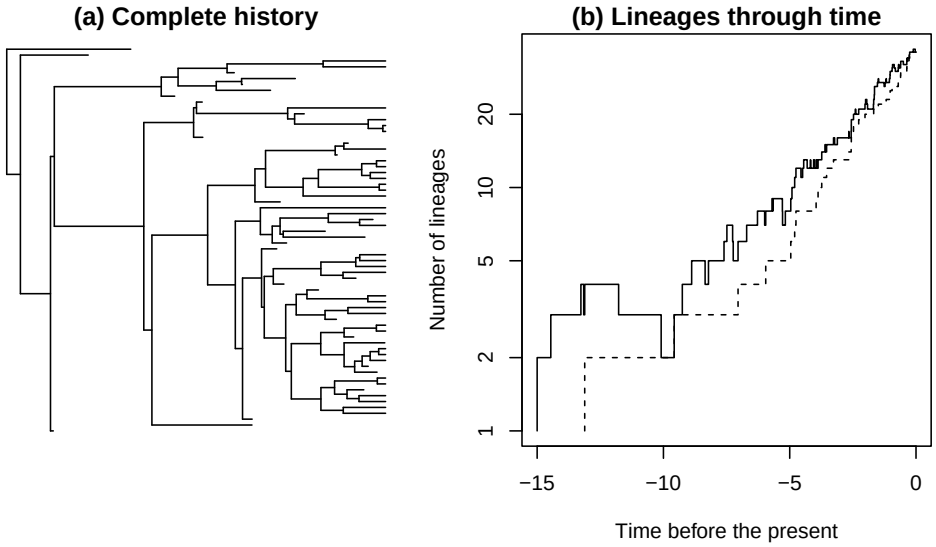


Figure A.3: A simulated birth-death history (splitting rate 0.4, extinction rate 0.2 per unit time). (a) The complete tree, including the lineages that died out. (b) Lineages through time on a logarithmic scale, for the complete history (solid) and for the reconstructed tree of surviving lineages only (dashed). Time is measured backwards from the present (0).

A.9 Brownian motion and the Ornstein-Uhlenbeck process

Some linguistic quantities vary continuously: the coordinates of a speech community, the logarithm of a phoneme inventory size, the proportion of head-final constructions in a corpus, or the logarithm of a rate of change. The simplest model of their drift is a random walk, moving each generation by a small random amount independent of earlier moves. Sums of many small independent moves are approximately normal (Section A.4.6), and in the limit of very many tiny steps the walk becomes *Brownian motion* (BM).

A.9.1 Brownian motion

A Brownian motion $X(t)$ with rate σ^2, started at $X(0) = x_0$, has independent, normally distributed increments:

$$X(t+s) - X(t) \sim \mathcal{N}(0, \sigma^2 s), \qquad \text{so} \qquad X(t) \sim \mathcal{N}(x_0, \sigma^2 t). \tag{A.24}$$

BM has no tendency to go anywhere: the expected value never changes. The variance grows *linearly* with time, so the spread grows with its square root; after 4,000 years it is twice, not four times, that after 1,000. And the future depends only on the present value. When a lineage splits, both daughters start from the same value and then wander independently, so two languages are correlated in proportion to the time they evolved together, the basis of the phylogenetic covariance matrix (Section A.10).

The book uses several variants: a *trend*, $X(t) \sim \mathcal{N}(x_0 + \mu t, \sigma^2 t)$, for directional expansion (Equation (11.8)); branch-specific rate multipliers in the *relaxed random walk* (Equation (11.7)); BM on the *logarithm* of a rate, *geometric Brownian motion*, for autocorrelated clocks (Equation (C.10)); and a declining σ^2 in the *early-burst* model of Chapter 13.

One consequence matters for homelands. Under BM the estimated root value is a weighted average of the tip values with positive weights (Section A.11), so a BM homeland estimate lies inside the

range of present-day locations and is pulled toward densely sampled regions. That is a property of the model, not a discovery (Chapter 11).

A.9.2 The Ornstein-Uhlenbeck process

Many linguistic quantities cannot wander arbitrarily far: inventory sizes and proportions are bounded, and functional pressures may pull some typological properties toward preferred values. The *Ornstein-Uhlenbeck* (OU) process adds such a pull:

$$dX(t) = \alpha\,[\theta - X(t)]\,dt + \sigma\,dW(t), \tag{A.25}$$

where $dW(t)$ is a standard Brownian increment. The first term pulls X toward the *optimum* θ with strength α; the second adds noise. From x_0,

$$\mathrm{E}[X(t)] = \theta + (x_0 - \theta)e^{-\alpha t}, \qquad \mathrm{Var}[X(t)] = \frac{\sigma^2}{2\alpha}\left(1 - e^{-2\alpha t}\right). \tag{A.26}$$

The mean approaches θ exponentially, moving halfway in the *phylogenetic half-life* $\ln 2/\alpha$, and the variance levels off at the *stationary variance* $\sigma^2/(2\alpha)$. As $\alpha \to 0$ the OU process becomes BM. On a tree, covariance decays with the time separating two languages; with the root drawn from the stationary distribution it is $\frac{\sigma^2}{2\alpha}e^{-\alpha d_{ij}}$, where d_{ij} is the path length between the tips (Equation (D.12)). OU parameters are hard to estimate from trees the size of language families: likelihood ratio tests favour OU over BM too often, and measurement error can mimic a pull toward an optimum (Cooper et al., 2016). An estimated α is weak evidence for a functional optimum without simulation under the fitted models (Chapter 13).

```
set.seed(8)
dt <- 0.01; times <- seq(0, 10, by = dt); n_steps <- length(times) - 1
bm <- replicate(5, c(0, cumsum(rnorm(n_steps, 0, sqrt(dt)))))
ou_path <- function(x0, alpha, theta, sigma) {
  a <- exp(-alpha * dt)                          # exact OU transition
  s <- sigma * sqrt((1 - a^2) / (2 * alpha))
  x <- numeric(length(times)); x[1] <- x0
  for (i in 2:length(times)) x[i] <- theta + (x[i - 1] - theta) * a +
    rnorm(1, 0, s)
  x
}
ou <- replicate(5, ou_path(x0 = 3, alpha = 1, theta = 0, sigma = 1))
op <- par(mfrow = c(1, 2), mar = c(4, 4, 1.5, 1), cex = 0.8)
matplot(times, bm, type = "l", lty = 1, col = "grey30", ylim = c(-7, 7),
        xlab = "Time", ylab = "Trait value", main = "(a) Brownian motion")
lines(times, 2 * sqrt(times), lty = 2); lines(times, -2 * sqrt(times), lty = 2)
matplot(times, ou, type = "l", lty = 1, col = "grey30", ylim = c(-7, 7),
        xlab = "Time", ylab = "Trait value", main = "(b) Ornstein-Uhlenbeck")
sd_ou <- sqrt((1 - exp(-2 * times)) / 2)
lines(times, 3 * exp(-times) + 2 * sd_ou, lty = 2)
lines(times, 3 * exp(-times) - 2 * sd_ou, lty = 2)
```

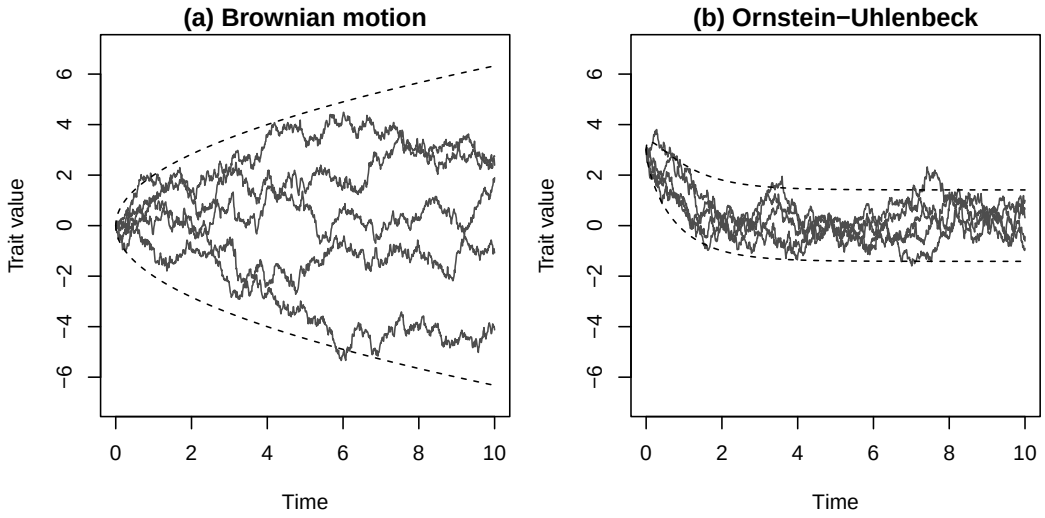


Figure A.4: Five simulated paths of (a) Brownian motion with rate one, started at zero, and (b) an Ornstein-Uhlenbeck process with optimum zero, alpha = 1 and sigma = 1, started at 3. The dashed lines show plus and minus two standard deviations: these widen without limit under Brownian motion, but settle at the stationary value under the Ornstein-Uhlenbeck process.

```
par(op)
```

Where this is used in the book. Brownian motion and its variants model the dispersal of languages in continuous phylogeography (Chapter 11) and the evolution of rates in autocorrelated relaxed clocks (Chapter 9). BM, OU and early-burst models are fitted and compared in Chapter 13 and applied to cultural traits in Chapter 14. Continuous ancestral states (Chapter 12) are estimated under BM.

A.10 The multivariate normal and phylogenetic covariance

A.10.1 The multivariate normal distribution

Continuous quantities that vary together, such as the latitude and longitude of a homeland or one trait in n related languages, are often modelled as *multivariate normal*. A vector $\mathbf{x}$ has the distribution $\mathcal{N}_n(\boldsymbol{\mu}, \Sigma)$, with mean vector $\boldsymbol{\mu}$ and covariance matrix Σ, if its density is

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{n/2}\det(\Sigma)^{1/2}} \exp\left[-\tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\top}\Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})\right]. \tag{A.27}$$

The diagonal of Σ holds variances and the off-diagonal entries covariances; the exponent is a squared distance from the mean that allows for the correlations. Four properties are used repeatedly:

1. Every component, and every subset of components, is normal, with the corresponding parts of $\boldsymbol{\mu}$ and Σ.
2. Linear combinations are normal: $\mathbf{Ax} \sim \mathcal{N}(\mathbf{A}\boldsymbol{\mu}, \mathbf{A}\Sigma\mathbf{A}^{\top})$. So if $\mathbf{z}$ holds independent standard normals and $\Sigma = \mathbf{L}\mathbf{L}^{\top}$, then $\boldsymbol{\mu} + \mathbf{Lz} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, which is how correlated values are simulated.

3. Conditional distributions are normal. Splitting $\mathbf{x}$ into $\mathbf{x}_1$ and $\mathbf{x}_2$,
$$\mathbf{x}_1 \mid \mathbf{x}_2 \sim \mathcal{N}\left(\boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2),\ \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}\right). \tag{A.28}$$
Knowing $\mathbf{x}_2$ shifts the mean of $\mathbf{x}_1$ and reduces its variance, the more so the stronger the covariance. The formula predicts a missing value from related languages, or an ancestral value from its descendants.
4. For multivariate normal variables (but not in general), zero covariance implies independence.

A.10.2 The phylogenetic covariance matrix

Let a trait evolve by Brownian motion with rate σ^2 on a tree, from the value μ at the root. The value at tip i is μ plus the independent normal increments on the branches from the root to i. Two tips share the increments from the root to their most recent common ancestor and are independent thereafter, so their covariance is σ^2 times the length of the shared path:

$$\mathbf{x} \sim \mathcal{N}_n\left(\mu\mathbf{1},\ \sigma^2\mathbf{C}\right), \qquad C_{ij} = \text{length of the path shared by tips } i \text{ and } j, \qquad C_{ii} = \text{root-to-tip length of } i \tag{A.29}$$

(Equation (13.2)). $\mathbf{C}$ depends only on the tree and is computed by ape's `vcv()`. Other models change its entries but not the logic: OU, Pagel's λ (which multiplies the off-diagonal entries by λ) and early burst each define their own matrix (Chapter 13). For two coordinates with 2×2 covariance Σ, all $2n$ values have covariance $\mathbf{C} \otimes \Sigma$ (Equation (D.3)).

The covariance matrix makes Galton's problem quantitative. For an ultrametric tree of height T, n languages estimate the root value as precisely as $n_{\text{eff}} = T\,\mathbf{1}^\top\mathbf{C}^{-1}\mathbf{1}$ independent languages would. This *effective number of languages* equals n for a star-shaped tree and is much smaller when long branches are shared. The example uses a small tree with Germanic names; the branch lengths are illustrative, not estimates.

```
germanic <- read.tree(text = paste0(
  "((((Dutch:0.5,Afrikaans:0.5):1,German:1.5):0.5,",
  "(English:1.5,Frisian:1.5):0.5):1,(Swedish:1,Danish:1):2);"))
C <- vcv(germanic)
C
```

```
#>           Dutch Afrikaans German English Frisian Swedish Danish
#> Dutch       3.0       2.5    1.5     1.0     1.0       0      0
#> Afrikaans   2.5       3.0    1.5     1.0     1.0       0      0
#> German      1.5       1.5    3.0     1.0     1.0       0      0
#> English     1.0       1.0    1.0     3.0     1.5       0      0
#> Frisian     1.0       1.0    1.0     1.5     3.0       0      0
#> Swedish     0.0       0.0    0.0     0.0     0.0       3      2
#> Danish      0.0       0.0    0.0     0.0     0.0       2      3
```

```
# simulate 5000 data sets under BM with mu = 0, sigma^2 = 1
set.seed(9)
L <- t(chol(C))                          # C = L %*% t(L)
sims <- L %*% matrix(rnorm(nrow(C) * 5000), nrow = nrow(C))
rownames(sims) <- rownames(C)
round(cov(t(sims))[1:3, 1:3], 2)         # close to C[1:3, 1:3]
```

```
#>           Dutch Afrikaans German
#> Dutch      2.99      2.47   1.49
#> Afrikaans  2.47      2.98   1.50
#> German     1.49      1.50   3.01
```

```
# effective number of independent languages for estimating the root
one <- rep(1, nrow(C))
c(n = nrow(C), n_eff = C[1, 1] * sum(solve(C, one)))
```

```
#>     n n_eff
#> 7.000 3.066
```

```
# predict Frisian from the other six languages (Equation math-mvn-conditional)
x <- sims[, 1]; i <- "Frisian"; o <- setdiff(rownames(C), i)
cond_mean <- C[i, o] %*% solve(C[o, o], x[o])
cond_var <- C[i, i] - C[i, o] %*% solve(C[o, o], C[o, i])
round(c(true = x[[i]], predicted = cond_mean, sd = sqrt(cond_var),
        sd_unconditional = sqrt(C[i, i])), 2)
```

```
#>            true       predicted              sd sd_unconditional
#>           -0.18           -0.81            1.45            1.73
```

Seven languages are worth only about three independent observations for estimating the root value. Knowing the other six reduces the standard deviation of Frisian's value from 1.73 to about 1.45. The reduction is modest because even Frisian's closest relative, English, has evolved independently of it for half the depth of the tree.

Where this is used in the book. The multivariate normal model with a phylogenetic covariance matrix is the likelihood of continuous phylogeography (Equation (D.3), Chapter 11) and of the Brownian, OU and λ models of Chapter 13 (Equation (D.10)). Its conditional distributions give continuous ancestral states (Chapter 12). Phylogenetic signal statistics and mixed models in Chapters 13, 14 and 17 are built on the same covariance matrices.

A.11 Least squares and generalized least squares

A.11.1 Ordinary least squares

Do languages with more speakers have simpler morphology (Chapter 13)? A regression of a complexity score y_i on log population size x_i, $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, is for all languages at once

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \tag{A.30}$$

where the *design matrix* $\mathbf{X}$ has a column of ones and a column of x_i. *Ordinary least squares* (OLS) minimizes the sum of squared residuals $\sum_i (y_i - \beta_0 - \beta_1 x_i)^2$, giving

$$\hat{\boldsymbol{\beta}}_{\text{OLS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}. \tag{A.31}$$

With only an intercept, $\hat{\beta}_0$ is the mean. If the errors are independent and normal with a common variance, OLS is the MLE and the standard errors and p-values of `lm()` are correct.

A.11.2 When errors are correlated

For languages, independent errors are the exception. Related languages resemble each other in morphology *and* population size for historical reasons outside the regression, so residuals are correlated, typically as $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{V}$ with $\mathbf{V} = \mathbf{C}$ (Section A.10.2). OLS estimates stay unbiased but are inefficient, and their standard errors are usually too small, because thirty closely related languages are treated as thirty independent pieces of evidence. *Generalized least squares* (GLS) minimizes $(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top\mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$ instead:

$$\hat{\boldsymbol{\beta}}_{\mathrm{GLS}} = (\mathbf{X}^\top\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{V}^{-1}\mathbf{y}, \qquad \mathrm{Var}(\hat{\boldsymbol{\beta}}_{\mathrm{GLS}}) = \sigma^2(\mathbf{X}^\top\mathbf{V}^{-1}\mathbf{X})^{-1} \tag{A.32}$$

(Equation (D.18)). To see why this works, write $\mathbf{V} = \mathbf{L}\mathbf{L}^\top$. Multiplying the model by $\mathbf{L}^{-1}$ gives $\mathbf{L}^{-1}\mathbf{y} = \mathbf{L}^{-1}\mathbf{X}\boldsymbol{\beta} + \mathbf{L}^{-1}\boldsymbol{\varepsilon}$, whose errors are independent with equal variance (Property 2 of Section A.10.1): GLS is OLS on "whitened" data. *Weighted least squares* is the special case of a diagonal $\mathbf{V}$.

With a phylogenetic $\mathbf{V}$, GLS is *phylogenetic generalized least squares* (PGLS); Felsenstein's independent contrasts reach the same slope by transforming n tip values into $n - 1$ independent differences (Chapter 13). With an intercept only, GLS gives the phylogenetically weighted mean $\hat{\mu} = (\mathbf{1}^\top\mathbf{C}^{-1}\mathbf{1})^{-1}\mathbf{1}^\top\mathbf{C}^{-1}\mathbf{x}$, the MLE of the root value under BM and, for coordinates, the BM homeland estimate of Equation (11.6). Least squares also fits trees: the Fitch-Margoliash method (Chapter 4) chooses branch lengths that minimize a weighted sum of squared differences between observed distances and path lengths through the tree.

In the simulation below, two traits evolve *independently* by BM on a random tree of 40 languages, so any association is spurious. In the vocabulary of Section 6.10, this checks the *robustness under misspecification* of OLS, whose assumption of independent languages the data violate; GLS is applied to data simulated under its own assumptions.

```
set.seed(21)
tree <- rcoal(40)                    # a random ultrametric tree
C <- vcv(tree)
gls_slope_p <- function(y, x, C) {
  X <- cbind(1, x)
  CiX <- solve(C, X)                 # C^{-1} X without inverting C
  XtCiX <- t(X) %*% CiX
  b <- solve(XtCiX, t(CiX) %*% y)    # Equation (math-gls)
  res <- y - X %*% b
  s2 <- drop(t(res) %*% solve(C, res)) / (length(y) - 2)
  se <- sqrt(diag(s2 * solve(XtCiX)))
  2 * pt(-abs(b[2] / se[2]), df = length(y) - 2)   # p-value of slope
}
p_values <- replicate(500, {
  x <- rTraitCont(tree, sigma = 1)   # two independent BM traits,
  y <- rTraitCont(tree, sigma = 1)   # in the same order as rows of C
  c(OLS = summary(lm(y ~ x))$coefficients[2, 4],
    GLS = unname(gls_slope_p(y, x, C)))
})
rowMeans(p_values < 0.05)            # false-positive rates; nominal 0.05
```

```
#>   OLS   GLS
#> 0.406 0.036
```

```
# GLS is OLS on whitened data: the two slopes agree
x <- rTraitCont(tree); y <- rTraitCont(tree)
L <- t(chol(C))
b_white <- coef(lm(solve(L, y) ~ solve(L, cbind(1, x)) - 1))
b_gls <- solve(t(cbind(1, x)) %*% solve(C, cbind(1, x)),
               t(cbind(1, x)) %*% solve(C, y))
rbind(whitened = unname(b_white), gls = unname(drop(b_gls)))
```

```
#>           [,1]   [,2]
#> whitened 0.041 0.1399
#> gls      0.041 0.1399
```

OLS declares the unrelated traits significantly associated in about 40% of the simulated data sets; GLS keeps the false-positive rate near 5%. The excess depends on the tree and is largest when a few deep splits divide the languages into large clades.

Where this is used in the book. GLS is the basis of PGLS and phylogenetic regression in Chapters 13, 14 and 17, and of the BM homeland estimate in Chapter 11. Least-squares fitting of trees to distances appears in Chapter 4.

A.12 Monte Carlo and Markov chain Monte Carlo

A.12.1 Monte Carlo

Many expectations are hard to compute exactly but easy to approximate by simulation. The *Monte Carlo* method draws S independent values $x_1, \dots, x_S$ from a distribution and approximates the expectation of a function g by

$$\mathrm{E}[g(X)] \approx \frac{1}{S} \sum_{s=1}^{S} g(x_s). \tag{A.33}$$

The average converges to the expectation (the law of large numbers) with standard error $\mathrm{sd}(g(X))/\sqrt{S}$, so quadrupling the draws halves the error. For example, let the replacement rate k of two languages' meanings vary as a gamma distribution with mean $\bar{k}$ and shape α. The expected proportion of shared cognates at time depth t, $\mathrm{E}[e^{-2kt}]$, has the closed form $(1 + 2\bar{k}t/\alpha)^{-\alpha}$ (Equation (C.7)), which Monte Carlo reproduces without algebra:

```
set.seed(5)
kbar <- 0.217; alpha <- 2; t <- 1.5          # rates per millennium
k_draws <- rgamma(1e5, shape = alpha, rate = alpha / kbar)
g <- exp(-2 * k_draws * t)
c(monte_carlo = mean(g), mc_se = sd(g) / sqrt(length(g)),
  exact = (1 + 2 * kbar * t / alpha)^(-alpha),
  constant_rate = exp(-2 * kbar * t))
```

```
#>   monte_carlo         mc_se         exact constant_rate
#>     0.5691396     0.0006563     0.5691681     0.5215240
```

Rate variation *raises* the expected sharing at a given depth, because slow meanings keep it high, so ignoring it underestimates time depth, a classic objection to glottochronology (Chapter 9).

Monte Carlo averaging also carries phylogenetic uncertainty into later analyses. If $\tau_1, \dots, \tau_S$ are trees sampled from a posterior, the posterior expectation of any tree-dependent quantity, such as the probability that an ancestor had a given word order or the slope of a phylogenetic regression, is approximated by its average over the sampled trees. Repeating an analysis on a few hundred posterior trees rather than one summary tree is thus a Monte Carlo integration over tree uncertainty, as Chapters 11, 12 and 13 recommend.

A.12.2 Markov chain Monte Carlo

A posterior $f(\theta \mid D) \propto f(D \mid \theta) f(\theta)$ can usually be evaluated only up to the unknown constant $f(D)$, and cannot be sampled directly. *Markov chain Monte Carlo* (MCMC) constructs a Markov chain whose stationary distribution (Section A.7.3) is the posterior. Run long enough, the chain visits values in proportion to their posterior probability, and averages over its path approximate posterior expectations. The *Metropolis-Hastings* algorithm (Metropolis et al., 1953; Hastings, 1970) proceeds from the current value θ as follows:

1. Propose θ' from a proposal distribution $q(\theta' \mid \theta)$, for example by adding a small random amount to θ or multiplying it by a random factor near 1.
2. Compute the acceptance probability

$$a(\theta, \theta') = \min\left(1,\ \frac{f(D \mid \theta')\, f(\theta')}{f(D \mid \theta)\, f(\theta)} \times \frac{q(\theta \mid \theta')}{q(\theta' \mid \theta)}\right). \tag{A.34}$$

3. Move to θ' with probability a; otherwise stay at θ, and record θ again.

The unknown $f(D)$ cancels in the ratio, which is the whole point. The *Hastings ratio* corrects for proposals that are easier to make in one direction: it is 1 for a symmetric proposal and m for a multiplier proposal $\theta' = m\theta$ (Equation (B.13)). The algorithm works because it satisfies detailed balance (Equation (A.22)) with respect to the posterior: reversibility, which a model of language change may or may not have, is here a design principle guaranteeing convergence to the right distribution. The example estimates the split time of Section A.3.1 with an exponential prior of mean 2 millennia; with one parameter, the exact posterior can be computed on a grid for comparison.

```
N <- 200; k <- 110; r <- -log(0.805)
log_post <- function(t) {
  if (t <= 0) return(-Inf)
  dbinom(k, N, exp(-2 * r * t), log = TRUE) +    # log-likelihood
    dexp(t, rate = 0.5, log = TRUE)              # log-prior, mean 2
}
set.seed(3)
n_iter <- 20000; t_cur <- 5; lp_cur <- log_post(t_cur)
draws <- numeric(n_iter); accepted <- 0
for (s in seq_len(n_iter)) {
  m <- exp(0.6 * (runif(1) - 0.5))               # multiplier proposal
  t_new <- t_cur * m
  lp_new <- log_post(t_new)
```

```
  if (log(runif(1)) < lp_new - lp_cur + log(m)) {  # Hastings ratio = m
    t_cur <- t_new; lp_cur <- lp_new; accepted <- accepted + 1
  }
  draws[s] <- t_cur
}
kept <- draws[-(1:1000)]                          # discard burn-in
# exact posterior on a grid, for comparison
grid <- seq(0.5, 2.5, by = 0.001)
dens <- exp(sapply(grid, log_post)); dens <- dens / sum(dens * 0.001)
ess <- function(x, max_lag = 500) {               # effective sample size
  rho <- acf(x, lag.max = max_lag, plot = FALSE)$acf[-1]
  cut <- which(rho < 0.05)[1]
  if (is.na(cut)) cut <- max_lag
  length(x) / (1 + 2 * sum(rho[seq_len(cut - 1)]))
}
round(c(acceptance = accepted / n_iter, mcmc_mean = mean(kept),
        grid_mean = sum(grid * dens * 0.001),
        mcmc_lower = quantile(kept, 0.025, names = FALSE),
        mcmc_upper = quantile(kept, 0.975, names = FALSE),
        ess = ess(kept)), 3)
```

```
#> acceptance  mcmc_mean  grid_mean mcmc_lower mcmc_upper        ess
#>      0.512      1.382      1.383      1.107      1.687   5012.956
```

```
op <- par(mfrow = c(1, 2), mar = c(4, 4, 1.5, 1), cex = 0.8)
plot(draws[1:2000], type = "l", xlab = "Iteration",
     ylab = "Time since split", main = "(a) Trace")
hist(kept, breaks = 50, freq = FALSE, col = "grey85", border = "white",
     xlab = "Time since split (millennia)", main = "(b) Posterior")
lines(grid, dens, lwd = 2)
```

```
par(op)
```

The MCMC estimates agree with the exact posterior. The chain started at 5 millennia, far from the high-probability region, and took a few dozen iterations to get there; these early iterations, the *burn-in*, are discarded. Successive values are correlated, so 19,000 retained iterations are worth fewer independent draws. The *effective sample size* (ESS), here $S/(1 + 2\sum_{k} \rho_k)$ from the chain's autocorrelations ρ_k, measures how many; a common rule of thumb asks for at least 200 per parameter, and programs such as Tracer use more careful estimators.

Real analyses have thousands of parameters, including the topology, and cycle through many kinds of proposal, but the diagnostics are the same: inspect traces, discard burn-in, check ESS, and above all run several independent chains from different starting points and confirm that they agree, for trees by comparing clade frequencies (Chapter 6). The book also meets Gibbs sampling (each parameter drawn in turn from its conditional distribution), Metropolis-coupled MCMC ("heated" chains that help the main chain cross valleys) and reversible-jump MCMC, which moves between models with different numbers of parameters (Green, 1995). Convergence shows only that the chain has explored the posterior defined by the software. Whether the software computes what the model says is a question of implementation validation, and whether the model fits the data is

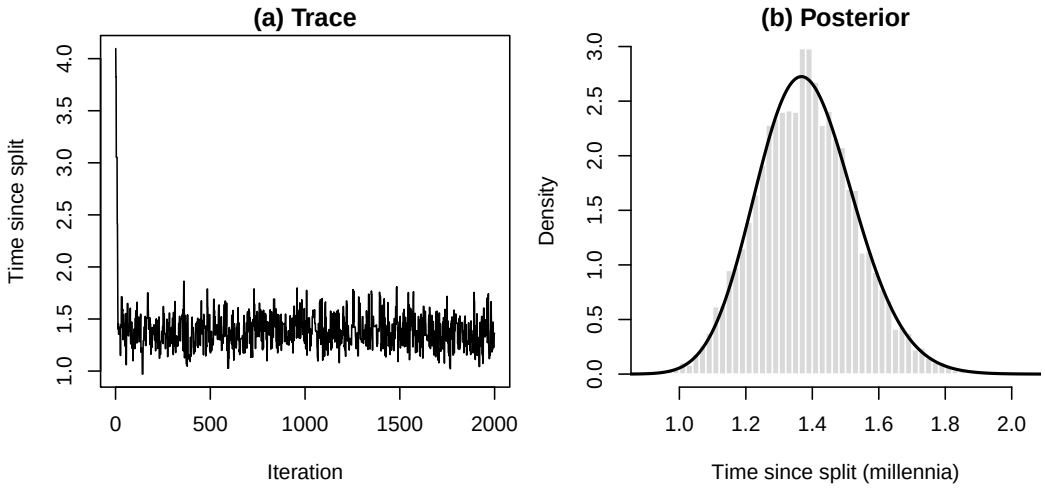


Figure A.5: Metropolis-Hastings sampling of the time since the split of two languages. (a) Trace of the first 2000 iterations, starting far from the posterior (the burn-in is visible at the left). (b) Histogram of the sampled values after burn-in, with the exact posterior density computed on a grid (line).

one of empirical adequacy (Section 6.10).

Where this is used in the book. MCMC is introduced in Chapter 6 and is the computational engine of every Bayesian analysis in the book, notably in Chapters 7, 9, 10, 11 and 12. Convergence diagnostics are covered in Chapters 6 and 17, and Monte Carlo integration over posterior samples of trees in Chapters 11, 12 and 13. Simulation studies of the kind used throughout Chapters 15 and 17 are Monte Carlo experiments.

- A phylogenetic analysis combines a *model of change*, which gives the probability of the data, with a *method of inference*, which turns that probability into estimates. Likelihood (Section A.3) is the link between them.
- Related languages are statistically dependent because they share ancestors. Given the ancestor, daughters change independently (Section A.1.4), and summing over unknown ancestral states with the law of total probability gives the likelihood of a tree.
- Bayes' theorem combines a prior with the likelihood. Posteriors become less sensitive to priors as data accumulate; marginal likelihoods and Bayes factors do not.
- Discrete characters such as cognate classes and word-order types are modelled as continuous-time Markov chains. A rate matrix $\mathbf{Q}$ gives transition probabilities through the matrix exponential, $\mathbf{P}(t) = e^{\mathbf{Q}t}$; the two-state chain has the closed form of Equation (A.21), and long branches lead to saturation.
- Only the product of rate and time enters $\mathbf{P}(t)$, so dating requires external information about rates or times.
- Poisson processes describe changes along a lineage, and birth-death processes describe the splitting and extinction of lineages; trees of surviving languages are biased samples of the full history.
- Continuous traits are modelled by Brownian motion or the Ornstein-Uhlenbeck process. On a tree these imply a multivariate normal distribution with a phylogenetic covariance matrix, which makes the non-independence of languages quantitative and leads to generalized least squares.

- Monte Carlo methods approximate expectations by averages of simulated values. MCMC builds a chain whose stationary distribution is the posterior; averaging over posterior samples of trees carries phylogenetic uncertainty into downstream analyses.

A.13 Further reading

- Blitzstein and Hwang (2019), *Introduction to Probability*. A clear introduction to probability, random variables, the standard distributions, Poisson processes and Markov chains, with many worked examples; the best single book for Sections A.1 to A.4.
- van Ravenzwaaij et al. (2018), "A simple introduction to Markov chain Monte–Carlo sampling". A short tutorial with R code for Metropolis and Metropolis-within-Gibbs samplers, with a clear discussion of burn-in, starting values, proposal width and correlated parameters; a gentle companion to Section A.12.
- McElreath (2020), *Statistical Rethinking*. A gentle, code-based introduction to Bayesian modelling and MCMC for social scientists, including Gaussian-process models of covariance between related units, close cousins of phylogenetic regression.
- Gelman et al. (2013), *Bayesian Data Analysis*. The standard reference on priors, hierarchical models, MCMC, convergence diagnostics and posterior predictive checks.
- Strang (2016), *Introduction to Linear Algebra*. An intuitive account of matrices, determinants, eigenvalues and the Cholesky decomposition, emphasizing applications over proofs.
- Norris (1997), *Markov Chains*. A rigorous but readable treatment of discrete- and continuous-time chains, Poisson and birth-death processes, with the proofs behind Sections A.7 and A.8.
- 1. A classic survey of how (and how not) to compute the matrix exponential of Section A.6.
- Felsenstein (2004), *Inferring Phylogenies*, and Yang (2014), *Molecular Evolution: A Statistical Approach*. The standard texts on statistical phylogenetics, developing Markov models, likelihood on trees, Bayesian inference and MCMC from this appendix's material; Yang's is the more mathematical.
- Harmon (2019), *Phylogenetic Comparative Methods*. An open-access textbook on Brownian motion, OU, phylogenetic covariance, PGLS, discrete-character and birth-death models, at about the level of this appendix.
- Drummond and Bouckaert (2015), *Bayesian Evolutionary Analysis with BEAST*. Clock models, tree priors and calibration densities as implemented in BEAST, with practical advice on priors and diagnostics.
- Semple and Steel (2003), *Phylogenetics*. The mathematics of trees, splits and tree spaces, for readers of Chapter 2 who want proofs.

Appendix B

Technical Notes: Trees and Inference

This appendix collects the technical material behind Chapters 2 to 6: derivations of results that the chapters state without proof, algorithms written out step by step, implementations from scratch in R, and checks of those implementations against established packages. The chapters can be read without it. It is meant for readers who want to see exactly how a number is computed, who need to adapt an algorithm, or who want to verify that a package does what its documentation says.

There is one section per chapter. Each section is self-contained: every code chunk recreates the trees, data and functions it needs, with the same random seeds as the chapter, so it can be run in a fresh R session. Most of the code checks a hand-written implementation against a package. In the vocabulary of Section 6.10, these checks are *implementation validation*: they show that the code computes what the model says, not that the model describes languages well. Background on probability, likelihood, matrices and continuous-time Markov chains is in Appendix A. Chapter 3 has no technical notes of its own.

B.1 Chapter 2: Trees, Networks and Tree Space

B.1.1 Counting trees

The number of unrooted binary trees in equation (2.1) follows from adding tips one at a time. There is only one unrooted tree for three tips. A fourth tip can be attached to any of the three branches of that tree, giving three trees. In general, an unrooted binary tree with k tips has $2k-3$ branches, so the $(k+1)$th tip can be attached in $2k-3$ ways, and every unrooted tree on $k+1$ tips arises in exactly one way (remove tip $k+1$ and its branch, and the tree on k tips and the attachment branch are recovered). Multiplying,

$$U(n) = \prod_{k=3}^{n-1} (2k-3) = 1 \times 3 \times 5 \times \cdots \times (2n-5) = (2n-5)!!.$$

A rooted tree with n tips corresponds one-to-one to an unrooted tree with $n+1$ tips, one of which marks the root: attach an extra tip at the root and unroot, or, conversely, delete the marker tip and make its point of attachment the root. Hence $R(n) = U(n+1) = (2n-3)!!$, and writing the double factorial with ordinary factorials gives the closed form in equation (2.2). Equivalently, a

rooted tree with n tips has $2n-2$ branches, plus the possibility of attaching a new tip above the root, so the $(k+1)$th tip can be added in $2k-1$ ways.

Bayesian tree priors such as the Yule and coalescent processes (Chapter 6) assign probabilities to *ranked* trees (also called *labelled histories*), in which the temporal order of the internal nodes is also specified. Their number follows from building the tree backwards in time: with j lineages, the next coalescence (looking backwards) joins one of $\binom{j}{2}$ pairs, so

$$H(n) = \prod_{j=2}^{n} \binom{j}{2} = \frac{n!\,(n-1)!}{2^{n-1}}. \tag{B.1}$$

For four tips there are 15 rooted trees but 18 ranked trees, because each of the three balanced topologies of the form ((A,B),(C,D)) admits two orders for its two cherries, while each of the twelve "caterpillar" topologies admits only one. The distinction matters for priors: a uniform distribution on ranked trees, which is what the Yule process implies, is not uniform on topologies, and gives balanced topologies more prior weight than caterpillars. The code below checks the formulas by brute force for small n, enumerating rooted topologies with **phangorn**'s `allTrees()`.

```
library(phangorn)
dfact <- function(m) if (m <= 1) 1 else prod(seq(m, 1, by = -2))
ranks <- function(tr) {     # number of rankings of a rooted binary topology
  # = (number of internal nodes)! / prod over internal nodes of subtree sizes
  # internal nodes in the subtree of a node = (tips below it) - 1
  sizes <- lengths(prop.part(tr)) - 1
  factorial(Nnode(tr)) / prod(sizes)
}
check <- t(sapply(3:6, function(n) {
  trs <- allTrees(n, rooted = TRUE)
  c(n = n, rooted_enumerated = length(trs), rooted_formula = dfact(2 * n - 3),
    ranked_enumerated = sum(sapply(trs, ranks)),
    ranked_formula = factorial(n) * factorial(n - 1) / 2^(n - 1))
}))
check
```

```
#>      n rooted_enumerated rooted_formula ranked_enumerated
#> [1,] 3                 3              3                 3
#> [2,] 4                15             15                18
#> [3,] 5               105            105               180
#> [4,] 6               945            945              2700
#>      ranked_formula
#> [1,]              3
#> [2,]             18
#> [3,]            180
#> [4,]           2700
```

The function `ranks()` uses the fact that a ranking of the internal nodes is a linear order consistent with the tree (every node younger than its parent), and the number of such orders is the factorial of the number of internal nodes divided by the product, over internal nodes, of the number of

internal nodes in the subtree that each one heads (for a binary tree, the number of tips below it minus one).

B.1.2 Tree distances: formulas and a quartet implementation

Writing $S(\tau)$ for the set of non-trivial splits of tree τ and $b_s(\tau)$ for the length of the branch inducing split s in τ (zero if s is absent), the *branch-score distance* of Kuhner and Felsenstein (1994) is

$$d_{KF}(\tau_1, \tau_2) = \sqrt{\sum_{s \in S(\tau_1) \cup S(\tau_2)} \big(b_s(\tau_1) - b_s(\tau_2)\big)^2}. \tag{B.2}$$

Path-length distances compare the matrices of distances between tips. If $d_{ij}(\tau)$ is the number of branches (or the sum of branch lengths) on the path between tips i and j,

$$d_{path}(\tau_1, \tau_2) = \sqrt{\sum_{i<j} \big(d_{ij}(\tau_1) - d_{ij}(\tau_2)\big)^2}. \tag{B.3}$$

In **phangorn** these are `KF.dist()` and `path.dist()`. The RF distance is the special case of the branch-score distance in which every branch has length one, squared: each split present in one tree only contributes 1^2.

The quartet distance of Chapter 2 can be computed from scratch for small trees. The topology a tree assigns to quartet $\{a, b, c, d\}$ is the pairing with the smallest sum of within-pair path lengths, which is the four-point condition of Chapter 4 in disguise; with branch lengths set to one, a quartet left unresolved by a polytomy has two (or three) equal minimal sums. The code recreates the chapter's toy tree and bootstrap sample with the same seed and checks the implementation against the **Quartet** package.

```
library(phangorn)
ie_nwk <- paste0(
  "((((((Italian:1.2,Spanish:1.2):0.3,French:1.5)Romance:2.5,",
  "(Irish:2.5,Welsh:2.5)Celtic:1.5)ItaloCeltic:0.5,",
  "((English:1.5,German:1.5):0.5,Swedish:2.0)Germanic:2.5):0.5,",
  "((Russian:1.2,Polish:1.2):1.8,Lithuanian:3.0)BaltoSlavic:2.0):1.5,",
  "(((Greek:4.8,Armenian:4.8):0.4,Albanian:5.2):0.6,",
  "(Hindi:3.8,Persian:3.8)IndoIranian:2.0):0.7)IndoEuropean;")
ie <- read.tree(text = ie_nwk)
set.seed(2024)
sim <- simSeq(ie, l = 400, type = "USER", levels = c("0", "1"), rate = 0.05)
boot_trees <- bootstrap.phyDat(sim, bs = 200,
  FUN = function(x) upgma(dist.hamming(x)))
boot_trees <- .compressTipLabel(boot_trees)

quartet_topology <- function(tree, quads) {
  tree$edge.length <- rep(1, Nedge(tree))   # count branches, not time
  d <- cophenetic(tree)
  apply(quads, 2, function(q) {
```

```
    s <- c(d[q[1], q[2]] + d[q[3], q[4]],    # ab|cd
           d[q[1], q[3]] + d[q[2], q[4]],    # ac|bd
           d[q[1], q[4]] + d[q[2], q[3]])    # ad|bc
    w <- which(s == min(s))
    if (length(w) == 1) w else 0L            # 0 = unresolved
  })
}
quartet_distance_hand <- function(t1, t2, generalized = FALSE) {
  quads <- combn(sort(t1$tip.label), 4)
  q1 <- quartet_topology(t1, quads); q2 <- quartet_topology(t2, quads)
  if (generalized) {          # GQD: only quartets resolved in t2 (reference)
    keep <- q2 != 0
    return(sum(q1[keep] != q2[keep]) / sum(keep))
  }
  sum(q1 != q2) / ncol(quads)
}
quartet_distance_pkg <- function(t1, t2, generalized = FALSE) {
  qs <- Quartet::QuartetStatus(t1, cf = t2)
  if (generalized) return(unname(qs[, "d"] / (qs[, "s"] + qs[, "d"])))
  unname((qs[, "d"] + qs[, "r1"] + qs[, "r2"]) / qs[, "Q"])
}
classification <- di2multi(ie, tol = 0.71)
rbind(
  hand = c(quartet_distance_hand(ie, boot_trees[[1]]),
           quartet_distance_hand(boot_trees[[1]], classification),
           quartet_distance_hand(boot_trees[[1]], classification, TRUE)),
  Quartet = c(quartet_distance_pkg(ie, boot_trees[[1]]),
              quartet_distance_pkg(boot_trees[[1]], classification),
              quartet_distance_pkg(boot_trees[[1]], classification, TRUE)))
```

```
#>            [,1]   [,2] [,3]
#> hand    0.08626 0.3099    0
#> Quartet 0.08626 0.3099    0
```

The direct implementation examines all $\binom{n}{4}$ quartets and computes a full distance matrix, so its cost grows as n^4; the algorithms used in **Quartet** (from the tqDist library) scale to thousands of tips.

B.1.3 Uniform random trees by Rémy's algorithm

Chapter 2 compares the Colless index of Yule trees with that of trees drawn from the uniform (PDA) distribution on rooted topologies. A short way to draw such trees is Rémy's algorithm, which grows a uniformly random rooted tree by repeatedly choosing a node (tip or internal) uniformly at random and inserting a new tip on the branch above it (or above the root). Each step multiplies the number of possible histories by the number of nodes, $2k-1$ for a tree with k tips, which is exactly the factor by which the number of rooted topologies grows (Section B.1.1), so every topology is equally likely. The implementation below represents a tree as nested lists and reproduces the comparison of the chapter.

```
colless <- function(tree) { b <- balance(tree); sum(abs(b[, 1] - b[, 2])) }
# Uniform (PDA) random rooted trees by Remy's algorithm, as nested lists
n_nodes <- function(t) if (is.character(t)) 1 else
  1 + n_nodes(t[[1]]) + n_nodes(t[[2]])
insert_tip <- function(t, k, lab) {
  if (k == 1) return(list(t, lab))
  k <- k - 1; c1 <- n_nodes(t[[1]])
  if (k <= c1) t[[1]] <- insert_tip(t[[1]], k, lab)
  else t[[2]] <- insert_tip(t[[2]], k - c1, lab)
  t
}
to_newick <- function(t) if (is.character(t)) t else
  paste0("(", to_newick(t[[1]]), ",", to_newick(t[[2]]), ")")
rpda <- function(n) {
  t <- "t1"
  for (i in 2:n) t <- insert_tip(t, sample.int(n_nodes(t), 1), paste0("t", i))
  read.tree(text = paste0(to_newick(t), ";"))
}
set.seed(1)
n_tips <- 30
colless_remy <- replicate(500, colless(rpda(n_tips)))
colless_tt <- replicate(500, colless(TreeTools::RandomTree(n_tips,
                                                           root = TRUE)))
rbind(Remy = summary(colless_remy), TreeTools = summary(colless_tt))
```

```
#>           Min. 1st Qu. Median  Mean 3rd Qu. Max.
#> Remy        49     129  166.5 168.0     204  311
#> TreeTools   48     122  155.5 160.4     194  341
```

```
# for 4 tips, all 15 rooted topologies should be about equally frequent
set.seed(2)
tab4 <- table(replicate(15000, write.tree(ladderize(rpda(4)))))
c(topologies = length(tab4), min = min(tab4), max = max(tab4))
```

```
#> topologies        min        max
#>         15        936       1044
```

B.1.4 The geometry of tree space

Distances defined by counting moves give tree space the structure of a graph. Billera et al. (2001) showed that the space of rooted trees *with branch lengths* has a much richer geometry. Fix a topology with n tips: its $n-2$ internal branch lengths are non-negative real numbers, so the trees with that topology form an *orthant*, a copy of the non-negative part of $(n-2)$-dimensional space. When an internal branch shrinks to length zero, the tree reaches the boundary of its orthant, where it is identical to a tree in each of the neighbouring orthants that differ from it by one NNI across that branch. Gluing all $(2n-3)!!$ orthants along these shared boundaries gives the *BHV tree space* (after Billera, Holmes and Vogtmann). In this space there is a unique shortest path (geodesic) between any two trees, which passes through lower-dimensional faces (trees with polytomies)

when the topologies differ, and the geodesic distance can be computed in polynomial time (Owen and Provan, 2011). The BHV geometry makes it possible to define genuinely geometric means and variances of tree samples, confidence regions, and principal component-like analyses of tree variation. Because geodesics between trees with different topologies pass through trees with polytomies, the geometric mean of a diffuse posterior sample is often a tree with polytomies, a natural expression of the uncertainty that an MCC tree hides. For rooted time trees, other metrics have been designed to capture differences in both topology and timing, such as the metric of Kendall and Colijn (2016), which compares the depths of the MRCAs of all pairs of tips and can be tuned to weight topology or branch lengths more heavily.

B.2 Chapter 4: Distance and Parsimony Methods

B.2.1 The neighbor-joining algorithm in detail

Neighbor joining (Section 4.5) joins at each step the pair of clusters i, j that minimizes $Q_{ij} = (n-2)\,d_{ij} - R_i - R_j$ (equation (4.7)), where n is the current number of clusters and $R_i = \sum_k d_{ik}$. When i and j are joined into a new node u, their branch lengths are

$$v_i = \frac{d_{ij}}{2} + \frac{R_i - R_j}{2(n-2)}, \qquad v_j = d_{ij} - v_i, \tag{B.4}$$

and the distances from u to each remaining cluster k are

$$d_{uk} = \tfrac{1}{2}\left(d_{ik} + d_{jk} - d_{ij}\right). \tag{B.5}$$

Equation (B.4) splits the distance d_{ij} between the two branches, giving the longer branch to the language that is on average farther from all others; equation (B.5) is the distance from the new node to k that would hold exactly on an additive tree, since $d_{ik} + d_{jk}$ counts the path from u to k twice plus the two new branches. The procedure repeats with $n-1$ clusters until three remain, which are joined to a final central node. In the four-language example of the chapter, $v_A = 5/2 + (16-22)/4 = 1$ and $v_B = 5 - 1 = 4$.

Why does the Q criterion work? Saitou and Nei (1987) motivated it as a greedy step toward the tree with the smallest total length (the *minimum evolution* principle). Studier and Keppler (1988) showed that the criterion as originally stated could be simplified to equation (4.7), and Gascuel and Steel (2006) later clarified that NJ is a greedy algorithm for the *balanced* minimum evolution criterion. The key theoretical properties are that NJ recovers the true tree from any additive matrix, and that it is robust to noise: Atteson (1999) proved that NJ returns the correct topology whenever every entry of the matrix is within half the length of the shortest internal branch of its true value. *BIONJ* (Gascuel, 1997) replaces the simple average in equation (B.5) by a weighted average whose weights minimize the variance of the new distances, under a model in which the variance of an estimated distance is proportional to the distance.

The code below computes the Q matrix for the five-language additive matrix `D5` of Section 4.3 and checks that the pair it selects, D and E, is the one NJ joins first; it then carries out the first joining step by hand.

```
true5 <- read.tree(text = "((A:2,B:3):2,C:4,(D:1,E:5):3);")
D5 <- cophenetic(true5)[LETTERS[1:5], LETTERS[1:5]]
nj_Q <- function(D) {
  n <- nrow(D); R <- rowSums(D)
  Q <- (n - 2) * D - outer(R, R, "+")
  diag(Q) <- NA
  Q
}
nj_Q(D5)
```

```
#>     A   B   C   D   E
#> A  NA -54 -46 -40 -40
#> B -54  NA -46 -40 -40
#> C -46 -46  NA -44 -44
#> D -40 -40 -44  NA -56
#> E -40 -40 -44 -56  NA
```

```
write.tree(nj(as.dist(D5)))      # identical to the generating tree (unrooted)
```

```
#> [1] "(B:3,A:2,((D:1,E:5):3,C:4):2);"
```

```
# the first joining step by hand: D and E
n <- nrow(D5); R <- rowSums(D5)
v_D <- D5["D", "E"] / 2 + (R["D"] - R["E"]) / (2 * (n - 2))
c(v_D = unname(v_D), v_E = unname(D5["D", "E"] - v_D))
```

```
#> v_D v_E
#>   1   5
```

```
d_u <- (D5[c("A", "B", "C"), "D"] + D5[c("A", "B", "C"), "E"] -
          D5["D", "E"]) / 2
d_u                              # distances from the new node to A, B, C
```

```
#> A B C
#> 7 8 7
```

The branch lengths of D and E (1 and 5) and the distances from their common ancestor to A, B and C (7, 8 and 7) are exactly those of the generating tree, as they must be for an additive matrix.

B.2.2 Fitch's algorithm in detail

For unordered characters with equal costs for all changes, the small parsimony problem on a rooted binary tree is solved by the algorithm of Fitch (1971), whose correctness was proved by Hartigan (1973). It works from the leaves to the root (a *postorder* traversal). Each node receives a set of states:

- A leaf receives the set containing its observed state.

- An internal node with children whose sets are S_1 and S_2 receives

$$S = \begin{cases} S_1 \cap S_2 & \text{if } S_1 \cap S_2 \neq \emptyset, \\ S_1 \cup S_2 & \text{otherwise, adding one step.} \end{cases} \tag{B.6}$$

The number of union operations is the parsimony score. The root set contains every state that some most-parsimonious reconstruction assigns to the root. A second, top-down pass recovers actual reconstructions at internal nodes: give the root any state in its set, and give each child its parent's state if that state is in the child's set, otherwise any state from the child's set. The sets computed in the first pass are not, in general, the sets of all states that some most-parsimonious reconstruction assigns to an internal node; recovering all optimal reconstructions requires Hartigan's more general algorithm or the Sankoff algorithm below. For a polytomy, the rule is applied to the children one at a time, as in the code.

The function below implements the first pass directly and is checked against **phangorn**'s `parsimony()`, on the 'hand' example of Section 4.9 and on 200 random characters on a random tree.

```
library(phangorn)
fitch_score <- function(tree, states) {
  tree <- reorder(tree, "postorder")
  ntip <- Ntip(tree)
  sets <- vector("list", ntip + tree$Nnode)
  for (i in seq_len(ntip)) sets[[i]] <- states[tree$tip.label[i]]
  score <- 0
  for (node in unique(tree$edge[, 1])) {          # internal nodes, postorder
    kids <- tree$edge[tree$edge[, 1] == node, 2]
    S <- sets[[kids[1]]]
    for (k in kids[-1]) {
      common <- intersect(S, sets[[k]])
      if (length(common) > 0) S <- common else {
        S <- union(S, sets[[k]]); score <- score + 1
      }
    }
    sets[[node]] <- S
  }
  list(score = score, root_set = sets[[ntip + 1]])
}

tr7 <- read.tree(text =
  "(((English,German),Swedish),((Italian,Spanish),(Russian,Polish)));")
hand <- c(English = "H", German = "H", Swedish = "H",
          Italian = "M", Spanish = "M", Russian = "R", Polish = "R")
fitch_score(tr7, hand)
```

```
#> $score
#> [1] 2
#>
#> $root_set
#> [1] "H" "M" "R"
```

```
hand_pd <- phyDat(matrix(hand, ncol = 1, dimnames = list(names(hand), NULL)),
                  type = "USER", levels = c("H", "M", "R"))
parsimony(tr7, hand_pd, method = "fitch")
```

```
#> [1] 2
```

```
# a larger check: 200 random four-state characters on a random 12-tip tree
set.seed(12)
tr12 <- rtree(12)
X <- matrix(sample(c("a", "b", "c", "d"), 12 * 200, replace = TRUE), 12,
            dimnames = list(tr12$tip.label, NULL))
hand_scores <- apply(X, 2, function(s) fitch_score(tr12, s)$score)
c(hand = sum(hand_scores),
  phangorn = parsimony(tr12, phyDat(X, type = "USER",
                                    levels = c("a", "b", "c", "d"))))
```

```
#>     hand phangorn
#>     1247     1247
```

B.2.3 The Sankoff algorithm

For a cost matrix $c(a, b)$ giving the cost of a change from state a to state b, the *Sankoff algorithm* (Sankoff, 1975) computes the minimum total cost by dynamic programming. For each node v and state a, let $L_v(a)$ be the minimum cost of the subtree below v given that v is in state a. For a leaf, $L_v(a) = 0$ if a is the observed state and ∞ otherwise. For an internal node with children u and w,

$$L_v(a) = \min_b \left[c(a, b) + L_u(b)\right] + \min_b \left[c(a, b) + L_w(b)\right], \tag{B.7}$$

and the tree's cost is $\min_a L_{\text{root}}(a)$. With unit costs it reproduces the Fitch score. Because the costs may be asymmetric, the tree must be rooted, and the root state matters: under a Dollo-like cost matrix, a root in state 1 (cognate present) followed by losses can be much cheaper than a root in state 0 followed by gains. Felsenstein's pruning algorithm (equation (5.8)) has exactly the same structure, with sums of products of probabilities in place of minima of sums of costs.

The implementation below evaluates the recursion on a binary cognate character, the presence of the Germanic *hand* cognate set in the seven languages of the chapter's example, under the Dollo-like cost matrix of Section 4.10 (gain costs 10, loss costs 1), and checks the result against **phangorn**.

```
sankoff_cost <- function(tree, states, cost) {
  tree <- reorder(tree, "postorder")
  ntip <- Ntip(tree); lev <- rownames(cost)
  L <- matrix(0, ntip + tree$Nnode, length(lev), dimnames = list(NULL, lev))
  for (i in seq_len(ntip)) {
    L[i, ] <- Inf; L[i, states[tree$tip.label[i]]] <- 0
  }
  for (e in seq_len(nrow(tree$edge))) {           # children before parents
    p <- tree$edge[e, 1]; ch <- tree$edge[e, 2]
```

```
    L[p, ] <- L[p, ] + sapply(lev, function(a) min(cost[a, ] + L[ch, ]))
  }
  L[ntip + 1, ]                                  # cost by root state
}
dollo_cost <- matrix(c(0, 10,
                       1,  0), nrow = 2, byrow = TRUE,
                     dimnames = list(c("0", "1"), c("0", "1")))
tr7r <- read.tree(text =
  "(((English,German),Swedish),((Italian,Spanish),(Russian,Polish)));")
germanic_hand <- c(English = "1", German = "1", Swedish = "1",
                   Italian = "0", Spanish = "0", Russian = "0", Polish = "0")
sankoff_cost(tr7r, germanic_hand, dollo_cost)
```

```
#>  0  1
#> 10  1
```

```
gh_pd <- phyDat(matrix(germanic_hand, ncol = 1,
                       dimnames = list(names(germanic_hand), NULL)),
                type = "USER", levels = c("0", "1"))
parsimony(tr7r, gh_pd, method = "sankoff", cost = dollo_cost)
```

```
#> [1] 1
```

With the root in state 0 (absent), the character needs one gain on the Germanic stem (cost 10); with the root in state 1 (present), a single loss on the branch leading to Romance and Slavic suffices (cost 1), which is the cheaper explanation. Under this rooting and cost matrix, the Germanic word is therefore reconstructed for the root language and lost in the rest of the family, which the comparative method tells us is wrong. Under a Dollo-like cost matrix, a cognate set shared by a subgroup is read as an old word lost elsewhere unless that requires many independent losses; the cost scheme, like a substitution model, builds assumptions about linguistic change into the inference, and the position of the root interacts with them.

B.2.4 Classical MDS, PCA and stress

Classical MDS (Section 4.7) rests on a simple identity. If n points $\mathbf{x}_1, \dots, \mathbf{x}_n$ have been centred, so that $\sum_i \mathbf{x}_i = \mathbf{0}$, their squared Euclidean distances $d_{ij}^2 = \|\mathbf{x}_i - \mathbf{x}_j\|^2$ determine the matrix of inner products $b_{ij} = \mathbf{x}_i^\top \mathbf{x}_j$ through

$$\mathbf{B} = -\tfrac{1}{2}\,\mathbf{J}\,\mathbf{D}^{(2)}\mathbf{J}, \qquad \mathbf{J} = \mathbf{I} - \tfrac{1}{n}\mathbf{1}\mathbf{1}^\top, \tag{B.8}$$

because expanding $d_{ij}^2 = b_{ii} + b_{jj} - 2b_{ij}$ and double-centring removes the first two terms. $\mathbf{B}$ is symmetric and positive semidefinite, so it can be written $\mathbf{B} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$, and the coordinates $\mathbf{V}\mathbf{\Lambda}^{1/2}$ reproduce the distances exactly. Keeping the k largest eigenvalues gives the best k-dimensional approximation of $\mathbf{B}$ in the least-squares sense. When the points are the rows of a centred data matrix $\mathbf{X}$, $\mathbf{B} = \mathbf{X}\mathbf{X}^\top$, whose eigenvectors scaled by the square roots of the eigenvalues are exactly the principal component scores of $\mathbf{X}$. If the input dissimilarities are not Euclidean distances, Equation (B.8) still defines $\mathbf{B}$, but some of its eigenvalues are negative, and no configuration of points reproduces the matrix.

Non-metric MDS minimizes Kruskal's *stress-1*,

$$S = \sqrt{\frac{\sum_{i<j}(d_{ij} - \hat{d}_{ij})^2}{\sum_{i<j} d_{ij}^2}}, \tag{B.9}$$

where d_{ij} are the distances in the configuration and the *disparities* $\hat{d}_{ij}$ are the monotone (isotonic) regression of the d_{ij} on the rank order of the input dissimilarities (Kruskal, 1964). **MASS**'s `isoMDS()` reports $100\,S$. The code below implements classical MDS and stress-1 from these formulas, checks them against `cmdscale()`, `prcomp()` and `isoMDS()`, and shows the negative eigenvalues produced by corrected binary distances.

```
library(phangorn)
library(MASS)
set.seed(11)
tr <- rtree(8, tip.label = paste0("L", 1:8))
tr$edge.length <- tr$edge.length * 0.4
sim <- simSeq(tr, l = 200, type = "USER", levels = c("0", "1"))
X <- matrix(as.integer(as.character(sim)), nrow = length(sim),
            dimnames = list(names(sim), NULL))

cmds <- function(D, k = 2) {                  # classical MDS by hand
  D2 <- as.matrix(D)^2
  n <- nrow(D2); J <- diag(n) - matrix(1 / n, n, n)
  e <- eigen(-0.5 * J %*% D2 %*% J, symmetric = TRUE)
  list(points = e$vectors[, 1:k] %*% diag(sqrt(e$values[1:k])),
       values = e$values)
}
DE <- dist(X)                                  # Euclidean distances
mine <- cmds(DE)
c(vs_cmdscale = isTRUE(all.equal(abs(mine$points),
                                 abs(unname(cmdscale(DE, k = 2))))),
  vs_prcomp = isTRUE(all.equal(abs(mine$points),
                               abs(unname(prcomp(X)$x[, 1:2])))))
```

```
#> vs_cmdscale   vs_prcomp
#>        TRUE        TRUE
```

```
# corrected binary distances are not Euclidean: negative eigenvalues
p <- as.matrix(dist.hamming(sim))
d_corr <- as.dist(-0.5 * log(1 - 2 * pmin(p, 0.499)))
round(cmds(d_corr)$values, 3)
```

```
#> [1]  2.137  0.480  0.223  0.072  0.047  0.000 -0.004 -0.539
```

```
stress1 <- function(delta, conf) {             # Kruskal's stress-1, in %
  delta <- as.vector(delta); d <- as.vector(dist(conf))
  o <- order(delta)
  dhat <- numeric(length(d)); dhat[o] <- isoreg(delta[o], d[o])$yf
  100 * sqrt(sum((d - dhat)^2) / sum(d^2))
```

```
}
D_tree <- as.dist(cophenetic(tr))              # no tied dissimilarities
sapply(1:2, function(k) {
  fit <- isoMDS(D_tree, k = k, trace = FALSE)
  c(k = k, isoMDS = fit$stress, by_hand = stress1(D_tree, fit$points))
})
```

```
#>             [,1]     [,2]
#> k       1.000000 2.000000
#> isoMDS  0.007634 0.009778
#> by_hand 0.007634 0.009778
```

The hand-coded classical MDS reproduces `cmdscale()` and the PCA scores up to the signs of the axes, and the corrected distances give negative eigenvalues. The stress check uses the patristic distances of the tree, which have no ties. When several input dissimilarities are equal, as they often are for Hamming proportions computed from a few hundred characters, implementations differ in how they treat the ties in the monotone regression, and a hand computation can differ slightly from `isoMDS()`. In the vocabulary of Section 6.10, these checks are implementation validation only.

B.3 Chapter 5: Maximum Likelihood

This section collects the derivations and the from-scratch implementations behind Chapter 5. The code is self-contained: it recreates the functions and simulated data it needs, with the same random seeds as the chapter.

```
library(phangorn)
binary_P <- function(t, pi1) {           # equation (likelihood-binaryP)
  pi0 <- 1 - pi1
  e <- exp(-t / (2 * pi0 * pi1))
  matrix(c(pi0 + pi1 * e, pi1 * (1 - e),
           pi0 * (1 - e), pi1 + pi0 * e), 2, byrow = TRUE)
}
binary_Q <- function(pi1) {
  pi0 <- 1 - pi1
  s <- 1 / (2 * pi0 * pi1)
  matrix(c(-s * pi1, s * pi1, s * pi0, -s * pi0), 2, byrow = TRUE)
}
sim_binary <- function(tree, nchar, pi1, rates = rep(1, nchar)) {
  tree <- reorder(tree, "cladewise")
  ntip <- Ntip(tree); pi0 <- 1 - pi1
  S <- matrix(NA_integer_, ntip + tree$Nnode, nchar)
  S[ntip + 1, ] <- rbinom(nchar, 1, pi1)
  for (e in seq_len(nrow(tree$edge))) {
    p <- tree$edge[e, 1]; ch <- tree$edge[e, 2]
    ex <- exp(-tree$edge.length[e] * rates / (2 * pi0 * pi1))
    prob1 <- ifelse(S[p, ] == 1, pi1 + pi0 * ex, pi1 * (1 - ex))
    S[ch, ] <- rbinom(nchar, 1, prob1)
```

```
  }
  out <- S[seq_len(ntip), , drop = FALSE]
  rownames(out) <- tree$tip.label
  out
}
```

B.3.1 From rates to transition probabilities

Section 5.2 stated that the transition probabilities of a CTMC are given by the matrix exponential $\mathbf{P}(t) = e^{\mathbf{Q}t}$ (equation (5.3)). The argument is short. By the Markov property, the probability of going from i to j in time $t+h$ is the probability of going from i to some intermediate state l in time t and then from l to j in time h, summed over l. In matrix form this is the *Chapman–Kolmogorov equation* $\mathbf{P}(t+h) = \mathbf{P}(t)\mathbf{P}(h)$. For small h, $\mathbf{P}(h) \approx \mathbf{I} + \mathbf{Q}h$, so

$$\frac{\mathbf{P}(t+h) - \mathbf{P}(t)}{h} \approx \mathbf{P}(t)\mathbf{Q} \quad \Longrightarrow \quad \frac{d}{dt}\mathbf{P}(t) = \mathbf{P}(t)\mathbf{Q}, \qquad \mathbf{P}(0) = \mathbf{I}. \tag{B.10}$$

This matrix differential equation (Kolmogorov's forward equation) has the solution $\mathbf{P}(t) = e^{\mathbf{Q}t} = \sum_{n \geq 0} (\mathbf{Q}t)^n / n!$, as can be checked by differentiating the series term by term.

The series is not a good way to compute the exponential. If $\mathbf{Q}$ can be diagonalized, $\mathbf{Q} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{-1}$ with $\mathbf{\Lambda}$ the diagonal matrix of eigenvalues $\lambda_1, \dots, \lambda_k$, then every power is $\mathbf{Q}^n = \mathbf{U}\mathbf{\Lambda}^n\mathbf{U}^{-1}$ and so

$$e^{\mathbf{Q}t} = \mathbf{U}\,\mathrm{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_k t})\,\mathbf{U}^{-1}.$$

For a reversible $\mathbf{Q}$ the eigenvalues are real. Because the rows of $\mathbf{Q}$ sum to zero, $\mathbf{Q}\mathbf{1} = \mathbf{0}$ and one eigenvalue is 0, with eigenvector $\mathbf{1}$; it gives the stationary part of $\mathbf{P}(t)$, to which every row converges. The other eigenvalues are negative and set how quickly the starting state is forgotten.

For the binary model (5.4) the eigenvalues are 0 and $-(\alpha+\beta)$. Writing $\mathbf{P}(t) = \mathbf{A} + \mathbf{B}e^{-(\alpha+\beta)t}$, the condition $\mathbf{P}(0) = \mathbf{I}$ gives $\mathbf{A} + \mathbf{B} = \mathbf{I}$, and the limit $t \to \infty$ gives $\mathbf{A}$ with both rows equal to π. Hence $\mathbf{B} = \mathbf{I} - \mathbf{A}$, which is exactly the closed form (5.5). The code checks the eigendecomposition route against the closed form and against **expm**. Sections A.5 and A.6 explain eigenvalues and the matrix exponential in more detail.

```
Q <- binary_Q(0.3)
eig <- eigen(Q)
round(eig$values, 6)                    # 0 and -(alpha + beta)
```

```
#> [1] -2.381  0.000
```

```
P_eigen <- function(t) {
  eig$vectors %*% diag(exp(eig$values * t)) %*% solve(eig$vectors)
}
c(eigen_vs_closed = all.equal(P_eigen(0.4), binary_P(0.4, 0.3)),
  expm_vs_closed  = all.equal(unname(expm::expm(Q * 0.4)),
                              binary_P(0.4, 0.3)))
```

```
#> eigen_vs_closed  expm_vs_closed
#>            TRUE            TRUE
```

B.3.2 A general pruning function

The function below implements the pruning algorithm of Section 5.4 for any tree and any binary data matrix, processing all characters at once: each node's conditional likelihoods are stored as a $2 \times m$ matrix, one column per character. It works on unrooted trees, because the node that **ape** stores as the root of an unrooted tree is simply a trifurcating node, and multiple children are handled by repeating the factor in equation (5.8). Missing data (`NA`) receive conditional likelihood 1 in both states.

```
prune_loglik <- function(tree, X, pi1, site = FALSE) {
  # tree: phylo with branch lengths; X: 0/1 matrix, rows named by tip
  # labels (NA = missing); pi1: equilibrium frequency of state 1
  tree <- reorder(tree, "postorder")
  ntip <- Ntip(tree)
  L <- vector("list", ntip + tree$Nnode)
  for (i in seq_len(ntip)) {                       # leaf vectors
    s <- X[tree$tip.label[i], ]
    L[[i]] <- rbind(as.numeric(s %in% c(0, NA)),
                    as.numeric(s %in% c(1, NA)))
  }
  for (v in (ntip + 1):(ntip + tree$Nnode)) L[[v]] <- matrix(1, 2, ncol(X))
  for (e in seq_len(nrow(tree$edge))) {            # postorder: children first
    parent <- tree$edge[e, 1]; child <- tree$edge[e, 2]
    L[[parent]] <- L[[parent]] *
      (binary_P(tree$edge.length[e], pi1) %*% L[[child]])
  }
  lik <- colSums(L[[ntip + 1]] * c(1 - pi1, pi1))  # equation (root)
  if (site) lik else sum(log(lik))
}

# the worked example of Section (likelihood-pruning)
Ps <- function(t) 0.5 * (1 + exp(-2 * t))
Pd <- function(t) 0.5 * (1 - exp(-2 * t))
Lv <- c(Ps(0.1) * Ps(0.3), Pd(0.1) * Pd(0.3))
L_hand <- 0.5 * Pd(0.1) * Pd(0.2) * (Ps(0.2) * Lv[1] + Pd(0.2) * Lv[2]) +
          0.5 * Ps(0.1) * Ps(0.2) * (Pd(0.2) * Lv[1] + Ps(0.2) * Lv[2])
tr4 <- read.tree(
  text = "((English:0.1,German:0.2):0.2,Italian:0.1,Spanish:0.3);")
x4 <- matrix(c(1, 1, 0, 0), ncol = 1,
             dimnames = list(c("English", "German", "Italian", "Spanish"),
                             NULL))
c(general_function = prune_loglik(tr4, x4, 0.5, site = TRUE),
  by_hand = L_hand)
```

```
#> general_function          by_hand
#>          0.05498          0.05498
```

To check the function on a less trivial case, we use the eight-language matrix of Section 5.5 (same seed) and compare our log-likelihood with the one computed by **phangorn**'s `pml()`. This is *implementation validation*: two independent programs, given the same model, tree and data, must

return the same number. We also check the pulley principle numerically by rerooting the tree, which splits one branch into two segments that sum to its original length.

```
set.seed(5)
langs8 <- c("Ana", "Bel", "Cor", "Dun", "Esk", "Fal", "Gor", "Hal")
tr8 <- rtree(8, rooted = FALSE, tip.label = langs8)
tr8$edge.length <- tr8$edge.length * 0.25
X8 <- sim_binary(tr8, 600, pi1 = 0.3)
pd8 <- phyDat(X8, type = "USER", levels = c(0, 1))

c(hand_coded = prune_loglik(tr8, X8, 0.3),
  phangorn   = pml(tr8, pd8, bf = c(0.7, 0.3))$logLik)
```

```
#> hand_coded   phangorn
#>      -2339      -2339
```

```
# the pulley principle: rerooting does not change the likelihood
c(original = prune_loglik(tr8, X8, 0.3),
  rerooted = prune_loglik(root(tr8, "Cor", resolve.root = TRUE), X8, 0.3))
```

```
#> original rerooted
#>    -2339    -2339
```

The two implementations agree, and rerooting leaves the likelihood unchanged.

B.3.3 Optimizing branch lengths

For a single branch with length t, the log-likelihood $\ell(t)$ is smooth, and its first and second derivatives can be computed cheaply because $\frac{d}{dt}\mathbf{P}(t) = \mathbf{Q}\mathbf{P}(t)$. Newton–Raphson iteration,

$$t^{(n+1)} = t^{(n)} - \frac{\ell'(t^{(n)})}{\ell''(t^{(n)})}, \tag{B.11}$$

then converges in a few steps from a reasonable starting value; programs guard against steps that leave the region where ℓ is concave, for example by halving the step or falling back to Brent's method. Figure B.1 shows the log-likelihood profile of one branch, which is the function Newton–Raphson climbs.

```
e_ana <- which(tr8$edge[, 2] == which(tr8$tip.label == "Ana"))
grid <- seq(0.005, 0.4, length.out = 120)
prof <- sapply(grid, function(b) {
  tr <- tr8; tr$edge.length[e_ana] <- b
  prune_loglik(tr, X8, 0.3)
})
par(mar = c(4, 4, 1, 1))
plot(grid, prof, type = "l", lwd = 2, xlab = "length of branch to Ana",
     ylab = "log-likelihood")
abline(v = tr8$edge.length[e_ana], lty = 2)
abline(v = grid[which.max(prof)], lty = 3)
```

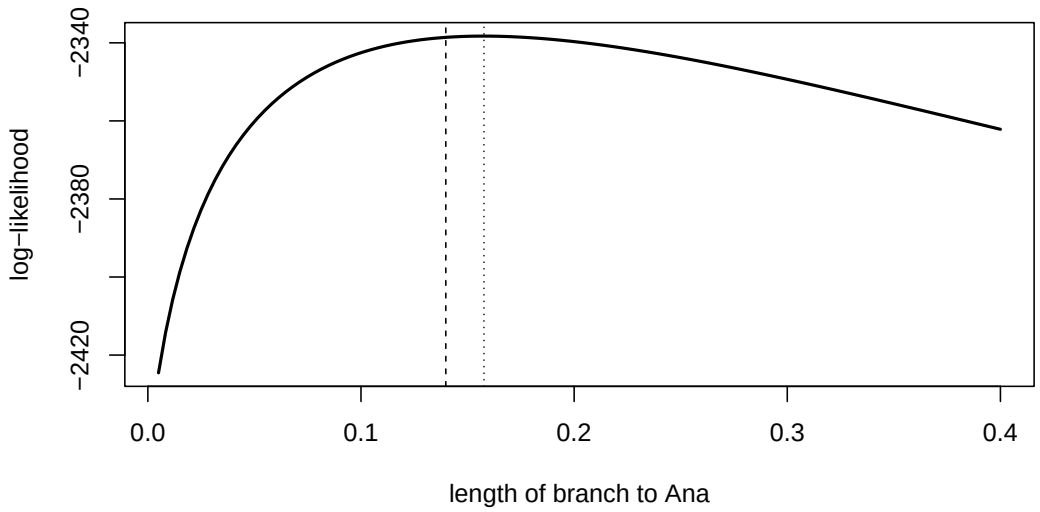


Figure B.1: Profile of the log-likelihood as a function of the length of one terminal branch (leading to Ana) in the eight-language example, with all other branch lengths held at their true values. The dashed line marks the true length and the dotted line the maximum.

We can also optimize all branch lengths at once with a general-purpose optimizer. The function below maximizes the hand-coded likelihood over all 13 branch lengths (on a log scale, which keeps them positive), starting from a tree in which every branch has length 0.1, and compares the result with **phangorn**'s `optim.pml()`. Its optional argument `correction` applies the all-absent ascertainment correction of equation (5.10), used in the next subsection.

```
fit_branch_lengths <- function(tree, X, pi1, correction = FALSE) {
  all_absent <- matrix(0, nrow(X), 1, dimnames = list(rownames(X), NULL))
  negll <- function(log_bl) {
    tree$edge.length <- exp(log_bl)
    ll <- prune_loglik(tree, X, pi1)
    if (correction)          # condition on each character being observable
      ll <- ll - ncol(X) *
        log(1 - prune_loglik(tree, all_absent, pi1, site = TRUE))
    -ll
  }
  o <- optim(log(tree$edge.length), negll, method = "L-BFGS-B",
             lower = -12, upper = 1.5)
  tree$edge.length <- exp(o$par)
  list(tree = tree, logLik = -o$value)
}
start8 <- tr8; start8$edge.length <- rep(0.1, Nedge(start8))
fit_hand <- fit_branch_lengths(start8, X8, 0.3)
fit_phangorn <- optim.pml(pml(start8, pd8, bf = c(0.7, 0.3)),
                          optEdge = TRUE, control = pml.control(trace = 0))
c(logLik_hand = fit_hand$logLik, logLik_phangorn = fit_phangorn$logLik,
  tree_length_hand = sum(fit_hand$tree$edge.length),
  tree_length_phangorn = sum(fit_phangorn$tree$edge.length),
  tree_length_true = sum(tr8$edge.length))
```

```
#>          logLik_hand      logLik_phangorn     tree_length_hand
#>            -2334.514            -2334.514                1.590
#> tree_length_phangorn     tree_length_true
#>                1.590                1.551
```

The hand-coded optimization and **phangorn** reach the same maximum (implementation validation), and the estimated tree length is close to the true value used in the simulation (parameter recovery under the model's own assumptions).

B.3.4 The all-absent ascertainment correction by hand

Section 5.6 demonstrated the Mkv correction, which conditions on characters being variable. Cognate matrices that keep sets present in every language (such as 'two') need instead the correction (5.10), which conditions only on a character not being absent everywhere. With the hand-coded functions we remove just the all-absent characters from the simulated matrix and fit branch lengths with and without the correction.

```
X8_obs <- X8[, colSums(X8) > 0]          # what a linguist could observe
c(characters_simulated = ncol(X8), characters_observed = ncol(X8_obs))
```

```
#> characters_simulated  characters_observed
#>                  600                  453
```

```
fit_uncorrected <- fit_branch_lengths(start8, X8_obs, 0.3,
                                      correction = FALSE)
fit_corrected   <- fit_branch_lengths(start8, X8_obs, 0.3,
                                      correction = TRUE)
c(true = sum(tr8$edge.length),
  uncorrected = sum(fit_uncorrected$tree$edge.length),
  corrected = sum(fit_corrected$tree$edge.length))
```

```
#>        true uncorrected   corrected
#>       1.551       2.144       1.524
```

Without the correction the tree length is inflated; with it, the estimate is back near the truth. Fewer characters are removed by this filter than by the variability filter (only the all-absent ones), so the bias of the uncorrected analysis is smaller here than in the Mkv example of the chapter, but it has the same direction.

B.3.5 Escaping local optima by stochastic search

Section 5.7 showed that NNI hill climbing from random starting trees stalls on poor local optima. The code below recreates that experiment (same seed and data) and then runs **phangorn**'s stochastic search, which alternates NNI hill climbing with random perturbations of the current tree in the spirit of IQ-TREE, from the worst of the random starts.

```
set.seed(42)
true14 <- rcoal(14, tip.label = paste0("L", sprintf("%02d", 1:14)))
true14$edge.length <- true14$edge.length * 0.3
site_rates <- rgamma(600, shape = 0.5, rate = 0.5)
X14 <- sim_binary(true14, 600, pi1 = 0.3, rates = site_rates)
```

```
cog14 <- phyDat(X14, type = "USER", levels = c(0, 1))
ctl <- pml.control(trace = 0)
nj_start <- nj(dist.hamming(cog14))
fit_nj_start <- optim.pml(pml(nj_start, cog14, k = 4), optNni = TRUE,
                          optBf = TRUE, optGamma = TRUE, control = ctl)
random_starts <- lapply(1:5, function(i) {
  st <- rtree(14, rooted = FALSE, tip.label = names(cog14))
  f <- optim.pml(pml(st, cog14, k = 4), optNni = TRUE, optBf = TRUE,
                 optGamma = TRUE, control = ctl)
  list(start = st, fit = f)
})
ll_random <- sapply(random_starts, function(r) r$fit$logLik)
worst <- random_starts[[which.min(ll_random)]]$start
fit_stochastic <- optim.pml(pml(worst, cog14, k = 4), optNni = TRUE,
                            optBf = TRUE, optGamma = TRUE,
                            rearrangement = "stochastic", control = ctl)
data.frame(
  search = c("NNI from NJ tree", "NNI from worst random tree",
             "stochastic from worst random tree"),
  logLik = c(fit_nj_start$logLik, min(ll_random), fit_stochastic$logLik),
  RF_to_true = c(RF.dist(fit_nj_start$tree, true14),
                 RF.dist(random_starts[[which.min(ll_random)]]$fit$tree,
                         true14),
                 RF.dist(fit_stochastic$tree, true14)))
```

```
#>                              search logLik RF_to_true
#> 1                  NNI from NJ tree  -1895          0
#> 2        NNI from worst random tree  -2347         12
#> 3 stochastic from worst random tree  -1895          0
```

The stochastic search escapes the local optimum in which NNI hill climbing from the same starting tree became stuck and ends close to the result obtained from the NJ starting tree. Perturbation, larger rearrangements and multiple independent searches are the standard remedies for rugged likelihood surfaces in tree space.

B.4 Chapter 6: Bayesian Phylogenetic Inference

This section contains the derivations, the from-scratch tree sampler and the marginal-likelihood estimators behind Chapter 6. The code is self-contained and uses the same random seeds as the chapter.

B.4.1 The Yule tree density

Under the *Yule process* each lineage splits at rate λ and never goes extinct. Starting with two lineages at the root, while there are j lineages the waiting time w_j to the next split is exponential with rate $j\lambda$, and the lineage that splits is chosen uniformly among the j, with probability $1/j$. The density of a particular split happening after a waiting time w_j is therefore $(1/j)\, j\lambda\, e^{-j\lambda w_j} = \lambda e^{-j\lambda w_j}$: the probability of choosing the particular lineage cancels the factor j in the rate. After the last split,

n lineages survive to the present for a time w_n with probability $e^{-n\lambda w_n}$. Multiplying these factors over the history of a tree with n leaves, the density of a particular labeled history (a topology with a ranking of its internal nodes) and its waiting times $w_2, \dots, w_n$ is

$$f(\tau, \mathbf{w} \mid \lambda) = \prod_{j=2}^{n-1} \lambda e^{-j\lambda w_j} \times e^{-n\lambda w_n} = \lambda^{n-2} \exp\left(-\lambda \sum_{j=2}^{n} j w_j\right) = \lambda^{n-2} e^{-\lambda T(\tau)}, \tag{B.12}$$

where $T(\tau) = \sum_j j w_j$ is the total length of all branches, since during the interval w_j exactly j branches are growing. The density depends only on the number of languages and the total tree length. For fixed n it is maximized over λ at $\hat{\lambda} = (n-2)/T$, the number of splits after the root divided by the total time at risk of splitting.

B.4.2 Why Metropolis–Hastings works, and Hastings ratios

A Markov chain with transition probabilities $P(x \to x')$ has the posterior π as its stationary distribution if it satisfies *detailed balance*, $\pi(x)P(x \to x') = \pi(x')P(x' \to x)$ for all x and x': summing both sides over x shows that a chain started in π stays in π. For the Metropolis–Hastings algorithm (equation (6.3)) and $x \neq x'$, $P(x \to x') = q(x' \mid x)\alpha(x, x')$. Suppose without loss of generality that $\pi(x')q(x \mid x') \leq \pi(x)q(x' \mid x)$. Then $\alpha(x, x') = \pi(x')q(x \mid x')/[\pi(x)q(x' \mid x)]$ and $\alpha(x', x) = 1$, so

$$\pi(x)\, q(x' \mid x)\, \alpha(x, x') = \pi(x')\, q(x \mid x') = \pi(x')\, q(x \mid x')\, \alpha(x', x),$$

which is detailed balance. If the chain can also reach every state from every other (*irreducibility*) and is not periodic, it converges to π from any starting point. Only ratios $\pi(x')/\pi(x)$ appear, so the normalizing constant $f(D)$ is never needed.

The multiplier proposal. For a positive parameter t, draw $u \sim \text{Uniform}(0, 1)$, set $m = e^{\lambda(u-1/2)}$ and propose $t' = mt$. The proposal is uniform on the log scale: $\ln t'$ is uniform on $[\ln t - \lambda/2, \ln t + \lambda/2]$, so by the change of variables from $\ln t'$ to t' the density of proposing t' from t is $1/(\lambda t')$ on its range. The reverse move from t' to t has density $1/(\lambda t)$, and the Hastings ratio is

$$\frac{q(t \mid t')}{q(t' \mid t)} = \frac{1/(\lambda t)}{1/(\lambda t')} = \frac{t'}{t} = m. \tag{B.13}$$

The tree scaler. Multiplying all B branch lengths by the same m changes a B-dimensional vector through a single random variable, so the proposal has no density on the full space and the Hastings ratio must be obtained from the Jacobian of the move, in the manner of Green (1995). Write $u = \ln m$, drawn from a distribution symmetric about zero, and view the move as the map $(t_1, \dots, t_B, u) \mapsto (e^u t_1, \dots, e^u t_B, -u)$, whose inverse is the same map. Its Jacobian matrix is triangular with diagonal $(m, \dots, m, -1)$, so the absolute determinant, and hence the Hastings ratio, is m^B. With $B = 1$ this reduces to the single-branch result (B.13).

NNI. The NNI move of the toy sampler picks an internal branch uniformly and one of two subtrees on each side uniformly; the reverse move is proposed with the same probability, so its Hastings ratio is 1.

Forgetting a Hastings ratio does not make a chain fail visibly; it makes it sample the wrong distribution. The code below runs the one-parameter sampler of Chapter 6 with and without the term

$\log m$ and compares both with the exact posterior mean computed on a grid. This is the kind of bug that implementation validation against an exact answer, or a prior-only run, is designed to catch.

```
loglik_t <- function(t, N = 100, k = 30) {
  p <- 0.5 * (1 - exp(-2 * t))
  k * log(0.5 * p) + (N - k) * log(0.5 * (1 - p))
}
mh_t <- function(niter, lambda = 1, t_start = 1, beta = 1, prior_rate = 1,
                 N = 100, k = 30, hastings = TRUE) {
  t <- t_start
  lp <- beta * loglik_t(t, N, k) + dexp(t, prior_rate, log = TRUE)
  out <- numeric(niter); accepted <- 0
  for (i in seq_len(niter)) {
    m <- exp(lambda * (runif(1) - 0.5))
    t_new <- t * m
    lp_new <- beta * loglik_t(t_new, N, k) +
      dexp(t_new, prior_rate, log = TRUE)
    logH <- if (hastings) log(m) else 0
    if (log(runif(1)) < lp_new - lp + logH) {
      t <- t_new; lp <- lp_new; accepted <- accepted + 1
    }
    out[i] <- t
  }
  structure(out, acceptance = accepted / niter)
}
grid <- seq(0.001, 2, length.out = 2000)
p <- exp(loglik_t(grid) - max(loglik_t(grid))) * dexp(grid, 1)
set.seed(11)
c(exact = sum(grid * p) / sum(p),
  with_hastings = mean(mh_t(50000)[-(1:1000)]),
  without_hastings = mean(mh_t(50000, hastings = FALSE)[-(1:1000)]))
```

```
#>            exact    with_hastings without_hastings
#>           0.5013           0.5010           0.4683
```

The chain without the Hastings term is shifted toward smaller values. The multiplier proposal is symmetric in $\ln t$, so without the correction the chain samples $\ln t$ as if its density were the posterior density of t; translated back to t, this is a density proportional to $\pi(t)/t$, which puts too much weight on small values. The error is small here because the likelihood is informative; with weak data and many branch lengths it can be large.

B.4.3 A toy Bayesian tree sampler

Chapter 6 analyses six simulated languages with a complete, if small, Bayesian sampler. Its model is a symmetric binary CTMC without rate variation; its priors are uniform on unrooted topologies and independent exponential with mean 0.1 on each branch length; and its proposals are a single-branch multiplier (probability 0.5), an NNI on a random internal branch (0.3), and a tree scaler (0.2), with the Hastings ratios derived above. The code is kept in the file `R/toy-tree-sampler.R`

so that the chapter and this appendix use exactly the same functions; it is listed and evaluated here.

```
# A toy Bayesian tree sampler for binary characters (Chapter 6).
# Listed and explained in the technical appendix (Section bayesian-toy-tree);
# sourced by Chapter 6. Model: symmetric binary CTMC, no rate variation,
# uniform prior on unrooted topologies, i.i.d. exponential branch lengths.
# Requires ape.

# transition probabilities of the symmetric binary model
Pt_sym <- function(t) {
  e <- exp(-2 * t)
  matrix(c(0.5 + 0.5 * e, 0.5 - 0.5 * e,
           0.5 - 0.5 * e, 0.5 + 0.5 * e), 2)
}

# simulate binary characters on a tree, optionally with per-character rates
sim_sym <- function(tree, nchar, rates = rep(1, nchar)) {
  tree <- reorder(tree, "cladewise"); ntip <- Ntip(tree)
  S <- matrix(NA_integer_, ntip + tree$Nnode, nchar)
  S[ntip + 1, ] <- rbinom(nchar, 1, 0.5)
  for (e in seq_len(nrow(tree$edge))) {
    p <- tree$edge[e, 1]; ch <- tree$edge[e, 2]
    flip <- rbinom(nchar, 1,
                   0.5 - 0.5 * exp(-2 * tree$edge.length[e] * rates))
    S[ch, ] <- abs(S[p, ] - flip)
  }
  out <- S[seq_len(ntip), , drop = FALSE]
  rownames(out) <- tree$tip.label
  out
}

# make_loglik(X) returns a function computing the log-likelihood of a tree
# for the 0/1 matrix X (rows named by language). Identical character
# patterns are compressed and weighted by their counts. The pruning
# recursion follows the edge matrix, whatever its order, which is
# convenient after NNI moves rearrange it.
make_loglik <- function(X) {
  pat <- apply(X, 2, paste, collapse = "")
  upat <- unique(pat)
  w <- as.numeric(table(factor(pat, levels = upat)))
  Xp <- X[, match(upat, pat), drop = FALSE]
  tip_partials <- lapply(rownames(Xp), function(l)
    rbind(as.numeric(Xp[l, ] == 0), as.numeric(Xp[l, ] == 1)))
  names(tip_partials) <- rownames(Xp)
  function(tree) {
    ntip <- Ntip(tree)
    kids <- split(seq_len(nrow(tree$edge)), tree$edge[, 1])
    partial <- function(node) {
```

```
      if (node <= ntip) return(tip_partials[[tree$tip.label[node]]])
      out <- 1
      for (e in kids[[as.character(node)]])
        out <- out * (Pt_sym(tree$edge.length[e]) %*%
                        partial(tree$edge[e, 2]))
      out
    }
    Lr <- partial(ntip + 1)
    sum(w * log(0.5 * Lr[1, ] + 0.5 * Lr[2, ]))
  }
}

# NNI: pick an internal branch at random and swap a subtree on one side
# with a subtree on the other; branch lengths travel with the subtrees.
# The move is symmetric, so its Hastings ratio is 1.
nni_move <- function(tree) {
  ntip <- Ntip(tree)
  internal <- which(tree$edge[, 2] > ntip)
  e <- internal[sample.int(length(internal), 1)]
  p <- tree$edge[e, 1]; c <- tree$edge[e, 2]
  sib <- setdiff(which(tree$edge[, 1] == p), e)
  sib <- sib[sample.int(length(sib), 1)]
  kid <- which(tree$edge[, 1] == c)
  kid <- kid[sample.int(length(kid), 1)]
  tree$edge[sib, 1] <- c; tree$edge[kid, 1] <- p     # swap the subtrees
  tree
}

# write a tree as Newick by recursion, whatever its edge order
to_newick <- function(tree) {
  ntip <- Ntip(tree)
  kids <- split(seq_len(nrow(tree$edge)), tree$edge[, 1])
  nw <- function(node) {
    if (node <= ntip) return(tree$tip.label[node])
    parts <- sapply(kids[[as.character(node)]], function(e)
      paste0(nw(tree$edge[e, 2]), ":", signif(tree$edge.length[e], 6)))
    paste0("(", paste(parts, collapse = ","), ")")
  }
  paste0(nw(ntip + 1), ";")
}

# non-trivial splits, each written as the side that excludes the
# alphabetically first language
split_keys <- function(tree) {
  ntip <- Ntip(tree)
  labels <- sort(tree$tip.label); ref <- labels[1]
  kids <- split(tree$edge[, 2], tree$edge[, 1])
  below <- function(node) if (node <= ntip) tree$tip.label[node] else
```

```
    unlist(lapply(kids[[as.character(node)]], below))
  keys <- sapply(setdiff(unique(tree$edge[, 2]), seq_len(ntip)),
                 function(node) {
    s <- sort(below(node))
    if (ref %in% s) s <- sort(setdiff(labels, s))
    paste(s, collapse = "+")
  })
  sort(keys)
}

# Metropolis-Hastings sampler. Proposals: multiplier on one branch length
# (probability 0.5), NNI (0.3), tree scaler on all branch lengths (0.2).
# Prior: exponential with rate prior_rate on each branch length.
run_mcmc <- function(niter, start, loglik, sample_every = 10,
                     use_data = TRUE, prior_rate = 10) {
  tree <- start
  ll <- if (use_data) loglik(tree) else 0
  lp <- sum(dexp(tree$edge.length, prior_rate, log = TRUE))
  ns <- niter %/% sample_every
  trace <- data.frame(iter = integer(ns), logLik = numeric(ns),
                      TL = numeric(ns), topology = character(ns))
  trees <- vector("list", ns)
  acc <- tries <- c(branch = 0, nni = 0, scaler = 0)
  for (i in seq_len(niter)) {
    u <- runif(1); new <- tree
    if (u < 0.5) {                                    # one branch length
      move <- "branch"; j <- sample.int(nrow(tree$edge), 1)
      m <- exp(1.0 * (runif(1) - 0.5))
      new$edge.length[j] <- tree$edge.length[j] * m; logH <- log(m)
    } else if (u < 0.8) {                             # topology
      move <- "nni"; new <- nni_move(tree); logH <- 0
    } else {                                          # all branch lengths
      move <- "scaler"; m <- exp(0.5 * (runif(1) - 0.5))
      new$edge.length <- tree$edge.length * m
      logH <- length(tree$edge.length) * log(m)       # Jacobian: m^B
    }
    tries[move] <- tries[move] + 1
    ll_new <- if (use_data) loglik(new) else 0
    lp_new <- sum(dexp(new$edge.length, prior_rate, log = TRUE))
    if (log(runif(1)) < (ll_new + lp_new) - (ll + lp) + logH) {
      tree <- new; ll <- ll_new; lp <- lp_new
      acc[move] <- acc[move] + 1
    }
    if (i %% sample_every == 0) {
      s <- i %/% sample_every
      trace[s, ] <- list(i, ll, sum(tree$edge.length),
                         paste(split_keys(tree), collapse = " | "))
      trees[[s]] <- read.tree(text = to_newick(tree))  # clean copy
```

```
    }
  }
  list(trace = trace, trees = trees, acceptance = acc / tries)
}

# a random unrooted starting tree with all branch lengths equal to bl
random_start <- function(labels, bl = 0.3) {
  st <- unroot(rtree(length(labels), tip.label = labels))
  st$edge.length <- rep(bl, Nedge(st))
  st
}
```

The pieces are:

- `Pt_sym()` gives the transition probabilities of the symmetric binary model and `sim_sym()` simulates characters on a tree, optionally with a different rate for each character (the chapter uses gamma-distributed rates to make the fitted model deliberately misspecified).
- `make_loglik()` compresses identical character patterns, stores the leaf vectors once, and returns a function that computes the log-likelihood of any tree by a recursive version of the pruning algorithm. The recursion follows the edge matrix, whatever its order, which is convenient after NNI moves rearrange it.
- `nni_move()` swaps a subtree on one side of a random internal branch with a subtree on the other; branch lengths travel with the subtrees.
- `to_newick()` and `split_keys()` write a tree as Newick and summarize it by its non-trivial splits, so that topologies and clades can be counted.
- `run_mcmc()` is the sampler. At each iteration it chooses a move, computes the log posterior ratio plus the log Hastings ratio, accepts or rejects, and records the state every `sample_every` iterations. With `use_data = FALSE` the likelihood is switched off and the chain samples the prior.

As a check of the implementation, the code below recreates the chapter's data (same seed), compares the likelihood with **phangorn**, and confirms that a short prior-only run reproduces the gamma distribution of total tree length implied by nine independent exponential branch-length priors with rate 10, whose mean is 0.9 and variance 0.09.

```
library(phangorn)
set.seed(7)
langs6 <- c("Ana", "Bel", "Cor", "Dun", "Esk", "Fal")
true6 <- read.tree(text = paste0("((Ana:0.10,Bel:0.15):0.04,Cor:0.25,",
  "((Dun:0.08,Esk:0.12):0.015,Fal:0.20):0.03);"))
X6 <- sim_sym(true6, 800, rates = rgamma(800, shape = 0.5, rate = 0.5))
loglik_tree <- make_loglik(X6)
c(toy = loglik_tree(true6),
  phangorn = pml(true6, phyDat(X6, type = "USER", levels = c(0, 1)))$logLik)
```

```
#>      toy phangorn
#>    -2158    -2158
```

```
set.seed(303)
prior_only <- run_mcmc(20000, random_start(langs6), loglik_tree,
                       use_data = FALSE)
TL <- prior_only$trace$TL[-(1:200)]
c(mean_TL = mean(TL), expected_mean = 0.9,
  var_TL = var(TL), expected_var = 0.09)
```

```
#>       mean_TL expected_mean        var_TL  expected_var
#>       0.89981       0.90000       0.09582       0.09000
```

B.4.4 Marginal likelihood estimators

The harmonic mean estimator of Newton and Raftery (1994) uses only the likelihoods of S posterior samples x_s:

$$\hat{f}(D)_{\mathrm{HM}} = \left[\frac{1}{S} \sum_{s=1}^{S} \frac{1}{f(D \mid x_s)} \right]^{-1}. \tag{B.14}$$

It follows from the identity $\mathbb{E}_{\mathrm{post}}[1/f(D \mid x)] = 1/f(D)$, so it is consistent, but its variance can be infinite: the average is dominated by the rare posterior samples with very low likelihood, and since posterior samples rarely visit the regions where the prior has most of its mass, it is almost insensitive to the prior (Lartillot and Philippe, 2006; Xie et al., 2011).

Path sampling (thermodynamic integration) defines the *power posterior* $f_\beta(x) \propto f(D \mid x)^\beta f(x)$ for β between 0 (the prior) and 1 (the posterior), with normalizing constant c_β; $c_0 = 1$ and $c_1 = f(D)$. Differentiating $\ln c_\beta$ with respect to β gives the expected log-likelihood under f_β, and integrating from 0 to 1 gives (Gelman and Meng, 1998; Lartillot and Philippe, 2006)

$$\ln f(D) = \int_0^1 \mathbb{E}_\beta \left[\ln f(D \mid x) \right] d\beta. \tag{B.15}$$

The expected log-likelihood is estimated by MCMC at a series of β values and the integral is computed numerically.

Stepping-stone sampling (Xie et al., 2011) writes the marginal likelihood as a product of ratios between successive power posteriors, $f(D) = \prod_j c_{\beta_{j+1}}/c_{\beta_j}$, and estimates each ratio by importance sampling from the power posterior at β_j:

$$\frac{c_{\beta_{j+1}}}{c_{\beta_j}} \approx \frac{1}{S} \sum_{s=1}^{S} f(D \mid x_s)^{\beta_{j+1}-\beta_j}, \qquad x_s \sim f_{\beta_j}. \tag{B.16}$$

The β values are usually spaced more densely near 0, where the power posterior changes fastest, for instance at the quantiles of a Beta(0.3, 1) distribution as recommended by Xie and colleagues.

The one-parameter model of Chapter 6 is small enough to compute the marginal likelihood exactly by numerical integration, which lets us test the estimators. The comparison is made for three priors on t, exponential with means 1, 10 and 100. Comparing estimators against an exact answer is implementation validation of the estimators themselves.

```
N <- 100; k <- 30
exact_logZ <- function(prior_rate) {
  ll_max <- loglik_t(-0.5 * log(1 - 2 * k / N))
  f <- function(t) exp(loglik_t(t) - ll_max) * dexp(t, prior_rate)
  log(integrate(f, 0, Inf, rel.tol = 1e-10)$value) + ll_max
}
log_mean_exp <- function(a) { m <- max(a); m + log(mean(exp(a - m))) }

harmonic_mean <- function(prior_rate, n = 5000) {
  s <- mh_t(n, lambda = 1, prior_rate = prior_rate)[-(1:500)]
  -log_mean_exp(-loglik_t(s))
}
stepping_stone <- function(prior_rate, K = 20, n = 1000) {
  b <- qbeta(seq(0, 1, length.out = K + 1), 0.3, 1)
  sum(sapply(1:K, function(j) {
    s <- if (b[j] == 0) rexp(n, prior_rate) else  # beta = 0: the prior
      mh_t(n, lambda = 1, beta = b[j], prior_rate = prior_rate,
           t_start = 0.5)[-(1:100)]
    log_mean_exp((b[j + 1] - b[j]) * loglik_t(s))
  }))
}
set.seed(4)
ml_table <- t(sapply(c(1, 0.1, 0.01), function(r) {
  hm <- replicate(8, harmonic_mean(r)); ss <- replicate(3, stepping_stone(r))
  c(prior_mean = 1 / r, exact = exact_logZ(r),
    HM_mean = mean(hm), HM_sd = sd(hm), SS_mean = mean(ss), SS_sd = sd(ss))
}))
knitr::kable(ml_table, digits = 2,
  caption = paste("Log marginal likelihood of the two-language model under",
                  "exponential priors with increasing means: exact value,",
                  "harmonic mean (HM) estimates from 8 runs, and",
                  "stepping-stone (SS) estimates from 3 runs (mean and",
                  "standard deviation)."))
```

Table B.1: Log marginal likelihood of the two-language model under exponential priors with increasing means: exact value, harmonic mean (HM) estimates from 8 runs, and stepping-stone (SS) estimates from 3 runs (mean and standard deviation).

prior_mean	exact	HM_mean	HM_sd	SS_mean	SS_sd
1	-132.1	-131.7	0.34	-132.1	0.05
10	-133.9	-132.7	1.33	-133.9	0.26
100	-136.1	-132.8	1.78	-136.1	0.44

The stepping-stone estimates track the exact values closely, including the drop of about two log units each time the prior becomes ten times more diffuse (approaching $\ln 10 \approx 2.3$ as the prior becomes vague relative to the likelihood). The harmonic mean estimates barely respond to the prior at all, are biased upward, and vary from run to run. In a real comparison between, say,

a model with a vague prior on an extra parameter and a model without it, this is exactly the behavior that leads the harmonic mean to favor the wrong model.

Appendix C

Technical Notes: Models of Change and Time

This appendix collects the technical material behind Chapters 7 to 10: derivations of the likelihoods and transition probabilities of the models of lexical change, the algebra of ascertainment corrections, hand-coded implementations and their checks against packages, the mathematics of clock models and calibration densities, and the longer simulations. Each section corresponds to one chapter and can be read, and run, on its own; the chapters point here wherever material has been moved. Readers who need the underlying probability, matrix algebra or Markov-chain theory will find it in Appendix A.

C.1 Chapter 7: Models of Lexical Evolution

This section collects the derivations and hand-coded implementations behind Chapter 7: the binary chain, the ascertainment corrections, the covarion model, the stochastic Dollo likelihood and its relation to the reversible model, the Mk model, and the algebra of within-meaning dependence. All code runs on its own; the trees and data are recreated with the same random seeds as in the chapter.

C.1.1 Simulation functions

The chapter uses two simulators. `sim_ctmc_binary()` simulates n independent binary characters under the reversible CTMC of Equation (7.1), drawing the root state from the stationary distribution and then, branch by branch from the root, the state at each child from the row of $\mathbf{P}(t)$ (Equation (7.2)) selected by the parent's state; with `shape` given, each character's branch lengths are multiplied by its own gamma-distributed rate. `sim_dollo()` simulates the stochastic Dollo process directly: it draws a Poisson number of traits at the root with the stationary mean λ/μ, thins inherited traits by their survival probability $e^{-\mu t}$ along each branch, adds a Poisson number of new traits born at uniform times along the branch, and keeps those that survive to its end. Only traits present in at least one leaf are returned, which reproduces the way cognate data are ascertained.

```
sim_ctmc_binary <- function(tree, n, pi1, rate = 1, shape = NULL) {
  # simulate n binary characters on 'tree' under the reversible binary CTMC;
```

```
  # if 'shape' is given, character rates are drawn from a Gamma(shape, shape)
  tree <- reorder(tree, "cladewise")          # parents before children
  ntip <- Ntip(tree)
  site_rate <- if (is.null(shape)) rep(1, n) else rgamma(n, shape, shape)
  states <- matrix(NA_integer_, ntip + tree$Nnode, n)
  states[ntip + 1, ] <- rbinom(n, 1, pi1)     # root drawn from stationarity
  pi0 <- 1 - pi1
  beta <- rate / (2 * pi0 * pi1)
  for (i in seq_len(nrow(tree$edge))) {
    p <- tree$edge[i, 1]; ch <- tree$edge[i, 2]
    e <- exp(-beta * tree$edge.length[i] * site_rate)
    p1 <- ifelse(states[p, ] == 1, pi1 + pi0 * e, pi1 * (1 - e))
    states[ch, ] <- rbinom(n, 1, p1)
  }
  out <- states[1:ntip, , drop = FALSE]
  rownames(out) <- tree$tip.label
  out
}

sim_dollo <- function(tree, lambda, mu) {
  # stochastic Dollo: new cognate classes are born at rate lambda per unit
  # time on every lineage; each existing class dies at rate mu; no rebirths
  tree <- reorder(tree, "cladewise")
  ntip <- Ntip(tree)
  present <- vector("list", ntip + tree$Nnode)
  n_root <- rpois(1, lambda / mu)             # stationary number at the root
  present[[ntip + 1]] <- seq_len(n_root)
  next_id <- n_root
  for (i in seq_len(nrow(tree$edge))) {
    p <- tree$edge[i, 1]; ch <- tree$edge[i, 2]; t <- tree$edge.length[i]
    inherited <- present[[p]]
    kept <- inherited[runif(length(inherited)) < exp(-mu * t)]
    nb <- rpois(1, lambda * t)                # births along this branch
    if (nb > 0) {
      born_at <- runif(nb, 0, t)
      alive <- runif(nb) < exp(-mu * (t - born_at))
      kept <- c(kept, next_id + which(alive))
      next_id <- next_id + nb
    }
    present[[ch]] <- kept
  }
  ids <- sort(unique(unlist(present[1:ntip])))  # only observable classes
  out <- t(vapply(present[1:ntip], function(z) as.integer(ids %in% z),
                  integer(length(ids))))
  rownames(out) <- tree$tip.label
  out
}
```

C.1.2 The binary chain: transition probabilities and scaling

Write $\mathbf{1}\pi$ for the 2×2 matrix whose rows are both equal to $\pi = (\pi_0, \pi_1)$. The rate matrix of Equation (7.1) is then $\mathbf{Q} = \beta(\mathbf{1}\pi - \mathbf{I})$, which is the "resampling" description of the chain: at rate β the state is redrawn from π. Because $\pi\mathbf{1} = 1$, the matrix $\mathbf{1}\pi$ is idempotent, $(\mathbf{1}\pi)^2 = \mathbf{1}\pi$, and therefore $e^{\beta t \mathbf{1}\pi} = \mathbf{I} + (e^{\beta t} - 1)\mathbf{1}\pi$. Since $\mathbf{I}$ commutes with everything,

$$\mathbf{P}(t) = e^{\mathbf{Q}t} = e^{-\beta t}\, e^{\beta t\, \mathbf{1}\pi} = e^{-\beta t}\, \mathbf{I} + \left(1 - e^{-\beta t}\right) \mathbf{1}\pi,$$

which is Equation (7.2) written in matrix form (Section A.6 gives the general method). The expected number of changes per unit time at stationarity is $\pi_0 q_{01} + \pi_1 q_{10} = \pi_0 \beta \pi_1 + \pi_1 \beta \pi_0 = 2\beta\pi_0\pi_1$, so measuring branch lengths in expected changes per character requires $2\beta\pi_0\pi_1 = 1$, the scaling used by `binary_P()`. The same argument applies to any chain of the form $\beta(\mathbf{1}\pi - \mathbf{I})$, including the K-state Mk model (Section C.1.8). The code below checks the closed form against a numerical matrix exponential computed by eigendecomposition; it is an implementation-validation check in the sense of Section 6.10.

```
pi1 <- 0.3
beta <- 1 / (2 * (1 - pi1) * pi1)
Q <- beta * matrix(c(-pi1, pi1, 1 - pi1, -(1 - pi1)), 2, 2, byrow = TRUE)
eg <- eigen(Q)
P_num <- eg$vectors %*% diag(exp(eg$values * 0.5)) %*% solve(eg$vectors)
all.equal(unname(binary_P(0.5, pi1)), P_num)
```

```
#> [1] TRUE
```

C.1.3 Ascertainment corrections

Suppose characters are generated independently, each with pattern probabilities $P(x \mid \theta)$ over all 2^n patterns of presence and absence among n languages, and that a character enters the dataset only if its pattern lies outside a set $\mathcal{U}$ of unobservable patterns. For an observed character, the probability of its pattern *given that it was recorded* is

$$P(D_i = x \mid D_i \notin \mathcal{U}) = \frac{P(x \mid \theta)}{P(D_i \notin \mathcal{U} \mid \theta)} = \frac{P(x \mid \theta)}{1 - \sum_{u \in \mathcal{U}} P(u \mid \theta)}, \qquad x \notin \mathcal{U},$$

and independence across characters gives Equation (7.4). The conditional likelihood is the likelihood of the observed patterns *given* the number N of observed characters. The full likelihood would also include the probability of observing exactly N characters out of an unknown total N^*, which depends on N^*; since N^*, the number of cognate classes that ever existed in the meaning slots, is unknowable, conditioning on N discards only information that could not be used anyway.

The denominators for the three filtering schemes of Chapter 7 are, writing $\mathbf{0}$ and $\mathbf{1}$ for the all-absent and all-present patterns and $\mathbf{e}_j$ for the pattern in which only language j has the cognate,

$$\begin{aligned} \text{no-absence:} \quad & 1 - P(\mathbf{0}), \\ \text{variable-only:} \quad & 1 - P(\mathbf{0}) - P(\mathbf{1}), \\ \text{informative-only:} \quad & 1 - P(\mathbf{0}) - P(\mathbf{1}) - \sum_{j=1}^{n} \left[P(\mathbf{e}_j) + P(\mathbf{1} - \mathbf{e}_j) \right], \end{aligned}$$

where the last line assumes $n \geq 4$ and binary characters, so that a pattern is parsimony-uninformative exactly when fewer than two languages have one of its states. Each probability is computed by the pruning algorithm, so the informative-only correction costs $2n + 2$ extra pattern likelihoods per evaluation.

With missing data the unobservable pattern differs among characters. Let S_i be the set of languages for which the meaning of character i is recorded. Character i would have gone unrecorded if its class were absent from every language in S_i, whatever the state in the other languages, so the no-absence denominator for character i is $1 - P(\mathbf{0}_{S_i})$, the probability of the pattern with 0 in S_i and "?" elsewhere. Characters of the same meaning share S_i and therefore share a denominator, which is why the correction must be computed per meaning (or per distinct pattern of missingness) rather than once for the whole matrix. Ryder and Nicholls (2011) work out the corresponding conditioning for the stochastic Dollo model.

The next chunk implements the pruning likelihood with an optional no-absence correction. The trick is to append one artificial all-zero column to the data, compute its probability along with the real columns, and use it in the denominator. In the vocabulary of Section 6.10, the simulation is a parameter-recovery experiment for the corrected model, and a demonstration of what goes wrong when the correction is omitted, which is a misspecification of the observation process.

```
loglik_binary <- function(tree, X, pi1, rate, correct = TRUE) {
  tree <- reorder(tree, "postorder")
  ntip <- Ntip(tree); nn <- ntip + tree$Nnode
  X <- cbind(X[tree$tip.label, , drop = FALSE], 0L)  # last column: all absent
  L0 <- L1 <- matrix(1, nn, ncol(X))
  L0[1:ntip, ] <- X == 0
  L1[1:ntip, ] <- X == 1
  for (i in seq_len(nrow(tree$edge))) {
    p <- tree$edge[i, 1]; ch <- tree$edge[i, 2]
    P <- binary_P(tree$edge.length[i], pi1, rate)
    L0[p, ] <- L0[p, ] * (P[1, 1] * L0[ch, ] + P[1, 2] * L1[ch, ])
    L1[p, ] <- L1[p, ] * (P[2, 1] * L0[ch, ] + P[2, 2] * L1[ch, ])
  }
  site <- (1 - pi1) * L0[ntip + 1, ] + pi1 * L1[ntip + 1, ]
  n <- ncol(X) - 1
  ll <- sum(log(site[1:n]))
  if (correct) ll <- ll - n * log(1 - site[n + 1])
  ll
}

fit_binary <- function(tree, X, correct) {
```

```
  nll <- function(p) -loglik_binary(tree, X, plogis(p[1]), exp(p[2]), correct)
  o <- optim(c(0, 0), nll)
  c(pi1 = plogis(o$par[1]), rate = exp(o$par[2]), logLik = -o$value)
}

set.seed(7)
tr <- rcoal(12, tip.label = paste0("L", 1:12))
tr$edge.length <- tr$edge.length / max(node.depth.edgelength(tr))  # root age 1
X_all <- sim_ctmc_binary(tr, 3000, pi1 = 0.25, rate = 1)
X_obs <- X_all[, colSums(X_all) > 0]  # all-absent columns are never observed
c(simulated = ncol(X_all), observed = ncol(X_obs))
```

```
#> simulated  observed
#>      3000      2459
```

```
est <- rbind(truth = c(pi1 = 0.25, rate = 1),
             uncorrected = fit_binary(tr, X_obs, FALSE)[1:2],
             corrected = fit_binary(tr, X_obs, TRUE)[1:2])
knitr::kable(est, digits = 3, caption = "Estimates of the stationary frequency of
presence and of the rate of change from simulated cognate data from which all-absent
characters were removed, with and without the ascertainment correction (true tree
fixed).")
```

Table C.1: Estimates of the stationary frequency of presence and of the rate of change from simulated cognate data from which all-absent characters were removed, with and without the ascertainment correction (true tree fixed).

	pi1	rate
truth	0.250	1.000
uncorrected	0.308	1.362
corrected	0.256	1.041

The uncorrected fit overestimates the frequency of presence, because it must explain why no all-absent column turned up, and it overestimates the rate as well. The corrected fit recovers both parameters closely. In a dated analysis the rate error translates directly into an error in branch lengths, and therefore in dates. How large the error is depends on the tree and on π_1, but it is not negligible for realistic cognate data, in which the proportion of cognate classes that have vanished without trace is high. The tree here is fixed at the truth, so the experiment shows that the correction is right when everything else is; it says nothing about how well π_1 and the rate can be estimated when the tree and dates are also unknown.

C.1.4 How many gamma categories?

The discrete gamma replaces the integral over a character's unknown rate by an average over K equiprobable categories, each represented by its mean rate (Section A.4.3). Whether $K = 4$ is enough is a numerical question that can be answered directly. The chunk below simulates 800 binary characters on a random tree of twelve languages with rates drawn from the *continuous*

gamma distribution with shape 0.4, the kind of strong heterogeneity found in basic vocabulary, and then fits the symmetric binary model by maximum likelihood on the true topology with $K = 1$ (no rate variation) and $K = 2$ to 32 categories, estimating branch lengths and the shape each time.

```
library(phangorn)
set.seed(17)
tr_k <- rtree(12, rooted = FALSE)
tr_k$edge.length <- tr_k$edge.length * 0.5
r_k <- rgamma(800, shape = 0.4, rate = 0.4)        # continuous gamma rates
X_k <- do.call(cbind, lapply(r_k, function(r)
  as.character(simSeq(tr_k, l = 1, type = "USER", levels = c("0", "1"),
                      rate = r))))
pd_k <- phyDat(X_k, type = "USER", levels = c("0", "1"))
ctl <- pml.control(trace = 0)
fit_k <- function(K) {
  f <- if (K == 1) optim.pml(pml(tr_k, pd_k), optEdge = TRUE, control = ctl)
       else optim.pml(pml(tr_k, pd_k, k = K, shape = 1), optEdge = TRUE,
                      optGamma = TRUE, control = ctl)
  c(K = K, logLik = f$logLik, shape = if (K == 1) NA else f$shape,
    tree_length = sum(f$tree$edge.length))
}
tab_k <- as.data.frame(t(sapply(c(1, 2, 4, 8, 16, 32), fit_k)))
knitr::kable(tab_k, digits = 3, caption = paste0(
  "Maximum-likelihood fits with K discrete gamma categories to 800 ",
  "characters simulated with continuous gamma rates (shape 0.4); true ",
  "tree length ", round(sum(tr_k$edge.length), 2), "."))
```

Table C.2: Maximum-likelihood fits with K discrete gamma categories to 800 characters simulated with continuous gamma rates (shape 0.4); true tree length 6.28.

K	logLik	shape	tree_length
1	-5001	NA	2.328
2	-4694	0.417	3.757
4	-4682	0.363	6.288
8	-4682	0.368	6.724
16	-4682	0.370	6.799
32	-4682	0.371	6.820

Going from no rate variation to two categories gains hundreds of log-likelihood units, two to four categories gains about ten more, and beyond four or eight categories the log-likelihood no longer changes appreciably: the approximation has converged, and the estimated shape settles a little below its true value of 0.4. The tree length is the quantity to watch. The single-rate model underestimates it by more than half, the saturation effect of Section 7.5, and it keeps changing as categories are added, here by 8 percent between 4 and 32 categories, presumably because more categories represent the fast tail of the gamma, whose characters carry much of the information about long branches, differently. For dating, where tree length becomes time, it can therefore be worth checking that a result does not move when K is raised from four to eight, a cheap

check that is rarely reported. As a numerical experiment on data simulated under the model this is implementation-level evidence about the approximation, not evidence that real cognate data are gamma-distributed (Section 6.10); the Bayesian counterpart, with free-weight and FreeRate alternatives, is script `17_rate_categories.Rev` (Section H.3.18).

C.1.5 The covarion model

For binary characters, the covarion model of Tuffley and Steel (1998) has four states: 0^+ and 1^+ for absent and present while on, 0^- and 1^- for absent and present while off. With switching rates s_{off} (from on to off) and s_{on} (from off to on), the rate matrix is

$$\mathbf{Q} = \begin{pmatrix} \cdot & \beta\pi_1 & s_{\text{off}} & 0 \\ \beta\pi_0 & \cdot & 0 & s_{\text{off}} \\ s_{\text{on}} & 0 & \cdot & 0 \\ 0 & s_{\text{on}} & 0 & \cdot \end{pmatrix}, \tag{C.1}$$

with rows and columns ordered $(0^+, 1^+, 0^-, 1^-)$ and diagonal entries chosen so that each row sums to zero. Changes of the visible state occur only in the on states; switching changes only the hidden rate class. The stationary distribution is $(\pi_0 p_{\text{on}}, \pi_1 p_{\text{on}}, \pi_0 p_{\text{off}}, \pi_1 p_{\text{off}})$ with $p_{\text{on}} = s_{\text{on}}/(s_{\text{on}} + s_{\text{off}})$, as one can check by verifying $\pi\mathbf{Q} = \mathbf{0}$ column by column. The mean rate of visible change at stationarity is $2\beta\pi_0\pi_1 p_{\text{on}}$, so the scaling that makes one unit of branch length equal one expected visible change is $\beta = 1/(2\pi_0\pi_1 p_{\text{on}})$, as in `covarion_Q()` below. The data record only the visible state, so at each leaf the likelihood for an observed 1 is the sum over 1^+ and 1^-, and similarly for 0; pruning then proceeds exactly as before with a four-by-four $\mathbf{P}(t)$. Tuffley and Steel (1998) discuss which of the covarion parameters are identifiable; in practice the switching rates are weakly determined by data from few taxa.

Figure 7.3 compares the expected proportion of differing characters between two languages separated for time t. The two languages are joined by a path of length $2t$, and for a stationary reversible chain the joint distribution of their states is $\mathrm{diag}(\pi)\,\mathbf{P}(2t)$. For the single-rate binary chain this gives $d(t) = 2\pi_0\pi_1\big(1 - e^{-2\beta t}\big)$. Under gamma-distributed rates with shape α and mean one, $d(t) = 2\pi_0\pi_1\big(1 - \mathrm{E}[e^{-2\beta\rho t}]\big)$, and the Laplace transform of the gamma distribution, $\mathrm{E}[e^{-s\rho}] = (1 + s/\alpha)^{-\alpha}$, gives $d(t) = 2\pi_0\pi_1\big[1 - (1 + 2\beta t/\alpha)^{-\alpha}\big]$. For the covarion model the joint distribution is computed numerically from the four-state matrix. The chapter's code for the figure is:

```
covarion_Q <- function(pi1, s_off, s_on, rate = 1) {
  pi0 <- 1 - pi1
  p_on <- s_on / (s_on + s_off)
  beta <- rate / (2 * pi0 * pi1 * p_on)   # mean rate of visible change = rate
  st <- c("0on", "1on", "0off", "1off")
  Q <- matrix(0, 4, 4, dimnames = list(st, st))
  Q["0on", "1on"] <- beta * pi1;  Q["1on", "0on"] <- beta * pi0
  Q["0on", "0off"] <- Q["1on", "1off"] <- s_off
  Q["0off", "0on"] <- Q["1off", "1on"] <- s_on
  diag(Q) <- -rowSums(Q)
  Q
```

```
}
expQ <- function(Q, t) {
  e <- eigen(Q)
  Re(e$vectors %*% diag(exp(e$values * t)) %*% solve(e$vectors))
}
Qc <- covarion_Q(pi1 = 0.5, s_off = 0.5, s_on = 0.5)
pi_c <- rep(0.25, 4)
p_diff_cov <- function(t) {
  J <- diag(pi_c) %*% expQ(Qc, 2 * t)   # joint distribution at the two tips
  vis <- c(0, 1, 0, 1)                  # visible state of each hidden state
  sum(J[outer(vis, vis, "!=")])
}
ts <- seq(0, 3, length.out = 121)
b <- 1 / (2 * 0.25)                     # beta for pi1 = 0.5, rate 1
d_single <- 0.5 * (1 - exp(-b * 2 * ts))
d_gamma  <- 0.5 * (1 - (1 + b * 2 * ts / 0.5)^(-0.5))
d_cov    <- sapply(ts, p_diff_cov)
matplot(ts, cbind(d_single, d_gamma, d_cov), type = "l", lty = 1, lwd = 2,
        col = c("#555555", "#1b6ca8", "#d1495b"), las = 1,
        xlab = "time since separation", ylab = "expected proportion differing")
legend("bottomright", c("single rate", "gamma (shape 0.5)", "covarion"),
       col = c("#555555", "#1b6ca8", "#d1495b"), lwd = 2, bty = "n")
```

The following check runs that code without drawing and confirms that the stated stationary distribution satisfies $\pi\mathbf{Q} = \mathbf{0}$ and that the mean rate of visible change is one (implementation validation).

```
cov <- new.env()
pdf(NULL)                               # run the figure code without drawing
eval(parse(text = chapter_code("07-lexical-models.Rmd",
                               "lexical-models-covfig")), envir = cov)
invisible(dev.off())
round(drop(cov$pi_c %*% cov$Qc), 12)    # stationarity: all zero
```

```
#>  0on  1on 0off 1off
#>    0    0    0    0
```

```
sum(cov$pi_c[1:2] * c(cov$Qc["0on", "1on"], cov$Qc["1on", "0on"]))
```

```
#> [1] 1
```

C.1.6 The stochastic Dollo likelihood

The likelihood of Nicholls and Gray (2008) is derived by thinking about where each trait was born. Number the nodes of the tree v, let t_v be the length of the branch above node v, and define for each node the weight

$$w_v = 1 - e^{-\mu t_v} \quad (v \neq \text{root}), \qquad w_{\text{root}} = 1. \tag{C.2}$$

The expected number of traits born on the branch above v that are still alive at v is $\int_0^{t_v} \lambda e^{-\mu(t_v - s)} ds = (\lambda/\mu) w_v$, and the expected number present at the root is $\lambda/\mu = (\lambda/\mu) w_{\text{root}}$. Now let x be a pattern of presences and absences across the leaves, and let $S_v(x)$ be the probability that a trait present at node v produces exactly the pattern x among the leaves descended from v. These probabilities are computed by a pruning recursion that has only deaths in it:

$$S_v(x) = \prod_{c \in \mathrm{ch}(v)} \left[e^{-\mu t_c} S_c(x) + \left(1 - e^{-\mu t_c}\right) \mathbb{1}\{x \text{ is all-absent below } c\} \right], \tag{C.3}$$

with $S_v(x) = x_v$ at a leaf. Either the trait survives along the branch to child c and must then produce the observed pattern below c, or it dies on that branch, which is consistent with the data only if no language below c has it. Finally, let $C_v(x) = 1$ if all the languages that have the trait in pattern x descend from v, and 0 otherwise. Because births form a Poisson process, the number of traits showing pattern x is Poisson with mean

$$\Lambda_x = \frac{\lambda}{\mu} \sum_v w_v \, C_v(x) \, S_v(x), \tag{C.4}$$

and the counts n_x of different patterns are independent. The likelihood of the data is therefore

$$L(\tau, \mathbf{t}, \lambda, \mu) = \prod_x \frac{\Lambda_x^{n_x} e^{-\Lambda_x}}{n_x!} = e^{-\Lambda} \prod_x \frac{\Lambda_x^{n_x}}{n_x!}, \tag{C.5}$$

where the product runs over observable (not all-absent) patterns and $\Lambda = \sum_x \Lambda_x$ is the expected total number of observable traits. Λ itself has a simple expression, $\Lambda = (\lambda/\mu) \sum_v w_v (1 - G_v)$, where G_v, the probability that a trait present at v dies out in every descendant, follows the recursion $G_v = \prod_c \left[e^{-\mu t_c} G_c + 1 - e^{-\mu t_c} \right]$ with $G = 0$ at leaves.

Because λ enters only as a multiplier of every Λ_x, its maximum-likelihood value given everything else is $\hat{\lambda} = N/(\Lambda/\lambda)$, where N is the number of observed traits, and conditional on N the pattern counts are multinomial with probabilities Λ_x/Λ, which do not depend on λ. Multiplying all branch lengths by a constant c and dividing μ by c leaves every w_v, S_v and G_v unchanged, and Λ_x then changes only through the factor λ/μ, which λ can absorb; this is the formal statement that only μt is identifiable without calibrations.

The following code implements Equations (C.2) to (C.5). For each character (column) it computes $S_v(x)$ by the recursion of Equation (C.3), counts how many presences fall below each node to evaluate $C_v(x)$, and sums the contributions. It returns $a_x = \Lambda_x/\lambda$ for every column and $A = \Lambda/\lambda$, from which both the full and the conditional log-likelihood follow. The simulation that follows is a parameter-recovery experiment with the tree fixed at the truth.

```
dollo_rates <- function(tree, X, mu) {
  # returns a = Lambda_x / lambda for each column of X, and A = Lambda / lambda
  tree <- reorder(tree, "postorder")
  ntip <- Ntip(tree); nn <- ntip + tree$Nnode; root <- ntip + 1
  X <- X[tree$tip.label, , drop = FALSE]
  S <- matrix(1, nn, ncol(X)); S[1:ntip, ] <- X       # S_v(x)
```

```
  cnt <- matrix(0, nn, ncol(X)); cnt[1:ntip, ] <- X   # presences below v
  G <- rep(1, nn); G[1:ntip] <- 0                     # P(dies out below v)
  w <- rep(1, nn)                                     # w_root = 1
  for (i in seq_len(nrow(tree$edge))) {
    p <- tree$edge[i, 1]; ch <- tree$edge[i, 2]
    s <- exp(-mu * tree$edge.length[i])               # survival on branch
    S[p, ] <- S[p, ] * (s * S[ch, ] + (1 - s) * (cnt[ch, ] == 0))
    cnt[p, ] <- cnt[p, ] + cnt[ch, ]
    G[p] <- G[p] * (s * G[ch] + 1 - s)
    w[ch] <- 1 - s
  }
  C <- sweep(cnt, 2, colSums(X), "==")                # all presences below v
  list(a = colSums(w * S * C) / mu, A = sum(w * (1 - G)) / mu)
}

dollo_loglik <- function(tree, X, lambda, mu) {
  # full log-likelihood (Equation dollo-lik), dropping the constant n_x! terms
  r <- dollo_rates(tree, X, mu)
  sum(log(lambda * r$a)) - lambda * r$A
}

set.seed(7)
Xd <- sim_dollo(tr, lambda = 100, mu = 0.8)   # same 12-language tree as above
dim(Xd)
```

```
#> [1]  12 430
```

```
mus <- seq(0.3, 2, by = 0.02)
prof <- sapply(mus, function(m) {
  lam_hat <- ncol(Xd) / dollo_rates(tr, Xd, m)$A    # profile out lambda
  dollo_loglik(tr, Xd, lam_hat, m)
})
mu_hat <- mus[which.max(prof)]
c(mu_hat = mu_hat, lambda_hat = ncol(Xd) / dollo_rates(tr, Xd, mu_hat)$A)
```

```
#>     mu_hat lambda_hat
#>       0.84     104.22
```

```
plot(mus, prof, type = "l", lwd = 2, col = "#1b6ca8", las = 1,
     xlab = expression("death rate " * mu), ylab = "profile log-likelihood")
abline(v = 0.8, lty = 2)
abline(h = max(prof) - 1.92, lty = 3)
```

The profile likelihood (Figure C.1) is sharply peaked near the true death rate, and the estimated birth rate is close to the true value of 100. With a fixed tree, a few hundred cognate classes carry plenty of information about μ. In real analyses the tree is unknown and the likelihood is embedded in an MCMC over trees, dates and rate parameters (Chapter 6), and μ is then estimable only jointly with the time scale set by the calibrations.

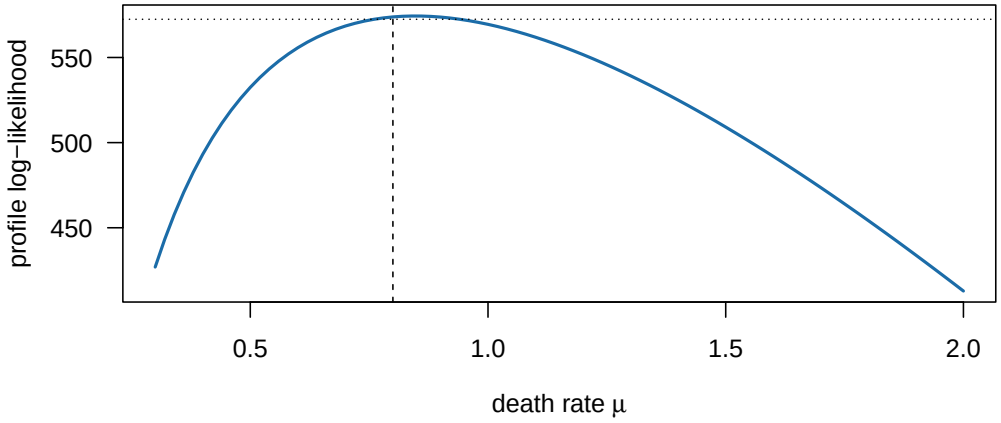


Figure C.1: Profile log-likelihood of the death rate mu under the stochastic Dollo model for data simulated on a known 12-language tree with mu = 0.8 and lambda = 100 (dashed line: true value; dotted line: approximate 95 percent likelihood interval, 1.92 log-units below the maximum).

C.1.7 The stochastic Dollo model as a limit of the reversible model

Write the reversible binary CTMC in terms of its gain rate $\beta\pi_1$ and loss rate $\beta\pi_0$, and let $\pi_1 \to 0$ while keeping the loss rate fixed at μ, so that $\beta \to \mu$. To first order in π_1, an observable pattern x arises in one of two ways. Either the root is in state 1, with probability π_1, and the trait then only dies (further gains are of order π_1), producing x with probability $C_{\text{root}}(x)S_{\text{root}}(x)$; or the root is in state 0 and a single gain occurs on the branch above some node v at time s and survives to v, with probability $\int_0^{t_v} \mu\pi_1 e^{-\mu(t_v-s)}\,ds = \pi_1(1-e^{-\mu t_v}) = \pi_1 w_v$, after which the trait produces x with probability $C_v(x)S_v(x)$. Summing,

$$P(x) \approx \pi_1 \sum_v w_v\, C_v(x)\, S_v(x) = \pi_1 \frac{\mu}{\lambda}\Lambda_x, \qquad 1 - P(\mathbf{0}) \approx \pi_1 \sum_v w_v(1-G_v) = \pi_1 \frac{\mu}{\lambda}\Lambda,$$

so the corrected pattern probabilities $P(x)/(1-P(\mathbf{0}))$ converge to Λ_x/Λ, the conditional pattern probabilities of the stochastic Dollo model. The stochastic Dollo model is thus a boundary case of the ascertainment-corrected reversible model.

We can check this numerically, and at the same time carry out a small model comparison. Both models define a probability distribution over observable patterns, so their conditional log-likelihoods (the multinomial part, conditioning on the number of characters) can be compared directly. The reversible model has two free parameters here (π_1 and the rate), the conditional Dollo model one (μ). In the vocabulary of Section 6.10, the two matched fits are parameter-recovery checks and the two mismatched fits are tests of robustness under misspecification.

```
set.seed(8)
Xr <- sim_ctmc_binary(tr, 3000, pi1 = 0.25, rate = 1)   # reversible data
Xr <- Xr[, colSums(Xr) > 0]
```

```
cond_dollo <- function(tree, X) {
  f <- function(m) { r <- dollo_rates(tree, X, m); sum(log(r$a / r$A)) }
  o <- optimize(f, c(0.01, 10), maximum = TRUE)
  c(logLik = o$objective, k = 1)
}
cond_ctmc <- function(tree, X) {
  c(logLik = fit_binary(tree, X, TRUE)[["logLik"]], k = 2)
}
fits <- list(
  "Dollo data, Dollo model"           = cond_dollo(tr, Xd),
  "Dollo data, reversible model"      = cond_ctmc(tr, Xd),
  "reversible data, Dollo model"      = cond_dollo(tr, Xr),
  "reversible data, reversible model" = cond_ctmc(tr, Xr))
cmp <- do.call(rbind, fits)
cmp <- cbind(cmp, AIC = -2 * cmp[, "logLik"] + 2 * cmp[, "k"])
knitr::kable(cmp, digits = 1, caption = "Conditional log-likelihoods and AIC for the
stochastic Dollo and the ascertainment-corrected reversible binary model, each
fitted to data simulated under each model on a known tree.")
```

Table C.3: Conditional log-likelihoods and AIC for the stochastic Dollo and the ascertainment-corrected reversible binary model, each fitted to data simulated under each model on a known tree.

	logLik	k	AIC
Dollo data, Dollo model	-1603	1	3208
Dollo data, reversible model	-1602	2	3209
reversible data, Dollo model	-15294	1	30591
reversible data, reversible model	-12612	2	25228

```
fit_binary(tr, Xd, TRUE)   # reversible fit to Dollo data: pi1 driven toward 0
```

```
#>        pi1       rate     logLik
#>  1.118e-02  1.898e-02 -1.602e+03
```

On the Dollo data, the reversible model reaches essentially the same likelihood as the Dollo model by pushing π_1 toward zero, exactly as the limit argument predicts, and the loss rate implied by its estimates, $\beta\pi_0 \approx \text{rate}/(2\pi_1)$, is close to the Dollo estimate of μ. AIC then slightly prefers the Dollo model because it achieves the same fit with one fewer parameter. On the reversible data, the Dollo model is far worse: it cannot explain characters that need repeated gains except by many parallel losses of a trait born deep in the tree. Real data, of course, are generated by neither model, and the question for practice is which model's failures matter less for the quantity of interest.

C.1.8 The Mk model for meaning slots

The Mk model of Lewis (2001) for a character with K states has equal rates between all pairs of states. Scaled to one expected change per unit time, every off-diagonal rate is $1/(K-1)$,

so that $\mathbf{Q} = \frac{K}{K-1}(\mathbf{1}\pi - \mathbf{I})$ with π uniform. The argument of Section C.1.2 then gives $\mathbf{P}(t) = e^{-Kt/(K-1)}\mathbf{I} + \left(1 - e^{-Kt/(K-1)}\right)\mathbf{1}\pi$, that is,

$$P_{ij}(t) = \begin{cases} \dfrac{1}{K} + \dfrac{K-1}{K}\exp\left(-\dfrac{Kt}{K-1}\right) & i = j, \\ \dfrac{1}{K} - \dfrac{1}{K}\exp\left(-\dfrac{Kt}{K-1}\right) & i \neq j. \end{cases} \tag{C.6}$$

For meaning slots, two features of this formula matter. The probability of "no change" depends on K, so that the same replacement rate implies different amounts of observable change for meanings with different numbers of classes; and the stationary probability $1/K$ of every class, including classes found only in distant relatives, is what licenses the implausible direct switches discussed in Chapter 7. If meanings in which every language has the same class have been dropped from the data, the variable-only correction applies, with the K constant patterns in $\mathcal{U}$.

C.1.9 Binary coding and within-meaning dependence

Consider a meaning with K cognate classes in the sample and binary columns $x_1, \ldots, x_K$. If each language has exactly one word for the meaning, the joint state of the K columns in a language is one of the K unit vectors $\mathbf{e}_1, \ldots, \mathbf{e}_K$, although K binary variables can take 2^K values. If synonyms are allowed, the possible joint states are the $2^K - 1$ non-empty subsets of classes. A meaning-slot model is a Markov chain on one of these constrained state spaces, which is exactly the structured-character construction of Tarasov (2019) applied to cognate classes.

An independent binary model ignores the constraint. At stationarity it assigns each language a joint state with probability $\prod_k \pi_1^{x_k}(1-\pi_1)^{1-x_k}$, so that the probability of the configurations that actually occur in synonym-free data, exactly one 1, is

$$P(\text{exactly one class present}) = K\pi_1(1-\pi_1)^{K-1},$$

while the probability that the language has no word for the meaning at all is $(1-\pi_1)^K$. The function below evaluates both for a few values of K, with $\pi_1 = 1/K$ (the value that makes the expected number of words per meaning equal to one).

```
p_exactly_one <- function(K, pi1) K * pi1 * (1 - pi1)^(K - 1)
K <- c(2, 4, 6, 8, 12)
round(data.frame(K = K, exactly_one = p_exactly_one(K, 1 / K),
                 none = (1 - 1 / K)^K), 3)
```

```
#>    K exactly_one  none
#> 1  2       0.500 0.250
#> 2  4       0.422 0.316
#> 3  6       0.402 0.335
#> 4  8       0.393 0.344
#> 5 12       0.384 0.352
```

With six classes, as for 'hand' in Table 7.1, the independent model gives only about 40 percent of its probability to configurations with exactly one word and about a third to languages with no word

at all; as K grows the two proportions approach $e^{-1} \approx 0.37$. The model is therefore spending most of its probability on data that cannot occur, and it explains the data that do occur as coincidences among independent columns. A replacement, which is a single event in a meaning-slot model, requires under the independent model a loss in one column and a gain in another, two events whose rates are estimated separately; the probability of both happening in the same short interval is of order dt^2, so the independent model can produce a replacement only by passing through a transient state with zero or two words. This is the sense in which the binary likelihood of Equation (7.3) is a composite likelihood: it multiplies terms that are not independent, and so counts the evidence of each replacement roughly twice and the evidence of each fast meaning many times.

C.2 Chapter 8: Borrowing, Contact and Reticulation

This section lists the code behind the simulations of Chapter 8, derives the quartet statistics used there, checks them against **phangorn**, and explains why lateral transfer makes likelihood computation hard.

C.2.1 Simulating lexical evolution with borrowing

`sim_borrowing()` implements the meaning-slot model of Section 8.3 by stepping through time in increments of length `dt`. At each step, every lineage alive at that moment replaces the word for each meaning by a brand-new cognate class with probability `innov * dt`, which approximates a Poisson process of innovations with rate r_{innov} per meaning; the approximation error is of order `dt`. Borrowing events are drawn for each recipient lineage as a Poisson number with mean `borrow * n_meanings * dt`; each event copies the donor's word for a randomly chosen meaning. For local borrowing, the donor is chosen with weight $e^{-d/s}$, where d is the patristic distance between recipient and donor at time t: two lineages alive at time t whose most recent common ancestor lived at time t_a are separated by $d = 2(t - t_a)$, which is what the weight `exp(-2 * (t - depth[anc]) / local_scale)` computes. When a branch begins, it inherits the words of its parent branch, so that daughters start identical. `meaning_dist()` returns the proportion of meanings for which two languages have words from different cognate classes, and `to_binary()` converts the meaning-slot matrix into binary presence/absence characters.

```
library(phangorn)
family <- read.tree(text = paste0(
  "((((A1:0.2,A2:0.2):0.2,(A3:0.25,A4:0.25):0.15):0.35,",
  "((B1:0.2,B2:0.2):0.25,(B3:0.3,B4:0.3):0.15):0.3):0.25,",
  "(((C1:0.15,C2:0.15):0.3,C3:0.45):0.25,(C4:0.4,C5:0.4):0.3):0.3);"))
```

```
sim_borrowing <- function(tree, n_meanings = 200, innov = 0.5, borrow = 0,
                          mode = c("global", "local", "pair"), pair = NULL,
                          window = c(0, Inf), local_scale = 0.2, dt = 0.01) {
  mode <- match.arg(mode)
  ntip <- Ntip(tree)
  depth <- node.depth.edgelength(tree)       # time of each node from the root
  E <- tree$edge
  start <- depth[E[, 1]]; end <- depth[E[, 2]]
  anc <- mrca(tree, full = TRUE)             # MRCA of every pair of nodes
  if (mode == "pair") {                      # lineages leading to two tips
```

```
  lineage <- function(tip) {
    k <- match(tip, tree$tip.label)
    which(E[, 2] %in% c(k, Ancestors(tree, k, "all")))
  }
  rec_edges <- lineage(pair[1]); don_edges <- lineage(pair[2])
}
state <- matrix(NA_integer_, nrow(E), n_meanings)   # words on each edge
opened <- rep(FALSE, nrow(E))
open_edges <- function(upto) {             # daughters inherit parent's words
  for (e in which(!opened & start <= upto + 1e-9)) {
    pe <- which(E[, 2] == E[e, 1])
    state[e, ] <<- if (length(pe)) state[pe, ] else seq_len(n_meanings)
    opened[e] <<- TRUE
  }
}
next_id <- n_meanings
for (t in seq(0, max(end) - dt / 2, by = dt)) {
  open_edges(t)
  active <- which(start <= t + 1e-9 & end > t + 1e-9)
  na <- length(active)
  # innovation: replace a word by a brand-new cognate class
  hit <- matrix(runif(na * n_meanings) < innov * dt, na)
  if (any(hit)) {
    sub <- state[active, , drop = FALSE]
    sub[hit] <- next_id + seq_len(sum(hit)); next_id <- next_id + sum(hit)
    state[active, ] <- sub
  }
  # borrowing: replace a word by the donor's word for the same meaning
  if (borrow > 0 && na > 1 && t >= window[1] && t < window[2]) {
    recips <- if (mode == "pair") intersect(active, rec_edges) else active
    for (rec in recips) {
      donors <- if (mode == "pair") intersect(active, don_edges) else
        setdiff(active, rec)
      nb <- rpois(1, borrow * n_meanings * dt)
      if (nb == 0 || length(donors) == 0) next
      w <- if (mode == "local") {
        exp(-2 * (t - depth[anc[E[rec, 2], E[donors, 2]]]) / local_scale)
      } else rep(1, length(donors))
      don <- donors[sample.int(length(donors), nb, replace = TRUE, prob = w)]
      m <- sample.int(n_meanings, nb, replace = TRUE)
      state[cbind(rec, m)] <- state[cbind(don, m)]
    }
  }
}
open_edges(Inf)
out <- state[match(seq_len(ntip), E[, 2]), ]
dimnames(out) <- list(tree$tip.label, paste0("m", seq_len(n_meanings)))
out
```

```
}

meaning_dist <- function(M) {
  # proportion of meanings for which two languages have non-cognate words
  n <- nrow(M)
  d <- matrix(0, n, n, dimnames = list(rownames(M), rownames(M)))
  for (i in 1:n) for (j in 1:n) d[i, j] <- mean(M[i, ] != M[j, ])
  as.dist(d)
}

to_binary <- function(M) {
  # binary presence/absence coding, one character per cognate class
  do.call(cbind, lapply(seq_len(ncol(M)), function(j) {
    cl <- unique(M[, j]); 1 * outer(M[, j], cl, "==")
  }))
}
```

The glottochronological estimate of the root age used in the experiment rests on the model without borrowing. Under that model, two languages whose lineages separated T units of time ago have cognate words for a meaning only if neither lineage has replaced the word since the split. Replacements occur at rate r on each of the two lineages, so the probability of no replacement on either is e^{-2rT}, and the expected proportion of non-cognate meanings is $d = 1 - e^{-2rT}$. Solving for T gives $\hat{T} = -\log(1-d)/(2r)$. The chapter's code computes $2\hat{T} = -\log(1-d)/r$ for every pair and builds a UPGMA tree from these values; because UPGMA places each node at half the distance between the groups it joins, the root height of that tree is an estimate of T for the deepest split. Borrowing makes d smaller than $1 - e^{-2rT}$, because shared words now arise by copying as well as by inheritance, and so $\hat{T}$ is biased downward.

C.2.2 Quartet statistics

For four languages w, x, y, z on a tree in which the internal edge, of length e, separates $\{w, x\}$ from $\{y, z\}$, the path from w to x and the path from y to z do not use the internal edge, whereas each of the other two pairings crosses it twice in total. With pendant edge lengths a_w, a_x, a_y, a_z,

$$s_1 = a_w + a_x + a_y + a_z, \qquad s_2 = s_3 = a_w + a_x + a_y + a_z + 2e,$$

in the notation of Equation (8.1). This is the four-point condition: the two largest sums are equal, and they exceed the smallest by twice the internal edge. Data that do not fit a tree exactly make the two largest sums differ, and the delta score (Equation (8.2)) measures the difference relative to the spread of the three sums, the Q-residual (Equation (8.3)) in absolute terms. The function `quartet_stats()` computes both for all $\binom{n}{4}$ quartets after rescaling the distances to mean one, and averages the delta scores over the quartets that contain each language to obtain taxon-specific scores.

```
quartet_stats <- function(d) {
  # delta score and Q-residual for every quartet of a distance matrix
  D <- as.matrix(d) / mean(d)                # rescale to mean distance 1
  q <- combn(nrow(D), 4)
```

```
  s <- cbind(D[cbind(q[1, ], q[2, ])] + D[cbind(q[3, ], q[4, ])],
             D[cbind(q[1, ], q[3, ])] + D[cbind(q[2, ], q[4, ])],
             D[cbind(q[1, ], q[4, ])] + D[cbind(q[2, ], q[3, ])])
  s <- t(apply(s, 1, sort))                  # s[, 1] <= s[, 2] <= s[, 3]
  delta <- ifelse(s[, 3] > s[, 1], (s[, 3] - s[, 2]) / (s[, 3] - s[, 1]), 0)
  qres <- (s[, 3] - s[, 2])^2
  taxon_delta <- sapply(seq_len(nrow(D)),
                        function(i) mean(delta[colSums(q == i) > 0]))
  names(taxon_delta) <- rownames(D)
  list(delta = mean(delta), qres = mean(qres), taxon_delta = taxon_delta,
       quartet_delta = delta)
}
```

The check below recreates the contact dataset of Section 8.4 and compares the overall delta score from `quartet_stats()` with that from `phangorn::delta.score()` on the binary coding. Because phangorn computes Hamming distances on the binary characters, and these are proportional to the meaning distances when each language has one word per meaning (each non-cognate meaning contributes exactly two differing binary columns), and because the delta score does not depend on the scale of the distances, the two should agree. This is an implementation-validation check in the sense of Section 6.10.

```
set.seed(5)
M_contact <- sim_borrowing(family, innov = 0.5, borrow = 1, mode = "pair",
                           pair = c("B4", "C5"), window = c(0.6, 1))
d_contact <- meaning_dist(M_contact)
qs_contact <- quartet_stats(d_contact)
pd_contact <- phyDat(to_binary(M_contact), type = "USER", levels = c(0, 1))
c(ours = qs_contact$delta, phangorn = delta.score(pd_contact))
```

```
#>     ours phangorn
#>   0.1439   0.1439
```

C.2.3 Listing conflicting splits

The helper used for Table 8.4 extracts the splits of a NeighborNet with `as.splits()`, labels each split canonically by the side that does not contain A1 (so that the same split from the network and from the true tree receives the same label), discards splits that occur in the true tree and trivial splits that separate a single language, and returns the heaviest remaining splits, each labeled by its smaller side.

```
conflicting_splits <- function(nnet, tree, top = 3) {
  sp <- as.splits(nnet)
  true_sp <- as.splits(tree)
  lab <- attr(sp, "labels")
  key <- function(s) {                    # canonical text label for a split
    side <- lab[s]
    if ("A1" %in% side) side <- setdiff(lab, side)
    paste(sort(side), collapse = ",")
  }
```

```
  k_true <- sapply(true_sp,
                   function(s) key(match(attr(true_sp, "labels")[s], lab)))
  k_net <- sapply(sp, key)
  w <- attr(sp, "weights")
  small <- sapply(sp, function(s) {
    side <- lab[s]
    if (length(side) > length(lab) / 2) side <- setdiff(lab, side)
    paste(sort(side), collapse = ",")
  })
  keep <- !(k_net %in% k_true) & lengths(sp) > 1 & lengths(sp) < length(lab) - 1
  out <- data.frame(split = small[keep], weight = w[keep])
  head(out[order(-out$weight), ], top)
}
```

C.2.4 Why lateral transfer breaks the pruning recursion

Felsenstein's pruning algorithm (Section 5.4), and the death-only recursion of the stochastic Dollo likelihood (Section C.1.6), rely on one property: once two lineages have split, what happens to a character on one of them is independent of what happens on the other. The likelihood can then be computed node by node from the leaves upward, combining independent contributions from the two children. Lateral transfer destroys this independence. Whether a trait reaches a lineage now depends on whether it is present in other lineages alive at the same time, so the state that must be tracked is not the presence or absence of the trait on one branch but its presence or absence on *every* branch alive at time t. With L contemporaneous lineages this joint state takes 2^L values (or $2^L - 1$, excluding extinction of the trait everywhere, which for an observed trait cannot have happened). Between consecutive branching events the number of lineages is fixed, and the probabilities of the joint states evolve according to a system of linear differential equations: deaths remove a lineage from the set of carriers, and transfers add one at a rate proportional to the number of carriers. At each branching event the carrier set is updated by duplicating the parent lineage's entry. Kelly and Nicholls (2017) showed how to organize this computation so that the likelihood can be evaluated exactly, given the tree, for moderate numbers of languages. The cost nevertheless grows quickly with the number of lineages alive at once, which is the main practical limitation of the model, and the reason why approximate methods, such as simulation-based inference (Chapter 16), are attractive for larger families.

C.3 Chapter 9: Clock Models and Rates of Language Change

This section collects the mathematics of the clock models of Chapter 9 and the code of its longer simulations. Each subsection can be run on its own after the setup chunk at the top of this appendix.

C.3.1 Rate variation among meanings and the glottochronological bias

Section 9.2.5 stated that rate variation among meanings biases glottochronological dates downward. Let the replacement rate k of a meaning be gamma distributed with shape α and rate $\alpha/\bar{k}$, so that its mean is $\bar{k}$ and its variance $\bar{k}^2/\alpha$. A meaning with rate k yields a shared cognate after a split t millennia ago with probability e^{-2kt}, since both lineages must retain the ancestral

word. Averaging over meanings uses the moment-generating function of the gamma distribution, $\mathrm{E}[e^{sk}] = (1 - s\bar{k}/\alpha)^{-\alpha}$ for $s < \alpha/\bar{k}$, here with $s = -2t$:

$$C(t) = \mathbb{E}\left[e^{-2kt}\right] = \left(1 + \frac{2\bar{k}t}{\alpha}\right)^{-\alpha}. \tag{C.7}$$

This curve decays more slowly than the exponential $e^{-2\bar{k}t}$ of Equation (9.3), and approaches it as $\alpha \to \infty$. A glottochronologist who knows $\bar{k}$ and applies Equation (9.4) computes $\hat{t} = -\ln C/(2\bar{k})$. Substituting Equation (C.7) gives

$$\hat{t} = \frac{\alpha}{2\bar{k}} \ln\left(1 + \frac{2\bar{k}t}{\alpha}\right). \tag{C.8}$$

Because $\ln(1 + x) < x$ for $x > 0$, $\hat{t} < t$ for every $t > 0$, and for large t the estimate grows only like $\ln t$. Strictly, Equation (C.8) is the estimate obtained from the *expected* proportion of shared cognates; with a finite list the observed proportion scatters around $C(t)$, and because the logarithm is concave the average estimate is slightly lower still. The check below compares Equation (C.7) with a Monte Carlo average over a million gamma-distributed rates, which is a small piece of implementation validation for the formula.

```
set.seed(1)
kbar <- -log(0.86)                      # mean rate per millennium
alpha <- 1
tt <- c(1, 3, 6, 10)                    # true depths (millennia)
k <- rgamma(1e6, shape = alpha, rate = alpha / kbar)
mc <- sapply(tt, function(t) mean(exp(-2 * k * t)))
exact <- (1 + 2 * kbar * tt / alpha)^(-alpha)          # Equation (clocks-gamma)
t_hat <- alpha / (2 * kbar) * log(1 + 2 * kbar * tt / alpha)
round(cbind(t = tt, C_monte_carlo = mc, C_formula = exact,
            t_hat = t_hat), 4)
```

```
#>       t C_monte_carlo C_formula t_hat
#> [1,]  1        0.7681    0.7683 0.874
#> [2,]  3        0.5247    0.5250 2.136
#> [3,]  6        0.3557    0.3559 3.425
#> [4,] 10        0.2488    0.2490 4.609
```

The code for Figure 9.2 is:

```
set.seed(11)
kbar <- -log(0.86)                      # mean replacement rate per millennium
n_mean <- 200
sim_pair <- function(t, alpha) {
  k <- rgamma(n_mean, shape = alpha, rate = alpha / kbar)
  keepA <- runif(n_mean) < exp(-k * t)
  keepB <- runif(n_mean) < exp(-k * t)
  mean(keepA & keepB)                   # proportion of shared cognates
}
true_t <- rep(seq(0.5, 10, by = 0.5), each = 10)
C_sim <- sapply(true_t, sim_pair, alpha = 1)
t_hat <- -log(pmax(C_sim, 1 / n_mean)) / (2 * kbar)
```

```
plot(true_t, t_hat, pch = 16, col = adjustcolor("grey30", 0.4),
     xlim = c(0, 10), ylim = c(0, 10), las = 1,
     xlab = "True time since split (millennia)",
     ylab = "Glottochronological estimate (millennia)")
abline(0, 1, lty = 3)
expected_hat <- function(t, alpha) {
  alpha / (2 * kbar) * log(1 + 2 * kbar * t / alpha)
}
alphas <- c(0.5, 1, 2, 5)
for (j in seq_along(alphas)) curve(expected_hat(x, alphas[j]), add = TRUE,
  lwd = 2, lty = j)
legend("topleft", bty = "n", lwd = 2, lty = seq_along(alphas),
       legend = paste("alpha =", alphas))
```

C.3.2 Relaxed-clock models

Uncorrelated lognormal clock. In the UCLN model of Drummond et al. (2006), the rate of branch j is

$$r_j = \bar{r}\,\epsilon_j, \qquad \ln \epsilon_j \sim \mathcal{N}\left(-\frac{\sigma^2}{2}, \sigma^2\right), \tag{C.9}$$

independently for every branch. If $X \sim \mathcal{N}(m, s^2)$ then $\mathrm{E}[e^X] = e^{m+s^2/2}$ and $\mathrm{Var}(e^X) = (e^{s^2} - 1)e^{2m+s^2}$ (Section A.4). With $m = -\sigma^2/2$ the multipliers therefore have mean one, so that $\bar{r}$ is the mean rate, and their coefficient of variation is $\sqrt{e^{\sigma^2} - 1}$, whatever $\bar{r}$. Some programs parameterize the lognormal by the mean and standard deviation of the rates themselves rather than of their logarithms; reported values of σ must be read with the program's convention in mind. The uncorrelated exponential clock replaces the lognormal by an exponential distribution with mean $\bar{r}$, whose coefficient of variation is one.

Autocorrelated (geometric Brownian motion) clock. In the model of Thorne et al. (1998) and Kishino et al. (2001), the logarithm of the rate performs a Brownian motion along the tree. For a branch of duration t from a parent node with rate r_{pa} to a child node with rate r_{ch},

$$\ln r_{\mathrm{ch}} \sim \mathcal{N}\left(\ln r_{\mathrm{pa}} - \frac{\nu t}{2},\ \nu t\right), \tag{C.10}$$

where ν is the variance of the log rate per unit time. By the same lognormal identity, $\mathrm{E}[r_{\mathrm{ch}} \mid r_{\mathrm{pa}}] = r_{\mathrm{pa}}$: the $-\nu t/2$ term keeps the expected rate constant along a lineage. The rate applied to a branch is usually the average of the rates at its two ends (as in the simulation of Section 9.5.6) or, more exactly, the time average of the rate process along the branch. Rates of sister branches are correlated through their shared parent, and the correlation decays as the branches lengthen. The Cox-Ingersoll-Ross process replaces the Brownian motion by a mean-reverting process, which keeps rates from drifting arbitrarily far from the mean over long times.

Random local clocks. In the model of Drummond and Suchard (2010), each branch j carries an indicator I_j of a rate change at its start and a multiplier m_j. The rate of branch j is the root rate multiplied by the multipliers of all branches with $I = 1$ on the path from the root to j, inclusive. The number of changes $\sum_j I_j$ has a prior, often a Poisson distribution truncated to the number of

branches, and the multipliers a prior centered on one. The strict clock ($\sum_j l_j = 0$) is nested within the model.

The following chunk checks the mean-one parameterization and the coefficient-of-variation formula of the UCLN model, and the mean-preserving property of Equation (C.10), by simulation.

```
set.seed(2)
sigma <- 0.6
eps <- rlnorm(1e6, meanlog = -sigma^2 / 2, sdlog = sigma)
round(c(mean_multiplier = mean(eps), cv_simulated = sd(eps) / mean(eps),
        cv_formula = sqrt(exp(sigma^2) - 1)), 3)
```

```
#> mean_multiplier    cv_simulated      cv_formula
#>           1.000           0.657           0.658
```

```
nu <- 0.15; dur <- 2                    # drift variance, branch duration
log_r <- rnorm(1e6, mean = 0 - nu * dur / 2, sd = sqrt(nu * dur))
round(c(mean_rate = mean(exp(log_r)), var_log_rate = var(log_r),
        nu_t = nu * dur), 3)
```

```
#>    mean_rate var_log_rate         nu_t
#>        0.999        0.300        0.300
```

C.3.3 The clock simulation of Section 9.5.6

The simulation in Section 9.5.6 draws a coalescent time tree of sixteen languages with a root age of six millennia, assigns rates to its branches under a strict clock, an uncorrelated lognormal clock (Equation (C.9)) and an autocorrelated clock (Equation (C.10)), and draws a Poisson number of lexical changes on each branch. The code is:

```
set.seed(9)
ntip <- 16
time_tree <- rcoal(ntip, tip.label = paste0("L", 1:ntip))
time_tree$edge.length <- time_tree$edge.length * 6 /
  max(node.depth.edgelength(time_tree))        # root age = 6 millennia
nb <- nrow(time_tree$edge)

# (1) strict clock
r_strict <- rep(1, nb)
# (2) uncorrelated lognormal, Equation (clocks-ucln), sigma = 0.6
sigma <- 0.6
r_ucln <- rlnorm(nb, meanlog = -sigma^2 / 2, sdlog = sigma)
# (3) autocorrelated geometric Brownian motion, Equation (clocks-gbm)
nu <- 0.15
logr_node <- numeric(ntip + time_tree$Nnode)   # root log-rate = 0
r_auto <- numeric(nb)
for (i in order(node.depth.edgelength(time_tree)[time_tree$edge[, 2]])) {
  pa <- time_tree$edge[i, 1]; ch <- time_tree$edge[i, 2]
  dt <- time_tree$edge.length[i]
  logr_node[ch] <- rnorm(1, logr_node[pa] - nu * dt / 2, sqrt(nu * dt))
```

```
  r_auto[i] <- exp((logr_node[pa] + logr_node[ch]) / 2)
}

changes_per_kyr <- 80
make_phylogram <- function(rates) {
  ph <- time_tree
  ph$edge.length <- rpois(nb, changes_per_kyr * rates *
                            time_tree$edge.length) + 0.5
  ph
}
phylograms <- list(strict = make_phylogram(r_strict),
                   uncorrelated = make_phylogram(r_ucln),
                   autocorrelated = make_phylogram(r_auto))
cv <- sapply(list(strict = r_strict, uncorrelated = r_ucln,
                  autocorrelated = r_auto),
             function(r) round(sd(r) / mean(r), 2))
cv
```

```
#>         strict   uncorrelated autocorrelated
#>           0.00           0.65           0.40
```

The four-panel Figure 9.3 was drawn with:

```
op <- par(mfrow = c(1, 4), mar = c(2.5, 0.5, 2, 0.5), cex = 1)
plot(time_tree, main = "Time tree", cex = 0.8, label.offset = 0.06)
axisPhylo(cex.axis = 0.8)
for (nm in names(phylograms)) {
  plot(phylograms[[nm]], main = nm, cex = 0.8,
       label.offset = 0.01 * max(node.depth.edgelength(phylograms[[nm]])))
  add.scale.bar(cex = 0.8)
}
par(op)
```

C.3.4 Penalized likelihood

Penalized likelihood (Sanderson, 2002) treats the number of changes x_j on branch j of an estimated phylogram as Poisson-distributed with mean $r_j t_j$, and maximizes

$$\ell_{\mathrm{PL}}(\mathbf{r}, \mathbf{t}) = \sum_{j} \left[x_j \ln(r_j t_j) - r_j t_j \right] - \lambda \sum_{(j,k)} (r_j - r_k)^2, \tag{C.11}$$

over the branch rates $\mathbf{r}$ and the node ages $\mathbf{t}$ that are not fixed by calibrations. The second sum runs over pairs of adjacent branches (a branch and its parent branch; the two branches descending from the root are usually treated as a pair or penalized against their mean). The first term is the Poisson log-likelihood up to a constant. The smoothing parameter λ sets how strongly rates of adjacent branches are pulled together. Sanderson's cross-validation chooses λ as follows: for each terminal branch in turn, remove it, estimate rates and ages from the remaining branches, predict the removed branch's length from the rate of its neighbor and its duration, and record the squared prediction error; the chosen λ minimizes the summed error.

In `ape::chronos()` (Paradis, 2013), the same idea is implemented for several rate models (`"correlated"`, `"relaxed"`, `"discrete"` and `"clock"`), and each fit carries the information criterion ΦIC of Paradis as `attr(fit, "PHIIC")$PHIIC`; smaller values are better. The chunk below uses the phylograms simulated in Section C.3.3 to show how ΦIC and the error in the node ages vary with λ for the uncorrelated scenario under the correlated-rates model.

```
true_ages <- branching.times(time_tree)
ph <- phylograms$uncorrelated
cal <- makeChronosCalib(ph, node = "root", age.min = 6, age.max = 6)
lams <- c(0.01, 0.1, 1, 10, 100)
pl_tab <- t(sapply(lams, function(l) {
  ch <- suppressWarnings(chronos(ph, lambda = l, model = "correlated",
                                 calibration = cal, quiet = TRUE))
  c(lambda = l, PhiIC = attr(ch, "PHIIC")$PHIIC,
    rmse = sqrt(mean((branching.times(ch) - true_ages)^2)))
}))
knitr::kable(pl_tab, digits = c(2, 1, 3),
  format.args = list(scientific = FALSE),
  caption = "PhiIC and root-mean-square error of node ages for chronos fits with
  different smoothing parameters (uncorrelated scenario, correlated-rates model,
  root fixed at its true age).")
```

Table C.4: PhiIC and root-mean-square error of node ages for chronos fits with different smoothing parameters (uncorrelated scenario, correlated-rates model, root fixed at its true age).

lambda	PhiIC	rmse
0.01	318.8	0.457
0.10	403.6	0.514
1.00	425.6	0.520
10.00	428.2	0.521
100.00	2620.5	0.251

The criterion and the true error need not agree, because ΦIC measures fit to the branch lengths, while the error in node ages also depends on how rate and time trade off on each branch. In this example they disagree sharply: ΦIC is smallest for $\lambda = 0.01$, whereas the node ages are most accurate for $\lambda = 100$, the strongest smoothing tried. In the vocabulary of Section 6.10, choosing λ by ΦIC is a model-selection step, not a validation of the dates.

C.3.5 The test for punctuational change

The simulation behind Figure 9.5 is listed below. It draws a Yule tree of 60 languages with `phytools::pbtree()`, adds Poisson numbers of gradual changes proportional to branch duration, optionally adds a Poisson burst of changes to every branch, and returns the root-to-tip path lengths.

```
set.seed(21)
yule_tree <- phytools::pbtree(n = 60, scale = 5)    # root age 5 millennia
ne <- nrow(yule_tree$edge)
nodes_on_path <- sapply(1:60, function(i)
```

```
  length(nodepath(yule_tree, from = 61, to = i)) - 1)

path_lengths <- function(burst) {
  # gradual change (40 per millennium) plus a Poisson burst on every branch
  ph <- yule_tree
  ph$edge.length <- rpois(ne, 40 * yule_tree$edge.length) + rpois(ne, burst)
  node.depth.edgelength(ph)[1:60]
}
op <- par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))
for (b in c(0, 8)) {
  pl <- path_lengths(b)
  fit <- lm(pl ~ nodes_on_path)
  plot(nodes_on_path, pl, pch = 16, col = "grey40", las = 1,
       xlab = "Nodes on root-to-tip path", ylab = "Path length")
  title(if (b == 0) "Gradual only" else "Gradual + bursts (8 per split)",
        line = 1.6, cex.main = 0.95)
  abline(fit, lwd = 2)
  mtext(sprintf("slope = %.1f changes per node", coef(fit)[2]),
        cex = 0.8, line = 0.3)
}
```

```
par(op)
```

Ordinary least squares treats the 60 path lengths as independent, but tips that share ancestral branches share the changes on them. Under purely gradual change the covariance of two path lengths is proportional to the time their paths share, which is the Brownian-motion covariance on the time tree, so phylogenetic generalized least squares with `ape::corBrownian()` (Section A.11) is the appropriate regression. The chunk below compares how often each method declares a significant slope ($p < 0.05$) when there are no bursts. In the vocabulary of Section 6.10 this checks the behavior of the test when its null hypothesis holds, under the test's own assumptions.

```
library(nlme)
dat <- data.frame(sp = yule_tree$tip.label, nodes = nodes_on_path)
p_values <- function(burst) {
  dat$pl <- path_lengths(burst)
  ols <- summary(lm(pl ~ nodes, data = dat))$coefficients["nodes", 4]
  bm <- corBrownian(1, yule_tree, form = ~ sp)
  gls_fit <- gls(pl ~ nodes, data = dat, correlation = bm)
  c(OLS = ols, PGLS = summary(gls_fit)$tTable["nodes", "p-value"])
}
set.seed(31)
null_p <- replicate(100, p_values(burst = 0))
burst_p <- replicate(100, p_values(burst = 8))
rbind(false_positive_rate = rowMeans(null_p < 0.05),
      power_with_bursts = rowMeans(burst_p < 0.05))
```

```
#>                       OLS PGLS
#> false_positive_rate 0.37 0.03
```

```
#> power_with_bursts   1.00 1.00
```

Ordinary least squares rejects the null far more often than the nominal 5 percent when there are no bursts, because it overstates the amount of independent information; phylogenetic GLS keeps the error rate close to nominal while retaining power to detect bursts of the size simulated here. Neither version addresses the node-density artefact, which arises when branch lengths are *estimated* rather than known (Exercise 4 of Chapter 9).

C.4 Chapter 10: Calibration and Divergence Times

This section gives the mathematics behind calibration densities and effective priors, the infinite-sites limit, root-to-tip regression and the identifiability of ancestry, together with the code of the longer simulations of Chapter 10.

C.4.1 The infinite-sites limit

Consider a strict clock and imagine infinitely many characters, so that the branch lengths, and hence the height h_i of every node above the tips in expected changes per character, are known exactly. Then the age of node i is $t_i = h_i / r$ and the root age is $T = h_{\mathrm{root}}/r$, so

$$t_i = \frac{h_i}{h_{\mathrm{root}}}\, T = c_i\, T,$$

with known constants c_i. All the remaining uncertainty is uncertainty about T, which only the calibrations inform. If the calibration gives T a posterior mean m and a 95 percent interval of width w, then t_i has mean $c_i m$ and interval width $c_i w$, so that every node falls on the line

$$\mathrm{width}_i = \frac{w}{m} \times \mathrm{mean}_i$$

through the origin. This is the *infinite-sites plot* of Yang and Rannala (2006) and Rannala and Yang (2007): with finite data the points lie above the line, and the closer they come to it, the less more data would help. With several calibrations the argument is the same, except that the posterior of the time scale is the product of the information from all of them; with a relaxed clock, the c_i are themselves uncertain, so the limit is never reached. The chunk below draws the limiting line for a random tree with a root calibration uniform between 5 and 7 millennia.

```
set.seed(12)
tr <- rcoal(20)
c_i <- branching.times(tr) / max(branching.times(tr))  # known with inf. data
T_draws <- runif(1e4, 5, 7)                            # root calibration
ages <- outer(T_draws, c_i)                            # t_i = c_i * T
mean_age <- colMeans(ages)
width <- apply(ages, 2, function(a) diff(quantile(a, c(0.025, 0.975))))
plot(mean_age, width, pch = 16, las = 1, xlim = c(0, 6.5),
     ylim = c(0, max(width) * 1.1),
     xlab = "Posterior mean age (millennia)",
     ylab = "Width of 95% interval (millennia)")
abline(0, coef(lm(width ~ mean_age - 1)), lty = 2)
```

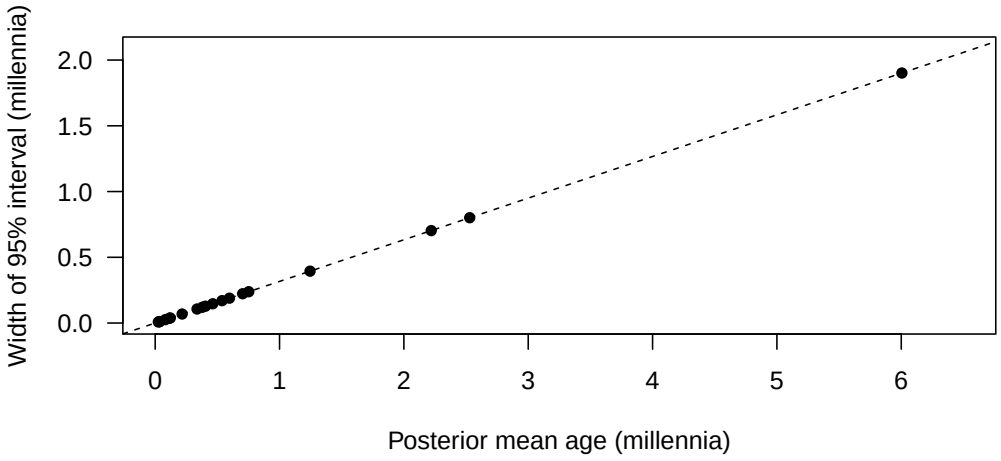


Figure C.2: The infinite-sites limit under a strict clock. With unlimited data, every node age is a fixed fraction of the root age, so the width of its 95 percent interval is proportional to its mean, with slope set by the relative width of the root calibration (here uniform between 5 and 7 millennia).

```
coef(lm(width ~ mean_age - 1))
```

```
#> mean_age
#>   0.3166
```

C.4.2 Calibration densities

Offset lognormal and exponential. If the minimum age a is treated as a hard bound and the excess $X = t - a$ has a lognormal distribution with parameters μ and σ (the mean and standard deviation of $\ln X$), the node age has density

$$X \sim \text{LogNormal}(\mu, \sigma^2),$$
$$p(t) = \frac{1}{(t-a)\,\sigma\sqrt{2\pi}} \exp\!\left(-\frac{(\ln(t-a)-\mu)^2}{2\sigma^2}\right), \quad t > a. \tag{C.12}$$

The median age is $a + e^{\mu}$ and the q quantile is $a + e^{\mu+\sigma z_q}$, where z_q is the standard normal quantile. To place the median at $t_{0.5}$ and the 97.5 percent quantile at an oldest plausible age u, choose $\mu = \ln(t_{0.5} - a)$ and $\sigma = [\ln(u-a) - \mu]/1.96$. For an offset exponential with rate λ, the q quantile is $a - \ln(1-q)/\lambda$, so placing the 97.5 percent quantile at u requires $\lambda = -\ln(0.025)/(u-a) \approx 3.69/(u-a)$. Programs differ in whether they parameterize the lognormal by μ or by the mean of X in real space, a frequent source of miscalibration when a published specification is transferred between programs.

Soft bounds. A soft-bounded uniform calibration places probability $1 - 2p$ uniformly on $[a, b]$, with density height $h = (1-2p)/(b-a)$, and probability p in each tail. With exponentially decaying tails, the lower tail has density $h e^{-\lambda(a-t)}$ for $t < a$, whose total mass is h/λ; setting this

equal to p gives $\lambda = h/p$, and likewise for the upper tail. The density is then continuous at the bounds. This is the construction used in Figure 10.1; programs differ in the exact shapes of the tails, so the construction should be checked against the documentation of the program used.

The functions below compute the parameters from a minimum age and an upper quantile, and check that the specified quantiles are reproduced.

```
offset_lnorm_par <- function(a, upper, median = a + 0.25 * (upper - a),
                             q = 0.975) {
  # offset lognormal with given median and q-quantile (ages > a)
  mu <- log(median - a)
  c(meanlog = mu, sdlog = (log(upper - a) - mu) / qnorm(q))
}
offset_exp_rate <- function(a, upper, q = 0.975) -log(1 - q) / (upper - a)
p_ln <- offset_lnorm_par(a = 2.5, upper = 4)
lam <- offset_exp_rate(a = 2.5, upper = 4)
round(c(p_ln, exp_rate = lam,
        check_ln = plnorm(4 - 2.5, p_ln[1], p_ln[2]),
        check_exp = pexp(4 - 2.5, lam)), 3)
```

```
#>   meanlog     sdlog  exp_rate  check_ln check_exp
#>    -0.981     0.707     2.459     0.975     0.975
```

The code for Figure 10.1, which also defines the function `soft_unif()`, is:

```
age <- seq(0, 7, by = 0.005)
a <- 2.5; b <- 4
d_unif <- dunif(age, a, b)
# soft uniform: 95% mass in [a, b], 2.5% exponential tails on each side
soft_unif <- function(x, a, b, p = 0.025) {
  h <- (1 - 2 * p) / (b - a)            # height of the flat part
  lam_lo <- h / p; lam_hi <- h / p      # tail rates giving mass p each side
  ifelse(x < a, h * exp(-lam_lo * (a - x)),
         ifelse(x > b, h * exp(-lam_hi * (x - b)), h))
}
d_soft <- soft_unif(age, a, b)
d_exp <- ifelse(age > a, dexp(age - a, rate = 1 / 0.6), 0)
d_lnorm <- ifelse(age > a, dlnorm(age - a, meanlog = log(0.6), sdlog = 0.8), 0)
plot(age, d_unif, type = "l", lwd = 2, ylim = c(0, 1.7), las = 1,
     xlab = "Node age (millennia before present)", ylab = "Prior density")
lines(age, d_soft, lwd = 2, col = "grey60")
lines(age, d_exp, lwd = 2, lty = 2)
lines(age, d_lnorm, lwd = 2, lty = 4, col = "grey35")
abline(v = a, lty = 3)
legend("topright", bty = "n", lwd = 2, lty = c(1, 1, 2, 4),
       col = c("black", "grey60", "black", "grey35"),
       legend = c("Uniform (hard)", "Uniform (soft bounds)",
                  "Offset exponential", "Offset lognormal"))
```

```
# the soft-bounded density integrates to one, with 2.5% in each tail
c(total = integrate(soft_unif, -Inf, Inf, a = 2.5, b = 4)$value,
  lower_tail = integrate(soft_unif, -Inf, 2.5, a = 2.5, b = 4)$value)
```

```
#>      total lower_tail
#>      1.000      0.025
```

C.4.3 Effective priors

In the multiplicative construction used by BEAST, the joint prior on node ages $\mathbf{t}$ is proportional to the tree prior times the calibration densities,

$$p(\mathbf{t}) \propto f_{\text{tree}}(\mathbf{t}) \prod_{c \in \mathscr{C}} f_c(t_c),$$

restricted to node ages that respect the order of the tree. The marginal density of a calibrated node under this joint prior is generally not f_c. In the conditional construction of MCMCTree (Yang and Rannala, 2006), the calibrated node ages receive the joint density built from the calibration densities, truncated to respect their order, and the tree prior supplies only the conditional density of the uncalibrated ages given the calibrated ones. Truncation by ordering therefore affects both constructions; interaction with the tree prior affects only the multiplicative one. Heled and Drummond (2012) showed how to modify the Yule prior so that the marginal of a calibrated node equals its calibration density.

For the two-node example of Section 10.3.2 there is no tree prior, and the effective densities can be written down exactly. With root density f_R (normal with mean 6 and standard deviation 0.5) and descendant density f_D (uniform on $[4, 7]$), with distribution functions F_R and F_D, the effective joint density is $f_R(x) f_D(y) \mathbb{1}\{y < x\}/Z$, with $Z = \Pr(D < R)$. Integrating out one age at a time gives the effective marginals

$$g_R(x) = \frac{f_R(x)\, F_D(x)}{Z}, \qquad g_D(y) = \frac{f_D(y)\,[1 - F_R(y)]}{Z}.$$

The chunk below computes them by numerical integration and compares their means with the rejection sampler used in the chapter. This is *implementation validation* in the sense of Section 6.10: two independent computations of the same prior must agree.

```
f_R <- function(x) dnorm(x, 6, 0.5); F_R <- function(x) pnorm(x, 6, 0.5)
f_D <- function(y) dunif(y, 4, 7);   F_D <- function(y) punif(y, 4, 7)
Z <- integrate(function(x) f_R(x) * F_D(x), -Inf, Inf)$value
g_R <- function(x) f_R(x) * F_D(x) / Z
g_D <- function(y) f_D(y) * (1 - F_R(y)) / Z
exact <- c(root = integrate(function(x) x * g_R(x), 3, 9)$value,
           desc = integrate(function(y) y * g_D(y), 4, 7)$value)
set.seed(3)                                  # same draws as in the chapter
m <- 2e5
root <- rnorm(m, 6, 0.5); desc <- runif(m, 4, 7); ok <- desc < root
sampler <- c(root = mean(root[ok]), desc = mean(desc[ok]))
round(rbind(exact = exact, rejection_sampler = sampler,
            specified = c(6, 5.5)), 3)
```

```
#>                     root  desc
#> exact             6.122 5.058
#> rejection_sampler 6.122 5.057
#> specified         6.000 5.500
```

C.4.4 Root-to-tip regression

Under a strict clock with rate r, a tip sampled at age a_i (zero for a living language) lies at the end of a path of duration $T - a_i$ from the root, so its expected root-to-tip distance is $r(T - a_i)$, which is Equation (10.2). The regression $d_i = \beta_0 + \beta_1 a_i + e_i$ therefore has $\beta_1 = -r$ and $\beta_0 = rT$, and the root age is estimated as $\hat{T} = -\hat{\beta}_0/\hat{\beta}_1$, the age at which the fitted line reaches zero distance. Two cautions apply. The residuals of different tips are correlated, because tips share ancestral branches, so the standard errors and p-values of ordinary regression are invalid, which is why the date-randomization test is used instead. And the root must be placed correctly: TempEst chooses the root that maximizes the fit of the regression, a choice that can be misled by rate variation.

The heterochronous family of Section 10.5.2 was simulated with the following code, which shortens the terminal branches of the eight longest-branched languages so that they end at their attestation ages:

```
set.seed(4)
n <- 24
het_tree <- phytools::pbtree(n = n, scale = 6, tip.label = paste0("L", 1:n))
term_len <- het_tree$edge.length[match(1:n, het_tree$edge[, 2])]
ancient <- order(term_len, decreasing = TRUE)[1:8]   # eight ancient languages
tip_age <- setNames(rep(0, n), het_tree$tip.label)
for (i in ancient) {
  e <- which(het_tree$edge[, 2] == i)
  tip_age[i] <- runif(1, 0.3, 0.9) * het_tree$edge.length[e]
  het_tree$edge.length[e] <- het_tree$edge.length[e] - tip_age[i]
}
nb <- nrow(het_tree$edge)
rates <- rlnorm(nb, -0.2^2 / 2, 0.2)
het_phylo <- het_tree
het_phylo$edge.length <- rpois(nb, 80 * rates * het_tree$edge.length) + 0.5
```

C.4.5 The calibration experiment

The phylogram `cal_phylo` used in Sections 10.7 and 10.8 was simulated with the following code. The three internal calibrations are placed on the three oldest nodes below the root (`cal_nodes`), with true ages `tc`.

```
set.seed(5)
n <- 20
cal_tree <- phytools::pbtree(n = n, scale = 6, tip.label = paste0("L", 1:n))
nb <- nrow(cal_tree$edge)
rates <- rlnorm(nb, -0.4^2 / 2, 0.4)
cal_phylo <- cal_tree
cal_phylo$edge.length <- rpois(nb, 80 * rates * cal_tree$edge.length) + 0.5
```

```
true_ages <- branching.times(cal_tree)
ord <- names(sort(true_ages, decreasing = TRUE))
cal_nodes <- as.integer(ord[2:4])          # three internal nodes
tc <- true_ages[ord[2:4]]                  # their true ages
```

C.4.6 Distinguishing an ancestor from a sister

Section 10.6.3 argued that ancestry is poorly identified for fragmentarily attested languages. Suppose an ancient language is attested for n meanings and lies on a side branch of length x millennia, from the point where it leaves the lineage of its relatives to its attestation. If each meaning is replaced independently at rate k per millennium, the probability that none of the n attested meanings has been replaced on the side branch is exactly $(e^{-kx})^n = e^{-nkx}$. A sampled ancestor ($x = 0$) and a sister on a side branch of length x therefore differ in the likelihood of "no unique changes" by the factor e^{-nkx}, and side branches with $e^{-nkx} > 0.05$, that is

$$x < \frac{-\ln 0.05}{nk} \approx \frac{3}{nk},$$

cannot be excluded at the 5 percent level. Conversely, if m meanings show apparent unique replacements, the maximum-likelihood length of the side branch is approximately $m/(nk)$ when m is small relative to n, whatever the source of those replacements: a literary register or a scribal convention is read as time on a side branch. In binary cognate coding, each such replacement appears as a unique loss and a unique gain. The calculation is optimistic, because it treats the state of the lineage at the time of attestation as known, whereas in practice it is reconstructed from the descendants with uncertainty.

```
max_side_branch <- function(n, retention = 0.86, conf = 0.95) {
  # longest side branch (millennia) not excluded when no unique
  # replacements are seen among n attested meanings
  k <- -log(retention)                  # replacements per meaning per kyr
  -log(1 - conf) / (n * k)
}
n_attested <- c(200, 100, 50, 20)
knitr::kable(data.frame(
  meanings = n_attested,
  years_0.80 = round(1000 * max_side_branch(n_attested, 0.80)),
  years_0.86 = round(1000 * max_side_branch(n_attested, 0.86)),
  years_0.90 = round(1000 * max_side_branch(n_attested, 0.90))),
  caption = "Longest side branch (years) that cannot be excluded at the 5 percent
  level when an ancient language attested for a given number of meanings shows no
  unique replacements, for three retention rates per millennium.")
```

Table C.5: Longest side branch (years) that cannot be excluded at the 5 percent level when an ancient language attested for a given number of meanings shows no unique replacements, for three retention rates per millennium.

meanings	years_0.80	years_0.86	years_0.90
200	67	99	142
100	134	199	284
50	269	397	569
20	671	993	1422

```
# apparent unique changes are read as time: two register-specific words
# among 100 attested meanings imply a side branch of about
round(1000 * 2 / (100 * -log(0.86)))
```

```
#> [1] 133
```

Appendix D

Technical Notes: Space, Ancestral States and Comparative Methods

This appendix collects the derivations, algorithms and longer pieces of code that support Chapters 11 to 14. Each section corresponds to one chapter and can be read alongside it. Every code chunk here is self-contained: the trees and data it needs are recreated with the same random seeds as in the chapter, so the objects are identical to those the chapter uses unless stated otherwise. The general background on likelihood, multivariate normal distributions, generalized least squares and continuous-time Markov chains is in Appendix A, especially Sections A.7, A.10 and A.11.

```
suppressPackageStartupMessages({
  library(ape)
  library(phytools)
  library(phangorn)
})
```

D.1 Chapter 11: Phylogeography and Homelands

D.1.1 Pruning for discrete locations

In discrete phylogeography (Section 11.4.1) the location of a lineage is one of K regions, evolving under a rate matrix $\mathbf{Q}$ with transition probabilities $\mathbf{P}(t) = e^{\mathbf{Q}t}$. The likelihood of the observed tip regions is computed by Felsenstein's pruning algorithm. For each node v and each region i, the partial likelihood $L_v(i)$ is the probability of the tip regions below v given that v was in region i. It is computed from the tips toward the root:

$$L_v(i) = \prod_{c \in \text{children}(v)} \sum_{j=1}^{K} P_{ij}(t_c)\, L_c(j), \tag{D.1}$$

with $L_v(i) = 1$ if tip v is in region i and 0 otherwise, and t_c the length of the branch leading to child c. The likelihood of the whole tree is $\sum_i \pi_i L_{\text{root}}(i)$, where π_i is a prior on the root region, and the posterior probability that the root was in region i is $\pi_i L_{\text{root}}(i) / \sum_j \pi_j L_{\text{root}}(j)$. Marginal probabilities at other internal nodes need a second, root-to-tips pass, or equivalently re-rooting at

each node in turn (Section 12.4.2). An implementation of the recursion for any discrete character is given in Section D.2.2.

D.1.2 Route Bayes factors in BSSVS

Under the BSSVS prior of Equation (11.3), the number of active symmetric routes is $K-1$ plus a Poisson variable with mean $\ln 2$, so the expected number of active routes is $\ln 2 + K - 1$ out of $K(K-1)/2$ possible ones. The prior inclusion probability of any one route is therefore $q = (\ln 2 + K - 1)/[K(K-1)/2]$. If p_{ij} is the posterior inclusion probability of route $i \leftrightarrow j$ (the posterior mean of δ_{ij}, read from the MCMC log), the Bayes factor for including the route is the ratio of posterior to prior odds:

$$\mathrm{BF}_{ij} = \frac{p_{ij}/(1-p_{ij})}{q/(1-q)}, \qquad q = \frac{\ln 2 + K - 1}{K(K-1)/2}. \tag{D.2}$$

The function below computes it. The posterior inclusion probabilities are invented, as if read from a BEAST log file for an analysis with four regions.

```
bssvs_bf <- function(post_incl, K, poisson_mean = log(2)) {
  q <- (poisson_mean + K - 1) / (K * (K - 1) / 2)  # prior inclusion probability
  post_incl <- pmin(post_incl, 1 - 1e-6)          # guard against p = 1
  (post_incl / (1 - post_incl)) / (q / (1 - q))
}
post <- c("R1-R2" = 0.99, "R1-R3" = 0.12, "R1-R4" = 0.55,
          "R2-R3" = 0.81, "R2-R4" = 0.08, "R3-R4" = 0.93)
round(bssvs_bf(post, K = 4), 1)
```

```
#> R1-R2 R1-R3 R1-R4 R2-R3 R2-R4 R3-R4
#>  61.8   0.1   0.8   2.7   0.1   8.3
```

```
# the same posterior inclusion probability with 4 and with 20 regions
round(c(K4 = bssvs_bf(0.81, K = 4), K20 = bssvs_bf(0.81, K = 20)), 1)
```

```
#>   K4  K20
#>  2.7 36.9
```

With four regions the prior inclusion probability is about 0.62, so a posterior inclusion probability of 0.81 gives a Bayes factor below 3; with twenty regions the prior inclusion probability is about 0.1 and the same posterior gives a Bayes factor above 30. A posterior inclusion probability that hits 1 in every sample gives an infinite Bayes factor; the guard in the function caps it, and in practice such routes are reported as "BF $>$ 1000" or similar. Because the Poisson prior can be changed by the analyst, and because q depends on K, route Bayes factors should always be reported together with K and the prior.

D.1.3 The multivariate normal model for coordinates

Under bivariate Brownian motion (Equation (11.5)), the location of each tip is the root location plus the sum of independent normal displacements along the branches from the root to the tip. Two tips i and j share the displacements on the path from the root to their most recent common

ancestor, so the covariance between any coordinate of i and any coordinate of j is the corresponding entry of Σ multiplied by the length of the shared path, C_{ij}. Writing $\mathbf{C}$ for the $n \times n$ phylogenetic covariance matrix (with C_{ii} the root-to-tip time of tip i) and stacking the coordinates of the n tips into a vector $\mathbf{X}$ of length $2n$ (tip by tip),

$$\mathbf{X} \sim \mathcal{N}_{2n}(\mathbf{1}_n \otimes \mathbf{x}_{\text{root}},\ \mathbf{C} \otimes \Sigma), \tag{D.3}$$

where $\otimes$ is the Kronecker product. The likelihood can be evaluated directly from this formula for small trees, or in time linear in the number of tips by a pruning algorithm analogous to Equation (D.1), in which the partial likelihoods are normal densities summarized by their means and variances (Felsenstein, 1985b).

For a single coordinate $\mathbf{x}$ the model is $\mathbf{x} \sim \mathcal{N}_n(x_{\text{root}}\mathbf{1}, \sigma^2\mathbf{C})$, a linear model with a single intercept and correlated errors. Maximizing the log-likelihood

$$\ell(x_{\text{root}}, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2}\log\det\mathbf{C} - \frac{(\mathbf{x} - x_{\text{root}}\mathbf{1})^\top \mathbf{C}^{-1}(\mathbf{x} - x_{\text{root}}\mathbf{1})}{2\sigma^2}$$

with respect to x_{root} gives the generalized least squares estimator of Equation (11.6) (Section A.11). Its variance follows from the general GLS result $\operatorname{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^\top\mathbf{C}^{-1}\mathbf{X})^{-1}$ with the design matrix $\mathbf{X} = \mathbf{1}$. Note that neither the weights $\mathbf{1}^\top\mathbf{C}^{-1}/\mathbf{1}^\top\mathbf{C}^{-1}\mathbf{1}$ nor the variance depends on the observed locations.

The chunk below recreates the tree and locations of Section 11.13.1 and checks that the GLS mean equals the maximum-likelihood root location returned by `fastAnc()`.

```
set.seed(11)
tree <- pbtree(n = 30, scale = 6)
tree$tip.label <- paste0("L", 1:30)
Sigma <- matrix(c(0.36, 0.10, 0.10, 0.36), 2)
XY <- sim.corrs(tree, vcv = Sigma, anc = c(0, 0), internal = TRUE)
tips <- XY[tree$tip.label, ]
colnames(tips) <- c("east", "north")
C <- vcv(tree)[rownames(tips), rownames(tips)]
one <- rep(1, nrow(C))
Ci <- solve(C)
w <- drop(one %*% Ci) / drop(one %*% Ci %*% one)   # GLS weights, sum to 1
c(gls = sum(w * tips[, "east"]),
  fastAnc = unname(fastAnc(tree, tips[, "east"])[1]))
```

```
#>     gls fastAnc
#> 0.09339 0.09339
```

```
range(w)                                          # all weights positive here
```

```
#> [1] 0.00589 0.12728
```

The weights are all positive, so the homeland estimate is a convex combination of the present-day locations and lies inside their convex hull.

D.1.4 Why standardized contrasts serve as an adequacy statistic

The adequacy check of Section 11.13.3 uses the standardized independent contrasts of a coordinate (Section D.3.4). Under Brownian motion on the correct tree, the $n-1$ standardized contrasts are independent and identically distributed $\mathcal{N}(0,\sigma^2)$, whatever the shape of the tree. Their kurtosis, $\overline{u^4}/(\overline{u^2})^2$, is invariant to σ^2, so its null distribution can be simulated from standard normal draws without reference to the tree. A relaxed random walk inflates the variance of the contrasts on some branches and not others, which produces a mixture of normals with heavier tails than any single normal, and so a larger kurtosis.

A directional trend leaves the contrasts unchanged when all tips are contemporaneous. Under BM with drift μ, a tip at height h has expected value $x_{\text{root}} + \mu h$, so on an ultrametric tree of depth T every tip has the same expectation, $x_{\text{root}} + \mu T$. The drift adds this constant to every tip and changes nothing else: the covariances are those of BM without drift. Every value used by the contrast algorithm is either a tip value or a weighted average of tip values with weights summing to one, so each has the same expectation, and every contrast, being a difference of two such values, has expectation zero and the same variance as without drift. The contrasts therefore have exactly the distribution they would have without drift. No statistic computed from the contrasts, and more generally no statistic that is invariant to a common shift of all tips, can detect a trend from contemporaneous tips. This is the formal reason why the adequacy check in Table 11.3 flags the trend scenario no more often than pure Brownian motion. With non-contemporaneous tips the cancellation fails and the drift becomes visible, which is what the drift regression of Section 11.13.4 exploits.

D.2 Chapter 12: Ancestral State Reconstruction

The code in this section uses the toy tree and word-order data of Section 12.15, recreated in the first chunk.

```
tree <- read.tree(text = paste0(
  "((((A1:1,A2:1):1.5,(A3:1.2,A4:1.2):1.3):1,",
  "((B1:0.8,B2:0.8):1.2,B3:2):1.5):1.5,",
  "(((C1:1,C2:1):1,C3:2):1.5,((D1:0.5,D2:0.5):2,(D3:1.5,D4:1.5):1):1):1.5);"))
wo <- c(A1 = "SOV", A2 = "SOV", A3 = "SOV", A4 = "SVO", B1 = "SVO", B2 = "SVO",
        B3 = "SOV", C1 = "VSO", C2 = "VSO", C3 = "SVO", D1 = "SVO", D2 = "SVO",
        D3 = "SOV", D4 = "SVO")
wo <- factor(wo[tree$tip.label])
root <- Ntip(tree) + 1
fitER <- fitMk(tree, wo, model = "ER")
Q <- unclass(as.Qmatrix(fitER))[levels(wo), levels(wo)]
pi_flat <- rep(1 / 3, 3)
```

D.2.1 Fitch and Sankoff algorithms

In Fitch's *downpass* (from tips to root), each node v with children c_1 and c_2 is assigned a set of candidate states:

$$S_v = \begin{cases} S_{c_1} \cap S_{c_2} & \text{if } S_{c_1} \cap S_{c_2} \neq \emptyset, \\ S_{c_1} \cup S_{c_2} & \text{otherwise (cost + 1).} \end{cases} \tag{D.4}$$

The total number of unions is the parsimony score. An *uppass* from the root then refines the sets into the most parsimonious reconstructions at each node.

Sankoff's generalization allows an arbitrary cost c_{ij} for a change from state i to state j. The downpass computes, for each node v and state i, the minimum cost $S_v(i)$ of the subtree below v given that v is in state i:

$$S_v(i) = \sum_{c \in \text{children}(v)} \min_j \left[c_{ij} + S_c(j) \right], \tag{D.5}$$

with $S_v(i) = 0$ if tip v is in state i and ∞ otherwise. The minimum over i of $S_{\text{root}}(i)$ is the parsimony score. The recursion has the structure of Equation (D.6) with minimum and addition in place of sum and product, which is why Sankoff parsimony and likelihood pruning are often presented together.

The function below implements the Fitch downpass and checks its score against **phangorn**.

```
fitch_down <- function(tree, x) {
  n <- Ntip(tree)
  sets <- vector("list", n + tree$Nnode)
  for (i in 1:n) sets[[i]] <- as.character(x[tree$tip.label[i]])
  cost <- 0
  parents <- rev(unique(rev(reorder(tree, "postorder")$edge[, 1])))
  for (v in parents) {                       # each node after all its children
    ch <- tree$edge[tree$edge[, 1] == v, 2]
    I <- Reduce(intersect, sets[ch])
    if (length(I) > 0) sets[[v]] <- I else {
      sets[[v]] <- Reduce(union, sets[ch]); cost <- cost + 1 }
  }
  list(sets = sets, cost = cost)
}
fd <- fitch_down(tree, wo)
pd <- phyDat(matrix(as.character(wo), dimnames = list(names(wo), "wo")),
             type = "USER", levels = levels(wo))
c(hand = fd$cost, phangorn = parsimony(tree, pd))
```

```
#>     hand phangorn
#>        5        5
```

```
fd$sets[[root]]                              # downpass set at the root
```

```
#> [1] "SOV" "SVO"
```

The downpass set at the root contains both SOV and SVO, matching the tie in the most-parsimonious reconstruction sets reported in Section 12.15.1.

D.2.2 Pruning and conditional likelihoods by hand

Felsenstein's pruning algorithm computes the conditional likelihood $L_v(i)$, the probability of the tip states below node v given that v is in state i, from the tips toward the root:

$$L_v(i) = \prod_{c \in \text{children}(v)} \sum_j P_{ij}(t_c)\, L_c(j), \qquad L(\theta) = \sum_i \pi_i\, L_{\text{root}}(i), \tag{D.6}$$

with $L_v(i) = 1$ for the observed state of a tip and 0 otherwise. The function below implements the recursion, with transition probabilities computed from an eigendecomposition of $\mathbf{Q}$ (Section A.6).

```
Pt <- function(Q, t) {
  e <- eigen(Q)
  Re(e$vectors %*% diag(exp(e$values * t)) %*% solve(e$vectors))
}
prune <- function(tree, x, Q) {
  tree <- reorder(tree, "postorder")
  n <- Ntip(tree); K <- ncol(Q)
  L <- matrix(1, n + tree$Nnode, K, dimnames = list(NULL, colnames(Q)))
  L[1:n, ] <- 0
  L[cbind(1:n, match(as.character(x[tree$tip.label]), colnames(Q)))] <- 1
  for (e in seq_len(nrow(tree$edge))) {
    par <- tree$edge[e, 1]; ch <- tree$edge[e, 2]
    L[par, ] <- L[par, ] * (Pt(Q, tree$edge.length[e]) %*% L[ch, ])
  }
  L
}
L <- prune(tree, wo, Q)
c(hand = log(sum(pi_flat * L[root, ])), fitMk = as.numeric(logLik(fitER)))
```

```
#>   hand  fitMk
#> -13.76 -13.76
```

```
round(pi_flat * L[root, ] / sum(pi_flat * L[root, ]), 3)   # root posterior
```

```
#>   SOV   SVO   VSO
#> 0.352 0.418 0.230
```

The hand-computed log-likelihood matches `fitMk()`, and the normalized root vector is the marginal posterior at the root. At other nodes the normalized conditional likelihoods are *not* marginal posteriors, because they ignore the rest of the tree. The chunk below shows that they are what `ace(..., marginal = TRUE)` returns, as stated in Section 12.15.3.

```
node <- 16                                      # ancestor of subgroups A and B
rbind(
  ancr_marginal     = ancr(fitER)$ace[as.character(node), ],
  ace_marginal_TRUE = ace(wo, tree, type = "discrete", model = "ER",
                          marginal = TRUE)$lik.anc[node - Ntip(tree), ],
  hand_subtree_only = L[node, ] / sum(L[node, ]) |> round(3)
```

```
#>                      SOV   SVO   VSO
#> ancr_marginal     0.494 0.367 0.139
#> ace_marginal_TRUE 0.533 0.324 0.143
#> hand_subtree_only 0.533 0.324 0.143
```

D.2.3 Joint reconstruction: the Pupko algorithm

Pupko et al. (2000) find the joint reconstruction of Equation (12.2) by dynamic programming,

the max-product analogue of pruning. For each non-root node c with parent a, and each possible parent state i, define

$$\Lambda_c(i) = \max_j \left[P_{ij}(t_c) \prod_{d \in \text{children}(c)} \Lambda_d(j) \right], \qquad C_c(i) = \arg\max_j [\cdots], \tag{D.7}$$

where for a tip c in observed state s, $\Lambda_c(i) = P_{is}(t_c)$. At the root, choose $\hat{x}_{\text{root}} = \arg\max_i \pi_i \prod_d \Lambda_d(i)$, and then trace back down the tree: each child takes the state $C_c(\hat{x}_a)$ recorded for its parent's chosen state. The implementation below works on the log scale and is checked against `ancr(..., type = "joint")`.

```
joint_pupko <- function(tree, x, Q, pi) {
  tree <- reorder(tree, "postorder")
  n <- Ntip(tree); K <- ncol(Q); states <- colnames(Q); N <- n + tree$Nnode
  S <- matrix(0, N, K)                    # sum of log Lambda over children
  best <- matrix(NA_integer_, N, K)       # C_c(i): best child state, parent i
  for (e in seq_len(nrow(tree$edge))) {
    par <- tree$edge[e, 1]; ch <- tree$edge[e, 2]
    lP <- log(Pt(Q, tree$edge.length[e]))
    if (ch <= n) {
      j <- match(as.character(x[tree$tip.label[ch]]), states)
      lv <- lP[, j]; b <- rep(j, K)
    } else {
      M <- lP + matrix(S[ch, ], K, K, byrow = TRUE)
      lv <- apply(M, 1, max); b <- apply(M, 1, which.max)
    }
    best[ch, ] <- b; S[par, ] <- S[par, ] + lv
  }
  r <- n + 1; st <- integer(N); st[r] <- which.max(log(pi) + S[r, ])
  pre <- reorder(tree, "cladewise")
  for (e in seq_len(nrow(pre$edge)))
    st[pre$edge[e, 2]] <- best[pre$edge[e, 2], st[pre$edge[e, 1]]]
  list(nodes = setNames(states[st[r:N]], r:N), logL = max(log(pi) + S[r, ]))
}
jp <- joint_pupko(tree, wo, Q, pi_flat)
jt <- ancr(fitER, type = "joint")
all(jp$nodes == as.character(jt$ace[names(jp$nodes)]))
```

```
#> [1] TRUE
```

```
c(hand = jp$logL, ancr = as.numeric(jt$logLik))
```

```
#>   hand   ancr
#> -17.45 -17.45
```

D.2.4 Sampling a stochastic map

A stochastic character map is drawn in three steps, using the conditional likelihoods $L_v(i)$ from a pruning pass.

1. **Root state.** Draw the root state i with probability $\pi_i L_{\text{root}}(i) / \sum_j \pi_j L_{\text{root}}(j)$.
2. **Node states, from root to tips.** For each child c of a node a whose state i has been drawn, draw the state j of c with probability

$$P(X_c = j \mid X_a = i, D) = \frac{P_{ij}(t_c)\, L_c(j)}{\sum_l P_{il}(t_c)\, L_c(l)}. \tag{D.8}$$

   Together, steps 1 and 2 draw a joint assignment of node states from its exact posterior.
3. **Branch histories.** For each branch, simulate the history of the character between the now-fixed start and end states, conditional on those endpoints. The simplest method is rejection sampling: simulate forward from the start state using exponential waiting times with rate $-q_{ii}$ in state i and jump probabilities $q_{ij}/(-q_{ii})$, and accept the history only if it ends in the required state. Rejection is inefficient on long branches and for improbable endpoints. *Uniformization* instead draws the number of potential changes on a branch from a Poisson distribution with a rate $\mu \geq \max_i(-q_{ii})$, and then the states at those events from the discrete-time chain with transition matrix $\mathbf{I} + \mathbf{Q}/\mu$, conditional on the endpoints; it never rejects. Irvahn and Minin (2014) built on uniformization a stochastic mapping algorithm that avoids matrix exponentiation altogether, which pays off when the state space is large.

Averages over many maps estimate posterior expectations of any function of the history, such as the number of changes of each type or the time spent in each state. The analytical alternative for expected counts and dwelling times is the method of Minin and Suchard (2008).

D.2.5 Hidden-rate models

With two hidden rate categories, slow (S) and fast (F), a character with observed states $1, \ldots, k$ has $2k$ hidden states, and the rate matrix has block structure:

$$\mathbf{Q} = \begin{pmatrix} \mathbf{Q}_S - \mathbf{D}_{SF} & \mathbf{D}_{SF} \\ \mathbf{D}_{FS} & \mathbf{Q}_F - \mathbf{D}_{FS} \end{pmatrix}, \tag{D.9}$$

where $\mathbf{Q}_S$ and $\mathbf{Q}_F$ are the rate matrices for observed-state changes within each category (with their own diagonals) and $\mathbf{D}_{SF}$, $\mathbf{D}_{FS}$ are diagonal matrices of switching rates between categories. At a tip with observed state s, the partial likelihood is 1 for both hidden states s_S and s_F and 0 otherwise, so the pruning algorithm sums over the hidden category automatically. Marginal reconstructions of the observed state are obtained by summing the reconstructed probabilities of s_S and s_F, as in the **corHMM** example of Section 12.15.8. Two identifiability issues are worth knowing. The labels of the categories can be swapped without changing the likelihood unless the model constrains, for example, the rates in $\mathbf{Q}_S$ to be smaller than those in $\mathbf{Q}_F$; and with one character on a small tree the switching rates are typically poorly estimated, so that the model can fit no better than the ordinary Mk model even when rates do vary.

D.2.6 Root treatments for a rare feature, computed exactly

Section 12.4.5 argued that, for a feature attested in only a few branches, the root reconstruction depends on the root treatment as much as on the data. For a binary character the Bayesian answer can be computed exactly, by integrating the pruning likelihood over the two rates on a grid, because the transition probabilities have the closed form $P_{01}(t) = \pi_1(1 - e^{-(q_{01}+q_{10})t})$ and $P_{10}(t) = \pi_0(1 - e^{-(q_{01}+q_{10})t})$, with $\pi_1 = q_{01}/(q_{01}+q_{10})$ the stationary frequency of state 1 (Section

A.7). The feature below is present in two languages of subgroup A and two of subgroup C, on opposite sides of the root of the toy tree; both rates have Exponential(1) priors.

```
rare <- setNames(rep(0, Ntip(tree)), tree$tip.label)
rare[c("A1", "A2", "C1", "C2")] <- 1
# root conditional likelihoods L0, L1 for vectors of rates (closed form)
root_L <- function(q01, q10) {
  tr <- reorder(tree, "postorder"); n <- Ntip(tr)
  L <- array(1, c(n + tr$Nnode, 2, length(q01)))
  for (i in 1:n) {
    L[i, , ] <- 0; L[i, rare[tr$tip.label[i]] + 1, ] <- 1 }
  a <- q01 + q10; p1 <- q01 / a
  for (e in seq_len(nrow(tr$edge))) {
    pa <- tr$edge[e, 1]; d <- tr$edge[e, 2]
    ex <- exp(-a * tr$edge.length[e])
    P01 <- p1 * (1 - ex); P10 <- (1 - p1) * (1 - ex)
    L[pa, 1, ] <- L[pa, 1, ] * ((1 - P01) * L[d, 1, ] + P01 * L[d, 2, ])
    L[pa, 2, ] <- L[pa, 2, ] * (P10 * L[d, 1, ] + (1 - P10) * L[d, 2, ])
  }
  list(L0 = L[n + 1, 1, ], L1 = L[n + 1, 2, ])
}
# integrate over q01, q10 ~ Exponential(1) on a grid in log q
u <- seq(log(1e-5), log(60), length.out = 300); du <- u[2] - u[1]
g <- expand.grid(a = exp(u), b = exp(u))
wt <- du^2 * g$a * g$b * dexp(g$a) * dexp(g$b)
rl <- root_L(g$a, g$b)
pi1 <- g$a / (g$a + g$b)                        # stationary P(present)
m0 <- sum(wt * rl$L0); m1 <- sum(wt * rl$L1)
m_stat <- sum(wt * ((1 - pi1) * rl$L0 + pi1 * rl$L1))
data.frame(
  root = c("flat", "stationary", "fixed absent", "fixed present"),
  lnML = round(log(c((m0 + m1) / 2, m_stat, m0, m1)), 3),
  P_present = round(c(m1 / (m0 + m1), sum(wt * pi1 * rl$L1) / m_stat,
                      0, 1), 3))
```

```
#>           root   lnML P_present
#> 1         flat -8.784     0.462
#> 2   stationary -8.836     0.272
#> 3  fixed absent -8.710     0.000
#> 4 fixed present -8.864     1.000
```

Three things are worth noticing. First, the marginal likelihoods differ by less than 0.2 log units, so the data cannot choose among the root treatments, and the Bayes factor for "present at the root" against "absent at the root" (fixing the root, as in the approach of Pagel et al. (2004)) is close to 1. Second, the posterior probability that the feature was present at the root is nevertheless 0.46 under the flat prior and 0.27 under the stationary prior: the stationary prior gives the root the frequencies of a process in which the rarely attested state is rare. Third, the flat-prior answer is exactly the Bayes factor turned into a probability, $m_1/(m_0 + m_1)$, because a flat root is an equal mixture of the two fixed roots.

An *estimated* root frequency adds nothing for a single character. With a uniform (Dirichlet(1, 1)) prior on π_1, the likelihood averaged over the prior is $\int_0^1 [\pi_1 L_1 + (1-\pi_1)L_0]\, d\pi_1 = (L_0 + L_1)/2$, the flat-prior likelihood, at any rates; and the posterior mean of π_1 is $(2L_1 + L_0)/3(L_0 + L_1)$, which can range only between 1/3 and 2/3. The chunk checks both at one pair of rates.

```
r1 <- root_L(0.2, 0.4)
p <- seq(0.0005, 0.9995, by = 0.001)           # grid over pi_1
lik_pi <- p * r1$L1 + (1 - p) * r1$L0
c(estimated = mean(lik_pi), flat = (r1$L0 + r1$L1) / 2)
```

```
#> estimated      flat
#> 0.0007825 0.0007825
```

```
c(posterior_mean_pi1 = sum(p * lik_pi) / sum(lik_pi),
  formula = (2 * r1$L1 + r1$L0) / (3 * (r1$L0 + r1$L1)))
```

```
#> posterior_mean_pi1            formula
#>             0.4876             0.4876
```

Maximum likelihood shows the same contrast. In **phytools**, `pi = "estimated"` sets the root prior to the stationary distribution of the fitted rate matrix.

```
x <- setNames(factor(rare), names(rare))
fit_flat <- fitMk(tree, x, model = "ARD", pi = "equal")
invisible(capture.output(                      # silence the message on pi
  fit_stat <- fitMk(tree, x, model = "ARD", pi = "estimated")))
rbind(flat = ancr(fit_flat)$ace[1, ],
      stationary = ancr(fit_stat)$ace[1, ]) |> round(3)
```

```
#>                0     1
#> flat       0.519 0.481
#> stationary 0.830 0.170
```

The RevBayes script of Section H.5.21 fits the same kinds of model to a simulated rare feature, adds informative rate priors and a loss-only model, and reproduces the exact values computed in this way.

D.2.7 A test for historical signal in one character

Harmon (2019) proposes a blunt check of whether a character carries any information about its history. The comparison model ignores the tree and draws the state of each language independently, with probability p_k for state k; its maximized log-likelihood is $\sum_k n_k \log(n_k/n)$, where n_k languages are in state k. As the rates of an Mk model grow, the tip states become independent draws from the stationary distribution, so the Mk likelihood approaches this value along a flat ridge. If the best Mk likelihood is no higher than the tree-free likelihood, the rates cannot be estimated and nothing can be said about ancestral states. The chunk compares two binary characters on the toy tree, one that follows the primary split and one scattered across the subgroups, and then the word-order data.

```
garbage <- function(x) { n <- table(x); sum(n * log(n / sum(n))) }
tips <- tree$tip.label
```

```
split_cd <- setNames(as.integer(substr(tips, 1, 1) %in% c("C", "D")), tips)
scattered <- setNames(as.integer(tips %in% c("A1", "A3", "B2", "C1", "C3",
                                             "D2", "D4")), tips)
sapply(list(follows_split = split_cd, scattered = scattered), function(y)
  c(Mk_ARD = as.numeric(logLik(fitMk(tree, y, model = "ARD"))),
    tree_free = garbage(y))) |> round(3)
```

```
#>           follows_split scattered
#> Mk_ARD           -4.001    -9.704
#> tree_free        -9.704    -9.704
```

```
c(word_order_ER = as.numeric(logLik(fitER)), tree_free = garbage(wo))
```

```
#> word_order_ER     tree_free
#>        -13.76        -13.89
```

For the scattered character the two likelihoods are identical: the Mk fit has run onto the ridge, and any reconstruction would only restate the root prior. The word-order data of Chapter 12 are better than that, but not by much: the equal-rates model improves on the tree-free model by about 0.13 log-likelihood units with one parameter fewer, a reminder that fourteen languages and one feature carry little historical information.

D.3 Chapter 13: Phylogenetic Comparative Methods

The code in this section recreates the simulated 60-language tree of Section 13.1 and the traits simulated from it, with the same seeds as in the chapter.

```
suppressPackageStartupMessages({
  library(nlme); library(caper); library(phylolm); library(MCMCglmm)
})
set.seed(13)
tree <- pbtree(n = 60, scale = 6)
tree$tip.label <- sprintf("L%02d", 1:60)
tips <- tree$tip.label
C <- vcv(tree)
set.seed(21)
x_bm <- fastBM(tree, sig2 = 1, internal = TRUE)    # as in pcms-traitgram
```

D.3.1 The Brownian likelihood

Under Brownian motion the tip values are multivariate normal, $\mathbf{x} \sim \mathcal{N}_n(\mu\mathbf{1}, \sigma^2\mathbf{C})$ (Equation (13.2)), so the likelihood is the multivariate normal density (Section A.10):

$$L(\mu, \sigma^2 \mid \mathbf{x}, \tau) = \frac{\exp\!\left[-\tfrac{1}{2}(\mathbf{x}-\mu\mathbf{1})^{\top}(\sigma^2\mathbf{C})^{-1}(\mathbf{x}-\mu\mathbf{1})\right]}{(2\pi)^{n/2}\,\det(\sigma^2\mathbf{C})^{1/2}}. \tag{D.10}$$

Setting the derivatives of its logarithm with respect to μ and σ^2 to zero gives the maximum-likelihood estimates

$$\hat{\mu} = (\mathbf{1}^\top \mathbf{C}^{-1}\mathbf{1})^{-1}\,\mathbf{1}^\top \mathbf{C}^{-1}\mathbf{x}, \qquad \hat{\sigma}^2 = \frac{(\mathbf{x}-\hat{\mu}\mathbf{1})^\top \mathbf{C}^{-1}(\mathbf{x}-\hat{\mu}\mathbf{1})}{n}. \tag{D.11}$$

The estimate $\hat{\mu}$ is the GLS mean, a weighted average of the tip values with weights $\mathbf{1}^\top\mathbf{C}^{-1}$ (the same estimator as the Brownian homeland of Equation (11.6)). The REML estimate of σ^2 uses the divisor $n-1$ instead of n, and it equals the mean squared standardized independent contrast (Section D.3.4).

```
x <- x_bm[tips]
Ci <- solve(C); one <- rep(1, length(x))
mu_hat <- drop(one %*% Ci %*% x) / drop(one %*% Ci %*% one)
s2_ml  <- drop(t(x - mu_hat) %*% Ci %*% (x - mu_hat)) / length(x)
c(mu_hat = mu_hat, fastAnc_root = unname(fastAnc(tree, x)[1]),
  s2_REML = s2_ml * length(x) / (length(x) - 1),
  mean_pic2 = mean(pic(x, tree)^2))
```

```
#>       mu_hat fastAnc_root      s2_REML    mean_pic2
#>       0.2138       0.2138       1.0072       1.0072
```

D.3.2 Covariances under the Ornstein-Uhlenbeck and early-burst models

For the OU process of Equation (13.3) on an ultrametric tree of depth T, with the root value drawn from the stationary distribution, the covariance between two tips separated by a patristic distance $d_{ij} = 2(T - s_{ij})$ is

$$\operatorname{Cov}(x_i, x_j) = \frac{\sigma^2}{2\alpha}\, e^{-\alpha d_{ij}}. \tag{D.12}$$

Covariance thus decays exponentially with the distance between languages, rather than linearly as under BM. As $\alpha \to 0$ the stationary variance $\sigma^2/(2\alpha)$ diverges, which is why software either fixes the root value or conditions on it when α is small; the two parameterizations are labelled `OUfixedRoot` and `OUrandomRoot` in **phylolm**.

For the early-burst model, in which the Brownian rate changes as $\sigma^2(t) = \sigma_0^2 e^{rt}$, the covariance between two tips is the integral of the rate over their shared history:

$$C_{ij} = \int_0^{s_{ij}} \sigma_0^2 e^{rt}\, dt = \sigma_0^2\, \frac{e^{r s_{ij}} - 1}{r}. \tag{D.13}$$

With $r < 0$ most of the variance is generated early, so deep splits dominate the covariance structure. On an ultrametric tree, the delta transformation of Section 13.3.4 maps C_{ij} to C_{ij}^{δ}; delta and EB are therefore alternative ways of concentrating change early or late.

D.3.3 Signal statistics

Blomberg's K. Let $\mathrm{MSE}_0 = (\mathbf{x}-\hat{\mu}\mathbf{1})^\top(\mathbf{x}-\hat{\mu}\mathbf{1})/(n-1)$ measure dispersion around the phylogenetic mean while ignoring the tree, and $\mathrm{MSE} = (\mathbf{x}-\hat{\mu}\mathbf{1})^\top\mathbf{C}^{-1}(\mathbf{x}-\hat{\mu}\mathbf{1})/(n-1)$ measure dispersion using the tree. Then

$$K = \frac{\mathrm{MSE}_0/\mathrm{MSE}}{\frac{1}{n-1}\left[\mathrm{tr}(\mathbf{C}) - \frac{n}{\mathbf{1}^\top\mathbf{C}^{-1}\mathbf{1}}\right]}, \tag{D.14}$$

where the denominator is the expected value of the ratio under BM, so that $K = 1$ is the BM expectation (Blomberg et al., 2003).

Moran's I. With weights w_{ij} (for example $w_{ij} = 1/d_{ij}$ for patristic distance d_{ij}),

$$I = \frac{n}{\sum_{i\neq j} w_{ij}} \cdot \frac{\sum_{i\neq j} w_{ij}(x_i-\bar{x})(x_j-\bar{x})}{\sum_i (x_i-\bar{x})^2}. \tag{D.15}$$

Its expected value under no autocorrelation is $-1/(n-1)$.

The D statistic. For a binary trait, Σd is the sum of differences between sister clades computed by a contrast-like algorithm. The observed value is compared with its mean under random reshuffling of the trait across tips ($\overline{\Sigma d_r}$) and under a Brownian threshold model ($\overline{\Sigma d_b}$):

$$D = \frac{\Sigma d_{\mathrm{obs}} - \overline{\Sigma d_b}}{\overline{\Sigma d_r} - \overline{\Sigma d_b}}. \tag{D.16}$$

The chunks below compute Moran's I for the BM trait of the chapter and for a noise trait (drawn here with its own seed, so it differs from the noise trait in Section 13.4), and D for a binary trait derived by thresholding the BM trait at its median.

```
set.seed(22)
noise <- setNames(rnorm(Ntip(tree)), tips)
w <- 1 / cophenetic(tree); diag(w) <- 0
Moran.I(x_bm[tips], w)$observed     # phylogenetic Moran's I, BM trait
```

```
#> [1] 0.2375
```

```
Moran.I(noise, w)$observed          # ... and for noise (expected ~ -1/(n-1))
```

```
#> [1] -0.007312
```

```
bin_dat <- data.frame(lang = tips,
                      feature = as.integer(x_bm[tips] > median(x_bm[tips])))
cd_bin <- comparative.data(tree, bin_dat, names.col = "lang")
d_res <- phylo.d(cd_bin, binvar = feature, permut = 200)
round(c(D = unname(d_res$DEstimate), P_random = d_res$Pval1,
        P_brownian = d_res$Pval0), 3)
```

```
#>          D   P_random P_brownian
#>     -0.247      0.000      0.805
```

The BM trait has a clearly positive Moran's I and the noise trait a value near its null expectation. The binary feature derived from the BM trait has D near or below zero: it is significantly more clumped than random, and not distinguishable from the Brownian threshold expectation.

D.3.4 Independent contrasts

Take two sister tips i and j with trait values x_i, x_j and branch lengths v_i, v_j to their common ancestor k.

1. Compute the standardized contrast

$$u_{ij} = \frac{x_i - x_j}{\sqrt{v_i + v_j}}. \tag{D.17}$$

   Under BM the raw difference has variance $\sigma^2(v_i + v_j)$, so the standardized contrast has variance σ^2 whatever the branch lengths.
2. Estimate the value at node k as the weighted average $x_k = (x_i/v_i + x_j/v_j)/(1/v_i + 1/v_j)$. The weights favor the descendant on the shorter branch.
3. Lengthen the branch below k by $v_i v_j/(v_i + v_j)$ to account for the error in the estimate of x_k.
4. Prune i and j, treat k as a tip, and repeat.

A tree with n tips yields $n - 1$ contrasts, independent and identically distributed $\mathcal{N}(0, \sigma^2)$ under BM. The regression of the contrasts of y on those of x through the origin gives exactly the PGLS slope under BM (Rohlf, 2001; Blomberg et al., 2012), as the chunk below confirms for the data of Section 13.5.

```
set.seed(31)
x <- fastBM(tree)
y <- 2 + 0.6 * x + fastBM(tree, sig2 = 0.5)
fit_pic <- lm(pic(y, tree) ~ pic(x, tree) - 1)
dat_pic <- data.frame(lang = tips, x = x[tips], y = y[tips])
m_pic_gls <- gls(y ~ x, data = dat_pic,
                 correlation = corBrownian(1, tree, form = ~lang))
c(PIC = unname(coef(fit_pic)), PGLS_BM = unname(coef(m_pic_gls)["x"]))
```

```
#>     PIC PGLS_BM
#>  0.6405  0.6405
```

D.3.5 Generalized least squares and the lambda profile

For the model of Equation (13.5), the generalized least squares estimator and its covariance are

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{y}, \qquad \operatorname{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1}, \tag{D.18}$$

with

$$\hat{\sigma}^2 = \frac{(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})^\top \mathbf{V}^{-1} (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})}{n - p}. \tag{D.19}$$

When $\mathbf{V} = \mathbf{I}$, these reduce to the familiar OLS formulas. GLS is OLS on transformed data. Factor $\mathbf{V} = \mathbf{L}\mathbf{L}^\top$, for example by a Cholesky decomposition, and premultiply the model by $\mathbf{L}^{-1}$:

$$\mathbf{L}^{-1}\mathbf{y} = \mathbf{L}^{-1}\mathbf{X}\boldsymbol{\beta} + \mathbf{L}^{-1}\boldsymbol{\varepsilon}, \qquad \mathrm{Var}(\mathbf{L}^{-1}\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}. \tag{D.20}$$

The transformed residuals are independent with equal variance, so OLS on the transformed data is optimal (the Gauss-Markov theorem; Section A.11). The Galton simulation in Section 13.1 uses the upper-triangular factor $\mathbf{W}$ with $\mathbf{W}^\top\mathbf{W} = \mathbf{V}^{-1}$, which is an equivalent whitening.

When λ is estimated jointly, $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}^2$ follow from Equations (D.18) and (D.19) for each value of λ, with $\mathbf{V} = \mathbf{C}(\lambda)$. Substituting them into the log-likelihood gives the profile log-likelihood of λ (Freckleton et al., 2002):

$$\ell(\lambda) = -\frac{n}{2}\log(2\pi\hat{\sigma}^2_\lambda) - \frac{1}{2}\log\det\mathbf{C}(\lambda) - \frac{n}{2}, \tag{D.21}$$

with $\hat{\sigma}^2_\lambda$ computed with divisor n for ML (or $n-p$, with an adjustment, for REML). Maximizing this one-dimensional function is fast, and the profile curve gives a likelihood interval for λ, as plotted in Figure 13.5.

D.3.6 Pagel's independent model

With the joint states of two binary characters ordered $(0,0), (0,1), (1,0), (1,1)$, the independent model gives each character its own gain and loss rates, α_X, β_X and α_Y, β_Y, regardless of the state of the other character:

$$\mathbf{Q}_{\text{ind}} = \begin{pmatrix} \cdot & \alpha_Y & \alpha_X & 0 \\ \beta_Y & \cdot & 0 & \alpha_X \\ \beta_X & 0 & \cdot & \alpha_Y \\ 0 & \beta_X & \beta_Y & \cdot \end{pmatrix}. \tag{D.22}$$

It is the special case of the dependent model of Equation (13.6) with $q_{12} = q_{34} = \alpha_Y$, $q_{21} = q_{43} = \beta_Y$, $q_{13} = q_{24} = \alpha_X$ and $q_{31} = q_{42} = \beta_X$: four constraints, hence the four degrees of freedom of the likelihood-ratio test. The zeros on the anti-diagonal encode the assumption that the two characters never change at exactly the same instant. Two caveats about the χ^2_4 reference distribution are worth knowing. It is an asymptotic approximation that can be poor for small trees and for rates near zero (on the boundary of the parameter space), and the test is a test of the model, not of the number of independent co-occurring changes (Section 13.7.3). A parametric bootstrap, simulating data under the fitted independent model and refitting both models, gives a better calibrated p-value.

D.3.7 Phylogenetic GLMM in brms

The Gaussian PGLMM of Section 13.9.2 can also be fitted in **brms**, which passes the model to Stan. The code below is not run here, because it requires compiling a Stan model. We ran it with brms 2.23 and rstan 2.32, which took a few minutes, most of them spent compiling. With the default `adapt_delta = 0.8`, the first model produced about 300 divergent transitions and an effective sample size below 100 for the residual standard deviation. Phylogenetic and residual variance are hard to separate with 60 languages, and the posterior has the funnel shape that Stan's sampler finds difficult. With `adapt_delta = 0.99`, as below, there were no divergent transitions and every effective sample size was above 350. The estimated slope (posterior mean -0.27, 95% interval -0.54 to 0.00) agrees with the PGLS and **MCMCglmm** estimates. The second model needs a spatial correlation matrix `S` for the languages of `tree100`, with row and column names matching `lang`.

```
library(brms)
A <- vcv(tree, corr = TRUE)
m_brms <- brm(complexity ~ logpop + (1 | gr(lang, cov = A)),
              data = langs, data2 = list(A = A), family = gaussian(),
              chains = 4, iter = 4000, cores = 4,
              control = list(adapt_delta = 0.99))
# binary response, with a spatial random effect alongside the phylogenetic one
# (S: a spatial correlation matrix, e.g. exp(-distance / range))
m_tone <- brm(tone ~ humid + (1 | gr(lang, cov = A)) + (1 | gr(lang2, cov = S)),
              data = transform(dtone, lang2 = lang),
              data2 = list(A = vcv(tree100, corr = TRUE), S = S),
              family = bernoulli())
```

D.3.8 The threshold model in MCMCglmm

The chunk below recreates the 100-language tree and the simulated tone data of Section 13.9.1 and fits a threshold model, a probit PGLMM with the residual variance fixed at 1, in **MCMCglmm**.

```
set.seed(7)
tree100 <- pbtree(n = 100, scale = 3)
set.seed(11)
humid <- rTrait(n = 1, phy = tree100, model = "BM",
                parameters = list(sigma2 = 0.4))
tone  <- rbinTrait(n = 1, phy = tree100, beta = c(-0.5, 1), alpha = 0.5,
                   X = cbind(1, humid))
dtone <- data.frame(lang = tree100$tip.label, humid = humid, tone = tone,
                    row.names = tree100$tip.label)
set.seed(6)
Ainv100 <- inverseA(tree100, nodes = "TIPS", scale = TRUE)$Ainv
prior_t <- list(G = list(G1 = list(V = 1, nu = 1000, alpha.mu = 0,
                                   alpha.V = 1)),
                R = list(V = 1, fix = 1))
m_thr <- MCMCglmm(tone ~ humid, random = ~lang, ginverse = list(lang = Ainv100),
                  family = "threshold", data = dtone, prior = prior_t,
                  nitt = 23000, burnin = 3000, thin = 20, verbose = FALSE)
summary(m_thr)$solutions
```

```
#>             post.mean l-95% CI u-95% CI eff.samp pMCMC
#> (Intercept)    -1.292  -3.1214   0.1431    861.5 0.104
#> humid           1.353   0.5693   2.4679    433.3 0.001
```

The coefficient of `humid` is on the liability (probit) scale. It is therefore not directly comparable with the logistic coefficients of Section 13.9.1, which are larger by a factor of roughly 1.6 to 1.8 for the same effect (the ratio of the standard deviations of the logistic and standard normal distributions is $\pi/\sqrt{3} \approx 1.81$, and approximations based on matching slopes at the center give about 1.6 to 1.7). The prior used here is a parameter-expanded prior on the phylogenetic variance; with the residual variance fixed, the latent scale is identified only up to that convention, so variance components from threshold and logistic models are not comparable either.

D.3.9 A spatiophylogenetic model by maximum likelihood

The chunk below recreates the simulated locations and trait of Section 13.10 and fits Equation (13.12) by maximum likelihood, holding the range ρ fixed, for three nested covariance structures: phylogeny with noise, space with noise, and all three components. The intercept is profiled out by GLS for each value of the variance components, which are optimized on the log scale.

```
set.seed(99)
lon <- fastBM(tree, sig2 = 1) + rnorm(Ntip(tree), 0, 0.3)
lat <- fastBM(tree, sig2 = 1) + rnorm(Ntip(tree), 0, 0.3)
D <- as.matrix(dist(cbind(lon, lat)))
S <- exp(-D / 1)                                # spatial correlation, range 1
Cs <- C / max(diag(C))                                 # phylogenetic correlation
L_s <- t(chol(S + diag(1e-8, nrow(S))))
a_phy <- fastBM(tree, sig2 = 1 / max(diag(C)))  # phylogenetic part, variance 1
s_sp  <- drop(L_s %*% rnorm(Ntip(tree)))               # spatial part, variance 1
trait <- 0.8 * a_phy + 0.8 * s_sp + rnorm(Ntip(tree), 0, 0.5)

negloglik <- function(logv, comps, y) {
  V <- Reduce(`+`, Map(`*`, exp(logv), comps))
  R <- chol(V); Xm <- matrix(1, length(y), 1)
  Ri <- backsolve(R, diag(length(y)))                  # V^-1 = Ri Ri'
  Vinv <- Ri %*% t(Ri)
  b <- solve(t(Xm) %*% Vinv %*% Xm, t(Xm) %*% Vinv %*% y)
  r <- y - Xm %*% b
  0.5 * (2 * sum(log(diag(R))) + t(r) %*% Vinv %*% r + length(y) * log(2 * pi))
}
I_n <- diag(Ntip(tree))
fits <- list(phylogeny_only = list(phylo = Cs, noise = I_n),
             space_only     = list(space = S, noise = I_n),
             both           = list(phylo = Cs, space = S, noise = I_n))
res <- t(sapply(fits, function(comps) {
  o <- optim(rep(log(0.3), length(comps)), negloglik, comps = comps,
             y = trait[tips])
  share <- setNames(rep(NA, 3), c("phylo", "space", "noise"))
  share[names(comps)] <- exp(o$par) / sum(exp(o$par))
  c(AIC = 2 * o$value + 2 * (length(comps) + 1), share)
```

```
}))
knitr::kable(res, digits = 2,
             caption = "Maximum-likelihood fits of phylogeny-only, space-only and
             spatiophylogenetic models to a simulated trait, with the estimated
             share of variance attributed to each component.")
```

Table D.1: Maximum-likelihood fits of phylogeny-only, space-only and spatiophylogenetic models to a simulated trait, with the estimated share of variance attributed to each component.

	AIC	phylo	space	noise
phylogeny_only	181.6	0.43	NA	0.57
space_only	181.0	NA	0.68	0.32
both	173.2	0.37	0.63	0.00

In this replicate the full model, with both phylogenetic and spatial components, is preferred by AIC (Table D.1). It apportions the variance between phylogeny and space, while the noise component shrinks toward zero; the true shares were about 0.42, 0.42 and 0.16. Maximum-likelihood estimates of variance components are biased downward in small samples and can sit on the boundary at zero, as the noise component nearly does here; the Bayesian fit in the chapter reports the full posterior instead, which makes the uncertainty in the shares visible.

D.4 Chapter 14: Language Trees as Scaffolds for Cultural Evolution

D.4.1 Borrowing as a transformation of the covariance matrix

In the horizontal transmission simulation of Section 14.14, each society i replaces a fraction h of its trait value with the value of a contact partner $p(i)$: $x_i' = (1-h)\,x_i + h\,x_{p(i)}$. In matrix form, $\mathbf{x}' = \mathbf{B}\mathbf{x}$ with

$$\mathbf{B} = (1-h)\,\mathbf{I} + h\,\mathbf{P}, \tag{D.23}$$

where $\mathbf{P}$ has a single 1 in each row, in the column of that society's partner. If $\mathbf{x} \sim \mathcal{N}(\mu\mathbf{1}, \sigma^2\mathbf{C})$ under Brownian motion on the tree, then $\mathbf{x}' \sim \mathcal{N}(\mu\mathbf{1}, \sigma^2\mathbf{B}\mathbf{C}\mathbf{B}^\top)$, because each row of $\mathbf{B}$ sums to one. A PGLS that uses $\mathbf{C}$ is therefore misspecified, and how badly depends on how different $\mathbf{B}\mathbf{C}\mathbf{B}^\top$ is from $\mathbf{C}$. When partners are close relatives, $C_{i,p(i)} \approx C_{ii}$ and the transformation changes little; when partners are distant, it adds covariance where $\mathbf{C}$ has almost none, and the regression treats similar societies as independent observations.

If the contact network and the borrowing fraction were known, a GLS with $\mathbf{V} = \mathbf{B}\mathbf{C}\mathbf{B}^\top$ would be correctly specified. The chunk below recreates the tree and the geographic partners of the chapter, and compares the false positive rates of PGLS with $\mathbf{C}$ and of GLS with $\mathbf{B}\mathbf{C}\mathbf{B}^\top$ at $h = 0.3$.

```
set.seed(12)
htree <- pbtree(n = 60, scale = 5)
htree$tip.label <- sprintf("S%02d", 1:60)
Ch <- vcv(htree)
geo <- as.matrix(dist(matrix(runif(120), 60, 2))); diag(geo) <- Inf
partner <- apply(geo, 1, which.min)                 # nearest neighbor in space
```

```
h <- 0.3
P <- matrix(0, 60, 60); P[cbind(1:60, partner)] <- 1
B <- (1 - h) * diag(60) + h * P
gls_p <- function(y, x, V) {
  Wv <- chol(solve(V))
  coef(summary(lm(Wv %*% y ~ Wv %*% cbind(1, x) - 1)))[2, 4]
}
V_borrow <- B %*% Ch %*% t(B)
set.seed(13)
p <- replicate(300, {
  x <- drop(B %*% fastBM(htree)); y <- drop(B %*% fastBM(htree))
  c(PGLS_tree = gls_p(y, x, Ch), GLS_borrowing = gls_p(y, x, V_borrow))
})
rowMeans(p < 0.05)                                  # false positive rates
```

```
#>     PGLS_tree GLS_borrowing
#>        0.6067        0.0600
```

The GLS that knows the contact structure has close to the nominal 5% false positive rate. In practice the network and h are unknown, so they have to be modeled: neighbor-graph models (Towner et al., 2012) and spatial covariance components (Section 13.10) are approximations to $\mathbf{BCB}^{\top}$ in which the contact structure is represented by geographic proximity and its strength is estimated from the data.

Appendix E

Technical Notes: Methods from Evolutionary Biology

This appendix collects the technical material behind Chapters 15 to 17: simulation code, derivations and complete input files that would interrupt the argument of those chapters. Each section corresponds to one chapter and can be read, and run, on its own; every code chunk recreates the data it needs. Chapter 18 has no technical notes.

E.1 Chapter 15: Critiques, Pitfalls and Controversies

E.1.1 A robustness study for the forty-language scenario

The worked scenario of Section 15.13 asks whether borrowing between two subgroups could move a disputed language from one to the other, and whether it changes the amount of change on the tree, on which dates depend. This section gives a small simulation study of that question. In the vocabulary of Section 6.10 it is a test of robustness under misspecification: the data are generated with borrowing and uneven missingness, and analyzed with methods that assume neither. In a real application the tree, the rates and the pattern of missing cells would be taken from the fitted model and the real data, and the inference would be the full Bayesian analysis; here, to keep the code fast, we simulate the tree and use neighbor-joining and maximum likelihood branch lengths on a fixed topology.

The simulator works on meanings, as in Chapter 17: each meaning has its own replacement rate, drawn from a gamma distribution, and every replacement creates a new cognate class. Missing cells are concentrated in twelve poorly documented languages. Borrowing copies the word of a random member of a donor subgroup into members of a recipient subgroup, with a higher rate for one *disputed* language of the recipient subgroup.

```
library(phangorn)
set.seed(1510)
n_lang <- 40
tree40 <- phytools::pbtree(n = n_lang, scale = 1,
                           tip.label = sprintf("L%02d", seq_len(n_lang)))
```

```
# two non-sister clades of 5 to 10 languages with a stem branch long
# enough to be recoverable: recipient A and donor B
clades <- lapply(Ntip(tree40) + seq_len(tree40$Nnode), function(nd)
  tree40$tip.label[unlist(Descendants(tree40, nd, "tips"))])
sizes <- lengths(clades)
nodes <- Ntip(tree40) + seq_len(tree40$Nnode)
stem <- tree40$edge.length[match(nodes, tree40$edge[, 2])]
cand <- clades[sizes >= 5 & sizes <= 10 & !is.na(stem) & stem > 0.05]
pick <- NULL
for (i in seq_along(cand)) for (j in seq_along(cand)) {
  if (is.null(pick) && i < j && !length(intersect(cand[[i]], cand[[j]])) &&
      !is.monophyletic(tree40, c(cand[[i]], cand[[j]])))
    pick <- c(i, j)
}
group_A <- cand[[pick[1]]]; group_B <- cand[[pick[2]]]
disputed <- group_A[1]
poor <- sample(setdiff(tree40$tip.label, c(group_A, group_B)), 12)

sim_meanings <- function(tree, n_mean = 200, mean_rate = 2, shape = 1) {
  tree <- reorder(tree, "cladewise")
  rates <- rgamma(n_mean, shape, shape / mean_rate)
  st <- matrix(0L, Ntip(tree) + tree$Nnode, n_mean)
  next_id <- 1L
  for (i in seq_len(nrow(tree$edge))) {
    p <- tree$edge[i, 1]; ch <- tree$edge[i, 2]
    hit <- rpois(n_mean, rates * tree$edge.length[i]) > 0
    st[ch, ] <- st[p, ]
    st[ch, hit] <- next_id + seq_len(sum(hit)) - 1L
    next_id <- next_id + sum(hit)
  }
  m <- st[seq_len(Ntip(tree)), ]
  rownames(m) <- tree$tip.label
  m
}
borrow <- function(m, b) {          # A copies from B; disputed copies more
  for (j in seq_len(ncol(m))) {
    for (l in group_A) {
      p <- if (l == disputed) 2 * b else b / 2
      if (runif(1) < p) m[l, j] <- m[sample(group_B, 1), j]
    }
  }
  m
}
add_missing <- function(m) {        # uneven documentation
  for (l in rownames(m)) {
    p <- if (l %in% poor) 0.5 else 0.1
    m[l, runif(ncol(m)) < p] <- NA
  }
```

```
  m
}
to_binary <- function(m) {          # one 0/1 column per cognate class
  cols <- lapply(seq_len(ncol(m)), function(j) {
    cl <- sort(unique(na.omit(m[, j])))
    x <- sapply(cl, function(k) ifelse(m[, j] == k, "1", "0"))
    x <- matrix(x, nrow(m))
    x[is.na(m[, j]), ] <- "?"
    x
  })
  b <- do.call(cbind, cols)
  rownames(b) <- rownames(m)
  b[, colSums(b == "1") < nrow(b), drop = FALSE]
}
```

For each level of borrowing, we simulate ten datasets, infer a neighbor-joining tree from Hamming distances, and record four quantities: the normalized RF distance to the true tree, whether the recipient subgroup is recovered as a clade, whether the disputed language is placed with the donor subgroup, and the total length of the tree under a binary model with branch lengths optimized on the inferred topology (a stand-in for the amount of change from which dates are derived).

```
one_run <- function(b) {
  m <- add_missing(borrow(sim_meanings(tree40), b))
  d <- phyDat(to_binary(m), type = "USER", levels = c("0", "1"),
              ambiguity = "?")
  nj <- NJ(dist.hamming(d))
  fit <- optim.pml(pml(nj, d), optEdge = TRUE, optNni = FALSE,
                   control = pml.control(trace = 0))
  c(b = b,
    RF = RF.dist(nj, tree40, normalize = TRUE, rooted = FALSE),
    A_is_clade = is.monophyletic(unroot(nj), group_A),
    disputed_with_B = is.monophyletic(unroot(nj), c(group_B, disputed)),
    tree_length = sum(fit$tree$edge.length))
}
res <- do.call(rbind, lapply(c(0, 0.1, 0.2, 0.3), function(b)
  t(replicate(10, one_run(b)))))
round(aggregate(cbind(RF, A_is_clade, disputed_with_B, tree_length) ~ b,
                data = as.data.frame(res), FUN = mean), 3)
```

```
#>     b    RF A_is_clade disputed_with_B tree_length
#> 1 0.0 0.392        1.0             0.0       1.702
#> 2 0.1 0.411        1.0             0.0       1.703
#> 3 0.2 0.419        0.9             0.0       1.743
#> 4 0.3 0.454        0.0             0.9       1.788
```

The output shows the pattern of Section 15.4 in a larger tree. The overall RF distance changes little. The recipient subgroup is recovered in every replicate without borrowing, but once borrowing is strong enough the disputed language is placed with the donors instead, and the recipient subgroup is no longer a clade. The total length of the tree also changes, here growing, because borrowed

words that conflict with the inferred tree must be explained as extra changes; in a dated analysis the effect on node ages depends on where the conflict falls, which is why the check must be run with the real model. In a real study, a borrowing rate of the size suggested by the flagged loans would be simulated, and if it moves the disputed language as often as the real analysis places it in the other subgroup, the lexical data cannot decide the question. The same design, with the Bayesian analysis in place of neighbor-joining, gives the corresponding check for dates.

E.2 Chapter 16: The Biologist's Toolbox

This section collects the simulators and derivations behind the discussion of sorting and borrowing in Chapter 16. Everything below runs in a fresh R session.

E.2.1 A coalescent simulator for concept trees

`sim_msc_tree()` simulates one concept-level genealogy inside a language tree whose branch lengths are in coalescent units. Each lineage is stored as a Newick string together with its height above the present. Working from the youngest internal node of the language tree to the oldest, the lineages entering each branch coalesce at rate $k(k-1)/2$ (Equation (16.1)) until the top of the branch is reached; the survivors are passed to the parent branch, and the root branch is open-ended, so that all lineages eventually coalesce.

```
library(phangorn)
library(phytools)
set.seed(1601)

# Coalesce a pool of lineages from height h_start until h_end (or until one
# lineage is left). Each lineage is a Newick string plus its height.
coalesce_branch <- function(pool, h_start, h_end) {
  t <- h_start
  while (length(pool) > 1) {
    k <- length(pool)
    t <- t + rexp(1, rate = k * (k - 1) / 2)
    if (t > h_end) break
    pick <- sample(k, 2)
    a <- pool[[pick[1]]]; b <- pool[[pick[2]]]
    merged <- list(nwk = sprintf("(%s:%.6f,%s:%.6f)", a$nwk, t - a$h,
                                 b$nwk, t - b$h), h = t)
    pool <- c(pool[-pick], list(merged))
  }
  pool
}

# Simulate one concept (gene) tree inside an ultrametric language tree `sp`
sim_msc_tree <- function(sp) {
  ntip <- Ntip(sp)
  height <- max(node.depth.edgelength(sp)) - node.depth.edgelength(sp)
  pools <- vector("list", ntip + sp$Nnode)
  for (i in seq_len(ntip))
```

```
    pools[[i]] <- list(list(nwk = sp$tip.label[i], h = 0))
  internal <- (ntip + 1):(ntip + sp$Nnode)
  for (node in internal[order(height[internal])]) {
    kids <- sp$edge[sp$edge[, 1] == node, 2]
    pools[[node]] <- do.call(c, lapply(kids, function(k)
      coalesce_branch(pools[[k]], height[k], height[node])))
  }
  root_pool <- coalesce_branch(pools[[ntip + 1]], height[ntip + 1], Inf)
  read.tree(text = paste0(root_pool[[1]]$nwk, ";"))
}
```

We check the simulator against Equation (16.2) for three languages. In the vocabulary of Section 6.10 this is implementation validation: it shows that the code computes what the coalescent says, and nothing about whether words follow the coalescent.

```
T_grid <- c(0.02, 0.1, 0.25, 0.5, 0.75, 1, 1.5, 2, 3)
p_sim <- sapply(T_grid, function(T) {
  sp3 <- read.tree(text = sprintf("((A:1,B:1):%f,C:%f);", T, 1 + T))
  mean(replicate(400, is.monophyletic(sim_msc_tree(sp3), c("A", "B"))))
})
curve(1 - 2/3 * exp(-x), from = 0, to = 3.2, ylim = c(0, 1), las = 1,
      xlab = "Internal branch length T (coalescent units)",
      ylab = "P(concept tree matches language tree)")
points(T_grid, p_sim, pch = 19)
abline(h = 1/3, lty = 2)
```

The simulated proportions lie on the analytical curve (Figure E.1). When the internal branch is short in coalescent units, concept trees are only weakly informative about the language tree.

E.2.2 The token model and its coalescent limit

The model. A speech community has N speakers, and each speaker uses one word, of some cognate class, for a given concept. In each generation every learner chooses a model in the previous generation and copies its word; with probability u the learner instead innovates a word of a new class (the infinitely-many-alleles assumption: every innovation is new). Let ν_i be the number of learners who copy speaker i, so that $\sum_i \nu_i = N$. Under neutral copying each learner chooses a model uniformly at random, $(\nu_1, \dots, \nu_N)$ is multinomial with equal probabilities, and the model is the Wright-Fisher model (Wright, 1931). More generally, if the joint distribution of the ν_i is exchangeable (no speaker is favoured *a priori*) and the same in every generation, the model is a Cannings model, which includes the Moran model (Moran, 1950) and models with highly variable numbers of imitators.

Pair coalescence. Follow two tokens back one generation. They have the same parent token with probability

$$c_N = \frac{\mathbb{E}[\nu_1(\nu_1 - 1)]}{N - 1}, \tag{E.1}$$

which is $1/N$ under neutral Wright-Fisher copying. The time back to their common ancestor is geometric with mean $1/c_N$ generations, and $N_e = 1/c_N$ is the coalescent effective size of the

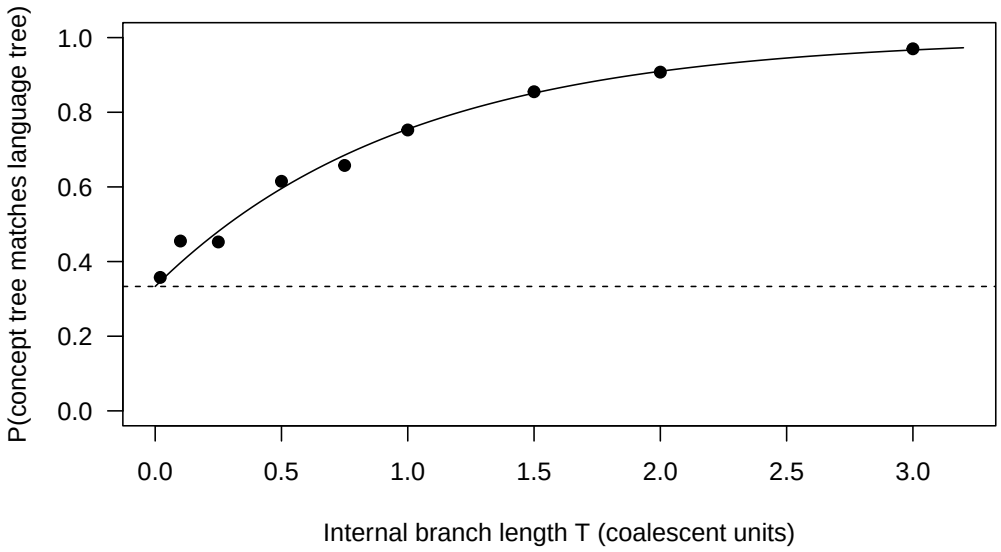


Figure E.1: Probability that a concept tree matches the language tree ((A,B),C) as a function of the internal branch length T in coalescent units. Points: proportion of 400 simulated concept trees with A and B as sisters. Curve: the analytical result of Equation
eqrefeq:biology-toolkit-ils. The dashed line marks 1/3, the value expected if the concept trees carried no information about the language tree.

community.

The Kingman limit. If $c_N \to 0$ as N grows and simultaneous mergers of three lineages are rare compared with mergers of two, in the sense that

$$\frac{\mathbb{E}[\nu_1(\nu_1-1)(\nu_1-2)]}{N^2 c_N} \to 0, \tag{E.2}$$

then the genealogy of a sample of n tokens, with time measured in units of $1/c_N$ generations, converges to Kingman's coalescent, in which each pair of lineages merges at rate 1 and the total rate with k lineages is $k(k-1)/2$ (Kingman, 1982; Möhle and Sagitov, 2001). Neutral Wright-Fisher and Moran copying satisfy the condition. If the condition fails because single speakers are occasionally copied by a non-negligible fraction of the community, the limit is a Λ-coalescent, in which several lineages can merge at once (Möhle and Sagitov, 2001; Eldon and Wakeley, 2006).

Innovation on the genealogy. Innovations occur on each lineage with probability u per generation, that is, at rate $uN_e = \theta/2$ per coalescent time unit, with $\theta = 2N_e u$. Because every innovation creates a new class, the cognate classes of a sample are the partition obtained by cutting the genealogy at the innovations, and their distribution is given by Ewens' sampling formula. Two consequences are used in the chapter. Two tokens belong to the same class with probability $1/(1+\theta)$, and the expected number of classes among n tokens is

$$\mathbb{E}[K_n] = \sum_{i=0}^{n-1} \frac{\theta}{\theta + i}. \tag{E.3}$$

The replacement rate. A new class starts in one speaker. Under neutral copying its probability of eventually being used by everyone is $1/N$, and about Nu new classes arise per generation, so the class that is fixed in a community is replaced at rate $Nu \times 1/N = u$ per generation, independently of N. If a new class has a fixed advantage s in copying, its fixation probability in the haploid Wright-Fisher model is approximately $(1-e^{-2s})/(1-e^{-2Ns})$, and the replacement rate $Nu(1-e^{-2s})/(1-e^{-2Ns})$ grows with N when $s > 0$ and shrinks with N when $s < 0$. Whether rates of lexical change depend on community size (Section 9.10) is therefore a question about whether replacement is neutral.

Several communities. A language split divides a community into two that copy only from themselves. Two tokens from different daughter communities cannot share a parent until their ancestry reaches the ancestral community, so, backward in time, lineages coalesce within each branch of the language tree according to the coalescent of that branch and merge into a common pool at the split. This is the multispecies coalescent (Rannala and Yang, 2003), with the length of a branch in coalescent units equal to its length in generations divided by its N_e (or $\int dt/N_e(t)$ if the size changes). The triplet result of Equation (16.2) follows as in the chapter. If a learner in community i copies a model in community j with probability m_{ij}, then, backward in time, a lineage in i moves to j at rate m_{ij} per generation, and the genealogy is a structured coalescent with migration rather than the MSC. A founder event, in which a daughter community starts from a few speakers, is a short branch of very small N_e, and therefore a burst of coalescence at the split.

Conformity. Under the conformist rule of the chapter, a learner meets a first model and two further models; if the two further models agree, the learner adopts their word with probability c, and otherwise copies the first model. For a class at frequency x, the probability that a learner

adopts it is

$$g(x) = c\,x^2 + \left(1 - c\left[x^2 + (1-x)^2\right]\right)x = x + c\,x(1-x)(2x-1), \tag{E.4}$$

when there are two classes. The term $c\,x(1-x)(2x-1)$ is frequency-dependent selection: it pushes a class below frequency $1/2$ down and a class above $1/2$ up. A new innovation starts at frequency $1/N$ and is selected against until it happens to drift towards $1/2$, so its fixation probability is far below $1/N$, polymorphism is lower than Ewens' formula predicts, and the replacement rate is lower than u. The genealogy of tokens is affected less, because the rule chooses among speakers of the same class, as the chapter's simulation shows. In general, when copying depends on the classes of the models, the genealogy and the classes are no longer independent, and the genealogy is not Kingman's coalescent; for selection it can be represented by the ancestral selection graph (Neuhauser and Krone, 1997).

Prestige. Under the prestige rule, in each generation with probability ε a fraction ψ of the learners copy one speaker. Two tokens then share a parent with probability

$$c_N \approx \frac{1}{N} + \varepsilon\psi^2, \tag{E.5}$$

and three share a parent with probability about $\varepsilon\psi^3$, which is not negligible compared with c_N when $\varepsilon\psi^2$ is comparable to $1/N$. Condition (E.2) fails, and the genealogy is the Λ-coalescent of Eldon and Wakeley (2006). Because one speaker is imitated for every concept at once, mergers in different concepts happen in the same generations, so concepts are not independent loci.

Checking the simulator. The functions below are those of the chapter. We run the neutral model for several innovation probabilities and compare the number of classes per concept among all N speakers with Equation (E.3), and we compare the prestige curve with the geometric prediction $1-(1-c_N)^t$ from Equation (E.5). Both are implementation checks.

```
# Speech community: speakers in rows, concepts in columns; `cls` holds the
# cognate class each speaker uses, `tag` the ancestor of the token at a
# reference time, so that genealogies can be followed forward in time.
choose_models <- function(cls, bias, conform = 0.5, eps = 0.1, psi = 0.5) {
  N <- nrow(cls); K <- ncol(cls)
  col <- rep(seq_len(K), each = N)
  draw <- function() sample.int(N, N * K, replace = TRUE)
  par <- draw()                                  # neutral: a random model
  if (bias == "conformist") {          # two further models that agree win
    p2 <- draw(); p3 <- draw()
    agree <- cls[cbind(p2, col)] == cls[cbind(p3, col)]
    use <- agree & runif(N * K) < conform
    par[use] <- p2[use]
  }
  if (bias == "prestige" && runif(1) < eps) {   # one speaker copied by many
    fans <- rep(runif(N) < psi, K)
    par[fans] <- sample.int(N, 1)
  }
  cbind(par, col)                      # (model, concept) for every learner
}
learn <- function(state, u, bias, ...) {
  idx <- choose_models(state$cls, bias, ...)
```

```
  d <- dim(state$cls)
  cls <- matrix(state$cls[idx], d[1], d[2])
  inn <- runif(length(cls)) < u                 # innovation: a new class
  cls[inn] <- state$next_id + seq_len(sum(inn))
  list(cls = cls, tag = matrix(state$tag[idx], d[1], d[2]),
       next_id = state$next_id + sum(inn))
}
pair_identity <- function(m) {    # P(two distinct speakers share a label)
  N <- nrow(m)
  mean(apply(m, 2, function(x) {
    n <- tabulate(match(x, unique(x)))
    sum(n * (n - 1)) / (N * (N - 1))
  }))
}
run_bias <- function(bias, N = 30, K = 400, u = 0.01, t_max = 2 * N,
                     every = 3, ...) {
  st <- list(cls = matrix(1L, N, K), tag = matrix(1L, N, K), next_id = 1L)
  for (i in seq_len(4 * N)) st <- learn(st, u, bias, ...)      # burn-in
  st$tag[] <- seq_len(N)                          # the reference time
  out <- NULL
  for (t in seq_len(t_max)) {
    st <- learn(st, u, bias, ...)
    if (t %% every == 0)
      out <- rbind(out, c(T = t / N, coal = pair_identity(st$tag)))
  }
  list(curve = out,
       classes = mean(apply(st$cls, 2, function(x) length(unique(x)))))
}
```

```
set.seed(1608)
N <- 30
ewens <- function(theta, n) sum(theta / (theta + 0:(n - 1)))
check_u <- sapply(c(0.002, 0.005, 0.01, 0.02), function(u) {
  s <- run_bias("neutral", N = N, K = 300, u = u, t_max = 3)
  c(u = u, simulated = s$classes, ewens = ewens(2 * N * u, N))
})
round(t(check_u), 3)
```

```
#>          u simulated ewens
#> [1,] 0.002     1.430 1.454
#> [2,] 0.005     2.027 2.069
#> [3,] 0.010     3.047 2.967
#> [4,] 0.020     4.640 4.456
```

```
prest <- run_bias("prestige", N = N, K = 300)
c_N <- 1 / N + 0.1 * 0.5^2
round(cbind(prest$curve, geometric = 1 - (1 - c_N)^(prest$curve[, "T"] * N),
            kingman_N = 1 - exp(-prest$curve[, "T"]))[c(5, 10, 15, 20), ], 3)
```

```
#>        T  coal geometric kingman_N
#> [1,] 0.5 0.536     0.594     0.393
#> [2,] 1.0 0.713     0.835     0.632
#> [3,] 1.5 0.930     0.933     0.777
#> [4,] 2.0 0.962     0.973     0.865
```

The neutral simulations reproduce Ewens' expectation, and the prestige curve follows the geometric prediction with $c_N \approx 0.058$, an effective size of about 17 for a community of 30 speakers, not the Kingman curve for $N = 30$. The occasional large jumps in the prestige curve are the multiple mergers.

E.2.3 The lexical D statistic

Null expectation. Take four languages with tree $(((P_1, P_2), P_3), O)$, one token sampled from each, and an infinitely-many-classes model of innovation. Write X_1, X_2, X_3, X_O for the classes. A meaning is an ABBA pattern if $X_2 = X_3 \neq X_1 = X_O$ and a BABA pattern if $X_1 = X_3 \neq X_2 = X_O$. Suppose that P_1 and P_2 are founded by the same split from the same ancestral community, follow the same copying rule after the split, are sampled in the same way, and exchange no words with P_3 or O. Then the joint distribution of (X_1, X_2, X_3, X_O) is unchanged when X_1 and X_2 are swapped, and the swap turns every ABBA pattern into a BABA pattern. Hence $\mathbb{E}[n_{\text{ABBA}}] = \mathbb{E}[n_{\text{BABA}}]$, and D of Equation (16.3) has expectation zero.

The argument needs exchangeability of P_1 and P_2, not neutral copying or a Kingman genealogy, which is why the test survives conformity and prestige when both sister languages are subject to them. It does not need equal replacement rates either: both patterns require that neither sampled token acquired a new class after the split of P_1 and P_2, so a difference in innovation rates multiplies both expected counts by the same factor. It fails if the sister languages differ in their copying rules (a conformist P_1 retains the commonest ancestral class more often), in the way they were documented (more synonyms recorded for P_2 give it more chances to share), if the ancestors of P_2 and P_3 were neighbouring dialects of a structured ancestral community, and, by design, if P_2 or P_1 borrowed from P_3.

Standard error. Under the null, each informative meaning is ABBA or BABA with probability $1/2$, so with $n = n_{\text{ABBA}} + n_{\text{BABA}}$ independent informative meanings, n_{ABBA} is binomial and D has standard error $1/\sqrt{n}$. Meanings linked by polysemy, by shared semantic fields or by prestige effects are not independent; a delete-one-block jackknife over B blocks of meanings then gives

$$\widehat{\text{SE}}(D)^2 = \frac{B-1}{B} \sum_{b=1}^{B} \left(D_{(-b)} - \bar{D}\right)^2, \tag{E.6}$$

where $D_{(-b)}$ is the statistic computed without block b and $\bar{D}$ is their mean.

A simulation. We simulate concept trees for four languages with `sim_msc_tree()`, place innovations on each concept tree as a Poisson process with rate $\theta/2 = 0.5$ per coalescent unit, and read off the classes. In a proportion b of meanings, P_2 borrows its word from P_3 (using the regrafting of the chapter). The function `lexical_D()` computes the patterns from a long table with the CLDF columns used in Chapter 17, so it can be applied unchanged to real data. This is a parameter-recovery exercise for the test.

```r
set.seed(1609)
lang4 <- read.tree(text = "(((P1:1,P2:1):0.5,P3:1.5):1,O:2.5);")

# classes at the tips: cut the concept tree at Poisson innovations
classes_on_tree <- function(g, rate = 0.5) {
  hit <- rpois(nrow(g$edge), rate * g$edge.length) > 0
  lab <- integer(Ntip(g) + g$Nnode)       # class 0 at the root
  for (i in seq_len(nrow(g$edge))) {      # edges in preorder
    lab[g$edge[i, 2]] <- if (hit[i]) g$edge[i, 2] else lab[g$edge[i, 1]]
  }
  setNames(lab[seq_len(Ntip(g))], g$tip.label)
}
borrow_into <- function(g, recipient, donor, t_contact = 0.3) {
  g2 <- drop.tip(g, recipient)
  bind.tip(g2, recipient, where = which(g2$tip.label == donor),
           position = t_contact, edge.length = t_contact)
}
simulate_cognate_table <- function(n_meanings, b) {
  do.call(rbind, lapply(seq_len(n_meanings), function(i) {
    g <- reorder(sim_msc_tree(lang4), "cladewise")
    if (runif(1) < b)
      g <- reorder(borrow_into(g, "P2", "P3"), "cladewise")
    x <- classes_on_tree(g)
    data.frame(Language_ID = names(x), Parameter_ID = sprintf("C%03d", i),
               Cognateset_ID = paste0(i, "-", x))
  }))
}
lexical_D <- function(cog, p1, p2, p3, o, n_blocks = 20) {
  w <- tapply(cog$Cognateset_ID, list(cog$Parameter_ID, cog$Language_ID),
              function(z) z[1])           # one form per language and meaning
  w <- w[stats::complete.cases(w[, c(p1, p2, p3, o)]), , drop = FALSE]
  abba <- w[, p2] == w[, p3] & w[, p1] == w[, o] & w[, p1] != w[, p2]
  baba <- w[, p1] == w[, p3] & w[, p2] == w[, o] & w[, p1] != w[, p2]
  d_of <- function(k) (sum(abba[k]) - sum(baba[k])) /
                      (sum(abba[k]) + sum(baba[k]))
  blocks <- cut(seq_len(nrow(w)), n_blocks, labels = FALSE)
  d_jack <- sapply(seq_len(n_blocks), function(b) d_of(blocks != b))
  se <- sqrt((n_blocks - 1) / n_blocks *
             sum((d_jack - mean(d_jack))^2))
  D <- d_of(rep(TRUE, nrow(w)))
  c(ABBA = sum(abba), BABA = sum(baba), D = D, SE = se, z = D / se)
}
round(rbind(
  `no borrowing` = lexical_D(simulate_cognate_table(1000, 0),
                             "P1", "P2", "P3", "O"),
  `b = 0.1` = lexical_D(simulate_cognate_table(1000, 0.1),
                        "P1", "P2", "P3", "O")), 3)
```

```
#>              ABBA BABA  D  SE   z
#> no borrowing    0    1 -1 NaN NaN
#> b = 0.1         3    0  1   0 Inf
```

Without borrowing the two patterns are about equally frequent and $|z|$ is small; with a tenth of the meanings borrowed from P_3 into P_2, ABBA patterns are in clear excess. Real applications need the robustness studies described in the chapter, because the simulation shares every assumption of the null model except the borrowing.

E.3 Chapter 17: A Complete Workflow in R

This section holds the code that Chapter 17 uses but does not print in full: two plotting helpers, an annotated excerpt of the BEAST 2 input file, and the complete RevBayes script. The input files and the output of every run are kept in the folder `software-runs/` of the book's repository, whose `README.md` lists the program versions, run times and diagnostics.

E.3.1 Plotting helpers

The chapter draws its trees with **ggtree** and its split networks with **phangorn**. Two helpers keep the labels of these figures readable. Both are pure functions, defined in hidden chunks of the chapter, and are printed here.

`net_labels()` labels the tips of a network drawn with `plot(..., show.tip.label = FALSE)`. It needs the object that `plot()` returns for a network, whose `.plot$vertices` element holds the coordinates of the vertices. Each label is first placed just outside its tip, on the line from the centre of the network through the tip. Overlapping labels are then pushed apart vertically, pair by pair, until no two overlap, and labels that had to move far from their tip are joined to it by a thin grey line.

```
# Tip labels for a network drawn with show.tip.label = FALSE: each label starts
# just outside its tip (away from the network's centre); overlapping labels are
# then pushed apart vertically and joined to their tips by thin grey lines.
net_labels <- function(pn, labels, col = "black", cex = 0.7) {
  v <- pn$.plot$vertices[seq_along(labels), , drop = FALSE]
  d <- sweep(v, 2, colMeans(pn$.plot$vertices))
  u <- d / sqrt(rowSums(d^2))
  w <- strwidth(labels, cex = cex, font = 3) * 1.1
  h <- strheight("Lg", cex = cex, font = 3) * 1.5
  lab <- v + u * (0.5 * w + 0.5 * h)[row(u)]
  for (iter in 1:500) {
    moved <- FALSE
    for (i in seq_len(nrow(lab) - 1)) for (j in (i + 1):nrow(lab)) {
      ox <- (w[i] + w[j]) / 2 - abs(lab[i, 1] - lab[j, 1])
      oy <- h - abs(lab[i, 2] - lab[j, 2])
      if (ox > 0 && oy > 0) {
        s <- if (lab[i, 2] >= lab[j, 2]) 1 else -1
        lab[i, 2] <- lab[i, 2] + s * (oy / 2 + 1e-9)
        lab[j, 2] <- lab[j, 2] - s * (oy / 2 + 1e-9)
        moved <- TRUE
```

```
      }
    }
    if (!moved) break
  }
  far <- sqrt(rowSums((lab - v)^2)) > 0.9 * (0.5 * w + 0.5 * h)
  segments(v[far, 1], v[far, 2], lab[far, 1] - u[far, 1] * w[far] / 2,
           lab[far, 2] - u[far, 2] * h / 3, col = "grey60", lwd = 0.6)
  text(lab[, 1], lab[, 2], labels, cex = cex, font = 3, col = col)
}
```

`support_layers()` returns two **ggtree** text layers for the node labels of a tree, such as bootstrap values or clade credibilities. Each value is placed at the right end of its stem branch: above the branch for the upper of two daughters and below it for the lower one, so that no value crosses the vertical line of its parent. Labels listed in `drop` (for example an empty root label) are skipped. It is used as `ggtree(tree) + support_layers(2.6)`.

```
# Support values: each one sits at the right end of its stem branch, above the
# branch for an upper daughter and below it for a lower daughter, so it never
# crosses the parent's vertical line (which spans only between the daughters).
support_layers <- function(size, drop = "") {
  pick <- function(up) function(d) {
    py <- d$y[match(d$parent, d$node)]
    d[!d$isTip & d$node != d$parent & !is.na(d$label) &
        !d$label %in% drop & (d$y >= py) == up, ]
  }
  list(geom_text(data = pick(TRUE), aes(label = label), size = size,
                 hjust = 1.1, vjust = -0.4, color = "grey30"),
       geom_text(data = pick(FALSE), aes(label = label), size = size,
                 hjust = 1.1, vjust = 1.4, color = "grey30"))
}
```

E.3.2 The BEAST 2 input file

The excerpt below shows the components of the BEAST 2.7 analysis of the toy family in Section 17.10.3: the restriction-site model with estimated frequencies and gamma rate variation, a strict clock, a Yule tree prior, the subgroup-D calibration and the root-age prior of the MrBayes block. Some operators, priors and log entries have been left out (`...`). The full file, about 140 lines long, is `software-runs/beast2/toy_beast.xml`, and it runs as it stands in BEAST 2.7.7. The data block is the one written by chunk `workflow-beast-data` of the chapter: one `sequence` element per language, preceded by a dummy all-absent column that the `ascertained`, `excludefrom` and `excludeto` attributes remove from the likelihood.

```
<beast version="2.7" namespace="beast.base.core:beast.base.inference:...">
  <data id="cognates" spec="Alignment" dataType="binary" ascertained="true"
        excludefrom="0" excludeto="1"> ... </data>
  <taxonset id="subgroupD" spec="TaxonSet">
    <taxon id="L21" spec="Taxon"/> ... </taxonset>
  <run id="mcmc" spec="MCMC" chainLength="3000000">
```

```
<!-- 1. State: everything the MCMC samples -->
<state id="state">
  <tree id="Tree" name="stateNode">
    <taxonset id="TaxonSet.cognates" spec="TaxonSet" alignment="@cognates"/>
  </tree>
  <parameter id="birthRate" name="stateNode" lower="0.0">1.0</parameter>
  <parameter id="clockRate" name="stateNode" lower="0.0">0.1</parameter>
  <parameter id="gammaShape" name="stateNode" lower="0.0">1.0</parameter>
  <parameter id="freqParameter" name="stateNode" dimension="2"
             lower="0.0" upper="1.0">0.5</parameter>
</state>
<init spec="RandomTree" initial="@Tree" taxa="@cognates" rootHeight="6.0">
  ... </init>
<!-- 2. Posterior = prior x likelihood -->
<distribution id="posterior" spec="CompoundDistribution">
  <distribution id="prior" spec="CompoundDistribution">
    <distribution id="YuleModel" spec="YuleModel" tree="@Tree"
                  birthDiffRate="@birthRate"/>
    <distribution id="clockRatePrior" spec="Prior" x="@clockRate">
      <distr
        spec="beast.base.inference.distribution.LogNormalDistributionModel"
        M="-2.3" S="1.0"/>
    </distribution>
    <!-- calibration: an MRCA prior on a set of taxa -->
    <distribution id="subgroupD.prior" spec="MRCAPrior" tree="@Tree"
                  taxonset="@subgroupD" monophyletic="true">
      <distr spec="beast.base.inference.distribution.Uniform"
             lower="0.1" upper="1.1"/>
    </distribution>
    ...
  </distribution>
  <distribution id="likelihood" spec="CompoundDistribution">
    <distribution id="treeLikelihood" spec="TreeLikelihood"
                  data="@cognates" tree="@Tree">
      <siteModel spec="SiteModel" gammaCategoryCount="4"
                 shape="@gammaShape">
        <parameter name="mutationRate" estimate="false">1.0</parameter>
        <!-- two-state model with estimated frequencies -->
        <substModel spec="GeneralSubstitutionModel">
          <parameter name="rates" dimension="2"
                     estimate="false">1.0</parameter>
          <frequencies spec="Frequencies" frequencies="@freqParameter"/>
        </substModel>
      </siteModel>
      <branchRateModel spec="StrictClockModel" clock.rate="@clockRate"/>
    </distribution>
  </distribution>
</distribution>
```

```
    <!-- 3. Operators: proposals for each state node -->
    <operator id="subtreeSlide" spec="SubtreeSlide" tree="@Tree" weight="15.0"/>
    <operator id="clockRateScaler" spec="ScaleOperator" parameter="@clockRate"
              scaleFactor="0.75" weight="5.0"/>
    ...
    <!-- 4. Loggers: parameter trace (.log) and sampled trees (.trees) -->
    <logger id="tracelog" fileName="toy.log" logEvery="2000"> ... </logger>
    <logger id="treelog" fileName="toy.trees" logEvery="2000" mode="tree">
      <log spec="TreeWithMetaDataLogger" tree="@Tree"/>
    </logger>
  </run>
</beast>
```

The `MRCAPrior` with `monophyletic="true"` both constrains subgroup D to be a clade and puts a uniform prior on its age. The `mutationRate` of the site model is fixed at 1, so that the clock rate alone sets the number of expected changes per unit of time. Replacing the strict clock or the substitution model changes only the `branchRateModel` or the `substModel` element, together with the parameters, priors and operators they need.

E.3.3 The RevBayes script

This is the complete script of Section 17.10.5, as it was run with RevBayes 1.4.1 on the file `toy_cognates_standard.nex` written by the chapter. It specifies an unrooted analysis under the symmetric two-state Mk model with gamma rate variation, conditioned on variable characters, with two independent runs.

```
# toy_cognates.Rev -- run with: rb toy_cognates.Rev
# the CHARSETs split the file into one matrix per meaning; rejoin them
data <- concatenate(readDiscreteCharacterData("toy_cognates_standard.nex"))
taxa <- data.taxa()
n_taxa <- data.ntaxa()
n_branches <- 2 * n_taxa - 3
moves = VectorMoves()
monitors = VectorMonitors()

# tree prior: uniform topology, exponential branch lengths
topology ~ dnUniformTopology(taxa)
moves.append( mvNNI(topology, weight = n_taxa) )
moves.append( mvSPR(topology, weight = n_taxa / 5.0) )
for (i in 1:n_branches) {
  br_lens[i] ~ dnExponential(10.0)
  moves.append( mvScale(br_lens[i], weight = 1.0) )
}
TL := sum(br_lens)
phylogeny := treeAssembly(topology, br_lens)

# substitution model: symmetric two-state Mk model with gamma rates
Q <- fnJC(2)
```

```
alpha ~ dnExponential(1.0)
moves.append( mvScale(alpha, weight = 2.0) )
site_rates := fnDiscretizeGamma(alpha, alpha, 4)

# likelihood, conditioned on variable characters (Mkv)
seq ~ dnPhyloCTMC(tree = phylogeny, Q = Q, siteRates = site_rates,
                  type = "Standard", coding = "variable")
seq.clamp(data)

mymodel = model(phylogeny)
monitors.append( mnModel(filename = "output/toy.log", printgen = 10) )
monitors.append( mnFile(filename = "output/toy.trees", printgen = 10,
                        phylogeny) )
monitors.append( mnScreen(printgen = 1000, TL) )
mymcmc = mcmc(mymodel, monitors, moves, nruns = 2, combine = "mixed")
mymcmc.run(generations = 10000)

trace = readTreeTrace("output/toy.trees", treetype = "non-clock")
map_tree = mapTree(trace, "output/toy_map.tre")
q()
```

The moves define the proposals of the MCMC: NNI and SPR moves on the topology, and scaling moves on every branch length and on the gamma shape. The monitors write the parameters to `output/toy.log` every 10 iterations, the trees to `output/toy.trees`, and a progress line to the screen. `readTreeTrace()` reads the sampled trees and `mapTree()` summarizes them as the maximum a posteriori tree. Run times and convergence diagnostics are reported in Section 17.10.5.

Appendix F

Software

This appendix is an annotated guide to the software used in linguistic phylogenetics. It covers Bayesian platforms and their extension packages, maximum-likelihood, parsimony and distance programs, network software, tools for summarizing and visualizing trees, tools for preparing linguistic data, and the R and Python libraries most often used alongside them. For each program we say what it does, what it is typically used for with language data, and where to obtain it. Software changes quickly: version numbers, package names and web addresses given here were correct at the time of writing, but readers should always check the current documentation and cite the version they actually used. The workflow chapter (Chapter 17) shows how several of these programs fit together in practice.

F.1 Choosing software

No single program does everything. Table F.1 maps the common tasks of a linguistic phylogenetic project to the programs most often used for them.

Table F.1: Common tasks in linguistic phylogenetics and the software most often used for them.

Task	Typical software
Building and curating cognate data	EDICTOR, LingPy, LingRex, spreadsheets with CLDF export, cldfbench/pylexibank
Automated cognate detection	LingPy (LexStat, SCA), LingRex, Python implementations of PMI-based methods
Exploratory analysis, networks	SplitsTree, phangorn (R), PhyloNetworks (Julia)
Distance and parsimony trees	phangorn, PAUP*, TNT, PHYLIP, FastME
Maximum-likelihood trees	IQ-TREE, RAxML-NG, phangorn
Bayesian dated trees	BEAST 2 (often via BEASTling), BEAST X, RevBayes, MrBayes, TraitLab
Model averaging and model comparison	bModelTest, nested sampling and path sampling packages for BEAST 2, RevBayes, MrBayes
Phylogeography	BEAST 2 (BEAST_CLASSIC, GEO_SPHERE, MASCOT), BEAST X, RevBayes, BioGeoBEARS
Ancestral states and correlated evolution	BayesTraits, RevBayes, phytools, corHMM, ape

Task	Typical software
Comparative regression	nlme, caper, phylolm, MCMCglmm, brms
Convergence diagnostics	Tracer, coda, RWTY
Summary trees and visualization	TreeAnnotator, phangorn, ggtree, FigTree, IcyTree, DensiTree, iTOL

F.2 Bayesian phylogenetic platforms

F.2.1 BEAST 2

BEAST 2 (Bayesian Evolutionary Analysis Sampling Trees) is a cross-platform Java program for Bayesian phylogenetic inference with an emphasis on time-calibrated trees (Bouckaert et al., 2014, 2019). It is the most widely used program for dated language phylogenies. Its core supports a wide range of substitution models (including binary, covarion and general discrete-state models suitable for cognate data), strict and relaxed clocks, tree priors from the Yule and birth-death families and the coalescent, node calibrations, and flexible priors on all parameters. Models are specified in an XML file, usually created with the graphical program BEAUti; the companion programs LogCombiner, TreeAnnotator and DensiTree combine runs, build summary trees and visualize tree sets. Much of BEAST 2's power comes from its *package* system: extensions are installed from within BEAUti through the package manager. The textbook by Drummond and Bouckaert (2015) is the standard introduction. Website: https://www.beast2.org.

F.2.2 BEAST 2 packages relevant to linguistics

BEASTling is not a BEAST 2 package but a standalone Python program that writes BEAST 2 XML files for linguistic analyses from short configuration files (Maurits et al., 2017). It reads data in CSV or CLDF format, links languages to Glottolog, can impose Glottolog subgroups as monophyly constraints, and supports models for lexical (cognate) and typological data, rate variation across features, relaxed clocks and calibrations. Its configuration files make analyses easy to publish and reproduce. However, BEASTling has not been updated since version 1.5.1 (2021). It writes XML for BEAST 2.6, which BEAST 2.7 cannot read without renaming classes to the new `beast.base` namespace. It also fails to start under Python 3.14, and it needs older versions of `clldutils` (2.x) and `pycldf` (below 1.15). Chapter 17 shows a working route. Source: https://github.com/lmaurits/BEASTling.

Babel is a BEAST 2 package developed by Remco Bouckaert specifically for linguistic data. It provides models and tools used in published language phylogenies, including stochastic Dollo-type and pseudo-Dollo covarion models for cognate data, ascertainment-correction schemes suitable for cognate presence/absence data, and utilities for preparing and summarizing linguistic analyses. Source: https://github.com/rbouckaert/Babel.

BEAST_CLASSIC ports to BEAST 2 several models originally developed in BEAST 1, most importantly discrete-trait phylogeography with Bayesian stochastic search variable selection (BSSVS) (Lemey et al., 2009) and continuous-trait diffusion models used in continuous phylogeography. Source: https://github.com/BEAST2-Dev/beast-classic.

GEO_SPHERE implements phylogeographic diffusion on the surface of a sphere and related landscape-aware diffusion models (Bouckaert, 2016). Spherical diffusion avoids the distortions

of planar diffusion models over large areas, which matters for language families spread across continents. Landscape-aware variants allow diffusion to depend on features such as land and water, an idea first explored for language data in the Indo-European analysis of Bouckaert et al. (2012).

SA (sampled ancestors) implements the fossilized birth-death process with sampled ancestors (Gavryushkina et al., 2014), which allows sampled taxa to lie directly on the lineages leading to other taxa rather than only at the tips. For linguistics, this allows ancient languages such as Latin or Vedic Sanskrit to be modeled as possible direct ancestors of modern languages, the approach taken by Heggarty et al. (2023). Source: https://github.com/CompEvol/sampled-ancestors.

ORC (optimized relaxed clock) provides a reparameterized uncorrelated relaxed clock with improved operators that make relaxed-clock analyses mix much faster (Douglas et al., 2021). Source: https://github.com/jordandouglas/ORC.

bModelTest performs Bayesian site-model averaging, sampling over substitution models and rate-heterogeneity options within the MCMC rather than selecting a single model in advance (Bouckaert and Drummond, 2017). It was designed for nucleotide data, but the approach of averaging over models rather than choosing one is relevant to any data type. Source: https://github.com/BEAST2-Dev/bModelTest.

MASCOT implements the marginal approximation of the structured coalescent, a phylogeographic model in which migration between discrete locations is modeled within a population-genetic framework rather than as a trait evolving on a fixed tree (Müller et al., 2018). It has, to our knowledge, rarely if ever been applied to language data (Chapter 11). Source: https://github.com/CompEvol/Mascot.

Model selection packages. The **NS** package implements nested sampling for estimating marginal likelihoods and their uncertainty (Maturana Russel et al., 2019); the **MODEL_SELECTION** package implements path sampling and stepping-stone sampling. Both are used to compute Bayes factors between models (Chapter 6).

CoupledMCMC implements Metropolis-coupled MCMC (parallel tempering) with adaptive temperatures for BEAST 2, which can help with multimodal posteriors (Müller and Bouckaert, 2020).

BDSKY implements the birth-death skyline model, in which birth and death rates change through time in a piecewise-constant way (Stadler et al., 2013). Developed for epidemiology, it could be used to study changes in the rate of language diversification through time (Chapter 16).

F.2.3 BEAST X and BEAST 1

BEAST 1, whose most widely cited release is BEAST 1.10 (Suchard et al., 2018), is the original BEAST, developed in parallel with BEAST 2 by a partly different team with a strong focus on viral phylodynamics and phylogeography. Its successor, BEAST X, extends the platform with new models and faster inference methods, including Hamiltonian Monte Carlo for many parameters (Baele et al., 2025). BEAST 1 and BEAST X provided the original implementations of discrete phylogeography with BSSVS (Lemey et al., 2009), relaxed random walk continuous phylogeography (Lemey et al., 2010) and phylogeographic generalized linear models with predictors. BEAST 1 and BEAST X files are not compatible with BEAST 2. Website: https://beast.community.

F.2.4 RevBayes

RevBayes (Höhna et al., 2016) is a Bayesian phylogenetics platform built around probabilistic graphical models. Models are written in Rev, an interpreted language resembling R, in which every random variable, deterministic function and constant is declared explicitly. This makes RevBayes the most flexible general-purpose platform: unusual combinations of models, priors and data types can be specified without programming new code. It supports discrete morphological (Mk) models and restriction-site models suitable for cognate data, relaxed clocks, a wide range of tree priors including the fossilized birth-death process, ancestral state reconstruction, stochastic character mapping, and marginal likelihood estimation by path sampling and stepping-stone sampling. The extensive tutorials (https://revbayes.github.io/tutorials/) are among the best learning resources in phylogenetics. The R package **RevGadgets** visualizes RevBayes output (Tribble et al., 2022). Website: https://revbayes.github.io.

F.2.5 MrBayes

MrBayes (Huelsenbeck and Ronquist, 2001; Ronquist et al., 2012b) is a long-established Bayesian phylogenetics program controlled by commands in a NEXUS block. It supports restriction-site (binary) data with ascertainment options such as `noabsencesites` and `variable`, standard discrete (morphological) data, covarion models, gamma rate variation, strict and relaxed clocks, node calibrations and tip dating, and Metropolis-coupled MCMC with multiple independent runs and a built-in convergence diagnostic (the average standard deviation of split frequencies). It also estimates marginal likelihoods by stepping-stone sampling. MrBayes is fast and well documented, and is a good choice for undated or simply dated analyses of cognate data. Website: https://nbisweden.github.io/MrBayes/.

F.2.6 TraitLab

TraitLab is a MATLAB package that implements the stochastic Dollo model for binary trait data, in which each cognate class is born once and then may be lost independently on different lineages (Nicholls and Gray, 2008). It also handles missing data and catastrophic rate changes on branches (Ryder and Nicholls, 2011), and lateral transfer between lineages (Kelly and Nicholls, 2017). TraitLab was developed specifically for language data and remains the reference implementation of these models (Chapter 7). It is distributed by its authors; current versions are available through code repositories linked from the publications.

F.2.7 BayesTraits

BayesTraits analyzes the evolution of traits on samples of phylogenetic trees (Pagel et al., 2004; Pagel and Meade, 2006). It implements discrete-trait models (including the test of correlated evolution between binary traits of Pagel (1994), with reversible-jump MCMC over model structures), continuous-trait models with Pagel's λ, κ and δ transformations, regression, variable-rates models, and ancestral state reconstruction. Because it integrates over a posterior sample of trees, it is a standard tool for comparative studies in linguistics and cultural evolution, including the word-order study of Dunn et al. (2011) (Chapter 13). It is distributed by the Evolution Group at the University of Reading.

F.2.8 Other Bayesian programs

PhyloBayes (Lartillot et al., 2009) implements the CAT family of site-heterogeneous profile mixture models. These models, developed for protein data, let different sites have different equilibrium frequencies. To our knowledge they have rarely if ever been applied to linguistic data, but they are relevant to the question of rate and pattern heterogeneity among meanings (Chapter 7). **BAli-Phy** (Redelings, 2021) jointly estimates sequence alignments and phylogenies; it is a natural model for future work on the joint alignment of word forms (Chapter 16). Website: https://www.bali-phy.org.

F.3 Maximum likelihood

IQ-TREE (Nguyen et al., 2015; Minh et al., 2020b) is a fast and widely used program for maximum-likelihood tree inference. It supports binary (`JC2`, `GTR2`) and general discrete morphological (`MK`) models with ascertainment-bias correction (`+ASC`), gamma and free-rate heterogeneity, partitioned analyses, automatic model selection with ModelFinder (Kalyaanamoorthy et al., 2017), the ultrafast bootstrap (Hoang et al., 2018), and tests of tree topology. Website: https://iqtree.github.io.

RAxML-NG (Kozlov et al., 2019) is the successor of RAxML, designed for speed and scalability. It supports binary data (`BIN`) and general multistate data, ascertainment-bias corrections (for example `+ASC_LEWIS`), and standard and transfer bootstrap support. Source: https://github.com/amkozlov/raxml-ng.

PhyML (Guindon et al., 2010) is a widely used maximum-likelihood program with fast SPR-based tree search and approximate likelihood-ratio tests of branch support.

phangorn (R; Schliep (2011)) provides maximum-likelihood inference for user-defined discrete data including binary cognate data, with ascertainment correction, model comparison and bootstrapping. It is slower than IQ-TREE or RAxML-NG but integrates seamlessly with the rest of the R ecosystem; it is used throughout this book.

F.4 Parsimony and distance methods

PAUP* (Swofford, 2003) (Phylogenetic Analysis Using Parsimony, and other methods) is a classic program for parsimony, distance and likelihood analysis, with powerful heuristic tree searches and support for character weighting, constraints and consensus methods. Website: https://paup.phylosolutions.com.

TNT (Tree analysis using New Technology) (Goloboff et al., 2008; Goloboff and Catalano, 2016) is the fastest program for parsimony analysis of large datasets, with sophisticated search strategies (ratchet, sectorial searches, tree drifting and fusing) and implied weighting. Website: https://www.lillo.org.ar/phylogeny/tnt/.

PHYLIP (Felsenstein, 1989) (Phylogeny Inference Package) is a long-standing collection of programs for distance, parsimony (including Dollo and Camin-Sokal parsimony) and likelihood methods. Its file format, the PHYLIP format, is still widely used for data exchange.

FastME (Lefort et al., 2015) implements balanced minimum-evolution tree construction from distance matrices, with tree rearrangements that improve on neighbor-joining.

phangorn and **ape** (R) provide UPGMA, neighbor-joining, BIONJ, FastME-style minimum evolution and parsimony (Fitch and Sankoff, with the parsimony ratchet), as illustrated in Chapters 4 and 17.

F.5 Networks

SplitsTree (Huson and Bryant, 2006) computes and displays split networks, including Neighbor-Net (Bryant and Moulton, 2004), split decomposition and median networks, together with statistics such as delta scores. It has been the standard tool for exploratory network analysis of language data. Recent versions (SplitsTree 6) have been rewritten with an interactive interface. Source: https://github.com/husonlab/splitstree6.

PhyloNet (Than et al., 2008; Wen et al., 2018) infers explicit phylogenetic networks, in which reticulation nodes represent hybridization or other horizontal events, from gene trees or sequence data, under parsimony, likelihood and Bayesian criteria.

PhyloNetworks (Solís-Lemus et al., 2017) is a Julia package for estimating and analyzing explicit phylogenetic networks, including SNaQ, a pseudolikelihood method that accounts for incomplete lineage sorting (Solís-Lemus and Ané, 2016). Source: https://github.com/JuliaPhylo/PhyloNetworks.jl.

Gabmap (Nerbonne et al., 2011) is a web application for dialectometry: it computes aggregate distances between dialect sites from transcriptions or categorical data, and visualizes them with maps, clustering and multidimensional scaling. It is useful for dialect continua where tree models are inappropriate.

phangorn (R) implements NeighborNet, consensus networks and delta scores, and plots split networks (Chapter 8).

F.6 Convergence diagnostics, summaries and visualization

Tracer (Rambaut et al., 2018) reads the parameter logs of BEAST, MrBayes and other programs, and displays traces, marginal posterior distributions, effective sample sizes and summary statistics. It is the standard first check on any Bayesian run. Source: https://github.com/beast-dev/tracer.

TreeAnnotator and **LogCombiner** are distributed with BEAST 2 and BEAST X. LogCombiner combines and resamples log and tree files from several runs, removing burn-in; TreeAnnotator builds a summary tree (maximum clade credibility or other) with node heights and highest posterior density intervals (Heled and Bouckaert, 2013).

DensiTree (Bouckaert, 2010), distributed with BEAST 2, draws all trees of a posterior sample on top of each other to visualize uncertainty in topology and node heights.

FigTree is a graphical viewer for trees, widely used for displaying annotated BEAST trees. Source: https://github.com/rambaut/figtree.

IcyTree (Vaughan, 2017) is a browser-based tree viewer that handles annotated trees and networks without installation. Website: https://icytree.org.

iTOL (Interactive Tree Of Life) (Letunic and Bork, 2021) is an online tool for displaying and annotating large trees with associated data. Website: https://itol.embl.de.

SpreaD3 (Bielejec et al., 2016) visualizes phylogeographic reconstructions from BEAST as animated maps.

RWTY (Warren et al., 2017) (R) provides convergence diagnostics specifically for tree topologies, including split-frequency comparisons between runs and visualizations of tree space.

F.7 Tools for linguistic data

LingPy (List et al., 2017, 2018b) is a Python library for quantitative historical linguistics. It provides phonetic alignment (including the SCA method), automated cognate detection (LexStat, SCA-based and edit-distance-based clustering, with Infomap or UPGMA-style partitioning), distance computation, and export to phylogenetic formats. Website: https://lingpy.org; source: https://github.com/lingpy/lingpy.

LingRex extends LingPy with methods for partial cognate detection, correspondence pattern inference and related tasks. Source: https://github.com/lingpy/lingrex.

EDICTOR (List, 2017) is a web-based interactive tool for creating, inspecting and editing etymological datasets: aligning word forms, assigning cognate sets (including partial cognates) and inspecting correspondence patterns. It runs in the browser and works with the same TSV formats as LingPy. Website: https://edictor.org.

cldfbench and pylexibank (Forkel and List, 2020) are Python packages for creating CLDF datasets reproducibly from raw sources. Each Lexibank dataset is a cldfbench project with a conversion script that downloads or reads the original data, maps concepts to Concepticon and languages to Glottolog, segments transcriptions, and writes validated CLDF. Sources: https://github.com/cldf/cldfbench and https://github.com/lexibank/pylexibank.

pycldf reads, writes and validates CLDF datasets in Python, and provides command-line tools for inspecting them. Source: https://github.com/cldf/pycldf. The CLDF specification itself is documented at https://cldf.clld.org (Forkel et al., 2018).

Other CLDF ecosystem tools. **pyglottolog** gives programmatic access to the Glottolog data repository (https://github.com/glottolog/pyglottolog); **pyconcepticon** does the same for Concepticon (https://github.com/concepticon/pyconcepticon); **pyclts** provides the Cross-Linguistic Transcription Systems reference catalog (https://github.com/cldf-clts/pyclts); and **segments** tokenizes transcriptions using orthography profiles (https://github.com/cldf/segments).

F.8 R packages

R is the most common environment for preparing data, analyzing trees and running comparative methods. Table F.2 lists the packages most relevant to linguistic phylogenetics. CRAN packages are available at `https://CRAN.R-project.org/package=<name>`; Bioconductor packages at `https://bioconductor.org/packages/<name>`.

Table F.2: R packages for linguistic phylogenetics.

Package	Purpose	Source
ape (Paradis and Schliep, 2019)	core tree data structures, reading and writing Newick/NEXUS, plotting, distances, NJ, ancestral states (`ace`), comparative methods	CRAN
phangorn (Schliep, 2011)	distance, parsimony and likelihood inference, bootstrap, consensus and MCC trees, networks, delta scores	CRAN
phytools (Revell, 2012, 2024)	comparative methods, stochastic character mapping, phylogenetic signal, Pagel's correlation test, simulation, tanglegrams	CRAN
geiger (Pennell et al., 2014)	fitting Brownian, OU, early-burst and discrete models; tree and data matching	CRAN
caper	phylogenetic independent contrasts and PGLS, the D statistic for binary traits	CRAN
nlme (Pinheiro and Bates, 2000)	generalized least squares with phylogenetic correlation structures (with ape)	CRAN
phylolm (Ho and Ané, 2014b)	fast phylogenetic linear and logistic regression	CRAN
MCMCglmm (Hadfield, 2010)	Bayesian phylogenetic generalized linear mixed models	CRAN
brms (Bürkner, 2017)	Bayesian multilevel models via Stan, including phylogenetic random effects	CRAN
corHMM (Beaulieu et al., 2013)	hidden-rates models for discrete characters	CRAN
diversitree (FitzJohn, 2012)	diversification models (BiSSE, MuSSE and relatives) and discrete-trait models	CRAN
castor (Louca and Doebeli, 2018)	fast comparative and simulation methods for very large trees	CRAN
phylopath (van der Bijl, 2018)	phylogenetic path analysis	CRAN
mvMORPH (Clavel et al., 2015)	multivariate models of continuous trait evolution	CRAN
bayou (Uyeda and Harmon, 2014)	Bayesian inference of shifting Ornstein-Uhlenbeck optima	CRAN/GitHub
BioGeoBEARS (Matzke, 2013)	historical biogeography models (DEC, DIVA-like, BayArea-like, founder events)	CRAN/GitHub
ggtree (Yu et al., 2017)	grammar-of-graphics plotting and annotation of trees	Bioconductor
treeio (Wang et al., 2020)	reading annotated tree output from BEAST, MrBayes, RevBayes and others	Bioconductor
coda (Plummer et al., 2006)	MCMC diagnostics (ESS, Gelman-Rubin, trace plots)	CRAN
RWTY (Warren et al., 2017)	convergence diagnostics for tree topologies	GitHub
treespace (Jombart et al., 2017)	exploring sets of trees with tree distances and ordination	CRAN
babette (Bilderbeek and Etienne, 2018)	creating and running BEAST 2 analyses from R	CRAN/GitHub
RevGadgets (Tribble et al., 2022)	plotting RevBayes output	CRAN/GitHub

Package	Purpose	Source
rcldf	reading CLDF datasets into R (https://github.com/SimonGreenhill/rcldf)	GitHub
lingtypology	access to Glottolog and typological databases, linguistic maps	CRAN
glottoTrees	building, pruning and combining Glottolog trees (https://github.com/erichround/glottoTrees)	GitHub

Packages can be installed with:

```r
install.packages(c("ape", "phangorn", "phytools", "geiger", "caper", "nlme",
                   "phylolm", "MCMCglmm", "brms", "corHMM", "coda"))
if (!requireNamespace("BiocManager", quietly = TRUE))
install.packages("BiocManager")
BiocManager::install(c("ggtree", "treeio"))
# GitHub packages, for example:
# remotes::install_github("SimonGreenhill/rcldf")
```

F.9 Python and other libraries

DendroPy (Sukumaran and Holder, 2010) is a Python library for reading, writing, manipulating and simulating trees, and for summarizing tree sets. **ETE** (Huerta-Cepas et al., 2016) provides tree manipulation and high-quality tree visualization in Python. **Biopython** (Cock et al., 2009) includes the `Bio.Phylo` module for tree input/output and basic manipulation. **baltic** is a lightweight Python library for plotting annotated BEAST trees (https://github.com/evogytis/baltic). **newick** is a small Python package from the CLDF ecosystem for reading and writing Newick trees, used by Glottolog's tools.

F.10 Citing software

Software is part of the method. A methods section should name every program and package used, with version numbers, and should cite the publications that describe them, as their authors request. For BEAST 2, the packages used (and their versions) should also be cited, since many of the models in a typical linguistic analysis come from packages rather than from the core. In R, `citation("<package>")` prints the preferred citation for any installed package, and `sessionInfo()` records the versions used. Archiving the input files (XML, Rev scripts, NEXUS files with command blocks, BEASTling configuration files) alongside the data allows others to reproduce the analysis exactly (Chapter 17).

Appendix G

Data Resources

This appendix describes the main databases used in linguistic phylogenetics and in the comparative study of language and culture. For each resource we describe what it contains, how it is organized, how it has been used in phylogenetic work, and where to find it. Many of these resources are now published in the Cross-Linguistic Data Formats (CLDF) (Forkel et al., 2018) and archived in versioned releases on Zenodo, which makes them easy to load with the tools described in Chapter 17 and Appendix F. Most of the databases that follow the CLDF model have a web interface built with the `clld` toolkit, usually at an address of the form `https://<name>.clld.org`, and a GitHub repository with the CLDF data. Two general points apply throughout. First, always cite the specific version of a database that was used, since databases are corrected and extended between releases. Second, read the documentation on how the data were collected and coded; every database embodies decisions (which concepts, which sources, which coding conventions) that affect what it can and cannot be used for (Chapters 3 and 15).

Table G.1 gives an overview.

Table G.1: Overview of major data resources. Coverage figures are approximate and change between releases.

Resource	Content	Coverage (approximate)	Web address
Glottolog	language catalogue, classification, bibliography	all known languages and dialects	https://glottolog.org
WALS	typological features	~2,600 languages, ~190 features	https://wals.info
Grambank	grammatical features	~2,400 languages, ~195 features	https://grambank.clld.org
PHOIBLE	phoneme inventories	~2,000+ languages	https://phoible.org
ASJP	40-item word lists in a simplified transcription	~5,000+ languages and dialects	https://asjp.clld.org
Lexibank	standardized word lists from many sources	100+ datasets, thousands of varieties	https://github.com/lexibank
Concepticon	links between concept lists	hundreds of concept lists	https://concepticon.clld.org
CLICS	colexifications (polysemies)	thousands of varieties	https://clics.clld.org

Resource	Content	Coverage (approximate)	Web address
CLTS	transcription systems and sound features	reference catalogue	https://clts.clld.org
IELex / IE-CoR	Indo-European cognate databases	100–160 Indo-European varieties	https://iecor.clld.org
ABVD	Austronesian basic vocabulary	1,000+ word lists	https://abvd.clld.org
STEDT	Sino-Tibetan etymologies	hundreds of varieties	https://stedt.berkeley.edu
NorthEuraLe	1,000-concept word lists	~100+ languages of northern Eurasia	http://northeuralex.org
IDS	dictionary-based word lists	300+ languages, ~1,300 concepts	https://ids.clld.org
WOLD	loanword status of vocabulary	41 languages	https://wold.clld.org
D-PLACE	cultural, environmental and linguistic data on societies	~1,400+ societies	https://d-place.org
Seshat	historical and archaeological data on polities	hundreds of polities	https://seshatdatabank.info
Chirila	Australian Aboriginal language data	most Australian languages	https://doi.org/10.60600/YU/XHPJXB

G.1 Language catalogues and classification

G.1.1 Glottolog

Glottolog (Hammarström et al., 2024) is a comprehensive catalogue of the world's languages, dialects and language families. Each languoid (a language, dialect or family) has a stable identifier, the *Glottocode* (for example `stan1293` for Standard English), together with its position in a genealogical classification, geographic coordinates, macro-area, endangerment status and an extensive bibliography of descriptive sources. Glottolog's classification is deliberately conservative: it recognizes only families whose unity has been demonstrated by the comparative method, and lists a large number of isolates and unclassified languages. This makes it a reliable scaffold, but means that its trees are mostly unresolved at higher levels and have no branch lengths.

Glottocodes have become the standard identifier for linking linguistic datasets: Lexibank, Grambank, WALS (via mappings), PHOIBLE, D-PLACE and many others use them, so data from different sources can be joined by Glottocode (Chapter 17). Glottolog's classification trees are distributed in Newick format and through its CLDF dataset (https://github.com/glottolog/glottolog-cldf), and are often used as topologies in global comparative analyses, with branch lengths assigned by rule (Chapter 13). Glottolog is published in numbered versions; analyses should cite the version used.

G.1.2 Other catalogues

Ethnologue is a commercial catalogue of the world's languages with information on speaker numbers, vitality and location, identified by ISO 639-3 codes. Its classification is less conservative than Glottolog's, and access to much of its data requires a subscription. ISO 639-3 codes, which Ethnologue maintains on behalf of the International Organization for Standardization, remain widely used and are mapped to Glottocodes by Glottolog.

G.2 Lexical databases

G.2.1 ASJP

The Automated Similarity Judgment Program (ASJP) database (Wichmann, 2022) contains short word lists for a 40-item subset of the Swadesh list, transcribed in a simplified phonetic alphabet (ASJPcode) for several thousand languages and dialects. The 40 items were chosen for their stability and their usefulness for classification (Holman et al., 2008). Because every list uses the same concepts and transcription, ASJP is the largest uniform lexical dataset available. It has been used for automated classification based on normalized Levenshtein distances (Brown et al., 2008), for dating language families with a calibrated distance-based method (Holman et al., 2011), and for global phylogenies based on automatically inferred cognates (Jäger, 2018). ASJP lists do not contain cognate judgments; cognacy must be inferred (Chapter 3).

G.2.2 Lexibank

Lexibank (List et al., 2022a) is a public repository of standardized word lists. It aggregates more than a hundred published datasets, each converted to CLDF by a reproducible script, with concepts linked to Concepticon, languages linked to Glottolog, and transcriptions standardized and segmented according to the CLTS reference catalogue. Many datasets include expert cognate judgments; others contain only word forms, for which Lexibank provides automatically computed features. Lexibank datasets are hosted on GitHub (https://github.com/lexibank) and archived on Zenodo, and include CLDF versions of several of the datasets described below (for example the ABVD and datasets used in published Sino-Tibetan and other phylogenies). Lexibank is the natural starting point for any new lexical phylogenetic analysis (Chapter 17).

G.2.3 Indo-European: IELex, IE-CoR and related databases

IELex (the Indo-European Lexical Cognacy Database) was assembled by Michael Dunn and collaborators, building on the Swadesh-list data of Dyen et al. (1992), and was used, with modifications, in several influential Bayesian analyses of Indo-European, including Bouckaert et al. (2012) and Chang et al. (2015). Its cognate judgments and meaning definitions were criticized and revised over time.

IE-CoR (Indo-European Cognate Relationships) (Heggarty et al., 2023) is a newer database designed to address these problems. It covers about 160 Indo-European languages, including ancient languages, for a list of 170 meanings, with explicit protocols for selecting the word for each meaning, for coding cognacy (including loanwords and parallel semantic shifts), and for documenting sources. It was developed from the earlier CoBL (Cognates in Basic Lexicon) project and underlies the analysis of Heggarty et al. (2023). Website: https://iecor.clld.org.

The data of Ringe et al. (2002), which code lexical, phonological and morphological characters for 24 Indo-European languages, remain an important resource for character-based analyses of Indo-European, and were used in comparisons of phylogenetic methods (Nakhleh et al., 2005b).

G.2.4 Austronesian Basic Vocabulary Database (ABVD)

The ABVD (Greenhill et al., 2008) contains word lists for 210 basic meanings for well over a thousand Austronesian language varieties, with cognate judgments for a large subset. It underlies many of the best-known phylogenetic studies of Austronesian, including Gray et al. (2009), and

has been used extensively as a scaffold for studies of cultural evolution in the Pacific (Chapter 14). The ABVD is available online (https://abvd.clld.org) and in CLDF through Lexibank. Sister databases built with the same software have been set up for some other families.

G.2.5 Bantu

Several lexical datasets underlie phylogenetic work on Bantu. The lexicostatistical study of Bastin et al. (1999) compiled basic vocabulary for several hundred Bantu languages, and has been reused in later work. Grollemund et al. (2015) assembled 100-meaning word lists with cognate judgments for over 400 Bantu and related languages, which were also used by Koile et al. (2022). The *Bantu Lexical Reconstructions* (BLR) database at the Royal Museum for Central Africa in Tervuren collects reconstructed Proto-Bantu roots, and is a resource for cognate identification. Several Bantu datasets are available in CLDF through Lexibank.

G.2.6 Sino-Tibetan

The Sino-Tibetan Etymological Dictionary and Thesaurus (STEDT) project at Berkeley (Matisoff, 2015) compiled a large database of lexical forms from several hundred Sino-Tibetan varieties, organized by etymological sets, and published a reconstruction of Proto-Tibeto-Burman. Website: https://stedt.berkeley.edu. The datasets of Sagart et al. (2019) and Zhang et al. (2019), which include cognate judgments for basic vocabulary, are available in CLDF.

G.2.7 Uralic: UraLex

UraLex is a basic-vocabulary dataset for the Uralic languages, with cognate judgments and documentation of sources, compiled to support phylogenetic analyses of the family (Syrjänen et al., 2013). It has been used to examine how the choice of meaning list affects inferred Uralic trees, and as the basis of dated analyses (Honkola et al., 2013).

G.2.8 Australia: Chirila

Chirila (Contemporary and Historical Resources for the Indigenous Languages of Australia) (Bowern, 2016) is a large database of lexical data for Australian Aboriginal languages, compiled from published and archival sources. It underlies phylogenetic studies of Pama-Nyungan (Bowern and Atkinson, 2012; Bouckaert et al., 2018) and studies of lexical change in Australian languages. Website: https://doi.org/10.60600/YU/XHPJXB.

G.2.9 NorthEuraLex

NorthEuraLex (Dellert et al., 2020) contains word lists for over 1,000 concepts for more than a hundred languages of northern Eurasia, from Uralic, Indo-European, Turkic, Mongolic, Tungusic and other families, all in a uniform phonetic transcription. Its long concept list makes it useful for automated cognate detection, studies of sound correspondences, and comparisons across many families. Website: http://northeuralex.org.

G.2.10 Intercontinental Dictionary Series and World Loanword Database

The Intercontinental Dictionary Series (IDS) (Key, 2015) provides word lists for a standard list of roughly 1,300 concepts for several hundred languages, compiled from dictionaries. The World

Loanword Database (WOLD) (Haspelmath, 2009) gives, for 41 languages from around the world, the vocabulary for a list of about 1,460 meanings together with expert judgments of whether each word is a loan, from which source, and how certain the judgment is. A companion handbook summarizes the findings (Haspelmath and Tadmor, 2009). WOLD's "borrowability" scores for meanings are relevant to choosing meaning lists for phylogenetic analysis, and its loanword data can be used to validate borrowing detection methods (Chapter 8). Websites: https://ids.clld.org and https://wold.clld.org.

G.2.11 Other lexical resources

The *Sound Comparisons* project (https://soundcomparisons.com) provides recorded audio and transcriptions of comparable word lists for dialects and languages of several families, useful for studies of phonetic variation and dialect continua. The *DiACL* database (Carling et al., 2018) (Diachronic Atlas of Comparative Linguistics) combines lexical cognate data and typological data for several language families, including Indo-European, with good coverage of ancient languages; it has been used in phylogenetic studies of grammatical change. The CLDF datasets accompanying many recent phylogenetic papers, such as those for Dravidian (Kolipakam et al., 2018) and Tupi-Guarani, are often available through Lexibank or from the authors' repositories.

G.3 Concepts, polysemy and sounds

G.3.1 Concepticon

Concepticon (List et al., 2016) links the concept lists used in linguistics (Swadesh lists, Leipzig-Jakarta list, ASJP list, questionnaires and the concept lists of individual datasets) to a common set of *concept sets* with stable identifiers and definitions. Linking a dataset's concepts to Concepticon makes it possible to combine datasets with different concept lists, to see which concepts are shared, and to compare the stability or borrowability of concepts across studies. Website: https://concepticon.clld.org.

G.3.2 CLICS

The Database of Cross-Linguistic Colexifications (CLICS) (List et al., 2018a; Rzymski et al., 2020) records cases in which a single word expresses two concepts (for example 'tree' and 'wood', or 'arm' and 'hand') across thousands of language varieties, aggregated from Lexibank datasets. Colexification networks reveal semantic associations and are relevant to modeling semantic change and to understanding why some cognate sets shift meaning. Website: https://clics.clld.org.

G.3.3 CLTS

The Cross-Linguistic Transcription Systems (CLTS) reference catalogue (Anderson et al., 2018) defines a standardized set of speech sounds with phonetic feature descriptions, and maps the transcription conventions of many datasets and databases (including PHOIBLE) to it. It underlies the standardization of transcriptions in Lexibank, which makes automated alignment and sound-correspondence analysis possible across datasets. Website: https://clts.clld.org.

G.3.4 PHOIBLE

PHOIBLE (Moran, 2019) is a database of phoneme inventories for more than two thousand languages, compiled from multiple sources (some languages have several inventories from different descriptions), with features for each segment. It is used in typological studies of sound systems and in phylogenetic comparative studies of phonological evolution (Chapters 12 and 13). Website: https://phoible.org.

G.4 Structural (typological) databases

G.4.1 WALS

The World Atlas of Language Structures (WALS) (Dryer, 2013) contains data on about 190 phonological, grammatical and lexical features for about 2,600 languages, each feature compiled by a specialist and documented in a chapter. WALS was the first large typological database freely available online, and has been used in a great many comparative studies, including early phylogenetic tests of word-order correlations and studies of the stability of features (Dediu, 2011; Wichmann and Holman, 2009b). Its main limitation for phylogenetic work is its sparsity: most languages are coded for only a minority of features, and different features are coded for different language samples. Website: https://wals.info.

G.4.2 Grambank

Grambank (Skirgård et al., 2023) codes about 195 grammatical features for more than 2,400 languages, with a design that addresses the main limitations of WALS: features are mostly binary, defined with detailed coding instructions, and coded for every language in the sample from the same reference grammars, so that coverage is much more complete. It was compiled by a large team using a common questionnaire and is published in CLDF. Grambank is now the standard resource for phylogenetic comparative studies of grammar and for global analyses of structural diversity (Chapter 13). Website: https://grambank.clld.org; data: https://github.com/grambank/grambank.

G.4.3 AUTOTYP

AUTOTYP (Bickel et al., 2022) is a typological database built on the principle of *autotypologizing*: categories are defined bottom-up from the data rather than imposed in advance, and complex phenomena are coded as many fine-grained variables. It covers several thousand languages for some variables, with particular depth in morphology, alignment and clause linkage, and includes a genealogical and areal classification. Data: https://github.com/autotyp/autotyp-data.

G.4.4 Other structural resources

The *Atlas of Pidgin and Creole Language Structures* (APiCS) (Michaelis et al., 2013) provides comparable structural data for 76 pidgin and creole languages, relevant to questions about contact languages and their place in (or outside) family trees (Chapter 8). Website: https://apics-online.info. Areal and family-specific structural databases, such as the *South American Indigenous Language Structures* database, and family-specific questionnaires used in studies such as Dunn et al. (2005), complement the global databases.

G.5 Cultural, environmental and historical data

G.5.1 D-PLACE

D-PLACE (Database of Places, Language, Culture and Environment) (Kirby et al., 2016) links cultural data on about 1,400 societies, drawn principally from the Ethnographic Atlas (Murdock, 1967) and other ethnographic datasets, to their languages (via Glottocodes), locations and environmental variables such as climate and ecology. Crucially for phylogenetic work, D-PLACE also collects published language phylogenies, so that cultural traits can be analyzed on a tree with the matching societies. It is the main data source for phylogenetic studies of cultural evolution (Chapter 14). Website: https://d-place.org; data: https://github.com/D-PLACE/dplace-cldf.

G.5.2 Pulotu

Pulotu (Watts et al., 2015b) is a database of supernatural beliefs and practices, together with social and environmental variables, for over one hundred Austronesian cultures, coded from ethnographic sources with careful attention to the historical period described. It was designed for phylogenetic analysis on Austronesian language trees and underlies studies of the coevolution of religion and political complexity (Watts et al., 2015a, 2016). It is now distributed in CLDF.

G.5.3 Seshat

Seshat: Global History Databank (Turchin et al., 2015) codes social complexity, warfare, religion, agriculture and many other variables for hundreds of historical polities, sampled through time from the Neolithic onward. It is used to study the dynamics of social complexity through time rather than across a tree, but can be combined with language trees for regions where polities are linked to languages. Website: https://seshatdatabank.info.

G.5.4 Kinbank

Kinbank (Passmore et al., 2023) is a global database of kinship terminology, recording the terms used for a standard set of kin relationships in more than a thousand languages, linked to Glottolog. It enables phylogenetic studies of the evolution of kinship systems, one of the classic topics of cultural phylogenetics (Chapter 14).

G.5.5 Genes and languages

The GeLaTo database (Genes and Languages Together) (Barbieri et al., 2022) pairs human population-genetic data with linguistic information for the same populations, and has been used to quantify how often genetic and linguistic histories match. Such resources are essential for formal joint models of genetic and linguistic history (Chapters 14 and 18).

G.6 Published trees

Phylogenetic comparative studies need trees, preferably full posterior samples rather than single summary trees. Several sources collect published language phylogenies:

- **D-PLACE** includes a collection of published language phylogenies (with posterior samples where available), matched to the societies in its cultural datasets.

- **Phlorest**, a collection of language phylogenies standardized in CLDF format by the CLDF team (https://github.com/phlorest), aims to make published trees uniformly accessible.
- **Glottolog** provides classification trees for every family, without branch lengths, which can be combined into a global tree.
- **Global lexical trees** have been inferred from ASJP data (Jäger, 2018) and are available from the authors' repositories.
- **Supplementary data** of individual papers, often deposited in Dryad, Zenodo, OSF or GitHub, remain the main source for many trees. Authors who deposit their full posterior tree samples, input files and data make their work far more useful to others (Chapter 17).

G.7 Data formats

The *Cross-Linguistic Data Formats* (CLDF) (Forkel et al., 2018) are the standard format for most of the resources in this appendix. A CLDF dataset consists of CSV tables (languages, parameters, forms, cognates, values, codes, and others) and a JSON metadata file that describes them, following the W3C recommendation for CSV on the web. Specifications and validation tools are documented at https://cldf.clld.org. Chapter 17 shows how to read CLDF data into R, and Appendix F lists the Python tools (`pycldf`, `cldfbench`, `pylexibank`) used to create and validate CLDF datasets. NEXUS, PHYLIP and Newick (Chapter 2) remain the formats for exchanging matrices and trees with phylogenetic software.

Appendix H

RevBayes Implementations

The Bayesian models described in this book are implemented in RevBayes (Höhna et al., 2016) in the folder `revbayes/` that accompanies the book. We call this collection **RevLingPhy**. RevBayes is well suited to the purpose because it expresses a model as an explicit probabilistic graphical model. Every rate matrix, prior, clock and tree process is written out in a few lines of the Rev language rather than hidden behind a menu. Each script can therefore be read as a precise statement of the model that the corresponding chapter describes in words and equations.

Every script was run with RevBayes 1.4.1 on a small simulated dataset, with between six and sixteen languages, for which the true parameter values are known. A companion R script compares the posterior with the truth. This appendix documents each model, shows its Rev code, and reports the result of that check.

H.1 Layout and use

The package has the following structure.

Table H.1: Folders of the RevLingPhy package.

Folder	Contents
`data/`	R scripts (`sim_<group>.R`) that simulate each test dataset, and the simulated data with its true parameter values
`analyses/<group>/`	One standalone Rev script per model, numbered within each group
`output/<group>/`	MCMC logs, sampled trees, ancestral states and marginal likelihoods
`tests/`	R scripts (`check_<group>.R`) that compare posteriors with the true values, and their results
`docs/`	The documentation reproduced in this appendix

The scripts are grouped by topic. The **lexical** group covers substitution models for cognate data, tree priors, model comparison and model adequacy (Chapters 5, 6 and 7). The **time** group covers clock models, calibration, tip-dating and diversification (Chapters 9, 10 and 16). The **traits** group covers phylogeography, ancestral states and comparative methods (Chapters 11, 12, 13 and 14).

To run a single analysis, change into the `revbayes/` folder and call RevBayes on the script. Chain lengths can be increased for real data by passing an argument after `--args`.

```
cd revbayes
rb analyses/lexical/01_binary_ctmc_asc.Rev
rb analyses/lexical/01_binary_ctmc_asc.Rev --args 50000   # longer chain
Rscript tests/check_lexical.R                              # compare with the truth
```

The scripts are written for clarity and for small test datasets. For real datasets, chains must be much longer, convergence must be checked with two or more independent runs (Section 6.5), and priors should be chosen for the problem at hand rather than copied from the test settings.

H.2 Verification summary

Table H.2: Verification of the RevBayes scripts on simulated data. For each script: the number of parameters compared with their true values, how many true values lie inside the 95% highest posterior density interval, the smallest effective sample size among them, and the overall status.

Group	Script	Model	Parame	Truth in 95% HPD	Min. ESS	Status
lexical	01_ctmc_ascertainment	binary CTMC, coding = all on no-absence data (uncorrected)	8	4/6	437	pass (bias shown), pass, info (MAP topology differs), info
lexical	02_ctmc_gamma	binary CTMC + Gamma(4), noabsencesites	5	3/3	479	pass, info (MAP topology differs), info
lexical	03_covarion	binary covarion (explicit no-absence correction)	6	4/4	65	pass, pass (low ESS), info (MAP topology differs)
lexical	04_stochastic_dollo	stochastic Dollo (eps-limit CTMC + Poisson N), root age fixed	5	2/2	406	pass, info (MAP topology differs), info
lexical	05_pseudo_dollo	pseudo-Dollo 3-state CTMC (ambiguity-coded), root age fixed	5	4/4	38	pass (low ESS), pass, info (MAP topology differs)
lexical	06_meaning_mk	meaning-slot Mk (K = 5), 3 rate partitions	5	4/4	592	pass, info (MAP topology differs)
lexical	07_frequency_regression	per-meaning Mk rates, log rho = a + b log f (time tree)	3	2/2	458	pass, info (MAP topology differs)

Group	Script	Model	Parame	Truth in 95% HPD	Min. ESS	Status
lexical	08_timetree_yule_bd	binary CTMC, strict clock, Yule prior, root ~ Lognormal(log 4, 0.05)	11	6/6	80	pass (low ESS), pass, info, info (MAP topology differs)
lexical	09_model_comparison	equal_rates	5	0/0	NA	info, pass
lexical	10_posterior_predictive	equal_rates fitted to Gamma(0.5) data	10	6/10	76	pass (inadequacy detected), info (not detected), info (agrees with R replicates), pass, info (RevBayes 1.4.1 simulator bias)
lexical	11_total_evidence	lexical binary CTMC + structural Mkv (K = 3), shared tree	4	3/3	464	pass, info (MAP topology differs)
lexical	12_meaning_rate_strata	sparse finite mixture (DP approximation) on meaning rates, Mk K = 5	7	4/5	300	pass, fail, info
lexical	13_mixture_pi_rate	2-class mixture over (pi1, rate), noabsencesites	6	6/6	185	pass
lexical	14_branch_length_priors	CTMC + Gamma, iid_exponential branch-length prior	4	0/4	281	info (prior sensitivity; covered = FALSE)
lexical	15_synonymy_slot_model	explicit replacement with synonymy (CTMC on subsets of 3 classes)	5	2/3	376	pass, fail, info (MAP topology differs), info
lexical	16_f81_beta_mixture	jc	13	3/5	776	info, pass, fail
lexical	17_rate_categories	gamma2	10	3/3	1081	info, pass
time	01_strict_clock_calibrations.Rev	strict clock + node calibrations	7	6/6	1248	pass, info
time	02_ucln_relaxed_clock.Rev	UCLN relaxed clock	7	7/7	198	pass
time	03_uce_relaxed_clock.Rev	UCE relaxed clock	6	6/6	370	pass
time	04_autocorrelated_lognormal_clock.Rev	autocorrelated lognormal (GBM) clock	6	6/6	57	pass, warn
time	05_random_local_clock.Rev	random local clock (RJ shifts)	7	7/7	119	pass

Group	Script	Model	Parame	Truth in 95% HPD	Min. ESS	Status
time	06_punctuational_bursts.Rev	punctuational bursts (data: punct)	10	8/8	27	pass, info, warn
time	07_tip_dating_serial_bd.Rev	tip-dating, serially sampled BD (no sampled ancestors)	5	4/5	77	pass, info-warn, info-fail
time	08_fbd_sampled_ancestors.Rev	fossilized birth-death with sampled ancestors	7	7/7	66	warn, pass
time	09_clock_model_comparison_ss.Rev	strict vs UCLN, data simulated under ucln	2	2/2	NA	pass
time	10_tree_prior_coalescent_vs_yule.Rev	strict clock, yule tree prior	6	4/4	199	pass, info-pass, info
time	11_multispecies_coalescent.Rev	MSC, true concept trees observed	12	7/12	51	pass, fail
time	12_episodic_birth_death_skyline.Rev	episodic birth-death (3 intervals)	5	5/5	719	pass
time	13_bisse_state_dependent_diversification.Rev	BiSSE (dnCDBDP)	7	6/7	193	pass, fail
time	14_effective_prior_calibrations.Rev	calibrations x Yule prior, no data	3	0/0	1355	info
time	15_gamma_rate_variation_dating_bias.Rev	strict clock + Gamma(4) among-meaning rates	6	5/6	676	pass, info-fail, info-pass
traits	01_discrete_phylogeo_bssvs	BSSVS discrete phylogeography	13	7/7	1753	pass, miss
traits	02_continuous_phylogeo_bm	bivariate BM diffusion	6	5/5	588	pass
traits	03_relaxed_random_walk	relaxed random walk (lognormal)	5	4/4	1078	pass
traits	04_dec	DEC	4	4/4	951	pass
traits	05_mk_asr	Mk ER	11	7/7	717	pass, miss
traits	06_stochastic_mapping	Mk ARD + stochastic map	7	7/7	951	pass
traits	07_hidden_rates	hidden-rates model	4	3/3	593	pass
traits	08_pagel_correlated	Pagel dependent vs independent	14	8/8	145	pass
traits	09_bm_ou	BM (data: BM)	6	4/4	366	pass
traits	10_early_burst	early burst	3	2/2	289	pass
traits	11_mvbm_correlation	bivariate BM	5	4/4	1901	pass
traits	12_pagel_lambda	BM with Pagel's lambda	2	2/2	1654	pass

Group	Script	Model	Parame	Truth in 95% HPD	Min. ESS	Status
traits	13_asr_tree_sample	ER Mk over tree sample	3	3/3	951	pass
traits	14_asr_joint_tree_trait	joint time tree + ER Mk	4	3/3	705	pass
traits	15_phylogeo_glm	phylogeographic GLM	6	4/4	951	pass, miss
traits	16_bm_trend_ancient_tips	(a) BM, modern tips	11	10/11	398	expected_miss, pass
traits	17_threshold_model	threshold model	2	1/1	592	pass
traits	18_ordered_political	ordered Mk: stepwise	6	2/2	476	pass, miss
traits	19_phylo_spatial	phylogeny + space	5	4/4	230	pass
traits	20_ou_multi_optimum	multi-optimum OU	5	4/4	70	pass, warn_ess
traits	21_root_treatment_asr	root treatment: FLAT	18	2/2	463	pass
traits	22_pathway_rj_tree_sample	reversible jump over rates	13	3/4	501	pass, miss

H.3 Models of lexical evolution

This section documents the RevBayes (Höhna et al., 2016) scripts for the models of lexical evolution of Chapter 7, together with the Bayesian devices of Chapter 6 that matter most for cognate data (tree-length priors, marginal likelihoods, posterior predictive checks). Every script was run on a small data set simulated by `revbayes/data/sim_lexical.R` with known parameter values (`revbayes/data/lexical/true_values.csv`), and `revbayes/tests/check_lexical.R` compares the posterior with the truth and, where possible, with an independent computation in R (the book's own likelihood code or **phangorn**).

Conventions. All data sets have ten languages (`L01` to `L10`). The true tree (`revbayes/data/lexical/true_time_tree.tre`) is a Yule tree re-scaled to a root age of 4 millennia; unrooted analyses use the same tree with branch lengths multiplied by a rate, so that branch lengths are expected changes per character. Binary cognate matrices never contain all-absent columns (every cognate class is attested somewhere), so the binary models use the no-absence ascertainment correction of Section 7.4 unless stated otherwise. Scripts are run from `revbayes/`:

```
cd revbayes
Rscript data/sim_lexical.R                                  # simulate all data sets
rb analyses/lexical/01_ctmc_ascertainment.Rev               # one analysis, default
settings
rb analyses/lexical/09_model_comparison.Rev --args 2 500    # one model, 500
generations per stone
bash tests/run_lexical.sh                                   # all analyses, recording
run times
Rscript tests/check_lexical.R                               # writes
tests/results_lexical.csv
```

In each verification table, *Truth* is the simulated value, *HPD* the 95% highest posterior density interval, and *ESS* the effective sample size after discarding 10% burn-in. *Status* is `pass` if the truth lies inside the HPD and ESS is at least 100, `pass (low ESS)` if the truth is covered with ESS below 100, `fail` otherwise, and `info` for descriptive rows (prior sensitivity, independent maximum-likelihood estimates, topology summaries). Rows marked `pass (bias shown)` belong to a deliberately misspecified model, for which missing the truth is the expected and desired outcome. Run times (`runtime_s` in the CSV) were measured on a shared 12-core machine with several other RevBayes jobs running and are therefore pessimistic.

Table H.3: Lexical-model scripts: number of verification rows by status and run time (seconds).

script	pass	fail	info	runtime_s
01_ctmc_ascertainment	6	0	2	84
02_ctmc_gamma	3	0	2	166
03_covarion	5	0	1	252
04_stochastic_dollo	3	0	2	81
05_pseudo_dollo	4	0	1	202
06_meaning_mk	4	0	1	91
07_frequency_regression	2	0	1	178
08_timetree_yule_bd	6	0	5	134
09_model_comparison	2	0	3	148
10_posterior_predictive	6	0	4	62
11_total_evidence	3	0	1	87
12_meaning_rate_strata	4	1	2	61
13_mixture_pi_rate	6	0	0	204
14_branch_length_priors	0	0	4	197
15_synonymy_slot_model	2	1	2	84
16_f81_beta_mixture	6	2	5	188
17_rate_categories	5	0	5	207

A note on topology. The simulated Yule tree has three very short internal branches (0.06 to 0.14 millennia) and four long ones. With ten languages and a few hundred to a thousand characters the four long splits are recovered with posterior probability close to 1 in every analysis, while the three short ones are genuinely uncertain (posterior probabilities mostly between 0.05 and 0.6). The topology rows in the tables therefore report the Robinson-Foulds distance of the most probable sampled topology to the truth, the posterior probability of the true topology and the support of each true split, with status `info`; they are a property of the simulated tree, not a failure of the models.

H.3.1 Implementation devices

Three devices recur, because RevBayes 1.4.1 has no built-in stochastic Dollo likelihood and offers ascertainment correction only for 2-state data.

1. **Hidden-state binary models** (covarion, stochastic Dollo, pseudo-Dollo) are written as CTMCs on 3 or 4 hidden states, and the observed binary matrix is expanded with `ex-`

`pandCharacters(2)`, which turns an observed 0 into the ambiguity `(0 2)` and a 1 into `(1 3)`.

2. **No-absence correction for hidden-state models.** The required likelihood is $\prod_i P(D_i)/(1-P_0)^N$, where P_0 is the probability that all languages lack the class (Equation (7.4)). We clamp a second copy `Z` of the CTMC to a single all-absent column, so that $\log P_0$ is available as `Z.lnProbability()`, cancel `Z`'s own contribution with a clamped Exponential(rate $1/P_0$) node at 0, and add $-N\log(1-P_0)$ with N clamped Exponential(rate $1/(1-P_0)$) nodes at 0 (an exponential density at 0 equals its rate). `check_lexical.R` recomputes the corrected likelihood in R for posterior samples; the two agree to better than 10^{-3} log-units.
3. **Two RevBayes 1.4.1 quirks** had to be worked around and are worth knowing: a deterministic node defined only as `Z.lnProbability()` is not re-evaluated when the tree or the rate matrix changes (its children keep stale probabilities, which silently removes the correction), so the scripts add `0 *` (a function of every parameter) to it; and `model()` crashes when entered from the tree in such a graph, so it is entered from the sink node (`model(asc)`). A user-defined Rev function computing P_0 by pruning works but is about a hundred times slower.

H.3.2 Binary CTMC and ascertainment correction

Script `01` fits the binary CTMC of Section 7.3 (restriction-site or Mk2 model with an estimated frequency of presence π_1, Equation (7.1), Felsenstein (1992)) on an unrooted tree with i.i.d. exponential branch lengths, three times: to data without all-absent columns with no correction (`coding="all"`), to the same data with the no-absence correction (`coding="noabsencesites"`), and to the variable-only version of the data with the Lewis correction (`coding="variable"`, Lewis (2001)). The uncorrected fit shows the bias described in Section 7.4: π_1 is inflated to 0.398 (truth 0.3) and the tree length to 2.82 (truth 1.85). Both corrected fits recover the truth, and agree with an independent maximum-likelihood fit with the book's corrected likelihood function.

```
################################################################################
# 01_ctmc_ascertainment.Rev
# Model   : binary CTMC (restriction-site / Mk2 with free stationary frequency
#           pi1) on cognate presence/absence, unrooted tree, i.i.d. exponential
#           branch-length prior.
# Book    : Chapter 7, Sections lexical-models-ctmc and lexical-models-ascertainment
#           (Equations lexical-models-Q and lexical-models-asc).
# Estimates: pi1, topology, branch lengths, tree length (expected changes per
#           cognate character), under three ascertainment schemes:
#   (A) "uncorrected": data without all-absent columns, coding="all"   (biased)
#   (B) "noabsence"  : same data, coding="noabsencesites"              (correct)
#   (C) "variable"   : data without all-absent AND all-present columns,
#                      coding="variable" (Lewis Mkv)                    (correct)
# Data    : data/lexical/d01_noabsence.nex, d01_variable.nex (truth: pi1 = 0.3,
#           TL = 1.854; see data/lexical/true_values.csv)
# Runtime : about 1.5 minutes for the three chains (default n_gen = 12000)
# Usage   : rb analyses/lexical/01_ctmc_ascertainment.Rev [--args n_gen]
################################################################################
seed(101)
n_gen = 12000
```

```
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/lexical/01_ctmc_ascertainment/"

schemes   = ["uncorrected", "noabsence", "variable"]
files     = ["data/lexical/d01_noabsence.nex", "data/lexical/d01_noabsence.nex",
             "data/lexical/d01_variable.nex"]
codings   = ["all", "noabsencesites", "variable"]

for (s in 1:3) {
    data <- readDiscreteCharacterData(files[s])
    taxa <- data.taxa()
    n_branches <- 2 * taxa.size() - 3
    moves    = VectorMoves()
    monitors = VectorMonitors()

    # --- tree: uniform topology, i.i.d. exponential branch lengths (mean 0.2)
    psi ~ dnUniformTopologyBranchLength(taxa, branchLengthDistribution =
dnExponential(5.0))
    moves.append( mvNNI(psi, weight = 10) )
    moves.append( mvSPR(psi, weight = 5) )
    moves.append( mvBranchLengthScale(psi, weight = n_branches) )
    TL := psi.treeLength()

    # --- binary CTMC: Q = beta * [[-pi1, pi1], [pi0, -pi0]], normalised so that
    #     one unit of branch length = one expected change per character
    pi ~ dnDirichlet(v(1, 1))
    moves.append( mvBetaSimplex(pi, alpha = 20, weight = 3) )
    pi1 := pi[2]
    Q := fnF81(pi)

    seq ~ dnPhyloCTMC(tree = psi, Q = Q, type = "Binary", coding = codings[s])
    seq.clamp(data)

    mymodel = model(psi)
    monitors.append( mnModel(filename = out_dir + schemes[s] + ".log", printgen =
    12) )
    monitors.append( mnFile(psi, filename = out_dir + schemes[s] + ".trees",
    printgen = 60) )
    monitors.append( mnScreen(printgen = 5000, pi1, TL) )
    mymcmc = mcmc(mymodel, monitors, moves, nruns = 1)
    mymcmc.burnin(generations = 500, tuningInterval = 100)
    mymcmc.run(generations = n_gen)

    trace = readTreeTrace(out_dir + schemes[s] + ".trees", treetype = "non-clock",
burnin = 0.1)
    map_tree = mapTree(trace, out_dir + schemes[s] + "_map.tre")
}
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
binary CTMC, coding = all on no-absence data (uncorrected)	pi1	0.300	0.398	0.382	0.413	901.0	pass (bias shown)
binary CTMC, coding = all on no-absence data (uncorrected)	TL	1.853	2.817	2.611	3.003	785.8	pass (bias shown)
binary CTMC, coding = noabsencesites	pi1	0.300	0.310	0.289	0.329	803.5	pass
binary CTMC, coding = noabsencesites	TL	1.853	1.990	1.823	2.161	514.3	pass
binary CTMC, coding = noabsencesites	topology_RF_MAP	0.000	6.000	NA	NA	NA	info (MAP topology differs)
binary CTMC, coding = variable (Mkv) on variable-only data	pi1	0.300	0.308	0.286	0.331	772.1	pass
binary CTMC, coding = variable (Mkv) on variable-only data	TL	1.853	1.997	1.822	2.186	436.6	pass
ML (R, true topology, branch-length scale free), noabsence correction	pi1	0.300	0.307	NA	NA	NA	info

H.3.3 Gamma rate variation among characters

Script `02` adds discrete-gamma rate variation (four categories, Equation (7.5), Yang (1994)), the model of Gray and Atkinson (2003). The data were simulated with rates drawn from the same 4-category discrete gamma ($\alpha = 0.5$), so that α is directly verifiable; with continuous-gamma data the discrete model estimates a smaller α, as the book's **phangorn** example also shows. The gamma shape and the tree length lie on a ridge (fast categories saturate), so the branch-length prior matters; script `02` uses the compound Dirichlet prior of Rannala et al. (2012) and Zhang et al. (2012) with tree length ~ Gamma(1, 0.2), and script `14` shows the sensitivity.

```
################################################################################
# 02_ctmc_gamma.Rev
# Model   : binary CTMC + discrete-gamma among-character rate variation
#           (4 equiprobable categories, mean rate 1), no-absence ascertainment
#           correction, unrooted tree with a gamma-Dirichlet branch-length prior.
# Book    : Chapter 7, Section lexical-models-rates (Equation lexical-models-gamma);
#           this is the "restriction site + Gamma" model of Gray & Atkinson (2003).
# Estimates: gamma shape alpha, pi1, topology, tree length.
# Data    : data/lexical/d02_gamma.nex (truth: alpha = 0.5, pi1 = 0.25, TL = 6.95;
#           rates drawn from the 4-category discrete gamma, i.e. the fitted model)
# Runtime : about 2.5 minutes (default n_gen = 10000)
# Usage   : rb analyses/lexical/02_ctmc_gamma.Rev [--args n_gen]
################################################################################
```

```
seed(202)
n_gen = 10000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/lexical/02_ctmc_gamma/"

data <- readDiscreteCharacterData("data/lexical/d02_gamma.nex")
taxa <- data.taxa()
n_branches <- 2 * taxa.size() - 3
moves    = VectorMoves()
monitors = VectorMonitors()

# gamma-Dirichlet (compound Dirichlet) branch-length prior, Rannala et al. (2012):
# TL ~ Gamma(1, 0.2) (mean 5), split among branches by a flat Dirichlet.  With
# strong rate variation the tree length and alpha lie on a ridge (fast categories
# saturate), so i.i.d. exponential priors pull TL towards their own prior mean
# (see 14_branch_length_priors.Rev); the explicit TL also allows a whole-tree move.
topology ~ dnUniformTopology(taxa)
moves.append( mvNNI(topology, weight = 10) )
moves.append( mvSPR(topology, weight = 5) )
TL ~ dnGamma(1.0, 0.2)
moves.append( mvScale(TL, lambda = 0.3, weight = 8) )
rel_bl ~ dnDirichlet(rep(1.0, n_branches))
moves.append( mvBetaSimplex(rel_bl, alpha = 50, weight = n_branches) )
moves.append( mvDirichletSimplex(rel_bl, alpha = 200, weight = 5) )
bl := rel_bl * TL
psi := fnTreeAssembly(topology, bl)

# binary CTMC with free stationary frequency of presence
pi ~ dnDirichlet(v(1, 1))
moves.append( mvBetaSimplex(pi, alpha = 20, weight = 3) )
pi1 := pi[2]
Q := fnF81(pi)

# among-character rate variation: rho_i ~ Gamma(alpha, alpha), 4 categories
alpha ~ dnExponential(1.0)
moves.append( mvScale(alpha, lambda = 0.5, weight = 5) )
site_rates := fnDiscretizeGamma(alpha, alpha, 4)

seq ~ dnPhyloCTMC(tree = psi, Q = Q, siteRates = site_rates, type = "Binary",
                  coding = "noabsencesites")
seq.clamp(data)

mymodel = model(topology)
monitors.append( mnModel(filename = out_dir + "gamma.log", printgen = 15, exclude =
["rel_bl", "bl"]) )
monitors.append( mnFile(psi, filename = out_dir + "gamma.trees", printgen = 75) )
monitors.append( mnScreen(printgen = 5000, alpha, pi1, TL) )
mymcmc = mcmc(mymodel, monitors, moves, nruns = 1)
```

```
mymcmc.burnin(generations = 1000, tuningInterval = 100)
mymcmc.run(generations = n_gen)

trace = readTreeTrace(out_dir + "gamma.trees", treetype = "non-clock", burnin = 0.1)
map_tree = mapTree(trace, out_dir + "gamma_map.tre")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
binary CTMC + Gamma(4), noabsencesites	alpha	0.50	0.485	0.372	0.597	601.0	pass
binary CTMC + Gamma(4), noabsencesites	pi1	0.25	0.246	0.234	0.259	601.0	pass
binary CTMC + Gamma(4), noabsencesites	TL	6.95	8.231	5.385	11.678	479.4	pass
binary CTMC + Gamma(4), noabsencesites	topology_RF_MAP	0.00	2.000	NA	NA	NA	info (MAP topology differs)
ML (phangorn, true topology, Lewis correction on variable columns)	alpha	0.50	0.212	NA	NA	NA	info

H.3.4 Binary covarion model

Script `03` implements the covarion model of Section 7.6 (Tuffley and Steel, 1998) with `fnCovarion`: hidden states ordered $(0^+, 1^+, 0^-, 1^-)$, visible change only in the "on" class, switching rates s_{off} and s_{on}, and the book's scaling (one expected visible change per unit branch length). The no-absence correction uses device 2 above. Switching rates and tree length are only weakly identified with ten languages (Table 7.3); joint up-down moves along these ridges keep the chain mixing, but the ESS of s_{on}, s_{off} and the tree length stays below 100 within the three-minute budget.

```
################################################################################
# 03_covarion.Rev
# Model   : binary covarion model (Tuffley & Steel 1998 type): every cognate
#           character carries a hidden on/off switch; visible gain/loss only in
#           the "on" class; switching off at rate s_off, on at rate s_on.
#           Hidden states ordered (0+, 1+, 0-, 1-) as in the book.
# Book    : Chapter 7, Section lexical-models-covarion (Equation
lexical-models-covarion).
#           Scaling as in the book's covarion_Q(): beta is chosen so that the
#           stationary mean rate of *visible* change is 1 per unit branch length,
#           i.e. the "on" class runs at rate (s_on + s_off)/s_on.
# Estimates: pi1, s_on, s_off, topology, tree length (gamma-Dirichlet prior).
# Ascertainment: RevBayes 1.4.1 supports coding="noabsencesites" only for the
#           2-state Binary/Restriction type, not for hidden-state (expanded) data,
#           so Equation lexical-models-asc is applied explicitly:
#            * Z = the same CTMC clamped to ONE all-absent column; lnP0 = log P(0...0)
#              * a clamped Exponential(rate = 1/P0) at 0 cancels Z's own term log P0
#              * N clamped Exponential(rate = 1/(1-P0)) at 0 add -N log(1 - P0)
```

```
#           (an Exponential density at 0 equals its rate), so the likelihood is
#           prod_i P(D_i) / (1 - P0)^N.  tests/check_lexical.R recomputes this
#           likelihood in R for posterior samples and checks it to 1e-4.
#           RevBayes 1.4.1 quirks handled here: (i) a node defined only as
#           Z.lnProbability() is not touched when the tree or Q change, so its
#           children would keep stale probabilities -- we add 0 * (a deterministic
#           function of every parameter) to force the dependency; (ii) model()
#           segfaults when entered from the tree in this DAG; entering it from the
#           sink node (model(asc)) works.
# Data    : data/lexical/d03_covarion.nex (truth: pi1 = 0.3, s_off = s_on = 0.5, TL
= 6.95)
# Runtime : about 3-4 minutes (default n_gen = 6000)
# Usage   : rb analyses/lexical/03_covarion.Rev [--args n_gen]
################################################################################
seed(303)
n_gen = 6000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/lexical/03_covarion/"

data  <- readDiscreteCharacterData("data/lexical/d03_covarion.nex")
# expand binary states to hidden classes: 0 -> {0+, 0-} = (0 2), 1 -> {1+, 1-} = (1 3)
data_h <- data.expandCharacters(2)
zero   <- readDiscreteCharacterData("data/lexical/zero_column.nex")
zero_h <- zero.expandCharacters(2)
N <- data.nchar()
taxa <- data.taxa()
n_branches <- 2 * taxa.size() - 3
moves    = VectorMoves()
monitors = VectorMonitors()

# compound-Dirichlet (gamma-Dirichlet) branch-length prior (Rannala et al. 2012):
# tree length TL ~ Gamma(1, 0.2) (mean 5), split among branches by a flat Dirichlet.
# The explicit TL parameter gives a whole-tree scaling move, which mixes much
# better than per-branch moves when rate heterogeneity makes TL weakly identified.
topology ~ dnUniformTopology(taxa)
moves.append( mvNNI(topology, weight = 4) )
moves.append( mvSPR(topology, weight = 2) )
TL ~ dnGamma(1.0, 0.2)
moves.append( mvScale(TL, lambda = 0.3, weight = 8) )
rel_bl ~ dnDirichlet(rep(1.0, n_branches))
moves.append( mvBetaSimplex(rel_bl, alpha = 50, weight = 8) )
moves.append( mvDirichletSimplex(rel_bl, alpha = 200, weight = 5) )
bl := rel_bl * TL
psi := fnTreeAssembly(topology, bl)
tree_len := psi.treeLength()

pi ~ dnDirichlet(v(1, 1))
moves.append( mvBetaSimplex(pi, alpha = 20, weight = 3) )
```

```
pi1 := pi[2]
Q_vis := fnF81(pi)                    # one expected visible change per unit time
when on

s_off ~ dnExponential(1.0)
s_on  ~ dnExponential(1.0)
moves.append( mvScale(s_off, lambda = 0.5, weight = 5) )
moves.append( mvScale(s_on,  lambda = 0.5, weight = 5) )
# joint moves along the ridges (both switching rates; switching rates vs tree length)
sw_updown = mvUpDownScale(lambda = 0.3, weight = 5)
sw_updown.addVariable(s_off, up = TRUE)
sw_updown.addVariable(s_on, up = TRUE)
moves.append( sw_updown )
tl_updown = mvUpDownScale(lambda = 0.3, weight = 5)
tl_updown.addVariable(TL, up = TRUE)
tl_updown.addVariable(s_off, up = FALSE)
tl_updown.addVariable(s_on, up = FALSE)
moves.append( tl_updown )
p_on := s_on / (s_on + s_off)
r_on := 1.0 / p_on                    # visible mean rate = p_on * r_on = 1
switch_rates := [[0.0, s_off], [s_on, 0.0]]
Q_cov := fnCovarion(RateMatrices = [Q_vis, Q_vis], RateScalars = v(r_on, 0.0),
                    SwitchRates = switch_rates, rescaled = FALSE)
p_off := s_off / (s_on + s_off)
root_freq := simplex(pi[1] * p_on, pi[2] * p_on, pi[1] * p_off, pi[2] * p_off)

seq ~ dnPhyloCTMC(tree = psi, Q = Q_cov, rootFrequencies = root_freq,
                  type = "NaturalNumbers", coding = "all")
seq.clamp(data_h)

# ---- explicit no-absence ascertainment correction (see header)
Z ~ dnPhyloCTMC(tree = psi, Q = Q_cov, rootFrequencies = root_freq,
                type = "NaturalNumbers", coding = "all")
Z.clamp(zero_h)
lnP0_raw := Z.lnProbability()
lnP0 := lnP0_raw + 0.0 * (tree_len + pi1 + s_off + s_on)   # forces re-evaluation
(see header)
p_obs := abs(1.0 - exp(lnP0))
asc_cancel ~ dnExponential(exp(-lnP0))
asc_cancel.clamp(0.0)
asc ~ dnIID(N, dnExponential(1.0 / p_obs))
asc.clamp(rep(0.0, N))

mymodel = model(asc)    # enter the DAG from the sink node (see header)
monitors.append( mnModel(filename = out_dir + "covarion.log", printgen = 4) )
monitors.append( mnFile(psi, filename = out_dir + "covarion.trees", printgen = 20) )
monitors.append( mnScreen(printgen = 1000, pi1, s_off, s_on, TL, lnP0) )
mymcmc = mcmc(mymodel, monitors, moves, nruns = 1)
```

```
mymcmc.burnin(generations = 500, tuningInterval = 100)
mymcmc.run(generations = n_gen)

trace = readTreeTrace(out_dir + "covarion.trees", treetype = "non-clock", burnin =
0.1)
map_tree = mapTree(trace, out_dir + "covarion_map.tre")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
binary covarion (explicit no-absence correction)	pi1	0.30	0.301	0.289	0.315	1351.0	pass
binary covarion (explicit no-absence correction)	s_off	0.50	0.383	0.190	0.615	96.2	pass (low ESS)
binary covarion (explicit no-absence correction)	s_on	0.50	0.442	0.212	0.749	80.7	pass (low ESS)
binary covarion (explicit no-absence correction)	TL	6.95	8.358	4.668	12.777	65.4	pass (low ESS)
binary covarion (explicit no-absence correction)	topology_RF_MAP	0.00	4.000	NA	NA	NA	info (MAP topology differs)
binary covarion (explicit no-absence correction)	lnL_RevBayes_minus_R	0.00	0.000	NA	NA	NA	pass

H.3.5 Stochastic Dollo model

RevBayes 1.4.1 has no stochastic Dollo distribution, so script `04` obtains the Nicholls and Gray (2008) likelihood (Section 7.7) as an exact limit of a CTMC. A 4-state chain $U \to P \to D$ ("unborn", "present", "dead", plus an unused copy of "present" for the ambiguity coding) with a vanishing birth rate $\varepsilon = 10^{-5}$, death rate μ and root frequency ε/μ for "present" gives $P(x) = \varepsilon\, a_x + O(\varepsilon^2)$ for every observable pattern x, with $a_x = \Lambda_x/\lambda$ (the expected number of classes with pattern x per unit birth rate), and $1 - P_0 = \varepsilon A$ with $A = \Lambda/\lambda$. Conditioning on observability (device 2) therefore yields the Dollo multinomial likelihood $\prod_x (a_x/A)^{n_x}$, with relative error of order $\varepsilon\times$ tree length, and a clamped Poisson node $N \sim \text{Poisson}(\lambda A)$ completes the full likelihood. This is the limit argument of Section 7.7.4 turned into an implementation. The time tree has its root age fixed at 4 millennia, because only μt is identifiable. The check script evaluates the book's own R implementation of the Dollo likelihood (`dollo_rates()`) on posterior samples: the RevBayes likelihood differs from it only by the constant $-\log N!$ (maximum deviation 0.0092 log-units), and the maximum-likelihood estimate on the true tree agrees with the posterior. Catastrophes and the missing-data extension of Ryder and Nicholls (2011) are not implemented.

```
#######################################################################
# 04_stochastic_dollo.Rev
# Model   : stochastic Dollo model of Nicholls & Gray (2008) on a time tree:
#           cognate classes are born once (Poisson process, rate lambda per
#           lineage per millennium; Poisson(lambda/mu) classes at the root) and
#           die independently at rate mu; all-absent classes are unobservable.
# Book    : Chapter 7, Section lexical-models-dollo (the Nicholls-Gray likelihood:
```

```
#           Poisson pattern counts with means Lambda_x; full likelihood
#           exp(-Lambda) prod_x Lambda_x^n_x / n_x!).
# Estimates: mu, lambda, tree topology and node ages (root age fixed at 4.0
#           millennia; only mu*t is identifiable without a calibration).
#
# IMPLEMENTATION (RevBayes 1.4.1 has no dnPhyloCTMCDollo):
#   The Dollo likelihood is obtained as an exact limit of a CTMC.  Consider the
#   4-state chain  U (unborn) --eps--> P (present) --mu--> D (dead), plus an
#   unused copy P' of "present" (state 4) that only serves the ambiguity coding.
#   Root frequencies: P = eps/mu, U = 1 - eps/mu.  Observed 0 = {U, D}, 1 = {P, P'}.
#   To first order in eps, P(pattern x) = eps * a_x with a_x = Lambda_x / lambda of
#   the book's Dollo section (birth on the branch above v with rate eps,
#   survival e^{-mu(t-s)}, root term eps/mu * S_root), and 1 - P(0...0) = eps * A.
#   Hence  prod_x [P(x) / (1 - P0)]^{n_x}  ->  prod_x (a_x / A)^{n_x}, the Dollo
#   multinomial likelihood conditional on N.  The relative error is O(eps * tree
#   length) ~ 2e-4 for eps = 1e-5.  The no-absence conditioning is done explicitly
#   (dummy all-absent column Z and the two RevBayes 1.4.1 work-arounds,
#   see 03_covarion.Rev), and the Poisson factor for the
#   number of observed classes, N ~ Poisson(lambda * A) with A = (1 - P0)/eps,
#   completes the full Nicholls-Gray likelihood (up to the
#   constant sum(log n_x!) - log N!).  The check script verifies this against the
#   book's own R implementation of the Dollo likelihood (dollo_rates()).
# Data    : data/lexical/d04_dollo.nex (truth: lambda = 20, mu = 0.3, root age 4)
# Runtime : about 2 minutes (default n_gen = 8000)
# Usage   : rb analyses/lexical/04_stochastic_dollo.Rev [--args n_gen]
################################################################################
seed(404)
n_gen = 8000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/lexical/04_stochastic_dollo/"
eps <- 1e-5                     # vanishing "birth" rate of the limiting CTMC
root_age <- 4.0

data  <- readDiscreteCharacterData("data/lexical/d04_dollo.nex")
zero  <- readDiscreteCharacterData("data/lexical/zero_column.nex")
data_h <- data.expandCharacters(2)   # 0 -> (0 2) = {U, D};  1 -> (1 3) = {P, P'}
zero_h <- zero.expandCharacters(2)
N <- data.nchar()
taxa <- data.taxa()
moves    = VectorMoves()
monitors = VectorMonitors()

# --- time tree, root age fixed (known calibration)
psi ~ dnUniformTimeTree(rootAge = root_age, taxa = taxa)
moves.append( mvNarrow(psi, weight = 10) )
moves.append( mvFNPR(psi, weight = 5) )
moves.append( mvNodeTimeSlideUniform(psi, weight = 15) )
moves.append( mvSubtreeScale(psi, weight = 5) )
```

```
tree_len := psi.treeLength()

# --- Dollo parameters
mu ~ dnLoguniform(0.01, 10.0)
lambda ~ dnLoguniform(0.1, 1000.0)
moves.append( mvScale(mu, lambda = 0.5, weight = 5) )
moves.append( mvScale(lambda, lambda = 0.5, weight = 3) )

Q := fnFreeK([[0.0, eps, 0.0, 0.0],
              [0.0, 0.0, mu,  0.0],
              [0.0, 0.0, 0.0, 0.0],
              [0.0, 0.0, 0.0, 0.0]], rescaled = FALSE)
root_freq := simplex(abs(1.0 - eps / mu), eps / mu, 0.0, 0.0)

seq ~ dnPhyloCTMC(tree = psi, Q = Q, rootFrequencies = root_freq,
                  type = "NaturalNumbers", coding = "all")
seq.clamp(data_h)

# --- conditioning on observability (no all-absent classes)
Z ~ dnPhyloCTMC(tree = psi, Q = Q, rootFrequencies = root_freq,
                type = "NaturalNumbers", coding = "all")
Z.clamp(zero_h)
lnP0_raw := Z.lnProbability()
lnP0 := lnP0_raw + 0.0 * (tree_len + mu)   # forces re-evaluation (03_covarion.Rev)
p_obs := abs(1.0 - exp(lnP0))              # = eps * A
asc_cancel ~ dnExponential(exp(-lnP0))  # cancels Z's own contribution log(P0)
asc_cancel.clamp(0.0)
asc ~ dnIID(N, dnExponential(1.0 / p_obs))   # adds -N log(1 - P0)
asc.clamp(rep(0.0, N))

# --- Poisson number of observed cognate classes
A := p_obs / eps                           # = Lambda / lambda (expected observable
classes per unit birth rate)
n_obs ~ dnPoisson(lambda * A)
n_obs.clamp(N)

mymodel = model(n_obs)   # enter the DAG from a sink node (see 03_covarion.Rev)
monitors.append( mnModel(filename = out_dir + "dollo.log", printgen = 5) )
monitors.append( mnFile(psi, filename = out_dir + "dollo.trees", printgen = 25) )
monitors.append( mnScreen(printgen = 1000, mu, lambda, A) )
mymcmc = mcmc(mymodel, monitors, moves, nruns = 1)
mymcmc.burnin(generations = 500, tuningInterval = 100)
mymcmc.run(generations = n_gen)

trace = readTreeTrace(out_dir + "dollo.trees", treetype = "clock", burnin = 0.1)
mcc = mccTree(trace, out_dir + "dollo_mcc.tre")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
stochastic Dollo (eps-limit CTMC + Poisson N), root age fixed	mu	0.3	0.269	0.226	0.321	412.2	pass
stochastic Dollo (eps-limit CTMC + Poisson N), root age fixed	lambda	20.0	18.363	15.18	21.52	406.5	pass
stochastic Dollo (eps-limit CTMC + Poisson N), root age fixed	topology_RF_MAP	0.0	6.000	NA	NA	NA	info (MAP topology differs)
ML (book's exact Dollo likelihood, true tree)	mu	0.3	0.276	NA	NA	NA	info
stochastic Dollo (eps-limit CTMC + Poisson N), root age fixed	lnL_RevBayes_minus_exact_Dollo_minus_const	0.0	0.009	NA	NA	NA	pass

H.3.6 Pseudo-Dollo model

Script `05` implements the pseudo-Dollo model of Bouckaert and Robbeets (2017) (Section 7.7.6) as in the BEAST 2 Babel package: a three-state chain "not yet born" → "present" → "lost" with rate matrix $b \cdot \left(\begin{smallmatrix} -1 & 1 & 0 \\ 0 & -d & d \\ 0 & 0 & 0 \end{smallmatrix}\right)$, $d \leq 1$, root frequencies $(1-\pi_P, \pi_P, 0)$, and binary observations through ambiguity coding (0 is "not yet born" or "lost"). The approximations relative to the stochastic Dollo model are stated in the script header: a class can be born independently in several lineages (which is what makes it "pseudo"), characters form a fixed pool rather than a Poisson birth process, so the number of observed classes carries no information, and the no-absence correction is applied explicitly. The birth rate trades off against the root frequency of "present" (Table 7.3), which shows up as low ESS for b, d and π_P; the well-identified death rate $b\,d$ mixes well.

```
################################################################################
# 05_pseudo_dollo.Rev
# Model   : pseudo-Dollo model of Bouckaert & Robbeets (2017) on a time tree:
#           an irreversible 3-state CTMC  U (not yet born) -> P (present) -> D (lost),
#              Q = b * | -1   1   0 |
#                      |  0  -d   d |      (as in BEAST 2 Babel's BirthDeathModel:
#                      |  0   0   0 |       relative birth rate 1, death d <= 1)
#           observed as binary through ambiguity coding: 0 = {U, D}, 1 = {P}.
# Book    : Chapter 7, Section lexical-models-pseudodollo.
# Estimates: birth rate b (per millennium), relative death rate d, death rate b*d,
#           root probability of "present" pi_P, topology, node ages (root age fixed
#           at 4.0 millennia).
# Approximations / deviations (documented, not hidden):
#   * Unlike the stochastic Dollo model, the same character can be born
#     independently in several lineages (birth is a finite-rate per-character
#     event, not a single Poisson birth); that is what makes it "pseudo".
```

```
#   * Characters are a fixed pool of exchangeable sites with stationary-free
#     root frequencies (pi_U, pi_P, 0) instead of a Poisson birth process; the
#     number of observed characters carries no information (conditional model).
#   * Implementation detail: the 3-state chain is embedded in 4 states so that
#     RevBayes' expandCharacters(2) produces the ambiguity coding (0 -> {U, D},
#     1 -> {P, P'}); state P' (4) is unreachable (no inflow, root frequency 0).
#   * Ascertainment (no all-absent characters) is applied explicitly with the
#     dummy all-absent column, as in 03_covarion.Rev (including the two
#     RevBayes 1.4.1 work-arounds explained there), because RevBayes 1.4.1 only
#     offers coding="noabsencesites" for 2-state data.
#   * Babel constrains d <= 1 (death no faster than birth); we use d ~ Beta(1,1).
# Data    : data/lexical/d05_pseudodollo.nex simulated under this very model
#           (truth: b = 0.25, d = 0.6, b*d = 0.15, pi_P = 0.3)
# Runtime : about 2.5 minutes (default n_gen = 9000)
# Usage   : rb analyses/lexical/05_pseudo_dollo.Rev [--args n_gen]
################################################################################
seed(505)
n_gen = 9000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/lexical/05_pseudo_dollo/"
root_age <- 4.0

data  <- readDiscreteCharacterData("data/lexical/d05_pseudodollo.nex")
zero  <- readDiscreteCharacterData("data/lexical/zero_column.nex")
data_h <- data.expandCharacters(2)
zero_h <- zero.expandCharacters(2)
N <- data.nchar()
taxa <- data.taxa()
moves    = VectorMoves()
monitors = VectorMonitors()

psi ~ dnUniformTimeTree(rootAge = root_age, taxa = taxa)
moves.append( mvNarrow(psi, weight = 4) )
moves.append( mvFNPR(psi, weight = 2) )
moves.append( mvNodeTimeSlideUniform(psi, weight = 8) )
moves.append( mvSubtreeScale(psi, weight = 3) )
tree_len := psi.treeLength()

b ~ dnLoguniform(0.001, 10.0)            # birth rate = clock rate
d ~ dnBeta(1, 1)                         # death rate relative to birth
moves.append( mvScale(b, lambda = 0.5, weight = 5) )
moves.append( mvBetaProbability(d, delta = 10, weight = 5) )
death := b * d
# joint move along the birth/relative-death ridge (keeps b*d roughly fixed)
bd_updown = mvUpDownScale(lambda = 0.3, weight = 5)
bd_updown.addVariable(b, up = TRUE)
bd_updown.addVariable(d, up = FALSE)
moves.append( bd_updown )
```

```
pi_P ~ dnBeta(1, 1)                       # root probability that the class is present
moves.append( mvBetaProbability(pi_P, delta = 10, weight = 3) )

Q := fnFreeK([[0.0, b,   0.0,   0.0],
              [0.0, 0.0, death, 0.0],
              [0.0, 0.0, 0.0,   0.0],
              [0.0, 0.0, 0.0,   0.0]], rescaled = FALSE)
root_freq := simplex(abs(1.0 - pi_P), pi_P, 0.0, 0.0)

seq ~ dnPhyloCTMC(tree = psi, Q = Q, rootFrequencies = root_freq,
                  type = "NaturalNumbers", coding = "all")
seq.clamp(data_h)

# explicit no-absence ascertainment (see 03_covarion.Rev)
Z ~ dnPhyloCTMC(tree = psi, Q = Q, rootFrequencies = root_freq,
                type = "NaturalNumbers", coding = "all")
Z.clamp(zero_h)
lnP0_raw := Z.lnProbability()
lnP0 := lnP0_raw + 0.0 * (tree_len + b + d + pi_P)   # forces re-evaluation (03_
covarion.Rev)
p_obs := abs(1.0 - exp(lnP0))
asc_cancel ~ dnExponential(exp(-lnP0))
asc_cancel.clamp(0.0)
asc ~ dnIID(N, dnExponential(1.0 / p_obs))
asc.clamp(rep(0.0, N))

mymodel = model(asc)
monitors.append( mnModel(filename = out_dir + "pseudodollo.log", printgen = 5) )
monitors.append( mnFile(psi, filename = out_dir + "pseudodollo.trees", printgen =
25) )
monitors.append( mnScreen(printgen = 1000, b, d, pi_P) )
mymcmc = mcmc(mymodel, monitors, moves, nruns = 1)
mymcmc.burnin(generations = 500, tuningInterval = 100)
mymcmc.run(generations = n_gen)

trace = readTreeTrace(out_dir + "pseudodollo.trees", treetype = "clock", burnin =
0.1)
mcc = mccTree(trace, out_dir + "pseudodollo_mcc.tre")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
pseudo-Dollo 3-state CTMC (ambiguity-coded), root age fixed	b	0.25	0.246	0.191	0.302	60.9	pass (low ESS)
pseudo-Dollo 3-state CTMC (ambiguity-coded), root age fixed	d	0.60	0.649	0.530	0.788	49.8	pass (low ESS)

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
pseudo-Dollo 3-state CTMC (ambiguity-coded), root age fixed	death	0.15	0.158	0.139	0.177	432.4	pass
pseudo-Dollo 3-state CTMC (ambiguity-coded), root age fixed	pi_P	0.30	0.398	0.112	0.654	37.7	pass (low ESS)
pseudo-Dollo 3-state CTMC (ambiguity-coded), root age fixed	topology_RF_MAP	0.00	2.000	NA	NA	NA	info (MAP topology differs)

H.3.7 Multistate meaning-slot Mk model with rate partitions

Script `06` treats each meaning as one character whose states are cognate classes (Section 7.8, the Mk model of Lewis (2001)), with $K = 5$ and three partitions of meanings (slow, medium, fast) that each have their own rate multiplier (Section 7.5.4); the multipliers have mean 1, so that branch lengths are in units of the average meaning. Constant meanings are observed, so no ascertainment correction is used.

```
################################################################################
# 06_meaning_mk.Rev
# Model   : multistate meaning-slot Mk model (Lewis 2001): one character per
#           meaning, states = cognate classes (K = 5), with concept-specific
#           rates via three rate partitions of meanings (e.g. slow / medium /
#           fast semantic domains), partition rates constrained to mean 1.
# Book    : Chapter 7, Sections lexical-models-multistate (Mk model, Lewis 2001)
#           and lexical-models-partitions.
# Estimates: partition rate multipliers r[1..3], topology, tree length.
# Ascertainment: none needed (coding = "all"): with multistate coding every
#           language has a state for every meaning, constant meanings are kept.
# Data    : data/lexical/d06_meanings.nex, 180 meanings (60 per class; class of
#           each meaning in d06_rate_class.txt). Truth: r = (0.237, 0.789, 1.974), TL
= 3.475
# Runtime : about 1.5 minutes (default n_gen = 15000)
# Usage   : rb analyses/lexical/06_meaning_mk.Rev [--args n_gen]
################################################################################
seed(606)
n_gen = 15000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/lexical/06_meaning_mk/"
K <- 5
n_part <- 3
part_start <- [1, 61, 121]
part_end   <- [60, 120, 180]

for (k in 1:n_part) {
    part[k] <- readDiscreteCharacterData("data/lexical/d06_meanings.nex")
    part[k].excludeAll()
```

```
    part[k].includeCharacter(part_start[k]:part_end[k])
}
taxa <- part[1].taxa()
n_branches <- 2 * taxa.size() - 3
moves    = VectorMoves()
monitors = VectorMonitors()

psi ~ dnUniformTopologyBranchLength(taxa, branchLengthDistribution =
dnExponential(5.0))
moves.append( mvNNI(psi, weight = 10) )
moves.append( mvSPR(psi, weight = 5) )
moves.append( mvBranchLengthScale(psi, weight = n_branches) )
TL := psi.treeLength()

# partition rates: Dirichlet-distributed proportions, weighted so that the
# mean rate over all meanings is 1 (partitions have equal sizes here)
rel ~ dnDirichlet(rep(1, n_part))
moves.append( mvBetaSimplex(rel, alpha = 10, weight = 5) )
moves.append( mvDirichletSimplex(rel, alpha = 10, weight = 3) )
for (k in 1:n_part) part_rate[k] := rel[k] * n_part

Q <- fnJC(K)
for (k in 1:n_part) {
    seq[k] ~ dnPhyloCTMC(tree = psi, Q = Q, branchRates = part_rate[k],
                         type = "Standard", coding = "all")
    seq[k].clamp(part[k])
}

mymodel = model(psi)
monitors.append( mnModel(filename = out_dir + "meaning_mk.log", printgen = 15) )
monitors.append( mnFile(psi, filename = out_dir + "meaning_mk.trees", printgen = 75)
)
monitors.append( mnScreen(printgen = 5000, part_rate, TL) )
mymcmc = mcmc(mymodel, monitors, moves, nruns = 1)
mymcmc.burnin(generations = 1000, tuningInterval = 100)
mymcmc.run(generations = n_gen)

trace = readTreeTrace(out_dir + "meaning_mk.trees", treetype = "non-clock", burnin =
0.1)
map_tree = mapTree(trace, out_dir + "meaning_mk_map.tre")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
meaning-slot Mk (K = 5), 3 rate partitions	part_rate[1]	0.237	0.254	0.183	0.332	788.6	pass
meaning-slot Mk (K = 5), 3 rate partitions	part_rate[2]	0.789	0.709	0.585	0.866	901.0	pass

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
meaning-slot Mk (K = 5), 3 rate partitions	part_rate[3]	1.974	2.037	1.868	2.196	791.5	pass
meaning-slot Mk (K = 5), 3 rate partitions	TL	3.475	3.415	2.949	3.890	591.9	pass
meaning-slot Mk (K = 5), 3 rate partitions	topology_RF_MAP	0.000	4.000	NA	NA	NA	info (MAP topology differs)

H.3.8 Word frequency as a rate predictor

Script `07` builds the regression $\log \rho_m = a + b \log f_m$ of Section 7.5 ("Word frequency as a rate predictor", Pagel et al. (2007)) directly into the phylogenetic likelihood: each of 150 meanings is a separate Mk partition whose rate is a deterministic function of its (simulated, standardised) log frequency. On a dated tree the intercept a (log rate per millennium of a word of average frequency) is identifiable as well as the slope b. The tree is fixed at the true time tree: with one likelihood object per meaning, every tree move recomputes 150 likelihoods, which is too slow for the time budget. The residual term ε_m of the book's regression is omitted.

```
################################################################################
# 07_frequency_regression.Rev
# Model   : meaning-specific replacement rates predicted by word frequency,
#           embedded in the phylogenetic likelihood:
#               log rho_m = a + b * log(freq_m)        (one partition per meaning)
#           Each meaning is a multistate (K = 5) Mk character on the dated tree
#           (root age 4 millennia), so that a (log rate per millennium of a word of
#           average log-frequency) is identifiable, not only b.
# Design  : the time tree is FIXED at the true tree.  With one CTMC object per
#           meaning (150 partitions) every tree move recomputes 150 likelihoods,
#           which is far too slow for the 3-minute budget (about 1.4 generations
#           per second in a test); conditioning on the tree isolates the
#           regression, which is what this script verifies.
# Book    : Chapter 7, Section "Word frequency as a rate predictor"
#           (the regression log rho_m = beta0 + beta1 log(frequency_m) + eps_m;
#           here eps_m = 0, i.e. the pure regression; Pagel et al. 2007).
# Estimates: intercept a, slope b (tree fixed).
# Data    : data/lexical/d07_frequency.nex (150 meanings) and
#           d07_logfreq.txt (centred, standardised log frequencies).
#           Truth: a = log(0.15) = -1.897, b = -0.6.
# Runtime : about 3 minutes (default n_gen = 3000)
# Usage   : rb analyses/lexical/07_frequency_regression.Rev [--args n_gen]
################################################################################
seed(707)
n_gen = 3000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/lexical/07_frequency_regression/"
K <- 5
```

```
data <- readDiscreteCharacterData("data/lexical/d07_frequency.nex")
lf_tab <- readDataDelimitedFile("data/lexical/d07_logfreq.txt", header = FALSE)
M <- data.nchar()
for (m in 1:M) logf[m] <- lf_tab[m][1]
taxa <- data.taxa()
moves    = VectorMoves()
monitors = VectorMonitors()

psi <- readTrees("data/lexical/true_time_tree.tre")[1]

a ~ dnNormal(0.0, 2.0)          # log rate (per millennium) at mean log frequency
b ~ dnNormal(0.0, 1.0)          # frequency effect
moves.append( mvSlide(a, delta = 0.2, weight = 1) )
moves.append( mvSlide(b, delta = 0.2, weight = 1) )

Q <- fnJC(K)
for (m in 1:M) {
    rho[m] := exp(a + b * logf[m])
    part[m] <- data
    part[m].excludeAll()
    part[m].includeCharacter(m)
    seq[m] ~ dnPhyloCTMC(tree = psi, Q = Q, branchRates = rho[m],
                         type = "Standard", coding = "all")
    seq[m].clamp(part[m])
}

mymodel = model(a)
monitors.append( mnModel(filename = out_dir + "frequency.log", printgen = 3,
                         exclude = ["rho"]) )
monitors.append( mnScreen(printgen = 500, a, b) )
mymcmc = mcmc(mymodel, monitors, moves, nruns = 1)
mymcmc.burnin(generations = 300, tuningInterval = 30)
mymcmc.run(generations = n_gen)
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
per-meaning Mk rates, log rho = a + b log f (time tree)	a	-1.897	-1.844	-1.952	-1.712	458.0	pass
per-meaning Mk rates, log rho = a + b log f (time tree)	b	-0.600	-0.519	-0.647	-0.399	555.9	pass
per-meaning Mk rates, log rho = a + b log f (time tree)	topology_RF_MAP	0.000	4.000	NA	NA	NA	info (MAP topology differs)

H.3.9 Dated analyses with Yule and birth-death tree priors

Script `08` dates a binary cognate matrix with a strict clock, the no-absence correction and a root calibration (root age ~ Lognormal(log 4, 0.05)), once under a Yule prior and once under a constant-

rate birth-death prior (Section 6.2). Tree height and π_1 are recovered precisely under both priors. The clock rate covers the truth but is imprecise and mixes slowly: in the true tree one language is the sister of all the others, so the root age only lengthens that language's own, nearly saturated branch, and the clock rate is pinned down mainly through the calibration and the internal node ages. This was confirmed by direct likelihood evaluation and is a property of this tree, not of the code. The birth and death rates have no single true value (the Yule tree was re-scaled to a root age of 4) and are reported for information.

```
################################################################################
# 08_timetree_yule_bd.Rev
# Model   : dated analysis of cognate data: binary CTMC (no-absence correction)
#           with a strict clock on a time tree, under two tree priors:
#             (A) Yule (pure birth)       dnBirthDeath(mu = 0)
#             (B) constant-rate birth-death, diversification/turnover
parameterisation
#           and a root-age calibration: root_age ~ Lognormal(log 4, 0.05)
#           (95% prior interval about 3.62 - 4.42 millennia), i.e. a dated root.
# Book    : Chapter 6 (Section bayesian-priors, tree priors for dated trees) and
#           Chapter 7 (binary CTMC + ascertainment); clocks in Chapter 9.
# Estimates: clock rate (expected changes per character per millennium), root age
#           (tree height), pi1, birth (and death) rates, topology and node ages.
# Data    : data/lexical/d08_dated.nex, simulated on true_time_tree.tre (root age 4)
#           with clock rate 0.25 and pi1 = 0.2.
# Note    : in the true tree one language (L10) is the sister of all others, so
#           the root age only lengthens L10's own (nearly saturated) branch: the
#           clock rate is identified mainly through the calibration and the
#           internal node ages, and its posterior is wide (an honest property of
#           this tree, not a coding error; checked by direct likelihood evaluation).
# Runtime : about 2.5 minutes for both chains (default n_gen = 12000)
# Usage   : rb analyses/lexical/08_timetree_yule_bd.Rev [--args n_gen]
################################################################################
seed(808)
n_gen = 12000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/lexical/08_timetree_yule_bd/"

data <- readDiscreteCharacterData("data/lexical/d08_dated.nex")
taxa <- data.taxa()
n_taxa <- taxa.size()
priors = ["yule", "bd"]

for (s in 1:2) {
    moves    = VectorMoves()
    monitors = VectorMonitors()

    # ---- root-age calibration
    root_age ~ dnLognormal(ln(4.0), 0.05)

    # ---- tree prior
```

```
if (priors[s] == "yule") {
    birth ~ dnExponential(1.0)
    moves.append( mvScale(birth, lambda = 0.5, weight = 3) )
    death <- 0.0
    psi ~ dnBirthDeath(lambda = birth, mu = death, rho = 1.0, rootAge = root_age,
                       samplingStrategy = "uniform", condition = "survival", taxa
                        = taxa)
} else {
    diversification ~ dnExponential(1.0)
    turnover ~ dnBeta(1.0, 1.0)
    moves.append( mvScale(diversification, lambda = 0.5, weight = 3) )
    moves.append( mvBetaProbability(turnover, delta = 10, weight = 3) )
    birth := diversification / abs(1.0 - turnover)
    death := birth * turnover
    psi ~ dnBirthDeath(lambda = birth, mu = death, rho = 1.0, rootAge = root_age,
                       samplingStrategy = "uniform", condition = "survival", taxa
                        = taxa)
}
moves.append( mvNarrow(psi, weight = 10) )
moves.append( mvFNPR(psi, weight = 5) )
moves.append( mvNodeTimeSlideUniform(psi, weight = 15) )
moves.append( mvSubtreeScale(psi, weight = 5) )
moves.append( mvTreeScale(psi, rootAge = root_age, delta = 0.5, tune = TRUE,
weight = 5) )

# ---- strict clock (log-uniform prior)
clock_rate ~ dnLoguniform(0.001, 10.0)
moves.append( mvScale(clock_rate, lambda = 0.5, weight = 5) )
updown = mvUpDownScale(lambda = 0.5, weight = 5)
updown.addVariable(clock_rate, up = TRUE)
updown.addVariable(psi, up = FALSE)
updown.addVariable(root_age, up = FALSE)
moves.append( updown )
moves.append( mvScale(root_age, lambda = 0.05, weight = 2) )

# ---- binary CTMC with no-absence correction
pi ~ dnDirichlet(v(1, 1))
moves.append( mvBetaSimplex(pi, alpha = 20, weight = 3) )
pi1 := pi[2]
Q := fnF81(pi)
seq ~ dnPhyloCTMC(tree = psi, Q = Q, branchRates = clock_rate, type = "Binary",
                  coding = "noabsencesites")
seq.clamp(data)

tree_height := psi.rootAge()
mymodel = model(psi)
monitors.append( mnModel(filename = out_dir + priors[s] + ".log", printgen = 8) )
monitors.append( mnFile(psi, filename = out_dir + priors[s] + ".trees", printgen
= 60) )
```

```
    monitors.append( mnScreen(printgen = 2000, clock_rate, tree_height, pi1) )
    mymcmc = mcmc(mymodel, monitors, moves, nruns = 1)
    mymcmc.burnin(generations = 500, tuningInterval = 100)
    mymcmc.run(generations = n_gen)

    trace = readTreeTrace(out_dir + priors[s] + ".trees", treetype = "clock", burnin
= 0.1)
    mcc = mccTree(trace, out_dir + priors[s] + "_mcc.tre")
}
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
binary CTMC, strict clock, Yule prior, root ~ Lognormal(log 4, 0.05)	clock_rate	0.25	0.449	0.163	0.911	99.5	pass (low ESS)
binary CTMC, strict clock, Yule prior, root ~ Lognormal(log 4, 0.05)	tree_height	4.00	3.997	3.600	4.386	1584.0	pass
binary CTMC, strict clock, Yule prior, root ~ Lognormal(log 4, 0.05)	pi1	0.20	0.202	0.192	0.212	822.7	pass
binary CTMC, strict clock, Yule prior, root ~ Lognormal(log 4, 0.05)	birth	NA	0.454	0.158	0.764	909.5	info
binary CTMC, strict clock, Yule prior, root ~ Lognormal(log 4, 0.05)	topology_ RF_MAP	0.00	4.000	NA	NA	NA	info (MAP topology differs)
binary CTMC, strict clock, birth-death prior, root ~ Lognormal(log 4, 0.05)	clock_rate	0.25	0.554	0.169	1.342	80.3	pass (low ESS)
binary CTMC, strict clock, birth-death prior, root ~ Lognormal(log 4, 0.05)	tree_height	4.00	4.006	3.620	4.439	1351.0	pass
binary CTMC, strict clock, birth-death prior, root ~ Lognormal(log 4, 0.05)	pi1	0.20	0.202	0.191	0.211	1244.5	pass
binary CTMC, strict clock, birth-death prior, root ~ Lognormal(log 4, 0.05)	birth	NA	0.717	0.157	1.627	358.6	info
binary CTMC, strict clock, birth-death prior, root ~ Lognormal(log 4, 0.05)	death	NA	0.385	0.002	1.197	223.2	info
binary CTMC, strict clock, birth-death prior, root ~ Lognormal(log 4, 0.05)	topology_ RF_MAP	0.00	4.000	NA	NA	NA	info (MAP topology differs)

H.3.10 Model comparison by stepping-stone and path sampling

Script `09` computes marginal likelihoods (Section 6.7) with `powerPosterior` (16 stones) and both `steppingStoneSampler` and `pathSampler` for four models fitted to the same data (simulated under the binary CTMC with gamma rates, $\alpha = 0.4$): equal rates, gamma, covarion, and the conditional (multinomial) form of the stochastic Dollo model, so that all four are likelihoods of the same thing, the observed patterns given that each class is observable. The dated tree is fixed at the truth, so that the comparison concerns the substitution models only. The generating gamma model wins, by about 14 log-units over the covarion model, which captures part of the rate heterogeneity, and by hundreds of log-units over the equal-rates and Dollo models (Section 7.9).

```
################################################################################
# 09_model_comparison.Rev
# Model   : Bayesian model comparison by stepping-stone and path sampling
#           (powerPosterior) among four models of cognate evolution fitted to
#           the SAME data (all conditioned on "no all-absent cognate classes"):
#             (1) equal-rates binary CTMC
#             (2) binary CTMC + discrete Gamma(4) rates    <- generating model
#             (3) binary covarion
#             (4) stochastic Dollo (conditional/multinomial form, eps-limit CTMC
#                 of 04_stochastic_dollo.Rev, without the Poisson term for N so
#                 that all four are likelihoods of the same thing: the observed
#                 patterns given that each class is observable)
# Book    : Chapter 6, Section bayesian-marginal (path sampling, stepping stones);
#           Chapter 7, Section lexical-models-comparison.
# Design  : the dated tree is FIXED at the true time tree (root age 4 millennia)
#           so that the comparison is among substitution models only and the
#           runtime stays small; models 1-3 have a strict clock rate, the Dollo
#           model a death rate mu per millennium.
# Data    : data/lexical/d09_modelcomp.nex, simulated under model (2) with
#           alpha = 0.4, clock rate 0.3, pi1 = 0.25.
# Output  : output/lexical/09_model_comparison/marginal_likelihoods.csv
# Runtime : about 3 minutes for all four models (default 16 stones x 300
generations);
#           run a single model with  --args <model 1-4> [generations per stone]
# Usage   : rb analyses/lexical/09_model_comparison.Rev [--args model n_per_stone]
################################################################################
seed(909)
models_to_run = [1, 2, 3, 4]
n_stone_gen = 300
if (args.size() > 1) models_to_run = [int(args[2])]
if (args.size() > 2) n_stone_gen = int(args[3])
n_cats = 16
out_dir = "output/lexical/09_model_comparison/"
model_names = ["equal_rates", "gamma", "covarion", "dollo"]
eps <- 1e-5

data  <- readDiscreteCharacterData("data/lexical/d09_modelcomp.nex")
zero  <- readDiscreteCharacterData("data/lexical/zero_column.nex")
```

```
data_h <- data.expandCharacters(2)
zero_h <- zero.expandCharacters(2)
N <- data.nchar()
psi <- readTrees("data/lexical/true_time_tree.tre")[1]

for (mi in models_to_run) {
    moves    = VectorMoves()
    monitors = VectorMonitors()

    if (mi <= 3) {
        clock_rate ~ dnLoguniform(0.001, 10.0)
        moves.append( mvScale(clock_rate, lambda = 0.5, weight = 3) )
        pi ~ dnDirichlet(v(1, 1))
        moves.append( mvBetaSimplex(pi, alpha = 20, weight = 3) )
        Q := fnF81(pi)
    }
    if (mi == 1) {
        seq ~ dnPhyloCTMC(tree = psi, Q = Q, branchRates = clock_rate, type =
"Binary",
                          coding = "noabsencesites")
        seq.clamp(data)
    }
    if (mi == 2) {
        alpha ~ dnExponential(1.0)
        moves.append( mvScale(alpha, lambda = 0.5, weight = 3) )
        site_rates := fnDiscretizeGamma(alpha, alpha, 4)
        seq ~ dnPhyloCTMC(tree = psi, Q = Q, branchRates = clock_rate, siteRates =
site_rates,
                          type = "Binary", coding = "noabsencesites")
        seq.clamp(data)
    }
    if (mi == 3) {
        s_off ~ dnExponential(1.0)
        s_on  ~ dnExponential(1.0)
        moves.append( mvScale(s_off, lambda = 0.5, weight = 3) )
        moves.append( mvScale(s_on,  lambda = 0.5, weight = 3) )
        p_on := s_on / (s_on + s_off)
        p_off := s_off / (s_on + s_off)
        Q_cov := fnCovarion(RateMatrices = [Q, Q], RateScalars = v(1.0 / p_on, 0.0),
                            SwitchRates = [[0.0, s_off], [s_on, 0.0]], rescaled = FALSE)
       root_freq := simplex(pi[1] * p_on, pi[2] * p_on, pi[1] * p_off, pi[2] * p_off)
        seq ~ dnPhyloCTMC(tree = psi, Q = Q_cov, branchRates = clock_rate,
                          rootFrequencies = root_freq, type = "NaturalNumbers",
                          coding = "all")
        seq.clamp(data_h)
        Z ~ dnPhyloCTMC(tree = psi, Q = Q_cov, branchRates = clock_rate,
                        rootFrequencies = root_freq, type = "NaturalNumbers", coding
                        = "all")
```

```
        Z.clamp(zero_h)
        lnP0_raw := Z.lnProbability()
        lnP0 := lnP0_raw + 0.0 * (clock_rate + pi[2] + s_off + s_on)  # see 03_
covarion.Rev
    }
    if (mi == 4) {
        mu ~ dnLoguniform(0.01, 10.0)
        moves.append( mvScale(mu, lambda = 0.5, weight = 5) )
        Q_d := fnFreeK([[0.0, eps, 0.0, 0.0], [0.0, 0.0, mu, 0.0],
                        [0.0, 0.0, 0.0, 0.0], [0.0, 0.0, 0.0, 0.0]], rescaled = FALSE)
        root_freq := simplex(abs(1.0 - eps / mu), eps / mu, 0.0, 0.0)
        seq ~ dnPhyloCTMC(tree = psi, Q = Q_d, rootFrequencies = root_freq,
                          type = "NaturalNumbers", coding = "all")
        seq.clamp(data_h)
        Z ~ dnPhyloCTMC(tree = psi, Q = Q_d, rootFrequencies = root_freq,
                        type = "NaturalNumbers", coding = "all")
        Z.clamp(zero_h)
        lnP0_raw := Z.lnProbability()
        lnP0 := lnP0_raw + 0.0 * mu
    }
    if (mi >= 3) {
        # explicit no-absence correction (03_covarion.Rev): prod P(D_i) / (1 - P0)^N
        p_obs := abs(1.0 - exp(lnP0))
        asc_cancel ~ dnExponential(exp(-lnP0))
        asc_cancel.clamp(0.0)
        asc ~ dnIID(N, dnExponential(1.0 / p_obs))
        asc.clamp(rep(0.0, N))
    }

    if (mi <= 2) mymodel = model(seq)
    if (mi >= 3) mymodel = model(asc)   # enter from the sink node (03_covarion.Rev)
    nm = model_names[mi]
    monitors.append( mnModel(filename = out_dir + nm + "_posterior.log", printgen =
    50) )
    pow_p = powerPosterior(mymodel, moves, monitors, out_dir + nm + "_powp.out",
                           cats = n_cats, sampleFreq = 5)
    pow_p.burnin(generations = 200, tuningInterval = 50)
    pow_p.run(generations = n_stone_gen)
    ss = steppingStoneSampler(file = out_dir + nm + "_powp.out", powerColumnName =
"power",
                              likelihoodColumnName = "likelihood")
    ps = pathSampler(file = out_dir + nm + "_powp.out", powerColumnName = "power",
                     likelihoodColumnName = "likelihood")
    ml_ss = ss.marginal()
    ml_ps = ps.marginal()
    print(nm + "  stepping-stone: " + ml_ss + "  path sampling: " + ml_ps)
    write(nm, mi, ml_ss, ml_ps, filename = out_dir + "ml_" + nm + ".txt", separator
    = ",")
```

```
}
q()
```

Model	Parameter	Truth	Post. mean
equal_rates	log_marginal_likelihood_SS	-4588	info
gamma	log_marginal_likelihood_SS	-4339	pass
covarion	log_marginal_likelihood_SS	-4353	info
dollo	log_marginal_likelihood_SS	-5346	info
all four	generating_model_wins	2	pass

```
#> path sampling -4588.36; log BF vs best (SS) -248.93
#> path sampling -4339.39; log BF vs best (SS) 0.00; generating model
#> path sampling -4353.48; log BF vs best (SS) -14.26
#> path sampling -5347.03; log BF vs best (SS) -1007.43
#> best by stepping-stone: gamma; by path sampling: gamma
```

H.3.11 Posterior predictive adequacy

Script `10` fits the equal-rates and the gamma models to the gamma-simulated data of script `02` and runs `posteriorPredictiveSimulation` (Sections 6.9 and 7.10, Bollback (2002), Höhna et al. (2018)). The test statistics, computed in `check_lexical.R`, are the proportion of cognate classes present in all languages, the proportion of singletons, the variance of cognate-set sizes (the site frequency spectrum) and the proportion of variable characters needing a single parsimony step. The equal-rates model is rejected: it predicts far too few universally shared cognates (slow characters) and too little spread in cognate-set sizes. The gamma model reproduces all four statistics. In the table, *Truth* is the observed statistic and the HPD columns give the 95% interval of the posterior predictive distribution.

A practical warning came out of this check. In RevBayes 1.4.1, the built-in simulator for a binary CTMC with `coding="noabsencesites"` *and* gamma rate categories rejects all-absent columns but keeps the rate category of the rejected column, so slow categories are over-represented and the replicates contain about twice the correct share of all-present cognates (the exact conditional expectation was computed analytically at posterior samples); an unconditioned (`coding="all"`) simulation with site rates was also wrong. The check script therefore draws the replicates in R from the RevBayes posterior samples (tree, π, α), with exact rejection of all-absent columns; for the equal-rates model the built-in replicates agree with them, and the biased built-in gamma replicates are shown in a separate row.

```
################################################################################
# 10_posterior_predictive.Rev
# Model   : posterior predictive model adequacy (Bollback 2002; Hoehna et al.
#           2018 "P3") for cognate data: the equal-rates binary CTMC and the
#           CTMC + Gamma model are fitted to data simulated WITH strong rate
#           heterogeneity (d02, alpha = 0.5); posteriorPredictiveSimulation()
#           simulates replicate datasets from posterior draws.  Test statistics
#           (computed in tests/check_lexical.R, ignoring simulated all-absent
#           columns): number of cognate classes present in all languages,
```

```
#           number of singletons, the variance of cognate-set sizes (the
#           site-frequency spectrum), and the proportion of variable characters
#           that need a single parsimony step on the true tree.
# Book    : Chapter 6, Section bayesian-ppc; Chapter 7, Section
lexical-models-adequacy.
# Expected: the equal-rates model fails (observed statistics in the tail of the
#           posterior predictive distribution); the gamma model passes.
# IMPORTANT (RevBayes 1.4.1): the built-in simulator is reliable for the
#           equal-rates model but NOT for the Gamma model: (i) with
#           coding="noabsencesites" it rejects all-absent columns but keeps their
#           rate category, so slow categories are over-represented (about twice the
#           correct share of all-present cognates, checked analytically); (ii) an
#           unconditioned (coding="all") Binary + siteRates simulation produced
#           almost no presences at all.  tests/check_lexical.R therefore draws the
#           replicates itself from the posterior samples written here (.var file:
#           tree, pi, alpha), with exact rejection of all-absent columns, and uses
#           the RevBayes replicates only as a cross-check (equal-rates: agree).
# Data    : data/lexical/d02_gamma.nex
# Runtime : about 2 minutes (default n_gen = 5000, 100 replicate datasets per model)
# Usage   : rb analyses/lexical/10_posterior_predictive.Rev [--args n_gen]
################################################################################
seed(1010)
n_gen = 5000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/lexical/10_posterior_predictive/"
data <- readDiscreteCharacterData("data/lexical/d02_gamma.nex")
taxa <- data.taxa()
n_branches <- 2 * taxa.size() - 3
fits = ["equal_rates", "gamma"]

for (f in 1:2) {
    moves    = VectorMoves()
    monitors = VectorMonitors()
    psi ~ dnUniformTopologyBranchLength(taxa, branchLengthDistribution =
dnExponential(1.0))
    moves.append( mvNNI(psi, weight = 5) )
    moves.append( mvSPR(psi, weight = 3) )
    moves.append( mvBranchLengthScale(psi, weight = n_branches) )
    TL := psi.treeLength()
    pi ~ dnDirichlet(v(1, 1))
    moves.append( mvBetaSimplex(pi, alpha = 20, weight = 3) )
    Q := fnF81(pi)
    if (f == 1) {
        seq ~ dnPhyloCTMC(tree = psi, Q = Q, type = "Binary", coding =
"noabsencesites")
    } else {
        alpha ~ dnExponential(1.0)
        moves.append( mvScale(alpha, lambda = 0.5, weight = 5) )
```

```
        site_rates := fnDiscretizeGamma(alpha, alpha, 4)
        seq ~ dnPhyloCTMC(tree = psi, Q = Q, siteRates = site_rates, type = "Binary",
                          coding = "noabsencesites")
    }
    seq.clamp(data)
    mymodel = model(psi)
    monitors.append( mnModel(filename = out_dir + fits[f] + ".log", printgen = 10) )
    monitors.append( mnStochasticVariable(filename = out_dir + fits[f] + ".var",
    printgen = 50) )
    mymcmc = mcmc(mymodel, monitors, moves, nruns = 1)
    mymcmc.burnin(generations = 500, tuningInterval = 100)
    mymcmc.run(generations = n_gen)

    # ---- posterior predictive simulation (built-in): one replicate per sampled
    state
    trace = readStochasticVariableTrace(out_dir + fits[f] + ".var", delimiter = TAB)
    pps = posteriorPredictiveSimulation(mymodel, directory = out_dir + fits[f] + "_
sims", trace)
    pps.run(thinning = 1)
}
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
equal_rates fitted to Gamma(0.5) data	PPS_n_all_present	0.054	0.007	0.004	0.014	91	pass (inadequacy detected)
equal_rates fitted to Gamma(0.5) data	PPS_n_singletons	0.241	0.240	0.211	0.270	91	info (not detected)
equal_rates fitted to Gamma(0.5) data	PPS_var_set_size	6.216	5.632	5.102	6.161	91	pass (inadequacy detected)
equal_rates fitted to Gamma(0.5) data	PPS_one_step	0.402	0.393	0.357	0.421	91	info (not detected)
equal_rates fitted to Gamma(0.5) data: RevBayes built-in simulator	PPS_n_all_present_RB_simulator	0.054	0.008	0.003	0.016	76	info (agrees with R replicates)
gamma fitted to Gamma(0.5) data	PPS_n_all_present	0.054	0.057	0.038	0.078	91	pass
gamma fitted to Gamma(0.5) data	PPS_n_singletons	0.241	0.232	0.203	0.265	91	pass
gamma fitted to Gamma(0.5) data	PPS_var_set_size	6.216	6.167	5.307	7.070	91	pass
gamma fitted to Gamma(0.5) data	PPS_one_step	0.402	0.397	0.361	0.436	91	pass
gamma fitted to Gamma(0.5) data: RevBayes built-in simulator	PPS_n_all_present_RB_simulator	0.054	0.121	0.081	0.167	76	info (RevBayes 1.4.1 simulator bias)

H.3.12 Total evidence: lexical and structural partitions

Script 11 combines a binary cognate partition (no-absence correction) and a partition of typological features (three-state Mk with the Mkv correction for variable-only features, Section 3.6) on one tree, with a rate multiplier for the structural partition relative to the lexical one (Section 7.5.4 and Section 16.3.2).

```
################################################################################
# 11_total_evidence.Rev
# Model   : total-evidence analysis on one shared tree with two partitions:
#             lexical    : binary cognate presence/absence, binary CTMC with
#                          no-absence correction (rate multiplier fixed at 1, so
#                          branch lengths are in lexical expected changes)
#             structural : typological features, 3-state Mk with the Lewis (2001)
#                          Mkv correction (coding = "variable"; only variable
#                          features are coded), own rate multiplier r_struct
# Book    : Chapter 7, Section lexical-models-partitions (separate data types);
#           Chapter 3, Section data-structural; Chapter 16 (total evidence).
# Estimates: r_struct (rate of structural relative to lexical change), pi1,
#           topology, tree length.
# Data    : data/lexical/d11_lexical.nex, d11_structural.nex
#           (truth: r_struct = 0.4, pi1 = 0.2, TL = 4.634)
# Runtime : about 1.5 minutes (default n_gen = 15000)
# Usage   : rb analyses/lexical/11_total_evidence.Rev [--args n_gen]
################################################################################
seed(1111)
n_gen = 15000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/lexical/11_total_evidence/"

lex    <- readDiscreteCharacterData("data/lexical/d11_lexical.nex")
struct <- readDiscreteCharacterData("data/lexical/d11_structural.nex")
taxa <- lex.taxa()
n_branches <- 2 * taxa.size() - 3
moves    = VectorMoves()
monitors = VectorMonitors()

psi ~ dnUniformTopologyBranchLength(taxa, branchLengthDistribution =
dnExponential(5.0))
moves.append( mvNNI(psi, weight = 10) )
moves.append( mvSPR(psi, weight = 5) )
moves.append( mvBranchLengthScale(psi, weight = n_branches) )
TL := psi.treeLength()

# lexical partition
pi ~ dnDirichlet(v(1, 1))
moves.append( mvBetaSimplex(pi, alpha = 20, weight = 3) )
pi1 := pi[2]
Q_lex := fnF81(pi)
```

```
seq_lex ~ dnPhyloCTMC(tree = psi, Q = Q_lex, type = "Binary", coding =
"noabsencesites")
seq_lex.clamp(lex)

# structural partition
r_struct ~ dnLognormal(0.0, 1.0)
moves.append( mvScale(r_struct, lambda = 0.5, weight = 5) )
Q_str <- fnJC(3)
seq_str ~ dnPhyloCTMC(tree = psi, Q = Q_str, branchRates = r_struct, type =
"Standard",
                      coding = "variable")
seq_str.clamp(struct)

mymodel = model(psi)
monitors.append( mnModel(filename = out_dir + "total_evidence.log", printgen = 15) )
monitors.append( mnFile(psi, filename = out_dir + "total_evidence.trees", printgen =
75) )
monitors.append( mnScreen(printgen = 5000, r_struct, pi1, TL) )
mymcmc = mcmc(mymodel, monitors, moves, nruns = 1)
mymcmc.burnin(generations = 1000, tuningInterval = 100)
mymcmc.run(generations = n_gen)

trace = readTreeTrace(out_dir + "total_evidence.trees", treetype = "non-clock",
burnin = 0.1)
map_tree = mapTree(trace, out_dir + "total_evidence_map.tre")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
lexical binary CTMC + structural Mkv (K = 3), shared tree	r_struct	0.400	0.404	0.292	0.514	757.2	pass
lexical binary CTMC + structural Mkv (K = 3), shared tree	pi1	0.200	0.206	0.193	0.219	554.3	pass
lexical binary CTMC + structural Mkv (K = 3), shared tree	TL	4.634	4.432	3.727	5.240	463.6	pass
lexical binary CTMC + structural Mkv (K = 3), shared tree	topology_ RF_MAP	0.000	4.000	NA	NA	NA	info (MAP topology differs)

H.3.13 Rate strata of meanings: a Dirichlet-process approximation

The biology box of Section 7.11 suggests a Dirichlet-process prior on meaning rates (Huelsenbeck and Suchard, 2007) to let the data sort meanings into stable and volatile strata. A direct `dnDPP` over per-meaning partitions is prohibitively slow in RevBayes 1.4.1 (the DPP vector is a single node of the model graph, so every allocation update recomputes all meanings). Script 12 therefore uses the

standard finite approximation to the Dirichlet process: a sparse mixture of four rate components with weights $\sim$ Dirichlet$(1/4, \ldots, 1/4)$ and rates drawn from the base distribution, in which each meaning-slot Mk character is a mixture over the components. Data: 50 meanings with rate 0.05 and 50 with rate 0.5 per millennium on the fixed true tree. The posterior puts most weight on two strata with equal weights; the slow rate is recovered, the fast rate (about 12 expected changes per character over the tree, so nearly saturated) is slightly underestimated, its HPD ending just below the truth. Per-meaning allocations are computed in the check script from the sampled rates and weights; 97% of meanings are assigned to the right stratum.

```
################################################################################
# 12_meaning_rate_strata.Rev
# Model   : clustering of meaning rates into an unknown number of rate strata
#           ("stable" vs "volatile" vocabulary), the Dirichlet-process idea of
#           Huelsenbeck & Suchard (2007) applied to meanings.
#           Implemented as a SPARSE FINITE MIXTURE (the Dirichlet-multinomial
#           approximation to the Dirichlet process, Ishwaran & Zarepour 2002):
#           K_max = 4 rate components, rates r_k ~ Gamma(1, 4) (the DP base
#           distribution), weights w ~ Dirichlet(alpha/K_max, ..., alpha/K_max)
#           with alpha = 1, so that superfluous components are emptied.  Each
#           meaning-slot Mk character (K = 5 states) is a mixture over the rate
#           components (siteRates / siteRatesProbs of dnPhyloCTMC).
# Why not dnDPP: dnDPP over per-meaning rates needs one CTMC object per meaning;
#           in RevBayes 1.4.1 the DPP vector is a single DAG node, so every
#           allocation update recomputes all M partitions (M^2 work per sweep) --
#           a 100-meaning test ran at well under 1 generation per second.  The
#           finite mixture integrates the allocations analytically in one CTMC.
# Book    : Chapter 7, "What biology offers" box (Dirichlet-process rates at the
#           level of meanings); Section lexical-models-partitions.
# Estimates: component rates and weights; the number of occupied strata and the
#           per-meaning rates are summarised in tests/check_lexical.R (posterior
#           allocation probabilities computed from the sampled r and w).
# Design  : time tree fixed at the true tree (root age 4 millennia), so rates are
#           per millennium.
# Data    : data/lexical/d12_dpp.nex, 100 meanings: 50 with rate 0.05 and 50 with
#           rate 0.5 per millennium (truth: 2 strata).
# Runtime : about 1 minute (default n_gen = 10000)
# Usage   : rb analyses/lexical/12_meaning_rate_strata.Rev [--args n_gen]
################################################################################
seed(1212)
n_gen = 10000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/lexical/12_meaning_rate_strata/"
K <- 5
K_max <- 4
dp_alpha <- 1.0

data <- readDiscreteCharacterData("data/lexical/d12_dpp.nex")
moves    = VectorMoves()
monitors = VectorMonitors()
```

```
psi <- readTrees("data/lexical/true_time_tree.tre")[1]

for (k in 1:K_max) {
    r[k] ~ dnGamma(1.0, 4.0)                    # DP base distribution
    moves.append( mvScale(r[k], lambda = 0.5, weight = 3) )
}
w ~ dnDirichlet(rep(dp_alpha / K_max, K_max))    # sparse symmetric Dirichlet
moves.append( mvDirichletSimplex(w, alpha = 20, weight = 5) )
moves.append( mvBetaSimplex(w, alpha = 20, weight = 5) )
moves.append( mvElementSwapSimplex(w, weight = 2) )

Q <- fnJC(K)
seq ~ dnPhyloCTMC(tree = psi, Q = Q, siteRates = r, siteRatesProbs = w,
                  type = "Standard", coding = "all")
seq.clamp(data)

mymodel = model(seq)
monitors.append( mnModel(filename = out_dir + "dpp.log", printgen = 10) )
monitors.append( mnScreen(printgen = 2500, r, w) )
mymcmc = mcmc(mymodel, monitors, moves, nruns = 1)
mymcmc.burnin(generations = 1000, tuningInterval = 100)
mymcmc.run(generations = n_gen)
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
sparse finite mixture (DP approximation) on meaning rates, Mk K = 5	n_occupied_strata_mode	2.00	2.000	2.000	3.000	300.0	pass
sparse finite mixture (DP approximation) on meaning rates, Mk K = 5	rate_slow_stratum	0.05	0.047	0.026	0.071	901.0	pass
sparse finite mixture (DP approximation) on meaning rates, Mk K = 5	rate_fast_stratum	0.50	0.374	0.248	0.499	901.0	fail
sparse finite mixture (DP approximation) on meaning rates, Mk K = 5	weight_slow_stratum	0.50	0.440	0.261	0.582	760.6	pass
sparse finite mixture (DP approximation) on meaning rates, Mk K = 5	weight_fast_stratum	0.50	0.457	0.279	0.611	793.3	pass
sparse finite mixture (DP approximation) on meaning rates, Mk K = 5	n_components_weight_gt_0.05	NA	2.654	2.000	4.000	NA	info
sparse finite mixture (DP approximation) on meaning rates, Mk K = 5	classification_accuracy	1.00	0.970	NA	NA	NA	info

H.3.14 Mixture over frequency-of-presence and rate classes

Script 13 implements the other suggestion of the biology box: a mixture over (π_1, rate) classes for binary cognates (Pagel and Meade, 2004), separating, for example, long-lived widespread cognates from short-lived local ones without assigning characters in advance (`siteMatrices` mixture in `dnPhyloCTMC`). Label switching is removed by ordering the two π_1 values. RevBayes applies the no-absence correction to the mixture as a whole, which is correct (checked against R). A random start can fall into a degenerate mode with a tiny class of saturated characters; the script starts from a sensible state and uses the compound Dirichlet tree-length prior, whose whole-tree move is essential for mixing.

```
################################################################################
# 13_mixture_pi_rate.Rev
# Model   : two-class mixture model for binary cognate characters: each class k
#           has its own stationary frequency of presence pi1[k] and relative rate
#           r[k] (mean rate over the mixture = 1); class memberships are unknown
#           and summed over (siteMatrices mixture), weights w estimated.  This is
#           the "mixture over (pi1, rate) classes" suggested for cognate data
#           (long-lived widespread vs short-lived local cognates).
# Book    : Chapter 7, "What biology offers" box (mixture and profile-mixture
models).
# Identifiability: label switching is removed by ordering pi1[1] < pi1[2]
#           (pi1[2] = pi1[1] + delta * (1 - pi1[1])); w ~ Dirichlet(2, 2) and a
#           sensible starting state keep the chain out of a degenerate mode.
# Ascertainment: coding = "noabsencesites"; RevBayes applies the correction to
#           the mixture likelihood as a whole (checked against R: global
#           1 - sum_k w_k P_k(0...0), not per component).
# Check   : tests/check_lexical.R recomputes the mixture likelihood in R (exact
match).
# Data    : data/lexical/d13_mixture.nex (truth: pi1 = (0.1, 0.4), r = (0.5, 1.5),
#           w2 = 0.5, TL = 6.95)
# Runtime : about 2.5 minutes (default n_gen = 12000)
# Usage   : rb analyses/lexical/13_mixture_pi_rate.Rev [--args n_gen]
################################################################################
seed(1313)
n_gen = 12000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/lexical/13_mixture_pi_rate/"

data <- readDiscreteCharacterData("data/lexical/d13_mixture.nex")
taxa <- data.taxa()
n_branches <- 2 * taxa.size() - 3
moves    = VectorMoves()
monitors = VectorMonitors()

# gamma-Dirichlet branch-length prior with an explicit tree length TL (mean 5):
# TL is weakly identified when one class saturates, and the TL scale move is
# essential for mixing (per-branch moves alone gave ESS ~ 10)
topology ~ dnUniformTopology(taxa)
```

```
moves.append( mvNNI(topology, weight = 5) )
moves.append( mvSPR(topology, weight = 3) )
TL ~ dnGamma(1.0, 0.2)
moves.append( mvScale(TL, lambda = 0.3, weight = 8) )
rel_bl ~ dnDirichlet(rep(1.0, n_branches))
moves.append( mvBetaSimplex(rel_bl, alpha = 50, weight = n_branches) )
moves.append( mvDirichletSimplex(rel_bl, alpha = 200, weight = 5) )
bl := rel_bl * TL
psi := fnTreeAssembly(topology, bl)

w ~ dnDirichlet(v(2, 2))                    # mixture weights (mildly away from 0)
moves.append( mvBetaSimplex(w, alpha = 20, weight = 3) )
pi1_low ~ dnBeta(1, 1)
delta ~ dnBeta(1, 1)
moves.append( mvBetaProbability(pi1_low, delta = 20, weight = 5) )
moves.append( mvBetaProbability(delta, delta = 20, weight = 5) )
pi1[1] := pi1_low
pi1[2] := pi1_low + delta * abs(1.0 - pi1_low)
rr ~ dnDirichlet(v(1, 1))                   # unnormalised relative rates
moves.append( mvBetaSimplex(rr, alpha = 20, weight = 5) )
mean_rr := w[1] * rr[1] + w[2] * rr[2]
# start away from the degenerate mode "tiny class of saturated sites" that a
# random start can fall into (seen in a test run with ESS ~ 10)
w.setValue(simplex(0.5, 0.5))
pi1_low.setValue(0.15)
delta.setValue(0.3)
rr.setValue(simplex(0.3, 0.7))
for (k in 1:2) {
    r[k] := rr[k] / mean_rr                  # sum_k w_k r_k = 1
    # binary F81-type matrix scaled to r[k] expected changes per unit branch length
    Qk[k] := fnFreeBinary([r[k] / (2.0 * abs(1.0 - pi1[k])), r[k] / (2.0 * pi1[k])],
                          rescaled = FALSE)
}

seq ~ dnPhyloCTMC(tree = psi, Q = Qk, siteMatrices = w, type = "Binary",
                  coding = "noabsencesites")
seq.clamp(data)

mymodel = model(topology)
monitors.append( mnModel(filename = out_dir + "mixture.log", printgen = 10, exclude
= ["rel_bl", "bl"]) )
monitors.append( mnFile(psi, filename = out_dir + "mixture.trees", printgen = 50) )
monitors.append( mnScreen(printgen = 2000, pi1, r, w, TL) )
mymcmc = mcmc(mymodel, monitors, moves, nruns = 1)
mymcmc.burnin(generations = 1000, tuningInterval = 100)
mymcmc.run(generations = n_gen)
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
2-class mixture over (pi1, rate), noabsencesites	pi1[1]	0.10	0.109	0.066	0.159	446.1	pass
2-class mixture over (pi1, rate), noabsencesites	pi1[2]	0.40	0.402	0.371	0.436	310.8	pass
2-class mixture over (pi1, rate), noabsencesites	r[1]	0.50	0.522	0.359	0.675	503.7	pass
2-class mixture over (pi1, rate), noabsencesites	r[2]	1.50	1.496	1.310	1.700	820.6	pass
2-class mixture over (pi1, rate), noabsencesites	w[2]	0.50	0.492	0.415	0.563	509.2	pass
2-class mixture over (pi1, rate), noabsencesites	TL	6.95	7.354	4.461	10.623	185.4	pass

H.3.15 Sensitivity to the branch-length prior

Script `14` refits the gamma model of script `02` under two branch-length priors (Sections 6.2 and 6.8): i.i.d. exponential branch lengths with mean 0.1 (prior tree length Gamma(17, 10), mean 1.7) and a compound Dirichlet prior with tree length ~ Gamma(1, 0.1) (mean 10). Because the tree length and α lie on a ridge, both posteriors are pulled towards their prior: the first gives a tree length of about 4.3 and the second about 16.3 (truth 6.95), with α moving in the opposite direction, and neither 95% interval covers the truth. The rows are marked `info`: the script demonstrates prior sensitivity, the warning of Rannala et al. (2012), and the reason why script `02` uses a compound Dirichlet prior with a moderate mean.

```
################################################################################
# 14_branch_length_priors.Rev
# Model   : sensitivity of the tree length to the branch-length prior for an
#           unrooted cognate analysis (binary CTMC + Gamma, no-absence correction):
#             (A) i.i.d. Exponential(rate 10) branch lengths (mean 0.1, the common
#                 default; prior tree length ~ Gamma(17, 10), mean 1.7)
#             (B) compound Dirichlet / gamma-Dirichlet prior (Rannala, Zhu & Yang
#                 2012; Zhang, Rannala & Yang 2012): TL ~ Gamma(1, 0.1),
#                 relative branch lengths ~ flat Dirichlet
#          The priors on TL are known analytically (Gamma(17, 10) and Gamma(1, 0.1));
#           tests/check_lexical.R compares them with the posteriors.
# Book    : Chapter 6, Section bayesian-priors ("Priors on topologies and branch
#           lengths for undated trees") and Section bayesian-sensitivity.
# Data    : data/lexical/d02_gamma.nex (true TL = 6.95, alpha = 0.5)
# Runtime : about 3 minutes for the two chains (default n_gen = 8000)
# Usage   : rb analyses/lexical/14_branch_length_priors.Rev [--args n_gen]
################################################################################
seed(1414)
n_gen = 8000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/lexical/14_branch_length_priors/"
```

```
data <- readDiscreteCharacterData("data/lexical/d02_gamma.nex")
taxa <- data.taxa()
n_branches <- 2 * taxa.size() - 3

# ---------------- (A) i.i.d. exponential branch lengths
moves    = VectorMoves()
monitors = VectorMonitors()
psiA ~ dnUniformTopologyBranchLength(taxa, branchLengthDistribution =
dnExponential(10.0))
moves.append( mvNNI(psiA, weight = 5) )
moves.append( mvSPR(psiA, weight = 3) )
moves.append( mvBranchLengthScale(psiA, weight = n_branches) )
TL := psiA.treeLength()
pi ~ dnDirichlet(v(1, 1))
moves.append( mvBetaSimplex(pi, alpha = 20, weight = 3) )
Q := fnF81(pi)
alpha ~ dnExponential(1.0)
moves.append( mvScale(alpha, lambda = 0.5, weight = 3) )
site_rates := fnDiscretizeGamma(alpha, alpha, 4)
seqA ~ dnPhyloCTMC(tree = psiA, Q = Q, siteRates = site_rates, type = "Binary",
                   coding = "noabsencesites")
seqA.clamp(data)
mymodel = model(psiA)
monitors.append( mnModel(filename = out_dir + "iid_exponential.log", printgen = 10)
)
mymcmc = mcmc(mymodel, monitors, moves, nruns = 1)
mymcmc.burnin(generations = 500, tuningInterval = 100)
mymcmc.run(generations = n_gen)

# ---------------- (B) gamma-Dirichlet (compound Dirichlet) prior
moves    = VectorMoves()
monitors = VectorMonitors()
topoB ~ dnUniformTopology(taxa)
moves.append( mvNNI(topoB, weight = 5) )
moves.append( mvSPR(topoB, weight = 3) )
TLB ~ dnGamma(1.0, 0.1)
moves.append( mvScale(TLB, lambda = 0.3, weight = 5) )
relB ~ dnDirichlet(rep(1.0, n_branches))
moves.append( mvBetaSimplex(relB, alpha = 50, weight = n_branches) )
moves.append( mvDirichletSimplex(relB, alpha = 200, weight = 3) )
blB := relB * TLB
psiB := fnTreeAssembly(topoB, blB)
TL := psiB.treeLength()
pi ~ dnDirichlet(v(1, 1))
moves.append( mvBetaSimplex(pi, alpha = 20, weight = 3) )
Q := fnF81(pi)
alpha ~ dnExponential(1.0)
moves.append( mvScale(alpha, lambda = 0.5, weight = 3) )
```

```
site_rates := fnDiscretizeGamma(alpha, alpha, 4)
seqB ~ dnPhyloCTMC(tree = psiB, Q = Q, siteRates = site_rates, type = "Binary",
                   coding = "noabsencesites")
seqB.clamp(data)
mymodel = model(topoB)
monitors.append( mnModel(filename = out_dir + "gamma_dirichlet.log", printgen = 10,
                         exclude = ["relB", "blB"]) )
mymcmc = mcmc(mymodel, monitors, moves, nruns = 1)
mymcmc.burnin(generations = 500, tuningInterval = 100)
mymcmc.run(generations = n_gen)
q()
```

Model	Paramete	Truth	Post. mean	HPD low	HPD high	ESS	Status
CTMC + Gamma, iid_ exponential branch-length prior	TL	6.95	4.271	3.524	5.086	281.3	info (prior sensitivity; covered = FALSE)
CTMC + Gamma, iid_ exponential branch-length prior	alpha	0.50	0.694	0.553	0.852	504.3	info (prior sensitivity; covered = FALSE)
CTMC + Gamma, gamma_ dirichlet branch-length prior	TL	6.95	16.339	10.571	23.271	320.9	info (prior sensitivity; covered = FALSE)
CTMC + Gamma, gamma_ dirichlet branch-length prior	alpha	0.50	0.291	0.246	0.332	349.7	info (prior sensitivity; covered = FALSE)

H.3.16 Explicit replacement with synonymy

Section 7.8.3 notes that a model in which each meaning slot holds a *set* of cognate classes, with entry of new words and competition-dependent loss, "can be written down as a CTMC on subsets and specified in general-purpose software such as RevBayes". Script 15 does so for $K = 3$ classes: seven states (the non-empty subsets of $\{A, B, C\}$), entry at rate γ for each absent class, loss of each word at rate δ times the number of its competitors (so a slot is never empty), and root frequencies equal to the stationary distribution, proportional to $(\gamma/\delta)^{|S|-1}/(|S|-1)!$. Each language's observed set of classes for a meaning is one 7-state character. The closed set of three classes (a lost class may re-enter) is a simplification of the open process. The ratio γ/δ, and with it the stationary frequency of synonymy, is sharply identified; the absolute rates trade off against the node ages, and the loss rate is underestimated, its HPD ending just below the truth (maximum likelihood on the true tree, computed in the check script, gives estimates closer to the truth).

```
################################################################################
# 15_synonymy_slot_model.Rev
# Model   : explicit model of replacement with synonymy for meaning slots: each
#           language's slot holds a SET of cognate classes (polymorphism allowed),
#           written as a CTMC on the non-empty subsets of K = 3 classes {A, B, C}:
#             states  1 {A}  2 {B}  3 {C}  4 {A,B}  5 {A,C}  6 {B,C}  7 {A,B,C}
#           * a new class enters the slot at rate g_in (for each absent class);
```

```
#           * each present word is lost at rate d_out x (number of competitors),
#             so a slot is never empty and loss accelerates with synonymy.
#           The chain is reversible with stationary frequencies proportional to
#           (g_in/d_out)^(|S|-1) / (|S|-1)!, used as root frequencies.
# Book    : Chapter 7, Section lexical-models-replacement (row 10 of Table
#           lexical-models-comparison: "small state spaces can be specified in
#           RevBayes").  Simplification: a closed set of K = 3 classes per meaning
#           (a class can re-enter after loss), not an open infinite-states process.
# Estimates: entry rate g_in and loss rate d_out (per millennium), topology and
#           node ages (root age fixed at 4 millennia).
# Data    : data/lexical/d15_synonymy.nex: 150 meanings, the observed subset of each
#           language coded as one 7-state character (truth: g_in = 0.3, d_out = 1.0).
# Runtime : about 2 minutes (default n_gen = 12000)
# Usage   : rb analyses/lexical/15_synonymy_slot_model.Rev [--args n_gen]
################################################################################
seed(1515)
n_gen = 12000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/lexical/15_synonymy_slot_model/"
root_age <- 4.0

data <- readDiscreteCharacterData("data/lexical/d15_synonymy.nex")
taxa <- data.taxa()
moves    = VectorMoves()
monitors = VectorMonitors()

psi ~ dnUniformTimeTree(rootAge = root_age, taxa = taxa)
moves.append( mvNarrow(psi, weight = 5) )
moves.append( mvFNPR(psi, weight = 3) )
moves.append( mvNodeTimeSlideUniform(psi, weight = 10) )
moves.append( mvSubtreeScale(psi, weight = 3) )

g_in ~ dnLoguniform(0.001, 100.0)    # entry rate of a new class into the slot
d_out ~ dnLoguniform(0.001, 100.0)    # loss rate per competitor
moves.append( mvScale(g_in, lambda = 0.5, weight = 5) )
moves.append( mvScale(d_out, lambda = 0.5, weight = 5) )
gd_updown = mvUpDownScale(lambda = 0.3, weight = 3)
gd_updown.addVariable(g_in, up = TRUE)
gd_updown.addVariable(d_out, up = TRUE)
moves.append( gd_updown )
d2 := 2.0 * d_out
# rows/columns: {A} {B} {C} {A,B} {A,C} {B,C} {A,B,C}
Q := fnFreeK([[0.0,   0.0,   0.0,   g_in,  g_in,  0.0,   0.0 ],
              [0.0,   0.0,   0.0,   g_in,  0.0,   g_in,  0.0 ],
              [0.0,   0.0,   0.0,   0.0,   g_in,  g_in,  0.0 ],
              [d_out, d_out, 0.0,   0.0,   0.0,   0.0,   g_in],
              [d_out, 0.0,   d_out, 0.0,   0.0,   0.0,   g_in],
              [0.0,   d_out, d_out, 0.0,   0.0,   0.0,   g_in],
```

```
              [0.0,   0.0,   0.0,   d2,    d2,    d2,    0.0 ]], rescaled = FALSE)
q_ratio := g_in / d_out
root_freq := simplex(1.0, 1.0, 1.0, q_ratio, q_ratio, q_ratio, 0.5 * q_ratio * q_
ratio)
p_synonymy := (3.0 * q_ratio + 0.5 * q_ratio * q_ratio) / (3.0 + 3.0 * q_ratio + 0.5
* q_ratio * q_ratio)

seq ~ dnPhyloCTMC(tree = psi, Q = Q, rootFrequencies = root_freq, type = "Standard",
                  coding = "all")
seq.clamp(data)

mymodel = model(psi)
monitors.append( mnModel(filename = out_dir + "synonymy.log", printgen = 10) )
monitors.append( mnFile(psi, filename = out_dir + "synonymy.trees", printgen = 60) )
monitors.append( mnScreen(printgen = 1000, g_in, d_out, p_synonymy) )
mymcmc = mcmc(mymodel, monitors, moves, nruns = 1)
mymcmc.burnin(generations = 500, tuningInterval = 100)
mymcmc.run(generations = n_gen)
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
explicit replacement with synonymy (CTMC on subsets of 3 classes)	g_in	0.300	0.250	0.194	0.323	375.5	pass
explicit replacement with synonymy (CTMC on subsets of 3 classes)	d_out	1.000	0.736	0.557	0.961	398.2	fail
explicit replacement with synonymy (CTMC on subsets of 3 classes)	p_ synonymy	0.262	0.265	0.243	0.289	1081.0	pass
explicit replacement with synonymy (CTMC on subsets of 3 classes)	topology_ RF_MAP	0.000	6.000	NA	NA	NA	info (MAP topology differs)
ML (R pruning, true tree)	d_out	1.000	0.866	NA	NA	NA	info

H.3.17 F81 mixtures across characters: JC, F81 and Beta mixtures compared

Section 7.3.1 gives every binary cognate character the same stationary frequency of presence π_1. Script 16 relaxes this in the way that morphological phylogenetics does for binary characters (Wright et al., 2016): each character belongs to one of four categories whose π_1 values are the mean-discretized quantile classes of a Beta distribution (`fnDiscretizeBeta`), each category has an F81 rate matrix rescaled to one expected change per unit branch length, and the likelihood sums over the unknown category of each character (`siteMatrices`). In morphology the Beta is usually symmetric, Beta(s, s), because the labels 0 and 1 are arbitrary; for cognate presence and absence they are not, so the script parameterizes the Beta by its mean m and concentration ϕ, Beta$(m\phi, (1-m)\phi)$. The data (925 observed cognate classes on the fixed true time tree, strict

clock 0.3) were simulated under this asymmetric mixture with Beta$(1, 3)$, and five models are compared by stepping-stone and path sampling (Section 6.7), all with the no-absence correction: the binary Jukes-Cantor (Mk) model with $\pi_1 = 1/2$, F81 with one estimated π_1, F81 with gamma rates, and the asymmetric and symmetric Beta mixtures. This is a parameter-recovery and model-recovery experiment in the sense of Section 6.10; it does not show that real cognate data contain this kind of heterogeneity.

The generating model wins clearly, by about 40 log-units over F81 with gamma rates. Gamma rate variation thus absorbs part of the heterogeneity in π_1, but only part, because characters with a low π_1 differ from the others in their pattern of presence and not only in their rate. The binary Jukes-Cantor model, which forces $\pi_1 = 1/2$, is worst by far, and the symmetric mixture, which keeps the mean of π_1 at one half, recovers only a fraction of the difference: for presence-absence data the mean frequency must be free. Parameter recovery under the generating model is mixed. The concentration, b and the clock rate are covered by their 95% HPD intervals, but the mean frequency of presence is overestimated (posterior mean 0.292, truth 0.25) and the interval for a ends just above the truth. The check script recomputes the mixture likelihood independently in R (agreement to 10^{-3} log-units at posterior samples) and finds the same estimate by maximum likelihood on the same data, while maximum-likelihood fits to four further simulated data sets of 3,000 classes give m between 0.245 and 0.254. The offset is therefore a property of this one simulated data set, a reminder that a single replicate can miss the truth by chance, and not a bias of the model or of RevBayes.

```
################################################################################
# 16_f81_beta_mixture.Rev
# Model   : F81 mixture across cognate characters ("Beta mixture" of stationary
#           frequencies): every character belongs to one of four categories,
#           each with its own frequency of presence pi1[k] taken from the four
#           mean-discretized quantile classes of a Beta distribution, and each
#           with an F81 rate matrix rescaled to one expected change per unit
#           branch length.  Category memberships are summed over (siteMatrices).
#           Compared by stepping-stone / path sampling with simpler models
#           fitted to the SAME data (all with the no-absence correction):
#             (1) JC / binary Mk: pi1 fixed at 0.5
#             (2) F81: one pi1 for all characters
#             (3) F81 + discrete Gamma(4) rates
#             (4) F81 Beta mixture, asymmetric Beta(m phi, (1 - m) phi)
#                 <- generating model
#             (5) F81 Beta mixture, symmetric Beta(s, s), the form used for
#                 morphological characters whose state labels are arbitrary
#                 (Wright et al. 2016); for presence/absence the labels are
#                 not arbitrary, so the mean frequency should be free
# Book    : Chapter 7, Sections lexical-models-ctmc-model (stationary
#           frequencies), lexical-models-rates and lexical-models-comparison;
#           Chapter 6, Section bayesian-marginal.
# Design  : the dated tree is FIXED at the true time tree (root age 4
#           millennia) with a strict clock rate, as in 09_model_comparison.Rev,
#           so that only the substitution models differ.  A short MCMC under
#           model (4) checks parameter recovery of m, phi and the clock rate.
# Data    : data/lexical/d16_beta_mixture.nex, simulated under model (4) with
```

```
#           Beta(1, 3) (m = 0.25, phi = 4), clock rate 0.3, 925 observed
#           cognate classes.
# Output  : output/lexical/16_f81_beta_mixture/  (ml_*.txt, beta_mixture.log)
# Runtime : about 6 minutes (default 16 stones x 300 generations per model);
#           run one model only with  --args <model 1-5> [generations per stone]
# Usage   : rb analyses/lexical/16_f81_beta_mixture.Rev
#               [--args model n_per_stone]
################################################################################
seed(1616)
models_to_run = [1, 2, 3, 4, 5]
n_stone_gen = 300
n_gen = 6000
run_mcmc = TRUE
if (args.size() > 1) models_to_run = [int(args[2])]
if (args.size() > 1) run_mcmc = FALSE
if (args.size() > 2) n_stone_gen = int(args[3])
n_cats = 16
n_pi_cats = 4
out_dir = "output/lexical/16_f81_beta_mixture/"
model_names = ["jc", "f81", "f81_gamma", "beta_mixture",
               "symmetric_beta_mixture"]

data <- readDiscreteCharacterData("data/lexical/d16_beta_mixture.nex")
psi <- readTrees("data/lexical/true_time_tree.tre")[1]
equal_w <- simplex(rep(1, n_pi_cats))

# ------------------------------------------------ parameter recovery, model 4
if (run_mcmc) {
    moves    = VectorMoves()
    monitors = VectorMonitors()
    clock_rate ~ dnLoguniform(0.001, 10.0)
    moves.append( mvScale(clock_rate, lambda = 0.5, weight = 3) )
    m ~ dnBeta(1, 1)                      # mean frequency of presence
    phi ~ dnExponential(0.1)              # concentration (a + b), prior mean 10
    moves.append( mvBetaProbability(m, delta = 20, weight = 3) )
    moves.append( mvScale(phi, lambda = 0.5, weight = 3) )
    beta_a := m * phi
    beta_b := abs(1.0 - m) * phi
    pi_cat := fnDiscretizeBeta(beta_a, beta_b, n_pi_cats)
    for (k in 1:n_pi_cats) {
        Qm[k] := fnF81(simplex(abs(1.0 - pi_cat[k]), pi_cat[k]))
    }
    seq ~ dnPhyloCTMC(tree = psi, Q = Qm, branchRates = clock_rate,
                      siteMatrices = equal_w, type = "Binary",
                      coding = "noabsencesites")
    seq.clamp(data)
    mymodel = model(clock_rate)
    monitors.append( mnModel(filename = out_dir + "beta_mixture.log",
```

```
                              printgen = 5) )
    monitors.append( mnScreen(printgen = 1000, m, phi, clock_rate) )
    mymcmc = mcmc(mymodel, monitors, moves, nruns = 1)
    mymcmc.burnin(generations = 500, tuningInterval = 50)
    mymcmc.run(generations = n_gen)
}

# ------------------------------------------------ marginal likelihoods
for (mi in models_to_run) {
    moves    = VectorMoves()
    monitors = VectorMonitors()
    clock_rate ~ dnLoguniform(0.001, 10.0)
    moves.append( mvScale(clock_rate, lambda = 0.5, weight = 3) )

    if (mi == 1) {
        Q1 <- fnJC(2)
        seq ~ dnPhyloCTMC(tree = psi, Q = Q1, branchRates = clock_rate,
                          type = "Binary", coding = "noabsencesites")
    }
    if (mi == 2 || mi == 3) {
        pi_f81 ~ dnDirichlet(v(1, 1))
        moves.append( mvBetaSimplex(pi_f81, alpha = 20, weight = 3) )
        Q2 := fnF81(pi_f81)
    }
    if (mi == 2) {
        seq ~ dnPhyloCTMC(tree = psi, Q = Q2, branchRates = clock_rate,
                          type = "Binary", coding = "noabsencesites")
    }
    if (mi == 3) {
        alpha ~ dnExponential(1.0)
        moves.append( mvScale(alpha, lambda = 0.5, weight = 3) )
        site_rates := fnDiscretizeGamma(alpha, alpha, 4)
        seq ~ dnPhyloCTMC(tree = psi, Q = Q2, branchRates = clock_rate,
                          siteRates = site_rates, type = "Binary",
                          coding = "noabsencesites")
    }
    if (mi == 4) {
        m ~ dnBeta(1, 1)
        phi ~ dnExponential(0.1)
        moves.append( mvBetaProbability(m, delta = 20, weight = 3) )
        moves.append( mvScale(phi, lambda = 0.5, weight = 3) )
        beta_a := m * phi
        beta_b := abs(1.0 - m) * phi
        pi_cat := fnDiscretizeBeta(beta_a, beta_b, n_pi_cats)
        for (k in 1:n_pi_cats) {
            Qm[k] := fnF81(simplex(abs(1.0 - pi_cat[k]), pi_cat[k]))
        }
        seq ~ dnPhyloCTMC(tree = psi, Q = Qm, branchRates = clock_rate,
```

```
                        siteMatrices = equal_w, type = "Binary",
                        coding = "noabsencesites")
    }
    if (mi == 5) {
        s_beta ~ dnExponential(1.0)
        moves.append( mvScale(s_beta, lambda = 0.5, weight = 3) )
        pi_sym := fnDiscretizeBeta(s_beta, s_beta, n_pi_cats)
        for (k in 1:n_pi_cats) {
            Qs[k] := fnF81(simplex(abs(1.0 - pi_sym[k]), pi_sym[k]))
        }
        seq ~ dnPhyloCTMC(tree = psi, Q = Qs, branchRates = clock_rate,
                        siteMatrices = equal_w, type = "Binary",
                        coding = "noabsencesites")
    }
    seq.clamp(data)

    mymodel = model(clock_rate)
    nm = model_names[mi]
    monitors.append( mnModel(filename = out_dir + nm + "_posterior.log",
                             printgen = 50) )
    pow_p = powerPosterior(mymodel, moves, monitors,
                           out_dir + nm + "_powp.out",
                           cats = n_cats, sampleFreq = 5)
    pow_p.burnin(generations = 200, tuningInterval = 50)
    pow_p.run(generations = n_stone_gen)
    ss = steppingStoneSampler(file = out_dir + nm + "_powp.out",
                              powerColumnName = "power",
                              likelihoodColumnName = "likelihood")
    ps = pathSampler(file = out_dir + nm + "_powp.out",
                     powerColumnName = "power",
                     likelihoodColumnName = "likelihood")
    ml_ss = ss.marginal()
    ml_ps = ps.marginal()
    print(nm + "  stepping-stone: " + ml_ss + "  path sampling: " + ml_ps)
    write(nm, mi, ml_ss, ml_ps, filename = out_dir + "ml_" + nm + ".txt",
          separator = ",")
}
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
jc	log_marginal_likelihood_SS	NA	-5379.200	NA	NA	NA	info
f81	log_marginal_likelihood_SS	NA	-5154.400	NA	NA	NA	info
f81_gamma	log_marginal_likelihood_SS	NA	-5135.700	NA	NA	NA	info

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
beta_mixture	log_marginal_likelihood_SS	NA	-5095.800	NA	NA	NA	pass
symmetric_beta_mixture	log_marginal_likelihood_SS	NA	-5302.400	NA	NA	NA	info
all five	generating_model_wins	4.00	4.000	NA	NA	NA	pass
F81 Beta mixture (4 categories), noabsencesites, true time tree	m	0.25	0.292	0.264	0.314	775.5	fail
F81 Beta mixture (4 categories), noabsencesites, true time tree	phi	4.00	4.799	3.676	6.089	1081.0	pass
F81 Beta mixture (4 categories), noabsencesites, true time tree	beta_a	1.00	1.407	1.018	1.890	853.3	fail
F81 Beta mixture (4 categories), noabsencesites, true time tree	beta_b	3.00	3.393	2.594	4.178	1081.0	pass
F81 Beta mixture (4 categories), noabsencesites, true time tree	clock_rate	0.30	0.292	0.257	0.331	1177.9	pass
F81 Beta mixture (4 categories), noabsencesites, true time tree	lnL_RevBayes_minus_R	0.00	0.000	NA	NA	NA	pass
ML (R pruning, true tree)	m	0.25	0.293	NA	NA	NA	info

H.3.18 How many rate categories, and of which kind?

The four equiprobable categories of the discrete gamma (script `02`) are a numerical convention, not a hypothesis about language. Script `17` asks what the choice costs. The data (809 observed cognate classes on the fixed true time tree, strict clock 0.3, $\pi_1 = 0.25$) were simulated with rates drawn from the *continuous* Gamma(0.4, 0.4) distribution, so that none of the five fitted models is the generating one: the discrete gamma with two, four and eight categories; four gamma categories with free weights (a Dirichlet prior on `siteRatesProbs`, which lets the mean rate depart from one, a departure absorbed by the clock rate); and a three-category *FreeRate* model with free rates and free weights whose weighted mean is fixed at one (Yang, 1995; Soubrier et al., 2012). All use the F81 binary CTMC with the no-absence correction and are compared by stepping-stone and path sampling (16 stones of 300 generations).

Two categories are clearly too few (about 17 log-units worse than four), while eight categories differ from four by about one log-unit, which is within the Monte Carlo error of estimates from runs of this length. Neither the free weights nor the FreeRate model improves on four gamma categories: with a few hundred characters, the extra parameters cost more in marginal likelihood than they gain in fit when the true rates are gamma-shaped. A short MCMC under the four-category model recovers the simulated values of α, π_1 and the clock rate within their 95% HPD intervals, and the check script recomputes the four-category likelihood independently in R at posterior samples. This is a model-recovery experiment on data that follow the gamma shape; FreeRate models are

worth considering precisely when real cognate data contain rate classes that the gamma cannot represent, such as a block of almost invariant meanings next to fast ones (Section 7.5).

```
###############################################################################
# 17_rate_categories.Rev
# Model   : How many rate categories, and of which kind?  Bayesian comparison by
#           stepping-stone and path sampling (powerPosterior) of five models of
#           among-character rate variation, all with the F81 binary CTMC and
#           the no-absence correction, fitted to the SAME data:
#             (1) discrete Gamma, 2 equiprobable categories
#             (2) discrete Gamma, 4 equiprobable categories (usual default)
#             (3) discrete Gamma, 8 equiprobable categories
#             (4) discrete Gamma, 4 categories, FREE category weights
#                 (Dirichlet prior on siteRatesProbs; the mean rate is then no
#                 longer one and is absorbed by the clock rate)
#             (5) FreeRate, 3 categories: free rates and free weights with the
#                 weighted mean rate fixed at one (Yang 1995;
#                 Soubrier et al. 2012)
#           None of the five is the generating model: the data were simulated
#           with rates drawn from the CONTINUOUS gamma, so the question is how
#           well each discrete approximation describes it.
# Book    : Chapter 7, Section lexical-models-rates ("The gamma model");
#           Chapter 6, Section bayesian-marginal.
# Design  : as 09_model_comparison.Rev, the dated tree is FIXED at the true time
#           tree (root age 4 millennia) with a strict clock, so that only the
#           rate-variation models differ.  A short MCMC under model (2) checks
#           recovery of alpha, pi1 and the clock rate.
# Data    : data/lexical/d17_rate_categories.nex, simulated with continuous
#           Gamma(0.4, 0.4) rates, pi1 = 0.25, clock rate 0.3 (809 observed
#           cognate classes).
# Output  : output/lexical/17_rate_categories/  (ml_*.txt, gamma4.log)
# Runtime : about 3.5 minutes (default 16 stones x 300 generations per model);
#           run one model only with
#           --args <model 1-5> [generations per stone]
# Usage   : rb analyses/lexical/17_rate_categories.Rev
#           [--args model n_per_stone]
###############################################################################
seed(1717)
models_to_run = [1, 2, 3, 4, 5]
n_stone_gen = 300
n_gen = 6000
run_mcmc = TRUE
if (args.size() > 1) models_to_run = [int(args[2])]
if (args.size() > 1) run_mcmc = FALSE
if (args.size() > 2) n_stone_gen = int(args[3])
n_cats = 16
out_dir = "output/lexical/17_rate_categories/"
model_names = ["gamma2", "gamma4", "gamma8", "gamma4_free_weights",
               "freerate3"]
```

```
data <- readDiscreteCharacterData("data/lexical/d17_rate_categories.nex")
psi <- readTrees("data/lexical/true_time_tree.tre")[1]

# parameter recovery under the usual default, Gamma(4)
if (run_mcmc) {
    moves    = VectorMoves()
    monitors = VectorMonitors()
    clock_rate ~ dnLoguniform(0.001, 10.0)
    moves.append( mvScale(clock_rate, lambda = 0.5, weight = 3) )
    pi ~ dnDirichlet(v(1, 1))
    moves.append( mvBetaSimplex(pi, alpha = 20, weight = 3) )
    pi1 := pi[2]
    Q := fnF81(pi)
    alpha ~ dnExponential(1.0)
    moves.append( mvScale(alpha, lambda = 0.5, weight = 3) )
    site_rates := fnDiscretizeGamma(alpha, alpha, 4)
    seq ~ dnPhyloCTMC(tree = psi, Q = Q, branchRates = clock_rate,
                      siteRates = site_rates, type = "Binary",
                      coding = "noabsencesites")
    seq.clamp(data)
    mymodel = model(clock_rate)
    monitors.append( mnModel(filename = out_dir + "gamma4.log",
                             printgen = 5) )
    monitors.append( mnScreen(printgen = 2000, alpha, pi1, clock_rate) )
    mymcmc = mcmc(mymodel, monitors, moves, nruns = 1)
    mymcmc.burnin(generations = 500, tuningInterval = 50)
    mymcmc.run(generations = n_gen)
}

for (mi in models_to_run) {
    moves    = VectorMoves()
    monitors = VectorMonitors()
    clock_rate ~ dnLoguniform(0.001, 10.0)
    moves.append( mvScale(clock_rate, lambda = 0.5, weight = 3) )
    pi ~ dnDirichlet(v(1, 1))
    moves.append( mvBetaSimplex(pi, alpha = 20, weight = 3) )
    Q := fnF81(pi)

    if (mi <= 4) {
        alpha ~ dnExponential(1.0)
        moves.append( mvScale(alpha, lambda = 0.5, weight = 3) )
    }
    if (mi == 1) site_rates := fnDiscretizeGamma(alpha, alpha, 2)
    if (mi == 2 || mi == 4) site_rates := fnDiscretizeGamma(alpha, alpha, 4)
    if (mi == 3) site_rates := fnDiscretizeGamma(alpha, alpha, 8)
    if (mi <= 3) {
        seq ~ dnPhyloCTMC(tree = psi, Q = Q, branchRates = clock_rate,
```

```
                        siteRates = site_rates, type = "Binary",
                        coding = "noabsencesites")
    }
    if (mi == 4) {
        # free weights on the four gamma categories (equal weights a priori)
        cat_w ~ dnDirichlet(rep(1, 4))
        moves.append( mvBetaSimplex(cat_w, alpha = 20, weight = 3) )
        moves.append( mvDirichletSimplex(cat_w, alpha = 20, weight = 2) )
        seq ~ dnPhyloCTMC(tree = psi, Q = Q, branchRates = clock_rate,
                        siteRates = site_rates, siteRatesProbs = cat_w,
                        type = "Binary", coding = "noabsencesites")
    }
    if (mi == 5) {
        # FreeRate: weights w and rate shares u, rate_k = u_k / w_k, so that
        # sum_k w_k rate_k = sum_k u_k = 1
        fr_w ~ dnDirichlet(rep(1, 3))
        fr_u ~ dnDirichlet(rep(1, 3))
        moves.append( mvBetaSimplex(fr_w, alpha = 20, weight = 3) )
        moves.append( mvBetaSimplex(fr_u, alpha = 20, weight = 3) )
        moves.append( mvDirichletSimplex(fr_u, alpha = 20, weight = 2) )
        for (k in 1:3) fr_rate[k] := fr_u[k] / fr_w[k]
        seq ~ dnPhyloCTMC(tree = psi, Q = Q, branchRates = clock_rate,
                        siteRates = fr_rate, siteRatesProbs = fr_w,
                        type = "Binary", coding = "noabsencesites")
    }
    seq.clamp(data)

    mymodel = model(clock_rate)
    nm = model_names[mi]
    monitors.append( mnModel(filename = out_dir + nm + "_posterior.log",
                             printgen = 50) )
    pow_p = powerPosterior(mymodel, moves, monitors,
                           out_dir + nm + "_powp.out",
                           cats = n_cats, sampleFreq = 5)
    pow_p.burnin(generations = 200, tuningInterval = 50)
    pow_p.run(generations = n_stone_gen)
    ss = steppingStoneSampler(file = out_dir + nm + "_powp.out",
                              powerColumnName = "power",
                              likelihoodColumnName = "likelihood")
    ps = pathSampler(file = out_dir + nm + "_powp.out",
                     powerColumnName = "power",
                     likelihoodColumnName = "likelihood")
    ml_ss = ss.marginal()
    ml_ps = ps.marginal()
    print(nm + "  stepping-stone: " + ml_ss + "  path sampling: " + ml_ps)
    write(nm, mi, ml_ss, ml_ps, filename = out_dir + "ml_" + nm + ".txt",
          separator = ",")
}
```

```
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
gamma2	log_marginal_likelihood_SS	NA	-4109.40(	NA	NA	NA	info
gamma4	log_marginal_likelihood_SS	NA	-4092.80(	NA	NA	NA	info
gamma8	log_marginal_likelihood_SS	NA	-4091.80(	NA	NA	NA	info
gamma4_free_weights	log_marginal_likelihood_SS	NA	-4095.30(	NA	NA	NA	info
freerate3	log_marginal_likelihood_SS	NA	-4096.40(	NA	NA	NA	info
gamma2 / gamma4 / gamma8	lnBF_gamma4_minus_gamma2	NA	16.604	NA	NA	NA	pass
F81 + discrete Gamma(4), noabsencesites, true time tree	alpha	0.40	0.384	0.322	0.460	1081	pass
F81 + discrete Gamma(4), noabsencesites, true time tree	pi1	0.25	0.253	0.238	0.269	1081	pass
F81 + discrete Gamma(4), noabsencesites, true time tree	clock_rate	0.30	0.259	0.209	0.321	1081	pass
F81 + discrete Gamma(4), noabsencesites, true time tree	lnL_RevBayes_minus_R	0.00	0.000	NA	NA	NA	pass

H.3.19 Models not implemented

Three rows of Table 7.2 have no script here. The lateral-transfer stochastic Dollo model (row 7) needs a likelihood over copying events between contemporaneous lineages that cannot be expressed with RevBayes' CTMC machinery; it belongs to Chapter 8. The infinite-states meaning-slot model (row 9) has no finite state space; the multistate Mk model (script `06`) and the subset model (script `15`) are its closest finite relatives. Catastrophes and the explicit missing-data treatment of the stochastic Dollo model (Ryder and Nicholls, 2011) are also not implemented.

H.4 Clocks, calibrations, tree priors and diversification

This section documents the RevBayes (Höhna et al., 2016) scripts for the models of Chapters 9 and 10, and for the time-tree models of the biologist's toolbox in Chapter 16: strict, relaxed and local clocks, punctuational change, node and tip calibration, sampled ancestors, tree priors, the multispecies coalescent and diversification models. Every script was run on a small data set simulated by `revbayes/data/sim_time.R` with known parameter values (`revbayes/data/time/true_values.csv`), and `revbayes/tests/check_time.R` compares the posterior with the truth.

Conventions. Time is measured in millennia (ky) before the present. Branch lengths are expected numbers of changes per binary cognate character, so clock rates are changes per character per millennium (true value 0.15 in most data sets). Unless stated otherwise the characters are binary cognate-presence characters evolving under a two-state F81 model with an estimated stationary

frequency of presence (true value 0.35), simulated without ascertainment (constant columns are kept, and every model uses `coding="all"`). Ascertainment correction and other lexical substitution models are covered by the scripts for Chapter 7. Scripts are run from `revbayes/`, for example

```
cd revbayes
rb analyses/time/01_strict_clock_calibrations.Rev           # default settings
rb analyses/time/06_punctuational_bursts.Rev --args 15000 punct0
bash analyses/time/run_time.sh                            # all analyses, 2 at a time
Rscript tests/check_time.R                                  # writes
tests/results_time.csv
```

In each verification table, *Truth* is the simulated value, *HPD* the 95% highest posterior density interval, and *ESS* the effective sample size after discarding 25% burn-in (summed over runs where two runs were made). *Status* is `pass` if the truth lies inside the HPD (or a model-specific criterion stated in the notes of `results_time.csv` is met) and ESS is at least 100, `warn` if the criterion is met with ESS below 100, `fail` otherwise, and `info` for descriptive rows. Rows prefixed `info-` belong to deliberately misspecified models, for which missing the truth is the expected outcome.

Five devices recur in the scripts.

- *Calibration densities as observations.* RevBayes has no "calibration prior" object. A density f on a node age `$t = $ tmrca(tree, clade)` is imposed by clamping an observation whose density, as a function of t, equals $f(t)$. For the offset lognormal of Equation (C.12), `obs ~ dnLognormal(mu, sigma, offset=-t)` clamped at $-a$ gives $(t-a) \sim \text{LogNormal}(\mu, \sigma^2)$. For the soft-bounded uniform, the density `dnSoftBoundUniformNormal(min=a-t, max=b-t, sd, p=0.95)` evaluated at 0 equals the soft uniform density of t on $[a, b]$, because the density is translation invariant. As in BEAST, the joint prior is the product of the tree prior and the calibration densities (Section 10.3.2), and script 14 shows the resulting effective prior.
- *Fixed topology.* The relaxed-clock, punctuational, tree-prior and gamma scripts date the true topology (node ages free, starting from the true topology with all ages multiplied by 1.5). This is a standard "dating a given tree" analysis, and it lets branch-specific rates be matched to their true values. Scripts 01, 07, 08 and 11 estimate the topology, starting from a UPGMA tree or a random draw from the prior.
- *Branch indices.* The relaxed-clock scripts write `branch_index.txt`, which maps each RevBayes branch index to its set of descendant languages. `check_time.R` uses it to match `branch_rates[k]` to the simulated rates.
- *Node order.* RevBayes numbers internal nodes so that a parent always has a larger index than its children, with the root at $2n-1$. Looping over branches from index $2n-2$ down to 1 therefore defines a parent's rate before its children's rates. The autocorrelated and random local clocks rely on this.
- *Runtimes.* The runtimes in the tables are wall-clock seconds recorded by `run_time.sh` on a shared 12-core machine with two RevBayes processes at once. On an idle core they are about half as long.

Table H.4: Verification status counts per script (rows of results_time.csv) and longest recorded runtime in seconds.

	pass	warn	fail	info	info-pass	info-warn	info-fail	runtime_s
01_strict_clock_calibrations.Rev	6	0	0	1	0	0	0	306

	pass	warn	fail	info	info-pass	info-warn	info-fail	runtime_s
02_ucln_relaxed_clock.Rev	7	0	0	0	0	0	0	205
03_uce_relaxed_clock.Rev	6	0	0	0	0	0	0	111
04_autocorrelated_lognormal_clock.Rev	5	1	0	0	0	0	0	219
05_random_local_clock.Rev	7	0	0	0	0	0	0	244
06_punctuational_bursts.Rev	6	2	0	2	0	0	0	418
07_tip_dating_serial_bd.Rev	3	0	0	0	0	1	1	238
08_fbd_sampled_ancestors.Rev	5	2	0	0	0	0	0	222
09_clock_model_comparison_ss.Rev	2	0	0	0	0	0	0	207
10_tree_prior_coalescent_vs_yule.Rev	2	0	0	2	2	0	0	70
11_multispecies_coalescent.Rev	7	0	5	0	0	0	0	181
12_episodic_birth_death_skyline.Rev	5	0	0	0	0	0	0	11
13_bisse_state_dependent_diversification.Rev	6	0	1	0	0	0	0	179
14_effective_prior_calibrations.Rev	0	0	0	3	0	0	0	24
15_gamma_rate_variation_dating_bias.Rev	4	0	0	0	1	0	1	187

H.4.1 Strict clock with node calibrations

Script `01_strict_clock_calibrations.Rev` implements the strict clock of Section 9.4 on twelve languages, with two internal calibrations of the kinds described in Section 10.3.1. C1 is a soft-bounded uniform with 95% of its mass in a 1,000-year window around the true age of a young split, as for a historically attested separation. C2 is an offset lognormal on an old split, with a hard minimum 400 years below the true age (the earliest attestation of a daughter) and a lognormal excess with median 600 years. The root is not calibrated. The tree prior is a Yule process with a uniform prior on the root age. Topology, node ages, rate and cognate frequency are estimated in two independent runs started from a UPGMA tree. `check_time.R` also reports the Robinson-Foulds distance of the maximum clade credibility tree to the true topology, and an independent ML estimate of the rate (phangorn tree length on the true topology divided by the true tree length in time).

```
################################################################################
# 01_strict_clock_calibrations.Rev
#
# Model    : strict clock + binary cognate CTMC (2-state F81, estimated
#            cognate frequency) + Yule tree prior + two internal node
#            calibrations of the kinds used for historically attested splits:
#              C1 soft-bounded uniform (95% in [min, max], normal tails)
#              C2 offset lognormal (hard minimum from attestation + LN excess)
#            The root is NOT calibrated: its age is inferred.
```

```
# Book     : Ch. 9 "The strict clock" (clocks-strict);
#            Ch. 10 "Calibration distributions" (divergence-densities)
# Estimates: topology, node ages, root age, clock rate, cognate frequency
# Data     : data/time/clock_strict.nex (12 languages, 600 binary characters)
# Truth    : data/time/true_values.csv (root age 6 ky, rate 0.15/char/ky)
# Runtime  : ~1-2 min (2 runs x 15000 generations)
# Usage    : cd revbayes && rb analyses/time/01_strict_clock_calibrations.Rev
[--args n_gen]
################################################################################

seed(101)
n_gen = 15000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/time/01_strict_clock_calibrations/"

data <- readDiscreteCharacterData("data/time/clock_strict.nex")
taxa <- data.taxa()
moves    = VectorMoves()
monitors = VectorMonitors()

#### tree prior: Yule (birth-death with mu = 0) with an unknown root age ######
root_age ~ dnUniform(0.5, 30.0)
birth_rate ~ dnExponential(1.0)
timetree ~ dnBDP(lambda=birth_rate, mu=0.0, rho=1.0, rootAge=root_age,
                 samplingStrategy="uniform", condition="time", taxa=taxa)

# start from a data-derived UPGMA tree (root age 9 ky), not from the truth
start_tree <- readTrees("data/time/clock_strict_upgma_start.tre")[1]
root_age.setValue(start_tree.rootAge())
timetree.setValue(start_tree)

moves.append( mvScale(birth_rate, lambda=1.0, tune=true, weight=1.0) )
moves.append( mvSlide(root_age, delta=0.5, tune=true, weight=3.0) )
moves.append( mvTreeScale(tree=timetree, rootAge=root_age, delta=0.5, tune=true,
weight=3.0) )
moves.append( mvNodeTimeSlideUniform(timetree, weight=15.0) )
moves.append( mvSubtreeScale(timetree, weight=3.0) )
moves.append( mvNarrow(timetree, weight=6.0) )
moves.append( mvNNI(timetree, weight=3.0) )
moves.append( mvFNPR(timetree, weight=3.0) )

#### node calibrations ########################################################
# defines clade_C1, C1_min, C1_max, C1_sd, clade_C2, C2_offset, C2_mu, C2_sd
source("data/time/calibrations_tree12.Rev")
age_C1 := tmrca(timetree, clade_C1)
age_C2 := tmrca(timetree, clade_C2)

# Soft-bounded uniform density on age_C1. The density is translation
```

```
# invariant, so evaluating an observation of 0 under bounds shifted by
# -age_C1 gives exactly f(age_C1; C1_min, C1_max).
obs_C1 ~ dnSoftBoundUniformNormal(min=C1_min - age_C1, max=C1_max - age_C1, sd=C1_
sd, p=0.95)
obs_C1.clamp(0.0)

# Offset lognormal on age_C2 (Equation divergence-lognormal):
# (age_C2 - C2_offset) ~ LogNormal(C2_mu, C2_sd). With offset = -age_C2 the
# clamped value -C2_offset gives x - offset = age_C2 - C2_offset.
obs_C2 ~ dnLognormal(mean=C2_mu, sd=C2_sd, offset=-age_C2)
obs_C2.clamp(-C2_offset)

#### strict clock ##################################################################
clock_rate ~ dnExponential(2.0)          # prior mean 0.5 changes/char/ky
moves.append( mvScale(clock_rate, lambda=0.5, tune=true, weight=5.0) )
# joint move: stretch the tree, shrink the rate (keeps branch lengths ~ fixed)
up_down = mvUpDownScale(lambda=0.3, tune=true, weight=5.0)
up_down.addVariable(root_age, up=true)
up_down.addVariable(timetree, up=true)
up_down.addVariable(clock_rate, up=false)
moves.append( up_down )

#### binary cognate substitution model #############################################
pi ~ dnDirichlet(v(1, 1))
pi1 := pi[2]                             # stationary frequency of "cognate present"
Q := fnF81(pi)
moves.append( mvBetaSimplex(pi, alpha=20, tune=true, weight=2.0) )

seq ~ dnPhyloCTMC(tree=timetree, Q=Q, branchRates=clock_rate, type="Standard")
seq.clamp(data)

#### MCMC ##########################################################################
mymodel = model(timetree)
monitors.append( mnModel(filename=out_dir + "model.log", printgen=10) )
monitors.append( mnFile(timetree, filename=out_dir + "trees.trees", printgen=50) )
monitors.append( mnScreen(printgen=5000, root_age, clock_rate, age_C1, age_C2) )

mymcmc = mcmc(mymodel, moves, monitors, nruns=2, combine="mixed")
mymcmc.run(generations=n_gen, tuningInterval=200)

# MCC tree of the combined sample (first 25% discarded)
trace = readTreeTrace(out_dir + "trees.trees", treetype="clock", burnin=0.25)
mcc = mccTree(trace, file=out_dir + "mcc.tre")

q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
strict clock + node calibrations	root_age	6.00	6.817	4.015	10.312	1541	pass
strict clock + node calibrations	clock_rate	0.15	0.133	0.098	0.168	1285	pass
strict clock + node calibrations	age_C1	1.46	1.548	1.105	1.973	1248	pass
strict clock + node calibrations	age_C2	4.13	4.375	3.896	5.055	2352	pass
strict clock + node calibrations	pi1	0.35	0.347	0.328	0.365	2335	pass
strict clock + node calibrations	RF_distance_MCC_vs_true	0.00	2.000	NA	NA	NA	pass
independent ML (phangorn)	clock_rate_ML_check	0.15	0.147	NA	NA	NA	info

All parameters are recovered, including the uncalibrated root (posterior mean 6.8 ky, HPD 4.0 to 10.3 ky; truth 6.0). The maximum clade credibility tree differs from the true topology by one split, which lies across an internal branch only 110 years long. The independent ML estimate of the rate (0.147) agrees with the posterior.

H.4.2 Uncorrelated lognormal relaxed clock

Script `02_ucln_relaxed_clock.Rev` implements Equation (C.9) (Drummond et al., 2006): each of the 22 branch rates is an independent lognormal draw with mean $\bar{r}$ and log-scale standard deviation σ, so the multipliers have mean one. The data were simulated with $\bar{r} = 0.15$ and $\sigma = 0.5$ (CV 0.53). Rates are sampled as continuous values rather than discretized categories. Mixing is helped by `mvRateAgeBetaShift`, which moves a node age and rescales the rates of the adjacent branches so that their lengths in changes stay the same. This is the joint rate-age proposal at the heart of the optimized relaxed clock (Section 9.5.4). An up-down scale move stretches the tree and shrinks all rates together. Besides $\bar{r}$, σ and the root age, the check reports what fraction of the true branch rates fall inside their branch-wise 95% HPD, and the correlation between posterior-mean and true log rates.

```
################################################################################
# 02_ucln_relaxed_clock.Rev
#
# Model    : uncorrelated lognormal (UCLN) relaxed clock (Drummond et al. 2006):
#              r_j = rbar * eps_j,  ln eps_j ~ N(-sigma^2/2, sigma^2)
#            (Equation clocks-ucln), binary cognate CTMC (2-state F81),
#            Yule tree prior, node calibrations C1 (soft uniform) and
#            C2 (offset lognormal) as in script 01. Topology fixed to the
#            true topology (a dating analysis on a given tree); node ages free.
# Book     : Ch. 9 "Uncorrelated relaxed clocks" (clocks-ucln)
# Estimates: mean rate rbar, sigma (and CV), 22 branch rates, node ages, root age
# Data     : data/time/clock_ucln.nex (12 languages, 600 characters)
# Runtime  : ~1-2 min
```

```
# Usage    : cd revbayes && rb analyses/time/02_ucln_relaxed_clock.Rev [--args n_gen]
################################################################################

seed(202)
n_gen = 15000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/time/02_ucln_relaxed_clock/"

data <- readDiscreteCharacterData("data/time/clock_ucln.nex")
taxa <- data.taxa()
n_taxa <- taxa.size()
n_branches <- 2 * n_taxa - 2
moves    = VectorMoves()
monitors = VectorMonitors()

#### tree prior; fixed topology (true topology, ages x1.5 as starting values) ##
root_age ~ dnUniform(0.5, 30.0)
birth_rate ~ dnExponential(1.0)
timetree ~ dnBDP(lambda=birth_rate, mu=0.0, rho=1.0, rootAge=root_age, taxa=taxa)
start_tree <- readTrees("data/time/tree12_start.tre")[1]
root_age.setValue(start_tree.rootAge())
timetree.setValue(start_tree)

moves.append( mvScale(birth_rate, lambda=1.0, tune=true, weight=1.0) )
moves.append( mvSlide(root_age, delta=0.5, tune=true, weight=3.0) )
moves.append( mvTreeScale(tree=timetree, rootAge=root_age, delta=0.5, tune=true,
weight=3.0) )
moves.append( mvNodeTimeSlideUniform(timetree, weight=10.0) )
moves.append( mvSubtreeScale(timetree, weight=5.0) )

# branch index -> descendant taxa (used by tests/check_time.R)
for (i in 1:n_branches) {
  write(i, timetree.getDescendantTaxa(i), filename=out_dir + "branch_index.txt",
        append=(i > 1), separator="\t")
  write("\n", filename=out_dir + "branch_index.txt", append=true)
}

#### calibrations (see script 01 for the construction) ########################
source("data/time/calibrations_tree12.Rev")
age_C1 := tmrca(timetree, clade_C1)
age_C2 := tmrca(timetree, clade_C2)
obs_C1 ~ dnSoftBoundUniformNormal(min=C1_min - age_C1, max=C1_max - age_C1, sd=C1_
sd, p=0.95)
obs_C1.clamp(0.0)
obs_C2 ~ dnLognormal(mean=C2_mu, sd=C2_sd, offset=-age_C2)
obs_C2.clamp(-C2_offset)

#### UCLN relaxed clock ########################################################
```

```
mean_rate ~ dnExponential(2.0)
sigma ~ dnExponential(1.0)
cv := sqrt(abs(exp(sigma * sigma) - 1.0))
moves.append( mvScale(mean_rate, lambda=0.5, tune=true, weight=5.0) )
moves.append( mvScale(sigma, lambda=0.5, tune=true, weight=5.0) )

for (j in 1:n_branches) {
  branch_rates[j] ~ dnLognormal(ln(mean_rate) - sigma * sigma / 2.0, sigma)
  moves.append( mvScale(branch_rates[j], lambda=0.5, tune=true, weight=1.0) )
}
realised_mean_rate := mean(branch_rates)
# joint rate-age proposals (the idea behind the optimized relaxed clock):
# move a node and change adjacent branch rates so branch lengths are preserved
moves.append( mvRateAgeBetaShift(tree=timetree, rates=branch_rates, tune=true,
weight=10.0) )
up_down = mvUpDownScale(lambda=0.3, tune=true, weight=5.0)
up_down.addVariable(root_age, up=true)
up_down.addVariable(timetree, up=true)
up_down.addVariable(mean_rate, up=false)
up_down.addVariable(branch_rates, up=false)
moves.append( up_down )

#### substitution model and likelihood ########################################
pi ~ dnDirichlet(v(1, 1))
pi1 := pi[2]
Q := fnF81(pi)
moves.append( mvBetaSimplex(pi, alpha=20, tune=true, weight=2.0) )
seq ~ dnPhyloCTMC(tree=timetree, Q=Q, branchRates=branch_rates, type="Standard")
seq.clamp(data)

#### MCMC #####################################################################
mymodel = model(timetree)
monitors.append( mnModel(filename=out_dir + "model.log", printgen=10) )
monitors.append( mnFile(timetree, filename=out_dir + "trees.trees", printgen=100) )
monitors.append( mnScreen(printgen=5000, root_age, mean_rate, sigma) )
mymcmc = mcmc(mymodel, moves, monitors)
mymcmc.run(generations=n_gen, tuningInterval=200)

q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
UCLN relaxed clock	root_age	6.000	5.906	4.033	8.942	1051	pass
UCLN relaxed clock	mean_rate	0.150	0.193	0.084	0.328	521	pass
UCLN relaxed clock	sigma	0.500	0.685	0.269	1.218	262	pass

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
UCLN relaxed clock	realised_mean_rate	0.156	0.183	0.113	0.277	873	pass
UCLN relaxed clock	age_C1	1.460	1.510	0.846	2.276	215	pass
UCLN relaxed clock	age_C2	4.130	4.343	3.884	4.925	1126	pass
UCLN relaxed clock	branch_rates_ coverage	0.950	0.955	NA	NA	198	pass

All parameters are recovered. With 22 branches, σ is only weakly identified (HPD 0.27 to 1.22), as expected. Of the 22 true branch rates, 21 lie inside their branch-wise HPDs.

H.4.3 Uncorrelated exponential relaxed clock

Script `03_uce_relaxed_clock.Rev` is the UCED variant of Section 9.5.1: branch rates are drawn from an exponential distribution with mean $\bar{r}$, so the coefficient of variation is fixed at one and many branches get very low rates.

```
################################################################################
# 03_uce_relaxed_clock.Rev
#
# Model    : uncorrelated exponential (UCE/UCED) relaxed clock (Drummond et
#            al. 2006): r_j ~ Exponential(mean = rbar), i.e. CV fixed at 1,
#            binary cognate CTMC (2-state F81),
#            Yule tree prior, node calibrations C1 (soft uniform) and
#            C2 (offset lognormal) as in script 01. Topology fixed to the
#            true topology (a dating analysis on a given tree); node ages free.
# Book     : Ch. 9 "Uncorrelated relaxed clocks" (clocks-ucln), UCED variant
# Estimates: mean rate rbar, 22 branch rates, node ages, root age
# Data     : data/time/clock_uce.nex (12 languages, 600 characters)
# Runtime  : ~1-2 min
# Usage    : cd revbayes && rb analyses/time/03_uce_relaxed_clock.Rev [--args n_gen]
################################################################################

seed(303)
n_gen = 15000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/time/03_uce_relaxed_clock/"

data <- readDiscreteCharacterData("data/time/clock_uce.nex")
taxa <- data.taxa()
n_taxa <- taxa.size()
n_branches <- 2 * n_taxa - 2
moves    = VectorMoves()
monitors = VectorMonitors()
```

```
#### tree prior; fixed topology (true topology, ages x1.5 as starting values) ##
root_age ~ dnUniform(0.5, 30.0)
birth_rate ~ dnExponential(1.0)
timetree ~ dnBDP(lambda=birth_rate, mu=0.0, rho=1.0, rootAge=root_age, taxa=taxa)
start_tree <- readTrees("data/time/tree12_start.tre")[1]
root_age.setValue(start_tree.rootAge())
timetree.setValue(start_tree)

moves.append( mvScale(birth_rate, lambda=1.0, tune=true, weight=1.0) )
moves.append( mvSlide(root_age, delta=0.5, tune=true, weight=3.0) )
moves.append( mvTreeScale(tree=timetree, rootAge=root_age, delta=0.5, tune=true,
weight=3.0) )
moves.append( mvNodeTimeSlideUniform(timetree, weight=10.0) )
moves.append( mvSubtreeScale(timetree, weight=5.0) )

# branch index -> descendant taxa (used by tests/check_time.R)
for (i in 1:n_branches) {
  write(i, timetree.getDescendantTaxa(i), filename=out_dir + "branch_index.txt",
        append=(i > 1), separator="\t")
  write("\n", filename=out_dir + "branch_index.txt", append=true)
}

#### calibrations (see script 01 for the construction) ########################
source("data/time/calibrations_tree12.Rev")
age_C1 := tmrca(timetree, clade_C1)
age_C2 := tmrca(timetree, clade_C2)
obs_C1 ~ dnSoftBoundUniformNormal(min=C1_min - age_C1, max=C1_max - age_C1, sd=C1_
sd, p=0.95)
obs_C1.clamp(0.0)
obs_C2 ~ dnLognormal(mean=C2_mu, sd=C2_sd, offset=-age_C2)
obs_C2.clamp(-C2_offset)

#### UCE relaxed clock ########################################################
mean_rate ~ dnExponential(2.0)
moves.append( mvScale(mean_rate, lambda=0.5, tune=true, weight=5.0) )

for (j in 1:n_branches) {
  branch_rates[j] ~ dnExponential(1.0 / mean_rate)
  moves.append( mvScale(branch_rates[j], lambda=0.5, tune=true, weight=1.0) )
}
realised_mean_rate := mean(branch_rates)
# joint rate-age proposals (the idea behind the optimized relaxed clock):
# move a node and change adjacent branch rates so branch lengths are preserved
moves.append( mvRateAgeBetaShift(tree=timetree, rates=branch_rates, tune=true,
weight=10.0) )
up_down = mvUpDownScale(lambda=0.3, tune=true, weight=5.0)
up_down.addVariable(root_age, up=true)
up_down.addVariable(timetree, up=true)
```

```
up_down.addVariable(mean_rate, up=false)
up_down.addVariable(branch_rates, up=false)
moves.append( up_down )

#### substitution model and likelihood ##################################
pi ~ dnDirichlet(v(1, 1))
pi1 := pi[2]
Q := fnF81(pi)
moves.append( mvBetaSimplex(pi, alpha=20, tune=true, weight=2.0) )
seq ~ dnPhyloCTMC(tree=timetree, Q=Q, branchRates=branch_rates, type="Standard")
seq.clamp(data)

#### MCMC ###############################################################
mymodel = model(timetree)
monitors.append( mnModel(filename=out_dir + "model.log", printgen=10) )
monitors.append( mnFile(timetree, filename=out_dir + "trees.trees", printgen=100) )
monitors.append( mnScreen(printgen=5000, root_age, mean_rate) )
mymcmc = mcmc(mymodel, moves, monitors)
mymcmc.run(generations=n_gen, tuningInterval=200)

q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
UCE relaxed clock	root_age	6.00	6.272	3.960	9.636	735	pass
UCE relaxed clock	mean_rate	0.15	0.165	0.070	0.279	540	pass
UCE relaxed clock	realised_mean_rate	0.17	0.153	0.097	0.241	370	pass
UCE relaxed clock	age_C1	1.46	1.808	1.113	2.432	607	pass
UCE relaxed clock	age_C2	4.13	4.303	3.880	4.861	1126	pass
UCE relaxed clock	branch_rates_ coverage	0.95	0.955	NA	NA	398	pass

H.4.4 Autocorrelated lognormal relaxed clock

Script `04_autocorrelated_lognormal_clock.Rev` implements the geometric Brownian motion model of Equation (C.10) (Thorne et al., 1998; Kishino et al., 2001). RevBayes 1.4.1 has no built-in distribution for autocorrelated rates, so the model is written out node by node. Each node's log rate is normal around its parent's log rate minus $vt/2$, with variance vt, where t is the branch duration, a deterministic function of the moving node ages (`timetree.branchLength(k)`). Each branch rate is the exponential of the mean of the log rates at its two ends, exactly as in the R simulation of Section 9.5.6. Besides single-node slides, an `mvUpDownSlide` move shifts all log rates together. The data were simulated with root rate 0.15 and $\nu = 0.15$ per millennium.

```
################################################################################
# 04_autocorrelated_lognormal_clock.Rev
#
# Model    : autocorrelated lognormal relaxed clock = geometric Brownian motion
#            of the log rate along the tree (Thorne et al. 1998; Kishino et al.
#            2001), Equation clocks-gbm of the book:
#              ln r_child ~ N(ln r_parent - nu*t/2, nu*t)
#            Branch rate = exp(mean of the log rates at its two ends).
#            Binary cognate CTMC (2-state F81), Yule prior, calibrations C1/C2.
#            Topology fixed to the true topology; node ages free.
#            RevBayes 1.4.1 has no built-in ACLN distribution: the model is
#            written node by node with dnNormal on the node log-rates, with the
#            variance a deterministic function of the (moving) branch durations.
# Book     : Ch. 9 "Autocorrelated relaxed clocks" (clocks-autocorrelated)
# Estimates: root rate, nu, node rates, node ages, root age
# Data     : data/time/clock_acln.nex (12 languages, 600 characters)
# Runtime  : ~2-3 min
# Usage    : cd revbayes && rb analyses/time/04_autocorrelated_lognormal_clock.Rev
[--args n_gen]
################################################################################

seed(404)
n_gen = 30000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/time/04_autocorrelated_lognormal_clock/"

data <- readDiscreteCharacterData("data/time/clock_acln.nex")
taxa <- data.taxa()
n_taxa <- taxa.size()
n_branches <- 2 * n_taxa - 2
moves    = VectorMoves()
monitors = VectorMonitors()

#### tree prior; fixed topology ################################################
root_age ~ dnUniform(0.5, 30.0)
birth_rate ~ dnExponential(1.0)
timetree ~ dnBDP(lambda=birth_rate, mu=0.0, rho=1.0, rootAge=root_age, taxa=taxa)
start_tree <- readTrees("data/time/tree12_start.tre")[1]
root_age.setValue(start_tree.rootAge())
timetree.setValue(start_tree)
root_index = 2 * n_taxa - 1

moves.append( mvScale(birth_rate, lambda=1.0, tune=true, weight=1.0) )
moves.append( mvSlide(root_age, delta=0.5, tune=true, weight=3.0) )
moves.append( mvTreeScale(tree=timetree, rootAge=root_age, delta=0.5, tune=true,
weight=3.0) )
moves.append( mvNodeTimeSlideUniform(timetree, weight=10.0) )
moves.append( mvSubtreeScale(timetree, weight=5.0) )
```

```
for (i in 1:n_branches) {
  write(i, timetree.getDescendantTaxa(i), filename=out_dir + "branch_index.txt",
        append=(i > 1), separator="\t")
  write("\n", filename=out_dir + "branch_index.txt", append=true)
}

#### calibrations ##################################################################
source("data/time/calibrations_tree12.Rev")
age_C1 := tmrca(timetree, clade_C1)
age_C2 := tmrca(timetree, clade_C2)
obs_C1 ~ dnSoftBoundUniformNormal(min=C1_min - age_C1, max=C1_max - age_C1, sd=C1_
sd, p=0.95)
obs_C1.clamp(0.0)
obs_C2 ~ dnLognormal(mean=C2_mu, sd=C2_sd, offset=-age_C2)
obs_C2.clamp(-C2_offset)

#### autocorrelated (GBM) clock ###################################################
ln_root_rate ~ dnNormal(ln(0.2), 2.0)
root_rate := exp(ln_root_rate)
nu ~ dnExponential(5.0)                          # prior mean 0.2 per ky
moves.append( mvSlide(ln_root_rate, delta=0.2, tune=true, weight=5.0) )
moves.append( mvScale(nu, lambda=0.5, tune=true, weight=5.0) )

# RevBayes numbers internal nodes so that a parent always has a larger index
# than its children (root = 2n-1); visiting nodes from high to low index
# therefore defines every parent log-rate before its children.
for (k in n_branches:1) {
  p = timetree.parent(k)
  bl[k] := timetree.branchLength(k)
  if (p == root_index) {
    node_lnrate[k] ~ dnNormal(ln_root_rate - nu * bl[k] / 2.0, sqrt(nu * bl[k]))
    branch_rates[k] := exp((node_lnrate[k] + ln_root_rate) / 2.0)
  } else {
    node_lnrate[k] ~ dnNormal(node_lnrate[p] - nu * bl[k] / 2.0, sqrt(nu * bl[k]))
    branch_rates[k] := exp((node_lnrate[k] + node_lnrate[p]) / 2.0)
  }
  moves.append( mvSlide(node_lnrate[k], delta=0.3, tune=true, weight=1.0) )
}
mean_branch_rate := mean(branch_rates)
# shift all log rates together (changes the overall rate level in one step)
shift_all = mvUpDownSlide(delta=0.1, tune=true, weight=5.0)
shift_all.addVariable(ln_root_rate, up=true)
for (k in 1:n_branches) shift_all.addVariable(node_lnrate[k], up=true)
moves.append( shift_all )

#### substitution model and likelihood ############################################
pi ~ dnDirichlet(v(1, 1))
```

```
pi1 := pi[2]
Q := fnF81(pi)
moves.append( mvBetaSimplex(pi, alpha=20, tune=true, weight=2.0) )
seq ~ dnPhyloCTMC(tree=timetree, Q=Q, branchRates=branch_rates, type="Standard")
seq.clamp(data)

#### MCMC ######################################################################
mymodel = model(timetree)
monitors.append( mnModel(filename=out_dir + "model.log", printgen=10) )
monitors.append( mnFile(timetree, filename=out_dir + "trees.trees", printgen=100) )
monitors.append( mnScreen(printgen=5000, root_age, root_rate, nu) )
mymcmc = mcmc(mymodel, moves, monitors)
mymcmc.run(generations=n_gen, tuningInterval=200)

q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
autocorrelated lognormal (GBM) clock	root_age	6.00	5.707	4.107	8.466	1130	pass
autocorrelated lognormal (GBM) clock	root_rate	0.15	0.149	0.054	0.276	640	pass
autocorrelated lognormal (GBM) clock	nu	0.15	0.144	0.018	0.346	309	pass
autocorrelated lognormal (GBM) clock	age_C1	1.46	1.706	1.051	2.299	136	pass
autocorrelated lognormal (GBM) clock	age_C2	4.13	4.356	3.875	4.939	1551	pass
autocorrelated lognormal (GBM) clock	branch_rates_ coverage	0.95	1.000	NA	NA	57	warn

All parameters are recovered. The branch-rate row is `warn` only because the slowest-mixing node rate has ESS 57.

H.4.5 Random local clock

Script `05_random_local_clock.Rev` implements the random local clock of Section 9.5.3 (Drummond and Suchard, 2010). RevBayes 1.4.1 has no dedicated distribution for it, so the script uses the reversible-jump indicator formulation that the book describes. Each branch has a multiplier that is exactly 1 (no shift) with prior probability $1 - q$, or a LogNormal(0, 1) draw (a shift) with probability q, via `dnReversibleJumpMixture` with `mvRJSwitch`. A branch's rate is its parent branch's rate times its own multiplier, so a shift is inherited by the whole clade below it until the next shift. Setting $q = 1 - 0.5^{1/22}$ makes the prior probability of no shift anywhere one half, matching the Poisson($\ln 2$) prior of Drummond and Suchard. The data contain a single threefold shift on the stem of a four-language clade. The checks require a posterior shift probability above 0.5 on that branch and below 0.5 on every other branch.

```
################################################################################
# 05_random_local_clock.Rev
#
# Model    : random local clock (Drummond & Suchard 2010). Each branch j has a
#            rate multiplier m_j that is exactly 1 (no shift) with probability
#            1-q or drawn from LogNormal(0, 1) (a shift) with probability q,
#            via a reversible-jump mixture. A branch inherits the rate of its
#            parent branch times its own multiplier, so a shift applies to the
#            whole subtree until the next shift:
#              r_j = r_parent(j) * m_j,  r for the root's children = base rate * m_j
#            q = 1 - 0.5^(1/22) gives P(no shift anywhere) = 0.5, as in the
#            Poisson(ln 2) prior on the number of shifts of Drummond & Suchard.
#            Binary cognate CTMC, Yule prior, calibrations C1/C2, fixed topology.
#            RevBayes 1.4.1 has no dedicated RLC distribution; this is the
#            standard RJ-indicator formulation built from dnReversibleJumpMixture.
# Book     : Ch. 9 "Local clocks and random local clocks" (clocks-local)
# Estimates: base rate, number and placement of shifts (posterior probability
#            per branch), shift multiplier, node ages, root age
# Data     : data/time/clock_rlc.nex (one x3 shift on the stem of a 4-language clade)
# Runtime  : ~2 min
# Usage    : cd revbayes && rb analyses/time/05_random_local_clock.Rev [--args n_gen]
################################################################################

seed(505)
n_gen = 15000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/time/05_random_local_clock/"

data <- readDiscreteCharacterData("data/time/clock_rlc.nex")
taxa <- data.taxa()
n_taxa <- taxa.size()
n_branches <- 2 * n_taxa - 2
moves    = VectorMoves()
monitors = VectorMonitors()

#### tree prior; fixed topology ################################################
root_age ~ dnUniform(0.5, 30.0)
birth_rate ~ dnExponential(1.0)
timetree ~ dnBDP(lambda=birth_rate, mu=0.0, rho=1.0, rootAge=root_age, taxa=taxa)
start_tree <- readTrees("data/time/tree12_start.tre")[1]
root_age.setValue(start_tree.rootAge())
timetree.setValue(start_tree)
root_index = 2 * n_taxa - 1

moves.append( mvScale(birth_rate, lambda=1.0, tune=true, weight=1.0) )
moves.append( mvSlide(root_age, delta=0.5, tune=true, weight=3.0) )
moves.append( mvTreeScale(tree=timetree, rootAge=root_age, delta=0.5, tune=true,
weight=3.0) )
```

```
moves.append( mvNodeTimeSlideUniform(timetree, weight=10.0) )
moves.append( mvSubtreeScale(timetree, weight=5.0) )

for (i in 1:n_branches) {
  write(i, timetree.getDescendantTaxa(i), filename=out_dir + "branch_index.txt",
        append=(i > 1), separator="\t")
  write("\n", filename=out_dir + "branch_index.txt", append=true)
}

#### calibrations ##########################################################
source("data/time/calibrations_tree12.Rev")
age_C1 := tmrca(timetree, clade_C1)
age_C2 := tmrca(timetree, clade_C2)
obs_C1 ~ dnSoftBoundUniformNormal(min=C1_min - age_C1, max=C1_max - age_C1, sd=C1_
sd, p=0.95)
obs_C1.clamp(0.0)
obs_C2 ~ dnLognormal(mean=C2_mu, sd=C2_sd, offset=-age_C2)
obs_C2.clamp(-C2_offset)

#### random local clock ####################################################
base_rate ~ dnExponential(2.0)
moves.append( mvScale(base_rate, lambda=0.5, tune=true, weight=5.0) )
q_shift <- 1.0 - 0.5^(1.0 / n_branches)

for (k in n_branches:1) {
  p = timetree.parent(k)
  rate_mult[k] ~ dnReversibleJumpMixture(constantValue=1.0,
                    baseDistribution=dnLognormal(0.0, 1.0), p=1.0 - q_shift)
  moves.append( mvRJSwitch(rate_mult[k], weight=2.0) )
  moves.append( mvScale(rate_mult[k], lambda=0.5, tune=true, weight=1.0) )
  shift[k] := ifelse(rate_mult[k] == 1.0, 0, 1)
  if (p == root_index) {
    branch_rates[k] := base_rate * rate_mult[k]
  } else {
    branch_rates[k] := branch_rates[p] * rate_mult[k]
  }
}
n_shifts := sum(shift)

#### substitution model and likelihood #####################################
pi ~ dnDirichlet(v(1, 1))
pi1 := pi[2]
Q := fnF81(pi)
moves.append( mvBetaSimplex(pi, alpha=20, tune=true, weight=2.0) )
seq ~ dnPhyloCTMC(tree=timetree, Q=Q, branchRates=branch_rates, type="Standard")
seq.clamp(data)

#### MCMC ##################################################################
```

```
mymodel = model(timetree)
monitors.append( mnModel(filename=out_dir + "model.log", printgen=10) )
monitors.append( mnFile(timetree, filename=out_dir + "trees.trees", printgen=100) )
monitors.append( mnScreen(printgen=5000, root_age, base_rate, n_shifts) )
mymcmc = mcmc(mymodel, moves, monitors)
mymcmc.run(generations=n_gen, tuningInterval=200)

q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
random local clock (RJ shifts)	root_age	6.00	5.261	4.052	7.128	792	pass
random local clock (RJ shifts)	base_rate	0.15	0.152	0.111	0.193	227	pass
random local clock (RJ shifts)	n_shifts	1.00	1.668	1.000	3.000	215	pass
random local clock (RJ shifts)	PP_shift_on_true_branch	1.00	1.000	NA	NA	NA	pass
random local clock (RJ shifts)	max_PP_shift_other_branches	0.00	0.211	NA	NA	NA	pass
random local clock (RJ shifts)	shift_multiplier	3.00	3.526	2.325	5.088	126	pass
random local clock (RJ shifts)	shifted_clade_rate	0.45	0.542	0.309	0.810	119	pass

The shift is found on the correct branch with posterior probability 1, and no other branch has a shift probability above 0.22.

H.4.6 Punctuational bursts at splits

Script `06_punctuational_bursts.Rev` implements the punctuational model of Section 9.8 (Atkinson et al., 2008) as a generative model rather than a regression. Every branch starts at a split and receives a burst of β expected changes per character on top of gradual change, so that $b_j = rt_j + \beta$ and the expected root-to-tip path is $rT + \beta n_i$ (Equation (9.5)). On a time tree this is written as the branch rate $r + \beta/t_j$. The parameter β has a reversible-jump prior that puts probability 0.5 on $\beta = 0$ (purely gradual change) and otherwise draws β from an exponential distribution with mean 0.05, so the posterior probability that $\beta > 0$ is directly a Bayes factor for punctuational change. The script runs on two data sets on the same 16-language tree: one with $\beta = 0.05$ and a negative control with $\beta = 0$. The tree is the most unbalanced of 200 Yule trees, with 1 to 11 splits between the root and a tip. It was chosen this way because β is identified only through differences in node counts among tips: bursts that were the same on every root-to-tip path could be absorbed by moving the node ages. A floor of one year on branch durations keeps β/t_j finite when a node slides onto its child. As an independent check, `check_time.R` regresses the ML (phangorn) branch lengths on the true topology against the true durations, $b = rt + \beta k$, using the short branches whose ML lengths are not saturated.

```
################################################################################
# 06_punctuational_bursts.Rev
#
# Model    : punctuational (speciational) change, after Atkinson et al. (2008):
#            every branch starts at a split and receives a burst of beta
#            expected changes per character in addition to gradual change,
#              b_j = r * t_j + beta,
#            so the expected root-to-tip path is r*T + beta*n_i (Equation
#            clocks-punct). Implemented with branch rates r + beta/t_j on a
#            time tree. beta has a reversible-jump prior: beta = 0 (purely
#            gradual) with probability 0.5, else beta ~ Exponential(mean 0.05);
#            the posterior probability of beta > 0 is then a Bayes factor
#            (prior odds 1). Strict clock otherwise; calibrations C1/C2;
#            fixed (true) topology of 16 languages; the tree is unbalanced
#            (1 to 11 splits between the root and a tip) to give the test power.
# Book     : Ch. 9 "Punctuational bursts at language splits" (clocks-punctuational)
# Estimates: beta, P(beta > 0 | data), clock rate r, node ages, root age
# Data     : data/time/punct.nex (beta = 0.05) by default;
#            data/time/punct0.nex (beta = 0, negative control) with --args n_gen
punct0
# Runtime  : ~1-2 min
# Usage    : cd revbayes && rb analyses/time/06_punctuational_bursts.Rev [--args n_
gen dataset]
################################################################################

seed(606)
n_gen = 15000
dataset = "punct"
if (args.size() > 1) n_gen = int(args[2])
if (args.size() > 2) dataset = args[3]
out_dir = "output/time/06_punctuational_bursts/" + dataset + "_"

data <- readDiscreteCharacterData("data/time/" + dataset + ".nex")
taxa <- data.taxa()
n_taxa <- taxa.size()
n_branches <- 2 * n_taxa - 2
moves    = VectorMoves()
monitors = VectorMonitors()

#### tree prior; fixed topology ################################################
root_age ~ dnUniform(0.5, 30.0)
birth_rate ~ dnExponential(1.0)
timetree ~ dnBDP(lambda=birth_rate, mu=0.0, rho=1.0, rootAge=root_age, taxa=taxa)
start_tree <- readTrees("data/time/tree16_start.tre")[1]
root_age.setValue(start_tree.rootAge())
timetree.setValue(start_tree)

moves.append( mvScale(birth_rate, lambda=1.0, tune=true, weight=1.0) )
```

```
moves.append( mvSlide(root_age, delta=0.5, tune=true, weight=3.0) )
moves.append( mvTreeScale(tree=timetree, rootAge=root_age, delta=0.5, tune=true,
weight=3.0) )
moves.append( mvNodeTimeSlideUniform(timetree, weight=15.0) )
moves.append( mvSubtreeScale(timetree, weight=5.0) )

#### calibrations ##################################################################
source("data/time/calibrations_tree16.Rev")
age_C1 := tmrca(timetree, clade_C1)
age_C2 := tmrca(timetree, clade_C2)
obs_C1 ~ dnSoftBoundUniformNormal(min=C1_min - age_C1, max=C1_max - age_C1, sd=C1_
sd, p=0.95)
obs_C1.clamp(0.0)
obs_C2 ~ dnLognormal(mean=C2_mu, sd=C2_sd, offset=-age_C2)
obs_C2.clamp(-C2_offset)

#### gradual clock + punctuational bursts ##########################################
clock_rate ~ dnExponential(2.0)
moves.append( mvScale(clock_rate, lambda=0.5, tune=true, weight=5.0) )

beta ~ dnReversibleJumpMixture(constantValue=0.0,
baseDistribution=dnExponential(20.0), p=0.5)
moves.append( mvRJSwitch(beta, weight=5.0) )
moves.append( mvScale(beta, lambda=0.5, tune=true, weight=5.0) )
burst_present := ifelse(beta > 0.0, 1, 0)

# The burst is spread over the branch as a rate beta/t_j. A floor of one year
# (0.001 ky) on t_j keeps the rate finite when a node slides onto its child;
# for any branch longer than a few years b_j = r*t_j + beta to within 0.1%.
for (k in 1:n_branches) {
  bl[k] := timetree.branchLength(k)
  branch_rates[k] := clock_rate + beta / (bl[k] + 0.001)
}
# total expected change per lineage from bursts, averaged over tips
tree_length_subs := clock_rate * timetree.treeLength() + beta * n_branches
burst_fraction := beta * n_branches / tree_length_subs

#### substitution model and likelihood #############################################
pi ~ dnDirichlet(v(1, 1))
pi1 := pi[2]
Q := fnF81(pi)
moves.append( mvBetaSimplex(pi, alpha=20, tune=true, weight=2.0) )
seq ~ dnPhyloCTMC(tree=timetree, Q=Q, branchRates=branch_rates, type="Standard")
seq.clamp(data)

#### MCMC ##########################################################################
mymodel = model(timetree)
monitors.append( mnModel(filename=out_dir + "model.log", printgen=10) )
```

```
monitors.append( mnScreen(printgen=5000, root_age, clock_rate, beta) )
mymcmc = mcmc(mymodel, moves, monitors)
mymcmc.run(generations=n_gen, tuningInterval=200)

q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
punctuational bursts (data: punct)	beta	0.05	0.052	0.037	0.065	294	pass
punctuational bursts (data: punct)	clock_rate	0.15	0.128	0.095	0.165	318	pass
punctuational bursts (data: punct)	root_age	6.00	6.325	4.447	8.660	367	pass
punctuational bursts (data: punct)	P(beta > 0)	1.00	1.000	NA	NA	NA	pass
independent ML regression (data: punct)	beta_ML_ regression	0.05	0.103	NA	NA	NA	info
punctuational bursts (data: punct0)	beta	0.00	0.000	0.000	0.004	27	warn
punctuational bursts (data: punct0)	clock_rate	0.15	0.172	0.120	0.226	63	warn
punctuational bursts (data: punct0)	root_age	6.00	6.462	4.370	9.626	267	pass
punctuational bursts (data: punct0)	P(beta > 0)	0.00	0.123	NA	NA	NA	pass
independent ML regression (data: punct0)	beta_ML_ regression	0.00	0.009	NA	NA	NA	info

With bursts in the data, β is estimated at 0.052 (HPD 0.037 to 0.065; truth 0.05) and $P(\beta > 0) = 1$. In the negative control $P(\beta > 0) = 0.12$, a Bayes factor of 0.14 in favour of purely gradual change. The ESS of β is low in the control because the sampled values are almost all exactly 0.

H.4.7 Tip-dating with ancient languages

Script `07_tip_dating_serial_bd.Rev` implements tip-dating (Section 10.5) with nine modern and three ancient languages attested 3.0, 2.2 and 1.4 ky ago. The ages come from `tipdate_taxa.tsv` through `readTaxonData`. The tree prior is the serially sampled birth-death process `dnBDSTP` with removal probability $r = 1$, so an ancient language cannot be an ancestor of another sample. No node is calibrated: the clock rate is informed only by the fact that the ancient languages have had less time to change (Drummond et al., 2003). The script can also be run on the sampled-ancestor data set of script 08 (`--args 15000 fbd`). This shows what happens, as discussed in Section 10.6, when a true ancestor (Latin) is forced onto a side branch of its own.

```
################################################################################
# 07_tip_dating_serial_bd.Rev
#
```

```
# Model    : tip-dating with ancient (non-contemporaneous) languages. Ancient
#            languages are tips with known attestation ages; the tree prior is
#            the serially sampled birth-death process (birth-death-sampling,
#            dnBDSTP) with sampling-and-removal (r = 1), i.e. ancient
#            languages may NOT be sampled ancestors (that is script 08).
#            Strict clock, binary cognate CTMC, no node calibrations at all:
#            the rate is informed only by the ages of the dated tips
#            (measurably evolving populations; Drummond et al. 2003).
# Book     : Ch. 10 "Tip-dating with ancient languages" (divergence-tipdating)
# Estimates: topology, node ages, root age, clock rate, lambda, mu, psi
# Data     : data/time/tipdate.nex + tipdate_taxa.tsv (9 modern + 3 ancient
#            languages, attested 3.0, 2.2, 1.4 ky ago)
#            Optional: --args n_gen fbd runs the same no-sampled-ancestor
#            model on the sampled-ancestor dataset of script 08 (Latin is a
#            true ancestor there) to show the distortion described in the book.
# Runtime  : ~2 min (2 runs)
# Usage    : cd revbayes && rb analyses/time/07_tip_dating_serial_bd.Rev [--args n_
gen dataset]
################################################################################

seed(707)
n_gen = 15000
dataset = "tipdate"
if (args.size() > 1) n_gen = int(args[2])
if (args.size() > 2) dataset = args[3]
out_dir = "output/time/07_tip_dating_serial_bd/"
if (dataset != "tipdate") out_dir = "output/time/07_tip_dating_serial_bd_" + dataset
+ "/"

data <- readDiscreteCharacterData("data/time/" + dataset + ".nex")
taxa <- readTaxonData("data/time/" + dataset + "_taxa.tsv")
moves    = VectorMoves()
monitors = VectorMonitors()

#### serially sampled birth-death tree prior ##################################
origin_time ~ dnUniform(3.5, 30.0)       # older than the oldest dated tip
lambda ~ dnExponential(1.0)
mu ~ dnExponential(2.0)
psi ~ dnExponential(5.0)                  # rate of sampling ancient languages
timetree ~ dnBDSTP(originAge=origin_time, lambda=lambda, mu=mu, phi=psi,
                   Phi=1.0, r=1.0, condition="survival", taxa=taxa)
root_age := timetree.rootAge()

moves.append( mvScale(lambda, lambda=1.0, tune=true, weight=1.0) )
moves.append( mvScale(mu, lambda=1.0, tune=true, weight=1.0) )
moves.append( mvScale(psi, lambda=1.0, tune=true, weight=1.0) )
moves.append( mvSlide(origin_time, delta=1.0, tune=true, weight=2.0) )
moves.append( mvRootTimeSlideUniform(timetree, origin_time, weight=5.0) )
```

```
moves.append( mvNodeTimeSlideUniform(timetree, weight=15.0) )
moves.append( mvNarrow(timetree, weight=6.0) )
moves.append( mvNNI(timetree, weight=3.0) )
moves.append( mvFNPR(timetree, weight=3.0) )

#### strict clock ##################################################################
clock_rate ~ dnExponential(2.0)
moves.append( mvScale(clock_rate, lambda=0.5, tune=true, weight=5.0) )

#### substitution model and likelihood #############################################
pi ~ dnDirichlet(v(1, 1))
pi1 := pi[2]
Q := fnF81(pi)
moves.append( mvBetaSimplex(pi, alpha=20, tune=true, weight=2.0) )
seq ~ dnPhyloCTMC(tree=timetree, Q=Q, branchRates=clock_rate, type="Standard")
seq.clamp(data)

#### MCMC ##########################################################################
mymodel = model(timetree)
monitors.append( mnModel(filename=out_dir + "model.log", printgen=10) )
monitors.append( mnFile(timetree, filename=out_dir + "trees.trees", printgen=50) )
monitors.append( mnScreen(printgen=5000, root_age, clock_rate) )
mymcmc = mcmc(mymodel, moves, monitors, nruns=2, combine="mixed")
mymcmc.run(generations=n_gen, tuningInterval=200)

trace = readTreeTrace(out_dir + "trees.trees", treetype="clock", burnin=0.25)
mcc = mccTree(trace, file=out_dir + "mcc.tre")

q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
tip-dating, serially sampled BD (no sampled ancestors)	root_age	6.00	5.999	4.445	7.984	131	pass
tip-dating, serially sampled BD (no sampled ancestors)	clock_rate	0.15	0.151	0.098	0.202	101	pass
tip-dating, serially sampled BD (no sampled ancestors)	RF_distance_MCC_vs_true	0.00	0.000	NA	NA	NA	pass
tip-dating without sampled ancestors on data WITH a sampled ancestor (Latin)	root_age	6.00	8.822	5.510	13.342	98	info-warn
tip-dating without sampled ancestors on data WITH a sampled ancestor (Latin)	clock_rate	0.15	0.100	0.064	0.142	77	info-fail

Dated tips alone recover the rate and the root age, and the MCC tree has the true topology. On the sampled-ancestor data set the rows are marked `info-`, because there Latin is forced onto a side

branch of its own. That moves the root older (8.8 ky) and makes the rate too slow (HPD 0.064 to 0.142, excluding the true 0.15), which is the distortion described in Section 10.6.4.

H.4.8 Fossilized birth-death process with sampled ancestors

Script `08_fbd_sampled_ancestors.Rev` implements the fossilized birth-death prior with sampled ancestors (Section 10.6) (Stadler, 2010; Heath et al., 2014; Gavryushkina et al., 2014) using `dnFBDP`. In the simulated family, "Latin" lies directly on the stem lineage of a three-language "Romance" clade (a true sampled ancestor, with a zero-length tip branch), while "Hittite" and "Gothic" sit on extinct side branches. `mvCollapseExpandFossilBranch` proposes moving ancient languages onto and off lineages as ancestors. The check counts, over the posterior trees, how often each ancient language is a sampled ancestor, and how often Latin is the direct ancestor of exactly the Romance clade.

```
################################################################################
# 08_fbd_sampled_ancestors.Rev
#
# Model    : fossilized birth-death process (Stadler 2010; Heath et al. 2014)
#            with sampled ancestors (Gavryushkina et al. 2014): ancient
#            languages are dated samples that may lie ON a lineage with
#            descendants (zero-length tip branch), as Latin does for Romance,
#            or on an extinct side branch, as Hittite. dnFBDP, strict clock,
#            binary cognate CTMC; mvCollapseExpandFossilBranch proposes moving
#            ancient languages on and off the tree as ancestors.
# Book     : Ch. 10 "Sampled ancestors and the fossilized birth-death process"
#            (divergence-fbd, divergence-ancestors); Ch. 16 total evidence
#            (biology-toolkit-total-evidence)
# Estimates: topology, node ages, root age, clock rate, lambda, mu, psi,
#            posterior probability that each ancient language is an ancestor
# Data     : data/time/fbd.nex + fbd_taxa.tsv (10 modern languages, "Latin"
#            = true sampled ancestor at 2.6 ky, "Hittite" 3.4 ky and "Gothic"
#            1.6 ky = extinct side lineages)
# Runtime  : ~2-3 min (2 runs)
# Usage    : cd revbayes && rb analyses/time/08_fbd_sampled_ancestors.Rev [--args n_
gen]
################################################################################

seed(808)
n_gen = 15000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/time/08_fbd_sampled_ancestors/"

data <- readDiscreteCharacterData("data/time/fbd.nex")
taxa <- readTaxonData("data/time/fbd_taxa.tsv")
moves    = VectorMoves()
monitors = VectorMonitors()

#### fossilized birth-death prior ##############################################
origin_time ~ dnUniform(3.5, 30.0)
```

```
lambda ~ dnExponential(1.0)
mu ~ dnExponential(2.0)
psi ~ dnExponential(5.0)
timetree ~ dnFBDP(originAge=origin_time, lambda=lambda, mu=mu, psi=psi, rho=1.0,
                  condition="survival", taxa=taxa)
root_age := timetree.rootAge()
num_SA := timetree.numSampledAncestors()

moves.append( mvScale(lambda, lambda=1.0, tune=true, weight=1.0) )
moves.append( mvScale(mu, lambda=1.0, tune=true, weight=1.0) )
moves.append( mvScale(psi, lambda=1.0, tune=true, weight=1.0) )
moves.append( mvSlide(origin_time, delta=1.0, tune=true, weight=2.0) )
moves.append( mvRootTimeSlideUniform(timetree, origin_time, weight=5.0) )
moves.append( mvNodeTimeSlideUniform(timetree, weight=15.0) )
moves.append( mvNarrow(timetree, weight=6.0) )
moves.append( mvFNPR(timetree, weight=4.0) )
moves.append( mvCollapseExpandFossilBranch(timetree, origin_time, weight=6.0) )

#### strict clock ##################################################################
clock_rate ~ dnExponential(2.0)
moves.append( mvScale(clock_rate, lambda=0.5, tune=true, weight=5.0) )

#### substitution model and likelihood #############################################
pi ~ dnDirichlet(v(1, 1))
pi1 := pi[2]
Q := fnF81(pi)
moves.append( mvBetaSimplex(pi, alpha=20, tune=true, weight=2.0) )
seq ~ dnPhyloCTMC(tree=timetree, Q=Q, branchRates=clock_rate, type="Standard")
seq.clamp(data)

#### MCMC ##########################################################################
mymodel = model(timetree)
monitors.append( mnModel(filename=out_dir + "model.log", printgen=10) )
monitors.append( mnFile(timetree, filename=out_dir + "trees.trees", printgen=20) )
monitors.append( mnScreen(printgen=5000, root_age, clock_rate, num_SA) )
mymcmc = mcmc(mymodel, moves, monitors, nruns=2, combine="mixed")
mymcmc.run(generations=n_gen, tuningInterval=200)

trace = readTreeTrace(out_dir + "trees.trees", treetype="clock", burnin=0.25)
mcc = mccTree(trace, file=out_dir + "mcc.tre")

q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
fossilized birth-death with sampled ancestors	root_age	6.00	7.392	4.854	11.119	66	warn

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
fossilized birth-death with sampled ancestors	clock_rate	0.15	0.123	0.071	0.164	68	warn
fossilized birth-death with sampled ancestors	num_SA	1.00	0.550	0.000	1.000	132	pass
fossilized birth-death with sampled ancestors	P(Latin is sampled ancestor)	1.00	0.543	NA	NA	NA	pass
fossilized birth-death with sampled ancestors	P(Hittite is sampled ancestor)	0.00	0.000	NA	NA	NA	pass
fossilized birth-death with sampled ancestors	P(Gothic is sampled ancestor)	0.00	0.000	NA	NA	NA	pass
fossilized birth-death with sampled ancestors	P(Latin ancestral to exactly the Romance clade)	1.00	0.543	NA	NA	NA	pass

Hittite and Gothic are never sampled ancestors. Latin is a sampled ancestor, and specifically of the Romance clade, with posterior probability 0.54, which passes the 0.5 criterion only narrowly. The root age and rate are covered, but their ESS is below 100 after 2 x 15,000 generations (`warn`).

H.4.9 Clock-model comparison by marginal likelihood

Script `09_clock_model_comparison_ss.Rev` estimates the log marginal likelihood of a strict or a UCLN clock by stepping-stone sampling (Xie et al., 2011) and path sampling (Lartillot and Philippe, 2006) from a single `powerPosterior` run with 20 power steps (Section 6.7). It is run four times: both models on the UCLN data of script 02 and both on the strict data of script 01. The log Bayes factor favours the UCLN when it is positive. The check passes if the generating model is chosen. In RevBayes, clamped calibration "observations" count as part of the likelihood that is raised to each power. The estimate is still the marginal likelihood of the full model, because the remaining prior (uniform root age, proper priors on everything else) is proper.

```
################################################################################
# 09_clock_model_comparison_ss.Rev
#
# Model    : Bayesian clock-model comparison by marginal likelihood. Fits a
#            strict clock or a UCLN relaxed clock (same tree prior, same node
#            calibrations, same binary cognate CTMC, fixed true topology) and
#            estimates log marginal likelihoods by stepping-stone sampling
#            (Xie et al. 2011) and path sampling (Lartillot & Philippe 2006)
#            from one powerPosterior run.
# Book     : Ch. 9 "Choosing a clock model" (clocks-choice) and "The strict
#            clock" (Bayes-factor comparison); Ch. 6 marginal likelihoods
# Estimates: log marginal likelihood of the chosen model on the chosen dataset
# Data     : data/time/clock_ucln.nex (generated under UCLN, sigma = 0.5) or
#            data/time/clock_strict.nex (generated under a strict clock)
# Runtime  : ~1-2.5 min per model/dataset combination
# Usage    : cd revbayes && rb analyses/time/09_clock_model_comparison_ss.Rev --args
MODEL DATASET [CATS]
#            MODEL in {strict, ucln}; DATASET in {clock_ucln, clock_strict}
```

```
#            The four combinations are run by analyses/time/run_time.sh.
################################################################################

seed(909)
model_name = "ucln"
dataset = "clock_ucln"
n_cats = 20
if (args.size() > 1) model_name = args[2]
if (args.size() > 2) dataset = args[3]
if (args.size() > 3) n_cats = int(args[4])
out_dir = "output/time/09_clock_model_comparison_ss/"
stem = out_dir + dataset + "_" + model_name

data <- readDiscreteCharacterData("data/time/" + dataset + ".nex")
taxa <- data.taxa()
n_taxa <- taxa.size()
n_branches <- 2 * n_taxa - 2
moves    = VectorMoves()
monitors = VectorMonitors()

#### tree prior; fixed topology ################################################
root_age ~ dnUniform(0.5, 30.0)
birth_rate ~ dnExponential(1.0)
timetree ~ dnBDP(lambda=birth_rate, mu=0.0, rho=1.0, rootAge=root_age, taxa=taxa)
start_tree <- readTrees("data/time/tree12_start.tre")[1]
root_age.setValue(start_tree.rootAge())
timetree.setValue(start_tree)
moves.append( mvScale(birth_rate, lambda=1.0, tune=true, weight=1.0) )
moves.append( mvSlide(root_age, delta=0.5, tune=true, weight=3.0) )
moves.append( mvTreeScale(tree=timetree, rootAge=root_age, delta=0.5, tune=true,
weight=3.0) )
moves.append( mvNodeTimeSlideUniform(timetree, weight=10.0) )
moves.append( mvSubtreeScale(timetree, weight=3.0) )

#### calibrations ##############################################################
source("data/time/calibrations_tree12.Rev")
age_C1 := tmrca(timetree, clade_C1)
age_C2 := tmrca(timetree, clade_C2)
obs_C1 ~ dnSoftBoundUniformNormal(min=C1_min - age_C1, max=C1_max - age_C1, sd=C1_
sd, p=0.95)
obs_C1.clamp(0.0)
obs_C2 ~ dnLognormal(mean=C2_mu, sd=C2_sd, offset=-age_C2)
obs_C2.clamp(-C2_offset)

#### clock model ###############################################################
mean_rate ~ dnExponential(2.0)
moves.append( mvScale(mean_rate, lambda=0.5, tune=true, weight=5.0) )
if (model_name == "strict") {
```

```
  branch_rates := mean_rate
} else {
  sigma ~ dnExponential(1.0)
  moves.append( mvScale(sigma, lambda=0.5, tune=true, weight=5.0) )
  for (j in 1:n_branches) {
    branch_rates[j] ~ dnLognormal(ln(mean_rate) - sigma * sigma / 2.0, sigma)
    moves.append( mvScale(branch_rates[j], lambda=0.5, tune=true, weight=1.0) )
  }
  moves.append( mvRateAgeBetaShift(tree=timetree, rates=branch_rates, tune=true,
  weight=10.0) )
}

#### substitution model and likelihood ########################################
pi ~ dnDirichlet(v(1, 1))
Q := fnF81(pi)
moves.append( mvBetaSimplex(pi, alpha=20, tune=true, weight=2.0) )
seq ~ dnPhyloCTMC(tree=timetree, Q=Q, branchRates=branch_rates, type="Standard")
seq.clamp(data)

mymodel = model(timetree)

#### power posterior, stepping-stone and path sampling ########################
pow_p = powerPosterior(mymodel, moves, monitors, stem + "_powp.out", cats=n_cats,
sampleFreq=10)
pow_p.burnin(generations=2000, tuningInterval=200)
pow_p.run(generations=500)

ss = steppingStoneSampler(file=stem + "_powp.out", powerColumnName="power",
                          likelihoodColumnName="likelihood")
ps = pathSampler(file=stem + "_powp.out", powerColumnName="power",
                 likelihoodColumnName="likelihood")
ml_ss = ss.marginal()
ml_ps = ps.marginal()
write("dataset,model,lnML_stepping_stone,lnML_path_sampling\n", filename=stem + "_
ml.csv")
write(dataset + "," + model_name + "," + ml_ss + "," + ml_ps + "\n", filename=stem +
"_ml.csv", append=true)
print("lnML (stepping stone) = " + ml_ss + "   lnML (path sampling) = " + ml_ps)

q()
```

Model	Parameter	Truth	Post. mean
strict vs UCLN, data simulated under ucln	lnBF_ucln_minus_strict (SS)	6.76	pass
strict vs UCLN, data simulated under strict	lnBF_ucln_minus_strict (SS)	-3.44	pass

model	notes
strict vs UCLN, data simulated under ucln	lnML SS: strict -3142.57, UCLN -3135.81; PS: strict -3143.32, UCLN -3135.86 (lnBF_PS = 7.46); selected ucln
strict vs UCLN, data simulated under strict	lnML SS: strict -3227.64, UCLN -3231.08; PS: strict -3228.02, UCLN -3230.99 (lnBF_PS = -2.97); selected strict

The generating model is selected in both directions. On UCLN data the log Bayes factor favours the UCLN by 6.8. On strict-clock data it favours the strict clock by 3.4, because the UCLN, which contains the strict clock as the limit $\sigma \to 0$, is penalized for its extra parameters. Stepping-stone and path-sampling estimates agree to within about 0.7 log units.

H.4.10 Tree prior sensitivity: coalescent versus Yule

Script `10_tree_prior_coalescent_vs_yule.Rev` reruns the calibrated strict-clock analysis of script 01 (on the true topology) under a Yule prior and under a constant-population Kingman coalescent prior (`dnCoalescent`), the alternative discussed in Section 10.4 (Rama, 2018; Kingman, 1982). The data come from a Yule tree, so the coalescent prior is misspecified. Because the coalescent concentrates nodes near the present, the question is how far it moves the uncalibrated root.

```
################################################################################
# 10_tree_prior_coalescent_vs_yule.Rev
#
# Model    : sensitivity of dates to the tree prior. The same strict-clock,
#            node-calibrated analysis (calibrations C1/C2, binary cognate
#            CTMC, fixed true topology) is run under
#              yule       : pure-birth prior with an unknown root age
#              coalescent : constant-population Kingman coalescent prior
#                           (dnCoalescent, population size theta estimated)
#            The data were simulated on a Yule tree, so the coalescent prior is
#            misspecified; the comparison shows how far it moves the root age.
# Book     : Ch. 10 "The tree prior" (divergence-treeprior); Ch. 6 tree priors
# Estimates: root age, clock rate, node ages under each prior
# Data     : data/time/clock_strict.nex
# Runtime  : ~1 min per prior
# Usage    : cd revbayes && rb analyses/time/10_tree_prior_coalescent_vs_yule.Rev
--args PRIOR [n_gen]
#            PRIOR in {yule, coalescent}
################################################################################

seed(1010)
prior_name = "yule"
n_gen = 15000
if (args.size() > 1) prior_name = args[2]
if (args.size() > 2) n_gen = int(args[3])
out_dir = "output/time/10_tree_prior_coalescent_vs_yule/" + prior_name + "_"
```

```
data <- readDiscreteCharacterData("data/time/clock_strict.nex")
taxa <- data.taxa()
moves    = VectorMoves()
monitors = VectorMonitors()
start_tree <- readTrees("data/time/tree12_start.tre")[1]

#### tree prior ################################################################
if (prior_name == "yule") {
  root_par ~ dnUniform(0.5, 30.0)
  birth_rate ~ dnExponential(1.0)
  timetree ~ dnBDP(lambda=birth_rate, mu=0.0, rho=1.0, rootAge=root_par, taxa=taxa)
  root_par.setValue(start_tree.rootAge())
  timetree.setValue(start_tree)
  moves.append( mvScale(birth_rate, lambda=1.0, tune=true, weight=1.0) )
  moves.append( mvSlide(root_par, delta=0.5, tune=true, weight=3.0) )
  moves.append( mvTreeScale(tree=timetree, rootAge=root_par, delta=0.5, tune=true,
  weight=3.0) )
} else {
  theta ~ dnLognormal(ln(2.0), 1.5)       # population size in ky
  timetree ~ dnCoalescent(theta=theta, taxa=taxa)
  timetree.setValue(start_tree)
  moves.append( mvScale(theta, lambda=1.0, tune=true, weight=2.0) )
  moves.append( mvRootTimeScaleBactrian(timetree, weight=5.0) )
}
root_age := timetree.rootAge()
moves.append( mvNodeTimeSlideUniform(timetree, weight=15.0) )
moves.append( mvSubtreeScale(timetree, weight=5.0) )

#### calibrations ##############################################################
source("data/time/calibrations_tree12.Rev")
age_C1 := tmrca(timetree, clade_C1)
age_C2 := tmrca(timetree, clade_C2)
obs_C1 ~ dnSoftBoundUniformNormal(min=C1_min - age_C1, max=C1_max - age_C1, sd=C1_
sd, p=0.95)
obs_C1.clamp(0.0)
obs_C2 ~ dnLognormal(mean=C2_mu, sd=C2_sd, offset=-age_C2)
obs_C2.clamp(-C2_offset)

#### strict clock ##############################################################
clock_rate ~ dnExponential(2.0)
moves.append( mvScale(clock_rate, lambda=0.5, tune=true, weight=5.0) )

#### substitution model and likelihood #########################################
pi ~ dnDirichlet(v(1, 1))
pi1 := pi[2]
Q := fnF81(pi)
moves.append( mvBetaSimplex(pi, alpha=20, tune=true, weight=2.0) )
seq ~ dnPhyloCTMC(tree=timetree, Q=Q, branchRates=clock_rate, type="Standard")
```

```
seq.clamp(data)

#### MCMC ######################################################################
mymodel = model(timetree)
monitors.append( mnModel(filename=out_dir + "model.log", printgen=10) )
monitors.append( mnFile(timetree, filename=out_dir + "trees.trees", printgen=100) )
monitors.append( mnScreen(printgen=5000, root_age, clock_rate) )
mymcmc = mcmc(mymodel, moves, monitors)
mymcmc.run(generations=n_gen, tuningInterval=200)

q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
strict clock, yule tree prior	root_age	6.00	7.038	4.143	10.622	863	pass
strict clock, yule tree prior	clock_rate	0.15	0.131	0.094	0.162	404	pass
strict clock, coalescent tree prior	root_age	6.00	11.727	4.386	25.400	1026	info-pass
strict clock, coalescent tree prior	clock_rate	0.15	0.134	0.105	0.170	199	info-pass
strict clock, coalescent tree prior	theta	NA	6.419	3.001	10.799	1000	info
Yule vs coalescent	root_age_shift_coalescent_minus_yule	NA	4.689	NA	NA	NA	info

Under the coalescent prior the root moves from 7.0 to 11.7 ky (posterior means), with an upper HPD limit of 25 ky. The calibrated nodes and the rate hardly change. The uncalibrated root, which is constrained mainly by the tree prior, is where the choice of prior matters.

H.4.11 Multispecies coalescent in a dialect continuum

Script `11_multispecies_coalescent.Rev` implements the multispecies coalescent of Section 16.2.1 (Rannala and Yang, 2003; Heled and Drummond, 2010). The five "species" are languages, each sampled at two local varieties ("individuals"). Eight "loci" are concept blocks, each with 200 binary cognate characters that share one concept tree (gene tree). The concept trees grow inside the language tree under `dnMultiSpeciesCoalescent`, with incomplete sorting of ancestral lexical variation across the short internal branches. The language-tree topology and ages and the effective population size N_e are the targets. The clock rate is fixed at its true value, because the absolute time scale and N_e cannot both be identified without a calibration. Treating a block of characters as one locus with a single history is an idealization: in real lexical data each meaning would be a separate, very short locus. In RevBayes 1.4.1 a pair of lineages coalesces at rate $2/N_e$ per unit of tree time, so the expected time to coalescence of two lineages is $N_e/2$ rather than the N_e implied by the units of Equation (16.1). We verified this by simulating from `dnMultiSpeciesCoalescent` and by recomputing its log density by hand, and the R simulator uses the same parameterization.

The script has two modes, and the table shows both.

- `genetrees` clamps the eight true concept trees. This tests the MSC density and the inference of the language tree and N_e on their own.
- `joint` (the default) estimates the concept trees from the cognate characters together with everything else.

Getting the joint mode to run at all took three changes, and all three are documented in the script.

1. *Runaway towards $N_e \to 0$.* With an exponential prior on N_e, the sampler repeatedly ran away towards $N_e \to 0$, with log densities above 10^{60}. This happened when root-age or subtree-scale proposals produced language trees with negative branch lengths, or concept trees incompatible with the language tree. A clamped "guard" observation now gives zero density to negative species-tree branches, `mvSpeciesSubtreeScaleBeta` is not used, and N_e has a Uniform(0.05, 5) prior.
2. *Starting tree.* The language tree starts from a data-derived UPGMA tree, because from a random start the joint sampler stayed on a wrong topology.
3. *Mixing.* Even with these changes the joint mode mixes poorly in a few minutes. The topology is recovered, but the ages stay near their starting values and their HPDs are far too narrow, so those rows fail. The `genetrees` mode recovers every parameter. Running the joint model reliably would need much longer chains or better joint species-gene moves.

```
################################################################################
# 11_multispecies_coalescent.Rev
#
# Model    : multispecies coalescent (Rannala & Yang 2003; Heled & Drummond
#            2010) for a dialect continuum: 5 languages ("species"), each
#            sampled at 2 local varieties ("individuals"); 8 independent
#            concept blocks ("loci"), each with its own concept tree (gene
#            tree) evolving inside the language tree with incomplete lineage
#            sorting of ancestral lexical variation. Each block is a set of
#            binary cognate characters sharing one history (an idealisation:
#            in real lexical data each meaning would be one short "locus").
#            Gene trees ~ dnMultiSpeciesCoalescent(language tree, Ne);
#            characters ~ binary CTMC on the gene trees with a known clock rate.
#            Note: in RevBayes 1.4.1 a pair of lineages coalesces at rate 2/Ne
#            (time units of the tree), i.e. E[pair coalescence time] = Ne/2;
#            data/sim_time.R simulates with the same parameterisation.
# Book     : Ch. 16 "The multispecies coalescent and incomplete lineage
#            sorting" (biology-toolkit-msc)
# Estimates: language (species) tree topology and ages, Ne, 8 concept trees
# Data     : data/time/msc_locus1..8.nex, msc_species_map.tsv, msc_setup.Rev
# Runtime  : ~2-3 min (one run; the MSC likelihood over 8 loci is the costly part)
# Usage    : cd revbayes && rb analyses/time/11_multispecies_coalescent.Rev [--args
n_gen MODE]
#            MODE joint (default): concept trees estimated from the characters
#            MODE genetrees: the true concept trees are clamped (tests the MSC
#                 density and the language-tree/Ne inference on their own)
################################################################################
```

```
seed(1111)
n_gen = 7000
mode = "joint"
if (args.size() > 1) n_gen = int(args[2])
if (args.size() > 2) mode = args[3]
out_dir = "output/time/11_multispecies_coalescent/" + mode + "_"

source("data/time/msc_setup.Rev")        # n_loci, msc_clock_rate, species_names
taxa <- readTaxonData("data/time/msc_species_map.tsv")
for (i in 1:species_names.size()) sp_taxa[i] = taxon(species_names[i])
moves    = VectorMoves()
monitors = VectorMonitors()

#### language (species) tree: Yule prior ######################################
sp_root_age ~ dnUniform(0.1, 20.0)
sp_lambda ~ dnExponential(1.0)
sptree ~ dnBDP(lambda=sp_lambda, mu=0.0, rho=1.0, rootAge=sp_root_age, taxa=sp_taxa)
moves.append( mvScale(sp_lambda, lambda=1.0, tune=true, weight=1.0) )
if (mode != "genetrees") {
  # joint mode: start from a data-derived UPGMA language tree (root 2.5 ky),
  # because the joint sampler cannot escape a wrong starting topology in a
  # few thousand generations (the concept trees adapt to it)
  sp_start <- readTrees("data/time/msc_species_start.tre")[1]
  sp_root_age.setValue(sp_start.rootAge())
  sptree.setValue(sp_start)
}

# Guard against negative species-tree branch lengths. A proposal on the root
# age can put the root below one of its children; with no character data on
# the species tree nothing else rejects such a state, and the MSC density of
# RevBayes 1.4.1 then becomes arbitrarily large (we observed the chain running
# away to Ne -> 0 with log densities > 1e60). The observation below has
# density 1/(min_bl + 1e6), constant to within 1e-6 for any valid tree, and
# zero when a branch is negative.
for (i in 1:(2 * species_names.size() - 2)) sp_bl[i] := sptree.branchLength(i)
sp_min_bl := min(sp_bl)
sp_guard ~ dnUniform(-1e6, sp_min_bl)
sp_guard.clamp(0.0)

#### effective population size (shared by all branches) ########################
# Uniform prior bounded away from 0: with an unbounded prior (Exponential(1))
# the joint sampler ran away to Ne -> 0 with an ever-increasing MSC density in
# RevBayes 1.4.1, even with the branch-length guard above.
Ne ~ dnUniform(0.05, 5.0)
Ne.setValue(1.0)
moves.append( mvScale(Ne, lambda=0.5, tune=true, weight=5.0) )

#### concept (gene) trees ######################################################
```

```
if (mode == "genetrees") {
  # verification of the MSC density alone: the true concept trees are observed
  true_gene_trees <- readTrees("data/time/msc_gene_trees_true.tre")
  for (l in 1:n_loci) {
    gene_tree[l] ~ dnMultiSpeciesCoalescent(speciesTree=sptree, Ne=Ne, taxa=taxa)
    gene_tree[l].clamp(true_gene_trees[l])
  }
} else {
  # full model: concept trees estimated from binary cognate characters
  pi ~ dnDirichlet(v(1, 1))
  Q := fnF81(pi)
  moves.append( mvBetaSimplex(pi, alpha=20, tune=true, weight=2.0) )
  for (l in 1:n_loci) {
    gene_tree[l] ~ dnMultiSpeciesCoalescent(speciesTree=sptree, Ne=Ne, taxa=taxa)
    locus_data[l] <- readDiscreteCharacterData("data/time/msc_locus" + l + ".nex")
    seq[l] ~ dnPhyloCTMC(tree=gene_tree[l], Q=Q, branchRates=msc_clock_rate,
type="Standard")
    seq[l].clamp(locus_data[l])
    moves.append( mvNNI(gene_tree[l], weight=1.0) )
    moves.append( mvNarrow(gene_tree[l], weight=1.0) )
    moves.append( mvFNPR(gene_tree[l], weight=0.5) )
    moves.append( mvNodeTimeSlideUniform(gene_tree[l], weight=2.0) )
    moves.append( mvSubtreeScale(gene_tree[l], weight=0.5) )
    moves.append( mvRootTimeScaleBactrian(gene_tree[l], weight=0.5) )
  }
}

#### species-tree moves ############################################################
# plain moves (gene trees held fixed; valid Metropolis-Hastings proposals)
moves.append( mvSlide(sp_root_age, delta=0.05, tune=true, weight=4.0) )
moves.append( mvNodeTimeSlideUniform(sptree, weight=4.0) )
moves.append( mvNarrow(sptree, weight=2.0) )
moves.append( mvFNPR(sptree, weight=1.0) )
if (mode != "genetrees") {
  # joint moves that also update the gene trees (mvSpeciesSubtreeScaleBeta
  # is not used: it produced the runaway state described above)
  mv_sp_narrow = mvSpeciesNarrow(speciesTree=sptree, weight=4.0)
  mv_sp_slide  = mvSpeciesNodeTimeSlideUniform(speciesTree=sptree, weight=4.0)
  mv_sp_scale  = mvSpeciesTreeScale(speciesTree=sptree, rootAge=sp_root_age,
weight=3.0)
  for (l in 1:n_loci) {
    mv_sp_narrow.addGeneTreeVariable(gene_tree[l])
    mv_sp_slide.addGeneTreeVariable(gene_tree[l])
    mv_sp_scale.addGeneTreeVariable(gene_tree[l])
  }
  moves.append( mv_sp_narrow )
  moves.append( mv_sp_slide )
  moves.append( mv_sp_scale )
```

```
}

age_AB  := tmrca(sptree, clade("A", "B"))
age_DE  := tmrca(sptree, clade("D", "E"))
age_CDE := tmrca(sptree, clade("C", "D", "E"))

#### MCMC ############################################################################
mymodel = model(sptree)
monitors.append( mnModel(filename=out_dir + "model.log", printgen=5) )
monitors.append( mnFile(sptree, filename=out_dir + "species.trees", printgen=10) )
monitors.append( mnScreen(printgen=1000, sp_root_age, Ne, sp_min_bl) )
mymcmc = mcmc(mymodel, moves, monitors)
mymcmc.run(generations=n_gen, tuningInterval=200)

trace = readTreeTrace(out_dir + "species.trees", treetype="clock", burnin=0.25)
mcc = mccTree(trace, file=out_dir + "species_mcc.tre")

q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
MSC, true concept trees observed	Ne	0.5	0.574	0.437	0.711	2299	pass
MSC, true concept trees observed	species_root_age	1.7	1.682	1.623	1.711	3001	pass
MSC, true concept trees observed	age_AB	1.2	1.249	1.178	1.285	401	pass
MSC, true concept trees observed	age_DE	0.9	0.948	0.881	0.985	472	pass
MSC, true concept trees observed	age_CDE	1.3	1.283	1.230	1.310	695	pass
MSC, true concept trees observed	PP(true language-tree topology)	1.0	1.000	NA	NA	NA	pass
MSC, concept trees estimated from cognate data	Ne	0.5	0.365	0.269	0.468	198	fail
MSC, concept trees estimated from cognate data	species_root_age	1.7	2.234	2.206	2.248	669	fail
MSC, concept trees estimated from cognate data	age_AB	1.2	1.293	1.233	1.318	51	fail
MSC, concept trees estimated from cognate data	age_DE	0.9	0.726	0.674	0.751	119	fail
MSC, concept trees estimated from cognate data	age_CDE	1.3	1.104	1.079	1.119	67	fail
MSC, concept trees estimated from cognate data	PP(true language-tree topology)	1.0	1.000	NA	NA	NA	pass

H.4.12 Episodic (skyline) birth-death diversification

Script `12_episodic_birth_death_skyline.Rev` fits a piecewise-constant birth-death model (Stadler et al., 2013) to a fixed dated tree of about 100 languages (Section 16.6.1). Language splitting has its own rate in each of three intervals (0 to 1.5, 1.5 to 3 and more than 3 ky ago); language death has one constant rate. The tree is larger than the others in this appendix because rates are estimated from branching times alone. Extinction is held constant to respect the non-identifiability of fully time-varying birth and death rates from trees of living languages (Louca and Pennell, 2020). The check asks whether the slowdown of splitting towards the present is detected.

```
################################################################################
# 12_episodic_birth_death_skyline.Rev
#
# Model    : episodic (piecewise-constant, "skyline") birth-death process on a
#            dated language-family tree (Stadler 2011, 2013): the speciation
#            (language-splitting) rate takes a different value in each of 3
#            time intervals (boundaries 1.5 and 3.0 ky), extinction (language
#            death) rate constant. dnEpisodicBirthDeath, tree fixed (clamped).
#            The tree is larger (about 100 languages) than in the other
#            scripts because diversification rates are estimated from
#            branching times alone. Extinction is held constant across
#            intervals to respect the non-identifiability of fully time-varying
#            lambda and mu from extant trees (Louca & Pennell 2020).
# Book     : Ch. 16 "Diversification models" (biology-toolkit-diversification)
#            and "Phylodynamics" (birth-death skyline)
# Estimates: lambda_1 (0-1.5 ky), lambda_2 (1.5-3 ky), lambda_3 (> 3 ky), mu;
#            posterior probability that splitting slowed towards the present
# Data     : data/time/skyline_tree.nex, skyline_setup.Rev
# Runtime  : < 1 min
# Usage    : cd revbayes && rb analyses/time/12_episodic_birth_death_skyline.Rev
[--args n_gen]
################################################################################

seed(1212)
n_gen = 20000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/time/12_episodic_birth_death_skyline/"

source("data/time/skyline_setup.Rev")      # sky_root_age, sky_times
observed_tree <- readTrees("data/time/skyline_tree.nex")[1]
taxa <- observed_tree.taxa()
moves    = VectorMoves()
monitors = VectorMonitors()

n_int = sky_times.size() + 1
for (i in 1:n_int) {
  speciation[i] ~ dnLognormal(ln(0.5), 1.0)
  moves.append( mvScale(speciation[i], lambda=0.5, tune=true, weight=2.0) )
```

```
}
extinction ~ dnExponential(5.0)
moves.append( mvScale(extinction, lambda=0.5, tune=true, weight=2.0) )
ext_vec := rep(extinction, n_int)

timetree ~ dnEpisodicBirthDeath(rootAge=observed_tree.rootAge(),
lambdaRates=speciation,
                                lambdaTimes=sky_times, muRates=ext_vec, muTimes=sky_
                                times,
                                rho=1.0, samplingStrategy="uniform",
                                condition="survival",
                                taxa=taxa)
timetree.clamp(observed_tree)

net_div := speciation - ext_vec
slowdown := ifelse(speciation[1] < speciation[n_int], 1, 0)
gamma_stat := observed_tree.gammaStatistic()

mymodel = model(timetree)
monitors.append( mnModel(filename=out_dir + "model.log", printgen=10) )
monitors.append( mnScreen(printgen=5000, speciation, extinction) )
mymcmc = mcmc(mymodel, moves, monitors)
mymcmc.run(generations=n_gen, tuningInterval=200)

q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
episodic birth-death (3 intervals)	lambda_1 (0-1.5 ky)	0.3	0.225	0.128	0.322	1337	pass
episodic birth-death (3 intervals)	lambda_2 (1.5-3 ky)	0.6	0.510	0.275	0.769	719	pass
episodic birth-death (3 intervals)	lambda_3 (>3 ky)	1.0	1.216	0.750	1.809	842	pass
episodic birth-death (3 intervals)	mu	0.1	0.151	0.000	0.399	719	pass
episodic birth-death (3 intervals)	P(lambda_1 < lambda_3) (slowdown)	1.0	1.000	NA	NA	NA	pass

All three splitting rates and the extinction rate are covered, and the slowdown towards the present is detected with posterior probability 1.

H.4.13 State-dependent diversification (BiSSE)

Script `13_bisse_state_dependent_diversification.Rev` implements BiSSE (Maddison et al., 2007) with the character-dependent birth-death process `dnCDBDP`, on a fixed tree of 150 languages simulated by **diversitree** (FitzJohn, 2012) with a binary trait under which state-1 languages

split twice as fast. The tree is clamped with `clamp` and the tip states with `clampCharData`. `check_time.R` compares the posterior with the diversitree maximum-likelihood estimates. HiSSE (Beaulieu and O'Meara, 2016) uses the same distribution with four states (two observed by two hidden) and ambiguous tip coding, and the character-independent null models it needs are built the same way. We have not run these, because a convincing test of HiSSE needs much larger trees than are practical here. The caution of Rabosky and Goldberg (2015) applies: a single clade with both the trait and a high rate can produce spurious support for state dependence.

All six rates are covered by their 95% HPDs. However, the posterior probability that $\lambda_1 > \lambda_0$ is only about 0.91, below the 0.95 threshold, so that row is reported as a fail. The diversitree ML estimates show the same weak contrast in this simulated tree ($\hat{\lambda}_0 = 0.64$, $\hat{\lambda}_1 = 0.78$, against true values 0.4 and 0.8). The shortfall is a property of this simulated data set rather than of the implementation: 150 languages are too few to detect a twofold difference in splitting rate reliably.

```
################################################################################
# 13_bisse_state_dependent_diversification.Rev
#
# Model    : BiSSE, binary-state speciation and extinction (Maddison et al.
#            2007): languages in state 0 / 1 of a binary trait (e.g. a
#            subsistence mode or a writing tradition) split at rates
#            lambda0 / lambda1 and die at rates mu0 / mu1; the trait changes
#            at rates q01 / q10. dnCDBDP (character-dependent birth-death),
#            tree fixed (clamped), tip states clamped with clampCharData.
#            Uses ~150 languages: SSE models need large trees for power.
#            A HiSSE extension uses the same machinery with 4 states (2
#            observed x 2 hidden) and ambiguous tip coding; it is not run
#            here (see docs/time.Rmd).
# Book     : Ch. 16 "Diversification models" (biology-toolkit-diversification):
#            BiSSE, HiSSE and the Rabosky & Goldberg caution
# Estimates: lambda0, lambda1, mu0, mu1, q01, q10; P(lambda1 > lambda0)
# Data     : data/time/bisse_tree.nex, data/time/bisse_trait.nex
# Runtime  : ~2 min
# Usage    : cd revbayes && rb analyses/time/13_bisse_state_dependent_
diversification.Rev [--args n_gen]
################################################################################

seed(1313)
n_gen = 4000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/time/13_bisse_state_dependent_diversification/"

observed_tree <- readTrees("data/time/bisse_tree.nex")[1]
trait <- readDiscreteCharacterData("data/time/bisse_trait.nex")
moves    = VectorMoves()
monitors = VectorMonitors()

n_states = 2
for (i in 1:n_states) {
  speciation[i] ~ dnLognormal(ln(0.5), 1.0)
```

```
  extinction[i] ~ dnExponential(5.0)
  moves.append( mvScale(speciation[i], lambda=0.5, tune=true, weight=2.0) )
  moves.append( mvScale(extinction[i], lambda=0.5, tune=true, weight=2.0) )
}
lambda0 := speciation[1]
lambda1 := speciation[2]
mu0 := extinction[1]
mu1 := extinction[2]
diff_lambda := lambda1 - lambda0
lambda1_gt_lambda0 := ifelse(lambda1 > lambda0, 1, 0)

q01 ~ dnExponential(10.0)
q10 ~ dnExponential(10.0)
moves.append( mvScale(q01, lambda=0.5, tune=true, weight=1.0) )
moves.append( mvScale(q10, lambda=0.5, tune=true, weight=1.0) )
rate_matrix := fnFreeK([q01, q10], rescaled=false)

root_freq ~ dnDirichlet(rep(1, n_states))
moves.append( mvBetaSimplex(root_freq, alpha=20, tune=true, weight=1.0) )

timetree ~ dnCDBDP(rootAge=observed_tree.rootAge(), speciationRates=speciation,
                   extinctionRates=extinction, Q=rate_matrix, delta=1.0,
                   pi=root_freq, rho=1.0, condition="time", nTimeSlices=50)
timetree.clamp(observed_tree)
timetree.clampCharData(trait)

mymodel = model(timetree)
monitors.append( mnModel(filename=out_dir + "model.log", printgen=3) )
monitors.append( mnScreen(printgen=1000, speciation, extinction) )
mymcmc = mcmc(mymodel, moves, monitors)
mymcmc.run(generations=n_gen, tuningInterval=200)

q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
BiSSE (dnCDBDP)	lambda0	0.4	0.507	0.265	0.759	238	pass
BiSSE (dnCDBDP)	lambda1	0.8	0.719	0.552	0.873	552	pass
BiSSE (dnCDBDP)	mu0	0.1	0.278	0.003	0.647	193	pass
BiSSE (dnCDBDP)	mu1	0.1	0.141	0.000	0.371	378	pass
BiSSE (dnCDBDP)	q01	0.1	0.114	0.018	0.245	327	pass
BiSSE (dnCDBDP)	q10	0.1	0.131	0.070	0.204	296	pass

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
BiSSE (dnCDBDP)	P(lambda1 > lambda0)	1.0	0.912	NA	NA	NA	fail

H.4.14 Effective priors of calibrated nodes

Script `14_effective_prior_calibrations.Rev` puts into practice the advice of Section 10.3.2 to always sample from the prior. It runs the model of script 01 without the cognate data and records the effective (marginal) prior of the two calibrated nodes and the root. For the calibrated nodes, *Truth* is the mean of the density the analyst specified, so the difference between it and the posterior-mean column measures how much the tree prior and the ordering constraints have shifted the effective prior (Heled and Drummond, 2012; Warnock et al., 2012). In RevBayes 1.4.1, `run(underPrior=true)` reported "no effect" for this model, so the script leaves out the `dnPhyloCTMC` node. That keeps the calibration densities and drops only the data.

The results illustrate two pitfalls. First, the soft-bounded uniform on C1 (95% of its mass in [0.96, 1.96] ky) becomes an effective prior centred near 1.9 ky, half a millennium older than the analyst specified, because of the Yule prior and the older calibration C2 above it. Second, the effective prior on the uncalibrated root is narrow and young, about 4.3 ky. With the topology free under the prior, the clade C2 (all languages except one) almost always contains the root, so the C2 calibration ends up dating the root itself. Once the data fix the topology, the root is no longer calibrated, and the posterior of script 01 is far wider (4.0 to 10.3 ky). A calibration on a large clade whose monophyly is not enforced can therefore act on a different node from the one intended. The `notes` column of `results_time.csv` reports how often this happens.

```
################################################################################
# 14_effective_prior_calibrations.Rev
#
# Model    : the model of script 01 (Yule prior x calibration densities C1, C2,
#            strict clock, binary CTMC) sampled WITHOUT the data
#            (mcmc(..., underPrior=TRUE)) to obtain the effective (marginal)
#            prior on calibrated and uncalibrated node ages. The book's advice
#            "always sample from the prior": the effective prior on a
#            calibrated node is the product of the calibration density, the
#            tree prior and the ordering constraints, and can differ from the
#            specified density. tests/check_time.R compares it with the
#            specified densities and with the posterior from script 01.
# Book     : Ch. 10 "How calibrations interact" (divergence-interaction)
# Estimates: effective prior of age_C1, age_C2 and the root age
# Data     : taxon names of data/time/clock_strict.nex only
# Runtime  : ~30 s
# Usage    : cd revbayes && rb analyses/time/14_effective_prior_calibrations.Rev
[--args n_gen]
################################################################################

seed(1414)
n_gen = 50000
if (args.size() > 1) n_gen = int(args[2])
```

```
out_dir = "output/time/14_effective_prior_calibrations/"

data <- readDiscreteCharacterData("data/time/clock_strict.nex")
taxa <- data.taxa()
moves    = VectorMoves()
monitors = VectorMonitors()

root_age ~ dnUniform(0.5, 30.0)
birth_rate ~ dnExponential(1.0)
timetree ~ dnBDP(lambda=birth_rate, mu=0.0, rho=1.0, rootAge=root_age,
                 samplingStrategy="uniform", condition="time", taxa=taxa)
start_tree <- readTrees("data/time/clock_strict_upgma_start.tre")[1]
root_age.setValue(start_tree.rootAge())
timetree.setValue(start_tree)

moves.append( mvScale(birth_rate, lambda=1.0, tune=true, weight=1.0) )
moves.append( mvSlide(root_age, delta=1.0, tune=true, weight=3.0) )
moves.append( mvTreeScale(tree=timetree, rootAge=root_age, delta=0.5, tune=true,
weight=3.0) )
moves.append( mvNodeTimeSlideUniform(timetree, weight=10.0) )
moves.append( mvSubtreeScale(timetree, weight=3.0) )
moves.append( mvNarrow(timetree, weight=3.0) )
moves.append( mvFNPR(timetree, weight=2.0) )

source("data/time/calibrations_tree12.Rev")
age_C1 := tmrca(timetree, clade_C1)
age_C2 := tmrca(timetree, clade_C2)
obs_C1 ~ dnSoftBoundUniformNormal(min=C1_min - age_C1, max=C1_max - age_C1, sd=C1_
sd, p=0.95)
obs_C1.clamp(0.0)
obs_C2 ~ dnLognormal(mean=C2_mu, sd=C2_sd, offset=-age_C2)
obs_C2.clamp(-C2_offset)

# No clock and no dnPhyloCTMC node: the cognate data are not part of this model. (In
RevBayes
# 1.4.1 run(underPrior=true) reports "no effect" here; also, clamped
# calibration "observations" count as likelihood terms, so leaving out the
# data node is the clean way to keep the calibrations and drop the data.)

mymodel = model(timetree)
monitors.append( mnModel(filename=out_dir + "model.log", printgen=20) )
monitors.append( mnScreen(printgen=10000, root_age, age_C1, age_C2) )
mymcmc = mcmc(mymodel, moves, monitors)
mymcmc.run(generations=n_gen, tuningInterval=200)

q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
calibrations x Yule prior, no data	age_C1 effective prior	1.46	1.939	1.205	2.597	1841	info
calibrations x Yule prior, no data	age_C2 effective prior	4.41	4.301	3.859	4.862	1748	info
calibrations x Yule prior, no data	root_age effective prior	6.00	4.317	3.857	4.899	1355	info

H.4.15 Rate variation among meanings biases dates

Script `15_gamma_rate_variation_dating_bias.Rev` is the Bayesian counterpart of the glottochronology bias of Section 9.2.5. The data were simulated with strong gamma-distributed rate variation among characters ($\alpha = 0.5$). They are dated with only the young calibration C1, so the root age is extrapolated, as in a glottochronological date. The script is run with discrete-gamma rate variation (four categories, α estimated) and without it. Ignoring the variation underestimates the amount of change on long paths and should make the deep dates too young (Equation (C.8)).

```
################################################################################
# 15_gamma_rate_variation_dating_bias.Rev
#
# Model    : strict clock dating with and without gamma-distributed rate
#            variation among meanings (cognate characters). Data simulated
#            with strong among-character rate variation (alpha = 0.5). Fitting
#            a model without it underestimates the amount of change on long
#            paths (saturation is mistaken for less change), which biases
#            deep node ages young: the Bayesian analogue of the glottochronology
#            bias of Equation clocks-gamma. Only the young calibration C1 is
#            used, so the root age is extrapolated as in a glottochronological
#            date. Fixed true topology.
#              gamma   : Gamma(alpha, alpha) site rates, 4 discrete categories
#              nogamma : all characters share one rate
# Book     : Ch. 9 "Rate variation among meanings biases dates"
#            (clocks-glotto-gamma); "Rate variation across meanings"
(clocks-meanings)
# Estimates: root age, clock rate, alpha
# Data     : data/time/clock_gamma.nex (12 languages, 600 characters, alpha = 0.5)
# Runtime  : ~1 min per model
# Usage    : cd revbayes && rb analyses/time/15_gamma_rate_variation_dating_bias.Rev
--args MODEL [n_gen]
#            MODEL in {gamma, nogamma}
################################################################################

seed(1515)
model_name = "gamma"
n_gen = 15000
if (args.size() > 1) model_name = args[2]
if (args.size() > 2) n_gen = int(args[3])
```

```
out_dir = "output/time/15_gamma_rate_variation_dating_bias/" + model_name + "_"

data <- readDiscreteCharacterData("data/time/clock_gamma.nex")
taxa <- data.taxa()
moves    = VectorMoves()
monitors = VectorMonitors()

root_age ~ dnUniform(0.5, 30.0)
birth_rate ~ dnExponential(1.0)
timetree ~ dnBDP(lambda=birth_rate, mu=0.0, rho=1.0, rootAge=root_age, taxa=taxa)
start_tree <- readTrees("data/time/tree12_start.tre")[1]
root_age.setValue(start_tree.rootAge())
timetree.setValue(start_tree)
moves.append( mvScale(birth_rate, lambda=1.0, tune=true, weight=1.0) )
moves.append( mvSlide(root_age, delta=0.5, tune=true, weight=3.0) )
moves.append( mvTreeScale(tree=timetree, rootAge=root_age, delta=0.5, tune=true,
weight=3.0) )
moves.append( mvNodeTimeSlideUniform(timetree, weight=15.0) )
moves.append( mvSubtreeScale(timetree, weight=5.0) )

# young calibration only
source("data/time/calibrations_tree12.Rev")
age_C1 := tmrca(timetree, clade_C1)
obs_C1 ~ dnSoftBoundUniformNormal(min=C1_min - age_C1, max=C1_max - age_C1, sd=C1_
sd, p=0.95)
obs_C1.clamp(0.0)

clock_rate ~ dnExponential(2.0)
moves.append( mvScale(clock_rate, lambda=0.5, tune=true, weight=5.0) )
up_down = mvUpDownScale(lambda=0.3, tune=true, weight=5.0)
up_down.addVariable(root_age, up=true)
up_down.addVariable(timetree, up=true)
up_down.addVariable(clock_rate, up=false)
moves.append( up_down )

pi ~ dnDirichlet(v(1, 1))
pi1 := pi[2]
Q := fnF81(pi)
moves.append( mvBetaSimplex(pi, alpha=20, tune=true, weight=2.0) )

if (model_name == "gamma") {
  alpha ~ dnExponential(1.0)
  site_rates := fnDiscretizeGamma(alpha, alpha, 4)
  moves.append( mvScale(alpha, lambda=0.5, tune=true, weight=3.0) )
  seq ~ dnPhyloCTMC(tree=timetree, Q=Q, branchRates=clock_rate, siteRates=site_
rates, type="Standard")
} else {
  seq ~ dnPhyloCTMC(tree=timetree, Q=Q, branchRates=clock_rate, type="Standard")
```

```
}
seq.clamp(data)

mymodel = model(timetree)
monitors.append( mnModel(filename=out_dir + "model.log", printgen=10) )
monitors.append( mnScreen(printgen=5000, root_age, clock_rate) )
mymcmc = mcmc(mymodel, moves, monitors)
mymcmc.run(generations=n_gen, tuningInterval=200)

q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
strict clock + Gamma(4) among-meaning rates	root_age	6.00	4.606	1.490	8.310	799	pass
strict clock + Gamma(4) among-meaning rates	clock_rate	0.15	0.201	0.082	0.391	873	pass
strict clock + Gamma(4) among-meaning rates	alpha	0.50	0.449	0.330	0.577	815	pass
strict clock, no among-meaning rate variation (misspecified)	root_age	6.00	2.309	0.726	3.947	676	info-fail
strict clock, no among-meaning rate variation (misspecified)	clock_rate	0.15	0.134	0.054	0.248	700	info-pass
gamma vs no gamma	root_age_bias_nogamma_minus_gamma	NA	-2.297	NA	NA	NA	pass

With gamma rate variation the root is recovered (posterior mean 4.6, HPD 1.5 to 8.3 ky; truth 6) and so is α (0.45; truth 0.5). Without it the root is far too young (2.3 ky, HPD 0.7 to 3.9 ky), which is the bias predicted by Equation (C.8).

H.4.16 Models not implemented

Three models from Chapters 9, 10 and 16 have no script here.

- *Penalized likelihood* and rate smoothing (Section 9.5.5) are not Bayesian. They are covered by `ape::chronos()` in the book itself.
- The optimized relaxed clock (Section 9.5.4) is a set of MCMC proposals for the UCLN model rather than a different model. Script 02 uses the corresponding RevBayes moves (`mvRateAgeBetaShift`, `mvUpDownScale`).
- Stratigraphic-range FBD models (`dnFBDRP`), skyline FBD models with a sampling rate that changes when writing appears (the `timeline` and `psiTimeline` arguments of `dnFBDP`), HiSSE and calibration cross-validation can all be expressed with the RevBayes 1.4.1 functions listed above. They are left as extensions, because verifying them would need either more ancient languages or much larger trees than these small test data sets have.

H.5 Phylogeography, ancestral states and comparative models

This section documents the RevBayes (Höhna et al., 2016) scripts for the trait models of Chapters 11, 12, 13 and 14: discrete and continuous phylogeography, range evolution, Mk models and ancestral states, stochastic character mapping, hidden rates, correlated evolution, Brownian motion and its relatives, phylogenetic regression and signal, threshold and spatial models. Each model has one standalone script in `revbayes/analyses/traits/`, run on a small data set simulated with known parameter values by `revbayes/data/sim_traits.R` (true values in `revbayes/data/traits/true_values.csv`); `revbayes/tests/check_traits.R` reads the output, computes posterior means, 95% highest-posterior-density (HPD) intervals and effective sample sizes (ESS, **coda**), checks whether the truth is covered and runs the model-specific checks described below, writing `revbayes/tests/results_traits.csv`.

Conventions. Most scripts condition on one fixed, known time tree of 16 languages (`data/traits/tree16.tre`, a Yule tree with root age 4; time is in millennia, so rates are per thousand years). Three scripts integrate over tree uncertainty instead (13, 14 and 22). Coordinates are planar, in units of 100 km. Scripts are run from `revbayes/`:

```
cd revbayes
Rscript data/sim_traits.R                                  # simulate all data sets
rb analyses/traits/05_mk_asr.Rev                           # one analysis, default
settings
rb analyses/traits/02_continuous_phylogeo_bm.Rev --args 20000   # longer chain
bash tests/run_traits.sh                                   # all analyses, 2 at a time
Rscript tests/check_traits.R                               # writes
tests/results_traits.csv
```

The first command-line argument after `--args` is the number of MCMC generations. Several scripts fit more than one model; each then also estimates the marginal likelihood of every model by stepping-stone and path sampling (`powerPosterior`, Chapter 6) and writes them to `marginal_likelihoods.csv` in its output folder.

Two RevBayes pitfalls shaped these scripts. First, `model(x)` collects *every* node connected to `x`, including through shared constant nodes (a tree object, a prior constant such as `rate <- 2.0`) and through the elements of one Rev vector (`x[1]` and `x[2]`). Two models built in the same script that share such a node are silently merged, and a power posterior then adds their likelihoods. Each model in a multi-model script therefore has its own tree copy and its own variable names. Second, inside a loop an index such as `i <- n - j` is a constant node that is re-evaluated when it is reassigned, so every distribution created with `tree.branchLength(i)` ends up using the last value; loops therefore run over the loop variable itself (`for (i in (n - 1):1)`).

The verification standard is the one used throughout this appendix: the script runs; the 95% HPD interval of each key parameter contains the simulated value (or the failure is explained); ESS is reported (100 or more counts as adequate); model comparisons are reported whether or not they pick the generating model; and, where possible, the Bayesian result is compared with an independent calculation (maximum likelihood from **ape**, **phytools**, **geiger** or **nlme**, or an exact numerical result). With 16 languages and one character, many parameters are only weakly identified: wide intervals that contain the truth are the expected outcome, and so are model comparisons that cannot separate similar models. The table summarizes the checks; “miss” marks a check that did not come out as the generating model would predict, “expected_miss” a failure

that the book itself predicts (script 16), and "warn_ess" an ESS below 100. The text explains each one.

Script	Checks	Pass	Other outcomes	Runtime (s)
01_discrete_phylogeo_bssvs	13	11	miss	46
02_continuous_phylogeo_bm	6	6		119
03_relaxed_random_walk	5	5		44
04_dec	4	4		71
05_mk_asr	11	10	miss	109
06_stochastic_mapping	7	7		8
07_hidden_rates	4	4		62
08_pagel_correlated	14	14		121
09_bm_ou	6	6		154
10_early_burst	3	3		60
11_mvbm_correlation	5	5		9
12_pagel_lambda	2	2		12
13_asr_tree_sample	3	3		47
14_asr_joint_tree_trait	4	4		82
15_phylogeo_glm	6	5	miss	26
16_bm_trend_ancient_tips	11	10	expected_miss	105
17_threshold_model	2	2		2
18_ordered_political	6	5	miss	69
19_phylo_spatial	5	5		139
20_ou_multi_optimum	5	4	warn_ess	153
21_root_treatment_asr	18	18		144
22_pathway_rj_tree_sample	13	12	miss	955

H.5.1 Discrete phylogeography with BSSVS

`01_discrete_phylogeo_bssvs.Rev` implements the discrete phylogeography of Section 11.4.1 with Bayesian stochastic search variable selection (Section 11.4.2, Equation (11.3)). Four regions A-D lie on a line, and the data were simulated with only the three adjacent routes (AB, BC, CD) active, at rate 0.3 per millennium, from a homeland in A. Each of the six symmetric route rates is a reversible-jump mixture of a point mass at zero (route absent) and an exponential rate, so the indicators δ_{ij} are the moves of a reversible-jump sampler (`dnReversibleJumpMixture` + `mvRJSwitch`). The book's prior puts Poisson($\ln 2$) plus $K-1$ on the number of active routes; RevBayes has no built-in prior on the sum of indicators, so the script uses independent indicators with the same expected number of active routes, i.e. prior inclusion probability $q = (\ln 2 + K - 1)/(K(K-1)/2) = 0.62$, which makes the book's Bayes factor formula exact. The root region is sampled with `mnJointConditionalAncestralState` (two runs).

```
################################################################################
# 01_discrete_phylogeo_bssvs.Rev
#
# Model:    Discrete phylogeography. The region of a lineage (K = 4 regions,
```

```
#           A-D) evolves as a symmetric CTMC whose route rates are switched on
#           and off by Bayesian stochastic search variable selection (BSSVS),
#           implemented with reversible-jump mixtures: each route rate q_ij is
#           0 (route absent, delta_ij = 0) or drawn from an exponential prior
#           (delta_ij = 1).
# Book:     Chapter 11, "Locations as states of a Markov chain"
#           (#phylogeography-ctmc) and "Bayesian stochastic search variable
#           selection" (#phylogeography-bssvs), Eqs. phylogeography-bssvs and
#           phylogeography-bf.
# Estimates: posterior inclusion probability of each route (-> Bayes factor),
#           route rates, and the posterior of the root (homeland) region.
# Data:     data/traits/regions.nex on the fixed time tree data/traits/tree16.tre
# Run:      cd revbayes && rb analyses/traits/01_discrete_phylogeo_bssvs.Rev [--args
n_gen]
# Runtime:  about 40 s (2 runs x 10,000 generations)
################################################################################

seed(1101)
n_gen = 10000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/traits/01_discrete_phylogeo_bssvs/"
t_start = time("seconds")

tree    <- readTrees("data/traits/tree16.tre")[1]
regions <- readDiscreteCharacterData("data/traits/regions.nex")

K        <- 4
n_routes <- K * (K - 1) / 2
# Prior inclusion probability of one route, q = (ln 2 + K - 1) / (K(K-1)/2).
# The book's BSSVS prior puts Poisson(ln 2) + (K - 1) on the number of active
# routes; we use independent indicators with the same expected number of
# active routes, so that the Bayes factor formula of the book applies exactly.
q_incl <- (ln(2) + K - 1) / n_routes

moves    = VectorMoves()
monitors = VectorMonitors()

# route rates in the order AB, AC, AD, BC, BD, CD
for (k in 1:n_routes) {
    r[k] ~ dnReversibleJumpMixture(0.0, dnExponential(1.0), p = 1.0 - q_incl)
    r[k].setValue(0.5)
    delta[k] := ifelse(r[k] > 0.0, 1, 0)
    moves.append( mvRJSwitch(r[k], weight = 1.0) )
    moves.append( mvScale(r[k], lambda = 0.5, weight = 1.0) )
}
n_active := sum(delta)

# symmetric 4 x 4 matrix; off-diagonal elements in row order
```

```
Q := fnFreeK([ r[1], r[2], r[3],
               r[1], r[4], r[5],
               r[2], r[4], r[6],
               r[3], r[5], r[6] ], rescaled = FALSE)

root_freq <- simplex(rep(1, K))         # flat prior on the homeland region
ctmc ~ dnPhyloCTMC(tree = tree, Q = Q, rootFrequencies = root_freq,
                   type = "Standard", nSites = 1)
ctmc.clamp(regions)

mymodel = model(Q)

monitors.append( mnModel(filename = out_dir + "model.log", printgen = 10) )
monitors.append( mnJointConditionalAncestralState(tree = tree, ctmc = ctmc,
                 type = "Standard", withTips = FALSE, withStartStates = FALSE,
                 filename = out_dir + "states.log", printgen = 10) )
monitors.append( mnScreen(printgen = 5000, n_active) )

mymcmc = mcmc(mymodel, monitors, moves, nruns = 2, combine = "none")
mymcmc.burnin(generations = 2000, tuningInterval = 100)
mymcmc.run(generations = n_gen, tuningInterval = 0)

write("runtime_s", time("seconds") - t_start, filename = out_dir + "runtime.txt",
separator = "\t")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
BSSVS discrete phylogeography	delta_AB	1.0	0.971	NA	NA	1902	pass
BSSVS discrete phylogeography	delta_AC	0.0	0.525	NA	NA	1902	pass
BSSVS discrete phylogeography	delta_AD	0.0	0.586	NA	NA	1881	pass
BSSVS discrete phylogeography	delta_BC	1.0	0.580	NA	NA	3581	miss
BSSVS discrete phylogeography	delta_BD	0.0	0.642	NA	NA	1828	miss
BSSVS discrete phylogeography	delta_CD	1.0	0.787	NA	NA	1902	pass
BSSVS discrete phylogeography	r[1]	0.3	1.335	0	3.301	1902	pass
BSSVS discrete phylogeography	r[2]	0.0	0.371	0	1.756	1753	pass
BSSVS discrete phylogeography	r[3]	0.0	0.442	0	1.916	1902	pass

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
BSSVS discrete phylogeography	r[4]	0.3	0.410	0	1.778	1902	pass
BSSVS discrete phylogeography	r[5]	0.0	0.530	0	2.189	2004	pass
BSSVS discrete phylogeography	r[6]	0.3	0.773	0	2.598	1902	pass
BSSVS discrete phylogeography	root_state	0.0	0.328	NA	NA	1902	pass

For indicators, the posterior mean is the posterior inclusion probability and a check passes when the Bayes factor points the right way (BF $>$ 1 for a true route, $<$ 1 for an absent one). Route AB is decisively supported (BF about 20) and CD positively (BF about 2), while BC and the absent route BD are both near the prior (BF 0.9 and 1.1): one location character on 16 tips carries little information about six routes, exactly the weak identifiability that Section 11.4.2 warns about. The root posterior puts the most mass on the true homeland A (0.33, with B close behind), in line with the maximum-likelihood (ML) marginal from `ape::ace`.

H.5.2 Continuous phylogeography: bivariate Brownian diffusion

`02_continuous_phylogeo_bm.Rev` implements Equation (11.5) (Section 11.5.1): the coordinates of every node are parameters, each child is bivariate normal around its parent with covariance $t_c\Sigma$, and the tip locations are clamped. The diffusion matrix is built from two rates and a correlation with `fnVarCovar`, and the root location (the homeland) has a vague bivariate normal prior. Since the internal locations are sampled, the same script gives ancestral locations for every proto-language.

```
################################################################################
# 02_continuous_phylogeo_bm.Rev
#
# Model:    Continuous phylogeography by bivariate Brownian diffusion of
#           projected coordinates (x = km east, y = km north, units of 100 km).
#           Every node location is a parameter (data augmentation):
#               x_c | x_a ~ N_2(x_a, t_c * Sigma),
#           Sigma = [[s_x^2, rho s_x s_y], [rho s_x s_y, s_y^2]].
#           Tip locations are clamped to the observed coordinates.
# Book:     Chapter 11, "Brownian diffusion" (#phylogeography-bm),
#           Eq. phylogeography-bm and phylogeography-mvn.
# Estimates: root location (homeland), diffusion rates s_x, s_y and rho;
#           ancestral locations of all internal nodes.
# Data:     data/traits/coords_bm.tsv on data/traits/tree16.tre
# Run:      cd revbayes && rb analyses/traits/02_continuous_phylogeo_bm.Rev [--args
n_gen]
# Runtime:  about 2 minutes (2 runs x 7,000 generations)
################################################################################

seed(1102)
```

```
n_gen = 7000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/traits/02_continuous_phylogeo_bm/"
t_start = time("seconds")

tree   <- readTrees("data/traits/tree16.tre")[1]
coords = readDataDelimitedFile("data/traits/coords_bm.tsv", header = TRUE, delimiter
= "\t")
n_tips  <- tree.ntips()
n_nodes <- tree.nnodes()
root_idx <- tree.getRootIndex()

# observed tip coordinates, indexed by RevBayes node index
for (r in 1:coords.size()) {
    obs[tree.tipIndex(coords[r][1])] <- [ coords[r][2], coords[r][3] ]
}

moves    = VectorMoves()
monitors = VectorMonitors()

# diffusion matrix
sigma_x ~ dnLognormal(0.0, 1.0)
sigma_y ~ dnLognormal(0.0, 1.0)
rho     ~ dnUniform(-1.0, 1.0)
Sigma := fnVarCovar([sigma_x, sigma_y], [rho])
moves.append( mvScale(sigma_x, lambda = 0.3, weight = 3.0) )
moves.append( mvScale(sigma_y, lambda = 0.3, weight = 3.0) )
moves.append( mvSlide(rho, delta = 0.2, weight = 3.0) )

# root location: vague bivariate normal prior (sd = 10, i.e. 1000 km)
# (mvVectorSlide element indices are 0-based)
loc[root_idx] ~ dnMultivariateNormal([0.0, 0.0], covariance = fnVarCovar([10.0,
10.0], [0.0]))
loc[root_idx].setValue([0.0, 0.0])
moves.append( mvVectorSlide(loc[root_idx], delta = 0.5, weight = 3.0) )
moves.append( mvVectorSlide(loc[root_idx], elements = [0], delta = 0.5, weight =
2.0) )
moves.append( mvVectorSlide(loc[root_idx], elements = [1], delta = 0.5, weight =
2.0) )

# all other nodes, from the root down (parents have larger indices than children)
# NB: the loop index must be the loop variable itself (a constant node such as
# `i <- n_nodes - j` would be re-evaluated when it is reassigned).
for (i in (n_nodes - 1):1) {
    loc[i] ~ dnMultivariateNormal(loc[tree.parent(i)], covariance = Sigma,
                                  scale = tree.branchLength(i))
    if (i <= n_tips) {
        loc[i].clamp(obs[i])
```

```
    } else {
        loc[i].setValue([0.0, 0.0])
        moves.append( mvVectorSlide(loc[i], elements = [0], delta = 0.3, weight =
        1.0) )
        moves.append( mvVectorSlide(loc[i], elements = [1], delta = 0.3, weight =
        1.0) )
    }
}

root_x := loc[root_idx][1]
root_y := loc[root_idx][2]

mymodel = model(Sigma)

monitors.append( mnFile(sigma_x, sigma_y, rho, root_x, root_y,
                        filename = out_dir + "model.log", printgen = 10) )
monitors.append( mnScreen(printgen = 5000, root_x, root_y, sigma_x) )

mymcmc = mcmc(mymodel, monitors, moves, nruns = 2, combine = "none")
mymcmc.burnin(generations = 1000, tuningInterval = 100)
mymcmc.run(generations = n_gen, tuningInterval = 0)

write("runtime_s", time("seconds") - t_start, filename = out_dir + "runtime.txt",
separator = "\t")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
bivariate BM diffusion	root_x	2.0	2.223	-0.176	4.337	694	pass
bivariate BM diffusion	root_y	-1.0	-2.342	-3.982	-0.576	588	pass
bivariate BM diffusion	sigma_x	1.0	1.166	0.761	1.624	739	pass
bivariate BM diffusion	sigma_y	0.7	0.815	0.555	1.124	863	pass
bivariate BM diffusion	rho	0.4	0.480	0.130	0.829	1200	pass
bivariate BM diffusion	root_vs_GLS	NA	0.108	NA	NA	NA	pass

All five parameters are covered. The posterior mean homeland agrees with the generalized least-squares root of Equation (11.6), computed independently in R, to within about 10 km (`root_vs_GLS`). That estimate lies about 140 km south of the simulated homeland, which sits near the edge of the 95% interval: the GLS root is a weighted average of this particular realization of the walk, and the interval is honest about it.

H.5.3 Relaxed random walk

`03_relaxed_random_walk.Rev` implements the lognormal relaxed random walk of Section 11.5.2 (Equation (11.7)). Each branch has a variance multiplier ϕ_b with mean one, and the internal locations are integrated out by `dnPhyloMultivariateBrownianREML` (the per-branch argument of

that distribution is a standard-deviation multiplier, so it receives $\sqrt{\phi_b}$). The multipliers use a non-centered parameterization, $\phi_b = \exp(-s^2/2 + s z_b)$ with $z_b \sim N(0,1)$. The model is unchanged, but a first, centered version got stuck in the "funnel" near $s = 0$ (ESS below 20). The script also reports a weighted diffusion rate.

```
################################################################################
# 03_relaxed_random_walk.Rev
#
# Model:    Relaxed random walk (RRW) for continuous coordinates. Bivariate
#           Brownian diffusion with diffusion matrix Sigma, where every branch b
#           has its own variance multiplier phi_b ~ Lognormal(-s^2/2, s) (mean 1):
#               x_c | x_a, phi_b ~ N_2(x_a, phi_b * t_c * Sigma).
#           The root and internal locations are integrated out analytically
#           (restricted likelihood, dnPhyloMultivariateBrownianREML).
# Book:     Chapter 11, "The relaxed random walk" (#phylogeography-rrw),
#           Eq. phylogeography-rrw (lognormal version).
# Estimates: sigma_x, sigma_y, rho, the spread s of the log multipliers,
#           branch multipliers phi_b, and the weighted diffusion rate.
# Data:     data/traits/coords_rrw.nex on data/traits/tree16.tre
# Run:      cd revbayes && rb analyses/traits/03_relaxed_random_walk.Rev [--args n_
gen]
# Runtime:  about 1 minute
################################################################################

seed(1103)
n_gen = 30000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/traits/03_relaxed_random_walk/"
t_start = time("seconds")

tree   <- readTrees("data/traits/tree16.tre")[1]
coords <- readContinuousCharacterData("data/traits/coords_rrw.nex")
n_branches <- tree.nnodes() - 1

moves    = VectorMoves()
monitors = VectorMonitors()

sigma_x ~ dnLognormal(0.0, 1.0)
sigma_y ~ dnLognormal(0.0, 1.0)
rho     ~ dnUniform(-1.0, 1.0)
Sigma := fnVarCovar([sigma_x, sigma_y], [rho])
moves.append( mvScale(sigma_x, lambda = 0.3, weight = 2.0) )
moves.append( mvScale(sigma_y, lambda = 0.3, weight = 2.0) )
moves.append( mvSlide(rho, delta = 0.2, weight = 2.0) )

# lognormal branch multipliers with mean one; s has an exponential prior.
# Non-centred parameterization, phi_b = exp(-s^2/2 + s z_b) with z_b ~ N(0, 1),
# which is the same model as phi_b ~ Lognormal(-s^2/2, s) but avoids the
```

```
# "funnel" between s and the phi_b that traps a centred sampler near s = 0.
s_phi ~ dnExponential(1.0)
moves.append( mvScale(s_phi, lambda = 0.3, weight = 3.0) )
m_phi := -0.5 * s_phi * s_phi
for (b in 1:n_branches) {
    z_phi[b] ~ dnNormal(0.0, 1.0)
    moves.append( mvSlide(z_phi[b], delta = 1.0, weight = 1.0) )
    phi[b] := exp(m_phi + s_phi * z_phi[b])
    # the REML likelihood takes per-branch standard-deviation multipliers
    br_sd[b] := sqrt(phi[b])
}

X ~ dnPhyloMultivariateBrownianREML(tree, branchRates = br_sd, rateMatrix = Sigma)
X.clamp(coords)

# weighted diffusion rate: expected squared displacement per unit time in the
# plane is (sigma_x^2 + sigma_y^2) times the time-weighted mean multiplier
for (b in 1:n_branches) bl[b] <- tree.branchLength(b)
total_time <- sum(bl)
for (b in 1:n_branches) w_phi[b] := phi[b] * bl[b] / total_time
mean_phi := sum(w_phi)
weighted_diffusion := (sigma_x * sigma_x + sigma_y * sigma_y) * mean_phi / 4.0

mymodel = model(Sigma)

monitors.append( mnModel(filename = out_dir + "model.log", printgen = 10) )
monitors.append( mnScreen(printgen = 5000, sigma_x, sigma_y, s_phi) )

mymcmc = mcmc(mymodel, monitors, moves, nruns = 1)
mymcmc.burnin(generations = 2000, tuningInterval = 100)
mymcmc.run(generations = n_gen, tuningInterval = 0)

write(tree, filename = out_dir + "tree_index.txt")
write("runtime_s", time("seconds") - t_start, filename = out_dir + "runtime.txt",
separator = "\t")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
relaxed random walk (lognormal)	sigma_x	0.8	0.756	0.444	1.145	2129	pass
relaxed random walk (lognormal)	sigma_y	0.8	0.665	0.387	0.999	2119	pass
relaxed random walk (lognormal)	rho	0.3	0.390	-0.112	0.800	2851	pass
relaxed random walk (lognormal)	s_phi	0.8	0.503	0.001	1.315	1078	pass

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
relaxed random walk (lognormal)	cor(true phi, posterior phi)	NA	0.172	NA	NA	NA	pass

The diffusion parameters and the spread s are covered. The individual branch multipliers are barely identified with 16 tips: the rank correlation between the realized and posterior-mean multipliers is weak but positive.

H.5.4 DEC range evolution

`04_dec.Rev` implements the dispersal-extinction-cladogenesis model described in the biologist's toolbox of Chapter 11 (Section 11.12; Ree et al. (2005); Ree and Smith (2008)), for three areas. The anagenetic rate matrix comes from `fnDECRateMatrix` (dispersal rate d per area pair, local extinction rate e; the null range is excluded, so a single-area range cannot be lost) and cladogenesis from `fnDECCladoProbs` with subset sympatry and allopatry (`probType = "class"`: a class is chosen with probability `clado_probs`, then a pattern uniformly within it, which is exactly how `sim_traits.R` simulates). Ranges are coded as presence/absence per area and converted with `formatDiscreteCharacterData(..., "DEC")`.

```
################################################################################
# 04_dec.Rev
#
# Model:    DEC (dispersal-extinction-cladogenesis) range evolution, three areas
#           A, B, C. Anagenetic CTMC on ranges (dispersal rate d per area pair,
#           local extinction rate e per area; the null range is excluded, so a
#           single-area range cannot be lost), plus cladogenetic range
#           inheritance at nodes: narrow sympatry for one-area ranges, and
#           subset sympatry ("s") or allopatry ("a") for wider ranges.
# Book:     Chapter 11, "From the biologist's toolbox" (#phylogeography-toolbox),
#           the DEC paragraph (Ree & Smith 2008), Bayesian version of Landis et al.
2013.
# Estimates: d, e, the cladogenetic class probability of subset sympatry, and
#           the ancestral range at the root.
# Data:     data/traits/ranges.nex (presence/absence in A, B, C) on tree16.tre
# Run:      cd revbayes && rb analyses/traits/04_dec.Rev [--args n_gen]
# Runtime:  about 1 minute
################################################################################

seed(1104)
n_gen = 10000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/traits/04_dec/"
t_start = time("seconds")

tree   <- readTrees("data/traits/tree16.tre")[1]
rng01  <- readDiscreteCharacterData("data/traits/ranges.nex")
n_areas  <- rng01.nchar()
```

```
n_states <- 2^n_areas
rng = formatDiscreteCharacterData(rng01, "DEC", n_states)

moves    = VectorMoves()
monitors = VectorMonitors()

dispersal   ~ dnExponential(2.0)
extirpation ~ dnExponential(2.0)
moves.append( mvScale(dispersal,   lambda = 0.5, weight = 3.0) )
moves.append( mvScale(extirpation, lambda = 0.5, weight = 3.0) )

for (i in 1:n_areas) {
    for (j in 1:n_areas) {
        dr[i][j] := dispersal
        er[i][j] <- 0.0
    }
}
for (i in 1:n_areas) er[i][i] := extirpation

Q_DEC := fnDECRateMatrix(dispersalRates = dr, extirpationRates = er, nullRange =
"Exclude")

# cladogenesis: probability of subset sympatry vs allopatry (class weights)
clado_probs ~ dnDirichlet([1, 1])
moves.append( mvSimplexElementScale(clado_probs, alpha = 10.0, weight = 2.0) )
p_sympatry := clado_probs[1]
P_DEC := fnDECCladoProbs(eventProbs = clado_probs, eventTypes = ["s", "a"],
                         numCharacters = n_areas, probType = "class")

# flat prior over the non-empty ranges at the root
rf <- rep(1, n_states)
rf[1] <- 0
root_freq <- simplex(rf)

m_bg ~ dnPhyloCTMCClado(tree = tree, Q = Q_DEC, cladoProbs = P_DEC,
                        rootFrequencies = root_freq, type = "NaturalNumbers", nSites
                        = 1)
m_bg.clamp(rng)

mymodel = model(m_bg)

monitors.append( mnModel(filename = out_dir + "model.log", printgen = 10) )
monitors.append( mnJointConditionalAncestralState(tree = tree, ctmc = m_bg, type =
"NaturalNumbers",
                 withTips = TRUE, withStartStates = TRUE,
                 filename = out_dir + "states.log", printgen = 10) )
monitors.append( mnScreen(printgen = 5000, dispersal, extirpation) )
```

```
mymcmc = mcmc(mymodel, monitors, moves, nruns = 1)
mymcmc.burnin(generations = 1000, tuningInterval = 100)
mymcmc.run(generations = n_gen, tuningInterval = 0)

write(tree, filename = out_dir + "tree_index.txt")
write("runtime_s", time("seconds") - t_start, filename = out_dir + "runtime.txt",
separator = "\t")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
DEC	dispersal	0.40	0.577	0.193	1.132	951	pass
DEC	extirpation	0.15	0.300	0.001	0.853	951	pass
DEC	p_sympatry	0.50	0.395	0.004	0.754	951	pass
DEC	root_state	0.00	0.176	NA	NA	NA	pass

d, *e* and the cladogenetic probability are covered. The check script also verifies the state coding (the sampled tip states equal the observed ranges). The root range is nearly uniform over the seven possible ranges. The true range {A} is in the 95% credible set but is not the most probable range: 16 tips and three areas say little about the root.

H.5.5 Mk models and ancestral states for word order

`05_mk_asr.Rev` fits the Mk models of Section 12.4.1 to a three-state word-order feature (SOV, SVO, VSO) simulated under an ordered model (SOV ↔ SVO ↔ VSO) from an SOV root: equal rates (ER), all rates different (ARD) and the ordered model, each with marginal ancestral states (Section 12.4.2) and stepping-stone marginal likelihoods.

```
################################################################################
# 05_mk_asr.Rev
#
# Model:    Mk models for a three-state feature, basic word order
#           (0 = SOV, 1 = SVO, 2 = VSO): equal rates (ER, one rate),
#           all rates different (ARD, six rates) and an ordered model
#           (SOV <-> SVO <-> VSO, four rates, no direct SOV <-> VSO change).
#           For each model: MCMC with marginal ancestral states at the root and
#           at every internal node, and the marginal likelihood by
#           stepping-stone and path sampling (power posteriors).
# Book:     Chapter 12, "The Mk model" (#asr-mk), "Marginal reconstruction"
#           (#asr-marginal), "Bayesian ASR" (#asr-bayes), Section #asr-wordorder;
#           Chapter 6 for marginal likelihoods.
# Estimates: rates, root state posterior, log marginal likelihoods of the three
#           models (the data were generated under the ordered model).
# Data:     data/traits/wordorder.nex on tree16.tre
# Run:      cd revbayes && rb analyses/traits/05_mk_asr.Rev [--args n_gen]
# Runtime:  about 1.5 minutes
################################################################################
```

```
seed(1105)
n_gen = 10000
if (args.size() > 1) n_gen = int(args[2])
n_cats  = 24          # power-posterior stones
pp_gen  = 1000        # generations per stone
out_dir = "output/traits/05_mk_asr/"
t_start = time("seconds")

# NB: every model gets its OWN copy of the tree (and of the root frequencies
# and prior constants):
# model() collects every node connected to its argument, including through
# shared constant nodes, so models that share a tree object would be merged
# into one graph and their likelihoods added in the power posteriors.
tree     <- readTrees("data/traits/tree16.tre")[1]
tree_er  <- readTrees("data/traits/tree16.tre")[1]
tree_ard <- readTrees("data/traits/tree16.tre")[1]
tree_ord <- readTrees("data/traits/tree16.tre")[1]
wo   <- readDiscreteCharacterData("data/traits/wordorder.nex")
# exponential prior on each rate, rate 2 (mean 0.5 per kyr); written inline in
# each model because a shared constant node would also join the model graphs

write("model,lnML_stepping_stone,lnML_path_sampling\n", filename = out_dir +
"marginal_likelihoods.csv")

############################## ER
q_er ~ dnExponential(2.0)
mv_er = VectorMoves()
mv_er.append( mvScale(q_er, lambda = 0.5, weight = 2.0) )
Q_er := fnFreeK([q_er, q_er, q_er, q_er, q_er, q_er], rescaled = FALSE)
ctmc_er ~ dnPhyloCTMC(tree = tree_er, Q = Q_er, rootFrequencies = simplex([1, 1,
1]), type = "Standard", nSites = 1)
ctmc_er.clamp(wo)
mod_er = model(Q_er)
mn_er = VectorMonitors()
mn_er.append( mnModel(filename = out_dir + "ER_model.log", printgen = 10) )
mn_er.append( mnJointConditionalAncestralState(tree = tree_er, ctmc = ctmc_er, type
= "Standard",
              withTips = FALSE, withStartStates = FALSE, filename = out_dir + "ER_
              states.log", printgen = 10) )
mcmc_er = mcmc(mod_er, mn_er, mv_er, nruns = 1)
mcmc_er.burnin(generations = 1000, tuningInterval = 100)
mcmc_er.run(generations = n_gen, tuningInterval = 0)
pp_er = powerPosterior(mod_er, mv_er, VectorMonitors(), out_dir + "ER_powp.log",
cats = n_cats, sampleFreq = 10)
pp_er.burnin(generations = 500, tuningInterval = 100)
pp_er.run(generations = pp_gen)
ss_er = steppingStoneSampler(out_dir + "ER_powp.log", "power", "likelihood",
"\t").marginal()
```

```
ps_er = pathSampler(out_dir + "ER_powp.log", "power", "likelihood", "\t").marginal()
write("ER", ss_er, ps_er, filename = out_dir + "marginal_likelihoods.csv", append =
TRUE, separator = ",")
write("\n", filename = out_dir + "marginal_likelihoods.csv", append = TRUE)

############################ ARD
for (k in 1:6) {
    q_ard[k] ~ dnExponential(2.0)
}
mv_ard = VectorMoves()
for (k in 1:6) mv_ard.append( mvScale(q_ard[k], lambda = 0.5, weight = 1.0) )
Q_ard := fnFreeK(q_ard, rescaled = FALSE)
ctmc_ard ~ dnPhyloCTMC(tree = tree_ard, Q = Q_ard, rootFrequencies = simplex([1, 1,
1]), type = "Standard", nSites = 1)
ctmc_ard.clamp(wo)
mod_ard = model(Q_ard)
mn_ard = VectorMonitors()
mn_ard.append( mnModel(filename = out_dir + "ARD_model.log", printgen = 10) )
mn_ard.append( mnJointConditionalAncestralState(tree = tree_ard, ctmc = ctmc_ard,
type = "Standard",
               withTips = FALSE, withStartStates = FALSE, filename = out_dir + "ARD_
               states.log", printgen = 10) )
mcmc_ard = mcmc(mod_ard, mn_ard, mv_ard, nruns = 1)
mcmc_ard.burnin(generations = 1000, tuningInterval = 100)
mcmc_ard.run(generations = n_gen, tuningInterval = 0)
pp_ard = powerPosterior(mod_ard, mv_ard, VectorMonitors(), out_dir + "ARD_powp.log",
cats = n_cats, sampleFreq = 10)
pp_ard.burnin(generations = 500, tuningInterval = 100)
pp_ard.run(generations = pp_gen)
ss_ard = steppingStoneSampler(out_dir + "ARD_powp.log", "power", "likelihood",
"\t").marginal()
ps_ard = pathSampler(out_dir + "ARD_powp.log", "power", "likelihood",
"\t").marginal()
write("ARD", ss_ard, ps_ard, filename = out_dir + "marginal_likelihoods.csv", append
= TRUE, separator = ",")
write("\n", filename = out_dir + "marginal_likelihoods.csv", append = TRUE)

############################ ordered (SOV <-> SVO <-> VSO)
q_sov_svo ~ dnExponential(2.0)
q_svo_sov ~ dnExponential(2.0)
q_svo_vso ~ dnExponential(2.0)
q_vso_svo ~ dnExponential(2.0)
mv_ord = VectorMoves()
mv_ord.append( mvScale(q_sov_svo, lambda = 0.5, weight = 1.0) )
mv_ord.append( mvScale(q_svo_sov, lambda = 0.5, weight = 1.0) )
mv_ord.append( mvScale(q_svo_vso, lambda = 0.5, weight = 1.0) )
mv_ord.append( mvScale(q_vso_svo, lambda = 0.5, weight = 1.0) )
Q_ord := fnFreeK([q_sov_svo, 0.0, q_svo_sov, q_svo_vso, 0.0, q_vso_svo], rescaled =
FALSE)
```

```
ctmc_ord ~ dnPhyloCTMC(tree = tree_ord, Q = Q_ord, rootFrequencies = simplex([1, 1,
1]), type = "Standard", nSites = 1)
ctmc_ord.clamp(wo)
mod_ord = model(Q_ord)
mn_ord = VectorMonitors()
mn_ord.append( mnModel(filename = out_dir + "ORD_model.log", printgen = 10) )
mn_ord.append( mnJointConditionalAncestralState(tree = tree_ord, ctmc = ctmc_ord,
type = "Standard",
               withTips = FALSE, withStartStates = FALSE, filename = out_dir + "ORD_
               states.log", printgen = 10) )
mcmc_ord = mcmc(mod_ord, mn_ord, mv_ord, nruns = 1)
mcmc_ord.burnin(generations = 1000, tuningInterval = 100)
mcmc_ord.run(generations = n_gen, tuningInterval = 0)
pp_ord = powerPosterior(mod_ord, mv_ord, VectorMonitors(), out_dir + "ORD_powp.log",
cats = n_cats, sampleFreq = 10)
pp_ord.burnin(generations = 500, tuningInterval = 100)
pp_ord.run(generations = pp_gen)
ss_ord = steppingStoneSampler(out_dir + "ORD_powp.log", "power", "likelihood",
"\t").marginal()
ps_ord = pathSampler(out_dir + "ORD_powp.log", "power", "likelihood",
"\t").marginal()
write("ORD", ss_ord, ps_ord, filename = out_dir + "marginal_likelihoods.csv", append
= TRUE, separator = ",")
write("\n", filename = out_dir + "marginal_likelihoods.csv", append = TRUE)

write(tree, filename = out_dir + "tree_index.txt")
write("runtime_s", time("seconds") - t_start, filename = out_dir + "runtime.txt",
separator = "\t")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
Mk ER	root_state	0.0	0.121	NA	NA	951	pass
Mk ARD	root_state	0.0	0.285	NA	NA	951	pass
Mk ORD	root_state	0.0	0.307	NA	NA	951	pass
Mk ordered (generating)	q_sov_svo	0.3	0.749	0.006	1.858	951	pass
Mk ordered (generating)	q_svo_sov	0.1	0.169	0.004	0.455	951	pass
Mk ordered (generating)	q_svo_vso	0.2	0.250	0.001	0.621	733	pass
Mk ordered (generating)	q_vso_svo	0.3	0.727	0.001	1.845	717	pass
Mk ER	lnML	NA	-12.531	NA	NA	NA	pass
Mk ARD	lnML	NA	-11.419	NA	NA	NA	pass
Mk ORD	lnML	NA	-11.642	NA	NA	NA	pass
Mk model comparison	best_model	NA	NA	NA	NA	NA	miss

Two exact checks are possible for the one-parameter ER model. Integrating the likelihood numerically over the rate gives a log marginal likelihood of -12.526; stepping-stone sampling

gives −12.531. The exact Bayesian root probabilities (0.11/0.78/0.11) match the MCMC values (0.12/0.76/0.12). The empirical-Bayes root from `ape::ace` (0.02/0.96/0.02) is much more confident, as Section 12.4.2 predicts, because it fixes the rate at its ML value. The tips are mostly SVO, so every model puts most mass on an SVO root even though the simulated history started in SOV; SOV stays in the 95% credible set. The three models' marginal likelihoods lie within 1.1 log units of each other (ARD −11.42, ordered −11.64, ER −12.53). One character on 16 tips cannot separate them, so the "best model" check is a miss, as expected.

H.5.6 Stochastic character mapping

`06_stochastic_mapping.Rev` fits a two-state Mk model and draws one complete character history per sample with `mnStochasticCharacterMap` (Section 12.6), so the rates are integrated over rather than fixed. `check_traits.R` parses the SIMMAP strings (which list the tip end of each branch first) to count changes of each type (Markov jumps) and to sum the time spent in each state (rewards; Section 11.4.3). It compares them with the history that actually generated the data and with `phytools::make.simmap`.

```
################################################################################
# 06_stochastic_mapping.Rev
#
# Model:    Two-state Mk model (all rates different, q01 and q10) for a binary
#           feature, with stochastic character mapping: at every sampled
#           iteration a complete history of the character along all branches
#           is drawn from its posterior given the tip states, the tree and the
#           current rates (the rates themselves are integrated over by MCMC).
# Book:     Chapter 12, "Stochastic character mapping" (#asr-simmap),
#           Eq. asr-simmap-node; Chapter 11, "Markov jumps and rewards"
#           (#phylogeography-jumps).
# Estimates: q01, q10; posterior distribution of the number of changes of each
#           type (Markov jumps) and of the time spent in each state (rewards,
#           dwell times); compared with the realized true history.
# Data:     data/traits/binary_map.nex on tree16.tre
# Run:      cd revbayes && rb analyses/traits/06_stochastic_mapping.Rev [--args n_
gen]
# Runtime:  about 10 s
################################################################################

seed(1106)
n_gen = 10000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/traits/06_stochastic_mapping/"
t_start = time("seconds")

tree <- readTrees("data/traits/tree16.tre")[1]
feat <- readDiscreteCharacterData("data/traits/binary_map.nex")

moves    = VectorMoves()
monitors = VectorMonitors()
```

```
q01 ~ dnExponential(2.0)
q10 ~ dnExponential(2.0)
moves.append( mvScale(q01, lambda = 0.5, weight = 2.0) )
moves.append( mvScale(q10, lambda = 0.5, weight = 2.0) )
Q := fnFreeK([q01, q10], rescaled = FALSE)

ctmc ~ dnPhyloCTMC(tree = tree, Q = Q, rootFrequencies = simplex([1, 1]),
                   type = "Standard", nSites = 1)
ctmc.clamp(feat)

mymodel = model(Q)

monitors.append( mnModel(filename = out_dir + "model.log", printgen = 10) )
monitors.append( mnStochasticCharacterMap(ctmc = ctmc, filename = out_dir +
"maps.log", printgen = 10) )
monitors.append( mnJointConditionalAncestralState(tree = tree, ctmc = ctmc, type =
"Standard",
                 withTips = TRUE, withStartStates = FALSE, filename = out_dir +
                 "states.log", printgen = 10) )
monitors.append( mnScreen(printgen = 5000, q01, q10) )

mymcmc = mcmc(mymodel, monitors, moves, nruns = 1)
mymcmc.burnin(generations = 1000, tuningInterval = 100)
mymcmc.run(generations = n_gen, tuningInterval = 0)

# summary map on the fixed tree (posterior probabilities of states along branches)
anc_trace = readAncestralStateTrace(out_dir + "maps.log")
characterMapTree(tree, anc_trace, character_file = out_dir + "map_summary_
character.tree",
                 posterior_file = out_dir + "map_summary_posterior.tree", burnin =
                 0.1, reconstruction = "marginal")

write(tree, filename = out_dir + "tree_index.txt")
write("runtime_s", time("seconds") - t_start, filename = out_dir + "runtime.txt",
separator = "\t")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
Mk ARD + stochastic map	q01	0.400	0.700	0.004	1.722	951	pass
Mk ARD + stochastic map	q10	0.250	0.221	0.018	0.560	951	pass
stochastic character mapping	n01	4.000	4.242	0.000	12.000	1045	pass
stochastic character mapping	n10	3.000	3.835	0.000	9.000	951	pass

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
stochastic character mapping	n	7.000	8.077	2.000	21.000	951	pass
stochastic character mapping	dwell_0	6.376	5.175	0.962	11.924	951	pass
stochastic character mapping	dwell_1	19.825	21.025	14.277	25.239	951	pass

The realized numbers of changes (4 gains, 3 losses) and dwell times lie well inside their posterior intervals, and the posterior means agree with the empirical-Bayes maps from **phytools** (8.1 vs 8.0 changes on average). The posterior expected number of changes exceeds the parsimony minimum, as the note in Section 12.6 anticipates.

H.5.7 Hidden-rates model

`07_hidden_rates.Rev` implements the hidden-rates model of Section 12.7 (Equation (D.9)) with a slow and a fast category. The data are expanded with `expandCharacters(2)` so that each observed state is ambiguous between its two hidden copies. The 4×4 matrix is built with `fnFreeK`, because in RevBayes 1.4.1 `fnHiddenStateRateMatrix` applies the same observed-state rates in every category. To avoid label switching, the fast rate is the slow rate plus a positive increment. One binary feature on 16 tips carries almost no information about hidden rates, so 40 independent features were simulated under the same process. The model is compared with a one-category Mk model by stepping-stone sampling.

```
################################################################################
# 07_hidden_rates.Rev
#
# Model:    Hidden-rates model (HRM; covarion-like) for binary features.
#           Each observed state (0/1) is paired with a hidden rate category,
#           slow (S) or fast (F), giving four states 0S, 1S, 0F, 1F with
#               Q = [ Q_S - D_SF   D_SF      ]
#                   [ D_FS        Q_F - D_FS ]
#           (Eq. asr-hrm). Q_S and Q_F are symmetric 2x2 matrices with rates
#           r_slow < r_fast (ordered to avoid label switching) and D_SF = D_FS
#           = switch * I. Only the observed state is known at the tips.
#           Compared with a one-category Mk model by stepping-stone sampling.
# Book:     Chapter 12, "Heterogeneous and hidden-rate models" (#asr-hidden);
#           Chapter 7 (covarion models).
# Estimates: r_slow, r_fast, switch; log marginal likelihood of HRM vs Mk.
# Data:     data/traits/hrm.nex (40 independent binary features) on tree16.tre
# Run:      cd revbayes && rb analyses/traits/07_hidden_rates.Rev [--args n_gen]
# Runtime:  about 1 minute
################################################################################

seed(1107)
n_gen = 10000
if (args.size() > 1) n_gen = int(args[2])
```

```
n_cats  = 20
pp_gen  = 500
out_dir = "output/traits/07_hidden_rates/"
t_start = time("seconds")

# NB: every model gets its OWN copy of the tree (and of the root frequencies):
# model() collects every node connected to its argument, including through
# shared constant nodes, so models that share a tree object would be merged
# into one graph and their likelihoods added in the power posteriors.
tree_h <- readTrees("data/traits/tree16.tre")[1]
tree_m <- readTrees("data/traits/tree16.tre")[1]
feat <- readDiscreteCharacterData("data/traits/hrm.nex")
# expand: observed 0 -> {0S, 0F} (states 0, 2); observed 1 -> {1S, 1F} (states 1, 3)
feat_hidden <- feat.expandCharacters(2)

write("model,lnML_stepping_stone,lnML_path_sampling\n", filename = out_dir +
"marginal_likelihoods.csv")

############################## HRM
r_slow ~ dnExponential(1.0)
r_diff ~ dnExponential(1.0)
r_fast := r_slow + r_diff
switch ~ dnExponential(1.0)
mv_h = VectorMoves()
mv_h.append( mvScale(r_slow, lambda = 0.5, weight = 2.0) )
mv_h.append( mvScale(r_diff, lambda = 0.5, weight = 2.0) )
mv_h.append( mvScale(switch, lambda = 0.5, weight = 2.0) )
# off-diagonal elements in row order (0S, 1S, 0F, 1F)
Q_h := fnFreeK([ r_slow, switch, 0.0,
                 r_slow, 0.0,    switch,
                 switch, 0.0,    r_fast,
                 0.0,    switch, r_fast ], rescaled = FALSE)
ctmc_h ~ dnPhyloCTMC(tree = tree_h, Q = Q_h, rootFrequencies = simplex(rep(1, 4)),
                     type = "NaturalNumbers", nSites = feat.nchar())
ctmc_h.clamp(feat_hidden)
mod_h = model(Q_h)
mn_h = VectorMonitors()
mn_h.append( mnModel(filename = out_dir + "HRM_model.log", printgen = 10) )
mn_h.append( mnScreen(printgen = 5000, r_slow, r_fast, switch) )
mcmc_h = mcmc(mod_h, mn_h, mv_h, nruns = 1)
mcmc_h.burnin(generations = 1000, tuningInterval = 100)
mcmc_h.run(generations = n_gen, tuningInterval = 0)
pp_h = powerPosterior(mod_h, mv_h, VectorMonitors(), out_dir + "HRM_powp.log", cats
= n_cats, sampleFreq = 10)
pp_h.burnin(generations = 500, tuningInterval = 100)
pp_h.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "HRM_powp.log", "power", "likelihood",
"\t").marginal()
```

```
ps = pathSampler(out_dir + "HRM_powp.log", "power", "likelihood", "\t").marginal()
write("HRM", ss, ps, filename = out_dir + "marginal_likelihoods.csv", append = TRUE,
separator = ",")
write("\n", filename = out_dir + "marginal_likelihoods.csv", append = TRUE)

############################ one-category Mk (symmetric)
q_mk ~ dnExponential(1.0)
mv_m = VectorMoves()
mv_m.append( mvScale(q_mk, lambda = 0.5, weight = 2.0) )
Q_m := fnFreeK([q_mk, q_mk], rescaled = FALSE)
ctmc_m ~ dnPhyloCTMC(tree = tree_m, Q = Q_m, rootFrequencies = simplex([1, 1]),
                     type = "Standard", nSites = feat.nchar())
ctmc_m.clamp(feat)
mod_m = model(Q_m)
mn_m = VectorMonitors()
mn_m.append( mnModel(filename = out_dir + "MK_model.log", printgen = 10) )
mcmc_m = mcmc(mod_m, mn_m, mv_m, nruns = 1)
mcmc_m.burnin(generations = 500, tuningInterval = 100)
mcmc_m.run(generations = 5000, tuningInterval = 0)
pp_m = powerPosterior(mod_m, mv_m, VectorMonitors(), out_dir + "MK_powp.log", cats =
n_cats, sampleFreq = 10)
pp_m.burnin(generations = 500, tuningInterval = 100)
pp_m.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "MK_powp.log", "power", "likelihood",
"\t").marginal()
ps = pathSampler(out_dir + "MK_powp.log", "power", "likelihood", "\t").marginal()
write("MK", ss, ps, filename = out_dir + "marginal_likelihoods.csv", append = TRUE,
separator = ",")
write("\n", filename = out_dir + "marginal_likelihoods.csv", append = TRUE)

write("runtime_s", time("seconds") - t_start, filename = out_dir + "runtime.txt",
separator = "\t")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
hidden-rates model	r_slow	0.05	0.053	0.000	0.122	593	pass
hidden-rates model	r_fast	1.50	1.577	0.587	2.657	951	pass
hidden-rates model	switch	0.15	0.223	0.031	0.438	599	pass
HRM vs Mk	lnBF_HRM_vs_MK	NA	5.244	NA	NA	NA	pass

All three rates are recovered, and the marginal likelihood prefers the generating hidden-rates model (ln BF = 5.2).

H.5.8 Pagel's test of correlated evolution

`08_pagel_correlated.Rev` implements Section 13.7: two binary features coded as one four-state character, with the independent model of Equation (D.22) and the dependent model of Equation (13.6), compared by stepping-stone sampling. A third, reversible-jump analysis samples over the dependence itself, in the spirit of BayesTraits. Each of the four pairs of rates that the independent model sets equal is written as a base rate and a multiplier that is exactly 1 with prior probability 1/2. The independent model is the state in which all four multipliers equal 1 (prior probability 1/16). The data follow the book's simulation (harmonic states stable, disharmonic states left ten times faster), with 20 independent feature pairs on the 16-tip tree so that the test has some power.

```
################################################################################
# 08_pagel_correlated.Rev
#
# Model:    Pagel's (1994) test of correlated evolution of two binary features
#           X (0 = OV, 1 = VO) and Y (0 = postpositions, 1 = prepositions),
#           coded jointly as one four-state character
#          0 = (0,0), 1 = (0,1), 2 = (1,0), 3 = (1,1); one feature changes at a time.
#           (a) independent model: 4 rates alpha_X, beta_X, alpha_Y, beta_Y
#               (Eq. pcms-pagel-ind);
#           (b) dependent model: 8 free rates q12 ... q43 (Eq. pcms-pagel-dep);
#           compared by stepping-stone / path-sampling marginal likelihoods; and
#           (c) reversible-jump MCMC over the dependence: for each of the four
#               pairs of rates that the independent model sets equal
#               (q12 = q34, q21 = q43, q13 = q24, q31 = q42), the second rate is
#               the first times a multiplier m that is exactly 1 (no dependence)
#               with prior probability 1/2 or Lognormal(0, 1.5) otherwise.
#               The posterior probability that all four m = 1 is the posterior
#               probability of the independent model.
# Book:     Chapter 13, "Correlated evolution of discrete characters"
#           (#pcms-pagel), Eqs. pcms-pagel-ind, pcms-pagel-dep; Bayes factors
#           and reversible jump as in BayesTraits; Chapter 14
(#cultural-traits-methods).
# Estimates: rates; ln marginal likelihoods; 2 ln BF; RJ inclusion probabilities.
# Data:     data/traits/pagel.nex: 20 independent feature pairs generated under
#           the dependent model of the book (harmonic states stable, rates 0.2
#           out of harmonic states and 2.0 out of disharmonic states), tree16.tre
# Run:      cd revbayes && rb analyses/traits/08_pagel_correlated.Rev [--args n_gen]
# Runtime:  about 2 minutes
################################################################################

seed(1108)
n_gen = 10000
if (args.size() > 1) n_gen = int(args[2])
n_cats  = 16
pp_gen  = 300
out_dir = "output/traits/08_pagel_correlated/"
t_start = time("seconds")
```

```
# NB: every model gets its OWN copy of the tree (and of the root frequencies):
# model() collects every node connected to its argument, including through
# shared constant nodes, so models that share a tree object would be merged
# into one graph and their likelihoods added in the power posteriors.
tree_i  <- readTrees("data/traits/tree16.tre")[1]
tree_d  <- readTrees("data/traits/tree16.tre")[1]
tree_rj <- readTrees("data/traits/tree16.tre")[1]
pag  <- readDiscreteCharacterData("data/traits/pagel.nex")
n_sites <- pag.nchar()

write("model,lnML_stepping_stone,lnML_path_sampling\n", filename = out_dir +
"marginal_likelihoods.csv")

############################## (a) independent
alpha_X ~ dnExponential(1.0)
beta_X  ~ dnExponential(1.0)
alpha_Y ~ dnExponential(1.0)
beta_Y  ~ dnExponential(1.0)
mv_i = VectorMoves()
mv_i.append( mvScale(alpha_X, lambda = 0.5, weight = 1.0) )
mv_i.append( mvScale(beta_X,  lambda = 0.5, weight = 1.0) )
mv_i.append( mvScale(alpha_Y, lambda = 0.5, weight = 1.0) )
mv_i.append( mvScale(beta_Y,  lambda = 0.5, weight = 1.0) )
Q_ind := fnFreeK([ alpha_Y, alpha_X, 0.0,
                   beta_Y,  0.0,     alpha_X,
                   beta_X,  0.0,     alpha_Y,
                   0.0,     beta_X,  beta_Y ], rescaled = FALSE)
ctmc_i ~ dnPhyloCTMC(tree = tree_i, Q = Q_ind, rootFrequencies = simplex(rep(1, 4)),
type = "Standard", nSites = n_sites)
ctmc_i.clamp(pag)
mod_i = model(Q_ind)
mn_i = VectorMonitors()
mn_i.append( mnModel(filename = out_dir + "IND_model.log", printgen = 10) )
mcmc_i = mcmc(mod_i, mn_i, mv_i, nruns = 1)
mcmc_i.burnin(generations = 500, tuningInterval = 100)
mcmc_i.run(generations = n_gen / 2, tuningInterval = 0)
pp_i = powerPosterior(mod_i, mv_i, VectorMonitors(), out_dir + "IND_powp.log", cats
= n_cats, sampleFreq = 10)
pp_i.burnin(generations = 500, tuningInterval = 100)
pp_i.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "IND_powp.log", "power", "likelihood",
"\t").marginal()
ps = pathSampler(out_dir + "IND_powp.log", "power", "likelihood", "\t").marginal()
write("IND", ss, ps, filename = out_dir + "marginal_likelihoods.csv", append = TRUE,
separator = ",")
write("\n", filename = out_dir + "marginal_likelihoods.csv", append = TRUE)

############################## (b) dependent
```

```
q12 ~ dnExponential(1.0)
q13 ~ dnExponential(1.0)
q21 ~ dnExponential(1.0)
q24 ~ dnExponential(1.0)
q31 ~ dnExponential(1.0)
q34 ~ dnExponential(1.0)
q42 ~ dnExponential(1.0)
q43 ~ dnExponential(1.0)
mv_d = VectorMoves()
mv_d.append( mvScale(q12, lambda = 0.5, weight = 1.0) )
mv_d.append( mvScale(q13, lambda = 0.5, weight = 1.0) )
mv_d.append( mvScale(q21, lambda = 0.5, weight = 1.0) )
mv_d.append( mvScale(q24, lambda = 0.5, weight = 1.0) )
mv_d.append( mvScale(q31, lambda = 0.5, weight = 1.0) )
mv_d.append( mvScale(q34, lambda = 0.5, weight = 1.0) )
mv_d.append( mvScale(q42, lambda = 0.5, weight = 1.0) )
mv_d.append( mvScale(q43, lambda = 0.5, weight = 1.0) )
Q_dep := fnFreeK([ q12, q13, 0.0,
                   q21, 0.0, q24,
                   q31, 0.0, q34,
                   0.0, q42, q43 ], rescaled = FALSE)
ctmc_d ~ dnPhyloCTMC(tree = tree_d, Q = Q_dep, rootFrequencies = simplex(rep(1, 4)),
type = "Standard", nSites = n_sites)
ctmc_d.clamp(pag)
mod_d = model(Q_dep)
mn_d = VectorMonitors()
mn_d.append( mnModel(filename = out_dir + "DEP_model.log", printgen = 10) )
mcmc_d = mcmc(mod_d, mn_d, mv_d, nruns = 1)
mcmc_d.burnin(generations = 500, tuningInterval = 100)
mcmc_d.run(generations = n_gen / 2, tuningInterval = 0)
pp_d = powerPosterior(mod_d, mv_d, VectorMonitors(), out_dir + "DEP_powp.log", cats
= n_cats, sampleFreq = 10)
pp_d.burnin(generations = 500, tuningInterval = 100)
pp_d.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "DEP_powp.log", "power", "likelihood",
"\t").marginal()
ps = pathSampler(out_dir + "DEP_powp.log", "power", "likelihood", "\t").marginal()
write("DEP", ss, ps, filename = out_dir + "marginal_likelihoods.csv", append = TRUE,
separator = ",")
write("\n", filename = out_dir + "marginal_likelihoods.csv", append = TRUE)

############################ (c) reversible jump over the dependence
# base rates (one per pair) and multipliers for the partner rate
for (k in 1:4) {
    base[k] ~ dnExponential(1.0)
    mult[k] ~ dnReversibleJumpMixture(1.0, dnLognormal(0.0, 1.5), p = 0.5)
    dep[k]  := ifelse(mult[k] == 1.0, 0, 1)
}
```

```
mv_rj = VectorMoves()
for (k in 1:4) {
    mv_rj.append( mvScale(base[k], lambda = 0.5, weight = 1.0) )
    mv_rj.append( mvRJSwitch(mult[k], weight = 1.0) )
    mv_rj.append( mvScale(mult[k], lambda = 0.5, weight = 1.0) )
}
# pairs: 1 = (q12, q34) gain of Y; 2 = (q21, q43) loss of Y;
#        3 = (q13, q24) gain of X; 4 = (q31, q42) loss of X
r12 := base[1]
r34 := base[1] * mult[1]
r21 := base[2]
r43 := base[2] * mult[2]
r13 := base[3]
r24 := base[3] * mult[3]
r31 := base[4]
r42 := base[4] * mult[4]
n_dep := sum(dep)
independent := ifelse(n_dep == 0, 1, 0)
Q_rj := fnFreeK([ r12, r13, 0.0,
                  r21, 0.0, r24,
                  r31, 0.0, r34,
                  0.0, r42, r43 ], rescaled = FALSE)
ctmc_rj ~ dnPhyloCTMC(tree = tree_rj, Q = Q_rj, rootFrequencies = simplex(rep(1,
4)), type = "Standard", nSites = n_sites)
ctmc_rj.clamp(pag)
mod_rj = model(Q_rj)
mn_rj = VectorMonitors()
mn_rj.append( mnModel(filename = out_dir + "RJ_model.log", printgen = 30) )
mn_rj.append( mnScreen(printgen = 5000, n_dep) )
mcmc_rj = mcmc(mod_rj, mn_rj, mv_rj, nruns = 1)
mcmc_rj.burnin(generations = 1000, tuningInterval = 100)
mcmc_rj.run(generations = 3 * n_gen, tuningInterval = 0)

write("runtime_s", time("seconds") - t_start, filename = out_dir + "runtime.txt",
separator = "\t")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
Pagel dependent vs independent	2lnBF_DEP_vs_IND	NA	194.430	NA	NA	NA	pass
Pagel dependent	q12	0.2	0.116	0.007	0.244	200	pass
Pagel dependent	q13	0.2	0.198	0.024	0.407	145	pass
Pagel dependent	q21	2.0	1.436	0.004	3.596	151	pass
Pagel dependent	q24	2.0	1.796	0.019	3.699	157	pass
Pagel dependent	q31	2.0	2.280	0.021	4.640	207	pass
Pagel dependent	q34	2.0	1.558	0.021	3.425	164	pass

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
Pagel dependent	q42	0.2	0.107	0.007	0.238	225	pass
Pagel dependent	q43	0.2	0.141	0.009	0.293	238	pass
Pagel reversible jump	P(independent)	0.0	0.000	NA	NA	NA	pass
Pagel reversible jump	dep[1]	1.0	0.601	NA	NA	NA	pass
Pagel reversible jump	dep[2]	1.0	0.678	NA	NA	NA	pass
Pagel reversible jump	dep[3]	1.0	0.942	NA	NA	NA	pass
Pagel reversible jump	dep[4]	1.0	0.975	NA	NA	NA	pass

The dependent model wins decisively ($2 \ln \mathrm{BF} \approx 194$). The reversible-jump chain never visited the independent model, and all eight dependent rates are covered. The reversible-jump posteriors also show which dependencies the data resolve. The rate of changing X (verb-object order) depends strongly on Y (posterior probabilities of dependence 0.94 and 0.97), whereas the evidence that changes in Y depend on X is weaker (0.60 and 0.68).

H.5.9 Brownian motion and Ornstein-Uhlenbeck

`09_bm_ou.Rev` fits BM (Equation (13.2)) and OU (Equation (13.3), root at the optimum) to two traits, one simulated under BM and one under OU (half-life 0.35 millennia), and compares them by stepping-stone sampling (Section 13.3.2). Both use full likelihoods with a proper prior on the root value or optimum (`dnPhyloBrownianMVN`, `dnPhyloOrnsteinUhlenbeckMVN`). The restricted (REML) BM likelihood integrates the root under an improper flat prior, so its marginal likelihood cannot be compared with an OU model. `dnPhyloOrnsteinUhlenbeckREML` was not used because its value did not match the analytic fixed-root OU likelihood in a test, whereas the MVN version did.

```
################################################################################
# 09_bm_ou.Rev
#
# Model:    Brownian motion (BM) and Ornstein-Uhlenbeck (OU) models for one
#           continuous trait on a fixed tree.
#             BM: x ~ N(mu 1, sigma2 C)                        (Eq. pcms-bm-mvn)
#             OU: dX = alpha (theta - X) dt + sigma dW, root at theta (Eq. pcms-ou)
#           Both are fitted to two simulated traits, one generated under BM and
#           one under OU, and compared by stepping-stone / path-sampling
#           marginal likelihoods. Full (not restricted) likelihoods are used so
#           that the two models' marginal likelihoods are comparable: the BM root
#           value mu and the OU optimum theta both get proper Uniform(-10, 10)
#           priors.
# Book:     Chapter 13, "Brownian motion" (#pcms-bm), "The Ornstein-Uhlenbeck
#           process" (#pcms-ou), "Fitting and comparing models".
# Estimates: sigma2 (BM); alpha, theta, sigma2, phylogenetic half-life (OU);
#           ln marginal likelihoods and the Bayes factor BM vs OU per trait.
# Data:     data/traits/trait_bm.nex, data/traits/trait_ou.nex on tree16.tre
# Run:      cd revbayes && rb analyses/traits/09_bm_ou.Rev [--args n_gen]
# Runtime:  about 2.5 minutes
################################################################################
```

```
seed(1109)
n_gen = 8000
if (args.size() > 1) n_gen = int(args[2])
n_cats  = 16
pp_gen  = 400
out_dir = "output/traits/09_bm_ou/"
t_start = time("seconds")

# NB: model() collects every node connected to its argument, including through
# shared constant nodes and through the elements of one Rev vector (x[1], x[2]).
# Each model therefore has its own tree object and its own variable names, so
# the four model graphs stay separate (otherwise power posteriors would add the
# likelihoods of the other models).
write("data,model,lnML_stepping_stone,lnML_path_sampling\n", filename = out_dir +
"marginal_likelihoods.csv")

############################## data: bm, model: BM
tree_bm_bm <- readTrees("data/traits/tree16.tre")[1]
dat_bm_bm  <- readContinuousCharacterData("data/traits/trait_bm.nex")
s2_bm_bm ~ dnLoguniform(1e-3, 100.0)
mu_bm_bm ~ dnUniform(-10.0, 10.0)
mu_bm_bm.setValue(0.0)
mv = VectorMoves()
mv.append( mvScale(s2_bm_bm, lambda = 0.5, weight = 1.0) )
mv.append( mvSlide(mu_bm_bm, delta = 0.5, weight = 1.0) )
x_bm_bm ~ dnPhyloBrownianMVN(tree_bm_bm, branchRates = sqrt(s2_bm_bm), rootStates =
mu_bm_bm, nSites = 1)
x_bm_bm.clamp(dat_bm_bm)
mod = model(s2_bm_bm)
mn = VectorMonitors()
mn.append( mnModel(filename = out_dir + "data_bm_BM_model.log", printgen = 10) )
mc = mcmc(mod, mn, mv, nruns = 1)
mc.burnin(generations = 500, tuningInterval = 100)
mc.run(generations = n_gen / 2, tuningInterval = 0)
pp = powerPosterior(mod, mv, VectorMonitors(), out_dir + "data_bm_BM_powp.log", cats
= n_cats, sampleFreq = 10)
pp.burnin(generations = 300, tuningInterval = 50)
pp.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "data_bm_BM_powp.log", "power", "likelihood",
"\t").marginal()
ps = pathSampler(out_dir + "data_bm_BM_powp.log", "power", "likelihood",
"\t").marginal()
write("bm", "BM", ss, ps, filename = out_dir + "marginal_likelihoods.csv", append =
TRUE, separator = ",")
write("\n", filename = out_dir + "marginal_likelihoods.csv", append = TRUE)

############################## data: bm, model: OU
```

```
tree_bm_ou <- readTrees("data/traits/tree16.tre")[1]
dat_bm_ou  <- readContinuousCharacterData("data/traits/trait_bm.nex")
alpha_bm_ou ~ dnExponential(1.0)
theta_bm_ou ~ dnUniform(-10.0, 10.0)
theta_bm_ou.setValue(0.0)
s2_bm_ou    ~ dnLoguniform(1e-3, 100.0)
half_life_bm_ou := ln(2.0) / alpha_bm_ou
mv = VectorMoves()
mv.append( mvScale(alpha_bm_ou, lambda = 0.5, weight = 1.0) )
mv.append( mvSlide(theta_bm_ou, delta = 0.5, weight = 1.0) )
mv.append( mvScale(s2_bm_ou, lambda = 0.5, weight = 1.0) )
# alpha and sigma2 are strongly correlated (stationary variance sigma2 / 2 alpha)
mv.append( mvUpDownScale(lambda = 0.3, weight = 1.0) )
mv[mv.size()].addVariable(alpha_bm_ou, up = TRUE)
mv[mv.size()].addVariable(s2_bm_ou, up = TRUE)
x_bm_ou ~ dnPhyloOrnsteinUhlenbeckMVN(tree_bm_ou, alpha = alpha_bm_ou, theta =
theta_bm_ou,
                                      sigma = sqrt(s2_bm_ou), rootStates = theta_bm_ou,
                                        nSites = 1)
x_bm_ou.clamp(dat_bm_ou)
mod = model(s2_bm_ou)
mn = VectorMonitors()
mn.append( mnModel(filename = out_dir + "data_bm_OU_model.log", printgen = 10) )
mc = mcmc(mod, mn, mv, nruns = 1)
mc.burnin(generations = 1000, tuningInterval = 100)
mc.run(generations = n_gen, tuningInterval = 0)
pp = powerPosterior(mod, mv, VectorMonitors(), out_dir + "data_bm_OU_powp.log", cats
= n_cats, sampleFreq = 10)
pp.burnin(generations = 300, tuningInterval = 50)
pp.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "data_bm_OU_powp.log", "power", "likelihood",
"\t").marginal()
ps = pathSampler(out_dir + "data_bm_OU_powp.log", "power", "likelihood",
"\t").marginal()
write("bm", "OU", ss, ps, filename = out_dir + "marginal_likelihoods.csv", append =
TRUE, separator = ",")
write("\n", filename = out_dir + "marginal_likelihoods.csv", append = TRUE)

############################ data: ou, model: BM
tree_ou_bm <- readTrees("data/traits/tree16.tre")[1]
dat_ou_bm  <- readContinuousCharacterData("data/traits/trait_ou.nex")
s2_ou_bm ~ dnLoguniform(1e-3, 100.0)
mu_ou_bm ~ dnUniform(-10.0, 10.0)
mu_ou_bm.setValue(0.0)
mv = VectorMoves()
mv.append( mvScale(s2_ou_bm, lambda = 0.5, weight = 1.0) )
mv.append( mvSlide(mu_ou_bm, delta = 0.5, weight = 1.0) )
x_ou_bm ~ dnPhyloBrownianMVN(tree_ou_bm, branchRates = sqrt(s2_ou_bm), rootStates =
mu_ou_bm, nSites = 1)
```

```
x_ou_bm.clamp(dat_ou_bm)
mod = model(s2_ou_bm)
mn = VectorMonitors()
mn.append( mnModel(filename = out_dir + "data_ou_BM_model.log", printgen = 10) )
mc = mcmc(mod, mn, mv, nruns = 1)
mc.burnin(generations = 500, tuningInterval = 100)
mc.run(generations = n_gen / 2, tuningInterval = 0)
pp = powerPosterior(mod, mv, VectorMonitors(), out_dir + "data_ou_BM_powp.log", cats
= n_cats, sampleFreq = 10)
pp.burnin(generations = 300, tuningInterval = 50)
pp.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "data_ou_BM_powp.log", "power", "likelihood",
"\t").marginal()
ps = pathSampler(out_dir + "data_ou_BM_powp.log", "power", "likelihood",
"\t").marginal()
write("ou", "BM", ss, ps, filename = out_dir + "marginal_likelihoods.csv", append =
TRUE, separator = ",")
write("\n", filename = out_dir + "marginal_likelihoods.csv", append = TRUE)

############################## data: ou, model: OU
tree_ou_ou <- readTrees("data/traits/tree16.tre")[1]
dat_ou_ou  <- readContinuousCharacterData("data/traits/trait_ou.nex")
alpha_ou_ou ~ dnExponential(1.0)
theta_ou_ou ~ dnUniform(-10.0, 10.0)
theta_ou_ou.setValue(0.0)
s2_ou_ou    ~ dnLoguniform(1e-3, 100.0)
half_life_ou_ou := ln(2.0) / alpha_ou_ou
mv = VectorMoves()
mv.append( mvScale(alpha_ou_ou, lambda = 0.5, weight = 1.0) )
mv.append( mvSlide(theta_ou_ou, delta = 0.5, weight = 1.0) )
mv.append( mvScale(s2_ou_ou, lambda = 0.5, weight = 1.0) )
# alpha and sigma2 are strongly correlated (stationary variance sigma2 / 2 alpha)
mv.append( mvUpDownScale(lambda = 0.3, weight = 1.0) )
mv[mv.size()].addVariable(alpha_ou_ou, up = TRUE)
mv[mv.size()].addVariable(s2_ou_ou, up = TRUE)
x_ou_ou ~ dnPhyloOrnsteinUhlenbeckMVN(tree_ou_ou, alpha = alpha_ou_ou, theta =
theta_ou_ou,
                                      sigma = sqrt(s2_ou_ou), rootStates = theta_ou_ou,
                                        nSites = 1)
x_ou_ou.clamp(dat_ou_ou)
mod = model(s2_ou_ou)
mn = VectorMonitors()
mn.append( mnModel(filename = out_dir + "data_ou_OU_model.log", printgen = 10) )
mc = mcmc(mod, mn, mv, nruns = 1)
mc.burnin(generations = 1000, tuningInterval = 100)
mc.run(generations = n_gen, tuningInterval = 0)
pp = powerPosterior(mod, mv, VectorMonitors(), out_dir + "data_ou_OU_powp.log", cats
= n_cats, sampleFreq = 10)
```

```
pp.burnin(generations = 300, tuningInterval = 50)
pp.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "data_ou_OU_powp.log", "power", "likelihood",
"\t").marginal()
ps = pathSampler(out_dir + "data_ou_OU_powp.log", "power", "likelihood",
"\t").marginal()
write("ou", "OU", ss, ps, filename = out_dir + "marginal_likelihoods.csv", append =
TRUE, separator = ",")
write("\n", filename = out_dir + "marginal_likelihoods.csv", append = TRUE)

write("runtime_s", time("seconds") - t_start, filename = out_dir + "runtime.txt",
separator = "\t")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
BM (data: BM)	s2_bm_bm	0.5	0.770	0.322	1.372	381	pass
OU (data: OU)	alpha_ou_ou	2.0	1.429	0.099	3.258	366	pass
OU (data: OU)	theta_ou_ou	1.0	1.105	0.711	1.492	669	pass
OU (data: OU)	s2_ou_ou	1.0	0.931	0.215	2.166	412	pass
BM vs OU (data: BM)	lnBF_OU_vs_BM	NA	-0.901	NA	NA	NA	pass
BM vs OU (data: OU)	lnBF_OU_vs_BM	NA	2.046	NA	NA	NA	pass

Both comparisons pick the generating model (ln BF -0.9 on the BM trait, $+2.0$ on the OU trait). The margins are modest, as the caution in Section 13.3.2 leads one to expect for 16 tips. Rates and the optimum are covered and agree with the ML estimates from `geiger::fitContinuous`.

H.5.10 Early burst

`10_early_burst.Rev` implements the EB model of Section 13.3.3. RevBayes 1.4.1 has `fnEarlyBurst`, but its rate parameter is restricted to be positive, so the model is written out exactly instead. The variance that accumulates on a branch running from time s_0 to s_1 after the root is $\sigma_0^2(e^{rs_1} - e^{rs_0})/r$, which reproduces the covariance of Equation (D.13). It enters `dnPhyloBrownianREML` as a per-branch multiplier computed from the node ages. Four independent traits were simulated with $r = -0.8$.

```
################################################################################
# 10_early_burst.Rev
#
# Model:    Early-burst (EB, ACDC) model: Brownian motion whose rate changes
#           exponentially with time since the root, sigma2(s) = sigma2_0 exp(r s)
#           (r < 0: fast early, slow late). RevBayes has no dedicated EB
#           distribution with r < 0 allowed, so the model is expressed exactly
#           through per-branch rate multipliers: the variance accumulated on a
#           branch from s0 to s1 (time since root) is
```

```
#               sigma2_0 * (exp(r s1) - exp(r s0)) / r,
#           which reproduces the EB covariance C_ij = sigma2_0 (exp(r s_ij) - 1) / r
#           (Eq. pcms-eb). The restricted (REML) likelihood integrates the root.
#           EB is compared with constant-rate BM by stepping-stone sampling
#           (both REML, so their marginal likelihoods are comparable).
# Book:     Chapter 13, "Early burst and related models" (#pcms-eb).
# Estimates: r, sigma2_0; ln marginal likelihoods of EB and BM.
# Data:     data/traits/trait_eb.nex (4 independent traits, r = -0.8,
#           sigma2_0 = 2) on tree16.tre
# Run:      cd revbayes && rb analyses/traits/10_early_burst.Rev [--args n_gen]
# Runtime:  about 1 minute
################################################################################

seed(1110)
n_gen = 10000
if (args.size() > 1) n_gen = int(args[2])
n_cats  = 32
pp_gen  = 1000
out_dir = "output/traits/10_early_burst/"
t_start = time("seconds")

# NB: every model gets its OWN copy of the tree (and of the root frequencies):
# model() collects every node connected to its argument, including through
# shared constant nodes, so models that share a tree object would be merged
# into one graph and their likelihoods added in the power posteriors.
tree    <- readTrees("data/traits/tree16.tre")[1]
tree_bm <- readTrees("data/traits/tree16.tre")[1]
eb   <- readContinuousCharacterData("data/traits/trait_eb.nex")
n_branches <- tree.nnodes() - 1
root_age <- tree.rootAge()

# branch start and end times, measured from the root
for (b in 1:n_branches) {
    s0[b] <- root_age - tree.nodeAge(tree.parent(b))
    s1[b] <- root_age - tree.nodeAge(b)
    bl[b] <- tree.branchLength(b)
}

write("model,lnML_stepping_stone,lnML_path_sampling\n", filename = out_dir +
"marginal_likelihoods.csv")

############################ EB
sigma2_0 ~ dnLoguniform(1e-3, 100.0)
r ~ dnUniform(-4.0, 1.0)
r.setValue(-0.1)
mv_eb = VectorMoves()
mv_eb.append( mvScale(sigma2_0, lambda = 0.5, weight = 2.0) )
mv_eb.append( mvSlide(r, delta = 0.3, weight = 3.0) )
```

```
for (b in 1:n_branches) {
    # mean rate multiplier over the branch, (exp(r s1) - exp(r s0)) / (r (s1 - s0))
    eb_mult[b] := abs( (exp(r * s1[b]) - exp(r * s0[b])) / (r * bl[b]) )
    br_eb[b]   := sqrt(sigma2_0 * eb_mult[b])
}
x_eb ~ dnPhyloBrownianREML(tree, branchRates = br_eb, nSites = eb.nchar())
x_eb.clamp(eb)
mod_eb = model(sigma2_0)
mn_eb = VectorMonitors()
mn_eb.append( mnFile(sigma2_0, r, filename = out_dir + "EB_model.log", printgen =
10) )
mn_eb.append( mnScreen(printgen = 5000, sigma2_0, r) )
mc = mcmc(mod_eb, mn_eb, mv_eb, nruns = 1)
mc.burnin(generations = 1000, tuningInterval = 100)
mc.run(generations = n_gen, tuningInterval = 0)
pp = powerPosterior(mod_eb, mv_eb, VectorMonitors(), out_dir + "EB_powp.log", cats =
n_cats, sampleFreq = 10)
pp.burnin(generations = 500, tuningInterval = 100)
pp.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "EB_powp.log", "power", "likelihood",
"\t").marginal()
ps = pathSampler(out_dir + "EB_powp.log", "power", "likelihood", "\t").marginal()
write("EB", ss, ps, filename = out_dir + "marginal_likelihoods.csv", append = TRUE,
separator = ",")
write("\n", filename = out_dir + "marginal_likelihoods.csv", append = TRUE)

############################ BM (r = 0)
sigma2_bm ~ dnLoguniform(1e-3, 100.0)
mv_bm = VectorMoves()
mv_bm.append( mvScale(sigma2_bm, lambda = 0.5, weight = 2.0) )
x_bm ~ dnPhyloBrownianREML(tree_bm, branchRates = sqrt(sigma2_bm), nSites =
eb.nchar())
x_bm.clamp(eb)
mod_bm = model(sigma2_bm)
mn_bm = VectorMonitors()
mn_bm.append( mnModel(filename = out_dir + "BM_model.log", printgen = 10) )
mc = mcmc(mod_bm, mn_bm, mv_bm, nruns = 1)
mc.burnin(generations = 500, tuningInterval = 100)
mc.run(generations = n_gen / 2, tuningInterval = 0)
pp = powerPosterior(mod_bm, mv_bm, VectorMonitors(), out_dir + "BM_powp.log", cats =
n_cats, sampleFreq = 10)
pp.burnin(generations = 500, tuningInterval = 100)
pp.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "BM_powp.log", "power", "likelihood",
"\t").marginal()
ps = pathSampler(out_dir + "BM_powp.log", "power", "likelihood", "\t").marginal()
write("BM", ss, ps, filename = out_dir + "marginal_likelihoods.csv", append = TRUE,
separator = ",")
```

```
write("\n", filename = out_dir + "marginal_likelihoods.csv", append = TRUE)

write("runtime_s", time("seconds") - t_start, filename = out_dir + "runtime.txt",
separator = "\t")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
early burst	r	-0.8	-0.653	-0.985	-0.325	289	pass
early burst	sigma2_0	2.0	1.123	0.246	2.465	300	pass
EB vs BM	lnBF_EB_vs_BM	NA	4.424	NA	NA	NA	pass

r and σ_0^2 are covered. The posterior mean of r (-0.65) agrees with a joint REML-ML fit of the same covariance in R (-0.66), and EB is preferred over BM ($\ln \mathrm{BF} = 4.4$). Because both models use the restricted likelihood, they treat the root identically and their marginal likelihoods are comparable.

H.5.11 Evolutionary correlation and phylogenetic regression

`11_mvbm_correlation.Rev` fits bivariate BM with an evolutionary correlation (`dnPhyloMultivariateBrownianREM`). The slope $\rho\sigma_2/\sigma_1$ is the phylogenetic regression slope that PGLS estimates (Section 13.6, Equation (D.18)).

```
################################################################################
# 11_mvbm_correlation.Rev
#
# Model:    Bivariate Brownian motion with an evolutionary correlation:
#               (x1, x2) evolve with rate matrix
#               Sigma = [[s1^2, rho s1 s2], [rho s1 s2, s2^2]],
#           i.e. X ~ N(1 (x) root, C (x) Sigma), with the root integrated out
#           (restricted likelihood, dnPhyloMultivariateBrownianREML). The
#           phylogenetic regression slope of x2 on x1 is beta = rho s2 / s1,
#           the quantity that PGLS estimates (the GLS slope equals the slope of
#           Sigma, Eq. pcms-pgls).
# Book:     Chapter 13, "Phylogenetic generalized least squares" (#pcms-pgls),
#           "Phylogenetically independent contrasts" (#pcms-pic), the multivariate
#           BM models of the chapter's toolbox box and Chapter 16
#           (#biology-toolkit-pfa).
# Estimates: s1, s2, rho, slope; compared in tests/check_traits.R with PGLS
#           (nlme::gls + ape::corBrownian) and with the non-phylogenetic OLS
#           slope and Pearson correlation.
# Data:     data/traits/corr_traits.nex on tree16.tre
# Run:      cd revbayes && rb analyses/traits/11_mvbm_correlation.Rev [--args n_gen]
# Runtime:  about 10 s
################################################################################

seed(1111)
n_gen = 20000
```

```
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/traits/11_mvbm_correlation/"
t_start = time("seconds")

tree <- readTrees("data/traits/tree16.tre")[1]
dat  <- readContinuousCharacterData("data/traits/corr_traits.nex")

moves    = VectorMoves()
monitors = VectorMonitors()

sigma_1 ~ dnLognormal(0.0, 1.0)
sigma_2 ~ dnLognormal(0.0, 1.0)
rho     ~ dnUniform(-1.0, 1.0)
moves.append( mvScale(sigma_1, lambda = 0.3, weight = 2.0) )
moves.append( mvScale(sigma_2, lambda = 0.3, weight = 2.0) )
moves.append( mvSlide(rho, delta = 0.2, weight = 3.0) )
Sigma := fnVarCovar([sigma_1, sigma_2], [rho])
slope := rho * sigma_2 / sigma_1

X ~ dnPhyloMultivariateBrownianREML(tree, rateMatrix = Sigma)
X.clamp(dat)

mymodel = model(Sigma)
monitors.append( mnFile(sigma_1, sigma_2, rho, slope, filename = out_dir +
"model.log", printgen = 10) )
monitors.append( mnScreen(printgen = 5000, rho, slope) )

mymcmc = mcmc(mymodel, monitors, moves, nruns = 1)
mymcmc.burnin(generations = 1000, tuningInterval = 100)
mymcmc.run(generations = n_gen, tuningInterval = 0)

write("runtime_s", time("seconds") - t_start, filename = out_dir + "runtime.txt",
separator = "\t")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
bivariate BM	sigma_1	1.000	1.215	0.828	1.663	1901	pass
bivariate BM	sigma_2	0.800	0.957	0.660	1.293	1901	pass
bivariate BM	rho	0.750	0.693	0.408	0.898	1901	pass
bivariate BM	slope	0.600	0.555	0.252	0.836	2046	pass
bivariate BM	slope_vs_PGLS	0.608	0.555	NA	NA	NA	pass

The correlation (true 0.75) and slope (true 0.6) are recovered. The posterior mean slope (0.55) is close to the PGLS estimate from `nlme::gls` with `corBrownian` (0.61). The non-phylogenetic estimates are inflated (OLS slope 0.74, Pearson $r = 0.89$), which is Galton's problem in miniature (Section 13.1.2).

H.5.12 Pagel's lambda

`12_pagel_lambda.Rev` estimates Pagel's λ (Equation (13.4)). For an ultrametric tree of height T, $\mathbf{C}(\lambda)$ is BM on a tree whose internal branches are multiplied by λ and whose terminal branches become $\lambda t_b + (1-\lambda)T$, so λ enters as deterministic per-branch rate multipliers with a uniform prior. The threshold model, the other half of this item, is script 17.

```
################################################################################
# 12_pagel_lambda.Rev
#
# Model:    Brownian motion on a lambda-transformed tree (Pagel's lambda),
#               C(lambda) = lambda C + (1 - lambda) diag(C)      (Eq. pcms-lambda).
#           For an ultrametric tree of height T this is BM on a tree whose
#           internal branches are multiplied by lambda and whose terminal
#           branches t_b become lambda t_b + (1 - lambda) T. We express it
#           through per-branch variance multipliers (lambda on internal branches,
#           lambda + (1 - lambda) T / t_b on terminal branches), so lambda is an
#           ordinary parameter with a Uniform(0, 1) prior. The root is
#           integrated out (REML).
#           A threshold model for a binary trait is in 17_threshold_model.Rev.
# Book:     Chapter 13, "Pagel's tree transformations" (#pcms-transforms),
#           "Phylogenetic signal" (#pcms-signal), "Estimating lambda jointly"
#           (#pcms-pgls-lambda).
# Estimates: lambda, sigma2; compared with the ML estimate of phytools::phylosig.
# Data:     data/traits/trait_lambda.nex (lambda = 0.4, sigma2 = 1) on tree16.tre
# Run:      cd revbayes && rb analyses/traits/12_pagel_lambda.Rev [--args n_gen]
# Runtime:  about 10 s
################################################################################

seed(1112)
n_gen = 20000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/traits/12_pagel_lambda/"
t_start = time("seconds")

tree <- readTrees("data/traits/tree16.tre")[1]
dat  <- readContinuousCharacterData("data/traits/trait_lambda.nex")
n_tips <- tree.ntips()
n_branches <- tree.nnodes() - 1
T <- tree.rootAge()
for (b in 1:n_branches) {
    bl[b] <- tree.branchLength(b)
    is_tip[b] <- ifelse(b <= n_tips, 1.0, 0.0)
}

moves    = VectorMoves()
monitors = VectorMonitors()

lambda ~ dnUniform(0.0, 1.0)
```

```
lambda.setValue(0.5)
sigma2 ~ dnLoguniform(1e-3, 100.0)
moves.append( mvSlide(lambda, delta = 0.2, weight = 3.0) )
moves.append( mvScale(sigma2, lambda = 0.5, weight = 2.0) )

for (b in 1:n_branches) {
    # internal: lambda; terminal: lambda + (1 - lambda) T / t_b
    lam_mult[b] := abs(lambda + is_tip[b] * (1.0 - lambda) * T / bl[b])
    br[b] := sqrt(sigma2 * lam_mult[b])
}

X ~ dnPhyloBrownianREML(tree, branchRates = br, nSites = 1)
X.clamp(dat)

mymodel = model(sigma2)
monitors.append( mnFile(lambda, sigma2, filename = out_dir + "model.log", printgen =
10) )
monitors.append( mnScreen(printgen = 5000, lambda, sigma2) )

mymcmc = mcmc(mymodel, monitors, moves, nruns = 1)
mymcmc.burnin(generations = 1000, tuningInterval = 100)
mymcmc.run(generations = n_gen, tuningInterval = 0)

write("runtime_s", time("seconds") - t_start, filename = out_dir + "runtime.txt",
separator = "\t")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
BM with Pagel's lambda	lambda	0.4	0.380	0.0	0.820	1654	pass
BM with Pagel's lambda	sigma2	1.0	0.929	0.3	1.797	1742	pass

The posterior of λ is wide but centered near the truth (mean 0.38, true 0.4). The ML estimate from `phytools::phylosig` sits on the boundary at 0 for these data, a reminder that point estimates of λ from 16 tips are fragile.

H.5.13 Ancestral states over a sample of trees

`13_asr_tree_sample.Rev` implements Equation (12.3) (Section 12.5) with the tree as a random variable over a fixed sample of 100 trees (`dnEmpiricalTree`, `mvEmpiricalTree`), as when a cultural or structural trait is analysed on the posterior trees of a lexical analysis (Sections 13.12.1 and 14.3). The sample stands in for such a posterior: UPGMA trees on bootstrap replicates of 150 simulated cognate characters, rescaled to the known root age. The same model is also run on the true tree.

```
################################################################################
# 13_asr_tree_sample.Rev
#
# Model:    Bayesian ancestral state reconstruction integrating over tree
#           uncertainty with a fixed sample of trees: the tree is a random
#           variable with an empirical distribution over 100 trees
#           (dnEmpiricalTree) and the MCMC jumps between them (mvEmpiricalTree)
#           while it samples the Mk rate. For comparison the same model is also
#           run on the single true tree.
#           The tree sample stands in for a posterior sample from a lexical
#           analysis: UPGMA trees on bootstrap replicates of 150 simulated
#           cognate characters, rescaled to the known root age (4 kyr).
# Book:     Chapter 12, "Bayesian ASR: integrating over trees and parameters"
#           (#asr-bayes, Eq. asr-bayes) and "Integrating over tree
#           uncertainty" (#asr-r-trees); Chapter 13 (#pcms-tree-uncertainty);
#           Chapter 14 (#cultural-traits-methods).
# Estimates: ER rate q; posterior probability of each state at the root, with
#           and without tree uncertainty.
# Data:     data/traits/asr_trait.nex, data/traits/tree_sample.trees,
#           data/traits/tree16.tre
# Run:      cd revbayes && rb analyses/traits/13_asr_tree_sample.Rev [--args n_gen]
# Runtime:  about 1 minute
################################################################################

seed(1113)
n_gen = 10000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/traits/13_asr_tree_sample/"
t_start = time("seconds")

trait     <- readDiscreteCharacterData("data/traits/asr_trait.nex")
tree_true <- readTrees("data/traits/tree16.tre")[1]
trees     <- readTrees("data/traits/tree_sample.trees")
tt = treeTrace(trees, burnin = 0)

############################## (a) over the tree sample
tree ~ dnEmpiricalTree(tt)
q_s ~ dnExponential(2.0)
mv_s = VectorMoves()
mv_s.append( mvEmpiricalTree(tree, weight = 3.0) )
mv_s.append( mvScale(q_s, lambda = 0.5, weight = 2.0) )
Q_s := fnFreeK([q_s, q_s], rescaled = FALSE)
ctmc_s ~ dnPhyloCTMC(tree = tree, Q = Q_s, rootFrequencies = simplex([1, 1]), type =
"Standard", nSites = 1)
ctmc_s.clamp(trait)
mod_s = model(Q_s)
mn_s = VectorMonitors()
mn_s.append( mnModel(filename = out_dir + "sample_model.log", printgen = 10) )
```

```
mn_s.append( mnJointConditionalAncestralState(tree = tree, ctmc = ctmc_s, type =
"Standard",
             withTips = FALSE, withStartStates = FALSE, filename = out_dir + "sample_
             states.log", printgen = 10) )
mcmc_s = mcmc(mod_s, mn_s, mv_s, nruns = 1)
mcmc_s.burnin(generations = 1000, tuningInterval = 100)
mcmc_s.run(generations = n_gen, tuningInterval = 0)

############################ (b) on the true tree only
q_t ~ dnExponential(2.0)
mv_t = VectorMoves()
mv_t.append( mvScale(q_t, lambda = 0.5, weight = 2.0) )
Q_t := fnFreeK([q_t, q_t], rescaled = FALSE)
ctmc_t ~ dnPhyloCTMC(tree = tree_true, Q = Q_t, rootFrequencies = simplex([1, 1]),
type = "Standard", nSites = 1)
ctmc_t.clamp(trait)
mod_t = model(Q_t)
mn_t = VectorMonitors()
mn_t.append( mnModel(filename = out_dir + "true_model.log", printgen = 10) )
mn_t.append( mnJointConditionalAncestralState(tree = tree_true, ctmc = ctmc_t, type
= "Standard",
             withTips = FALSE, withStartStates = FALSE, filename = out_dir + "true_
             states.log", printgen = 10) )
mcmc_t = mcmc(mod_t, mn_t, mv_t, nruns = 1)
mcmc_t.burnin(generations = 1000, tuningInterval = 100)
mcmc_t.run(generations = n_gen, tuningInterval = 0)

write("runtime_s", time("seconds") - t_start, filename = out_dir + "runtime.txt",
separator = "\t")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
ER Mk over tree sample	q_s	0.35	0.344	0.014	0.975	951	pass
ER Mk on true tree	q_t	0.35	0.436	0.057	1.117	951	pass
ER Mk over tree sample	root_state	1.00	0.641	NA	NA	NA	pass

The root reconstruction shifts from $P(\text{root} = 1) = 0.53$ on the true tree to 0.64 over the tree sample. The empirical-Bayes values from `ape::ace` move the same way (0.56 on the true tree, 0.70 averaged over the sample), which illustrates the point of Section 12.5: tree uncertainty changes the reconstruction.

H.5.14 Joint inference of the tree and an ancestral state

`14_asr_joint_tree_trait.Rev` estimates the time tree (uniform time-tree prior, root age fixed at 4 millennia) from the 150 binary cognate characters together with the structural feature on the

same tree, so the root reconstruction integrates over topology, branch lengths and both rates in one MCMC.

```
################################################################################
# 14_asr_joint_tree_trait.Rev
#
# Model:    Joint Bayesian inference of a time tree and an ancestral state on
#           small data: the topology and node ages are estimated from 150
#           binary cognate characters (symmetric two-state CTMC, rate mu_c),
#           with a uniform time-tree prior and the root age fixed at the known
#           4 kyr; on the same tree, an ER Mk model for a binary structural
#           feature gives the posterior of the state at the root. The root
#           reconstruction therefore integrates over the tree, its branch
#           lengths and both rates at once.
# Book:     Chapter 12, "Bayesian ASR: integrating over trees and parameters"
#           (#asr-bayes); Chapter 13, "Integrating over tree uncertainty"
#           (#pcms-tree-uncertainty); Chapter 7 for the binary cognate model.
# Estimates: mu_c, q, the tree (clade posterior probabilities; compared with the
#           true tree), the root-state posterior.
# Data:     data/traits/cognates.nex, data/traits/asr_trait.nex
# Run:      cd revbayes && rb analyses/traits/14_asr_joint_tree_trait.Rev [--args n_
gen]
# Runtime:  about 1.5 minutes
################################################################################

seed(1114)
n_gen = 15000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/traits/14_asr_joint_tree_trait/"
t_start = time("seconds")

cog   <- readDiscreteCharacterData("data/traits/cognates.nex")
trait <- readDiscreteCharacterData("data/traits/asr_trait.nex")
taxa  <- cog.taxa()
n_taxa <- taxa.size()

moves    = VectorMoves()
monitors = VectorMonitors()

# time tree with the root age fixed (the family age is assumed known)
root_age <- 4.0
tree ~ dnUniformTimeTree(rootAge = root_age, taxa = taxa)
moves.append( mvNNI(tree, weight = n_taxa / 2.0) )
moves.append( mvFNPR(tree, weight = n_taxa / 4.0) )
moves.append( mvNodeTimeSlideUniform(tree, weight = n_taxa) )
moves.append( mvSubtreeScale(tree, weight = n_taxa / 4.0) )

# lexical (cognate) model
mu_c ~ dnExponential(2.0)
```

```
moves.append( mvScale(mu_c, lambda = 0.3, weight = 3.0) )
Q_c := fnFreeK([mu_c, mu_c], rescaled = FALSE)
ctmc_c ~ dnPhyloCTMC(tree = tree, Q = Q_c, rootFrequencies = simplex([1, 1]),
                     type = "Standard", nSites = cog.nchar())
ctmc_c.clamp(cog)

# structural feature (ER Mk) on the same tree
q ~ dnExponential(2.0)
moves.append( mvScale(q, lambda = 0.5, weight = 2.0) )
Q_t := fnFreeK([q, q], rescaled = FALSE)
ctmc_t ~ dnPhyloCTMC(tree = tree, Q = Q_t, rootFrequencies = simplex([1, 1]),
                     type = "Standard", nSites = 1)
ctmc_t.clamp(trait)

mymodel = model(tree)

monitors.append( mnModel(filename = out_dir + "model.log", printgen = 10) )
monitors.append( mnFile(tree, filename = out_dir + "trees.trees", printgen = 20) )
monitors.append( mnJointConditionalAncestralState(tree = tree, ctmc = ctmc_t, type =
"Standard",
                 withTips = FALSE, withStartStates = FALSE, filename = out_dir +
                 "states.log", printgen = 10) )
monitors.append( mnScreen(printgen = 5000, mu_c, q) )

mymcmc = mcmc(mymodel, monitors, moves, nruns = 1)
mymcmc.burnin(generations = 1500, tuningInterval = 100)
mymcmc.run(generations = n_gen, tuningInterval = 0)

trace = readTreeTrace(out_dir + "trees.trees", treetype = "clock", burnin = 0.1)
mccTree(trace, file = out_dir + "mcc.tre")

write("runtime_s", time("seconds") - t_start, filename = out_dir + "runtime.txt",
separator = "\t")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
joint time tree + ER Mk	mu_c	0.25	0.219	0.160	0.293	705	pass
joint time tree + ER Mk	q	0.35	0.387	0.026	1.015	1426	pass
joint time tree + ER Mk	true_clade_support	1.00	0.496	0.031	1.000	NA	pass
joint time tree + ER Mk	root_state	1.00	0.572	NA	NA	NA	pass

The cognate rate and the trait rate are covered. The 150 characters nearly saturate over eight

millennia of independent history, so the backbone is uncertain. The posterior probabilities of the 14 true clades range from 0.03 to 1 (mean 0.50), and they track the bootstrap support of the same clades closely (correlation above 0.99), which is the check used here. The joint root reconstruction (0.57) lies between the true-tree and tree-sample values of script 13.

H.5.15 Phylogeographic GLM

`15_phylogeo_glm.Rev` implements the phylogeographic generalized linear model of Section 11.4.4 (Equation (11.4)). The log of each route rate is a baseline plus coefficients for two pairwise predictors, standardized log distance and an irrelevant random predictor. Each coefficient has a spike-and-slab prior (a reversible-jump mixture of exactly 0 and Normal(0, 1)).

```
################################################################################
# 15_phylogeo_glm.Rev
#
# Model:    Phylogeographic generalized linear model (GLM) for discrete
#           phylogeography: the symmetric dispersal rate between areas i and j is
#               log q_ij = beta_0 + sum_k beta_k delta_k x_ijk     (Eq.
phylogeography-glm)
#           with two pairwise predictors, x_1 = standardized log distance between
#           area centroids and x_2 = an irrelevant random predictor. Predictor
#           selection by reversible jump: each beta_k is exactly 0 (delta_k = 0)
#           with prior probability 1/2, otherwise Normal(0, 1).
# Book:     Chapter 11, "Explaining dispersal: phylogeographic generalized
#           linear models" (#phylogeography-glm).
# Estimates: beta_0, beta_dist, beta_noise, inclusion probabilities (Bayes
#           factors), root area.
# Data:     data/traits/regions_glm.nex, data/traits/glm_predictors.tsv, tree16.tre
# Run:      cd revbayes && rb analyses/traits/15_phylogeo_glm.Rev [--args n_gen]
# Runtime:  about 30 s
################################################################################

seed(1115)
n_gen = 10000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/traits/15_phylogeo_glm/"
t_start = time("seconds")

tree    <- readTrees("data/traits/tree16.tre")[1]
regions <- readDiscreteCharacterData("data/traits/regions_glm.nex")
pred = readDataDelimitedFile("data/traits/glm_predictors.tsv", header = TRUE,
delimiter = "\t")
n_pairs <- pred.size()          # 6 pairs, in the order AB, AC, AD, BC, BD, CD
for (k in 1:n_pairs) {
    x_dist[k]  <- pred[k][3]
    x_noise[k] <- pred[k][4]
}

moves    = VectorMoves()
```

```
monitors = VectorMonitors()

beta_0 ~ dnNormal(0.0, 2.0)
moves.append( mvSlide(beta_0, delta = 0.5, weight = 3.0) )
beta_dist  ~ dnReversibleJumpMixture(0.0, dnNormal(0.0, 1.0), p = 0.5)
beta_noise ~ dnReversibleJumpMixture(0.0, dnNormal(0.0, 1.0), p = 0.5)
moves.append( mvRJSwitch(beta_dist, weight = 2.0) )
moves.append( mvSlide(beta_dist, delta = 0.5, weight = 2.0) )
moves.append( mvRJSwitch(beta_noise, weight = 2.0) )
moves.append( mvSlide(beta_noise, delta = 0.5, weight = 2.0) )
delta_dist  := ifelse(beta_dist  == 0.0, 0, 1)
delta_noise := ifelse(beta_noise == 0.0, 0, 1)

for (k in 1:n_pairs) {
    q[k] := exp(beta_0 + beta_dist * x_dist[k] + beta_noise * x_noise[k])
}
Q := fnFreeK([ q[1], q[2], q[3],
               q[1], q[4], q[5],
               q[2], q[4], q[6],
               q[3], q[5], q[6] ], rescaled = FALSE)

ctmc ~ dnPhyloCTMC(tree = tree, Q = Q, rootFrequencies = simplex(rep(1, 4)),
                   type = "Standard", nSites = 1)
ctmc.clamp(regions)

mymodel = model(Q)
monitors.append( mnFile(beta_0, beta_dist, beta_noise, delta_dist, delta_noise,
                        filename = out_dir + "model.log", printgen = 10) )
monitors.append( mnJointConditionalAncestralState(tree = tree, ctmc = ctmc, type =
"Standard",
                 withTips = FALSE, withStartStates = FALSE, filename = out_dir +
                 "states.log", printgen = 10) )
monitors.append( mnScreen(printgen = 5000, beta_dist, beta_noise) )

mymcmc = mcmc(mymodel, monitors, moves, nruns = 1)
mymcmc.burnin(generations = 1000, tuningInterval = 100)
mymcmc.run(generations = n_gen, tuningInterval = 0)

write("runtime_s", time("seconds") - t_start, filename = out_dir + "runtime.txt",
separator = "\t")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
phylogeographic GLM	beta_0	-0.916	-0.833	-2.362	0.722	1050	pass
phylogeographic GLM	beta_dist	-1.200	-0.037	-1.402	1.364	1020	pass

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
phylogeographic GLM	beta_noise	0.000	0.216	-0.789	1.663	951	pass
phylogeographic GLM	delta_dist	1.000	0.472	NA	NA	NA	miss
phylogeographic GLM	delta_noise	0.000	0.473	NA	NA	NA	pass
phylogeographic GLM	root_state	0.000	0.253	NA	NA	NA	pass

All coefficients are covered, but neither predictor's inclusion probability moves from its prior of 0.5 (BF 0.9 for both). The distance effect ($\beta = -1.2$) is real but cannot be detected from one location character on 16 tips, so its indicator check is a miss. Phylogeographic GLMs earn their keep with many tips and many areas.

H.5.16 Trend, ancient languages and identifiability

`16_bm_trend_ancient_tips.Rev` reproduces the argument of Section 11.5.5 (Equation (11.8)) in a Bayesian setting. The data are a simulated eastward expansion from a homeland at (0, 0) with drift 0.8 per millennium; four languages are ancient, sampled 0.9-1.8 millennia ago. With the internal nodes integrated out, each coordinate is multivariate normal with mean $x_{\text{root}} + \mu h_i$ and covariance $\sigma^2 \mathbf{C}$. That is the GLS regression on tip height described in the book. There are three analyses: (a) undirected BM on the 12 modern languages; (b) BM with trend on the modern languages; (c) BM with trend on all 16. Analysis (b) is sampled in the equivalent parameterization $x_{\text{end}} = x_{\text{root}} + \mu T$, which has the same prior but lets the chain move along the ridge that contemporaneous tips cannot resolve.

```
################################################################################
# 16_bm_trend_ancient_tips.Rev
#
# Model:    Brownian diffusion with a directional trend (drift) and the role of
#           ancient (non-contemporaneous) languages in identifying it:
#               x_c | x_a ~ N_2(x_a + mu t_c, t_c sigma^2 I)   (Eq.
phylogeography-trend).
#           Integrating out the internal nodes, each coordinate of the tips is
#           multivariate normal with mean x_root + mu h_i (h_i = time from the
#           root to tip i) and covariance sigma^2 C (C = shared path lengths),
#           which is what the script uses: the GLS-regression form of the book.
#           Three analyses of the same simulated eastward expansion (true root
#           (0, 0), drift mu = (0.8, 0) per kyr):
#             (a) undirected BM, 12 contemporary languages only;
#             (b) BM + trend, 12 contemporary languages only (root and drift
#                 not separately identifiable: only root + mu T is, so the
#                 posterior follows the prior along a ridge);
#             (c) BM + trend, the 12 contemporary plus 4 ancient languages
#                 (sampled 0.9-1.8 kyr ago), which restores identifiability.
# Book:     Chapter 11, "Trend, ancient languages and identifiability"
```

```
#           (#phylogeography-trend) and its R section (#phylogeography-r-ancient).
# Estimates: root location (homeland), drift mu, sigma, in each analysis.
# Note:     the covariance is passed as sigma^2 * C (a deterministic matrix); the
#           `scale` argument of dnMultivariateNormal is not used because a
#           stochastic scale did not enter the likelihood in RevBayes 1.4.1 tests.
# Data:     data/traits/trend_{modern,ancient}.tsv (x, y, depth) and
#           data/traits/trend_C_{modern,ancient}.tsv (covariance matrices)
# Run:      cd revbayes && rb analyses/traits/16_bm_trend_ancient_tips.Rev [--args
n_gen]
# Runtime:  about 2 minutes
################################################################################

seed(1116)
n_gen = 6000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/traits/16_bm_trend_ancient_tips/"
t_start = time("seconds")

# NB: model() collects every node connected to its argument, including through
# shared constant nodes and through the elements of one Rev vector. The three
# analyses therefore use separate data objects and separate variable names.

############################ data
dm = readDataDelimitedFile("data/traits/trend_modern.tsv", header = TRUE, delimiter
= "\t")
Cm = readDataDelimitedFile("data/traits/trend_C_modern.tsv", header = TRUE,
delimiter = "\t")
da = readDataDelimitedFile("data/traits/trend_ancient.tsv", header = TRUE, delimiter
= "\t")
Ca = readDataDelimitedFile("data/traits/trend_C_ancient.tsv", header = TRUE,
delimiter = "\t")
nm <- dm.size()
na <- da.size()
for (i in 1:nm) {
    xm[i] <- dm[i][2]
    ym[i] <- dm[i][3]
    hm[i] <- dm[i][4]
    sdm[i] <- sqrt(Cm[i][i + 1])
}
for (i in 1:na) {
    xa[i] <- da[i][2]
    ya[i] <- da[i][3]
    ha[i] <- da[i][4]
    sda[i] <- sqrt(Ca[i][i + 1])
}
k = 0
for (i in 1:(nm - 1)) {
    for (j in (i + 1):nm) {
```

```
        k = k + 1
        rm[k] <- Cm[i][j + 1] / (sdm[i] * sdm[j])
    }
}
k = 0
for (i in 1:(na - 1)) {
    for (j in (i + 1):na) {
        k = k + 1
        ra[k] <- Ca[i][j + 1] / (sda[i] * sda[j])
    }
}

############################## (a) undirected BM, modern tips
C_a <- fnVarCovar(sdm, rm)
sigma_a ~ dnLognormal(0.0, 1.0)
root_x_a ~ dnNormal(0.0, 10.0)
root_y_a ~ dnNormal(0.0, 10.0)
mv = VectorMoves()
mv.append( mvScale(sigma_a, lambda = 0.3, weight = 2.0) )
mv.append( mvSlide(root_x_a, delta = 0.5, weight = 2.0) )
mv.append( mvSlide(root_y_a, delta = 0.5, weight = 2.0) )
s2_a := sigma_a * sigma_a
V_a := s2_a * C_a
for (i in 1:nm) {
    mx_a[i] := root_x_a
    my_a[i] := root_y_a
}
X_a ~ dnMultivariateNormal(mx_a, covariance = V_a)
Y_a ~ dnMultivariateNormal(my_a, covariance = V_a)
X_a.clamp(xm)
Y_a.clamp(ym)
mod = model(C_a)
mn = VectorMonitors()
mn.append( mnFile(root_x_a, root_y_a, sigma_a, filename = out_dir + "a_bm_
modern.log", printgen = 10) )
mc = mcmc(mod, mn, mv, nruns = 1)
mc.burnin(generations = 1000, tuningInterval = 100)
mc.run(generations = n_gen, tuningInterval = 0)

############################## (b) BM + trend, modern tips
C_b <- fnVarCovar(sdm, rm)
sigma_b ~ dnLognormal(0.0, 1.0)
# Priors: root ~ N(0, 10^2) per axis, drift mu ~ N(0, 1) per axis (100 km per kyr).
# Sampled in the equivalent parameterization end = root + mu T (T = 4 kyr, the
# root age), i.e. end | mu ~ N(mu T, 10^2) and root := end - mu T, which has the
# same joint prior (unit Jacobian) but lets the sampler move along the ridge
# root + mu T = const that contemporaneous tips cannot resolve.
T_b <- 4.0
```

```
mu_x_b ~ dnNormal(0.0, 1.0)
mu_y_b ~ dnNormal(0.0, 1.0)
end_x_b ~ dnNormal(mu_x_b * T_b, 10.0)
end_y_b ~ dnNormal(mu_y_b * T_b, 10.0)
end_x_b.setValue(0.0)
end_y_b.setValue(0.0)
root_x_b := end_x_b - mu_x_b * T_b
root_y_b := end_y_b - mu_y_b * T_b
mv = VectorMoves()
mv.append( mvScale(sigma_b, lambda = 0.3, weight = 2.0) )
mv.append( mvSlide(end_x_b, delta = 0.5, weight = 2.0) )
mv.append( mvSlide(end_y_b, delta = 0.5, weight = 2.0) )
mv.append( mvSlide(mu_x_b, delta = 0.5, weight = 2.0) )
mv.append( mvSlide(mu_y_b, delta = 0.5, weight = 2.0) )
s2_b := sigma_b * sigma_b
V_b := s2_b * C_b
for (i in 1:nm) {
    mx_b[i] := root_x_b + mu_x_b * hm[i]
    my_b[i] := root_y_b + mu_y_b * hm[i]
}
X_b ~ dnMultivariateNormal(mx_b, covariance = V_b)
Y_b ~ dnMultivariateNormal(my_b, covariance = V_b)
X_b.clamp(xm)
Y_b.clamp(ym)
mod = model(C_b)
mn = VectorMonitors()
mn.append( mnFile(root_x_b, root_y_b, mu_x_b, mu_y_b, sigma_b, filename = out_dir +
"b_trend_modern.log", printgen = 10) )
mc = mcmc(mod, mn, mv, nruns = 1)
mc.burnin(generations = 1000, tuningInterval = 100)
mc.run(generations = n_gen, tuningInterval = 0)

############################## (c) BM + trend, modern + ancient tips
C_c <- fnVarCovar(sda, ra)
sigma_c ~ dnLognormal(0.0, 1.0)
# Priors: root ~ N(0, 10^2) per axis, drift mu ~ N(0, 1) per axis (100 km per kyr).
# Sampled in the equivalent parameterization end = root + mu T (T = 4 kyr, the
# root age), i.e. end | mu ~ N(mu T, 10^2) and root := end - mu T, which has the
# same joint prior (unit Jacobian) but lets the sampler move along the ridge
# root + mu T = const that contemporaneous tips cannot resolve.
T_c <- 4.0
mu_x_c ~ dnNormal(0.0, 1.0)
mu_y_c ~ dnNormal(0.0, 1.0)
end_x_c ~ dnNormal(mu_x_c * T_c, 10.0)
end_y_c ~ dnNormal(mu_y_c * T_c, 10.0)
end_x_c.setValue(0.0)
end_y_c.setValue(0.0)
root_x_c := end_x_c - mu_x_c * T_c
```

```
root_y_c := end_y_c - mu_y_c * T_c
mv = VectorMoves()
mv.append( mvScale(sigma_c, lambda = 0.3, weight = 2.0) )
mv.append( mvSlide(end_x_c, delta = 0.5, weight = 2.0) )
mv.append( mvSlide(end_y_c, delta = 0.5, weight = 2.0) )
mv.append( mvSlide(mu_x_c, delta = 0.5, weight = 2.0) )
mv.append( mvSlide(mu_y_c, delta = 0.5, weight = 2.0) )
s2_c := sigma_c * sigma_c
V_c := s2_c * C_c
for (i in 1:na) {
    mx_c[i] := root_x_c + mu_x_c * ha[i]
    my_c[i] := root_y_c + mu_y_c * ha[i]
}
X_c ~ dnMultivariateNormal(mx_c, covariance = V_c)
Y_c ~ dnMultivariateNormal(my_c, covariance = V_c)
X_c.clamp(xa)
Y_c.clamp(ya)
mod = model(C_c)
mn = VectorMonitors()
mn.append( mnFile(root_x_c, root_y_c, mu_x_c, mu_y_c, sigma_c, filename = out_dir +
"c_trend_ancient.log", printgen = 10) )
mc = mcmc(mod, mn, mv, nruns = 1)
mc.burnin(generations = 1000, tuningInterval = 100)
mc.run(generations = n_gen, tuningInterval = 0)

write("runtime_s", time("seconds") - t_start, filename = out_dir + "runtime.txt",
separator = "\t")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
(a) BM, modern tips	root_x_a	0.0	2.850	1.936	3.833	571	expected_ miss
(a) BM, modern tips	root_y_a	0.0	0.355	-0.583	1.441	571	pass
(a) BM, modern tips	sigma_a	0.5	0.464	0.344	0.614	506	pass
(b) BM + trend, modern tips	root_x_b	0.0	2.309	-4.763	9.591	571	pass
(b) BM + trend, modern tips	root_y_b	0.0	0.580	-7.525	7.267	571	pass
(b) BM + trend, modern tips	mu_x_b	0.8	0.123	-1.667	1.921	571	pass
(b) BM + trend, modern tips	sigma_b	0.5	0.467	0.336	0.595	571	pass
(c) BM + trend, modern + ancient tips	root_x_c	0.0	0.344	-1.666	1.904	571	pass
(c) BM + trend, modern + ancient tips	root_y_c	0.0	0.614	-0.869	2.879	733	pass
(c) BM + trend, modern + ancient tips	mu_x_c	0.8	0.611	0.146	1.050	398	pass

(c) BM + trend, modern + ancient tips	sigma_c	0.5	0.475	0.360	0.601	571	pass

The results follow the book. (a) Undirected BM places the homeland in the middle of the present range, with a tight interval that excludes the true homeland: this miss is the intended demonstration. (b) Adding a trend without ancient languages gives intervals for the root and the drift that are about as wide as their priors: the data inform only root + μT. (c) With four ancient languages the homeland and the drift are both recovered.

H.5.17 Threshold model

`17_threshold_model.Rev` implements the threshold model of Section 13.9.3 (Equation (13.11)). A Brownian liability with unit rate (which fixes the scale) is sampled at the internal nodes, and the tip liabilities are integrated out analytically: given the parent's liability l_a and branch length t, a tip is in state 1 with probability $\Phi(l_a/\sqrt{t})$ (`pnorm`). The check script computes the exact posterior of the root liability from multivariate-normal orthant probabilities (`mvtnorm::pmvnorm`) on a grid.

```
################################################################################
# 17_threshold_model.Rev
#
# Model:    Threshold model for a binary feature (Felsenstein 2005, 2012): an
#           unobserved liability l evolves by Brownian motion (rate fixed at 1,
#           which sets the liability scale) and the observed state is 1 when
#           l > 0:  y_i = 1(l_i > 0),  l ~ N(mu_root 1, C)   (Eq. pcms-threshold).
#           Liabilities at internal nodes are sampled (data augmentation); the
#           tip liabilities are integrated out analytically, because given its
#           parent value l_a and branch length t,
#               P(y_i = 1 | l_a) = Phi(l_a / sqrt(t)).
# Book:     Chapter 13, "The threshold model" (#pcms-threshold), Chapter 16
#           (#biology-toolkit-threshold); Pagel's lambda
#           (the other option of this item) is 12_pagel_lambda.Rev.
# Estimates: root liability mu_root, P(root state = 1), liabilities at internal
#           nodes; compared in check_traits.R with the exact posterior of
#           mu_root computed from multivariate-normal orthant probabilities.
# Data:     data/traits/threshold.tsv (true mu_root = 0.5) on tree16.tre
# Run:      cd revbayes && rb analyses/traits/17_threshold_model.Rev [--args n_gen]
# Runtime:  about 5 s
################################################################################

seed(1117)
n_gen = 20000
if (args.size() > 1) n_gen = int(args[2])
out_dir = "output/traits/17_threshold_model/"
t_start = time("seconds")

tree <- readTrees("data/traits/tree16.tre")[1]
ydat = readDataDelimitedFile("data/traits/threshold.tsv", header = TRUE, delimiter =
"\t")
```

```
n_tips   <- tree.ntips()
n_nodes  <- tree.nnodes()
root_idx <- tree.getRootIndex()
for (r in 1:ydat.size()) y_obs[tree.tipIndex(ydat[r][1])] <- ifelse(ydat[r][2] >
0.5, 1, 0)

moves    = VectorMoves()
monitors = VectorMonitors()

mu_root ~ dnNormal(0.0, 2.0)
moves.append( mvSlide(mu_root, delta = 0.5, weight = 5.0) )
# internal-node liabilities, stored at position (node index - n_tips)
liab[root_idx - n_tips] := mu_root

for (i in (n_nodes - 1):1) {
    if (i > n_tips) {
        liab[i - n_tips] ~ dnNormal(liab[tree.parent(i) - n_tips],
sqrt(tree.branchLength(i)))
        liab[i - n_tips].setValue(0.0)
        moves.append( mvSlide(liab[i - n_tips], delta = 0.5, weight = 1.0) )
    } else {
        p_one[i] := pnorm(liab[tree.parent(i) - n_tips], 0.0,
sqrt(tree.branchLength(i)))
        y[i] ~ dnBernoulli(p_one[i])
        y[i].clamp(y_obs[i])
    }
}
root_state := ifelse(mu_root > 0.0, 1, 0)

mymodel = model(mu_root)
monitors.append( mnFile(mu_root, root_state, filename = out_dir + "model.log",
printgen = 10) )
monitors.append( mnScreen(printgen = 5000, mu_root) )

mymcmc = mcmc(mymodel, monitors, moves, nruns = 1)
mymcmc.burnin(generations = 1000, tuningInterval = 100)
mymcmc.run(generations = n_gen, tuningInterval = 0)

write("runtime_s", time("seconds") - t_start, filename = out_dir + "runtime.txt",
separator = "\t")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
threshold model	mu_root	0.5	0.520	-1.319	2.51	592	pass
threshold model	P(root state = 1)	1.0	0.711	NA	NA	NA	pass

The MCMC posterior agrees with the exact calculation: posterior mean root liability 0.52 vs 0.49, and $P(\text{root liability} > 0) = 0.71$ vs 0.68.

H.5.18 Constrained models for political complexity

`18_ordered_political.Rev` compares the three constrained rate structures for a four-level political-complexity trait from the worked example of Chapter 14 (Section 14.5): stepwise; up one step, down any number of steps (the generating model); and any jump. Each has one rate for increases and one for decreases, and the three are compared by marginal likelihood, as in the Bayes-factor comparisons of Currie et al. (2010a).

```
################################################################################
# 18_ordered_political.Rev
#
# Model:    Constrained Mk models for a four-level ordered cultural trait,
#           political complexity (0 = acephalous, 1 = simple chiefdom,
#           2 = complex chiefdom, 3 = state). Three rate structures, each with
#           one rate u for increases and one rate d for decreases:
#             1 stepwise:          up and down one level at a time;
#             2 up-step-down-any:  up one level at a time, down any number of
#                                  levels (the generating model);
#             3 any-jump:          any increase, any decrease.
#           Compared by stepping-stone / path-sampling marginal likelihoods, as
#           in the Bayes-factor comparisons of Currie et al. (2010).
# Book:     Chapter 14, "Sequential evolution of political complexity"
#           (#cultural-traits-political); Chapter 12, ordered Mk models (#asr-mk).
# Estimates: u, d under each model; ln marginal likelihoods.
# Data:     data/traits/political.nex on tree16.tre
# Run:      cd revbayes && rb analyses/traits/18_ordered_political.Rev [--args n_gen]
# Runtime:  about 1 minute
################################################################################

seed(1118)
n_gen = 5000
if (args.size() > 1) n_gen = int(args[2])
n_cats  = 24
pp_gen  = 500
out_dir = "output/traits/18_ordered_political/"
t_start = time("seconds")

# NB: model() collects every node connected to its argument, including through
# shared constant nodes and through the elements of one Rev vector, so each
# model has its own tree object and its own variable names.
pol  <- readDiscreteCharacterData("data/traits/political.nex")
write("model,lnML_stepping_stone,lnML_path_sampling\n", filename = out_dir +
"marginal_likelihoods.csv")
# off-diagonal elements in row order: (1,2) (1,3) (1,4) | (2,1) (2,3) (2,4) | (3,1)
(3,2) (3,4) | (4,1) (4,2) (4,3)
```

```
############################ model: stepwise
tree_stepwise <- readTrees("data/traits/tree16.tre")[1]
u_stepwise ~ dnExponential(2.0)
d_stepwise ~ dnExponential(2.0)
mv = VectorMoves()
mv.append( mvScale(u_stepwise, lambda = 0.5, weight = 2.0) )
mv.append( mvScale(d_stepwise, lambda = 0.5, weight = 2.0) )
Q_stepwise := fnFreeK([ u_stepwise, 0.0, 0.0,  d_stepwise, u_stepwise, 0.0,  0.0, d_
stepwise, u_stepwise,  0.0, 0.0, d_stepwise ], rescaled = FALSE)
ctmc_stepwise ~ dnPhyloCTMC(tree = tree_stepwise, Q = Q_stepwise, rootFrequencies =
simplex(rep(1, 4)),
                        type = "Standard", nSites = 1)
ctmc_stepwise.clamp(pol)
mod = model(Q_stepwise)
mn = VectorMonitors()
mn.append( mnFile(u_stepwise, d_stepwise, filename = out_dir + "stepwise_model.log",
printgen = 10) )
mc = mcmc(mod, mn, mv, nruns = 1)
mc.burnin(generations = 500, tuningInterval = 100)
mc.run(generations = n_gen, tuningInterval = 0)
pp = powerPosterior(mod, mv, VectorMonitors(), out_dir + "stepwise_powp.log", cats =
n_cats, sampleFreq = 10)
pp.burnin(generations = 500, tuningInterval = 100)
pp.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "stepwise_powp.log", "power", "likelihood",
"\t").marginal()
ps = pathSampler(out_dir + "stepwise_powp.log", "power", "likelihood",
"\t").marginal()
write("stepwise", ss, ps, filename = out_dir + "marginal_likelihoods.csv", append =
TRUE, separator = ",")
write("\n", filename = out_dir + "marginal_likelihoods.csv", append = TRUE)

############################ model: up_step_down_any
tree_up_step_down_any <- readTrees("data/traits/tree16.tre")[1]
u_up_step_down_any ~ dnExponential(2.0)
d_up_step_down_any ~ dnExponential(2.0)
mv = VectorMoves()
mv.append( mvScale(u_up_step_down_any, lambda = 0.5, weight = 2.0) )
mv.append( mvScale(d_up_step_down_any, lambda = 0.5, weight = 2.0) )
Q_up_step_down_any := fnFreeK([ u_up_step_down_any, 0.0, 0.0,  d_up_step_down_any,
u_up_step_down_any, 0.0,  d_up_step_down_any, d_up_step_down_any, u_up_step_down_
any,  d_up_step_down_any, d_up_step_down_any, d_up_step_down_any ], rescaled =
FALSE)
ctmc_up_step_down_any ~ dnPhyloCTMC(tree = tree_up_step_down_any, Q = Q_up_step_
down_any, rootFrequencies = simplex(rep(1, 4)),
                        type = "Standard", nSites = 1)
ctmc_up_step_down_any.clamp(pol)
```

```
mod = model(Q_up_step_down_any)
mn = VectorMonitors()
mn.append( mnFile(u_up_step_down_any, d_up_step_down_any, filename = out_dir + "up_
step_down_any_model.log", printgen = 10) )
mc = mcmc(mod, mn, mv, nruns = 1)
mc.burnin(generations = 500, tuningInterval = 100)
mc.run(generations = n_gen, tuningInterval = 0)
pp = powerPosterior(mod, mv, VectorMonitors(), out_dir + "up_step_down_any_
powp.log", cats = n_cats, sampleFreq = 10)
pp.burnin(generations = 500, tuningInterval = 100)
pp.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "up_step_down_any_powp.log", "power",
"likelihood", "\t").marginal()
ps = pathSampler(out_dir + "up_step_down_any_powp.log", "power", "likelihood",
"\t").marginal()
write("up_step_down_any", ss, ps, filename = out_dir + "marginal_likelihoods.csv",
append = TRUE, separator = ",")
write("\n", filename = out_dir + "marginal_likelihoods.csv", append = TRUE)

############################ model: any_jump
tree_any_jump <- readTrees("data/traits/tree16.tre")[1]
u_any_jump ~ dnExponential(2.0)
d_any_jump ~ dnExponential(2.0)
mv = VectorMoves()
mv.append( mvScale(u_any_jump, lambda = 0.5, weight = 2.0) )
mv.append( mvScale(d_any_jump, lambda = 0.5, weight = 2.0) )
Q_any_jump := fnFreeK([ u_any_jump, u_any_jump, u_any_jump,  d_any_jump, u_any_jump,
u_any_jump,  d_any_jump, d_any_jump, u_any_jump,  d_any_jump, d_any_jump, d_any_jump
], rescaled = FALSE)
ctmc_any_jump ~ dnPhyloCTMC(tree = tree_any_jump, Q = Q_any_jump, rootFrequencies =
simplex(rep(1, 4)),
                        type = "Standard", nSites = 1)
ctmc_any_jump.clamp(pol)
mod = model(Q_any_jump)
mn = VectorMonitors()
mn.append( mnFile(u_any_jump, d_any_jump, filename = out_dir + "any_jump_model.log",
printgen = 10) )
mc = mcmc(mod, mn, mv, nruns = 1)
mc.burnin(generations = 500, tuningInterval = 100)
mc.run(generations = n_gen, tuningInterval = 0)
pp = powerPosterior(mod, mv, VectorMonitors(), out_dir + "any_jump_powp.log", cats =
n_cats, sampleFreq = 10)
pp.burnin(generations = 500, tuningInterval = 100)
pp.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "any_jump_powp.log", "power", "likelihood",
"\t").marginal()
ps = pathSampler(out_dir + "any_jump_powp.log", "power", "likelihood",
"\t").marginal()
```

```
write("any_jump", ss, ps, filename = out_dir + "marginal_likelihoods.csv", append =
TRUE, separator = ",")
write("\n", filename = out_dir + "marginal_likelihoods.csv", append = TRUE)

write("runtime_s", time("seconds") - t_start, filename = out_dir + "runtime.txt",
separator = "\t")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
ordered Mk: stepwise	lnML	NA	-20.441	NA	NA	NA	pass
ordered Mk: up_step_down_any	lnML	NA	-20.875	NA	NA	NA	pass
ordered Mk: any_jump	lnML	NA	-20.760	NA	NA	NA	pass
ordered Mk comparison	best_model	2.00	1.000	NA	NA	NA	miss
ordered Mk: up_step_down_any	u_up_step_down_any	0.40	0.764	0.020	1.793	476	pass
ordered Mk: up_step_down_any	d_up_step_down_any	0.15	0.212	0.003	0.527	476	pass

The marginal likelihoods differ by less than 0.5 log units, so the 16 tips do not discriminate between the models (the book's example uses 84 societies). The stepwise model comes out marginally best, and the check is a miss. The rates of the generating model are covered.

H.5.19 Phylogeny and space together

`19_phylo_spatial.Rev` implements Equation (13.12) (Section 13.10): a Gaussian trait with phylogenetic, spatial and independent variance components. The phylogenetic correlation matrix and the spatial correlation matrix $S_{ij} = \exp(-d_{ij})$ are read from files and assembled in Rev with `fnVarCovar`, and the total covariance $\sigma_p^2\mathbf{C} + \sigma_s^2\mathbf{S} + \sigma_e^2\mathbf{I}$ is a deterministic matrix passed to `dnMultivariateNormal`. The full model is compared with phylogeny-only and space-only models.

```
################################################################################
# 19_phylo_spatial.Rev
#
# Model:    Gaussian trait with separate phylogenetic, spatial and independent
#           components (Freckleton & Jetz 2009 style):
#               y = mu 1 + a + s + e,  a ~ N(0, s2_phy C),  s ~ N(0, s2_space S),
#               e ~ N(0, s2_err I)                              (Eq. pcms-spatphylo)
#           with C the phylogenetic correlation matrix (vcv scaled to unit
#           diagonal) and S_ij = exp(-d_ij / rho) a spatial correlation matrix
#           (range rho = 1, fixed, as in the book's example). Three models are
#           compared by stepping-stone sampling: phylogeny + space, phylogeny
#           only, space only. The matrices are built in Rev from the files with
#           fnVarCovar (unit standard deviations, off-diagonal correlations).
# Book:     Chapter 13, "Space and phylogeny together" (#pcms-spatial);
#           Chapter 13 PGLMM (#pcms-pglmm) for the variance-components view.
```

```
# Estimates: mu, s2_phy, s2_space, s2_err and their shares of the variance;
#           ln marginal likelihoods of the three models.
# Data:     data/traits/spatial_trait.tsv, spatial_Cphy.tsv, spatial_S.tsv
# Run:      cd revbayes && rb analyses/traits/19_phylo_spatial.Rev [--args n_gen]
# Runtime:  about 2.5 minutes (each 16 x 16 multivariate normal density is slow)
################################################################################

seed(1119)
n_gen = 4000
if (args.size() > 1) n_gen = int(args[2])
n_cats  = 12
pp_gen  = 250
out_dir = "output/traits/19_phylo_spatial/"
t_start = time("seconds")

ydat = readDataDelimitedFile("data/traits/spatial_trait.tsv", header = TRUE,
delimiter = "\t")
Cdat = readDataDelimitedFile("data/traits/spatial_Cphy.tsv", header = TRUE,
delimiter = "\t")
Sdat = readDataDelimitedFile("data/traits/spatial_S.tsv", header = TRUE, delimiter =
"\t")
n <- ydat.size()
for (i in 1:n) y_obs[i] <- ydat[i][2]      # rows of all three files are in the same
taxon order
# upper-triangle (row-major) correlations
k = 0
for (i in 1:(n - 1)) {
    for (j in (i + 1):n) {
        k = k + 1
        c_phy[k] <- Cdat[i][j + 1]
        c_sp[k]  <- Sdat[i][j + 1]
    }
}
# NB: model() collects every node connected to its argument, including through
# shared constant nodes and through the elements of one Rev vector, so each
# model has its own constant matrices and its own variable names.
write("model,lnML_stepping_stone,lnML_path_sampling\n", filename = out_dir +
"marginal_likelihoods.csv")

############################ model: phylo_space
C_phylo_space <- fnVarCovar(rep(1.0, n), c_phy)
S_phylo_space <- fnVarCovar(rep(1.0, n), c_sp)
I_phylo_space <- diagonalMatrix(n)
n_phylo_space <- n
mv = VectorMoves()
mu_phylo_space ~ dnNormal(0.0, 10.0)
mv.append( mvSlide(mu_phylo_space, delta = 0.5, weight = 1.0) )
s2_err_phylo_space ~ dnExponential(1.0)
```

```
mv.append( mvScale(s2_err_phylo_space, lambda = 0.5, weight = 1.0) )
s2_phy_phylo_space ~ dnExponential(1.0)
mv.append( mvScale(s2_phy_phylo_space, lambda = 0.5, weight = 1.0) )
s2_space_phylo_space ~ dnExponential(1.0)
mv.append( mvScale(s2_space_phylo_space, lambda = 0.5, weight = 1.0) )
V_phylo_space := s2_phy_phylo_space * C_phylo_space + s2_space_phylo_space * S_
phylo_space + s2_err_phylo_space * I_phylo_space
total_phylo_space := s2_phy_phylo_space + s2_space_phylo_space + s2_err_phylo_space
share_phy_phylo_space := s2_phy_phylo_space / total_phylo_space
share_space_phylo_space := s2_space_phylo_space / total_phylo_space
mvec_phylo_space := rep(mu_phylo_space, n_phylo_space)
y_phylo_space ~ dnMultivariateNormal(mvec_phylo_space, covariance = V_phylo_space)
y_phylo_space.clamp(y_obs)
mod = model(V_phylo_space)
mn = VectorMonitors()
mn.append( mnFile(mu_phylo_space, s2_phy_phylo_space, s2_space_phylo_space, s2_err_
phylo_space, share_phy_phylo_space, share_space_phylo_space, filename = out_dir +
"phylo_space_model.log", printgen = 10) )
mc = mcmc(mod, mn, mv, nruns = 1)
mc.burnin(generations = 500, tuningInterval = 50)
mc.run(generations = n_gen, tuningInterval = 0)
pp = powerPosterior(mod, mv, VectorMonitors(), out_dir + "phylo_space_powp.log",
cats = n_cats, sampleFreq = 10)
pp.burnin(generations = 200, tuningInterval = 50)
pp.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "phylo_space_powp.log", "power", "likelihood",
"\t").marginal()
ps = pathSampler(out_dir + "phylo_space_powp.log", "power", "likelihood",
"\t").marginal()
write("phylo_space", ss, ps, filename = out_dir + "marginal_likelihoods.csv", append
= TRUE, separator = ",")
write("\n", filename = out_dir + "marginal_likelihoods.csv", append = TRUE)

############################ model: phylo_only
C_phylo_only <- fnVarCovar(rep(1.0, n), c_phy)
S_phylo_only <- fnVarCovar(rep(1.0, n), c_sp)
I_phylo_only <- diagonalMatrix(n)
n_phylo_only <- n
mv = VectorMoves()
mu_phylo_only ~ dnNormal(0.0, 10.0)
mv.append( mvSlide(mu_phylo_only, delta = 0.5, weight = 1.0) )
s2_err_phylo_only ~ dnExponential(1.0)
mv.append( mvScale(s2_err_phylo_only, lambda = 0.5, weight = 1.0) )
s2_phy_phylo_only ~ dnExponential(1.0)
mv.append( mvScale(s2_phy_phylo_only, lambda = 0.5, weight = 1.0) )
V_phylo_only := s2_phy_phylo_only * C_phylo_only + s2_err_phylo_only * I_phylo_only
total_phylo_only := s2_phy_phylo_only + s2_err_phylo_only
share_phy_phylo_only := s2_phy_phylo_only / total_phylo_only
```

```
mvec_phylo_only := rep(mu_phylo_only, n_phylo_only)
y_phylo_only ~ dnMultivariateNormal(mvec_phylo_only, covariance = V_phylo_only)
y_phylo_only.clamp(y_obs)
mod = model(V_phylo_only)
mn = VectorMonitors()
mn.append( mnFile(mu_phylo_only, s2_phy_phylo_only, s2_err_phylo_only, share_phy_
phylo_only, filename = out_dir + "phylo_only_model.log", printgen = 10) )
mc = mcmc(mod, mn, mv, nruns = 1)
mc.burnin(generations = 500, tuningInterval = 50)
mc.run(generations = n_gen, tuningInterval = 0)
pp = powerPosterior(mod, mv, VectorMonitors(), out_dir + "phylo_only_powp.log", cats
= n_cats, sampleFreq = 10)
pp.burnin(generations = 200, tuningInterval = 50)
pp.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "phylo_only_powp.log", "power", "likelihood",
"\t").marginal()
ps = pathSampler(out_dir + "phylo_only_powp.log", "power", "likelihood",
"\t").marginal()
write("phylo_only", ss, ps, filename = out_dir + "marginal_likelihoods.csv", append
= TRUE, separator = ",")
write("\n", filename = out_dir + "marginal_likelihoods.csv", append = TRUE)

############################ model: space_only
C_space_only <- fnVarCovar(rep(1.0, n), c_phy)
S_space_only <- fnVarCovar(rep(1.0, n), c_sp)
I_space_only <- diagonalMatrix(n)
n_space_only <- n
mv = VectorMoves()
mu_space_only ~ dnNormal(0.0, 10.0)
mv.append( mvSlide(mu_space_only, delta = 0.5, weight = 1.0) )
s2_err_space_only ~ dnExponential(1.0)
mv.append( mvScale(s2_err_space_only, lambda = 0.5, weight = 1.0) )
s2_space_space_only ~ dnExponential(1.0)
mv.append( mvScale(s2_space_space_only, lambda = 0.5, weight = 1.0) )
V_space_only := s2_space_space_only * S_space_only + s2_err_space_only * I_space_
only
total_space_only := s2_space_space_only + s2_err_space_only
share_space_space_only := s2_space_space_only / total_space_only
mvec_space_only := rep(mu_space_only, n_space_only)
y_space_only ~ dnMultivariateNormal(mvec_space_only, covariance = V_space_only)
y_space_only.clamp(y_obs)
mod = model(V_space_only)
mn = VectorMonitors()
mn.append( mnFile(mu_space_only, s2_space_space_only, s2_err_space_only, share_
space_space_only, filename = out_dir + "space_only_model.log", printgen = 10) )
mc = mcmc(mod, mn, mv, nruns = 1)
mc.burnin(generations = 500, tuningInterval = 50)
mc.run(generations = n_gen, tuningInterval = 0)
```

```
pp = powerPosterior(mod, mv, VectorMonitors(), out_dir + "space_only_powp.log", cats
= n_cats, sampleFreq = 10)
pp.burnin(generations = 200, tuningInterval = 50)
pp.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "space_only_powp.log", "power", "likelihood",
"\t").marginal()
ps = pathSampler(out_dir + "space_only_powp.log", "power", "likelihood",
"\t").marginal()
write("space_only", ss, ps, filename = out_dir + "marginal_likelihoods.csv", append
= TRUE, separator = ",")
write("\n", filename = out_dir + "marginal_likelihoods.csv", append = TRUE)

write("runtime_s", time("seconds") - t_start, filename = out_dir + "runtime.txt",
separator = "\t")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
phylogeny + space	mu_phylo_space	0.5	0.823	-0.915	2.297	381	pass
phylogeny + space	s2_phy_phylo_space	1.0	1.628	0.001	3.938	230	pass
phylogeny + space	s2_space_phylo_space	0.8	1.143	0.016	2.827	288	pass
phylogeny + space	s2_err_phylo_space	0.2	0.698	0.001	1.963	292	pass
phylogeny + space comparison	best_model	NA	NA	NA	NA	NA	pass

All variance components are covered but broad: with 16 languages, descent and space are hard to separate. The generating model has the highest marginal likelihood, by only about 0.8 log units over the phylogeny-only model.

H.5.20 Multi-optimum Ornstein-Uhlenbeck

`20_ou_multi_optimum.Rev` fits an OU model with two optima painted on the tree in advance (Section 13.3.2; Butler and King (2004)). One clade of eight languages, together with its stem branch, is pulled toward θ_2; every other branch is pulled toward θ_1. The per-branch optima are passed to `dnPhyloOrnsteinUhlenbeckMVN`, and the model is compared with a single-optimum OU model.

```
################################################################################
# 20_ou_multi_optimum.Rev
#
# Model:    Multi-optimum Ornstein-Uhlenbeck model (OUM; Hansen 1997, Butler &
#           King 2004) with regimes painted on the tree in advance: branches in
#           one clade (8 languages and its stem) are pulled toward theta_2, all
#           other branches toward theta_1; alpha and sigma2 are shared. The
#           optimum is given per branch to dnPhyloOrnsteinUhlenbeckMVN; the root
#           starts at theta_1. Compared with a single-optimum OU (OU1) by
```

```
#           stepping-stone sampling.
# Book:     Chapter 13, "The Ornstein-Uhlenbeck process" (#pcms-ou), multiple
#           optima, and the toolbox box on multi-optimum OU models.
# Estimates: theta_1, theta_2, alpha, sigma2; ln marginal likelihoods OUM vs OU1.
# Data:     data/traits/trait_oum.nex, data/traits/oum_regime2_tips.txt, tree16.tre
# Run:      cd revbayes && rb analyses/traits/20_ou_multi_optimum.Rev [--args n_gen]
# Runtime:  about 2.5 minutes
################################################################################

seed(1120)
n_gen = 6000
if (args.size() > 1) n_gen = int(args[2])
n_cats  = 12
pp_gen  = 300
out_dir = "output/traits/20_ou_multi_optimum/"
t_start = time("seconds")

# separate tree objects keep the two model graphs apart (see 05_mk_asr.Rev)
tree_m <- readTrees("data/traits/tree16.tre")[1]
tree_1 <- readTrees("data/traits/tree16.tre")[1]
dat    <- readContinuousCharacterData("data/traits/trait_oum.nex")
rt = readDataDelimitedFile("data/traits/oum_regime2_tips.txt", header = FALSE)
for (r in 1:rt.size()) reg2_names[r] <- rt[r][1]
reg2_clade <- clade(reg2_names)
n_branches <- tree_m.nnodes() - 1
for (b in 1:n_branches) in_reg2[b] <- ifelse(tree_m.isContainedInClade(b, reg2_
clade), 1.0, 0.0)

write("model,lnML_stepping_stone,lnML_path_sampling\n", filename = out_dir +
"marginal_likelihoods.csv")

############################ OUM
theta_1 ~ dnUniform(-10.0, 10.0)
theta_2 ~ dnUniform(-10.0, 10.0)
theta_1.setValue(0.0)
theta_2.setValue(0.0)
alpha   ~ dnExponential(1.0)
sigma2  ~ dnLoguniform(1e-3, 100.0)
mv = VectorMoves()
mv.append( mvSlide(theta_1, delta = 0.5, weight = 1.0) )
mv.append( mvSlide(theta_2, delta = 0.5, weight = 1.0) )
mv.append( mvSlide(theta_2, delta = 2.0, weight = 1.0) )   # theta_2 has a long tail
when alpha is small
mv.append( mvScale(alpha, lambda = 0.5, weight = 1.0) )
mv.append( mvScale(sigma2, lambda = 0.5, weight = 1.0) )
mv.append( mvUpDownScale(lambda = 0.3, weight = 1.0) )
mv[mv.size()].addVariable(alpha, up = TRUE)
mv[mv.size()].addVariable(sigma2, up = TRUE)
```

```
for (b in 1:n_branches) theta_b[b] := theta_1 + in_reg2[b] * (theta_2 - theta_1)
half_life := ln(2.0) / alpha
x ~ dnPhyloOrnsteinUhlenbeckMVN(tree_m, alpha = alpha, theta = theta_b, sigma =
sqrt(sigma2),
                                rootStates = theta_1, nSites = 1)
x.clamp(dat)
mod = model(alpha)
mn = VectorMonitors()
mn.append( mnFile(theta_1, theta_2, alpha, sigma2, half_life, filename = out_dir +
"OUM_model.log", printgen = 20) )
mn.append( mnScreen(printgen = 5000, theta_1, theta_2) )
mc = mcmc(mod, mn, mv, nruns = 1)
mc.burnin(generations = 1000, tuningInterval = 100)
mc.run(generations = 3 * n_gen, tuningInterval = 0)
pp = powerPosterior(mod, mv, VectorMonitors(), out_dir + "OUM_powp.log", cats = n_
cats, sampleFreq = 10)
pp.burnin(generations = 300, tuningInterval = 50)
pp.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "OUM_powp.log", "power", "likelihood",
"\t").marginal()
ps = pathSampler(out_dir + "OUM_powp.log", "power", "likelihood", "\t").marginal()
write("OUM", ss, ps, filename = out_dir + "marginal_likelihoods.csv", append = TRUE,
separator = ",")
write("\n", filename = out_dir + "marginal_likelihoods.csv", append = TRUE)

############################ OU1
theta_s ~ dnUniform(-10.0, 10.0)
theta_s.setValue(0.0)
alpha_s ~ dnExponential(1.0)
sigma2_s ~ dnLoguniform(1e-3, 100.0)
mv1 = VectorMoves()
mv1.append( mvSlide(theta_s, delta = 0.5, weight = 1.0) )
mv1.append( mvScale(alpha_s, lambda = 0.5, weight = 1.0) )
mv1.append( mvScale(sigma2_s, lambda = 0.5, weight = 1.0) )
mv1.append( mvUpDownScale(lambda = 0.3, weight = 1.0) )
mv1[mv1.size()].addVariable(alpha_s, up = TRUE)
mv1[mv1.size()].addVariable(sigma2_s, up = TRUE)
x1 ~ dnPhyloOrnsteinUhlenbeckMVN(tree_1, alpha = alpha_s, theta = theta_s, sigma =
sqrt(sigma2_s),
                                 rootStates = theta_s, nSites = 1)
x1.clamp(dat)
mod1 = model(alpha_s)
mn1 = VectorMonitors()
mn1.append( mnFile(theta_s, alpha_s, sigma2_s, filename = out_dir + "OU1_model.log",
printgen = 10) )
mc1 = mcmc(mod1, mn1, mv1, nruns = 1)
mc1.burnin(generations = 1000, tuningInterval = 100)
mc1.run(generations = n_gen / 2, tuningInterval = 0)
```

```
pp1 = powerPosterior(mod1, mv1, VectorMonitors(), out_dir + "OU1_powp.log", cats =
n_cats, sampleFreq = 10)
pp1.burnin(generations = 300, tuningInterval = 50)
pp1.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "OU1_powp.log", "power", "likelihood",
"\t").marginal()
ps = pathSampler(out_dir + "OU1_powp.log", "power", "likelihood", "\t").marginal()
write("OU1", ss, ps, filename = out_dir + "marginal_likelihoods.csv", append = TRUE,
separator = ",")
write("\n", filename = out_dir + "marginal_likelihoods.csv", append = TRUE)

write("runtime_s", time("seconds") - t_start, filename = out_dir + "runtime.txt",
separator = "\t")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
multi-optimum OU	theta_1	0.0	0.150	-0.384	0.610	744	pass
multi-optimum OU	theta_2	3.0	3.535	2.624	6.549	70	warn_ess
multi-optimum OU	alpha	1.5	1.067	0.066	2.521	310	pass
multi-optimum OU	sigma2	0.5	0.347	0.078	0.805	574	pass
OUM vs OU1	lnBF_OUM_vs_OU1	NA	7.461	NA	NA	NA	pass

The two-optimum model is strongly preferred (ln BF of about 7.5), and all parameters are covered. θ_2 mixes slowly (ESS below 100 even with a three-times longer chain), because when α is small the regime-2 optimum is only weakly constrained and its posterior has a long upper tail. A longer run, or an informative prior on the phylogenetic half-life, fixes this.

H.5.21 Root treatments for a rare feature

`21_root_treatment_asr.Rev` fits a two-rate Mk model to one binary feature that is present in only five of the 16 languages, in both primary clades, under seven treatments of the root and the rate priors (Sections 12.4.4 and 12.4.5): flat, stationary (the RevBayes default when `rootFrequencies` is omitted), estimated with a Dirichlet(1, 1) prior, fixed to “absent”, fixed to “present”, flat with informative asymmetric rate priors scaled to the tree length (about one gain expected on the whole tree, losses ten times faster), and a loss-only model. The feature was simulated with a gain rate of 0.1, a loss rate of 0.6 and a root in which it was present, the pattern of a feature such as the Indo-European augment. The design follows exercises from a graduate course on Bayesian reconstruction. Because the model has only two rates, the check script computes every marginal likelihood and root probability exactly, by integrating the likelihood over the rates on a grid.

```
################################################################################
# 21_root_treatment_asr.Rev
#
# Model:    Two-rate Mk model (gain q01, loss q10) for a rare binary
#           innovation that is present in only a few languages, fitted under
#           different treatments of the root and of the rate priors:
#             FLAT  root frequencies (1/2, 1/2);
#             STAT  root at the stationary distribution of Q (the RevBayes
#                   default when rootFrequencies is not given);
#             EST   root frequencies estimated, Dirichlet(1, 1) prior;
#             ROOT0 root fixed to "absent";  ROOT1 root fixed to "present";
#             INFO  flat root, informative asymmetric priors scaled by the
#                   tree length (about one gain expected on the whole tree,
#                   losses ten times faster);
#             LOSS  irreversible loss-only model (q01 = 0), flat root.
#           Every model has the same Exponential(1) rate priors except INFO
#           and LOSS. Marginal likelihoods by stepping-stone / path sampling;
#           root-state posteriors from joint ancestral-state samples. The
#           ROOT0/ROOT1 marginal likelihoods give a Bayes factor for the root
#           state ("fossilizing" the root, as in Pagel et al. 2004).
#           The design (an augment-like feature, asymmetric gain and loss
#           priors, estimated root frequencies, marginal likelihoods) follows
#           exercises from a graduate course on Bayesian reconstruction.
# Book:     Chapter 12, "The root prior" (#asr-root), "Constraints and priors
#           that decide the answer" (#asr-constraints); Appendix
#           Section #tech-asr-rootprior.
# Estimates: q01, q10 under each model; P(root = present); ln marginal
#           likelihoods; posterior of the root frequency under EST.
# Data:     data/traits/rare_feature.nex on tree16.tre
# Run:      cd revbayes && rb analyses/traits/21_root_treatment_asr.Rev
#           [--args n_gen]
# Runtime:  about 3 minutes
################################################################################

seed(1121)
n_gen = 10000
if (args.size() > 1) n_gen = int(args[2])
n_cats  = 24
pp_gen  = 500
out_dir = "output/traits/21_root_treatment_asr/"
t_start = time("seconds")
ml_file = out_dir + "marginal_likelihoods.csv"
write("model,lnML_stepping_stone,lnML_path_sampling\n", filename = ml_file)

rare <- readDiscreteCharacterData("data/traits/rare_feature.nex")

# Each model has its own tree copy and its own variable names: model() merges
# models that share any node (a tree object, a prior constant, two elements of
```

```
# one Rev vector), and reassigning a variable name in Rev replaces the node in
# place, so that the previous model's children are carried over (see
# 05_mk_asr.Rev).

############################## FLAT
tr_f <- readTrees("data/traits/tree16.tre")[1]
mv_f = VectorMoves()
q01_f ~ dnExponential(1.0)
q10_f ~ dnExponential(1.0)
mv_f.append( mvScale(q01_f, lambda = 0.5, weight = 2.0) )
mv_f.append( mvScale(q10_f, lambda = 0.5, weight = 2.0) )
Q_f := fnFreeK([q01_f, q10_f], rescaled = FALSE)
rf_f <- simplex(v(1.0, 1.0))
ctmc_f ~ dnPhyloCTMC(tree = tr_f, Q = Q_f, rootFrequencies = rf_f,
                     type = "Standard", nSites = 1)
ctmc_f.clamp(rare)
mod_f = model(Q_f)
mn_f = VectorMonitors()
mn_f.append( mnModel(filename = out_dir + "FLAT_model.log", printgen = 10) )
mn_f.append( mnJointConditionalAncestralState(tree = tr_f, ctmc = ctmc_f,
             type = "Standard", withTips = FALSE, withStartStates = FALSE,
             filename = out_dir + "FLAT_states.log", printgen = 10) )
run_f = mcmc(mod_f, mn_f, mv_f, nruns = 1)
run_f.burnin(generations = 1000, tuningInterval = 100)
run_f.run(generations = n_gen, tuningInterval = 0)
pp_f = powerPosterior(mod_f, mv_f, VectorMonitors(), out_dir + "FLAT_powp.log",
                      cats = n_cats, sampleFreq = 10)
pp_f.burnin(generations = 500, tuningInterval = 100)
pp_f.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "FLAT_powp.log", "power", "likelihood",
                          "\t").marginal()
ps = pathSampler(out_dir + "FLAT_powp.log", "power", "likelihood",
                 "\t").marginal()
write("FLAT", ss, ps, filename = ml_file, append = TRUE, separator = ",")
write("\n", filename = ml_file, append = TRUE)

############################## STAT
tr_s <- readTrees("data/traits/tree16.tre")[1]
mv_s = VectorMoves()
q01_s ~ dnExponential(1.0)
q10_s ~ dnExponential(1.0)
mv_s.append( mvScale(q01_s, lambda = 0.5, weight = 2.0) )
mv_s.append( mvScale(q10_s, lambda = 0.5, weight = 2.0) )
Q_s := fnFreeK([q01_s, q10_s], rescaled = FALSE)
# no rootFrequencies: the stationary distribution of Q is used
ctmc_s ~ dnPhyloCTMC(tree = tr_s, Q = Q_s, type = "Standard",
                     nSites = 1)
ctmc_s.clamp(rare)
```

```
mod_s = model(Q_s)
mn_s = VectorMonitors()
mn_s.append( mnModel(filename = out_dir + "STAT_model.log", printgen = 10) )
mn_s.append( mnJointConditionalAncestralState(tree = tr_s, ctmc = ctmc_s,
             type = "Standard", withTips = FALSE, withStartStates = FALSE,
             filename = out_dir + "STAT_states.log", printgen = 10) )
run_s = mcmc(mod_s, mn_s, mv_s, nruns = 1)
run_s.burnin(generations = 1000, tuningInterval = 100)
run_s.run(generations = n_gen, tuningInterval = 0)
pp_s = powerPosterior(mod_s, mv_s, VectorMonitors(), out_dir + "STAT_powp.log",
                      cats = n_cats, sampleFreq = 10)
pp_s.burnin(generations = 500, tuningInterval = 100)
pp_s.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "STAT_powp.log", "power", "likelihood",
                          "\t").marginal()
ps = pathSampler(out_dir + "STAT_powp.log", "power", "likelihood",
                 "\t").marginal()
write("STAT", ss, ps, filename = ml_file, append = TRUE, separator = ",")
write("\n", filename = ml_file, append = TRUE)

############################ EST
tr_e <- readTrees("data/traits/tree16.tre")[1]
mv_e = VectorMoves()
q01_e ~ dnExponential(1.0)
q10_e ~ dnExponential(1.0)
mv_e.append( mvScale(q01_e, lambda = 0.5, weight = 2.0) )
mv_e.append( mvScale(q10_e, lambda = 0.5, weight = 2.0) )
Q_e := fnFreeK([q01_e, q10_e], rescaled = FALSE)
rf_e ~ dnDirichlet(v(1.0, 1.0))
mv_e.append( mvBetaSimplex(rf_e, weight = 2.0) )
ctmc_e ~ dnPhyloCTMC(tree = tr_e, Q = Q_e, rootFrequencies = rf_e,
                     type = "Standard", nSites = 1)
ctmc_e.clamp(rare)
mod_e = model(Q_e)
mn_e = VectorMonitors()
mn_e.append( mnModel(filename = out_dir + "EST_model.log", printgen = 10) )
mn_e.append( mnJointConditionalAncestralState(tree = tr_e, ctmc = ctmc_e,
             type = "Standard", withTips = FALSE, withStartStates = FALSE,
             filename = out_dir + "EST_states.log", printgen = 10) )
run_e = mcmc(mod_e, mn_e, mv_e, nruns = 1)
run_e.burnin(generations = 1000, tuningInterval = 100)
run_e.run(generations = n_gen, tuningInterval = 0)
pp_e = powerPosterior(mod_e, mv_e, VectorMonitors(), out_dir + "EST_powp.log",
                      cats = n_cats, sampleFreq = 10)
pp_e.burnin(generations = 500, tuningInterval = 100)
pp_e.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "EST_powp.log", "power", "likelihood",
                          "\t").marginal()
```

```
ps = pathSampler(out_dir + "EST_powp.log", "power", "likelihood",
                 "\t").marginal()
write("EST", ss, ps, filename = ml_file, append = TRUE, separator = ",")
write("\n", filename = ml_file, append = TRUE)

############################ ROOT0
tr_r0 <- readTrees("data/traits/tree16.tre")[1]
mv_r0 = VectorMoves()
q01_r0 ~ dnExponential(1.0)
q10_r0 ~ dnExponential(1.0)
mv_r0.append( mvScale(q01_r0, lambda = 0.5, weight = 2.0) )
mv_r0.append( mvScale(q10_r0, lambda = 0.5, weight = 2.0) )
Q_r0 := fnFreeK([q01_r0, q10_r0], rescaled = FALSE)
rf_r0 <- simplex(v(1.0, 0.0))
ctmc_r0 ~ dnPhyloCTMC(tree = tr_r0, Q = Q_r0, rootFrequencies = rf_r0,
                      type = "Standard", nSites = 1)
ctmc_r0.clamp(rare)
mod_r0 = model(Q_r0)
mn_r0 = VectorMonitors()
mn_r0.append( mnModel(filename = out_dir + "ROOT0_model.log", printgen = 10) )
mn_r0.append( mnJointConditionalAncestralState(tree = tr_r0, ctmc = ctmc_r0,
              type = "Standard", withTips = FALSE, withStartStates = FALSE,
              filename = out_dir + "ROOT0_states.log", printgen = 10) )
run_r0 = mcmc(mod_r0, mn_r0, mv_r0, nruns = 1)
run_r0.burnin(generations = 1000, tuningInterval = 100)
run_r0.run(generations = n_gen, tuningInterval = 0)
pp_r0 = powerPosterior(mod_r0, mv_r0, VectorMonitors(),
                       out_dir + "ROOT0_powp.log",
                       cats = n_cats, sampleFreq = 10)
pp_r0.burnin(generations = 500, tuningInterval = 100)
pp_r0.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "ROOT0_powp.log", "power", "likelihood",
                          "\t").marginal()
ps = pathSampler(out_dir + "ROOT0_powp.log", "power", "likelihood",
                 "\t").marginal()
write("ROOT0", ss, ps, filename = ml_file, append = TRUE, separator = ",")
write("\n", filename = ml_file, append = TRUE)

############################ ROOT1
tr_r1 <- readTrees("data/traits/tree16.tre")[1]
mv_r1 = VectorMoves()
q01_r1 ~ dnExponential(1.0)
q10_r1 ~ dnExponential(1.0)
mv_r1.append( mvScale(q01_r1, lambda = 0.5, weight = 2.0) )
mv_r1.append( mvScale(q10_r1, lambda = 0.5, weight = 2.0) )
Q_r1 := fnFreeK([q01_r1, q10_r1], rescaled = FALSE)
rf_r1 <- simplex(v(0.0, 1.0))
ctmc_r1 ~ dnPhyloCTMC(tree = tr_r1, Q = Q_r1, rootFrequencies = rf_r1,
```

```
                    type = "Standard", nSites = 1)
ctmc_r1.clamp(rare)
mod_r1 = model(Q_r1)
mn_r1 = VectorMonitors()
mn_r1.append( mnModel(filename = out_dir + "ROOT1_model.log", printgen = 10) )
mn_r1.append( mnJointConditionalAncestralState(tree = tr_r1, ctmc = ctmc_r1,
             type = "Standard", withTips = FALSE, withStartStates = FALSE,
             filename = out_dir + "ROOT1_states.log", printgen = 10) )
run_r1 = mcmc(mod_r1, mn_r1, mv_r1, nruns = 1)
run_r1.burnin(generations = 1000, tuningInterval = 100)
run_r1.run(generations = n_gen, tuningInterval = 0)
pp_r1 = powerPosterior(mod_r1, mv_r1, VectorMonitors(),
                       out_dir + "ROOT1_powp.log",
                       cats = n_cats, sampleFreq = 10)
pp_r1.burnin(generations = 500, tuningInterval = 100)
pp_r1.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "ROOT1_powp.log", "power", "likelihood",
                          "\t").marginal()
ps = pathSampler(out_dir + "ROOT1_powp.log", "power", "likelihood",
                 "\t").marginal()
write("ROOT1", ss, ps, filename = ml_file, append = TRUE, separator = ",")
write("\n", filename = ml_file, append = TRUE)

############################## INFO
tr_i <- readTrees("data/traits/tree16.tre")[1]
mv_i = VectorMoves()
# a priori about one gain on the whole tree, losses 10 times faster
q01_i ~ dnExponential(tr_i.treeLength())
q10_i ~ dnExponential(tr_i.treeLength() / 10.0)
mv_i.append( mvScale(q01_i, lambda = 0.5, weight = 2.0) )
mv_i.append( mvScale(q10_i, lambda = 0.5, weight = 2.0) )
Q_i := fnFreeK([q01_i, q10_i], rescaled = FALSE)
rf_i <- simplex(v(1.0, 1.0))
ctmc_i ~ dnPhyloCTMC(tree = tr_i, Q = Q_i, rootFrequencies = rf_i,
                     type = "Standard", nSites = 1)
ctmc_i.clamp(rare)
mod_i = model(Q_i)
mn_i = VectorMonitors()
mn_i.append( mnModel(filename = out_dir + "INFO_model.log", printgen = 10) )
mn_i.append( mnJointConditionalAncestralState(tree = tr_i, ctmc = ctmc_i,
             type = "Standard", withTips = FALSE, withStartStates = FALSE,
             filename = out_dir + "INFO_states.log", printgen = 10) )
run_i = mcmc(mod_i, mn_i, mv_i, nruns = 1)
run_i.burnin(generations = 1000, tuningInterval = 100)
run_i.run(generations = n_gen, tuningInterval = 0)
pp_i = powerPosterior(mod_i, mv_i, VectorMonitors(), out_dir + "INFO_powp.log",
                      cats = n_cats, sampleFreq = 10)
pp_i.burnin(generations = 500, tuningInterval = 100)
```

```
pp_i.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "INFO_powp.log", "power", "likelihood",
                          "\t").marginal()
ps = pathSampler(out_dir + "INFO_powp.log", "power", "likelihood",
                 "\t").marginal()
write("INFO", ss, ps, filename = ml_file, append = TRUE, separator = ",")
write("\n", filename = ml_file, append = TRUE)

############################ LOSS
tr_l <- readTrees("data/traits/tree16.tre")[1]
mv_l = VectorMoves()
q10_l ~ dnExponential(1.0)
mv_l.append( mvScale(q10_l, lambda = 0.5, weight = 2.0) )
Q_l := fnFreeK([0.0, q10_l], rescaled = FALSE)     # no gains
rf_l <- simplex(v(1.0, 1.0))
ctmc_l ~ dnPhyloCTMC(tree = tr_l, Q = Q_l, rootFrequencies = rf_l,
                     type = "Standard", nSites = 1)
ctmc_l.clamp(rare)
mod_l = model(Q_l)
mn_l = VectorMonitors()
mn_l.append( mnModel(filename = out_dir + "LOSS_model.log", printgen = 10) )
mn_l.append( mnJointConditionalAncestralState(tree = tr_l, ctmc = ctmc_l,
             type = "Standard", withTips = FALSE, withStartStates = FALSE,
             filename = out_dir + "LOSS_states.log", printgen = 10) )
run_l = mcmc(mod_l, mn_l, mv_l, nruns = 1)
run_l.burnin(generations = 1000, tuningInterval = 100)
run_l.run(generations = n_gen, tuningInterval = 0)
pp_l = powerPosterior(mod_l, mv_l, VectorMonitors(), out_dir + "LOSS_powp.log",
                      cats = n_cats, sampleFreq = 10)
pp_l.burnin(generations = 500, tuningInterval = 100)
pp_l.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "LOSS_powp.log", "power", "likelihood",
                          "\t").marginal()
ps = pathSampler(out_dir + "LOSS_powp.log", "power", "likelihood",
                 "\t").marginal()
write("LOSS", ss, ps, filename = ml_file, append = TRUE, separator = ",")
write("\n", filename = ml_file, append = TRUE)

write("runtime_s", time("seconds") - t_start,
      filename = out_dir + "runtime.txt", separator = "\t")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
root treatment: FLAT	lnML	NA	-7.826	NA	NA	NA	pass
root treatment: STAT	lnML	NA	-7.806	NA	NA	NA	pass
root treatment: EST	lnML	NA	-7.906	NA	NA	NA	pass

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
root treatment: ROOT0	lnML	NA	-7.756	NA	NA	NA	pass
root treatment: ROOT1	lnML	NA	-7.984	NA	NA	NA	pass
root treatment: INFO	lnML	NA	-7.287	NA	NA	NA	pass
root treatment: LOSS	lnML	NA	-9.483	NA	NA	NA	pass
root treatment: FLAT	P(root = present)	1.0	0.457	NA	NA	NA	pass
root treatment: STAT	P(root = present)	1.0	0.223	NA	NA	NA	pass
root treatment: EST	P(root = present)	1.0	0.468	NA	NA	NA	pass
root treatment: ROOT0	P(root = present)	1.0	0.000	NA	NA	NA	pass
root treatment: ROOT1	P(root = present)	1.0	1.000	NA	NA	NA	pass
root treatment: INFO	P(root = present)	1.0	0.392	NA	NA	NA	pass
root treatment: LOSS	P(root = present)	1.0	1.000	NA	NA	NA	pass
root fixed present vs absent	lnBF_ROOT1_vs_ ROOT0	NA	-0.228	NA	NA	NA	pass
root treatment: EST	rf_e[2]	NA	0.491	0	0.944	649	pass
root treatment: FLAT	q01_f	0.1	0.351	0	0.977	463	pass
root treatment: FLAT	q10_f	0.6	0.946	0	2.371	497	pass

All stepping-stone marginal likelihoods lie within 0.12 log units of the exact values, and all root probabilities within 0.03. The posterior probability that the feature was present at the root is 0.46 under the flat root, 0.47 with an estimated root frequency, 0.39 with the informative priors and 0.22 under the stationary root, and it is fixed at 1 by the loss-only model. The marginal likelihoods of the flat, stationary, estimated and fixed-root models differ by less than 0.25 log units, so the data cannot choose among them, and the Bayes factor between the two fixed roots is close to 1. The true root state (present) is never strongly supported. The loss-only model, which makes the root certain, has a marginal likelihood about 1.6 log units lower than the flat model. Two further checks follow from the theory of Section D.2.6: the flat-root probability equals $m_1/(m_0 + m_1)$ computed from the fixed-root marginal likelihoods, and the estimated root frequency barely moves from its prior (posterior mean 0.49, exact 0.48). The informative priors did not favor the true state: they make gains rare, but they also make the many losses that a present root requires expensive.

H.5.22 Pathways of acquisition over a tree sample

`22_pathway_rj_tree_sample.Rev` implements the pathway models of Section 13.7.4: two binary features coded as one four-state character (Equation (13.6)), gains only, root fixed at $(0, 0)$, analysed over the same sample of 100 trees as script 13. It samples the model structure in two ways: (a) reversible jump, in which each of the four rates is exactly 0 with prior probability 1/2; and (b) a categorical model indicator over four models (four free rates; X first, $q_{24} = 0$; Y first, $q_{34} = 0$; independent gains), with the rates shared across models. As an independent check, (c) fits the four models separately and estimates their marginal likelihoods. The data are 30 feature pairs simulated under "X first" ($q_{12} = 0.15$, $q_{13} = 0.3$, $q_{34} = 0.4$, $q_{24} = 0$). The design follows exercises from a graduate course on Bayesian reconstruction. Integrating over trees makes every generation expensive, so this is the slowest script in the group.

```
################################################################################
# 22_pathway_rj_tree_sample.Rev
#
# Model:    Order of acquisition of two binary features X and Y (for example
#           two steps of an implicational hierarchy or a grammaticalization
#           cline), coded jointly as one four-state character as in Pagel's
#           test: 0 = (0,0), 1 = (0,1), 2 = (1,0), 3 = (1,1). Changes are gains
#           only and the root is fixed at (0,0), so four rates remain:
#           q12 (Y alone), q13 (X alone), q24 (X after Y), q34 (Y after X).
#           All analyses integrate over a sample of 100 trees
#           (dnEmpiricalTree, mvEmpiricalTree).
#           (a) reversible jump over the rates: each rate is exactly 0 with
#               prior probability 1/2, or Exponential(1) otherwise; the
#               posterior probability that a rate is 0 says whether that
#               route is used at all;
#           (b) model averaging over four pathway models with a categorical
#               model indicator (uniform prior): DEP (four free rates),
#               X_FIRST (q24 = 0: (1,1) only through (1,0)), Y_FIRST
#               (q34 = 0) and IND (independent gains: q13 = q24, q12 = q34);
#               rates are shared across models, so a rate that the current
#               model does not use is drawn from its prior;
#           (c) the four models fitted separately, with marginal likelihoods
#               by stepping-stone and path sampling, as an independent check
#               of the model probabilities in (b).
#           With the root fixed at (0,0) and gains only, the root state is
#           an assumption, not an inference (Chapter 12, #asr-constraints).
#           The design (gain-only correlated model with the root fixed, RJ
#           over rates, model averaging over a tree sample) follows exercises
#           from a graduate course on Bayesian reconstruction.
# Book:     Chapter 13, "Correlated evolution of discrete characters"
#           (#pcms-pagel, Eq. pcms-pagel-dep, #pcms-pathways); Chapter 14,
#           color terms (#cultural-traits-color-number); Chapter 12
#           (#asr-constraints).
# Estimates: posterior probability that each rate is 0; posterior model
#           probabilities; ln marginal likelihoods; rates.
# Data:     data/traits/pathway.nex: 30 independent feature pairs generated
#           under X_FIRST on tree16.tre; data/traits/tree_sample.trees
# Run:      cd revbayes && rb analyses/traits/22_pathway_rj_tree_sample.Rev
#           [--args n_gen]
# Runtime:  about 15 minutes
################################################################################

seed(1122)
n_gen = 5000
if (args.size() > 1) n_gen = int(args[2])
n_cats  = 12
pp_gen  = 300
out_dir = "output/traits/22_pathway_rj_tree_sample/"
```

```
t_start = time("seconds")
ml_file = out_dir + "marginal_likelihoods.csv"
write("model,lnML_stepping_stone,lnML_path_sampling\n", filename = ml_file)

pw <- readDiscreteCharacterData("data/traits/pathway.nex")
n_sites <- pw.nchar()

# NB: every model gets its own copy of the tree sample, tree and root
# frequencies, and its own variable names (see 21_root_treatment_asr.Rev).

############################## (a) reversible jump over the four rates
ts_rj <- readTrees("data/traits/tree_sample.trees")
tt_rj = treeTrace(ts_rj, burnin = 0)
tree_rj ~ dnEmpiricalTree(tt_rj)
mv_rj = VectorMoves()
mv_rj.append( mvEmpiricalTree(tree_rj, weight = 3.0) )
q12_rj ~ dnReversibleJumpMixture(0.0, dnExponential(1.0), p = 0.5)
q13_rj ~ dnReversibleJumpMixture(0.0, dnExponential(1.0), p = 0.5)
q24_rj ~ dnReversibleJumpMixture(0.0, dnExponential(1.0), p = 0.5)
q34_rj ~ dnReversibleJumpMixture(0.0, dnExponential(1.0), p = 0.5)
# start from all routes open (a start with every rate 0 has likelihood 0)
q12_rj.setValue(0.2)
q13_rj.setValue(0.2)
q24_rj.setValue(0.2)
q34_rj.setValue(0.2)
z12 := ifelse(q12_rj == 0.0, 1, 0)       # 1 = route closed
z13 := ifelse(q13_rj == 0.0, 1, 0)
z24 := ifelse(q24_rj == 0.0, 1, 0)
z34 := ifelse(q34_rj == 0.0, 1, 0)
mv_rj.append( mvRJSwitch(q12_rj, weight = 2.0) )
mv_rj.append( mvRJSwitch(q13_rj, weight = 2.0) )
mv_rj.append( mvRJSwitch(q24_rj, weight = 2.0) )
mv_rj.append( mvRJSwitch(q34_rj, weight = 2.0) )
mv_rj.append( mvScale(q12_rj, lambda = 0.5, weight = 1.0) )
mv_rj.append( mvScale(q13_rj, lambda = 0.5, weight = 1.0) )
mv_rj.append( mvScale(q24_rj, lambda = 0.5, weight = 1.0) )
mv_rj.append( mvScale(q34_rj, lambda = 0.5, weight = 1.0) )
Q_rj := fnFreeK([ q12_rj, q13_rj, 0.0,
                  0.0,    0.0,    q24_rj,
                  0.0,    0.0,    q34_rj,
                  0.0,    0.0,    0.0 ], rescaled = FALSE)
rf_rj <- simplex(v(1.0, 0.0, 0.0, 0.0))
ctmc_rj ~ dnPhyloCTMC(tree = tree_rj, Q = Q_rj, rootFrequencies = rf_rj,
                      type = "Standard", nSites = n_sites)
ctmc_rj.clamp(pw)
mod_rj = model(Q_rj)
mn_rj = VectorMonitors()
mn_rj.append( mnModel(filename = out_dir + "RJ_model.log", printgen = 10) )
```

```
mcmc_rj = mcmc(mod_rj, mn_rj, mv_rj, nruns = 1)
mcmc_rj.burnin(generations = 1000, tuningInterval = 100)
mcmc_rj.run(generations = 2 * n_gen, tuningInterval = 0)

############################ (b) model averaging with a model indicator
ts_ma <- readTrees("data/traits/tree_sample.trees")
tt_ma = treeTrace(ts_ma, burnin = 0)
tree_ma ~ dnEmpiricalTree(tt_ma)
mv_ma = VectorMoves()
mv_ma.append( mvEmpiricalTree(tree_ma, weight = 3.0) )
g12 ~ dnExponential(1.0)
g13 ~ dnExponential(1.0)
g24 ~ dnExponential(1.0)
g34 ~ dnExponential(1.0)
mv_ma.append( mvScale(g12, lambda = 0.5, weight = 1.0) )
mv_ma.append( mvScale(g13, lambda = 0.5, weight = 1.0) )
mv_ma.append( mvScale(g24, lambda = 0.5, weight = 1.0) )
mv_ma.append( mvScale(g34, lambda = 0.5, weight = 1.0) )
Q_dep_ma := fnFreeK([ g12, g13, 0.0,  0.0, 0.0, g24,
                      0.0, 0.0, g34,  0.0, 0.0, 0.0 ], rescaled = FALSE)
Q_xf_ma  := fnFreeK([ g12, g13, 0.0,  0.0, 0.0, 0.0,
                      0.0, 0.0, g34,  0.0, 0.0, 0.0 ], rescaled = FALSE)
Q_yf_ma  := fnFreeK([ g12, g13, 0.0,  0.0, 0.0, g24,
                      0.0, 0.0, 0.0,  0.0, 0.0, 0.0 ], rescaled = FALSE)
Q_ind_ma := fnFreeK([ g12, g13, 0.0,  0.0, 0.0, g13,
                      0.0, 0.0, g12,  0.0, 0.0, 0.0 ], rescaled = FALSE)
Q_vec := v(Q_dep_ma, Q_xf_ma, Q_yf_ma, Q_ind_ma)
# 1 = DEP, 2 = X_FIRST, 3 = Y_FIRST, 4 = IND
model_indicator ~ dnCategorical(simplex(1, 1, 1, 1))
model_indicator.setValue(1)
mv_ma.append( mvRandomGeometricWalk(model_indicator, weight = 5.0,
                                    tune = FALSE) )
Q_ma := Q_vec[model_indicator]
rf_ma <- simplex(v(1.0, 0.0, 0.0, 0.0))
ctmc_ma ~ dnPhyloCTMC(tree = tree_ma, Q = Q_ma, rootFrequencies = rf_ma,
                      type = "Standard", nSites = n_sites)
ctmc_ma.clamp(pw)
mod_ma = model(Q_ma)
mn_ma = VectorMonitors()
mn_ma.append( mnModel(filename = out_dir + "MA_model.log", printgen = 10) )
mcmc_ma = mcmc(mod_ma, mn_ma, mv_ma, nruns = 1)
mcmc_ma.burnin(generations = 1000, tuningInterval = 100)
mcmc_ma.run(generations = 2 * n_gen, tuningInterval = 0)

############################ (c) the four models separately
# DEP
ts_dep <- readTrees("data/traits/tree_sample.trees")
tt_dep = treeTrace(ts_dep, burnin = 0)
```

```
tree_dep ~ dnEmpiricalTree(tt_dep)
mv_dep = VectorMoves()
mv_dep.append( mvEmpiricalTree(tree_dep, weight = 3.0) )
a12_dep ~ dnExponential(1.0)
mv_dep.append( mvScale(a12_dep, lambda = 0.5, weight = 1.0) )
a13_dep ~ dnExponential(1.0)
mv_dep.append( mvScale(a13_dep, lambda = 0.5, weight = 1.0) )
a24_dep ~ dnExponential(1.0)
mv_dep.append( mvScale(a24_dep, lambda = 0.5, weight = 1.0) )
a34_dep ~ dnExponential(1.0)
mv_dep.append( mvScale(a34_dep, lambda = 0.5, weight = 1.0) )
Q_dep := fnFreeK([ a12_dep, a13_dep, 0.0,  0.0, 0.0, a24_dep,
                 0.0, 0.0, a34_dep,  0.0, 0.0, 0.0 ], rescaled = FALSE)
rf_dep <- simplex(v(1.0, 0.0, 0.0, 0.0))
ctmc_dep ~ dnPhyloCTMC(tree = tree_dep, Q = Q_dep, rootFrequencies = rf_dep,
                    type = "Standard", nSites = n_sites)
ctmc_dep.clamp(pw)
mod_dep = model(Q_dep)
pp_dep = powerPosterior(mod_dep, mv_dep, VectorMonitors(),
                      out_dir + "DEP_powp.log", cats = n_cats,
                      sampleFreq = 10)
pp_dep.burnin(generations = 1000, tuningInterval = 100)
pp_dep.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "DEP_powp.log", "power",
                          "likelihood", "\t").marginal()
ps = pathSampler(out_dir + "DEP_powp.log", "power", "likelihood",
                 "\t").marginal()
write("DEP", ss, ps, filename = ml_file, append = TRUE,
      separator = ",")
write("\n", filename = ml_file, append = TRUE)

# X_FIRST
ts_xf <- readTrees("data/traits/tree_sample.trees")
tt_xf = treeTrace(ts_xf, burnin = 0)
tree_xf ~ dnEmpiricalTree(tt_xf)
mv_xf = VectorMoves()
mv_xf.append( mvEmpiricalTree(tree_xf, weight = 3.0) )
a12_xf ~ dnExponential(1.0)
mv_xf.append( mvScale(a12_xf, lambda = 0.5, weight = 1.0) )
a13_xf ~ dnExponential(1.0)
mv_xf.append( mvScale(a13_xf, lambda = 0.5, weight = 1.0) )
a34_xf ~ dnExponential(1.0)
mv_xf.append( mvScale(a34_xf, lambda = 0.5, weight = 1.0) )
Q_xf := fnFreeK([ a12_xf, a13_xf, 0.0,  0.0, 0.0, 0.0,
                 0.0, 0.0, a34_xf,  0.0, 0.0, 0.0 ], rescaled = FALSE)
rf_xf <- simplex(v(1.0, 0.0, 0.0, 0.0))
ctmc_xf ~ dnPhyloCTMC(tree = tree_xf, Q = Q_xf, rootFrequencies = rf_xf,
                    type = "Standard", nSites = n_sites)
```

```
ctmc_xf.clamp(pw)
mod_xf = model(Q_xf)
pp_xf = powerPosterior(mod_xf, mv_xf, VectorMonitors(),
                       out_dir + "X_FIRST_powp.log", cats = n_cats,
                       sampleFreq = 10)
pp_xf.burnin(generations = 1000, tuningInterval = 100)
pp_xf.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "X_FIRST_powp.log", "power",
                          "likelihood", "\t").marginal()
ps = pathSampler(out_dir + "X_FIRST_powp.log", "power", "likelihood",
                 "\t").marginal()
write("X_FIRST", ss, ps, filename = ml_file, append = TRUE,
      separator = ",")
write("\n", filename = ml_file, append = TRUE)

# Y_FIRST
ts_yf <- readTrees("data/traits/tree_sample.trees")
tt_yf = treeTrace(ts_yf, burnin = 0)
tree_yf ~ dnEmpiricalTree(tt_yf)
mv_yf = VectorMoves()
mv_yf.append( mvEmpiricalTree(tree_yf, weight = 3.0) )
a12_yf ~ dnExponential(1.0)
mv_yf.append( mvScale(a12_yf, lambda = 0.5, weight = 1.0) )
a13_yf ~ dnExponential(1.0)
mv_yf.append( mvScale(a13_yf, lambda = 0.5, weight = 1.0) )
a24_yf ~ dnExponential(1.0)
mv_yf.append( mvScale(a24_yf, lambda = 0.5, weight = 1.0) )
Q_yf := fnFreeK([ a12_yf, a13_yf, 0.0,  0.0, 0.0, a24_yf,
                  0.0, 0.0, 0.0,  0.0, 0.0, 0.0 ], rescaled = FALSE)
rf_yf <- simplex(v(1.0, 0.0, 0.0, 0.0))
ctmc_yf ~ dnPhyloCTMC(tree = tree_yf, Q = Q_yf, rootFrequencies = rf_yf,
                      type = "Standard", nSites = n_sites)
ctmc_yf.clamp(pw)
mod_yf = model(Q_yf)
pp_yf = powerPosterior(mod_yf, mv_yf, VectorMonitors(),
                       out_dir + "Y_FIRST_powp.log", cats = n_cats,
                       sampleFreq = 10)
pp_yf.burnin(generations = 1000, tuningInterval = 100)
pp_yf.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "Y_FIRST_powp.log", "power",
                          "likelihood", "\t").marginal()
ps = pathSampler(out_dir + "Y_FIRST_powp.log", "power", "likelihood",
                 "\t").marginal()
write("Y_FIRST", ss, ps, filename = ml_file, append = TRUE,
      separator = ",")
write("\n", filename = ml_file, append = TRUE)

# IND
```

```
ts_ind <- readTrees("data/traits/tree_sample.trees")
tt_ind = treeTrace(ts_ind, burnin = 0)
tree_ind ~ dnEmpiricalTree(tt_ind)
mv_ind = VectorMoves()
mv_ind.append( mvEmpiricalTree(tree_ind, weight = 3.0) )
a12_ind ~ dnExponential(1.0)
mv_ind.append( mvScale(a12_ind, lambda = 0.5, weight = 1.0) )
a13_ind ~ dnExponential(1.0)
mv_ind.append( mvScale(a13_ind, lambda = 0.5, weight = 1.0) )
Q_ind := fnFreeK([ a12_ind, a13_ind, 0.0,  0.0, 0.0, a13_ind,
                 0.0, 0.0, a12_ind,  0.0, 0.0, 0.0 ], rescaled = FALSE)
rf_ind <- simplex(v(1.0, 0.0, 0.0, 0.0))
ctmc_ind ~ dnPhyloCTMC(tree = tree_ind, Q = Q_ind, rootFrequencies = rf_ind,
                    type = "Standard", nSites = n_sites)
ctmc_ind.clamp(pw)
mod_ind = model(Q_ind)
pp_ind = powerPosterior(mod_ind, mv_ind, VectorMonitors(),
                      out_dir + "IND_powp.log", cats = n_cats,
                      sampleFreq = 10)
pp_ind.burnin(generations = 1000, tuningInterval = 100)
pp_ind.run(generations = pp_gen)
ss = steppingStoneSampler(out_dir + "IND_powp.log", "power",
                          "likelihood", "\t").marginal()
ps = pathSampler(out_dir + "IND_powp.log", "power", "likelihood",
                 "\t").marginal()
write("IND", ss, ps, filename = ml_file, append = TRUE,
      separator = ",")
write("\n", filename = ml_file, append = TRUE)
write("runtime_s", time("seconds") - t_start,
      filename = out_dir + "runtime.txt", separator = "\t")
q()
```

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
reversible jump over rates	P(q12 = 0)	0.00	0.000	NA	NA	NA	pass
reversible jump over rates	P(q13 = 0)	0.00	0.000	NA	NA	NA	pass
reversible jump over rates	P(q24 = 0)	1.00	0.997	NA	NA	501	pass
reversible jump over rates	P(q34 = 0)	0.00	0.000	NA	NA	NA	pass
reversible jump over rates	q12_rj	0.15	0.163	0.121	0.207	951	pass
reversible jump over rates	q13_rj	0.30	0.246	0.188	0.300	951	miss
reversible jump over rates	q24_rj	0.00	0.000	0.000	0.000	951	pass

Model	Parameter	Truth	Post. mean	HPD low	HPD high	ESS	Status
reversible jump over rates	q34_rj	0.40	0.340	0.264	0.421	951	pass
model averaging (indicator)	P(X_FIRST)	1.00	0.997	NA	NA	NA	pass
pathway model: DEP	lnML	NA	-419.990	NA	NA	NA	pass
pathway model: X_FIRST	lnML	NA	-413.830	NA	NA	NA	pass
pathway model: Y_FIRST	lnML	NA	-497.630	NA	NA	NA	pass
pathway model: IND	lnML	NA	-444.230	NA	NA	NA	pass

Reversible jump closes the absent route with posterior probability 0.997 (a Bayes factor of about 300 against a prior of 1/2) and never closes the three routes that were used. The model indicator gives "X first" a posterior probability of 0.997, in agreement with the probabilities computed from the stepping-stone marginal likelihoods of the four separately fitted models (0.998); "X first" beats the four-rate model by 6.2 log units, independent gains by 30 and "Y first" by 84. The rate q_{13} is slightly underestimated: its 95% HPD interval (0.19 to 0.30) ends just below the true value of 0.3, so that check is a miss. These are parameter-recovery results for 30 replicate feature pairs that evolved under the analysed model; Section 13.7.4 discusses why a single pair of features would be far less informative.

H.5.23 Models not implemented as stated

Every model on the list was implemented. Four were expressed through general building blocks rather than a dedicated RevBayes function, and are labelled as such in their scripts:

- the BSSVS prior on the number of active routes (independent indicators with the same expected number, script 01);
- the early-burst model (exact per-branch multipliers, script 10);
- Pagel's λ (exact per-branch multipliers, script 12);
- the hidden-rates matrix (`fnFreeK` rather than `fnHiddenStateRateMatrix`, script 07).

Spherical diffusion (Section 11.5.3), landscape-aware models and phylogenetic logistic regression were not attempted. RevBayes 1.4.1 has no spherical diffusion distribution. Phylogenetic logistic regression in the sense of Ives and Garland (2010) is a penalized-likelihood method; its Bayesian counterpart is the threshold model (script 17) with covariates in the liability.

Appendix I

Glossary

This glossary defines the technical terms used in the book, from both historical linguistics and evolutionary biology. Definitions are brief; the chapter indicated in parentheses, where given, treats the term in detail. Terms in *italics* within a definition have their own entries.

A

Acceptance probability In the Metropolis-Hastings algorithm, the probability with which a proposed new state of the Markov chain is accepted. It depends on the ratio of posterior densities and on the proposal distribution (Chapter 6).

ACCTRAN "Accelerated transformation", a rule for placing character changes on a parsimony tree as close to the root as possible, favoring reversals over parallel changes. Its counterpart is *DELTRAN* (Chapter 4).

Ancestral state reconstruction (ASR) Inference of the character states (for example, a cognate class, a word order, or a phoneme) of ancestral nodes of a tree, by parsimony, maximum likelihood or Bayesian methods (Chapter 12).

Areal feature A linguistic feature shared by languages in a geographic area because of contact rather than common descent, as in a *Sprachbund*.

Ascertainment bias A bias arising because some kinds of characters cannot be observed. In cognate data, a cognate class absent from every sampled language is never coded, so models must condition on observable patterns; this is done by an *ascertainment correction* (Chapter 7).

ASJP The Automated Similarity Judgment Program, a database of 40-item word lists in a simplified transcription for thousands of languages, and the distance-based methods developed with it (Appendix G).

Autapomorphy A derived character state unique to a single terminal taxon; in cognate data, a *singleton* cognate class. Autapomorphies carry no information about branching order.

B

Basic vocabulary Meanings thought to be universal, culturally neutral and resistant to borrowing, such as body parts, low numerals, pronouns and basic actions, used to build word lists for comparison. The *Swadesh list* is the classic example (Chapter 3).

Bayes factor The ratio of the *marginal likelihoods* of two models, used to compare them in Bayesian inference. Values above about 20 (or a log Bayes factor above about 3) are conventionally taken as strong evidence (Chapter 6).

Bayesian inference Statistical inference that combines a *prior* distribution with the *likelihood* of the data to obtain a *posterior* distribution of parameters, including trees (Chapter 6).

BEAST A family of programs (BEAST 2, BEAST 1 and BEAST X) for Bayesian inference of time-calibrated phylogenies (Appendix F).

Bipartition See *split*.

Birth-death process A model of diversification in which lineages split at rate λ (birth) and go extinct at rate μ (death); used as a *tree prior* (Chapter 6).

Bootstrap A resampling procedure that estimates the support for clades by re-inferring the tree from many pseudo-replicate datasets formed by sampling characters with replacement (Chapter 4).

Borrowing The adoption of a word or structural feature from another language through contact. Borrowing is the linguistic analogue of *horizontal gene transfer* (Chapter 8).

Branch length The length of an edge of a tree, measured either in time or in expected amount of change.

Brownian motion A model of continuous trait evolution in which changes are normally distributed with variance proportional to time, so that trait covariances between species are proportional to shared branch length (Chapter 13).

Burn-in The initial portion of an MCMC run, before the chain has reached its stationary distribution, which is discarded before summarizing the posterior.

C

Calibration Information about the age of a node (or a tip) used to convert relative branch lengths into absolute time, often expressed as a prior distribution on the node's age (Chapter 10).

Character A feature of the languages under study that can take different states, such as the presence or absence of a cognate class, a word-order type, or a sound correspondence.

Chronogram A tree whose branch lengths represent time, so that all contemporary tips are equidistant from the root. Compare *phylogram*.

Clade A group consisting of an ancestor and all of its descendants; a *monophyletic* group.

Clade credibility The posterior probability of a clade, estimated as the proportion of trees in a posterior sample that contain it.

CLDF Cross-Linguistic Data Formats, a standard for publishing linguistic data as CSV tables with JSON metadata (Appendix G).

Clock model A model of how rates of change vary across branches of a tree. A *strict clock* assumes a single rate; *relaxed clocks* allow rates to vary (Chapter 9).

Coalescent A population-genetic model that describes the ancestry of a sample of gene copies backward in time within a population; also used as a tree prior.

Cognate A word that descends from the same ancestral word as a word in another related language, for example English *two*, German *zwei* and Latin *duo*.

Cognate class (cognate set) The set of words, across languages, that are cognate with each other for a given meaning.

Comparative method The method of historical linguistics that establishes genetic relationship and reconstructs proto-languages by identifying systematic sound correspondences among cognates.

Compatibility A criterion for tree inference that seeks the tree on which the largest number (or weight) of characters can evolve without homoplasy (Chapter 4).

Consensus tree A tree that summarizes a set of trees, such as a *strict consensus* (clades present in all trees) or *majority-rule consensus* (clades present in more than half) (Chapter 2).

Consistency index (CI) A measure of homoplasy on a parsimony tree: the minimum possible number of changes divided by the observed number. A value of 1 indicates no homoplasy.

Continuous-time Markov chain (CTMC) A stochastic process in which a character moves among discrete states at rates given by a *rate matrix*; the basis of most substitution models (Chapter 5).

Convergence In MCMC, the state in which a chain is sampling from the target posterior distribution; assessed with diagnostics such as *ESS*, trace plots and comparisons between independent runs.

Covarion model A substitution model in which each character switches between "on" (evolving) and "off" (invariant) states along the tree, so that rates vary through time as well as among characters (Chapter 7).

Credible interval An interval containing a specified proportion (often 95%) of the posterior probability of a parameter. See *HPD interval*.

D

Delta score A measure of how far a distance matrix departs from tree-likeness, computed over quartets of taxa. Low values indicate tree-like data (Chapter 8).

DELTRAN "Delayed transformation", a rule for placing character changes on a parsimony tree as far from the root as possible, favoring parallel changes over reversals.

DensiTree A visualization that draws all trees of a posterior sample superimposed, so that uncertainty in topology and node heights is visible.

Dialect continuum A chain of geographically adjacent varieties in which neighbors are mutually intelligible but the ends are not, typically poorly described by a tree.

Discrete phylogeography Phylogeographic inference that treats location as a discrete character evolving on the tree under a CTMC (Chapter 11).

Distance method A tree-building method that works from a matrix of pairwise distances between taxa, such as *UPGMA* or *neighbor-joining* (Chapter 4).

Diversification rate The net rate at which lineages multiply, the speciation (splitting) rate minus the extinction rate.

Dollo's law The principle that a complex character, once lost, cannot be regained. In linguistics, a cognate class is born only once. See *stochastic Dollo model*.

E

Edge A branch of a tree.

Effective sample size (ESS) The number of independent samples that an autocorrelated MCMC sample is equivalent to; values above about 200 are commonly required.

Equilibrium frequencies See *stationary distribution*.

F

Felsenstein's pruning algorithm The dynamic-programming algorithm that computes the likelihood of a tree efficiently by working from the tips toward the root (Chapter 5).

Fossilized birth-death (FBD) process A tree prior that includes the sampling of extinct lineages (fossils) through time, allowing them to be placed in the tree as tips or *sampled ancestors* (Chapter 10).

G

Galton's problem The problem that cross-cultural or cross-linguistic observations are not independent because of shared ancestry and contact, named after Francis Galton's objection to an early comparative study (Chapter 13).

Gamma rate heterogeneity Modeling variation in rates among characters by drawing each character's rate from a gamma distribution, usually approximated with a few discrete categories (Chapter 5).

Glottochronology The method, developed by Swadesh and Lees, of estimating the time since two languages separated from the proportion of shared basic vocabulary, assuming a constant rate of replacement (Chapter 1).

Glottocode The stable identifier assigned to every languoid in Glottolog, used to link linguistic datasets.

Glottolog A comprehensive catalogue and classification of the world's languages, dialects and families (Appendix G).

H

Highest posterior density (HPD) interval The shortest interval containing a specified proportion of the posterior probability; commonly reported for node ages.

Homeland The geographic area in which the ancestral language (proto-language) of a family was spoken (Chapter 11).

Homology Similarity due to common ancestry. Cognacy is the linguistic counterpart of homology.

Homoplasy Similarity not due to common ancestry, arising from parallel change, convergence or reversal; in language data, also from undetected borrowing or chance resemblance.

Horizontal transmission Transfer of traits between lineages other than by descent; in biology, horizontal gene transfer or hybridization; in linguistics, borrowing.

I

Incomplete lineage sorting Persistence of ancestral variation through several speciation events, so that gene trees differ from the species tree. A linguistic analogue is the persistence of synonyms or variants across language splits.

Independent contrasts Felsenstein's method of transforming trait values on a tree into independent differences for regression and correlation analysis (Chapter 13).

Ingroup The set of taxa under study, as opposed to an *outgroup*.

Innovation A change shared by a group of languages that defines a subgroup, in contrast to a *retention*. Shared innovations are the basis of subgrouping in the comparative method.

J

Joint reconstruction Ancestral state reconstruction that finds the most probable combination of states at all nodes simultaneously, in contrast to *marginal reconstruction*.

K

Kendall-Colijn metric A tree distance that compares rooted trees by the depths of the most recent common ancestors of pairs of tips, optionally weighting topology and branch lengths (Chapter 2).

L

Languoid In Glottolog, any language, dialect or family, treated as a unit of classification.

Lateral transfer See *horizontal transmission*.

Levenshtein distance The minimum number of insertions, deletions and substitutions needed to transform one string into another; normalized versions are used to compare word forms (Chapter 4).

Lexicostatistics The quantitative comparison of languages by the proportions of cognates they share in basic vocabulary lists (Chapter 1).

Likelihood The probability of the observed data given a model and its parameter values, viewed as a function of the parameters (Chapter 5).

Lineage-specific Describing a pattern or dependency that holds within one family or lineage but not universally.

Loanword A word adopted from another language. See *borrowing*.

Long-branch attraction The tendency of some methods, notably parsimony, to group long branches together erroneously because of convergent changes (Chapter 4).

M

Majority-rule consensus A consensus tree containing all clades found in more than 50% of a set of trees.

Marginal likelihood The probability of the data under a model, averaged over all parameter values weighted by their prior; the quantity compared in *Bayes factors*. It is estimated by methods such as *path sampling*, *stepping-stone sampling* or *nested sampling* (Chapter 6).

Marginal reconstruction Ancestral state reconstruction that gives the probability of each state at each node separately, integrating over states at other nodes.

Markov chain Monte Carlo (MCMC) A family of algorithms that sample from a probability distribution by constructing a Markov chain whose stationary distribution is that distribution (Chapter 6).

Maximum clade credibility (MCC) tree The tree in a posterior sample that maximizes the product of the posterior probabilities of its clades; a common summary tree.

Maximum likelihood (ML) The method of choosing parameter values (including the tree) that maximize the *likelihood* (Chapter 5).

Meaning slot A meaning (concept) in a word list, considered as a multistate character whose states are the cognate classes used to express it (Chapter 7).

Metropolis-coupled MCMC (MC3) An MCMC variant that runs several "heated" chains in parallel and swaps states among them, to help the cold chain move between peaks of the posterior.

Midpoint rooting Rooting an unrooted tree at the midpoint of the longest path between two tips.

Missing data Cells of a data matrix for which the state is unknown, coded as "?". In cognate data, a meaning with no recorded word in a language.

Mk model A model for discrete characters with k states in which all changes occur at equal rates; with correction for unobserved invariant characters it is called Mkv (Chapter 7).

Model adequacy Whether a model can reproduce key features of the observed data, assessed for example by *posterior predictive simulation* (Chapter 15).

Monophyletic Describing a group that contains an ancestor and all of its descendants. See *clade*.

Multistate character A character with more than two possible states.

N

Neighbor-joining (NJ) A distance method that builds an unrooted tree by successively joining pairs of taxa that minimize the total tree length (Chapter 4).

NeighborNet A method that builds a planar split network from a distance matrix, displaying conflicting signal as boxes (Chapter 8).

Nested sampling A method for estimating marginal likelihoods, together with an estimate of their uncertainty, by sampling from progressively constrained regions of the prior.

Newick format A text format for trees using nested parentheses, for example `((A,B),C);` (Chapter 2).

NEXUS format A block-structured file format for phylogenetic data, trees and commands, used by many programs (Chapter 2).

Node A point in a tree: an internal node represents an ancestor, a terminal node (tip, leaf) an observed taxon.

Node density effect The correlation between the number of nodes on a root-to-tip path and the amount of change along it, used as evidence of punctuational change (Chapter 9).

O

Ornstein-Uhlenbeck (OU) model A model of continuous trait evolution in which the trait is pulled toward an optimum while also drifting randomly (Chapter 13).

Outgroup A taxon known to lie outside the group of interest, used to root the tree.

P

Paraphyletic Describing a group that contains an ancestor and some, but not all, of its descendants.

Parsimony The criterion that prefers the tree requiring the fewest character changes (Chapter 4).

Partial cognate A word that is cognate with a word in another language in only some of its morphemes, as in compounds (Chapter 3).

Partitioning Dividing the characters into subsets that evolve under separate models or parameters.

Path sampling A method for estimating marginal likelihoods by sampling along a path between the prior and the posterior. Compare *stepping-stone sampling*.

PGLS Phylogenetic generalized least squares, regression that models residual covariance among taxa according to the tree (Chapter 13).

Phylogenetic comparative methods (PCMs) Statistical methods that account for shared ancestry when analyzing traits across taxa (Chapter 13).

Phylogenetic network A graph that represents evolutionary history with reticulation. *Explicit networks* represent specific events such as hybridization or borrowing, while *implicit (split) networks* display conflicting signal (Chapter 8).

Phylogenetic signal The tendency of related taxa to resemble each other more than random taxa, measured for example by Pagel's λ or Blomberg's K (Chapter 13).

Phylogeography The study of the geographic history of lineages using phylogenies (Chapter 11).

Phylogram A tree whose branch lengths represent amounts of change. Compare *chronogram*.

Polytomy A node with more than two descendant branches. It may be "hard" (simultaneous splitting) or "soft" (unresolved).

Posterior distribution The probability distribution of parameters given the data, obtained by Bayes' theorem.

Posterior predictive simulation Simulating new datasets from parameters drawn from the posterior and comparing them with the observed data, to assess model adequacy.

Prior distribution The probability distribution assigned to parameters before observing the data.

Proto-language The reconstructed common ancestor of a language family, such as Proto-Indo-European.

Pseudo-Dollo model A model for cognate data that approximates the stochastic Dollo assumption (each cognate class born once) within a CTMC framework, often combined with a covarion process (Chapter 7).

Punctuational change Change concentrated at speciation events or language splits rather than spread evenly over time (Chapter 9).

Q

Q-matrix (rate matrix) The matrix $\mathbf{Q}$ of instantaneous rates of change between states in a CTMC. Transition probabilities over time t are given by $\mathbf{P}(t) = e^{\mathbf{Q}t}$.

Quartet A set of four taxa; the smallest unit for which unrooted tree topologies differ, used in quartet-based distances and delta scores.

R

Rate heterogeneity Variation in rates of change among characters, among lineages, or through time.

Relaxed clock A clock model that allows rates of change to vary among branches, either independently (*uncorrelated*) or with rates inherited from parent branches (*autocorrelated*) (Chapter 9).

Restriction-site model A binary substitution model originally developed for restriction-site data and used for cognate presence/absence data in MrBayes and RevBayes.

Retention A feature inherited unchanged from the proto-language; shared retentions do not define subgroups.

Retention index (RI) A measure of the fit of characters to a parsimony tree that, unlike the consistency index, is not affected by autapomorphies.

Reticulation Network-like evolution in which lineages exchange material, such as hybridization, horizontal gene transfer or borrowing.

Robinson-Foulds (RF) distance A tree distance that counts the splits present in one tree but not in the other (Chapter 2).

Root The node representing the most recent common ancestor of all taxa in a rooted tree.

Root-meaning coding A coding of lexical data in which characters represent etyma (roots) regardless of the meaning they now express (Chapter 3).

S

Sampled ancestor A sampled taxon (such as an ancient language) that lies directly on the lineage leading to other taxa, rather than at the tip of a separate branch (Chapter 10).

Singleton A character (such as a cognate class) present in only one taxon. See *autapomorphy*.

Sister group The closest relative of a clade; two clades that share an immediate common ancestor.

Sound correspondence A regular, recurring match between sounds in cognate words of related languages, such as Latin *p* corresponding to Germanic *f* (*pater*, *father*).

Split A bipartition of the taxa into two groups, defined by an edge of an unrooted tree. Trees can be described by their sets of splits.

Split network A network that displays a set of splits, some of which may be incompatible with any single tree. See *NeighborNet*.

Sprachbund A linguistic area in which languages from different families share features through prolonged contact.

Stationary distribution The long-run frequencies of states in a CTMC, denoted π.

Stepping-stone sampling A method for estimating marginal likelihoods using a series of distributions between prior and posterior.

Stochastic character mapping Simulation of complete histories of character change on a tree, conditional on the observed tip states, to estimate the number and timing of changes (Chapter 12).

Stochastic Dollo model A model for binary trait data in which each trait (cognate class) is born once on the tree and may be lost independently on any lineage (Chapter 7).

Strict clock A clock model in which the rate of change is the same on all branches.

Subgroup A set of languages within a family that share a more recent common ancestor with each other than with other members, established by shared innovations.

Substitution model A model of how character states change over time, typically a CTMC defined by a rate matrix.

Swadesh list A list of basic meanings (in versions of 100 or 200 items) compiled by Morris Swadesh for lexicostatistics and still widely used.

Synapomorphy A shared derived character state, the evidence for a clade; the biological counterpart of a shared *innovation*.

T

Taxon A unit of analysis at the tips of a tree; in linguistics, a language or dialect.

Tip dating Dating a tree using the known ages of sampled tips (for example ancient languages) rather than, or in addition to, node calibrations.
Topology The branching pattern of a tree, ignoring branch lengths.
Tree prior The prior probability distribution on tree topologies and branch lengths or node ages, such as the Yule or birth-death prior.
Tree space The set of all possible trees for a given set of taxa, with a notion of distance or neighborhood among them (Chapter 2).
Treelikeness The degree to which data are consistent with a single tree rather than a network.

U

Ultrametric tree A rooted tree in which all tips are equidistant from the root, as expected when branch lengths represent time and all tips are contemporary.
Unrooted tree A tree that specifies relationships without a root, and therefore without a direction of time.
UPGMA Unweighted Pair Group Method with Arithmetic mean, a clustering method that produces an ultrametric tree and assumes a constant rate of change (Chapter 4).

W

Wave model The model of language diversification, associated with Johannes Schmidt (*Wellentheorie*), in which innovations spread across overlapping areas of a dialect network rather than splitting lineages cleanly (Chapter 1).
Word list A list of words for a fixed set of meanings in a set of languages, the basic data of lexical phylogenetics.

Y

Yule process A pure-birth model of diversification in which lineages split at a constant rate and never go extinct; used as a tree prior.

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